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Theta Kernels, Weil Positivity, and the Mathematical Theory Atlas

Complete research compilation

Foundations, full checkpoint calculations,
energy constructions, and 301 atlas entries

Updated through Checkpoint 20

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This edition extends the 78-page consolidated manuscript through Checkpoint 8. It includes the complete subsequent mathematical notes, the theory atlas with mathematical representations, the earlier supporting derivations, and the continuing research record. Proved local statements, conditional implications, and unresolved global estimates retain their stated scope.

Current mathematical position

The primary objective is unconditional positivity of the complete Weil form,

$$\mathfrak{W}(q, q) \geq 0, \quad q \in C_c^\infty(\mathbb{R}),$$

with the original Suzuki metric and every arithmetic, polar, archimedean, and boundary contribution retained. The work collected here establishes representation, transfer, domain, local positivity, and comparison results. The global signed estimate remains open.

The strongest currently checked support result in this edition is the Checkpoint 20 inequality

$$Q_a^-(q) > \frac{5278}{10^6} \|q\|_2^2 > \frac{1}{200} \|q\|_2^2, \quad 0 < a \leq \frac{3}{8},$$

for nonzero odd inputs in the closed form domain. The complete first-prime reflection is retained. Its proof includes the exact rational concentration certificate and every stated approximation, archimedean, and polar payment; the executable certificate and verification record accompany the source archive. This is a local support theorem. The atlas and CP19 also record an older half-unit certificate whose executable verification was not authenticated in the current run; that historical record is preserved with its qualification.

The original Suzuki defect criterion remains a mathematically motivated sufficient route:

$$\mathcal{E}_N = \|G_N^{-1/2} Z_N G_N^{-1/2}\|_F^2, \quad \liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1.$$

Its threshold comes from the proved negative-witness transfer. A negative local witness yields a nonzero compact smooth test and a permanent floor

$$\mathcal{E}_N \geq \left(1 + \frac{\delta}{6\mathcal{S}(q, q)}\right)^2 > 1 \quad (N \geq N_0), \quad \delta > 0.$$

Thus an independently established unit ceiling on an unbounded subsequence would contradict that witness. The ceiling is not proved in this edition. Direct all-support Weil positivity is another sufficient route to the same intended conclusion.

The most recent complete finite-family energy identity is

$$\frac{1}{\tau_R} K_a(I - K_a) = \eta_p(a) W_a^* W_a + \mathbf{N}_a + \sum_{n \geq 2} b_n \mathbf{P}_{a, \log n}, \quad \eta_p(a) > 0, \quad \mathbf{N}_a \succeq 0.$$

The paired-prime matrices include their signed exterior terms. Checkpoint 20 supplies exact continuous covariance identities, joint reflection-path spectra, prime-square savings, and graph-residual payments. The all-support comparison needed to absorb the full signed prime contribution remains unresolved.

How this edition is organized

Part I: Foundations

The original consolidated manuscript is retained, including Sections 1–20, Appendices A–G, its theorem labels, and bibliography. Its equivalences and conditional implications retain their hypotheses.

Part II: Checkpoints 9–20

Twelve complete mathematical continuations carry the argument from the direct defect witness through the global Suzuki domain, first-null analysis, odd support certificates, boundary evolution, recovered energy machinery, and the Checkpoint 20 covariance/reflection certificate.

Part III: Mathematical atlas

All 301 indexed entries are included: energy; Möbius modifiers, storage, recurrence and compensation; prime DNA and colors; spirals, Fibonacci, angles and phase; light cones and causal/operator methods; quadratic, infinity and cross-plane interfaces; theta, amplitude and Weil–Suzuki constructions. Each entry keeps its purpose, mathematical representation or recorded content, present implication, and sources. Related entries describe different uses of the same object; the count is not a claim of 301 independent theorems.

Part IV: Earlier mathematical supplements

Ten complete notes preserve older calculations that were absent from or compressed in the previous PDF. These include the compact color transfers, balanced prime–pole matching, mixed-energy compensation, causal two-plane feedback, spiral representations, and the original local-amplitude derivation.

Part V: Research record

The intervening research directives and the continuing ledger are included. A directive records a requested calculation, not a proved result. Historical status statements describe the time at which their note was written.

Equation navigation. Original Part I numbering is retained. Every later numbered formula has a source prefix: for example, (CP20:2.7), (H6:12), or (D15:1). Within a source unit, a prose reference such as “(2.7)” refers to that unit’s original local number. This prevents identical local tags in different notes from being mistaken for the same formula. PDF bookmarks provide deeper navigation than the compact contents.

Editorial scope. The compilation preserves complete source arguments; it is not a new independent audit of every historical claim. The atlas uses “proved” and “exact” for the derivation recorded in its named source. Later corrections are stated where recovered. The source archive retains the original Markdown, the earlier manuscript, and a manifest so presentation changes can be checked against source bytes. Two missing backslashes in CP19 equation (4.7) are repaired consistently with the identical formula (3.32). Opaque application identifiers are removed from the printed edition, while human-readable filenames and source versions remain.

Chronology and notation that matter

Development	Current reading
CP9–11	The negative-witness floor, raw extraction cap, and Suzuki-relative extraction bound concern different quantities and metrics. The full inverse Gram metric remains in the defect.
CP12–14	CP12 proves fixed-support injectivity for the null-domain passage. CP14 later gives the logarithmic-square graph bound, finite positive Suzuki norm of the null vector, and strong convergence of the selected same-parity core sequence. The earlier domain uncertainty is superseded; the signed estimate is still open.
CP15–16	CP15’s initial interval is $a \leq 1/8$, with $H_a > \frac{1}{6}I$ and $A_a^- > \frac{253}{1536}I$. CP16 reaches $a \leq 699/2000 + e^{-2500}$ with floor $1/5000$.
CP17–18	CP17’s scaled regularized matrix has a Dini estimate. CP18 fixes the regularizer in physical coordinates and proves monotonicity, Lipschitz continuity, and an almost-everywhere boundary Gram derivative for that physical matrix. These are distinct coordinate constructions.
CP19–20	Complete energy maps and response certificates feed the covariance/reflection comparisons. CP20 enlarges the certified odd interval to $3/8$. No all-support positivity conclusion is asserted.

Letters such as F, G, H, A, B, K are local notation. The entire function $F(z) = \xi(1/2 + z)$, a source map F_a , a pole moment, and a regulated product are different objects. The scalar extraction constant K_0 is distinct from the modified Bessel function $K_0(z)$ in CP14. The symbol $D_v(s)$ for dilation energy is distinct from the regulator allowance D_v^2 defined later in that note. Each definition controls its subsequent formulas.

The Fourier and sesquilinear conventions also remain local. The original theta Fourier convention, the later plus-sign transform, and the sign $V = -L_0$ are recorded where used. CP11 distinguishes feature-indexed Gram entries from the standard coefficient matrix. CP19 explicitly switches to conjugate-linear-first form notation. No transpose, conjugation, parity-channel swap, or metric replacement is made silently in this edition.

For the square map $w = z^2$, the two sheets and both parity measures remain part of the transformation. A “plus” coefficient channel is not automatically an even ungauged function channel. Likewise, a positive scalar arithmetic energy, a Hilbert norm, an operator lower bound, and a squared Hilbert–Schmidt norm are separate quantities until an exact map or a sufficient comparison connects them.

A mathematical guide to the atlas

The atlas records both the constructions and what they were designed to measure. The following equations provide entry points; their full definitions, qualifications, and derivations appear in the atlas and the checkpoint/supplement units.

Factor energy and arithmetic source. For $n = \prod_p p^{a_p} > 1$,

$$L(n) = \sum_p a_p \left(\frac{\log p}{\log n} \right)^2, \quad C(n) = 1 - \sum_p \left(\frac{a_p \log p}{\log n} \right)^2.$$

The repeated-prime part, $L(n)$, and $C(n)$ sum to one. The prime-power source is recovered exactly by

$$\Lambda(n) = L(n) \log n \mathbf{1}_{\{C(n)=0\}}.$$

These scalar energies explain how factor information reaches the arithmetic weights; the later Weil comparison retains the full signed operator.

Möbius history and Green storage. With $M(t) = \sum_{n \leq t} \mu(n)$, the complete finite history satisfies

$$\sum_{m,n \leq N} \frac{\mu(m)\mu(n)}{\max(m,n)} = \int_1^N \frac{M(t)^2}{t^2} dt + \frac{M(N)^2}{N}.$$

The endpoint term is part of the norm. The storage, color, Fibonacci-packet, selected/residual, and compensation entries track how cancellation is retained or paid when a history is regrouped.

Causal arithmetic and discrepancy energy. Writing $e(x) = \psi(x) - x$ and $F_{\text{disc}}(u) = e^{-u/2} e(e^u)$ gives

$$(\partial_u + \tfrac{1}{2})F_{\text{disc}} = \sum_{p,k \geq 1} \frac{\log p}{p^{k/2}} \delta_{k \log p} - e^{u/2} du.$$

Its block energy separates into centered fluctuation and inherited mean:

$$B(X, Y) = \frac{1}{X^2} \int_X^Y |e - \bar{e}|^2 dx + \frac{Y - X}{X^2} |\bar{e}|^2.$$

The light-cone, recurrence, gap-reserve, and transport constructions organize this forced history and its propagation. Dropping the mean channel changes the energy.

Quadratic and infinity representations. The finite defect tensor lift retains its inverse metric:

$$\mathcal{E} = \text{operatorname{vec}}(Z)^* (\overline{G^{-1}} \otimes G^{-1}) \text{vec}(Z).$$

The quadratic map separates parity channels, while the infinity constructions control explicit tails, boundary contributions, and introduced poles. A change of representation preserves the complete object only with those terms retained. Spirals and angles provide arithmetic coordinates and observation maps; their source formulas and exact implications are catalogued alongside these analytic interfaces.

The present forced family. For the physical regularizer and its source map,

$$J_a = A_a^- + \tau_R F_a F_a^* \geq g_R I, \quad W_a = J_a^{-1} F_a, \quad K_a = \tau_R F_a^* W_a, \\ Q_a^-(W_a c) = \frac{1}{\tau_R} c^* K_a (I - K_a) c.$$

This family is where the recovered energy and comparison machinery is applied in CP19–20. The complete positive continuous part and the complete signed paired-prime part remain together throughout the calculation.

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Part I

Theta kernels and local-amplitude foundations

Source edition: 22 September 2026, through Checkpoint 8. The original Sections 1–20 and Appendices A–G follow with their equation and theorem labels retained. Their hypotheses remain part of each result. In particular, Theorem 14.4 uses the full spatial positivity premise (14.16); its conclusion is conditional on that premise. Later developments begin in Part II.

Original abstract. We develop a complete analytic framework for the centered Riemann function $F(z) = \xi(\frac{1}{2} + z)$. Starting from the Jacobi transformation, we construct its positive even theta density, prove the entire transform representation, and derive the full complex translation laws of its derivative Gram lift. An exact Fourier identity identifies the complete coupled theta kernel with the transverse derivative of $|F|^2$. We prove that positive definiteness of the complete coupled kernel, nonnegativity of the transverse derivative, and critical-line localization are equivalent statements. An arbitrary off-axis zero of arbitrary multiplicity forces a negative neighboring derivative and an indefinite finite kernel matrix. Additional equivalent formulations use a fixed compact integral operator, a two-point kernel, and a denominator-free second-order differential inequality. We also derive a signed moment criterion, strict positive regions from degree-two Jensen inequalities, complete autocorrelation identities, and exact theta-tail cancellation formulas. The equivalence theorem concerns the entire zero divisor and adopts no coupled-kernel positivity axiom. Positive definiteness of the transformed kernel is proved separately. The arguments use no enumeration of zero ordinates. The arithmetic section incorporates the earlier prime-indexed determining family and positive Suzuki metric. An exact completed-Weil identity and a smooth-approximation lemma connect that framework to the fixed theta operator, with the metric-identification premise stated explicitly. The supporting prime-ladder, Suzuki-metric, and accumulated-positivity derivations are included in the appendices, together with the explicit full-line limiting passages used in the spatial transfer. The present consolidation also includes the complete prime-color and Smith–Jordan square decompositions, the local reciprocal-zero criterion at a known zero, its exact compression from the spatial kernel, and the two-trace polynomial restriction that preserves every possible off-axis zero. All supporting local, primitive, density, and boundary derivations are supplied within this paper. The assembled results establish the equivalence and transfer theorems with their stated hypotheses; the nonnegativity of the complete restricted local form is not asserted as an independently proved consequence.

1 Coordinates, definitions, and principal equivalence

For $\operatorname{Re} s > 1$, set $\zeta(s) = \sum_{n \geq 1} n^{-s}$ and continue meromorphically. The completed and centered functions are

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad F(z) = \xi\left(\frac{1}{2} + z\right). \quad (1.1)$$

All removable singularities in ξ are filled. Write $z = a + it$. The original critical line corresponds to $a = 0$. The alternative spectral convention is

$$f(w) = \Xi(w) = F(iw), \quad z = iw, \quad w = x + iy \implies (a, t) = (-y, x). \quad (1.2)$$

In particular, derivatives in the two conventions satisfy $f^{(j)}(w) = i^j F^{(j)}(iw)$.

The full density, constructed in Section 2, is

$$\eta(u) = \sum_{n=1}^{\infty} \left(4\pi^2 n^4 e^{9u/2} - 6\pi n^2 e^{5u/2} \right) e^{-\pi n^2 e^{2u}}, \quad u \geq 0, \quad (1.3)$$

with its complete even continuation to $\mathbb{R}$. For $a > 0$ define

$$C_a(y) = 2 \int_{\mathbb{R}} x \sinh(2ax) \eta(x+y) \eta(x-y) dx. \quad (1.4)$$

All quantities in (1.4) use the complete series (1.3). Our Fourier convention is

$$\widehat{q}(\omega) = \int_{\mathbb{R}} q(y) e^{-i\omega y} dy, \quad q(y) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{q}(\omega) e^{i\omega y} d\omega \quad (1.5)$$

when both sides are integrable. The classical completion conventions agree with [2]. Related positive-kernel criteria for the xi function are developed in [6]; the horizontal monotonicity formulation appears in [4].

Definition 1.1 (Positive definiteness). A function $q : \mathbb{R} \rightarrow \mathbb{C}$ is positive definite if, for every $N \geq 1$, every $y_1, \dots, y_N \in \mathbb{R}$, and every $v_1, \dots, v_N \in \mathbb{C}$,

$$\sum_{j,k=1}^N \overline{v_j} v_k q(y_j - y_k) \geq 0. \quad (1.6)$$

Denote by P the statement that C_a is positive definite for every $a > 0$.

Theorem 1.2 (Principal equivalence). *For the functions defined above, the following statements are equivalent:*

- (i) every zero z of F satisfies $\operatorname{Re} z = 0$, equivalently every nontrivial zero ρ of ζ satisfies $\operatorname{Re} \rho - \frac{1}{2} = 0$;
- (ii) P ;
- (iii) $\operatorname{Re}(F'(a+it)\overline{F'(a+it)}) \geq 0$ for every $0 < a < 1/2$ and $t \in \mathbb{R}$.

Independently of these equivalent statements, $t \mapsto \operatorname{Re}(F'(a+it)\overline{F'(a+it)})$ is positive definite for every $a > 0$.

The theorem proves equivalence; none of (i)–(iii) is adopted as an axiom or asserted independently. Its final assertion is proved directly in Theorem 4.4. The equivalence proof is completed in Section 20. Sections 4 and 5 establish the central implication for one arbitrary hypothetical zero. Sections 6–8 give operator and differential versions of the same structural theorem.

For reference, the principal objects and their domains are

Object	Definition	Domain
F	$\xi(\frac{1}{2} + z)$	$z \in \mathbb{C}$
U	$ F(a + it) ^2$	$(a, t) \in \mathbb{R}^2$
G_{mn}	$F^{(m)}\overline{F^{(n)}}/(m!n!)$	$z \in \mathbb{C}$
K	$\operatorname{Re}(F'\overline{F})$	$(a, t) \in \mathbb{R}^2$
T	$FF'' - (F')^2$	$z \in \mathbb{C}$
S	$K^2 - a^2 T ^2$	$(a, t) \in \mathbb{R}^2$
$\mathcal{B}$	$(F'\overline{F} + F\overline{F'})/(z + \overline{w})$	$z, w \in \mathbb{H}$
J	$\int_0^\infty (x + y + 2r)\eta(x + r)\eta(y + r) \, dr$	$(x, y) \in \mathbb{R}^2$

Here $\mathbb{H} = \{z \in \mathbb{C} : \operatorname{Re} z > 0\}$. In the row for $\mathcal{B}$, unbarred factors are evaluated at z and barred factors at w .

2 The complete theta density and its transform

2.1 Modular construction and positivity

Put

$$\Theta(v) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 v}, \quad \psi(v) = \frac{1}{2}(\Theta(v) - 1), \quad \operatorname{Re} v > 0. \quad (2.1)$$

The Jacobi transformation is

$$\Theta(v) = v^{-1/2} \Theta(1/v), \quad (2.2)$$

with the branch positive for $v > 0$ [1]. For $|\operatorname{Im} u| < \pi/4$, define

$$g_0(u) = e^{u/2} \psi(e^{2u}), \quad L(u) = e^{u/2} \Theta(e^{2u}), \quad \eta(u) = g_0''(u) - \frac{1}{4}g_0(u). \quad (2.3)$$

The notation g_0 distinguishes this theta primitive from the logarithmic derivative $g = F'/F$ used later.

Lemma 2.1. *The function η is holomorphic on $|\operatorname{Im} u| < \pi/4$, even, and strictly positive on $\mathbb{R}$. Its real derivatives are smooth and rapidly decreasing. More precisely, for $j \geq 0$, $q > 0$, and $0 < \theta < \pi/4$, there is $M_{j,q,\theta} < \infty$ such that*

$$|\eta^{(j)}(u + iv)| \leq M_{j,q,\theta} e^{-qu^2} \quad (u \in \mathbb{R}, |v| \leq \theta). \quad (2.4)$$

Proof. Equation (2.2) gives $L(-u) = L(u)$ and

$$g_0(-u) = g_0(u) + \sinh(u/2). \quad (2.5)$$

The operator $\partial_u^2 - 1/4$ annihilates $\sinh(u/2)$ and $e^{u/2}$. Consequently $\eta(-u) = \eta(u)$ and $\eta = \frac{1}{2}(\partial_u^2 - 1/4)L$. Differentiating (2.5) at zero yields $g_0'(0) = -1/4$.

For $u \geq 0$, let $\beta_n = \pi n^2 e^{2u}$. Termwise differentiation gives (1.3), whose n th term is $e^{u/2 - \beta_n} (4\beta_n^2 - 6\beta_n) > 0$, since $\beta_n \geq \pi > 3/2$. Evenness gives positivity on the negative half-line.

On $|\operatorname{Im} u| \leq \theta$, with $\operatorname{Re} u \geq 0$,

$$\operatorname{Re}(\pi n^2 e^{2u}) \geq \pi n^2 e^{2\operatorname{Re} u} \cos(2\theta).$$

Every fixed differentiated summand is $e^{u/2 - \pi n^2 e^{2u}}$ times a fixed polynomial in $\pi n^2 e^{2u}$. Absorbing this polynomial into half the exponential gives

$$|\eta^{(j)}(u + iv)| \leq C_{j,\theta} e^{u/2} \sum_{n \geq 1} \exp(-c_\theta n^2 e^{2u}), \quad u \geq 0. \quad (2.6)$$

The series converges normally on compact subsets of the strip, and its right side decreases faster than e^{-qu^2} for every fixed q . Reflection supplies the estimate for $u \leq 0$. This also justifies all termwise derivatives used above. $\square$

2.2 Mellin calculation and endpoint terms

For $\operatorname{Re} s > 1$, absolute convergence gives

$$\pi^{-s/2} \Gamma(s/2) \zeta(s) = \int_0^\infty \psi(v) v^{s/2} \frac{dv}{v}. \quad (2.7)$$

Split at $v = 1$, apply (2.2) on $(0, 1)$, and use $v \mapsto 1/v$. The elementary part is

$$\frac{1}{2} \int_0^1 (v^{(s-1)/2} - v^{s/2}) \frac{dv}{v} = \frac{1}{s-1} - \frac{1}{s} = \frac{1}{s(s-1)}.$$

Thus

$$\xi(s) = \frac{1}{2} + \frac{s(s-1)}{2} \int_1^\infty (v^{s/2} + v^{(1-s)/2}) \psi(v) \frac{dv}{v}. \quad (2.8)$$

The integral is entire in s , so (2.8) provides the continuation and proves $\xi(s) = \xi(1-s)$ and $\xi(0) = \xi(1) = 1/2$.

Theorem 2.2 (Exact entire transform). *For every $z \in \mathbb{C}$ and $j \geq 0$,*

$$F(z) = \int_{\mathbb{R}} \eta(u) e^{zu} du, \quad F^{(j)}(z) = \int_{\mathbb{R}} u^j \eta(u) e^{zu} du. \quad (2.9)$$

In particular $F(-z) = F(z)$, $F(\bar{z}) = \overline{F(z)}$, and $F(0) > 0$.

Proof. Substitute $s = 1/2 + z$ and $v = e^{2u}$ in (2.8). Then

$$F(z) = \frac{1}{2} + 2(z^2 - \frac{1}{4}) \int_0^\infty g_0(u) \cosh(zu) du. \quad (2.10)$$

Two integrations by parts give

$$\int_0^\infty g_0''(u) \cosh(zu) du = -g_0'(0) + z^2 \int_0^\infty g_0(u) \cosh(zu) du, \quad (2.11)$$

$$2 \int_0^\infty \eta(u) \cosh(zu) du = \frac{1}{2} + 2(z^2 - \frac{1}{4}) \int_0^\infty g_0(u) \cosh(zu) du. \quad (2.12)$$

All boundary terms at positive infinity vanish by the theta decay; the boundary at zero uses $g_0'(0) = -1/4$. Evenness now proves the first identity in (2.9). On every compact subset of $\mathbb{C}$, Lemma 2.1 gives an integrable majorant for each differentiated integrand, proving the remaining assertions. $\square$

2.3 Growth, strip decay, and zero confinement

Proposition 2.3. *F has order one. For $A < \infty$, $j \geq 0$, and $0 < \theta < \pi/4$,*

$$|F^{(j)}(a + it)| \leq C_{A,j,\theta} e^{-\theta|t|} \quad (|a| \leq A, t \in \mathbb{R}). \quad (2.13)$$

Proof. Positivity gives $|F(z)| \leq F(r)$ for $|z| \leq r$, and equality holds at $z = r$. From (1.1), Stirling's formula, and $\zeta(r + 1/2) \rightarrow 1$,

$$\log F(r) = \frac{r}{2} \log r - \frac{r}{2} (1 + \log(2\pi)) + O(\log r). \quad (2.14)$$

Therefore $\log \log \max_{|z| \leq r} |F(z)| / \log r \rightarrow 1$. For (2.13), shift the integration contour in (2.9) to $u + i\theta \operatorname{sgn}(t)$. The factor $e^{(a+it)\theta \operatorname{sgn}(t)}$ has modulus $e^{-\theta|t|}$. Estimate (2.4) dominates the horizontal integrals uniformly for $|a| \leq A$ and makes the closing vertical integrals vanish as their real endpoints tend to infinity. The polynomial factor u^j is absorbed by the same estimate. $\square$

Proposition 2.4. *The zeros of F are precisely the centered nontrivial zeros of ζ . Every such zero ρ_0 satisfies $|\operatorname{Re} \rho_0| < 1/2$.*

Proof. The absolutely convergent Euler product excludes zeros of ζ in $\operatorname{Re} s > 1$. For $\sigma > 1$ its logarithm gives, for real t ,

$$3 \log \zeta(\sigma) + 4 \log |\zeta(\sigma + it)| + \log |\zeta(\sigma + 2it)| \quad (2.15)$$

$$= \sum_p \sum_{k \geq 1} \frac{p^{-k\sigma}}{k} (3 + 4 \cos(kt \log p) + \cos(2kt \log p)) \geq 0, \quad (2.16)$$

since the bracket equals $2(1 + \cos(kt \log p))^2$. If $t \neq 0$ and $\zeta(1 + it)$ had multiplicity $m \geq 1$, the corresponding product of absolute values would be $O((\sigma - 1)^{4m-3}) \rightarrow 0$, whereas the displayed inequality makes it at least one. Here $\zeta(\sigma + 2it)$ stays bounded and the pole at one has order one. Thus ζ has no zeros on $\operatorname{Re} s = 1$. The gamma factor has no zeros. The values $\xi(0) = \xi(1) = 1/2$ and the functional equation transfer the zero-free assertion to $\operatorname{Re} s \leq 0$. In $0 < \operatorname{Re} s < 1$ the other factors in (1.1) are finite and nonzero, so the zeros agree with those of ζ , including multiplicities. $\square$

3 Entire derivative lifts and their positive matrices

3.1 Sequence space and exact translations

Let

$$\mathcal{E} = \left\{ c = (c_j)_{j \geq 0} : p_R(c) = \sum_{j \geq 0} |c_j| R^j < \infty \text{ for each } R > 0 \right\}, \quad c_j(z) = \frac{F^{(j)}(z)}{j!}. \quad (3.1)$$

The map $c \mapsto \sum c_j w^j$ identifies $\mathcal{E}$ with the entire functions with locally uniform convergence: on radius R the supremum is at most $p_R(c)$, and Cauchy's estimate at radius $2R$ gives the converse bound $p_R(c) \leq 2 \max_{|w| \leq 2R} |\sum c_j w^j|$.

Proposition 3.1 (Complete translation law). *For $h \in \mathbb{C}$,*

$$(T_h c)_m = \sum_{k \geq m} \binom{k}{m} h^{k-m} c_k, \quad c(z+h) = T_h c(z), \quad T_h T_k = T_{h+k}, \quad T_h^{-1} = T_{-h}. \quad (3.2)$$

Each T_h is continuous on $\mathcal{E}$, with $p_R(T_h c) \leq p_{R+|h|}(c)$.

Proof. Expand $F(z+h+w) = \sum_k c_k(z)(h+w)^k$ and collect the coefficient of w^m . Absolute convergence follows from

$$\sum_m R^m \sum_{k \geq m} \binom{k}{m} |h|^{k-m} |c_k| = \sum_k |c_k| (R + |h|)^k.$$

The group identities follow by translating the entire functions represented by the sequences. $\square$

3.2 The full-plane positive lift

Define $G_{mn}(z) = c_m(z) \overline{c_n(z)}$. For every finitely supported v ,

$$v^* G(z) v = \left| \sum_m \overline{v_m} c_m(z) \right|^2 \geq 0, \quad z \in \mathbb{C}. \quad (3.3)$$

The complete matrix has rank one because $F \not\equiv 0$; a finite principal block can have rank zero or one. Since $c \in \mathcal{E}$, each weighted matrix $(R^{m+n} G_{mn})$ is a positive trace-class operator on ℓ^2 , with trace $\sum_m R^{2m} |c_m|^2$. The map $z \mapsto c(z)$ is holomorphic with values in $\mathcal{E}$; $G = cc^*$ is a real-analytic Hermitian matrix field defined on $\mathbb{C}$. Use the Wirtinger derivatives $\partial_z = (\partial_a - i\partial_t)/2$ and $\partial_{\bar{z}} = (\partial_a + i\partial_t)/2$.

Writing $U(z) = |F(z)|^2$, the entire Taylor series gives

$$U(z+w) = \sum_{m,n \geq 0} G_{mn}(z) w^m \overline{w}^n, \quad G_{mn}(z) = \frac{\partial_z^m \partial_{\bar{z}}^n U(z)}{m!n!}. \quad (3.4)$$

Thus the complete squared-amplitude germ determines the matrix and the matrix determines the analytic germ up to one constant phase. Explicitly, if $G_{rr} > 0$, then $c_m^\circ = G_{mr}/\sqrt{G_{rr}}$ is a representative.

The differential and translation identities are

$$\partial_a G_{mn} = (m+1)G_{m+1,n} + (n+1)G_{m,n+1}, \quad (3.5)$$

$$\partial_t G_{mn} = i(m+1)G_{m+1,n} - i(n+1)G_{m,n+1}, \quad (3.6)$$

$$G(z+h) = T_h G(z) T_h^*. \quad (3.7)$$

The last formula denotes the absolutely convergent coefficientwise congruence furnished by (3.2); it does not require that T_h be bounded on an unweighted ℓ^2 space.

Proposition 3.2 (Recovery of the base point). *For the actual F , the map $z \mapsto G(z)$ is injective. Its differential has trivial real kernel.*

Proof. The nonconstant even entire function F of order one has a zero: otherwise Hadamard factorization would give $F(z) = e^{A+Bz}$, and evenness would give $B = 0$. If $G(z_1) = G(z_2)$, rank-one reconstruction and Taylor convergence give $F(v+d) = \lambda F(v)$ for all v , where $d = z_2 - z_1$ and $|\lambda| = 1$. The translates of any zero by kd , $k \in \mathbb{Z}$, stay zeros. Proposition 2.4 forces $\operatorname{Re} d = 0$. Write $d = i\tau$. If $\tau \neq 0$, then $|F(ki\tau)| = F(0)$, contradicting (2.13). Hence $d = 0$.

For the differential, let $d \neq 0$ be a real tangent represented by a complex number and let $v_j = d(j+1)c_{j+1}$. Cauchy estimates give $v \in \mathcal{E} \subset \ell^2$. If $vc^* + cv^* = 0$, projection onto the orthogonal complement of c first gives $v = \alpha c$; the same identity then gives $\alpha = i\tau$ for real τ . Taylor reconstruction implies $dF' = i\tau F$ as entire functions. Its exponential solutions are incompatible with evenness and nonconstancy. Thus $dG(d) \neq 0$. $\square$

3.3 Amplitude and phase identities

From (3.5),

$$U_a = 2 \operatorname{Re} G_{10}, \quad U_t = -2 \operatorname{Im} G_{10}, \quad \Delta U = 4G_{11}, \quad U\Delta U = |\nabla U|^2. \quad (3.8)$$

Away from zeros an analytic logarithm gives

$$\partial_a \log |F| = \partial_t \arg F = \operatorname{Re} \frac{F'}{F}, \quad \partial_t \log |F| = -\partial_a \arg F = -\operatorname{Im} \frac{F'}{F}. \quad (3.9)$$

At a zero of multiplicity m , $F(z_0 + h) = h^m h_0(h)$ with $h_0(0) \neq 0$. Consequently, in distributions,

$$\Delta \log U = 4\pi \sum_{F(z_0)=0} m_{z_0} \delta_{z_0}. \quad (3.10)$$

This follows locally from $\Delta \log |h| = 2\pi \delta_0$ and the harmonicity of $\log |h_0|$.

The contraction governing the derivative has the exact finite form

$$K = \operatorname{Re} G_{10} = \operatorname{tr} \left[\begin{pmatrix} 0 & 1/2 \\ 1/2 & 0 \end{pmatrix} \begin{pmatrix} |F|^2 & F\overline{F'} \\ F'\overline{F} & |F'|^2 \end{pmatrix} \right]. \quad (3.11)$$

The coefficient matrix in (3.11) has eigenvalues $1/2, -1/2$. Thus positive semidefiniteness of the Gram lift in (3.3) does not, by the positivity of a matrix contraction, imply $K \geq 0$: the contracting matrix is indefinite. For any $\ell > 0$ the same contraction is

$$K = \frac{|F' + \ell F|^2 - |F' - \ell F|^2}{4\ell}. \quad (3.12)$$

4 Complete kernel identities and Fourier positivity

4.1 Autocorrelation, derivative, and increment

For real a let $q_a(u) = e^{au}\eta(u)$ and

$$A_a(y) = \int_{\mathbb{R}} q_a(x+y)q_a(x-y) dx = \int_{\mathbb{R}} e^{2ax}\eta(x+y)\eta(x-y) dx. \quad (4.1)$$

It is a positive-definite autocorrelation: with $v = x - y$, $A_a(y) = \int q_a(v+2y)q_a(v) dv$, and finite translation forms are integrals of squared absolute values. Moreover

$$\partial_a A_a(y) = 2 \int_{\mathbb{R}} x e^{2ax} \eta(x+y)\eta(x-y) dx = C_a(y), \quad (4.2)$$

$$\widehat{A}_a(2t) = \frac{1}{2} |F(a+it)|^2. \quad (4.3)$$

In (4.2), the product of densities is even in x , so the $x \cosh(2ax)$ part integrates to zero. The change of variables $(u, v) = (x + y, x - y)$, with $du dv = 2 dx dy$, proves (4.3). For $0 \leq a < b$ set

$$D_{a,b} = A_b - A_a = \int_a^b C_r dr, \quad H_{a,b}(t) = |F(b + it)|^2 - |F(a + it)|^2 = 2\widehat{D}_{a,b}(2t). \quad (4.4)$$

The spatial formula is

$$D_{a,b}(y) = \int_{\mathbb{R}} [\cosh(2bx) - \cosh(2ax)] \eta(x + y) \eta(x - y) dx. \quad (4.5)$$

Lemma 4.1. *For each $a > 0$, C_a is real, even, strictly positive on $\mathbb{R}$, and belongs to the Schwartz class. Its derivatives with respect to a are locally uniformly Schwartz. All integrals above can be differentiated and reordered.*

Proof. The factor $x \sinh(2ax)$ is even and nonnegative, and is positive off $x = 0$. Positivity of η gives strict positivity of $C_a(y)$. Symmetry under $y \mapsto -y$ gives evenness. For each $q > 0$, Lemma 2.1 bounds every required derivative of the density by a constant times e^{-qu^2} . Since $(x + y)^2 + (x - y)^2 = 2x^2 + 2y^2$, each differentiated integrand is bounded by

$$C(1 + |x|)^M e^{2A|x|} e^{-2q(x^2 + y^2)}$$

when a ranges over a compact interval $[0, A]$. This bound is integrable in x , has arbitrary Gaussian decay in y , and permits every fixed parameter and spatial derivative. $\square$

Theorem 4.2 (Exact complete-kernel identity). *For $a > 0$ and $t \in \mathbb{R}$,*

$$\widehat{C}_a(2t) = K(a, t) := \operatorname{Re} (F'(a + it) \overline{F(a + it)}) = \tfrac{1}{2} \partial_a |F(a + it)|^2. \quad (4.6)$$

Proof. Differentiate (4.3) and apply (4.2), justified by Lemma 4.1. Equivalently, symmetrization in u, v gives

$$K(a, t) = \frac{1}{2} \iint_{\mathbb{R}^2} (u + v) \eta(u) \eta(v) e^{a(u+v)} \cos(t(u - v)) du dv \quad (4.7)$$

$$= 2 \iint_{\mathbb{R}^2} x e^{2ax} \eta(x + y) \eta(x - y) \cos(2ty) dx dy. \quad (4.8)$$

Oddness removes $x \cosh(2ax)$, leaving (4.6). $\square$

4.2 A direct proof of the Fourier sign equivalence

Lemma 4.3. *For a real even Schwartz function C , the following are equivalent:*

- (i) $[C(y_j - y_k)]_{j,k=1}^N \succeq 0$ for every finite real node set;
- (ii) $\widehat{C}(\omega) \geq 0$ for every $\omega \in \mathbb{R}$.

Proof. Assume (i). For $T > 0$, approximate the double integral below by finite quadratic-form Riemann sums to obtain

$$0 \leq \int_{-T}^T \int_{-T}^T C(u - v) e^{-i\omega(u-v)} du dv \quad (4.9)$$

$$= \int_{-2T}^{2T} (2T - |y|) C(y) e^{-i\omega y} dy. \quad (4.10)$$

Divide by $2T$ and use dominated convergence with majorant $|C(y)|$. The limit is $\widehat{C}(\omega) \geq 0$. Conversely, Fourier inversion gives

$$\sum_{j,k} \overline{v_j} v_k C(y_j - y_k) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{C}(\omega) \left| \sum_k v_k e^{-i\omega y_k} \right|^2 d\omega \geq 0. \quad (4.11)$$

□

Combining Lemma 4.3 with Theorem 4.2 gives the exact equivalence

$$\mathbf{P} \iff K(a, t) \geq 0 \quad (a > 0, t \in \mathbb{R}). \quad (4.12)$$

In particular, the proof applies to an arbitrary pair (a, t) without an enumeration of pairs or an ordinate cutoff.

4.3 Unconditional positive definiteness of the transform

Theorem 4.4 (The transformed kernel). *For each $a > 0$, the function $t \mapsto K(a, t)$ is positive definite. More precisely, for any finite real nodes $t_1, \dots, t_N$ and $v_1, \dots, v_N \in \mathbb{C}$,*

$$\sum_{j,k=1}^N \overline{v_j} v_k K(a, t_j - t_k) = \int_{\mathbb{R}} C_a(y) \left| \sum_{k=1}^N v_k e^{2it_k y} \right|^2 dy \geq 0. \quad (4.13)$$

For pairwise distinct nodes, this quadratic form is strictly positive unless every coefficient vanishes.

Proof. Insert (4.6) into the finite sum and interchange the sum and integral. The exponential factors combine into the displayed square. Lemma 4.1 proves $C_a(y) > 0$. If the quadratic form is zero, the exponential polynomial is zero almost surely with respect to $C_a(y) dy$, hence identically zero by continuity. Its first N derivatives at zero yield the Vandermonde system $\sum_k v_k (2it_k)^n = 0$, $0 \leq n < N$. Distinctness makes this system invertible, so every $v_k = 0$. □

The matrices in (4.13) have entries $K(a, t_j - t_k)$. The statement $\mathbf{P}$ concerns the matrices with entries $C_a(y_j - y_k)$. Theorem 4.4 supplies a proof of the former statement directly from the positive theta density. The latter statement is characterized by Theorem 1.2. In particular, positive definiteness of a function and nonnegativity of its scalar values are different properties: $\cos t$ is positive definite, since it is the transform of $(\delta_{-1} + \delta_1)/2$, and $\cos \pi = -1$.

4.4 Cumulative positivity and the polynomial test class

Proposition 4.5 (Exact distinction between integral and pointwise signs). *For each $\alpha > 0$, there exists a strictly positive, strictly log-concave, even Schwartz function C_* such that $k_*(t) = \widehat{C}_*(2t)$ is strictly positive definite at distinct finite nodes, has absolute exponential moments of every order, and satisfies*

$$\int_0^T k_*(t) dt > 0 \quad (T > 0), \quad k_*(2\sqrt{\alpha}) < 0. \quad (4.14)$$

Its first two nonconstant even moments are

$$\frac{1}{2\pi} \int_{\mathbb{R}} t^2 k_*(t) dt = \frac{11\alpha}{80} > 0, \quad \frac{1}{2\pi} \int_{\mathbb{R}} t^4 k_*(t) dt = -\frac{11\alpha^2}{32} < 0. \quad (4.15)$$

Proof. Take the exact Fourier pair

$$C_*(y) = e^{-\alpha y^2} \left(1 - \frac{1}{20} \cos(4\sqrt{\alpha}y)\right), \quad (4.16)$$

$$k_*(t) = \sqrt{\frac{\pi}{\alpha}} \left(e^{-t^2/\alpha} - \frac{1}{40} e^{-(t/\sqrt{\alpha}-2)^2} - \frac{1}{40} e^{-(t/\sqrt{\alpha}+2)^2} \right). \quad (4.17)$$

The Gaussian Fourier integral gives (4.17) with precisely the convention (1.5). Moreover $C_*(y) \geq (19/20)e^{-\alpha y^2} > 0$. The proof of Theorem 4.4 gives strict finite Gram positivity of k_* . All asserted decay and exponential integrability follow from the Gaussian formulas. Write $h(y) = 1 - \cos(4\sqrt{\alpha}y)/20$. Since $h \geq 19/20$ and $h'' \leq 4\alpha/5$,

$$(\log C_*)'' = -2\alpha + \frac{h''}{h} - \left(\frac{h'}{h}\right)^2 \leq -\frac{22\alpha}{19} < 0. \quad (4.18)$$

At $t = 2\sqrt{\alpha}$,

$$k_*(2\sqrt{\alpha}) = \sqrt{\frac{\pi}{\alpha}} \left(e^{-4} - \frac{1}{40} - \frac{1}{40} e^{-16} \right) < 0. \quad (4.19)$$

Indeed $e > 8/3$, so $e^4 > 4096/81 > 40$.

For $R > 0$, let $M(R) = \int_0^R e^{-u^2} du$ and $J_R(b) = \int_{-R}^R e^{-(u-b)^2} du$. For $b > 0$,

$$J'_R(b) = e^{-(R+b)^2} - e^{-(R-b)^2} < 0. \quad (4.20)$$

Thus $J_R(2) < 2M(R)$. Taking $R = T/\sqrt{\alpha}$ gives

$$\int_0^T k_*(t) dt = \sqrt{\pi} \left(M(R) - \frac{1}{40} J_R(2) \right) > \frac{19\sqrt{\pi}}{20} M(R) > 0. \quad (4.21)$$

Finally, direct differentiation gives $C_*''(0) = -11\alpha/10$ and $C_*^{(4)}(0) = -11\alpha^2$. Differentiating Fourier inversion yields

$$\frac{1}{2\pi} \int_{\mathbb{R}} t^{2n} k_*(t) dt = \frac{(-1)^n C_*^{(2n)}(0)}{2 \cdot 4^n},$$

which proves (4.15). □

This proposition concerns an auxiliary Gaussian Fourier pair, not the particular Jacobi-theta kernel. It proves that positive prefix integrals, evenness, Schwartz regularity, strict log-concavity, a positive spatial density, and positive definiteness of the transform do not together imply its pointwise nonnegativity. The polynomial test $P(t) = t^2$ fails in (4.15). For the actual theta function, Theorem 9.2 specifies the full polynomial inequalities that are equivalent to the desired pointwise sign. The autocorrelation proved in (4.3) is A_a ; (4.2) identifies C_a as its parameter derivative and supplies no autocorrelation factorization of C_a .

5 Local zeros, sign reversal, and analytic exclusion

Theorem 5.1 (Universal local sign reversal). *Let a nonzero holomorphic function F have a zero $z_0 = a_0 + it_0$ of multiplicity $m \geq 1$. Write $F(z_0 + h) = ch^m + O(h^{m+1})$, $c \neq 0$. For real $\delta \rightarrow 0$,*

$$K(a_0 + \delta, t_0) = m|c|^2 \delta^{2m-1} + O(|\delta|^{2m}). \quad (5.1)$$

Hence K is negative immediately to the left of the zero and positive immediately to its right.

Proof. Differentiate the local factorization and multiply:

$$F'(z_0 + \delta)\overline{F(z_0 + \delta)} = (mc\delta^{m-1} + O(|\delta|^m))(\overline{c}\delta^m + O(|\delta|^{m+1})).$$

Its leading coefficient is real and positive, proving (5.1). Since $2m - 1$ is odd, the asserted signs follow for sufficiently small nonzero δ . $\square$

Where $F \neq 0$, (3.9) identifies the corresponding phase velocity

$$\Omega(a, t) = \partial_t \arg F(a + it) = \frac{K(a, t)}{|F(a + it)|^2}. \quad (5.2)$$

With $F(z) = (z - z_0)^m h_0(z)$, it has the sharper local expression

$$\Omega(a_0 + \delta, t_0) = \frac{m}{\delta} + \operatorname{Re} \frac{h'_0(z_0 + \delta)}{h_0(z_0 + \delta)}. \quad (5.3)$$

The second term is bounded in a sufficiently small disk.

Theorem 5.2 (Local exclusion under a structural sign). *Let $F \not\equiv 0$ be holomorphic on a connected domain D . If $K \geq 0$ on D , then F has no zero in D whose sufficiently small horizontal two-sided neighborhood lies in D .*

Proof. Such a zero would contradict Theorem 5.1 on its left. There is also a division-free proof. If $F(a_0 + it_0) = 0$, then for small $\varepsilon > 0$,

$$0 \leq |F(a_0 - \varepsilon + it_0)|^2 \leq |F(a_0 + it_0)|^2 = 0,$$

because $\partial_a |F|^2 = 2K \geq 0$. The same argument applies to every point of a left interval. The identity theorem would give $F \equiv 0$. $\square$

For the actual F , evenness transfers exclusion in $\mathbb{H}$ to the left half-plane. The boundary $a = 0$ remains admissible: the local sign change at a zero on that boundary places its negative side in $a < 0$.

6 A fixed integral operator for the complete kernel

Define, for $x, y \in \mathbb{R}$,

$$J(x, y) = \int_0^\infty (x + y + 2r)\eta(x + r)\eta(y + r) \, dr. \quad (6.1)$$

Put $m = (x + y)/2$ and $d = (x - y)/2$. Substitution $v = m + r$ and evenness give

$$J(x, y) = 2 \int_m^\infty v\eta(v + d)\eta(v - d) \, dv = 2 \int_{|m|}^\infty v\eta(v + d)\eta(v - d) \, dv > 0. \quad (6.2)$$

If $\eta(u) \leq M_q e^{-qu^2}$, then

$$0 < J(x, y) \leq 2M_q^2 e^{-2qd^2} \int_{|m|}^\infty v e^{-2qv^2} \, dv = \frac{M_q^2}{2q} e^{-q(x^2 + y^2)}. \quad (6.3)$$

Continuity follows by dominated convergence locally in x, y . Consequently

$$(\mathcal{J}h)(x) = \int_{\mathbb{R}} J(x, y)h(y) \, dy \quad (6.4)$$

is a bounded compact self-adjoint Hilbert–Schmidt operator on $L^2(\mathbb{R})$.

Theorem 6.1 (Double-transform representation). *For $z, w \in \mathbb{H}$,*

$$\mathcal{B}(z, w) := \frac{F'(z)\overline{F(w)} + F(z)\overline{F'(w)}}{z + \overline{w}} = \iint_{\mathbb{R}^2} e^{zx + \overline{w}y} J(x, y) \, dx \, dy. \quad (6.5)$$

Its diagonal is

$$\mathcal{B}(a + it, a + it) = \frac{K(a, t)}{a}. \quad (6.6)$$

Proof. Insert (6.1), substitute $u = x + r$, $v = y + r$, and integrate first in r . Absolute convergence follows from

$$\int_0^\infty e^{-(\operatorname{Re} z + \operatorname{Re} w)r} \, dr \iint |u + v| \eta(u) \eta(v) e^{(\operatorname{Re} z)u + (\operatorname{Re} w)v} \, du \, dv < \infty.$$

The r integral is $(z + \overline{w})^{-1}$ and the two terms $u + v$ give the numerator of (6.5). The diagonal identity is immediate. $\square$

Theorem 6.2 (Fixed operator equivalence). *The following are equivalent:*

- (i) $\mathcal{J} \succeq 0$ on $L^2(\mathbb{R})$;
- (ii) $[\mathcal{B}(z_j, z_k)] \succeq 0$ for every finite node set in $\mathbb{H}$;
- (iii) for one fixed $a_* > 0$, $[\mathcal{B}(a_* + it_j, a_* + it_k)] \succeq 0$ for every finite real node set.

Proof. To prove (i) $\Rightarrow$ (ii), use compact cutoffs on a finite sum of exponentials in (6.5). Their quadratic forms are nonnegative by (i). Bound (6.3) permits removal of the cutoffs by dominated convergence, giving the desired finite form. Assertion (ii) implies (iii).

For (iii) $\Rightarrow$ (i), set $J_*(x, y) = e^{a_*(x+y)} J(x, y)$. It is a continuous Hilbert–Schmidt kernel and belongs to $L^1(\mathbb{R}^2)$ by (6.3). Its double Fourier transform is $\mathcal{B}(a_* + it, a_* + is)$. For $\varphi \in C_c^\infty(\mathbb{R})$, finite Riemann sums in (iii) give

$$\iint \overline{\varphi(t)} \varphi(s) \mathcal{B}(a_* + it, a_* + is) \, dt \, ds \geq 0.$$

Fubini expresses this as the quadratic form of J_* on $p(x) = \int \varphi(t) e^{-itx} \, dt$. Such p form a dense subspace of L^2 by the Fourier isometry and density of C_c^∞ . Boundedness therefore gives nonnegativity of the J_* form on all L^2 . For any compactly supported $h \in L^2$, insert $p(x) = e^{-a_* x} h(x)$. The resulting form equals $\langle h, \mathcal{J} h \rangle$. Finally use density of compactly supported L^2 functions and boundedness of $\mathcal{J}$. $\square$

6.1 Exterior products retain every interaction

Suppose a positive kernel has a representation $B(z, w) = \int \phi_z(r) \overline{\phi_w(r)} \, d\mu(r)$, $\mu \geq 0$. Expanding both products and exchanging the integration variables gives

$$\det[B(z_j, z_k)]_{j,k=1}^2 = \frac{1}{2} \iint |\phi_{z_1}(r) \phi_{z_2}(s) - \phi_{z_1}(s) \phi_{z_2}(r)|^2 \, d\mu(r) \, d\mu(s). \quad (6.7)$$

Cauchy–Schwarz supplies the absolute integrability. Thus a complete Gram factorization automatically pays the two-point determinant with all cross terms retained. Theorem 6.2 states the exact operator property corresponding to such positivity for (6.5).

7 Finite determinants and a differential exclusion law

7.1 Two points at a hypothetical zero

Theorem 7.1. *Let $F \not\equiv 0$ be holomorphic near $z_0 = a_0 + it_0$, where $a_0 > 0$. If the two-point matrices of $\mathcal{B}$ are positive semidefinite for all sufficiently close pairs in a neighborhood of z_0 , then $F(z_0) \neq 0$.*

Proof. Suppose $F(z_0 + h) = ch^m + O(h^{m+1})$, $c \neq 0$, $m \geq 1$, and put $d = 2a_0$. Choose $z_j = z_0 + j\varepsilon$, $j = 1, 2$, with $\varepsilon > 0$. For $u, v \in \{1, 2\}$,

$$\mathcal{B}(z_0 + \varepsilon u, z_0 + \varepsilon v) = \frac{m|c|^2}{d} \varepsilon^{2m-1} u^{m-1} v^{m-1} (u + v) + O(\varepsilon^{2m}). \quad (7.1)$$

The leading coefficient matrix is

$$\frac{m|c|^2}{d} \begin{pmatrix} 2 & 3 \cdot 2^{m-1} \\ 3 \cdot 2^{m-1} & 2^{2m} \end{pmatrix},$$

whose determinant is $-m^2|c|^4 2^{2m-2}/d^2$. Hence

$$\det[\mathcal{B}(z_j, z_k)]_{j,k=1}^2 = -\frac{m^2|c|^4}{d^2} 2^{2m-2} \varepsilon^{4m-2} + O(\varepsilon^{4m-1}) < 0 \quad (7.2)$$

for sufficiently small ε , a contradiction. $\square$

7.2 An exact derivative determinant

For a zero of multiplicity m put $A = F^{(m)}(z_0)$ and define

$$B_{jk}^{(z_0)} = \partial_z^j \partial_{\bar{w}}^k \mathcal{B}(z, w) \Big|_{z=w=z_0}.$$

The product rule, with all derivatives of F below order m vanishing, gives

$$B_{m-1, m-1}^{(z_0)} = 0, \quad B_{m-1, m}^{(z_0)} = B_{m, m-1}^{(z_0)} = |A|^2/d, \quad (7.3)$$

$$B_{m, m}^{(z_0)} = \frac{2 \operatorname{Re}(F^{(m+1)}(z_0) \bar{A})}{d} - \frac{2m|A|^2}{d^2}. \quad (7.4)$$

Therefore the principal matrix on derivative orders $m-1, m$ has

$$\det \begin{pmatrix} 0 & |A|^2/d \\ |A|^2/d & B_{m, m}^{(z_0)} \end{pmatrix} = -|A|^4/d^2 < 0. \quad (7.5)$$

For a positive holomorphic kernel, derivative matrices are positive: replace each order- j derivative by

$$h^{-j} \sum_{r=0}^j (-1)^{j-r} \binom{j}{r} \phi(z_0 + rh).$$

Each finite-difference matrix is a congruence of a finite kernel matrix. As $h \downarrow 0$, analyticity and the closedness of the finite positive cone give the assertion. Thus (7.5) is a second exact contradiction under kernel positivity. Factorial normalization divides this determinant by $((m-1)!m!)^2$ and preserves its sign.

7.3 The denominator-free discriminant

Define

$$T(z) = F(z)F''(z) - F'(z)^2, \quad S(a, t) = K(a, t)^2 - a^2|T(a + it)|^2. \quad (7.6)$$

Where $F \neq 0$, put $g = F'/F$ and $u = \operatorname{Re} g$. Then $T = F^2 g'$ and $K = |F|^2 u$. For $z = a + it$, $w = b + it$ direct expansion gives

$$\det \left[\frac{g(z_j) + \overline{g(z_k)}}{z_j + \overline{z_k}} \right]_{j,k=1}^2 = \frac{(a-b)^2 u(z)u(w) - ab|g(z) - g(w)|^2}{ab(a+b)^2}. \quad (7.7)$$

Indeed, the determinant is $u(z)u(w)/(ab) - |g(z) + \overline{g(w)}|^2/(a+b)^2$, and $|g(z) + \overline{g(w)}|^2 = 4u(z)u(w) + |g(z) - g(w)|^2$. Multiplication by $|F(z)F(w)|^2$ gives the determinant of $\mathcal{B}$. Setting $b = a + \varepsilon$ therefore yields

$$\det[\mathcal{B}(z_j, z_k)]_{j,k=1}^2 = \frac{\varepsilon^2}{4a^4} S(a, t) + O(\varepsilon^3). \quad (7.8)$$

The coefficient on the left is continuous in the base point, so the identity of second-order coefficients extends to zeros of F by density of its nonzero set.

Theorem 7.2 (Differential exclusion). *If $S(a, t) \geq 0$ for $0 < a < 1/2$ and $t \in \mathbb{R}$, every zero of the actual F lies on $\operatorname{Re} z = 0$.*

Proof. At a hypothetical zero $z_0 = a_0 + it_0$, $0 < a_0 < 1/2$, the factorization in Theorem 5.1 gives

$$K(a_0 + \delta, t_0) = m|c|^2 \delta^{2m-1} + O(\delta^{2m}), \quad (7.9)$$

$$T(z_0 + \delta) = -mc^2 \delta^{2m-2} + O(\delta^{2m-1}). \quad (7.10)$$

It follows that

$$S(a_0 + \delta, t_0) = -a_0^2 m^2 |c|^4 \delta^{4m-4} + O(\delta^{4m-3}) < 0 \quad (7.11)$$

for sufficiently small positive δ . For $m = 1$ the value at the zero itself is $-a_0^2 |F'(z_0)|^4 < 0$; for $m > 1$ the neighboring point in (7.11) supplies the contradiction. Evenness and Proposition 2.4 finish the assertion. $\square$

This theorem takes $S \geq 0$ as its sole additional inequality; the sign of K is not an additional premise in this version.

8 Canonical products and exact equivalences

8.1 The paired product

By Proposition 2.3, Hadamard factorization applies with genus at most one. Since $F(0) > 0$, no factor at zero occurs. Pair each nonzero zero ρ_0 with $-\rho_0$, including multiplicities. The genus-one factors satisfy

$$(1 - z/\rho_0)e^{z/\rho_0}(1 + z/\rho_0)e^{-z/\rho_0} = 1 - z^2/\rho_0^2.$$

Moreover $\sum m_{\rho_0}/|\rho_0|^2 < \infty$ over one representative of each pair. Evenness makes the remaining linear exponential constant. Evaluation at zero gives

$$F(z) = F(0) \prod_{\{\rho_0, -\rho_0\}} (1 - z^2/\rho_0^2)^{m_{\rho_0}}. \quad (8.1)$$

The product converges normally on compact sets. Logarithmic differentiation away from zeros gives

$$\frac{F'(z)}{F(z)} = \sum_{\{\rho_0, -\rho_0\}} m_{\rho_0} \left(\frac{1}{z - \rho_0} + \frac{1}{z + \rho_0} \right). \quad (8.2)$$

Pairing is retained in this series. The standard xi product and its horizontal modulus consequences are also treated in [4].

Proposition 8.1 (Exterior half-plane). *For the actual F ,*

$$K(a, t) > 0 \quad (a \geq 1/2, t \in \mathbb{R}). \quad (8.3)$$

Proof. For a zero $\rho_0 = \beta + i\gamma$, Proposition 2.4 gives $|\beta| < 1/2$. At $z = a + it$, $a \geq 1/2$, the real parts of the two terms in (8.2) are

$$\frac{a - \beta}{|z - \rho_0|^2} > 0, \quad \frac{a + \beta}{|z + \rho_0|^2} > 0.$$

Their sum over zeros converges and is positive. The function has no zero there, so multiplication by $|F(z)|^2$ gives (8.3). $\square$

8.2 Consequences of a localized zero divisor

Suppose in this subsection that every zero of F is imaginary. Write the positive ordinates as $\gamma > 0$ with multiplicities m_γ . Equations (8.1)–(8.2) become

$$F(z) = F(0) \prod_{\gamma > 0} (1 + z^2/\gamma^2)^{m_\gamma}, \quad (8.4)$$

$$g(z) = \sum_{\gamma > 0} m_\gamma \left(\frac{1}{z - i\gamma} + \frac{1}{z + i\gamma} \right). \quad (8.5)$$

For $a > 0$,

$$\operatorname{Re} g(a + it) = a \sum_{\gamma > 0} m_\gamma \left[\frac{1}{a^2 + (t - \gamma)^2} + \frac{1}{a^2 + (t + \gamma)^2} \right] > 0. \quad (8.6)$$

Also

$$|g'(a + it)| \leq \sum_{\gamma > 0} m_\gamma \left[\frac{1}{a^2 + (t - \gamma)^2} + \frac{1}{a^2 + (t + \gamma)^2} \right] = \frac{\operatorname{Re} g(a + it)}{a}. \quad (8.7)$$

All terms in the differentiated series are locally absolutely summable. Thus $K > 0$ and $S \geq 0$ in $\mathbb{H}$ under this subsection's premise.

For a full kernel factorization, the identity

$$\frac{(z - i\gamma)^{-1} + (\bar{w} + i\gamma)^{-1}}{z + \bar{w}} = \frac{1}{(z - i\gamma)(\bar{w} + i\gamma)}$$

gives

$$\begin{aligned} \mathcal{B}(z, w) = \sum_{\gamma > 0} m_\gamma & \left[\frac{F(z)}{z - i\gamma} \frac{\overline{F(w)}}{w - i\gamma} \right. \\ & \left. + \frac{F(z)}{z + i\gamma} \frac{\overline{F(w)}}{w + i\gamma} \right]. \end{aligned} \quad (8.8)$$

The series converges locally absolutely by inverse-square summability. It is a Gram representation, so every finite kernel matrix is positive semidefinite. The premise about the zero divisor is part of this converse construction.

Theorem 8.2 (Equivalent structural formulations). *For the actual function F the following statements are equivalent:*

- (a) every zero of F is imaginary;
- (b) C_a is positive definite for every $a > 0$ (statement P);
- (c) C_a is positive definite for each $0 < a < 1/2$;
- (d) $K(a, t) \geq 0$ for $a > 0, t \in \mathbb{R}$;
- (e) $K(a, t) \geq 0$ for $0 < a < 1/2, t \in \mathbb{R}$;
- (f) $\mathcal{B}$ is a positive-definite kernel on $\mathbb{H}$;
- (g) the fixed operator $\mathcal{J}$ is positive semidefinite;
- (h) $S(a, t) \geq 0$ for $a > 0, t \in \mathbb{R}$;
- (i) $S(a, t) \geq 0$ for $0 < a < 1/2, t \in \mathbb{R}$;
- (j) $D_{a,b}$ is positive definite for every $0 \leq a < b$;
- (k) for each $t \in \mathbb{R}$, the function $a \mapsto |F(a + it)|^2$ is nondecreasing on $a > 0$.

Proof. Lemma 4.3 and (4.6) give (b) $\Leftrightarrow$ (d) and (c) $\Leftrightarrow$ (e). Theorem 5.2, reflection, and Proposition 2.4 give (e) $\Rightarrow$ (a). Equation (8.6) gives (a) $\Rightarrow$ (d), and (d) implies (e). Equation (8.8) gives (a) $\Rightarrow$ (f); (6.6) gives (f) $\Rightarrow$ (d). Theorem 6.2 gives (f) $\Leftrightarrow$ (g). Equation (8.7) gives (a) $\Rightarrow$ (h), and (h) implies (i). Theorem 7.2 gives (i) $\Rightarrow$ (a). Integrating positive finite forms in (4.4) gives (b) $\Rightarrow$ (j). Conversely, divide the finite forms of $D_{a,a+h}$ by h and let $h \downarrow 0$; Lemma 4.1 gives (b). Finally (4.6) and differentiability give (d) $\Leftrightarrow$ (k). $\square$

9 All-degree signed moments and complete increments

9.1 A density lemma

Lemma 9.1. *If μ is a finite positive measure on $\mathbb{R}$ and $\int e^{\epsilon|t|} d\mu(t) < \infty$ for some $\epsilon > 0$, complex polynomials are dense in $L^2(\mu)$.*

Proof. Let $h \in L^2(\mu)$ be orthogonal to all polynomials. After conjugating h if necessary, $\int t^n h(t) d\mu(t) = 0$ for every $n \geq 0$. For $0 < c < \epsilon/2$, Cauchy–Schwarz gives

$$\int e^{c|t|} |h(t)| d\mu(t) \leq \|h\|_{L^2(\mu)} \left(\int e^{2c|t|} d\mu(t) \right)^{1/2} < \infty.$$

The Fourier transform of the finite complex measure $h\mu$ is therefore holomorphic on $|\operatorname{Im} z| < \epsilon/2$. Its derivatives at zero all vanish, so the identity theorem makes the transform zero on that strip. Fourier uniqueness gives $h\mu = 0$, hence $h = 0$ μ -almost everywhere. For completeness, Fourier uniqueness here follows by testing against Schwartz functions, applying Fourier inversion and Fubini, and then using density of compactly supported smooth functions in continuous compactly supported test functions. The orthogonal complement of the polynomials is zero. $\square$

Theorem 9.2 (Signed-density criterion). *Let w be real, continuous and even, with $\int e^{|t|}|w(t)| dt < \infty$. Define*

$$\nu_n = \frac{1}{2\pi} \int_{\mathbb{R}} t^{2n} w(t) dt, \quad A_N = (\nu_{j+k})_{j,k=0}^N, \quad B_N = (\nu_{j+k+1})_{j,k=0}^N. \quad (9.1)$$

Then each of the following is equivalent to $w(t) \geq 0$ for $t \in \mathbb{R}$:

- (i) $\int |P(t)|^2 w(t) dt \geq 0$ for all $P \in \mathbb{C}[t]$;
- (ii) $A_N \succeq 0$ and $B_N \succeq 0$ for every N ;
- (iii) $A_N \succeq 0$ for every N ;
- (iv) $B_N \succeq 0$ for every N .

Proof. Set $d\mu = |w(t)| dt / (2\pi)$ and $Q(h) = \int \operatorname{sgn}(w) |h|^2 d\mu$. This form is continuous on $L^2(\mu)$ because

$$|Q(h) - Q(k)| \leq \|h - k\|_2 (\|h\|_2 + \|k\|_2).$$

Lemma 9.1 therefore extends nonnegativity on polynomials to all $L^2(\mu)$. Inserting the indicator of $\{w < 0\}$ forces this set to have zero μ measure. Continuity gives $w \geq 0$.

For $P(t) = p(t^2) + tq(t^2)$ the mixed term is odd, and its integral vanishes. Thus $Q(P) = u^* A_N u + v^* B_N v$, proving the two-block equivalence. For (iii), the measure is even, and symmetrization of polynomial approximations proves density of even polynomials in the even subspace. The negative-set indicator is even, so the same argument applies using A_N alone. For (iv), apply (iii) to the even density $t^2 w(t)$; it has an exponential absolute moment with any smaller positive exponent. The conclusion $t^2 w(t) \geq 0$ gives $w(t) \geq 0$ off zero and then at zero by continuity. The converse implications are immediate. $\square$

9.2 Exact application to the complete theta kernels

Estimate (2.13) gives, for every $\epsilon < \pi/2$,

$$\int_{\mathbb{R}} e^{\epsilon|t|} |K(a, t)| dt < \infty. \quad (9.2)$$

Thus Theorem 9.2 applies directly to $K(a, \cdot)$. Fourier inversion in (4.6) reads

$$C_a(y) = \frac{1}{\pi} \int_{\mathbb{R}} K(a, t) e^{2ity} dt. \quad (9.3)$$

Consequently its exact moments are

$$\frac{1}{2\pi} \int_{\mathbb{R}} t^{2n} K(a, t) dt = \frac{(-1)^n}{2 \cdot 4^n} C_a^{(2n)}(0). \quad (9.4)$$

Likewise (4.4) gives

$$\frac{1}{2\pi} \int_{\mathbb{R}} t^{2n} H_{a,b}(t) dt = \frac{(-1)^n}{4^n} D_{a,b}^{(2n)}(0). \quad (9.5)$$

The factor $1/2$ distinguishes (9.4) from (9.5).

To compare the cumulative sign with the polynomial tests, put

$$I_a(T) = \int_0^T K(a, t) dt, \quad M_a = \int_0^\infty K(a, t) dt = \frac{\pi}{2} C_a(0). \quad (9.6)$$

For $n \geq 1$, integration by parts against the rapidly decreasing tail $M_a - I_a(t) = \int_t^\infty K(a, u) du$ gives the exact identity

$$\int_0^\infty t^{2n} K(a, t) dt = 2n \int_0^\infty t^{2n-1} [M_a - I_a(t)] dt. \quad (9.7)$$

The established inequality $I_a(T) > 0$ gives no sign for $M_a - I_a(t)$. For a full polynomial test, Theorem 9.2 requires positivity of the entire moment quadratic form, including all mixed coefficients. For example, the first even two-node block requires

$$C_a(0)C_a^{(4)}(0) - C_a''(0)^2 \geq 0. \quad (9.8)$$

Proposition 4.5 demonstrates why cumulative positivity cannot be substituted for the polynomial condition.

With $\delta = b^2 - a^2$, the normalized midpoint weight has the exact formula

$$\frac{\cosh(2bx) - \cosh(2ax)}{b^2 - a^2} = \int_0^1 2x^2 \operatorname{sinhc}\left(2x\sqrt{a^2 + \theta\delta}\right) d\theta, \quad (9.9)$$

where $\operatorname{sinhc}(v) = \sinh(v)/v$ and $\operatorname{sinhc}(0) = 1$. This supplies the continuous normalized diagonal limit, including $a = 0$. In particular, for $a > 0$,

$$\lim_{b \downarrow a} \frac{H_{a,b}(t)}{b^2 - a^2} = \frac{K(a, t)}{a}. \quad (9.10)$$

The differentiated density bounds give dominated convergence in every fixed polynomial moment, locally uniformly in bounded a, b .

9.3 Schur complements and all retained cross terms

For a Hermitian block matrix $M = \begin{pmatrix} A & b \\ b^* & c \end{pmatrix}$ with $A \succ 0$,

$$M \succeq 0 \iff c - b^* A^{-1} b \geq 0. \quad (9.11)$$

For any trial vector v the exact decomposition is

$$c - b^* A^{-1} b = (-v, 1)^* M (-v, 1) - (b - Av)^* A^{-1} (b - Av). \quad (9.12)$$

Both assertions follow by completing the square in $x^* Ax + 2 \operatorname{Re}(x^* by) + c|y|^2$. If $A \succeq 0$ is singular, the corresponding criterion is $b \in \operatorname{Ran} A$ and $c - b^* A^\dagger b \geq 0$; this follows by diagonalizing A and testing its nullspace first.

The full hierarchy in Theorem 9.2 concerns every finite degree. Each finite block in (9.11) retains all ordered coefficient interactions of its polynomial form. The theorem connects this complete hierarchy to the specific signed density by the proved density argument, rather than by selecting another representing measure.

9.4 Complete autocorrelation factorization

For $q \in L^2(\mathbb{R})$ define

$$A[q](y) = \int_{\mathbb{R}} q(x+y) \overline{q(x-y)} dx.$$

If $q \in L^1 \cap L^2$, then $\widehat{A[q]}(2t) = \frac{1}{2}|\widehat{q}(t)|^2$. Suppose that for each $0 \leq a < b$ there is a family $q_j \in L^1 \cap L^2$ with $\sum_j (\|q_j\|_1^2 + \|q_j\|_2^2) < \infty$ and

$$D_{a,b}(y) = \sum_j A[q_j](y). \quad (9.13)$$

The series converges uniformly because $|A[q_j](y)| \leq \|q_j\|_2^2$, and in L^1 because $\|A[q_j]\|_1 \leq \|q_j\|_1^2/2$. Every finite form is a limit of nonnegative forms, so $D_{a,b}$ is positive definite. Theorem 8.2 then applies.

Conversely, if $H_{a,b} \geq 0$, define an L^2 function q by $\widehat{q}(t) = \sqrt{H_{a,b}(t)}$ using the Fourier isometry. Since $H_{a,b} \in L^1$, this is legitimate. The L^2 autocorrelation identity and (4.4) give $D_{a,b} = A[q]$. This converse guarantees an L^2 factor. In the present setting that factor also belongs to L^1 , as the following lemma verifies.

Lemma 9.3. *If $h \geq 0$ is a Schwartz function on $\mathbb{R}$, then $\sqrt{h} \in H^1(\mathbb{R})$. Its inverse Fourier transform belongs to $L^1 \cap L^2$.*

Proof. Let $M(t) = \sup_{|u-t| \leq 1} |h''(u)|$ and $p = |h'(t)|$. Taylor's theorem at $t - \delta \operatorname{sgn}(h'(t))$ gives $0 \leq h(t) - p\delta + M(t)\delta^2/2$ for $0 \leq \delta \leq 1$. If $0 < p \leq M(t)$, choose $\delta = p/M(t)$; if $p > M(t)$, choose $\delta = 1$. In both cases, at points where $h(t) > 0$,

$$\frac{|h'(t)|^2}{h(t)} \leq 2(M(t) + |h'(t)|).$$

The right side is integrable because h is Schwartz. At zeros of h , its derivative vanishes. The regularized functions $r_\varepsilon = \sqrt{h + \varepsilon} - \sqrt{\varepsilon}$ satisfy $|r_\varepsilon|^2 \leq h$ and $|r'_\varepsilon|^2 \leq (M + |h'|)/2$ almost everywhere. Dominated convergence in L^2 and weak compactness in H^1 prove $\sqrt{h} \in H^1$. If q is its inverse Fourier transform, Parseval gives $q, xq \in L^2$, and Cauchy–Schwarz gives

$$\|q\|_1^2 \leq \left(\int_{\mathbb{R}} \frac{dx}{1+x^2} \right) \int_{\mathbb{R}} (1+x^2) |q(x)|^2 dx < \infty.$$

□

Applying the lemma to $h = H_{a,b}$ proves the full single-factor form of (9.13) whenever $H_{a,b} \geq 0$. Thus the complete autocorrelation condition with its stated summability is equivalent to condition (j) of Theorem 8.2.

10 Strict positivity on the real ray and an explicit complex region

10.1 The degree-two coefficient theorem

Define the entire function H by $F(z) = H(z^2)$, so that

$$H(x) = \sum_{n \geq 0} \gamma_n \frac{x^n}{n!}, \quad \gamma_n = \frac{n!}{(2n)!} \int_{\mathbb{R}} u^{2n} \eta(u) du > 0. \quad (10.1)$$

The degree-two Jensen theorem, as contained in [5, Theorem 2], gives

$$\gamma_{n+1}^2 \geq \gamma_n \gamma_{n+2} \quad (n \geq 0). \quad (10.2)$$

The normalization is exact: the function expanded in that reference is $(-1+4z^2)\pi^{-(1/2+z)/2}\Gamma((1/2+z)/2)\zeta(1/2+z) = 8F(z)$, so its coefficients are $8\gamma_n$. The common factor cancels in (10.2). This section uses that cited coefficient theorem as input.

Theorem 10.1. *For real $a > 0$, with $g(a) = F'(a)/F(a)$,*

$$0 < ag'(a) < g(a), \quad S(a, 0) > 0. \quad (10.3)$$

Proof. For $x > 0$ put

$$w_n(x) = \frac{\gamma_n x^n}{n!H(x)}, \quad r_n = \frac{\gamma_{n+1}}{\gamma_n}.$$

Then $w_n > 0$, $\sum_n w_n = 1$, and $0 < r_{n+1} \leq r_n \leq r_0$. Differentiating the entire series gives

$$\frac{H'}{H} = \sum_n w_n r_n, \quad w'_n = w_n \left(\frac{n}{x} - \frac{H'}{H} \right), \quad \sum_n n w_n = x \frac{H'}{H}.$$

For $L = \log H$ on $x > 0$, symmetrization yields

$$L''(x) = \frac{1}{2x} \sum_{j,k \geq 0} w_j w_k (r_j - r_k)(j - k) < 0. \quad (10.4)$$

Absolute convergence follows from $|r_j - r_k| \leq r_0$ and $\sum_n n w_n < \infty$. Each summand is nonpositive. If all ratios were equal to $r > 0$, then $F(z) = \gamma_0 e^{rz^2}$, contradicting Proposition 2.3. Some pair is therefore strictly negative, proving strictness in (10.4).

Since $g(a) = 2aL'(a^2)$,

$$g(a) - ag'(a) = -4a^3 L''(a^2) > 0. \quad (10.5)$$

Also $F'(a) = 2 \int_0^\infty u \eta(u) \sinh(au) du > 0$, and

$$T(a) = \frac{1}{2} \iint_{\mathbb{R}^2} (u-v)^2 \eta(u) \eta(v) e^{a(u+v)} du dv > 0. \quad (10.6)$$

Thus $g'(a) = T(a)/F(a)^2 > 0$. Finally

$$S(a, 0) = F(a)^4 [g(a) - ag'(a)][g(a) + ag'(a)] > 0.$$

□

Corollary 10.2. *For distinct positive real a, b , the matrix $[\mathcal{B}(a_j, a_k)]_{j,k=1}^2$, $a_1 = a$, $a_2 = b$, is positive definite.*

Proof. Suppose $a < b$. Theorem 10.1 gives $0 < g'/g < 1/a$ along the positive ray. Integration yields $1 < g(b)/g(a) < b/a$. Put $x = b/a$ and $r = g(b)/g(a)$ in the numerator of (7.7). It becomes

$$a^2 g(a)^2 [(x-1)^2 r - x(r-1)^2] = -a^2 g(a)^2 x(r-x)(r-x^{-1}) > 0.$$

The diagonal entries are positive, giving positive definiteness. □

10.2 An exact band with strict differential positivity

Lemma 10.3. *For $a > 0$ and $t \neq 0$, $|T(a + it)| < T(a)$.*

Proof. Formula (10.6) holds with a replaced by $a + it$. The absolute value is bounded by its positive integral at a . Equality in the integral triangle inequality would require $e^{it(u+v)}$ to have one constant phase almost everywhere relative to $(u - v)^2 \eta(u) \eta(v) e^{a(u+v)} du dv$. This density is strictly positive off the diagonal, and $u + v$ varies on an open interval in each suitable open rectangle. Such a constant phase is impossible for $t \neq 0$. $\square$

Define, using real derivatives at $a > 0$,

$$R(a) = \frac{aT(a)}{F(a)F'(a)}, \quad Q(a) = \frac{F(a)F'''(a) - F'(a)F''(a)}{F(a)F'(a)}, \quad W(a) = \frac{\arccos R(a)}{\sqrt{Q(a)}}. \quad (10.7)$$

Theorem 10.1 implies $0 < R(a) < 1$. Twice differentiating (4.6) in t gives

$$Q(a) = 4 \frac{\int_{\mathbb{R}} y^2 C_a(y) dy}{\int_{\mathbb{R}} C_a(y) dy} > 0. \quad (10.8)$$

Indeed $K_{tt}(a, 0) = F'(a)F''(a) - F(a)F'''(a)$ and $K(a, 0) = F(a)F'(a)$.

Theorem 10.4 (Explicit strict region). *For every $a > 0$,*

$$|t| \leq W(a) \implies S(a, t) > 0. \quad (10.9)$$

Proof. Use the positive normalized weight $p(y) = C_a(y)/K(a, 0)$ and set $r(t) = K(a, t)/K(a, 0) = \int p(y) \cos(2ty) dy$. Weighted Cauchy–Schwarz and $\int p = 1$ give

$$|r'(t)|^2 \leq 4 \int y^2 p(y) dy \int \sin^2(2ty) p(y) dy \quad (10.10)$$

$$\leq Q(a)(1 - r(t)^2). \quad (10.11)$$

Apply this to $(1 - \varepsilon)r$, differentiate its arccosine, and integrate from zero. Letting $\varepsilon \downarrow 0$ gives

$$K(a, t) \geq K(a, 0) \cos(t\sqrt{Q(a)}), \quad |t|\sqrt{Q(a)} \leq \pi. \quad (10.12)$$

For $|t| \leq W(a)$, the right side is at least $K(a, 0)R(a) = aT(a) > 0$. If $t \neq 0$, Lemma 10.3 makes this strictly larger than $a|T(a + it)|$. Squaring yields $S(a, t) > 0$. At $t = 0$, use Theorem 10.1. $\square$

11 Exact positive factorizations and residual identities

11.1 A complete quartic-density calculation

Consider temporarily an auxiliary density $\eta_{c,b}(x) = e^{-cx^4 - bx^2}$, where $c > 0$ and $b \in \mathbb{R}$, and construct $J_{c,b}$ by (6.1).

Proposition 11.1. *The operator with kernel $J_{c,b}$ is positive semidefinite.*

Proof. In (6.2) substitute $r = v^2 - m^2$. Since $2v \, dv = dr$, expansion of the two quartics gives

$$J_{c,b}(x, y) = \eta_{c,b}(x)\eta_{c,b}(y) \int_0^\infty e^{-2cr^2-2br} e^{-4cr(x^2+y^2-xy)} \, dr. \quad (11.1)$$

Expand e^{4crxy} to obtain

$$J_{c,b}(x, y) = \int_0^\infty e^{-2cr^2-2br} \sum_{k \geq 0} \phi_{r,k}(x) \phi_{r,k}(y) \, dr, \quad (11.2)$$

$$\phi_{r,k}(x) = \eta_{c,b}(x) e^{-4crx^2} \frac{(4cr)^{k/2} x^k}{\sqrt{k!}}. \quad (11.3)$$

For fixed finitely many x, y , absolute summability under the integral follows by replacing xy by $|xy|$; the exponent $-4cr(x^2 + y^2 - |xy|)$ is nonpositive and e^{-2cr^2-2br} is integrable. Every finite quadratic form is consequently an integral of a sum of squares. Positivity on compactly supported test functions follows by approximation and then on L^2 by boundedness of the Gaussian-dominated kernel. This proves the operator assertion. $\square$

For the Gaussian density e^{-bx^2} , $b > 0$, the same substitution gives the rank-one kernel $J(x, y) = e^{-bx^2} e^{-by^2} / (2b)$.

11.2 The exact general integrand

For a positive even smooth density with the decay in Lemma 2.1, write $\eta = e^{-V}$. Then

$$J(x, y) = \eta(x)\eta(y) \int_0^\infty e^{-D_V(x,y,r)} \, dr, \quad (11.4)$$

where

$$D_V(x, y, r) = V(\sqrt{m^2 + r} + d) + V(\sqrt{m^2 + r} - d) - V(x) - V(y).$$

Evenness gives $D_V(x, y, 0) = 0$. Differentiation at zero yields the continuous extension

$$A_V(x, y) = \partial_r D_V(x, y, 0) = \frac{V'(x) + V'(y)}{x + y}, \quad A_V(x, -x) = V''(x). \quad (11.5)$$

At the nodes $(-h, 0, h)$ with coefficients $(1, -2, 1)$,

$$\sum_{j,k=1}^3 v_j v_k A_V(x_j, x_k) = 2V''(h) - \frac{6V'(h)}{h} + 4V''(0). \quad (11.6)$$

For the actual theta density, the first term of the full series and its normally convergent differentiated tail give

$$V'(h) = 2\pi e^{2h} - \frac{9}{2} + O(e^{-2h}), \quad (11.7)$$

$$V''(h) = 4\pi e^{2h} + O(e^{-2h}) \quad (h \rightarrow +\infty). \quad (11.8)$$

Indeed

$$\eta(h) = 4\pi^2 e^{9h/2 - \pi e^{2h}} \left(1 - \frac{3}{2\pi} e^{-2h} + O(e^{-3\pi e^{2h}}) \right),$$

If $R(h)$ denotes the relative remainder in this formula, then $R^{(j)}(h) = O_j(e^{2jh}e^{-3\pi e^{2h}})$ for $j = 1, 2$. These terms are $O(e^{-2h})$, as required in the displayed logarithmic derivatives. Thus (11.6) is positive for sufficiently large h . Since $\sum v_j = 0$, Taylor expansion gives

$$\sum_{j,k} v_j v_k e^{-D_V(x_j, x_k, r)} = -r \left(2V''(h) - \frac{6V'(h)}{h} + 4V''(0) \right) + O(r^2) < 0 \quad (11.9)$$

for sufficiently small positive r . This specifies the sign of these individual integrand matrices; (11.4) retains the complete integral over r .

11.3 The exact-mass autocorrelation residual

For the actual density and $a > 0$, put

$$w_a(x) = 2x \sinh(2ax), \quad q(x) = \sqrt{w_a(x)} \eta(x), \quad E_a = C_a - A[q]. \quad (11.10)$$

The square root behaves as $2\sqrt{a}|x|$ at zero, so $q \in L^1 \cap H^1$. Direct evaluation gives $E_a(0) = 0$.

Proposition 11.2. *The residual satisfies*

$$E_a''(0) = \int_{\mathbb{R}} \left[\frac{2 \sinh(2ax)}{x} + \frac{8a^2 x}{\sinh(2ax)} \right] \eta(x)^2 dx > 0, \quad (11.11)$$

where the continuous bracket value at zero is $8a$.

Proof. Differentiating the shifted density product twice and integrating by parts gives

$$C_a''(0) = -4 \int w_a(\eta')^2 + \int w_a'' \eta^2.$$

The H^1 autocorrelation identity gives $A[q]''(0) = -4\|q'\|_2^2$. Expanding the weak derivative of q and integrating the mixed term gives

$$E_a''(0) = \int \left(\frac{(w_a')^2}{w_a} - w_a'' \right) \eta^2.$$

The possible quotient singularity at zero is removable. Direct differentiation of w_a reduces its bracket to the bracket in (11.11), whose terms are positive. Decay eliminates the boundary terms; the factors $w_a'(0) = 0$ also eliminate the split-origin boundary term. $\square$

Every continuous positive-definite function E obeys $|E(y)| \leq E(0)$, as follows from its two-node matrix. Hence a positive-definite residual with $E(0) = 0$ must vanish identically. Proposition 11.2 fully determines why this particular residual cannot be represented by additional positive autocorrelation squares.

12 Complete theta sums and their exact boundary cancellations

For $u \geq 0$ let

$$f_n(u) = e^{u/2 - \beta_n e^{2u}} (4\beta_n^2 e^{4u} - 6\beta_n e^{2u}), \quad \beta_n = \pi n^2, \quad F_n(z) = 2 \int_0^\infty f_n(u) \cosh(zu) du. \quad (12.1)$$

Then $F = \sum_n F_n$ with locally normal convergence of every fixed derivative. For any fixed A, j , and M ,

$$\sum_n \int_0^\infty (1+u)^M e^{Au} |f_n^{(j)}(u)| du < \infty. \quad (12.2)$$

To verify it, substitute $v = \pi n^2 e^{2u}$ in each differentiated term. The resulting polynomial and logarithmic factors multiply e^{-v} on $v \geq \pi n^2$, giving a summable exponential tail in n^2 .

Lemma 12.1 (Boundary derivative asymptotic). *Let f be real smooth on $[0, \infty)$ with every derivative integrable against every fixed positive exponential weight. Suppose its first nonzero odd derivative at zero is $d = f^{(2r-1)}(0)$, $r \geq 1$. For $G(z) = 2 \int_0^\infty f(u) \cosh(zu) du$ and fixed $a > 0$,*

$$\operatorname{Re}(G'(a+it)\overline{G(a+it)}) = -\frac{8rad^2}{(a^2+t^2)^{2r+1}} + O_{a,f}(|t|^{-4r-4}). \quad (12.3)$$

Proof. Integrate the two exponential integrals by parts to sufficiently high order. Even boundary derivatives cancel, leaving

$$G(z) = 2dz^{-2r} + 2f^{(2r+1)}(0)z^{-2r-2} + \dots$$

Every desired finite remainder bound follows from the assumed integrable derivatives, uniformly on compact intervals of $\operatorname{Re} z$. Differentiate using the same integrations for $uf(u)$; equivalently use Cauchy estimates after taking additional remainder orders. The leading product is $-8rd^2/(z|z|^{4r})$, whose real part is the displayed leading term. The next cross terms have odd total inverse degree $4r+3$ with real coefficients. They are purely imaginary at $z = it$ and have real part $O_a(|t|^{-4r-4})$ at $a+it$. All further terms are smaller. $\square$

For the theta terms,

$$d_n = f'_n(0) = e^{-\beta_n} \beta_n (-8\beta_n^2 + 30\beta_n - 15), \quad \sum_{n \geq 1} d_n = \eta'(0) = 0. \quad (12.4)$$

For $n \geq 2$, $d_n < 0$. Hence $d^{[N]} = \sum_{n \leq N} d_n > 0$ for every finite N . If $F^{[N]} = \sum_{n \leq N} F_n$, Lemma 12.1 gives

$$\operatorname{Re}((F^{[N]})'(a+it)\overline{F^{[N]}(a+it)}) = -\frac{8a(d^{[N]})^2}{(a^2+t^2)^3} + O_{a,N}(|t|^{-8}). \quad (12.5)$$

The threshold implicit in this asymptotic can depend on N .

For the infinite diagonal and off-diagonal contributions, define

$$D(a, t) = \sum_n \operatorname{Re}(F'_n \overline{F_n}), \quad O(a, t) = \sum_{m \neq n} \operatorname{Re}(F'_m \overline{F_n}),$$

with evaluation at $a+it$. Equation (12.2) justifies absolute summation of the repeated integration-by-parts bounds. Thus

$$D(a, t) = -8at^{-6} \sum_n d_n^2 + O_a(|t|^{-8}), \quad (12.6)$$

$$O(a, t) = +8at^{-6} \sum_n d_n^2 + O_a(|t|^{-8}). \quad (12.7)$$

For the second identity, the complete smooth density on $\mathbb{R}$ gives $F(a+it), F'(a+it) = O_{a,M}((1+|t|)^{-M})$ for every M by integration by parts. Since $K = D + O$, subtract (12.6). Algebraically the leading cancellation is also $(\sum_n d_n)^2 = 0$. The complete kernel therefore preserves interactions that are altered by taking a finite reflected sum before the spectral transform.

13 Heat deformation and differential continuation

For real λ define

$$\eta_\lambda(u) = e^{\lambda u^2} \eta(u), \quad F_\lambda(z) = \int_{\mathbb{R}} \eta_\lambda(u) e^{zu} du. \quad (13.1)$$

Superexponential decay makes these definitions and all fixed derivatives locally uniform for (λ, z) in compact sets. Construct $C_{a,\lambda}$ and K_λ by the earlier formulas. Direct differentiation gives

$$\partial_\lambda F_\lambda = F''_\lambda, \quad (13.2)$$

$$\partial_\lambda C_{a,\lambda} = \frac{1}{2} \partial_a^2 C_{a,\lambda} + 2y^2 C_{a,\lambda}, \quad (13.3)$$

$$\partial_\lambda K_\lambda = \frac{1}{2} (\partial_a^2 - \partial_t^2) K_\lambda. \quad (13.4)$$

The second identity uses $(x+y)^2 + (x-y)^2 = 2x^2 + 2y^2$; Fourier transformation gives the third. If

$$\gamma_\lambda(n) = \frac{n!}{(2n)!} \int u^{2n} e^{\lambda u^2} \eta(u) du, \quad J_\lambda^{d,n}(X) = \sum_{j=0}^d \binom{d}{j} \gamma_\lambda(n+j) X^j,$$

then

$$\partial_\lambda \gamma_\lambda(n) = (4n+2) \gamma_\lambda(n+1), \quad (13.5)$$

$$\partial_\lambda J_\lambda^{d,n} = (4n+2) J_\lambda^{d,n+1} + 4X \partial_X J_\lambda^{d,n+1}. \quad (13.6)$$

These equations retain the exact coefficients of both the width and frequency derivatives.

Proposition 13.1 (A bounded-domain continuation theorem). *Let D be a bounded domain with $\overline{D} \subset \mathbb{H}$. Suppose g is holomorphic near $\overline{D}$ and $u = \operatorname{Re} g > 0$ there. If $a|g'(a+it)|/u(a,t) \leq 1$ on ∂D , the same inequality holds in D .*

Proof. Where $g' \neq 0$, set $q = \log(a|g'|/u)$. Since u is harmonic and $|\nabla u|^2 = |g'|^2$,

$$\Delta q = -\frac{1}{a^2} + \frac{|g'|^2}{u^2} = \frac{e^{2q} - 1}{a^2}. \quad (13.7)$$

If the continuous ratio $a|g'|/u$ exceeded one inside, it would attain a maximum greater than one at an interior point. At that point $g' \neq 0$, q is smooth, and $\Delta q \leq 0$, contradicting (13.7). Zeros of g' have ratio zero and cannot be such a maximum. $\square$

For $g = F'/F$, the hypotheses include nonvanishing of F on the specified neighborhood and positivity of $\operatorname{Re}(F'/F)$. Theorem 7.2 provides the independent local exclusion argument directly from S .

14 Exact arithmetic reconstruction and source correspondence

14.1 Prime coordinates and the complete scalar function

For $n \geq 2$, unique factorization gives

$$n = \prod_p p^{\nu_p(n)}, \quad \Omega(n) = \sum_p \nu_p(n), \quad \log n = \sum_p \nu_p(n) \log p. \quad (14.1)$$

The prime-exponent vector is finite for each n . Its scalar transform is

$$n^{-s} = \exp\left(-s \sum_p \nu_p(n) \log p\right).$$

In the prime-coordinate reconstruction let $\Delta W_n(s) = \Omega(n)n^{-s}$. The unit and remainder are retained in the exact identity, for integer $N \geq 1$, $\operatorname{Re} s > 0$, $s \neq 1$,

$$\zeta(s) = 1 + \sum_{n=2}^N \frac{\Delta W_n(s)}{\Omega(n)} + \frac{N^{1-s}}{s-1} - s \int_N^\infty \{u\} u^{-s-1} du. \quad (14.2)$$

For $\operatorname{Re} s > 1$ this follows by Stieltjes integration of the counting function on (N, ∞) and writing $[u] = u - \{u\}$. The remainder integral is absolutely convergent for $\operatorname{Re} s > 0$ and locally holomorphic there, giving the stated continuation. Substitution in (1.1) gives the same completed function as Theorem 2.2 by analytic continuation.

For a partition $\mathcal{P}$ of a finite ordered index set into nonempty blocks, with $\operatorname{right}(B) = \max B$, define

$$\mathbf{M}_{\mathcal{P}} a = \sum_{B \in \mathcal{P}} \left(\sum_{n \in B} a(n) \right) e_{\operatorname{right}(B)}, \quad \mathbf{R}_{\mathcal{P}} a = a - \mathbf{M}_{\mathcal{P}} a.$$

Then $a = \mathbf{M}_{\mathcal{P}} a + \mathbf{R}_{\mathcal{P}} a$ exactly. Any linear transform of the indexed arithmetic data retains this identity by applying the transform to both terms. The theta and Gram constructions above are attached to the complete function reconstructed in (14.2).

For $\operatorname{Re} s > 1$, the prime phase identity is

$$\frac{\zeta'(s)}{\zeta(s)} = - \sum_{n \geq 1} \frac{\Lambda(n)}{n^s}, \quad \Lambda(p^k) = \log p, \quad \Lambda(n) = 0 \text{ otherwise.} \quad (14.3)$$

The logarithmic derivative of the completion is therefore

$$\frac{\xi'(s)}{\xi(s)} = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \frac{\Gamma'(s/2)}{\Gamma(s/2)} - \sum_{n \geq 1} \frac{\Lambda(n)}{n^s}, \quad \operatorname{Re} s > 1. \quad (14.4)$$

Here each prime-power phase is $e^{-it \log n}$. Formula (14.4) records its convergence domain; the complete theta formula supplies the entire representation used in the proofs.

14.2 The completed prime form on one vertical line

This subsection identifies the complete arithmetic form defined below and developed in Appendix A directly with the two-point kernel of Section 6. It retains the prime-power, pole, and gamma terms together. Fix $b > 1/2$, put $z_j = b + it_j$, and set

$$f(u) = \mathbf{1}_{[0, \infty)}(u) \sum_{j=1}^N d_j e^{-z_j u}, \quad R_f(v) = \int_{\mathbb{R}} f(u+v) \overline{f(u)} du, \quad E_f = \|f\|_2^2. \quad (14.5)$$

All coefficients $d_j \in \mathbb{C}$ and frequencies $t_j \in \mathbb{R}$ are allowed. For any compact smooth q , use the same definitions of R_q and E_q . The completed distribution in Appendix A gives

$$W(h) = \mathcal{F}_h(0) + \mathcal{F}_h(1) - P(h) - A_\infty(h), \quad \mathfrak{W}(q, q) = W(h),$$

with $h = q * q^\star$ and $q^\star(u) = \overline{q(-u)}$. The convolution and correlation definitions give

$$h(v) = R_q(v), \quad h(-v) = \overline{R_q(v)}, \quad h(0) = E_q.$$

Splitting the endpoint integrals at the origin and pairing the two prime-power evaluations therefore yields

$$\begin{aligned} \mathcal{F}_h(0) + \mathcal{F}_h(1) &= 2 \int_0^\infty (e^{v/2} + e^{-v/2}) \operatorname{Re} R_q(v) \, dv, \\ P(h) &= 2 \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \operatorname{Re} R_q(\log n), \\ A_\infty(h) &= (\log(4\pi) + \gamma) E_q + 2 \int_0^\infty \frac{e^{-v/2} \operatorname{Re} R_q(v) - e^{-v} E_q}{1 - e^{-2v}} \, dv. \end{aligned}$$

Substitution in the completed distribution gives

$$\begin{aligned} \mathfrak{W}(q, q) &= 2 \int_0^\infty (e^{v/2} + e^{-v/2}) \operatorname{Re} R_q(v) \, dv - 2 \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \operatorname{Re} R_q(\log n) \\ &\quad - (\log(4\pi) + \gamma) E_q - 2 \int_0^\infty \frac{e^{-v/2} \operatorname{Re} R_q(v) - e^{-v} E_q}{1 - e^{-2v}} \, dv. \end{aligned} \quad (14.6)$$

Here γ is Euler's constant. The Hermitian form on different tests is obtained by polarization. The paired numerator in the last integral is part of the formula and is kept intact at zero.

Proposition 14.1 (Exact arithmetic-to-kernel identification). *The expression (14.6) converges for every packet (14.5); denote its value by $\mathfrak{W}_{\text{ext}}[f]$. Let $g = F'/F$ on $\operatorname{Re} z > 1/2$. Then*

$$\mathfrak{W}_{\text{ext}}[f] = \sum_{j,k=1}^N d_j \overline{d_k} \frac{g(z_j) + \overline{g(z_k)}}{z_j + \overline{z_k}}. \quad (14.7)$$

Proof. The ordinary Euler half-plane gives $F(z) \neq 0$ for $\operatorname{Re} z > 1/2$. The finite correlation expansion is

$$R_f(v) = \sum_{j,k} d_j \overline{d_k} \frac{e^{-z_j v}}{z_j + \overline{z_k}} \quad (v \geq 0). \quad (14.8)$$

Insert (14.4) in the right side of (14.7). The Laplace representations of $(z + 1/2)^{-1}$ and $(z - 1/2)^{-1}$ give the first integral in (14.6). The prime series gives its second term. For the gamma factor use the convergent paired integral

$$\frac{1}{2} \psi\left(\frac{z + 1/2}{2}\right) = -\frac{\gamma}{2} + \int_0^\infty \frac{e^{-2v} - e^{-(z+1/2)v}}{1 - e^{-2v}} \, dv, \quad \psi = \Gamma'/\Gamma, \quad (14.9)$$

which follows from [3]. Its contribution, including the factor $\pi^{-s/2}$, is

$$-(\log \pi + \gamma) E_f + 2 \int_0^\infty \frac{e^{-2v} E_f - e^{-v/2} \operatorname{Re} R_f(v)}{1 - e^{-2v}} \, dv. \quad (14.10)$$

This agrees exactly with the last two terms of (14.6), since

$$\int_0^\infty \frac{e^{-v} - e^{-2v}}{1 - e^{-2v}} \, dv = \int_0^\infty \frac{e^{-v}}{1 + e^{-v}} \, dv = \log 2.$$

Thus the normalizations of the earlier completed form and the present phase calculation coincide.

For convergence, $R_f(v) = O_f(e^{-bv})$ and $R_f(v) = E_f + O_f(v)$ near zero. The paired gamma numerator is therefore $O_f(v)$ at zero and has an integrable tail. The pole integrals converge because $b > 1/2$, and the arithmetic series converges absolutely by $\sum_{n \geq 2} \Lambda(n)n^{-b-1/2} < \infty$. These estimates justify the substitutions and prove the proposition. $\square$

Lemma 14.2 (Transfer from compact smooth tests). *For every packet (14.5), there are $q_m \in C_c^\infty((0, \infty))$ such that*

$$\mathfrak{W}(q_m, q_m) \longrightarrow \mathfrak{W}_{\text{ext}}[f]. \quad (14.11)$$

In particular, positivity of the completed form on compact smooth tests passes to the packets in Proposition 14.1.

Proof. Write $f_+(u) = \sum_j d_j e^{-z_j u}$ on $u \geq 0$, and put

$$M = \sum_j |d_j|, \quad D = \int_0^\infty |f'_+(u)| du < \infty.$$

Choose smooth cutoffs $0 \leq \chi_m \leq 1$, vanishing near zero and outside a compact interval, which converge pointwise to one on $(0, \infty)$ and satisfy $\int |\chi'_m| \leq 2$. Set $q_m = \chi_m f_+$ and extend by zero. Then $q_m \rightarrow f$ in $L^2(\mathbb{R})$, while

$$\|q_m\|_\infty \leq M, \quad \text{Var}(q_m) \leq D + 2M, \quad |R_{q_m}(v)| \leq \frac{M^2}{2b} e^{-bv} \quad (v \geq 0).$$

The energies and correlations converge to those of f . To control the origin of the gamma integral uniformly, translation on the full line gives

$$\begin{aligned} 0 \leq E_{q_m} - \text{Re } R_{q_m}(v) &= \frac{1}{2} \|q_m(\cdot + v) - q_m\|_2^2 \\ &\leq \frac{1}{2} \|q_m(\cdot + v) - q_m\|_\infty \|q_m(\cdot + v) - q_m\|_1 \\ &\leq M(D + 2M)v. \end{aligned} \quad (14.12)$$

The last inequality follows by integrating $|q'_m|$ over translated intervals and applying Fubini. The numerator of the last integral in (14.6) is consequently bounded by a constant times v on $0 < v \leq 1$, independently of m . Its tail is uniformly dominated by exponentials. The pole terms have the same property since $b > 1/2$. Finally,

$$\frac{\Lambda(n)}{\sqrt{n}} |R_{q_m}(\log n)| \leq \frac{M^2}{2b} \Lambda(n) n^{-b-1/2},$$

whose sum is finite. Dominated convergence in each integral and in the series proves (14.11). This argument includes the boundary at zero of the half-line packet. $\square$

14.3 Minimal insertion of the prime-indexed positivity theorem

Theorem A.1, Lemma A.5, and Theorem B.1 give the definitions and proofs of the prime-indexed construction used here, including the following facts. Let $\mathcal{D} = C_c^\infty(\mathbb{R}; \mathbb{C})$ and $\alpha_k = \frac{1}{2} \log p_k$, where p_k is the k th prime.

- (i) The spaces $\mathcal{D}_{\alpha_k} = C_c^\infty((-\alpha_k, \alpha_k))$ contain every compact smooth test at some stage. Positivity on a later full space restricts to every smaller support space.

- (ii) There are nested finite-dimensional spaces $X_k \subset \mathcal{D}_{\alpha_k}$ whose union is determining in the fixed-support test-function topology.
- (iii) Suzuki's source-defined linear map L_0 supplies the positive metric $\mathcal{S}(q, r) = \langle L_0 q, L_0 r \rangle$, with $\mathcal{S}(q, q) > 0$ for $q \neq 0$ in $\mathcal{D}$. The positive successor theorem holds for the entire prime-indexed family in this metric.

The prime-power packets $\{\ell^m : \ell \text{ prime}, m \geq 1, p_k \leq \ell^m < p_{k+1}\}$ partition all prime powers, by Corollary A.4. Thus the complete arithmetic series in (14.6) is covered by the existing index.

The coefficient quantifier is already absorbed by the metric identity: for every finite family $q_j \in \mathcal{D}$,

$$\mathcal{S}\left(\sum_j c_j q_j, \sum_j c_j q_j\right) = \left\| \sum_j c_j L_0 q_j \right\|^2 \geq 0. \quad (14.13)$$

No separate enumeration of prime-node combinations enters this theorem.

Theorem 14.3 (Direct reuse of a positive completed metric). *Use the determining family X_k and positive metric $\mathcal{S}$ just described. Suppose the following identification is established on that family:*

$$\mathfrak{W}(q, r) = \mathcal{S}(q, r) \quad (q, r \in \bigcup_k X_k). \quad (14.14)$$

Then $\mathcal{J} \succeq 0$ and the complete theta kernels C_a satisfy Hypothesis H for every $a > 0$. More generally, it suffices that the authentic completed form $\mathfrak{W}$ be nonnegative on every X_k .

Proof. Identity (14.14) and (14.13) give positivity of $\mathfrak{W}$ on $\bigcup_k X_k$. The form is continuous in the fixed-support smooth topology: the prime sum is finite on a fixed compact support, the pole terms are continuous integrals, and the paired gamma numerator is uniformly $O(v)$ at zero under such convergence. The determining property therefore gives $\mathfrak{W}(q, q) \geq 0$ for every $q \in \mathcal{D}$.

Fix any $b > 1/2$. Lemma 14.2 and Proposition 14.1 give positive semidefiniteness on the line $\text{Re } z = b$ of

$$P_g(z, w) = \frac{g(z) + \overline{g(w)}}{z + \overline{w}}.$$

The displayed quadratic convention in (14.7) uses $d_j \overline{d_k}$; conjugating the arbitrary coefficient vector gives the usual conjugate-first convention. On this line,

$$\mathcal{B}(z, w) = F(z) \overline{F(w)} P_g(z, w). \quad (14.15)$$

Thus every finite $\mathcal{B}$ matrix there is a diagonal congruence of a positive matrix. Theorem 6.2 gives $\mathcal{J} \succeq 0$, and Theorem 8.2 gives Hypothesis H. The proof uses only nonnegativity on the determining family, which proves the more general assertion as well. $\square$

Theorem 14.4 (Transfer from the full spatial matrix inequality). *Use the following full matrix statement, as supplied for this transfer: for every pairwise distinct finite set $x_1, \dots, x_N \in \mathbb{R}$ and every nonzero $c \in \mathbb{C}^N$,*

$$\sum_{j,k=1}^N \overline{c_j} c_k J(x_j, x_k) > 0. \quad (14.16)$$

Then $\mathcal{J} \succeq 0$ on $L^2(\mathbb{R})$ and Hypothesis H holds for every $a > 0$. No zero expansion or RH-dependent orthogonality is used.

Proof. For a compactly supported step function h , approximate $\langle h, \mathcal{J}h \rangle$ by the corresponding finite Riemann sums. After coincident nodes are combined, every sum is nonnegative by (14.16). Continuity and the Gaussian bound (6.3) give the integral limit. Compactly supported step functions are dense in $L^2(\mathbb{R})$, while $\mathcal{J}$ is bounded, so $\mathcal{J} \succeq 0$ on all of $L^2(\mathbb{R})$.

Theorem 6.2 now gives positive definiteness of $\mathcal{B}$ on $\mathbb{H}$. On its diagonal, (6.6) gives

$$0 \leq \mathcal{B}(a + it, a + it) = K(a, t)/a, \quad a > 0, t \in \mathbb{R}.$$

Thus $K(a, t) \geq 0$ everywhere in the target half-plane. The exact Fourier identity (4.6) and Lemma 4.3 imply that C_a is positive definite for every $a > 0$. This is Hypothesis H. $\square$

Appendix E gives the finite Riemann sums, exponential cutoffs, and full-line remainder bound used in this transfer.

Corollary 14.5 (Spatial route to Theorem 14.3). *Assuming the full spatial matrix inequality (14.16), the conclusion of Theorem 14.3 holds without taking (14.14) as a premise.*

Proof. The preceding theorem supplies directly the intermediate conclusion $\mathcal{J} \succeq 0$ and then Hypothesis H. Hence the Weil–Suzuki interface in Theorem 14.3 is no longer needed to reach its conclusion. $\square$

Remark 14.6 (Role of the inserted identification). The exact arithmetic match, the approximation lemma, the determining family, and Suzuki’s positive metric are established results. Equation (14.14) is the additional hypothesis of the prime-metric route in Theorem 14.3. Equations (B.13)–(B.14) state the exact prime-successor identities. Theorem C.4 and Corollary C.5 include the defect-energy vanishing criterion and its proof. These statements retain the metric identification and the zero-defect bound as the conditions specified there. The insertion identifies exactly how a proof of that equality would enter the prime-metric route. Theorem 14.4 reaches the target through the spatial kernel instead, so this equality is not a premise of that conditional spatial proof of H. Its input is the full matrix inequality (14.16).

14.4 Location of the included derivations

The manuscript uses one density normalization and one centered coordinate throughout. The following table locates the supporting mathematics inside this paper.

Construction	Exact object	Internal location
Modular density	$\phi = \eta$, modular density and positivity	Section 2
Full-plane derivative lift	entire Taylor sequence, $G = cc^*$, translations	Section 3
Complete kernels and increments	$A_a, C_a, D_{a,b}$, full Fourier identities	Section 4
Transformed-kernel positivity	positive matrices $[K(a, t_j - t_k)]$	Theorem 4.4
All-degree signed moments	exponential density, both parity blocks	Section 9
Structural exclusion	$\mathcal{B}, J, S$, finite obstruction	Sections 6–8
Jensen and complex bands	γ_n, R, Q, W	Section 10
Prime reconstruction	full prime-exponent data, unit and tail	Section 14
Prime-anchored Weil ladder	determining spaces and positive Suzuki metric	Appendices A–B
Completed prime-phase identity	exact Weil form and compact-test transfer	Section 14.2

The full-plane lift theorem (3.3), the pointwise density theorem of Lemma 2.1, and the transformed-kernel theorem (4.13) are unconditional results. The principal theorem proves that $\mathbf{P}$ is equivalent to critical-line localization; it does not adopt $\mathbf{P}$ as a premise of the manuscript. The real-ray and complex-band results additionally use the established coefficient theorem (10.2). No finite numerical moment certificate is a premise of the principal equivalence proof.

The complete transformed-kernel proof is Theorem 4.4. The accumulated-positivity proof is included in Appendix D; its conclusion has the exact form

$$\int_0^T K(a, t) dt > 0 \quad (a > 0, T > 0). \quad (14.17)$$

The normalized quantities used in that derivation obey $\mathcal{K}_a(y) = C_a(y)/(2a)$ and $Q(a, t) = K(a, t)/a$. Equation (14.17) records that result with its original integral quantifier; it is not used in the equivalence proof.

The mechanism of that proof also identifies the distinct pointwise test. For $t, T > 0$, evenness and integration by parts give

$$I_a(T) = \int_0^\infty \frac{C_a(y)}{y} \sin(2Ty) dy, \quad (14.18)$$

$$K(a, t) = \frac{1}{t} \int_0^\infty h_a(y) \sin(2ty) dy, \quad h_a = -C'_a. \quad (14.19)$$

Appendix D proves the decrease of $C_a(y)/y$ used in the first formula and includes the theta log-concavity calculation giving

$$C''_a(0) = 4 \int_{\mathbb{R}} x \sinh(2ax) \eta(x)^2 (\log \eta)''(x) dx < 0. \quad (14.20)$$

Consequently $h_a(0) = 0$ and $h'_a(0) > 0$: the weight in the second formula is increasing near zero. Choose $\delta > 0$ on which it is strictly increasing and let $b = \pi/(2t)$ satisfy $2b < \delta$. The first paired sine half-waves then contribute

$$\int_0^b [h_a(y) - h_a(y + b)] \sin(2ty) dy < 0. \quad (14.21)$$

The remaining half-waves determine the total sign. This calculation concerns the actual theta kernel and shows why the proved cumulative argument does not give a termwise positive proof of $K(a, t) \geq 0$.

15 Positive prime-color forms and exact recombination

This section supplies the positive arithmetic forms, their complete prime-power factors, and the exact relation between additive and tensor recombination. All finite coefficient combinations are covered by the displayed identities. The final subsection identifies the scalar phase orbit on which the completed theta form is evaluated.

15.1 The one-prime valuation operator and its positive factor

Fix a nonempty finite set of primes $\mathcal{P}$ and $\sigma > 0$. Write

$$r_p = p^{-\sigma}, \quad c_p = \frac{1 - r_p}{1 + r_p}, \quad \Gamma_{\mathcal{P}} = \mathbb{Z}^{\mathcal{P}}. \quad (15.1)$$

On $\ell^2(\Gamma_{\mathcal{P}})$ use the inner product $\langle u, v \rangle = \sum_{\nu} \overline{u_{\nu}} v_{\nu}$ and the commuting unitary shifts $(U_p v)(\nu) = v(\nu - e_p)$. The Fourier transform

$$\widehat{v}(\theta) = \sum_{\nu \in \Gamma_{\mathcal{P}}} v_{\nu} e^{i\nu \cdot \theta}$$

is unitary onto $L^2(\mathbb{T}^{\mathcal{P}}, dm)$, with normalized Haar measure dm , and sends U_p to multiplication by $e^{i\theta_p}$. The same notation on a single coordinate means an operator on $\ell^2(\mathbb{Z})$.

Define the complete one-prime covariance by the operator-norm convergent series

$$G_{p,\sigma} = I + \sum_{k \geq 1} r_p^k (U_p^k + U_p^{*k}). \quad (15.2)$$

Its matrix on the exponent line is $r_p^{|a-b|}$, and its multiplier is

$$P_{r_p}(\theta) = \sum_{k \in \mathbb{Z}} r_p^{|k|} e^{ik\theta} = \frac{1 - r_p^2}{|1 - r_p e^{i\theta}|^2}, \quad \min_{\theta} P_{r_p}(\theta) = c_p. \quad (15.3)$$

Proposition 15.1 (Complete local factorization). *Put*

$$\begin{aligned} B_{p,\sigma} &= \sqrt{1 - r_p^2} (I - r_p U_p)^{-1}, \\ X_{p,\sigma} &= \sqrt{\frac{r_p(1 - r_p)}{1 + r_p}} (I + U_p)(I - r_p U_p)^{-1}. \end{aligned} \quad (15.4)$$

Then

$$G_{p,\sigma} = B_{p,\sigma}^* B_{p,\sigma} = c_p I + X_{p,\sigma}^* X_{p,\sigma}. \quad (15.5)$$

In particular, $A_{p,\sigma} := G_{p,\sigma} - c_p I$ is positive.

Proof. The inverse in (15.4) is the norm-convergent Neumann series $\sum_{a \geq 0} r_p^a U_p^a$. On the circle its first factor has squared modulus P_{r_p} . Direct subtraction gives

$$P_r(\theta) - \frac{1 - r}{1 + r} = \frac{r(1 - r)}{1 + r} \frac{|1 + e^{i\theta}|^2}{|1 - r e^{i\theta}|^2},$$

which is the squared modulus of the second factor. The unitary Fourier transform proves the operator identities. $\square$

The two complementary filters are

$$V_p = \frac{I - r_p U_p}{1 + r_p}, \quad W_p = \frac{\sqrt{r_p}(I + U_p)}{1 + r_p}. \quad (15.6)$$

Expansion of their squares gives

$$V_p^* V_p + W_p^* W_p = I, \quad V_p B_{p,\sigma} = \sqrt{c_p} I, \quad W_p B_{p,\sigma} = X_{p,\sigma}. \quad (15.7)$$

These identities retain all prime powers. Indeed the spectral factor is

$$\frac{\sqrt{1 - r_p^2}}{1 - r_p z} = \sum_{a \geq 0} \sqrt{1 - r_p^2} r_p^a z^a. \quad (15.8)$$

At $\sigma = 1/2$ the squared coefficient at depth a is $(1 - p^{-1})p^{-a}$; the positive-depth sum is $1/p$, and the depth-zero coefficient has squared modulus $1 - p^{-1}$.

The first identity in (15.7) also holds with any isometry S_p in place of U_p . In particular, the delay

$$(S_p f)(u) = \begin{cases} f(u - \log p), & u \geq \log p, \\ 0, & 0 \leq u < \log p, \end{cases}$$

is an isometry on $L^2(0, \infty)$. For a finite ordered list of primes, set $f_0 = f$, $f_j = V_{p_j} f_{j-1}$, and $h_j = W_{p_j} f_{j-1}$, storing each h_j in its own orthogonal channel. Telescoping gives

$$\|f_n\|_2^2 + \sum_{j=1}^n \|h_j\|_2^2 = \|f\|_2^2. \quad (15.9)$$

This statement concerns the full half-line norm and a finite iteration.

15.2 Additive and tensor recombination

Embed the local factors in their corresponding coordinates, and define

$$\mathcal{K}_{\mathcal{P},\sigma}^{\log} = \sum_{p \in \mathcal{P}} (\log p) A_{p,\sigma}^{(p)}, \quad \mathcal{K}_{\mathcal{P},\sigma}^{\text{val}} = \bigotimes_{p \in \mathcal{P}} G_{p,\sigma}. \quad (15.10)$$

For $S \subseteq \mathcal{P}$, let $X_S = \bigotimes_{p \in S} X_{p,\sigma} \otimes \bigotimes_{q \notin S} I$, with $X_\emptyset = I$.

Theorem 15.2 (Every retained color is a square). *For every $v \in \ell^2(\Gamma_{\mathcal{P}})$,*

$$\langle v, \mathcal{K}_{\mathcal{P},\sigma}^{\log} v \rangle = \sum_{p \in \mathcal{P}} (\log p) \|X_{p,\sigma}^{(p)} v\|_2^2, \quad (15.11)$$

$$\langle v, \mathcal{K}_{\mathcal{P},\sigma}^{\text{val}} v \rangle = \sum_{S \subseteq \mathcal{P}} \left(\prod_{q \notin S} c_q \right) \|X_S v\|_2^2 \geq \left(\prod_{p \in \mathcal{P}} c_p \right) \|v\|_2^2. \quad (15.12)$$

Proof. Equation (15.11) follows from $A_p = X_p^* X_p$. Expand every factor of $\bigotimes_p (c_p I + X_p^* X_p)$. Operators on different coordinates commute, and the term indexed by S is $(\prod_{q \notin S} c_q) X_S^* X_S$. The empty-set term gives the displayed lower bound. All products and sums here are finite. $\square$

The tensor covariance has matrix entries

$$K_{\mathcal{P},\sigma}^{\text{val}}(\nu, \mu) = \prod_{p \in \mathcal{P}} p^{-\sigma|\nu_p - \mu_p|}. \quad (15.13)$$

The additive covariance has diagonal $\sum_{p \in \mathcal{P}} 2(\log p)r_p/(1 + r_p)$ and off-diagonal entries

$$K_{\mathcal{P},\sigma}^{\log}(\nu, \mu) = \begin{cases} (\log p)r_p^{|k|}, & \nu - \mu = ke_p, \ k \neq 0, \\ 0, & \nu - \mu \text{ has at least two nonzero coordinates.} \end{cases} \quad (15.14)$$

Its multiplier is

$$D_{\mathcal{P},\sigma}(\theta) = \sum_{p \in \mathcal{P}} (\log p)(P_{r_p}(\theta_p) - c_p) \geq 0. \quad (15.15)$$

It vanishes only at $(\pi, \dots, \pi)$, a Haar-null set. Thus its quadratic form is strictly positive on every nonzero vector, in particular on every nonzero finite coefficient array. The tensor floor in (15.12) is asserted for the stated finite prime set; no positive lower bound independent of that set is used.

15.3 The logarithmic and exponential relation

Set

$$c_{\mathcal{P},\sigma} = \prod_{p \in \mathcal{P}} c_p, \quad V_{\mathcal{P},\sigma} = \prod_{p \in \mathcal{P}} \frac{I - r_p U_p}{1 + r_p}.$$

All factors are commuting invertible Fourier multipliers. The local identity (15.7) implies

$$V_{\mathcal{P},\sigma}^* V_{\mathcal{P},\sigma} = c_{\mathcal{P},\sigma} (\mathcal{K}_{\mathcal{P},\sigma}^{\text{val}})^{-1}. \quad (15.16)$$

Proposition 15.3 (Scale integration reconstructs the full tensor). *On the full exponent lattice,*

$$\mathcal{K}_{\mathcal{P},\sigma}^{\log} = \partial_{\sigma} \log(V_{\mathcal{P},\sigma}^* V_{\mathcal{P},\sigma}) = -\partial_{\sigma} \log(c_{\mathcal{P},\sigma}^{-1} \mathcal{K}_{\mathcal{P},\sigma}^{\text{val}}), \quad (15.17)$$

$$c_{\mathcal{P},\sigma}^{-1} \mathcal{K}_{\mathcal{P},\sigma}^{\text{val}} = \exp \left(\int_{\sigma}^{\infty} \mathcal{K}_{\mathcal{P},u}^{\log} du \right). \quad (15.18)$$

The integral converges in operator norm.

Proof. For one prime, differentiating the logarithm of its positive multiplier gives

$$\partial_{\sigma} \log \frac{|1 - r_p e^{i\theta}|^2}{(1 + r_p)^2} = (\log p) \left(\frac{1 - r_p^2}{|1 - r_p e^{i\theta}|^2} - c_p \right).$$

Summing proves (15.17). For a fixed finite prime set, each summand has norm $O(p^{-u} \log p)$ as $u \rightarrow \infty$; thus the integral converges. The normalized tensor tends to I in norm at infinity. Integrating the logarithmic identity and exponentiating gives (15.18). $\square$

In convolution notation let $H_{\mathcal{P},\sigma} = \int_{\sigma}^{\infty} \mathcal{K}_{\mathcal{P},u}^{\log} du$. It is an $\ell^1(\Gamma_{\mathcal{P}})$ kernel, with

$$H_{\mathcal{P},\sigma}(0) = 2 \sum_{p \in \mathcal{P}} \log(1 + r_p), \quad H_{\mathcal{P},\sigma}(\pm ke_p) = \frac{r_p^k}{k} \quad (k \geq 1), \quad (15.19)$$

and zero at all other nonzero displacements. Indeed $\int_{\sigma}^{\infty} (\log p) p^{-ku} du = p^{-k\sigma}/k$. Consequently

$$K_{\mathcal{P},\sigma}^{\text{val}} = c_{\mathcal{P},\sigma} \exp_{*\Gamma_{\mathcal{P}}} H_{\mathcal{P},\sigma}, \quad \exp_* H = \delta_0 + \sum_{j \geq 1} \frac{H^{*j}}{j!}. \quad (15.20)$$

The last series converges in ℓ^1 . Its convolution powers generate every mixed-color displacement. This identity is established before finite matrix compression; it is not an entrywise exponential.

15.4 The integer carrier and its prime-power tangent

For $z \in \mathbb{C}$ define

$$J_z(n) = \sum_{d|n} \mu(d) (n/d)^z = n^z \prod_{p|n} (1 - p^{-z}), \quad J_z(1) = 1. \quad (15.21)$$

Here J_z denotes the arithmetic Jordan function, distinct from the spatial kernel $J(x, y)$. Dirichlet convolution is $(a *_D b)(n) = \sum_{d|n} a(d)b(n/d)$.

For each prime, its formal local series is

$$\sum_{a \geq 0} J_z(p^a) X^a = \frac{1 - X}{1 - p^z X}, \quad \log \frac{1 - X}{1 - p^z X} = \sum_{a \geq 1} \frac{p^{az} - 1}{a} X^a.$$

Unique factorization therefore gives the coefficientwise identities

$$J_z = \exp_{*D} \beta_z, \quad \beta_z(p^a) = \frac{p^{az} - 1}{a}, \quad \beta_z(n) = 0 \text{ otherwise}, \quad \partial_z \beta_z(n) = n^z \Lambda(n). \quad (15.22)$$

In particular,

$$J_z = \exp_{*D} \left(\int_0^z n^u \Lambda(n) du \right). \quad (15.23)$$

The function inside the exponential has value zero at $n = 1$. For each fixed $n > 1$, its j th convolution power vanishes when $2^j > n$; hence these coefficient identities involve only finitely many terms and require no Dirichlet-series convergence assumption.

Theorem 15.4 (Smith–Jordan positivity). *For real $z > 0$, let*

$$K_z(m, n) = \frac{\gcd(m, n)^z}{(mn)^{z/2}}. \quad (15.24)$$

Every finitely supported complex sequence x satisfies

$$\sum_{m,n} \overline{x_m} K_z(m, n) x_n = \sum_{d \geq 1} J_z(d) \left| \sum_{d|n} \frac{x_n}{n^{z/2}} \right|^2 \geq 0. \quad (15.25)$$

The inequality is strict when $x \neq 0$.

Proof. Möbius inversion gives $r^z = \sum_{d|r} J_z(d)$. Substitute $r = \gcd(m, n)$ and interchange the finite sums to obtain (15.25). All $J_z(d)$ are positive for $z > 0$. If n_0 is the largest index for which $x_{n_0} \neq 0$, the $d = n_0$ square has the single nonzero term $x_{n_0}/n_0^{z/2}$, proving strictness. $\square$

The parameter match with (15.13) is exact:

$$K_z(m, n) = \prod_p p^{-\frac{z}{2}|\nu_p(m) - \nu_p(n)|}. \quad (15.26)$$

Thus $z = 2\sigma$ identifies the integer carrier with the tensor valuation kernel on nonnegative exponent vectors. Any finite integer packet uses finitely many primes, so Theorem 15.4 covers the full positive-integer index without an infinite product of operator lower bounds.

Theorem 15.5 (Prime-power positivity on the vacuum-null space). *For a finitely supported complex sequence x satisfying $\sum_n x_n = 0$, put $\mathcal{Q}_z[x] = \sum_{m,n} \overline{x_m} K_z(m, n) x_n$. Then*

$$\partial_z \mathcal{Q}_z[x]|_{z=0} = \sum_{q=p^a} (\log p) \left| \sum_{q|n} x_n \right|^2 \geq 0. \quad (15.27)$$

Proof. The finite sums in (15.25) can be differentiated at zero. The $d = 1$ term is $|\sum_n x_n n^{-z/2}|^2$; its derivative vanishes at zero because $\sum_n x_n = 0$. For $d > 1$, $J_0(d) = 0$. If $d = p^a$, then $J_z(d) = p^{az} - p^{(a-1)z}$ and $J'_0(d) = \log p$. If d has at least two distinct prime factors, the product in (15.21) has a zero of order at least two, so $J'_0(d) = 0$. These are precisely the terms in (15.27). $\square$

The tangent need not be strictly positive: the coefficients $(1, -1, -1, 1)$ on $(1, 2, 3, 6)$ have zero sum and every prime-power divisor sum vanishes. The full tensor retains the higher mixed-color terms for this vector through (15.25).

15.5 The scalar phase orbit and its completed weight

The coherent scalar phase is

$$\iota_{\mathcal{P}}(t) = (e^{-it \log p})_{p \in \mathcal{P}}, \quad L_{\mathcal{P}}(\nu) = \frac{1}{2} \sum_{p \in \mathcal{P}} \nu_p \log p. \quad (15.28)$$

Unique factorization implies $\sum_p \nu_p \log p \neq 0$ for every nonzero integer vector ν . Integration of each nonconstant character therefore shows, for every trigonometric polynomial A , that

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |A(\iota_{\mathcal{P}}(t))|^2 dt = \int_{\mathbb{T}^{\mathcal{P}}} |A(z)|^2 dm(z). \quad (15.29)$$

The same argument, followed by uniform trigonometric approximation, gives the Haar average for every continuous function and implies that the orbit is dense.

Writing $G_{\mathcal{P}, \sigma}(\theta) = \prod_p P_{r_p}(\theta_p)$ for the tensor multiplier, restriction gives the exact finite-prime formulas

$$G_{\mathcal{P}, \sigma}(\iota_{\mathcal{P}}(t)) = \prod_{p \in \mathcal{P}} (1 - p^{-2\sigma}) \left| \prod_{p \in \mathcal{P}} (1 - p^{-\sigma - it})^{-1} \right|^2, \quad (15.30)$$

$$D_{\mathcal{P}, \sigma}(\iota_{\mathcal{P}}(t)) = \partial_{\sigma} \log \left| \prod_{p \in \mathcal{P}} \frac{1 - p^{-\sigma - it}}{1 + p^{-\sigma}} \right|^2 \geq 0. \quad (15.31)$$

The arguments on the left denote the corresponding circle coordinates.

For the completed function use

$$F(z) = \xi(\tfrac{1}{2} + z), \quad K(a, t) = \operatorname{Re}(F'(a + it)\overline{F(a + it)}), \quad a > 0,$$

and the complete theta transform convention

$$C_a(y) = \frac{1}{\pi} \int_{\mathbb{R}} K(a, t) e^{-2ity} dt. \quad (15.32)$$

The rapid theta decay makes $K(a, \cdot)$ integrable. Define the finite signed measure

$$\mu_{a, \mathcal{P}} = (\iota_{\mathcal{P}})_* \left(\frac{1}{\pi} K(a, t) dt \right). \quad (15.33)$$

For finitely supported $b = (b_\nu)$, substitution in (15.32) gives

$$\sum_{\nu, \mu} \overline{b_\nu} b_\mu C_a(L_{\mathcal{P}}(\nu - \mu)) = \int_{\mathbb{T}^{\mathcal{P}}} \left| \sum_{\nu} b_\nu z^{-\nu} \right|^2 d\mu_{a, \mathcal{P}}(z). \quad (15.34)$$

This is the exact completed weight on the color orbit. The arithmetic forms above instead integrate against the positive Haar densities $D_{\mathcal{P}, \sigma}$ and $G_{\mathcal{P}, \sigma}$. When $|\mathcal{P}| \geq 2$, the orbit has Haar measure zero: each compact time interval has a Lipschitz image, and a finite-length curve has zero volume in a torus of dimension at least two; take the countable union of these images. Thus an orbit average such as (15.29) does not identify the measure in (15.33) with either Haar density.

The same sign bookkeeping is visible directly in the completed source. For $s = \sigma + it$, $\sigma > 1$, write

$$A_{\text{arch}}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi(s/2), \quad \frac{\xi'(s)}{\xi(s)} = A_{\text{arch}}(s) - \sum_p (\log p) \frac{p^{-s}}{1 - p^{-s}}. \quad (15.35)$$

Here $\psi = \Gamma'/\Gamma$; the prime-power series converges absolutely and locally uniformly in this half-plane. For the finite-prime version $g_{\mathcal{P}}(s)$ of the right side, direct real-part evaluation yields

$$\operatorname{Re} g_{\mathcal{P}}(s) = \operatorname{Re} A_{\text{arch}}(s) + \sum_{p \in \mathcal{P}} \frac{(\log p) r_p}{1 + r_p} - \frac{1}{2} D_{\mathcal{P}, \sigma}(\iota_{\mathcal{P}}(t)). \quad (15.36)$$

Indeed $P_r(\theta) = 1 + 2 \operatorname{Re}\{r e^{i\theta}/(1 - r e^{i\theta})\}$ and $1 - c_p = 2r_p/(1 + r_p)$. Absolute convergence permits the increasing finite-prime limit in (15.35); no such Euler-series limit in the critical strip is invoked here.

The square identities (15.11), (15.12), (15.25), and (15.27) prove positivity of their stated forms for their whole domains. Applying one of them to the local form in Theorem 17.1 requires a proved sign-preserving implication between the identified forms. An exact norm identity is one sufficient way to obtain that implication; a valid nonnegative lower comparison would also suffice.

16 Complete local amplitude data at one known zero

We use the normalization

$$f(w) = \Xi(w) = \xi(1/2 + iw) = F(iw) = \int_{\mathbb{R}} \eta(x) e^{iwx} dx. \quad (16.1)$$

Here η is the complete even positive theta density. Its established decay bounds give, for every $q > 0$, a constant $M_q < \infty$ such that $\eta(x) \leq M_q e^{-qx^2}$. In this coordinate, the critical line corresponds to the real axis. Fix one established real zero γ of f , of multiplicity $m \geq 1$, and remove precisely its vanishing factor:

$$h(z) = f(\gamma + z) = z^m g(z), \quad g(0) = \frac{f^{(m)}(\gamma)}{m!} \neq 0. \quad (16.2)$$

The quotient g is a real entire function of order one. Its normalized local squared amplitude is

$$\mathcal{A}_\gamma(u) = \frac{|f(\gamma + u)|^2}{u^{2m} |f^{(m)}(\gamma)/m!|^2} = \frac{g(u)^2}{g(0)^2} \quad (u \in \mathbb{R}). \quad (16.3)$$

The value at $u = 0$ is removable and equals 1. Thus $\mathcal{A}_\gamma > 0$ near zero and its logarithm is real analytic there. Define

$$s_n = -\frac{(\log \mathcal{A}_\gamma)^{(n+2)}(0)}{2(n+1)!} = -\frac{(\log g)^{(n+2)}(0)}{(n+1)!}, \quad n \geq 0. \quad (16.4)$$

The second expression is independent of the local choice of a complex logarithm. Let $\mathcal{L}_\gamma$ be the real polynomial functional determined by

$$\mathcal{L}_\gamma(X^n) = s_n, \quad H_N(\gamma) = [s_{j+k}]_{j,k=0}^N. \quad (16.5)$$

We extend $\mathcal{L}_\gamma$ complex-linearly when complex coefficients occur.

16.1 The complete reciprocal-zero identity

Let ρ range over the distinct zeros of f other than γ , with multiplicities m_ρ , and set

$$\beta_\rho = \rho - \gamma, \quad \lambda_\rho = \beta_\rho^{-1}. \quad (16.6)$$

Zero isolation and the order-one canonical product imply

$$r_\gamma := \sup_{\rho \neq \gamma} |\lambda_\rho| < \infty, \quad C_\gamma := \sum_{\rho \neq \gamma} m_\rho |\lambda_\rho|^2 < \infty. \quad (16.7)$$

Indeed, 0 is not a zero of g , so its zeros stay a positive distance from 0; order one gives convergence of the reciprocal squares. The canonical product has an exponential factor of degree at most one. Taking logarithmic derivatives of order at least two removes that factor and gives

$$\boxed{s_n = \sum_{\rho \neq \gamma} m_\rho \lambda_\rho^{n+2}.} \quad (16.8)$$

Every sum in (16.8) converges absolutely, with

$$|s_n| \leq C_\gamma r_\gamma^n. \quad (16.9)$$

Conjugate symmetry makes the moments real.

Write $\ell(z) = g'(z)/g(z)$. The moment generating function is

$$\begin{aligned} \mathcal{M}_\gamma(z) &:= -\frac{\ell(z) - \ell(0)}{z} \\ &= \sum_{\rho \neq \gamma} \frac{m_\rho \lambda_\rho^2}{1 - \lambda_\rho z} = \sum_{n \geq 0} s_n z^n, \quad |z| < r_\gamma^{-1}. \end{aligned} \quad (16.10)$$

Its singularity at 0 is removable. The zero sum converges normally on compact sets away from its poles: the finitely many large λ_ρ can be separated, and the remaining terms are bounded by a constant times $m_\rho |\lambda_\rho|^2$. At every shifted zero,

$$\operatorname{Res}_{z=\beta_\rho} \mathcal{M}_\gamma(z) = -\frac{m_\rho}{\beta_\rho} \neq 0. \quad (16.11)$$

Thus each nonreal zero gives a genuine nonreal pole in a function determined by the complete local amplitude derivatives.

17 The finite local witness theorem

The following is the classical reciprocal-zero Hankel criterion in the present local coordinates; see [8]. A complete proof is included to specify the infinite sums, the operator domain, and the retained zero contributions.

Theorem 17.1 (Local positivity criterion at one known zero). *For the fixed zero γ , put*

$$D_0 = \gamma^2 + \frac{1}{4}, \quad \chi_\gamma(X) = D_0 X^2 + 2\gamma X + 1. \quad (17.1)$$

The following statements are equivalent:

1. *Every zero of Ξ is real.*
2. *$H_N(\gamma) \succeq 0$ for every $N \geq 0$.*
3. *$\mathcal{L}_\gamma((\chi_\gamma R)^2) \geq 0$ for every real polynomial R .*

Under the first statement, every $H_N(\gamma)$ is strictly positive definite. If a nonreal zero exists, then at every chosen established real zero γ there are a finite N and real coefficients $c \neq 0$ for which

$$c^T H_N(\gamma) c < 0. \quad (17.2)$$

There is also a real polynomial R for which $\mathcal{L}_\gamma((\chi_\gamma R)^2) < 0$.

The theorem is an equivalence. It does not assert that its positivity conditions have been proved unconditionally for Ξ .

17.1 Bounded multiplication from the complete moment bounds

Lemma 17.2 (Moment positivity and a bounded resolvent). *Let $(u_n)_{n \geq 0}$ be real and satisfy $|u_n| \leq Cr^n$, where $0 < r < \infty$. Let $\mathcal{U}(X^n) = u_n$, and assume $\mathcal{U}(P^2) \geq 0$ for every real polynomial P .*

Then there are a Hilbert space, a bounded self-adjoint operator T with $\|T\| \leq r$, and a vector v such that

$$u_n = \langle T^n v, v \rangle, \quad \sum_{n \geq 0} u_n z^n = \langle (I - zT)^{-1} v, v \rangle \quad (|z| < r^{-1}). \quad (17.3)$$

The expression on the right is holomorphic on each open half-plane.

Proof. The positive semidefinite form $\langle P, Q \rangle = \mathcal{U}(PQ)$ on real polynomials satisfies Cauchy–Schwarz. For a fixed polynomial P , put

$$a_k = \mathcal{U}(X^{2k} P^2) \geq 0.$$

Cauchy–Schwarz applied to $X^{k-1}P$ and $X^{k+1}P$ gives

$$a_k^2 \leq a_{k-1} a_{k+1} \quad (k \geq 1). \quad (17.4)$$

If $P^2 = \sum_{\ell=0}^d b_\ell X^\ell$, the moment bound gives

$$0 \leq a_k \leq C \left(\sum_{\ell=0}^d |b_\ell| r^\ell \right) r^{2k}. \quad (17.5)$$

If $a_0 = 0$, (17.4) at $k = 1$ gives $a_1 = 0$. If $a_0 > 0$ and $a_1 = 0$, the inequality $a_1 \leq r^2 a_0$ is immediate. Otherwise (17.4) implies inductively that all $a_k > 0$ and their successive ratios are nondecreasing. Hence

$$a_k \geq a_0 (a_1/a_0)^k.$$

Comparison with (17.5) forces $a_1/a_0 \leq r^2$. In every case,

$$\mathcal{U}(X^2 P^2) \leq r^2 \mathcal{U}(P^2). \quad (17.6)$$

Quotient by the zero-seminorm polynomials and complete. Multiplication $T[P] = [XP]$ is well defined by (17.6), is bounded by r , and is symmetric since $\mathcal{U}(XPQ) = \mathcal{U}(PXQ)$. After complexification, it is a bounded self-adjoint operator on the completed Hilbert space. With $v = [1]$, one obtains $u_n = \langle T^n v, v \rangle$, and the norm-convergent Neumann series proves (17.3). For nonreal z , $1/z$ is nonreal and cannot belong to the spectrum of T . Thus $I - zT$ is invertible; its inverse is holomorphic there. $\square$

17.2 Proof of the criterion and the finite witness

Proof of Theorem 17.1. For $P(X) = \sum_{j=0}^N c_j X^j$, absolute convergence of (16.8) permits the finite expansion

$$c^T H_N c = \mathcal{L}_\gamma(P^2) = \sum_{\rho \neq \gamma} m_\rho \lambda_\rho^2 P(\lambda_\rho)^2. \quad (17.7)$$

If all zeros are real, each summand is nonnegative. There are infinitely many distinct zeros of Ξ , so a nonzero polynomial cannot vanish at every λ_ρ . The quadratic form is then strictly positive. This proves the first statement implies the second, and the second plainly implies the third.

Assume the second statement. Apply Lemma 17.2 to s_n , using (16.9). Its resolvent agrees with $\mathcal{M}_\gamma$ near zero in each half-plane. The identity theorem for meromorphic functions makes them equal on that half-plane. The resolvent is holomorphic, whereas a nonreal shifted zero would give the nonzero residue (16.11). Therefore every zero is real.

It remains to prove that restricting to the ideal generated by χ_γ loses no zero. Define

$$t_n = \mathcal{L}_\gamma(X^n \chi_\gamma(X)^2) = \sum_{\rho \neq \gamma} m_\rho \lambda_\rho^{n+2} \chi_\gamma(\lambda_\rho)^2. \quad (17.8)$$

The bound $|\lambda_\rho| \leq r_\gamma$ and (16.7) give

$$|t_n| \leq C_\chi r_\gamma^n, \quad C_\chi = \sum_{\rho \neq \gamma} m_\rho |\lambda_\rho|^2 |\chi_\gamma(\lambda_\rho)|^2 < \infty.$$

If the third statement holds, the functional $\mathcal{U}(Q) = \mathcal{L}_\gamma(\chi_\gamma^2 Q)$ is nonnegative on every polynomial square. The same lemma supplies a bounded self-adjoint representation of t_n . The corresponding generating function is

$$\sum_{n \geq 0} t_n z^n = \sum_{\rho \neq \gamma} \frac{m_\rho \lambda_\rho^2 \chi_\gamma(\lambda_\rho)^2}{1 - \lambda_\rho z}. \quad (17.9)$$

It has normally convergent meromorphic continuation away from its poles, by the same tail estimate used for (16.10). At $\beta_\rho = \rho - \gamma$, its residue is

$$-\frac{m_\rho}{\rho - \gamma} \chi_\gamma((\rho - \gamma)^{-1})^2. \quad (17.10)$$

Direct substitution gives

$$\chi_\gamma((\rho - \gamma)^{-1}) = \frac{\rho^2 + 1/4}{(\rho - \gamma)^2}. \quad (17.11)$$

Its only possible vanishing locations are $\rho = \pm i/2$. Independently,

$$\Xi(i/2) = \xi(0) = \frac{1}{2}, \quad \Xi(-i/2) = \xi(1) = \frac{1}{2}. \quad (17.12)$$

Neither is a zero. Thus every residue in (17.10) is nonzero. Agreement with the holomorphic resolvent on each half-plane excludes every nonreal pole, proving the third statement implies the first.

Finally, if a nonreal zero exists, the second statement fails. Failure of positive semidefiniteness of some finite real symmetric H_N supplies a real vector satisfying (17.2). Failure of the third statement supplies a real polynomial R with a strictly negative filtered form. The argument used an arbitrary fixed established real zero γ , so both conclusions hold at every such choice. $\square$

The witness H_N uses derivatives of $\log \mathcal{A}_\gamma$ through order $2N + 2$. These are determined by the regularized squared-amplitude jet through that order, or by the original squared-amplitude jet through order $2m + 2N + 2$. The theorem supplies a finite witness if an off-critical zero exists, without a uniform degree cutoff and without truncating the other zeros.

17.3 Connection to the amplitude coefficient hierarchy

For a real entire function a , define $L_n[a]$ by

$$|a(x + iy)|^2 = \sum_{n \geq 0} L_n[a](x) y^{2n}. \quad (17.13)$$

Multiplication of the two entire Taylor series gives

$$L_n[a](x) = (-1)^n \sum_{j=0}^{2n} \frac{(-1)^j a^{(j)}(x) a^{(2n-j)}(x)}{j!(2n-j)!}. \quad (17.14)$$

The product identity follows from multiplication of the amplitude series:

$$L_n[ab](x) = \sum_{j=0}^n L_j[a](x) L_{n-j}[b](x). \quad (17.15)$$

At the known zero, $h(z) = z^m g(z)$ gives

$$L_{m+n}[f](\gamma) = L_n[g](0), \quad L_m[f](\gamma) = g(0)^2 > 0. \quad (17.16)$$

Since $L_1[g](0) = g'(0)^2 - g(0)g''(0)$, the first reciprocal-zero moment is

$$s_0 = \frac{L_{m+1}[f](\gamma)}{L_m[f](\gamma)}. \quad (17.17)$$

For a simple zero this specializes to

$$s_0 = \frac{f''(\gamma)^2}{4f'(\gamma)^2} - \frac{f'''(\gamma)}{3f'(\gamma)}, \quad L_2[f](\gamma) = f'(\gamma)^2 s_0. \quad (17.18)$$

These are exact identifications with the previous amplitude coefficients.

18 Exact compression of the complete spatial kernel

Retain the spatial kernel and its Hermitian form, linear in the first argument,

$$\begin{aligned} J(x, y) &= \int_0^\infty (x + y + 2r) \eta(x + r) \eta(y + r) dr, \\ \mathcal{B}_J(a, b) &= \iint_{\mathbb{R}^2} J(x, y) a(x) \overline{b(y)} dx dy. \end{aligned} \quad (18.1)$$

For $c = (x + y)/2$ and $d = (x - y)/2$, evenness of η gives

$$J(x, y) = 2 \int_{|c|}^\infty v \eta(v + d) \eta(v - d) dv. \quad (18.2)$$

The integrand before the reduction from c to $|c|$ is odd in v ; its integral over the symmetric segment cancels. Consequently

$$0 < J(x, y) \leq \frac{M_q^2}{2q} e^{-q(x^2 + y^2)}. \quad (18.3)$$

In particular, the form and all polynomial-exponential derivatives below converge absolutely. The scalar inequality in (18.3) is not a statement of matrix positivity.

18.1 The complete two-variable transform

For a real entire function a , put

$$W_a(z, w) = \frac{a(z)\overline{a'(w)} - a'(z)\overline{a(w)}}{z - \overline{w}}. \quad (18.4)$$

The apparent singularity at $z = \overline{w}$ is removable, since the numerator vanishes there.

Proposition 18.1 (Complete spatial transform). *With h from (16.2),*

$$\mathcal{B}_J(e^{i(\gamma+z)x}, e^{i(\gamma+w)x}) = W_h(z, w). \quad (18.5)$$

Proof. Differentiation of (18.1), equivalently translation of its lower integration endpoint, gives

$$(\partial_x + \partial_y)J(x, y) = -(x + y)\eta(x)\eta(y). \quad (18.6)$$

Set $\alpha = \gamma + z$ and $\beta = \gamma + \overline{w}$. The transform of the left side against $e^{i\alpha x - i\beta y}$ is $-i(\alpha - \beta)$ times the transform of J . Evenness of f , differentiation of (16.1), and the right side of (18.6) give the transform

$$-i[f(\alpha)f'(\beta) - f'(\alpha)f(\beta)].$$

The rapid decay justifies integration by parts and eliminates the boundary terms. Dividing for $\alpha \neq \beta$ proves (18.5); the removable diagonal follows by analytic continuation. $\square$

Substitution of $h(z) = z^m g(z)$ into (18.4) yields

$$W_h(z, w) = (z\overline{w})^{m-1} \left[mg(z)\overline{g(w)} + z\overline{w} W_g(z, w) \right]. \quad (18.7)$$

Thus the removable quotient $\mathcal{R}_m = W_h/(z\overline{w})^{m-1}$ has positive pivot $\mathcal{R}_m(0, 0) = mg(0)^2$, and its exact Schur residual is

$$\mathcal{R}_m(z, w) - \frac{\mathcal{R}_m(z, 0)\mathcal{R}_m(0, w)}{\mathcal{R}_m(0, 0)} = z\overline{w} W_g(z, w). \quad (18.8)$$

This identity retains the known zero with its full multiplicity. The positivity theorem for Schur complements would additionally require positivity of the complete tested kernel.

18.2 The explicit local test functions

Define

$$\begin{aligned} e_k(x) &= e^{i\gamma x} \frac{(ix)^k}{k!}, \\ E_z(x) &= \frac{e^{i(\gamma+z)x} - \sum_{k=0}^{m-2} z^k e_k(x)}{z^{m-1}}. \end{aligned} \quad (18.9)$$

The sum is empty when $m = 1$, and the value at $z = 0$ is $E_0 = e_{m-1}$. Put

$$v(x) = \frac{e_{m-1}(x)}{g(0)}, \quad U_z(x) = \frac{1}{z} \left(\frac{E_z(x)}{g(z)} - v(x) \right) \quad (18.10)$$

in a zero-free neighborhood of the origin for g . The numerator vanishes at 0, so U_z is holomorphic there.

The factor $(z\bar{w})^{m-1}$ in (18.7) says that the e_k -rows for $k < m - 1$ pair to zero with every polynomial-exponential jet. Moreover, $\mathcal{B}_J(E_z, E_w) = \mathcal{R}_m(z, w)$. Dividing by $g(z)\overline{g(w)}$ gives a constant term m and the residual $z\bar{w}W_g/(g(z)\overline{g(w)})$. Subtracting v in each argument removes precisely the constant term. Hence

$$\begin{aligned}\mathcal{B}_J(U_z, U_w) &= \frac{W_g(z, w)}{g(z)\overline{g(w)}} \\ &= \frac{-\ell(z) + \overline{\ell(w)}}{z - \bar{w}} = \sum_{j,k \geq 0} s_{j+k} z^j \bar{w}^k.\end{aligned}\tag{18.11}$$

For the last equality, use $\ell(z) = \ell(0) - \sum_{n \geq 0} s_n z^{n+1}$ from (16.10) and $(z^{n+1} - u^{n+1})/(z - u) = \sum_{j=0}^n z^j u^{n-j}$.

Write $1/g(z) = \sum_{r \geq 0} b_r z^r$. Coefficient extraction in (18.10) gives the explicit functions

$$U_z(x) = \sum_{j \geq 0} a_j(x) z^j, \quad a_j(x) = \sum_{r=0}^{j+1} b_r e_{m+j-r}(x).\tag{18.12}$$

The Gaussian majorants justify differentiating under $\mathcal{B}_J$. For $P(X) = \sum_{j=0}^N c_j X^j$, put $A_P = \sum_{j=0}^N c_j a_j$. We have

$$\boxed{\mathcal{B}_J(a_j, a_k) = s_{j+k}, \quad \mathcal{B}_J(A_P, A_P) = \mathcal{L}_\gamma(P\bar{P}).}\tag{18.13}$$

The bar in the polynomial denotes conjugation of coefficients. For real P , the last expression is $\mathcal{L}_\gamma(P^2)$. Both infinite spatial tails and all mixed coefficients are included.

18.3 The full jet block at the known zero

Proposition 18.2 (Exact block decomposition). *Let $V_N = \text{span}_{\mathbb{C}}\{e_0, \dots, e_{m+N}\}$. In the basis*

$$e_0, \dots, e_{m-2}, \quad v, \quad a_0, \dots, a_N,$$

its $\mathcal{B}_J$ -Gram matrix is

$$0_{m-1} \oplus [m] \oplus H_N(\gamma).\tag{18.14}$$

The zero block is absent for $m = 1$.

Proof. Equations (18.7)–(18.11) give

$$\mathcal{B}_J(e_k, e_\ell) = 0 \quad (k < m - 1, \ell \geq 0), \quad \mathcal{B}_J(v, v) = m, \quad \mathcal{B}_J(v, a_j) = 0.$$

Equation (18.13) gives the remaining block. The list is a basis because a_j has nonzero leading coefficient $1/g(0)$ on e_{m+j} , making the change of basis invertible and triangular. $\square$

The coefficient transformation can also be written explicitly. If $g(z) = \sum_{r \geq 0} g_r z^r$ and $q = \sum_{k=0}^{m+N} d_k e_k$, put

$$c_j = \sum_{\ell=j}^N g_{\ell-j} d_{m+\ell}, \quad \tau = \sum_{r=0}^{N+1} g_r d_{m-1+r}.\tag{18.15}$$

Multiplication of the reciprocal series for g verifies

$$q = \sum_{k=0}^{m-2} d_k e_k + \tau v + A_P, \quad \mathcal{B}_J(q, q) = m|\tau|^2 + c^* H_N(\gamma) c. \quad (18.16)$$

Thus the local matrix retains every possible negative direction of the full jet form on V_N . The proved coordinate change preserves its negative index.

19 The primitive identity and the exact two-trace class

Let B be the complete even positive modular primitive constructed in Appendix F.2; equation (F.10) fixes its normalization. In the present notation,

$$L = \partial_x^2 - \frac{1}{4}, \quad \eta = -LB, \quad \int_{\mathbb{R}} B(x) e^{iwx} dx = \frac{f(w)}{w^2 + 1/4}. \quad (19.1)$$

The transform identity holds initially for $|\operatorname{Im} w| < 1/2$. The primitive and its derivatives have the exponential decay proved in Appendix F.2; these bounds suffice for all polynomial-exponential tests below. Define

$$J_B(x, y) = \int_0^\infty (x + y + 2r) B(x + r) B(y + r) dr. \quad (19.2)$$

Proposition 19.1 (Complete primitive correction). *The kernels satisfy*

$$J = L_x L_y J_B + 2B'(x)B'(y) - \frac{1}{2}B(x)B(y). \quad (19.3)$$

For the local functions A_P ,

$$\mathcal{B}_J(A_P, A_P) = \mathcal{B}_{J_B}(LA_P, LA_P) + 2|\tau_{B'}(A_P)|^2 - \frac{1}{2}|\tau_B(A_P)|^2, \quad \tau_b(a) = \int_{\mathbb{R}} b(x)a(x) dx. \quad (19.4)$$

Proof. Apply $L_x L_y$ to (19.2). Product differentiation gives

$$L_x L_y J_B = J + 2 \int_0^\infty \{ (LB)(x+r)B'(y+r) + B'(x+r)(LB)(y+r) \} dr.$$

The integrand in braces is the r -derivative of

$$B'(x+r)B'(y+r) - \frac{1}{4}B(x+r)B(y+r).$$

Exponential decay eliminates its upper boundary. Evaluating the lower boundary gives (19.3). Integrating by parts in x and y transfers $L_x L_y$ to A_P and its conjugate, proving (19.4). The boundary terms vanish because the tests grow polynomially and B and its derivatives decay exponentially. The double integrals are absolute: evenness gives the analogue of (18.2), and $|B(t)| \leq C e^{-|t|/2}$ bounds J_B by an exponentially decaying function times a polynomial in $|x| + |y|$. $\square$

19.1 The traces and their simultaneous cancellation

Write

$$D(z) = \frac{1}{4} + (\gamma + z)^2. \quad (19.5)$$

By (19.1), the B -transform of the exponential in (18.9) is $z^m g(z)/D(z)$. Its coefficients below degree m vanish. The B' -transform is $-i(\gamma + z)z^m g(z)/D(z)$, with the same vanishing property. Substitution in (18.10) therefore gives

$$\tau_B(U_z) = \frac{1}{D(z)}, \quad \tau_{B'}(U_z) = -\frac{i(\gamma + z)}{D(z)}. \quad (19.6)$$

This computation retains both traces exactly.

Set

$$\lambda_+ = \frac{1}{-\gamma + i/2}, \quad Z_P = \lambda_+ P(\lambda_+). \quad (19.7)$$

For real P , partial fractions in (19.6) give

$$\boxed{\tau_B(A_P) = -2 \operatorname{Im} Z_P, \quad \tau_{B'}(A_P) = i \operatorname{Re} Z_P.} \quad (19.8)$$

Indeed, the coefficient of z^j in $1/D(z)$ is $-2 \operatorname{Im}(\lambda_+^{j+1})$; the coefficient in the second rational function is $i \operatorname{Re}(\lambda_+^{j+1})$. Equations (18.13) and (19.4) now give

$$\mathcal{L}_\gamma(P^2) = \mathcal{B}_{J_B}(LA_P, LA_P) + 2 \operatorname{Re}(Z_P^2). \quad (19.9)$$

The polynomial from (17.1) can be written

$$\chi_\gamma(X) = (1 + \gamma X)^2 + \frac{1}{4}X^2.$$

Its roots are the distinct conjugates λ_+ and $\overline{\lambda_+}$. Since P is real and $\lambda_+ \neq 0$, equation (19.8) proves

$$\tau_B(A_P) = \tau_{B'}(A_P) = 0 \iff P = \chi_\gamma R \text{ for a real polynomial } R. \quad (19.10)$$

Thus the exact local family in the two-trace class satisfies

$$\boxed{\mathcal{L}_\gamma((\chi_\gamma R)^2) = \mathcal{B}_J(A_{\chi_\gamma R}, A_{\chi_\gamma R}) = \mathcal{B}_{J_B}(LA_{\chi_\gamma R}, LA_{\chi_\gamma R}).} \quad (19.11)$$

Both primitive corrections vanish individually. Theorem 17.1 proves that this restriction retains the entire off-critical-zero detector. In particular, its complete zero sum is

$$\mathcal{L}_\gamma((\chi_\gamma R)^2) = \sum_{\rho \neq \gamma} m_\rho \frac{(\rho^2 + 1/4)^2}{(\rho - \gamma)^6} R((\rho - \gamma)^{-1})^2, \quad (19.12)$$

with absolute convergence. Establishing nonnegativity of (19.11) for every real polynomial R at this one fixed known zero remains the sign assertion required by this criterion; the identities above do not assume it.

20 Completion of the principal equivalence

Theorem 20.1 (Complete-kernel characterization of the critical line). *Let ξ , F , η , and C_a be given by (1.1), (1.3), and (1.4). Then C_a is positive definite for every $a > 0$ if and only if every nontrivial zero ρ of ζ satisfies $\operatorname{Re} \rho - \frac{1}{2} = 0$.*

Proof. Theorem 2.2 proves that F is a nonzero even entire function and that $F(0) = \int_{\mathbb{R}} \eta(u) du > 0$. Lemma 4.1 proves that each C_a is an even real Schwartz function.

Positive definiteness implies localization. If C_a is positive definite for every $a > 0$, Lemma 4.3 gives

$$\widehat{C}_a(2t) \geq 0 \quad (a > 0, t \in \mathbb{R}). \quad (20.1)$$

The exact identity of Theorem 4.2 now gives

$$\partial_a |F(a + it)|^2 = 2K(a, t) = 2\widehat{C}_a(2t) \geq 0. \quad (20.2)$$

Suppose $F(z_0) = 0$ for $z_0 = a_0 + it_0$ with $a_0 > 0$. Choose any $\varepsilon \in (0, a_0)$. For every $a \in [a_0 - \varepsilon, a_0]$,

$$0 \leq |F(a + it_0)|^2 \leq |F(a_0 + it_0)|^2 = 0. \quad (20.3)$$

Hence F vanishes on that interval. The identity theorem contradicts $F(0) > 0$. Thus F has no zero in $\operatorname{Re} z > 0$. Evenness excludes a zero in $\operatorname{Re} z < 0$ as well, since $F(z) = 0$ implies $F(-z) = 0$. All zeros of F therefore have real part zero.

For a nontrivial zero ρ of ζ , Proposition 2.4 identifies $z = \rho - 1/2$ as a zero of F , and hence $\operatorname{Re} \rho - \frac{1}{2} = 0$.

Localization implies positive definiteness. Conversely, if every nontrivial zero has real part $1/2$, the zeros of F are $\pm i\gamma$, $\gamma > 0$, with multiplicities m_γ . The paired canonical product (8.4), justified by the order-one growth and inverse-square summability proved above, yields

$$K(a, t) = a|F(a + it)|^2 \sum_{\gamma > 0} m_\gamma \left(\frac{1}{a^2 + (t - \gamma)^2} + \frac{1}{a^2 + (t + \gamma)^2} \right) > 0 \quad (a > 0, t \in \mathbb{R}). \quad (20.4)$$

By (4.6), $\widehat{C}_a(\omega) = K(a, \omega/2) > 0$. Fourier inversion therefore gives, for arbitrary finite nodes and coefficients,

$$\sum_{j,k} \overline{v_j} v_k C_a(y_j - y_k) = \frac{1}{2\pi} \int_{\mathbb{R}} K(a, \omega/2) \left| \sum_k v_k e^{-i\omega y_k} \right|^2 d\omega \geq 0. \quad (20.5)$$

This is positive definiteness of C_a for every $a > 0$ and proves the converse. The equivalence with the half-strip inequality in Theorem 1.2 follows from Theorem 8.2; its independent transformed-kernel assertion is Theorem 4.4. The resulting unconditional equivalence is

$[C_a \text{ is positive definite for every } a > 0]$
$\iff$
$\left[\forall \rho \in \mathbb{C} : \quad (\zeta(\rho) = 0 \wedge 0 < \operatorname{Re} \rho < 1) \implies \operatorname{Re} \rho - \frac{1}{2} = 0 \right].$

□

20.1 Assembly with the complete local criterion

Theorem 17.1 now places the known-zero argument inside the same proof. Fix a real zero γ and use the complete functions and forms defined there. The preceding principal equivalence and the local criterion give

$$\begin{array}{l} K(a, t) \geq 0 \quad (a > 0, t \in \mathbb{R}) \\ \iff \text{all zeros of } \Xi \text{ are real} \\ \iff \mathcal{B}_J(A_{\chi_\gamma R}, A_{\chi_\gamma R}) \geq 0 \quad \text{for every } R \in \mathbb{R}[X]. \end{array} \quad (20.6)$$

The primitive identity identifies the final form exactly with $\mathcal{B}_{J_B}(LA_{\chi_\gamma R}, LA_{\chi_\gamma R})$. Both primitive traces vanish, and no zero is hidden by this restriction, because the only potential lost locations are $\pm i/2$, where $\Xi = 1/2$. The full proof, including the moment bound and self-adjoint resolvent argument, is part of Theorem 17.1.

Thus a hypothetical off-critical zero supplies a negative finite quadratic form in this very family. A proof of its nonnegativity would contradict that witness. The positive prime forms and their complete square decompositions are included above; applying them here needs a proved implication connecting their sign to this local form. An exact norm identity is one sufficient implementation, and a valid lower comparison also suffices. Neither a prime-coordinate representation nor separate positivity of every term in a spatial decomposition is an additional condition in (20.6).

The original spatial closing step is Theorem 14.4: it starts with (14.16) and proves H. Its proof supplies the transfer from that inequality; it does not derive the inequality from the theta log-concavity theorem. All conclusions in this assembled proof retain precisely the hypotheses stated at their point of use.

A Prime support spaces and the completed successor form

A.1 Support convention and the completed distribution

Let $\mathcal{D} = C_c^\infty(\mathbb{R}; \mathbb{C})$ and $g^*(x) = \overline{g(-x)}$. In this appendix the Hermitian forms are linear in their first argument. For $h \in \mathcal{D}$, put

$$\begin{aligned} \mathcal{F}_h(s) &= \int_{\mathbb{R}} h(x) e^{(s-1/2)x} dx, \\ P(h) &= \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} (h(\log n) + h(-\log n)), \\ A_\infty(h) &= (\log(4\pi) + \gamma)h(0) + \int_0^\infty \frac{h(x) + h(-x) - 2e^{-x/2}h(0)}{e^x - e^{-x}} e^{x/2} dx. \end{aligned} \quad (\text{A.1})$$

The completed distribution and form are

$$W(h) = \mathcal{F}_h(0) + \mathcal{F}_h(1) - P(h) - A_\infty(h), \quad \mathfrak{W}(f, g) = W(f * g^*). \quad (\text{A.2})$$

For $h = q * q^*$ this is exactly (14.6), including its paired gamma numerator. The prime sum is finite on any fixed compact support. The numerator in the archimedean integral is $O(x)$ at the origin. Thus W is continuous on each fixed support space; continuity of convolution gives continuity of $\mathfrak{W}$ on each pair of such spaces.

For $a > 0$, define

$$\mathcal{D}_a = C_c^\infty((-a, a); \mathbb{C}), \quad q_a = \mathfrak{W}|_{\mathcal{D}_a \times \mathcal{D}_a}. \quad (\text{A.3})$$

Compact support is strictly inside the open interval. Consequently

$$f, g \in \mathcal{D}_a \implies \text{supp}(f * g^*) \subseteq (-2a, 2a). \quad (\text{A.4})$$

Here $K \Subset U$ means that K is compact and contained in the open set U . Let

$$2 = p_1 < p_2 < \cdots, \quad \alpha_k = \frac{1}{2} \log p_k. \quad (\text{A.5})$$

Theorem A.1 (Prime support exhaustion). *The following statements are equivalent:*

- (i) $\mathfrak{W}(f, f) \geq 0$ for every $f \in \mathcal{D}$;
- (ii) $q_a \succeq 0$ for every $a > 0$;
- (iii) $q_{\alpha_k} \succeq 0$ for every $k \geq 1$.

Proof. The first two implications are restrictions. Since the ordinary primes are unbounded,

$$\alpha_k \longrightarrow \infty, \quad \bigcup_{k \geq 1} \mathcal{D}_{\alpha_k} = \mathcal{D}.$$

Given $f \in \mathcal{D}$, compactness of its support supplies k with $f \in \mathcal{D}_{\alpha_k}$. Under (iii), $\mathfrak{W}(f, f) = q_{\alpha_k}(f, f) \geq 0$, proving (i). $\square$

Corollary A.2 (Intermediate support spaces). *If $\alpha_k \leq a \leq \alpha_{k+1}$ and $q_{\alpha_{k+1}} \succeq 0$, then $q_a \succeq 0$.*

Proof. The inclusion $\mathcal{D}_a \subseteq \mathcal{D}_{\alpha_{k+1}}$ makes q_a a restriction of $q_{\alpha_{k+1}}$. $\square$

A.2 Prime-power packets

Define the finite packet

$$\mathcal{A}_k = \{\ell^m : \ell \text{ prime}, m \geq 1, p_k \leq \ell^m < p_{k+1}\}. \quad (\text{A.6})$$

Proposition A.3 (Exact activation interval). *The newly available von Mangoldt atoms in the support transition $\alpha_k \rightarrow \alpha_{k+1}$ are exactly the members of $\mathcal{A}_k$.*

Proof. For $f, g \in \mathcal{D}_{\alpha_k}$, (A.4) gives

$$\text{supp}(f * g^*) \subseteq (-\log p_k, \log p_k).$$

Its values at $\pm \log q$ vanish for $q \geq p_k$. At the next anchor only $q < p_{k+1}$ can occur, so the difference is precisely $p_k \leq q < p_{k+1}$. The distribution P has atoms only at prime powers. Conversely, for $q \in \mathcal{A}_k$ one has $\frac{1}{2} \log q < \alpha_{k+1}$. Choose small smooth bumps centered at $\frac{1}{2} \log q$ and $-\frac{1}{2} \log q$, supported in the next interval, so that $f * g^*$ is nonzero at $\log q$. Thus each member of the packet is potentially active. $\square$

Corollary A.4 (Exhaustion by disjoint packets). *The packets $\mathcal{A}_k$ are finite and pairwise disjoint, and*

$$\bigcup_{k \geq 1} \mathcal{A}_k = \{\ell^m : \ell \text{ prime}, m \geq 1\}. \quad (\text{A.7})$$

Proof. The half-open intervals $[p_k, p_{k+1})$ partition $[2, \infty)$; their intersections with the integers are finite. Intersecting this partition with the prime powers proves the statement. $\square$

The left endpoint p_k belongs to the new packet because the old convolution support is strictly inside its boundary. The new endpoint p_{k+1} belongs to the next packet. The formula $\Lambda(n) = 0$ for integers with at least two distinct prime divisors leaves no additional arithmetic atom to assign.

A.3 A determining sequence of finite-dimensional spaces

Lemma A.5 (Determining prime ladder). *There exist nested finite-dimensional spaces*

$$X_1 \subseteq X_2 \subseteq \cdots, \quad X_k \subset \mathcal{D}_{\alpha_k}, \quad (\text{A.8})$$

such that, for every $A > 0$, elements of $\bigcup_k X_k$ supported compactly inside $(-A, A)$ are dense in $C_c^\infty((-A, A))$ in the test-function topology.

Proof. For each positive rational r , choose a countable dense subset $E_r \subset C_c^\infty((-r, r))$. Enumerate $\bigcup_r E_r$ as $\{\phi_m\}_{m \geq 1}$. Choose an integer $\kappa(m) \geq m$ such that

$$\text{supp } \phi_m \subseteq (-\alpha_{\kappa(m)}, \alpha_{\kappa(m)}), \quad X_k = \text{span}\{\phi_m : \kappa(m) \leq k\}.$$

The inequality $\kappa(m) \geq m$ makes X_k finite-dimensional. The support choices give (A.8). For $f \in C_c^\infty((-A, A))$, choose a positive rational $r < A$ such that $\text{supp } f \subseteq (-r, r)$. Approximation by E_r followed by the continuous inclusion of its test-function space into $C_c^\infty((-A, A))$ proves the density assertion. $\square$

Choose an algebraic complement Y_k and coordinate maps

$$X_{k+1} = X_k \oplus Y_k, \quad \Phi_k : \mathbb{C}^{d_k} \longrightarrow X_k, \quad \Theta_k : \mathbb{C}^{r_k} \longrightarrow Y_k, \quad (\text{A.9})$$

with $\Phi_{k+1}(x, y) = \Phi_k x + \Theta_k y$. Zero-dimensional increments are allowed. These spaces, maps and bases are fixed before any sign is examined. Define

$$\begin{aligned} (K_k)_{rs} &= \mathfrak{W}(\Phi_k e_s, \Phi_k e_r), \\ (B_k)_{ra} &= \mathfrak{W}(\Theta_k e_a, \Phi_k e_r), & (C_k)_{ab} &= \mathfrak{W}(\Theta_k e_b, \Theta_k e_a). \end{aligned} \quad (\text{A.10})$$

Thus

$$K_{k+1} = \begin{pmatrix} K_k & B_k \\ B_k^* & C_k \end{pmatrix}. \quad (\text{A.11})$$

B The completed Suzuki map and metric transport

B.1 Explicit feature and two-sided convention

For the spectral coordinate w define

$$A_\xi(w) = \xi(\tfrac{1}{2} - iw), \quad E_\xi(w) = A_\xi(w) + iA'_\xi(w), \quad \Theta_\xi(w) = \frac{E_\xi^\sharp(w)}{E_\xi(w)}, \quad H^\sharp(w) = \overline{H(\overline{w})}. \quad (\text{B.1})$$

The derivative in A'_ξ is with respect to w . Thus $E_\xi(w) = \xi(\tfrac{1}{2} - iw) + \xi'(\tfrac{1}{2} - iw)$. Write $\psi_\Gamma = \Gamma'/\Gamma$ and $\Phi(v, s, b) = \sum_{n \geq 0} v^n (n + b)^{-s}$ for the Hurwitz–Lerch function. The completed positive-time feature is

$$\begin{aligned} \mathfrak{P}_t(w) &= \frac{4(e^{t/2} - 1)}{1 + 2iw} + \frac{4(e^{-t/2} - 1)}{1 - 2iw} \\ &\quad + \frac{e^{-iwt} - 1}{iw} \frac{\zeta'(\tfrac{1}{2} - iw)}{\zeta(\tfrac{1}{2} - iw)} + \sum_{n \leq e^t} \frac{\Lambda(n)}{\sqrt{n}} \frac{e^{-iw(t - \log n)} - 1}{iw} \\ &\quad - \frac{1}{2iw} \left[\psi_\Gamma\left(\tfrac{1}{4} - \frac{iw}{2}\right) - \psi_\Gamma\left(\tfrac{1}{4}\right) \right] \\ &\quad - \frac{e^{-t/2}}{2iw} \left[\Phi(e^{-2t}, 1, \tfrac{1}{4} - \frac{iw}{2}) - \Phi(e^{-2t}, 1, \tfrac{1}{4}) \right], \quad t \geq 0. \end{aligned} \quad (\text{B.2})$$

Paired expressions at removable singularities are taken by their limits. This is Suzuki's completed pole–prime–gamma feature [7, equations (1.3)–(1.8)]. The two-sided convention is

$$\mathfrak{P}_t(w) = \mathfrak{P}_{-t}(-w) \quad (t < 0), \quad \mathfrak{S}_t(w) = \frac{iA_\xi(w)}{E_\xi^\sharp(w)} \mathfrak{P}_t(w) = \frac{i(1 + \Theta_\xi^\sharp(w))}{2} \mathfrak{P}_t(w). \quad (\text{B.3})$$

Evenness of A_ξ and $E_\xi^\sharp(-w) = E_\xi(w)$ give

$$\mathfrak{S}_t(w) = \Theta_\xi^\sharp(w) \mathfrak{S}_{-t}(-w) \quad (t < 0). \quad (\text{B.4})$$

On the real axis $|\Theta_\xi^\sharp| = 1$, so reflection and this multiplier preserve the L^2 norm. Suzuki's positive-time L^2 bound therefore extends to all real times with this convention.

Define

$$\widehat{\mathcal{P}_\phi}(w) = \int_{\mathbb{R}} \mathfrak{S}_t^\#(w) \phi(t) dt, \quad Df = if', \quad L_0 f = \pi^{-1/2} \widehat{\mathcal{P}_{Df}}. \quad (\text{B.5})$$

Suzuki's norm theorem [7, equation (3.9), Lemma 3.2] states that $\|L_0 f\|_2$ is a norm on $\mathcal{D}$. In particular,

$$\mathcal{S}(f, g) = \langle L_0 f, L_0 g \rangle_2, \quad \mathcal{S}(f, f) > 0 \quad (0 \neq f \in \mathcal{D}). \quad (\text{B.6})$$

For fixed support, local uniformity of the feature bound gives

$$M_A := \sup_{|t| \leq A} \|\mathfrak{S}_t\|_2 < \infty, \quad \|L_0 f\|_2 \leq \pi^{-1/2} M_A \|f'\|_1 \quad (\text{supp } f \subseteq [-A, A]). \quad (\text{B.7})$$

The inequality is Minkowski's inequality applied to (B.5). It proves fixed-support continuity of $\mathcal{S}$. The metric criterion in [7, Theorem 1.4], polarized, is

$$\text{RH} \iff \mathfrak{W}(f, g) = \mathcal{S}(f, g) \quad \text{for all } f, g \in \mathcal{D}. \quad (\text{B.8})$$

B.2 The positive successor in the Suzuki metric

Using the coordinate maps (A.9), set

$$\begin{aligned} G_k &= L_0 \Phi_k, & H_k &= L_0 \Theta_k, \\ \tilde{K}_k &= G_k^* G_k, & \tilde{B}_k &= G_k^* H_k, & \tilde{C}_k &= H_k^* H_k. \end{aligned} \quad (\text{B.9})$$

The metric is positive definite on an injectively coordinated space, so $\tilde{K}_k$ is invertible. Put

$$\tilde{R}_k = \tilde{K}_k^{-1} \tilde{B}_k, \quad \Pi_k^{(0)} = G_k \tilde{K}_k^{-1} G_k^*, \quad D_k^{(0)} = (I - \Pi_k^{(0)}) H_k. \quad (\text{B.10})$$

On the zero-dimensional space these are the unique empty maps.

Theorem B.1 (Unconditional Suzuki successor identity). *For every k ,*

$$\tilde{B}_k = \tilde{K}_k \tilde{R}_k, \quad \tilde{C}_k = \tilde{R}_k^* \tilde{K}_k \tilde{R}_k + (D_k^{(0)})^* D_k^{(0)}. \quad (\text{B.11})$$

Proof. The first identity is the definition of $\tilde{R}_k$. Orthogonal projection onto $\text{Ran } G_k$ gives

$$H_k = G_k \tilde{R}_k + D_k^{(0)}, \quad G_k^* D_k^{(0)} = 0.$$

Taking the Gram matrix of this orthogonal decomposition proves the second identity. $\square$

For any finite family of tests the same metric yields

$$\mathcal{S}\left(\sum_j c_j f_j, \sum_j c_j f_j\right) = \left\| \sum_j c_j L_0 f_j \right\|_2^2 \geq 0. \quad (\text{B.12})$$

B.3 Exact transport statement

The transport equations on the fixed determining sequence are

$$K_1 = \tilde{K}_1, \tag{B.13}$$

$$B_k = K_k \tilde{R}_k, \quad C_k = \tilde{R}_k^* K_k \tilde{R}_k + (D_k^{(0)})^* D_k^{(0)} \quad (k \geq 1). \tag{B.14}$$

Theorem B.2 (Determining-family transport equivalence). *The following statements are equivalent:*

- (i) Equations (B.13)–(B.14) hold;
- (ii) $K_k = \tilde{K}_k$ for every k ;
- (iii) $\mathfrak{W}(f, g) = \mathcal{S}(f, g)$ for every $f, g \in \mathcal{D}$;
- (iv) RH.

Proof. Assume (i). The first equation is the induction base. If $K_k = \tilde{K}_k$, then

$$B_k = K_k \tilde{R}_k = \tilde{K}_k \tilde{R}_k = \tilde{B}_k.$$

The second transport equation and (B.11) give $C_k = \tilde{C}_k$. Thus $K_{k+1} = \tilde{K}_{k+1}$, proving (ii). Equality on all finite spaces gives equality on their union. Lemma A.5, continuity of the completed distribution and (B.7) extend that equality to all of $\mathcal{D}$, proving (iii). Restriction of (iii), followed by Theorem B.1, proves (i). Finally (iii) and (iv) are equivalent by (B.8). $\square$

C The prime-indexed metric defect

C.1 One-direction determining spaces

Let

$$\Delta(f, g) = \mathfrak{W}(f, g) - \mathcal{S}(f, g). \tag{C.1}$$

To construct a determining independent sequence explicitly, use the smooth bump

$$b(x) = \begin{cases} \exp(-(1-x^2)^{-1}), & |x| < 1, \\ 0, & |x| \geq 1. \end{cases}$$

Enumerate all nonzero functions

$$P(x)b((x-c)/r), \quad c \in \mathbb{Q}, \quad r \in \mathbb{Q}_{>0}, \quad P \in (\mathbb{Q} + i\mathbb{Q})[x], \tag{C.2}$$

retaining a function precisely when it is outside the span of those already retained. Denote the retained sequence by (ψ_n) .

Lemma C.1 (Explicit fixed-support determining family). *The sequence (ψ_n) is linearly independent. For every $A > 0$, the elements of its span with compact support in $(-A, A)$ are dense in $C_c^\infty((-A, A))$.*

Proof. Pruning preserves the algebraic span and gives independence. For a given test f , choose rational c, r such that $\text{supp } f \subseteq (c-r, c+r) \subseteq (-A, A)$. Dividing f by the positive bump on this interval and extending by zero near its endpoints gives a smooth function on the closed interval. Polynomial approximation of its derivatives, a diagonal argument and rational approximation of coefficients give tests from (C.2) converging to f with a common compact support inside $(-A, A)$. $\square$

Gram–Schmidt in the positive metric $\mathcal{S}$ gives

$$u_n = \psi_n - \sum_{j < n} \mathcal{S}(\psi_n, \rho_j) \rho_j, \quad \rho_n = \frac{u_n}{\sqrt{\mathcal{S}(u_n, u_n)}}, \quad \mathcal{S}(\rho_i, \rho_j) = \delta_{ij}. \quad (\text{C.3})$$

The vectors u_n are nonzero by independence and strict positivity. Gram–Schmidt preserves every finite span, hence the determining property. Choose s_n greater than the support radius of ψ_n and schedule

$$k_0 = 1, \quad k_n = \min\{k > k_{n-1} : \alpha_k > s_n\}, \quad X_k = \text{span}\{\rho_n : k_n \leq k\}. \quad (\text{C.4})$$

These minima exist by cofinality. Each ρ_n is a linear combination of $\psi_1, \dots, \psi_n$, and

$$\text{supp } \rho_n \subseteq \bigcup_{j \leq n} \text{supp } \psi_j \subseteq (-\alpha_{k_n}, \alpha_{k_n}).$$

Therefore

$$X_1 = \{0\}, \quad X_k \subset \mathcal{D}_{\alpha_k}, \quad X_k \subseteq X_{k+1}, \quad \dim(X_{k+1}/X_k) \leq 1. \quad (\text{C.5})$$

The union is determining. At the initial space both metric matrices are empty, and every empty successor preserves them.

C.2 Successor equations in orthonormal coordinates

Write $V_N = \text{span}\{\rho_1, \dots, \rho_N\}$. At the successor adjoining ρ_n the Suzuki blocks satisfy

$$\tilde{K} = I_{n-1}, \quad \tilde{B} = 0, \quad \tilde{C} = 1, \quad \tilde{R} = 0, \quad D^{(0)}1 = L_0\rho_n, \quad \|D^{(0)}1\|_2^2 = 1. \quad (\text{C.6})$$

Theorem C.2 (Orthonormal successor identity). *Assume that $\mathfrak{W} = \mathcal{S}$ on V_{n-1} . They agree on V_n if and only if*

$$\mathfrak{W}(\rho_n, \rho_j) = 0 \quad (j < n), \quad \mathfrak{W}(\rho_n, \rho_n) = 1. \quad (\text{C.7})$$

Equivalently, the corresponding entries $\Delta(\rho_n, \rho_j)$, $j \leq n$, all vanish.

Proof. In (B.14) the cross-block equation becomes $B = 0$, and the new diagonal equation becomes $C = \|L_0\rho_n\|_2^2 = 1$. These are precisely (C.7). The reformulation in terms of Δ uses (C.3). $\square$

C.3 Exact defect-energy law

Set

$$\mathbf{D}_N = [\Delta(\rho_j, \rho_i)]_{i,j=1}^N, \quad \mathcal{E}_N = \|\mathbf{D}_N\|_F^2 = \sum_{i,j \leq N} |\Delta(\rho_i, \rho_j)|^2 \geq 0. \quad (\text{C.8})$$

The completed Weil matrix is $K_N = I_N + \mathbf{D}_N$.

Theorem C.3 (Noncancellation law). *For*

$$d_N = (\Delta(\rho_{N+1}, \rho_i))_{i=1}^N, \quad \delta_N = \Delta(\rho_{N+1}, \rho_{N+1}) \in \mathbb{R},$$

one has

$$\mathbf{D}_{N+1} = \begin{pmatrix} \mathbf{D}_N & d_N \\ d_N^* & \delta_N \end{pmatrix}, \quad (\text{C.9})$$

and

$$\mathcal{E}_{N+1} - \mathcal{E}_N = 2\|d_N\|_2^2 + |\delta_N|^2 \geq 0. \quad (\text{C.10})$$

Proof. Equation (C.9) is the next compression of the same Hermitian form. The squared Frobenius norm sums the squared moduli of all entries. The new column and its adjoint contribute twice, and the new diagonal entry contributes once. $\square$

Let $\nu(k) = \#\{n : k_n \leq k\}$ and $\mathcal{E}_k = \mathcal{E}_{\nu(k)}$. This is the ordinary-prime version of the same non-decreasing sequence. Its increment vanishes at empty successors and equals (C.10) at a nonempty successor.

For any basis $(v_1, \dots, v_N)$ of a fixed finite-dimensional space V , define

$$G = [\mathcal{S}(v_j, v_i)]_{i,j}, \quad H = [\Delta(v_j, v_i)]_{i,j}, \quad \mathcal{E}(V) = \|G^{-1/2} H G^{-1/2}\|_F^2 = \text{tr}(G^{-1} H G^{-1} H). \quad (\text{C.11})$$

Here $G \succ 0$ and $H = H^*$. A basis change by T sends G to $T^* G T$ and H to $T^* H T$; cyclicity of the trace proves that $\mathcal{E}(V)$ is intrinsic to the fixed space. In the basis $(\rho_1, \dots, \rho_N)$ it equals $\mathcal{E}_N$.

Theorem C.4 (Prime-indexed defect criterion). *The following statements are equivalent:*

- (i) $\mathcal{E}_N = 0$ for every N ;
- (ii) the transport identity holds at every nonempty prime successor;
- (iii) $\Delta(f, g) = 0$ for every $f, g \in \bigcup_N V_N$;
- (iv) $\mathfrak{W}(f, g) = \mathcal{S}(f, g)$ for all $f, g \in \mathcal{D}$;
- (v) RH .

Proof. The sum of squares in (C.8) vanishes exactly when every entry of D_N is zero. The successor equations and induction therefore prove the equivalence of (i)–(iii). The determining union is dense on every fixed support interval. Continuity of convolution and of W proves continuity of $\mathfrak{W}$ there. The estimate (B.7) proves continuity of $\mathcal{S}$. Consequently (iii) and (iv) are equivalent. Equation (B.8) gives the equivalence of (iv) and (v). $\square$

Corollary C.5 (Shrinking-ceiling criterion). *Suppose there are $N_r \rightarrow \infty$ and nonnegative $\varepsilon_r \rightarrow 0$ such that*

$$\mathcal{E}_{N_r} \leq \varepsilon_r. \quad (\text{C.12})$$

Then RH holds. Conversely RH implies (C.12) with $\varepsilon_r = 0$ for every sequence $N_r \rightarrow \infty$.

Proof. For fixed N and all sufficiently large r , monotonicity gives

$$0 \leq \mathcal{E}_N \leq \mathcal{E}_{N_r} \leq \varepsilon_r.$$

The limit forces $\mathcal{E}_N = 0$. Apply Theorem C.4. Its reverse implication gives the converse assertion. $\square$

Thus the inherited transport and defect statements have the exact form

$$\mathfrak{W} = \mathcal{S} \iff \mathcal{E}_N = 0 \ (N \geq 1) \iff \lim_{N \rightarrow \infty} \mathcal{E}_N = 0. \quad (\text{C.13})$$

D Theta log-concavity and accumulated transverse positivity

This appendix gives the complete accumulated-positivity argument in the normalization of (1.3) and (4.6). Throughout, $a \geq 0$ is the horizontal displacement and $t \in \mathbb{R}$ is the ordinate. The theta density η , its entire transform F , and the complete coupled kernel C_a are those of Sections 2 and 4. All density indices and both spatial half-lines are retained.

D.1 The log-concavity input and normalization

The classical theta log-concavity theorem applies to the density in this paper. To record its normalization, write

$$\Phi_C(v) = \sum_{n=1}^{\infty} \pi n^2 (2\pi n^2 e^{4v} - 3) \exp(5v - \pi n^2 e^{4v}), \quad v \geq 0, \quad (\text{D.1})$$

and extend it evenly. Comparison with (1.3) gives

$$\eta(u) = 2\Phi_C(u/2), \quad u \in \mathbb{R}. \quad (\text{D.2})$$

Theorem 4.5 of [6], using its Theorem 4.2 and Remark 4.3(a), states the strict logarithmic concavity of Φ_C . Consequently, for $\ell = \log \eta$,

$$\ell''(u) < 0 \quad (u \neq 0), \quad \eta(u)\eta''(u) - \eta'(u)^2 = \eta(u)^2 \ell''(u) < 0 \quad (u \neq 0). \quad (\text{D.3})$$

Evenness and smoothness imply $\ell'(0) = 0$. The derivative ℓ' is strictly decreasing on all of $\mathbb{R}$: it is strictly decreasing on each open half-line, and its values on the two half-lines have opposite signs. At zero, continuity gives $\ell''(0) \leq 0$.

D.2 The normalized complete coupled kernel

Define the continuous parameter weight

$$w_a(x) = \begin{cases} x \sinh(2ax)/a, & a > 0, \\ 2x^2, & a = 0, \end{cases} \quad \mathcal{K}_a(y) = \int_{\mathbb{R}} \eta(x+y)\eta(x-y)w_a(x) \, dx. \quad (\text{D.4})$$

For every $a \geq 0$, w_a is even and nonnegative, and it is strictly positive when $x \neq 0$. The Gaussian bounds in Lemma 2.1 justify all fixed spatial and parameter derivatives below. Indeed, for $0 \leq a \leq A$, their integrands admit bounds of the form

$$C(1 + |x|)^M e^{2A|x|} e^{-2q(x^2+y^2)}, \quad q > 0, \quad (\text{D.5})$$

as in the proof of Lemma 4.1.

The normalization of the complete transverse derivative is

$$C_a(y) = 2a\mathcal{K}_a(y), \quad Q(a, t) = \frac{K(a, t)}{a} = 2 \int_{\mathbb{R}} \mathcal{K}_a(y) \cos(2ty) \, dy \quad (a > 0). \quad (\text{D.6})$$

At $a = 0$, the last integral defines the continuous value $Q(0, t)$. To see the complete change of variables explicitly, evenness of η and the transform formula give

$$|F(a + it)|^2 = 2 \iint_{\mathbb{R}^2} \eta(x+y)\eta(x-y) \cosh(2ax) \cos(2ty) \, dx \, dy. \quad (\text{D.7})$$

Here $(u, v) = (x + y, x - y)$ has Jacobian 2, and the odd sine and hyperbolic-sine parts integrate to zero. Differentiating (D.7), dividing by $2a$, and using (4.6) proves (D.6). The same majorant permits the limit $a \downarrow 0$ through the integral.

Proposition D.1 (Strict decrease of the associated kernel). *For every $a \geq 0$, the function $\mathcal{K}_a$ is positive, even, smooth, rapidly decreasing, and satisfies*

$$\mathcal{K}'_a(y) < 0 \quad (y > 0). \quad (\text{D.8})$$

Proof. The positive density and weight in (D.4) give strict positivity. The product of densities is unchanged by $y \mapsto -y$, which gives evenness. Smoothness and decay follow from (D.5). For $y > 0$, logarithmic differentiation of the product gives the exact identity

$$\begin{aligned} \partial_y \{ \eta(x+y)\eta(x-y) \} &= \eta(x+y)\eta(x-y) \{ \ell'(x+y) - \ell'(x-y) \} \\ &< 0. \end{aligned} \quad (\text{D.9})$$

The inequality holds for every real x because $x+y > x-y$ and ℓ' is strictly decreasing. Multiplication by $w_a(x)$ and integration therefore yield (D.8); the weight vanishes only at the single point $x = 0$. $\square$

D.3 Sine pairing with the full infinite tail

Theorem D.2 (Accumulated transverse positivity). *For every $a \geq 0$ and $T > 0$,*

$$\int_0^T Q(a, t) \, dt > 0. \quad (\text{D.10})$$

In particular,

$$\int_0^T K(a, t) \, dt > 0 \quad (a > 0, \, T > 0). \quad (\text{D.11})$$

There is an explicit strictly positive lower bound obtained from the first paired sine half-waves.

Proof. Put $\omega = 2T$, $b = \pi/\omega$, and

$$g_a(y) = \frac{\mathcal{K}_a(y)}{y}, \quad y > 0. \quad (\text{D.12})$$

Proposition D.1 gives

$$g'_a(y) = \frac{\mathcal{K}'_a(y)}{y} - \frac{\mathcal{K}_a(y)}{y^2} < 0.$$

Integrating (D.6) over $[0, T]$, then using evenness, yields

$$\int_0^T Q(a, t) \, dt = 2 \int_0^\infty g_a(y) \sin(\omega y) \, dy. \quad (\text{D.13})$$

The original interchange is justified by the integrable kernel and the finite t interval. The last integral is absolutely convergent. Near zero, $\sin(\omega y)/y$ is bounded; at infinity, the rapid decay of $\mathcal{K}_a$ supplies an integrable majorant.

Absolute convergence permits subdivision into all consecutive pairs of half-waves. Translating each pair to $[0, b]$ gives

$$\begin{aligned} \int_0^\infty g_a(y) \sin(\omega y) dy &= \sum_{k=0}^\infty \int_0^b [g_a(y + 2kb) - g_a(y + (2k+1)b)] \sin(\omega y) dy \\ &\geq \int_0^b [g_a(y) - g_a(y+b)] \sin(\omega y) dy > 0. \end{aligned} \quad (\text{D.14})$$

On $(0, b)$, both the difference and the sine are strictly positive. The possible $1/y$ factor at the left endpoint is canceled by the sine, as above. Thus the retained first-pair lower bound is

$$\int_0^T Q(a, t) dt \geq 2 \int_0^b \left[\frac{\mathcal{K}_a(y)}{y} - \frac{\mathcal{K}_a(y+b)}{y+b} \right] \sin(2Ty) dy > 0. \quad (\text{D.15})$$

All later pairs in (D.14) are retained and nonnegative. Finally $K(a, t) = aQ(a, t)$ for $a > 0$, which proves (D.11). $\square$

D.4 Curvature at the origin and the pointwise sine identity

Proposition D.3 (Exact curvature formula). *For every $a \geq 0$,*

$$\mathcal{K}_a''(0) = 2 \int_{\mathbb{R}} w_a(x) [\eta(x)\eta''(x) - \eta'(x)^2] dx < 0. \quad (\text{D.16})$$

Consequently, for $a > 0$,

$$C_a''(0) = 4 \int_{\mathbb{R}} x \sinh(2ax) \eta(x)^2 \ell''(x) dx < 0. \quad (\text{D.17})$$

Proof. Twice differentiating the product under the integral in (D.4) gives

$$\partial_y^2 \{ \eta(x+y)\eta(x-y) \} \big|_{y=0} = 2\eta(x)\eta''(x) - 2\eta'(x)^2.$$

This proves the equality in (D.16). The integrand is strictly negative away from $x = 0$ by (D.3) and the positivity of w_a there. The Gaussian majorant ensures integrability. Multiplication by $2a$ and substitution of w_a give (D.17). $\square$

For completeness, the two sine identities in the main text follow directly from the same normalization. If $I_a(T) = \int_0^T K(a, t) dt$, then

$$I_a(T) = \int_0^\infty \frac{C_a(y)}{y} \sin(2Ty) dy. \quad (\text{D.18})$$

For a fixed $t > 0$, let $h_a(y) = -C_a'(y)$. Integration by parts in $K(a, t) = 2 \int_0^\infty C_a(y) \cos(2ty) dy$ gives

$$K(a, t) = \frac{1}{t} \int_0^\infty h_a(y) \sin(2ty) dy. \quad (\text{D.19})$$

The boundary at zero vanishes because the sine vanishes there, and the boundary at infinity vanishes by Lemma 4.1. Evenness and (D.17) give

$$h_a(0) = 0, \quad h_a'(0) = -C_a''(0) > 0. \quad (\text{D.20})$$

Hence h_a is strictly increasing on some interval $[0, \delta]$. Whenever $b = \pi/(2t)$ satisfies $2b < \delta$, its first paired sine half-waves obey

$$\int_0^b [h_a(y) - h_a(y + b)] \sin(2ty) \, dy < 0. \quad (\text{D.21})$$

Equations (D.18) and (D.19) record the respective weights of the accumulated and pointwise identities with their complete tails.

E Finite forms, compact cutoffs, and the full-line limit

This appendix writes out the limiting passages used in Theorems 6.2 and 14.4. The symbols J , $\mathcal{J}$, $\mathcal{B}$, and K retain their definitions in Sections 4 and 6.

E.1 From finite spatial matrices to the integral form

For $h \in C_c(\mathbb{R})$ with support in $[-R, R]$, take tagged partitions $\mathcal{P}$ of this interval, with tags x_j and lengths $\Delta_j > 0$. Continuity gives the exact limit

$$\langle h, \mathcal{J}h \rangle = \lim_{\|\mathcal{P}\| \rightarrow 0} \sum_{j,k} \overline{h(x_j)\Delta_j} h(x_k)\Delta_k J(x_j, x_k). \quad (\text{E.1})$$

When the finite forms in (14.16) are nonnegative, each sum is nonnegative after coincident nodes are combined. Consequently the limit is nonnegative. Density of $C_c(\mathbb{R})$ in $L^2(\mathbb{R})$ and boundedness of $\mathcal{J}$ give the same inequality for every $h \in L^2(\mathbb{R})$.

E.2 The exponential cutoff and conjugation convention

For $z_1, \dots, z_N \in \mathbb{H}$ and $c \in \mathbb{C}^N$, put

$$h_R(x) = \mathbf{1}_{[-R,R]}(x) \sum_{j=1}^N c_j e^{\overline{z_j}x}. \quad (\text{E.2})$$

Let $A = \max_j \operatorname{Re} z_j$. The Gaussian bound (6.3) bounds the absolute quadratic integrand, independently of R , by a constant times

$$e^{-q(x^2+y^2)+A(|x|+|y|)} \in L^1(\mathbb{R}^2).$$

The double-transform formula therefore gives

$$\sum_{j,k} \overline{c_j} c_k \mathcal{B}(z_j, z_k) = \lim_{R \rightarrow \infty} \langle h_R, \mathcal{J}h_R \rangle. \quad (\text{E.3})$$

The conjugate on z_j in (E.2) yields the conjugate-first matrix convention displayed in (E.3).

For the diagonal, symmetry of J cancels the sine part and gives

$$\boxed{\begin{aligned} \frac{1}{2} \partial_a |\xi(\tfrac{1}{2} + a + it)|^2 &= \widehat{C}_a(2t) = K(a, t) = a \mathcal{B}(a + it, a + it) \\ &= a \iint_{\mathbb{R}^2} J(x, y) e^{a(x+y)} \cos(t(x-y)) \, dx \, dy. \end{aligned}} \quad (\text{E.4})$$

This is the combination of (4.6), (6.5), and (6.6), with the common normalization retained.

E.3 An explicit remainder for the infinite line

Define

$$Q_R(a, t) = \iint_{[-R,R]^2} J(x, y) e^{a(x+y)} \cos(t(x-y)) \, dx \, dy. \quad (\text{E.5})$$

The preceding cutoff argument gives $Q_R(a, t) \geq 0$ whenever $\mathcal{J} \succeq 0$. For all $a > 0$, the unconditional limiting identity and bound are

$$\begin{aligned} K(a, t) &= a \lim_{R \rightarrow \infty} Q_R(a, t), \\ \left| \frac{K(a, t)}{a} - Q_R(a, t) \right| &\leq \frac{M_q^2}{2q} \iint_{\mathbb{R}^2 \setminus [-R, R]^2} e^{-q(x^2+y^2)+a(x+y)} \, dx \, dy \longrightarrow 0. \end{aligned} \quad (\text{E.6})$$

The majorant is integrable for every $a > 0$ and is independent of t . Thus positivity of the compact forms passes to the full-line diagonal.

Finally, Fourier inversion gives the complete coefficient identity

$$\sum_{j,k} \overline{v_j} v_k C_a(y_j - y_k) = \frac{1}{2\pi} \int_{\mathbb{R}} K(a, \omega/2) \left| \sum_k v_k e^{-i\omega y_k} \right|^2 \, d\omega. \quad (\text{E.7})$$

This is the finite form in Lemma 4.3, written with (4.6); its coefficients and nodes are arbitrary.

F Complete source classes, primitive traces, and the retained mixed form

This section assembles the existing whole-class identities in a common normalization. We use

$$F(z) = \xi(\tfrac{1}{2} + z) = \int_{\mathbb{R}} \eta(u) e^{zu} du, \quad J(x, y) = \int_0^\infty (x + y + 2r) \eta(x + r) \eta(y + r) dr, \quad (\text{F.1})$$

where the complete theta density is even and positive. Its derivatives have Gaussian decay of every fixed rate. Throughout this section the sesquilinear form is linear in its first argument:

$$\mathcal{B}_J(f, g) = \iint_{\mathbb{R}^2} J(x, y) f(x) \overline{g(y)} dx dy, \quad Q_J[f] = \mathcal{B}_J(f, f). \quad (\text{F.2})$$

For atomic sources the integrals mean the corresponding finite sums. All sums of spatial sources below retain every pair of indices.

F.1 A common dense core and simultaneous source convergence

Define, initially for compact sources,

$$(Sf)(r) = \int_{\mathbb{R}} f(x) \eta(x + r) dx, \quad (Af)(r) = \int_{\mathbb{R}} (x + r) f(x) \eta(x + r) dx, \quad r \geq 0. \quad (\text{F.3})$$

Young's convolution inequality and restriction to the half-line give

$$\|Sf\|_2 \leq \|\eta\|_1 \|f\|_2, \quad \|Af\|_2 \leq \|u\eta(u)\|_1 \|f\|_2. \quad (\text{F.4})$$

Fubini's theorem therefore gives, first for compact sources and then by continuity on unweighted $L^2(\mathbb{R})$,

$$\mathcal{B}_J(f, g) = \int_0^\infty ((Af) \overline{Sg} + (Sf) \overline{Ag}) dr, \quad (\text{F.5})$$

$$Q_J[f] = 2 \operatorname{Re} \int_0^\infty (Af) \overline{Sf} dr, \quad |Q_J[f]| \leq 2 \|\eta\|_1 \|u\eta(u)\|_1 \|f\|_2^2. \quad (\text{F.6})$$

The half-line expression is used only where its integrals converge; weighted extensions below are defined by the double kernel integral.

Proposition F.1 (Common smooth prime-shift core). *Let*

$$f_L(x) = \sqrt{2/L} \sin(2\pi x/L) \mathbf{1}_{[-L/2, L/2]}(x), \quad L > 0, \quad \Gamma = \mathbb{Z} \log 2 + \mathbb{Z} \log 3. \quad (\text{F.7})$$

*Choose any nonzero $\chi \in C_c^\infty(\mathbb{R})$, put $w = f_L * \chi$, and set*

$$\mathcal{C} = \operatorname{span}_{\mathbb{C}} \{w(\cdot - \gamma) : \gamma \in \Gamma\}. \quad (\text{F.8})$$

Then $\mathcal{C}$ is dense in $H^s(\mathbb{R})$ for every finite real s . In particular it is a common core for the pair (S, A) in (F.3). Any finite collection of continuous linear constraints can be imposed exactly on the approximants.

Proof. Unique factorization shows that $\log 2 / \log 3$ is irrational. Pigeonhole approximation supplies nonzero elements of Γ arbitrarily close to zero; their integer multiples approximate every real number. Thus Γ is dense in $\mathbb{R}$.

The compact smooth function w is nonzero: both factors in $\widehat{w} = \widehat{f}_L \widehat{\chi}$ are nonzero entire functions. Their product has a discrete real zero set and is nonzero almost everywhere. If $v \in H^s$ is orthogonal to every translate in (F.8), the integrable function

$$(1 + \omega^2)^s \widehat{v}(\omega) \overline{\widehat{w}(\omega)}$$

has Fourier transform zero on Γ . Integrability follows from Cauchy–Schwarz in the H^s norm. Its Fourier transform is continuous, so density of Γ makes it zero everywhere. Fourier uniqueness then gives $\widehat{v} \widehat{w} = 0$ almost everywhere, hence $v = 0$. This proves density. Equation (F.4) proves simultaneous convergence of both source coordinates under L^2 approximation.

For the constraint statement, let $\tau : E \rightarrow \mathbb{C}^k$ be continuous on a normed space E in which $\mathcal{C}$ is dense. The image $\tau(\mathcal{C})$ is a linear subspace dense in the finite-dimensional space $\tau(E)$, hence equals that space. Choose fixed $u_1, \dots, u_d \in \mathcal{C}$ whose trace vectors form a basis of $\tau(E)$. For $v \in \ker \tau$ and $v_n \in \mathcal{C}$ converging to v , subtract the unique linear combination of the u_j with trace $\tau(v_n)$. Its coefficients tend to zero. The corrected sequence lies in $\mathcal{C} \cap \ker \tau$ and still converges to v . $\square$

F.2 The primitive identity and weighted continuity

Put

$$B(u) = \frac{1}{2}e^{-u/2} - e^{u/2} \sum_{n \geq 1} e^{-\pi n^2 e^{2u}}, \quad \mathcal{D} = \partial_u^2 - \frac{1}{4}. \quad (\text{F.9})$$

Jacobi’s theta relation makes B even. A strictly decreasing Gaussian sum is smaller than its integral from zero to infinity, which proves $B(u) > 0$. For $u \geq 0$ the displayed formula also gives $B(u) < \frac{1}{2}e^{-u/2}$; evenness supplies the other half-line. Thus

$$0 < B(u) < \frac{1}{2}e^{-|u|/2}, \quad \eta = -\mathcal{D}B, \quad |B'(u)| \leq Ce^{-|u|/2}. \quad (\text{F.10})$$

The derivative bound follows by termwise differentiation on the positive half-line and parity on the negative half-line. Let J_B be the kernel in (F.1) with B in place of η , and write $\tau_b(f) = \int b(x)f(x) dx$.

Proposition F.2 (Exact primitive boundary identity). *For compact smooth f ,*

$$J(x, y) = \mathcal{D}_x \mathcal{D}_y J_B(x, y) + 2B'(x)B'(y) - \frac{1}{2}B(x)B(y), \quad (\text{F.11})$$

$$Q_J[f] = Q_{J_B}[\mathcal{D}f] + 2|\tau_{B'}(f)|^2 - \frac{1}{2}|\tau_B(f)|^2. \quad (\text{F.12})$$

Proof. Write $u = x + r$, $v = y + r$. Applying both differential operators to $(u + v)B(u)B(v)$ gives

$$(u + v)(\mathcal{D}B)(u)(\mathcal{D}B)(v) + 2(B'(u)(\mathcal{D}B)(v) + (\mathcal{D}B)(u)B'(v)).$$

The expression in the second pair of parentheses is

$$\frac{d}{dr}(B'(x + r)B'(y + r) - \frac{1}{4}B(x + r)B(y + r)).$$

Its integral is minus its value at $r = 0$, by (F.10). Since $\mathcal{D}B = -\eta$, this proves (F.11). Two integrations by parts in each spatial variable give (F.12); compact support removes the spatial boundary terms. $\square$

Lemma F.3 (Weighted extension and exact traces). *For every $\epsilon > 0$, the form $\mathcal{B}_J$ is continuous on*

$$\mathcal{H}_\epsilon = L^2(\mathbb{R}, e^{-2\epsilon|x|} dx). \quad (\text{F.13})$$

If $0 < \epsilon < 1/2$, both τ_B and $\tau_{B'}$ are continuous on this space, and

$$\mathcal{C} \cap \ker \tau_B \cap \ker \tau_{B'}$$

is dense in its corresponding closed two-trace subspace.

Proof. For any even b satisfying $|b(u)| \leq Ce^{-a|u|}$, put $m = (x + y)/2$, $d = (x - y)/2$. The function $b(t + d)b(t - d)$ is even in t , so cancellation of the odd integrand gives

$$J_b(x, y) = 2 \int_{|m|}^{\infty} t b(t + d)b(t - d) dt. \quad (\text{F.14})$$

Set $R = \max(|m|, |d|) = (|x| + |y|)/2$. Since $|t + d| + |t - d| = 2 \max(t, |d|)$ for $t \geq 0$, splitting the integral at R gives

$$|J_b(x, y)| \leq 2C^2 e^{-2aR} \left(\frac{R^2}{2} + \frac{R}{2a} + \frac{1}{4a^2} \right). \quad (\text{F.15})$$

The theta density satisfies this estimate for every $a > 0$. Choose $a > \epsilon$. Conjugating the kernel by $e^{\epsilon|x|}e^{\epsilon|y|}$ leaves a square-integrable kernel, which defines a bounded form on unweighted L^2 . This proves the first claim. Cauchy–Schwarz and (F.10) prove continuity of both traces when $\epsilon < 1/2$.

To approximate an element of $\mathcal{H}_\epsilon$, first truncate it to a compact interval, then approximate the truncation in unweighted L^2 by Proposition F.1. Since the weight is at most one, this also approximates it in $\mathcal{H}_\epsilon$. Finally apply the exact finite trace correction from Proposition F.1. The two traces are independent because B is a nonzero even function and B' is a nonzero odd function. $\square$

F.3 The existing two-trace theorem with its full hypothesis

Theorem F.4 (Sufficiency of the primitive two-trace class). *Suppose that*

$$Q_{J_B}[\mathcal{D}f] \geq 0 \quad \text{for every } f \in \mathcal{C} \text{ satisfying } \tau_B(f) = \tau_{B'}(f) = 0. \quad (\text{F.16})$$

Then F has no zeros in $\operatorname{Re} z > 0$ and

$$\operatorname{Re}(F'(a + it)\overline{F(a + it)}) \geq 0 \quad (a > 0, t \in \mathbb{R}). \quad (\text{F.17})$$

Consequently Hypothesis H holds.

Proof. By (F.12), assumption (F.16) says that Q_J is nonnegative on the indicated core. Lemma F.3 extends this nonnegativity to the two-trace subspace of $\mathcal{H}_\epsilon$ for each $0 < \epsilon < 1/2$.

The actual completed zeta function has no zeros with $\operatorname{Re} z \geq 1/2$. Suppose that z_0 is a zero with $0 < \operatorname{Re} z_0 < 1/2$, of multiplicity m . Integration by parts in (F.10) gives

$$\widehat{B}(z) := \int_{\mathbb{R}} B(x)e^{zx} dx = \frac{F(z)}{1/4 - z^2}, \quad |\operatorname{Re} z| < 1/2. \quad (\text{F.18})$$

Set $f_0(x) = x^{m-1}e^{z_0x}$. The function $\widehat{B}$ has a zero of order m at z_0 , so both its derivative of order $m - 1$ and the corresponding derivative of $-z\widehat{B}(z)$ vanish. These are exactly the two traces of f_0 . Changing variables $u = x + r$ in (F.3) and using the multiplicity conditions gives

$$Sf_0 = 0, \quad (Af_0)(r) = F^{(m)}(z_0)e^{-z_0r}, \quad Q_J[f_0] = 0. \quad (\text{F.19})$$

The half-line integrals here converge since $\operatorname{Re} z_0 > 0$.

Choose distinct $w_1, w_2, w_3 \in (0, 1/2)$, and define

$$a_i = \frac{1}{\prod_{j \neq i} (w_i - w_j)}, \quad f_1(x) = \sum_{i=1}^3 \frac{a_i e^{w_i x}}{\widehat{B}(w_i)}. \quad (\text{F.20})$$

All denominators are positive. Lagrange interpolation of constants and linear functions gives $\sum_i a_i = \sum_i a_i w_i = 0$. Hence $\tau_B(f_1) = \tau_{B'}(f_1) = 0$. In the linear-first convention of (F.2), the mixed pairing is

$$\begin{aligned} \mathcal{B}_J(f_0, f_1) &= F^{(m)}(z_0) \sum_{i=1}^3 \frac{a_i(1/4 - w_i^2)}{z_0 + w_i} \\ &= F^{(m)}(z_0) \frac{1/4 - z_0^2}{\prod_{i=1}^3 (z_0 + w_i)} \neq 0. \end{aligned} \quad (\text{F.21})$$

Indeed, $(1/4 - w^2)/(z_0 + w) = -w + z_0 + (1/4 - z_0^2)/(z_0 + w)$; the polynomial terms vanish under the coefficients a_i , and partial fractions give $\sum_i a_i/(z_0 + w_i) = 1/\prod_i (z_0 + w_i)$.

Choose $\max(\operatorname{Re} z_0, w_1, w_2, w_3) < \epsilon < 1/2$. Both tests belong to $\mathcal{H}_\epsilon$ and have zero traces. Their mixed half-line integral is absolutely convergent and agrees with the continuous kernel form. Put $b = \mathcal{B}_J(f_0, f_1) \neq 0$ and $t = -\delta b$ with $\delta > 0$. Then

$$Q_J[f_0 + t f_1] = -2\delta |b|^2 + \delta^2 |b|^2 Q_J[f_1] < 0$$

for sufficiently small δ , contradicting the extended nonnegativity. Thus the open right half-plane is zero-free.

Evenness and real-entire symmetry put all zeros at imaginary pairs $\pm i\gamma$, with multiplicities m_γ . The order-one canonical product, paired using this symmetry, gives for $z = a + it$, $a > 0$,

$$\operatorname{Re} \frac{F'(z)}{F(z)} = \sum_{\gamma > 0} m_\gamma \left(\frac{a}{a^2 + (t - \gamma)^2} + \frac{a}{a^2 + (t + \gamma)^2} \right) \geq 0. \quad (\text{F.22})$$

The paired sum converges locally by the order-one zero summability. Multiplication by $|F(z)|^2$ proves (F.17), and the Fourier equivalence of the main text gives Hypothesis H. $\square$

F.4 The all-prime variance and its exact derivative component

Fix $\alpha > 0$ and $\sigma > (1 + \alpha)/2$, keeping α fixed under differentiation. For $s = \sigma + it$, define

$$\begin{aligned} J_\alpha(d) &= d^\alpha \prod_{p|d} (1 - p^{-\alpha}), & a_d(s) &= \sqrt{J_\alpha(d)} d^{-s}, \\ R &= \sum_{d \geq 1} |a_d|^2 = \frac{\zeta(2\sigma - \alpha)}{\zeta(2\sigma)}, & e &= a/\sqrt{R}. \end{aligned} \quad (\text{F.23})$$

The Euler product proves the displayed expression for R . All logarithmic moments converge locally uniformly in this strict half-plane, since $0 < J_\alpha(d) \leq d^\alpha$. Set

$$\mu(d) = |e_d|^2, \quad \lambda = \mathbb{E}_\mu \log d, \quad V = \operatorname{Var}_\mu(\log d). \quad (\text{F.24})$$

Proposition F.5 (Complete positive color variance). *For every finite or countable partition $\{C_j\}$ of the positive integers into nonempty cells, writing $p_j = \mu(C_j)$ and denoting the conditional mean and variance by λ_j, V_j , one has*

$$V = \sum_j p_j V_j + \sum_j p_j (\lambda_j - \lambda)^2, \quad (\text{F.25})$$

$$V = \sum_p (\log p)^2 \left(\frac{p^{\alpha-2\sigma}}{(1-p^{\alpha-2\sigma})^2} - \frac{p^{-2\sigma}}{(1-p^{-2\sigma})^2} \right) > 0, \quad (\text{F.26})$$

$$V \geq \frac{(2^\alpha - 1)2^{-2\sigma}(\log 2)^2}{R^2} > 0. \quad (\text{F.27})$$

Let $\Psi = \xi(s)a(s)$ and $X = \Psi/\sqrt{R} = \xi(s)e(s)$. With inner products linear in the first slot,

$$\operatorname{Re}(X_\sigma, X) = \operatorname{Re}(\xi'(s)\overline{\xi(s)}), \quad (\text{F.28})$$

$$\|X_\sigma\|^2 = |\xi'(s)|^2 + |\xi(s)|^2 V. \quad (\text{F.29})$$

If $\xi(s) \neq 0$, the derivative Gram complement is

$$\|\Psi_\sigma\|^2 - \frac{|\langle \Psi_\sigma, \Psi \rangle|^2}{\|\Psi\|^2} = R|\xi(s)|^2 V. \quad (\text{F.30})$$

Proof. Expand $\log d - \lambda = (\log d - \lambda_j) + (\lambda_j - \lambda)$ on each cell. Conditional expectation removes the cross term. Summing proves (F.25); the finite second moment justifies a countable partition.

Logarithmic differentiation gives $\lambda = -\frac{1}{2}\partial_\sigma \log R$ and $V = -\frac{1}{2}\partial_\sigma \lambda$. Differentiating the absolutely convergent Euler product yields (F.26). Each summand is positive because $x/(1-x)^2$ increases on $(0, 1)$ and $p^{\alpha-2\sigma} > p^{-2\sigma}$. The identity

$$V = \frac{1}{2} \sum_{d,e \geq 1} \mu(d)\mu(e)(\log d - \log e)^2$$

and the two ordered pairs $(1, 2), (2, 1)$ give (F.27).

Direct differentiation yields $e_\sigma = (\lambda - \log d)e$, so $(e_\sigma, e) = 0$ and $\|e_\sigma\|^2 = V$. Differentiate $X = \xi e$ and take its radial component and squared norm to obtain (F.28)–(F.29). For $\Psi = \xi\sqrt{R}e$, projection perpendicular to e leaves $\xi\sqrt{R}e_\sigma$, whose squared norm is $R|\xi|^2 V$. This is (F.30). $\square$

The positive variance thus occupies a precisely identified component of the derivative. The scalar radial component required by H is exactly (F.28); it is retained in that equality.

F.5 Complete score compensation and its prime-power assembly

For any fixed $\kappa > 0$, define

$$q(u) = -\frac{\eta'(u)}{\eta(u)} - \kappa u. \quad (\text{F.31})$$

For a finite spatial source (x_j, c_j) put

$$\begin{aligned} h(r) &= \sum_j c_j \eta(x_j + r), \\ k(r) &= \sum_j c_j q(x_j + r) \eta(x_j + r), \\ \ell(r) &= \sum_j c_j (x_j + r) \eta(x_j + r). \end{aligned} \quad (\text{F.32})$$

Then $h' = -\kappa\ell - k$. Since $h(\infty) = 0$, integration by parts in the complete half-line gives

$$\kappa Q_J[c] = |h(0)|^2 - 2 \operatorname{Re} \int_0^\infty k(r) \overline{h(r)} dr. \quad (\text{F.33})$$

The same proof holds for compactly supported integrable source functions, with the sums replaced by integrals. If the theta curvature bound is written as $-(\log \eta)'' \geq \kappa$, its consequence here is $q' \geq 0$; evenness of η gives oddness of q . The exact remaining whole-class comparison in these coordinates is

$$2 \operatorname{Re} \int_0^\infty k \bar{h} dr \leq |h(0)|^2. \quad (\text{F.34})$$

In particular, on the class $h(0) = 0$, the required score correlation is nonpositive. Equation (F.34) is the sign condition in (F.33), with its mixed term retained.

The score term also has an exact complete arithmetic assembly. Define

$$g(u) = (4\pi^2 e^{9u/2} - 6\pi e^{5u/2}) e^{-\pi e^{2u}}, \quad T_a v(u) = v(u + a), \quad (\text{F.35})$$

and, on this seed and its differentiated translates, define

$$Z = \sum_{n \geq 1} n^{-1/2} T_{\log n}, \quad P = \sum_{n \geq 2} \Lambda(n) n^{-1/2} T_{\log n}, \quad \mathcal{E}_\kappa = -\partial_u - \kappa u. \quad (\text{F.36})$$

The theta series is exactly $\eta = Zg$. Moreover,

$$[u, Z] = -ZP, \quad \mathcal{E}_\kappa Z = Z\mathcal{E}_\kappa + \kappa ZP. \quad (\text{F.37})$$

Indeed $[u, T_{\log n}] = -(\log n) T_{\log n}$, and the coefficient of $T_{\log n}$ in ZP is

$$n^{-1/2} \sum_{d|n} \Lambda(d) = n^{-1/2} \log n.$$

Absolute locally uniform convergence of the differentiated Gaussian series justifies these identities; on every half-line bounded below the same tails justify the integrations used here.

For the packet in (F.32), let

$$H_g(r) = \sum_j c_j g(x_j + r), \quad K_g(r) = \sum_j c_j (\mathcal{E}_\kappa g)(x_j + r).$$

Equation (F.37) gives the complete identities

$$h = ZH_g, \quad k = ZK_g + \kappa Ph. \quad (\text{F.38})$$

Consequently, on $h(0) = 0$, the same sign condition is

$$\operatorname{Re} \int_0^\infty (ZK_g) \bar{h} dr + \kappa \operatorname{Re} \int_0^\infty (Ph) \bar{h} dr \leq 0. \quad (\text{F.39})$$

Both the original source term and every prime power remain in this formula.

F.6 Positive color covariance through the actual spatial operation

Fix finitely many spatial nodes, put $x_* = \min_j x_j$, and for $r \geq 0$ define real vectors and positive weights by

$$\begin{aligned} (f_n(r))_j &= n^{-1/2} g(x_j + r + \log n), \\ \rho_n(r) &= e^{(r+x_*)/2 - \pi n^2 e^{2(r+x_*)}}, \quad a_n = f_n / \rho_n, \quad R_\rho = \sum_n \rho_n. \end{aligned} \quad (\text{F.40})$$

Writing $\eta_r = (\eta(x_j + r))_j = \sum_n \rho_n a_n$, the complete covariance is

$$C_r = R_\rho \sum_n \rho_n a_n a_n^* - \eta_r \eta_r^* = \sum_{m < n} \rho_m \rho_n (a_m - a_n)(a_m - a_n)^* \succeq 0. \quad (\text{F.41})$$

To prove the identity, expand the right side, pair its ordered cross terms, and collect its diagonal terms. All rearrangements are justified: $\|f_n\|^2 / \rho_n$ is a polynomial in n times a decaying Gaussian, because $x_j \geq x_*$. The same estimate controls the integrated identities below.

Let $D_r = \text{diag}(x_j + r)$. The actual spatial matrix equals

$$\begin{aligned} [J(x_j, x_k)]_{j,k} &= \int_0^\infty (D_r \eta_r \eta_r^* + \eta_r \eta_r^* D_r) dr \\ &= \int_0^\infty \left[R_\rho \sum_n \rho_n (D_r a_n a_n^* + a_n a_n^* D_r) - (D_r C_r + C_r D_r) \right] dr. \end{aligned} \quad (\text{F.42})$$

This is the positive covariance applied to the precise spatial operation, including its sign and every mixed color. For a rank-one term vv^* , the determinant on two nodes is exactly

$$\det[(D_r vv^* + vv^* D_r)_{\{i,j\}}] = -(x_i - x_j)^2 |v_i v_j|^2. \quad (\text{F.43})$$

The formula follows from the diagonal entries $2(x_i + r)|v_i|^2$, $2(x_j + r)|v_j|^2$ and the off-diagonal entry $(x_i + x_j + 2r)v_i \bar{v}_j$. Thus a proof using (F.42) must keep the displayed complete operation; its covariance positivity (F.41) alone does not assign a sign to the resulting spatial matrix.

F.7 Dependency statement for the assembled proof

The source core, its exact trace corrections, the primitive identity, the all-prime variance, and the complete spatial covariance identities above are proved statements. The implication in Theorem F.4 is also proved with its explicit hypothesis (F.16). The identities (F.33) and (F.42) retain the whole-class sign required in that hypothesis; they do not remove it by relabeling a positive variance or covariance. The smaller known-zero family treated elsewhere in this paper is another sufficient test family for the same conclusion. Its nonnegativity remains the precise sign statement to which these assembled identities may be applied.

G Complete density and spatial feature identities

The identities in this section supply the complete transformed densities and boundary terms for Theorem 17.1; its nonnegativity premise concerns the complete local form. Throughout, γ is a real zero of $f = \Xi$ of multiplicity m , and

$$h(u) = f(\gamma + u) = u^m g(u), \quad g(u) = g_\gamma(u), \quad g_0 = g(0) \neq 0, \quad \ell(u) = \frac{g'(u)}{g(u)}. \quad (\text{G.1})$$

The density is normalized by $f(w) = \int_{\mathbb{R}} \eta(t) e^{iwt} dt$. The local moments and their polynomial functional are

$$s_n = -\frac{(\log g)^{(n+2)}(0)}{(n+1)!}, \quad \mathcal{L}_\gamma(X^n) = s_n. \quad (\text{G.2})$$

Local logarithmic derivatives in this formula are independent of the branch of the logarithm. We extend $\mathcal{L}_\gamma$ complex-linearly when its argument has complex coefficients.

G.1 The complete density after removal of a known zero

Proposition G.1 (Zero-removed Fourier density). *Define*

$$\psi_{\gamma,m}(t) = \frac{(-i)^m}{(m-1)!} \int_{-\infty}^t (t-v)^{m-1} e^{i\gamma v} \eta(v) dv \quad (\text{G.3})$$

$$= \frac{i^m}{(m-1)!} \int_t^{\infty} (v-t)^{m-1} e^{i\gamma v} \eta(v) dv. \quad (\text{G.4})$$

Both expressions agree, have rapid Gaussian decay at their respective ends, and satisfy

$$\psi_{\gamma,m}^{(m)}(t) = (-i)^m e^{i\gamma t} \eta(t), \quad g(u) = \int_{\mathbb{R}} \psi_{\gamma,m}(t) e^{iut} dt. \quad (\text{G.5})$$

The transform identity holds for every complex u .

Proof. The vanishing of the first m Taylor coefficients of h is precisely

$$\int_{\mathbb{R}} v^j e^{i\gamma v} \eta(v) dv = 0, \quad 0 \leq j < m. \quad (\text{G.6})$$

Expanding $(t-v)^{m-1}$ and applying (G.6) shows that its full-line integral against $e^{i\gamma v} \eta(v)$ vanishes. Splitting that integral at t , and using $(t-v)^{m-1} = (-1)^{m-1} (v-t)^{m-1}$ on the right half-line, gives (G.4) from (G.3). Differentiating the left primitive m times gives the first identity in (G.5).

For each $q > 0$, the complete theta bound $\eta(v) \leq M_q e^{-qv^2}$ bounds the right primitive and each of its first $m-1$ derivatives by a Gaussian times a polynomial as $t \rightarrow +\infty$. The left primitive gives the corresponding bounds as $t \rightarrow -\infty$. Decreasing the Gaussian exponent absorbs every such polynomial. In particular, these functions and their products with e^{iut} have zero boundary values for every fixed complex u . Integrating by parts m times yields

$$(-iu)^m \int_{\mathbb{R}} \psi_{\gamma,m}(t) e^{iut} dt = (-i)^m \int_{\mathbb{R}} \eta(t) e^{i(\gamma+u)t} dt = (-i)^m h(u).$$

For $u \neq 0$, division gives the second identity in (G.5). Gaussian domination makes its left side entire in u , so the removable value at $u = 0$ follows as well. $\square$

Set

$$\mu_\gamma(t) = \frac{\psi_{\gamma,m}(t)}{g_0}. \quad (\text{G.7})$$

Then $\int_{\mathbb{R}} \mu_\gamma = 1$ and $\mu_\gamma(-t) = \overline{\mu_\gamma(t)}$. For the latter identity, substitute $v \mapsto -v$ in (G.3), use evenness of η , and compare with the conjugate of (G.4); g_0 is real. The function μ_γ is complex-valued. Indeed, if $\psi_{\gamma,m}$ were real everywhere, then its m th derivative would be real everywhere, whereas $(-i)^m e^{i\gamma t} \eta(t)$ is not real for all t . Here $\eta(t) > 0$ and $\gamma \neq 0$, since $f(0) = \int_{\mathbb{R}} \eta(t) dt > 0$.

Define the cumulants of this complete density by

$$\kappa_r = \frac{d^r}{dv^r} \log \int_{\mathbb{R}} e^{vt} \mu_\gamma(t) dt \Big|_{v=0}. \quad (\text{G.8})$$

The mass-one normalization makes the logarithm analytic near $v = 0$. Equation (G.5) gives

$$\log \frac{g(u)}{g_0} = \log \int_{\mathbb{R}} e^{iut} \mu_\gamma(t) dt, \quad s_n = \frac{i^n \kappa_{n+2}}{(n+1)!}. \quad (\text{G.9})$$

The reflection identity for μ_γ implies $\overline{\kappa_r} = (-1)^r \kappa_r$. Thus even cumulants are real and odd cumulants are purely imaginary. In particular, direct expansion of the second cumulant gives the exact formula

$$s_0 = \kappa_2 = \frac{1}{2} \iint_{\mathbb{R}^2} (t-v)^2 \mu_\gamma(t) \mu_\gamma(v) dt dv. \quad (\text{G.10})$$

The two factors in this formula occur without complex conjugation.

G.2 The completed Euler expression and its individual terms

Put $s_* = \frac{1}{2} + i\gamma$ and define the completion contribution

$$A(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \frac{\Gamma'(s/2)}{\Gamma(s/2)}. \quad (\text{G.11})$$

Logarithmic differentiation of the completed zeta function, together with the absolutely convergent Euler derivative for $\text{Re } s > 1$, gives

$$\ell(u) = iA(s_* + iu) - i \sum_{n \geq 2} \Lambda(n) n^{-s_* - iu} - \frac{m}{u}, \quad \text{Im } u < -\frac{1}{2}. \quad (\text{G.12})$$

Here Λ is the von Mangoldt function. Define the holomorphic two-variable version of the local Hermitian kernel by

$$\mathcal{Q}_\gamma(u, v) = -\frac{\ell(u) - \ell(v)}{u - v}, \quad (\text{G.13})$$

with its removable diagonal value. If both u and v lie in the half-plane in (G.12), their joining segment also lies there. Termwise differentiation on that compact segment is justified by absolute locally uniform convergence. The fundamental theorem of calculus therefore gives

$$\begin{aligned} \mathcal{Q}_\gamma(u, v) = \int_0^1 [A'(s_* + i[\tau u + (1-\tau)v]) \\ + \sum_{n \geq 2} \Lambda(n) \log n n^{-s_* - i[\tau u + (1-\tau)v]}] d\tau - \frac{m}{uv}. \end{aligned} \quad (\text{G.14})$$

The signs follow directly from $\ell'(u) = -A'(s_* + iu) - \sum \Lambda(n) \log n n^{-s_* - iu} + m/u^2$ and $\int_0^1 (\tau u + (1 - \tau)v)^{-2} d\tau = 1/(uv)$. Meromorphic continuation of the complete expression agrees with (G.13). Since $g(0) \neq 0$, it is analytic near $(u, v) = (0, 0)$; thus its apparent singularities there cancel. This continuation applies to the completed expression, with the Euler series used in its convergence domain.

For completeness, the sign of an individual finite Euler logarithm term can be extracted exactly. Fix a prime power n , a real $\sigma > 1$, and put

$$L_n = \log n, \quad \theta_n = \gamma L_n, \quad w_{\sigma,n} = \Lambda(n) L_n n^{-\sigma} > 0. \quad (\text{G.15})$$

The real term of the Euler logarithm on this line is

$$a_n(u) = \frac{\Lambda(n) n^{-\sigma}}{L_n} \cos(\theta_n + L_n u).$$

Applying the extraction $-a_n^{(r+2)}(0)/(r+1)!$ for $r = 0, 1, 2$ yields the matrix

$$T_n = w_{\sigma,n} \begin{pmatrix} \cos \theta_n & -L_n \sin \theta_n/2 \\ -L_n \sin \theta_n/2 & -L_n^2 \cos \theta_n/6 \end{pmatrix}. \quad (\text{G.16})$$

Consequently

$$\det T_n = -w_{\sigma,n}^2 L_n^2 \left(\frac{\cos^2 \theta_n}{6} + \frac{\sin^2 \theta_n}{4} \right) < 0. \quad (\text{G.17})$$

More generally, for a real polynomial $P(X) = \sum_{j=0}^N c_j X^j$, define its exponential coefficient polynomial

$$\mathcal{E}_P(z) = \sum_{j=0}^N c_j \frac{z^j}{j!}. \quad (\text{G.18})$$

The full contribution of this one term is

$$w_{\sigma,n} \operatorname{Re} \left[e^{i\theta_n} \int_0^1 \mathcal{E}_P(iL_n \tau) \mathcal{E}_P(iL_n(1 - \tau)) d\tau \right]. \quad (\text{G.19})$$

Indeed, $\int_0^1 \tau^j (1 - \tau)^k d\tau = j!k!/(j+k+1)!$ gives every cross coefficient. Equations (G.17)–(G.19) concern the individual finite term; the complete local form is given by the completion and continuation in (G.14).

G.3 Forms carried by the analytic, quadratic, and prime spaces

The infinite analytic lift has coefficient space

$$\mathcal{E} = \left\{ c = (c_j) : p_R(c) = \sum_{j \geq 0} |c_j| R^j < \infty \text{ for every } R > 0 \right\}. \quad (\text{G.20})$$

Its Fréchet seminorms encode the entire translated function. The quadratic amplitude lift of a coefficient vector is the outer product $G = cc^*$, and for every finitely supported complex q it satisfies

$$q^* G q = |c^* q|^2 \geq 0. \quad (\text{G.21})$$

When a weighted coefficient vector lies in ℓ^2 , the corresponding outer-product operator is positive and trace class. Its translation law $G(z+h) = T_h G(z) T_h^*$ preserves this quadratic form.

The prime two-plane Hilbert construction carries the conserved norm

$$\|V_p u\|_2^2 + \|W_p u\|_2^2 = \|u\|_2^2, \quad (\text{G.22})$$

on its stated L^2 inputs. Iteration retains fresh hidden channels and the terminal reserve. The current local form, in contrast, is specified by the divided difference

$$\mathcal{B}_J(U_z, U_w) = \frac{-\ell(z) + \overline{\ell(w)}}{z - \overline{w}}, \quad \mathcal{B}_J(A_P, A_P) = \mathcal{L}_\gamma(P\overline{P}). \quad (\text{G.23})$$

Here $\overline{P}$ conjugates coefficients, and A_P is the local spatial feature defined below. Equations (G.21) and (G.22) identify the forms for which positivity is already present in those spaces; (G.23) identifies the complete form appearing in Theorem 17.1.

G.4 Spatial feature norms and the complete translation boundary

We use

$$J(x, y) = \int_0^\infty (x + y + 2r) \eta(x+r) \eta(y+r) dr, \quad \mathcal{B}_J(a, b) = \iint_{\mathbb{R}^2} J(x, y) a(x) \overline{b(y)} dx dy. \quad (\text{G.24})$$

Define the two spatial features

$$(Sq)(r) = \int_{\mathbb{R}} \eta(x+r) q(x) dx, \quad (Aq)(r) = \int_{\mathbb{R}} (x+r) \eta(x+r) q(x) dx. \quad (\text{G.25})$$

For compactly supported $q \in L^2(\mathbb{R})$ both features lie in $L^2(0, \infty)$. Absolute integration, followed by expansion of the two squares, gives

$$\begin{aligned} \mathcal{B}_J(q, q) &= 2 \operatorname{Re} \int_0^\infty (Aq)(r) \overline{(Sq)(r)} dr \\ &= \frac{1}{2} (\|Sq + Aq\|_2^2 - \|Sq - Aq\|_2^2). \end{aligned} \quad (\text{G.26})$$

For example, compact support and Cauchy–Schwarz bound the integrals in (G.25) by Gaussian tails times a polynomial in r , which proves their asserted L^2 membership and justifies Fubini.

To specify the local family, set

$$\begin{aligned} e_k(x) &= e^{i\gamma x} \frac{(ix)^k}{k!}, & v(x) &= \frac{e_{m-1}(x)}{g_0}, \\ E_z(x) &= \frac{e^{i(\gamma+z)x} - \sum_{k=0}^{m-2} z^k e_k(x)}{z^{m-1}}, & U_z(x) &= \frac{1}{z} \left(\frac{E_z(x)}{g(z)} - v(x) \right). \end{aligned} \quad (\text{G.27})$$

The sum is empty for $m = 1$, and all removable values are filled. Write

$$U_z(x) = \sum_{j \geq 0} a_j(x) z^j, \quad A_P(x) = \sum_{j=0}^N c_j a_j(x). \quad (\text{G.28})$$

For a fixed real r , translation in the Fourier integral gives

$$S(e^{i(\gamma+z)\cdot})(r) = e^{-i(\gamma+z)r} h(z) = e^{-i(\gamma+z)r} z^m g(z).$$

In particular $Se_k = 0$ for $k < m$, so

$$(SU_z)(r) = e^{-i\gamma r} e^{-izr}, \quad (SA_P)(r) = e^{-i\gamma r} \mathcal{E}_P(-ir). \quad (\text{G.29})$$

For nonzero P , $\mathcal{E}_P$ is a nonzero polynomial; hence $SA_P \notin L^2(0, \infty)$. The exact feature identity for this family is therefore expressed on a finite interval with its complete boundary term.

Proposition G.2 (Complete finite-interval feature identity). *For every $b > 0$ and every polynomial P ,*

$$\begin{aligned} \mathcal{L}_\gamma(P\overline{P}) &= 2 \operatorname{Re} \int_0^b (AA_P)(r) \overline{(SA_P)(r)} \, dr \\ &\quad + m|\tau_b|^2 + \mathcal{L}_\gamma(P_b\overline{P_b}), \end{aligned} \quad (\text{G.30})$$

where

$$\begin{aligned} c_k(b) &= \sum_{j=k}^N c_j \frac{(-ib)^{j-k}}{(j-k)!}, \quad P_b(X) = \sum_{k=0}^N c_k(b) X^k, \\ \tau_b &= \sum_{j=0}^N c_j \frac{(-ib)^{j+1}}{(j+1)!}. \end{aligned} \quad (\text{G.31})$$

The bar in these formulas conjugates coefficients.

Proof. Splitting the defining integral for J at b gives exactly

$$J(x, y) = \int_0^b (x + y + 2r) \eta(x + r) \eta(y + r) \, dr + J(x + b, y + b). \quad (\text{G.32})$$

Every A_P is a polynomial times $e^{i\gamma x}$. The Gaussian bound for η , and the corresponding bound for J , justify absolute integration of (G.32) against $A_P(x) \overline{A_P(y)}$. Fubini applies to the integral over $0 \leq r \leq b$. Translating the variables in the last term gives

$$\begin{aligned} \mathcal{L}_\gamma(P\overline{P}) &= 2 \operatorname{Re} \int_0^b (AA_P)(r) \overline{(SA_P)(r)} \, dr \\ &\quad + \mathcal{B}_J(A_P(\cdot - b), A_P(\cdot - b)). \end{aligned} \quad (\text{G.33})$$

Let $\mathcal{R} = \operatorname{span}\{e_0, \dots, e_{m-2}\}$, with $\mathcal{R} = \{0\}$ when $m = 1$. The local block decomposition gives

$$\mathcal{B}_J(\mathcal{R}, e_k) = 0 \quad (k \geq 0), \quad \mathcal{B}_J(v, v) = m, \quad \mathcal{B}_J(v, a_j) = 0. \quad (\text{G.34})$$

Translation preserves $\mathcal{R}$. Modulo this radical, $E_z(x - b) = e^{-i\gamma b} e^{-izb} E_z(x)$ and $v(x - b) = e^{-i\gamma b} v(x)$, so

$$U_z(x - b) \equiv e^{-i\gamma b} \left(e^{-izb} U_z(x) + \frac{e^{-izb} - 1}{z} v(x) \right) \pmod{\mathcal{R}}. \quad (\text{G.35})$$

Taking the coefficient of z^j in this identity and summing against c_j yields

$$A_P(x - b) \equiv e^{-i\gamma b} (A_{P_b}(x) + \tau_b v(x)) \pmod{\mathcal{R}}.$$

The common phase has modulus one, the radical contributes zero, and the cross terms vanish by (G.34). Hence

$$\mathcal{B}_J(A_P(\cdot - b), A_P(\cdot - b)) = m|\tau_b|^2 + \mathcal{L}_\gamma(P_b\overline{P_b}). \quad (\text{G.36})$$

Combining (G.33) and (G.36) proves (G.30). $\square$

The primitive-balanced restriction is $P = \chi_\gamma R$, where

$$\chi_\gamma(X) = (\gamma^2 + \tfrac{1}{4})X^2 + 2\gamma X + 1. \quad (\text{G.37})$$

It eliminates the two primitive traces in the local criterion. The translation boundary in (G.36) has its own exact coefficient formula. In particular,

$$\frac{d\tau_b}{db} = -i\mathcal{E}_P(-ib). \quad (\text{G.38})$$

Thus τ_b is not identically zero for nonzero P . If $\deg P = N$ with $c_N \neq 0$, then $m|\tau_b|^2$ has leading term

$$\frac{m|c_N|^2}{((N+1)!)^2} b^{2N+2}, \quad (\text{G.39})$$

whereas $\mathcal{L}_\gamma(P_b \overline{P_b})$ has degree at most $2N$ in the real variable b . The full boundary in (G.36) consequently remains a nonzero polynomial. The complete relation to the local criterion is (G.30), including this positive known-zero contribution and the translated local form.

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Part II

Complete developments through Checkpoint 20

This part contains the complete twelve continuations following the last consolidated PDF. Each unit retains its calculations, proofs, domains, boundary terms, source citations, and stated unresolved estimates. The prefixed equation tags distinguish identical local numbers in different notes.

Checkpoint 9: Heat, defect, and two-trace intersection

Source: *Michaels_Heat_Defect_Two_Trace_Intersection.md*. Equation tags carry the source prefix CP9; references within this unit retain their original local numbering.

Analytical continuation of the Michaels consolidated local-amplitude manuscript
22 September 2026 · Research checkpoint 9

The three requested investigations were carried out in parallel and compared on their common domains. The resulting direct bridge is

$$Q_\gamma(R) = -\delta < 0 \implies \mathcal{E}_N \geq \left(1 + \frac{\delta}{6\mathcal{S}(q, q)}\right)^2 > 1 \quad (N \geq N_0), \quad (\text{CP9:I1})$$

where the construction produces a nonzero compact smooth Weil test q , with $\mathcal{S}(q, q) > 0$. Thus a specific negative local witness reaches a permanent completed arithmetic defect. The August archive already supplies the closing *criterion* $\mathcal{E}_N \leq 1$ on an unbounded subsequence; it does not supply an independent proof of that upper bound.

The other two tests determine what their proposed operations actually preserve. The Gaussian limiting kernel has rank one, and its positive limit coexists with nonreal zeros at every finite negative theta deformation. Two-trace projection cancels the primitive boundary correction exactly, but a real polynomial in the specified trace-free local family still has a negative spatial integrand on a short interval. That interval is not a negative value of the complete form.

Proof status: the direct witness transport and the route tests below are analytical results. The restricted nonnegativity and the Riemann Hypothesis are not established by this continuation. No numerical zero tests are used.

The one target and its established identification

Use the normalization of the consolidated manuscript:

$$F(z) = \xi(\tfrac{1}{2} + z), \quad f(w) = F(iw) = \Xi(w) = \int_{\mathbb{R}} \eta(x) e^{iwx} dx.$$

Here η is the actual even positive theta density, with superexponential decay. Fix a known real zero γ of multiplicity m , and write

$$f(\gamma + z) = z^m g(z), \quad g_0 = g(0) \neq 0, \quad s_n = -\frac{(\log g)^{(n+2)}(0)}{(n+1)!}.$$

The real linear functional $\mathcal{L}_\gamma$ is defined by $\mathcal{L}_\gamma(X^n) = s_n$. The established feature construction is

$$e_k(x) = e^{i\gamma x} \frac{(ix)^k}{k!}, \quad E_z(x) = \frac{e^{i(\gamma+z)x} - \sum_{k=0}^{m-2} z^k e_k(x)}{z^{m-1}},$$

$$U_z(x) = \frac{1}{z} \left(\frac{E_z(x)}{g(z)} - \frac{e_{m-1}(x)}{g_0} \right) = \sum_{j \geq 0} a_j(x) z^j, \quad A_P = \sum_j c_j a_j \quad \left(P = \sum_j c_j X^j \right).$$

An empty sum is zero. These formulas are analytic at $z = 0$. In particular, every A_P is a polynomial times $e^{i\gamma x}$.

For $L = \partial_x^2 - 1/4$, the decaying even positive primitive satisfies $\eta = -LB$. Define

$$J(x, y) = \int_0^\infty (x + y + 2r)\eta(x + r)\eta(y + r) dr,$$

and define J_B by the same formula with B in place of η . The forms are linear in their first argument:

$$\mathcal{B}_J(a, b) = \iint J(x, y)a(x)\overline{b(y)} dx dy.$$

Write

$$\chi_\gamma(X) = 1 + 2\gamma X + (\gamma^2 + \frac{1}{4})X^2.$$

The complete existing identification is

$$\boxed{Q_\gamma(R) := \mathcal{L}_\gamma((\chi_\gamma R)^2) = \mathcal{B}_J(A_{\chi_\gamma R}, A_{\chi_\gamma R}) = \mathcal{B}_{J_B}(LA_{\chi_\gamma R}, LA_{\chi_\gamma R})} \quad (\text{CP9:I2})$$

for every real polynomial R . Both primitive traces of $A_{\chi_\gamma R}$ are zero. The local moment/resolvent theorem already proves that $Q_\gamma(R) \geq 0$ for all real R , at this one fixed zero, is equivalent to all nontrivial zeta zeros lying on the critical line. The factor χ_γ loses no possible zero: its only potential blind locations are $\Xi(\pm i/2) = 1/2$.

Comparison across the three routes

Route	Exact retained structure	Result relevant to the target
Heat and complex plane	Complete theta deformation, moving primitive and zero anchor	The PDE is correct; positive Gaussian convergence does not preserve the global sign at finite deformation.
Local witness and prime ladder	Complete Weil form, exact packet coefficients, Suzuki normalization	Every negative local witness forces a permanent defect strictly greater than one.
Quadratic form and two traces	Weighted orthogonal projection and the full spatial mixed operation	The boundary correction vanishes; adverse integrand contributions remain even for real R .
Positive prime-color norms	Existing positive Hilbert and Smith-Jordan forms	Their positivity is preserved in their own metrics. The routes above do not identify the remaining compressed spatial form with those norms.

The proofs that follow keep these objects distinct. A Gaussian probability limit, a normalized radical boundary matrix, and the unnormalized Suzuki defect are different quantities. Their signs or decay rates cannot be exchanged without an exact comparison.

A. Heat deformation and its Gaussian limit

A.1 Recovered source results

The latest consolidated manuscript, `Michaels_Consolidated_Theta_Local_Amplitude_Proof.tex`, lines 1550-1578, already proves the exact heat equation quoted in the directive. It is correct.

The earlier `Michaels_Hypothesis_H_Combined_Routes.pdf`, section 5, lines 590-698, supplies the full moment evolution, a positive degree-lowering map, an exact Gaussian averaging transfer, and the remaining signed Schur-complement costs.

The August 11 `Arithmetic_Light_and_the_Coherent_Closure_Conjecture_v1.1.tex`, lines 958-989, already records the de Bruijn-Newman threshold, Rodgers-Tao nonnegativity, and explicitly describes its occupied-edge principle as heuristic.

The September 5 `Prime_DNA_Diffusion_Investigation.md`, lines 23-45, establishes exact finite divisor-source evolution while retaining the completion and coordinate qualifications separating that evolution from the actual completed heat flow. Its finite reversal experiment does not establish a spectral invariant cone.

Primary source independently checked: B. Rodgers and T. Tao, *The de Bruijn-Newman constant is non-negative*, arXiv:1801.05914v5, <https://arxiv.org/html/1801.05914v5>. Equations (1), (3), (4) and introduction lines 57-78 specify normalization and the all-real-zero threshold. Theorem 1, lines 94-97, proves that threshold is nonnegative.

A.2 The exact flow and what it preserves

With the manuscript's even positive density,

$$\eta_\lambda(u) = e^{\lambda u^2} \eta(u), \quad F_\lambda(z) = \int_{\mathbb{R}} \eta_\lambda(u) e^{zu} du, \quad K_\lambda(a, t) = \Re(F'_\lambda(a + it) \overline{F_\lambda(a + it)}),$$

superexponential tails justify all fixed derivatives locally uniformly. Then

$$\partial_\lambda F_\lambda = F''_\lambda, \quad \partial_\lambda K_\lambda = \frac{1}{2}(\partial_a^2 - \partial_t^2) K_\lambda. \quad (\text{CP9:H1})$$

For completeness, differentiating the product gives

$$\partial_a^2 K = \Re(F''' \overline{F} + 2F'' \overline{F'} + F' \overline{F''}),$$

$$\partial_t^2 K = \Re(-F''' \overline{F} + 2F'' \overline{F'} - F' \overline{F''}).$$

The mixed products cancel in the difference, proving (H1). The generator has one positive and one negative second-derivative coefficient; it is not a positive heat semigroup on arbitrary nonnegative fields in the two variables.

For the complete degree-N moment matrix

$$M_N(\lambda, a)_{jk} = \frac{1}{2\pi} \int_{\mathbb{R}} t^{j+k} K_\lambda(a, t) dt, \quad (D_N)_{jk} = k \mathbf{1}_{j=k-1},$$

the recovered exact equation is

$$\partial_\lambda M_N = \frac{1}{2} \partial_a^2 M_N - D_N^* M_N D_N - \frac{1}{2} [(D_N^2)^* M_N + M_N D_N^2]. \quad (\text{CP9:H2})$$

In backward parameter time $\tau = \lambda_* - \lambda$, the degree terms alone generate the positive map

$$T_\tau(X) = e^{\tau(D_N^2)^*/2} \left[\sum_{k=0}^N \frac{\tau^k}{k!} (D_N^k)^* X D_N^k \right] e^{\tau D_N^2/2}. \quad (\text{CP9:H3})$$

The full backward equation also contains $-\frac{1}{2} \partial_a^2 M_N$. Omitting this term changes the equation. For a positive lower block A, the source's exact Schur-complement formula is

$$\partial_\lambda h_N = \frac{1}{2} \partial_a^2 h_N + v_a^* A^{-1} v_a - Q_\lambda(P'_N). \quad (\text{CP9:H4})$$

Thus the positive backward lower-degree contribution is retained together with the horizontal-curvature and inverse-matrix contributions. The archive supplies no uniform estimate eliminating these costs.

The exact averaging identity is

$$\int_{\mathbb{R}} G_\tau(b-a) K_{\lambda_*-\tau}(a, t) da = \int_{\mathbb{R}} G_\tau(t-s) K_{\lambda_*}(b, s) ds, \quad (\text{CP9:H5})$$

where $G_\tau(x) = (2\pi\tau)^{-1/2} e^{-x^2/(2\tau)}$. Starting at the classical positive value $\lambda_* = 1/8$ and taking $\tau = 1/8$ proves the corresponding positive Gaussian average of the undeformed field. The variance remains fixed at $1/8$; this identity does not give pointwise positivity by taking that variance to zero while keeping the same certified endpoint.

A.3 The actual Gaussian limit is rigorous, and its scope is exact

Set $T > 0$ and

$$G_T(z) = \frac{F_{-T}(\sqrt{T}z)}{F_{-T}(0)}.$$

The substitution $u = v/\sqrt{T}$ gives

$$G_T(z) = \frac{\int_{\mathbb{R}} e^{-v^2} \eta(v/\sqrt{T}) e^{zv} dv}{\int_{\mathbb{R}} e^{-v^2} \eta(v/\sqrt{T}) dv} \longrightarrow e^{z^2/4}. \quad (\text{CP9:H6})$$

This convergence is locally uniform in the complex plane and holds for every fixed derivative. Indeed η is bounded and continuous with $\eta(0) > 0$; on $|z| \leq R$, every derivative is dominated by a constant times $|v|^k e^{-v^2+R|v|}$. The denominator tends to $\eta(0)\sqrt{\pi} > 0$.

Consequently the rescaled amplitude derivative obeys

$$\frac{\sqrt{T}}{F_{-T}(0)^2} K_{-T}(\sqrt{T}a, \sqrt{T}t) \longrightarrow \frac{a}{2} e^{(a^2-t^2)/2} > 0 \quad (a > 0). \quad (\text{CP9:H7})$$

The endpoint spatial form has an additional exact feature. For the normalized limiting density $\rho_G(x) = \pi^{-1/2} e^{-x^2}$,

$$J_G(x, y) = \int_0^\infty (x + y + 2r) \rho_G(x + r) \rho_G(y + r) dr = \frac{1}{2\pi} e^{-x^2 - y^2}. \quad (\text{CP9:H8})$$

The equality follows by integrating the derivative of $e^{-(x+r)^2 - (y+r)^2}$. Therefore

$$\mathcal{B}_{J_G}(a, a) = \frac{1}{2\pi} \left| \int_{\mathbb{R}} e^{-x^2} a(x) dx \right|^2.$$

This Gaussian form has rank one. It has infinitely many zero directions on polynomial test spaces, so its positivity supplies no uniform strictly positive margin for the full polynomial family.

The Gaussian limit is zero-free. By locally uniform convergence and Hurwitz's theorem, zeros of the scaled F_{-T} cannot remain in any fixed compact set as $T \rightarrow \infty$. A local family anchored at a zero therefore has no fixed finite zero anchor at this limiting Gaussian endpoint.

A.4 Exact comparison with the de Bruijn-Newman family

Rodgers-Tao use

$$H_t(z) = \int_0^\infty e^{tu^2} \Phi(u) \cos(zu) du, \quad H_0(z) = \frac{1}{8} \Xi(z/2).$$

In the manuscript normalization, $\eta(2u) = 2\Phi(u)$. Writing $\Xi_\lambda(w) = F_\lambda(iw)$ therefore gives the exact conversion

$$\boxed{\Xi_\lambda(w) = 8H_{4\lambda}(2w)}. \quad (\text{CP9:H9})$$

The all-real-zero region is $4\lambda \geq \Lambda$, and Rodgers-Tao proves $\Lambda \geq 0$. It follows unconditionally that, for every finite $\lambda < 0$, Ξ_λ has nonreal zeros. Reflection gives a zero $a_0 + it_0$ of F_λ with $a_0 > 0$. If its multiplicity is m and $F_\lambda(z) = (z - z_0)^m h(z)$, then

$$K_\lambda(a_0 - \varepsilon, t_0) = -m|h(z_0)|^2 \varepsilon^{2m-1} + O(\varepsilon^{2m}) < 0$$

for sufficiently small positive $\varepsilon < a_0$.

Thus (H6)-(H7) coexist with a negative complete amplitude derivative at every finite negative deformation parameter. The exact theta family itself demonstrates why positive Gaussian convergence on compacts does not imply global positivity before the endpoint.

At positive deformation, the classical value $\lambda = 1/8$ has only real zeros. Preservation all the way from that endpoint to $\lambda = 0$ would require $\Lambda \leq 0$; together with the known lower bound this is precisely $\Lambda = 0$. The archive's positive degree map and positive averaging identity do not independently establish this reverse-preservation statement.

A.5 Compatibility with the primitive and the two-trace family

Keep $L = \partial_x^2 - \frac{1}{4}$. The exact deformed positive primitive is

$$B_\lambda(x) = \int_{\mathbb{R}} e^{-|x-y|/2} \eta_\lambda(y) dy, \quad -LB_\lambda = \eta_\lambda. \quad (\text{CP9:H10})$$

This follows from the Green kernel of $-\partial_x^2 + \frac{1}{4}$. The primitive is even and strictly positive. At zero deformation it agrees with the manuscript's decaying modular primitive, by uniqueness of the decaying solution.

The exact commutator is

$$-L(x^2 B_\lambda) = x^2 \eta_\lambda - 4x B'_\lambda - 2B_\lambda,$$

and hence

$$\boxed{\partial_\lambda B_\lambda = x^2 B_\lambda + (-L)^{-1}(4x B'_\lambda + 2B_\lambda).} \quad (\text{CP9:H11})$$

Consequently $e^{\lambda x^2} B_0$ is not the primitive for η_λ . The trace complement must move with B_λ : for a differentiable test family a_λ ,

$$\frac{d}{d\lambda} \tau_{B_\lambda} a_\lambda = \tau_{\partial_\lambda B_\lambda} a_\lambda + \tau_{B_\lambda} \partial_\lambda a_\lambda, \quad (\text{CP9:H12})$$

and there is the analogous identity for B'_λ . Two traces vanishing at one parameter do not, by themselves, make the two derivatives in (H12) vanish.

Whenever a deformed real zero γ_λ and the associated zero-removal family are defined, the Fourier multiplier

$$\widehat{B}_\lambda(w) = \Xi_\lambda(w)/(w^2 + 1/4)$$

retains the same algebraic annihilator

$$\chi_{\gamma_\lambda}(X) = (\gamma_\lambda^2 + \frac{1}{4})X^2 + 2\gamma_\lambda X + 1.$$

But the zero, inverse germ, test vectors and primitive all depend on the deformation parameter. At a simple real zero, implicit differentiation gives $\gamma'_\lambda = \Xi''_\lambda(\gamma_\lambda)/\Xi'_\lambda(\gamma_\lambda)$, because $\partial_\lambda \Xi_\lambda = -\Xi''_\lambda$. This formula is local to a simple-zero branch, not a global collision-exclusion theorem.

Finally, negative heat deformation strengthens the already established logarithmic curvature:

$$-(\log \eta_{-T})'' = -(\log \eta)'' + 2T. \quad (\text{CP9:H13})$$

Thus strict positive density, stronger curvature, a positive primitive, and exact two-trace algebra all coexist within the actual deformed theta family with nonreal zeros. An argument completing the undeformed theorem must use additional information special to its completed arithmetic structure. This does not obstruct a valid such additional identity; it specifies what the recovered generic positivity statements do not yet provide.

A.6 Route result

The proposed heat PDE is verified. The Gaussian endpoint limit, its rank-one spatial form, and the exact transport equations are now explicit. The recovered source contains genuine positive finite-degree maps and averaged inequalities. Neither a preserved full sign nor the target restricted local inequality follows from those maps. No unconditional closure was found in this route.

B. The direct local-witness to prime-defect transfer

B.1 Result

There is a direct, sign-preserving passage from a **specific negative finite local form** to a negative compact completed-Weil test. It uses the established spatial kernel identity, Gaussian weighted density of exponential packets, and the manuscript's existing compact-packet approximation. It does not infer the compact witness solely by invoking RH equivalence.

If

$$\mathcal{L}_\gamma((\chi_\gamma R)^2) = \mathcal{B}_J(A_{\chi_\gamma R}, A_{\chi_\gamma R}) = -\delta < 0,$$

the construction below produces a compact smooth test q with

$$\mathfrak{W}(q, q) = -\eta \leq -\delta/2, \quad s := \mathcal{S}(q, q) > 0.$$

It then produces a finite determining-ladder index N_0 such that

$$\boxed{\mathcal{E}_N \geq \left(1 + \frac{\eta}{3s}\right)^2 \geq \left(1 + \frac{\delta}{6s}\right)^2 > 1 \quad (N \geq N_0).} \quad (\text{CP9:D1})$$

All constants refer to the constructed fixed witness. No bound on s , or on the required index N_0 , uniform over all witnesses is asserted.

An independently established bound $\liminf_N \mathcal{E}_N \leq 1$ would collide with (D1). The August 25 collection already proves this threshold-one criterion. A shrinking bound $\mathcal{E}_N \rightarrow 0$ is sufficient but is not needed for this contradiction. No upper bound on this same $\mathcal{E}_N$ was recovered in the checked asymptotic sources.

B.2 The exact existing defect

The manuscript uses the same completed forms as the original August 23 defect note:

$$\Delta(f, g) = \mathfrak{W}(f, g) - \mathcal{S}(f, g), \quad \mathcal{S}(f, g) = \langle L_0 f, L_0 g \rangle.$$

Suzuki positivity gives $\mathcal{S}(q, q) > 0$ for every nonzero compact smooth test. Let (ρ_n) be its orthonormal fixed-support determining sequence, $V_N = \text{span}\{\rho_1, \dots, \rho_N\}$, and

$$\mathbf{D}_N = [\Delta(\rho_j, \rho_i)]_{i,j \leq N}, \quad \mathcal{E}_N = \|\mathbf{D}_N\|_F^2, \quad \mathbf{W}_N = I_N + \mathbf{D}_N.$$

The exact same-form compression law is

$$\mathbf{D}_{N+1} = \begin{pmatrix} \mathbf{D}_N & d_N \\ d_N^* & \delta_N \end{pmatrix}, \quad \boxed{\mathcal{E}_{N+1} - \mathcal{E}_N = 2\|d_N\|_2^2 + |\delta_N|^2 \geq 0.} \quad (\text{CP9:D2})$$

This is the unnormalized Frobenius defect in Suzuki-orthonormal coordinates. Its invariance is under basis change on a fixed seat. It is not independent of the chosen determining ladder.

B.3 Direct transfer from the local witness to an Euler-line packet

Write $A = A_{\chi_\gamma R}$. The current local theorem makes A a polynomial times $e^{i\gamma x}$. The actual complete theta density satisfies, for each $a > 0$,

$$\eta(x) \leq M_a e^{-ax^2}, \quad 0 < J(x, y) \leq C_a e^{-a(x^2+y^2)}, \quad C_a = \frac{M_a^2}{2a}. \quad (\text{CP9:D3})$$

Fix one such a , and set

$$\mathcal{H}_a = L^2(\mathbb{R}, e^{-ax^2} dx), \quad \kappa_a = C_a \sqrt{\frac{\pi}{a}}.$$

Multiplication by $e^{-ax^2/2}$ identifies this space with ordinary L^2 . The resulting integral kernel is

$$\tilde{J}_a(x, y) = J(x, y) e^{a(x^2+y^2)/2}.$$

Equation (D3) gives

$$\|\tilde{J}_a\|_{\text{HS}}^2 \leq C_a^2 \iint e^{-a(x^2+y^2)} dx dy = C_a^2 \frac{\pi}{a}.$$

Thus the Hermitian form obeys

$$|\mathcal{B}_J(u, v)| \leq \kappa_a \|u\|_{\mathcal{H}_a} \|v\|_{\mathcal{H}_a}. \quad (\text{CP9:D4})$$

In particular, $A \in \mathcal{H}_a$.

Fix any $b > 1/2$. The family

$$\{e^{(b+it)x} : t \in \mathbb{R}\}$$

has dense linear span in $\mathcal{H}_a$. To prove this, suppose h is orthogonal to all these functions. The measure

$$h(x) e^{bx} e^{-ax^2} dx$$

has finite total variation by Cauchy-Schwarz, since $\int e^{2bx-ax^2} dx < \infty$. Its Fourier transform vanishes at every real frequency. Fourier uniqueness gives $h = 0$ almost everywhere. This proves density.

Let $A_0 = \|A\|_{\mathcal{H}_a}$. Choose

$$0 < \varepsilon \leq \min \left\{ 1, \frac{\delta}{4\kappa_a(2A_0 + 1)} \right\}.$$

Density supplies a finite packet

$$v(x) = \sum_{j=1}^m a_j e^{z_j x}, \quad z_j = b + it_j, \quad \|v - A\|_{\mathcal{H}_a} \leq \varepsilon.$$

Using (D4),

$$|\mathcal{B}_J(v, v) - \mathcal{B}_J(A, A)| \leq \kappa_a \varepsilon (2A_0 + \varepsilon) \leq \delta/4,$$

so

$$\mathcal{B}_J(v, v) \leq -3\delta/4. \quad (\text{CP9:D5})$$

On this Euler-convergent line, $F(z_j) \neq 0$. Put

$$d_j = a_j F(z_j), \quad f(u) = \mathbf{1}_{u \geq 0} \sum_{j=1}^m d_j e^{-z_j u}, \quad g = F'/F.$$

The established double Laplace identity and the established completed arithmetic phase identity give, with all pole, prime-power, gamma and scalar terms retained,

$$\begin{aligned} \mathcal{B}_J(v, v) &= \sum_{j,k} a_j \bar{a}_k \frac{F'(z_j) \overline{F'(z_k)} + F(z_j) \overline{F(z_k)}}{z_j + \bar{z}_k} \\ &= \sum_{j,k} d_j \bar{d}_k \frac{g(z_j) + \overline{g(z_k)}}{z_j + \bar{z}_k} \\ &= \mathfrak{W}_{\text{ext}}[f]. \end{aligned} \quad (\text{CP9:D6})$$

This is the explicit coefficient transformation between the two packets. The Gaussian approximation step comes before this exact finite equality; no globally bounded map between the two Hilbert spaces is claimed.

The manuscript's compact-packet approximation lemma supplies $q_m \in C_c^\infty((0, \infty))$ with

$$\mathfrak{W}(q_m, q_m) \longrightarrow \mathfrak{W}_{\text{ext}}[f].$$

Its proof retains the gamma boundary at zero and controls the complete prime series using $b > 1/2$. Choose one $q = q_m$ for which the difference is at most $\delta/4$. Equations (D5)-(D6) then yield

$$\mathfrak{W}(q, q) \leq -\delta/2. \quad (\text{CP9:D7})$$

This carries the original negative witness to the genuine compact Weil domain.

B.4 Quantitative fixed-support passage to the prime ladder

Let $\text{supp } q \subseteq (-A, A)$, $\eta = -\mathfrak{W}(q, q) > 0$, and $s = \mathcal{S}(q, q) > 0$. The letter η in this subsection denotes the negative margin, not the theta density.

Here is an explicit continuity constant sufficient for this passage. For $h, k \in C_c^\infty((-A, A))$, use

$$\|h\|_{H^1} = (\|h\|_2^2 + \|h'\|_2^2)^{1/2}.$$

Define the finite absolute constants

$$I_1 = \int_0^1 \frac{ve^{-v/2}}{1 - e^{-2v}} dv, \quad I_2 = \int_1^\infty \frac{e^{-v/2} + e^{-v}}{1 - e^{-2v}} dv,$$

and

$$C_A^W = 8 \sinh A + 2 \sum_{2 \leq n \leq e^{2A}} \frac{\Lambda(n)}{\sqrt{n}} + |\log(4\pi) + \gamma_E| + 3I_1 + 2I_2. \quad (\text{CP9:D8})$$

The paired completed Weil formula proves

$$|\mathfrak{W}(h, k)| \leq C_A^W \|h\|_{H^1} \|k\|_{H^1}. \quad (\text{CP9:D9})$$

Indeed the correlation $R(v) = \int h(u+v) \overline{k(u)} du$ obeys

$$|R(v)| \leq \|h\|_2 \|k\|_2, \quad |R(v) - R(0)| \leq |v| \|h'\|_2 \|k\|_2.$$

Its support lies in $[-2A, 2A]$. These bounds give the first two terms of (D8) for poles and prime powers. For $0 < v \leq 1$, the paired gamma numerator is at most

$$ve^{-v/2} (2\|h'\|_2 \|k\|_2 + \|h\|_2 \|k\|_2),$$

and its tail is at most $2(e^{-v/2} + e^{-v})\|h\|_2 \|k\|_2$. This proves (D9).

The original Suzuki continuity bound is

$$\|L_0 h\|_2 \leq \pi^{-1/2} M_A \|h'\|_1 \leq k_A \|h\|_{H^1}, \quad k_A = \sqrt{2A/\pi} M_A, \quad M_A = \sup_{|t| \leq A} \|\mathfrak{S}_t\|_2 < \infty. \quad (\text{CP9:D10})$$

Choose a fixed-support determining approximation $w \in V_{N_0} \cap C_c^\infty((-A, A))$ such that

$$\|w - q\|_{H^1} \leq \varepsilon_1,$$

where

$$0 < \varepsilon_1 \leq \min \left\{ 1, \frac{\eta}{2C_A^W(2\|q\|_{H^1} + 1)}, \frac{(\sqrt{3/2} - 1)\sqrt{s}}{k_A} \right\}. \quad (\text{CP9:D11})$$

Equations (D9)-(D11) imply

$$\mathfrak{W}(w, w) \leq -\eta/2, \quad 0 < \mathcal{S}(w, w) \leq 3s/2. \quad (\text{CP9:D12})$$

Write $w = \sum_{j=1}^{N_0} c_j p_j$. Since $\mathcal{S}(w, w) = \|c\|_2^2$, the normalized vector $u = c/\sqrt{\mathcal{S}(w, w)}$ has unit norm and

$$u^* \mathbf{D}_{N_0} u = \frac{\mathfrak{W}(w, w)}{\mathcal{S}(w, w)} - 1 \leq -1 - \frac{\eta}{3s}.$$

Hence

$$\mathcal{E}_{N_0} \geq \|\mathbf{D}_{N_0}\|_{\text{op}}^2 \geq \left(1 + \frac{\eta}{3s}\right)^2.$$

Monotonicity (D2) proves (D1) for all later indices. The ordinary-prime index merely repeats this value at empty successors.

For any already finite ladder vector w the sharper exact expression is

$$\boxed{\mathcal{E}_N \geq \left(1 + \frac{-\mathfrak{W}(w, w)}{\mathcal{S}(w, w)}\right)^2 \quad (N \geq N_0), \quad \mathfrak{W}(w, w) < 0.} \quad (\text{CP9:D13})$$

B.5 What upper estimate would collide with this floor?

The original August 23 note proves the conditional shrinking-ceiling criterion

$$\mathcal{E}_{N_r} \leq \epsilon_r, \quad N_r \rightarrow \infty, \quad \epsilon_r \rightarrow 0.$$

The August 25 collection, in *Prime-Power Impulse Gate*, Theorem 2.1 and Corollary 2.2, already strengthens the usable criterion to

$$\boxed{\mathcal{E}_{N_r} \leq 1 \quad \text{on any unbounded subsequence.}} \quad (\text{CP9:D14})$$

Equivalently, because the sequence is nondecreasing, $\liminf_N \mathcal{E}_N \leq 1$ suffices. The proof is the normalized Rayleigh argument above, followed by Weil's criterion and the independently stated Suzuki equivalence. Under RH the defect is zero, so the source gives an equivalence, not an independently established bound.

For the given witness, any proved upper ceiling strictly below the right side of (D1) would suffice. Neither convergence of new increments to zero nor a bound on $\mathcal{E}_N/N$ conflicts with a positive permanent floor. Summable increments may converge to a nonzero total as well.

B.6 Checked asymptotic sources and their actual quantities

Build Log 050, 14 August 2026 The actual proved estimates read

$$\|e_\lambda\|_2 = O(\lambda^{-1/2}),$$

$$\|F_{\lambda, h_\lambda} - F_h\|_{L^2(I_\lambda, d^*u)} = O(\lambda^{-1/2} \log \lambda) + \text{poly}(\lambda) e^{-c\lambda^2},$$

with a weighted Mellin observation estimate of order $O_\alpha(\lambda^{-\alpha} \log \lambda)$ for $0 < \alpha < 1/2$. These are approximation errors of a particular radical vector and its forward source. The source itself retains the projected completed graph-norm condition

$$\frac{C_\alpha(\lambda)}{\delta_{\text{gap},\lambda}^+} \|(I - P_{k_\lambda^+})A_\lambda k_\lambda^+\|_2 \longrightarrow 0$$

as unproved, and explains that ordinary L^2 convergence does not control the unbounded archimedean multiplier or the pole-moment family. No comparison to the entire Suzuki-normalized defect matrix D_N is provided. These estimates are therefore compatible with (D1).

Build Log 052A, 14 August 2026 For each fixed Hermite radical flag the raw tail matrix is eventually positive, and after boundary-value normalization its leading term is

$$(D_N^\pm)^{-1} W_{\sqrt{N}}^\pm (D_N^\pm)^{-1} = \frac{\log N}{2\pi N} \mathbf{11}^* + O_m(N^{-1}).$$

Here $D_N^\pm$ is a diagonal matrix of boundary values, **not** the Suzuki-Weil defect D_N . The authentic effective matrix retains

$$\mathfrak{F}_\lambda^\pm(b) = W_\lambda^\pm - bG_\lambda^\pm - (C_\lambda^\pm)^*(H_\lambda^\pm(b))^{-1}C_\lambda^\pm.$$

Its threshold and subtracted self-energy are not controlled by raw-tail positivity. The statement is neither an upper bound for $\mathcal{E}_N$ nor a sign theorem for the present local witness.

The checked archive therefore gives a rigorous permanent lower floor and several genuine decays in different norms. It does not currently force their collision. The exact missing assertion for this route is an independent upper bound for the **same unnormalized Suzuki defect** that falls below the permanent floor; (D14) is one sufficient universal version.

B.7 Source record and scope

1. Current unified TeX: `consolidated_proof/output/Michaels_Consolidated_Theta_Local_Amplitude_Proof.tex`, spatial double transform, exact arithmetic-to-kernel identification (`eq:weil-phase-match`), compact-packet approximation (`lem:weil-packet-approximation`), Suzuki continuity (`app:ladder-L0-bound`), and defect Appendix C (`app:defect-energy` through `app:defect-ceiling`). These exact formulas were read locally.
2. `Christophe_Michaels_Prime_Anchored_Suzuki_Weil_Defect_Energy.docx`, lib, file, read all 312 extracted lines. Its extraction drops most mathematical runs; the current TeX and collected PDF restore their formulas. The text explicitly retains the unnormalized shrinking ceiling as a condition.
3. `Prime_DNA_Collected_Papers_Official_Edition_2026(2).pdf`, lib, file, read lines 1850-2000: *Prime-Power Impulse Gate*, Theorem 2.1 and Corollary 2.2, threshold-one defect criterion and successor law. PDF pages 47-49, printed collection pages 37-39. Find snippets additionally located the original shrinking-ceiling discussion at lines 8082-8099 and 8277-8298.
4. `Build_Log_050_Global_Radical_Subtraction_Arithmetic_Interface_Graph_and_Constrained_Schur.md`, lib, file, read lines 1065-1192 and 1470-1511; exact equations (9.4), (9.7)-(9.10), (14.1), and the retained graph-norm and gap conditions.
5. `Build_Log_052A_Finite_Radical_Flag_Boundary_Feshbach_Pencil.md`, lib, file, read lines 1-125 and find snippets at 810-839, 994-1010, 1092-1108, 1215-1231. Exact leading matrix and effective-pencil definitions establish the distinction of quantities. A September 9 audit ledger was used only to locate this original, not as the proof of its estimates.

The Gaussian-density witness transport and explicit continuity constants (D3)-(D13) are the analytical continuation supplied in this draft. The permanent-defect principle and the threshold-one criterion were already present in the prior proofs. No equality between the old approximation errors and the present defect energy has been introduced.

C. Two-trace projection and the retained spatial operation

C.1 Sources and normalization

Checked the current consolidated manuscript's Appendix F: the primitive identity and weighted continuity (TeX lines 4190-4240); the two-trace theorem with its explicit sign hypothesis (the section beginning at line 4284); and the positive color covariance through the actual spatial operation (lines 4528-4578). Also checked the archived two-trace theorem in `Michaels_Density_Recovery_and_Parallel_Route_Comparison.md`, Section 6 (lines 314-407). The one-trace theorem in the earlier balanced-class note is not substituted for it.

Use

$$f(w) = \Xi(w) = \int_{\mathbb{R}} \eta(x) e^{iwx} dx, \quad L = \partial_x^2 - \frac{1}{4}, \quad \eta = -LB.$$

The actual B is positive and even, B' is odd, and the source formula gives

$$B(x) \sim \frac{1}{2} e^{-x/2}, \quad B'(x) \sim -\frac{1}{4} e^{-x/2} \quad (x \rightarrow +\infty).$$

The complete η and $x\eta$ decay faster than every exponential. The recovered identity is

$$J = L_x L_y J_B + 2B'(x)B'(y) - \frac{1}{2}B(x)B(y). \quad (\text{CP9:T1})$$

C.2 The correct orthogonal projector

Fix $0 < \epsilon < 1/2$, set $w = e^{-2\epsilon|x|}$, and use

$$\mathcal{H}_\epsilon = L^2(\mathbb{R}, w dx), \quad \langle a, b \rangle_\epsilon = \int a \bar{b} w dx$$

linear in the first argument. This space contains every local polynomial-exponential test A_P . Its two trace representers are

$$b_0 = B/w, \quad b_1 = B'/w, \quad \tau_B(a) = \langle a, b_0 \rangle_\epsilon, \quad \tau_{B'}(a) = \langle a, b_1 \rangle_\epsilon.$$

They are orthogonal by parity and have finite positive squared norms κ_j . Therefore the orthogonal projection onto both zero traces is exactly

$$\boxed{\Pi a = a - \frac{\tau_B(a)}{\kappa_0} b_0 - \frac{\tau_{B'}(a)}{\kappa_1} b_1.} \quad (\text{CP9:T2})$$

Let the bounded self-adjoint K_ϵ represent $\mathcal{B}_J$:

$$(K_\epsilon a)(y) = w(y)^{-1} \int J(x, y) a(x) dx.$$

The proved Gaussian bound on J gives this bounded realization. With $(u \otimes v)a = \langle a, v \rangle_\epsilon u$, put

$$C_\epsilon = K_\epsilon - 2b_1 \otimes b_1 + \frac{1}{2}b_0 \otimes b_0.$$

On the smooth core this is the $L_x L_y J_B$ part of (T1); the formula defines its continuous extension. The exact compression is

$$\boxed{\Pi K_\epsilon \Pi = \Pi C_\epsilon \Pi.} \quad (\text{CP9:T3})$$

Thus the projection deletes precisely the rank-two boundary correction. A sign for the remaining compressed operator is not supplied by (T3).

C.3 Actual theta test of the projected mixed operation

For finite $r \geq 0$, let

$$u_r(x) = \eta(x+r)/w(x), \quad v_r(x) = (x+r)\eta(x+r)/w(x).$$

These are continuous $\mathcal{H}_\epsilon$ -valued functions of r . The signed spatial integrand is represented by

$$K_r = u_r \otimes v_r + v_r \otimes u_r.$$

At $r = 0$, its exact compression is

$$\boxed{\Pi K_0 \Pi = p \otimes q + q \otimes p, \quad p = \Pi(\eta/w), \quad q = \Pi(x\eta/w).} \quad (\text{CP9:T4})$$

The vector p is even and q is odd, so they are orthogonal. Both are nonzero: $p = 0$ would give $\eta = cB$ by parity, while $q = 0$ would give $x\eta = cB'$. The different asymptotic rates force $c = 0$ in either case, contradicting $\eta > 0$.

Consequently (T4) has the two eigenvalues

$$\boxed{+\|p\|_\epsilon \|q\|_\epsilon, \quad -\|p\|_\epsilon \|q\|_\epsilon.} \quad (\text{CP9:T5})$$

The trace-free unit vector

$$a_- = \frac{1}{\sqrt{2}} (p/\|p\|_\epsilon - q/\|q\|_\epsilon)$$

has a negative signed integrand at zero and, by continuity, over a small initial interval.

This uses the actual theta density and its actual two-trace projection. It proves that the projection does not annihilate all indefinite spatial cross operations. It does not prove that the complete J -form is negative: the remaining spatial integral and the exact retained boundary may compensate. All operations here are at finite spatial parameter; no divergent half-line feature norm is invoked.

C.4 The same effect within the specified local family

Keep the known real zero γ , its multiplicity m , the zero-removed g , $\ell = g'/g$, and

$$\chi_\gamma(X) = 1 + 2\gamma X + D_0 X^2, \quad D_0 = \gamma^2 + \frac{1}{4}.$$

Define the spatial features

$$(Sa)(r) = \int \eta(x+r)a(x) dx, \quad (Aa)(r) = \int (x+r)\eta(x+r)a(x) dx.$$

The already explicit generator U_z satisfies

$$SU_z(r) = e^{-i\gamma r} e^{-izr},$$

$$AU_z(r) = -ie^{-i\gamma r} \left[m \frac{e^{-izr} - 1}{z} + e^{-izr} \ell(z) \right]. \quad (\text{CP9:T6})$$

Changing variables in the complete theta integral and using $h(z) = z^m g(z)$ proves these formulas. Since $\ell(z) = \ell(0) - \sum_{n \geq 0} s_n z^{n+1}$, at $r = 0$ they give

$$SA_P(0) = c_0, \quad AA_P(0) = -i\ell(0)c_0 + i \sum_{j \geq 1} c_j s_{j-1}, \quad P(X) = \sum_j c_j X^j. \quad (\text{CP9:T7})$$

For $k \geq 1$, set

$$d_k = s_{k-1} + 2\gamma s_k + D_0 s_{k+1} = \sum_{\rho \neq \gamma} m_\rho \lambda_\rho^{k+1} \chi_\gamma(\lambda_\rho). \quad (\text{CP9:T8})$$

There is a $k \geq 2$ with $d_k \neq 0$. Otherwise the normally convergent meromorphic function

$$\sum_{\rho \neq \gamma} \frac{m_\rho \lambda_\rho^2 \chi_\gamma(\lambda_\rho)}{1 - \lambda_\rho z}$$

would be the constant d_1 near zero. Its residue at each shifted zero is nonzero: $\chi_\gamma(\lambda_\rho) \neq 0$ because the only possible lost zero locations are $\rho = \pm i/2$, where $\Xi = 1/2$. The identity theorem gives a contradiction.

Choose such $k \geq 2$, and set

$$\alpha = \frac{1 - d_1}{d_k}, \quad R(X) = X + \alpha X^k, \quad P = \chi_\gamma R, \quad a = A_P.$$

Every coefficient is real, and both primitive traces vanish. The coefficients of P satisfy

$$c_0 = 0, \quad c_1 = 1, \quad \sum_{j \geq 1} c_j s_{j-1} = d_1 + \alpha d_k = 1.$$

Consequently (T6)-(T7) give

$$Sa(0) = 0, \quad (Sa)'(0) = -i, \quad Aa(0) = i.$$

For the real-valued signed integrand $I_a(r) = 2\Re[(Aa)(r)\overline{Sa(r)}]$, this proves

$$\boxed{I_a(0) = 0, \quad I'_a(0) = -2, \quad I_a(r) = -2r + O(r^2) < 0 \quad (0 < r \ll 1).} \quad (\text{CP9:T9})$$

Differentiation is justified by the complete theta decay. This demonstrates nonannihilation inside the real-polynomial target class itself. The existence argument selects a finite polynomial analytically and requires no numerical degree search.

Equation (T9) is a statement about the signed spatial integrand. It does not give the sign of the complete J -form; the exact translation boundary and the remaining integral must still be retained.

C.5 Exact relation to the positive color covariance

For finite real nodes x_j , put $x_* = \min_j x_j$, $D_r = \text{diag}(x_j + r)$, and

$$\psi(u) = (4\pi^2 e^{9u/2} - 6\pi e^{5u/2}) e^{-\pi e^{2u}}, \quad (f_n(r))_j = n^{-1/2} \psi(x_j + r + \log n),$$

$$\rho_n(r) = e^{(r+x_*)/2 - \pi n^2 e^{2(r+x_*)}}, \quad a_n = f_n/\rho_n, \quad R_\rho = \sum_n \rho_n, \quad \eta_r = \sum_n \rho_n a_n.$$

All second moments converge absolutely. The positive covariance is exactly

$$C_r = R_\rho \sum_n \rho_n a_n a_n^* - \eta_r \eta_r^* = \sum_{m < n} \rho_m \rho_n (a_m - a_n)(a_m - a_n)^* \succeq 0.$$

The archived finite-node identity retains the full symmetrized spatial operation:

$$J_{\text{nodes}} = \int_0^\infty \left[R_\rho \sum_n \rho_n (D_r a_n a_n^* + a_n a_n^* D_r) - (D_r C_r + C_r D_r) \right] dr, \quad C_r \succeq 0.$$

For any orthogonal P , with $Q = I - P$, direct expansion gives

$$\begin{aligned} P(D_r C_r + C_r D_r)P &= (PD_r P)(PC_r P) + (PC_r P)(PD_r P) \\ &\quad + PD_r Q C_r P + PC_r Q D_r P. \end{aligned} \tag{CP9:T10}$$

Although $PC_r P$ is positive, both its symmetrized coordinate operation and the displayed cross terms remain.

For a real rank-one feature the compressed contribution is

$$(PD_r v)(Pv)^* + (Pv)(PD_r v)^*.$$

For $Pv \neq 0$, it is positive semidefinite exactly when $PD_r v = \alpha Pv$ with a real $\alpha \geq 0$. The potentially nonzero eigenvalues are

$$\langle Pv, PD_r v \rangle \pm \|Pv\| \|PD_r v\|.$$

This is a precise alignment condition for termwise positivity, not an additional necessary condition for proving positivity of the complete integral by compensation or another method.

C.6 The actual heat deformation moves the trace complement

The heat route uses

$$\eta_\lambda = e^{\lambda x^2} \eta, \quad B_\lambda = (-L)^{-1} \eta_\lambda = e^{-|\cdot|^2/2} * \eta_\lambda.$$

The correctly normalized primitive is generally different from $e^{\lambda x^2} B$. Its exact derivative is

$$\boxed{\partial_\lambda B_\lambda = x^2 B_\lambda + (-L)^{-1} (4x B'_\lambda + 2B_\lambda).} \tag{CP9:T11}$$

Indeed $(-L)(x^2 B_\lambda) = x^2 \eta_\lambda - 4x B'_\lambda - 2B_\lambda$.

The weighted projector Π_λ defined by (T2) with B_λ, B'_λ must vary. If $\partial_\lambda B_\lambda = c B_\lambda$, applying $-L$ would give $x^2 \eta_\lambda = c \eta_\lambda$, impossible since $\eta_\lambda > 0$ on the line. Parity excludes a B'_λ term. Therefore the even trace representer moves outside its original span.

For $a \in \text{ran } \Pi_\lambda$, differentiating the exact projector gives

$$(\partial_\lambda \Pi_\lambda) a = - \sum_{j=0}^1 \frac{\langle a, \partial_\lambda b_{j,\lambda} \rangle_\epsilon}{\kappa_{j,\lambda}} b_{j,\lambda}, \quad b_{0,\lambda} = B_\lambda/w, \quad b_{1,\lambda} = B'_\lambda/w. \tag{CP9:T12}$$

Consequently any transported test must satisfy

$$\tau_{B_\lambda}(\partial_\lambda a_\lambda) + \tau_{\partial_\lambda B_\lambda}(a_\lambda) = 0$$

and the analogous equation for B'_λ . The compressed operator derivative is exactly

$$\partial_\lambda(\Pi_\lambda K_\lambda \Pi_\lambda) = \Pi_\lambda(\partial_\lambda K_\lambda) \Pi_\lambda + (\partial_\lambda \Pi_\lambda) K_\lambda \Pi_\lambda + \Pi_\lambda K_\lambda (\partial_\lambda \Pi_\lambda). \quad (\text{CP9:T13})$$

The last two transport terms do not disappear from the two original trace equalities.

At a real zero γ_λ where the zero-removal construction is available, recomputing $A_{P,\lambda}$ and χ_{γ_λ} gives exact cancellation at that parameter. This does not make the original complement fixed, and it does not delete (T13). No real zero is assumed to continue for every heat parameter.

C.7 Result and exact scope

The orthogonal two-trace compression is explicit and deletes the primitive boundary operator exactly. The actual spatial mixed operation survives that projection, including within the specified local polynomial family. The moving primitive complement also contributes a nonzero heat-transport derivative.

No negative value of the complete J -form has been deduced. These results neither exclude a positive complete compensation nor exclude an arithmetic sign proof specific to the undeformed density. They establish that two-trace cancellation alone does not turn the retained spatial operation into termwise unconditional prime norms.

D. What the intersection establishes

The direct transfer in Part B starts with the actual local form (I2). It uses a Gaussian weighted space only to approximate its negative value by a finite exponential packet. It does not transport the primitive traces to that packet or identify the Gaussian weighted space with the trace Hilbert space in Part C. Such trace preservation is unnecessary: the complete Weil criterion admits the resulting compact test without those restrictions.

The coefficient map $d_j = a_j F(b + it_j)$ then passes to the genuine completed arithmetic form. Every pole, prime-power, archimedean and scalar term is retained. Finally, the Suzuki-orthonormal ladder fixes the normalization in which the threshold is one. The original witness, its compact image, and its finite ladder approximation are linked by explicit sign margins, rather than by an assumed equality between unrelated positive norms.

The heat limit and two-trace results supply no upper bound for this same defect. Their exact transport formulas also explain why their separate positive structures cannot be combined by keeping the metric or the trace complement fixed. In the actual negative heat family, positive density and stronger curvature coexist with nonreal zeros. In the undeformed local family, exact cancellation of both traces coexists with a negative short-interval spatial contribution. Neither fact decides the sign of the complete undeformed form.

The contradiction available from the recovered arithmetic criterion is now explicit:

$$\begin{aligned} \exists R \in \mathbb{R}[X] : Q_\gamma(R) < 0 &\implies \exists \varepsilon > 0, N_0 : \mathcal{E}_N \geq 1 + \varepsilon \quad (N \geq N_0), \\ \liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1 &\implies Q_\gamma(R) \geq 0 \quad \text{for every real polynomial } R. \end{aligned} \quad (\text{CP9:I3})$$

The second line has its displayed hypothesis. The checked archive does not establish that hypothesis independently; its August 25 *Prime-Power Impulse Gate* explicitly leaves the all-knot arithmetic inequality unproved. A proved upper bound below each hypothetical witness's permanent floor would also suffice. There is no requirement to establish positivity separately for every prime contribution, to delete every tail, or to construct an exact prime-norm representation.

Archive coverage and file record

This was a targeted recovery of relevant sources from the last two months, guided by prior-context records and direct source searches. Directly checked sources include the August 11 heat-threshold discussion; August 14 Builds 050 and 052A; August 23 Suzuki-Weil defect note; August 25 collected threshold-one theorem and its closing status; September 5 diffusion note; September 21 combined heat, density, covariance and prime-transfer notes; and the September 22 consolidated manuscript. The source records within Parts A-C give the specific locations used. This is not a claim to have read every page in every historical file.

The full main manuscript remains at checkpoint 8. This continuation, the updated running ledger, and checkpoint 9 preserve the new results with that paper and all previously packaged research notes. The three working route drafts are intermediate assembly material; this document is their consolidated deliverable.

We are aiming to prove $\mathcal{B}_{J_B}(LA_{\chi_\gamma R}, LA_{\chi_\gamma R}) \geq 0$ for every real polynomial R , at one fixed known zero, thereby forcing the backward-propagation contradiction.

Checkpoint 10: Uniform defect-cap tests

Source: *Michaels_Uniform_Defect_Cap_Analytical_Tests.md*. Equation tags carry the source prefix CP10; references within this unit retain their original local numbering.

Christophe Michaels research continuation · 22 September 2026 · Checkpoint 10

This note tests the three proposed sources of an unconditional ceiling for

$$\boxed{\liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1, \quad \mathcal{E}_N = \|\mathbf{D}_N\|_F^2.} \quad (\text{CP10:U1})$$

It uses the existing completed Weil form $\mathfrak{W}$, positive Suzuki metric $\mathcal{S}$, and the same determining ladder. The preceding continuation already transports every hypothetical negative local witness into a permanent defect strictly above one. The issue here is the upper bound on that same defect.

The extraction analysis yields a new bound valid at every polynomial degree:

$$\boxed{\text{tr } X_m \leq \frac{1}{175\pi}, \quad \|X_m\|_F^2 \leq \frac{1}{(175\pi)^2}.} \quad (\text{CP10:U2})$$

Its operator and metric are specified below. The PNT calculation and Parseval calculation also have exact outcomes. None of these three tests establishes (U1). All computations below are symbolic analytical arguments; no numerical zero or degree ladder is used.

1. Fixing the quantity throughout the argument

Let $\mathcal{D} = C_c^\infty(\mathbb{R})$, with the source's complex Hermitian forms linear in their first argument. Write

$$\Delta(q, r) = \mathfrak{W}(q, r) - \mathcal{S}(q, r), \quad \mathcal{S}(q, r) = \langle L_0 q, L_0 r \rangle_2.$$

Let (ρ_j) be the established Suzuki-orthonormal determining sequence, and $V_N = \text{span}\{\rho_1, \dots, \rho_N\}$. Then

$$\mathbf{D}_N = [\Delta(\rho_j, \rho_i)]_{i,j \leq N}, \quad \mathbf{W}_N = I_N + \mathbf{D}_N,$$

$$\mathcal{E}_{N+1} - \mathcal{E}_N = 2 \sum_{i \leq N} |\Delta(\rho_{N+1}, \rho_i)|^2 + |\Delta(\rho_{N+1}, \rho_{N+1})|^2 \geq 0. \quad (\text{CP10:U3})$$

In particular, (U1) means $\mathcal{E}_N \leq 1$ for every N . Once a finite prefix exceeds one, subsequent dimensions retain it.

For any other basis (q_j) of the same finite space, the exact invariant is

$$G_N = [\mathcal{S}(q_j, q_i)], \quad Z_N = [\Delta(q_j, q_i)], \quad \boxed{\mathcal{E}(V_N) = \|G_N^{-1/2} Z_N G_N^{-1/2}\|_F^2.} \quad (\text{CP10:U4})$$

This is the normalization used in every comparison below.

2. Finite extraction energy: an actual uniform theorem

The earlier theta extraction uses $0 \leq a < b < 1/2$, $\delta = b^2 - a^2$, and

$$H_0 = F(b)^2 - F(a)^2, \quad F(y) = \Xi(iy).$$

The archived complete-theta bounds give

$$0 < \frac{H_0}{\delta} < \frac{1}{80}, \quad 0 < T \leq 1, \quad b_T(x) = (1 - x^2/T^2)_+^4. \quad (\text{CP10:U5})$$

Here H_0 is the scalar extraction amplitude. Its parameter is unrelated to the de Bruijn-Newman function bearing that subscript. The source obtains the amplitude estimate from the positive even Taylor coefficients of F : $s \mapsto F(\sqrt{s})^2$ is convex and its derivative at $s = 1/4$ is $2F(1/2)F'(1/2) = F'(1/2) < 1/80$. The original source supplies the complete bound on this last integral. Normalized diagonal limits retain (U5).

For the full monomial vector $v_m(x) = (1, x, \dots, x^m)^T$, define the extraction matrix

$$X_m = \frac{H_0}{2\pi\delta} \int_{-T}^T v_m(x) v_m(x)^* b_T(x) dx \succeq 0. \quad (\text{CP10:U6})$$

The symbol X_m distinguishes this matrix from the defect matrix D_N . Since $T \leq 1$, one has $b_T(x) \leq (1 - x^2)_+^4$. Consequently

$$\begin{aligned} \text{tr } X_m &\leq \frac{1}{160\pi} \int_{-1}^1 \left(\sum_{j=0}^m x^{2j} \right) (1 - x^2)^4 dx \\ &\leq \frac{1}{160\pi} \int_{-1}^1 (1 - x^2)^3 dx = \frac{1}{160\pi} \frac{32}{35} = \frac{1}{175\pi}. \end{aligned} \quad (\text{CP10:U7})$$

The endpoints have measure zero. Positivity gives $\|X_m\|_F \leq \text{tr } X_m$, proving (U2). Equivalently, the infinite extraction matrix is trace class on ordinary ℓ^2 of monomial coefficients. This proof handles the entire degree class at once.

The metric transition is explicit. If a given test parametrization has monomial coefficient map T_N , its extraction contribution in Suzuki-orthonormal coordinates is

$$Y_N = G_N^{-1/2} T_N^* X_m T_N G_N^{-1/2}, \quad \|Y_N\|_F \leq \frac{\|T_N G_N^{-1/2}\|_{\text{op}}^2}{175\pi}. \quad (\text{CP10:U8})$$

The archive does not identify the entire completed defect with this extraction contribution or bound this metric transition uniformly. Strict positivity of each finite G_N gives an inverse at that dimension, but supplies no uniform lower eigenvalue across the ladder. Even a raw numerator bound would yield only

$$\|G_N^{-1/2} Z_N G_N^{-1/2}\|_F^2 \leq \lambda_{\min}(G_N)^{-2} \|Z_N\|_F^2. \quad (\text{CP10:U9})$$

3. What the complete exponential tails control

The recovered height-tail parameter is

$$\tau_L(\rho) = \sum_{\ell > L} (\ell + 1) \rho^\ell = \frac{\rho^{L+1} [(L+2) - (L+1)\rho]}{(1-\rho)^2} \leq 2^{-L}, \quad 0 \leq \rho \leq \frac{1}{4}. \quad (\text{CP10:U10})$$

The September 20 bound retains every theta index and both spatial tails:

$$|e_L(x)| \leq C_s \tau_L e^{-\alpha|x|}, \quad C_s = 18874368, \quad \alpha = \frac{6}{5}. \quad (\text{CP10:U11})$$

In the Hilbert space $\mathcal{H}_\alpha = L^2(e^{-\alpha|x|}dx/(2\pi))$, this proves

$$\left| \frac{1}{2\pi} \int e_L(x) |P(x)|^2 dx \right| \leq C_s \tau_L \|P\|_{\mathcal{H}_\alpha}^2. \quad (\text{CP10:U12})$$

Thus the relative remainder is a bounded Hermitian operator, with norm at most $C_s \tau_L$, on the polynomial closure. On any m -dimensional orthonormal family in that metric, its compression satisfies

$$\|T_{L,m}\|_F^2 \leq m(C_s \tau_L)^2. \quad (\text{CP10:U13})$$

This estimate retains dimension explicitly. In the original monomial coordinates the positive reference has diagonal entries

$$\frac{1}{2\pi} \int_{\mathbb{R}} x^{2j} e^{-\alpha|x|} dx = \frac{(2j)!}{\pi \alpha^{2j+1}}. \quad (\text{CP10:U14})$$

These are the entries of the majorant, not a claim about growth of the signed defect.

On the full weighted space, the relative remainder acts by multiplication by $e_L(x)e^{\alpha|x|}$. If that multiplier is nonzero on a set of positive measure, the operator is not compact: an infinite orthonormal sequence supported where its modulus is at least a fixed $c > 0$ has images with norms at least c . A nonzero bounded multiplier on this nonatomic space therefore need not be Hilbert-Schmidt, regardless of decay in the original density.

The exact centered identity remains

$$k - k_L = e_L - e_L(0)b_T. \quad (\text{CP10:U15})$$

Letting the height cutoff L grow controls this approximation error. It neither signs nor removes the retained complete core k_L . The height cutoff L , polynomial dimension m , and prime-ladder index N are distinct parameters.

4. Spatial Hilbert-Schmidt decay and subtraction of the metric

The complete spatial kernel satisfies

$$0 < J(x, y) \leq \frac{M_a^2}{2a} e^{-a(x^2+y^2)} \quad (a > 0).$$

Therefore, as an operator on ordinary spatial L^2 ,

$$\|\mathcal{J}\|_{\text{HS}}^2 \leq \frac{\pi M_a^4}{8a^3} < \infty. \quad (\text{CP10:U16})$$

This is a finite total kernel energy. The target $\Delta = \mathfrak{W} - \mathcal{S}$ uses a different relative operator, with the Suzuki metric represented by the identity.

The distinction can be tested exactly. Suppose a bounded map U from the infinite-dimensional Suzuki completion identified the *whole* Weil form with $K = U^* \mathcal{J} U$. Then K would be compact. For the determining projections P_N , compactness gives $\|K \rho_j\| \rightarrow 0$, and hence

$$\begin{aligned} \|P_N(K - I)P_N\|_F^2 &= N - 2\Re \text{tr}(P_N K P_N) + \|P_N K P_N\|_F^2, \\ \frac{1}{N} \|P_N(K - I)P_N\|_F^2 &\longrightarrow 1. \end{aligned} \quad (\text{CP10:U17})$$

For the last term, use $\|P_N K P_N\|_F^2 \leq \sum_{j \leq N} \|K \rho_j\|^2$, then take Cesàro means. This is a conditional test of that proposed bounded factorization, not a divergence theorem about the actual defect. The existing finite-packet witness transfer asserts no bounded operator U of this kind.

Thus the usable compactness statement would have to concern the completed *difference* in its own metric. Compactness of the whole spatial operator alone does not provide it.

5. Exact prime-discrepancy substitution

For $q \in C_c^\infty((-A, A))$, set

$$R_q(v) = \int q(u+v) \overline{q(u)} du, \quad r(v) = \Re R_q(v).$$

Then r is smooth and supported in $[-2A, 2A]$. Define

$$E_0(x) = \psi(x) - x + 1, \quad a(v) = e^{-v/2} E_0(e^v). \quad (\text{CP10:U18})$$

The $+1$ retains the lower endpoint: $E_0(1) = 0$. Stieltjes integration by parts gives the exact formula

$$\begin{aligned} \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} r(\log n) &= \int_1^\infty x^{-1/2} r(\log x) d\psi(x) \\ &= \int_0^{2A} e^{v/2} r(v) dv - \int_0^{2A} a(v) (r'(v) - \tfrac{1}{2} r(v)) dv. \end{aligned} \quad (\text{CP10:U19})$$

Both boundary terms vanish by compact support and $E_0(1) = 0$. In the completed Weil formula, the continuum term cancels the corresponding pole term, leaving the exact prime-discrepancy contribution

$$2 \int_0^{2A} a(v) (r'(v) - \tfrac{1}{2} r(v)) dv. \quad (\text{CP10:U20})$$

The other pole, gamma term, scalar term, and Suzuki subtraction remain part of the same completed defect.

Fiori-Kadiri-Swidinsky, *Sharper bounds for the Chebyshev function $\psi(x)$* , Corollary 1.4 in arXiv:2204.02588v3, proves unconditionally that, for $x > 2$,

$$|\psi(x) - x| < 9.22022 x (\log x)^{3/2} e^{-0.8476836 \sqrt{\log x}}. \quad (\text{CP10:U21})$$

Consequently, for $v > \log 2$,

$$|a(v)| \leq e^{-v/2} + 9.22022 v^{3/2} e^{v/2 - 0.8476836 \sqrt{v}}. \quad (\text{CP10:U22})$$

On $0 \leq v < \log 2$, its exact value is $a(v) = e^{-v/2} - e^{v/2}$. The large- v majorant in (U22) grows. This statement concerns the majorant; it asserts no lower growth rate for the actual prime discrepancy.

For example, the direct absolute estimate on (U20) is

$$\left| 2 \int_0^{2A} a(v) (r'(v) - \tfrac{1}{2} r(v)) dv \right| \leq 2 (\|q'\|_2 \|q\|_2 + \tfrac{1}{2} \|q\|_2^2) \int_0^{2A} |a(v)| dv. \quad (\text{CP10:U23})$$

It is finite on a fixed support. It provides no uniform bound on Suzuki-unit tests across expanding supports. The known continuity estimate $\|L_0 q\|_2 \leq \pi^{-1/2} M_A \|q'\|_1$ is an upper bound for the Suzuki norm; it supplies no upper bound for the derivative norm when $\mathcal{S}(q, q) = 1$.

One can retain the complete finite-matrix estimate. Let $\mathfrak{P}_{\text{err}}$ be the form in (U20), and put

$$\begin{aligned}\mathfrak{W}_0(q, q) &= 2 \int_0^\infty e^{-v/2} r(v) dv - (\log(4\pi) + \gamma_E) \|q\|_2^2 \\ &\quad - 2 \int_0^\infty \frac{e^{-v/2} r(v) - e^{-v} \|q\|_2^2}{1 - e^{-2v}} dv.\end{aligned}$$

The paired integral is finite at zero. Polarization gives the exact identities

$$\mathfrak{W} = \mathfrak{W}_0 + \mathfrak{P}_{\text{err}}, \quad \Delta = (\mathfrak{W}_0 - \mathcal{S}) + \mathfrak{P}_{\text{err}}. \quad (\text{CP10:U23a})$$

Choose A_N containing the first N test supports, and define

$$U_N = \sum_{j \leq N} \|\rho_j\|_2^2, \quad V_N = \sum_{j \leq N} \|\rho'_j\|_2^2, \quad \mathbf{B}_N = [(\mathfrak{W}_0 - \mathcal{S})(\rho_j, \rho_i)].$$

Let $\widehat{I}_A$ be the integral over $[0, 2A]$ of the majorant for $|a(v)|$ in (U22), using its exact absolute value below $\log 2$. The polarized correlation estimate is

$$|\mathfrak{P}_{\text{err}}(f, g)| \leq \widehat{I}_A(\|f'\|_2 \|g\|_2 + \|f\|_2 \|g'\|_2 + \|f\|_2 \|g\|_2).$$

Taking the Frobenius norm of the corresponding three rank-one majorant matrices proves

$$\boxed{\sqrt{\mathcal{E}_N} \leq \|\mathbf{B}_N\|_F + \widehat{I}_{A_N}(2\sqrt{U_N V_N} + U_N).} \quad (\text{CP10:U23b})$$

Every quantity belongs to the same finite Suzuki-orthonormal test family. The right side has no proved uniform bound by one; in particular, its remaining matrix $\mathbf{B}_N$ has not been shown to vanish.

There is a genuine decay statement for each fixed Euler packet from the prior witness transfer. For $f(u) = \mathbf{1}_{u \geq 0} \sum_j d_j e^{-(b+it_j)u}$, $b > 1/2$, its correlation and derivative are bounded by constants times e^{-bv} . Its prime-discrepancy tail past V is therefore bounded by a packet-dependent constant times

$$\int_V^\infty \left[e^{-(b+1/2)v} + v^{3/2} e^{-(b-1/2)v - 0.8476836\sqrt{v}} \right] dv \longrightarrow 0.$$

This controls the tail of that fixed packet. It does not bound the complete prefix or provide a uniform energy budget over Suzuki-normalized packets.

The scale also explains the limitation of the cited zero-free-region estimates. For every fixed $\beta < 1$ and $c > 0$,

$$\frac{x^\beta}{x e^{-c\sqrt{\log x}}} = e^{-(1-\beta)\log x + c\sqrt{\log x}} \longrightarrow 0. \quad (\text{CP10:U24})$$

A possible zero contribution of size x^β , with $1/2 < \beta < 1$, is compatible with that error scale. The same comparison holds for relative bounds $x \exp(-o(\log x))$. This calculation diagnoses the proposed use of the pointwise PNT bound; it does not exclude an argument using additional signed correlations in the complete form.

6. Orthogonal saturation and the exact content of Parseval

Let $\mathcal{H}_S$ be the Hilbert completion of $\mathcal{D}$ for the Suzuki norm. The determining sequence is complete there, because its fixed-support approximation implies Suzuki-norm approximation. Then

$$\boxed{\sup_N \mathcal{E}_N < \infty \iff \Delta(q, r) = \langle Dq, r \rangle_S \text{ for a Hilbert-Schmidt self-adjoint } D.} \quad (\text{CP10:U25})$$

For the forward direction, (U3) gives $\sum_{i,j \geq 1} |\Delta(\rho_j, \rho_i)|^2 < \infty$. This square-summable Hermitian matrix defines a Hilbert-Schmidt operator on ℓ^2 , hence on $\mathcal{H}_S$. Its form agrees with Δ on the determining union. To obtain agreement on all compact smooth tests, approximate each test on fixed support: both the original forms and the constructed bounded form converge, using the established fixed-support continuity. The converse follows by Parseval. In either case,

$$\lim_{N \rightarrow \infty} \mathcal{E}_N = \|D\|_{\text{HS}}^2 = \sum_{j \geq 1} \|D\rho_j\|_S^2. \quad (\text{CP10:U26})$$

For $Q_N = I - P_N$, the remainder has the exact block expression

$$\|D\|_{\text{HS}}^2 - \mathcal{E}_N = 2\|Q_N D P_N\|_{\text{HS}}^2 + \|Q_N D Q_N\|_{\text{HS}}^2. \quad (\text{CP10:U27})$$

This is saturation of a finite total, with no restriction on the numerical value of that total. Ordinary boundedness gives only $\mathcal{E}_N \leq N\|D\|_{\text{op}}^2$, and does not imply a finite limit.

A rank-one example settles the proposed inference from a stable total above one to an infinite reserve. On ℓ^2 , set

$$Du = -2\langle u, e_1 \rangle e_1.$$

Then D is self-adjoint, finite rank, and Hilbert-Schmidt, but

$$\|P_N D P_N\|_F^2 = 4 \quad (N \geq 1). \quad (\text{CP10:U28})$$

There is one persistent defect direction, and every later increment is zero. This is an operator-theoretic example, not a claim about the actual zeta forms. It shows exactly what exhaustion and Parseval alone allow.

For the actual defect, (U1) is therefore equivalent to the quantitative statement $\|D\|_{\text{HS}}^2 \leq 1$, including existence of the extension. A finite bound with an unspecified constant does not provide this threshold. Hilbert-Schmidt compactness would give at most finitely many negative directions for $I + D$, since negative directions correspond to eigenvalues of D below -1 ; it does not eliminate those directions.

The actual Suzuki zero-feature expansion was also checked against its primary source. Equations (3.5) and (3.8) supply an unconditional synthesis, while Proposition 4.1 assumes RH when asserting orthonormality of those zero features. That conditional orthonormality is not used here. Parseval in (U26) uses the unconditionally Suzuki-orthonormal determining sequence (ρ_j) . The checked formulas do not establish that a single off-critical zero would force an infinite Hilbert-Schmidt defect; the proved consequence remains the permanent floor above one.

7. Outcome and source scope

Proposed bound	Established result	Relation to the target
Finite extraction energy	Uniform trace and Frobenius cap (U2), over every degree	Controls the extraction in monomial coordinates; the complete Suzuki defect and inverse metric remain to be compared.
Complete theta tails	Uniform operator bound (U12), with explicit Frobenius dimension factor	Controls a retained approximation error in its specified exponential metric.

Proposed bound	Established result	Relation to the target
Explicit PNT bounds	Exact completed substitution (U19)-(U20) and fixed-support estimate (U23)	The tested absolute bound supplies no uniform cap on Suzuki-unit tests.
Orthogonal saturation	Exact equivalence (U25) and limit (U26)	A finite total must be proved, and its size must meet the threshold one.

The prior negative-witness transfer still gives

$$Q_\gamma(R) = -\delta < 0 \implies \mathcal{E}_N \geq \left(1 + \frac{\delta}{6\mathcal{S}(q, q)}\right)^2 > 1 \quad (N \geq N_0). \quad (\text{CP10:U29})$$

The present tests establish no contradictory upper bound for that same energy. They establish the uniform extraction theorem, the precise effect of normalization, the prime-discrepancy formula, and the saturation theorem. The target (U1) remains unproved.

Sources used: the current consolidated manuscript’s completed Weil formula, Suzuki-map appendix, and defect-energy appendix; *Michaels_Heat_Defect_Two_Trace_Intersection.md*, Part B; *MATHEMATICS(7).md*, U11-U13, for the original extraction; *MATHEMATICS(20260918-115624).md*, JT2-JT4 and CE5-CE6, for the complete height tails; *RUNNING_MATHEMATICS(2).md*, EH3-TAIL and EH3-ERROR, for the improved full exterior allowance; and Fiori-Kadiri-Swidinsky, arXiv:2204.02588v3, Corollary 1.4, for the explicit unconditional PNT estimate. The original scalar certificates are inherited from the checked source statements; they were not numerically replayed here.

The source for the distinction between unconditional synthesis and RH-dependent zero-feature orthonormality is Masatoshi Suzuki, *On the Hilbert space derived from the Weil distribution*, arXiv:2301.00421v3, Section 3 and Proposition 4.1.

We are aiming to establish $\liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1$ independently of RH, so the permanent floor from any hypothetical negative local witness becomes a contradiction.

Checkpoint 11: Complete Gram correction and relative extraction

Source: *Michaels_Weil_Suzuki_Gram_Correction_Bridge.md*. Equation tags carry the source prefix CP11; references within this unit retain their original local numbering.

Christophe Michaels research continuation · 22 September 2026 · Checkpoint 11

This note carries out the selected bridge calculation using the actual Suzuki features, with every real/nonreal Gram term retained. It proves an explicit correction formula for the completed Weil-Suzuki difference and the following uniform comparison for the central extraction:

$$\mathcal{X}(P_{S,c}) \leq K_* c^* G_S c, \quad K_* := \frac{3567}{509894} < \frac{1}{140}. \quad (\text{CP11:1})$$

Here S is any finite distinct-zero set closed under sign and conjugation, and $P_{S,c}$ is the exact normalized polynomial numerator defined in Section 7. The comparison is independent of the size of S . Its proof uses the previously established scalar theta bound and elementary analytic inequalities; no zero computation or increasing-degree experiment is used.

The complete defect is a different, explicitly identified form. Its required Hilbert-Schmidt ceiling remains unproved. This note does not claim unconditional RH closure.

1. Target and normalizations

Let $\mathcal{D} = C_c^\infty(\mathbb{R})$, with inner products linear in the first variable, and

$$\widehat{q}(z) = \int_{\mathbb{R}} q(t) e^{izt} dt, \quad \Delta(q, r) = \mathfrak{W}(q, r) - \mathcal{S}(q, r).$$

Use the same positive metric $\mathcal{S}(q, r) = \langle L_0 q, L_0 r \rangle$ and the same Suzuki-orthonormal determining sequence (ρ_j) as in the consolidated manuscript. The fixed target is

$$D_N = [\Delta(\rho_j, \rho_i)]_{i,j \leq N}, \quad \mathcal{E}_N = \|D_N\|_F^2, \quad \boxed{\liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1}. \quad (\text{CP11:2})$$

The established monotone-increment formula makes this equivalent to $\mathcal{E}_N \leq 1$ for every N . The earlier local-witness transfer supplies a permanent lower floor above one if a negative local witness exists. That transfer is reused here; it is not rederived.

The earlier extraction uses

$$0 \leq a < b < 1/2, \quad \delta = b^2 - a^2, \quad H_0 = F(b)^2 - F(a)^2, \quad F(y) = \xi(1/2 + y),$$

$$0 < H_0/\delta < 1/80, \quad F(1/2) = 1/2, \quad F'(1/2) < 1/80, \quad b_T(x) = (1 - x^2/T^2)_+^4, \quad 0 < T \leq 1. \quad (\text{CP11:3})$$

These are the archived scalar bounds used in checkpoint 10. Section 8 states every consequence needed from them. The source's RH-dependent zero-feature orthonormality is not used.

2. Actual features and real-axis weight

Write

$$A(z) = \xi(1/2 - iz), \quad E(z) = A(z) + iA'(z), \quad h(z) = \frac{A(z)}{E(z)}.$$

The real entire even function A has zero set Γ ; let m_γ be the multiplicity of γ . The source feature is

$$F_\gamma(z) = \sqrt{\frac{m_\gamma}{\pi}} \frac{ih(z)}{z - \gamma}. \quad (\text{CP11:4})$$

On the real axis define

$$\omega(x) = |h(x)|^2 = \frac{A(x)^2}{A(x)^2 + A'(x)^2} = \operatorname{Re} h(x). \quad (\text{CP11:5})$$

At common zeros the quotient is interpreted by removal. It is real analytic near every real point, satisfies $0 \leq \omega \leq 1$, and is positive away from the discrete real zero set of A .

At any zero γ , real or nonreal, local factorization gives

$$h(z) = -\frac{i}{m_\gamma}(z - \gamma) + O((z - \gamma)^2), \quad h^\sharp(z) = \frac{i}{m_\gamma}(z - \gamma) + O((z - \gamma)^2). \quad (\text{CP11:6})$$

Thus $F_\gamma(\gamma) = 1/\sqrt{\pi m_\gamma}$ is removable. In particular, at a real zero

$$\omega(x) = \frac{(x - \gamma)^2}{m_\gamma^2} + O((x - \gamma)^3). \quad (\text{CP11:7})$$

Each F_γ belongs to $L^2(\mathbb{R})$: local removal handles real nodes, nonreal denominators have no real singularity, and $|h| \leq 1$ gives the integrable squared $O(1/x)$ tail.

3. A single Cauchy transform gives every Gram entry

Define, initially for $z \notin \mathbb{R}$,

$$C(z) = \frac{1}{\pi} \int_{\mathbb{R}} \omega(x) \left(\frac{1}{x - z} - \frac{x}{1 + x^2} \right) dx. \quad (\text{CP11:8})$$

The integral is absolutely convergent. Its imaginary part is

$$\boxed{\operatorname{Im} C(z) = \frac{\operatorname{Im} z}{\pi} \int_{\mathbb{R}} \frac{\omega(x)}{|x - z|^2} dx.} \quad (\text{CP11:9})$$

It has strictly the sign of $\operatorname{Im} z$. Also $C(\bar{z}) = \overline{C(z)}$. At a real zero γ , the integral in (8) itself is absolutely convergent by (7), and defines a real value. Its derivative there is

$$C'(\gamma) = \frac{1}{\pi} \int_{\mathbb{R}} \frac{\omega(x)}{(x - \gamma)^2} dx. \quad (\text{CP11:10})$$

The derivative is understood through the analytic upper or lower boundary continuation; these derivatives agree at the zero.

The exact Gram entry, linear in the first feature, is

$$\begin{aligned} G_{\gamma\delta} &:= \langle F_\gamma, F_\delta \rangle_{L^2} \\ &= \frac{\sqrt{m_\gamma m_\delta}}{\pi} \int_{\mathbb{R}} \frac{\omega(x)}{(x - \gamma)(x - \bar{\delta})} dx. \end{aligned} \quad (\text{CP11:11})$$

The integral is absolute, including real-node removals. Partial fractions therefore give

$$\boxed{G_{\gamma\delta} = \sqrt{m_\gamma m_\delta} \frac{C(\gamma) - C(\bar{\delta})}{\gamma - \bar{\delta}}} \quad (\gamma \neq \bar{\delta}). \quad (\text{CP11:12})$$

For the coincident case $\gamma = \bar{\delta}$, the exact replacement is

$$\boxed{G_{\gamma\delta} = \sqrt{m_\gamma m_\delta} C'(\gamma)}. \quad (\text{CP11:13})$$

This covers distinct conjugate nonreal nodes as well as the real diagonal. For a nonreal diagonal,

$$G_{\delta\delta} = m_\delta \frac{\text{Im } C(\delta)}{\text{Im } \delta} > 0. \quad (\text{CP11:14})$$

4. Coupling to every critical-line feature

Let γ be any real zero and δ any nonreal zero. The value $C(\gamma)$ is real, whereas $\text{Im } C(\bar{\delta}) \neq 0$. Therefore

$$\boxed{G_{\gamma\delta} \neq 0}. \quad (\text{CP11:15})$$

Quantitatively,

$$|G_{\gamma\delta}| \geq \sqrt{\frac{m_\gamma}{m_\delta}} \frac{|\text{Im } \delta|}{|\gamma - \bar{\delta}|} G_{\delta\delta} > 0. \quad (\text{CP11:16})$$

Thus, if a nonreal zero exists, its actual Suzuki feature is coupled to every real-zero feature. This is an explicit unconditional Gram calculation under that hypothetical zero configuration.

For two distinct real zeros, orthogonality is equivalent to $C(\gamma) = C(\delta)$. Unit norm at a real zero is equivalent to $C'(\gamma) = 1/m_\gamma$. Neither assertion is supplied merely by reality of the node.

5. The exact meromorphic correction

The boundary values of (8) satisfy

$$C_+(x) - C_-(x) = 2i\omega(x). \quad (\text{CP11:17})$$

This can be obtained locally without a global contour limit. Split the integral near a real point into a finite interval and its complement, and in the local integral write

$$\frac{\omega(t)}{t - z} = \frac{\omega(t) - \omega(z)}{t - z} + \frac{\omega(z)}{t - z}.$$

The first term is analytic across the interval; the second has the elementary logarithmic jump $2\pi i$. The remaining tail is locally analytic. This also proves local analytic continuation of each boundary branch. Since $\omega(\gamma) = \omega'(\gamma) = 0$, their values and first derivatives agree at a real zero, justifying (10).

Define the correction piecewise by

$$\mathcal{R}(z) = \begin{cases} C(z) - ih(z), & \text{Im } z > 0, \\ C(z) + ih^\sharp(z), & \text{Im } z < 0. \end{cases} \quad (\text{CP11:18})$$

On the real axis $h + h^\sharp = 2\omega$. Equation (17) shows that these definitions glue meromorphically across the real axis. Equivalently, the upper expression has real boundary values and the lower expression is its Schwarz reflection. Thus $\mathcal{R}^\sharp = \mathcal{R}$. The lower expression is specifically $C + ih^\sharp$, not $C - ih$.

At every zero γ of A , (6) yields

$$C(\gamma) = \mathcal{R}(\gamma), \quad C'(\gamma) = \frac{1}{m_\gamma} + \mathcal{R}'(\gamma). \quad (\text{CP11:19})$$

Both formulas hold in the upper and lower half-planes and at real nodes. A common zero of A, E creates no pole of h .

Define the divided difference by

$$\mathcal{R}[z, u] = \begin{cases} (\mathcal{R}(z) - \mathcal{R}(u))/(z - u), & z \neq u, \\ \mathcal{R}'(z), & z = u. \end{cases}$$

Let $J_{\gamma\delta} = \mathbf{1}_{\{\gamma=\bar{\delta}\}}$, the conjugation-pairing matrix with multiplicities absorbed into the feature coefficients. Equations (12)-(13) and (19) prove the exact finite-feature identity (using the linear-first entries just defined)

$$G_{\gamma\delta} = J_{\gamma\delta} + \sqrt{m_\gamma m_\delta} \mathcal{R}[\gamma, \bar{\delta}]. \quad (\text{CP11:20})$$

There is no omitted real-real block and no omitted real/nonreal block.

In particular, on the real-zero block the correction is the divided-difference matrix of $\mathcal{R}$:

$$G_{\gamma\delta} = \begin{cases} \sqrt{m_\gamma m_\delta} (\mathcal{R}(\gamma) - \mathcal{R}(\delta))/(\gamma - \delta), & \gamma \neq \delta, \\ 1 + m_\gamma \mathcal{R}'(\gamma), & \gamma = \delta. \end{cases} \quad (\text{CP11:21})$$

5.1. Finite contour formula with the boundary retained

In the upper half-plane, the poles of $\mathcal{R}$ are exactly the nonremovable poles of $h = A/E$. At a simple zero a of E with $A(a) \neq 0$,

$$\text{Res}_{z=a} \mathcal{R}(z) = -i \frac{A(a)}{E'(a)}. \quad (\text{CP11:22})$$

The lower pole and principal part are the Schwarz reflections. Higher-order zeros contribute the complete principal part of $-ih$.

Let Ω be a bounded conjugation-symmetric contour domain containing the finitely many nodes being tested, with no pole on its boundary. For a pole a , let $P_a(z)$ denote its full principal part. The residue theorem gives exactly

$$\mathcal{R}(z) = H_\Omega(z) + \sum_{a \in \Omega} P_a(z), \quad H_\Omega(z) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{\mathcal{R}(\zeta)}{\zeta - z} d\zeta. \quad (\text{CP11:23})$$

The function H_Ω is holomorphic in Ω . It is retained: no estimate has been imposed that would allow it to be discarded as the contour expands.

For a simple pole of residue r_a ,

$$P_a[z, u] = -\frac{r_a}{(z-a)(u-a)}. \quad (\text{CP11:24})$$

Substitution of (23)-(24) into (20) is an exact contour correction formula for every tested Gram entry, containing all enclosed pole terms and the complete holomorphic boundary term. Finite poles alone do not justify setting the latter to zero.

6. The complete compact-test defect

For $q \in C_c^\infty(\mathbb{R})$, put

$$a_\gamma(q) = \sqrt{m_\gamma} \widehat{q}(\gamma).$$

With the corrected two-sided convention recorded in the existing manuscript, define the synthesis map $V = -L_0$. Its identities are

$$Vq = \sum_{\gamma \in \Gamma} a_\gamma(q) F_\gamma = -L_0 q, \quad \mathfrak{W}(q, r) = \sum_{\gamma \in \Gamma} a_\gamma(q) \overline{a_{\bar{\gamma}}(r)}. \quad (\text{CP11:25})$$

The zero sum in the second expression is absolutely convergent, since the zeros lie in $|\operatorname{Im} \gamma| < 1/2$, their number in a unit real window is $O(\log(2 + |\operatorname{Re} \gamma|))$, and compact smooth Fourier transforms decay rapidly in that strip.

For clarity, a valid L^2 limiting convention for the first expression is smooth symmetric height cutoff. Choose an even real smooth χ , equal to one on $[-1, 1]$, supported in $[-2, 2]$, and put

$$a_{\gamma, H}(q) = \chi(\operatorname{Re} \gamma / H) a_\gamma(q).$$

The sums are finite and converge to Vq in L^2 . The global sign cancels in $\mathcal{S}(q, r) = \langle Vq, Vr \rangle$. Here is a direct unconditional justification of the norm convergence.

The local partial-fraction estimate used in the source's unconditional L^2 proof gives, on the real axis with removable limits understood,

$$\frac{A'}{A}(x) = \sum_{|\operatorname{Re} \gamma - x| \leq 1} \frac{m_\gamma}{x - \gamma} + O(\log(2 + |x|)). \quad (\text{CP11:26})$$

The source disk window can be enlarged to the displayed real-coordinate window: the added terms have distance greater than one and number $O(\log(2 + |x|))$, so they are absorbed in the remainder. Also $|h(x)| \leq 1$ and

$$ih(x) \frac{A'}{A}(x) = 1 - h(x). \quad (\text{CP11:27})$$

For the cutoff error put

$$b_{\gamma, H} = (1 - \chi(\operatorname{Re} \gamma / H)) \widehat{q}(\gamma), \quad b_H(x) = (1 - \chi(x/H)) \widehat{q}(x).$$

Rapid strip decay of $\widehat{q}, \widehat{q}'$ gives numbers $\varepsilon_H \rightarrow 0$ such that

$$|b_{\gamma, H}| \leq \varepsilon_H (1 + |\operatorname{Re} \gamma|)^{-2}, \quad |b_H(x)| \leq \varepsilon_H (1 + |x|)^{-2},$$

and, for $|\operatorname{Re} \gamma - x| \leq 1$,

$$|b_{\gamma,H} - b_H(x)| \leq C\varepsilon_H |\gamma - x|(1 + |x|)^{-2}.$$

These bounds can be made uniform by writing

$$b_H(s + iy) = (1 - \chi(s/H))\widehat{q}(s + iy), \quad |y| \leq 1/2.$$

For every integer $M > 2$, the weighted real C^1 norm of this function is $O_{q,M}(H^{2-M})$. The derivatives in s, y use rapid strip decay of $\widehat{q}$ and $\widehat{q}'$, together with the factor $H^{-1}\chi'$. Integrating those derivatives on the segment from x to γ proves the displayed Lipschitz estimate. Holomorphicity of the cutoff is not required.

In the error sum

$$\frac{ih(x)}{\sqrt{\pi}} \sum_{\gamma} \frac{m_{\gamma} b_{\gamma,H}}{x - \gamma},$$

subtract $b_H(x)$ from the nearby coefficients. Equations (26)-(27) bound the common nearby term; the last displayed Lipschitz bound controls the remaining nearby terms. The far terms are absolutely bounded by

$$\varepsilon_H \sum_{|\operatorname{Re} \gamma - x| > 1} \frac{m_{\gamma}}{(1 + |\operatorname{Re} \gamma|)^2 |x - \gamma|} \leq C\varepsilon_H \frac{\log^2(2 + |x|)}{1 + |x|}.$$

For large $|x|$, split the far sum into $|\Re \gamma| \leq |x|/2$, comparable heights, and $|\Re \gamma| \geq 2|x|$; the unit-window zero count bounds these pieces by the displayed majorant. Bounded x is immediate from absolute convergence. The final majorant lies in $L^2(\mathbb{R})$. The nearby terms satisfy the stronger bound $C\varepsilon_H \log(2 + |x|)/(1 + |x|)^2$. This proves L^2 convergence without invoking orthogonality of the zero features.

It is now legitimate to apply (20) first to finite cutoff sums and then take their L^2 limits. The result is the completed defect identity

$$\boxed{\Delta(q, r) = - \lim_{H \rightarrow \infty} \sum_{\gamma, \delta} a_{\gamma,H}(q) \overline{a_{\delta,H}(r)} \sqrt{m_{\gamma} m_{\delta}} \mathcal{R}[\gamma, \bar{\delta}].} \quad (\text{CP11:28})$$

The two-variable sum is finite at each H . Its displayed limit is justified by the convergence of both completed forms. Absolute convergence of an uncut double correction series is not asserted.

7. Exact polynomial numerator and its metric

For a nonempty finite set $S = \{\gamma_1, \dots, \gamma_n\}$ of distinct zeros, use the standard coefficient matrix

$$(G_S)_{ij} = \langle F_{\gamma_j}, F_{\gamma_i} \rangle, \quad c^* G_S c = \left\| \sum_{\gamma \in S} c_{\gamma} F_{\gamma} \right\|_2^2. \quad (\text{CP11:29})$$

This transpose convention is needed because the inner product is linear in its first argument. Write $w = \omega$. The matrix G_S is strictly positive definite: zero norm would force the rational function $\sum \sqrt{m_{\gamma}} c_{\gamma} / (x - \gamma)$ to vanish almost everywhere where $w > 0$, hence identically. Its distinct-pole residues then give $c = 0$.

Choose a nonzero scalar normalization of

$$Q_S(x) = \prod_{\gamma \in S} (x - \gamma), \quad P_{S,c}(x) = Q_S(x) \sum_{\gamma \in S} \frac{\sqrt{m_\gamma} c_\gamma}{x - \gamma}. \quad (\text{CP11:30})$$

The degree of P is at most $|S| - 1$, and the map $c \mapsto P_{S,c}$ is invertible, with

$$c_\gamma = \frac{P_{S,c}(\gamma)}{\sqrt{m_\gamma} Q'_S(\gamma)}.$$

It gives exactly

$$\sum c_\gamma F_\gamma(x) = \frac{iA(x)}{\sqrt{\pi}E(x)} \frac{P_{S,c}(x)}{Q_S(x)},$$

$$\boxed{c^* G_S c = \frac{1}{\pi} \int_{\mathbb{R}} \frac{w(x)}{|Q_S(x)|^2} |P_{S,c}(x)|^2 dx.} \quad (\text{CP11:31})$$

For polynomials of these degrees the integral converges: real zeros of Q are canceled by w , and its large- x decay is at least x^{-2} . This is the exact positive polynomial metric furnished by the zero features.

In particular the original extraction form applied to the same polynomial is

$$\mathcal{X}(P) = \frac{H_0}{2\pi\delta} \int_{-T}^T b_T(x) |P(x)|^2 dx. \quad (\text{CP11:32})$$

The comparison is between the measures $b_T(x)dx$ and $w(x)|Q_S(x)|^{-2}dx$, with their displayed constants. No renaming identifies these measures.

8. Explicit uniform extraction comparison

The theta density η is even, positive, and has the moments used in the earlier extraction. Its Fourier-Laplace pair is $A(z) = \int \eta(u) \cos(zu) du$ and $F(y) = A(iy)$. The prior extraction theorem gives $F'(1/2) < 1/80$ and $F(1/2) = 1/2$, where $F(y) = \int \eta(u) \cosh(yu) du$. Positivity of its even Taylor coefficients makes F' increasing for $y \geq 0$. Therefore

$$A(0) = F(0) > \frac{1}{2} - \frac{1}{2} \cdot \frac{1}{80} = \frac{79}{160}.$$

Let $\mu_2 = \int u^2 \eta(u) du$. The pointwise inequality $u \sinh(u/2) \geq u^2/2$ gives

$$\mu_2 \leq 2F'(1/2) < \frac{1}{40}.$$

For real $|x| \leq 1$, use $\cos(xu) \geq 1 - x^2 u^2/2$ and $|\sin(xu)| \leq |xu|$:

$$A(x) > \frac{77}{160}, \quad |A'(x)| < \frac{4}{160}, \quad \boxed{w(x) \geq w_* := \frac{5929}{5945}.} \quad (\text{CP11:33})$$

This proves the needed central zero-free band analytically, without zero computations or an RH assumption.

Now require S also to be closed under $\gamma \mapsto -\gamma$ and normalize

$$Q_S(x) = \prod_{\{\gamma, -\gamma\} \subset S} \left(1 - \frac{x^2}{\gamma^2}\right), \quad Q_S(0) = 1. \quad (\text{CP11:34})$$

The zeros here are distinct; their multiplicities still occur in (30). The complete function has $\sum_{\gamma \in \Gamma} m_\gamma / |\gamma|^2 < \infty$, and $A(0) > 0$. Hence

$$C_\Gamma := \sum_{\{\gamma, -\gamma\} \subset \Gamma} |\gamma|^{-2} < \infty.$$

For $|x| \leq 1$, the elementary product bound gives

$$|Q_S(x)|^2 \leq \prod_{\{\gamma, -\gamma\} \subset S} (1 + |\gamma|^{-2})^2 \leq e^{2C_\Gamma}. \quad (\text{CP11:35})$$

The constant in (35) can also be bounded analytically, using the same archived scalar amplitude bound. First,

$$F(1) = \xi(3/2) = \frac{3}{8} \pi^{-3/4} \Gamma(3/4) \zeta(3/2) < \frac{9}{8\sqrt{\pi}} < \frac{3}{4}. \quad (\text{CP11:36})$$

Indeed Cauchy-Schwarz in the gamma integral gives $\Gamma(3/4) \leq \sqrt{\Gamma(1/2)\Gamma(1)} = \pi^{1/4}$, and the integral test gives $\zeta(3/2) < 1 + \int_1^\infty x^{-3/2} dx = 3$. The last inequality uses $\pi > 9/4$. Positive theta moments give, for $|z| \leq 1$,

$$|A(z) - A(0)| \leq F(1) - A(0) < A(0),$$

because $A(0) > 79/160$. Hence A has no zero in the closed unit disk. This argument uses neither zero computations nor RH.

Every zero $\gamma = u + iv$ also satisfies $|v| < 1/2$. Put $r = |\gamma| > 1$. Then

$$\Re \frac{1}{\gamma^2 + 1/4} \geq \frac{r^2 - 1/4}{(r^2 + 1/4)^2} \geq \frac{12}{25r^2}. \quad (\text{CP11:37})$$

For the last inequality set $t = 1/(4r^2) \leq 1/4$ and use $(1-t)/(1+t)^2 \geq 12/25$. The even order-one Hadamard product, paired under $\gamma \mapsto -\gamma$, gives the absolutely convergent identity

$$\frac{F'(1/2)}{F(1/2)} = \sum_{\{\gamma, -\gamma\}} \frac{m_\gamma}{\gamma^2 + 1/4} < \frac{1}{40}.$$

Taking real parts term by term in this convergent sum and using (37) proves

$$C_\Gamma \leq \sum_{\{\gamma, -\gamma\}} \frac{m_\gamma}{|\gamma|^2} < \frac{5}{96}, \quad e^{2C_\Gamma} < e^{5/48} < \frac{48}{43}. \quad (\text{CP11:38})$$

The last inequality follows from $e^x < \sum_{n \geq 0} x^n = (1-x)^{-1}$ for $0 < x < 1$.

Combining (31)-(38), $T \leq 1$, and $H_0/\delta < 1/80$, proves the dimension-independent theorem

$$\boxed{\mathcal{X}(P_{S,c}) \leq K_* c^* G_S c, \quad K_* := \frac{3567}{509894} < \frac{1}{140}.} \quad (\text{CP11:39})$$

The exact comparison constant is at most $H_0 e^{2C_T} / (2\delta w_*)$, which is strictly below K_* : indeed

$$\frac{5945}{160 \cdot 5929} \frac{48}{43} = \frac{3567}{509894} < \frac{1}{140},$$

where the last comparison is the integer inequality $3567 \cdot 140 = 499380 < 509894$. Thus this is an explicit relative operator bound strictly below one, valid for the extraction component over every finite symmetric zero-feature set.

Equivalently the explicitly constructed extraction map on each feature span,

$$U_S f = \sqrt{\frac{H_0}{2\pi\delta}} b_T^{1/2} P_{S,c}, \quad f = \sum_{\gamma \in S} c_\gamma F_\gamma,$$

has operator norm at most $\sqrt{K_*}$, uniformly over S . This proves a uniform norm comparison for the explicitly specified map. No commutation identity is asserted.

9. Compatibility, trace, and complete-defect scope

For nested sets $S \subset S'$ and the same finite synthesis vector, the numerators obey

$$P_{S',c} = \frac{Q_{S'}}{Q_S} P_{S,c}. \quad (\text{CP11:40})$$

Thus these particular finite polynomial maps are not a single unchanged map on nested sets. The normalized products converge uniformly on the central band to a fixed canonical product Q_∞ , by the convergent squared-reciprocal sum. There is consequently a compatible bounded limiting map on ordinary L^2 , and hence on the actual Suzuki image, namely multiplication on the band by

$$U_\infty f(x) = \sqrt{\frac{H_0}{2\delta}} b_T(x)^{1/2} \frac{Q_\infty(x) E(x)}{iA(x)} f(x). \quad (\text{CP11:41})$$

The denominator is nonzero on this band by (33), and the multiplier has squared modulus at most K_* . For a fixed finite synthesis vector f , $U_{S'} f \rightarrow U_\infty f$ in the band L^2 norm along any exhausting nested symmetric sets. This limit controls the canonical-product numerator; it is no longer the degree- m monomial extraction matrix with its fixed coefficient metric.

On an n -dimensional feature span, (39) yields an operator bound for the relative extraction matrix and the consequent estimates

$$0 \preceq Y_S \preceq K_* I_n, \quad \text{tr } Y_S \leq nK_*, \quad \|Y_S\|_F^2 \leq nK_*^2. \quad (\text{CP11:42})$$

Here B_S is the coefficient matrix of $c \mapsto \mathcal{X}(P_{S,c})$, and $Y_S = G_S^{-1/2} B_S G_S^{-1/2}$. Its exact trace is

$$\text{tr } Y_S = \frac{H_0}{2\pi\delta} \int_{-T}^T b_T(x) v_{n-1}(x)^* H_S^{-1} v_{n-1}(x) dx, \quad (\text{CP11:43})$$

where $v_{n-1}(x) = (1, x, \dots, x^{n-1})^T$, and H_S is the polynomial Gram matrix of the measure $w(x)|Q_S(x)|^{-2}dx/\pi$ in (31). Equation (43) retains the exact inverse metric. The original raw monomial trace bound $1/(175\pi)$ does not remove that inverse.

The exact finite coefficient defect, for conjugation-closed S , is

$$D_S = G_S^{-1/2}(J_S - G_S)G_S^{-1/2}, \quad (J_S c)_\gamma = c_{\bar{\gamma}}. \quad (\text{CP11:44})$$

The completed defect on compact tests is the limit in (28). Neither expression has been identified with the extraction form in (32) or the multiplier in (41).

The map $c \mapsto P_{S,c}$, including its normalization $Q_S(0) = 1$, is explicit. Matching it to a polynomial parametrization occurring elsewhere in the manuscript requires the corresponding identity; the map is not silently substituted for an arbitrary earlier T_N . In particular, this result does not transport the raw monomial trace bound $1/(175\pi)$ unchanged into the Suzuki metric.

10. Direct test against the signed reflection form

The new upper bound has a definite scope. For a finite conjugation-closed S containing a nonreal pair, the involution J_S has a negative eigenvalue. Congruence by the positive $G_S^{-1/2}$ preserves inertia, so

$$G_S^{-1/2} J_S G_S^{-1/2} \quad \text{has an eigenvalue } \mu < 0.$$

Consequently the exact relative defect in (44) has an eigenvalue $\mu - 1 < -1$, and

$$\|D_S\|_F^2 > 1. \quad (\text{CP11:45})$$

This algebraic test shows precisely why the small positive extraction form cannot by itself be a bound for the entire signed reflection defect under a hypothetical nonreal configuration.

For an isolated pair $\gamma = \alpha + i\beta, \bar{\gamma}$, with common multiplicity m , write

$$G_S = \begin{pmatrix} d & c \\ \bar{c} & d \end{pmatrix}, \quad 0 < d = \frac{m}{\pi} \int \frac{w(x)}{(x - \alpha)^2 + \beta^2} dx \leq \frac{m}{|\beta|}, \quad |c| < d.$$

The vector $(1, -1)^T/\sqrt{2}$ has reflection energy -1 and Gram energy $d - \Re c$. Its relative Rayleigh quotient yields

$$\|D_S\|_F^2 \geq \left(1 + \frac{1}{d - \Re c}\right)^2 \geq \left(1 + \frac{|\beta|}{2m}\right)^2 > 1. \quad (\text{CP11:46})$$

This is a finite coefficient statement. Arbitrary finitely supported zero arrays are not asserted to be Fourier samples of compact smooth tests. The actual compact-test defect has already been treated by the justified limit in Section 6.

The nonzero coupling in Section 4 gives a rigorous form of the proposed interaction with known critical-line zeros. It does not imply that their locations change. Indeed, for a nonempty finite real-zero set with Gram matrix G_R , and the cross column $b_i = \langle F_\delta, F_{\gamma_i} \rangle$ against a nonreal feature F_δ , using convention (29), orthogonal projection gives

$$0 < b^* G_R^{-1} b < \|F_\delta\|_2^2. \quad (\text{CP11:47})$$

The strict upper bound follows from independence of the distinct-pole rational functions. As real features are added, these projection energies increase and remain bounded by the same norm. Thus interaction with every real-zero feature is compatible with finite total projection energy; it does not itself contradict an existing zero.

11. What the calculation establishes

The completed correction is now explicit: Section 5 gives its exact meromorphic divided-difference kernel, and Section 6 proves its compact-test limiting formula. Its real-real, real/nonreal, and nonreal/nonreal blocks are retained. Section 8 proves the all-size central extraction comparison (1) in the specified Gram metric.

The remaining statement is a bound on the *complete* correction in the determining metric. Writing $a_{\gamma,H}(q) = \chi(\Re \gamma/H) \sqrt{m_\gamma} \hat{q}(\gamma)$, the exact entries to be bounded are

$$\Delta(\rho_j, \rho_i) = - \lim_{H \rightarrow \infty} \sum_{\gamma, \delta} a_{\gamma,H}(\rho_j) \overline{a_{\delta,H}(\rho_i)} \sqrt{m_\gamma m_\delta} \mathcal{R}[\gamma, \bar{\delta}]. \quad (\text{CP11:48})$$

The squared moduli of these entries must sum to at most one across every finite determining block to reach the stated target. The bounded extraction map does not currently supply this estimate. In particular, an operator norm bound for a positive component does not imply the required Hilbert-Schmidt bound for the full signed difference.

12. Sources and provenance

Source	Material used
Current consolidated manuscript, “The completed Suzuki map and metric transport”	Actual two-sided convention, normalization, positive metric, and determining space
Prior Suzuki-Weil audit and Build v13, coefficient convention	Global minus in $L_0 q = -Vq$; it cancels in the metric. RH-dependent orthogonality is excluded.
MATHEMATICS(7).md, U11-U13; MATHEMATICS(20260918-115624).md, JT2	Original scalar amplitude derivative bound and compact extraction profile
Michaels_Uniform_Defect_Cap_Analytical_Tests.md, checkpoint 10	Exact target normalization, scalar extraction inputs, original monomial trace bound, and Hilbert-Schmidt scope
Masatoshi Suzuki, <i>On the Hilbert space derived from the Weil distribution</i> , arXiv:2301.00421v3, Section 3	Feature normalization, strip facts, local partial fractions, and unconditional norm construction

Primary source: <https://arxiv.org/html/2301.00421v3>. The corrected two-sided convention from the existing manuscript is retained explicitly; the source’s printed RH-dependent Proposition 4.1 is not an input. The cutoff passage, Cauchy correction, and relative extraction estimate are derived in this note.

This continuation is added to checkpoint 11 with the running ledger and checksum manifest. Earlier paper and note contents are preserved. Its proved scope is the exact completed correction and the explicit extraction comparison; the unconditional full-defect cap remains open.

We are aiming to establish $\liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1$ independently of RH for the same completed Suzuki-normalized defect.

Checkpoint 12: Four closure groups and first-null domains

Source: *Michaels_Four_Closure_Groups_Checkpoint_12.md*. Equation tags carry the source prefix CP12; references within this unit retain their original local numbering.

Christophe Michaels research continuation · 22 September 2026 · Checkpoint 12

This round starts from checkpoint 11 and preserves the exact target

$$\boxed{\mathcal{E}_N = \|[\Delta(\rho_j, \rho_i)]_{i,j \leq N}\|_F^2, \quad \Delta = \mathfrak{W} - \mathcal{S}, \quad \liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1.} \quad (\text{CP12:0.1})$$

The ρ_j are the existing Suzuki-orthonormal determining tests. All maps below carry their metric; the target is never replaced by an unnormalized matrix norm.

The strongest completed auxiliary results are an exact full-defect factorization through extraction, a sharper rational extraction constant, a closed and injective fixed-support Suzuki extension, the complete first-null-vector equation, two exact parity correction kernels, and an exact Hardy projection norm budget including its mixed terms. The complete signed remainder estimate needed to close the target is still open. Section 5 combines the groups, attempts that narrower estimate, and states its precise quantified form.

The supplied extraction constant is retained:

$$K_* = \frac{3567}{509894} < \frac{1}{140}.$$

The same analytic inputs sharpen it to

$$K_0 = \frac{17607901}{2531730432}, \quad \varepsilon_* = K_* - K_0 = \frac{4429025}{108864408576} > 0.$$

No numerical zero computations or increasing-degree tests are used. The arithmetic here consists of exact rational inequalities and analytic estimates. The four groups distinguish proved auxiliary statements, conditional closing implications, and the remaining independent estimate.

Notation is local where unavoidable: G_S is a finite zero-feature Gram matrix; the G_a in Group 2 is a continuous convolution operator. $\mathcal{R}$ is the meromorphic correction from checkpoint 11; $\mathcal{T} = \Delta + \mathcal{X}_\infty$ is the complete retained extraction remainder. The shifted metric called S_a in Suzuki's new paper is carried separately from our fixed $\mathcal{S}$.

1. Closure group 1: Complete factorization and the shortest closing inequality

1.1. Domain and the shortest complete implication

Put $\mathcal{D} = C_c^\infty(\mathbb{R})$, $V = -L_0$, and $\mathcal{S}(q, r) = \langle Vq, Vr \rangle$. Inner products are linear in their first argument. The normalization is the one in checkpoint 11 and the consolidated manuscript: $L_0q = \pi^{-1/2} \widehat{\mathcal{P}_{Dq}}$, $Dq = iq'$, with the recorded two-sided Suzuki feature convention.

Closing implication. If, independently of RH,

$$\Delta(q, q) \geq -\mathcal{S}(q, q) \quad (q \in \mathcal{D}), \quad (\text{CP12:1.1})$$

then $\mathfrak{W}(q, q) \geq 0$ on the complete Weil test class. Weil's positivity criterion gives RH. Suzuki's correctly normalized identity under RH then gives

$$\mathfrak{W}(q, r) = \langle L_0q, L_0r \rangle = \mathcal{S}(q, r) \quad (q, r \in \mathcal{D}).$$

Consequently $\Delta = 0$, every $D_N = 0$, and

$$\boxed{\mathcal{E}_N = 0 \text{ for every } N.} \quad (\text{CP12:1.2})$$

The invocation of Suzuki's identity occurs after Weil's criterion has supplied RH. No RH-dependent orthogonality is used to prove the antecedent.

Thus the lower form bound is a sufficient closing inequality. Its independent proof is the work still required; the implication by itself is a reduction.

1.2. One compatible extraction and a stronger explicit constant

Use the canonical product Q_∞ over distinct signed pairs of zeros, normalized by $Q_\infty(0) = 1$, exactly as in checkpoint 11. Let $\omega = |A/E|^2$. For fixed admissible a, b, T , write

$$k_T(x) = \frac{H_0}{2\delta} b_T(x) \frac{|Q_\infty(x)|^2}{\omega(x)}, \quad \mathcal{X}_\infty(q, r) = \int_{-T}^T k_T(x) Vq(x) \overline{Vr(x)} dx. \quad (\text{CP12:1.3})$$

There are no real zeros of A on this band. This is precisely $\langle U_\infty Vq, U_\infty Vr \rangle$, including the phase in the checkpoint-11 multiplier. Its diagonal is positive and $k_T \leq K_*$.

The same established estimates yield a slightly stronger, explicit rational constant. For $0 \leq x < 3$, the inequality $n! \geq 2 \cdot 3^{n-2}$, $n \geq 2$, gives

$$e^x \leq 1 + x + \frac{x^2}{2(1 - x/3)}.$$

At $x = 5/48$, this is

$$e^{5/48} \leq \frac{14809}{13344}.$$

Replacing only the last exponential majorant in checkpoint 11 gives

$$0 \leq \mathcal{X}_\infty(q, q) \leq K_0 \mathcal{S}(q, q), \quad \boxed{K_0 = \frac{17607901}{2531730432} < K_* = \frac{3567}{509894}.} \quad (\text{CP12:1.4})$$

This also holds on every finite symmetric zero-feature span. Every previously retained term remains present; this calculation strengthens the scalar constant only. The exact gap is

$$\boxed{\varepsilon_* = K_* - K_0 = \frac{4429025}{108864408576} > 0.} \quad (\text{CP12:1.5})$$

1.3. Exact complete remainder, including all mixed entries

For the smooth symmetric zero cutoffs from checkpoint 11, put

$$a_{\gamma, H}(q) = \chi(\Re \gamma / H) \sqrt{m_\gamma} \hat{q}(\gamma).$$

Write

$$\mathcal{K}_H(q, r) = \sum_{\gamma, \delta} a_{\gamma, H}(q) \overline{a_{\delta, H}(r)} \sqrt{m_\gamma m_\delta} \mathcal{R}[\gamma, \bar{\delta}]. \quad (\text{CP12:1.6})$$

Each sum is finite. Checkpoint 11 proves convergence of the completed forms and $\Delta(q, r) = -\lim_H \mathcal{K}_H(q, r)$. Define the following complete remainder rather than assigning it a sign:

$$\boxed{\mathcal{T}(q, r) := -\lim_H \mathcal{K}_H(q, r) + \mathcal{X}_\infty(q, r).} \quad (\text{CP12:1.7})$$

Then the exact extraction factorization with remainder is

$$\boxed{\Delta = -\mathcal{X}_\infty + \mathcal{T}, \quad \mathfrak{W} = \mathcal{S} - \mathcal{X}_\infty + \mathcal{T}.} \quad (\text{CP12:1.8})$$

This identity makes no assumption about the sign of the meromorphic correction. In (1.6), every ordered pair of nodes remains, including real/nonreal, conjugate-pair and nonconjugate-pair terms.

For example, on a two-dimensional test span $q = \alpha q_1 + \beta q_2$, all three forms in (1.8) contain the diagonal contributions and

$$2\Re(\alpha\bar{\beta}[-\mathcal{X}_\infty(q_1, q_2) + \mathcal{T}(q_1, q_2)]).$$

No diagonal-only estimate is substituted for this mixed term.

The meromorphic decomposition retains its boundary term within this same remainder. If Ω encloses the cutoff nodes and $\mathcal{R} = H_\Omega + \sum_{a \in \Omega} P_a$, then

$$\mathcal{K}_H = \mathcal{K}_{H, H_\Omega} + \sum_{a \in \Omega} \mathcal{K}_{H, P_a},$$

where each form is obtained by substituting its displayed function for $\mathcal{R}$ in (1.6). In particular, the simple-pole term is exactly

$$\mathcal{K}_{H, P_a}(q, r) = -r_a \left(\sum_\gamma \frac{a_{\gamma, H}(q) \sqrt{m_\gamma}}{\gamma - a} \right) \left(\sum_\delta \frac{\overline{a_{\delta, H}(r)} \sqrt{m_\delta}}{\bar{\delta} - a} \right). \quad (\text{CP12:1.9})$$

A conjugate pole must be paired with its reflected residue. The second factor in (1.9) is not the conjugate of the first factor at the same nonreal pole; replacing it by a squared norm would change the form. Higher-order poles and the complete boundary contribution remain as their divided differences. Group 4 treats the norm-square interactions of these contributions.

1.4. The exact constant that would close the null argument

Let

$$r_* = 1 - K_* = \frac{506327}{509894}.$$

If the independent remainder inequality

$$\boxed{\mathcal{T}(q, q) \geq -r_* \mathcal{S}(q, q) \quad (q \in \mathcal{D})} \quad (\text{CP12:1.10})$$

holds, then (1.4) and (1.8) imply the strict lower bound

$$\boxed{\mathfrak{W}(q, q) \geq (K_* - K_0) \mathcal{S}(q, q) = \varepsilon_* \mathcal{S}(q, q) > 0 \quad (0 \neq q \in \mathcal{D}).} \quad (\text{CP12:1.11})$$

This would prove (1.1), and hence the complete target by Section 1.1.

For a genuine Weil null vector v in a domain on which the forms and their limiting identities are defined, its null equation gives the exact relation

$$\mathcal{T}(v, v) = -\mathcal{S}(v, v) + \mathcal{X}_\infty(v, v).$$

Using the same complete expression, (1.10) would force

$$0 < \mathcal{S}(v, v) \leq (r_* + K_0)\mathcal{S}(v, v) = \theta_*\mathcal{S}(v, v),$$

with the actual constant

$$\theta_* = \frac{108859979551}{108864408576} < 1. \quad (\text{CP12:1.12})$$

This is the requested absorption constant, derived from the recorded extraction and an explicitly identified full-remainder estimate. It is a conditional calculation: (1.10) has not been established by the extraction estimate.

If v is only in the closed Weil form domain, this argument can instead run on an approximating sequence $q_n \in \mathcal{D}$ supported in its fixed interval. Under (1.10), (1.11) implies $\mathcal{S}(q_n, q_n) \rightarrow 0$ whenever $\mathfrak{W}(q_n, q_n) \rightarrow 0$. The required domain step is then the implication

$$q_n \rightarrow v \text{ in } L^2 \text{ on a fixed interval, } \mathcal{S}(q_n, q_n) \rightarrow 0 \implies v = 0.$$

Group 2 proves this precise implication and the complete endpoint terms. No Suzuki norm is assigned to a null vector before that extension is justified.

1.5. An exact finite factorization on the actual determining spaces

The factorization through extraction can also be made literal, without a remainder, on every actual determining space. This calculation uses compact tests rather than arbitrary finite zero arrays.

Let $q_c = \sum_{j=1}^N c_j \rho_j$, where the ρ_j are Suzuki-orthonormal. Define

$$(X_N)_{ij} = \mathcal{X}_\infty(\rho_j, \rho_i), \quad (D_N)_{ij} = \Delta(\rho_j, \rho_i).$$

Then $0 < X_N \leq K_0 I$. Strict positivity follows as follows. The zero synthesis for Vq is meromorphic and locally normally convergent near the central real band; compact-test Fourier coefficients decay rapidly in the zero strip. If $\mathcal{X}_\infty(q, q) = 0$, it vanishes on an open subinterval. The identity theorem makes Vq identically zero, and the unconditional norm property gives $q = 0$.

Define

$$B_N = X_N^{-1/2} D_N X_N^{-1/2}. \quad (\text{CP12:1.13})$$

The complete defect factors exactly as

$$D_N = X_N^{1/2} B_N X_N^{1/2}. \quad (\text{CP12:1.14})$$

Equivalently, if $U_N c = U_\infty V q_c$, then $W_N := U_N X_N^{-1/2}$ is an isometry, and

$$D_N = U_N^* (W_N B_N W_N^*) U_N.$$

All mixed entries of the complete correction occur in B_N . Thus the operation connecting the two forms exists exactly; its middle factor is now explicit.

A sufficient uniform closing bound would be

$$B_N \succeq -K_*^{-1} I, \quad K_*^{-1} = \frac{509894}{3567}, \quad N \geq 1. \quad (\text{CP12:1.15})$$

It would give

$$D_N \succeq -K_*^{-1} X_N \succeq -(K_0/K_*) I \succ -I.$$

On a determining ladder this supplies Weil positivity, followed by Section 1.1. The exact entry formula (1.6) and the positive Cauchy Gram identity do not give (1.15): they identify its left side but leave its negative part unbounded. This is the precise stopping line of the literal factorization route.

An HS alternative is $\sup_N \|B_N\|_F \leq 1/K_0$, which would give $\|D_N\|_F \leq K_0 \|B_N\|_F \leq 1$. That is a stronger norm demand than the lower spectral bound needed for Weil positivity. Group 1 therefore selects the lower-bound route.

1.6. Testing actual xi information rather than a generic matrix example

There is an exact decomposition of the actual Weil form into conjugation channels. Let $\Gamma_{\mathbb{R}}$ be the real zeros and choose one representative from each nonreal conjugate pair. For a nonreal $\gamma = u + ib$, put

$$c_{\gamma}^{+}(q) = \frac{\widehat{q}(\gamma) + \widehat{q}(\bar{\gamma})}{\sqrt{2}}, \quad c_{\gamma}^{-}(q) = \frac{\widehat{q}(\gamma) - \widehat{q}(\bar{\gamma})}{\sqrt{2}}.$$

Absolute convergence of the compact-test zero sum gives

$$\boxed{\mathfrak{W}(q, q) = \sum_{\gamma \in \Gamma_{\mathbb{R}}} m_{\gamma} |\widehat{q}(\gamma)|^2 + \sum_{\{\gamma, \bar{\gamma}\}} m_{\gamma} (|c_{\gamma}^{+}(q)|^2 - |c_{\gamma}^{-}(q)|^2).} \quad (\text{CP12:1.16})$$

In the time variable,

$$c_{\gamma}^{+}(q) = \sqrt{2} \int q(t) e^{iut} \cosh(bt) dt, \quad c_{\gamma}^{-}(q) = -\sqrt{2} \int q(t) e^{iut} \sinh(bt) dt. \quad (\text{CP12:1.17})$$

These are the actual extra identities furnished by conjugation and the zero strip. The pointwise inequality $|\sinh(bt)| \leq \cosh(bt)$ does not compare the two Fourier integrals for arbitrary complex tests: cancellation occurs inside each integral. An additional inequality for their complete sampled sums is needed. The extraction comparison controls a central norm of Vq ; it has not bounded the negative sum in (1.16) by the positive sum.

A hypothetical actual compact negative witness, normalized by $\mathcal{S}(q, q) = 1$ and $\mathfrak{W}(q, q) = -\eta$, would satisfy

$$\mathcal{T}(q, q) = -1 - \eta + \mathcal{X}_{\infty}(q, q) \leq -(1 - K_0 + \eta). \quad (\text{CP12:1.18})$$

This is an exact consequence for the actual forms, not a generic counterexample. It shows quantitatively which part of (1.10) must exclude the hypothetical negative witness: it must close a gap of at least $\varepsilon_{*} + \eta$.

1.7. A localization calculation for extraction-only factorizations

For a fixed nonzero compact test, define its analytic numerator near zero by

$$n_q(x) = \sqrt{\pi} \frac{Q_{\infty}(x)}{ih(x)} Vq(x).$$

It is nonzero as an analytic function. Suppose its order of vanishing at zero is ℓ , so $n_q(x) = a_{\ell} x^{\ell} + O(x^{\ell+1})$, $a_{\ell} \neq 0$. Then the actual extraction has the exact small-band asymptotic

$$\boxed{\mathcal{X}_{\infty, T}(q, q) = \frac{H_0}{2\pi\delta} |a_{\ell}|^2 B(\ell + \tfrac{1}{2}, 5) T^{2\ell+1} + O_q(T^{2\ell+3}).} \quad (\text{CP12:1.19})$$

Proof: substitute $x = Tt$, expand $|n_q(Tt)|^2$, and integrate against $(1 - t^2)^4$. The next odd term integrates to zero; the leading integral is the stated beta value.

Consequently an extraction-only identity $\Delta(q, r) = \langle C_T U_T V q, U_T V r \rangle$ with $\sup_{T \downarrow 0} \|C_T\| < \infty$ would force $\Delta(q, q) = 0$ for every compact test and hence $\Delta = 0$ by polarization. If a fixed test has nonzero defect, any exact such factor has

$$\|C_T\| \geq \frac{|\Delta(q, q)|}{\mathcal{X}_{\infty, T}(q, q)} \asymp_q T^{-2\ell-1}.$$

This calculation pinpoints how the middle factor carries the missing global information. For closure at the fixed band $T = 1$, it remains legitimate to seek (1.10) or (1.15); shrinking the extraction alone supplies no bound for them.

1.8. Closure statement and one quantified missing lemma

Closed auxiliary results: the exact full-remainder factorization (1.7)-(1.9); the actual rational strengthening K_0 ; the conditional absorption constant $\theta_* < 1$; literal complete finite factorization (1.13)-(1.14); the conjugation-channel identity (1.16); and the analytic localization law (1.19).

Selected missing lemma, fixed band: Fix any archived admissible a, b , take $T = 1$, and use the canonical product, smooth cutoffs, and actual meromorphic correction specified above. Prove independently of RH that for every $q \in C_c^\infty(\mathbb{R})$,

$$-\lim_{H \rightarrow \infty} \sum_{\gamma, \delta} a_{\gamma, H}(q) \overline{a_{\delta, H}(q)} \sqrt{m_\gamma m_\delta} \mathcal{R}[\gamma, \bar{\delta}] + \frac{H_0}{2\delta} \int_{-1}^1 (1 - x^2)^4 \frac{|Q_\infty(x)|^2}{\omega(x)} |Vq(x)|^2 dx \geq -\frac{506327}{509894} \|Vq\|_2^2.$$

(CP12:1.20)

The cutoff topology and complete pair sum are those already justified in checkpoint 11. No contour term is discarded. The derivation stops at obtaining this lower bound for the explicitly retained remainder. Proving it would yield (1.11), then Weil positivity, RH, Suzuki equality, and $\mathcal{E}_N = 0$ for all N . It has not been proved in this group.

Group 1 closure record.

1. **Statement and domain:** exact $\Delta = -\mathcal{X}_\infty + \mathcal{T}$ on $C_c^\infty(\mathbb{R})$, with literal finite factorizations on the actual determining spaces.
2. **Outcome:** Sections 1.2-1.7 prove the constants, factorization, conjugation channels, and localization law. The lower bound for $\mathcal{T}$ remains open.
3. **Return to target:** proving that lower bound yields Weil positivity, then normalized Suzuki equality and $\mathcal{E}_N = 0$.
4. **Quantified missing lemma and stop:** equation (1.20), with threshold 506327/509894; the derivation stops at independently estimating its complete meromorphic sum plus extraction.

2. Closure group 2: The first zero crossing, with its metric and domain

2.1. Source normalization and exact transport

Primary source: M. Suzuki, *Weil's quadratic form via the screw function*, arXiv:2606.09096v2, especially §§8.2-8.5, <https://arxiv.org/html/2606.09096v2>. Its Theorems 1.1, 1.3, 1.4 and Corollary 1.2 supply the following inputs: the localized Weil form has a Friedrichs operator with compact resolvent; its lowest eigenvalue is continuous in the interval parameter and positive on

sufficiently small intervals; compact smooth tests form a form core. The source's shifted metric denoted $\mathbf{S}_a$ is not the fixed Suzuki metric $\mathcal{S}$ in our target. The source's factorization in §7.7 assumes RH; it is not used here.

Write $I_a = (-a, a)$, and let Q_a be the closed localized Weil form on $L^2(I_a)$, with zero extension understood. Let A_a be its associated self-adjoint operator. Let g be the real even continuous screw-function candidate from the explicit formula. Define

$$(P_a f)(x) = f(x) - (2a)^{-1} \int_{-a}^a f(y) dy, \quad G_a = P_a(g*)P_a, \quad D = i\partial_x.$$

The name G_a here denotes the continuous convolution operator, not the finite zero-feature Gram matrix G_S .

For $u \in L_0^2(I_a)$, the exact inverse derivative is

$$(J_a u)(x) = D^{-1}u(x) = -i \int_{-a}^x u(t) dt. \quad (\text{CP12:2.1})$$

Both endpoint values vanish because $\int u = 0$. Direct Fubini gives

$$\|J_a u\|_2^2 = \iint (a - \max\{x, y\}) u(y) \overline{u(x)} dy dx.$$

On mean-zero functions, the separable differences vanish, leaving

$$\|J_a u\|_2^2 = \langle K_a u, u \rangle, \quad K_a(x, y) = \frac{x^2 + y^2}{4a} - \frac{|x - y|}{2} + \frac{a}{6}. \quad (\text{CP12:2.2})$$

Differentiating this kernel proves $-\partial_x^2 K_a = \delta_y - (2a)^{-1}$, its two Neumann boundary derivatives vanish, and its x -integral is zero. Thus K_a is exactly the inverse Neumann Laplacian on the mean-zero subspace.

The corresponding Weil identity, first on the core and then on $H_0^1(I_a)$, is

$$Q_a(J_a u, J_a w) = \langle G_a u, w \rangle. \quad (\text{CP12:2.3})$$

The original Suzuki metric must be transported separately:

$$\boxed{\Sigma_a(u, w) := \mathcal{S}(J_a u, J_a w)}. \quad (\text{CP12:2.4})$$

Initially take u, w in $D(C_c^\infty(I_a))$. This is positive definite there. It is not replaced by K_a or by $G_a - \lambda K_a$. In this representation

$$\Delta(J_a u, J_a w) = \langle G_a u, w \rangle - \Sigma_a(u, w). \quad (\text{CP12:2.5})$$

A finite determining family q_j transports to $u_j = Dq_j$ with exactly the same Suzuki Gram entries and exactly the same defect entries. Hence its normalized Frobenius energy is unchanged.

For the spectral problem normalized in $L^2(I_a)$, the generalized equation is

$$G_a u = \lambda K_a u. \quad (\text{CP12:2.6})$$

When the eigenvector belongs only to the closed Weil form domain, this equation is interpreted in the derivative completion, not asserted as an equation between ordinary L^2 functions. For a pencil normalized in the original Suzuki metric, the weak equation instead reads

$$\langle G_a u, w \rangle = \mu \Sigma_a(u, w). \quad (\text{CP12:2.7})$$

Its zero equation is the same, but its nonzero eigenvalues need not match those of (2.6).

2.2. First zero crossing and its complete null equation

Let $\lambda(a) = \inf Q_a(v)/\|v\|^2$. Inclusion of core spaces makes $\lambda(a)$ nonincreasing. If any negative compact witness exists, $\lambda(a_1) < 0$ for some finite a_1 . Positivity for small a , continuity, and compact resolvent therefore produce a finite $a_* > 0$ satisfying

$$\lambda(a_*) = 0, \quad Q_{a_*} \geq 0, \quad 0 \neq v \in \mathfrak{D}(A_{a_*}), \quad A_{a_*}v = 0. \quad (\text{CP12:2.8})$$

This is an attained zero of the unbounded Friedrichs problem. It is not inferred from the bottom of the ordinary spectrum of compact G_a , which can already equal zero without a null vector.

The complete null equation can be written with the continuous kernel without imposing an unproved endpoint regularity. For $v \in D(A_a)$, let

$$f_v(x) = \int_{-a}^a g(x-y)v(y) dy.$$

On interior compact tests, $A_a v = -\partial_x^2 f_v$ in distributions. Therefore

$$A_a v = 0 \iff \int_{-a}^a g(x-y)v(y) dy = \alpha x + \beta \quad (-a < x < a), \quad (\text{CP12:2.9})$$

where the converse is understood for v in the closed form domain. Indeed, the distributional equation annihilates every core test and therefore every form-domain test by core density; the representation theorem then places v in $D(A_a)$ with $A_a v = 0$.

The constants are retained explicitly:

$$\beta = \int_{-a}^a g(-y)v(y) dy, \quad \alpha = \int_{-a}^a g'(-y)v(y) dy. \quad (\text{CP12:2.10})$$

These are defined: g is continuous, while g' is locally L^2 , its origin singularity being logarithmic and its prime-power jumps finite on each bounded interval. Convolution with the compact L^2 function v makes f_v continuously differentiable, so (2.10) is valid.

For $u = i\partial(1_{I_a}v)$ in distributions, equation (2.9) gives

$$g * u = i\alpha \quad \text{inside } I_a, \quad (\text{CP12:2.11})$$

whose mean-zero projection vanishes. If v happens to lie in $H^1(I_a)$, the derivative contains precisely

$$u = i1_{I_a}v' + iv(-a)\delta_{-a} - iv(a)\delta_a. \quad (\text{CP12:2.12})$$

Thus the corresponding convolution is

$$i \int_{-a}^a g(x-y)v'(y) dy + iv(-a)g(x+a) - iv(a)g(x-a) = i\alpha. \quad (\text{CP12:2.13})$$

No endpoint value is set to zero unless $v \in H_0^1$ has been proved. For a general form-domain vector, (2.9) itself is the boundary-complete formulation; traces are unnecessary.

2.3. Mixed terms simplify before estimates

For a genuine null vector, $Q_a(v, z) = 0$ for every form-domain z . If $v = v_0 + v_1$ with both pieces in that domain, put $d = Q_a(v_0, v_0)$. Testing against each piece gives the exact identities

$$Q_a(v_0, v_1) = Q_a(v_1, v_0) = -d, \quad Q_a(v_1, v_1) = d \geq 0. \quad (\text{CP12:2.14})$$

Thus both cross terms are retained and determined before any absolute values are taken. Their sum cancels the two diagonal terms. In an extension where the original Suzuki norm is finite,

$$\Delta(v, v) = -\mathcal{S}(v, v). \quad (\text{CP12:2.15})$$

This equality supplies the precise null-vector normalization. It supplies no strict absorption constant on its own.

2.4. A closed auxiliary theorem: a fixed-support limit theorem

The following closes the domain issue needed for a contradiction using core approximants; it does not assume that every Friedrichs eigenvector already has finite Suzuki norm.

Theorem. Fix $a > 0$. For $q_n \in C_c^\infty(I_a)$,

$$q_n \longrightarrow v \text{ in } L^2(I_a), \quad \mathcal{S}(q_n, q_n) \longrightarrow 0 \implies v = 0. \quad (\text{CP12:2.16})$$

Proof. Use the same corrected two-sided Suzuki kernel as checkpoint 11. If

$$P_t(z) = \sum_{\gamma \in \Gamma} m_\gamma \frac{e^{-i\gamma t} - 1}{\gamma(z - \gamma)},$$

its exact negative-time continuation is $P_{-t}(z) = P_t(-z)$, by $\gamma \mapsto -\gamma$. Define $h = A/E$ and

$$\mathcal{T}_t(z) = -ih(z)P_t^\sharp(z).$$

The finite-support synthesis $V = -L_0$ from checkpoint 11 has the integral form

$$(Vq)(z) = \frac{i}{\sqrt{\pi}} \int_{-a}^a \partial_t \mathcal{T}_t(z) q(t) dt, \quad (\text{CP12:2.17})$$

obtained by integrating by parts on a compact smooth test. Its sign agrees with $Vq = \sum_\gamma \sqrt{m_\gamma} \hat{q}(\gamma) F_\gamma$.

Here is the local bound needed to extend (2.17), rather than assuming it. For positive t , the explicit Suzuki formula consists of elementary exponential terms, finitely many prime-power packets on $[0, a]$, and one Lerch difference. For z in a compact set avoiding its genuine meromorphic poles, the only potentially singular differentiated term has the form

$$L_b(t) = e^{-t/2} \sum_{n \geq 0} e^{-2nt} \left(\frac{1}{n+b} - \frac{1}{n+1/4} \right), \quad b = 1/4 - iz/2.$$

The coefficient difference is $O_K((n+1)^{-2})$; its differentiated summands are $O_K(e^{-2nt}/(n+1))$. Since

$$\sum_{n \geq 1} \frac{e^{-2nt}}{n} = -\log(1 - e^{-2t}),$$

we get, locally uniformly in z ,

$$|\partial_t \mathcal{T}_t(z)| \leq C_{a,K}(1 + |\log t|), \quad 0 < t \leq a. \quad (\text{CP12:2.18})$$

The finite prime packets vanish at their switch points and have bounded one-sided derivatives, so differentiating introduces no point mass. The kernel is continuous at $t = 0$, and the logarithmic derivative is integrable, so no point mass appears there either. The negative-time relation gives the same estimate for $t < 0$. Apparent z -poles at zeros of A and at elementary normalization points are removed by the exact kernel identity; the bound extends across such points using a small surrounding circle and the maximum principle. Thus (2.18) holds on every compact set avoiding genuine poles of the completed meromorphic kernel.

The series defining P_t is normally convergent on compact z -sets away from Γ , uniformly for $|t| \leq a$, because its terms are bounded by $C_{a,K}m_\gamma/|\gamma|^2$ for large γ . This also proves the asserted continuity at $t = 0$ directly.

Because $1 + |\log |t||$ belongs to $L^2(I_a)$, formula (2.17) defines a continuous map from $L^2(I_a)$ to locally meromorphic functions, with locally uniform convergence away from genuine poles. Call this extension V_a^{loc} . In particular, $q_n \rightarrow v$ in L^2 implies $Vq_n(x) \rightarrow V_a^{\text{loc}}v(x)$ for every real x after removable values are filled in. The hypothesis $\|Vq_n\|^2 = \mathcal{S}(q_n, q_n) \rightarrow 0$ supplies a subsequence converging to zero almost everywhere on the real line. Hence $V_a^{\text{loc}}v = 0$ there and thus identically by analytic continuation.

The local expansion $h(z) = -i(z - \gamma)/m_\gamma + O((z - \gamma)^2)$ gives

$$\mathcal{T}_t(\gamma) = -\frac{e^{i\gamma t} - 1}{\gamma}, \quad \partial_t \mathcal{T}_t(\gamma) = -ie^{i\gamma t},$$

and therefore

$$(V_a^{\text{loc}}v)(\gamma) = \frac{\widehat{v}(\gamma)}{\sqrt{\pi}}. \quad (\text{CP12:2.19})$$

Consequently $\widehat{v}$ vanishes at every distinct zero γ of A . A nonzero Fourier transform of an L^2 function supported in I_a has exponential type at most a ; Jensen's formula and $|\widehat{v}(z)| \leq \sqrt{2a}\|v\|_2 e^{a|\text{Im } z|}$ bound its disk zero count by $O_v(R)$. Unconditionally, the distinct real zeros of A already have count $\gg R \log R$, using the established positive proportion of simple critical-line zeros. Thus $\widehat{v}$ must vanish identically, and $v = 0$. $\square$

Equivalently, define the maximal operator by

$$\mathfrak{D}(V_a) = \{v \in L^2(I_a) : V_a^{\text{loc}}v|_{\mathbb{R}} \in L^2(\mathbb{R})\}.$$

It is densely defined, closed, and injective. To see closedness, combine the locally uniform convergence from (2.18) with an almost-everywhere subsequence whenever the spectral images converge in L^2 . Injectivity is exactly the zero-evaluation argument above. Consequently every fixed-support sequence $q_n \rightarrow v \neq 0$ in L^2 satisfies $\liminf_n \mathcal{S}(q_n, q_n) > 0$, allowing an infinite $\liminf$. This is a statement about each convergent sequence; a uniform coercivity constant on all inputs is not asserted.

For the final density input one may use the original primary paper J. B. Conrey, *More than two fifths of the zeros of the Riemann zeta function are on the critical line*, J. reine angew. Math. 399 (1989), 1-26, <https://aimath.org/~kaur/publications/24.pdf>. Only a positive proportion of simple zeros is needed. Multiplicity-counting alone would not justify this last step.

The kernel formula underlying this auxiliary theorem is Suzuki, arXiv:2301.00421v3, (1.5)-(1.7), (3.2), (3.5)-(3.9), with the corrected two-sided time convention already adopted in checkpoint 11. No RH-dependent orthogonality is used.

A further consequence gives the exact maximal domain. Define

$$\mathfrak{D}(V_a) = \{v \in L^2(I_a) : V_a^{\text{loc}} v|_{\mathbb{R}} \in L^2(\mathbb{R})\}. \quad (\text{CP12:2.20})$$

Then V_a is a closed, injective operator from $L^2(I_a)$ to $L^2(\mathbb{R})$. Indeed, if $v_n \rightarrow v$ and $V_a v_n \rightarrow f$ in the respective L^2 spaces, the locally uniform kernel limit and an almost-everywhere subsequence identify f with $V_a^{\text{loc}} v$. Injectivity was proved above. Fatou also gives

$$\|V_a v\|^2 \leq \liminf_n \|V_a v_n\|^2 \quad (\text{CP12:2.21})$$

whenever $v_n \rightarrow v$ in the fixed-support L^2 space and the right side is finite. This defines the closed extended Suzuki metric precisely, without asserting that its domain equals the Friedrichs form domain.

2.5. A quantitatively strict conditional null contradiction

Let $X(q, q)$ be the compatible completed extraction from Group 1, and define the entire retained remainder by

$$\mathcal{T}(q, r) = \Delta(q, r) + X(q, r). \quad (\text{CP12:2.22})$$

This is an exact definition including every pole, boundary, and mixed contribution. Then

$$Q_a(q, q) = \mathcal{S}(q, q) - X(q, q) + \mathcal{T}(q, q). \quad (\text{CP12:2.23})$$

Group 1 refines the rational bound using the same analytical input:

$$X(q, q) \leq K_0 \mathcal{S}(q, q), \quad K_0 = \frac{17607901}{2531730432} < K_* = \frac{3567}{509894}.$$

The exact difference is

$$\varepsilon = K_* - K_0 = \frac{4429025}{108864408576} > 0. \quad (\text{CP12:2.24})$$

If the independent remainder bound

$$\boxed{\mathcal{T}(q, q) \geq -(1 - K_*) \mathcal{S}(q, q) = -\frac{506327}{509894} \mathcal{S}(q, q)} \quad (\text{CP12:2.25})$$

holds on the relevant compact core, then (2.23) yields

$$Q_a(q, q) \geq \varepsilon \mathcal{S}(q, q). \quad (\text{CP12:2.26})$$

For a null vector with finite Suzuki norm, the actual absorption factor would be

$$\theta = (1 - K_*) + K_0 = 1 - \varepsilon < 1, \quad 0 < \mathcal{S}(v, v) \leq \theta \mathcal{S}(v, v). \quad (\text{CP12:2.27})$$

For the general first null vector, take core $q_n \rightarrow v$ in the form norm. Then $q_n \rightarrow v$ in L^2 and $Q_a(q_n) \rightarrow 0$. Bound (2.26) forces $\mathcal{S}(q_n, q_n) \rightarrow 0$. The proved theorem (2.16) gives $v = 0$, contradicting (2.8). Thus the argument has a rigorous domain-complete version without asserting a nonexistent Dirichlet condition or a priori Suzuki norm for v .

2.6. Exact stopping line and target implication

The null equation (2.9) determines complete mixed terms and boundary terms, and the domain passage in (2.16) is closed. The actual equations do not presently prove (2.25). Substituting the null equation into the definition of the remainder instead gives $\mathcal{T}(v, v) = -\mathcal{S}(v, v) + X(v, v)$, exactly the value which strict absorption would exclude. Treating this identity as an independent bound would repeat the desired conclusion.

Fully quantified remaining lemma for Groups 1-2. For every $a > 0$ for which Q_a is nonnegative and $\ker A_a$ is nonzero, there exists a form-core approximating sequence $q_n \rightarrow v$ for each $0 \neq v \in \ker A_a$ such that (2.25) holds for every n . A stronger, cleaner sufficient version is that (2.25) holds for all $q \in C_c^\infty(I_a)$ for every such a . The needed threshold is exactly 506327/509894 and the resulting strict margin is exactly (2.24).

More precisely, for any hypothetical first-null core approximants $q_n \rightarrow v \neq 0$, the closed injective realization gives $\liminf \mathcal{S}(q_n, q_n) > 0$. Hence $Q_a(q_n)/\mathcal{S}(q_n, q_n) \rightarrow 0$. The complete retained remainder obeys the actual identity and estimate

$$\frac{\mathcal{T}(q_n, q_n)}{\mathcal{S}(q_n, q_n)} = \frac{Q_a(q_n, q_n)}{\mathcal{S}(q_n, q_n)} - 1 + \frac{X(q_n, q_n)}{\mathcal{S}(q_n, q_n)} \leq -1 + K_0 + o(1). \quad (\text{CP12:2.28})$$

Thus it would eventually lie at least $\varepsilon/2$ below the proposed threshold $-1 + K_*$. This is the exact quantitative collision which (2.25) must rule out, now formulated entirely on the original compact-test domain.

The derivation stops at the attempted independent proof of (2.25); all later implications are now explicit. If this lemma is proved, first degeneracy is impossible; continuity rules out a negative localized Weil form; Weil positivity gives RH; the correctly normalized Suzuki identity then gives $\Delta = 0$ on the original compact test domain and therefore $\mathcal{E}_N = 0$ for every original determining N . Consequently the unchanged target $\liminf \mathcal{E}_N \leq 1$ follows. At present, this group closes the transport, null equation, endpoint bookkeeping, and core-to-null domain argument, while leaving the explicitly quantified remainder estimate open.

Group 2 closure record.

1. **Statement and domain:** the transported form uses $\Sigma_a(u, w) = \mathcal{S}(D^{-1}u, D^{-1}w)$; the first-null equation holds for the Friedrichs operator domain, and the closed-map theorem applies to fixed-support L^2 limits of compact tests.
2. **Outcome:** the complete affine null equation, endpoint bookkeeping, mixed-term cancellation, and domain passage are proved. The displayed absorption factor is conditional on the remainder estimate.
3. **Return to target:** excluding first degeneracy excludes negative compact Weil tests and gives $\mathcal{E}_N = 0$ through the correctly ordered criterion and identity.
4. **Quantified missing lemma and stop:** equation (2.25) on the stated core approximants; Section 5 narrows this further to the limiting inequality (5.6). The stop is the independent remainder bound, not the domain passage.

3. Closure group 3: Tensor and square-coordinate transports

This calculation uses the definitions and complete cutoff convergence of checkpoint 11. All assertions below concern the actual function $A(z) = \xi(1/2 - iz)$, unless another scope is explicitly stated. Inner products are linear in the first variable; coefficient matrices use the usual conjugate-linear-first matrix convention. No orthogonality of the zero features is assumed.

3.1. Exact tensor lift, including the inverse metric

Let $q_1, \dots, q_n \in C_c^\infty(\mathbb{R})$ be linearly independent in the Suzuki completion, and set

$$G_{ij} = \mathcal{S}(q_j, q_i), \quad Z_{ij} = \Delta(q_j, q_i), \quad H = G^{-1}, \quad D = G^{-1/2} Z G^{-1/2}.$$

Here $G > 0$ and $Z = Z^*$. For column-major vectorization,

$$\|D\|_F^2 = \text{vec}(Z)^* (\overline{G^{-1}} \otimes G^{-1}) \text{vec}(Z) = \sum_{i,j,k,l=1}^n \overline{Z_{ij}} H_{ik} Z_{kl} H_{lj}. \quad (\text{CP12:3.1})$$

Indeed $\text{vec}(AZB) = (B^T \otimes A) \text{vec}(Z)$; applying this with $A = B = G^{-1/2}$ proves the identity. Equivalently this is $\text{tr}(Z^* H Z H)$.

If an exact decomposition, including an extraction term, pole terms and a contour term, is written as $Z = \sum_{\alpha=1}^r Z_\alpha$, define $D_\alpha = G^{-1/2} Z_\alpha G^{-1/2}$. Then

$$\|D\|_F^2 = \sum_{\alpha} \|D_\alpha\|_F^2 + 2 \text{Re} \sum_{\alpha < \beta} \text{tr}(D_\alpha^* D_\beta). \quad (\text{CP12:3.2})$$

This formula retains every pole-pole, pole-boundary, extraction-remainder and boundary-boundary contribution. The exact cross terms can have either sign. The sufficient triangle budget $\sum_{\alpha} \|D_\alpha\|_F \leq 1$ would close this finite compression; individual diagonal budgets alone would not.

Vectorization is invertible and loses no witness. If $c^*(G + Z)c < 0$, then with $u = G^{1/2}c$,

$$u^* D u < -\|u\|^2, \quad \|D\|_F > 1. \quad (\text{CP12:3.3})$$

The second statement follows from $|u^* D u| \leq \|D\|_{\text{op}} \|u\|^2 \leq \|D\|_F \|u\|^2$. Thus the tensor norm detects a negative witness through a strict floor above one. Its positivity as a tensor quadratic form does not supply an upper bound on that norm.

For the established extraction matrix Y , the transported estimate remains

$$0 \preceq Y \preceq K_* I_n, \quad \|Y\|_F^2 \leq n K_*^2, \quad K_* = \frac{3567}{509894}. \quad (\text{CP12:3.4})$$

The tensor lift leaves this dimension factor visible. An estimate of the complete tensor expression (3.2), rather than positivity of its metric, is required for the target.

3.2. Squaring the coordinate: exact measure and two channels

Evenness gives a unique real entire function B with

$$A(z) = B(z^2), \quad A'(z) = 2z B'(z^2).$$

For $s > 0$, put

$$W(s) = \frac{B(s)^2}{B(s)^2 + 4s B'(s)^2}, \quad d\mu(s) = \frac{W(s)}{\pi \sqrt{s}} ds. \quad (\text{CP12:3.5})$$

At common zeros use the removable real-axis value. This measure is positive almost everywhere. The powers $s^{-1/2}$ and $s^{1/2}$ below are the exact Jacobian factors; the point $s = 0$ creates no atom.

First remove the feature phase by the unitary gauge

$$f(x) \mapsto g(x) = \frac{\sqrt{\pi}f(x)}{ih(x)}, \quad L^2(dx) \longrightarrow L^2(\omega(x)dx/\pi), \quad h = \frac{A}{A + iA'}. \quad (\text{CP12:3.6})$$

The value at the discrete zeros of h is immaterial to the Hilbert spaces. The inverse is multiplication by $ih/\sqrt{\pi}$. Now define

$$g_e(s) = \frac{g(\sqrt{s}) + g(-\sqrt{s})}{2}, \quad g_o(s) = \frac{g(\sqrt{s}) - g(-\sqrt{s})}{2\sqrt{s}}.$$

The exact norm identity is

$$\boxed{\frac{1}{\pi} \int_{\mathbb{R}} \omega(x) |g(x)|^2 dx = \int_0^\infty |g_e(s)|^2 d\mu(s) + \int_0^\infty |g_o(s)|^2 s d\mu(s).} \quad (\text{CP12:3.7})$$

Consequently the gauge followed by squaring is unitary onto $L^2(d\mu) \oplus L^2(s d\mu)$. Its inverse is $g(x) = g_e(x^2) + x g_o(x^2)$. This retains both sheets and both parity channels. Ordinary complex conjugation acts componentwise on (g_e, g_o) .

For a finite symmetric zero set choose one γ from each pair $\{\gamma, -\gamma\}$, and let $\lambda = \gamma^2$. For the actual xi function one may choose $\text{Re } \gamma > 0$: positivity of the theta density gives $A(iy) = F(y) > 0$ for every real y , so there are no imaginary-axis zeros. This choice is preserved by conjugation. Then

$$d_{+,\lambda} = \frac{c_\gamma + c_{-\gamma}}{\sqrt{2}}, \quad d_{-,\lambda} = \frac{c_\gamma - c_{-\gamma}}{\sqrt{2}}$$

gives the explicit transformed rational features

$$g_o(s) = \sum_{\lambda} \frac{\sqrt{2m_\gamma} d_{+,\lambda}}{s - \lambda}, \quad g_e(s) = \sum_{\lambda} \frac{\sqrt{2m_\gamma} \gamma d_{-,\lambda}}{s - \lambda}. \quad (\text{CP12:3.8})$$

Thus the plus coefficient channel is the odd rational-function channel. This distinction fixes the parity labels unambiguously.

Squaring preserves off-critical information: a nonreal zero with nonzero real part maps to a nonreal λ , with conjugation $\lambda \mapsto \bar{\lambda}$. A hypothetical purely imaginary zero would map to a negative real λ ; it would not become a positive real node. Before using the actual exclusion just proved, the conjugation-pairing form acts as $+1$ on its plus channel and -1 on its minus channel. The sheet difference therefore retains that witness as well. For the actual function, RH is exactly the assertion that all zeros of B lie on $(0, \infty)$.

3.3. The complete correction becomes two divided-difference kernels

The regularized Cauchy transform in checkpoint 11 obeys $C(-z) = -C(z)$, because ω is even. Also $h(-z) = h^\sharp(z)$. Using the upper and lower definitions of $\mathcal{R}$ therefore proves

$$\mathcal{R}(-z) = -\mathcal{R}(z), \quad \mathcal{R}^\sharp = \mathcal{R}.$$

It follows that there is a single-valued real meromorphic function r such that

$$\boxed{\mathcal{R}(z) = z r(z^2).} \quad (\text{CP12:3.9})$$

There is no pole at zero: $\mathcal{R}$ is holomorphic and odd there. Write $f[u, v] = (f(u) - f(v))/(u - v)$, with $f'(u)$ at coincidence. If $\lambda = \gamma^2$, $\nu = \bar{\delta}^2$, elementary common-denominator algebra gives

$$\mathcal{R}[\gamma, \bar{\delta}] + \mathcal{R}[\gamma, -\bar{\delta}] = 2(wr(w))[\lambda, \nu],$$

$$\mathcal{R}[\gamma, \bar{\delta}] - \mathcal{R}[\gamma, -\bar{\delta}] = 2\gamma\bar{\delta}r[\lambda, \nu]. \quad (\text{CP12:3.10})$$

The formulas extend to coincident nodes by holomorphic continuation. Therefore the complete correction matrix has parity blocks

$$\boxed{K_+(\lambda, \mu) = 2\sqrt{m_\gamma m_\delta} (wr(w))[\lambda, \bar{\mu}],}$$

$$\boxed{K_-(\lambda, \mu) = 2\sqrt{m_\gamma m_\delta} \gamma\bar{\delta} r[\lambda, \bar{\mu}].} \quad (\text{CP12:3.11})$$

The opposite-parity blocks are exactly zero. This is an actual simplification of the full correction, including all pole and contour contributions; it follows from symmetry before any estimate. Within each block, every distinct-node and conjugate-node term remains.

For compact tests the cutoff in checkpoint 11 is even in $\text{Re } \gamma$, so it commutes with the pair transform. Its proved L^2 convergence carries (3.11) to the complete compact-test defect. An absolutely convergent uncut double series is not needed. Arbitrary finite zero coefficients are not silently identified with complete compact-test zero samples.

The displayed two-variable kernels use the linear-first convention; their coefficient matrices have entries with the two node labels transposed, as in checkpoint 11. The full Suzuki Gram kernels in these coordinates are also explicit. Define the Stieltjes transform

$$K(w) = \int_0^\infty \frac{d\mu(s)}{s - w}, \quad w \in \mathbb{C} \setminus [0, \infty). \quad (\text{CP12:3.12})$$

This integral converges, since $0 \leq W \leq 1$. Pairing x and $-x$ in the Cauchy integral gives $C(z) = zK(z^2)$. Consequently

$$\begin{aligned} G_+(\lambda, \mu) &= 2\sqrt{m_\gamma m_\delta} \int_0^\infty \frac{s d\mu(s)}{(s - \lambda)(s - \bar{\mu})}, \\ G_-(\lambda, \mu) &= 2\sqrt{m_\gamma m_\delta} \gamma\bar{\delta} \int_0^\infty \frac{d\mu(s)}{(s - \lambda)(s - \bar{\mu})}. \end{aligned} \quad (\text{CP12:3.13})$$

Real-node removals give the same formulas at real positive zero nodes. These kernels are positive and retain the original Suzuki metric exactly. With the conjugation-pairing matrices $J_\pm$, they satisfy $G_\pm = J_\pm + K_\pm$.

3.4. Exact branch formula and an actual xi estimate at infinity

Let

$$\Omega(w) = \frac{B(w)^2}{B(w)^2 + 4wB'(w)^2}, \quad \Lambda(w) = \frac{2B(w)B'(w)}{B(w)^2 + 4wB'(w)^2}.$$

Then $h(z) = \Omega(z^2) - iz\Lambda(z^2)$. On $\mathbb{C} \setminus [0, \infty)$, let $\sqrt{w}_+$ denote the square root with positive imaginary part. The exact expression is

$$\boxed{r(w) = K(w) - \Lambda(w) - \frac{i\Omega(w)}{\sqrt{w}_+}.} \quad (\text{CP12:3.14})$$

The apparent branch jump cancels: on the positive real axis the jump of K is $2iW(s)/\sqrt{s}$, exactly the jump of the last term with its sign. Equation (3.9) supplies the meromorphic continuation across the cut and removal at zero. Keeping only the Stieltjes term would omit the explicit last two terms.

There is a useful unconditional estimate on the negative ray, specific to the actual theta function. For $y > 0$, $F(y) > 0$, $F'(y) > 0$, and

$$E(iy) = F(y) + F'(y) > 0, \quad h(iy) = \frac{F(y)}{F(y) + F'(y)}.$$

Moreover

$$0 < yK(-y^2) = \frac{1}{\pi} \int_{\mathbb{R}} \frac{y\omega(x)}{x^2 + y^2} dx < 1.$$

Both inequalities are strict because $\omega > 0$ almost everywhere and $\omega < 1$ on a set of positive measure. Therefore

$$-\frac{F(y)}{y(F(y) + F'(y))} < r(-y^2) < \frac{F'(y)}{y(F(y) + F'(y))},$$

$$\boxed{|r(-y^2)| < \frac{1}{y}, \quad |\mathcal{R}(iy)| < 1.} \quad (\text{CP12:3.15})$$

In particular the actual correction has no imaginary-axis poles, its square transform has no negative-axis poles, and its transformed negative-ray boundary tends to zero. This is a closed auxiliary bound on the actual xi correction. It supplies no bound on the rest of an expanding contour; that additional boundary estimate is explicitly retained in Group 4.

For illustration of the exact pole transport, a pair of simple poles $a, -a$ of the odd correction has equal residue r_a , and

$$\frac{1}{z} \left(\frac{r_a}{z-a} + \frac{r_a}{z+a} \right) = \frac{2r_a}{w-a^2}. \quad (\text{CP12:3.16})$$

Thus both poles become one squared-coordinate pole with its full factor two. Higher principal parts are transported by the same even rational expression $(P_a(z) + P_{-a}(z))/z$; none are discarded. Conjugate poles remain conjugate in the new coordinate.

3.5. A complete boundary estimate and the metric under rescaling

This further calculation controls the full correction on the real boundary, with an explicit scale. For $a > 0$, set

$$d\nu_a(x) = \frac{a}{\pi(x^2 + a^2)} dx, \quad m_a = \int_{\mathbb{R}} \omega(x) d\nu_a(x).$$

This is the pullback of normalized circle measure under the Cayley map centered at ia . The real part of C is the mean-zero circle Hilbert transform of ω : the identity follows from the Poisson formula for $\text{Im } C$, while $\text{Re } C(ia) = 0$ follows from parity. Consequently, with H_a denoting that Hilbert transform,

$$\mathcal{R} = H_a \omega + \text{Im } h, \quad \|H_a \omega\|_{L^2(\nu_a)}^2 = \int \omega^2 d\nu_a - m_a^2,$$

$$\|\text{Im } h\|_{L^2(\nu_a)}^2 = m_a - \int \omega^2 d\nu_a.$$

The second identity uses $|h|^2 = \operatorname{Re} h = \omega$. It follows, with the mixed term retained and then estimated by Cauchy-Schwarz, that

$$\|\mathcal{R}\|_{L^2(\nu_a)}^2 \leq 2m_a(1 - m_a) \leq \frac{1}{2}. \quad (\text{CP12:3.17})$$

An equivalent exact Fourier statement is useful. Let u_a be the circle boundary function corresponding to h , and P_- its strictly negative Fourier projection. Parity gives $(u_a)_0 = m_a \in \mathbb{R}$, and comparison with the analytic Fourier series of C gives

$$\mathcal{R} \circ \text{Cayley}_a^{-1} = 2 \operatorname{Im} P_- u_a, \quad \|\mathcal{R}\|_{L^2(\nu_a)}^2 = 2 \sum_{n \geq 1} |(u_a)_{-n}|^2. \quad (\text{CP12:3.18})$$

Parseval yields $\sum_{n \neq 0} |(u_a)_n|^2 = m_a - m_a^2$, proving (3.17) again. Negative Fourier truncations converge in boundary L^2 , with the exact tail

$$\|\mathcal{R} - \mathcal{R}_M\|_{L^2(\nu_a)}^2 = 2 \sum_{n > M} |(u_a)_{-n}|^2 \longrightarrow 0.$$

This is boundary-norm convergence; transferring it to the original inverse-metric norm requires the additional bound discussed below.

The actual central estimates sharpen the bound. For $|x| \leq 1$,

$$1 - \omega(x) \leq \frac{A'(x)^2}{A(x)^2} \leq \frac{16x^2}{5929}.$$

On the complement use $1 - \omega \leq 1$. Integration gives

$$\begin{aligned} 1 - m_a &\leq \frac{2}{\pi} \left(\frac{16}{5929} \int_0^1 \frac{ax^2}{x^2 + a^2} dx + \int_1^\infty \frac{a}{x^2 + a^2} dx \right) \\ &\leq \frac{2a}{\pi} \frac{5945}{5929}, \end{aligned}$$

and hence

$$\|\mathcal{R}\|_{L^2(\nu_a)}^2 \leq \min \left\{ \frac{1}{2}, \frac{4a}{\pi} \frac{5945}{5929} \right\}. \quad (\text{CP12:3.19})$$

This complete boundary norm tends to zero when the Cayley center approaches the origin. The Hilbert-space metric of a Hankel representation must accompany this change. For the actual boundary symbol,

$$\|H_{u_a}\|_{\text{HS}}^2 = \sum_{n \geq 1} n |(u_a)_{-n}|^2 \quad (\text{CP12:3.20})$$

(with both sides allowed to be infinite), obtained by summing the squared Hankel entries in the circle monomial bases. Equation (3.18) is the unweighted Fourier sum; (3.20) retains the full multiplicity of each anti-diagonal.

More precisely, a disk automorphism ϕ induces a unitary weighted-composition operator U_ϕ on circle L^2 , with the analytic Hardy half-density factor. It maps H^2 onto itself, and therefore its orthogonal complement onto itself. Multiplication and the two projections satisfy

$$U_\phi M_u U_\phi^* = M_{u \circ \phi}, \quad H_{u \circ \phi} = (U_\phi)_- H_u (U_\phi)_+^*.$$

Changing the Cayley scale is such an automorphism. The Hankel operator norm and Hilbert-Schmidt norm are therefore invariant, including the possibility of an infinite Hilbert-Schmidt norm. Their invariance also follows from the conformal invariance of the weighted Fourier/Dirichlet sum in (3.20). The small complete boundary norm in (3.19) consequently supplies a genuine scalar boundary estimate while leaving the required metric-weighted sum to be controlled. Fourier mass may move to higher indices as the scale changes.

3.6. Extraction, return map, and the quantified stopping line

For a symmetric finite set, $Q_S(x) = \tilde{Q}_S(x^2)$ and $P(x) = p_e(x^2) + xp_o(x^2)$. The extraction is exactly

$$\mathcal{X}(P) = \frac{H_0}{2\pi\delta} \int_0^{T^2} (1 - s/T^2)^4 (|p_e(s)|^2 + s|p_o(s)|^2) \frac{ds}{\sqrt{s}}. \quad (\text{CP12:3.21})$$

The Suzuki polynomial metric is

$$\int_0^\infty \frac{|p_e(s)|^2 + s|p_o(s)|^2}{|\tilde{Q}_S(s)|^2} d\mu(s). \quad (\text{CP12:3.22})$$

The pointwise measure comparison from checkpoint 11 gives the same constant K_* in each channel separately. Symmetry has removed opposite-parity mixed terms exactly. It has not bounded the within-channel correction matrices (3.11).

For clarity, the reflection on the ungauged Suzuki image is twisted. If $\check{q}(t) = q(-t)$, then direct substitution in the complete synthesis gives

$$V\check{q}(x) = -\frac{h(x)}{h(-x)} Vq(-x) = -\Theta(x) Vq(-x), \quad \Theta = E^\sharp/E.$$

The multiplier has modulus one on the real axis and $\Theta(x)\Theta(-x) = 1$. Applying the gauge (3.6) changes this into $g(x) \mapsto -g(-x)$, so the exact Suzuki norm is preserved.

On compact tests, reflection $q(t) \mapsto q(-t)$ preserves both forms by substitution $\gamma \mapsto -\gamma$. Its isometry for $\mathcal{S}$ extends to a unitary involution on the Suzuki completion. The mixed defect vanishes on compact tests from its opposite parity eigenspaces, and on any later continuous extension of that form. No bounded global defect operator is assumed at this step. In the gauge (3.6), these are the rational parity channels above, with the sign interchange in (3.8).

A precise sufficient return lemma, stated entirely for these explicit kernels, is the following.

Square-channel complete norm lemma (open). On the two closures of the images of C_c^∞ under (3.6)-(3.8), the cutoff limits of the forms with kernels $-K_+$ and $-K_-$ extend to Hilbert-Schmidt operators D_+ , D_- , and there are nonnegative constants b_+ , b_- such that

$$\|D_+\|_{\text{HS}} \leq b_+, \quad \|D_-\|_{\text{HS}} \leq b_-, \quad b_+^2 + b_-^2 \leq 1. \quad (\text{CP12:3.23})$$

Here the kernels, their complete cutoff convention, their actual measures, and their exact domains have all been specified; a form on unrestricted finite zero arrays is not substituted for them. An equivalent check is that for every pair of finite Suzuki-orthonormal systems in these respective domains, the two sums of squared complete kernel entries have total at most one.

If (3.23) is proved, the unitary inverse produces $D = D_+ \oplus D_-$ on the original Suzuki completion and

$$\mathcal{E}_N = \|P_N D P_N\|_{\text{HS}}^2 \leq \|D\|_{\text{HS}}^2 = \|D_+\|_{\text{HS}}^2 + \|D_-\|_{\text{HS}}^2 \leq 1 \quad \text{for every } N. \quad (\text{CP12:3.24})$$

Equivalently, one may enlarge any original finite test span V_N to $V_N + \check{V}_N$. This is a finite parity-invariant span containing the original one, preserves any original negative witness, and the original normalized defect is the compression of the enlarged normalized form under the Suzuki isometric inclusion. Its Frobenius norm cannot exceed that of the enlarged form.

This holds for the original determining sequence even when its finite spans are not invariant under parity. In that case its individual finite compressions must not be silently split into a sum of parity energies; the global inequality in (3.24) is the valid return map.

The present derivation stops at estimating the within-channel complete correction in (3.23). The extraction gives (3.21)-(3.22) with constant K_* , while (3.15) gives actual negative-ray decay and (3.17)-(3.19) give the complete boundary-symbol norm. Neither calculation presently controls the full inverse-metric pole-boundary budgets in (3.2). This group therefore closes the unitary transport, exact parity cancellation, witness preservation and negative-ray bound; the complete norm ceiling remains the quantified missing estimate.

Group 3 closure record.

1. **Statement and domain:** equations (3.1)-(3.2) hold on every finite actual test span; the square-coordinate map is unitary on the two weighted spaces, with the compact-test images and their closures explicitly retained.
2. **Outcome:** both parity kernels, exact opposite-parity cancellation, witness preservation, boundary estimates, and the metric return are proved.
3. **Return to target:** the inverse unitary map and finite compression give (3.24) for the original determining sequence.
4. **Quantified missing lemma and stop:** equation (3.23), $\|D_+\|_{\text{HS}}^2 + \|D_-\|_{\text{HS}}^2 \leq 1$; the stop is the complete within-channel bound, including its residue and boundary contributions.

4. Closure group 4: Complete poles, Hardy operators, and infinity

4.1. Exact finite-contour correction, with every interaction

Let S be a finite conjugation-closed set of distinct zeros of A , let m_γ denote multiplicities, and use the standard matrix convention $G_{ij} = \langle F_{\gamma_j}, F_{\gamma_i} \rangle$. Put

$$(M_S f)_{ij} = \sqrt{m_i m_j} f[\gamma_j, \bar{\gamma}_i], \quad D_S = -G_S^{-1/2} M_S(\mathcal{R}) G_S^{-1/2}. \quad (\text{CP12:4.1})$$

This finite-feature D_S is distinguished from the determining-test matrix D_N ; subsection 4.1.1 gives the latter directly.

For a conjugation-symmetric contour domain Ω containing S and avoiding poles on its boundary, checkpoint 11 gives

$$\mathcal{R} = H_\Omega + \sum_{a \in \Omega} P_a, \quad P_a(z) = \sum_{k=1}^{d_a} r_{a,k} (z - a)^{-k}. \quad (\text{CP12:4.2})$$

The exact elementary identity is

$$P_a[z, u] = - \sum_{k=1}^{d_a} r_{a,k} \sum_{l=1}^k (z - a)^{-l} (u - a)^{-(k+1-l)}. \quad (\text{CP12:4.3})$$

Define

$$b_a^{(l)} = (\sqrt{m_i}(\bar{\gamma}_i - a)^{-l})_i, \quad u_a^{(l)} = G_S^{-1/2} b_a^{(l)}, \quad B_\Omega = -G_S^{-1/2} M_S(H_\Omega) G_S^{-1/2}.$$

Then the full relative defect is exactly

$$D_S = B_\Omega + \sum_{a,k,l} r_{a,k} u_a^{(k+1-l)} (u_a^{(l)})^*. \quad (\text{CP12:4.4})$$

Conjugate pole terms occur together, making the complete matrix Hermitian. No positivity of an individual complex residue is asserted.

Write each summand as $\alpha_n x_n y_n^*$. The complete norm budget is

$$\begin{aligned} \|D_S\|_F^2 &= \|B_\Omega\|_F^2 + 2\Re \sum_n \alpha_n y_n^* B_\Omega^* x_n \\ &\quad + \sum_{n,k} \alpha_n \bar{\alpha}_k (y_n^* y_k) (x_k^* x_n). \end{aligned} \quad (\text{CP12:4.5})$$

The second line includes all pole-pole crosses, including crosses between distinct poles and different orders. The first line includes every pole-boundary cross.

The boundary matrix itself retains its full contour:

$$B_\Omega = -\frac{1}{2\pi i} \int_{\partial\Omega} \mathcal{R}(\zeta) u_\zeta^{(1)} (u_\zeta^{(1)})^* d\zeta. \quad (\text{CP12:4.6})$$

Thus a completely explicit sufficient, though potentially wasteful, majorant is

$$\|D_S\|_F \leq \frac{1}{2\pi} \int_{\partial\Omega} |\mathcal{R}(\zeta)| \|u_\zeta^{(1)}\| \|u_\zeta^{(1)}\| |d\zeta| + \sum_{a,k,l} |r_{a,k}| \|u_a^{(k+1-l)}\| \|u_a^{(l)}\|. \quad (\text{CP12:4.7})$$

Formula (4.5), rather than (4.7), is the appropriate expression when cancellations matter.

At a simple noncommon zero a of $E = A + iA'$, let $m(z) = A'(z)/A(z)$. Since $m(a) = i$,

$$r_{a,1} = -iA(a)/E'(a) = -1/m'(a). \quad (\text{CP12:4.8})$$

This is an actual xi residue formula. The scalar theta bounds on the central interval do not presently give a global sign or size for the complex values $m'(a)$.

4.1.1. The complete compact-test version For the smooth symmetric zero cutoff from checkpoint 11, set

$$L_{q,H}(\zeta) = \sum_{\gamma} \frac{m_\gamma \chi(\Re \gamma / H) \hat{q}(\gamma)}{\gamma - \zeta}.$$

The sums are finite. Direct substitution in the divided-difference contour identity gives

$$\begin{aligned} \Delta_H(q, r) &= -\frac{1}{2\pi i} \int_{\partial\Omega} \mathcal{R}(\zeta) L_{q,H}(\zeta) \overline{L_{r,H}(\bar{\zeta})} d\zeta \\ &\quad + \sum_{a \in \Omega} \sum_{k=1}^{d_a} \frac{r_{a,k}}{(k-1)!} \frac{d^{k-1}}{d\zeta^{k-1}} \{L_{q,H}(\zeta) \overline{L_{r,H}(\bar{\zeta})}\} \Big|_{\zeta=a}. \end{aligned} \quad (\text{CP12:4.9})$$

Here Ω contains the cutoff zero set. Differentiation is complex differentiation of the analytic function displayed; conjugation of the reflected variable makes the second factor analytic. The product rule reproduces every mixed higher-pole term in (4.3).

The already proved L^2 synthesis convergence gives $\Delta_H(q, r) \rightarrow \Delta(q, r)$ for fixed compact smooth q, r . For each fixed N , applying (4.9) to (ρ_j, ρ_i) therefore gives convergence in the finite-dimensional Frobenius norm to the actual D_N . This is the justified topology. Uniformity in N does not follow from fixed-test convergence.

For precision, in the determining basis define

$$u_{a,H}^{(l)} = (\sum_{\gamma} m_{\gamma} \chi(\Re \gamma / H) \hat{\rho}_i(\gamma) / (\gamma - \bar{a})^l)_{i \leq N}.$$

Replacing u by these vectors in (4.4)-(4.6) gives the exact $D_{N,H}$, with no additional G inverse because ρ_i are $\mathcal{S}$ -orthonormal. Consequently (4.5) is the full original norm budget after the fixed- N H limit.

4.2. A closed actual Hardy theorem

Use the Cayley map

$$z(\zeta) = i \frac{1 + \zeta}{1 - \zeta}, \quad x(t) = -\cot(t/2), \quad dx = \frac{1 + x^2}{2} dt.$$

Let $u(e^{it}) = h(x(t))$, $\theta(e^{it}) = \Theta(x(t)) = 2u - 1$. The exact boundary identity gives

$$|\theta| = 1, \quad |u|^2 = \Re u. \quad (\text{CP12:4.10})$$

Let P_+ be the circle Hardy projection, $H_u = (I - P_+)M_uP_+$. Because the constant term has no Hankel part,

$$\boxed{H_u = \frac{1}{2}H_{\theta}, \quad \|H_u\| \leq \frac{1}{2}.} \quad (\text{CP12:4.11})$$

This is an unconditional operator estimate for the actual xi symbol on H^2 , not an assumed pole sum.

Writing $u = \sum u_n e^{int}$, Parseval and (4.10) give the stronger precise boundary energy identity

$$\sum_{n \neq 0} |u_n|^2 = \Re u_0 - |u_0|^2 = \frac{1}{4} - |u_0 - \frac{1}{2}|^2 \leq \frac{1}{4}. \quad (\text{CP12:4.12})$$

Evenness of A implies $h(-x) = \overline{h(x)}$, so $u(-t) = \overline{u(t)}$; hence all u_n are real, including u_0 . Checkpoint 11's complete correction formula is then

$$\mathcal{R}(x(t)) = i \sum_{n \geq 1} u_{-n} (e^{int} - e^{-int}) \quad \text{in } L^2(dt/2\pi). \quad (\text{CP12:4.13})$$

In particular,

$$\boxed{\|\mathcal{R} \circ x\|_{L^2(dt/2\pi)}^2 = 2 \sum_{n \geq 1} |u_{-n}|^2 \leq \frac{1}{2}.} \quad (\text{CP12:4.14})$$

The L^2 remainder after degree M has exact norm squared $2 \sum_{n > M} |u_{-n}|^2$, which tends to zero. This proves a genuine complete boundary tail statement without discarding infinity.

The Hilbert-Schmidt Hankel norm, when finite, is exactly

$$\boxed{\|H_u\|_{\text{HS}}^2 = \sum_{n \geq 1} n |u_{-n}|^2.} \quad (\text{CP12:4.15})$$

Indeed its matrix, in positive and negative Fourier bases, is $(u_{-(j+k+1)})_{j,k \geq 0}$; each coefficient u_{-n} appears exactly n times. The weighted tail criterion

$$\sum_{n>M} n|u_{-n}|^2 \longrightarrow 0 \quad (\text{CP12:4.16})$$

is precisely HS convergence of truncated Hankel symbols. Equations (4.12)-(4.14) prove the unweighted tail but do not establish (4.16) for the actual symbol.

The same boundary calculation admits a scale-sensitive actual-xi refinement. At the Cayley center ia with $a > 0$, use $d\nu_a = a dx / [\pi(x^2 + a^2)]$ and $m_a = \int w d\nu_a$. The constant of C 's normalization does not change: w is even, so $\text{Re } C(ia) = 0$, and all relevant odd real constants vanish. The mean-zero circle Hilbert transform of w is $\text{Re } C$, while $\mathcal{R} = \text{Re } C + \text{Im } h$ on the boundary. Hence

$$\|\mathcal{R}\|_{L^2(\nu_a)}^2 \leq 2m_a(1 - m_a) \leq \frac{1}{2}. \quad (\text{CP12:4.17})$$

Indeed the two squared norms before the triangle inequality are $\int w^2 - m_a^2$ and $m_a - \int w^2$, respectively. Checkpoint 11's pointwise bounds give $1 - w(x) \leq 16x^2/5929$ for $|x| \leq 1$. Integrating the central interval and using the bound 1 outside it yields

$$1 - m_a \leq \frac{2a}{\pi} \frac{5945}{5929}, \quad \boxed{\|\mathcal{R}\|_{L^2(\nu_a)}^2 \leq \frac{4a}{\pi} \frac{5945}{5929}}. \quad (\text{CP12:4.18})$$

Thus the entire real-axis correction has an explicitly small weighted boundary norm as a decreases. The changing probability weight concentrates at zero; this estimate does not identify it with the Suzuki norm or the HS Hankel weight.

4.3. Exact difference-of-projections representation and its mixed terms

On $L^2(\mathbb{R})$, let P denote the upper Hardy orthogonal projection, let $U = M_\Theta$ (unitary by (4.10)), and set

$$Q = UPU^*, \quad K = P - Q, \quad H = (I - P)UP. \quad (\text{CP12:4.19})$$

The same objects are unitarily transported to the circle by

$$(\mathcal{C}f)(\zeta) = \frac{2\sqrt{\pi}}{1-\zeta} f\left(i \frac{1+\zeta}{1-\zeta}\right). \quad (\text{CP12:4.20})$$

The Jacobian verifies its L^2 isometry; its analytic prefactor carries the Hardy subspaces correctly. Multiplication by Θ is transported into multiplication by θ , so H is the actual Hankel operator H_θ in this representation.

K is a selfadjoint contraction. Its distribution kernel off the diagonal is

$$K(x, y) = \frac{i}{2\pi} \frac{1 - \Theta(x)\overline{\Theta(y)}}{x - y}. \quad (\text{CP12:4.21})$$

The singular distribution parts cancel, and smoothness of the actual real-axis symbol supplies the continuous diagonal value on finite intervals.

Direct multiplication of the projections proves

$$\boxed{K - K^2 = -\{HU^* + UH^*\}}. \quad (\text{CP12:4.22})$$

Thus the indefinite correction is an explicit Hermitian pair of the negative Hardy component. Every mixed term survives. Spectral calculus for the contraction K also gives

$$-2I \preceq K - K^2 \preceq \frac{1}{4}I. \quad (\text{CP12:4.23})$$

If H is HS, its exact complete norm budget is

$$\boxed{\|K - K^2\|_{\text{HS}}^2 = 2\|H\|_{\text{HS}}^2 + 2\|H\|_{S_4}^4.} \quad (\text{CP12:4.24})$$

Proof: in the splitting P plus $I - P$, the lower-right block of Q is HH^* , and its off-diagonal block obeys $Q_{-+}Q_{+-} = HH^* - (HH^*)^2$, because Q is a projection. The blocks of $K - K^2$ are 0, minus the two Q off-diagonal blocks, and $-2HH^*$. Summing their squared norms yields $2\text{tr}(HH^*) + 2\text{tr}((HH^*)^2)$, including rather than estimating away the mixed terms. Conversely a finite left-hand HS norm forces H to be HS by the same nonnegative formula, interpreted through finite-rank trace approximations.

Equation (4.24) provides a concrete closed auxiliary theorem: the ambient complete projection correction is HS exactly when the negative Hardy part is HS, with the displayed exact budget. It is stronger than using the triangle inequality in (4.22).

4.3.1. Exact relation to the zero-feature correction and the metric For a real zero r of multiplicity m_r , $\Theta(r) = -1$ and $\Theta'(r) = -2i/m_r$. Consequently

$$K(\cdot, r) = F_r/\sqrt{\pi m_r}, \quad K(r, s) = \frac{\mathbf{1}_{r=s}}{\pi\sqrt{m_r m_s}}. \quad (\text{CP12:4.25})$$

Since these columns belong to L^2 , the kernel of K^2 at the real zero nodes is defined by their L^2 Gram integral. Therefore for a finite real-zero set,

$$\begin{aligned} \pi\sqrt{m_r m_s}K^2(r, s) &= (G_S)_{rs}, \\ \pi\sqrt{m_r m_s}(K - K^2)(r, s) &= (I - G_S)_{rs}. \end{aligned} \quad (\text{CP12:4.26})$$

The complete relative matrix is the sampling in (4.26) followed by congruence with $G_S^{-1/2}$, exactly as in (4.1). Thus (4.22) genuinely represents the actual real-zero correction rather than an unrelated boundary operator.

The corresponding full meromorphic node kernel has reflected pairing J , and checkpoint 11's $\mathcal{R}$ identity gives its rigorous finite and cutoff formulas. Analytic continuation to nonreal nodes is not a bounded point-evaluation map on ambient L^2 ; it crosses possible poles. Formula (4.26) must therefore not be extended into an unconditional bounded sampling assertion. Even real-node delta evaluation requires the explicit Gram normalization in (4.26). The actual estimate needed to transfer (4.23) or (4.24) is an estimate for that sampling/continuation map in the Suzuki metric. Formula (4.4) identifies the corresponding pole evaluation vectors explicitly.

Also, $K \geq 0$ holds exactly when $Q \leq P$, or equivalently when multiplication by Θ maps upper H^2 into itself. In that case K is a projection and (4.22) vanishes. Boundary unimodularity alone gave the contraction; the analytic Hardy inclusion is the additional exact structural condition. No such inclusion is assumed here.

4.4. Infinity and its exact boundary terms

The Cauchy transform C itself has a completely controlled tail. In disk coordinates it is

$$C(z(\zeta)) = \frac{i}{2\pi} \int_0^{2\pi} w(x(t)) \frac{e^{it} + \zeta}{e^{it} - \zeta} dt. \quad (\text{CP12:4.27})$$

The arc corresponding to $|x| > L$ has length $\ell_L = 4 \arctan(1/L)$. Since $0 \leq w \leq 1$, its contribution C_{tail} satisfies, for $|\zeta| \leq r < 1$,

$$|C_{\text{tail}}| \leq \frac{\ell_L}{2\pi} \frac{1+r}{1-r}, \quad |\partial_\zeta^k C_{\text{tail}}| \leq \frac{\ell_L}{\pi} \frac{k!}{(1-r)^{k+1}} \quad (k \geq 1). \quad (\text{CP12:4.28})$$

This proves $O(1/L)$ uniform convergence with every derivative on compact disk sets. There is no hidden Herglotz mass atom at infinity: the measure in (4.27) is absolutely continuous with density between zero and one. The statement concerns C ; the complete $\mathcal{R} = C - ih$ still retains the meromorphic h contribution.

For the contour boundary term in (4.2), let s be the Cayley coordinate of z and t that of the integration variable. An exact change of variables gives

$$\frac{dz(t)}{z(t) - z(s)} = \left(\frac{1}{t-s} + \frac{1}{1-t} \right) dt. \quad (\text{CP12:4.29})$$

Thus the second, infinity-associated term is an s -independent constant. It cancels in divided differences. The remaining complete boundary correction is

$$H_\Omega[z(s), z(v)] = \frac{(1-s)(1-v)}{2i} \frac{1}{2\pi i} \int_{\partial\tilde{\Omega}} \frac{\mathcal{R}(z(t))}{(t-s)(t-v)} dt. \quad (\text{CP12:4.30})$$

The complete transformed contour remains in this formula. Its arc near $t = 1$ may not be omitted merely because the explicit constant canceled. On the real-line boundary, the proved L^2 statement (4.14) bounds pairing with any L^2 contour test and implies smallness on shrinking arcs for that boundary pairing. Replacing interior meromorphic contour limits by those boundary pairings additionally requires uniform integrability while poles approach $t = 1$; no such unproved passage is made.

Group 3 supplies a further actual- ξ infinity-ray bound: with $F(y) = A(iy) > 0$ and $F'(y) > 0$, $E(iy) = F + F' > 0$, and $|\mathcal{R}(iy)| < 1$. This removes poles on the imaginary axis and controls this ray. An entire expanding contour and the normalized evaluation vectors in (4.6) require their own estimates; a bound on this single ray does not bound those expressions.

4.5. A precise sufficient missing lemma and the stopping line

The following is a fully quantified sufficient contour closure lemma, stated directly on the unchanged determining spaces. It avoids assuming a coordinate-free defect bound and displays the actual functions that must be estimated.

For each finite N and cutoff H choose any conjugation-symmetric pole-free-boundary contour $\Omega_{N,H}$ containing the cutoff zeros, and form $B_{N,H,\Omega}$ and $u_{a,H}^{(l)}$ exactly as in (4.4)-(4.9). If

$$\sup_{N \geq 1} \limsup_{H \rightarrow \infty} \left\{ \frac{1}{2\pi} \int_{\partial\Omega_{N,H}} |\mathcal{R}(\zeta)| \|u_{\zeta,H}^{(1)}\| \|u_{\bar{\zeta},H}^{(1)}\| |d\zeta| + \sum_{a,k,l} |r_{a,k}| \|u_{a,H}^{(k+1-l)}\| \|u_{\bar{a},H}^{(l)}\| \right\} \leq 1, \quad (\text{CP12:4.31})$$

then every actual $\|D_N\|_F \leq 1$, by (4.7) and the fixed-N limit justified after (4.9). Hence the unchanged liminf target follows. An improved lemma may replace the triangle budget in braces by the square root of the exact cross budget (4.5), but then cancellations must be proved using the actual residues and boundary matrix.

The derivation currently stops before (4.31): checkpoint 11's extraction bound controls multiplication on the central interval, whereas the explicit quantities here are Suzuki-normalized Cauchy evaluation vectors at E poles and on the contour. Their size is not bounded by the known constant K_* in the recovered equations. The actual boundary identities prove (4.11), (4.14), (4.22), (4.24), and (4.28), but do not yet supply the uniform residue-evaluation budget (4.31).

The stronger intermediate Hardy possibility is now also precise: prove $\sum_{n \geq 1} n|u_n|^2$ finite with an explicit upper bound, and a bounded continuation/sampling transfer for the full node family in the actual Suzuki metric. If that transfer were an operator of norm C taking the ambient correction of (4.24) to the full relative defect, the sufficient budget would be

$$C^4 (8\|H_u\|_{HS}^2 + 32\|H_u\|_{S_4}^4) \leq 1. \quad (\text{CP12:4.32})$$

Both the domain and the transfer identity would have to be proved; (4.32) is not asserted as an established comparison.

Group outcome. Closed auxiliary theorems: complete contour factorization with every cross term; actual xi negative-Hardy contraction and L^2 correction cap; exact difference-of-projections correction with complete HS budget; explicit compact-set Cauchy tail and exact Cayley infinity term. The original defect target remains open at the quantified normalized residue-evaluation estimate (4.31). The next useful calculation is to estimate those explicitly displayed Cauchy evaluation vectors, or to use the exact cross budget to cancel them before estimating their sizes.

4.6. Direct comparison with the combined first-null target

The combined Groups 1-2 argument asks, on a specified first-null form-core sequence q_n , for

$$\liminf_n \frac{\mathcal{T}(q_n)}{\mathcal{S}(q_n)} \geq -\frac{506327}{509894}. \quad (\text{CP12:4.33})$$

Substitution of (4.9) writes that exact remainder as its extraction contribution plus the complete contour-and-residue expression. Equations (4.11), (4.18), and (4.23) then control the boundary symbol or ambient Hardy operator. The next necessary inequality would bound the Cauchy factors $L_{q_n, H}$ on the contour and at the enclosed E poles by the same $\mathcal{S}(q_n)$, uniformly in cutoff and along this first-null sequence. That inequality is precisely where this attempted proof stops: the proven extraction comparison controls a central numerator integral; it supplies no uniform bound for these meromorphic evaluation factors in the equations currently available. Taking absolute values after the exact null-vector cancellation must therefore await that remaining metric comparison. No numerical θ can be derived from the presently proved Group-4 constants alone.

Group 4 closure record.

1. **Statement and domain:** the finite contour identities hold on every specified finite feature or determining-test span; the Hardy identities act on ordinary Hardy spaces with the given unitary transport.
2. **Outcome:** all pole and boundary cross terms, the actual Hardy contraction, the exact ambient Hilbert-Schmidt budget, and the stated compact-set and boundary-tail limits are proved.

3. **Return to target:** the uniform determining-metric contour budget (4.31), followed by the proved fixed- N cutoff limit, gives $\mathcal{E}_N \leq 1$ for every N .
4. **Quantified missing lemma and stop:** equation (4.31); its normalized evaluation vectors remain unbounded by the established estimates. Section 5 tests these factors against the narrower first-null lemma.

5. Combining the groups and attempting the narrower estimate

5.1. The complete norm calculation from Groups 3-4

The tensor identity (3.1) and the pole-boundary expansion (4.5) are the same norm calculation in two representations. The square-coordinate map splits it into the two complete parity forms (3.11), preserving their measures and every within-channel cross term. The original determining matrix is obtained by inverse unitary transport and finite compression, even if its original span is not parity invariant.

The actual boundary symbol additionally satisfies

$$\|H_h\| \leq \frac{1}{2}, \quad \|\mathcal{R}\|_{L^2(\nu_\eta)}^2 \leq \min \left\{ \frac{1}{2}, \frac{4\eta}{\pi} \frac{5945}{5929} \right\}, \quad \nu_\eta(dx) = \frac{\eta dx}{\pi(x^2 + \eta^2)}. \quad (\text{CP12:5.1})$$

The exact ambient Hardy correction has norm budget (4.24). The transport back to the original target still requires the Cauchy evaluation vectors and their metric from (4.4), or the corresponding complete signed contour expression (4.9). Those factors remain inside the calculation.

One can calculate precisely what happens when the small boundary norm in (5.1) is used in a Cauchy evaluation. Let

$$u_\eta(e^{it}) = h(-\eta \cot(t/2)), \quad F_\eta(\zeta) = \sum_{n \geq 1} \overline{(u_\eta)_{-n}} \zeta^n.$$

Here the first symbol is u_η , a boundary function; it is distinct from the probability measure ν_η . Parseval and the complete boundary estimate give

$$\|F_\eta\|_{H^2}^2 \leq \frac{2\eta}{\pi} \frac{5945}{5929}. \quad (\text{CP12:5.2})$$

For a point $\gamma = \alpha + i\beta$, $\beta > 0$, write

$$\zeta_\gamma = \frac{\gamma - i\eta}{\gamma + i\eta}, \quad 1 - |\zeta_\gamma|^2 = \frac{4\eta\beta}{|\gamma + i\eta|^2}.$$

Cauchy-Schwarz on the Taylor coefficients gives

$$|F_\eta(\zeta_\gamma)|^2 \leq \|F_\eta\|_{H^2}^2 \frac{|\zeta_\gamma|^2}{1 - |\zeta_\gamma|^2} \leq \boxed{\frac{5945}{2\pi \cdot 5929} \frac{|\gamma - i\eta|^2}{\beta}}. \quad (\text{CP12:5.3})$$

This is an actual-symbol calculation. The factor η that made the full boundary norm small has canceled against the evaluation metric. The resulting upper bound has no uniform control in the height or in $\beta \downarrow 0$. At real nodes, the ordinary H^2 point-evaluation norm is unbounded. The actual zero-feature removals remain valid, but require their specific coupled identities. This calculation proves no divergence of the actual defect; it identifies exactly where this attempted absorption estimate loses its small constant.

Cayley rescaling gives another exact check: its weighted composition operator intertwines the Hardy projections and multiplication, so the Hankel operator and Hilbert-Schmidt norms are invariant under the rescaling. Thus the small unweighted boundary norm cannot, through this return map, become a shrinking Hilbert-Schmidt budget. The weighted Fourier tail and the complete normalized evaluation factors in Group 4 remain the estimates to establish.

5.2. A narrower first-null calculation from Groups 1-2

The full norm ceiling is stronger than the one-sided inequality needed for Weil positivity. We therefore select the first-null route as the narrower remaining calculation.

Fix the extraction once, with any previously admissible amplitude parameters and $T = 1$. Set

$$r_* := 1 - K_* = \frac{506327}{509894}.$$

Suppose that a first degenerate interval I_{a_*} exists. Take $0 \neq v \in \ker A_{a_*}$, and any form-core sequence

$$q_n \in C_c^\infty(I_{a_*}), \quad q_n \rightarrow v \quad \text{in the closed Weil form norm.}$$

Since $Q_{a_*} \geq 0$, this implies $q_n \rightarrow v$ in L^2 and $Q_{a_*}(q_n, q_n) \rightarrow 0$. Put

$$s_n = \mathcal{S}(q_n, q_n), \quad x_n = \mathcal{X}_\infty(q_n, q_n), \quad t_n = \mathcal{T}(q_n, q_n).$$

The closed, injective map proved in Group 2 yields $\liminf_n s_n > 0$. All denominators below are therefore positive for large n . The null equation and the complete mixed-term identity give the exact ratio

$$\frac{t_n}{s_n} = \frac{Q_{a_*}(q_n, q_n)}{s_n} - 1 + \frac{x_n}{s_n}. \quad (\text{CP12:5.4})$$

The first ratio tends to zero, and $x_n/s_n \leq K_0$. Consequently

$$\limsup_{n \rightarrow \infty} \frac{t_n}{s_n} \leq -(1 - K_0) = -r_* - \varepsilon_*. \quad (\text{CP12:5.5})$$

This is a complete calculation for any hypothetical first-null core sequence. It includes the original Suzuki metric and does not require a finite Suzuki norm or endpoint trace for v .

The narrower missing statement is now a single precise lemma.

First-null remainder lemma – open. Fix the extraction described above. For every finite $a > 0$ for which the closed localized Weil form Q_a is nonnegative, and every $0 \neq v \in \ker A_a$, there exists a sequence $q_n \in C_c^\infty(I_a)$ converging to v in the $Q_a + \|\cdot\|_2^2$ form norm such that

$$\liminf_{n \rightarrow \infty} \frac{-\lim_{H \rightarrow \infty} \mathcal{K}_H(q_n, q_n) + \mathcal{X}_\infty(q_n, q_n)}{\mathcal{S}(q_n, q_n)} \geq -\frac{506327}{509894}. \quad (\text{CP12:5.6})$$

The finite-cutoff form $\mathcal{K}_H$ is explicitly (1.6), with the same zero cutoffs, meromorphic correction, and all ordered cross terms. Its inner limit is the fixed-test limit already proved in checkpoint 11. The outer limit is along the stated form-core sequence. No interchange of these limits is assumed.

This lemma is narrower than the global remainder bound (1.20): it asks for one approximating sequence at each hypothetical null vector, and only its one-sided limiting ratio. The positive margin ε_* makes (5.6) incompatible with (5.5).

5.3. The attempted proof of the selected lemma and its exact stopping line

Substituting the full contour identity (4.9) into (5.6) expresses its numerator as the joint cutoff limit of the contour integral and every pole term, plus the controlled extraction. The null equation has already simplified the form's mixed terms in (2.14) and produced (5.4). Parity eliminates only the opposite-parity blocks, as proved in Group 3; all terms within each parity block persist.

The actual Hardy estimate (5.1), the exact Hilbert-Schmidt identity (4.24), and the explicit infinity tail (4.28) were then tested against these remaining factors. The calculation stops when their scalar contour weights must be paired with

$$\frac{L_{q_n,H}(\zeta) \overline{L_{q_n,H}(\bar{\zeta})}}{s_n} \quad \text{and its derivatives at the poles of } \mathcal{R}. \quad (\text{CP12:5.7})$$

Neither the recovered extraction comparison nor the derived Hardy identities bounds these quantities uniformly in the cutoff and the null sequence, with the signed cancellation required by (5.6). Estimate (5.3) displays the loss in the proposed small-boundary-norm route. The boundary contour remains in (4.30); the value $r_{a,1} = -1/(A'/A)'(a)$ leaves actual complex residues rather than positive prime norms. An independent bound for these normalized coupled evaluations is the exact line still unproved.

This is a completed analytical attempt at the selected lemma. The retained expressions have been calculated; their needed sign has not been obtained. No generic matrix counterexample is used to pronounce the actual xi estimate impossible.

5.4. Complete closing implication, conditional on the selected lemma

If (5.6) is established independently, (5.5) rules out every first-null vector. The continuity and small-interval positivity in Group 2 then rule out a negative localized Weil form. Every compact smooth test lies in one finite interval, so $\mathfrak{W}(q, q) \geq 0$ for all $q \in C_c^\infty(\mathbb{R})$. Weil's criterion gives RH. Only at that stage is Suzuki's correctly normalized identity invoked:

$$\mathfrak{W}(q, r) = \pi^{-1} \langle \widehat{\mathcal{P}_{Dq}}, \widehat{\mathcal{P}_{Dr}} \rangle = \mathcal{S}(q, r).$$

Thus $\Delta = 0$, and the original target would hold in the stronger form

$$\mathcal{E}_N = 0 \quad \text{for every original determining } N. \quad (\text{CP12:5.8})$$

At present the antecedent (5.6) remains open. The strongest justified closing statement for this round is the collection of auxiliary theorems above together with the exact first-null separation (5.5). The next round has one quantified inequality, one specified family of approximants, and the complete normalized contour factors (5.7) to estimate.

6. Sources, checks, and preserved scope

- Checkpoint 11, *The Completed Weil-Suzuki Correction and a Uniform Extraction Comparison*: actual features, full meromorphic correction, fixed-test cutoff convergence, and the extraction comparison.
- Masatoshi Suzuki, *Weil's quadratic form via the screw function*, arXiv:2606.09096v2, §§8.2-8.5 and the earlier operator/domain results specified in Group 2. The RH-dependent factorization in §7.7 is not used.

- Masatoshi Suzuki, *On the Hilbert space derived from the Weil distribution*, arXiv:2301.00421v3: completed feature formulas and the correctly normalized identity. The existing manuscript's corrected two-sided convention is retained.
- J. B. Conrey, *More than two fifths of the zeros of the Riemann zeta function are on the critical line*, J. reine angew. Math. 399 (1989), 1-26: only the unconditional positive proportion of distinct simple critical-line zeros is used in the fixed-support injectivity argument.

The null-vector domain lemma and the main Hardy identities received independent analytical review. Formula signs, matrix entry conventions, endpoint contributions, parity return maps, and limit topologies were checked. This continuation is a research proof note with accurately stated open lemmas; it does not claim an unconditional RH proof. Earlier paper and note contents remain preserved in checkpoint 12.

We are aiming to establish $\liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1$ independently of RH for the unchanged completed Suzuki-normalized defect.

Checkpoint 13: First-null arithmetic and endpoint cancellation

Source: *Michaels_First_Null_Arithmetic_Closure_Checkpoint_13.md*. Equation tags carry the source prefix *CP13*; references within this unit retain their original local numbering.

Christophe Michaels research continuation · 23 September 2026 · Checkpoint 13

0. Preserved target and completed inputs

This continuation starts from checkpoint 12 and preserves

$$\boxed{\mathcal{E}_N = \|[\Delta(\rho_j, \rho_i)]_{i,j \leq N}\|_F^2, \quad \Delta = \mathfrak{W} - \mathcal{S}, \quad \liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1.} \quad (\text{CP13:0.1})$$

The ρ_j are the original Suzuki-orthonormal determining tests. The positive metric is $\mathcal{S}(q, r) = \langle Vq, Vr \rangle$, $V = -L_0$, with the corrected two-sided convention retained from checkpoints 11-12. Fourier transforms are $\hat{q}(z) = \int q(t)e^{izt}dt$, and inner products are linear in their first entry.

The already completed first-null domain passage is retained: if $q_n \rightarrow v \neq 0$ on fixed support in L^2 , then $\liminf_n \mathcal{S}(q_n, q_n) > 0$. No finite global Suzuki norm is assigned to a general null vector. The exact correction, all poles, all mixed terms, and the original order of limits are retained:

$$\Delta = -\mathcal{X}_\infty + \mathcal{T}, \quad 0 \leq \mathcal{X}_\infty(q, q) \leq K_0 \mathcal{S}(q, q),$$

$$K_0 = \frac{17607901}{2531730432} < K_* = \frac{3567}{509894}, \quad \varepsilon_* = K_* - K_0 = \frac{4429025}{108864408576} > 0. \quad (\text{CP13:0.2})$$

The new closed results are: the exact arithmetic multiplier matrix on the full form domain; signed endpoint substitutions; absolute compact- L^2 Cauchy synthesis; a null-specific translation identity; endpoint-complete exterior transforms; a quantitative near-null error bound from form Cauchy-Schwarz; exact cancellation of the translation factors in the complete quadratic pairing; and stronger logarithmic regularity for an exact null vector. The complete signed inequality needed to exclude a first null remains unproved. The round ends with its attempted calculation, quantified stopping line, and conditional return to the unchanged target.

Notation warning: the archimedean density in Group 1 is denoted h_{ar} ; the meromorphic Suzuki factor is $h = A/(A + iA')$. The function called J in Group 2 is an archimedean primitive, not an earlier spatial matrix kernel. Since $\xi(s) = \xi(1-s)$, the two occurrences $A(z) = \xi(1/2 \pm iz)$ denote the same even entire function.

1. Closure group 1 – Null multipliers and the complete arithmetic form

1.1. Exact kernel and convention

Let $I_a = (-a, a)$, with zero extension to the line, and let Q_a be the closed localized Weil form, A_a its associated operator. Assume hypothetically

$$Q_a \geq 0, \quad 0 \neq v \in \ker A_a. \quad (\text{CP13:1.1})$$

The Fourier convention is $\hat{v}(z) = \int v(x)e^{izx}dx$.

Source: Suzuki, *Weil's quadratic form via the screw function*, arXiv:2606.09096v2, <https://arxiv.org/html/2606.09096v2>, equations (1.3), (2.3), (2.5), (2.9) and the form-core result. All calculations below retain the kernel on the whole fixed interval.

Put $b_n = \Lambda(n)/\sqrt{n}$, $\ell_n = \log n$, $d = \psi(1/4) - \log \pi$. For $t \geq 0$,

$$\begin{aligned} g(t) = & -8(\cosh(t/2) - 1) + \sum_{\ell_n \leq t} b_n(t - \ell_n) - \frac{d}{2}t \\ & - \frac{1}{4}\{\Phi(1, 2, 1/4) - e^{-t/2}\Phi(e^{-2t}, 2, 1/4)\}, \end{aligned} \quad (\text{CP13:1.2})$$

with even continuation. Define

$$h_{\text{ar}}(t) = \frac{e^{-t/2}}{1 - e^{-2t}}, \quad k(t) = h_{\text{ar}}(t) - 2 \cosh(t/2) \quad (t > 0). \quad (\text{CP13:1.3})$$

The term $-2 \cosh(t/2)$ comes from the completed polar factor.

With $A_0 = (\log(2\pi) + \gamma_E - 1)/2$, the exact distribution is

$$g'' = \frac{1}{2} \text{Pf} |t|^{-1} + (2A_0 + 1)\delta_0 + \sum_{n \geq 2} b_n(\delta_{\ell_n} + \delta_{-\ell_n}) + r''(t), \quad r''(t) = k(|t|) - \frac{1}{2|t|} \quad (t \neq 0). \quad (\text{CP13:1.4})$$

The function r'' extends continuously through zero with value $-7/4$, and is bounded on $[-2a, 2a]$. This defines the exact remainder globally on that interval, rather than substituting a local expansion. The finite part convention is

$$\langle \text{Pf} |t|^{-1}, F \rangle = \lim_{\epsilon \downarrow 0} \left\{ \int_{|t| > \epsilon} \frac{F(t)}{|t|} dt + 2 \log \epsilon F(0) \right\}.$$

On compact smooth tests $Q_a(u, w) = \langle -g'' * u, w \rangle$.

1.2. Real parity and smooth multiplication on the closed domain

Conjugation and reflection preserve the core and its form norm because the kernel is real and even. They preserve the closed domain and commute with A_a . Taking a nonzero real or imaginary part, then a nonzero even or odd part, permits the choice

$$v \text{ real}, \quad v(-x) = \varepsilon v(x), \quad \varepsilon \in \{1, -1\}. \quad (\text{CP13:1.5})$$

This does not assert simplicity or a pointwise sign.

The exact form on its core is

$$\begin{aligned} Q_a(u) = & \frac{1}{4} \iint_{I_a^2} \frac{|u(x) - u(y)|^2}{|x - y|} dx dy - \frac{1}{2} \int_{I_a} \log(a^2 - x^2) |u(x)|^2 dx \\ & - (\log(2\pi) + \gamma_E) \|u\|_2^2 - 2 \sum_{\ell_n < 2a} b_n \Re \int u(x + \ell_n) \overline{u(x)} dx \\ & - \iint_{I_a^2} r''(x - y) u(y) \overline{u(x)} dx dy. \end{aligned} \quad (\text{CP13:1.6})$$

The last two terms are bounded on $L^2(I_a)$. After adding a sufficiently large L^2 multiple, the norm is equivalent to the logarithmic difference seminorm, the L^2 norm, and a nonnegative endpoint logarithmic weight. This description applies to the form-core completion.

For smooth ϕ on a neighborhood of $[-a, a]$, set $M = \sup |\phi|$, $L = \sup |\phi'|$. Then

$$|\phi(x)u(x) - \phi(y)u(y)|^2 \leq 2M^2|u(x) - u(y)|^2 + 2L^2|x - y|^2|u(y)|^2.$$

Since $\sup_y \int_{I_a} |x - y| dx \leq 2a^2$, the second term contributes at most $a^2 L^2 \|u\|_2^2$ to the seminorm with coefficient $1/4$. The endpoint weight is multiplied by at most M^2 . Therefore smooth multiplication is bounded in the closed form norm. It preserves core approximation, so

$$\phi v, \phi^2 v \in \mathfrak{D}(Q_a), \quad Q_a(v, \phi^2 v) = 0. \quad (\text{CP13:1.7})$$

The same argument covers bounded Lipschitz multipliers.

1.3. Double commutator: sign, regularization and passage from the core

For real core q and real smooth ϕ , symmetry gives

$$Q_a(\phi q) - Q_a(q, \phi^2 q) = \frac{1}{2} \iint g''(x - y)(\phi(x) - \phi(y))^2 q(x)q(y) dx dy. \quad (\text{CP13:1.8})$$

The sign is plus in front of g'' , since the form kernel is $-g''$.

The origin delta term vanishes exactly. The finite-part subtraction also vanishes, while its remaining integral is absolutely convergent: the square difference is at most $L^2|x - y|^2$, reducing $|x - y|^{-1}$ to an integrable $O(|x - y|)$ factor. The remaining density is the ordinary $h_{\text{ar}}(t)$; the nonzero prime deltas and completed polar term survive.

For complex multipliers define

$$J_{\phi, \psi}(t; v) = \int [\phi(y + t) - \phi(y)][\overline{\psi(y + t) - \psi(y)}] v(y + t)v(y) dy.$$

The complete polarized identity is

$$Q_a(\phi v, \psi v) = \int_0^{2a} k(t) J_{\phi, \psi}(t; v) dt + \sum_{\ell_n < 2a} b_n J_{\phi, \psi}(\ell_n; v). \quad (\text{CP13:1.9})$$

For a direct core-limit bound set

$$\mathfrak{c}_a = \int_0^{2a} |k(t)| t^2 dt + \sum_{\ell_n < 2a} b_n \ell_n^2 < \infty.$$

The absolute value of the right side is at most

$$\mathfrak{c}_a \text{Lip}(\phi) \text{Lip}(\psi) \|v\|_2^2. \quad (\text{CP13:1.10})$$

The bilinear expression in the two copies of v is consequently continuous under L^2 convergence. On the left, bounded multiplication gives form convergence; the null term tends to zero. This proves (1.9) on the full closed domain without derivative or endpoint traces of v .

1.4. Linear multiplier: full prime-shift weights

Let $C_v(t) = \int v(x+t)v(x)dx$. It is real, even, continuous, supported in $[-2a, 2a]$, with $C_v(0) = \|v\|_2^2$. Its nonzero-shift values are signed. Formula (1.9) gives

$$\boxed{0 \leq Q_a(xv) = \int_0^{2a} \frac{t^2 e^{-t/2}}{1 - e^{-2t}} C_v(t) dt - 2 \int_0^{2a} t^2 \cosh(t/2) C_v(t) dt + \sum_{n < e^{2a}} \frac{\Lambda(n)}{\sqrt{n}} (\log n)^2 C_v(\log n).} \quad (\text{CP13:1.11})$$

The prime sum is exactly

$$\sum_{p^r < e^{2a}} r^2 (\log p)^3 p^{-r/2} C_v(r \log p).$$

When $n = e^{2a}$, its correlation is zero, so inclusion is immaterial.

An exact origin split valid on the whole interval is

$$Q_a(xv) = \frac{1}{2} \int_0^{2a} t C_v(t) dt + \int_0^{2a} t^2 \left[k(t) - \frac{1}{2t} \right] C_v(t) dt + \sum_{\ell_n < 2a} b_n \ell_n^2 C_v(\ell_n). \quad (\text{CP13:1.12})$$

The middle integral is the complete remainder and has not been given the sign of its value at zero.

For the polar term let $F_j = \int x^j e^{x/2} v(x) dx$. Parity gives $\int x^j e^{-x/2} v(x) dx = \varepsilon (-1)^j F_j$. The polar contribution to (1.11) is exactly

$$\boxed{-2\varepsilon(F_0 F_2 + F_1^2).} \quad (\text{CP13:1.13})$$

No sign for $F_0 F_2$ follows from parity.

1.5. Paired trigonometric multipliers with exact remainder

Since the two squared differences sum to $2[1 - \cos(\tau(x - y))]$,

$$\begin{aligned} 0 \leq \mathfrak{F}_v(\tau) &:= Q_a(\cos(\tau \cdot) v) + Q_a(\sin(\tau \cdot) v) \\ &= \int_{-2a}^{2a} g''(t) [1 - \cos(\tau t)] C_v(t) dt \\ &= 2 \int_0^{2a} h_{\text{ar}}(t) [1 - \cos(\tau t)] C_v(t) dt - 4 \int_0^{2a} \cosh(t/2) [1 - \cos(\tau t)] C_v(t) dt \\ &\quad + 2 \sum_{\ell_n < 2a} b_n [1 - \cos(\tau \ell_n)] C_v(\ell_n). \end{aligned} \quad (\text{CP13:1.14})$$

The distributional notation equals the absolutely convergent expanded expression.

For $R_\tau(t) = 1 - \cos(\tau t) - \tau^2 t^2 / 2$, one has $-\tau^4 t^4 / 24 \leq R_\tau(t) \leq 0$ for all real t, τ . The exact expansion is

$$\mathfrak{F}_v(\tau) = \tau^2 Q_a(xv) + 2 \int_0^{2a} k(t) R_\tau(t) C_v(t) dt + 2 \sum_{\ell_n < 2a} b_n R_\tau(\ell_n) C_v(\ell_n), \quad (\text{CP13:1.15})$$

with bound

$$|\mathfrak{F}_v(\tau) - \tau^2 Q_a(xv)| \leq \frac{\tau^4}{12} \|v\|_2^2 \left[\int_0^{2a} |k(t)| t^4 dt + \sum_{\ell_n < 2a} b_n \ell_n^4 \right]. \quad (\text{CP13:1.16})$$

The sign of R_τ does not determine the signed correction.

There is also a closed analytical large-frequency calculation:

$$\boxed{\lim_{|\tau| \rightarrow \infty} \mathfrak{F}_v(\tau) / \log |\tau| = \|v\|_2^2.} \quad (\text{CP13:1.17})$$

This holds for every real v in the fixed-interval form domain. Indeed $\mathfrak{F}_v(\tau) = Q_a(e^{i\tau} v)$. The exact archimedean Fourier multiplier is $m(\xi) = \Re \psi(1/4 + i\xi/2) - \log \pi$. It is bounded below, satisfies $m(\xi) = \log |\xi| - \log(2\pi) + o(1)$, and is bounded in modulus by $C + \log(2 + |\xi|)$. The domain has finite integral of $\log(2 + |\xi|) |\widehat{v}(\xi)|^2$. After translating ξ , dominated convergence gives (1.17). The finite prime-shift contribution and rank-two polar contribution remain bounded. Thus this family's high-frequency limit has a positive leading term; a negative direction detected by it would occur at finite frequency.

1.6. Complete multiplier matrix and polynomial determinant

For a finite family ϕ_j , put $M_{jk} = Q_a(\phi_j v, \phi_k v)$. Every cross term is retained in

$$\begin{aligned} \sum_{j,k} c_j \overline{c_k} M_{jk} &= \int_0^{2a} k(t) \int \left| \sum_j c_j [\phi_j(y+t) - \phi_j(y)] \right|^2 v(y+t) v(y) dy dt \\ &\quad + \sum_{\ell_n < 2a} b_n \int \left| \sum_j c_j [\phi_j(y+\ell_n) - \phi_j(y)] \right|^2 v(y+\ell_n) v(y) dy \geq 0. \end{aligned} \quad (\text{CP13:1.18})$$

Nonnegativity here follows from the hypothetical nonnegative form. The displayed arithmetic expression is the new representation to be estimated independently; its signed product is not replaced by absolute values.

For explicit low multipliers define midpoint correlations

$$C_j(t) = \int (y + t/2)^j v(y+t) v(y) dy, \quad H_{r,s} = \int_0^{2a} k(t) t^r C_s(t) dt + \sum_{\ell_n < 2a} b_n \ell_n^r C_s(\ell_n). \quad (\text{CP13:1.19})$$

All uses below have $r \geq 2$. Definite parity makes $C_{2j+1} = 0$. The exact matrix for (x, x^2, x^3) is

$$\boxed{M(v) = \begin{pmatrix} H_{2,0} & 0 & 3H_{2,2} + H_{4,0}/4 \\ 0 & 4H_{2,2} & 0 \\ 3H_{2,2} + H_{4,0}/4 & 0 & 9H_{2,4} + 3H_{4,2}/2 + H_{6,0}/16 \end{pmatrix}.} \quad (\text{CP13:1.20})$$

Proof: with $r_0 = (x+y)/2$, $t = x-y$, the three differences are t , $2r_0 t$, $t(3r_0^2 + t^2/4)$. Integrating their pairwise products gives the matrix. Thus

$$Q_a(xv, x^3 v) = \frac{3}{4} Q_a(x^2 v) + \frac{1}{4} H_{4,0}.$$

The determinant of the coupled odd-multiplier block is

$$\boxed{\mathfrak{D}_a(v) = H_{2,0} \left(9H_{2,4} + \frac{3}{2}H_{4,2} + \frac{1}{16}H_{6,0} \right) - \left(3H_{2,2} + \frac{1}{4}H_{4,0} \right)^2 \geq 0.} \quad (\text{CP13:1.21})$$

This exposes the cross term that separate scalar estimates would miss. The polar part of $Q_a(x^2v)$ is $-2\varepsilon(F_0F_4 - F_2^2)$.

For the exponential family define $\mathcal{C}_v(t; \lambda) = \int e^{i\lambda(y+t/2)}v(y+t)v(y)dy$. For real definite-parity v it is real and even in λ . The full entry for $e^{i\sigma x}, e^{i\tau x}$ is

$$\begin{aligned} M(\sigma, \tau) = & 4 \int_0^{2a} k(t) \sin(\sigma t/2) \sin(\tau t/2) \mathcal{C}_v(t; \sigma - \tau) dt \\ & + 4 \sum_{\ell_n < 2a} b_n \sin(\sigma \ell_n/2) \sin(\tau \ell_n/2) \mathcal{C}_v(\ell_n; \sigma - \tau). \end{aligned} \quad (\text{CP13:1.22})$$

Its diagonal is (1.14).

1.7. Inserted endpoint cancellation in the polar term

The endpoint calculation in Group 2 can be inserted before estimating (1.13). Define the archimedean primitive

$$j(s) = -\frac{d}{2}s - \frac{1}{4}\{\Phi(1, 2, 1/4) - e^{-s/2}\Phi(e^{-2s}, 2, 1/4)\}, \quad s \geq 0,$$

and the signed packet sums

$$\begin{aligned} P_0 &= \sum_{\ell_n < 2a} b_n \int_{-a}^{a-\ell_n} v(y) dy, \quad P_1 = \sum_{\ell_n < 2a} b_n \int_{-a}^{a-\ell_n} (a - \ell_n - y) v(y) dy, \\ J_0 &= \int_{-a}^a j(a - y) v(y) dy, \quad J_1 = \int_{-a}^a j'(a - y) v(y) dy. \end{aligned}$$

These are absolutely defined: j' has only a logarithmic endpoint singularity. If v is even, $\alpha = 0$ gives

$$F_0 = \frac{P_0 + J_1}{4 \sinh(a/2)}.$$

Consequently the complete arithmetic identity (1.11) becomes

$$Q_a(xv) = \int_0^{2a} h_{\text{ar}}(t) t^2 C_v(t) dt + \sum b_n \ell_n^2 C_v(\ell_n) - 2F_1^2 - \frac{(P_0 + J_1)F_2}{2 \sinh(a/2)}. \quad (\text{CP13:1.23})$$

If v is odd, $\beta = \int v = 0$. Eliminating α between the endpoint value and derivative gives

$$d_a F_0 = P_1 - aP_0 + J_0 - aJ_1, \quad d_a = 4a \cosh(a/2) - 8 \sinh(a/2) > 0,$$

where positivity follows from $x \cosh x - \sinh x > 0$ for $x > 0$. Thus

$$Q_a(xv) = \int_0^{2a} h_{\text{ar}}(t) t^2 C_v(t) dt + \sum b_n \ell_n^2 C_v(\ell_n) + 2F_1^2 + \frac{2(P_1 - aP_0 + J_0 - aJ_1)F_2}{d_a}. \quad (\text{CP13:1.24})$$

These equations use the endpoint conditions before absolute values. The surviving cross terms are explicit signed packet-moment products. The endpoints do not set F_2 , F_1 , or the shifted correlations to zero.

1.8. Closure record

Theorem and domain. Under (1.1), a real definite-parity null vector exists, smooth multiplication preserves its full closed form domain, and all multiplier identities above hold. Their origin regularization retains every prime, archimedean, and completed polar term. The commutator bound (1.10) and frequency asymptotic (1.17) are closed auxiliary estimates.

Actual attempt and stopping point. The low-degree calculation produces the complete matrix (1.20), determinant (1.21), and endpoint-substituted identities (1.23)-(1.24). No sign for the determinant, its diagonal entries, or the displayed packet-moment products has been established independently of the hypothetical form positivity. These are explicit actual-kernel expressions, rather than a generic counterexample or omission of a tail.

Quantified possible missing lemma. A sufficient statement would be: for every finite $a > 0$, there is $\eta_a > 0$ such that every nonzero real definite-parity $v \in \mathfrak{D}(Q_a)$ with $g * v = \alpha x + \beta$ on I_a satisfies

$$\min_{c \in \mathbb{R}^3, |c|=1} c^T M(v) c \leq -\eta_a \|v\|_2^2, \quad (\text{CP13:1.25})$$

where $M(v)$ is the actual matrix (1.20). This lemma is unproved, and this specific three-dimensional multiplier family may be insufficient. The complete finite-frequency alternative is (1.22).

Return to the unchanged target. An independently established negative direction would contradict $Q_a \geq 0$, exclude a first null, and, through checkpoint 12's domain and continuity results, imply Weil positivity, RH, the correctly normalized Suzuki equality $\Delta = 0$, and $\mathcal{E}_N = 0$ for every N . The proved identities alone have not established the missing negative direction.

2. Closure group 2 – Endpoint packets, an absolute resolvent, and exact exterior cancellation

The Fourier convention is $\widehat{v}(z) = \int v(t) e^{izt} dt$. Let Γ be the zero set of $A(z) = \xi(1/2 + iz)$, counted by multiplicities m_γ . Thus $|\Im \gamma| < 1/2$, with the usual conjugation and sign symmetries. Let $a > 0$, extend $v \in L^2(-a, a)$ by zero, and suppose $v \in \ker A_a$. Throughout this group, a claimed spectral identity is understood with the completed xi zeros, not a critical-line-only sum.

2.1. Endpoint values and derivatives require no traces of the null vector

The actual screw function is

$$g(t) = -4(e^{t/2} + e^{-t/2} - 2) + \sum_{n \geq 2} b_n (|t| - \ell_n)_+ + J(|t|), \quad b_n = \frac{\Lambda(n)}{\sqrt{n}}, \quad \ell_n = \log n,$$

where

$$J(s) = -\frac{s}{2} \{\psi(1/4) - \log \pi\} - \frac{1}{4} \{\Phi(1, 2, 1/4) - e^{-s/2} \Phi(e^{-2s}, 2, 1/4)\}.$$

For $s > 0$,

$$J'(s) = -\frac{1}{2} \{\psi(1/4) - \log \pi\} - \frac{1}{2} \sum_{k \geq 0} \frac{e^{-2(k+1/4)s}}{k + 1/4}, \quad J''(s) = \frac{e^{-s/2}}{1 - e^{-2s}}.$$

In particular $J'(s) = O(1 + |\log s|)$ near zero. Therefore $g' \in L^2_{\text{loc}}(\mathbb{R})$; its finite prime-packet jumps on a bounded interval are harmless for this assertion. Translation continuity in local L^2 , followed

by Cauchy-Schwarz against compactly supported v , proves

$$f_v = g * v \in C^1(\mathbb{R}), \quad f'_v = g' * v. \quad (\text{CP13:2.1})$$

This also follows by distributional differentiation and the continuity of the displayed convolution. No value of v at $\pm a$ has been used.

The null equation supplies $f_v(t) = \alpha t + \beta$ for $-a < t < a$. By (2.1), it extends with its first derivative to both endpoints:

$$\boxed{f_v(\pm a) = \pm a\alpha + \beta, \quad f'_v(\pm a) = \alpha.} \quad (\text{CP13:2.2})$$

If v is real and has parity $\varepsilon \in \{1, -1\}$, then f_v has the same parity. Thus $\alpha = 0$ in the even channel and $\beta = 0$ in the odd channel.

Here is the full arithmetic content of the right endpoint. Put

$$V_0 = \int_{-a}^a v(y) dy, \quad F_0 = \int_{-a}^a e^{y/2} v(y) dy,$$

$$I_n(v) = \int_{-a}^{a-\ell_n} (a - \ell_n - y) v(y) dy, \quad J_n(v) = \int_{-a}^{a-\ell_n} v(y) dy,$$

with both prime-packet quantities zero when $\ell_n > 2a$. Define also

$$U_a(v) = \int_{-a}^a J(a - y) v(y) dy, \quad V_a(v) = \int_{-a}^a J'(a - y) v(y) dy.$$

The integral defining V_a is legitimate by its logarithmic endpoint singularity and Cauchy-Schwarz. Direct substitution into (2.2) gives

$$\boxed{a\alpha + \beta = -4\{(\varepsilon e^{a/2} + e^{-a/2})F_0 - 2V_0\} + \sum_{\ell_n \leq 2a} b_n I_n(v) + U_a(v),} \quad (\text{CP13:2.3})$$

$$\boxed{\alpha = -2(\varepsilon e^{a/2} - e^{-a/2})F_0 + \sum_{\ell_n \leq 2a} b_n J_n(v) + V_a(v).} \quad (\text{CP13:2.4})$$

The left endpoint equations are their parity reflections, with no additional traces. Every active prime power remains in the finite sums, including the endpoint value $\ell_n = 2a$, whose packet integral is zero.

In the even channel, (2.4) becomes the exact signed identity

$$\boxed{F_0 = \frac{\sum_{\ell_n \leq 2a} b_n J_n(v) + V_a(v)}{4 \sinh(a/2)}.} \quad (\text{CP13:2.5})$$

In the odd channel, $V_0 = \beta = 0$; elimination of α from (2.3)-(2.4) yields

$$\boxed{F_0 = \frac{\sum_{\ell_n \leq 2a} b_n \{I_n(v) - aJ_n(v)\} + U_a(v) - aV_a(v)}{4a \cosh(a/2) - 8 \sinh(a/2)}.} \quad (\text{CP13:2.6})$$

The denominator in (2.6) is positive for every $a > 0$, since $u \cosh u - \sinh u > 0$ for $u > 0$. These equations can be inserted into the polar moment terms of Group 1 before any estimate. The packet integrals and the archimedean functionals are signed. Neither formula independently controls the higher moments $F_1, F_2, \dots$ entering the multiplier matrix.

2.2. A new absolute-resolvent theorem on the entire compact L^2 domain

A useful improvement over merely distributional cutoff convergence is available. The unconditional local zero-count bound gives a finite constant C_Γ such that, for every real x ,

$$\sum_{|\Re \gamma - x| \leq 1} m_\gamma \leq C_\Gamma \log(2 + |x|). \quad (\text{CP13:2.7})$$

This is the standard immediate local consequence of the Riemann-von Mangoldt formula with its $O(\log T)$ error. It involves all zeros, not RH.

Let $F = \widehat{v}$. The analytic mean-value inequality on disks of radius one centered at γ , followed by (2.7), bounds the overlap of the disks after weighting by $\log(2 + |\gamma|)^{-1}$. All disks lie in $|\Im z| < 3/2$. Thus, with a finite explicit overlap constant C_1 determined by C_Γ ,

$$\sum_{\gamma} \frac{m_\gamma |F(\gamma)|^2}{\log(2 + |\gamma|)} \leq C_1 \int_{-3/2}^{3/2} \int_{\mathbb{R}} |F(x + iy)|^2 dx dy \leq 6\pi C_1 e^{3a} \|v\|_2^2. \quad (\text{CP13:2.8})$$

The last inequality is ordinary Plancherel applied to $e^{-yt}v(t)$. Denote the final constant by C_a .

Since (2.7) also implies

$$\sum_{\gamma} \frac{m_\gamma \log(2 + |\gamma|)}{1 + |\gamma|^2} < \infty,$$

weighted Cauchy-Schwarz proves

$$\boxed{\sum_{\gamma} \frac{m_\gamma |\widehat{v}(\gamma)|}{1 + |\gamma|} < \infty.} \quad (\text{CP13:2.9})$$

The finite set of bounded zeros causes no issue, and $A(0) \neq 0$. Consequently

$$\boxed{L_{v,b}(z) = \sum_{\gamma} \frac{m_\gamma \widehat{v}(\gamma) e^{-ib\gamma}}{\gamma - z}} \quad (\text{CP13:2.10})$$

converges absolutely, locally uniformly for $z \notin \Gamma$, uniformly for b in a bounded real interval. It defines a meromorphic function with residue

$$\text{Res}_{z=\gamma} L_{v,b}(z) = -m_\gamma \widehat{v}(\gamma) e^{-ib\gamma}. \quad (\text{CP13:2.11})$$

All derivatives in z converge locally uniformly as well.

Here are quantitative tails, sufficient to specify the topology completely. For any compact $K \subset \mathbb{C} \setminus \Gamma$, any $B < \infty$, and any integer $j \geq 0$, there is a finite $C_{a,B,K,j}$ such that, for $R \geq 2(1 + \sup_K |z|)$,

$$\sup_{|b| \leq B, z \in K} \left| \partial_z^j \sum_{|\gamma| > R} \frac{m_\gamma \widehat{v}(\gamma) e^{-ib\gamma}}{\gamma - z} \right| \leq C_{a,B,K,j} \|v\|_2 \frac{\log(2 + R)}{R^{j+1/2}}. \quad (\text{CP13:2.12})$$

Indeed the second Cauchy-Schwarz factor is bounded by $j! e^{B/2} \{\sum_{|\gamma| > R} m_\gamma \log(2 + |\gamma|) / |\gamma - z|^{2j+2}\}^{1/2}$, and (2.7) bounds its square by $O(\log^2(2 + R) / R^{2j+1})$. This is an actual compact-support uniform tail theorem. Its constants depend on a and the evaluation compact set; it is not a Suzuki-relative estimate on growing contours.

2.3. The completed zero distribution and two-primitive regularization

The twice integrated completed explicit formula gives

$$g(t) = \sum_{\gamma} m_{\gamma} \frac{e^{-i\gamma t} - 1}{\gamma^2}. \quad (\text{CP13:2.13})$$

The series is absolutely locally uniform, because $|\Im \gamma| < 1/2$ and $\sum m_{\gamma}/|\gamma|^2 < \infty$. Equivalently, obtain (2.13) by integrating the completed zero distribution twice: the difference from the actual arithmetic g has zero second derivative, and evenness together with $g(0) = 0$ removes its affine ambiguity.

Put $c_{\gamma} = m_{\gamma} \widehat{v}(\gamma)$ and $s_2 = \sum m_{\gamma}/\gamma^2$. Convolution gives

$$f_v(t) = \sum_{\gamma} \frac{c_{\gamma} e^{-i\gamma t}}{\gamma^2} - \widehat{v}(0) s_2, \quad f'_v(t) = -i \sum_{\gamma} \frac{c_{\gamma} e^{-i\gamma t}}{\gamma}. \quad (\text{CP13:2.14})$$

Both series now converge absolutely locally uniformly, the derivative series by (2.9). In particular its differentiation from the primitive is legitimate. The next derivative is taken in $\mathcal{D}'(\mathbb{R})$:

$$\boxed{Z_v(t) := \sum_{\gamma} c_{\gamma} e^{-i\gamma t} = -f''_v(t) = -\mathcal{B}_v''(t), \quad \mathcal{B}_v = f_v - \alpha t - \beta.} \quad (\text{CP13:2.15})$$

Testing the sum against C_c^{∞} makes it absolutely convergent through rapid horizontal Fourier decay and the fixed zero strip. The completed operator agrees with this convolution on its localized operator domain by core approximation. Therefore $Z_v = 0$ on $(-a, a)$.

We work here in distributions of exponential growth. Continuity of g alone would not justify calling it tempered. The unconditional estimates needed below are

$$|\mathcal{B}_v(t)| + |\mathcal{B}'_v(t)| \leq C_v(1 + |t|)e^{|t|/2}, \quad (\text{CP13:2.16})$$

which follow immediately from (2.9), (2.14), and the affine subtraction.

2.4. Endpoint cancellation yields an exact meromorphic support-gap identity

Define the entire difference

$$E_{v,b}(z) = L_{v,b}(z) - e^{-ibz} L_{v,0}(z) = \sum_{\gamma} c_{\gamma} \frac{e^{-ib\gamma} - e^{-ibz}}{\gamma - z}. \quad (\text{CP13:2.17})$$

Its node singularities are removable, and (2.9) makes the series locally uniform, including after filling in those points. It is continuous in real b . Differentiate in the sense of distributions in b , which is legitimate by testing against a compact smooth function:

$$\partial_b L_{v,b}(z) = -iz L_{v,b}(z) - iZ_v(b).$$

Hence

$$\partial_b E_{v,b}(z) + iz E_{v,b}(z) = -iZ_v(b), \quad E_{v,0}(z) = 0. \quad (\text{CP13:2.18})$$

Since $Z_v = 0$ on $(-a, a)$, the elementary distributional ODE has the unique solution zero there. Continuity in b includes both endpoints. Analytic continuation in z then proves the central new identity:

$$\boxed{L_{v,b}(z) = e^{-ibz} L_v(z) \quad (|b| \leq a), \quad L_v := L_{v,0}.} \quad (\text{CP13:2.19})$$

This is an independent identity extracted from the null equation. It cancels the entire translation correction; no individual spectral residue has been discarded.

The same cancellation can be checked directly from (2.14) by integrating twice and using (2.2). At the right endpoint, for $u \geq 0$,

$$\mathcal{B}_v(a+u) = \sum_{\gamma} c_{\gamma} e^{-ia\gamma} \frac{e^{-i\gamma u} - 1 + i\gamma u}{\gamma^2}. \quad (\text{CP13:2.20})$$

At the left endpoint,

$$\mathcal{B}_v(-a-u) = \sum_{\gamma} c_{\gamma} e^{ia\gamma} \frac{e^{i\gamma u} - 1 - i\gamma u}{\gamma^2}. \quad (\text{CP13:2.21})$$

The constant and linear subtractions are exactly the endpoint value and derivative conditions. Their series are justified by (2.9). They are present before integration, rather than silently dropped.

For a core sequence $q_n \rightarrow v$ in fixed-support L^2 , (2.17) also supplies a quantitative approximate identity:

$$\sup_{|b| \leq a, z \in K} |\partial_z^j E_{q_n, b}(z)| \leq C_{a, K, j} \|q_n - v\|_2 \rightarrow 0, \quad (\text{CP13:2.22})$$

for every compact $K \subset \mathbb{C}$, including filled-in zero nodes. This follows from the continuous linear dependence on the compact L^2 input, followed by (2.19). Thus the new cancellation is compatible with the preserved checkpoint-12 domain passage.

2.5. The two exterior Fourier-Laplace representations, with signs and normalization poles

Set $B_+(t) = \mathbf{1}_{[a, \infty)}(t) \mathcal{B}_v(t)$ and $B_-(t) = \mathbf{1}_{(-\infty, -a]}(t) \mathcal{B}_v(t)$. Because $\mathcal{B}_v$ and $\mathcal{B}'_v$ vanish at both endpoints, these functions are C^1 , and their distributional second derivatives contain no endpoint delta or delta-prime term. Define $Z_{\pm} = -B''_{\pm}$; then $Z_v = Z_+ + Z_-$.

Initially, for $\Im z > 1/2$,

$$\mathcal{Z}_+(z) := \widehat{Z}_+(z) = z^2 \int_a^{\infty} \mathcal{B}_v(t) e^{izt} dt. \quad (\text{CP13:2.23})$$

For $\Im z < -1/2$,

$$\mathcal{Z}_-(z) := \widehat{Z}_-(z) = z^2 \int_{-\infty}^{-a} \mathcal{B}_v(t) e^{izt} dt. \quad (\text{CP13:2.24})$$

The integration-by-parts boundary terms at $\pm a$ are zero by (2.2); the terms at infinity vanish by (2.16). These formulas define the Fourier-Laplace transforms of the exponential-growth distributions, without assuming global temperedness.

For the right regularized summand, the exact elementary integral is

$$\int_0^{\infty} \frac{e^{-i\gamma u} - 1 + i\gamma u}{\gamma^2} e^{izu} du = \frac{i}{z^2(z - \gamma)}. \quad (\text{CP13:2.25})$$

Its convergence and summation interchange follow from (2.9), (2.16), and $\Im z > 1/2$. The left integral has the opposite sign. Substitution of (2.20)-(2.21), followed by (2.19), gives

$$\boxed{\mathcal{Z}_+(z) = -ie^{iaz} L_{v, a}(z) = -iL_v(z), \quad \Im z > 1/2,} \quad (\text{CP13:2.26})$$

$$\boxed{\mathcal{L}_-(z) = ie^{-iaz}L_{v,-a}(z) = iL_v(z), \quad \Im z < -1/2.} \quad (\text{CP13:2.27})$$

Equivalently,

$$\boxed{L_v(z) = iz^2 \int_a^\infty \mathcal{B}_v(t)e^{izt}dt = -iz^2 \int_{-\infty}^{-a} \mathcal{B}_v(t)e^{izt}dt,} \quad (\text{CP13:2.28})$$

where the first equality is initially in the upper half-plane and the second in the lower half-plane. Equality elsewhere means their unique meromorphic continuations through the zero strip, supplied by (2.10). One must not read (2.28) as asserting convergence of either improper integral at a real zero.

The Cauchy-denominator sign is important: $\int_0^\infty e^{i(z-\gamma)t}dt = i/(z-\gamma) = -i/(\gamma-z)$. It is responsible for the signs in (2.26)-(2.28).

The apparently introduced poles at zero also have explicit bookkeeping. From (2.14) at zero,

$$L_v(0) = i\alpha, \quad L'_v(0) = \beta + \widehat{v}(0)s_2. \quad (\text{CP13:2.29})$$

Thus the continued right primitive transform is

$$\int_a^\infty \mathcal{B}_v(t)e^{izt}dt = -\frac{iL_v(z)}{z^2} = \frac{\alpha}{z^2} - \frac{i\{\beta + \widehat{v}(0)s_2\}}{z} + O(1). \quad (\text{CP13:2.30})$$

The left primitive has opposite principal parts. These are the retained constant and linear regularization terms. In the even channel $\alpha = 0$, and in the odd channel $\beta = \widehat{v}(0) = 0$. No value at zero is omitted through an informal division by z^2 .

2.6. Return to the exact correction, and what cancellation has and has not been proved

For compact smooth tests, checkpoint 12 gives $Vq = -ihL_q/\sqrt{\pi}$. Both sides are continuous from fixed-support L^2 into locally meromorphic functions; the left side has the completed domain passage already proved, and (2.8)-(2.12) supply it on the right. Therefore

$$Vv(z) = -\frac{i}{\sqrt{\pi}}h(z)L_v(z) \quad (\text{CP13:2.31})$$

holds meromorphically for the null vector. At a xi zero γ ,

$$L_v(z) = -\frac{m_\gamma \widehat{v}(\gamma)}{z - \gamma} + O(1), \quad h(z) = -\frac{i(z - \gamma)}{m_\gamma} + O((z - \gamma)^2),$$

so their product removes the apparent singularity and

$$\boxed{Vv(\gamma) = \widehat{v}(\gamma)/\sqrt{\pi}.} \quad (\text{CP13:2.32})$$

The support-gap identity does not force this value to be zero: its exponential factors cancel at the residue itself, leaving exactly $-m_\gamma \widehat{v}(\gamma)$ in (2.11). This is an actual calculation for the xi structure.

For $\Im z > 1/2$, (2.19) and weighted Cauchy-Schwarz give the explicit support-gap estimate

$$|L_v(z)| \leq e^{-a\Im z} \sqrt{C_a} \|v\|_2 \left\{ \sum_\gamma \frac{m_\gamma \log(2 + |\gamma|) e^{2a\Im \gamma}}{|\gamma - z|^2} \right\}^{1/2}. \quad (\text{CP13:2.33})$$

A reflected estimate holds below the strip. This includes the new cancellation, but the remaining Cauchy sum still detects proximity to real zero nodes. It cannot be replaced by an interval-independent Suzuki-normalized constant using (2.8) alone.

For each fixed bounded contour avoiding nodes and correction poles, the core convergence and (2.12), (2.22) allow the exterior representations to be inserted into the complete residue-boundary pairing. They preserve every residue and every contour term. The unresolved quantity in the growing-contour passage is the signed paired expression

$$-\frac{1}{2\pi i} \int_{\partial\Omega} \mathcal{R}(z) L_{q_n, H}(z) L_{q_n, H}^\sharp(z) dz + \sum_{p \in \Omega} \text{Res}_{z=p} \{ \mathcal{R}(z) L_{q_n, H}(z) L_{q_n, H}^\sharp(z) \}, \quad (\text{CP13:2.34})$$

normalized by $\mathcal{S}(q_n, q_n)$, and then completed in the required cutoff order. Pole terms here mean poles of $\mathcal{R}$ with the appropriate higher derivatives, exactly as in checkpoint 12. Fixed-contour convergence does not furnish uniformity as the contour exhausts all nodes and poles.

There is an exact further cancellation at the level where the complete defect is paired. For any null vector (even without imposing real parity), sharp reflection of (2.19) gives

$$L_{v, b}^\sharp(z) = e^{ibz} L_v^\sharp(z), \quad \boxed{L_{v, b}(z) L_{v, b}^\sharp(z) = L_v(z) L_v^\sharp(z)}. \quad (\text{CP13:2.35})$$

Thus every derivative of these products, and hence every higher-pole residue, agrees as well. For real parity ε , the more explicit relation is $L_{v, b}^\sharp = \varepsilon L_{v, -b}$, and the common product is εL_v^2 . The support-gap exponential factor cancels exactly in this complete quadratic pairing. This calculation explains the lack of a small norm constant without discarding the improved linear identity.

At a xi node γ , write $L_v(z) = -c_\gamma/(z - \gamma) + \ell_\gamma + O(z - \gamma)$. Filling in the finite part of (2.19) yields

$$\boxed{\sum_{\delta \neq \gamma} \frac{c_\delta e^{-ib\delta}}{\delta - \gamma} = e^{-ib\gamma} \left\{ \sum_{\delta \neq \gamma} \frac{c_\delta}{\delta - \gamma} + ibc_\gamma \right\}}. \quad (\text{CP13:2.36})$$

These sums are absolutely convergent by (2.9); the finitely many neighboring nodes are distinct and separated from the fixed node. The ibc_γ term is the retained interaction between the pole and the translating exponential. It must remain when inserting the removable-zero factors into Group 3.

2.7. Additional exact-null regularity and a sharper Cauchy tail

There is a further independent estimate for the null vector itself. Let $\mathcal{L} = \log |D|$ be the whole-line Fourier multiplier $\log |\xi|$. Compact support gives bounded $\widehat{v}$ near frequency zero, so $\mathcal{L}v \in H^{-\epsilon}(\mathbb{R})$ for every $\epsilon > 0$. The exact kernel formula in Group 1 gives, on I_a ,

$$\mathcal{L}v = \mathcal{B}_a v, \quad \mathcal{B}_a v = \log(2\pi)v + \sum_{\ell_n \leq 2a} b_n \{v(\cdot + \ell_n) + v(\cdot - \ell_n)\} + \int_{-a}^a r''(\cdot - y)v(y) dy. \quad (\text{CP13:2.37})$$

All translations use the zero extension. This is a bounded operator on $L^2(I_a)$, with the explicit upper bound

$$\|\mathcal{B}_a\| \leq C_a^{\text{op}} := \log(2\pi) + 2 \sum_{\ell_n \leq 2a} b_n + \int_{-2a}^{2a} |r''(t)| dt < \infty. \quad (\text{CP13:2.38})$$

Outside the support there is no finite-part term:

$$(\mathcal{L}v)(t) = -\frac{1}{2} \int_{-a}^a \frac{v(y)}{|t-y|} dy, \quad |t| > a. \quad (\text{CP13:2.39})$$

For $t = a + u$, $y = a - s$, this is one half of a Carleman operator with kernel $1/(u + s)$. Its norm is at most π : the Schur test with weight $s^{-1/2}$ uses $\int_0^\infty s^{-1/2}(u + s)^{-1} ds = \pi u^{-1/2}$. The other endpoint is identical. The piecewise function specified by (2.37) and (2.39) is therefore in $L^2(\mathbb{R})$. Its difference from the distribution $\mathcal{L}v$ is supported at $\{-a, a\}$. It belongs to $H^{-\epsilon}$ for $0 < \epsilon < 1/2$, whereas no nonzero distribution supported on finitely many points belongs to these spaces. Hence this difference vanishes and

$$\|\log |D|v\|_2^2 \leq ((C_a^{\text{op}})^2 + \pi^2/2) \|v\|_2^2. \quad (\text{CP13:2.40})$$

This proves logarithmic-square Fourier regularity, including the possible endpoint behavior of v , without assuming endpoint traces.

For completeness, this improves the actual-null zero tail. Multiplication by a compact smooth function equal to e^{-yt} on $[-a, a]$, uniformly for $|y| \leq 3/2$, is bounded on the Fourier space with weight $\log^2(2 + |\xi|)$. This follows by Young's inequality and $\log(2 + |\xi|) \leq \log(2 + |\xi - u|) + \log(1 + |u|)$. Apply the disk-mean proof of (2.8), now with a positive logarithmic weight. The local zero-count overlap contributes one additional logarithm, so (2.40) yields

$$\sum_{\gamma} m_{\gamma} \log(2 + |\Re \gamma|) |\widehat{v}(\gamma)|^2 \leq C_a^{\text{null}} \|v\|_2^2. \quad (\text{CP13:2.41})$$

For a compact K avoiding the nodes and every integer $j \geq 0$, weighted Cauchy-Schwarz therefore gives

$$\sup_{z \in K} \sum_{|\Re \gamma| > R} \frac{m_{\gamma} |\widehat{v}(\gamma)|}{|\gamma - z|^{j+1}} \leq C_{a,K,j}^{\text{null}} \|v\|_2 R^{-j-1/2}. \quad (\text{CP13:2.42})$$

The reciprocal logarithm in the second Cauchy-Schwarz factor cancels the local zero-density logarithm. The corresponding z -derivative has the factor $j!$; bounded real shifts contribute at most $e^{|b|/2}$. These statements strengthen the exact null representation. They do not assert uniformly bounded operator graph norms on an arbitrary form-core approximating sequence, or uniformity for growing evaluation contours.

Closure record.

1. **Exact theorem and domain.** For every finite a and every $v \in \ker A_a \subset L^2(-a, a)$, the convolution is C^1 , satisfies (2.2), and its completed resolvent obeys the absolute convergence theorem (2.8)-(2.12), the exact gap identity (2.19), the two exterior representations (2.28), and the exact-null regularity and tail bounds (2.40)-(2.42). These statements require no endpoint traces of v , no RH, and no assumption that $\mathcal{S}(v, v)$ is finite.
2. **Proof status.** All these identities and compact-contour tails are proved above. The endpoint prime expansions (2.3)-(2.6) are exact and signed. The origin regularization terms are displayed in (2.30). The support gap removes the entire translation correction, a new structural cancellation beyond its exponential factor.
3. **Downstream consequence.** The completed defect can now be expressed through one common meromorphic exterior resolvent on both sides. On null core sequences the entire correction tends to zero with the quantitative estimate (2.22). This advances the substitution into the checkpoint-12 pairing while preserving its original metric and every pole contribution.

4. **Remaining quantified estimate.** The group does not establish a bound on (2.34) uniform in the growing contours and normalized by $\mathcal{S}(q_n, q_n)$. A sufficient missing statement is that, for every nonzero first-null parity vector, some form-core sequence satisfies

$$\liminf_n \frac{-\lim_H \mathcal{K}_H(q_n, q_n) + \mathcal{X}_\infty(q_n, q_n)}{\mathcal{S}(q_n, q_n)} \geq -1 + K_*.$$

The newly available inputs for this estimate are (2.3)-(2.6), (2.19), (2.22), and (2.28). The precise unresolved line is the passage from compact-contour convergence to the signed normalized exhaustion in (2.34), not the existence or sign of the Cauchy denominator.

Sources: M. Suzuki, *Weil's quadratic form via the screw function*, arXiv:2606.09096v2, equation (1.3), §§2.5 and 8.2-8.5. The proofs here use its actual kernel and retained checkpoint-12 domain results; the compact-support sampling, exact gap ODE, and exterior formulas are derived in this group.

3. Closure group 3 – Complete signed pairing and the remaining estimate

This group uses the unchanged conventions of checkpoints 11-12. Put $A(z) = \xi(1/2 - iz)$, $E = A + iA'$, $h = A/E$, and $\mathcal{R} = C - ih$ in the upper half-plane, with its Schwarz-reflected lower continuation. Write $V = -L_0$, $\mathcal{S}(q, q) = \|Vq\|_2^2$, and $\Delta = \mathfrak{W} - \mathcal{S} = -\mathcal{X}_\infty + \mathcal{T}$. All zero sums below count distinct zeros with multiplicity m_γ .

3.1. An exact additional identity forced by the null equation

For a compactly supported L^2 function v , define

$$L_{v,b}(z) = \sum_\gamma \frac{m_\gamma \widehat{v}(\gamma) e^{-ib\gamma}}{\gamma - z}, \quad L_v = L_{v,0}. \quad (\text{CP13:3.1})$$

Group 2 proves absolute local uniform convergence away from the zeros, using the weighted sampling inequality and Cauchy-Schwarz. The convergence is locally uniform also in bounded real b . Consequently the sums define genuine meromorphic functions, not merely formal transforms.

Let $Q_a \geq 0$ and let $v \neq 0$ be a real first-null vector of parity $\epsilon \in \{1, -1\}$. Since $Z_v = -f_v''$ vanishes on $(-a, a)$, differentiation in distributions in the variable b gives

$$\partial_b L_{v,b}(z) = -iz L_{v,b}(z) - iZ_v(b).$$

Solving this elementary distributional ODE and using continuity in b proves the exact identity

$$\boxed{L_{v,b}(z) = e^{-ibz} L_v(z), \quad |b| \leq a.} \quad (\text{CP13:3.2})$$

This includes the endpoints by continuity. No endpoint trace of v occurs. The entire divided-exponential correction is zero on the exact null vector:

$$\sum_\gamma m_\gamma \widehat{v}(\gamma) \frac{e^{-ib\gamma} - e^{-ibz}}{\gamma - z} = 0. \quad (\text{CP13:3.3})$$

The identity is global after meromorphic continuation. It preserves every node residue; indeed both sides of (3.2) have residue $-m_\gamma \widehat{v}(\gamma) e^{-ib\gamma}$ at a zero γ .

By Group 2's endpoint-complete exterior transform calculation, if $\mathcal{B}_v = f_v - \alpha t - \beta$, then

$$\boxed{L_v(z) = iz^2 \int_a^\infty \mathcal{B}_v(t) e^{izt} dt, \quad \Im z > 1/2,} \quad (\text{CP13:3.4})$$

$$\boxed{L_v(z) = -iz^2 \int_{-\infty}^{-a} \mathcal{B}_v(t) e^{izt} dt, \quad \Im z < -1/2.} \quad (\text{CP13:3.5})$$

The terms at the finite endpoints vanish because $\mathcal{B}_v(\pm a) = \mathcal{B}'_v(\pm a) = 0$. The affine subtractions remain inside these formulas. Continuing either exterior expression through the zero strip means continuing the meromorphic function (3.1); it does not mean that its original Laplace integral converges throughout that strip.

3.2. Quantitative control of the entire correction for null core sequences

There is a useful improvement over mere local convergence. For $0 < b \leq a$ set

$$r_{b,z}(t) = \mathbf{1}_{(0,b)}(t) e^{-i\bar{z}t}.$$

This compactly supported piecewise smooth function belongs to the closed logarithmic Weil form domain. Its Fourier transform is $O((1 + |\xi|)^{-1})$ and therefore has finite logarithmic energy. This also applies when $b = a$: smooth inward truncations followed by mollification converge in the logarithmic form norm. The bounded prime and pole terms converge by L^2 convergence. Thus its use requires no trace of an arbitrary form-domain vector.

For a compact smooth q in $(-a, a)$, distributional integration of the ODE gives

$$\boxed{\mathcal{E}_b(q; z) := L_{q,b}(z) - e^{-ibz} L_q(z) = -ie^{-ibz} Q_a(q, r_{b,z}).} \quad (\text{CP13:3.6})$$

Since $Q_a \geq 0$, its form Cauchy-Schwarz inequality yields

$$\boxed{|\mathcal{E}_b(q; z)|^2 \leq e^{2b\Im z} Q_a(q, q) Q_a(r_{b,z}, r_{b,z}).} \quad (\text{CP13:3.7})$$

Here the test energy can be written with all boundary terms. In distributions,

$$r'_{b,z} = \delta_0 - e^{-i\bar{z}b} \delta_b - i\bar{z} r_{b,z}.$$

Since $g(0) = 0$, the exact expression is

$$\begin{aligned} Q_a(r_{b,z}) = & -2g(b) \Re(e^{izb}) \\ & + 2\Re \left[-i\bar{z} \int_0^b e^{-i\bar{z}y} \{g(y) - e^{izb} g(b-y)\} dy \right] \\ & + |z|^2 \int_0^b \int_0^b g(x-y) e^{izx-i\bar{z}y} dy dx. \end{aligned} \quad (\text{CP13:3.8})$$

The formula follows first by smooth approximation and then by weak convergence of the derivative measures against the continuous function $g(x-y)$. In particular,

$$Q_a(\mathbf{1}_{(0,b)}) = -2g(b) \geq 0. \quad (\text{CP13:3.9})$$

Translating the indicator shows that $g(b) \leq 0$ for every $0 \leq b \leq 2a$ at a nonnegative interval. This is a necessary condition, not an asserted strict exclusion.

For real $z = \tau$, (3.8) becomes

$$Q_a(r_{b,\tau}) = -2g(b) \cos(\tau b) - 4\tau \int_0^b g(t) \sin(\tau t) dt + 2\tau^2 \int_0^b (b-t)g(t) \cos(\tau t) dt. \quad (\text{CP13:3.10})$$

An explicit compact estimate follows from the total variation of r' : with $M_a = \sup_{|t| \leq 2a} |g(t)|$,

$$0 \leq Q_a(r_{b,z}) \leq M_a(2+b|z|)^2 e^{2b|\Im z|}. \quad (\text{CP13:3.11})$$

For a valid first-null core sequence $q_n \rightarrow v$, put $s_n = \mathcal{S}(q_n, q_n)$. Checkpoint 12 gives $\liminf s_n > 0$, while $Q_a(q_n) \rightarrow 0$. Equations (3.7)-(3.11) therefore prove

$$\sup_{z \in K} \frac{|\mathcal{E}_b(q_n; z)|}{\sqrt{s_n}} \leq C_{a,b,K} \sqrt{\frac{Q_a(q_n)}{s_n}} \rightarrow 0 \quad (\text{CP13:3.12})$$

on every compact K . Both sides of (3.6) have removable singularities at the zero nodes. Derivatives satisfy corresponding compact estimates by Cauchy's formula on slightly larger compact sets. This controls the error in replacing a core resolvent by its null-gap form locally, including any fixed finite set of correction poles.

It does not give a bound uniform over an expanding contour: the explicit factor in (3.11) grows with $|z|$. This distinction remains inside the limiting calculation.

3.3. Parity and the complete signed pole pairing

For real v of parity ϵ ,

$$\widehat{v}(-z) = \epsilon \widehat{v}(z), \quad \widehat{v}^\sharp(z) = \epsilon \widehat{v}(z),$$

and reindexing the absolutely convergent sums gives

$$\boxed{L_v(-z) = -\epsilon L_v(z), \quad L_v^\sharp(z) = \epsilon L_v(z).} \quad (\text{CP13:3.13})$$

The symmetric smooth cutoffs preserve these identities. The complete contour integrand is thus the analytic function

$$\mathcal{R}(z) L_{v,H}(z) L_{v,H}^\sharp(z) = \epsilon \mathcal{R}(z) L_{v,H}(z)^2. \quad (\text{CP13:3.14})$$

This retains the phase at nonreal points. It is not $\mathcal{R}(z) |L_{v,H}(z)|^2$.

For every compact smooth real parity core function q , the exact finite-cutoff identity is

$$\boxed{\Delta_H(q, q) = \epsilon \left[-\frac{1}{2\pi i} \int_{\partial\Omega_H} \mathcal{R}(z) L_{q,H}(z)^2 dz + \sum_{p \in \Omega_H} \sum_{k=1}^{d_p} \frac{r_{p,k}}{(k-1)!} \partial_z^{k-1} L_{q,H}(z)^2 \Big|_{z=p} \right]}. \quad (\text{CP13:3.15})$$

The domain Ω_H contains the cutoff zero set; its boundary avoids all nodes and correction poles. The full defect is the fixed-test $H \rightarrow \infty$ limit, exactly as in checkpoint 12.

At a simple correction pole p , a conjugate pair contributes

$$2\epsilon \Re\{r_{p,1} L_{q,H}(p)^2\}. \quad (\text{CP13:3.16})$$

Since $\mathcal{R}$ is odd and L^2 is even, a nondegenerate quartet $p, -p, \bar{p}, -\bar{p}$ contributes

$$4\epsilon\Re\{r_{p,1}L_{q,H}(p)^2\}. \quad (\text{CP13:3.17})$$

The associated contour pieces must be paired at the same time. There is no asserted sign for the real part in (3.16).

Substitution of (3.4) on the upper continuation gives

$$\epsilon\mathcal{R}(z)L_v(z)^2 = -\epsilon z^4\mathcal{R}(z) \left(\int_a^\infty \mathcal{B}_v(t)e^{izt} dt \right)^2. \quad (\text{CP13:3.18})$$

This is a square of a complex Laplace integral with its phase retained, and must be continued meromorphically when used inside the strip. The lower formula is the same with the left exterior transform. Although the endpoint integrations removed their value and derivative contributions, they have not changed (3.18) into a positive norm square at nonreal poles.

3.4. Explicit removable-zero calculations

Let γ be a zero of A of multiplicity m , write $s = z - \gamma$, and factor $A(z) = s^m a(s)$, $a(0) \neq 0$. Put $c_0 = a'(0)/a(0)$. Then

$$\frac{A'}{A} = \frac{m}{s} + c_0 + O(s), \quad h(z) = -\frac{i}{m}s + \frac{1+ic_0}{m^2}s^2 + O(s^3). \quad (\text{CP13:3.19})$$

The lower reflected factor similarly has expansion $h^\sharp(z) = is/m + (1-ic_0)s^2/m^2 + O(s^3)$. Set

$$b_\gamma = m\widehat{v}(\gamma), \quad \ell_\gamma = \sum_{\delta \neq \gamma} \frac{m_\delta \widehat{v}(\delta)}{\delta - \gamma}.$$

Absolute local convergence proves

$$L_v(z) = -\frac{b_\gamma}{s} + \ell_\gamma + O(s). \quad (\text{CP13:3.20})$$

For compact smooth tests, and then for the local L^2 extension,

$$Vv(z) = -\frac{i}{\sqrt{\pi}}h(z)L_v(z), \quad Vv(\gamma) = \frac{\widehat{v}(\gamma)}{\sqrt{\pi}}. \quad (\text{CP13:3.21})$$

At a real zero, the full original metric has the exact removal

$$\frac{\omega(x)|L_v(x)|^2}{\pi} \longrightarrow \frac{|\widehat{v}(\gamma)|^2}{\pi}. \quad (\text{CP13:3.22})$$

Thus the generic Hardy point-evaluation singularity is not a pointwise singularity of the actual Suzuki feature at a real node. This is an exact local improvement. It supplies no uniform bound for all normalized node evaluations.

The correction integrand still records the node through its residue. Write $\mathcal{R}(z) = R_\gamma + R'_\gamma s + O(s^2)$, where

$$R_\gamma = C(\gamma), \quad R'_\gamma = C'(\gamma) - 1/m.$$

Using parity,

$$\text{Res}_{z=\gamma}\{\mathcal{R}(z)L_v(z)L_v^\sharp(z)\} = \epsilon\{R'_\gamma b_\gamma^2 - 2R_\gamma b_\gamma \ell_\gamma\}. \quad (\text{CP13:3.23})$$

The regular coefficient ℓ_γ contains every other zero. Equation (3.23) explicitly retains the mixed-node term. Removing the feature singularity does not remove that interaction.

Now let p be a simple noncommon zero of E in the upper half-plane, and put $m_A = A'/A$. Then $m_A(p) = i$,

$$\operatorname{Res}_p h = -\frac{i}{m'_A(p)}, \quad \operatorname{Res}_p \mathcal{R} = -\frac{1}{m'_A(p)}, \quad h^\sharp(p) = \frac{1}{2}. \quad (\text{CP13:3.24})$$

The resolvents $L_v, L_v^\sharp$ are holomorphic at p . Although Vv shares the pole of h ,

$$L_v L_v^\sharp = \pi \frac{Vv(Vv)^\sharp}{hh^\sharp} \quad (\text{CP13:3.25})$$

has a finite value there. The remaining complete correction residue is exactly

$$\boxed{\operatorname{Res}_p \{\mathcal{R} L_v L_v^\sharp\} = -\frac{\epsilon L_v(p)^2}{m'_A(p)}}. \quad (\text{CP13:3.26})$$

Neither (3.2) nor the endpoint equations forces $L_v(p) = 0$. They instead give $L_v(p) = e^{ibp} L_{v,b}(p)$, an exact relation preserving its value. Higher poles retain all derivatives of L_v^2 as in (3.15). Lower-half-plane correction poles and their complete principal parts are the Schwarz reflections; they correspond to $E^\sharp$, so a lower-half-plane zero of E alone is not asserted to be a pole of $\mathcal{R}$.

There is a stronger exact cancellation in the completed quadratic pairing. For real parity v ,

$$L_{v,b}^\sharp(z) = \epsilon L_{v,-b}(z).$$

The two null-gap identities therefore imply

$$\boxed{L_{v,b}(z) L_{v,b}^\sharp(z) = L_v(z) L_v^\sharp(z), \quad |b| \leq a.} \quad (\text{CP13:3.27})$$

Thus the exponential gained from one exterior continuation cancels exactly against the exponential from its reflected partner. The identity holds meromorphically, so it persists under every derivative entering a higher pole contribution and on the full contour. This proves precisely that the endpoint gap alone supplies no exponential small factor in the completed signed energy.

For a same-parity core function, retain the entire correction $\mathcal{E}_b$ from (3.6). The exact difference is

$$L_{q,b} L_{q,b}^\sharp - L_q L_q^\sharp = e^{-ibz} L_q \mathcal{E}_b^\sharp + e^{ibz} L_q^\sharp \mathcal{E}_b + \mathcal{E}_b \mathcal{E}_b^\sharp. \quad (\text{CP13:3.28})$$

All mixed terms are displayed. Equations (3.7) and (3.11), applied on a reflection-symmetric compact set, bound this expression divided by s_n by $C_{a,b,K}[\sqrt{Q_a(q_n)/s_n} + Q_a(q_n)/s_n]$. This is the actual quadratic substitute for an uncoupled estimate of either exterior evaluation.

3.5. Substitution into the full budget and the exact remaining estimate

Choose the core approximants real and of the same parity as v . This is legitimate by projecting a form-core sequence under conjugation and reflection, which preserve the form and its domain. Keep the cutoff limit inside the sequence limit. Define

$$C_n := \frac{\epsilon}{s_n} \lim_{H \rightarrow \infty} \left[-\frac{1}{2\pi i} \int_{\partial\Omega_H} \mathcal{R}(z) L_{q_n,H}(z)^2 dz + \sum_{p \in \Omega_H} \sum_{k=1}^{d_p} \frac{r_{p,k}}{(k-1)!} \partial_z^{k-1} (L_{q_n,H}^2)(p) \right]. \quad (\text{CP13:3.29})$$

This is the complete signed contour-residue expression, not an estimate of its separate summands. Exactly,

$$\mathcal{C}_n = \frac{\Delta(q_n, q_n)}{s_n} = -1 + \frac{Q_a(q_n, q_n)}{s_n} \longrightarrow -1. \quad (\text{CP13:3.30})$$

The new identities (3.2), (3.7), and the full multiplier matrix from Group 1 constrain the functions in (3.29). The local removals (3.19)-(3.26) prove that the apparent real-node singularity cancels in the actual metric. They also show exactly which signed terms remain: the phase-sensitive quantities $-\epsilon L_v(p)^2/m'_A(p)$, their higher-pole derivatives, and their complete accompanying contour.

An independently proved estimate sufficient to close the round is the following quantified lemma:

Null-gap signed correction lemma (open). For every finite a with $Q_a \geq 0$, every nonzero real parity null vector obeying the endpoint equations, the gap identity (3.2), and the full multiplier positivity identities, there exists a same-parity smooth form-core sequence $q_n \rightarrow v$ for which

$$\liminf_{n \rightarrow \infty} \left\{ \mathcal{C}_n + \frac{\mathcal{X}_\infty(q_n, q_n)}{s_n} \right\} \geq -1 + \frac{3567}{509894}. \quad (\text{CP13:3.31})$$

The extraction comparison gives the opposite upper bound

$$\limsup_n \left\{ \mathcal{C}_n + \frac{\mathcal{X}_\infty(q_n, q_n)}{s_n} \right\} \leq -1 + \frac{17607901}{2531730432}, \quad (\text{CP13:3.32})$$

so (3.31) would exclude the null vector with the explicit positive margin

$$\varepsilon_* = \frac{3567}{509894} - \frac{17607901}{2531730432} = \frac{4429025}{108864408576}. \quad (\text{CP13:3.33})$$

The new calculation attempted here replaces core resolvents by their exact exterior forms before estimating. The error is bounded by (3.7), and vanishes after normalization on every fixed compact contour. On an exhaustion containing every cutoff node and pole, that estimate is not uniform: (3.11) grows with the contour, and the pole evaluations and signed contour contribution remain coupled. Consequently compact convergence has not justified exchanging the exhaustion with the null-sequence limit or established (3.31).

Here is an explicit paired version of that calculation. For a fixed contour domain Ω , place disjoint small circles c_p around all enclosed correction poles, avoiding every zero node. Set

$$\mathcal{P}_\Omega[F] = \frac{1}{2\pi i} \left[- \int_{\partial\Omega} \mathcal{R}(z)F(z) dz + \sum_p \int_{c_p} \mathcal{R}(z)F(z) dz \right], \quad M_\Omega = \frac{1}{2\pi} \left[\int_{\partial\Omega} |\mathcal{R}| |dz| + \sum_p \int_{c_p} |\mathcal{R}| |dz| \right]. \quad (\text{CP13:3.34})$$

This pairs each residue with the original contour before making an estimate, and includes every higher-order derivative automatically. Let U be the union of these curves, let $Y = \sup_U |\Im z|$, and use $L_q = e^{ibz}(L_{q,b} - \mathcal{E}_b)$. Then

$$\frac{|\mathcal{P}_\Omega[L_q^2 - e^{2ibz}L_{q,b}^2]|}{\mathcal{S}(q, q)} \leq M_\Omega e^{2bY} \left[2 \frac{\sup_U |L_{q,b}|}{\sqrt{\mathcal{S}(q, q)}} \frac{\sup_U |\mathcal{E}_b|}{\sqrt{\mathcal{S}(q, q)}} + \frac{\sup_U |\mathcal{E}_b|^2}{\mathcal{S}(q, q)} \right]. \quad (\text{CP13:3.35})$$

On a null core sequence, weighted sampling bounds the first ratio by a finite constant depending on a, b, U , because $\|q_n\|_2$ is bounded and $\liminf s_n > 0$. Formula (3.7) therefore yields an explicit finite $C_{a,b,\Omega,B_{\text{seq}}}$, where $B_{\text{seq}} = \sup_{n \geq n_0} \|q_n\|_2 / \sqrt{s_n} < \infty$ such that

$$\frac{|\mathcal{P}_\Omega[L_{q_n}^2 - e^{2ibz}L_{q_n,b}^2]|}{s_n} \leq C_{a,b,\Omega,B_{\text{seq}}} \left(\sqrt{\frac{Q_a(q_n)}{s_n}} + \frac{Q_a(q_n)}{s_n} \right) \longrightarrow 0. \quad (\text{CP13:3.36})$$

For finite cutoffs on this fixed collection of curves, add the explicit uniform weighted-sampling tail $O_a((\log H)/\sqrt{H})\|q_n\|_2$ for each resolvent; it tends to zero uniformly in the core sequence. In (3.29), however, Ω_H itself must contain the entire cutoff zero set. No bound independent of that growing domain has been proved for M_{Ω_H} , the normalized shifted resolvents on its curves, or their signed cancellation. Equation (3.36) is a completed local pairing theorem and displays the exact uniform estimate still required by this attempted exhaustion.

There is a second precise possible loss. Compact continuity proves $\mathcal{X}_\infty(q_n, q_n) \rightarrow \mathcal{X}_\infty(v, v) < \infty$, using its local integral formula, without assigning a finite global Suzuki norm to v . If a subsequence has $s_n \rightarrow \infty$, its extraction ratio tends to zero and its remainder ratio tends to -1 . If s_n stays bounded along a subsequence, Fatou and weak compactness place Vv in L^2 and give

$$s_n = \|Vv\|_2^2 + \|Vq_n - Vv\|_2^2 + o(1). \quad (\text{CP13:3.37})$$

The last nonnegative term describes possible spectral energy escaping every fixed compact set. These statements identify the required uniformity; they do not turn its absence into a counterexample to the actual xi estimate.

Exact theorem and domain. Equations (3.2), (3.13), and (3.19)-(3.26) hold for the compact L^2 first-null vector as locally meromorphic identities. Equations (3.6)-(3.12) hold on the nonnegative closed form and its smooth core. The complete defect identity (3.15) and its limit (3.29) are asserted for compact smooth core functions, with fixed-test cutoff convergence.

Proof status. The additional null-gap identity, quantitative near-null error estimate, parity pairing, removable-zero expansions, and exact surviving correction residues are proved. The independent signed lower bound (3.31) remains open.

Downstream consequence. If (3.31) is established, the positive margin (3.33) excludes every finite first-null interval. The checkpoint-12 continuity and small-interval positivity then give Weil positivity on every compact smooth test. Weil's criterion gives RH, after which the correctly normalized Suzuki identity yields $\Delta = 0$ and hence $\mathcal{E}_N = 0$ for every N . No step of that implication is invoked before the independent missing inequality.

Stopping line. The unresolved object is the full signed expression (3.29) on contours exhausting the zero and pole sets, under the endpoint, multiplier, and gap constraints. Individual actual node removals are proved; a uniform signed pole-boundary estimate in the original Suzuki metric is not.

4. Combined calculation and one selected remaining lemma

4.1. What the explicit signed comparison now says

The direct multiplier attempt produced the exact matrix (1.20), including the off-diagonal $3H_{2,2} + H_{4,0}/4$, and the determinant (1.21). Its entries contain the actual prime, archimedean and polar terms on the entire interval. The endpoint-substituted linear-multiplier identities (1.23)-(1.24) exhibit the remaining signed packet-moment products explicitly. Neither these products nor the determinant has acquired an independently proved negative sign. The trigonometric attempt also has a completed endpoint: its large-frequency leading term is positive, by (1.17), so that limit cannot furnish a negative direction.

The exterior attempt proves more than a support gap. Its entire translation correction vanishes exactly on a null vector, and its core error is controlled by the vanishing Weil energy. Its complete

quadratic consequence is nevertheless exact invariance:

$$L_{v,b}L_{v,b}^\# = L_vL_v^\#. \quad (\text{CP13:4.1})$$

Thus the two exterior exponential factors cancel at the operation where the complete defect is formed. The added logarithmic regularity improves the exact-null Cauchy tail to $O(R^{-1/2})$ on compact evaluation sets. It does not change the phase of a correction-pole residue, or supply a bound uniform over an expanding contour.

At an upper correction pole the remaining simple-pole term is

$$-\varepsilon \frac{L_v(p)^2}{(A'/A)'(p)}. \quad (\text{CP13:4.2})$$

Its reflected partner and contour are included in the same pairing. At an A -zero the full Suzuki factor removes the point singularity, but the residue of the completed correction still contains

$$\varepsilon\{R'_\gamma c_\gamma^2 - 2R_\gamma c_\gamma \ell_\gamma\}. \quad (\text{CP13:4.3})$$

The second term explicitly couples the chosen zero to every other zero. Equations (4.2)-(4.3), together with their associated contour, identify the actual signed quantities whose estimate remains open.

4.2. The completed near-null calculation, before any exchange of limits

For each real same-parity core q_n , let $s_n = \mathcal{S}(q_n, q_n)$, $x_n = \mathcal{X}_\infty(q_n, q_n)$, and set $b = a$. Use the same symmetric zero cutoff as checkpoint 12 and define

$$U_{n,H}(z) = e^{iaz} L_{q_n,H,a}(z), \quad D_{n,H}(z) = U_{n,H}(z) - L_{q_n,H}(z). \quad (\text{CP13:4.4})$$

These are finite meromorphic expressions. Let $\mathcal{P}_{\Omega_H}$ be the paired contour functional from Group 3: its negative outer contour and the small positively oriented correction-pole circles are taken together, with Ω_H enclosing every cutoff node. Then the exact complete defect is

$$\Delta(q_n, q_n) = \lim_{H \rightarrow \infty} \mathcal{P}_{\Omega_H} \left[U_{n,H} U_{n,H}^\# - U_{n,H} D_{n,H}^\# - D_{n,H} U_{n,H}^\# + D_{n,H} D_{n,H}^\# \right]. \quad (\text{CP13:4.5})$$

This is an exact insertion into the full residue-boundary expression, including every cross term. For fixed n , the inner limit is the completed defect already justified in checkpoint 12.

On a fixed contour, the cutoff tails converge uniformly. The continuous-cutoff limit of $D_{n,H}$ is $e^{iaz} \mathcal{E}_a(q_n; z)$, and the independent estimate from Group 3 gives

$$|e^{iaz} \mathcal{E}_a(q_n; z)|^2 \leq Q_a(q_n) Q_a(r_{a,z}). \quad (\text{CP13:4.6})$$

Notice the exact cancellation of the two factors $e^{\pm 2a\Im z}$ here. After the original normalization, the entire difference between the paired expressions tends to zero on any fixed contour at rate

$$C_{\Omega,a,B_{\text{seq}}} \left\{ \sqrt{Q_a(q_n)/s_n} + Q_a(q_n)/s_n \right\}. \quad (\text{CP13:4.7})$$

Every quantity in this bound is defined in Group 3; the test energy $Q_a(r_{a,z})$ is explicitly calculated there. The same estimates apply to the derivatives needed by higher poles using slightly larger compact curves.

The complete exhaustion requires Ω_H to grow with H . The available constants include the integral of $|\mathcal{R}|$ on these curves and the actual normalized Cauchy evaluation factors. No uniform bound for their product, and no signed cancellation replacing it, has been proved. Hence (4.7) does not justify moving the n -limit inside (4.5).

Even a successful uniform error passage would still leave the principal signed term $\mathcal{P}[UU^\sharp]$. On an exact null vector (4.1) shows that it has the original full size. Therefore uniform error control alone is not presented as a sufficient closing lemma.

4.3. Selected quantified lemma and its exact margin

The selected missing statement uses the full principal term and the three calculated correction terms in (4.5).

Gap-paired remainder lemma – open. For every finite $a > 0$ with $Q_a \geq 0$, and every nonzero real parity $v \in \ker A_a$, there exists a same-parity sequence $q_n \in C_c^\infty(-a, a)$ converging to v in the $Q_a + \|\cdot\|_2^2$ form norm such that, with all objects defined in (4.4)-(4.5),

$$\liminf_{n \rightarrow \infty} \frac{\lim_{H \rightarrow \infty} \mathcal{P}_{\Omega_H} \left[U_{n,H} U_{n,H}^\sharp - U_{n,H} D_{n,H}^\sharp - D_{n,H} U_{n,H}^\sharp + D_{n,H} D_{n,H}^\sharp \right] + x_n}{s_n} \geq -1 + K_*. \quad (\text{CP13:4.8})$$

The limiting null vector obeys the exact endpoint prime identities and the complete multiplier matrix. Its core sequence obeys the quantitative regularizer estimate and converges to those identities in the proved topologies. The lemma asks for an independent estimate of this actual coupled expression. Its proof is not supplied by naming these constraints.

The margin is exactly $\varepsilon_* = 4429025/108864408576 > 0$: every hypothetical null core sequence already satisfies

$$\limsup_n \frac{\Delta(q_n, q_n) + x_n}{s_n} \leq -1 + K_0. \quad (\text{CP13:4.9})$$

Indeed $Q_a(q_n) \rightarrow 0$, $\liminf s_n > 0$, and $(\Delta + x_n)/s_n = Q_a(q_n)/s_n - 1 + x_n/s_n$. An interval-dependent lower threshold $-1 + K_0 + \eta_a$, with any independently proved $\eta_a > 0$, would also suffice; its contradiction margin would be precisely η_a . No value of such an η_a is asserted.

4.4. Strongest justified closing statement

This round proves the auxiliary theorems listed in the three closure records and the exact combined calculation (4.5)-(4.7). The unproved line is the lower estimate (4.8) for the complete signed pairing. Equivalently, the direct multiplier route still requires a negative direction in its explicitly calculated arithmetic matrix or a larger fully retained multiplier family.

If (4.8), or the stated strictly positive interval-dependent margin, is proved independently, the full dependency chain is

$$\text{exclusion of every finite first-null vector} \implies \mathfrak{W}(q, q) \geq 0 \quad (q \in C_c^\infty) \implies \text{RH} \implies \mathfrak{W} = \mathcal{S} \implies \mathcal{E}_N = 0 \quad (N \geq 1). \quad (\text{CP13:4.10})$$

The first implication uses the already retained small-interval positivity, continuity of the lowest eigenvalue, and attained first crossing. Weil's criterion gives the next implication. Only then is the correctly normalized Suzuki identity invoked. No RH-dependent orthogonality is used as an input to the missing estimate.

At present (4.8) is open. The target has not been proved by this round.

5. Source and verification scope

- Current checkpoint 12, read in full, including its preserved full correction and domain passage.
- `Michaels_Checkpoint_12_Null_Equation_Followup.md`, the archived follow-up calculation directive, recovered and included in the new checkpoint.
- M. Suzuki, *Weil's quadratic form via the screw function*, arXiv:2606.09096v2, especially the exact kernel (1.3), the localized logarithmic form and distribution formulas in Section 2, and the form-core/first-crossing framework. Its conditional Section 7 factorization is not an unconditional input.
- The unconditional local zero-count estimate from the Riemann-von Mangoldt formula is used only to prove explicitly displayed weighted sampling and tail estimates.

The multiplier signs, prime factors, endpoint eliminations, Laplace denominator signs, weighted sampling, regularizer estimate, upper/lower correction-pole distinction, and Carleman regularity were checked independently across the parallel derivations. Verification is analytical; no numerical zero experiments or increasing-degree computational proof ladder is used.

The preserved target is $\liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1$, independently of RH.

Checkpoint 14: Global Suzuki domain and signed dilations

Source: *Michaels_Global_Suzuki_Domain_and_Four_Closures_Checkpoint_14.md*. Equation tags carry the source prefix *CP14*; references within this unit retain their original local numbering.

23 September 2026

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0. Contract, normalization, and the unchanged target
1. The global Suzuki domain at fixed support
2. Finite inward dilation with every arithmetic term retained
3. Three-dimensional extension and complete renormalized energy
4. A quantified infinity regulator with the complete correction retained
5. Combined closing calculation and the exact remaining lemma
6. Sources, dependency record, and file record

0. Contract, normalization, and the unchanged target

This calculated continuation preserves the completed domain passage, actual arithmetic kernel, endpoint-complete exterior formulas, meromorphic correction, and original Suzuki metric from checkpoint 13. Its strongest new theorem is the global fixed-support Suzuki graph estimate. It also proves the finite arithmetic dilation identity and its exterior-collar constraint, the strong three-dimensional trace and renormalized energy, and an exponentially small budget for all poles introduced by the infinity regulator. The remaining complete signed estimate is not established; the original target remains open.

The unchanged target and its coordinate-free normalization are

$$\boxed{\mathcal{E}_N = \|[\Delta(\rho_j, \rho_i)]_{i,j \leq N}\|_F^2, \quad \Delta = \mathfrak{W} - \mathcal{S}, \quad \liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1.} \quad (\text{CP14:0.1})$$

Here (ρ_i) is the original Suzuki-orthonormal determining ladder; it is not a new monomial or prime-color metric. For unnormalized coordinates $q_1, \dots, q_N$ spanning the same determining subspace as $\rho_1, \dots, \rho_N$, set

$$G_N = [\mathcal{S}(q_j, q_i)], \quad Z_N = [\Delta(q_j, q_i)], \quad D_N = G_N^{-1/2} Z_N G_N^{-1/2}. \quad (\text{CP14:0.2})$$

All restrictions are to the positive metric quotient if needed. With column vectorization,

$$\mathcal{E}_N = \text{tr}(G_N^{-1} Z_N G_N^{-1} Z_N^*) = \text{vec}(Z_N)^* (\overline{G_N^{-1}} \otimes G_N^{-1}) \text{vec}(Z_N). \quad (\text{CP14:0.3})$$

No estimate below replaces either inverse metric by the identity.

Fix a hypothetical first-null interval $I_a = (-a, a)$, $a > 0$, with $Q_a \geq 0$ and $0 \neq v \in \ker A_a$. Here A_a is the closed localized operator, distinct from the entire xi function $A(z)$. As already proved, a real null vector of definite parity may be selected: $v(-x) = \varepsilon v(x)$, $\varepsilon \in \{1, -1\}$. All functions are extended by zero, and $\widehat{q}(\xi) = \int q(x) e^{i\xi x} dx$.

The exact data retained are

$$\begin{aligned}
 A(z) &= \xi(1/2 - iz) = \xi(1/2 + iz), \quad E = A + iA', \quad h = A/E, \quad \mathcal{S}(q, q) = \|Vq\|_{L^2(\mathbb{R})}^2, \\
 L_q(z) &= \sum_{\gamma} \frac{m_{\gamma} \hat{q}(\gamma)}{\gamma - z}, \quad Vq(z) = -\frac{i}{\sqrt{\pi}} h(z) L_q(z), \\
 C(z) &= \frac{1}{\pi} \int_{\mathbb{R}} |h(x)|^2 \left(\frac{1}{x - z} - \frac{x}{1 + x^2} \right) dx, \\
 \mathcal{R} &= C - ih \quad (\Im z > 0), \quad \mathcal{R} = C + ih^{\sharp} \quad (\Im z < 0).
 \end{aligned} \tag{CP14:0.4}$$

The last two definitions glue meromorphically; $f^{\sharp}(z) = \overline{f(\bar{z})}$. Actual upper correction poles are noncommon zeros of E , with the reflected lower correction. Real-node removals are made before evaluation. The corrected time convention is $P_{-t}(z) = P_t(-z)$.

The real-shift identity is preserved as an invariant:

$$L_{v,b}(z) = e^{-ibz} L_v(z), \quad L_{v,b} L_{v,b}^{\sharp} = L_v L_v^{\sharp}, \quad |b| \leq a. \tag{CP14:0.5}$$

It supplies exterior estimates for individual factors and does not give a decaying factor for the complete central quadratic product.

For each core input and finite zero cutoff H , the complete defect formula is

$$\begin{aligned}
 \Delta_H(q, r) &= -\frac{1}{2\pi i} \oint_{\partial\Omega_H} \mathcal{R}(z) L_{q,H}(z) L_{r,H}^{\sharp}(z) dz \\
 &\quad + \sum_{p \in P_{\Omega_H}} \sum_{k=1}^{d_p} \frac{r_{p,k}}{(k-1)!} \partial_z^{k-1} [L_{q,H}(z) L_{r,H}^{\sharp}(z)] \Big|_{z=p},
 \end{aligned} \tag{CP14:0.6}$$

where Ω_H encloses every active finite-cutoff node, every actual principal part $\sum_{k=1}^{d_p} r_{p,k}(z-p)^{-k}$ inside it is retained, and the boundary avoids every node and pole. The established core limit is taken in its original order. For the new graph-core sequence q_n , the defect at v is defined by the justified form passage $\lim_n \lim_H \Delta_H(q_n, q_n)$. Local convergence of resolvents does not authorize interchanging these limits with an infinite signed pole sum.

The already established extraction constants and margin are

$$K_* = \frac{3567}{509894} < \frac{1}{140}, \quad K_0 = \frac{17607901}{2531730432}, \quad \varepsilon_* = K_* - K_0 = \frac{4429025}{108864408576} > 0, \tag{CP14:0.7}$$

$$0 \leq \mathcal{X}_{\infty}(q, q) \leq K_0 \mathcal{S}(q, q), \quad \mathcal{T} = \Delta + \mathcal{X}_{\infty}. \tag{CP14:0.8}$$

The sharper constant K_0 is the retained checkpoint-12 comparison; K_* is the original required threshold. These bounds do not themselves estimate the complete signed remainder.

1. Closing group 1 – The global Suzuki domain at fixed support

1.1. Statement, conventions, and an explicit constant

Let $a > 0$, and let $q \in C_c^{\infty}(-a, a)$, extended by zero. Set

$$\hat{q}(x) = \int_{\mathbb{R}} q(t) e^{ixt} dt, \quad \mathcal{L} = \log |D|, \quad \|q\|_{\mathcal{H}}^2 = \frac{1}{2\pi} \int_{\mathbb{R}} [1 + \log^2(2 + |x|)] |\hat{q}(x)|^2 dx. \tag{CP14:1.1}$$

The metric is exactly $\mathcal{S}(q, q) = \|Vq\|_2^2$, with $V = -L_0$, $A(z) = \xi(1/2 - iz) = \xi(1/2 + iz)$, $E = A + iA'$, and $h = A/E$. The corrected time convention is $P_{-t}(z) = P_t(-z)$. No RH-dependent orthogonality is used.

Fixed-support graph theorem. There is an explicitly finite constant C_a for which

$$\boxed{\|Vq\|_2^2 \leq C_a \|q\|_{\mathcal{H}}^2} \quad (\text{CP14:1.2})$$

One permissible constant is the following. Put

$$\begin{aligned} c_\psi &= |\psi(1/4)| + 4 + \log 4, & d_\gamma &= 5 + \frac{1}{2} \log \pi + \frac{1}{2} c_\psi, & D_\gamma &= (d_\gamma^2 + 1/4)^{1/2}, \\ B_a^{\text{prime}} &= \sum_{\log n \leq a} \frac{\Lambda(n)}{\sqrt{n}}, & P_a^{\text{pol}} &= \sqrt{e^a - 1} + \sqrt{1 - e^{-a}}, \\ D_a &= \sqrt{2} \left(D_\gamma + B_a^{\text{prime}} + \frac{\pi}{2} + 1 + P_a^{\text{pol}} \right), \\ H_a &= 1 + \log 3 + \frac{\pi}{2} + \sqrt{1 + 4a/\pi}, & \boxed{C_a} &= 4D_a^2 H_a^2. \end{aligned} \quad (\text{CP14:1.3})$$

These constants are upper bounds; no smallness is asserted or needed for the domain theorem.

The starting formula is the actual Suzuki transform, arXiv:2301.00421v3, equations (1.5)-(1.7), with the corrected two-sided convention already fixed in checkpoints 11-13. The proof below differentiates its explicit Lerch term before estimating any operator.

1.2. Differentiation of the complete positive-time kernel

Write $\lambda_j = 2j + 1/2$, $\ell_n = \log n$, and $b_n = \Lambda(n)/\sqrt{n}$. The Lerch difference in P_t , for $t > 0$, equals

$$-\sum_{j=0}^{\infty} \frac{e^{-\lambda_j t}}{\lambda_j(\lambda_j - iz)}.$$

Indeed, the difference of the two Lerch denominators is $2/(\lambda_j - iz) - 2/\lambda_j = 2iz/[\lambda_j(\lambda_j - iz)]$. Its differentiated sharp is therefore

$$\partial_t(P_t^{\text{Lerch}})^{\sharp}(x) = \sum_{j=0}^{\infty} \frac{e^{-\lambda_j t}}{\lambda_j + ix} = \int_0^{\infty} \kappa(t+u) e^{-ixu} du, \quad \kappa(r) = \frac{e^{-r/2}}{1 - e^{-2r}}. \quad (\text{CP14:1.4})$$

The gamma difference in the source formula is independent of t , hence differentiates to zero. Its role reappears exactly when the zeta logarithmic derivative is completed below; it has not been omitted.

The full differentiated kernel is

$$\begin{aligned} \partial_t P_t^{\sharp}(x) &= \frac{2e^{t/2}}{1 - 2ix} - \frac{2e^{-t/2}}{1 + 2ix} - \frac{\zeta'}{\zeta}(1/2 + ix) e^{ixt} \\ &\quad - \sum_{\ell_n \leq t} b_n e^{ix(t - \ell_n)} + \int_0^{\infty} \kappa(t+u) e^{-ixu} du. \end{aligned} \quad (\text{CP14:1.5})$$

Every finite prime packet has zero value at its switching point before differentiation, so differentiating produces no delta mass there. The locally logarithmic singularity at $t = 0$ is integrable; the original kernel is continuous at zero and contributes no delta there either.

For g supported in $(0, a)$, define

$$\begin{aligned}
 F_g(x) &= \int_0^a g(t) e^{ixt} dt, & Hg(u) &= \int_0^a \kappa(t+u) g(t) dt \quad (u > 0), \\
 T_n g(u) &= \mathbf{1}_{u>0} g(u + \ell_n), & G_{\pm}(g) &= \int_0^a e^{\pm t/2} g(t) dt, \\
 P_g(x) &= \frac{2G_+(g)}{1-2ix} - \frac{2G_-(g)}{1+2ix}, & & \text{(CP14:1.6)} \\
 Rg(x) &= Pg(x) - \sum_{\ell_n \leq a} b_n \widehat{T_n g}(x) + \widehat{Hg}(-x), \\
 Dg(x) &= -\frac{\zeta'}{\zeta} (1/2 + ix) F_g(x) + Rg(x).
 \end{aligned}$$

The hats on positive-half-line functions mean their zero-extended Fourier transforms. Integrating (1.5) gives precisely

$$V(g\mathbf{1}_{t>0})(x) = \frac{h(x)}{\sqrt{\pi}} Dg(x). \quad \text{(CP14:1.7)}$$

For general q , let $g_+(t) = q(t)$, $g_-(t) = q(-t)$ for $t > 0$. The corrected negative-time derivative supplies a minus sign:

$$Vq(x) = \frac{h(x)}{\sqrt{\pi}} [Dg_+(x) - Dg_-(-x)]. \quad \text{(CP14:1.8)}$$

This is the complete operator decomposition, including the negative-time convention. On the real axis $|h(-x)| = |h(x)|$, so the reflected half-line has exactly the same norm bounds.

1.3. Carleman bound, completed gamma multiplier, and the commutator

For $r > 0$,

$$0 < \kappa(r) \leq \frac{1}{2r} + e^{-r/2}.$$

The half-line Carleman operator $Cf(u) = \int_0^\infty f(t)/(u+t)dt$ has norm π , by the Mellin transform (or the sharp Schur weight $t^{-1/2}$). The rank-one kernel $e^{-(t+u)/2}$ has norm one. Consequently

$$\|H\|_{L^2(0,a) \rightarrow L^2(0,\infty)} \leq \pi/2 + 1. \quad \text{(CP14:1.9)}$$

The domination applies to absolute values, so it controls complex inputs as well.

Put $s = 1/2 + ix$. Completion of the logarithmic derivative gives the exact real-axis identity

$$-h(x) \frac{\zeta'}{\zeta}(s) = (1 - h(x)) + h(x) d_0(x), \quad d_0(x) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi(s/2). \quad \text{(CP14:1.10)}$$

The relation $ihA'/A = 1 - h$ proves the first term; at a real zero, the displayed product is interpreted by removal. Since A, A' are real there,

$$|h| \leq 1, \quad |1 - h| \leq 1.$$

Thus no bound on ζ'/ζ near its real poles is required.

Here is an elementary sufficient bound for the gamma factor. The digamma series gives, with $c = 1/4$, $Y = |x|/2$,

$$|\psi(c + iY) - \psi(c)| \leq 4 + \sum_{n \geq 1} \frac{Y}{(n+c)\sqrt{(n+c)^2 + Y^2}} \leq 4 + \operatorname{arsinh}(4Y).$$

The last step integrates the decreasing summand from c to infinity. Since $\operatorname{arsinh}(4Y) \leq \log(1+8Y)$,

$$|\psi(1/4 + ix/2)| \leq c_\psi + \log(2 + |x|).$$

Using $|1/s| + |1/(s-1)| \leq 4$, (1.10) yields

$$\left| -h \frac{\zeta'}{\zeta}(s) \right| \leq d_\gamma + \frac{1}{2} \log(2 + |x|) \leq D_\gamma \sqrt{1 + \log^2(2 + |x|)}. \quad (\text{CP14:1.11})$$

The half-line truncation must also preserve logarithmic regularity. Let P_+ be multiplication by $\mathbf{1}_{(0,\infty)}$. For the distributional kernel of $\mathcal{L}$, only the off-diagonal term $-1/(2|x-y|)$ survives in the commutator. With the negative half-line reflected to $(0, \infty)$,

$$\boxed{\mathbf{C} := [\mathcal{L}, P_+] = \frac{1}{2} \begin{pmatrix} 0 & C \\ -C & 0 \end{pmatrix}, \quad \|\mathbf{C}\| = \frac{\pi}{2}.} \quad (\text{CP14:1.12})$$

The equality follows first for smooth rapidly decreasing inputs: their half-line truncations have at most a jump, whose Fourier decay is sufficient for the logarithmic graph norm. The finite-part diagonal and every diagonal delta cancel. The Carleman formula extends boundedly to all of L^2 . Approximating in the Fourier multiplier graph then gives

$$P_+ D(\mathcal{L}) \subset D(\mathcal{L}), \quad \mathcal{L} P_+ q = P_+ \mathcal{L} q + \mathbf{C} q. \quad (\text{CP14:1.13})$$

The same holds for $P_- = I - P_+$, with the opposite commutator sign. This proof does not require a trace of q at zero.

For a function supported in $[-a, a]$, $|\widehat{q}(x)| \leq \sqrt{2a} \|q\|_2$. Thus

$$\frac{1}{2\pi} \int_{|x| < 1} \log^2 |x| |\widehat{q}(x)|^2 dx \leq \frac{4a}{\pi} \|q\|_2^2, \quad \|\mathcal{L} q\|_2 \leq \sqrt{1 + 4a/\pi} \|q\|_{\mathcal{H}}. \quad (\text{CP14:1.14})$$

Also $\log(2 + |x|) \leq \log 3 + |\log |x||$, so (1.12)-(1.14) imply

$$\|P_\pm q\|_{\mathcal{H}} \leq H_a \|q\|_{\mathcal{H}}. \quad (\text{CP14:1.15})$$

Reflection is isometric in this norm.

Plancherel, (1.9), and (1.11) bound the gamma-completed term by $\sqrt{2} D_\gamma \|g\|_{\mathcal{H}}$, all finite prime packets by $\sqrt{2} B_a^{\text{prime}} \|g\|_2$, and the Lerch operator by $\sqrt{2}(\pi/2+1) \|g\|_2$. Since $\|(1 \pm 2ix)^{-1}\|_2 = \sqrt{\pi/2}$, the complete polar term is bounded by $\sqrt{2} P_a^{\text{pol}} \|g\|_2$. Therefore

$$\|Vq\|_2 \leq D_a (\|g_+\|_{\mathcal{H}} + \|g_-\|_{\mathcal{H}}) \leq 2D_a H_a \|q\|_{\mathcal{H}}.$$

This proves (1.2). The exact identity (1.8) retains all interactions. Triangle inequalities in this domain estimate bound those interactions; they do not redefine the Suzuki energy as a sum of separate energies.

1.4. Same-parity graph approximation and uniform spectral tightness

Assume the hypothetical first-null hypotheses $Q_a \geq 0$, $0 \neq v \in \ker A_a$, with v real and of definite parity as justified in checkpoint 13. Its equation (2.40) establishes

$$\mathcal{L}v \in L^2(\mathbb{R}), \quad \|\mathcal{L}v\|_2^2 \leq ((C_a^{\text{op}})^2 + \pi^2/2) \|v\|_2^2. \quad (\text{CP14:1.16})$$

In particular $v \in \mathcal{H}$.

For $s > 0$, define $U_s v(t) = e^{s/2} v(e^s t)$. Its support lies in $[-ae^{-s}, ae^{-s}]$, and Fourier scaling gives the exact graph identity

$$\mathcal{L}U_s v = U_s(\mathcal{L}v + sv). \quad (\text{CP14:1.17})$$

The strong continuity of dilations on L^2 proves $U_s v \rightarrow v$ in the $\mathcal{L}$ -graph norm as $s \downarrow 0$. This implies convergence in $\mathcal{H}$ by the pointwise bound used above.

Choose a real even smooth mollifier η of mass one. For fixed $s > 0$, convolution with η_ϵ converges in the same graph norm by dominated convergence of its bounded Fourier multipliers. Choose $s_n \downarrow 0$ and $0 < \epsilon_n < a(1 - e^{-s_n})/2$ diagonally, and set

$$q_n = \eta_{\epsilon_n} * U_{s_n} v \in C_c^\infty(-a, a). \quad (\text{CP14:1.18})$$

The sequence is real and has the same parity as v . It converges to v in $\mathcal{H}$. It is also a form-core sequence: the exact arithmetic expression $Q_a(u, w) = \langle \mathcal{L}u, w \rangle - \langle \mathcal{B}_a u, w \rangle$ for logarithmically regular zero-extended inputs, with $\mathcal{B}_a$ bounded, makes the graph convergence imply form-norm convergence. This agrees with the already established closed form and its core, rather than selecting a new extension.

The estimate (1.2) makes Vq_n Cauchy in L^2 . Checkpoint 12's local meromorphic continuity identifies its limit with the existing maximal transform Vv . That maximal transform is injective by the completed domain theorem. Consequently

$$\boxed{Vq_n \longrightarrow Vv \text{ in } L^2(\mathbb{R}), \quad 0 < \mathcal{S}(v, v) = \|Vv\|_2^2 < \infty, \quad \mathcal{S}(q_n, q_n) \rightarrow \mathcal{S}(v, v).} \quad (\text{CP14:1.19})$$

Furthermore,

$$\boxed{\lim_{R \rightarrow \infty} \sup_n \int_{|x| > R} |Vq_n(x)|^2 dx = 0.} \quad (\text{CP14:1.20})$$

Indeed, for sufficiently large n , use the small global L^2 distance to Vv ; the remaining finitely many transforms have uniformly vanishing tails. This is uniform spectral tightness for the constructed sequence. It does not assert uniform control of arbitrary meromorphic evaluations off the real line.

1.5. Insert the null arithmetic equation into the complete transform

The equation supplies more than abstract finiteness: it can be substituted into an exact operator decomposition. Let $J = P_+ - P_-$, and define

$$d_1(x) = d_0(x) - \frac{1}{2} \log |x|, \quad q_\pm = P_\pm q.$$

Using $\mathcal{L}Jq = J\mathcal{L}q + 2Cq$, the complete identity (1.8) becomes

$$\boxed{\begin{aligned} \sqrt{\pi} Vq &= (1 - h)\widehat{q} + \frac{h}{2} \widehat{J\mathcal{L}q + 2Cq} \\ &\quad + h\{d_1(x)\widehat{q}_+(x) - d_1(-x)\widehat{q}_-(x)\} \\ &\quad + h\{Rg_+(x) - Rg_-(-x)\}. \end{aligned}} \quad (\text{CP14:1.21})$$

For example the coefficient of $\widehat{q}$ follows from oddness of A'/A ; it is $1 - h$ in both half-line channels. The d_1 terms are interpreted on fixed-support inputs: their logarithmic low-frequency singularity is square integrable against their bounded Fourier transforms. The sum in (1.21) equals the nonsingular completed expression (1.8).

Define the exterior function

$$(\mathbf{E}v)(t) = -\frac{1}{2}\mathbf{1}_{|t|>a}\int_{-a}^a\frac{v(y)}{|t-y|}dy.$$

Checkpoint 13's exact equation, including its endpoint-distribution audit, is

$$\mathcal{L}v = \mathbf{1}_{I_a}\mathcal{B}_av + \mathbf{E}v, \quad \mathcal{B}_av = \log(2\pi)v + \sum_{\ell_n \leq 2a} b_n\{v(\cdot + \ell_n) + v(\cdot - \ell_n)\} + (R * v)|_{I_a}, \quad R = r''. \quad (\text{CP14:1.22})$$

Substitute it into (1.21):

$$\boxed{\begin{aligned} \sqrt{\pi}Vv &= (1-h)\widehat{v} + \frac{h}{2}J\mathbf{1}_{I_a}\widehat{\mathcal{B}_av} + \widehat{J\mathbf{E}v} + 2Cv \\ &\quad + h\{d_1(x)\widehat{v}_+(x) - d_1(-x)\widehat{v}_-(x)\} + h\{Rg_+(x) - Rg_-(-x)\}. \end{aligned}} \quad (\text{CP14:1.23})$$

Every translation in $\mathcal{B}_a$, each half-line prime packet in $\mathbf{R}$, the polar terms, the gamma multiplier, the Carleman operator, and the exterior endpoint contributions remain explicit. There is no assumed cancellation between different appearances of the prime packets. Squaring (1.23) retains all pairwise real cross terms: if its four displayed summands are $Y_1, \dots, Y_4$, then

$$\pi\mathcal{S}(v, v) = \sum_{j=1}^4 \|Y_j\|_2^2 + 2\Re \sum_{1 \leq j < k \leq 4} \langle Y_j, Y_k \rangle. \quad (\text{CP14:1.24})$$

This is a finite norm identity on the exact null vector. A sign bound on the cross terms has not been inferred from the separate norm estimates. In particular, a useful explicit consequence of (1.2) and (1.16) is

$$\mathcal{S}(v, v) \leq C_a \left[1 + \left(\log 3 + \sqrt{(C_a^{\text{op}})^2 + \pi^2/2} \right)^2 \right] \|v\|_2^2.$$

This upper bound depends on the finite interval and still makes no comparison in the opposite direction.

1.6. Exact-null remainder and closure record

The graph-core sequence gives $Q_a(q_n) \rightarrow 0$, $\mathcal{S}(q_n) \rightarrow \mathcal{S}(v) > 0$, and, by the bounded extraction multiplier, $\mathcal{X}_\infty(q_n) \rightarrow \mathcal{X}_\infty(v)$. Thus the complete defect and retained remainder extend along this sequence with

$$\boxed{\Delta(v, v) = -\mathcal{S}(v, v), \quad \mathcal{T}(v, v) = -\mathcal{S}(v, v) + \mathcal{X}_\infty(v, v).} \quad (\text{CP14:1.25})$$

These equalities use the complete correction by its already established form identity; no separate uniform pole-sum limit is claimed here. The complete cutoff formulas can be evaluated on the approximants in their original order of limits.

The extraction comparison now acts on the exact null vector:

$$\mathcal{T}(v, v) \leq -(1 - K_0)\mathcal{S}(v, v). \quad (\text{CP14:1.26})$$

The required opposite estimate

$$\boxed{\mathcal{T}(v, v) \geq -(1 - K_*)\mathcal{S}(v, v)} \quad (\text{CP14:1.27})$$

would contradict (1.26) with margin $(K_* - K_0)\mathcal{S}(v, v) > 0$. It has not followed from the graph theorem. In fact the graph estimates have positive, interval-dependent constants and contain no independent lower estimate for the signed cross terms in (1.24) or the complete correction.

1. **Exact theorem and domain.** The unconditional fixed-support estimate (1.2) holds on $C_c^\infty(-a, a)$, extends to all supported inputs in $\mathcal{H}$, and supplies (1.19)-(1.20) for every hypothetical real parity null vector.
2. **Completed calculation.** The differentiated Lerch, completed gamma and polar terms, finite prime packets, negative-time sign, exact commutator, graph core, tightness, and null substitution are proved above.
3. **Consequence downstream.** Groups 2-4 may use the exact null vector with finite, strictly positive original Suzuki norm, and the same-parity core sequence converges in that norm. No approximation mass can escape along this sequence.
4. **Remaining inequality and failed implication.** Equation (1.27) is still independent. The attempted norm estimate establishes only finiteness and the explicit unsigned upper bound (1.2); it gives no lower sign to the complete paired expression or its mixed terms. The original target $\liminf_N \mathcal{E}_N \leq 1$ remains unproved by this group.

2. Closure group 2 – Finite inward dilation, with every arithmetic term retained

2.1. Hypotheses, exact form, and graph-domain passage

Fix $a > 0$, write $I_a = (-a, a)$, and extend all functions by zero to $\mathbb{R}$. The Fourier convention is $\widehat{u}(\xi) = \int u(x)e^{i\xi x} dx$. Let $\mathcal{L} = \log |D|$, $\ell_n = \log n$, $b_n = \Lambda(n)/\sqrt{n}$, and retain the exact checkpoint-13 remainder

$$R(t) = \frac{e^{-|t|/2}}{1 - e^{-2|t|}} - 2 \cosh(t/2) - \frac{1}{2|t|} \quad (t \neq 0), \quad R(0) = -\frac{7}{4}. \quad (\text{CP14:2.1})$$

It is even, continuous and locally Lipschitz. Its derivative on $(0, \infty)$ has a finite right limit at zero. No Taylor truncation replaces R below.

The exact localized form, initially on $C_c^\infty(I_a)$, is

$$\begin{aligned} Q_a(u) = & \frac{1}{2\pi} \int_{\mathbb{R}} \log |\xi| |\widehat{u}(\xi)|^2 d\xi - \log(2\pi) \|u\|_2^2 \\ & - 2 \sum_{\ell_n < 2a} b_n \Re C_u(\ell_n) - \int_{-2a}^{2a} R(t) C_u(t) dt, \quad C_u(t) = \int u(x+t) \overline{u(x)} dx. \end{aligned} \quad (\text{CP14:2.2})$$

For a real u , C_u is real and even. Formula (2.2) follows from the full finite-part kernel in checkpoint 13; in particular the coefficient of the Fourier multiplier is exactly one, and the constant is $-\log(2\pi)$. All completed polar terms are already in R .

Assume, hypothetically,

$$Q_a \geq 0, \quad 0 \neq v \in \ker A_a, \quad v \text{ real}, \quad v(-x) = \varepsilon v(x), \quad \varepsilon \in \{1, -1\}. \quad (\text{CP14:2.3})$$

Checkpoint 13 proves

$$v \in D(\mathcal{L}), \quad \|\mathcal{L}v\|_2^2 \leq \kappa_a^2 \|v\|_2^2, \quad \kappa_a^2 = (C_a^{\text{op}})^2 + \frac{\pi^2}{2}. \quad (\text{CP14:2.4})$$

Here and below $D(\mathcal{L})$ is the whole-line multiplier domain, with the compact support making the negative logarithmic frequency singularity integrable.

For $s \geq 0$, define $U_s v(x) = e^{s/2} v(e^s x)$. Direct Fourier transformation gives

$$\widehat{U_s v}(\xi) = e^{-s/2} \widehat{v}(e^{-s} \xi), \quad \mathcal{L}U_s v = U_s \mathcal{L}v + sU_s v. \quad (\text{CP14:2.5})$$

Consequently $U_s v \in D(\mathcal{L})$, its support is contained in $[-ae^{-s}, ae^{-s}]$, it has the same parity, and

$$U_s v \longrightarrow v \quad \text{in the graph norm of } \mathcal{L} \quad (s \downarrow 0). \quad (\text{CP14:2.6})$$

Indeed dilation is strongly continuous on L^2 , and (2.5) treats the logarithmic part. For $s > 0$, mollification with a real even compactly supported approximate identity of radius less than $a(1-e^{-s})$ produces same-parity $C_c^\infty(I_a)$ approximants converging in that graph norm. Their Fourier multipliers converge pointwise to one and are uniformly bounded, so dominated convergence applies to $(1 + |\log |\xi||^2) |\widehat{U_s v}|^2$. The bounded prime and R -operators then imply closed-form convergence. Thus all subsequent uses of (2.2) on $U_s v$ are legitimate.

No derivative of v , no endpoint trace of v , and no differentiability in s in the form norm is asserted.

2.2. Exact finite-dilation identity and its mixed term

Put $M = \|v\|_2^2$. Unitarity and (2.5) give

$$\langle \mathcal{L}U_s v, U_s v \rangle = \langle \mathcal{L}v, v \rangle + sM, \quad C_{U_s v}(t) = C_v(e^s t). \quad (\text{CP14:2.7})$$

Changing variables in the complete R -integral and subtracting $Q_a(v) = 0$ yields

$$\boxed{D_v(s) := Q_a(U_s v) = sM - 2 \sum_{\ell_n < 2a} b_n \{C_v(e^s \ell_n) - C_v(\ell_n)\} - \int_{-2a}^{2a} \{e^{-s} R(e^{-s} t) - R(t)\} C_v(t) dt.} \quad (\text{CP14:2.8})$$

This is the proposed identity with its normalization verified from the original form. Every prime power in the fixed interval remains present. Because $C_v(t) = 0$ for $|t| \geq 2a$, a term disappears continuously when $e^s \ell_n$ reaches $2a$. There is no unrecorded threshold term.

The complete mixed term is also fixed exactly. The null equation says $Q_a(v, w) = 0$ for every $w \in \mathfrak{D}(Q_a)$. Hence

$$\boxed{D_v(s) = Q_a(U_s v - v).} \quad (\text{CP14:2.9})$$

This follows by expanding all four terms of the sesquilinear form; both mixed terms vanish by the null equation. In particular $D_v(s) \rightarrow 0$ as $s \downarrow 0$, and the hypothetical nonnegative form forces $D_v(s) \geq 0$. A proof that the arithmetic expression in (2.8) is strictly negative at one finite scale would contradict (2.3). Formula (2.9) supplies no stationarity assertion: graph continuity of dilation does not give $Q_a(U_s v - v) = o(s)$.

2.3. Exact endpoint substitution in the completed polar contribution

For $t > 0$, write

$$A_{\text{ar}}(t) = \frac{e^{-t/2}}{1 - e^{-2t}} - \frac{1}{2t}, \quad A_{\text{ar}}(0) = \frac{1}{4}, \quad R(t) = A_{\text{ar}}(|t|) - 2 \cosh(t/2).$$

Let

$$F(r) = \int_{-a}^a e^{rx} v(x) dx, \quad p = e^{-s}.$$

Since $F(-r) = \varepsilon F(r)$,

$$\int_{-2a}^{2a} \cosh(rt) C_v(t) dt = \varepsilon F(r)^2. \quad (\text{CP14:2.10})$$

The complete finite-scale expression therefore equals

$$\boxed{\begin{aligned} D_v(s) = sM - 2 \sum_{\ell_n < 2a} b_n \{C_v(\ell_n/p) - C_v(\ell_n)\} \\ - \int_{-2a}^{2a} \{pA_{\text{ar}}(p|t|) - A_{\text{ar}}(|t|)\} C_v(t) dt \\ + 2\varepsilon \{pF(p/2)^2 - F(1/2)^2\}. \end{aligned}} \quad (\text{CP14:2.11})$$

This identity keeps the polar terms coupled and signed before taking any absolute value.

Here is the exact insertion of checkpoint 13's endpoint equations. Let

$$J(t) = -\frac{t}{2} \{\psi(1/4) - \log \pi\} - \frac{1}{4} \{\Phi(1, 2, 1/4) - e^{-t/2} \Phi(e^{-2t}, 2, 1/4)\}, \quad t \geq 0,$$

$$I_n = \int_{-a}^{a-\ell_n} (a - \ell_n - y) v(y) dy, \quad J_n = \int_{-a}^{a-\ell_n} v(y) dy,$$

$$U_a = \int_{-a}^a J(a - y) v(y) dy, \quad V_a = \int_{-a}^a J'(a - y) v(y) dy.$$

Empty prime intervals contribute zero. All integrals are finite because J' has only a logarithmic singularity. Define the two actual signed endpoint functionals

$$P_e(v) = \sum_{\ell_n < 2a} b_n J_n + V_a, \quad P_o(v) = \sum_{\ell_n < 2a} b_n (I_n - aJ_n) + U_a - aV_a,$$

$$d_a = 4a \cosh(a/2) - 8 \sinh(a/2) > 0.$$

The endpoint-complete null equation gives

$$F(1/2) = \frac{P_e(v)}{4 \sinh(a/2)} \quad (\varepsilon = 1), \quad F(1/2) = \frac{P_o(v)}{d_a} \quad (\varepsilon = -1). \quad (\text{CP14:2.12})$$

Thus the last line of (2.11) is exactly

$$\begin{cases} 2pF(p/2)^2 - \frac{P_e(v)^2}{8 \sinh^2(a/2)}, & \varepsilon = 1, \\ -2pF(p/2)^2 + \frac{2P_o(v)^2}{d_a^2}, & \varepsilon = -1. \end{cases} \quad (\text{CP14:2.13})$$

Every mixed term inside the squared endpoint functional is retained: for example P_e^2 includes $\sum_{m,n} b_m b_n J_m J_n$, $2V_a \sum b_n J_n$, and V_a^2 . The new scaled moment $F(p/2)$ remains. Endpoint equations alone neither replace it by $F(1/2)$ nor assign signs to the shifted autocorrelations in (2.11).

2.4. A second exact null identity: finite dilation and the exterior collar

The full equation provides more than its diagonal pairing. Let $\mathcal{B}_a$ be the bounded operator on $L^2(I_a)$

$$(\mathcal{B}_a v)(x) = \log(2\pi)v(x) + \sum_{\ell_n < 2a} b_n \{v(x + \ell_n) + v(x - \ell_n)\} + \int_{-a}^a R(x - y)v(y) dy. \quad (\text{CP14:2.14})$$

Extend $\mathcal{B}_a v$ by zero. The complete whole-line logarithmic equation is

$$\mathcal{L}v = \mathbf{1}_{I_a} \mathcal{B}_a v + E_v, \quad E_v(t) = -\frac{\mathbf{1}_{|t| > a}}{2} \int_{-a}^a \frac{v(y)}{|t - y|} dy. \quad (\text{CP14:2.15})$$

The two Carleman estimates of checkpoint 13 give $\|E_v\|_2 \leq \pi \|v\|_2 / \sqrt{2}$.

Put $A_v(s) = \langle U_s v, v \rangle$. Self-adjointness of the whole-line multiplier, (2.5), and the null equation tested against $U_s v$ yield

$$\boxed{sA_v(s) = \langle \mathcal{B}_a v, U_s v \rangle - \langle U_s(\mathbf{1}_{I_a} \mathcal{B}_a v), v \rangle + C_v(s)}, \quad (\text{CP14:2.16})$$

where the entire exterior term is

$$\boxed{C_v(s) = \frac{e^{s/2}}{2} \int_{ae^{-s} < |x| < a} \int_{-a}^a \frac{v(x)v(y)}{|e^s x - y|} dy dx}. \quad (\text{CP14:2.17})$$

The integral is absolutely defined by the same Carleman estimate applied to $|v|$. Its useful bound is

$$|C_v(s)| \leq \frac{\pi}{\sqrt{2}} \|v\|_2 \|\mathbf{1}_{ae^{-s} < |x| < a} v\|_2 \longrightarrow 0. \quad (\text{CP14:2.18})$$

There is no assumed boundary value of v .

For complete arithmetic bookkeeping, the two $\log(2\pi)$ terms in (2.16) cancel. The prime part left on its right is

$$\begin{aligned} \sum_{\ell_n < 2a} b_n \left[\int_{-a}^a \{v(x + \ell_n) + v(x - \ell_n)\} U_s v(x) dx \right. \\ \left. - e^{s/2} \int_{|x| < ae^{-s}} v(x) \{v(e^s x + \ell_n) + v(e^s x - \ell_n)\} dx \right], \end{aligned}$$

and the remaining R -part is

$$\int_{I_a^2} R(x - y)v(y)U_s v(x) dy dx - e^{s/2} \int_{|x| < ae^{-s}} \int_{I_a} R(e^s x - y)v(y)v(x) dy dx. \quad (\text{CP14:2.19})$$

Equations (2.16)-(2.19) are an endpoint-complete finite-scale use of the full null equation. The collar is a signed bilinear pairing; parity does not make its two factors pointwise nonnegative. Replacing it by a positive boundary value would be an additional, unproved assertion.

2.5. Quantitative attempted estimate near the original scale

The logarithmic regularity does give an explicit correlation modulus. Fourier inversion and $|e^{iu} - 1| \leq \min(2, |u|)$ imply, for every $T > 1$,

$$|C_v(t+h) - C_v(t)| \leq |h|TM + \frac{2\|\mathcal{L}v\|_2^2}{\log^2 T}, \quad t \in \mathbb{R}. \quad (\text{CP14:2.20})$$

Only $|\xi| > T$ is used in the second term, so the zero-frequency logarithmic singularity causes no problem. For $0 < |h| < e^{-2}$, set $T = |h|^{-1/2}$ and use (2.4):

$$|C_v(t+h) - C_v(t)| \leq M \left(\sqrt{|h|} + \frac{8\kappa_a^2}{\log^2(1/|h|)} \right). \quad (\text{CP14:2.21})$$

The estimate is uniform in the base point t , including every prime threshold.

Define the finite exact-kernel constant

$$D_R(a) = 4a \sup_{0 \leq u \leq 2a} \{|R(u)| + u|R'_+(u)|\}.$$

The derivative at zero is the right derivative; the product $uR'_+(u)$ vanishes there. Differentiating the smooth scalar function $e^{-s}R(e^{-s}t)$ for $t \neq 0$, then applying the mean value integral, gives

$$\left| \int_{-2a}^{2a} \{e^{-s}R(e^{-s}t) - R(t)\}C_v(t) dt \right| \leq sD_R(a)M. \quad (\text{CP14:2.22})$$

This differentiates only the explicit kernel, not v , C_v , or the dilation orbit in the form domain.

Let $h_n(s) = (e^s - 1)\ell_n$, and take $s > 0$ small enough that every active $h_n(s) < e^{-2}$. Then the actual attempted upper/lower estimate is

$$\left| \frac{D_v(s)}{M} - s \right| \leq 2 \sum_{\ell_n < 2a} b_n \left(\sqrt{h_n(s)} + \frac{8\kappa_a^2}{\log^2(1/h_n(s))} \right) + sD_R(a). \quad (\text{CP14:2.23})$$

The bound tends to zero, as required by the closed form argument. Its right side need not be $o(s)$; in fact this bound has terms of size $1/\log^2(1/s)$. It therefore proves neither stationarity nor the signed excess needed to reverse the positive term sM . This is the precise failure of the estimate based solely on the completed logarithmic regularity. It is not a claim that the actual signed increment has this size.

2.6. The very small-support limit has a positive leading term

Let

$$s_p = \max\{0, \log(2a/\log 2)\}, \quad A_v = 2 \sum_{\ell_n < 2a} b_n C_v(\ell_n) + \int_{-2a}^{2a} R(t)C_v(t) dt.$$

For $s \geq s_p$, all shifted prime correlations $C_v(e^s \ell_n)$ vanish exactly. Set $m_0 = \int v$ and $L_R(a) = \sup_{0 < u \leq 2a} |R'(u)|$. The retained remainder in the expansion at zero is bounded by the Lipschitz constant, giving

$$D_v(s) = sM + A_v + \frac{7}{4}e^{-s}m_0^2 + \mathfrak{r}_v(s), \quad |\mathfrak{r}_v(s)| \leq 4a^2 L_R(a)e^{-2s}M. \quad (\text{CP14:2.24})$$

Thus $D_v(s)/s \rightarrow M > 0$. In the odd channel $m_0 = 0$; no claim about the sign of the retained remainder is needed.

A fully explicit finite cutoff is also available. Put

$$A_a^{\text{abs}} = 2 \sum_{\ell_n < 2a} b_n + \int_{-2a}^{2a} |R(t)| dt, \quad R_a^{\text{max}} = \sup_{|t| \leq 2a} |R(t)|,$$

$$S_a = \max\{s_p, A_a^{\text{abs}} + 4aR_a^{\text{max}} + 1\}. \quad (\text{CP14:2.25})$$

For every $s \geq S_a$, direct use of (2.8) gives $D_v(s) \geq M$. Therefore any negative inward-dilation direction, if the required independent arithmetic estimate can produce one, must occur in the finite interval $0 < s < S_a$. This rules out obtaining a negative direction merely by shrinking the support indefinitely.

2.7. Closure record and a fully quantified remaining lemma

Exact theorem and hypotheses. For every finite $a > 0$ and every real definite-parity hypothetical null vector (2.3), the graph-domain passage (2.5)-(2.6), finite-dilation identity (2.8), complete mixed cancellation (2.9), endpoint-substituted polar formulas (2.11)-(2.13), exterior-collar identity (2.16)-(2.19), explicit modulus (2.20)-(2.23), and positive large-scale asymptotic (2.24) hold. The only input about null regularity is checkpoint 13's proved estimate (2.40).

Completed calculation and proof status. All these statements are proved. Inward dilation is admissible and its normalization is exactly sM . The pole, archimedean and prime terms have not been discarded. Neither the endpoint substitution nor the exterior-collar identity has independently assigned the needed sign to their surviving coupled terms.

Consequence for the next group. Group 3's renormalized three-dimensional energy has scale decrement exactly $-sM$. Its boundary calculation must therefore control the same signed arithmetic increment

$$\mathfrak{A}_v(s) = 2 \sum_{\ell_n < 2a} b_n \{C_v(e^s \ell_n) - C_v(\ell_n)\} + \int_{-2a}^{2a} \{e^{-s} R(e^{-s} t) - R(t)\} C_v(t) dt. \quad (\text{CP14:2.26})$$

The endpoint-complete version is (2.11)-(2.13), and the additional cross constraint is (2.16). No change of the Suzuki metric has been made.

One quantified missing lemma. A sufficient finite-scale statement is: for every $a > 0$, every nonzero real parity vector satisfying (2.3), and the explicit S_a in (2.25), there exist $s \in (0, S_a)$ and $\eta > 0$ such that

$$\boxed{\mathfrak{A}_v(s) \geq (s + \eta) \|v\|_2^2}. \quad (\text{CP14:2.27})$$

The margin may depend on the interval and the vector. A stronger uniform version would use $\eta = \eta_a > 0$. Either version implies $Q_a(U_s v) \leq -\eta \|v\|_2^2 < 0$, contradicting nonnegativity and excluding that null vector. The attempted independent estimate (2.23) controls only the absolute size and does not establish (2.27). The exact signed expressions carrying the unresolved sign are (2.26), equivalently (2.11)-(2.13), subject to the cross identity (2.16)-(2.19).

If (2.27) were proved independently for every hypothetical first-null vector, checkpoint 12's first-crossing and domain results would give Weil positivity, hence RH by Weil's criterion, then the correctly normalized Suzuki identity $\mathfrak{W} = \mathcal{S}$, and finally $\mathcal{E}_N = 0$ for every N . This group has proved the auxiliary statements, not the sign lemma or the original target.

3. Closure group 3 – Three-dimensional extension and complete renormalized energy

3.1. Exact hypotheses and arithmetic equation

Set $I_a = (-a, a)$, extend every function by zero, and use

$$\widehat{v}(\xi) = \int v(x) e^{i\xi x} dx, \quad \mathcal{L} = \log |D|, \quad L_\Delta = 2\mathcal{L}, \quad c_0 = \log 2 - \gamma_E. \quad (\text{CP14:3.1})$$

Checkpoint 13 proves that every $v \in \ker A_a$ satisfies

$$v \in L^2, \quad \text{supp } v \subset [-a, a], \quad \mathcal{L}v \in L^2(\mathbb{R}). \quad (\text{CP14:3.2})$$

The trace and energy theorems below apply to any v satisfying (3.2). For a null vector the actual arithmetic equation is

$$\mathcal{L}v = \mathcal{B}_a v = \log(2\pi)v + \sum_{\ell_n \leq 2a} b_n \{v(\cdot + \ell_n) + v(\cdot - \ell_n)\} + \int_{-a}^a R(\cdot - t)v(t) dt \quad \text{on } I_a, \quad (\text{CP14:3.3})$$

where

$$\ell_n = \log n, \quad b_n = \Lambda(n)/\sqrt{n}, \quad R(t) = \frac{e^{-|t|/2}}{1 - e^{-2|t|}} - 2 \cosh(t/2) - \frac{1}{2|t|}, \quad R(0) = -7/4. \quad (\text{CP14:3.4})$$

This includes the full archimedean remainder and the completed polar term. Equation (3.3) is local: outside the support,

$$(\mathcal{L}v)(x) = -\frac{1}{2} \int_{-a}^a \frac{v(t)}{|x - t|} dt, \quad |x| > a. \quad (\text{CP14:3.5})$$

The local right side of (3.3) cannot be extended by zero to obtain the whole-line operator.

The cited extension theorem is Chen-Hauer-Weth, *Extension problems for the logarithmic Laplacian*, arXiv:2312.15689v1, Theorems 1.2 and 1.6. The stronger trace topology and complete energy formula used here are proved below.

3.2. Line source and strong L^2 trace

Define

$$W_v(x, y) = \frac{1}{2} \int_{-a}^a \frac{v(t)}{\sqrt{(x - t)^2 + |y|^2}} dt, \quad \rho = |y| > 0. \quad (\text{CP14:3.6})$$

It is locally integrable in three dimensions and smooth away from the source segment. Applying $-\Delta(1/(4\pi|X|)) = \delta_0$ under the integral, tested against a compact smooth function, proves

$$\boxed{-\Delta_{\mathbb{R}^3} W_v = 2\pi v(x) dx \otimes \delta_0(y).} \quad (\text{CP14:3.7})$$

No derivative of the zero extension of v occurs, and no endpoint point masses are created.

The Fourier representation is

$$\widehat{W}_v(\xi, \rho) = K_0(\rho|\xi|)\widehat{v}(\xi), \quad -\widehat{\rho\partial_\rho W_v}(\xi, \rho) = (\rho|\xi|)K_1(\rho|\xi|)\widehat{v}(\xi). \quad (\text{CP14:3.8})$$

One derivation writes the Newton kernel as a Laplace integral of Gaussians, Fourier transforms in x , and obtains $K_0(z) = \frac{1}{2} \int_0^\infty u^{-1} e^{-u - z^2/(4u)} du$. Differentiation gives $K'_0 = -K_1$.

At zero, $K_0(z) + \log z \rightarrow c_0$, $zK_1(z) \rightarrow 1$. Also $0 < zK_1(z) \leq 1$, since its derivative is $-zK_0(z) < 0$. Thus

$$-\rho \partial_\rho W_v \rightarrow v \quad \text{strongly in } L^2(\mathbb{R}). \quad (\text{CP14:3.9})$$

For $0 < \rho \leq 1$, split $z = \rho|\xi|$ into $z \leq 1$ and $z > 1$. Boundedness of $K_0(z) + \log z$ in the first region, and $|\log \rho| \leq \log |\xi|$ in the second, yield

$$|K_0(\rho|\xi|) + \log \rho| \leq C + |\log |\xi||.$$

The majorant times $\hat{v}$ is in L^2 by (3.2). Compact support also gives bounded $\hat{v}$ near frequency zero. Plancherel and dominated convergence prove

$$\boxed{W_v(\cdot, \rho) + v \log \rho \rightarrow H_v := c_0 v - \mathcal{L}v \quad \text{strongly in } L^2(\mathbb{R}).} \quad (\text{CP14:3.10})$$

This proves the requested distributional trace and a stronger topology. On the null interval,

$$\boxed{H_v|_{I_a} = [(\log 2 - \gamma_E)I - \mathcal{B}_a]v.} \quad (\text{CP14:3.11})$$

The global trace retains

$$H_v(x) = \frac{1}{2} \int_{-a}^a \frac{v(t)}{|x-t|} dt \quad (|x| > a), \quad (\text{CP14:3.12})$$

which is in L^2 by the two Carleman estimates from checkpoint 13. No endpoint trace of v is assumed.

3.3. All boundaries and the logarithmic subtraction

For $\epsilon > 0$, $R > \epsilon$, $T > 0$, Green's formula in the cylinder $|x| < T$, $\epsilon < \rho < R$ gives

$$\begin{aligned} \int_{-T}^T \int_{\epsilon}^R \rho (|\partial_x W_v|^2 + |\partial_\rho W_v|^2) d\rho dx &= R \Re \int_{-T}^T \overline{W_v(x, R)} \partial_\rho W_v(x, R) dx \\ &\quad - \epsilon \Re \int_{-T}^T \overline{W_v(x, \epsilon)} \partial_\rho W_v(x, \epsilon) dx \\ &\quad + \Re \int_{\epsilon}^R \rho [\overline{W_v(T, \rho)} \partial_x W_v(T, \rho) - \overline{W_v(-T, \rho)} \partial_x W_v(-T, \rho)] d\rho. \end{aligned} \quad (\text{CP14:3.13})$$

The physical three-dimensional energy is 2π times this expression. Set $T = a$ to display both physical endpoint caps. They are not zero by the affine equation for $g * v$, since W_v is a different extension. Adding the two exterior x -regions cancels the caps exactly for each positive ϵ , before taking any singular limit. No separate limit of an endpoint cap is needed.

For the outer boundary, one can instead exhaust by balls of radius $L > a$, with the infinite tube $\rho \leq \epsilon$ removed. On their outer sphere,

$$|W_v| \leq \frac{\|v\|_1}{2(L-a)}, \quad |\nabla W_v| \leq \frac{\|v\|_1}{2(L-a)^2}.$$

The absolute outer flux is bounded by

$$\frac{\pi L^2 \|v\|_1^2}{(L-a)^3} = O(L^{-1}). \quad (\text{CP14:3.14})$$

It vanishes uniformly for bounded L^1 data at fixed support. The inner boundary is the complete tube, including the parts beyond $x = \pm a$. The order is: cancel finite caps; exhaust the outer boundary; then let $\epsilon \downarrow 0$.

Define the energy divided by 2π :

$$\mathcal{D}_\epsilon(v) = \frac{1}{2\pi} \int_{\mathbb{R} \times \{|y| > \epsilon\}} |\nabla W_v|^2.$$

The exact Green and Fourier identities are

$$\mathcal{D}_\epsilon(v) = -\epsilon \Re \int_{\mathbb{R}} \overline{W_v(x, \epsilon)} \partial_\rho W_v(x, \epsilon) dx = \frac{1}{2\pi} \int_{\mathbb{R}} z K_0(z) K_1(z) |\widehat{v}(\xi)|^2 d\xi, \quad z = \epsilon |\xi|. \quad (\text{CP14:3.15})$$

They also follow directly from the Bessel equation:

$$\int_\epsilon^\infty \rho |\xi|^2 [K_0(\rho |\xi|)^2 + K_1(\rho |\xi|)^2] d\rho = z K_0(z) K_1(z).$$

These integrals are finite for fixed positive ϵ .

Let $M = \|v\|_2^2$. The bound

$$|z K_0(z) K_1(z) + \log \epsilon| \leq C + |\log |\xi|| \quad (0 < \epsilon \leq 1)$$

follows by the same two-region argument. At zero, $z K_0(z) K_1(z) + \log z$ remains bounded; for $z \geq 1$ the Bessel product is bounded. Dominated convergence proves

$$\mathcal{D}_{\text{ren}}(v) := \lim_{\epsilon \downarrow 0} [\mathcal{D}_\epsilon(v) + M \log \epsilon] = c_0 M - \langle \mathcal{L}v, v \rangle. \quad (\text{CP14:3.16})$$

Equivalently the unscaled physical energy, after adding $2\pi M \log \epsilon$, tends to $2\pi[c_0 M - \langle \mathcal{L}v, v \rangle]$.

Polarization keeps all mixed terms. For compactly supported $u, v \in D(\mathcal{L})$,

$$\mathcal{D}_{\text{ren}}(u, v) = c_0 \langle u, v \rangle - \langle \mathcal{L}u, v \rangle, \quad \mathcal{D}_{\text{ren}}(u + v) = \mathcal{D}_{\text{ren}}(u) + \mathcal{D}_{\text{ren}}(v) + 2\Re \mathcal{D}_{\text{ren}}(u, v). \quad (\text{CP14:3.17})$$

This is the polarized Fourier integral. The finite part is signed even though the unrenormalized bulk energy is nonnegative.

3.4. Exact arithmetic energy and finite dilation

For real v , put $C_v(t) = \int v(x+t)v(x) dx$, and

$$\mathcal{A}_a(v) = 2 \sum_{\ell_n < 2a} b_n C_v(\ell_n) + \int_{-2a}^{2a} R(t) C_v(t) dt. \quad (\text{CP14:3.18})$$

The threshold term at $\ell_n = 2a$ vanishes by support. Pairing the null equation (3.3) with v gives $\langle \mathcal{L}v, v \rangle = \log(2\pi)M + \mathcal{A}_a(v)$. Consequently

$$\mathcal{D}_{\text{ren}}(v) = -(\gamma_E + \log \pi)M - 2 \sum_{\ell_n < 2a} b_n C_v(\ell_n) - \int_{-2a}^{2a} R(t) C_v(t) dt. \quad (\text{CP14:3.19})$$

All autocorrelations remain signed. For complex functions use sesquilinear polarization, retaining the two opposite shifts separately. The R -term includes the complete $-2 \cosh(t/2)$ contribution, without replacing it by a local series.

For $U_s v(x) = e^{s/2} v(e^s x)$, $s \geq 0$,

$$\widehat{U_s v}(\xi) = e^{-s/2} \widehat{v}(e^{-s} \xi), \quad \mathcal{L} U_s v = U_s \mathcal{L} v + s U_s v. \quad (\text{CP14:3.20})$$

Thus $U_s v \in D(\mathcal{L})$, and its support remains in I_a . Changing variables directly in (3.6) yields

$$W_{U_s v}(x, y) = e^{s/2} W_v(e^s x, e^s y), \quad \mathcal{D}_\epsilon(U_s v) = \mathcal{D}_{e^s \epsilon}(v).$$

After the logarithmic subtraction,

$$\boxed{\mathcal{D}_{\text{ren}}(U_s v) - \mathcal{D}_{\text{ren}}(v) = -sM.} \quad (\text{CP14:3.21})$$

This is a finite-dilation identity. It assumes no support derivative, endpoint value, or stationarity at $s = 0$.

For a supported $w \in D(\mathcal{L})$, the actual Weil form is $Q_a(w) = \langle \mathcal{L} w, w \rangle - \log(2\pi) \|w\|_2^2 - \mathcal{A}_a(w)$. Using $Q_a(v) = 0$, (3.21) gives

$$\boxed{Q_a(U_s v) = sM - [\mathcal{A}_a(U_s v) - \mathcal{A}_a(v)].} \quad (\text{CP14:3.22})$$

Since $C_{U_s v}(t) = C_v(e^s t)$, this is exactly

$$\begin{aligned} Q_a(U_s v) = sM - 2 \sum_{\ell_n < 2a} b_n [C_v(e^s \ell_n) - C_v(\ell_n)] \\ - \int_{-2a}^{2a} [e^{-s} R(e^{-s} t) - R(t)] C_v(t) dt. \end{aligned} \quad (\text{CP14:3.23})$$

Zero extension handles every prime threshold crossing. The extension returns to the original form and leaves the original Suzuki metric unchanged.

3.5. Calculated boundary flux and its surviving sign

The finite-tube energy has the exact derivative, for every $\epsilon > 0$,

$$\boxed{\frac{d}{d \log \epsilon} [\mathcal{D}_\epsilon(v) + M \log \epsilon] = \frac{1}{2\pi} \int_{\mathbb{R}} \{1 - z^2 [K_0(z)^2 + K_1(z)^2]\} |\widehat{v}(\xi)|^2 d\xi, \quad z = \epsilon |\xi|.} \quad (\text{CP14:3.24})$$

This differentiates the smooth extension at positive height, not the one-sided support variation. It follows from $[zK_0K_1]' = -z(K_0^2 + K_1^2)$. The resulting multiplier is bounded, justifying differentiation locally uniformly for positive ϵ .

This explicitly locates the sign left after subtraction. Write $L(z) = \log(2/z) - \gamma_E$. The convergent Bessel series give

$$z^2(K_0(z)^2 + K_1(z)^2) = 1 + z^2[L(z)^2 - L(z) - 1/2] + O(z^4(1 + L(z)^2)). \quad (\text{CP14:3.25})$$

It exceeds one for sufficiently small positive z , and tends to zero at infinity. Thus the braces in (3.24) are negative in a small-frequency range and positive in a large-frequency range. The integral

against the actual $|\widehat{v}|^2$ is the signed flux to be controlled. Positivity of the bulk supplies no sign for this finite part. The series and decay used here are recorded in NIST DLMF §§10.31 and 10.40.

A sufficient signed boundary/arithmetic estimate is: for each hypothetical nonzero null vector there exist finite $s > 0$ and $\eta_{a,v} > 0$ such that

$$2 \sum_{\ell_n < 2a} b_n [C_v(e^s \ell_n) - C_v(\ell_n)] + \int_{-2a}^{2a} [e^{-s} R(e^{-s} t) - R(t)] C_v(t) dt \geq (s + \eta_{a,v}) M. \quad (\text{CP14:3.26})$$

It compares the arithmetic increment with the exact energy drop sM , and would give $Q_a(U_s v) \leq -\eta_{a,v} M < 0$, contradicting $Q_a \geq 0$.

The attempted estimates do not prove (3.26). Cauchy-Schwarz gives

$$|\mathcal{A}_a(U_s v) - \mathcal{A}_a(v)| \leq M \left[4 \sum_{\ell_n < 2a} b_n + \int_{-2a}^{2a} |e^{-s} R(e^{-s} t) - R(t)| dt \right]. \quad (\text{CP14:3.27})$$

This controls magnitude without a positive lower bound for the signed increment. Logarithmic-square regularity also yields

$$|C_v(t+h) - C_v(t)| \leq |h| T M + \frac{2 \|\mathcal{L}v\|_2^2}{\log^2 T}, \quad T > e. \quad (\text{CP14:3.28})$$

Split the Fourier integral at $|\xi| = T$ to prove this. Taking $T = |h|^{-1/2}$, for $0 < |h| < e^{-2}$, gives

$$|C_v(t+h) - C_v(t)| \leq |h|^{1/2} M + \frac{8 \|\mathcal{L}v\|_2^2}{\log^2(1/|h|)}.$$

This proves continuity, including at the finitely many prime thresholds. It does not give a linear-in- h expansion or justify differentiating (3.23) at zero. Inequality (3.26) is the precise unresolved signed step.

3.6. Unique continuation: both hypotheses checked

Chen-Hauer-Weth's Theorem 1.7 requires $v = 0$ and $L_\Delta v = 0$ on the same nonempty open set. Compact support supplies the first condition outside I_a . Equation (3.3) is an interior equation, so it does not supply the second exterior condition.

Indeed, for any nonzero compactly supported L^2 function, (3.5) cannot vanish on an open interval of either exterior half-line. On $x > a$, $\int_{-a}^a v(t)/(x-t) dt$ is real analytic. Vanishing on an open interval makes it zero throughout $x > a$. Its convergent expansion at infinity then gives $\int t^k v(t) dt = 0$ for every $k \geq 0$. Polynomial density in $L^2([-a, a])$ forces $v = 0$. The other half-line is identical. Therefore a hypothetical nonzero null vector does not meet the second exterior hypothesis. The completed equation $Q_a v = 0$ cannot replace that hypothesis.

3.7. Group 3 closure record

1. **Theorem and hypotheses.** For each compactly supported $v \in L^2 \cap D(\log |D|)$, the line-source identity (3.7), global strong traces (3.9)-(3.10), boundary identity (3.13), renormalized energy (3.16)-(3.17), and finite dilation (3.21) hold. Every actual null vector has these hypotheses by checkpoint 13, and satisfies the complete arithmetic substitutions (3.11), (3.19), (3.23).

2. **Calculation and proof status.** The normalization, logarithmic subtraction, both endpoint caps, outer-boundary decay, every prime translation, and polarized mixed energies are proved. The flux derivative (3.24) exposes the remaining signed frequency integral. No unproved endpoint value or support derivative is used.
3. **Downstream consequence.** The three-dimensional representation returns exactly to Group 2's finite signed dilation expression. It provides a legitimate local harmonic and renormalized boundary formulation with the original metric intact. It does not yet yield an independent positive remainder margin.
4. **Quantified missing lemma and stopping line.** Inequality (3.26) for every hypothetical parity null vector, with some finite $s > 0$ and positive $\eta_{a,v}$, would exclude the null. The completed estimates (3.27)-(3.28) establish finiteness and continuity but not its required signed lower bound. If (3.26) were proved at every first crossing, the chain would be no first null, Weil positivity, RH, the correctly normalized Suzuki identity, and $\mathcal{E}_N = 0$ for all N . That chain remains conditional here.

4. Closure group 4 – A quantified infinity regulator with the complete correction retained

4.1. Domain, notation, and what is being regulated

Fix a finite hypothetical first-null interval and a nonzero real parity null vector $v \in \ker A_a$, with parity $\varepsilon \in \{1, -1\}$. Retain the completed checkpoint-13 definitions

$$L(z) = \sum_{\gamma} \frac{c_{\gamma}}{\gamma - z}, \quad c_{\gamma} = m_{\gamma} \widehat{v}(\gamma), \quad L^{\sharp} = \varepsilon L,$$

$$\mathcal{R} = C - ih \quad (\Im z > 0), \quad h = \frac{A}{A + iA'}, \quad F = \mathcal{R}LL^{\sharp} = \varepsilon \mathcal{R}L^2. \quad (\text{CP14:4.1})$$

The lower definition of $\mathcal{R}$ is its Schwarz reflection. Thus $\mathcal{R}^{\sharp} = \mathcal{R}$, $F^{\sharp} = F$, and $F(-z) = -F(z)$. Every occurrence of F below is this complete product. In particular, L^2 on a nonreal contour is not replaced by $|L|^2$.

The logarithmic regularity from checkpoint 13 gives

$$S_{\log}(v) := \sum_{\gamma} m_{\gamma} \log(2 + |\Re \gamma|) |\widehat{v}(\gamma)|^2 \leq C_a^{\text{null}} \|v\|_2^2 < \infty. \quad (\text{CP14:4.2})$$

No global Suzuki norm is required for the estimates through Section 4.7. When Group 1 establishes that norm, its strong approximation can subsequently be used; it does not change the meromorphic factors in (4.1).

Take

$$M_{\tau}(z) = \text{sech}(\tau z), \quad p_k = \frac{i\pi(k + 1/2)}{\tau} \quad (k \in \mathbb{Z}), \quad 0 < \tau \leq \pi/2. \quad (\text{CP14:4.3})$$

4.2. The actual function has no poles on the regulator axis

Write $F_{\theta}(y) = A(iy)$ to distinguish the positive theta Laplace transform from the complete product F . The actual positive theta density gives $F_{\theta}(y) > 0$, $F'_{\theta}(y) > 0$ for $y > 0$, and

$$E(iy) = F_{\theta}(y) + F'_{\theta}(y) > 0, \quad h(iy) = \frac{F_{\theta}(y)}{F_{\theta}(y) + F'_{\theta}(y)} \in (0, 1).$$

There are no xi nodes on the imaginary axis. Also

$$C(iy) = \frac{i}{\pi} \int_{\mathbb{R}} \frac{y\omega(x)}{x^2 + y^2} dx, \quad 0 < \omega(x) < 1 \text{ a.e.},$$

so $|\mathcal{R}(iy)| < 1$. Reflection handles $-iy$. Consequently F is holomorphic at every introduced pole p_k ; no coincidence with an actual node or an actual correction pole is being assumed away.

There is a useful derivative estimate specific to the actual zero strip. For $y > 1/2$, the symmetrically paired Hadamard logarithmic derivative obeys

$$\Im \frac{A'}{A}(x + iy) = - \sum_{\gamma} m_{\gamma} \frac{y - \Im \gamma}{|x + iy - \gamma|^2} < 0. \quad (\text{CP14:4.4})$$

The imaginary-part sum is absolutely convergent, and the paired regularization constants cancel. Therefore $|h(z)| \leq 1$ and h is holomorphic in $\Im z > 1/2$. Schwarz's lemma after mapping this half-plane and the unit disk to the disk gives

$$|h'(z)| \leq \frac{1 - |h(z)|^2}{2(\Im z - 1/2)}.$$

Meanwhile differentiation of the regularized Cauchy transform is absolutely legitimate and gives $|C'(x + iy)| \leq 1/y$. Thus

$$\boxed{|\mathcal{R}(iy)| < 1, \quad |\mathcal{R}'(iy)| \leq \frac{2}{y} \quad (y \geq 1).} \quad (\text{CP14:4.5})$$

These estimates include all actual xi structure and use no RH. They also show that actual upper correction poles lie in $0 < \Im z \leq 1/2$, with the reflected statement below. For later contours the simple bound

$$|C(x + iy)| \leq 1 + |x|/y, \quad |\mathcal{R}(x + iy)| \leq 2 + |x|/y \quad (y > 1/2) \quad (\text{CP14:4.6})$$

follows by integrating C' horizontally from iy .

4.3. Endpoint-complete estimates for the resolvent and its derivative

Let

$$\mathfrak{c}_{\Gamma} = \sup_{j \in \mathbb{Z}} \sum_{j \leq \Re \gamma < j+1} \frac{m_{\gamma}}{\log(2 + |\Re \gamma|)} < \infty.$$

Finiteness follows from the unconditional unit-interval zero-count bound. Put

$$D_v^2 = e^a \mathfrak{c}_{\Gamma} (8 + 2\pi) S_{\log}(v), \quad d = \sqrt{\frac{32 + 4\pi}{8 + 2\pi}}. \quad (\text{CP14:4.7})$$

Checkpoint 13 gives the endpoint-complete identities

$$L(z) = e^{iaz} L_{v,a}(z) = iz^2 \int_a^{\infty} \mathcal{B}_v(t) e^{izt} dt \quad (\Im z > 1/2). \quad (\text{CP14:4.8})$$

Their affine subtractions and vanishing endpoint value and derivative are retained. Estimate the absolutely convergent Cauchy sum in the first equality with (4.2); differentiate that equality before estimating. Since $|e^{-ia\gamma}| \leq e^{a/2}$,

$$\sum_{\gamma} \frac{m_{\gamma}}{\log(2 + |\Re \gamma|) |\gamma - iy|^2} \leq \frac{(8 + 2\pi) \mathfrak{c}_{\Gamma}}{y},$$

$$\sum_{\gamma} \frac{m_{\gamma}}{\log(2 + |\Re \gamma|)|\gamma - iy|^4} \leq \frac{(32 + 4\pi)\mathfrak{c}_{\Gamma}}{y^3} \quad (y \geq 1).$$

For example the first follows by summing $2 \sum_{j \geq 0} (j^2 + (y - 1/2)^2)^{-1}$, bounded by $2/(y - 1/2)^2 + \pi/(y - 1/2)$. The fourth-power estimate uses $2/(y - 1/2)^4 + \pi/[2(y - 1/2)^3]$. Consequently

$$\boxed{|L(iy)| \leq D_v e^{-ay} y^{-1/2}, \quad |L'(iy)| \leq D_v e^{-ay} (ay^{-1/2} + dy^{-3/2}).} \quad (\text{CP14:4.9})$$

Reflection gives the same moduli below. The factor aL in the derivative is explicitly retained. It is the differentiated endpoint exponential and cannot be dropped as a lower-order regularization term.

The same bin argument, with at most twice the displayed constants, is uniform in $\Re z$ on each half-plane $\Im z \geq \eta > 1/2$. In particular, $|L(x + iy)| \leq C_{a,\eta} \|v\|_2 e^{-ay}$ there, with corresponding derivative bounds. This is used only away from the actual zero strip.

4.4. Exact artificial-pole residues and a complete absolute budget

At $z = p_k + w$,

$$M_{\tau}(z)^2 = -\frac{1}{\tau^2 w^2} + \frac{1}{3} + O(\tau^2 w^2).$$

The missing simple Laurent term matters: since F is regular at p_k ,

$$\boxed{\text{Res}_{p_k}(M_{\tau}^2 F) = -\tau^{-2} F'(p_k).} \quad (\text{CP14:4.10})$$

Using (4.5) and (4.9) in $F' = \varepsilon(\mathcal{R}'L^2 + 2\mathcal{R}LL')$ yields

$$|F'(iy)| \leq D_v^2 e^{-2ay} \left(\frac{2a}{y} + \frac{2+2d}{y^2} \right), \quad y \geq 1. \quad (\text{CP14:4.11})$$

Thus, including both imaginary half-axes and every artificial pole,

$$\boxed{\sum_{k \in \mathbb{Z}} |\text{Res}_{p_k}(M_{\tau}^2 F)| \leq D_v^2 \frac{e^{-a\pi/\tau}}{1 - e^{-2a\pi/\tau}} \left\{ \frac{8a}{\pi\tau} + \frac{16(1+d)}{\pi^2} \right\}.} \quad (\text{CP14:4.12})$$

This follows by putting $y_k = \pi(k + 1/2)/\tau$, $k \geq 0$, replacing y_k^{-j} by $(2\tau/\pi)^j$, and summing the resulting geometric series. It is an unconditional $O_a(\tau^{-1} e^{-a\pi/\tau}) \|v\|_2^2$ budget, uniformly for null vectors on this fixed support. After Group 1, it can also be divided by the fixed positive $\mathcal{S}(v, v)$; no interval-uniform coercivity is asserted.

There is also uniformity over the full nullspace of this one interval. Compact resolvent makes $\ker A_a$ finite-dimensional. Group 1 and checkpoint-13 (2.40) make V continuous on this space, and the retained injectivity theorem makes it injective. Therefore, whenever the nullspace is nonzero,

$$\sigma_a := \min\{\mathcal{S}(w, w) : w \in \ker A_a, \|w\|_2 = 1\} > 0.$$

The normalized right side of (4.12) is then bounded uniformly over its nonzero null vectors by replacing $D_v^2/\mathcal{S}(v, v)$ with $e^a \mathfrak{c}_{\Gamma}(8 + 2\pi)C_a^{\text{null}}/\sigma_a$. This is fixed-interval uniformity; it proves neither a lower bound on σ_a uniform in a nor the central signed estimate.

4.5. Multiplication by the regulator is not coefficient regulation

Define the weighted-coefficient synthesis and its complete correction by

$$L_\tau^{\text{coeff}}(z) = \sum_\gamma \frac{c_\gamma M_\tau(\gamma)}{\gamma - z}, \quad E_\tau(z) = M_\tau(z)L(z) - L_\tau^{\text{coeff}}(z) = \sum_\gamma c_\gamma \frac{M_\tau(z) - M_\tau(\gamma)}{\gamma - z}. \quad (\text{CP14:4.13})$$

For $0 < \tau < \pi$, $|M_\tau(\gamma)| \leq \sec(\tau/2)$ throughout the actual zero strip. Thus these sums converge absolutely and locally uniformly away from their displayed poles. All xi-node singularities of E_τ are removable. It retains regulator poles, with

$$\text{Res}_{p_k} E_\tau = \frac{(-1)^k}{i\tau} L(p_k). \quad (\text{CP14:4.14})$$

In particular their total absolute residue budget is at most

$$2D_v \sqrt{\frac{2}{\pi\tau}} \frac{e^{-a\pi/(2\tau)}}{1 - e^{-a\pi/\tau}}.$$

The exact full product is

$$\boxed{M_\tau^2 L L^\sharp = L_\tau^{\text{coeff}}(L_\tau^{\text{coeff}})^\sharp + L_\tau^{\text{coeff}} E_\tau^\sharp + E_\tau(L_\tau^{\text{coeff}})^\sharp + E_\tau E_\tau^\sharp.} \quad (\text{CP14:4.15})$$

Thus a replacement by weighted zero coefficients must carry both cross terms and the correction square. Its scalar multiplier is the Fourier transform of $(2\tau)^{-1} \text{sech}(\pi t/(2\tau))$. Convolution with that kernel changes its support, so the original localized null equation does not automatically apply to the resulting function.

The xi-node expansion is particularly explicit. Write $L = -c_\gamma/w + \ell_\gamma + O(w)$, $w = z - \gamma$. Then

$$E_\tau(\gamma) = -c_\gamma M'_\tau(\gamma) + \sum_{\delta \neq \gamma} c_\delta \frac{M_\tau(\gamma) - M_\tau(\delta)}{\delta - \gamma}. \quad (\text{CP14:4.16})$$

Every mixed node is present. The residue at γ of the difference between the left side of (4.15) and its first term, after multiplication by $\mathcal{R}$, is

$$-2\varepsilon \mathcal{R}(\gamma) M_\tau(\gamma) c_\gamma E_\tau(\gamma). \quad (\text{CP14:4.17})$$

At a real xi node the Suzuki factor still removes the feature pole; the finite value in (4.16), and hence this mixed-node interaction, remains.

On each fixed compact set, filling in the removable xi nodes, $E_\tau \rightarrow 0$ with all fixed derivatives as $\tau \downarrow 0$. This follows by dominated convergence in (4.13), (4.2), and the absolutely convergent Cauchy bound, followed by Cauchy's formula. It does not furnish a uniform estimate over infinitely many nodes or correction poles.

4.6. Actual poles, mixed derivatives, and exact finite contours

If p is an actual correction pole with principal part $\sum_{j=1}^{d_p} r_{p,j}(z-p)^{-j}$, its retained contribution is

$$\boxed{\text{Res}_p(M_\tau^2 F) = \sum_{j=1}^{d_p} \frac{r_{p,j}}{(j-1)!} \partial_z^{j-1}(M_\tau^2 L L^\sharp) \Big|_{z=p}.} \quad (\text{CP14:4.18})$$

Leibniz expansion retains derivatives of the regulator and of both resolvents. Equivalently substitute all four terms of (4.15). In the simple upper-pole case this is $-\varepsilon M_\tau(p)^2 L(p)^2 / (A'/A)'(p)$; its phase is retained.

For a finite domain Ω whose boundary avoids every pole, let Γ_Ω , P_Ω , and $I_{\tau,\Omega}$ denote its xi nodes, actual correction poles, and introduced regulator poles. The exact residue statement is

$$\boxed{-\sum_{\gamma \in \Gamma_\Omega} \text{Res}_\gamma(M_\tau^2 F) = -\frac{1}{2\pi i} \oint_{\partial\Omega} M_\tau^2 F dz + \sum_{p \in P_\Omega} \text{Res}_p(M_\tau^2 F) + \sum_{p_k \in I_{\tau,\Omega}} \text{Res}_{p_k}(M_\tau^2 F).} \quad (\text{CP14:4.19})$$

The same formula holds with finite zero cutoffs throughout. All actual poles and boundary integrals are retained. The left side is the regulated node-residue functional; identifying it with coefficient regulation requires the explicit correction (4.15)-(4.17).

4.7. Proven infinity limits and their order

Fix $\eta > 1/2$, then take $0 < \tau < \min\{\pi/2, \pi/(2\eta)\}$. In $\Im z > \eta$, the only poles of $M_\tau^2 F$ are the introduced ones. Along the lines $\Im z = J\pi/\tau$, $|M_\tau(x + iy)| = \text{sech}(\tau x)$. By (4.6) and the uniform version of (4.9), the top integrals tend to zero as $J \rightarrow \infty$. The side integrals tend to zero first as their real coordinate tends to infinity: $|M_\tau(x + iy)|^2 \leq C_\tau e^{-2\tau|x|}$ on sufficiently far vertical lines, uniformly in y , and the remaining integrand is bounded by a polynomial in $|x|$ times e^{-2ay} .

Absolute local convergence first permits removing the zero cutoff on each fixed contour. Then, for this fixed τ , the side limit, the top-height limit, and the absolutely convergent artificial-pole series give

$$\boxed{\frac{1}{2\pi i} \int_{\mathbb{R}+i\eta} M_\tau(z)^2 F(z) dz = -\tau^{-2} \sum_{\Im p_k > \eta} F'(p_k).} \quad (\text{CP14:4.20})$$

The lower-half-plane version, with its horizontal integral also oriented left to right, is

$$\boxed{\frac{1}{2\pi i} \int_{\mathbb{R}-i\eta} M_\tau(z)^2 F(z) dz = +\tau^{-2} \sum_{\Im p_k < -\eta} F'(p_k).} \quad (\text{CP14:4.21})$$

Only after these fixed- τ limits is $\tau \downarrow 0$ taken; (4.12) proves that these paired exterior objects tend to zero, with its displayed exponential rate. No exchange with an unregulated infinite signed residue sum is used.

The missing uniformity concerns the original zero strip, which contains the xi nodes and the actual correction poles. Finite-strip rectangles retain their vertical boundary terms through (4.19). Neither (4.20) nor the small artificial-pole budget establishes a sign for those interior terms. In particular, it does not prove absolute summability of all actual pole costs or uniform convergence of (4.17) over every node.

Once Group 1 proves $Vv \in L^2(\mathbb{R})$, ordinary dominated convergence gives

$$\|M_\tau Vv - Vv\|_2 \rightarrow 0, \quad \int M_\tau(x)^2 |Vv(x)|^2 dx \rightarrow \mathcal{S}(v, v). \quad (\text{CP14:4.22})$$

This is a norm statement about the multiplied Suzuki output. Equation (4.15) remains necessary when replacing that output by weighted Cauchy coefficients. Strong output convergence does not itself justify an unregulated contour-residue exchange.

4.8. Closure record and the remaining signed estimate

Exact theorem and hypotheses. For any finite-support real parity null vector satisfying checkpoint-13 logarithmic regularity, the actual axis is free of correction and node poles; (4.5), (4.9), the residue formula (4.10), the complete exponential budget (4.12), and the ordered exterior contour identities (4.20)-(4.21) hold unconditionally. The weighted-coefficient correction is exactly (4.13)-(4.18).

Completed calculation and status. Infinity on the two exterior half-planes is controlled with an explicit $\tau^{-1}e^{-a\pi/\tau}$ cost. Every actual pole, mixed-node residue, and contour term remains in (4.19). The invariant $L_{v,b}L_{v,b}^\# = LL^\#$ has not been used to create a false small factor in the central signed expression.

Downstream consequence. These estimates allow the artificial poles to be included and subsequently removed in the stated order. They do not establish the original Suzuki-relative sign. With Group 1, the exact null identity still reads $\mathcal{T}(v, v) = -\mathcal{S}(v, v) + \mathcal{X}_\infty(v, v)$.

Explicit residual inequality and attempted estimate. Let $\mathfrak{P}(v)$ denote the complete defect pairing obtained in the preserved order $\lim_n \lim_H$ along the Group-1 strong graph-core sequence. Thus $\mathfrak{P}(v) = \Delta(v, v)$. For each core vector and each finite cutoff, its contour realization is the right side of (4.19) at $\tau = 0$, with the original finite zero cutoffs and graph-core limits retained in checkpoint order. The local convergence to the exact-null resolvent L justifies fixed-contour identities; it does not by itself identify an exhaustion of F with $\mathfrak{P}(v)$. In particular, no raw zero-cutoff convergence $V_H v \rightarrow Vv$ in L^2 is asserted. The strong graph-core limit and the exterior regulator limits are separate, explicitly ordered statements. A sufficient still-unproved estimate is

$$\boxed{\mathfrak{P}(v) + \mathcal{X}_\infty(v, v) \geq (-1 + K_*)\mathcal{S}(v, v).} \quad (\text{CP14:4.23})$$

Its regulator calculation must retain (4.15) and the mixed node residues (4.17), or work directly with (4.19). Equations (4.12) and (4.20) control the introduced outer costs. They provide no lower bound for the surviving signed actual-pole-node-contour pairing in the zero strip. Estimating the outer costs by their positive absolute budget therefore stops before (4.23); the surviving term is of the same complete sign-indefinite form as (4.18)-(4.19). No contradiction to a null vector is claimed.

5. Combined closing calculation and the exact remaining lemma

5.1. What has closed and what each limit means

Group	Proved statement on the actual kernel	Remaining signed operation
1	Fixed-support global Suzuki graph bound; same-parity core converges strongly in the original metric; energy is positive and finite; spectral tails are uniformly tight	Lower-bound the complete remainder, including all cross terms
2	Full finite dilation, prime thresholds, polar endpoint substitution, and signed exterior-collar identity; every possible negative scale lies in an explicit finite interval	Prove a signed arithmetic increment greater than $s\ v\ _2^2$

Group	Proved statement on the actual kernel	Remaining signed operation
3	Line source, global strong trace, renormalized energy with every boundary retained, exact scale decrement, and explicit sign-changing flux	Relate the signed flux or collar to the actual arithmetic increment with a positive margin
4	Safe artificial poles, derivative bounds, exponentially small complete artificial-pole budget, and ordered exterior contour exhaustion	Control the coupled actual-pole, node, and boundary expression within the original zero strip

The topology record is precise. The core sequence converges in the whole-line logarithmic graph norm, in the localized closed-form norm, and under V strongly in L^2 . The three-dimensional trace converges globally in L^2 ; tube energies converge after their displayed logarithmic subtraction. Cauchy synthesis and its derivatives converge locally off their poles. Regulated exterior contours converge in the order fixed in Section 4.7. Uniform spectral tightness of Vq_n is not asserted to be uniform control of all nonreal residues, nor to prove raw zero-cutoff convergence $V_H v \rightarrow Vv$.

5.2. The extension and collar identities give the same complete constraint

The new strong trace makes their compatibility exact, without a formal endpoint derivative. Write

$$H_v = (\log 2 - \gamma_E)v - \mathcal{L}v.$$

Group 3 and Fourier scaling give

$$H_{U_s v} = U_s H_v - s U_s v. \quad (\text{CP14:5.1})$$

Self-adjointness of $\mathcal{L}$, valid on the proved whole-line domain, supplies reciprocity $\langle H_{U_s v}, v \rangle = \langle U_s v, H_v \rangle$. Therefore, for real v ,

$$\boxed{s \langle U_s v, v \rangle = \langle U_s H_v, v \rangle - \langle U_s v, H_v \rangle.} \quad (\text{CP14:5.2})$$

Insert the complete trace

$$H_v = (\log 2 - \gamma_E)v - \mathbf{1}_{I_a} \mathcal{B}_a v - E_v.$$

The constant terms cancel and the second exterior pairing vanishes only because $U_s v$ is supported inside I_a . The first exterior pairing survives. It is exactly

$$-\langle U_s E_v, v \rangle = \frac{e^{s/2}}{2} \int_{ae^{-s} < |x| < a} \int_{-a}^a \frac{v(x)v(y)}{|e^s x - y|} dy dx = \mathcal{C}_v(s). \quad (\text{CP14:5.3})$$

Thus (5.2) becomes (2.16), including both prime-translation cross terms and both R -cross terms in (2.19). This is an exact compatibility calculation. The positive bulk energy has not created a new nonnegative boundary quantity: the surviving boundary object is the signed collar (5.3).

5.3. The attempted combined estimate and its stopping line

Normalize $M = \|v\|_2^2 > 0$. Define the full signed arithmetic increment

$$\begin{aligned} \mathfrak{A}_v(s) = 2 \sum_{\ell_n < 2a} b_n [C_v(e^s \ell_n) - C_v(\ell_n)] \\ + \int_{-2a}^{2a} [e^{-s} R(e^{-s} t) - R(t)] C_v(t) dt. \end{aligned} \quad (\text{CP14:5.4})$$

The combined exact identities are

$$Q_a(U_s v) = sM - \mathfrak{A}_v(s) = Q_a(U_s v - v), \quad \mathcal{D}_{\text{ren}}(U_s v) - \mathcal{D}_{\text{ren}}(v) = -sM. \quad (\text{CP14:5.5})$$

The endpoint equations give the exact even/odd substitutions (2.12)-(2.13); the trace equation gives (5.2)-(5.3). These are constraints on the actual v , not generic counterexamples or assumptions about arbitrary kernels.

The independent estimates obtained from these constraints are

$$\frac{|\mathcal{C}_v(s)|}{M} \leq \frac{\pi}{\sqrt{2}} \left(\frac{\int_{ae^{-s} < |x| < a} |v(x)|^2 dx}{M} \right)^{1/2}, \quad (\text{CP14:5.6})$$

and, with $h_n(s) = (e^s - 1)\ell_n < e^{-2}$,

$$\frac{|\mathfrak{A}_v(s)|}{M} \leq 2 \sum_{\ell_n < 2a} b_n \left(\sqrt{h_n(s)} + \frac{8\kappa_a^2}{\log^2(1/h_n(s))} \right) + sD_R(a). \quad (\text{CP14:5.7})$$

They come respectively from the endpoint Carleman operator and the whole-line logarithmic regularity. They establish convergence and explicit absolute control. Neither inequality gives a positive lower bound on $\mathfrak{A}_v(s) - sM$. Equation (5.2) is a mixed pairing, not the diagonal increment (5.4); replacing its collar by an unsigned positive energy would lose its signed factors.

The three-dimensional flux attempt is equally explicit:

$$\mathfrak{F}_\epsilon(v) = \frac{1}{2\pi} \int [1 - (\epsilon|\xi|)^2(K_0(\epsilon|\xi|)^2 + K_1(\epsilon|\xi|)^2)] |\widehat{v}(\xi)|^2 d\xi. \quad (\text{CP14:5.8})$$

Section 3.5 proves analytically that its multiplier takes both signs. No identity derived here forces the actual spectral weight to occupy the sign needed in (5.4). This diagnoses the exact failed inference; it does not rule out a further xi-specific identity.

Large inward dilation does not supply the missing sign: (2.24) shows $Q_a(U_s v)/s \rightarrow M > 0$, and $Q_a(U_s v) \geq M$ for every $s \geq S_a$ from (2.25). The next sign problem is therefore restricted to $0 < s < S_a$, with all prime powers still present through their zero-extended correlations.

5.4. Exact-null signed budget in the original Suzuki metric

Group 1 now permits the exact-null statement

$$0 < \mathcal{S}(v, v) < \infty, \quad \frac{\mathcal{T}(v, v)}{\mathcal{S}(v, v)} = -1 + \frac{\mathcal{X}_\infty(v, v)}{\mathcal{S}(v, v)} \leq -1 + K_0. \quad (\text{CP14:5.9})$$

Consequently the independent inequality

$$\boxed{\frac{\mathcal{T}(v, v)}{\mathcal{S}(v, v)} \geq -1 + K_*} \quad (\text{CP14:5.10})$$

would give the positive contradiction margin $\varepsilon_* \mathcal{S}(v, v)$. More generally an interval-dependent lower bound $-1 + K_0 + \eta_a$, $\eta_a > 0$, would suffice. Finiteness, approximation, and artificial-pole removal do not independently prove either lower bound.

The regulator calculation does remove one quantitative obstacle. With D_v, d as in (4.7), its complete introduced-pole budget is

$$B_\tau(v) = D_v^2 \frac{e^{-a\pi/\tau}}{1 - e^{-2a\pi/\tau}} \left(\frac{8a}{\pi\tau} + \frac{16(1+d)}{\pi^2} \right). \quad (\text{CP14:5.11})$$

For any fixed nonzero null v and any $\delta > 0$, sufficiently small $\tau > 0$ gives $B_\tau(v) \leq \delta \mathcal{S}(v, v)$. In fact this is uniform over the fixed interval's nullspace: the retained compact-resolvent theorem makes that nullspace finite-dimensional; (1.2), (1.16), and injectivity give

$$\sigma_a := \min_{\substack{v \in \ker A_a \\ \|v\|_2=1}} \mathcal{S}(v, v) > 0$$

whenever the nullspace is nonzero. Thus $\|v\|_2^2 \leq \mathcal{S}(v, v)/\sigma_a$, so the explicit exponential bound is uniform after division by $\mathcal{S}$. For a general complex null vector, the same pole budget follows by estimating $L(iy)$ and $L^\sharp(iy) = \overline{L(-iy)}$, and their derivatives, separately using the two endpoint shifts; the product rule gives the same bound (4.11) without requiring $L^\sharp = \varepsilon L$. This is an interval-dependent existence bound, not an explicit numerical value or a new signed estimate. What survives is the complete actual-pole, node, and contour pairing of (4.19), including (4.15)-(4.18). Its sign has not become positive when the artificial cost tends to zero. The preserved limit defining $\mathcal{T}(v, v)$ remains the graph-core limit of the complete correction.

5.5. One quantified next lemma and the conditional dependency chain

The narrower next attempt can be formulated without any new infinite contour exhaustion. Use the explicit finite S_a from (2.25):

Finite-scale arithmetic excess lemma – still unproved. For every $a > 0$ and every nonzero real definite-parity v in the hypothetical first-null space $Q_a \geq 0$, $A_a v = 0$, there exist

$$0 < s < S_a, \quad \eta_{a,v} > 0$$

such that the complete expression (5.4), subject to the exact endpoint equations (2.12) and trace-collar equation (5.2), satisfies

$$\boxed{\mathfrak{A}_v(s) \geq (s + \eta_{a,v})\|v\|_2^2}. \quad (\text{CP14:5.12})$$

No uniform positive margin in a or v is required. The missing operation is a signed lower estimate for this explicit finite-scale expression. The actual attempted bounds are (5.6)-(5.8); the derivation stops when they would have to imply (5.12). The null and endpoint identities retain signed prime, polar, archimedean, and collar couplings at that line.

If (5.12) were independently established, the complete chain would be:

1. Equation (5.5) gives $Q_a(U_s v) \leq -\eta_{a,v}\|v\|_2^2 < 0$, contradicting $Q_a \geq 0$.
2. Every finite first-null interval is excluded.
3. The preserved small-interval positivity, domain passage, continuity, and first-crossing theorem from checkpoint 12 give Weil positivity on every compact smooth test.
4. Weil's criterion gives RH.
5. Only then the correctly normalized Suzuki identity yields $\mathfrak{W} = \mathcal{S}$ on the determining test space.
6. Hence $\Delta = 0$, $\mathcal{E}_N = 0$ for every N , and the original target follows.

An independent proof of (5.10) would supply the alternate contradiction with the exact margin in (0.7), followed by the same chain. Neither antecedent has been proved here. The new global domain theorem, endpoint-compatible finite dilation, renormalized extension, and complete artificial-pole budget are closed auxiliary results. The unconditional target is not closed.

6. Sources, dependency record, and file record

The primary sources checked for this continuation are:

- Masatoshi Suzuki, *On the Hilbert space derived from the Weil distribution*, arXiv:2301.00421v3, the explicit transform and Lerch formulas. The corrected two-sided convention is the previously audited one in the archived manuscript and checkpoints. RH-dependent orthogonality is not used as an unconditional input.
- Masatoshi Suzuki, *Weil's quadratic form via the screw function*, arXiv:2606.09096, as used in checkpoints 12-13 for the exact arithmetic kernel and localized operator. This continuation preserves that established normalization and domain passage.
- Huyuan Chen, Daniel Hauer, and Tobias Weth, *Extension problems for the logarithmic Laplacian*, arXiv:2312.15689v1, especially Theorems 1.2, 1.6, and 1.7. The present note proves the stronger L^2 trace topology and the energy statements it uses.
- NIST Digital Library of Mathematical Functions, §10.31 and §10.40, the Bessel series and large-argument behavior used in the explicit flux calculation.

The source checkpoint is `Michaels_First_Null_Arithmetic_Closure_Checkpoint_13.md`. Its follow-up directive `Michaels_Checkpoint_13_Graph_Dilation_Extension_Followup.md` was recovered unchanged. This continuation makes no changes to any previous manuscript or note. The current note and that recovered directive are added to a new checkpoint packet; the continuing file ledger is updated, and its manifest records the byte sizes and SHA-256 hashes of all content members. All prior non-ledger content is preserved byte-for-byte.

Analytical checks covered the differentiated Lerch signs, the negative-time convention, the completed gamma multiplier, Carleman constants, the null-domain passages, dilation normalization, endpoint and exterior cross terms, the three-dimensional 2π factors, Bessel flux, actual-pole regularity, regulator residue signs, and the order of limits. No numerical sampling of zeros or computational ladder is used as a proof step.

The unchanged target is $\liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1$, independently of RH.

Checkpoint 15: Odd prime packets and the polar resolvent

Source: *Michaels_Odd_Prime_Packets_and_Polar_Resolvent_Checkpoint_15.md*. Equation tags carry the source prefix *CP15*; references within this unit retain their original local numbering.

23 September 2026

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0. Objective, retained normalization, and status

The primary objective is unconditional Weil positivity:

$$\boxed{\mathfrak{W}(q, q) \geq 0 \quad (q \in C_c^\infty(\mathbb{R}))}. \quad (\text{CP15:0.1})$$

The original Suzuki metric and defect criterion remain

$$\mathcal{S}(q, q) = \|Vq\|_2^2, \quad \Delta = \mathfrak{W} - \mathcal{S}, \quad \mathcal{E}_N = \|[\Delta(\rho_j, \rho_i)]_{i,j \leq N}\|_F^2, \quad \liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1. \quad (\text{CP15:0.2})$$

Here (ρ_i) is the original Suzuki-orthonormal determining ladder. A direct Hilbert-Schmidt estimate is optional. Proving (0.1) allows the established Weil criterion and correctly normalized Suzuki identity to imply $\Delta = 0$, a stronger conclusion than the defect criterion.

This continuation proves the odd-sector first-crossing framework, the complete mixed integer-dilation kernel, explicit prime-power composition and mixed coefficient signs, and an actual inverse estimate giving strict odd positivity on $0 < a \leq 1/8$. It calculates the full two- and three-scale signed interactions and their exact relation to the polar resolvent. The global signed exclusion of every possible odd first null remains open. Accordingly, both (0.1) and the success criterion in (0.2) remain unproved.

All functions on $I_a = (-a, a)$ are zero extended. The conventions retained from the complete checkpoint 14 note are

$$\begin{aligned} \widehat{q}(\xi) &= \int_{\mathbb{R}} q(x) e^{i\xi x} dx, \quad \mathcal{L} = \log |D|, \quad \ell_n = \log n, \quad b_n = \frac{\Lambda(n)}{\sqrt{n}}, \\ R(t) &= \frac{e^{-|t|/2}}{1 - e^{-2|t|}} - \frac{1}{2|t|} - 2 \cosh(t/2), \quad R(0) = -\frac{7}{4}. \end{aligned} \quad (\text{CP15:0.3})$$

The full real quadratic form is

$$\boxed{Q_a(q) = \langle \mathcal{L}q, q \rangle - \log(2\pi) \|q\|_2^2 - 2 \sum_{\ell_n < 2a} b_n C_q(\ell_n) - \int_{-2a}^{2a} R(t) C_q(t) dt, \quad C_q(t) = \int q(x+t) q(x) dx.} \quad (\text{CP15:0.4})$$

Complex inputs use its Hermitian polarization. The operator A_a is the self-adjoint operator associated with this closed form; it is distinct from the entire function $A(z) = \xi(1/2 - iz)$.

The complete Suzuki transform, corrected negative-time convention, meromorphic correction, all real-node removals, and all actual pole contributions from checkpoint 14 are preserved. The global graph estimate gives a strong odd core for each odd null vector, with positive finite original Suzuki norm. The original order of limits for the completed correction is retained. No RH-dependent feature orthogonality is used as an unconditional input.

For a finite family u_i , testing its signed Weil matrix does not change the metric. With

$$K = [Q_a(u_j, u_i)], \quad G = [\mathcal{S}(u_j, u_i)], \quad \tilde{K} = G^{-1/2} K G^{-1/2}, \quad (\text{CP15:0.5})$$

negative directions of K and $\tilde{K}$ are equivalent wherever $G > 0$; otherwise take the original positive metric quotient. On the retained injective domain of V , the kernel of G consists exactly of coefficient dependencies $\sum_i c_i u_i = 0$, on which the form matrix also vanishes. Thus this quotient preserves every energy direction. All entries and all mixed terms are kept. The L^2 realization used for the polar resolvent is an operator representation of the same signed form.

1. Closing group 1 – Odd-sector reduction and exact null domain

1.1. The criterion and its normalization

Use the Fourier convention $\hat{q}(\xi) = \int q(x) e^{i\xi x} dx$, the original Suzuki metric $\mathcal{S}(q) = \|Vq\|_2^2$, and the complete arithmetic form $Q_a = \mathfrak{W}|_{I_a}$. Suzuki, *Weil's quadratic form via the screw function*, arXiv:2606.09096v2, Introduction, records Yoshida's Proposition 1: nonnegativity of the Weil form on every nonzero odd compact smooth test implies RH. We use this established criterion. The parity and first-crossing calculations below are supplied explicitly.

The normalization agrees exactly: Suzuki uses $Q_W(q, r) = W(q * \tilde{r})$, $\tilde{r}(x) = \overline{r(-x)}$, and the same Fourier sign. On the core our form is

$$Q_a(q, r) = \langle -g'' * q, r \rangle = Q_W(q, r), \quad q, r \in C_c^\infty(I_a). \quad (\text{CP15:1.1})$$

Here g is the complete screw function retained in checkpoint 14. Its origin finite part gives the multiplier $\log|\xi|$ and constant $-\log(2\pi)$; its prime delta masses give $-b_n$ at each shift $\pm \log n$; and its completed polar term is $-8(\cosh(t/2) - 1)$. These are the original arithmetic coefficients. Multiplying ξ by the constant $1/2$, as in our definition, changes neither its logarithmic derivative nor $h = A/(A + iA')$ and hence changes neither form. No positive or negative normalization factor is introduced. In particular,

$$[Q_a(q) \geq 0 \text{ for every } a > 0 \text{ and every odd } q \in C_c^\infty(I_a)] \implies \text{RH} \implies [Q_a(q) \geq 0 \text{ for all compact smooth } q]. \quad (\text{CP15:1.2})$$

For this real kernel, complex odd positivity and real odd positivity are equivalent: if $q = u + iw$ with u, w real and odd, then $Q_a(q) = Q_a(u) + Q_a(w)$. Thus a real negative odd witness exists whenever an odd negative witness exists.

1.2. The odd closed domain and parity reduction

Let $Pq(x) = q(-x)$, $P_\pm = (I \pm P)/2$, and $\mathcal{H}_a^- = P_- L^2(I_a)$. The exact closed form has the domain obtained by completing $C_c^\infty(I_a)$ in a norm equivalent to

$$\|q\|_2^2 + \frac{1}{2\pi} \int_{\mathbb{R}} \log(2 + |\xi|) |\hat{q}(\xi)|^2 d\xi. \quad (\text{CP15:1.3})$$

All inputs in this formula are zero extended. The negative part of $\log|\xi|$ is bounded on fixed-support L^2 , since $|\widehat{q}(\xi)| \leq \sqrt{2a}\|q\|_2$. The remaining prime and remainder operators are bounded on $L^2(I_a)$. Thus (1.3) describes precisely the same closure as the endpoint-logarithmic form used in checkpoints 13-14. Define

$$\mathfrak{D}_a^- = \mathfrak{D}(Q_a) \cap \mathcal{H}_a^-, \quad Q_a^- = Q_a|_{\mathfrak{D}_a^-}, \quad A_a^- \text{ its associated operator.} \quad (\text{CP15:1.4})$$

The core, the form, and its closure are invariant under reflection and conjugation. Hence $P_\pm$ are bounded projections on the form domain, the odd core is dense in (1.4), and

$$Q_a(q_+, q_-) = 0, \quad A_a = A_a^+ \oplus A_a^-. \quad (\text{CP15:1.5})$$

Indeed reflection invariance gives $Q_a(q_+, q_-) = Q_a(Pq_+, Pq_-) = -Q_a(q_+, q_-)$. This proof works for the complete form, including every prime translation and both polar channels.

Suppose now only that

$$Q_a^- \geq 0, \quad 0 \neq v \in \ker A_a^-. \quad (\text{CP15:1.6})$$

Conjugation permits a nonzero real choice of v , which remains odd. For every full-domain w ,

$$Q_a(v, w) = Q_a(v, P_-w) = 0. \quad (\text{CP15:1.7})$$

The representation theorem for the full lower-bounded form therefore gives $v \in \mathfrak{D}(A_a)$, $A_av = 0$. Thus the restricted null vector is a full-operator null vector. Equation (1.7) does not require the even sector to be nonnegative.

1.3. Odd lowest-eigenvalue continuity and the attained first crossing

This step does not infer continuity of one parity branch from continuity of the minimum of both branches. Define

$$\lambda_-(a) = \inf_{0 \neq q \in \mathfrak{D}_a^-} \frac{Q_a(q)}{\|q\|_2^2}. \quad (\text{CP15:1.8})$$

Transport to $(-1, 1)$ by $q(x) = a^{-1/2}w(x/a)$. Write $\ell_n = \log n$, $b_n = \Lambda(n)/\sqrt{n}$, and retain

$$R(t) = \frac{e^{-|t|/2}}{1 - e^{-2|t|}} - \frac{1}{2|t|} - 2 \cosh(t/2), \quad R(0) = -\frac{7}{4}. \quad (\text{CP15:1.9})$$

The transported odd domain $\mathfrak{D}_1^-$ is independent of a , and the form is

$$\begin{aligned} \tilde{Q}_a(w) &= \frac{1}{2\pi} \int \log|\xi| |\widehat{w}(\xi)|^2 d\xi - \log(2\pi a) \|w\|_2^2 \\ &\quad - \sum_n b_n \langle w(\cdot + \ell_n/a) + w(\cdot - \ell_n/a), w \rangle \\ &\quad - \iint_{(-1,1)^2} a R(a(x-y)) w(y) \overline{w(x)} dy dx. \end{aligned} \quad (\text{CP15:1.10})$$

On any compact parameter interval $a \in [a_0/2, 2a_0]$, one may keep the single finite list $\ell_n \leq 4a_0$; all other translations vanish by zero extension. Equality at a threshold is harmless: translates with separation exactly two have overlap of measure zero. The prime operators are uniformly bounded and strongly continuous in a . The last kernel is continuous on its compact parameter set, so its operator is norm continuous in a . No norm continuity of translation operators is assumed.

The common positive logarithmic form embeds compactly into $L^2(-1, 1)$. To see this directly, its bounded sets have fixed compact support and Fourier tails bounded by $C/\log(2+T)$ outside $[-T, T]$. On that bounded frequency interval, the translation multiplier tends to one uniformly. These two statements give uniform translation continuity, and the Fréchet-Kolmogorov criterion gives compactness. The odd subspace is closed, so its embedding is compact as well. Consequently $\lambda_-(a)$ is attained.

For upper semicontinuity, insert a fixed normalized odd minimizer at a_0 into (1.10). Strong continuity of its finite prime translates and continuity of the remainder imply

$$\limsup_{a \rightarrow a_0} \lambda_-(a) \leq \lambda_-(a_0). \quad (\text{CP15:1.11})$$

For lower semicontinuity, choose normalized odd minimizers w_j for $a_j \rightarrow a_0$. A fixed odd trial function bounds their energies above. Uniform boundedness of all perturbations then bounds w_j in the common logarithmic form norm. After passage to a subsequence realizing the liminf, $w_j \rightarrow w$ strongly in L^2 , with $\|w\|_2 = 1$ and w odd. Uniform boundedness and strong continuity of the prime operators imply convergence of their pairings even with the changing vectors. The continuous remainder pairings converge as well. Lower semicontinuity of the common closed logarithmic form gives

$$\liminf_{j \rightarrow \infty} \lambda_-(a_j) \geq \tilde{Q}_{a_0}(w) \geq \lambda_-(a_0). \quad (\text{CP15:1.12})$$

Thus λ_- is continuous. It is also nonincreasing, by zero extension of the odd cores from smaller intervals into larger intervals.

There is an explicit unconditional positive starting interval. For odd w supported in $[-1, 1]$, $\int w = 0$, and Cauchy-Schwarz gives

$$|\widehat{w}(\xi)|^2 \leq \frac{2}{3} \xi^2 \|w\|_2^2, \quad \frac{1}{2\pi} \int \log |\xi| |\widehat{w}(\xi)|^2 d\xi \geq -\frac{2}{27\pi} \|w\|_2^2. \quad (\text{CP15:1.13})$$

The constant follows from $\int_0^1 (-\log t) t^2 dt = 1/9$. For $t > 0$, $-1/4 \leq e^{-t/2}/(1 - e^{-2t}) - 1/(2t) \leq 1$. Therefore $R(t) < 0$ and $|R(t)| \leq 1/4 + 2 \cosh a$ for $|t| \leq 2a$. If $2a < \log 2$, there are no active prime shifts, and (1.10) proves

$$\boxed{\lambda_-(a) \geq \log \frac{1}{2\pi a} - \frac{2}{27\pi} - a(1 + 8 \cosh a).} \quad (\text{CP15:1.14})$$

In particular the displayed lower bound exceeds one for $0 < a \leq 1/100$. This needs no numerical zero information: use $2 < \pi < 4$, $\cosh(1/100) < 2$, and $\log(100/8) > 2$.

If RH fails, the criterion (1.2) supplies a compact smooth odd negative witness, so $\lambda_-(a_1) < 0$ for some finite a_1 . Continuity, (1.14), and attainment give a finite first crossing a_* with

$$a_* > 1/100, \quad \lambda_-(a_*) = 0, \quad Q_{a_*}^- \geq 0, \quad 0 \neq v \in \ker A_{a_*}^-, \quad v \text{ real and odd.} \quad (\text{CP15:1.15})$$

Excluding these precise odd null vectors therefore suffices for unconditional Weil positivity.

1.4. Audit of the null identities and positivity uses

By (1.7), the full distributional equation $-g'' * v = 0$ holds on I_a . Every null-vector conclusion in checkpoints 13-14 that used this equation rather than positivity on arbitrary tests remains available. In particular,

$$\log |D|v = \mathcal{B}_a v \quad \text{on } I_a, \quad \mathcal{B}_a v = \log(2\pi)v + \sum_{\ell_n < 2a} b_n [v(\cdot + \ell_n) + v(\cdot - \ell_n)] + R * v. \quad (\text{CP15:1.16})$$

Outside I_a ,

$$\log |D|v(x) = -\frac{1}{2} \int_{-a}^a \frac{v(y)}{|x-y|} dy, \quad \|\log |D|v\|_2^2 \leq [(C_a^{\text{op}})^2 + \pi^2/2] \|v\|_2^2. \quad (\text{CP15:1.17})$$

Here $C_a^{\text{op}} = \log(2\pi) + 2 \sum_{\ell_n < 2a} b_n + \int_{-2a}^{2a} |R(t)| dt$. The exterior Carleman estimates, endpoint-distribution removal, and this logarithmic-square regularity use no even-sector sign.

Since g is even and v odd, $f_v = g * v$ is odd. Its affine restriction is therefore $f_v(t) = \alpha t$, and the proved convolution traces are

$$f_v(\pm a) = \pm a\alpha, \quad f'_v(\pm a) = \alpha. \quad (\text{CP15:1.18})$$

The endpoint-complete exterior formulas retain $\mathcal{B}_v(t) = f_v(t) - \alpha t$, with $\mathcal{B}_v(\pm a) = \mathcal{B}'_v(\pm a) = 0$. There is no need to assign endpoint values to v . The shifted resolvent identities and their invariant product remain exact.

The global Suzuki estimate and inward dilation followed by real even mollification now yield an odd real sequence $q_j \in C_c^\infty(I_a)$ such that

$$q_j \rightarrow v \text{ in the } \log |D| \text{ graph norm, } Vq_j \rightarrow Vv \text{ in } L^2, \quad 0 < \mathcal{S}(v) < \infty, \quad \mathcal{S}(q_j) \rightarrow \mathcal{S}(v), \quad Q_a(q_j) \rightarrow 0. \quad (\text{CP15:1.19})$$

The spectral tightness and exact complete correction are retained in the same established order of limits. No raw zero-cutoff norm convergence is added.

Smooth multiplication is still bounded on the full form domain, and (1.7) permits $Q_a(v, \phi^2 v) = 0$ for any smooth ϕ . Thus the complete double-commutator identity remains valid for all such multipliers. Its nonnegative-energy consequences now apply when the resulting tests are odd. Since

$$P_-(\phi v) = \phi_{\text{even}} v, \quad (\text{CP15:1.20})$$

the multiplier matrix is guaranteed positive semidefinite for even multipliers. In particular $\cos(\tau x)v$ is an admissible nonnegative-energy test. The identities for xv and $\sin(\tau x)v$ remain exact, but those two vectors are even; their energies and the sum of the cosine and sine energies acquire no sign from (1.6). Every dilation $U_s v$ is odd, so the entire mixed-dilation matrix in Groups 2-3 is admissible.

For any odd $q \in \mathfrak{D}_a^-$ and any full-domain r , the correct form Cauchy-Schwarz statement is

$$|Q_a(q, r)|^2 = |Q_a(q, P_- r)|^2 \leq Q_a(q) Q_a(P_- r). \quad (\text{CP15:1.21})$$

The projected energy retains a reflection cross term:

$$Q_a(P_- r) = \frac{1}{2} \{Q_a(r) - \Re Q_a(r, P r)\}. \quad (\text{CP15:1.22})$$

Using $Q_a(r)$ alone in (1.21) would not be justified under the present hypotheses.

1.5. A complete projected test calculation

For $0 < b \leq a$, $z \in \mathbb{C}$, set

$$r_{b,z}(t) = \mathbf{1}_{(0,b)}(t) e^{-izt}, \quad r_{b,z}^- = P_- r_{b,z}, \quad \mu_z = \delta_0 - e^{-izb} \delta_b - i\bar{z} \mathbf{1}_{(0,b)}(t) e^{-izt} dt. \quad (\text{CP15:1.23})$$

These piecewise smooth functions belong to the logarithmic form domain. The measure μ_z is the distributional derivative of $r_{b,z}$. Reflection changes the derivative's sign, so the exact projected energy, including every reflected prime and polar contribution, is

$$Q_a(r_{b,z}^-) = \frac{1}{2} \int_{[0,b]^2} [g(x-y) + g(x+y)] d\mu_z(y) \overline{d\mu_z(x)}. \quad (\text{CP15:1.24})$$

One proves the identity first by smoothing and then by weak convergence of derivative measures against the continuous kernel. It includes both endpoint atoms and all mixed endpoint-density terms. With $M_a^g = \sup_{|t| \leq 2a} |g(t)|$, its total variation estimate gives

$$0 \leq Q_a(r_{b,z}^-) \leq M_a^g (2 + b|z|)^2 e^{2b|\Im z|}. \quad (\text{CP15:1.25})$$

Consequently the checkpoint-13 core error identity becomes, for odd q ,

$$\mathcal{E}_b(q; z) = -ie^{-ibz} Q_a(q, r_{b,z}^-), \quad |\mathcal{E}_b(q; z)|^2 \leq e^{2b\Im z} Q_a(q) Q_a(r_{b,z}^-). \quad (\text{CP15:1.26})$$

Thus the compact-contour core-error estimates survive with the corrected projected test and the same finite upper bound. This calculation does not supply uniformity on an expanding contour.

At $z = 0$, (1.24) gives a new explicit necessary signed condition at an odd nonnegative interval:

$$\boxed{Q_a\left(\frac{1}{2}[\mathbf{1}_{(0,b)} - \mathbf{1}_{(-b,0)}]\right) = \frac{1}{2}g(2b) - 2g(b) \geq 0 \quad (0 < b \leq a).} \quad (\text{CP15:1.27})$$

Define the exact Lerch difference $D(t) = \Phi(1, 2, 1/4) - e^{-t/2} \Phi(e^{-2t}, 2, 1/4)$, and $d = \psi(1/4) - \log \pi$. Expanding every term in (1.27) yields

$$\begin{aligned} 0 \leq g(2b) - 4g(b) = & -16[\cosh(b/2) - 1]^2 + db + D(b) - \frac{1}{4}D(2b) \\ & + \sum_{\ell_n < 2b} b_n \{(2b - \ell_n)_+ - 4(b - \ell_n)_+\}. \end{aligned} \quad (\text{CP15:1.28})$$

This is the actual prime-archimedean-polar packet on the whole required interval. The earlier full-sector consequence $g(b) \leq 0$ cannot be imported into the odd-only argument. Equation (1.28) is its correctly projected replacement. It retains signed prime coefficients even though each b_n is positive.

1.6. Closing statement

Exact theorem and hypotheses. The unchanged complete Weil form has the odd closed domain (1.4); its lowest odd eigenvalue is attained, continuous and nonincreasing. It obeys the explicit positive lower bound (1.14). Failure of RH therefore produces exactly the finite odd first-null configuration (1.15). In that configuration the restricted null is a full-operator null and enjoys (1.16)-(1.19), with the original Suzuki norm.

Completed calculation and proof status. The parity transfer, independent odd continuity proof, quantitative small-interval sign, corrected Cauchy-Schwarz projection, and reflected test identity (1.24)-(1.28) are proved. Multipliers that leave the odd sector have been separated from legitimate positivity consequences; their exact identities remain available.

Consequence for Weil positivity. It is sufficient to exclude nonzero real odd vectors satisfying (1.6). All inward dilation combinations preserve the sector and can be tested by the full arithmetic form in Groups 2-3. Exclusion at every finite first crossing gives nonnegativity on all odd compact smooth tests, then Yoshida's criterion gives RH and Weil positivity for all tests. Only after that step may the normalized Suzuki identity be invoked to give $\Delta = 0$ and $\mathcal{E}_N = 0$ for every N .

Remaining signed expression and stopping line. The missing conclusion is a negative odd energy, not another domain passage. One precisely quantified sufficient lemma is: for every

$a > 1/100$ and every nonzero real odd v satisfying (1.6), there are finitely many $s_j > 0$, real coefficients c_j , and $\eta > 0$ such that

$$\sum_{j,k} c_j c_k Q_a(U_{s_j} v, U_{s_k} v) \leq -\eta \|v\|_2^2 \sum_j c_j^2 < 0. \quad (\text{CP15:1.29})$$

Its precise arithmetic entries are calculated in Group 2. The attempted general estimates (1.21) and (1.25) provide upper control of absolute values under the hypothetical positivity assumption; they supply no independent negative sign. The separate scalar packet (1.28) is exact and necessary, but its sign does not by itself rule out a null vector. Neither unconditional Weil positivity nor the original defect criterion is established by this group alone.

2. Closing group 2 – Complete mixed dilation and prime-power packets

2.1. Domain, signed cross correlations, and the mixed identity

Assume the hypothetical odd first crossing from Group 1: $a > 0$, $Q_a^- \geq 0$, and $0 \neq v \in \ker A_a^-$, chosen real and odd. Functions are extended by zero from $I_a = (-a, a)$. Group 1 identifies this vector with a full-operator null vector and supplies the checkpoint-14 regularity $v \in D(\mathcal{L})$, where $\mathcal{L} = \log |D|$. In particular

$$Q_a(v, w) = 0 \quad (w \in \mathfrak{D}(Q_a)), \quad U_s v = e^{s/2} v(e^s \cdot) \in D(\mathcal{L}) \cap L_{\text{odd}}^2(I_a), \quad s \geq 0. \quad (\text{CP15:2.1})$$

The latter assertion follows from $\mathcal{L}U_s = U_s(\mathcal{L} + s)$. Inward dilation and an even mollifier give the same-parity graph core already established. Thus the mixed calculations below pass from smooth functions without a derivative or endpoint trace of v .

Use the exact remainder and its exact archimedean-polar splitting

$$\begin{aligned} R(h) &= A_{\text{ar}}(|h|) - 2 \cosh(h/2), \\ A_{\text{ar}}(u) &= \frac{e^{-u/2}}{1 - e^{-2u}} - \frac{1}{2u} = \frac{e^{u/2}}{2 \sinh u} - \frac{1}{2u} \quad (u > 0), \\ A_{\text{ar}}(0) &= \frac{1}{4}, \quad R(0) = -\frac{7}{4}. \end{aligned} \quad (\text{CP15:2.2})$$

No truncation of this kernel is used. Write $M = \|v\|_2^2$. For $r \geq 0$, define

$$\begin{aligned} w_r &= U_r v, \quad \delta = e^{-r}, \quad A_v(r) = \langle v, w_r \rangle, \\ C_r(h) &= \int_{\mathbb{R}} v(x+h) w_r(x) dx, \quad J_r(h) = C_r(h) + C_r(-h). \end{aligned} \quad (\text{CP15:2.3})$$

These are real continuous functions. Both factors are odd, so $C_r(-h) = C_r(h)$, $J_r = 2C_r$, and

$$C_r(0) = A_v(r), \quad |C_r(h)| \leq M, \quad C_r(h) = 0 \quad \text{if } |h| \geq H_r := a(1 + \delta). \quad (\text{CP15:2.4})$$

The support endpoint equality follows by continuity, or directly from the zero measure of the possible overlap. Neither oddness nor (2.4) makes C_r nonnegative.

The real polarized form in the original normalization is

$$Q_a(u, w) = \langle \mathcal{L}u, w \rangle - \log(2\pi) \langle u, w \rangle - \sum_{\ell_n < 2a} b_n \{C_{u,w}(\ell_n) + C_{u,w}(-\ell_n)\} - \int_{-2a}^{2a} R(h) C_{u,w}(h) dh, \quad (\text{CP15:2.5})$$

where $\ell_n = \log n$, $b_n = \Lambda(n)/\sqrt{n}$. For $0 \leq s \leq t$, put $r = t - s$. Unitarity gives

$$\langle \mathcal{L}U_s v, U_t v \rangle = \langle \mathcal{L}v, U_r v \rangle + sA_v(r), \quad C_{U_s v, U_t v}(h) = C_r(e^s h). \quad (\text{CP15:2.6})$$

Subtract $Q_a(v, U_r v) = 0$, and change variables in the R -integral. The complete answer is

$$\begin{aligned} K_v(s, t) := Q_a(U_s v, U_t v) &= sA_v(r) \\ &\quad - \sum_{\ell_n < 2a} b_n [J_r(e^s \ell_n) - J_r(\ell_n)] \\ &\quad - \int_{-2a}^{2a} [e^{-s} R(e^{-s} h) - R(h)] C_r(h) dh. \end{aligned} \quad (\text{CP15:2.7})$$

Thus the proposed mixed formula has the stated normalization. On the diagonal $r = 0$, $J_0 = 2C_v$, and (2.7) is checkpoint 14's finite-dilation identity. At $s = 0$, every increment vanishes and $K_v(0, t) = 0$. For arbitrary $s, t \geq 0$, define (2.7) using $\min(s, t)$ and $r = |t - s|$; the real symmetry $K_v(s, t) = K_v(t, s)$ then follows both from the original form and from this formula.

No boundary term has been suppressed: (2.6) is a whole-line multiplier identity, zero extension handles every support threshold, and the test $U_r v$ lies inside the interval where the complete null equation holds. The $-\log(2\pi)$ terms cancel exactly in the subtraction. The complete polar and archimedean terms remain in R . The exterior part of $\mathcal{L}v$ pairs with zero in $Q_a(v, U_r v)$. This is an equality of legitimate closed forms, without a one-sided stationarity assumption.

2.2. Exact integer prime-power regrouping

For an integer $m \geq 2$, define

$$\mathcal{P}_m(r) = \sum_{\ell_n < 2a} b_n [J_r(m\ell_n) - J_r(\ell_n)], \quad w_m(p, k) = \mathbf{1}_{m|k} p^{-k/(2m)} - p^{-k/2}. \quad (\text{CP15:2.8})$$

Since $\Lambda(p^j) = \log p$, substitute $k = mj$ in the first part. Terms with $k \log p \geq 2a$ have zero correlation, while every remaining first-part term has $j \log p < 2a$. Therefore

$$\mathcal{P}_m(r) = \sum_{\substack{p \text{ prime}, k \geq 1 \\ k \log p < 2a}} (\log p) w_m(p, k) J_r(k \log p). \quad (\text{CP15:2.9})$$

The sharper effective cutoff is $k \log p < H_r$, but retaining the common $2a$ cutoff is useful when comparing packets. This regrouping retains prime powers of every admissible exponent. The coefficient is positive on multiples of m and negative on the other exponents; the correlation multiplying it remains signed.

Introduce the complete archimedean and polar companions

$$\begin{aligned} \mathcal{H}_m(r) &= \int_{-2a}^{2a} [m^{-1} A_{\text{ar}}(|h|/m) - A_{\text{ar}}(|h|)] C_r(h) dh, \\ F(b) &= \int_{-a}^a e^{bx} v(x) dx = \int_{-a}^a \sinh(bx) v(x) dx. \end{aligned} \quad (\text{CP15:2.10})$$

Because $\int e^{bx} U_r v(x) dx = \sqrt{\delta} F(\delta b)$ and both functions are odd,

$$\int \cosh(bh) C_r(h) dh = -\sqrt{\delta} F(b) F(\delta b). \quad (\text{CP15:2.11})$$

Consequently, with

$$\mathcal{Z}_m(r) = 2\sqrt{\delta} \left[F(1/2)F(\delta/2) - \frac{1}{m}F(1/(2m))F(\delta/(2m)) \right], \quad (\text{CP15:2.12})$$

the complete mixed packet is

$$\boxed{\mathcal{K}_m(r) := K_v(\log m, \log m + r) = (\log m)A_v(r) - \mathcal{P}_m(r) - \mathcal{H}_m(r) + \mathcal{Z}_m(r).} \quad (\text{CP15:2.13})$$

Here are all three requested packets. In the table, the displayed coefficient is multiplied by $(\log p)J_r(k \log p)$, and the displayed archimedean kernel is integrated against C_r .

Factor m	Prime coefficient $w_m(p, k)$	Archimedean kernel	Polar term
2	$\mathbf{1}_{2 k}p^{-k/4} - p^{-k/2}$	$\frac{1}{2}A_{\text{ar}}(h /2) - A_{\text{ar}}(h)$	$2\sqrt{\delta} [F(1/2)F(\delta/2) - \frac{1}{2}F(1/4)F(\delta/4)]$
3	$\mathbf{1}_{3 k}p^{-k/6} - p^{-k/2}$	$\frac{1}{3}A_{\text{ar}}(h /3) - A_{\text{ar}}(h)$	$2\sqrt{\delta} [F(1/2)F(\delta/2) - \frac{1}{3}F(1/6)F(\delta/6)]$
6	$\mathbf{1}_{6 k}p^{-k/12} - p^{-k/2}$	$\frac{1}{6}A_{\text{ar}}(h /6) - A_{\text{ar}}(h)$	$2\sqrt{\delta} [F(1/2)F(\delta/2) - \frac{1}{6}F(1/12)F(\delta/12)]$

For example, the $(2, 3)$ matrix entry has $m = 2$, $r = \log(3/2)$, $\delta = 2/3$; it uses $F(1/2)F(1/3)$ and $\frac{1}{2}F(1/4)F(1/6)$. The $(2, 6)$ entry has $m = 2$, $\delta = 1/3$, while the $(3, 6)$ entry has $m = 3$, $\delta = 1/2$. These are different cross correlations and moments. They cannot be replaced by the diagonal packets.

2.3. Endpoint substitution and odd cancellations before estimating

Retain checkpoint 14's exact endpoint functionals:

$$\begin{aligned} I_n &= \int_{-a}^{a-\ell_n} (a - \ell_n - y)v(y) dy, & J_n &= \int_{-a}^{a-\ell_n} v(y) dy, \\ U_a &= \int_{-a}^a J_{\text{ar}}(a - y)v(y) dy, & V_a &= \int_{-a}^a J'_{\text{ar}}(a - y)v(y) dy, \\ P_o &= \sum_{\ell_n < 2a} b_n(I_n - aJ_n) + U_a - aV_a, & d_a &= 4a \cosh(a/2) - 8 \sinh(a/2) > 0, \end{aligned} \quad (\text{CP15:2.14})$$

where

$$J_{\text{ar}}(t) = -\frac{t}{2}[\psi(1/4) - \log \pi] - \frac{1}{4}\{\Phi(1, 2, 1/4) - e^{-t/2}\Phi(e^{-2t}, 2, 1/4)\}, \quad t \geq 0. \quad (\text{CP15:2.15})$$

The odd affine endpoint equation $f_v(t) = \alpha t$ gives $F(1/2) = P_o/d_a$. Insert it into (2.12) before estimating:

$$\boxed{\mathcal{Z}_m(r) = 2\sqrt{\delta} \left[\frac{P_o}{d_a}F(\delta/2) - \frac{1}{m}F(1/(2m))F(\delta/(2m)) \right].} \quad (\text{CP15:2.16})$$

For $r = 0$, both occurrences of $F(1/2)$ are replaced. If $E_n = I_n - aJ_n$ and $E_{\text{ar}} = U_a - aV_a$, the endpoint contribution is exactly

$$\mathcal{Z}_m(0) = \frac{2}{d_a^2} \left[\sum_{n,j} b_n b_j E_n E_j + 2E_{\text{ar}} \sum_n b_n E_n + E_{\text{ar}}^2 \right] - \frac{2}{m} F(1/(2m))^2. \quad (\text{CP15:2.17})$$

Thus the prime-prime and prime-archimedean endpoint cross terms are retained. The scaled function $U_r v$ has not been asserted to be a null vector, so no endpoint-null identity is applied to its independent moment $F(\delta/2)$.

Oddness supplies two further exact cancellations:

$$\int C_r(h) dh = 0, \quad \int h^2 C_r(h) dh = -2\delta^{3/2} \left(\int_{-a}^a x v(x) dx \right)^2 \leq 0. \quad (\text{CP15:2.18})$$

Both follow by integrating $h = x - y$ in the product $v(x)U_r v(y)$. In particular the constant $A_{\text{ar}}(0) = 1/4$ drops out of the entire archimedean packet. Let

$$L_{\text{ar}}(a) = \sup_{0 \leq u \leq 2a} |A'_{\text{ar}}(u)| < \infty.$$

Then the exact kernel remaining in $\mathcal{H}_m(r)$ is

$$B_m(h) = m^{-1} [A_{\text{ar}}(|h|/m) - 1/4] - [A_{\text{ar}}(|h|) - 1/4]. \quad (\text{CP15:2.19})$$

To bound it without a monotonicity assumption on A'_{ar} , set $f(t) = t[A_{\text{ar}}(t|h|) - 1/4]$, $t \in [1/m, 1]$. The mean value bound yields $|f'(t)| \leq 2t|h|L_{\text{ar}}(a)$. Integration in t , followed by (2.4), proves

$$|B_m(h)| \leq (1 - m^{-2})L_{\text{ar}}(a)|h|, \quad |\mathcal{H}_m(r)| \leq (1 - m^{-2})L_{\text{ar}}(a)H_r^2 M. \quad (\text{CP15:2.20})$$

This is an odd-sector improvement of the unsigned archimedean budget. It is not a sign estimate for the complete kernel. It differentiates only the explicit scalar kernel, never the unknown vector.

There is also a useful exact signed leading component. Put $G(x) = \int_{-\infty}^x v(y) dy$, which is compactly supported because $\int v = 0$, and set

$$E_{\text{ar}}^{(2)}(u) = A_{\text{ar}}(u) - \frac{1}{4} + \frac{u}{48}.$$

Here $A'_{\text{ar}}(0) = -1/48$ and $E_{\text{ar}}^{(2)}(u) = O(u^2)$, but the following formula retains the whole function:

$$\begin{aligned} \mathcal{H}_m(r) = & -\frac{1 - m^{-2}}{24} \delta \langle G, U_r G \rangle \\ & + \int \left[m^{-1} E_{\text{ar}}^{(2)}(|h|/m) - E_{\text{ar}}^{(2)}(|h|) \right] C_r(h) dh. \end{aligned} \quad (\text{CP15:2.21})$$

Indeed, $\int |h| C_r(h) dh = -2\delta \langle G, U_r G \rangle$, by two distributional integrations by parts using $(|h|)'' = 2\delta_0$. For $r = 0$, the displayed leading contribution to $-\mathcal{H}_m(0)$ is nonnegative. The remainder is still present. For example, with $B_{\text{ar}}(a) = \frac{1}{2} \sup_{[0, 2a]} |A''_{\text{ar}}|$, its absolute value is at most $\frac{2}{3} B_{\text{ar}}(a)(1 + m^{-3})H_r^3 M$. Derivatives here are on the nonnegative scalar variable, including their finite right limits at zero.

In the zero-polar-moment branch $F(1/2) = 0$, (2.17) gives the independent sign

$$\mathcal{Z}_m(0) = -\frac{2}{m} F(1/(2m))^2 \leq 0. \quad (\text{CP15:2.22})$$

This sign concerns precisely the polar increment. The prime and archimedean contributions in (2.13) remain coupled to it.

2.4. Complete local prime symbols and the 2-by-3 interaction

For each active prime $p < e^{2a}$, put $q = p^{-1/2}$ and

$$P_q(\theta) = \frac{1 - q^2}{1 - 2q \cos \theta + q^2} = 1 + 2 \sum_{j \geq 1} q^j \cos(j\theta). \quad (\text{CP15:2.23})$$

The series is absolutely convergent. Since $C_r(k \log p) = 0$ past the support cutoff, extending each exponent sum in (2.9) to infinity is exact. Plancherel gives the complete local formula

$$\sum_{k \geq 1} w_m(p, k) J_r(k \log p) = \frac{1}{2\pi} \int_{\mathbb{R}} [P_q(m\xi \log p) - P_q(\xi \log p)] \widehat{v}(\xi) \overline{\widehat{U_r v}(\xi)} d\xi. \quad (\text{CP15:2.24})$$

For real odd inputs the Fourier product is real and even; it need not have a fixed sign when $r > 0$. The symbol in brackets has mean zero on its phase circle and is nonzero for $m > 1$, since its frequency-one coefficient is $-2q$. Thus each actual prime symbol has both signs. This is an exact property of the prime packet under investigation. It does not prove that the actual null vector places sufficient mass on either sign.

The Poisson range yields the explicit bound

$$|P_q(m\theta) - P_q(\theta)| \leq \frac{4q}{1 - q^2}, \quad |\mathcal{P}_m(r)| \leq M \sum_{p < e^{2a}} (\log p) \frac{4p^{-1/2}}{1 - p^{-1}}. \quad (\text{CP15:2.25})$$

This uses the complete exponent series and the genuine L^2 Cauchy-Schwarz inequality; it does not treat signed cross correlations as positive measures.

The integer packets obey an exact composition law:

$$w_{mn}(p, k) = w_m(p, k) + \mathbf{1}_{m|k} w_n(p, k/m) = w_n(p, k) + \mathbf{1}_{n|k} w_m(p, k/n). \quad (\text{CP15:2.26})$$

For the scales 2, 3, 6, set

$$\eta(p, k) = w_6(p, k) - w_2(p, k) - w_3(p, k) = \mathbf{1}_{6|k} p^{-k/12} - \mathbf{1}_{2|k} p^{-k/4} - \mathbf{1}_{3|k} p^{-k/6} + p^{-k/2}. \quad (\text{CP15:2.27})$$

This exposes an arithmetic sign pattern that individual scale energies do not display:

- If k is divisible by neither 2 nor 3, then $\eta = p^{-k/2} > 0$.
- If k is divisible by exactly one of 2, 3, then $\eta < 0$.
- If $6 \mid k$, put $u = p^{-k/12}$. Then

$$\eta = u - u^2 - u^3 + u^6 = u(1 - u)(1 - u^2 - u^3 - u^4). \quad (\text{CP15:2.28})$$

This is positive except for $(p, k) = (2, 6)$, whose coefficient is $(2\sqrt{2} - 3)/8 < 0$. Indeed, in every other case $u \leq 1/\sqrt{3}$, so $u^2 + u^3 + u^4 \leq (4 + \sqrt{3})/9 < 1$. The exceptional term occurs only if $6 \log 2 < 2a$.

The archimedean and polar companions of this arithmetic interaction must also be included. Define

$$\begin{aligned} \mathcal{A}_{2,3}(h) &= \frac{1}{6} A_{\text{ar}}(|h|/6) - \frac{1}{2} A_{\text{ar}}(|h|/2) - \frac{1}{3} A_{\text{ar}}(|h|/3) + A_{\text{ar}}(|h|), \\ \mathcal{Z}_{2,3}(r) &= 2\sqrt{\delta} \left[-F(1/2)F(\delta/2) - \frac{1}{6} F(1/12)F(\delta/12) \right. \\ &\quad \left. + \frac{1}{2} F(1/4)F(\delta/4) + \frac{1}{3} F(1/6)F(\delta/6) \right]. \end{aligned} \quad (\text{CP15:2.29})$$

Then $\log 6 - \log 2 - \log 3 = 0$ cancels the full logarithmic term and leaves the new exact combined identity

$$\boxed{\mathcal{K}_6(r) - \mathcal{K}_2(r) - \mathcal{K}_3(r) = - \sum_{\substack{p,k \\ k \log p < 2a}} (\log p) \eta(p, k) J_r(k \log p) - \int_{-2a}^{2a} \mathcal{A}_{2,3}(h) C_r(h) dh + \mathcal{Z}_{2,3}(r).} \quad (\text{CP15:2.30})$$

Its prime symbol is $P_q(6\theta) - P_q(2\theta) - P_q(3\theta) + P_q(\theta)$. The constant $\mathcal{A}_{2,3}(0) = 1/12$ integrates to zero by (2.18). The endpoint substitution $F(1/2) = P_o/d_a$ applies to (2.29) exactly as in (2.16); for $r = 0$ its squared endpoint term retains the three classes of cross terms in (2.17).

Identity (2.30) compares three entries at the same relative scale r . The 3-by-3 matrix on factors 2, 3, 6 uses three distinct relative scales. Equating those correlations would be an additional assertion; no such step is made.

2.5. Closing statement and the remaining signed interaction

For every real odd first-null vector with the domain in (2.1), equations (2.7), (2.9), (2.13), (2.16)-(2.18), (2.21), (2.24), and (2.26)-(2.30) are proved exact identities. The explicit bounds (2.20), (2.25), and the stated exact-remainder bound are proved independently of RH. Every prime translation, completed polar term, and endpoint cross term is retained. These formulas supply the entries needed for the optimized matrix in Group 3 while preserving the original form and Suzuki metric.

The strongest general signed facts obtained here are the odd moment sign in (2.18), the diagonal polar sign on the zero-moment branch (2.22), and the exact arithmetic coefficient classification (2.28). None assigns a sign to the full matrix.

For clarity, a one-scale contradiction would require the actual coupled expression to satisfy, for some finite integer $m \geq 2$,

$$\mathcal{P}_m(0) + \mathcal{H}_m(0) - \mathcal{Z}_m(0) > (\log m)M. \quad (\text{CP15:2.31})$$

An optimized mixed witness instead requires the determinant inequality calculated in Group 3. The attempted bounds (2.20) and (2.25) estimate absolute magnitudes. They do not establish (2.31) or the determinant sign: the prime symbols have both signs, their cross correlations are signed, and the endpoint substitutions constrain $F(1/2)$ while retaining the independent scaled moments.

The narrowed arithmetic object is now explicit: the prime-power divisibility combination in (2.27), with the complete companions (2.29), or its three relative-scale versions in the matrix. Proving an independent strict sign for an admissible combination would exclude that odd null vector and feed the odd-test Weil-positivity route of Group 1. That strict sign remains open in this group.

3. Closing group 3 – Full mixed-dilation optimization

3.1. Domain and complete three-scale matrix

Assume the odd first-null hypotheses of Group 1. Thus v is real, odd, nonzero, supported in $[-a, a]$, belongs to the whole-line domain of $\mathcal{L} = \log|D|$, and satisfies the complete null equation. All

forms in this group use the original arithmetic form; the Suzuki metric is unchanged. Set

$$M = \|v\|_2^2, \quad u_m = U_{\log m} v, \quad F(b) = \int_{-a}^a e^{bx} v(x) dx, \quad A(r) = \langle v, U_r v \rangle. \quad (\text{CP15:3.1})$$

The established graph-domain passage permits every u_m , its finite linear combinations, and all mixed pairings. For $m \leq n$ write $r_{mn} = \log(n/m)$. The exact Group-2 calculation has the following four-component form:

$$K_{mn} = Q_a(u_m, u_n) = T_{mn} - P_{mn} - B_{mn} + \Pi_{mn}, \quad T_{mn} = (\log m) A(r_{mn}). \quad (\text{CP15:3.2})$$

The complete prime packet and archimedean companion are

$$\begin{aligned} P_{mn} &= \sum_{p, k: k \log p < 2a} (\log p) w_m(p, k) J_{v, r_{mn}}(k \log p), \\ w_m(p, k) &= \mathbf{1}_{m|k} p^{-k/(2m)} - p^{-k/2}, \\ B_{mn} &= \int_{-2a}^{2a} [m^{-1} A_{\text{ar}}(|h|/m) - A_{\text{ar}}(|h|)] C_{v, r_{mn}}(h) dh, \\ A_{\text{ar}}(h) &= \frac{e^{-h/2}}{1 - e^{-2h}} - \frac{1}{2h}, \quad A_{\text{ar}}(0) = \frac{1}{4}. \end{aligned} \quad (\text{CP15:3.3})$$

Here $J_{v, r} = 2C_{v, r}$ because both real inputs are odd. All correlations retain their signs. The actual support cutoff is $|h| < a(1 + e^{-r})$; using the larger fixed cutoff $2a$ adds zero terms only. Define $F_0 = F(1/2)$. The completed polar entry is

$$\Pi_{mn} = 2\sqrt{\frac{m}{n}} \left[F_0 F\left(\frac{m}{2n}\right) - \frac{1}{m} F\left(\frac{1}{2m}\right) F\left(\frac{1}{2n}\right) \right]. \quad (\text{CP15:3.4})$$

The expression is specified for $m \leq n$ and extended symmetrically. No assertion that its two summands separately define positive matrices is used.

For the requested factors, every polar entry is explicitly

$$\begin{aligned} \Pi_{22} &= 2F_0^2 - F(1/4)^2, & \Pi_{33} &= 2F_0^2 - \frac{2}{3}F(1/6)^2, \\ \Pi_{66} &= 2F_0^2 - \frac{1}{3}F(1/12)^2, & \Pi_{23} &= 2\sqrt{2/3}F_0F(1/3) - \sqrt{2/3}F(1/4)F(1/6), \\ \Pi_{26} &= \frac{2}{\sqrt{3}}F_0F(1/6) - \frac{1}{\sqrt{3}}F(1/4)F(1/12), & \Pi_{36} &= \sqrt{2}F_0F(1/4) - \frac{\sqrt{2}}{3}F(1/6)F(1/12). \end{aligned} \quad (\text{CP15:3.5})$$

These include the scaled mixed moments, notably $F(1/3)$, which an endpoint equation for F_0 alone does not determine.

3.2. Endpoint substitution before optimization

Use the exact endpoint functional from checkpoint 14:

$$F_0 = \frac{P_o(v)}{d_a}, \quad d_a = 4a \cosh(a/2) - 8 \sinh(a/2) > 0, \quad (\text{CP15:3.6})$$

$$\begin{aligned} P_o(v) &= \sum_{\ell_n < 2a} b_n D_n + E_a, \\ D_n &= \int_{-a}^{a-\ell_n} (-\ell_n - y) v(y) dy, \\ E_a &= \int_{-a}^a [J(a-y) - aJ'(a-y)] v(y) dy. \end{aligned} \quad (\text{CP15:3.7})$$

Here J is the exact Lerch/digamma primitive of checkpoint 14, not the correlation $J_{v,r}$. Its logarithmic derivative singularity is integrable against this compact L^2 input. In every entry of (3.5), insert (3.6) before estimating. In particular

$$P_o(v)^2 = \sum_{n,j} b_n b_j D_n D_j + 2E_a \sum_n b_n D_n + E_a^2. \quad (\text{CP15:3.8})$$

Thus the endpoint substitution retains prime-prime and prime-archimedean interactions. Off-diagonal polar entries contain the signed products $d_a^{-1}(\sum b_n D_n + E_a)F(b)$, which remain coupled.

3.3. Exact two-scale Schur residual and all cross terms

Write $d_2 = K_{22}$, $d_3 = K_{33}$, and $h = K_{23}$. When $d_3 > 0$, direct minimization over real c gives

$$\inf_{c \in \mathbb{R}} Q_a(u_2 - cu_3) = d_2 - \frac{h^2}{d_3}, \quad c_{\text{opt}} = \frac{h}{d_3}. \quad (\text{CP15:3.9})$$

Consequently a strict negative residual is equivalent to $h^2 > d_2 d_3$. This is a genuine mixed-direction test, with no contribution from the original null vector's zero row.

For any symmetric component arrays X, Y , define

$$\mathcal{D}(X) = X_{22}X_{33} - X_{23}^2, \quad \mathcal{M}(X, Y) = X_{22}Y_{33} + Y_{22}X_{33} - 2X_{23}Y_{23}. \quad (\text{CP15:3.10})$$

Then the full determinant, with every interaction retained, is

$$\begin{aligned} d_2 d_3 - h^2 &= \mathcal{D}(T) + \mathcal{D}(P) + \mathcal{D}(B) + \mathcal{D}(\Pi) \\ &\quad - \mathcal{M}(T, P) - \mathcal{M}(T, B) + \mathcal{M}(T, \Pi) \\ &\quad + \mathcal{M}(P, B) - \mathcal{M}(P, \Pi) - \mathcal{M}(B, \Pi). \end{aligned} \quad (\text{CP15:3.11})$$

For example the complete prime-prime piece is

$$\begin{aligned} \mathcal{D}(P) &= \sum_{p,k} \sum_{q,j} (\log p)(\log q) [w_2(p, k)w_3(q, j)J_{v,0}(k \log p)J_{v,0}(j \log q) \\ &\quad - w_2(p, k)w_2(q, j)J_{v,\log(3/2)}(k \log p)J_{v,\log(3/2)}(j \log q)]. \end{aligned} \quad (\text{CP15:3.12})$$

The prime-archimedean and prime-polar pieces are exactly

$$\begin{aligned} \mathcal{M}(P, B) &= \sum_{p,k} (\log p) \{w_2 J_0 B_{33} + w_3 J_0 B_{22} - 2w_2 J_{\log(3/2)} B_{23}\}, \\ \mathcal{M}(P, \Pi) &= \sum_{p,k} (\log p) \{w_2 J_0 \Pi_{33} + w_3 J_0 \Pi_{22} - 2w_2 J_{\log(3/2)} \Pi_{23}\}. \end{aligned} \quad (\text{CP15:3.13})$$

The arguments of each w_m and J_r in (3.13) are respectively (p, k) and $k \log p$. Inserting (3.5)-(3.8) expands every remaining endpoint and polar product; none is discarded or bounded separately at this stage.

If $d_3 < 0$, u_3 already supplies a negative direction. If $d_3 = 0$ and $h \neq 0$, the polynomial $d_2 - 2ch$ is negative for an appropriate finite c . If $d_3 = h = 0$, this pair gives no new direction beyond the sign of d_2 . The actual *first-crossing* hypotheses strengthen this bookkeeping: the earlier interval $I_{a/2}$ has strictly positive odd lowest eigenvalue $\lambda_1^-(a/2) > 0$. Therefore

$$[K_{mn}]_{m,n \in \{2,3,6\}} \succeq \lambda_1^-(a/2)[\langle u_m, u_n \rangle]_{m,n \in \{2,3,6\}} > 0. \quad (\text{CP15:3.14})$$

The strict final inequality follows from linear independence of finitely many distinct inward dilates of a nonzero compactly supported function: order their essential support radii and restrict a purported dependence to the annulus between the largest two; its largest-support coefficient vanishes, and induction finishes. Thus zero denominators cannot occur at a genuine first crossing. Equation (3.14) is a consequence of the *hypothetical first crossing*, not an independent exclusion of it. An arithmetic estimate contradicting (3.14) would exclude that hypothesis.

3.4. Three-scale determinant and a composed arithmetic direction

The symmetric three-scale matrix is completely specified by (3.2)-(3.5). Its determinant is

$$\det K = K_{22}K_{33}K_{66} + 2K_{23}K_{26}K_{36} - K_{22}K_{36}^2 - K_{33}K_{26}^2 - K_{66}K_{23}^2. \quad (\text{CP15:3.15})$$

For explicit all-component expansion, take $X^0 = T, X^1 = P, X^2 = B, X^3 = \Pi$, with signs $\epsilon = (1, -1, -1, 1)$. Then

$$\begin{aligned} \det K = \sum_{\alpha, \beta, \gamma=0}^3 \epsilon_\alpha \epsilon_\beta \epsilon_\gamma [& X_{22}^\alpha X_{33}^\beta X_{66}^\gamma + 2X_{23}^\alpha X_{26}^\beta X_{36}^\gamma \\ & - X_{22}^\alpha X_{36}^\beta X_{36}^\gamma - X_{33}^\alpha X_{26}^\beta X_{26}^\gamma - X_{66}^\alpha X_{23}^\beta X_{23}^\gamma]. \end{aligned} \quad (\text{CP15:3.16})$$

This finite sum includes all cubic prime, archimedean, and polar interactions. If all one- and two-dimensional principal minors are nonnegative, a negative determinant gives a negative eigenvalue. More generally a real symmetric matrix is positive semidefinite exactly when every principal minor is nonnegative; a negative principal minor is sufficient for the desired negative witness.

A particularly natural composed direction is

$$w = (U_{\log 2} - I)(U_{\log 3} - I)v = u_6 - u_2 - u_3 + v. \quad (\text{CP15:3.17})$$

Using the full null identity, its energy is exactly

$$Q_a(w) = K_{66} + K_{22} + K_{33} + 2K_{23} - 2K_{26} - 2K_{36} = T_* - P_* - B_* + \Pi_*. \quad (\text{CP15:3.18})$$

The logarithmic contribution and full prime contribution are

$$T_* = 2 \log 6 M + 2 \log 2 A(\log(3/2)) - 2 \log 2 A(\log 3) - 2 \log 3 A(\log 2), \quad (\text{CP15:3.19})$$

$$\begin{aligned} P_* = \sum_{p,k} (\log p) \{ & \eta(p, k) J_0 + 2w_2(p, k) [J_0 + J_{\log(3/2)} - J_{\log 3}] \\ & + 2w_3(p, k) [J_0 - J_{\log 2}] \}, \quad \eta = w_6 - w_2 - w_3. \end{aligned} \quad (\text{CP15:3.20})$$

Every J in (3.20) is evaluated at $k \log p$. Group 2's exact divisibility signs for η , including its exceptional $p = 2, k = 6$ coefficient, therefore do not collapse this direction to a single signed packet. The other relative-scale correlations remain.

The archimedean contribution is the exact combination $B_* = B_{66} + B_{22} + B_{33} + 2B_{23} - 2B_{26} - 2B_{36}$. Its full polar companion simplifies algebraically to

$$\begin{aligned} \Pi_* = \frac{P_o(v)}{d_a} \left[\frac{6P_o(v)}{d_a} + 4\sqrt{2/3}F(1/3) - \frac{4}{\sqrt{3}}F(1/6) - 2\sqrt{2}F(1/4) \right] \\ - 2 \left[\frac{F(1/12)}{\sqrt{6}} - \frac{F(1/4)}{\sqrt{2}} - \frac{F(1/6)}{\sqrt{3}} \right]^2. \end{aligned} \quad (\text{CP15:3.21})$$

The last square has a proved nonpositive sign. The preceding endpoint-coupled term has no sign supplied by the endpoint identity. Formula (3.21) keeps all mixed moment products, including the ones inside $P_o(v)^2$. It is the concrete remaining polar contribution to this composed witness.

3.5. Exact link to the polar inverse, with the complement retained

Let $H_a = A_a^- + 2|s_a\rangle\langle s_a|$, as in Group 4, with $s_a(x) = \sinh(x/2)$. On any linearly independent finite dilation family set

$$f_i = \langle s_a, u_{m_i} \rangle = m_i^{-1/2} F(1/(2m_i)), \quad D_{ij} = \langle H_a u_{m_i}, u_{m_j} \rangle. \quad (\text{CP15:3.22})$$

The domain permits these operator pairings, or equivalently their closed-form pairings. Directly from the complete odd polar term,

$$\boxed{K = D - 2ff^\top}. \quad (\text{CP15:3.23})$$

All prime translations and archimedean terms are still in D . For $D > 0$, the matrix determinant lemma and congruence give

$$\det K = \det D (1 - \beta_{\text{block}}), \quad \beta_{\text{block}} = 2f^\top D^{-1}f, \quad K \succeq 0 \iff \beta_{\text{block}} \leq 1. \quad (\text{CP15:3.24})$$

For the two-scale block the exact formula is

$$\det K = \det D - 2\{D_{33}f_2^2 + D_{22}f_3^2 - 2D_{23}f_2f_3\}. \quad (\text{CP15:3.25})$$

The quartic polar terms cancel identically because the completed odd polar correction has rank one. The terms D_{ij} still contain every arithmetic contribution, so this cancellation does not remove prime-polar or archimedean-polar coupling. It identifies the mixed-dilation sign test as a finite-dimensional instance of Group 4's actual inverse problem, without altering the Suzuki metric.

If $H_a > 0$ is invertible on the full odd space, set $z = H_a^{-1}s_a$, and let P_D be orthogonal projection onto the chosen finite dilation span in the H_a -energy inner product. Galerkin's identity yields

$$\beta_{\text{full}} = 2\langle s_a, H_a^{-1}s_a \rangle = \beta_{\text{block}} + 2\|(I - P_D)z\|_{H_a}^2. \quad (\text{CP15:3.26})$$

Thus a finite-block value greater than one would prove a negative direction. A finite-block value below one is a lower bound on the full scalar, and requires the full nonnegative tail correction in (3.26) for an upper-bound argument. At the hypothetical first crossing, (3.14) already forces $\beta_{\text{block}} < 1$; this is again a constraint under that hypothesis. Group 4 retains the zero-moment kernel branch when the full inverse is unavailable.

3.6. Independent bounds and the precise sign obstruction

Several genuinely unconditional constraints are available, with the null equation used only to supply its established logarithmic regularity constant κ_a .

First, the logarithmic scale matrix is strictly positive. For $s_i > 0$,

$$T_{ij} = \min(s_i, s_j) \langle U_{s_i}v, U_{s_j}v \rangle = \int_0^\infty \langle \mathbf{1}_{u < s_i} U_{s_i}v, \mathbf{1}_{u < s_j} U_{s_j}v \rangle du. \quad (\text{CP15:3.27})$$

This Gram representation proves positivity without arithmetic assumptions. For factors 2, 3,

$$\mathcal{D}(T) = (\log 2)(\log 3)M^2 - (\log 2)^2 A(\log(3/2))^2 \geq (\log 2) \log(3/2)M^2 > 0. \quad (\text{CP15:3.28})$$

For factors 2, 3, 6, the Brownian matrix $[\min(s_i, s_j)]$ has inverse trace $3/\log 2 + 2/\log(3/2)$. The Schur product theorem, or (3.27), gives

$$T \succeq \frac{M}{3/\log 2 + 2/\log(3/2)} I, \quad T_* \geq \frac{3M}{3/\log 2 + 2/\log(3/2)} > 0. \quad (\text{CP15:3.29})$$

Thus a negative mixed direction requires the *complete* arithmetic increment to overcome a strictly positive logarithmic contribution.

Second, legitimate translation-Gram inequalities preserve signed correlations:

$$|C_{v,r}(h)| \leq M, \quad |C_{v,r}(h) - C_{v,r}(k)|^2 \leq 2M\{M - C_v(h - k)\}. \quad (\text{CP15:3.30})$$

The second follows by Cauchy-Schwarz applied to $(T_h - T_k)v$ and $U_r v$. It makes no assertion that $C_{v,r}(h)$ is nonnegative.

The full one-prime packet from Group 2 has Poisson symbol

$$P_{p^{-1/2}}(m\theta) - P_{p^{-1/2}}(\theta), \quad P_q(\theta) = \frac{1 - q^2}{1 - 2q \cos \theta + q^2}. \quad (\text{CP15:3.31})$$

It has zero average and takes both signs for $m > 1$. Its operator norm is at most $4p^{-1/2}/(1 - p^{-1})$; hence, with the complete finite active prime set,

$$|P_{mn}| \leq \mathfrak{p}_a M, \quad \mathfrak{p}_a = 4 \sum_{p < e^{2a}} \frac{(\log p)p^{-1/2}}{1 - p^{-1}}. \quad (\text{CP15:3.32})$$

Alternatively the exact finite coefficients give $|P_{mn}| \leq 2M \sum_{p,k} (\log p) |w_m(p, k)|$, and the smaller bound may be used entrywise. This is an operator-size bound with both signs; it supplies no positive lower bound on P_* .

For the moments, oddness gives the exact Gram formulas

$$\begin{aligned} |F(b)| &\leq \sqrt{M\nu_a(b)}, \quad \nu_a(b) = \frac{\sinh(2ab)}{2b} - a, \\ \langle \sinh(b \cdot), \sinh(c \cdot) \rangle &= \frac{\sinh((b+c)a)}{b+c} - \frac{\sinh((b-c)a)}{b-c}. \end{aligned} \quad (\text{CP15:3.33})$$

At $b = c$, the second quotient is interpreted as a . These formulas can bound the *whole* square in (3.21) using its exact moment Gram matrix. The endpoint insertion also gives $|F_0| \leq \|P_{-p_a}\|_2 \sqrt{M}/d_a$, where the explicit density p_a is the integrand in (3.7). Taking the smaller of this bound and (3.33) still leaves the sign of the first term of (3.21) undetermined.

For a fully explicit matrix bound put

$$\begin{aligned} \beta_m(a) &= \int_{-2a}^{2a} |m^{-1} A_{\text{ar}}(|h|/m) - A_{\text{ar}}(|h|)| dh, \\ \pi_{mn}(a) &= 2\sqrt{m/n} \left[\sqrt{\nu_a(1/2)\nu_a(m/(2n))} + m^{-1} \sqrt{\nu_a(1/(2m))\nu_a(1/(2n))} \right], \\ e_{mn}(a) &= \mathfrak{p}_a + \beta_m(a) + \pi_{mn}(a) \quad (m \leq n). \end{aligned} \quad (\text{CP15:3.34})$$

These yield $|K_{mn} - T_{mn}| \leq e_{mn}(a)M$, and by symmetry define the lower-triangular values. For example,

$$\lambda_{\min}(K) \geq \frac{M}{3/\log 2 + 2/\log(3/2)} - M \max_m \sum_{n \in \{2,3,6\}} e_{mn}(a). \quad (\text{CP15:3.35})$$

This inequality is an independently valid size estimate. It is a *lower* bound and, when unfavorable, gives no negative eigenvalue. Reversing it to infer negativity would be invalid. In the two-scale determinant, the corresponding absolute-value estimate bounds the correction to the strictly positive quantity (3.28); it supplies no signed upper bound below zero. The exact unresolved expression is (3.11), or the more structured (3.18)-(3.21).

3.7. Why indefinitely shrinking these fixed ratios cannot force a negative direction

Fix distinct nonnegative base scales $s_1, \dots, s_d$, put $z_i = U_{s_i}v$, and let $G = [\langle z_i, z_j \rangle]$. The essential-support argument above gives $G > 0$. For a common further inward scale λ , all prime shifts vanish once $e^\lambda \log 2 \geq 2a$. The complete form then gives

$$Q_a(U_\lambda z_i, U_\lambda z_j) = \lambda G_{ij} + B_{ij}^0 - \int e^{-\lambda} R(e^{-\lambda} h) C_{z_i, z_j}(h) dh, \quad B_{ij}^0 = \langle \mathcal{L} z_i, z_j \rangle - \log(2\pi) G_{ij}. \quad (\text{CP15:3.36})$$

Both inputs are odd, so $\int C_{z_i, z_j} = 0$. With $L_R(a) = \sup_{0 < h < 2a} |R'(h)| < \infty$, subtracting $R(0)$ inside the last integral proves

$$\left| \int e^{-\lambda} R(e^{-\lambda} h) C_{z_i, z_j}(h) dh \right| \leq 4a^2 L_R(a) e^{-2\lambda} M. \quad (\text{CP15:3.37})$$

Since $\|\mathcal{L} z_i\|_2 \leq (\kappa_a + s_i) \sqrt{M}$,

$$\lambda_{\min}[Q_a(U_\lambda z_i, U_\lambda z_j)] \geq \lambda \lambda_{\min}(G) - d(\kappa_a + \max_i s_i + \log(2\pi))M - 4da^2 L_R(a) e^{-2\lambda} M. \quad (\text{CP15:3.38})$$

This is strictly positive for all sufficiently large explicitly bounded λ . It is an unconditional large-scale sign statement for these inputs, using only their proved graph bound; every prime term is accounted for by its exact vanishing threshold. It does not exclude a first null at the original scale.

3.8. Closure record and precise remaining inequality

Exact theorem and hypotheses. Under the odd first-null hypotheses, the full matrices, Schur optimization, zero-denominator alternatives, endpoint-substituted determinant expansions, and composed direction (3.1)-(3.21) hold on the established graph domain. Equations (3.27)-(3.38) provide independently proved Gram, packet-size, moment, and large-common-scale estimates. No derivative of the dilation orbit or endpoint value of v is used.

Completed calculation and proof status. Every displayed identity and stated bound is proved by the operations given. The negative polar square in (3.21) is a new signed interaction retained together with its endpoint companion. The actual complete determinant or composed energy has not been proved negative.

Consequence for Weil positivity. If an independent estimate made any of these admissible odd directions negative at every hypothetical first crossing, that crossing would be excluded. Odd-test positivity would then follow, yielding the return chain in Group 1. The current size estimates do not establish that exclusion.

One quantified missing lemma using the new calculation. A sufficient concrete composed-packet statement is: for each nonzero odd first-null vector at each finite a , there exists $\eta_{a,v} > 0$ such that

$$\boxed{P_* + B_* - \Pi_* \geq T_* + \eta_{a,v} M.} \quad (\text{CP15:3.39})$$

It would give $Q_a(w) \leq -\eta_{a,v}M < 0$. Equivalently, the two-scale route can seek $(T_{23} - P_{23} - B_{23} + \Pi_{23})^2 > (T_{22} - P_{22} - B_{22} + \Pi_{22})(T_{33} - P_{33} - B_{33} + \Pi_{33})$. Neither fixed family is asserted to detect every possible first null. The attempted estimates (3.30)-(3.35) stop at two-sided packet sizes and moment norms; they do not lower-bound the signed expression in (3.39). The surviving arithmetic terms are precisely the three correlation channels in (3.20), together with the complete archimedean combination and endpoint-coupled polar expression (3.21).

4. Odd polar resolvent, a quantitative inverse estimate, and the reflected prime obstruction

4.1. Domain, normalization, and the exact rank-one removal

Let $I_a = (-a, a)$, and let $\mathcal{H}_a^-$ be the full odd subspace of $L^2(I_a)$, with zero extension to the line. The Fourier transform is $\widehat{q}(\xi) = \int q(x)e^{i\xi x} dx$. Its closed form domain is

$$\mathcal{D}_a^- = \left\{ q \in \mathcal{H}_a^- : \int_{\mathbb{R}} \log(2 + |\xi|) |\widehat{q}(\xi)|^2 d\xi < \infty \right\}. \quad (\text{CP15:4.1})$$

The low-frequency negative part of $\log |\xi|$ is bounded on fixed-support L^2 inputs. All prime translation and bounded remainder terms preserve this domain. Define

$$\kappa(t) = \frac{e^{-t/2}}{1 - e^{-2t}}, \quad \alpha(t) = \kappa(|t|) - \frac{1}{2|t|}, \quad \alpha(0) = \frac{1}{4}, \quad R(t) = \alpha(t) - 2 \cosh(t/2). \quad (\text{CP15:4.2})$$

Here α is a kernel, not the affine slope in the endpoint equations. Write $s_a(x) = \sinh(x/2)$, and use the original L^2 pairing, not a changed metric. For odd q, r , the complete polar pairing is

$$\iint \cosh((x-y)/2) q(y) \overline{r(x)} dy dx = -\langle q, s_a \rangle \langle s_a, r \rangle. \quad (\text{CP15:4.3})$$

Indeed the $\cosh(x/2) \cosh(y/2)$ term vanishes by oddness; the sinh-product remains with its minus sign. Consequently the exact operator and form identities are

$$H_a = A_a^- + 2|s_a\rangle\langle s_a|, \quad A_a^- = H_a - 2|s_a\rangle\langle s_a|, \quad (\text{CP15:4.4})$$

$$\begin{aligned} h_a(q, r) &= \frac{1}{2\pi} \int \log |\xi| \widehat{q}(\xi) \overline{\widehat{r}(\xi)} d\xi - \log(2\pi) \langle q, r \rangle \\ &\quad - \sum_{\ell_n < 2a} b_n \int [q(x + \ell_n) + q(x - \ell_n)] \overline{r(x)} dx \\ &\quad - \iint_{I_a^2} \alpha(x-y) q(y) \overline{r(x)} dy dx. \end{aligned} \quad (\text{CP15:4.5})$$

This retains every prime packet and the entire archimedean kernel. The removed term is exactly the completed odd polar term. In particular

$$\|s_a\|_2^2 = \sinh a - a, \quad \|2|s_a\rangle\langle s_a|\| = 2(\sinh a - a). \quad (\text{CP15:4.6})$$

The factor two in the second identity is essential.

The multiplier requested in the question has the following exact normalization:

$$h_a(q, r) = \frac{1}{2\pi} \int_{\mathbb{R}} \Psi_a(\xi) \widehat{q}(\xi) \overline{\widehat{r}(\xi)} d\xi, \quad \Psi_a(\xi) = \Re \psi \left(\frac{1}{4} + \frac{i\xi}{2} \right) - \log \pi - 2 \sum_{\ell_n < 2a} b_n \cos(\xi \ell_n). \quad (\text{CP15:4.7})$$

One way to verify the archimedean constant and coefficient is to Abel-regularize the Fourier transform of α . For $\xi > 0$, differentiating its transform gives

$$\frac{d}{d\xi} \widehat{\alpha}(\xi) = \frac{1}{\xi} - 4 \sum_{j \geq 0} \frac{(2j+1/2)\xi}{((2j+1/2)^2 + \xi^2)^2}. \quad (\text{CP15:4.8})$$

The sum is the derivative of $\Re \psi(1/4 + i\xi/2)$. The constant is fixed at infinity by $\psi(z) = \log z + O(1/z)$ and the vanishing Fourier transform of the differentiated, integrable kernel. Thus, distributionally, $\widehat{\alpha}(\xi) = \log |\xi| - \log 2 - \Re \psi(1/4 + i\xi/2)$, which proves (4.7). The bounded-support pairings justify removal of the Abel regulator. In particular the coefficient of $\Re \psi$ is one.

The operator H_a is the Dirichlet compression of this multiplier to I_a . Its inverse is not multiplication by $1/\Psi_a$: zero extension and restriction remain part of the operator. The logarithmically growing form has compact embedding into $L^2(I_a)$, by uniform Fourier tightness and fixed support. Hence A_a^- and H_a have compact resolvent.

4.2. Resolvent criterion with both first-null branches retained

If $H_a \geq \lambda I$ for some $\lambda > 0$, set $w_a = H_a^{-1} s_a$. The exact forced equation is

$$(\log |D| w_a)(x) - \log(2\pi) w_a(x) - \sum_{\ell_n < 2a} b_n [w_a(x + \ell_n) + w_a(x - \ell_n)] - \int_{-a}^a \alpha(x-y) w_a(y) dy = \sinh(x/2), \quad x \in I_a. \quad (\text{CP15:4.9})$$

The equation holds distributionally on I_a , with zero extension in every shift, and in the operator sense for $w_a \in \text{Dom } H_a$. There is no endpoint trace prescribed on w_a . Factoring in the form sense gives

$$A_a^- = H_a^{1/2} \left[I - 2|H_a^{-1/2} s_a\rangle \langle H_a^{-1/2} s_a| \right] H_a^{1/2}. \quad (\text{CP15:4.10})$$

It follows, with all constants fixed, that

$$A_a^- \geq 0 \iff \sigma_a := 2\langle s_a, H_a^{-1} s_a \rangle \leq 1. \quad (\text{CP15:4.11})$$

If $A_a^- v = 0$, put $m_v = \langle v, s_a \rangle$. The full forced equation is $H_a v = 2m_v s_a$. Invertibility makes $m_v \neq 0$ for every nonzero null vector and forces $v = 2m_v w_a$ and $\sigma_a = 1$.

At a hypothetical first odd null, only $A_a^- \geq 0$ is known. It implies $H_a \geq 0$, and positivity of both summands gives the exact kernel equality

$$\mathcal{K}_a := \ker H_a = \ker A_a^- \cap s_a^\perp. \quad (\text{CP15:4.12})$$

Compact resolvent gives a positive spectral gap for H_a on $\mathcal{K}_a^\perp$. Since $s_a \perp \mathcal{K}_a$, its generalized inverse $H_a^\dagger s_a$ exists. Formula (4.11) remains valid on $\mathcal{K}_a^\perp$ with H_a^{-1} replaced by $H_a^\dagger$. More explicitly, writing $\tilde{\sigma}_a = 2\langle s_a, H_a^\dagger s_a \rangle$, the first-null possibilities are

$$\ker A_a^- = \begin{cases} \mathcal{K}_a, & \tilde{\sigma}_a < 1, \\ \mathcal{K}_a \oplus \text{span}\{H_a^\dagger s_a\}, & \tilde{\sigma}_a = 1. \end{cases} \quad (\text{CP15:4.13})$$

Thus a strict resolvent inequality alone excludes the nonzero-moment branch. An independent proof $\mathcal{K}_a = \{0\}$ is also necessary to exclude a zero-moment null vector.

For an odd null vector the inherited endpoint identity is especially explicit. In checkpoint 14's notation,

$$m_v = F_v(1/2) = \frac{P_o(v)}{d_a}, \quad d_a = 4a \cosh(a/2) - 8 \sinh(a/2) > 0. \quad (\text{CP15:4.14})$$

Here $P_o(v) = \sum b_n(I_n - aJ_n) + U_a - aV_a$ retains the complete prime and archimedean endpoint functionals defined there. Therefore

$$H_a v = \frac{2P_o(v)}{d_a} s_a. \quad (\text{CP15:4.15})$$

The zero-moment branch satisfies $P_o(v) = 0$. The saturation branch gives $1 = 2\langle s_a, H_a^\dagger s_a \rangle$. The endpoint equation supplies these exact compatibility conditions, not an independent strict bound on that scalar.

4.3. An actual independent inverse bound from odd support uncertainty

For every odd $q \in \mathcal{D}_a^-$, with $M = \|q\|_2^2$,

$$\widehat{q}(\xi) = i \int_{-a}^a q(x) \sin(\xi x) dx, \quad |\widehat{q}(\xi)|^2 \leq \frac{2a^3}{3} \xi^2 M. \quad (\text{CP15:4.16})$$

For any $T > 0$, split the logarithmic integral at $|\xi| = T$, retain the nonnegative contribution above T , and bound the deficit below it using (4.16). Since $\int_0^T \xi^2 \log(T/\xi) d\xi = T^3/9$, this gives

$$\langle \log |D| q, q \rangle \geq \left[\log T - \frac{2a^3 T^3}{27\pi} \right] M. \quad (\text{CP15:4.17})$$

Optimizing at $T^3 = 9\pi/(2a^3)$ proves the closed odd-support inequality

$$\boxed{\langle \log |D| q, q \rangle \geq [-\log a + c_{\text{odd}}] \|q\|_2^2, \quad c_{\text{odd}} = \frac{1}{3} \log \frac{9\pi}{2} - \frac{1}{3}.} \quad (\text{CP15:4.18})$$

No null equation or RH input is used.

Let

$$B_a = 2 \sum_{\ell_n < 2a} b_n + \int_{-2a}^{2a} |\alpha(t)| dt, \quad \lambda_a = -\log a + c_{\text{odd}} - \log(2\pi) - B_a. \quad (\text{CP15:4.19})$$

The shift operators have norm at most one and Young's inequality controls the complete α -convolution. Therefore

$$H_a \geq \lambda_a I. \quad (\text{CP15:4.20})$$

Whenever $\lambda_a > 0$, (4.9) has a unique solution and the genuine inverse estimate is

$$\boxed{\|w_a\|_2 \leq \frac{\sqrt{\sinh a - a}}{\lambda_a}, \quad 0 < \sigma_a = 2\langle s_a, w_a \rangle \leq \frac{2(\sinh a - a)}{\lambda_a}.} \quad (\text{CP15:4.21})$$

In particular $\lambda_a > 2(\sinh a - a)$ gives strict odd Weil positivity with the explicit gap

$$A_a^- \geq [\lambda_a - 2(\sinh a - a)] I > 0. \quad (\text{CP15:4.22})$$

This is an estimate for the actual inverse of the compressed operator. It controls every prime translation through its exact coefficient and does not replace the inverse by a pointwise Fourier reciprocal.

A simple entirely analytic interval certificate follows. Since

$$\frac{e^{-t/2}}{2t} \leq \kappa(t) \leq \frac{1}{2t} + e^{-t/2}, \quad -\frac{1}{4} \leq \alpha(t) \leq 1, \quad (\text{CP15:4.23})$$

we have $\int_{-2a}^{2a} |\alpha| \leq 4a$. For $0 < a \leq 1/8$, no prime packet occurs: $2a \leq 1/4 < \log 2$. Hence

$$\lambda_a \geq \frac{1}{3} \log \frac{288}{\pi^2} - \frac{5}{6} > \frac{1}{6}. \quad (\text{CP15:4.24})$$

For the last strict inequality, $\pi^2 < 10$ and $e < 3$ imply $288/\pi^2 > 28.8 > 27 > e^3$. Also $\sinh a - a \leq a^3 e^a/6$, so $2(\sinh a - a) < a^3 \leq 1/512$. Consequently

$$0 < a \leq \frac{1}{8} \implies H_a > \frac{1}{6}I, \quad 2\langle s_a, H_a^{-1}s_a \rangle < \frac{3}{256}, \quad A_a^- > \frac{253}{1536}I. \quad (\text{CP15:4.25})$$

These strict bounds are unconditional, use no zero computations, and close an actual resolvent estimate on a fixed interval. They establish a positive starting interval for the odd first-crossing argument.

The same calculation gives a convergent inverse construction whenever $d_a^0 = -\log a + c_{\text{odd}} - \log(2\pi) > B_a$. Let D_a be the odd Dirichlet compression of $\log|D| - \log(2\pi)$ and let B_a be the full sum of prime shifts and α -convolution in (4.9). Then

$$w_a = \sum_{j=0}^{\infty} (D_a^{-1}B_a)^j D_a^{-1}s_a, \quad \left\| w_a - \sum_{j=0}^J (D_a^{-1}B_a)^j D_a^{-1}s_a \right\|_2 \leq \frac{\|s_a\|_2}{d_a^0} \frac{(B_a/d_a^0)^{J+1}}{1 - B_a/d_a^0}. \quad (\text{CP15:4.26})$$

Every power of B_a includes all ordered prime-prime, prime-archimedean and archimedean interactions. The displayed tail is an operator-norm tail; no interaction has been dropped or given an assumed favorable sign.

4.4. The reflected prime terms block a positivity-improvement shortcut

Under the unitary identification $u(x) = \sqrt{2}q(x)$, $0 < x < a$, the prime part of the odd operator is exactly

$$- \sum_{\ell_n < 2a} b_n [\mathbf{1}_{x+\ell_n < a} u(x + \ell_n) + \mathbf{1}_{x > \ell_n} u(x - \ell_n) - \mathbf{1}_{0 < \ell_n - x < a} u(\ell_n - x)]. \quad (\text{CP15:4.27})$$

Thus ordinary translations have negative coefficients and reflected translations have positive coefficients. For disjoint positive-half-line test supports, the archimedean off-diagonal kernel is

$$-\kappa(|x - y|) + \kappa(x + y) < 0, \quad x, y > 0, \quad (\text{CP15:4.28})$$

because κ is strictly decreasing. Its favorable sign does not remove the positive prime antidiagonal $+b_n \delta(x + y - \ell_n)$.

This is an actual obstruction for the present operator. If a packet $0 < \ell < 2a$ is present, choose distinct $x_0, y_0 \in (0, a)$ with $x_0 + y_0 = \ell$, avoiding the finite set of all other ordinary and

reflected prime relations. Take nonnegative, disjoint, L^2 -normalized smooth bumps f_ϵ, g_ϵ around those points, with matched reflected profiles. Their reflected-prime pairing is $+b$. All other prime pairings vanish for small ϵ . Since their supports stay separated, the continuous archimedean kernel is bounded there and its pairing is $O(\epsilon)$. Therefore

$$h_a(f_\epsilon, g_\epsilon) = b + O(\epsilon) > 0. \quad (\text{CP15:4.29})$$

The notation here means the unitarily transported odd form. Adding the smooth polar rank-one term changes only $O(\epsilon)$. The off-diagonal nonpositivity necessary for positivity preservation of the associated real semigroup fails. In particular one cannot invoke a Perron theorem requiring positivity preservation on this half-line cone once $\log 2 < 2a$. This conclusion uses the actual arithmetic coefficients and reflected geometry; it does not by itself determine the sign pattern or multiplicity of the lowest eigenfunction.

4.5. Complete Schur correction and a rigorous operator-norm tail

The finite dilation Gram matrix used in Group 3 is a compression of the same identity (4.4). For arbitrary odd trial vectors u_j , put $D_{jk} = h_a(u_j, u_k)$ and $f_j = \langle u_j, s_a \rangle$. Then its complete Weil matrix is $K = D - 2ff^*$. A positive finite D and a scalar rank-one test concern the chosen span. When $H_a > 0$, the scalar $2f^*D^{-1}f$ is the squared H_a -norm of the projection of $H_a^{-1}s_a$, times two, and hence is a lower bound for the full resolvent scalar. Group 3 retains the exact complementary projection energy. The following calculation records a different exact correction, with an operator-norm tail estimate, for the full inverse.

Let $\mathbf{C}_a$ be the supported odd logarithmic operator, with increasing eigenvalues $\mu_j \rightarrow \infty$, counted with multiplicity. Its semiboundedness follows from (4.18); its compact resolvent follows from the fixed-support compact form embedding described above. Let P_d project onto its first d eigenvectors and $Q_d = I - P_d$. Set

$$b_a = \log(2\pi) + 2 \sum_{\ell_n < 2a} b_n + 4a, \quad \delta_d = \mu_{d+1} - b_a. \quad (\text{CP15:4.30})$$

The complete bounded remainder $\mathbf{B}$ in $H_a = \mathbf{C}_a - \mathbf{B}$ satisfies $\|\mathbf{B}\| \leq b_a$. Consequently, whenever $\delta_d > 0$, the tail operator and cross block satisfy

$$H_{QQ} \geq \delta_d I, \quad \|H_{QQ}^{-1}\| \leq \delta_d^{-1}, \quad \|H_{PQ}\| = \|P_d \mathbf{B} Q_d\| \leq b_a. \quad (\text{CP15:4.31})$$

The equality for the cross block uses that P_d is a spectral projection of $\mathbf{C}_a$. All statements are on the corresponding operator or form domains; the off-diagonal block extends boundedly.

Define the exact Schur complement

$$S_d = H_{PP} - H_{PQ} H_{QQ}^{-1} H_{QP}. \quad (\text{CP15:4.32})$$

Then

$$0 \leq H_{PQ} H_{QQ}^{-1} H_{QP} \leq \frac{\|P_d \mathbf{B} Q_d\|^2}{\delta_d} I \leq \frac{b_a^2}{\mu_{d+1} - b_a} I. \quad (\text{CP15:4.33})$$

This is a proved uniform operator-norm tail bound at fixed a , tending to zero as $d \rightarrow \infty$. It uses no assumed numerical spectrum and no discarded prime interaction. The spectral sequence exists and diverges because of the compact embedding: bounded form norm gives uniform Fourier tails, while fixed support gives translation tightness, hence L^2 precompactness.

When $S_d > 0$, write $s_P = P_d s_a$, $s_Q = Q_d s_a$, and $z_d = H_{PQ} H_{QQ}^{-1} s_Q$. Completing the square yields the full inverse identity

$$\boxed{\langle s_a, H_a^{-1} s_a \rangle = \langle s_Q, H_{QQ}^{-1} s_Q \rangle + \langle s_P, S_d^{-1} s_P \rangle - 2\Re \langle s_P, S_d^{-1} z_d \rangle + \langle z_d, S_d^{-1} z_d \rangle.} \quad (\text{CP15:4.34})$$

The mixed term and both positive correction terms are retained. If a finite calculation independently certifies $H_{PP} \geq \rho_d I$ with $\rho_d > \kappa_d^2 / \delta_d$, where $\kappa_d = \|P_d B Q_d\|$, then

$$2\langle s_a, H_a^{-1} s_a \rangle \leq \frac{2\|s_Q\|^2}{\delta_d} + \frac{2(\|s_P\| + \kappa_d \|s_Q\| / \delta_d)^2}{\rho_d - \kappa_d^2 / \delta_d}. \quad (\text{CP15:4.35})$$

This is a valid conditional finite-block upper bound. In the attempted general first-null argument, there is no independently established ρ_d and resulting right side strictly below one. In particular the limit $\delta_d \rightarrow \infty$ does not keep the other denominator uniformly away from zero. Dropping the negative Schur correction would decrease the computed inverse and would not furnish the needed upper bound.

4.6. Closing statement and the precise remaining inverse estimate

Exact theorem and hypotheses. Equations (4.4)-(4.9) hold on the full odd closed form domain (4.1), with the original complete arithmetic normalization. Under $H_a > 0$, (4.11) is the exact resolvent criterion. At an odd first null, the two branches (4.12)-(4.15) hold. The independent inverse bounds (4.18)-(4.26) and strict sign theorem (4.25) require no RH hypothesis.

Completed calculation and proof status. The complete polar rank, digamma multiplier, forced equation, kernel branch, saturation branch, quantitative odd uncertainty, actual inverse bound, and reflected-prime obstruction are proved. The interval certificate excludes every odd null for $a \leq 1/8$. It does not cover arbitrary finite first-crossing intervals.

Consequence for Weil positivity. The strict small-interval theorem initializes the odd first-crossing argument. For a general hypothetical first-null interval, exclusion by this route requires both of the following independently quantified statements:

$$\boxed{\ker H_a = \{0\}, \quad 2\langle s_a, H_a^{-1} s_a \rangle \leq 1 - \eta_a \quad \text{for some } \eta_a > 0.} \quad (\text{CP15:4.36})$$

The first statement removes the zero-moment branch; the second contradicts resolvent saturation in the nonzero-moment branch. A uniform margin in a is unnecessary.

Remaining signed expression and the stopping line. The complete scalar is

$$2\langle s_a, w_a \rangle, \quad \log |D| w_a - \log(2\pi) w_a - \sum_{\ell_n < 2a} b_n(w_a(\cdot + \ell_n) + w_a(\cdot - \ell_n)) - \alpha * w_a = s_a \quad \text{on } I_a. \quad (\text{CP15:4.37})$$

The attempted estimate is (4.21). Its denominator λ_a ceases to be positive as a grows, since it uses the absolute norm of the prime sum and the full archimedean remainder. Where it is positive but at most $2(\sinh a - a)$, it gives no strict gap below one. The endpoint relation (4.15) and the exact null equation retain the missing signed cancellations; neither supplies an independent estimate of them. The positive reflected-prime pairing (4.29) rules out a direct positivity-improvement replacement for that estimate.

If (4.36) were established at every hypothetical finite odd first crossing, there would be odd Weil positivity on every compact support. Yoshida's odd-test criterion would then imply RH, and the

correctly normalized Suzuki identity would give $\mathfrak{W} = \mathcal{S}$ and $\mathcal{E}_N = 0$ for every N . Only the stated small-interval sign and auxiliary resolvent results have been proved here; the global antecedent remains open.

5. Combined calculation, remaining estimate, and return chain

5.1. The four groups connect through one actual polar term

The odd reduction permits a hypothetical first null only after the independently positive starting interval. Every such vector is a full-operator null, and every inward dilation remains an admissible odd test. Its complete mixed entries are the formulas of Group 2. The integer prime regrouping introduces divisibility weights while keeping all archimedean and polar companions.

Let $u_i = U_{\log m_i} v$, $m_i > 1$, and retain the original G from (0.5). Write

$$F(b) = \int_{-a}^a e^{bx} v(x) dx, \quad f_i = \langle u_i, s_a \rangle = m_i^{-1/2} F(1/(2m_i)), \quad s_a(x) = \sinh(x/2). \quad (\text{CP15:5.1})$$

Because v is odd, F is odd. Define the full polar-free matrix

$$D_{ij} = \mathfrak{h}_a(u_j, u_i), \quad \mathfrak{h}_a(u, w) = Q_a(u, w) + 2\langle u, s_a \rangle \langle s_a, w \rangle.$$

Then the exact intersection of Groups 3 and 4 is

$$\boxed{K = D - 2ff^\top, \quad \tilde{K} = G^{-1/2}DG^{-1/2} - 2(G^{-1/2}f)(G^{-1/2}f)^*}. \quad (\text{CP15:5.2})$$

For the real family here the matrices are real symmetric. Neither D nor the dilation L^2 Gram matrix replaces G . Matrix D contains the complete digamma multiplier and every finite prime translation.

If $D > 0$, congruence by $D^{-1/2}$ proves

$$K \geq 0 \iff 2f^\top D^{-1}f \leq 1. \quad (\text{CP15:5.3})$$

Thus the mixed-dilation determinant and the odd polar inverse calculation test the same rank-one subtraction on different domains. This identity retains the mixed prime-prime, prime-archimedean, and polar interactions expanded in Group 3.

5.2. Exact finite-block inverse correction

Suppose $H_a > 0$, and write $w = H_a^{-1}s_a$. In the Hilbert space with inner product $\mathfrak{h}_a$, let $P_{\mathcal{U}}^H$ be orthogonal projection onto the finite span $\mathcal{U} = \text{span}\{u_i\}$. The projection coefficients solve $Dc = f$. Therefore

$$\boxed{2\langle s_a, H_a^{-1}s_a \rangle = 2f^\top D^{-1}f + 2\|(I - P_{\mathcal{U}}^H)w\|_{\mathfrak{h}_a}^2}. \quad (\text{CP15:5.4})$$

This is an exact energy decomposition with the complete complement retained. A finite-block scalar is a lower approximation to the full inverse scalar. An upper bound for the full scalar must include the last term, or the equivalent exact Schur correction in Group 4.

For example, in an L^2 -orthogonal block decomposition for which the displayed inverses exist, write

$$H_a = \begin{pmatrix} D & B \\ B^* & C \end{pmatrix}, \quad s_a = (f, g).$$

Completing the square gives

$$\begin{aligned} \langle s_a, H_a^{-1} s_a \rangle &= \langle f, D^{-1} f \rangle \\ &+ \langle g - B^* D^{-1} f, (C - B^* D^{-1} B)^{-1} (g - B^* D^{-1} f) \rangle. \end{aligned} \quad (\text{CP15:5.5})$$

All cross terms and the complement resolvent survive. Group 4 supplies a proved tail estimate when it uses such a decomposition. No finite compression alone is asserted to cap the full inverse.

5.3. The composed 2, 3, 6 direction

The arithmetic composition law has a natural admissible test:

$$u = (U_{\log 2} - I)(U_{\log 3} - I)v = U_{\log 6}v - U_{\log 2}v - U_{\log 3}v + v.$$

Since v is a full null, its complete energy is

$$\boxed{Q_a(u) = K_{66} + K_{22} + K_{33} + 2K_{23} - 2K_{26} - 2K_{36}.} \quad (\text{CP15:5.6})$$

Group 3 expands this expression with all relative correlations and its completed polar square. In the prime coefficients, writing

$$w_m(p, k) = \mathbf{1}_{m|k} p^{-k/(2m)} - p^{-k/2},$$

the mixed composition coefficient is

$$\eta(p, k) = w_6(p, k) - w_2(p, k) - w_3(p, k). \quad (\text{CP15:5.7})$$

Group 2 proves its exact divisibility signs, including the single negative case $(p, k) = (2, 6)$ among exponents divisible by six. Equation (5.6) also contains the different relative correlations for 2 versus 3, 2 versus 6, and 3 versus 6. Their coefficients are not combined as if these correlations were identical. The exceptional coefficient is therefore one exact component of the complete signed calculation.

The attempted estimates keep valid translation-Gram constraints and the exact endpoint substitution. The complete Poisson symbols for each prime take both signs; the mixed spectral weights also retain their signs. The bounds in Groups 2-3 control the absolute packet and its archimedean companion, but do not show that the whole expression (5.6) is negative.

At a genuine first odd crossing, every test in $\text{span}\{u_2, u_3, u_6\}$ is supported in $I_{a/2}$. The composed vector $u = u_6 - u_2 - u_3 + v$ generally retains full support, but its energy equals that of $u_6 - u_2 - u_3$ by nullness. The already positive earlier odd interval consequently gives a positive lower bound for the three-dilation compression under the hypothetical first-crossing assumptions. This is a constraint any independent contradictory arithmetic estimate must overcome. It is not itself an exclusion of that hypothetical first crossing. A fixed finite dilation family has not been proved to detect every hypothetical null.

5.4. Strongest independently proved sign and its scope

The actual inverse estimate in Group 4 establishes, without a first-null assumption,

$$\boxed{H_a > \frac{1}{6}I, \quad 2\langle s_a, H_a^{-1} s_a \rangle < \frac{3}{256}, \quad A_a^- > \frac{253}{1536}I \quad (0 < a \leq 1/8).} \quad (\text{CP15:5.8})$$

The inequalities follow from odd Fourier cancellation, the exact complete kernel bound, and the absence of prime shifts in this interval. All constants refer to the original arithmetic normalization. They are proved analytically; no zero sampling or numerical spectral ladder is used.

Thus unconditional odd Weil positivity holds on the stated support interval, with a strict positive margin. Every hypothetical odd first crossing satisfies $a > 1/8$. The finite-scale kinetic matrix and the prime-coefficient classifications provide additional independently proved sign information, with the scopes recorded in Groups 2-3.

For a general odd first null, Group 4 retains both branches:

$$\ker H_a = \ker A_a^- \cap s_a^\perp, \quad (\text{CP15:5.9})$$

and, if $\ker H_a = \{0\}$,

$$2\langle s_a, H_a^{-1} s_a \rangle = 1. \quad (\text{CP15:5.10})$$

Its forced equation and exact Schur formula are calculated. The lower bound used for (5.8) ceases to supply a useful general inverse ceiling as the interval grows and the complete prime sum enters. The reflected prime translations also prevent importing the proposed positivity-improvement argument.

5.5. Quantified closing condition and separate status statements

One sufficient remaining lemma is:

For every hypothetical first odd crossing $a > 1/8$, independently prove

$$\boxed{\ker H_a = \{0\}, \quad 2\langle s_a, H_a^{-1} s_a \rangle \leq 1 - \eta_a \quad \text{for some } \eta_a > 0.} \quad (\text{CP15:5.11})$$

The margin may depend on a . The first clause excludes the zero-moment kernel branch. The second contradicts the saturation branch (5.10). The actual attempted inverse estimates and their stopping point are in Group 4; the missing bound is not inferred from the resolvent criterion.

An alternative sufficient closure is an independently negative mixed-dilation energy, with the exact entries and endpoint substitutions of Groups 2-3. It can be used for any remaining branch. The two- and three-scale calculations here give explicit candidates and full mixed expressions, but establish no universal negative witness.

If either complete exclusion is proved, the return chain is:

1. No finite odd first crossing exists.
2. Continuity and the positive starting interval imply nonnegativity on every compact smooth odd test.
3. Yoshida's criterion yields RH.
4. Weil's criterion yields positivity on every compact smooth test, the primary objective.
5. The correctly normalized Suzuki identity, now under its justified hypotheses, gives $\mathfrak{W} = \mathcal{S}$.
6. Hence $\Delta = 0$, $\mathcal{E}_N = 0$ for every N , and the original defect success criterion follows.

The independently proved result in this round is the stated local strict sign, together with the exact odd reduction, signed mixed-packet calculations, inverse identities, and their proved estimates. The universal exclusion (5.11) or its mixed-energy alternative remains open.

Weil positivity: proved on the explicit odd support interval $0 < a \leq 1/8$; positivity on all compact tests remains open.

Original defect criterion: $\liminf_N \mathcal{E}_N \leq 1$ remains open. This checkpoint makes no conclusion that $\mathcal{E}_N = 0$.

6. Sources and file record

- Masatoshi Suzuki, *Weil's quadratic form via the screw function*, arXiv:2606.09096v2: Introduction for Yoshida's odd-test criterion; the complete localized form, closed domain, and parity discussion in Sections 2-4. The odd-branch continuity argument needed here is explicitly proved in Group 1.
- Masatoshi Suzuki, *On the Hilbert space derived from the Weil distribution*, arXiv:2301.00421v3: original metric as already normalized and audited in checkpoints 11-14. RH-dependent orthogonality is used only in the conditional return chain, after RH.
- `Michaels_Global_Suzuki_Domain_and_Four_Closures_Checkpoint_14.md`, read in full in its current version: the global Suzuki graph theorem, complete arithmetic form, null equation, endpoint and collar identities, and the complete correction.
- `Michaels_Checkpoint_14_Odd_Prime_Packet_Followup.md`, recovered unchanged: the archived directive for this continuation.

This note compiles all four analytical groups, their signed estimates, and the combined return calculation. The running ledger is updated to record the primary objective and separate proof statuses. The new packet preserves every previous non-ledger content file byte-for-byte, adds this note and the recovered follow-up, and refreshes the ledger and checksum manifest. Earlier manuscripts and checkpoints retain their original contents.

We are aiming to prove unconditional Weil positivity, with the original Suzuki defect criterion preserved.

Checkpoint 16: Prime-active support positivity

Source: *Michaels_Support_Growth_and_Complete_Positivity_Budget_Checkpoint_16.md*. Equation tags carry the source prefix CP16; references within this unit retain their original local numbering.

Checkpoint 16 – 23 September 2026

This note continues the complete Checkpoint 15, preserves its original arithmetic normalization and Suzuki metric, and changes the direction of the calculation from inward dilation to support enlargement. Every fixed family of strict inward dilations at a hypothetical first odd crossing already belongs to an earlier positive interval. Positivity of that family does not exclude the crossing.

The primary objective remains unconditional Weil positivity. The retained defect criterion and its conditional zero-defect return are

$$\Delta = \mathfrak{W} - \mathcal{S}, \quad \mathcal{E}_N = \left\| [\Delta(\rho_j, \rho_i)]_{i,j \leq N} \right\|_F^2, \quad \liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1. \quad (\text{CP16:0.1})$$

Here the determining vectors and their normalization are those of the original Suzuki metric. In arbitrary determining coordinates the defect is still normalized by $G_N^{-1/2}$, where $G_N = [\mathcal{S}(u_j, u_i)]$. None of the following L^2 operator estimates replaces that metric. They concern the sign of the same complete Weil form.

Contents

1. A certified odd support interval with the first prime packet present.
2. The exact support-collar decomposition and inverse-energy increment.
3. Complete collar estimates, explicit residuals, and a certified enlargement.
4. Finite-endpoint continuation and the independent kernel branch.
5. Combined result, exact remaining estimates, and the conditional return chain.

Retained definitions and scope

Use $\widehat{q}(\xi) = \int_{\mathbb{R}} q(x) e^{i\xi x} dx$, zero extension outside $(-a, a)$, and

$$\mathcal{D}_a^- = \left\{ q \in L^2(-a, a) : q(-x) = -q(x), \quad \int_{\mathbb{R}} \log(2 + |\xi|) |\widehat{q}(\xi)|^2 d\xi < \infty \right\}. \quad (\text{CP16:0.2})$$

The form is sesquilinear; real inputs suffice for detecting a negative direction, and the estimates extend to complex inputs. Write

$$\ell_n = \log n, \quad b_n = \frac{\Lambda(n)}{\sqrt{n}}, \quad \kappa(t) = \frac{e^{-t/2}}{1 - e^{-2t}} \quad (t > 0), \quad \alpha(t) = \kappa(|t|) - \frac{1}{2|t|}, \quad \alpha(0) = \frac{1}{4}, \quad R(t) = \alpha(t) - 2 \cosh(t/2). \quad (\text{CP16:0.3})$$

For $C_q(t) = \int q(x+t) \overline{q(x)} dx$, the complete form is

$$Q_a(q) = \langle \log |D| q, q \rangle - \log(2\pi) \|q\|_2^2 - 2 \sum_{\ell_n < 2a} b_n \Re C_q(\ell_n) - \int_{-2a}^{2a} R(t) C_q(t) dt. \quad (\text{CP16:0.4})$$

The operator associated with its odd restriction is A_a^- . With $s_a(x) = \sinh(x/2)$ on the full odd interval,

$$H_a = A_a^- + 2|s_a\rangle\langle s_a|, \quad \|s_a\|_2^2 = \sinh a - a. \quad (\text{CP16:0.5})$$

Under the unitary map $q \mapsto \sqrt{2}q|_{(0,a)}$, the same symbol s_a denotes $\sqrt{2}\sinh(x/2)$. The source norm and the factor two in (0.5) are unchanged.

The established global Suzuki domain, strong same-parity core convergence, endpoint-complete convolution identities, and full null equation are retained. The new estimates below are established on the whole indicated odd form domain, rather than only on hypothetical null vectors. Sharp spatial truncation is justified again in Group 2 because it is essential to the support split; no endpoint trace of an arbitrary form-domain vector is assumed.

Checkpoint 15 certified $H_a > \frac{1}{6}I$ and $A_a^- > \frac{253}{1536}I$ for $0 < a \leq 1/8$. This checkpoint independently extends the certified range across $\log 2/2$, calculates the exact cost of a further collar, and identifies the quantitative obstacle to continuation through an arbitrary finite endpoint. Local sign theorems and global closure are stated separately.

Source and dependency record. The complete Checkpoint 15, its current ledger, and the archived support-continuation follow-up were recovered before calculation. Suzuki, *Weil's quadratic form via the screw function*, arXiv:2606.09096v2, Introduction and Sections 2-4, supplies the inherited Weil normalization and records Yoshida's odd-test criterion. The new concentration, prime-localization, collar, and finite-rank estimates below are calculations of this checkpoint, not statements imported from that paper. No RH-dependent orthogonality or factorization is used. The Connes-Consani support-and-moment theorem mentioned in the follow-up is not needed or used as unrestricted odd positivity.

1. Prime-active odd positivity: an explicit support certificate

1.1 Exact domain and retained operator

Let $\mathcal{D}_a^-$ be the odd closed Weil form domain of checkpoint 15, with every function zero extended. The unitary half-interval scaling is

$$u(x) = \sqrt{2a}q(ax), \quad 0 < x < 1, \quad \|u\|_2 = \|q\|_2. \quad (\text{CP16:1.1})$$

Put $\alpha(t) = e^{-|t|/2}/(1 - e^{-2|t|}) - 1/(2|t|)$, with $\alpha(0) = 1/4$. The operator denoted H_a below is the same odd Dirichlet compression as in checkpoint 15. When $\log 2/2 < a < \min(\log 2, \log 3/2)$, its complete scaled form is

$$\begin{aligned} h_a(u) &= \langle \log |D|q, q \rangle - \log(2\pi)\|u\|^2 - \langle A_a u, u \rangle + b_2 \langle J_E u, u \rangle, \\ Q_a^-(q) &= h_a(u) - 2|\langle \varsigma_a, u \rangle|^2, \\ A_a(x, y) &= a[\alpha(a|x - y|) - \alpha(a(x + y))], \\ \varsigma_a(x) &= \sqrt{2a} \sinh(ax/2), \quad \|\varsigma_a\|^2 = \sinh a - a. \end{aligned} \quad (\text{CP16:1.2})$$

Here $b_2 = \log 2/\sqrt{2}$, $L = \log 2/a$, $E = (L - 1, 1)$, and $J_E u(x) = \mathbf{1}_E(x)u(L - x)$. The original Suzuki metric has not been replaced: this is a unitary coordinate representation of the original arithmetic form, used to estimate its sign. On smaller intervals without an active prime, set $E = \emptyset$ and $J_E = 0$.

The coefficient $\alpha(0)$ cancels **exactly** in A_a . The polar term in (1.2) has its full source factor $\sqrt{2a}$. There is precisely one active prime packet in the range eventually certified below. Because $a < \log 2$, no ordinary positive-half-interval translation occurs; the complete first packet is the reflected translation $+b_2 J_E$.

1.2 Weighted concentration and an explicit remainder

Let $Su(\xi) = \sqrt{2/\pi} \int_0^1 u(x) \sin(\xi x) dx$. Plancherel gives the exact split

$$\begin{aligned} \langle \log |D|q, q \rangle &= (-\log a + \log \Omega) \|u\|^2 - \langle T_\Omega u, u \rangle + P_\Omega(u), \\ T_\Omega(x, y) &= \frac{2}{\pi} \int_0^\Omega \log(\Omega/\xi) \sin(\xi x) \sin(\xi y) d\xi, \\ P_\Omega(u) &= \int_\Omega^\infty \log(\xi/\Omega) |Su(\xi)|^2 d\xi \geq 0. \end{aligned} \quad (\text{CP16:1.3})$$

The identity first holds on the odd smooth core; truncation of the Fourier integrals and closed-form convergence extend it to $\mathcal{D}_a^-$. In particular,

$$\langle \log |D|q, q \rangle \geq [-\log a + \log \Omega - \|T_\Omega\|] \|q\|^2. \quad (\text{CP16:1.4})$$

Here is a completely specified finite-rank certificate at $\Omega = 8$. Define

$$p(z) = \sum_{j=0}^{15} \frac{(-1)^j z^{2j+1}}{(2j+1)!}, \quad |\sin z - p(z)| \leq \frac{|z|^{33}}{33!} \quad (z \in \mathbb{R}). \quad (\text{CP16:1.5})$$

The latter is Taylor's theorem through degree 32; that polynomial equals the displayed odd polynomial. Let $A, A_p : L^2(0, 8) \rightarrow L^2(0, 1)$ have respective kernels $\sqrt{(2/\pi) \log(8/\xi)} \sin(\xi x)$ and the same expression with p . Then $T_8 = AA^*$, $T_p = A_p A_p^*$, and

$$\begin{aligned} \|A - A_p\|_{HS}^2 &\leq \frac{2}{\pi} \frac{8^{67}}{(33!)^2 67^3} < 10^{-14}, \\ \|A\|_{HS}^2 &\leq 16/\pi < 6, \\ \|T_8 - T_p\| &\leq \|A - A_p\|_{HS} (2\|A\|_{HS} + \|A - A_p\|_{HS}) < 10^{-5}. \end{aligned} \quad (\text{CP16:1.6})$$

For $0 \leq i, j \leq 15$, set

$$D_{ij} = \frac{(-1)^{i+j} 8^{2i+2j+3}}{(2i+1)!(2j+1)!(2i+2j+3)^2}, \quad G_{ij} = \frac{1}{2i+2j+3}, \quad \pi_0 = \frac{314159}{100000}. \quad (\text{CP16:1.7})$$

All entries are rational. Direct finite rational expansion gives the two certificate inequalities

$$\text{Tr}((DG)^4) < \frac{2857}{200}, \quad \left(\frac{2}{\pi_0}\right)^4 \frac{2857}{200} < \left(\frac{123799}{100000}\right)^4. \quad (\text{CP16:1.8})$$

Since $T_p \geq 0$, $\text{Tr } T_p^4 = (2/\pi)^4 \text{Tr}((DG)^4)$, and $\pi > \pi_0$, (1.6)-(1.8) prove

$$\boxed{\|T_8\| < \frac{619}{500} =: t_0.} \quad (\text{CP16:1.9})$$

Thus the approximation has an explicit operator-norm remainder; no numerical eigenvalue assumption is used. The finite rational calculation is reproduced at the end of this group.

1.3 Exact prime channels and their mixed companions

Put $P = \mathbf{1}_E$, $P_{\pm} = (P \pm J_E)/2$, and $P_0 = I - P$. The two prime channels satisfy

$$J_E = P_+ - P_-, \quad b_2 \langle J_E u, u \rangle = b_2 \|P_+ u\|^2 - b_2 \|P_- u\|^2 \geq -b_2 \|Pu\|^2. \quad (\text{CP16:1.10})$$

This bound covers the full negative reflection channel rather than presuming the prime autocorrelation has a sign.

For $x, y \in E$, write $jx = L - x$. If K denotes either T_8 's kernel or A_a 's kernel, its complete channel coupling is

$$(P_{\epsilon} K P_{\eta})(x, y) = \frac{1}{4} [K(x, y) + \eta K(x, jy) + \epsilon K(jx, y) + \epsilon \eta K(jx, jy)], \quad \epsilon, \eta \in \{+1, -1\}. \quad (\text{CP16:1.11})$$

The entries with P_0 are the corresponding one-sided projections and are retained as well. With D_a denoting the scaled logarithmic operator minus $\log(2\pi)I$, the full three-channel block matrix is

$$(Q_a^-)_{\epsilon\eta} = P_{\epsilon}(D_a - A_a)P_{\eta} + b_2 \epsilon \delta_{\epsilon\eta} P_{\epsilon} - 2|\varsigma_{\epsilon}\rangle\langle\varsigma_{\eta}|, \quad \epsilon, \eta \in \{0, +, -\}, \quad (\text{CP16:1.12})$$

where the prime term is zero when either index is 0. This includes each prime-archimedean interaction and each polar cross term without assuming the blocks commute.

The polar projections on the prime collar can be calculated explicitly:

$$\begin{aligned} \varsigma_+(x) &= \sqrt{2a} \sinh(\log 2/4) \cosh(a(x - L/2)/2), \\ \varsigma_-(x) &= \sqrt{2a} \cosh(\log 2/4) \sinh(a(x - L/2)/2), \quad x \in E. \end{aligned} \quad (\text{CP16:1.13})$$

If $d = |E|$, then

$$\begin{aligned} \|\varsigma_+\|^2 &= \sinh^2(\log 2/4) [ad + 2 \sinh(ad/2)], \\ \|\varsigma_-\|^2 &= \cosh^2(\log 2/4) [2 \sinh(ad/2) - ad]. \end{aligned} \quad (\text{CP16:1.14})$$

In particular the negative prime channel has a small but nonzero polar component. Its mixed polar block in (1.12) is $-2|\varsigma_+\rangle\langle\varsigma_-|$, not zero. The norm estimates below bound the complete quadratic form for every vector, and therefore retain these couplings.

1.4 Absorbing the first prime packet using its actual collar

A useful positive-operator lemma avoids charging the prime coefficient on the entire support. If $0 \leq T \leq t_0 I$, P is a projection, and $\|PTP\| \leq d_T$, then

$$\|T + bP\| \leq \Lambda(t_0, b, d_T) := \frac{t_0 + b + \sqrt{(t_0 - b)^2 + 4bd_T}}{2}. \quad (\text{CP16:1.15})$$

Indeed, for $\lambda > t_0$, functional calculus gives

$$P(\lambda - T)^{-1}P \leq \frac{1}{\lambda} \left(1 + \frac{d_T}{\lambda - t_0} \right) P. \quad (\text{CP16:1.16})$$

If $(\lambda - b)(\lambda - t_0) > bd_T$, the right side times b is strictly below P ; the usual positive square-root factorization then gives $\lambda I - T - bP > 0$. Taking the infimum of such λ proves (1.15). This proof includes all inner-collar cross terms of T .

The diagonal of the actual concentration kernel is

$$T_8(x, x) = \frac{8}{\pi} - \frac{\text{Si}(16x)}{2\pi x}. \quad (\text{CP16:1.17})$$

An integration by parts in the sine integral tail gives $\text{Si}(z) \geq \pi/2 - 2/z$ for $z > 0$. Thus for $E = (c, 1)$, $c = \log 2/a - 1 > 0$,

$$\|PT_8P\| \leq \text{Tr}(PT_8P) \leq (1 - c) \left[\frac{8}{\pi} - \frac{1}{4} + \frac{1}{16\pi c^2} \right] =: d_a. \quad (\text{CP16:1.18})$$

The trace is legitimate because T_8 is positive trace class with its continuous integral kernel. Combining (1.3), (1.10) and (1.15), while retaining the positive surplus, gives

$$\langle \log |D|q, q \rangle + b_2 \langle J_E u, u \rangle \geq [-\log a + \log 8 - \Lambda(t_0, b_2, d_a)] \|q\|^2 + P_8(u). \quad (\text{CP16:1.19})$$

This is the additional estimate that crosses the first prime threshold. It uses the small measure of the prime's actual reflected collar and a resolvent estimate for the bounded concentration operator.

1.5 Odd archimedean cancellation with a uniform kernel bound

For $0 < t \leq 7/10$, put $d_0(t) = 1/t - 1/\sinh t$. Then

$$\alpha(t) = \frac{e^{t/2} - 1}{2t} - \frac{e^{t/2}}{2} d_0(t), \quad 0 \leq d_0(t) \leq t/6, \quad 0 \leq d'_0(t) \leq 2/\pi^2. \quad (\text{CP16:1.20})$$

The bound on d_0 follows by expanding $(1 - t^2/6) \sinh t \leq t$, whose coefficients after the linear one are nonpositive. For the derivative, use the absolutely locally convergent partial fraction identity

$$d_0(t) = 2t \sum_{n \geq 1} \frac{(-1)^{n+1}}{t^2 + \pi^2 n^2}. \quad (\text{CP16:1.21})$$

After differentiation the summands have magnitudes $(\pi^2 n^2 - t^2)/(\pi^2 n^2 + t^2)^2$, positive and decreasing in n on this interval. The alternating-series bounds prove the last assertion of (1.20). For a self-contained derivation of (1.21), expand $\cosh(tx)$ in its cosine series on $(0, 1)$. Its constant coefficient is $\sinh t/t$, and its coefficient of $\cos(n\pi x)$ is $2t \sinh t (-1)^n / (t^2 + \pi^2 n^2)$. The coefficients are absolutely summable. Evaluation at $x = 0$, followed by division by $\sinh t$, gives (1.21). The differentiated rational summands converge uniformly on compact t -intervals here, justifying the derivative used above.

Writing $(e^{t/2} - 1)/(2t) = \frac{1}{4} \int_0^1 e^{ut/2} du$, using $e^{7/20} < 3/2$ and $\pi^2 > 48/5$, gives

$$-\frac{11}{80} \leq \alpha'(t) \leq \frac{3}{32}, \quad 0 < t \leq 7/10. \quad (\text{CP16:1.22})$$

For example the lower bound is $1/16 - (3/4)(7/120 + 5/24) = -11/80$. Therefore the origin constant cancels and

$$|A_a(x, y)| \leq \frac{11a^2}{40} \min(x, y), \quad \boxed{\|A_a\| \leq \frac{11a^2}{10\pi^2}} \quad (2a \leq 7/10). \quad (\text{CP16:1.23})$$

The norm of the kernel $\min(x, y)$ on $(0, 1)$ is $4/\pi^2$, as follows by solving its elementary Dirichlet-Neumann Green eigenvalue equation. No absolute estimate of the nonzero constant $\alpha(0)$ enters (1.23).

1.6 A certified support interval, inverse, and strict Weil margin

Take

$$a_1 = \frac{699}{2000} = 0.3495 > \frac{\log 2}{2}. \quad (\text{CP16:1.24})$$

The following inequalities are rational certificates, with logarithms evaluated by their convergent positive atanh series and hyperbolic functions by Taylor series with a geometric remainder:

$$\begin{aligned} \frac{314159}{100000} < \pi < \frac{355}{113}, \quad \frac{693147}{10^6} < \log 2 < \frac{693148}{10^6}, \quad \sqrt{2} > \frac{1414213}{10^6}, \\ b_2 < \frac{49013}{100000}, \quad d_{a_1} < \frac{38804}{10^6}, \quad \Lambda(t_0, b_2, d_{a_1}) < \frac{126263}{100000}, \\ \log \frac{4}{\pi a_1} > \frac{129281}{100000}, \quad \frac{11a_1^2}{10\pi^2} < \frac{13615}{10^6}, \quad 2(\sinh a_1 - a_1) < \frac{14318}{10^6}. \end{aligned} \quad (\text{CP16:1.25})$$

For the concentration-prime eigenvalue, the strict rational comparison is specifically

$$(\lambda_0 - t_0)(\lambda_0 - b_0) - b_0 d_0 = \frac{95881}{12500000000} > 0, \quad \lambda_0 = \frac{126263}{100000}, \quad b_0 = \frac{49013}{100000}, \quad d_0 = \frac{38804}{10^6}. \quad (\text{CP16:1.26})$$

All other comparisons, including their convergent remainders, are specified in the reproducer below. Equations (1.19), (1.23) and (1.25) prove

$$\begin{aligned} H_a &> \frac{3313}{200000} I, \\ 2\langle s_a, H_a^{-1} s_a \rangle &< \frac{14318}{16565} < 1, \\ \mu_a = 1 - 2\langle s_a, H_a^{-1} s_a \rangle &> \frac{2247}{16565}, \\ A_a^- &> \frac{2247}{10^6} I > \frac{1}{500} I, \quad 0 < a \leq a_1. \end{aligned} \quad (\text{CP16:1.27})$$

The bounds are uniform throughout the stated interval. Below the first threshold the prime collar is empty and its cost is zero. Above it, $1 - c$, c^{-2} , d_a , $\Lambda(t_0, b_2, d_a)$, a^2 , and $\sinh a - a$ are increasing with a , whereas $\log(4/(\pi a))$ is decreasing. Consequently the worst case in this proof is $a = a_1$.

This is an actual bound for the compressed inverse, not a reciprocal of the full-line Fourier multiplier. Existence follows from $H_a \geq gI$, $g = 3313/200000$, and the bound uses $\|H_a^{-1}\| \leq g^{-1}$ and the complete polar norm. In particular this interval has no H_a kernel and no odd Weil null vector.

Exact theorem and hypotheses. Equation (1.27) holds unconditionally for the complete odd form on $\mathcal{D}_a^-$, for every $0 < a \leq 699/2000$, with the original arithmetic coefficients and the original Suzuki metric preserved.

Completed calculation and consequence. The first reflected prime packet is present. Both of its sign channels, its full concentration coupling, the odd archimedean kernel, and its polar couplings have been calculated and bounded. Every hypothetical first odd crossing satisfies $a > 699/2000$. This gives a strictly larger initial interval for the support-continuation argument.

Exact remaining estimate. For later supports the complete lower-bound method reads

$$\log \frac{4}{\pi a} - \Lambda(t_0, b_2, d_a) - \frac{11a^2}{10\pi^2} - 2(\sinh a - a) > 0 \quad (\text{CP16:1.28})$$

while only the first packet is active and $2a \leq 7/10$. It ceases to certify positivity by $a = 7/20$: substitution gives an approximately -0.00328 bound (only a diagnostic, not a claim that the actual form is negative). More primes and larger support require the full collar budget in Groups 2-4. No estimate proving positivity at every finite support has been derived in this group.

1.7 Finite exact-arithmetic certificate

The following is a reproducible certificate for (1.6), (1.8), and the nontrivial scalar comparisons in (1.25). It evaluates finite rational expressions and explicit series bounds; it does not test zeros or infer positivity from floating-point samples.

```
from fractions import Fraction as F
from math import factorial
n=16
D=[F((-1)**(i+j)*8**(2*i+2*j+3),
    factorial(2*i+1)*factorial(2*j+1)*(2*i+2*j+3)**2)
    for j in range(n)] for i in range(n)]
G=[F(1,2*i+2*j+3) for j in range(n)] for i in range(n)]
def mm(A,B):
    return [[sum((A[i][k]*B[k][j] for k in range(n)),F())
        for j in range(n)] for i in range(n)]
V=mm(D,G); W=mm(V,V)
R4=sum((W[i][j]*W[j][i] for i in range(n) for j in range(n)),F())
p0,p1=F(314159,100000),F(355,113)
assert R4<F(2857,200)
assert (2/p0)**4*F(2857,200)<F(123799,100000)**4
assert 2/p0*F(8**67,factorial(33)**2*67**3)<F(1,10**14)
assert F(1,10**7)*(5+F(1,10**7))<F(1,10**5)
# pi bounds may equivalently be checked by Machin's identity and
# alternating Taylor bounds for 16 atan(1/5)-4 atan(1/239).
def atan_bounds(x,N):
    s=sum((-1)**j*x**(2*j+1)/F(2*j+1) for j in range(N)),F())
    nxt=(-1)**N*x**(2*N+1)/F(2*N+1)
    return min(s,s+nxt),max(s,s+nxt)
a5,b5=atan_bounds(F(1,5),12); a239,b239=atan_bounds(F(1,239),4)
assert p0<16*a5-4*b239 and 16*b5-4*a239<p1
# log2 from log((1+h)/(1-h)), h=1/3.
h=F(1,3); N=8
lo=2*sum((h**(2*j+1)/F(2*j+1) for j in range(N)),F())
hi=lo+2*h**(2*N+1)/((2*N+1)*(1-h*h))
assert F(693147,10**6)<lo and hi<F(693148,10**6)
a=F(699,2000); rt=F(1414213,10**6)
assert rt*rt<2
b=F(49013,100000)
assert b*rt>F(693148,10**6)
c=F(693147,10**6)/a-1
d=(1-c)*(8/p0-F(1,4)+1/(16*p0*c*c))
assert d<F(38804,10**6)
t=F(619,500); lam=F(126263,100000)
assert (lam-t)*(lam-b)-b*F(38804,10**6)==F(95881,12500000000)
x=4/(p1*a); h=(x-1)/(x+1)
loglo=2*sum((h**(2*j+1)/F(2*j+1) for j in range(10)),F())
assert loglo>F(129281,100000)
assert 11*a/(10*p0*p0)<F(13615,10**6)
s=sum((a**(2*j+1)/F(factorial(2*j+1)) for j in range(1,6)),F())
s+=a**13/F(factorial(13))/(1-a*a/F(14*15))
```

```
assert 2*s<F(14318,10**6)
assert F(129281,100000)-lam-F(13615,10**6)==F(3313,200000)
assert F(3313,200000)-F(14318,10**6)==F(2247,10**6)
```

2. Exact enlargement: form domains, complete blocks, and the positivity cost

This group proves the enlargement identities. It supplies no unconditional upper bound for the resulting scalar cost. The original Suzuki metric is unchanged: the map to a positive half-interval below is an L^2 -unitary representation of the odd Weil operator, not a replacement for $\mathcal{S}$.

2.1. The actual domain survives a sharp interval split

Let $0 < a < b < \infty$, $\delta = b - a$, and use the Fourier normalization $\widehat{q}(\xi) = \int_{\mathbb{R}} q(x) e^{i\xi x} dx$. On the odd, zero-extended space the form domain is

$$\mathcal{D}_b^- = \left\{ q \in L^2(-b, b) : q(-x) = -q(x), \int_{\mathbb{R}} \log(2 + |\xi|) |\widehat{q}(\xi)|^2 d\xi < \infty \right\}. \quad (\text{CP16:2.1})$$

The unitary map is $u(x) = \sqrt{2}q(x)$, $0 < x < b$. Write Eu for its odd zero-extension divided by $\sqrt{2}$. Let $I = (0, a)$ and $J = (a, b)$. The inner and collar form domains are

$$\mathcal{D}_I = \{u \in L^2(I) : Eu \in \mathcal{D}_b^-\}, \quad \mathcal{D}_J = \{z \in L^2(J) : Ez \in \mathcal{D}_b^-\}, \quad (\text{CP16:2.2})$$

where extension by zero outside the indicated interval is understood. The first is the transported domain at support a .

Multiplication by a half-line indicator preserves these domains. Here is a justification that does not impose endpoint traces. For $P_c = \mathbf{1}_{(c, \infty)}$, the distribution kernel of $[\log |D|, P_c]$, away from $x = y$, is

$$-\frac{\mathbf{1}_{y>c} - \mathbf{1}_{x>c}}{2|x - y|}. \quad (\text{CP16:2.3})$$

After reflecting one half-line about c , its two nonzero off-diagonal blocks are signed copies of half the Carleman operator. Therefore

$$\|[\log |D|, P_c]\|_{2 \rightarrow 2} \leq \frac{\pi}{2}. \quad (\text{CP16:2.4})$$

This distributional identity first holds for smooth compactly supported functions; the same kernel calculation shows their cutoffs belong to the logarithmic graph domain. Graph closure then proves that P_c is bounded on $\text{Dom } \log |D|$. Interpolation between that graph domain and L^2 proves boundedness on $\text{Dom}(1 + (\log |D|)^2)^{1/4}$. For functions with support in a fixed bounded interval the low-frequency part of this last norm is bounded by the L^2 norm: use $|\widehat{q}(\xi)| \leq \sqrt{2b} \|q\|_2$ and $\int_{|\xi|<1} |\log |\xi|| d\xi < \infty$. Thus this interpolated norm is equivalent, on these supported functions, to the norm in (2.1). The two cutoffs at $\pm a$ commute with parity when combined as $\mathbf{1}_{\{|x|<a\}}$.

Consequently the sharp decomposition is legitimate:

$$\mathcal{D}_b^- \simeq \mathcal{D}_I \oplus \mathcal{D}_J, \quad Eu = E(\mathbf{1}_I u) + E(\mathbf{1}_J u). \quad (\text{CP16:2.5})$$

This is a direct sum of vector spaces with equivalent form norms; it does not assert orthogonality for the arithmetic form. In particular, a jump at the interface is not replaced by a Dirichlet trace condition.

2.2. An exact archimedean representation with the boundary weights present

Put

$$\kappa(h) = \frac{e^{-h/2}}{1 - e^{-2h}}, \quad d_0 = \psi(1/4) - \log \pi, \quad h > 0. \quad (\text{CP16:2.6})$$

The elementary digamma difference identity, obtained by integrating its geometric series, gives

$$\Re \psi(1/4 + i\xi/2) - \log \pi = d_0 + 2 \int_0^\infty \kappa(h)(1 - \cos(\xi h)) dh. \quad (\text{CP16:2.7})$$

Thus the whole-line archimedean form is exactly

$$h_{\text{ar}}(q) = d_0 \|q\|_2^2 + \int_0^\infty \kappa(h) \|q(\cdot + h) - q\|_2^2 dh. \quad (\text{CP16:2.8})$$

All integrals of differences are nonnegative, so Tonelli and then form closure justify (2.8) on the domain. Under the odd half-line representation, with u zero outside its support in $(0, \infty)$,

$$\begin{aligned} h_{\text{ar}}^+(u) &= d_0 \|u\|_2^2 \\ &+ \frac{1}{2} \iint_{(0, \infty)^2} \kappa(|x - y|) |u(x) - u(y)|^2 dy dx \\ &+ \frac{1}{2} \iint_{(0, \infty)^2} \kappa(x + y) |u(x) + u(y)|^2 dy dx. \end{aligned} \quad (\text{CP16:2.9})$$

The last term includes, rather than discards, the positive reflected Gram operator

$$\kappa(x + y) = \sum_{j=0}^{\infty} e^{-(2j+1/2)x} e^{-(2j+1/2)y}. \quad (\text{CP16:2.10})$$

Its pairing $\iint \kappa(x + y) u(y) \overline{u(x)} dy dx$ is nonnegative whenever finite; the assertion follows first for finite sums and then by the Laplace representation and approximation.

For an interval $K = (c, d) \subset (0, \infty)$, all exterior interactions can instead be displayed explicitly:

$$\begin{aligned} h_{\text{ar}, K}^+(u) &= d_0 \|u\|_2^2 + \frac{1}{2} \iint_{K^2} \kappa(|x - y|) |u(x) - u(y)|^2 dy dx \\ &+ \frac{1}{2} \iint_{K^2} \kappa(x + y) |u(x) + u(y)|^2 dy dx + \int_K \beta_K(x) |u(x)|^2 dx, \\ \beta_K(x) &= \int_{(0, \infty) \setminus K} [\kappa(|x - y|) + \kappa(x + y)] dy. \end{aligned} \quad (\text{CP16:2.11})$$

No unspecified boundary term remains in this formula. If $\mathcal{K}(d) = \int_d^\infty \kappa(h) dh$, then

$$\begin{aligned} \beta_I(x) &= \mathcal{K}(a - x) + \mathcal{K}(a + x), & 0 < x < a, \\ \beta_J(x) &= \mathcal{K}(x - a) - \mathcal{K}(x + a) + \mathcal{K}(b - x) + \mathcal{K}(b + x), & a < x < b. \end{aligned} \quad (\text{CP16:2.12})$$

These logarithmically divergent endpoint weights belong to the form itself. There is no point evaluation of u or its derivative at $0, a, b$.

2.3. Every prime block and every threshold

Write $\ell_n = \log n$ and $b_n = \Lambda(n)/\sqrt{n}$. In the half-interval representation the complete prime operator is

$$(P_b u)(x) = - \sum_{\ell_n < 2b} b_n [\mathbf{1}_{0 < x + \ell_n < b} u(x + \ell_n) + \mathbf{1}_{0 < x - \ell_n < b} u(x - \ell_n) - \mathbf{1}_{0 < \ell_n - x < b} u(\ell_n - x)]. \quad (\text{CP16:2.13})$$

It is a finite sum of bounded, selfadjoint translation/reflection pairs. Its distribution kernel on $(0, b)^2$ is

$$\sum_{\ell_n < 2b} b_n [\delta(x + y - \ell_n) - \delta(x - y - \ell_n) - \delta(x - y + \ell_n)]. \quad (\text{CP16:2.14})$$

The prime quadratic form on $K = (c, d)$, with zero extension, is therefore

$$-2 \sum_{\ell_n < d-c} b_n \Re \int_c^{d-\ell_n} u(x + \ell_n) \overline{u(x)} dx + \sum_{2c < \ell_n < 2d} b_n \int_{\substack{x \in K \\ \ell_n - x \in K}} u(\ell_n - x) \overline{u(x)} dx. \quad (\text{CP16:2.15})$$

The reflected pairing is real, but need not be nonnegative.

Let H_b be the odd polar-removed operator from Checkpoint 15, transported unitarily. Combining (2.11) and (2.15) gives both diagonal blocks exactly. In particular its inner restriction is the original H_a , and its collar restriction will be denoted C . Their off-diagonal block $B : L^2(J) \rightarrow L^2(I)$ is

$$\begin{aligned} (Bz)(x) &= \int_a^b [-\kappa(y - x) + \kappa(x + y)] z(y) dy \\ &\quad - \sum_{\ell_n < b} b_n \mathbf{1}_{a < x + \ell_n < b} z(x + \ell_n) \\ &\quad + \sum_{a < \ell_n < a+b} b_n \mathbf{1}_{a < \ell_n - x < b} z(\ell_n - x), \quad 0 < x < a. \end{aligned} \quad (\text{CP16:2.16})$$

All indicators in this formula matter. For completeness, its adjoint is

$$\begin{aligned} (B^* u)(y) &= \int_0^a [-\kappa(y - x) + \kappa(x + y)] u(x) dx \\ &\quad - \sum_{\ell_n < b} b_n \mathbf{1}_{0 < y - \ell_n < a} u(y - \ell_n) \\ &\quad + \sum_{a < \ell_n < a+b} b_n \mathbf{1}_{0 < \ell_n - y < a} u(\ell_n - y), \quad a < y < b. \end{aligned} \quad (\text{CP16:2.17})$$

The exact location cutoffs are:

Block	Ordinary shifts	Reflected shifts
Inner H_a	$0 < \ell_n < a$	$0 < \ell_n < 2a$
Cross B, B^*	$0 < \ell_n < b$	$a < \ell_n < a + b$
Collar C	$0 < \ell_n < \delta$	$2a < \ell_n < 2b$

A threshold equal to an extreme endpoint has only a measure-zero contact and contributes zero to the L^2 operator. The table is valid without assuming $\delta < a$. Every new global prime threshold $\ell_n \in [2a, 2b)$ is retained in its applicable cross or collar block; no tail of prime powers is omitted.

2.4. The complete cross block is bounded

The origin behavior is $\kappa(h) = 1/(2h) + \alpha(h)$, with the established bound $|\alpha(h)| \leq 1$. The simpler triangle estimate gives

$$\|B_{\text{ar}}\| \leq \frac{\pi}{2} + [1 + \kappa(a)]\sqrt{a\delta}. \quad (\text{CP16:2.18})$$

There is a better estimate retaining the reflected cancellation. Since κ is decreasing and $y - x < x + y$ for $0 < x < a < y < b$,

$$0 \leq \kappa(y - x) - \kappa(x + y) \leq \frac{1}{2(y - x)} + 1.$$

Pointwise domination, the truncated Carleman estimate, and the norm of the constant rectangular kernel give

$$\|B\| \leq c_{a,\delta} := \frac{\pi}{2} + \sqrt{a\delta} + \sum_{\ell_n < b} b_n + \sum_{a < \ell_n < a+b} b_n. \quad (\text{CP16:2.19})$$

The positive reflected archimedean kernel has been included before estimating. Both prime sign channels remain explicitly in (2.16); (2.19) charges the absolute norm of each rather than asserting their signs agree.

The boundedness of B , together with (2.5), proves the form and operator block descriptions

$$h_b(u, z) = h_a(u) + 2\Re\langle Bz, u \rangle + c(z), \quad H_b = \begin{pmatrix} H_a & B \\ B^* & C \end{pmatrix}. \quad (\text{CP16:2.20})$$

Here the form domain is $\mathcal{D}_I \oplus \mathcal{D}_J$, and the operator domain is $\text{Dom } H_a \oplus \text{Dom } C$. The diagonal forms are closed and bounded below; the cross form is bounded on $L^2(I) \oplus L^2(J)$. These assertions specify the meaning of the unbounded block matrix.

To make the mixed arithmetic content of its Schur correction explicit, write $B = B_0 + \sum_n b_n T_n$, where B_0 is the archimedean integral in (2.16), $O_n z(x) = \mathbf{1}_{a < x + \ell_n < b} z(x + \ell_n)$, $F_n z(x) = \mathbf{1}_{a < \ell_n - x < b} z(\ell_n - x)$, and $T_n = -O_n + F_n$; an inapplicable operator is zero. Whenever $H_a > 0$, its complete correction is

$$\begin{aligned} B^* H_a^{-1} B &= B_0^* H_a^{-1} B_0 \\ &\quad + \sum_n b_n (B_0^* H_a^{-1} T_n + T_n^* H_a^{-1} B_0) + \sum_{m,n} b_m b_n T_m^* H_a^{-1} T_n, \\ T_m^* H_a^{-1} T_n &= O_m^* H_a^{-1} O_n - O_m^* H_a^{-1} F_n - F_m^* H_a^{-1} O_n + F_m^* H_a^{-1} F_n. \end{aligned} \quad (\text{CP16:2.21})$$

Thus the Schur correction contains all archimedean-prime, ordinary-reflected, and prime-prime pairs, including pairs belonging to different prime powers. Equation (2.19) bounds their combined operator; it does not delete any mixed pair.

2.5. Exact inverse increment

Assume now, explicitly, $H_a \geq gI$ for some $g > 0$. Then

$$S = C - B^* H_a^{-1} B \quad (\text{CP16:2.22})$$

is selfadjoint on $\text{Dom } C$, with form domain $\mathcal{D}_J$. Its subtracted operator is bounded and nonnegative. The exact completing-square identity is

$$h_b(u, z) = h_a(u + H_a^{-1} Bz) + \langle Sz, z \rangle. \quad (\text{CP16:2.23})$$

Thus $H_b > 0$ in the coercive sense if and only if $S > 0$ in that sense. For the reverse implication the triangular map in (2.23) and its inverse are bounded. If $H_b \geq g_b I$, minimization in u also directly yields $S \geq g_b I$.

The transported sources must be

$$s_a^+(x) = \sqrt{2} \sinh(x/2) \quad (0 < x < a), \quad t(x) = \sqrt{2} \sinh(x/2) \quad (a < x < b). \quad (\text{CP16:2.24})$$

In particular $\|s_a^+\|^2 = \sinh a - a$ and $\|t\|^2 = \sinh b - \sinh a - \delta$, with no extra or missing factor of two. In this subsection write $s = s_a^+$,

$$p = H_a^{-1} s, \quad d = B^* p, \quad r = t - d, \quad \sigma_a = 2\langle s, H_a^{-1} s \rangle. \quad (\text{CP16:2.25})$$

If, additionally, $S \geq \gamma I > 0$, the solution of $H_b(u, z) = (s, t)$ is

$$z = S^{-1} r, \quad u = p - H_a^{-1} B S^{-1} r. \quad (\text{CP16:2.26})$$

Pairing with the complete source gives the exact increment

$$\boxed{\sigma_b - \sigma_a = 2\langle r, S^{-1} r \rangle \geq 0, \quad \sigma_b = 2\langle (s, t), H_b^{-1}(s, t) \rangle.} \quad (\text{CP16:2.27})$$

This is a proved identity under the two stated coercivity assumptions. In particular, it is not legitimate to write (2.27) with an ordinary inverse at a collar or inner kernel.

For use in a concrete residual calculation, the complete source residual is already explicit:

$$\begin{aligned} r(y) &= \sqrt{2} \sinh(y/2) + \int_0^a [\kappa(y-x) - \kappa(x+y)] p(x) dx \\ &+ \sum_{\ell_n < b} b_n \mathbf{1}_{0 < y - \ell_n < a} p(y - \ell_n) - \sum_{a < \ell_n < a+b} b_n \mathbf{1}_{0 < \ell_n - y < a} p(\ell_n - y), \quad a < y < b. \end{aligned} \quad (\text{CP16:2.28})$$

The mixed polar-archimedean and polar-prime contributions are precisely the pairings between the first line's source and the remaining terms of this formula. They cannot be omitted when bounding (2.27).

2.6. Complete Schur complement of the Weil operator

Let $\mu_a = 1 - \sigma_a > 0$. The full Weil block matrix, retaining the completed polar term, is

$$A_b^- = \begin{pmatrix} H_a - 2|s\rangle\langle s| & B - 2|s\rangle\langle t| \\ B^* - 2|t\rangle\langle s| & C - 2|t\rangle\langle t| \end{pmatrix}. \quad (\text{CP16:2.29})$$

The inner inverse is the bounded rank-one identity

$$(A_a^-)^{-1} = H_a^{-1} + \frac{2}{\mu_a} |p\rangle\langle p|. \quad (\text{CP16:2.30})$$

Taking the complete Schur complement of (2.29), not merely the Schur complement of H_b , gives

$$\boxed{S_A = S - \frac{2}{\mu_a} |r\rangle\langle r|.} \quad (\text{CP16:2.31})$$

For clarity, the unsimplified expansion is

$$S_A = S - 2(1 + \sigma_a)|t\rangle\langle t| + 2|d\rangle\langle t| + 2|t\rangle\langle d| - \frac{2}{\mu_a}|d - \sigma_a t\rangle\langle d - \sigma_a t|. \quad (\text{CP16:2.32})$$

Expanding the final square and using $\mu_a = 1 - \sigma_a$ yields (2.31). Equation (2.32) displays both polar cross terms and the additional inverse correction, including its own mixed terms. No sign estimate entered this algebra.

If $S > 0$, the rank-one criterion applied to (2.31) is

$$A_b^- \geq 0 \iff 2\langle r, S^{-1}r \rangle \leq \mu_a. \quad (\text{CP16:2.33})$$

A strict inequality gives coercivity of the collar Schur complement and then of A_b^- ; equality allows a null vector. Combining (2.27) and (2.33) recovers the exact scalar condition $\sigma_b \leq 1$.

2.7. Closing statement for Group 2

Exact theorem and hypotheses. For every $0 < a < b < \infty$, the domain decomposition (2.5), complete boundary formulas (2.11)-(2.12), arithmetic blocks (2.13)-(2.17), and bounded cross estimate (2.19) hold on the actual odd Weil form domain. If $H_a \geq gI > 0$ and $S \geq \gamma I > 0$, the inverse increment is exactly (2.27). If also $\mu_a > 0$, the complete Weil Schur complement is (2.31).

Proof status. These are completed identities and independently proved cross-block estimates. They account for all prime thresholds, both prime sign channels, the reflected archimedean term, logarithmic interface weights, and the full polar rank one.

Consequence for Weil positivity. A certified collar cost below μ_a propagates strict odd Weil positivity from a to b . This would be a local extension. Repeating it to every finite endpoint still requires the independent uniformity estimate in Group 4.

Exact unresolved inequality. The new scalar that must be bounded is

$$2 \left\langle \sqrt{2} \sinh(\cdot/2) - B^* H_a^{-1} s_a^+, (C - B^* H_a^{-1} B)^{-1} \left[\sqrt{2} \sinh(\cdot/2) - B^* H_a^{-1} s_a^+ \right] \right\rangle < 1 - 2\langle s_a^+, H_a^{-1} s_a^+ \rangle. \quad (\text{CP16:2.34})$$

For an independently certified lower bound $C \geq LI$, (2.19) supplies only

$$S \geq \left(L - \frac{c_{a,\delta}^2}{g} \right) I. \quad (\text{CP16:2.35})$$

Even if this lower bound is positive, the elementary source bound yields

$$2\langle r, S^{-1}r \rangle \leq \frac{2(\|t\| + c_{a,\delta}\|H_a^{-1}s_a^+\|)^2}{L - c_{a,\delta}^2/g}. \quad (\text{CP16:2.36})$$

This is a valid absolute budget, with all terms specified. It is not the relative budget required at a possible finite positivity endpoint: its denominator depends on the independently known inner spectral gap, and its numerator has not been shown to acquire the factor μ_a . Group 3 can improve the residual evaluation; Group 4 must prove the missing relative control rather than infer it from the exact identities.

3. A complete, certified collar cost

3.1. Domain and complete blocks

Use the unitary odd half-interval representation $u(x) = \sqrt{2}q(x)$, $x > 0$. The inner space is $L^2(0, a)$, the collar is $L^2(a, b)$, and $b = a + \delta$. These L^2 lower bounds test the sign of the original form; they do not replace the Suzuki metric. The letter S below denotes the collar Schur complement, not the Suzuki form $\mathcal{S}$. The source is

$$s(x) = \sqrt{2} \sinh(x/2), \quad 0 < x < a, \quad t(x) = \sqrt{2} \sinh(x/2), \quad a < x < b. \quad (\text{CP16:3.1})$$

In particular $\|s\|^2 = \sinh a - a$; there is no lost factor of two.

The closed form domains are the zero-extended logarithmic form domains. Interval multiplication preserves this form domain: the commutator of $\log |D|$ with a half-line characteristic function is an off-diagonal Carleman operator of norm $\pi/2$, and the corresponding bounded cross form extends from the smooth core. Thus the decomposition in Group 2 is a legitimate form decomposition. The block $B : L^2(a, b) \rightarrow L^2(0, a)$ is bounded; the diagonal block C is the self-adjoint operator associated with the collar form. No endpoint trace of a general form-domain input is used.

Write $\ell_n = \log n$, $b_n = \Lambda(n)/\sqrt{n}$, and

$$\kappa(h) = \frac{e^{-h/2}}{1 - e^{-2h}}, \quad h > 0, \quad \alpha(h) = \kappa(|h|) - \frac{1}{2|h|}, \quad \alpha(0) = \frac{1}{4}. \quad (\text{CP16:3.2})$$

The complete cross block is

$$\begin{aligned} (Bz)(x) &= \int_a^b [-\kappa(y-x) + \kappa(x+y)]z(y) dy \\ &\quad - \sum_{\ell_n < b} b_n \mathbf{1}_{\{a < x + \ell_n < b\}} z(x + \ell_n) \\ &\quad + \sum_{a < \ell_n < a+b} b_n \mathbf{1}_{\{a < \ell_n - x < b\}} z(\ell_n - x). \end{aligned} \quad (\text{CP16:3.3})$$

Every indicator is retained. The first sum is the ordinary prime shift, the second the odd reflected prime shift. The opposite-sign conventions are essential.

For z in the collar form domain, the complete diagonal form is

$$\begin{aligned} \langle Cz, z \rangle &= \langle \log |D| \tilde{z}, \tilde{z} \rangle - \log(2\pi) \|z\|^2 - \int_a^b \int_a^b \alpha(x-y) z(y) \overline{z(x)} dy dx \\ &\quad + \int_a^b \int_a^b \kappa(x+y) z(y) \overline{z(x)} dy dx \\ &\quad - 2 \sum_{\ell_n < \delta} b_n \Re \int_a^{b-\ell_n} z(x + \ell_n) \overline{z(x)} dx \\ &\quad + \sum_{2a < \ell_n < 2b} b_n \int_{\max(a, \ell_n - b)}^{\min(b, \ell_n - a)} z(\ell_n - x) \overline{z(x)} dx. \end{aligned} \quad (\text{CP16:3.4})$$

Here $\tilde{z}$ is its ordinary zero extension from (a, b) to the line. The reflected archimedean term is positive:

$$\int_a^b \int_a^b \kappa(x+y) z(y) \overline{z(x)} dy dx = \sum_{j \geq 0} \left| \int_a^b e^{-(2j+1/2)x} z(x) dx \right|^2 \geq 0. \quad (\text{CP16:3.5})$$

Since $a > 0$, the geometric series is uniformly convergent on the collar square, so (3.5) includes its full tail. The two eigenvalue channels of each reflected prime involution have signs $+b_n$ and $-b_n$; they are not replaced by a positivity assertion.

3.2. Quantitative diagonal and cross bounds

For every function z supported in an interval of length δ ,

$$\langle \log |D| \tilde{z}, \tilde{z} \rangle \geq [\log(\pi/\delta) - 1] \|z\|^2. \quad (\text{CP16:3.6})$$

Indeed $|\hat{z}(\xi)|^2 \leq \delta \|z\|^2$, and for every $T > 0$,

$$\frac{1}{2\pi} \int_{|\xi| < T} \log(T/|\xi|) |\hat{z}(\xi)|^2 d\xi \leq \frac{\delta T}{\pi} \|z\|^2.$$

Taking $T = \pi/\delta$ proves (3.6), first on the core and then on the closed domain.

The already established $|\alpha| \leq 1$, the positive term (3.5), and the norm-one reflected restrictions give, when $\delta < \log 2$,

$$C \geq L_{a,\delta} I, \quad L_{a,\delta} = \log \frac{1}{2\delta} - 1 - \delta - \sum_{2a < \ell_n < 2b} b_n. \quad (\text{CP16:3.7})$$

There is no ordinary collar prime in this range of δ . Formula (3.7) does charge every reflected packet newly meeting the collar, including its negative channel.

The cross estimate can be improved slightly over the candidate in the followup. Since κ is decreasing,

$$0 \leq \kappa(y-x) - \kappa(x+y) \leq \kappa(y-x) \leq \frac{1}{2(y-x)} + 1, \quad 0 < x < a < y < b.$$

Pointwise domination of the absolute integral operator, followed by the Carleman norm bound, gives

$$\|B\| \leq c_{a,\delta} := \frac{\pi}{2} + \sqrt{a\delta} + \sum_{\ell_n < b} b_n + \sum_{a < \ell_n < a+b} b_n. \quad (\text{CP16:3.8})$$

Thus no additional $\kappa(a)\sqrt{a\delta}$ charge is needed. This estimate keeps the reflected archimedean contribution in the exact block before using domination. Truncating the Carleman kernel on either side only decreases its positive-operator majorant norm.

If $H_a \geq gI > 0$, the exact Schur complement satisfies

$$S = C - B^* H_a^{-1} B \geq \gamma I, \quad \gamma = L_{a,\delta} - c_{a,\delta}^2/g. \quad (\text{CP16:3.9})$$

This is an independently proved lower bound whenever its displayed right side is positive. No inverse of an unknown nonpositive operator has been used.

3.3. Full residual identity and all mixed terms

Put $w = H_a^{-1}s$, $r = t - B^*w$. For any $z \in D(C) = D(S)$, let

$$e = r - Sz = t - Cz - B^* H_a^{-1}(s - Bz). \quad (\text{CP16:3.10})$$

If $S \geq \gamma I > 0$, direct completion of the square proves the exact identity

$$\begin{aligned} 2\langle r, S^{-1}r \rangle &= 4\Re\langle r, z \rangle - 2\langle Sz, z \rangle + 2\langle e, S^{-1}e \rangle \\ &\leq 4\Re\langle r, z \rangle - 2\langle Sz, z \rangle + \frac{2\|e\|^2}{\gamma}. \end{aligned} \quad (\text{CP16:3.11})$$

All interactions occur in (3.10), including the polar source and its interaction with every prime and archimedean component of B . To make the cross terms explicit, write

$$B = B_{\text{ar}} - \sum_n b_n P_n + \sum_n b_n F_n, \quad r = t - B_{\text{ar}}^* w + \sum_n b_n P_n^* w - \sum_n b_n F_n^* w. \quad (\text{CP16:3.12})$$

The sums have exactly the cutoffs in (3.3). If these finitely many summands of r are denoted r_ν , then

$$2\langle r, S^{-1}r \rangle = 2 \sum_\nu \langle r_\nu, S^{-1}r_\nu \rangle + 4\Re \sum_{\nu < \eta} \langle r_\nu, S^{-1}r_\eta \rangle. \quad (\text{CP16:3.13})$$

This displays every prime-prime, prime-archimedean, reflected-ordinary, and source cross term. The estimates below use a bound on the complete sum, not an omission of these terms.

The Schur complement itself also retains all cross terms. If $B = \sum_j \varepsilon_j B_j$, where $\varepsilon_j \in \{1, -1\}$ records the exact signs in (3.12), then

$$S = C - \sum_{j,k} \varepsilon_j \varepsilon_k B_j^* H_a^{-1} B_k.$$

In particular the mixed ordinary-reflected terms have not been absorbed by replacing S with the uncorrected collar operator C .

The zero trial already gives the rigorous general budget

$$2\langle r, S^{-1}r \rangle \leq \frac{2}{\gamma} \left[\sqrt{2\delta} \sinh(b/2) + \frac{c_{a,\delta}}{g} \sqrt{\sinh a - a} \right]^2. \quad (\text{CP16:3.14})$$

The next subsection certifies an actual nonzero trial as well.

3.4. A concrete enlargement beyond the first prime

Use the independently certified Group 1 seed

$$a = \frac{699}{2000}, \quad H_a > \frac{33}{2000} I, \quad A_a^- > \frac{1}{500} I. \quad (\text{CP16:3.15})$$

Set

$$\delta = e^{-2500}, \quad b = \frac{699}{2000} + e^{-2500}, \quad L_0 = 2500. \quad (\text{CP16:3.16})$$

The only active prime is 2. No new threshold enters between a and b ; no ordinary cross prime occurs because $b < \log 2$. The cross block does contain the reflected packet $+b_2 F_2$, and this is charged below. No reflected prime lies entirely in the collar, since $\log 2 < 2a < 2b < \log 3$.

Elementary bounds from the exponential series give

$$\sinh a - a < \frac{9}{1250}, \quad \|s\| < \frac{17}{200}, \quad \|t\| \leq \frac{\sqrt{\delta}}{4}, \quad c_{a,\delta} < \frac{21}{10}. \quad (\text{CP16:3.17})$$

For example $b < 7/20$, $\sqrt{2} \sinh(7/40) < 1/4$, and

$$c_{a,\delta} = \frac{\pi}{2} + \sqrt{a\delta} + \frac{\log 2}{\sqrt{2}} < \frac{21}{10}.$$

The exact prime coefficient has not been replaced by zero. Equations (3.7)-(3.9) give

$$C > 2498I, \quad \frac{c_{a,\delta}^2}{g} < 268, \quad S > 2230I. \quad (\text{CP16:3.18})$$

The inner inverse scalar has the independent estimate

$$\sigma_a = 2\langle s, H_a^{-1}s \rangle < \frac{2(9/1250)}{33/2000} = \frac{48}{55}, \quad \mu_a > \frac{7}{55}. \quad (\text{CP16:3.19})$$

Choose the explicit, nonzero trial

$$z = \frac{t}{2500}. \quad (\text{CP16:3.20})$$

This is not the unknown solution $S^{-1}r$. It belongs to $D(C)$: the only endpoint singularities in the logarithmic operator applied to a smooth interval-restricted function are logarithms, which lie in L^2 . More explicitly, for $a < x < b$,

$$\begin{aligned} (Ct)(x) = & \left[-\gamma_E - \log(2\pi) - \frac{1}{2} \log((x-a)(b-x)) \right] t(x) \\ & + \frac{1}{2} \int_a^b \frac{t(x) - t(y)}{|x-y|} dy - \int_a^b \alpha(x-y)t(y) dy + \int_a^b \kappa(x+y)t(y) dy. \end{aligned} \quad (\text{CP16:3.21})$$

There is no prime term in C for this particular collar; (3.3) still includes the reflected prime in both Bt and $B^*H_a^{-1}Bt$. Formula (3.21) contains both collar endpoints and the entire reflected archimedean series. It follows either from the distributional Fourier kernel of $\log|D|$, whose constant is $-\gamma_E$, or by differentiating the fractional-Laplacian kernel at exponent zero.

For an explicit remainder bound, put $u = (x-a)/\delta$. Then

$$-\frac{1}{2} \log((x-a)(b-x)) = 2500 - \frac{1}{2} \log(u(1-u)), \quad \int_0^1 \left[\frac{1}{2} \log(u(1-u)) \right]^2 du = 2 - \frac{\pi^2}{12} < 2. \quad (\text{CP16:3.22})$$

Since $\|t\|_\infty < 1/4$, $\|t'\|_\infty < 3/4$, $|\alpha| \leq 1$, and $\kappa(2a) < 1$, (3.21) gives the deliberately conservative bound

$$\|Ct\| < 628\sqrt{\delta}, \quad \|t - Ct/2500\| < \sqrt{\delta}. \quad (\text{CP16:3.23})$$

The complete residual is

$$e = t - \frac{Ct}{2500} - B^*H_a^{-1} \left(s - \frac{Bt}{2500} \right). \quad (\text{CP16:3.24})$$

Thus, retaining the source and all cross blocks,

$$\|r\| < 11, \quad \|e\| \leq \sqrt{\delta} + \frac{21/10}{33/2000} \left(\frac{17}{200} + \frac{21\sqrt{\delta}}{100000} \right) < 11. \quad (\text{CP16:3.25})$$

The constant part of the right side is $119/11 < 11$. The exact trial expression in (3.11) is

$$\begin{aligned} 2\langle r, S^{-1}r \rangle = & \frac{4}{2500} \|t\|^2 - \frac{4}{2500} \Re \langle H_a^{-1}s, Bt \rangle - \frac{2}{2500^2} \langle Ct, t \rangle \\ & + \frac{2}{2500^2} \langle H_a^{-1}Bt, Bt \rangle + 2\langle e, S^{-1}e \rangle. \end{aligned} \quad (\text{CP16:3.26})$$

In particular the mixed polar-prime term has not disappeared. Using (3.11), the favorable sign of $-2\langle Sz, z \rangle$, and $\|z\| \leq \sqrt{\delta}/10000$, we obtain the certified upper bound

$$2\langle r, S^{-1}r \rangle < \frac{242}{2230} + \frac{1}{10^6} < \frac{1}{9}. \quad (\text{CP16:3.27})$$

The tiny term bounds $4\|r\|\|z\|$; it follows, for example, from $e^{1250} > 4400$. All estimates are analytical bounds with explicit remainders, not a numerical truncation of the zero set or the archimedean series.

For the resulting operator lower bound, the scalar block comparison proves

$$H_b > \frac{7}{500}I : \quad \left(\frac{33}{2000} - \frac{7}{500} \right) \left(2498 - \frac{7}{500} \right) > \left(\frac{21}{10} \right)^2. \quad (\text{CP16:3.28})$$

The enlargement identity from Group 2 and (3.19), (3.27) now yield

$$\mu_b = 1 - \sigma_b > \frac{7}{55} - \frac{1}{9} = \frac{8}{495}, \quad A_b^- \geq \mu_b H_b > \frac{1}{5000}I. \quad (\text{CP16:3.29})$$

The last implication follows from $|\langle s_b, u \rangle|^2 \leq \langle s_b, H_b^{-1} s_b \rangle \langle H_b u, u \rangle$, with the original odd source. Therefore the certified interval is

$$0 < a' \leq \frac{699}{2000} + e^{-2500}, \quad Q_{a'}^-(q) \geq \frac{1}{5000} \|q\|^2. \quad (\text{CP16:3.30})$$

Restriction to smaller support preserves this lower bound. It is a local support gain beyond a prime-active seed; its small size reflects the conservative Carleman and inverse estimates.

3.5. Closing statement and remaining uniformity estimate

Exact theorem and hypotheses. For $a > 0$, $\delta < \log 2$, and $H_a \geq gI > 0$, (3.3)-(3.14) are complete closed-form/block estimates. If $\gamma > 0$, they give a computable sufficient collar budget. With the independently proved seed (3.15), the nonzero trial (3.20) certifies (3.27) and hence the explicit support interval and lower bound (3.30).

Proof status and consequence. The actual collar estimate closes for this specified extension, including the reflected first prime and all polar and archimedean couplings. It proves unconditional odd Weil positivity on the displayed finite support interval. It does not prove positivity on all finite supports.

Exact unresolved estimate. To iterate without finite accumulation, the complete signed expression (3.13) would need a uniform relative estimate such as

$$2 \sum_{\nu, \eta} \langle r_\nu, S^{-1} r_\eta \rangle \leq (1 - e^{-\int_a^b c_R(u) du}) \mu_a, \quad b \leq R, \quad (\text{CP16:3.31})$$

with an independently bounded integrable c_R , not containing the reciprocal of an unknown margin. The attempted bound (3.14) supplies only an absolute budget. With the nonvanishing Carleman majorant $c_{a,\delta} \rightarrow \pi/2 + \text{prime charges}$, its available decay is only of order $1/\log(1/\delta)$, with constants involving g^{-2} . This does not yield a locally linear relative rate, and making it less than μ_a can require exponentially tiny steps depending on that unknown margin. The exact point of failure is the attempted replacement of the right side of (3.14) by a uniform multiple of $\mu_a \delta$. No such inequality has been proved here. The kernel branch where H_a^{-1} is unavailable remains the separate issue treated in Group 4.

4. Finite endpoints, a quantitative tail, and the kernel branch

All operators in this group act on the zero-extended odd subspace of $L^2(-a, a)$, with the closed logarithmic form domain retained from checkpoint 15. The source $s_a(x) = \sinh(x/2)$ has squared norm $\sinh a - a$. Under the unitary half-interval identification it is $\sqrt{2} \sinh(x/2)$; these are the same Hilbert-space normalization.

The L^2 spectral projections below are auxiliary devices for testing the sign of the complete Weil form. They do not replace the original Suzuki metric or its Gram matrices.

4.1. What a relative collar budget would prove

Suppose, on $a_0 \leq a < b \leq R$, the hypotheses making Group 2's inverse identity legitimate hold: $H_a > 0$, the complete collar Schur operator $S > 0$, and $\mu_a = 1 - \sigma_a > 0$. Then

$$\mu_b = \mu_a - 2\langle r, S^{-1}r \rangle. \quad (\text{CP16:4.1})$$

Consequently the proposed independently quantified estimate

$$2\langle r, S^{-1}r \rangle \leq \left(1 - \exp \left\{ - \int_a^b c_R(u) du \right\}\right) \mu_a, \quad c_R \in L^1(a_0, R), \quad c_R \geq 0, \quad (\text{CP16:4.2})$$

would give

$$\mu_b \geq \mu_a \exp \left\{ - \int_a^b c_R(u) du \right\}. \quad (\text{CP16:4.3})$$

Iterating over any permitted partition gives

$$\mu_{a_j} \geq \mu_{a_0} \exp \left\{ - \int_{a_0}^{a_j} c_R(u) du \right\} \geq \mu_{a_0} e^{-\|c_R\|_{L^1(a_0, R)}} > 0. \quad (\text{CP16:4.4})$$

Thus, provided H_a remains invertible with the requisite limiting domain control, (4.2) would prevent resolvent saturation at a finite accumulation endpoint. The possible loss of invertibility of H_a is an additional branch; (4.4) alone does not exclude it.

For comparison, let the actual complete collar estimates of Group 3 give

$$H_a \geq gI, \quad \|B\| \leq c, \quad C \geq LI, \quad \gamma = L - c^2/g > 0. \quad (\text{CP16:4.5})$$

For $0 < \delta < \log 2$, Group 3 supplies the explicit admissible choices

$$L = \log \frac{1}{2\delta} - 1 - \delta - \sum_{2a < \ell_n < 2b} b_n,$$

$$c = \frac{\pi}{2} + \sqrt{a\delta} + \sum_{\ell_n < b} b_n + \sum_{a < \ell_n < a+b} b_n.$$

The two sums in c respectively retain ordinary and reflected prime cross-blocks; the sum in L retains reflected packets within the collar. The reflected archimedean kernel remains in the exact operators. Its sign permits the cross-kernel estimate $0 \leq \kappa(y-x) - \kappa(x+y) \leq \kappa(y-x)$ for $0 < x < a < y < b$, which gives the displayed Carleman bound without an extra $\kappa(a)\sqrt{a\delta}$ loss. Then $S \geq \gamma I$. Keeping

the exact residual $r = t - B^* H_a^{-1} s_a$, its elementary size estimate is

$$\begin{aligned} \|r\| &\leq \|t\| + \frac{c}{g} \|s_a\|, \\ \|t\|^2 &= \sinh b - b - (\sinh a - a) = 2 \int_a^b \sinh^2(x/2) dx \leq 2\delta \sinh^2(b/2), \\ 2\langle r, S^{-1}r \rangle &\leq F_{a,\delta}(g, c, L) := \frac{2}{\gamma} \left[\sqrt{2\delta} \sinh(b/2) + \frac{c}{g} \sqrt{\sinh a - a} \right]^2. \end{aligned} \quad (\text{CP16:4.6})$$

This is an unconditional consequence of the stated lower and cross-block bounds; it does not replace the sharper signed residual evaluation in Group 3. The complete Carleman estimate has a nonvanishing cross-block bound as $\delta \downarrow 0$, while L grows only logarithmically. Accordingly this particular norm estimate gives, at fixed $a, g > 0$, at best a bound of order $1/\log(1/\delta)$. It is an absolute collar bound, not the requested relative bound.

Even if a sharper calculation improves the right side of (4.6) to $K_R \delta$, this alone would prove (4.2) only after inserting a lower bound for μ_a : the sufficient rate would involve K_R/μ_a . More exactly, whenever $0 < F_{a,\delta} < \mu_a$, the rate manufactured from (4.6) is

$$c_{a,\delta}^{\text{manufactured}} = -\frac{1}{\delta} \log \left(1 - \frac{F_{a,\delta}}{\mu_a} \right). \quad (\text{CP16:4.7})$$

It explicitly divides by the unknown margin and therefore does not meet the requested hypothesis. This identifies the failed estimate, rather than treating continuity of σ_a as prevention of a finite first crossing. A relation tying the actual signed residual to μ_a , rather than only to $\|s_a\|$ and $\|H_a^{-1}\|$, is still required.

4.2. A completely specified finite-rank regularization

The inverse-unavailable branch can be put into a uniformly well-defined finite-dimensional test without assuming H_a invertible. This does not by itself supply its strict sign, but its complementary estimate can be made explicit.

Let C_a denote the supported odd logarithmic operator, with increasing eigenvalues $\nu_j(a)$. Its compact resolvent and closed form domain were established in checkpoint 15. Set

$$\begin{aligned} \Pi_a q &= \sum_{\ell_n < 2a} b_n (q(\cdot + \ell_n) + q(\cdot - \ell_n)) \Big|_{(-a,a)}, \\ (K_{\alpha,a} q)(x) &= \int_{-a}^a \alpha(x-y) q(y) dy, \\ D_a &= \log(2\pi) I + \Pi_a + K_{\alpha,a} + 2|s_a\rangle\langle s_a|. \end{aligned} \quad (\text{CP16:4.8})$$

The zero extension is part of Π_a . Thus (4.8) includes both ordinary and reflected prime shifts after passage to the half interval. The completed Weil operator is exactly

$$A_a^- = C_a - D_a. \quad (\text{CP16:4.9})$$

The full polar term, with its original sign in A_a^- , is retained. Young's inequality and the translation norm bound give

$$\|D_a\| \leq M_a := \log(2\pi) + 2 \sum_{\ell_n < 2a} b_n + 4a + 2(\sinh a - a). \quad (\text{CP16:4.10})$$

Here the established complete-kernel estimate $\int_{-2a}^{2a} |\alpha(t)| dt \leq 4a$ suffices. It is deliberately not a sign assertion about the prime or polar terms.

An explicit eigenvalue lower bound is

$$\boxed{\nu_N(a) \geq \log \frac{\pi N}{2a} - 1 \quad (N \geq 1).} \quad (\text{CP16:4.11})$$

To prove it, let $f_1, \dots, f_N$ be orthonormal odd eigenfunctions for the first N eigenvalues, and put $F(\xi) = \sum_{j=1}^N |\widehat{f_j}(\xi)|^2$. Bessel's inequality, applied to the odd part of the exponential on $(-a, a)$, yields

$$0 \leq F(\xi) \leq \int_{-a}^a \sin^2(\xi x) dx = a - \frac{\sin(2a\xi)}{2\xi} \leq 2a, \quad \frac{1}{2\pi} \int_{\mathbb{R}} F = N. \quad (\text{CP16:4.12})$$

Among functions with these mass and height constraints, the increasing weight $\log |\xi|$ has its smallest integral by filling the interval $|\xi| \leq L$ at height $2a$, with $L = \pi N/(2a)$. The usual subtraction proof is direct: on both this interval and its complement, $(\log |\xi| - \log L)(F - 2a\mathbf{1}_{[-L, L]}) \geq 0$. The negative logarithmic singularity is integrable against the bound $2a$, so truncation justifies the calculation. Therefore

$$\sum_{j=1}^N \nu_j(a) = \frac{1}{2\pi} \int \log |\xi| F(\xi) d\xi \geq N(\log L - 1).$$

Since ν_N is at least this average, (4.11) follows.

Fix a finite $R > 0$. In (4.10), replace a by R to define M_R , including every prime packet with $\ell_n < 2R$. Put

$$c_{\text{odd}} = \frac{1}{3} \log \frac{9\pi}{2} - \frac{1}{3}, \quad h_R = -\log R + c_{\text{odd}}, \quad (\text{CP16:4.13})$$

$$d_R = \max \left\{ 1, \left\lceil \frac{2R}{\pi} e^{M_R+2} \right\rceil \right\}, \quad \tau_R = 1 + \max\{0, M_R + M_R^2 - h_R\}. \quad (\text{CP16:4.14})$$

These constants are finite and explicit. For each $0 < a \leq R$, let $P = P_{d_R}(a)$ be the spectral projection of $\mathbb{C}_a$ onto its first d_R eigenvectors, and put $Q = I - P$. Then

$$D := QA_a^- Q \geq I, \quad \|PA_a^- Q\| = \|PD_a Q\| \leq M_R. \quad (\text{CP16:4.15})$$

Indeed (4.11) gives $\nu_{d_R+1}(a) > M_R + 1$. The form-domain decomposition is legitimate because P is a finite spectral projection of $\mathbb{C}_a$, and D_a is bounded. The tail inverse D^{-1} is consequently bounded on QL^2 .

Let

$$F_a = PA_a^- P - PA_a^- Q D^{-1} QA_a^- P. \quad (\text{CP16:4.16})$$

The earlier odd logarithmic estimate gives $\mathbb{C}_a \geq h_R I$, so

$$F_a \geq \lambda_R I_P, \quad \lambda_R := h_R - M_R - M_R^2. \quad (\text{CP16:4.17})$$

Thus the finite-rank regularized full operator

$$\boxed{H_{\tau_R, a} := A_a^- + \tau_R P} \quad (\text{CP16:4.18})$$

is strictly positive for every $a \leq R$, without any positivity assumption on A_a^- or invertibility assumption on H_a . In fact

$$\boxed{H_{\tau_R, a} \geq \frac{1}{\max\{2, 1 + 2M_R^2\}} I.} \quad (\text{CP16:4.19})$$

For the proof, write a form vector as $u + w$, $u \in PL^2$, $w \in Q\mathcal{D}$. Completing the square gives

$$\langle H_{\tau_R, a}(u + w), u + w \rangle = \langle (F_a + \tau_R I)u, u \rangle + \langle D(w + D^{-1}QA_a^-Pu), w + D^{-1}QA_a^-Pu \rangle. \quad (\text{CP16:4.20})$$

By (4.14), (4.17), and (4.15), the right side is at least $\|u\|^2 + \|w + D^{-1}QA_a^-Pu\|^2$. The squared norm of $u + w$ is at most $(1 + 2M_R^2)\|u\|^2 + 2\|w + D^{-1}QA_a^-Pu\|^2$. This proves (4.19), including operator-domain invertibility by the representation theorem.

4.3. Every null branch is retained; the attempted strict estimate

For the positive regularization just constructed,

$$\boxed{\ker A_a^- \neq \{0\} \iff 1 \in \text{spec}(\tau_R PH_{\tau_R, a}^{-1} P|_{PL^2}).} \quad (\text{CP16:4.21})$$

If $A_a^- v = 0$, then $H_{\tau_R, a} v = \tau_R P v$. Moreover $P v \neq 0$, because otherwise (4.15) and the tail equation would imply $v = 0$. Applying $PH_{\tau_R, a}^{-1}$ gives the eigenvalue-one equation. Conversely, an eigenvector $u \neq 0$ for eigenvalue one gives the null vector $v = \tau_R H_{\tau_R, a}^{-1} u$, with $P v = u$. This includes a null vector in $s_a^\perp$, so the zero-moment branch has not disappeared.

The block inverse is exact:

$$PH_{\tau_R, a}^{-1} P = (\tau_R I + F_a)^{-1}. \quad (\text{CP16:4.22})$$

Thus the sufficient strict estimate

$$\|\tau_R PH_{\tau_R, a}^{-1} P\| < 1 \quad (\text{CP16:4.23})$$

is precisely $F_a > 0$, equivalently $A_a^- > 0$ because the complementary block is positive. At a hypothetical first null, $A_a^- \geq 0$, so $F_a \geq 0$ with a zero eigenvalue, and the norm in (4.23) equals one.

For completeness the *calculated* estimate from the certified tail is

$$\|\tau_R PH_{\tau_R, a}^{-1} P\| \leq \frac{\tau_R}{\tau_R + \lambda_R}. \quad (\text{CP16:4.24})$$

The denominator is at least one. Whenever $\lambda_R < 0$, the displayed upper bound exceeds one; it supplies invertibility of the regularization but fails at exactly the needed strict sign. Increasing d improves the term M_R^2 in (4.17) to M_R^2/δ_d , with

$$\delta_d \geq \log \frac{\pi(d+1)}{2R} - 1 - M_R, \quad 0 \leq PA_a^- QD^{-1}QA_a^- P \leq \frac{M_R^2}{\delta_d} I. \quad (\text{CP16:4.25})$$

It does not independently prove positivity of the uncorrected finite block as d grows.

There is a strict verified application on every interval already certified by Group 1: if $A_a^- \geq m_* I$, $m_* > 0$, then minimizing its full quadratic form over w in (4.20) gives $F_a \geq m_* I$, hence

$$\|\tau_R PH_{\tau_R, a}^{-1} P\| \leq \frac{\tau_R}{\tau_R + m_*} < 1. \quad (\text{CP16:4.26})$$

For example checkpoint 15 alone gives $m_* = 253/1536$ when $a \leq 1/8$. Group 1's larger certified interval and its own margin can replace this example. This is a genuine local exclusion of both null branches; it is not a global continuation estimate.

No arithmetic cross term has been removed from (4.16). In explicit finite-block notation, put

$$E_\pi = P\Pi_a Q, \quad E_\alpha = PK_{\alpha,a} Q, \quad E_{\text{pol}} = 2|Ps_a\rangle\langle Qs_a|.$$

The constant term has zero cross block. Then

$$\boxed{F_a = PC_a P - \log(2\pi)P - P\Pi_a P - PK_{\alpha,a} P - 2|Ps_a\rangle\langle Ps_a| - \sum_{X,Y \in \{\pi, \alpha, \text{pol}\}} E_X D^{-1} E_Y^*} \quad (\text{CP16:4.27})$$

The last line contains three diagonal corrections and all six mixed corrections. The tail D itself contains the complete arithmetic and polar operator. A fully quantified sufficient missing estimate is, for each fixed finite R , a choice d with $\delta_d > 0$ and a number $\eta_R > 0$ such that, uniformly over the support range in question,

$$\lambda_{\min} \left[PC_a P - \log(2\pi)P - P\Pi_a P - PK_{\alpha,a} P - 2|Ps_a\rangle\langle Ps_a| - \sum_{X,Y} E_X D^{-1} E_Y^* \right] \geq \eta_R. \quad (\text{CP16:4.28})$$

An interval-dependent positive bound at every hypothetical first crossing would also suffice. The known attempt is (4.17)/(4.25); its deficit is the possible nonpositivity of the resulting lower bound, not a missing tail definition.

4.4. Closing statement

Exact theorem and domain. The relative-budget implication (4.1)-(4.4) holds whenever the complete interval Schur inverse exists on the closed odd form domain. The explicit tail estimate (4.11), complete regularization (4.18)-(4.20), null equivalence (4.21), and full cross-term expansion (4.27) hold unconditionally at every fixed finite R , with all arithmetic packets and completed polar terms retained.

Completed calculation and proof status. A positive complementary block and an everywhere positive finite-rank regularization have been constructed with explicit constants. Every null vector, including the inverse-unavailable zero-moment branch, is retained by the eigenvalue-one test. The estimates (4.6) and (4.24) have been calculated; neither supplies the required global strict sign.

Consequence for Weil positivity. Group 1's local certificate excludes both branches on its certified support range. Outside that range, finite-endpoint failure and a zero-moment kernel have not been excluded. No global odd Weil positivity or unconditional identity $\mathfrak{W} = \mathcal{S}$ follows from the regularization alone.

Exact unresolved estimate. Either establish (4.2) with an independently bounded integrable c_R , together with control of the kernel branch, or establish the complete finite-matrix inequality (4.28). The attempted absolute collar estimate stops at the missing factor μ_a ; the attempted finite-rank bound stops at the possible nonpositive lower bound in (4.17). If either route closes with the stated additional requirements, the established odd Weil criterion gives RH, and the correctly normalized Suzuki identity then gives $\mathfrak{W} = \mathcal{S}$ and $\mathcal{E}_N = 0$ for every N . Those antecedents remain unproved here.

5. Combined result and the exact remaining global estimate

5.1. The strongest independently proved sign

Group 1 crosses the first arithmetic threshold by estimating the concentration operator and its coupling to the actual reflected prime collar. The strict seed is

$$a_1 = \frac{699}{2000} > \frac{\log 2}{2}, \quad H_a > \frac{3313}{200000}I, \quad \sigma_a < \frac{14318}{16565}, \quad A_a^- > \frac{2247}{10^6}I \quad (0 < a \leq a_1). \quad (\text{CP16:5.1})$$

The concentration certificate includes an explicit operator-norm approximation remainder; the scalar comparisons have exact rational certificates. The archimedean estimate uses the cancellation of $\alpha(0)$ in the odd sector before any absolute estimate. Both reflection signs and the complete polar source are present.

Groups 2-3 prove and quantitatively use the exact support-enlargement identity. They give the largest support interval certified in this round:

$$a_* := \frac{699}{2000} + e^{-2500}, \quad Q_a^-(q) \geq \frac{1}{5000} \|q\|_2^2 \quad (0 < a \leq a_*, \quad q \in \mathcal{D}_a^-). \quad (\text{CP16:5.2})$$

For the collar from a_1 to a_* , the complete estimates are

$$C > 2498I, \quad \|B\| < \frac{21}{10}, \quad S > 2230I, \quad \sigma_{a_*} - \sigma_{a_1} < \frac{1}{9}, \quad (\text{CP16:5.3})$$

$$H_{a_*} > \frac{7}{500}I, \quad 1 - \sigma_{a_*} > \frac{8}{495}. \quad (\text{CP16:5.4})$$

The nonzero approximate solution is explicitly $z = t/2500$; its residual includes both endpoint logarithms and every cross-block contribution. The collar is very small because the uniform Carleman norm and the inverse estimate are conservative. Its purpose is to certify a complete support-enlargement calculation. Its size is not evidence for continuation to infinity.

These inequalities exclude both kinds of odd null vector on the displayed support range. In particular H_a cannot have a zero-moment kernel there. They concern arbitrary odd form-domain inputs on these supports, not a sampled family of vectors or a finite collection of zeta zeros.

5.2. The surviving signed interaction

The exact obstruction to a general support step is the full effective collar form

$$\begin{aligned} \langle S_A z, z \rangle &= \langle C z, z \rangle - \langle B z, H_a^{-1} B z \rangle \\ &\quad - \frac{2}{\mu_a} \left| \langle t, z \rangle - \langle H_a^{-1} s_a, B z \rangle \right|^2, \quad \mu_a = 1 - \sigma_a. \end{aligned} \quad (\text{CP16:5.5})$$

Here B contains the ordinary and reflected prime shifts and the full kernel $-\kappa(y-x) + \kappa(x+y)$. The second term contains every prime-prime and prime-archimedean Schur correction. The last square contains the source, its coupling to those same operators, and the original polar normalization. The positive reflected archimedean Gram kernel remains inside C . No source norm bound makes this square acquire the factor μ_a automatically.

The proved general absolute estimate is

$$\sigma_b - \sigma_a \leq \frac{2}{L_{a,\delta} - c_{a,\delta}^2/g} \left[\sqrt{2\delta} \sinh(b/2) + \frac{c_{a,\delta}}{g} \sqrt{\sinh a - a} \right]^2, \quad (\text{CP16:5.6})$$

provided $H_a \geq gI > 0$ and the denominator is positive. It proves a local step when its right side is below a known μ_a . At a possible finite first crossing, neither a persistent positive g nor a bound tying the numerator to μ_a has been established. Repeating an absolute estimate with ever smaller steps does not resolve this issue.

5.3. A quantified remaining lemma, including the kernel branch

One route remains the relative estimate (4.2), with an independently bounded integrable c_R , accompanied by a persistent lower bound for H_a . Its missing feature is exactly the factor $1 - \sigma_a$; dividing the absolute bound by that unknown factor does not prove it.

The following single finite-rank alternative retains both null branches. For each finite $R > a_*$, take the explicitly defined M_R, d_R, τ_R of (4.10), (4.14), and the corresponding projections $P = P_{d_R}(a)$, $Q = I - P$. The complementary operator $D = QA_a^-Q$ is independently known to satisfy $D \geq I$. What remains to prove, uniformly for $a \in [a_*, R]$, is a number $\eta_R > 0$ such that

$$F_a = PA_a^-P - PA_a^-QD^{-1}QA_a^-P \geq \eta_R I_P. \quad (\text{CP16:5.7})$$

The complete nine-term correction inside this matrix is written explicitly in (4.27). Equivalently, the required quantified strict estimate is

$$\|\tau_R P H_{\tau_R, a}^{-1} P\| \leq \frac{\tau_R}{\tau_R + \eta_R} < 1. \quad (\text{CP16:5.8})$$

This is recorded as an open, equivalent sign estimate, not as a newly proved closure. The attempted independent calculation gives only

$$F_a \geq \lambda_R I_P, \quad \lambda_R = h_R - M_R - M_R^2, \quad \|\tau_R P H_{\tau_R, a}^{-1} P\| \leq \frac{\tau_R}{\tau_R + \lambda_R}. \quad (\text{CP16:5.9})$$

Where $\lambda_R \leq 0$, this fails the needed strict inequality. The sharper, rigorously vanishing tail M_R^2/δ_d does not assign a positive sign to the remaining finite block. Thus the exact stopping line is the upgrade from (5.9), or from the signed collar expression (5.5), to the independently strict estimate (5.7). This identifies the required calculation while preserving all arithmetic and polar interactions.

5.4. Status and conditional return to the original target

Question	Checkpoint 16 result
Largest support interval certified here	Every odd test in $\mathcal{D}_a^-$ for $0 < a \leq 699/2000 + e^{-2500}$.
Explicit Weil lower bound on that interval	$Q_a^-(q) \geq \ q\ _2^2/5000$; the seed up to 699/2000 has the stronger bound $2247\ q\ _2^2/10^6$.
Complete collar identities and estimates	Proved; the specified nonzero-trial collar cost is strictly below 1/9.
Relative budget reaching every finite endpoint	Not proved. The available estimate is absolute and depends on the inner inverse gap.

Question	Checkpoint 16 result
Kernel branch	Excluded locally; retained exactly by the uniformly positive finite-rank regularization, but not excluded globally.
Unconditional Weil positivity on all compact tests	Not proved.
Original defect criterion $\liminf_N \mathcal{E}_N \leq 1$	Not proved.

If the remaining global sign estimate were established, the dependency chain would be

$$\begin{aligned}
 & \text{strict odd Weil positivity on every finite support} \\
 \implies & \text{nonnegativity on every compactly supported smooth odd test} \\
 & \xRightarrow{\text{Yoshida}} \text{RH} \xRightarrow{\text{Weil}} \mathfrak{W}(q, q) \geq 0 \text{ on all compact smooth tests} \quad (\text{CP16:5.10}) \\
 & \xRightarrow{\text{normalized Suzuki identity}} \mathfrak{W} = \mathcal{S} \implies \Delta = 0 \implies \mathcal{E}_N = 0 \text{ for every } N.
 \end{aligned}$$

The strong Suzuki-domain passage and original metric remain available for that return. The antecedent covering all finite supports remains open; the local theorem (5.2) does not assert it. Earlier checkpoint notes are preserved unchanged, and this checkpoint records its independent local gain and unresolved global estimate separately.

We are aiming to prove unconditional Weil positivity with the original Suzuki metric preserved.

Checkpoint 17: Boundary costs and first crossings

Source: *Michaels_Boundary_Cost_and_Unified_First_Crossing_Checkpoint_17.md*. Equation tags carry the source prefix *CP17*; references within this unit retain their original local numbering.

Checkpoint 17 – 23 September 2026

This note continues the complete Checkpoint 16 and its current ledger. The primary objective remains unconditional Weil positivity. The independently proved starting interval is preserved:

$$a_0 = \frac{699}{2000} + e^{-2500}, \quad Q_a^-(q) \geq \frac{1}{5000} \|q\|_2^2 \quad (0 < a \leq a_0). \quad (\text{CP17:0.1})$$

Here Q_a^- is the restriction of the complete Weil form to odd, zero-extended tests supported in $(-a, a)$. Its closed form domain is

$$\mathcal{D}_a^- = \left\{ q \in L^2(-a, a) : q(-x) = -q(x), \quad \int_{\mathbb{R}} \log(2 + |\xi|) |\widehat{q}(\xi)|^2 d\xi < \infty \right\}, \quad \widehat{q}(\xi) = \int q(x) e^{i\xi x} dx. \quad (\text{CP17:0.2})$$

The original Suzuki metric and determining ladder remain unchanged. The retained defect criterion is

$$\Delta = \mathfrak{W} - \mathcal{S}, \quad \mathcal{E}_N = \left\| [\Delta(\rho_j, \rho_i)]_{i,j \leq N} \right\|_F^2, \quad \liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1. \quad (\text{CP17:0.3})$$

In other coordinates the original Suzuki Gram normalization remains part of this definition. The auxiliary L^2 projections and unitary support rescalings below preserve the sign of the original complete form; they do not replace $\mathcal{S}$.

Contents

1. A fixed-space regularization detecting both first-null branches.
2. Boundedness of the actual forced solutions, boundary regularity, and the complete residual.
3. The actual linear collar cost, averaged boundary coefficient, and full matrix variation through prime interfaces.
4. Exact endpoint substitution and the attempted Osgood estimate.
5. Combined conclusions, precise remaining estimate, and conditional return to zero defect.

Retained arithmetic and boundary normalization

Use

$$\begin{aligned} \ell_n &= \log n, & b_n &= \Lambda(n)/\sqrt{n}, & \kappa(t) &= \frac{e^{-t/2}}{1 - e^{-2t}} \quad (t > 0), \\ \alpha(t) &= \kappa(|t|) - \frac{1}{2|t|}, & \alpha(0) &= \frac{1}{4}, & R(t) &= \alpha(t) - 2 \cosh(t/2). \end{aligned} \quad (\text{CP17:0.4})$$

With zero extension in every translation, the complete odd operator is

$$A_a^- = H_a - 2|s_a\rangle\langle s_a|, \quad H_a = (\log|D|)_{(-a,a)} - \log(2\pi)I - \Pi_a - K_{a,a}, \quad s_a(x) = \sinh(x/2), \quad \|s_a\|_2^2 = \sinh a - a. \quad (\text{CP17:0.5})$$

All ordinary/reflected prime terms are included in Π_a ; both channels appear explicitly on the positive half-interval. The unitary half-interval source is $\sqrt{2} \sinh(x/2)$. Accordingly a boundary

coefficient for the half-interval solution is $\sqrt{2}$ times the coefficient for the full odd solution. The distinction is carried through the collar-rate calculation.

The established global Suzuki-domain passage, strong core convergence, full null equation, and endpoint-complete convolution identities are retained. This round additionally calculates the inhomogeneous endpoint relation for the regularized near-null family. Whole-line exterior contributions remain in the supported logarithmic operator; a weak interior equation is never extended to a zero exterior right-hand side.

Scope of the new results. The bounded-solution and bounded-forcing hypotheses for the logarithmic boundary theorem are established for the actual operators. They yield a linear collar cost wherever $H_a > 0$, and a quantitative formula using a signed averaged boundary coefficient without a pointwise trace assumption. The complete regularized matrix has a uniform Dini modulus through every prime threshold. These are independent auxiliary results. The margin-dependent signed estimate needed to exclude every finite first crossing remains a separate calculation in Group 4.

Sources checked. The complete Checkpoint 16, current ledger, and archived boundary/first-crossing follow-up were read before calculation. The boundary theorem used here is Hernández-Santamaría, López Ríos and Saldaña, *Optimal boundary regularity and a Hopf-type lemma for Dirichlet problems involving the logarithmic Laplacian*, arXiv:2401.18033v2, Theorem 1.1. Its bounded-data hypotheses are proved here before application. Its statement is an upper boundary estimate; this note makes no attribution of a normalized trace limit to it. The inherited odd Weil criterion and original normalization remain those recorded in Suzuki, *Weil's quadratic form via the screw function*, arXiv:2606.09096v2. No RH-dependent orthogonality or global positivity theorem is used.

1. One fixed-space calculation retaining every first-null branch

1.1. Normalization, complete operator, and common domain

Set

$$a_0 = \frac{699}{2000} + e^{-2500}, \quad m_0 = \frac{1}{5000}, \quad a_0 \leq a \leq R < \infty. \quad (\text{CP17:1.1})$$

Checkpoint 16 proves $A_a^- \geq m_0 I$ for $0 < a \leq a_0$. Throughout this group the original Suzuki form is carried by pullback:

$$\tilde{\mathcal{S}}_a(u, v) = \mathcal{S}(U_a^{-1}u, U_a^{-1}v). \quad (\text{CP17:1.2})$$

The auxiliary L^2 projection used below tests the sign of the complete Weil operator; it does not substitute an L^2 metric for (1.2).

Let $\mathcal{H} = L^2_{\text{odd}}(-1, 1)$, and let $E : \mathcal{H} \rightarrow L^2(\mathbb{R})$ be zero extension. On supported functions write C_1 for the self-adjoint operator associated with the closed logarithmic form

$$\mathfrak{c}_1(u, v) = \frac{1}{2\pi} \int_{\mathbb{R}} \log |\xi| \widehat{Eu}(\xi) \overline{\widehat{Ev}(\xi)} d\xi. \quad (\text{CP17:1.3})$$

Its form domain is the odd, supported logarithmic form domain inherited from Checkpoint 16. The integral near $\xi = 0$ is well defined by compact support and the local integrability of $\log |\xi|$. Let

$(T_h f)(x) = f(x + h)$, with zero extension understood, and put

$$\begin{aligned}\tilde{\Pi}_a &= \sum_{\ell_n < 2R} b_n E^*(T_{\ell_n/a} + T_{-\ell_n/a}) E, \\ (\tilde{K}_{\alpha,a} u)(x) &= \int_{-1}^1 a \alpha(a(x-y)) u(y) dy, \\ S_a(x) &= \sqrt{a} \sinh(ax/2), \\ \tilde{D}_a &= \log(2\pi) I + \tilde{\Pi}_a + \tilde{K}_{\alpha,a} + 2|S_a\rangle\langle S_a|.\end{aligned}\tag{CP17:1.4}$$

Here

$$\ell_n = \log n, \quad b_n = \Lambda(n)/\sqrt{n}, \quad \alpha(t) = \frac{e^{-|t|/2}}{1 - e^{-2|t|}} - \frac{1}{2|t|}, \quad \alpha(0) = \frac{1}{4}.\tag{CP17:1.5}$$

Every sum in this group is finite. Including all $\ell_n < 2R$ at every a is exact: the compressed shift is zero when $\ell_n/a \geq 2$. At equality any remaining endpoint value is supported on a measure-zero set and defines the zero L^2 operator. This convention retains every prime-power threshold without adding a discontinuous indicator to the operator formula.

For $U_a q(x) = \sqrt{a} q(ax)$, the Fourier change of variables gives

$$U_a C_a U_a^{-1} = C_1 - \log a I.$$

Consequently the complete scaled Weil operator is

$$\boxed{\tilde{A}_a = C_1 - \log a I - \tilde{D}_a}.\tag{CP17:1.6}$$

Both prime directions, the full archimedean remainder, and the polar rank-one operator occur in (1.6). Compression and zero extension are part of C_1 ; they retain the exterior boundary contribution of the logarithmic operator. No periodic or spectral Dirichlet replacement is made.

All $\tilde{D}_a$ are bounded and self-adjoint. Thus

$$\text{Dom } \tilde{A}_a = \text{Dom } C_1, \quad \text{Dom } |\tilde{A}_a + cI|^{1/2} = \text{Dom } |C_1 + c'I|^{1/2}\tag{CP17:1.7}$$

for sufficiently large shifts, with the same underlying form domain. This is bounded-perturbation invariance, not an assertion that every form-domain function has pointwise endpoint traces.

The full bounded difference on this common domain is

$$\begin{aligned}\tilde{A}_b - \tilde{A}_a &= -\log(b/a) I \\ &\quad - \sum_{\ell_n < 2R} b_n E^*(T_{\ell_n/b} + T_{-\ell_n/b} - T_{\ell_n/a} - T_{-\ell_n/a}) E \\ &\quad - \tilde{K}_{\alpha,b} + \tilde{K}_{\alpha,a} - 2(|S_b\rangle\langle S_b| - |S_a\rangle\langle S_a|).\end{aligned}\tag{CP17:1.8}$$

If $\delta S = S_b - S_a$, the last difference is exactly

$$|S_b\rangle\langle S_b| - |S_a\rangle\langle S_a| = |\delta S\rangle\langle S_a| + |S_a\rangle\langle \delta S| + |\delta S\rangle\langle \delta S|.\tag{CP17:1.9}$$

The two linear cross terms and the quadratic term are all retained.

1.2. A fixed projection and a margin-independent inverse

The constants certified in Checkpoint 16 are

$$\begin{aligned}
 M_R &= \log(2\pi) + 2 \sum_{\ell_n < 2R} b_n + 4R + 2(\sinh R - R), \\
 c_{\text{odd}} &= \frac{1}{3} \log(9\pi/2) - \frac{1}{3}, \quad h_R = -\log R + c_{\text{odd}}, \\
 d_R &= \max \left\{ 1, \left\lceil \frac{2R}{\pi} e^{M_R+2} \right\rceil \right\}, \\
 \tau_R &= 1 + \max\{0, M_R + M_R^2 - h_R\}, \\
 g_R &= \frac{1}{\max\{2, 1 + 2M_R^2\}}.
 \end{aligned} \tag{CP17:1.10}$$

Fix one ordered orthonormal eigenbasis of $\mathbf{C}_1$, with eigenvalues repeated according to multiplicity, and let P project onto exactly its first d_R vectors. In a terminal repeated eigenspace one fixed choice is made once; it is independent of a and commutes with $\mathbf{C}_1$. This is a fixed orthogonal projection commuting with the reference operator; a Borel spectral projection onto an entire repeated eigenspace is not required. Its rank remains exactly d_R in all subsequent kernel and source bounds. Write $Q = I - P$.

Logarithmic scaling shifts every reference eigenvalue by the same scalar and leaves these eigenspaces unchanged. Therefore Checkpoint 16's complementary estimate and regularization apply on this fixed space:

$$Q\tilde{A}_aQ \geq I, \quad \|P\tilde{A}_aQ\| \leq M_R, \quad \boxed{\mathcal{H}_a := \tilde{A}_a + \tau_R P \geq g_R I.} \tag{CP17:1.11}$$

In particular

$$\|\mathcal{H}_a^{-1}\| \leq g_R^{-1} \tag{CP17:1.12}$$

holds for every $a \in [a_0, R]$, regardless of the sign of $\tilde{A}_a$, and without assuming invertibility of the unregularized polar operator H_a .

For completeness, the finite-dimensional Schur operator

$$F_a = P\tilde{A}_aP - P\tilde{A}_aQ(Q\tilde{A}_aQ)^{-1}Q\tilde{A}_aP \tag{CP17:1.13}$$

satisfies $F_a \geq (h_R - M_R - M_R^2)I_P$. Completing the complementary square proves (1.11), exactly as in Checkpoint 16, with neither an arithmetic truncation nor a polar omission. It also gives

$$K_a := \tau_R P \mathcal{H}_a^{-1} P|_{P\mathcal{H}} = \tau_R (\tau_R I_P + F_a)^{-1} > 0. \tag{CP17:1.14}$$

Hence, with

$$\varepsilon(a) = 1 - \lambda_{\max}(K_a), \tag{CP17:1.15}$$

one has the exact equivalences

$$\begin{aligned}
 \tilde{A}_a > 0 &\iff F_a > 0 \iff \varepsilon(a) > 0, \\
 \tilde{A}_a \geq 0 &\iff F_a \geq 0 \iff \varepsilon(a) \geq 0.
 \end{aligned} \tag{CP17:1.16}$$

Strict positivity here means a positive spectral lower bound; compact resolvent makes it equivalent to positivity of the quadratic form on every nonzero vector. On the positive branch, the congruence

$$\tilde{A}_a = \mathcal{H}_a^{1/2} (I - \tau_R \mathcal{H}_a^{-1/2} P \mathcal{H}_a^{-1/2}) \mathcal{H}_a^{1/2}$$

also yields $\tilde{A}_a \geq \varepsilon(a)g_R I$.

Every null branch is preserved:

$$\ker \tilde{A}_a \neq \{0\} \iff 1 \in \text{spec } K_a. \quad (\text{CP17:1.17})$$

Indeed $v \in \ker \tilde{A}_a$ implies $\mathcal{H}_a v = \tau_R P v$. Its projection $P v$ cannot vanish because the complementary block in (1.11) is positive. Conversely $K_a u = u$, $u \neq 0$, gives the nonzero null vector $\mathcal{H}_a^{-1} u$. At a first crossing $\tilde{A}_a \geq 0$, so (1.17) says precisely $\varepsilon(a) = 0$. Away from that hypothesis, a nonmaximal eigenvalue of K_a may equal one while $\varepsilon(a) < 0$; the first-crossing qualification is essential. No moment against S_a is required, so the zero-moment branch is included.

The established local certificate supplies the explicit initial condition

$$\boxed{\varepsilon(a_0) \geq \frac{m_0}{\tau_R + m_0} > 0.} \quad (\text{CP17:1.18})$$

To check it, minimize the complete quadratic form over the complementary block: $\tilde{A}_{a_0} \geq m_0 I$ implies $F_{a_0} \geq m_0 I_P$, and then use (1.14).

1.3. Exact finite differences and near-null equations

Since (1.8) is bounded and the operator domains agree, the second resolvent identity is an identity of bounded operators:

$$\boxed{K_b - K_a = -\tau_R P \mathcal{H}_b^{-1} (\tilde{A}_b - \tilde{A}_a) \mathcal{H}_a^{-1} P.} \quad (\text{CP17:1.19})$$

The expression contains the entire difference (1.8). It is valid across all prime thresholds. It does not require differentiability in a , nor norm continuity of the translation operators themselves.

Choose a unit eigenvector $u_a \in P\mathcal{H}$ for the largest eigenvalue of K_a and put $v_a = \mathcal{H}_a^{-1} u_a$. Then $v_a \in \text{Dom } C_1$, and the two exact identities are

$$\boxed{\tilde{A}_a v_a = \varepsilon(a) u_a, \quad P v_a = \frac{1 - \varepsilon(a)}{\tau_R} u_a.} \quad (\text{CP17:1.20})$$

The proof is algebraic: $\mathcal{H}_a v_a = u_a$, while the eigenvalue equation says $\tau_R P v_a = (1 - \varepsilon(a)) u_a$. The associated complete energies and norm bounds are

$$\begin{aligned} \langle \tilde{A}_a v_a, v_a \rangle &= \frac{\varepsilon(a)(1 - \varepsilon(a))}{\tau_R}, \\ \langle \mathcal{H}_a v_a, v_a \rangle &= \frac{1 - \varepsilon(a)}{\tau_R}, \\ \frac{1 - \varepsilon(a)}{\tau_R} &\leq \|v_a\|_2 \leq g_R^{-1}. \end{aligned} \quad (\text{CP17:1.21})$$

Thus v_a remains nonzero as $\varepsilon(a) \rightarrow 0$, while the inverse used to define it remains uniformly bounded. These identities remain valid for every vector in a multiple top eigenspace. They do not select a differentiable eigenvector through an eigenvalue collision.

1.4. Uniform graph control and continuity across thresholds

This paragraph establishes regularity used for the variation calculation. Set

$$L_R = \max\{|\log a_0|, |\log R|\}, \quad J_R = 1 + \frac{L_R + M_R + \tau_R}{g_R}, \quad G_R = \left(J_R^2 + \frac{\pi^2}{2g_R^2}\right)^{1/2}. \quad (\text{CP17:1.22})$$

For $\|f\|_2 \leq 1$, let $z_a = \mathcal{H}_a^{-1}f$. Its full interior equation is

$$\mathbf{C}_1 z_a = f + (\log a I + \tilde{\mathbf{D}}_a - \tau_R P) z_a, \quad \|\mathbf{C}_1 z_a\|_2 \leq J_R. \quad (\text{CP17:1.23})$$

Outside $(-1, 1)$, the distributional whole-line logarithmic operator satisfies

$$(\log |D| E z_a)(x) = -\frac{1}{2} \int_{-1}^1 \frac{z_a(y)}{|x-y|} dy. \quad (\text{CP17:1.24})$$

Each exterior half-line is controlled by half the Carleman operator, of norm $\pi/2$, hence

$$\|(\log |D| E z_a)|_{\mathbb{R} \setminus [-1, 1]}\|_2 \leq \frac{\pi}{\sqrt{2}} \|z_a\|_2.$$

There is no hidden boundary distribution. Indeed $\log |D|$ maps a supported L^2 function into $H^{-\eta}(\mathbb{R})$ for every $\eta > 0$; near zero the logarithmic singularity is square integrable against the bounded Fourier transform. Subtracting the interior and exterior L^2 representatives leaves a distribution supported at $\{-1, 1\}$. No nonzero distribution with that support belongs to $H^{-\eta}$ for $0 < \eta < 1/2$. This proves the global identification and the uniform bound

$$\boxed{\|E z_a\|_2 \leq g_R^{-1}, \quad \|\log |D| E z_a\|_2 \leq G_R.} \quad (\text{CP17:1.25})$$

The same argument is valid first on the smooth core and then by the distributional operator equation; no trace of z_a is assumed.

For $0 < h < 1$, splitting the Fourier integral at $T = h^{-1/2}$ gives

$$\|T_h E z_a - E z_a\|_2 \leq \frac{\sqrt{h}}{g_R} + \frac{4G_R}{\log(1/h)}. \quad (\text{CP17:1.26})$$

At low frequency use $|e^{-ih\xi} - 1| \leq h|\xi|$, and above T use $|e^{-ih\xi} - 1| \leq 2$ and the logarithmic graph norm. Thus the prime terms in (1.8), applied to all unit-forced solutions, tend to zero uniformly. The other terms in (1.8) are norm-continuous: α is Lipschitz on each finite interval, and $a \mapsto S_a$ is continuously differentiable in L^2 . Inserting these facts in the resolvent identity proves norm continuity of $\mathcal{H}_a^{-1}$, K_a , and $\varepsilon(a)$ throughout $[a_0, R]$, including all prime thresholds.

The bilinear version, used in Group 3, is stronger. If $z_b = \mathcal{H}_b^{-1}f$, $z_a = \mathcal{H}_a^{-1}g$, $\|f\|, \|g\| \leq 1$, then

$$|\langle z_b, (T_h - T_k) z_a \rangle| \leq \frac{\sqrt{\eta}}{g_R^2} + \frac{8G_R^2}{\log^2(1/\eta)}, \quad \eta = |h - k| \in (0, 1). \quad (\text{CP17:1.27})$$

The high-frequency estimate now uses both logarithmic graph norms. Summing both directions and all finite packets gives a Dini modulus for the complete finite matrix, as calculated in Group 3. This proves continuity sufficient to define a hypothetical first zero of ε . It does not prove absolute continuity or a differential Osgood inequality.

1.5. Closing statement for Group 1

Theorem proved. On every fixed finite range $[a_0, R]$, (1.6)-(1.12) put the complete odd Weil family on one common operator domain and one fixed finite-dimensional reference space, with the independent bound $\mathcal{H}_a \geq g_R I$. The finite-difference identity (1.19), the near-null identities (1.20)-(1.21), the graph estimate (1.25), and continuity through every prime threshold hold unconditionally. Every possible first null, including a zero-moment null, is represented by $\varepsilon = 0$. The initial margin is bounded below by (1.18).

Consequence for Weil positivity. A bound preventing ε from reaching zero on every finite $[a_0, R]$ would exclude both first-null branches in the original odd Weil form. No separate invertibility hypothesis for H_a would then be needed.

Exact remaining estimate. The identities leave the signed scalar or top-eigenspace compression of

$$\tau_R P \mathcal{H}_b^{-1} (\tilde{A}_b - \tilde{A}_a) \mathcal{H}_a^{-1} P \quad (\text{CP17:1.28})$$

to be controlled with a vanishing factor in $\varepsilon(a)$, or an equivalent integrated comparison. The proven graph bounds control its magnitude by a modulus independent of that margin; they do not yet supply the factor required for an Osgood argument. The equality $\langle \tilde{A}_a v_a, v_a \rangle = \varepsilon(a)(1 - \varepsilon(a))/\tau_R$ is retained for that calculation and is not itself a bound on the signed variation.

2. Actual boundary regularity and the complete collar residual

This group uses the closed, zero-extended odd logarithmic form domain retained from Checkpoints 15-16. Fourier transforms are $\hat{q}(\xi) = \int q(x) e^{i\xi x} dx$, and $L_\Delta = 2 \log |D|$. The half-interval map is always unitary: $W(x) = \sqrt{2} w(x)$ for $0 < x < a$. In particular the half-interval source is $\sqrt{2} \sinh(x/2)$.

2.1 A boundedness estimate for the actual forced solutions

The passage to a boundary theorem requires bounded solutions and bounded complete forcing. Both hypotheses can be established independently of the unknown Weil positivity margin.

The semigroup domination below is carried out on full $L^2(I)$ for the parity-preserving selfadjoint extension given by the displayed odd operator formula. Its restriction to the odd invariant subspace is exactly the required operator. The even polar source annihilates odd inputs; the odd polar source is retained explicitly. We make no positivity-preserving assertion about the odd-restricted semigroup.

Let $B = \frac{1}{2} \log(1 - \partial_x^2)$ on the line. The elementary identity

$$\frac{1}{2} \log(1 + \xi^{-2}) = \int_0^1 \frac{t}{\xi^2 + t^2} dt$$

gives the exact distributional decomposition

$$\log |D| = B - J, \quad (Jq)(x) = \int j(x-y) q(y) dy, \quad j(h) = \frac{1 - e^{-|h|}}{2|h|}, \quad j(0) = \frac{1}{2}. \quad (\text{CP17:2.1})$$

Thus $0 \leq j \leq 1/2$. On an interval of radius at most R , its kernel is dominated by $\frac{1}{2} \mathbf{1}_{[-2R, 2R]}$, of mass $2R$.

The semigroup of $\mathbf{B}$ is a positive convolution semigroup. This follows directly from

$$(1 + \xi^2)^{-t/2} = \frac{1}{\Gamma(t/2)} \int_0^\infty u^{t/2-1} e^{-u} e^{-u\xi^2} du. \quad (\text{CP17:2.2})$$

For its exterior-Dirichlet compression $\mathbf{B}_I$, add the nonnegative potential $n\mathbf{1}_{I^c}$ to $\mathbf{B}$. The Trotter product formula proves positivity and domination by the free semigroup for every finite n . Monotone convergence of the closed forms as $n \rightarrow \infty$ gives the exterior-Dirichlet operator on the supported subspace. Strong L^2 convergence and an almost-everywhere subsequence preserve domination, proving

$$|e^{-t\mathbf{B}_I} q| \leq e^{-t\mathbf{B}} |q|. \quad (\text{CP17:2.3})$$

This statement concerns the exterior-Dirichlet compression, not a spectral logarithm of the ordinary Dirichlet Laplacian.

Suppose a real selfadjoint bounded perturbation D has the pointwise majorant

$$|Dq| \leq \mu * |q| \quad \text{on } I, \quad \mu \geq 0, \quad M = \mu(\mathbb{R}) < \infty. \quad (\text{CP17:2.4})$$

The Dyson expansion of $T = \mathbf{B}_I - D$, (2.3), and commutation of whole-line convolution operators imply

$$|e^{-tT} q| \leq e^{-t\mathbf{B}} e^{t\mu*} |q|. \quad (\text{CP17:2.5})$$

Each term of the expansion is dominated separately; its simplex has volume $t^k/k!$. Convergence holds in L^2 , and monotone convergence of the positive majorants justifies the pointwise inequality. Therefore

$$\|e^{-tT}\|_{2 \rightarrow \infty} \leq e^{tM} \left[\frac{\Gamma(t-1/2)}{2\sqrt{\pi}\Gamma(t)} \right]^{1/2} \quad (t > 1/2), \quad \|e^{-tT}\|_{\infty \rightarrow \infty} \leq e^{tM}. \quad (\text{CP17:2.6})$$

At $t = 1$ the square bracket is $1/2$.

For any solution $Tu = f \in L^2(I) \cap L^\infty(I)$ in the operator domain, the semigroup identity

$$u = e^{-T} u + \int_0^1 e^{-tT} f dt$$

now gives

$$\|u\|_\infty \leq \frac{e^M}{\sqrt{2}} \|u\|_2 + E(M) \|f\|_\infty, \quad E(M) = \frac{e^M - 1}{M}, \quad E(0) = 1. \quad (\text{CP17:2.7})$$

No sign assumption on D , no positivity-preserving assertion about the odd Weil operator, and no unknown spectral margin is used in (2.7).

For H_a , the polar-removed operator, all actual perturbations in (2.1)-(2.4) are retained. Writing

$$P_R^{\text{ar}} = \sum_{\log n < 2R} \frac{\Lambda(n)}{\sqrt{n}}, \quad M_H(R) = \log(2\pi) + 2P_R^{\text{ar}} + 6R, \quad (\text{CP17:2.8})$$

is sufficient: the scalar gives an atom at zero, the two translations of every prime power give atoms of total mass $2P_R^{\text{ar}}$, the bounded α -kernel costs at most $4R$, and j costs at most $2R$. These are majorants of the complete operator; they do not discard the reflected or ordinary prime contributions.

The finite-rank term also admits an independent bounded kernel. Let ϕ_j be the $L^2(-1,1)$ -normalized odd eigenfunctions of the supported reference $\log |D|$, with eigenvalues ν_j . Equations (2.5)-(2.6) for the reference operator on $(-1,1)$ use $M = 2$. Hence

$$\|\phi_j\|_\infty \leq \frac{e^{\nu_j+2}}{\sqrt{2}}.$$

On the span of the first j odd Dirichlet sine functions, Jensen's inequality in Fourier space gives $\langle \log |D| q, q \rangle \leq \log(j\pi) \|q\|^2$. The min-max principle therefore yields $\nu_j \leq \log(j\pi)$. Consequently the explicit constant

$$J_d = \sum_{j=1}^d \frac{e^4 \pi^2 j^2}{2} = \frac{e^4 \pi^2 d(d+1)(2d+1)}{12} \quad (\text{CP17:2.9})$$

bounds $\sum_{j \leq d} \|\phi_j\|_\infty^2$. The physical projection has kernel

$$P_a(x, y) = \frac{1}{a} \sum_{j=1}^d \phi_j(x/a) \overline{\phi_j(y/a)}, \quad |P_a(x, y)| \leq J_d/a. \quad (\text{CP17:2.10})$$

It is therefore dominated by a convolution density of mass $4J_d$, independently of a .

For the full finite-rank regularization $H_{\tau,a} = A_a^- + \tau P_a$, a complete uniform majorant is

$$M_{\text{reg}}(R) = M_H(R) + 8R \sinh^2(R/2) + 4\tau J_d. \quad (\text{CP17:2.11})$$

The penultimate term retains the polar kernel $2 \sinh(x/2) \sinh(y/2)$; its sign has not been changed in the operator, only in this norm majorant.

Take $d = d_R, \tau = \tau_R$ from Checkpoint 16 and its independently proved lower bound $H_{\tau,a} \geq g_R I > 0$. For any unit $f \in \text{ran } P_a$,

$$\|f\|_\infty \leq F_R := a_0^{-1/2} \sqrt{J_{d_R}}, \quad \|H_{\tau,a}^{-1} f\|_2 \leq g_R^{-1},$$

and thus

$$\|H_{\tau,a}^{-1} f\|_\infty \leq B_R := \frac{e^{M_{\text{reg}}(R)}}{\sqrt{2} g_R} + E(M_{\text{reg}}(R)) F_R. \quad (\text{CP17:2.12})$$

This estimate is uniform for $a_0 \leq a \leq R$, including the near-null family of Group 1, and contains no reciprocal of $\varepsilon(a)$.

For $w_a = H_a^{-1} s_a$, the same argument applies whenever the inverse exists. If $H_a \geq g_a I > 0$, then, more explicitly,

$$\|w_a\|_\infty \leq \frac{e^{M_H(a)}}{\sqrt{2} g_a} \sqrt{\sinh a - a} + E(M_H(a)) \sinh(a/2). \quad (\text{CP17:2.13})$$

If H_a is invertible but indefinite, replace g_a^{-1} by $\|H_a^{-1}\|$. Positivity of H_a is needed later for its scalar Schur budget, not for this boundedness assertion.

2.2 The complete bounded forcing and the endpoint upper bound

Define the full-line prime translation sum on zero extensions by

$$(\Pi_a u)(x) = \sum_{\ell_n < 2a} b_n [u(x + \ell_n) + u(x - \ell_n)].$$

The actual equations are

$$L_\Delta w_a = 2\{s_a + \log(2\pi)w_a + \Pi_a w_a + K_{\alpha,a} w_a\}, \quad (\text{CP17:2.14})$$

and

$$L_\Delta u = 2\{f + \log(2\pi)u + \Pi_a u + K_{\alpha,a} u + 2s_a \langle s_a, u \rangle - \tau_R P_a u\}. \quad (\text{CP17:2.15})$$

Here every equation holds weakly against compact smooth tests supported inside $(-a, a)$, of both parities; the odd and even sectors commute, so the odd equation supplies the interior equation against both test sectors. No equality with this right-hand side is asserted on the exterior, where the logarithmic operator retains its nonlocal tail. The polar term is absent from (2.14) because H_a is defined by removing it, and is explicitly present in (2.15).

Both right sides belong to L^∞ . For the regularized solutions, put

$$D_R^\infty = \log(2\pi) + 2P_R^{\text{ar}} + 4R + 4R \sinh^2(R/2) + 2\tau_R J_{d_R}.$$

Then the right side of (2.15), denoted F_u , satisfies the independent bound

$$\|F_u\|_\infty \leq 2(F_R + D_R^\infty B_R). \quad (\text{CP17:2.16})$$

Translations have norm at most one on bounded zero extensions even at a threshold. The estimates include both translated arguments; no prime packet is removed.

For a unit eigenvector of Group 1, write its physical source as f , its regularized inverse as v , and its matrix margin as ε . The exact identity $\tau_R P_a v = (1 - \varepsilon)f$ cancels the two finite-rank source terms in (2.15), giving the stronger actual equation

$$L_\Delta v = 2\{\varepsilon f + \log(2\pi)v + \Pi_a v + K_{\alpha,a} v + 2s_a \langle s_a, v \rangle\}. \quad (\text{CP17:2.17})$$

Thus the forcing norm can be bounded by $2\{\varepsilon F_R + [\log(2\pi) + 2P_R^{\text{ar}} + 4R + 4R \sinh^2(R/2)]B_R\}$ on a positive interval, where $0 \leq \varepsilon \leq 1$. This uses the near-null equation before taking absolute values. It eliminates the large finite-rank forcing charge, but the retained prime, archimedean, and polar terms still have a bound of order one as $\varepsilon \downarrow 0$.

Theorem 1.1 of Hernández-Santamaría, López Ríos and Saldaña, *Optimal boundary regularity and a Hopf-type lemma for Dirichlet problems involving the logarithmic Laplacian*, arXiv:2401.18033v2, applies to bounded weak solutions with bounded forcing on domains satisfying an exterior sphere condition. It gives continuity of the zero extension and the upper decay $\ell(d)^{1/2}$, where $\ell(d) = |\log \min(d, 0.1)|^{-1}$. The interval satisfies the geometric hypothesis; (2.12)-(2.16) establish the analytic hypotheses here.

For clarity about uniform constants, let C_{BL} be the norm of the fixed-interval boundary solution embedding

$$\sup_{-1 < x < 1} \frac{|u(x)|}{\sqrt{\ell(1 - |x|)}} \leq C_{\text{BL}}(\|u\|_\infty + \|L_\Delta u\|_{L^\infty(-1,1)}). \quad (\text{CP17:2.18})$$

It is finite. One can read this linear dependence from the barrier proof, or obtain it rigorously by the closed graph theorem: the graph norm on the right makes the weak-solution space Banach, since the closed logarithmic form and its lower bound control the form norm by the graph norm. The boundary theorem places every graph element in the weighted supremum space, and convergence there and in L^∞ identifies the same limit. This argument proves a fixed finite constant without supplying a numerical value. In particular C_{BL} is independent of the spectral margin, a , and the forcing vector.

The scaling identity is

$$L_\Delta(U_a u) = U_a(L_\Delta u) + 2 \log a U_a u. \quad (\text{CP17:2.19})$$

Let $L_R = \max\{|\log a_0|, |\log R|\}$, and choose $d_0 = \min\{a_0/10, a_0^2, 1/10\}$. For $0 < d < d_0$, scaling (2.18) gives the uniform endpoint estimate

$$|u(\pm(a-d))| \leq \frac{B_{\partial,R}}{\sqrt{\log(1/d)}}, \quad B_{\partial,R} = \sqrt{2}C_{\text{BL}}\{(1+2L_R)B_R + 2(F_R + D_R^\infty B_R)\}. \quad (\text{CP17:2.20})$$

The same argument, with (2.13)-(2.14), gives a finite constant $B_{\partial,a}$ for w_a . Thus the requested logarithmic boundary upper estimate is established for the actual solutions; it is not an additional hypothesis in the collar-rate calculation.

The theorem and this derivation do **not** establish convergence of $\sqrt{\log(1/d)}w_a(a-d)$. The normalized boundary trace remains a separate question. The two-sided torsion comparison in the same source states bounds and a positive liminf; it cannot be imported as a trace-limit theorem for a signed Weil solution.

2.3 Complete residual and its singular coefficient

Write $W_a = \sqrt{2}w_a|_{(0,a)}$, and let the collar source be $t(y) = \sqrt{2} \sinh(y/2)$. The complete residual for $y = a + h$, $0 < h < \delta$, is

$$\begin{aligned} r_{a,\delta}(y) &= t(y) + \int_0^a [\kappa(y-x) - \kappa(x+y)]W_a(x) dx \\ &\quad + \sum_{\ell_n < a+\delta} b_n \mathbf{1}_{0 < y-\ell_n < a} W_a(y-\ell_n) \\ &\quad - \sum_{a < \ell_n < 2a+\delta} b_n \mathbf{1}_{0 < \ell_n-y < a} W_a(\ell_n-y). \end{aligned} \quad (\text{CP17:2.21})$$

This is $t - B^*W_a$ with the exact cross block of Checkpoint 16, including its ordinary and reflected signs. Repeated locations carry the complete von Mangoldt coefficient. There is no exterior integral omitted from the archimedean kernel.

Choose $0 < \eta < \min\{a, e^{-2}\}$ so that

$$|W_a(a-d)| \leq B/\sqrt{\log(1/d)} \quad (0 < d < \eta),$$

where $B = \sqrt{2}B_{\partial,a}$ is admissible. Since $\kappa(h+d) = 1/[2(h+d)] + \alpha(h+d)$, the only unbounded part of (2.21) is

$$I_a(h) = \frac{1}{2} \int_0^\eta \frac{W_a(a-d)}{h+d} dd. \quad (\text{CP17:2.22})$$

For $0 < h < \eta$, split the integral at $d = h$. On $(0, h)$, its absolute value is at most $B/(2\sqrt{\log(1/h)})$. On (h, η) , integrate $1/[d\sqrt{\log(1/d)}]$. This gives

$$|I_a(h)| \leq B\sqrt{\log(1/h)} + \frac{B}{2\sqrt{\log(1/h)}} \leq \frac{3}{2}B\sqrt{\log(1/h)}. \quad (\text{CP17:2.23})$$

The last inequality uses $h < e^{-2}$; its constant is deliberately generous.

All other terms are uniformly bounded for $a + \delta \leq R$. One completely specified majorant is

$$D_{a,R} = \sqrt{2} \sinh(R/2) + \sqrt{a} \|W_a\|_2 \left[\frac{1}{2\eta} + 1 + \kappa(a) \right] + 2P_R^{\text{ar}} \|W_a\|_\infty. \quad (\text{CP17:2.24})$$

The three contributions in the middle bracket bound the nonsingular tail of $1/[2(h+d)]$, the full α -term, and the reflected $\kappa(x+y)$ -term. The last summand retains every ordinary and reflected arithmetic packet. Therefore

$$|r_{a,\delta}(a+h)| \leq \frac{3}{2} B \sqrt{\log(1/h)} + D_{a,R}. \quad (\text{CP17:2.25})$$

This bound is valid at a prime threshold as well. For example, if $\ell_n = 2a$, its cross-block reflected value is $-b_n W_a(a-h)$, of size $O(1/\sqrt{\log(1/h)})$. If $\ell_n = a$, the new ordinary argument is h ; continuity and oddness give $W_a(h) \rightarrow 0$. Every other fixed packet samples an interior continuous value or is zero. Uniform boundedness also covers thresholds occurring within the collar, with no assertion that their signs agree.

2.4 Conditional normalized trace and exact factor

Assume additionally that the half-interval trace

$$\beta_a^{\text{half}} := \lim_{d \downarrow 0} \sqrt{\log(1/d)} W_a(a-d) \quad (\text{CP17:2.26})$$

exists. Inserting $W_a(a-d) = (\beta_a^{\text{half}} + o(1))/\sqrt{\log(1/d)}$ into (2.22) yields

$$I_a(h) = \beta_a^{\text{half}} \sqrt{\log(1/h)} + o(\sqrt{\log(1/h)}). \quad (\text{CP17:2.27})$$

Here is the constant calculation. On (h, η) , replacing $(h+d)^{-1}$ by d^{-1} changes the integral by $O(1/\sqrt{\log(1/h)})$. The principal integral is

$$\frac{1}{2} \int_h^\eta \frac{dd}{d\sqrt{\log(1/d)}} = \sqrt{\log(1/h)} - \sqrt{\log(1/\eta)}.$$

The part $(0, h)$ has order $1/\sqrt{\log(1/h)}$; a fixed small η bounds the coefficient error by any prescribed multiple of $\sqrt{\log(1/h)}$, after which $\eta \downarrow 0$. Equations (2.21), (2.24), and (2.27) prove the conditional identity

$$r_{a,\delta}(a+h) = \beta_a^{\text{half}} \sqrt{\log(1/h)} + o(\sqrt{\log(1/h)}). \quad (\text{CP17:2.28})$$

There is no prime, polar, or reflected-archimedean contribution to this leading singular coefficient; those contributions have been retained in (2.21) and bounded in (2.24).

If the symbol β_a instead denotes the trace of the original full odd solution w_a , then

$$\beta_a^{\text{half}} = \sqrt{2} \beta_a^{\text{full}}. \quad (\text{CP17:2.29})$$

Consequently a collar derivative written as $2|\beta_a^{\text{half}}|^2$ equals $4|\beta_a^{\text{full}}|^2$. This normalization is essential when comparing the proposed formula with $\sigma_a = 2\langle s_a, H_a^{-1} s_a \rangle$ on the full odd space.

2.5 Closing statement for Group 2

Theorem proved. Every actual $H_a^{-1}s_a$ for which the inverse exists is bounded and has the endpoint upper bound $O((\log(1/d))^{-1/2})$. Every finite-family regularized inverse of Group 1 satisfies this estimate uniformly on $[a_0, R]$, with (2.9)-(2.12), (2.16), and (2.20) independent of ε . The complete actual collar residual satisfies (2.21) and (2.25).

Constants and domain. All solutions lie in the exterior-Dirichlet logarithmic operator domain and all prime sums are finite for fixed R . The explicit semigroup and arithmetic constants are given above. The fixed-domain boundary constant C_{BL} is proved finite by the cited boundary theorem and the closed graph argument; no numerical value for it is certified in this round. For the scalar inverse, the bound can depend on $\|H_a^{-1}\|$; for the regularized family it depends on the independently certified g_R^{-1} , not on an unknown Weil margin.

Consequence. Group 3 can use the actual boundary upper bound to prove a linear $O(\delta)$ collar budget. The conditional trace computation identifies its sharp proposed coefficient and its full/half-interval conversion.

Exact remaining estimate. Existence of (2.26) has not been proved. A sufficient missing lemma is: for every actual forced solution considered here, the function $d \mapsto \sqrt{\log(1/d)}W_a(a-d)$ is Cauchy as $d \downarrow 0$. Even existence would not supply a vanishing factor in ε . For that further step the needed estimate is, for the regularized near-null family, a uniform bound of the form $|\beta_a^{\text{half}}|^2 \leq C_R \omega(\varepsilon(a))$ together with control of the remaining complete signed variation. The proved upper bound supplies only $\limsup_{d \downarrow 0} \log(1/d)|W_a(a-d)|^2 \leq B^2$; it contains no factor tending to zero with ε .

3. The actual collar cost: its rate and its conditional boundary coefficient

This section uses the unitary odd half-interval convention throughout. Thus the source is $s_a^+(x) = \sqrt{2}\sinh(x/2)$, and $w = H_a^{-1}s_a^+$ is the half-interval solution. If $w_{\text{full}} = H_a^{-1}\sinh(\cdot/2)$ denotes the solution on the full odd space, then $w(x) = \sqrt{2}w_{\text{full}}(x)$ for $x > 0$. This factor changes the proposed derivative constant when a boundary trace is defined before the unitary map.

3.1. Exact scaled Schur operator, with all bounded corrections

Fix $a > 0$, an independently positive lower bound $H_a \geq gI > 0$, and $R > a$. Choose

$$0 < \delta_0 < \min(1, \log 2, R - a), \quad 0 < \delta \leq \delta_0, \quad b = a + \delta, \quad L_\delta = \log(1/\delta). \quad (\text{CP17:3.1})$$

Write $P_R = \sum_{\log n < 2R} \Lambda(n)/\sqrt{n}$. The prime sum is finite but includes every prime power, with its actual von Mangoldt coefficient. In the notation of Checkpoint 16,

$$S_\delta = C_\delta - B_\delta^* H_a^{-1} B_\delta, \quad r_\delta = t_\delta - B_\delta^* w, \quad t_\delta(y) = \sqrt{2}\sinh(y/2). \quad (\text{CP17:3.2})$$

All inverses and Schur complements in this subsection concern H , not the Suzuki metric. The original Suzuki metric is unchanged.

Define the unitary collar rescaling $D_\delta : L^2(a, a + \delta) \rightarrow L^2(0, 1)$ by

$$(D_\delta z)(x) = \sqrt{\delta} z(a + \delta x). \quad (\text{CP17:3.3})$$

The complete operator identity is

$$\widehat{S}_\delta := D_\delta S_\delta D_\delta^{-1} = L_\delta I + T + E_\delta, \quad T = (\log |D|)_{(0,1)} - \log(2\pi)I. \quad (\text{CP17:3.4})$$

Here $(\log |D|)_{(0,1)}$ means compression on zero-extended functions, with the actual logarithmic form domain. Its domain is independent of δ . The equality follows from the exact Fourier scaling of $\log |D|$; a translation of the support has no effect on the whole-line logarithmic multiplier.

For completeness the bounded remainder in (3.4) is exactly

$$\begin{aligned} E_\delta &= -\delta A_\delta + \delta K_{a,\delta} + \sum_{2a < \ell_n < 2a+2\delta} b_n F_{n,a,\delta} \\ &\quad - D_\delta B_\delta^* H_a^{-1} B_\delta D_\delta^{-1}, \\ (A_\delta f)(x) &= \int_0^1 \alpha(\delta(x-y)) f(y) dy, \\ (K_{a,\delta} f)(x) &= \int_0^1 \kappa(2a + \delta(x+y)) f(y) dy, \\ (F_{n,a,\delta} f)(x) &= \mathbf{1}_{\{0 < (\ell_n - 2a)/\delta - x < 1\}} f((\ell_n - 2a)/\delta - x). \end{aligned} \tag{CP17:3.5}$$

There are no ordinary prime translations wholly inside this collar because $\delta < \log 2$. The reflected prime operators in (3.5), each of norm at most one, include both sign channels. Equality at a support endpoint means only measure-zero contact. No reflected prime is dropped on the grounds that its unscaled support has small measure.

The positive reflected archimedean kernel remains in $K_{a,\delta}$; its complete series is

$$\kappa(2a + \delta(x+y)) = \sum_{j \geq 0} e^{-(2j+1/2)(a+\delta x)} e^{-(2j+1/2)(a+\delta y)}. \tag{CP17:3.6}$$

The cross block entering the last term of (3.5) is precisely the sum of the archimedean integral, ordinary negative prime shifts, and reflected positive prime shifts from Checkpoint 16. In particular, if $B = B_0 + \sum_n b_n(-O_n + F_n)$, then

$$\begin{aligned} B^* H_a^{-1} B &= B_0^* H_a^{-1} B_0 + \sum_n b_n \{ B_0^* H_a^{-1} (-O_n + F_n) + (-O_n + F_n)^* H_a^{-1} B_0 \} \\ &\quad + \sum_{m,n} b_m b_n \{ O_m^* H_a^{-1} O_n - O_m^* H_a^{-1} F_n - F_m^* H_a^{-1} O_n + F_m^* H_a^{-1} F_n \}. \end{aligned} \tag{CP17:3.7}$$

Every mixed prime and archimedean term is therefore included in (3.4).

Set

$$c = \frac{\pi}{2} + \sqrt{a\delta_0} + 2P_R, \quad E_* := \delta_0[1 + \kappa(2a)] + P_R + c^2/g, \quad M_* := 1 + \log 2 + \delta_0 + P_R + c^2/g. \tag{CP17:3.8}$$

The complete Carleman cross estimate gives $\|B_\delta\| \leq c$, hence $\|E_\delta\| \leq E_*$. The more favorable lower bound retains the positive reflected archimedean term and gives

$$S_\delta \geq (L_\delta - M_*)I. \tag{CP17:3.9}$$

This follows from $(\log |D|)_{(0,1)} \geq (\log \pi - 1)I$, $|\alpha| \leq 1$, the full reflected prime charge, and $\|B_\delta^* H_a^{-1} B_\delta\| \leq c^2/g$. For $L_\delta > M_*$, S_δ and $H_{a+\delta}$ are consequently positive and invertible, regardless of the sign of the Weil rank-one subtraction.

There is a strong, rather than operator-norm, inverse limit:

$$L_\delta \widehat{S}_\delta^{-1} \longrightarrow I \quad \text{strongly on } L^2(0,1). \tag{CP17:3.10}$$

Indeed for $f \in D(T)$,

$$\|(L_\delta \widehat{S}_\delta^{-1} - I)f\| \leq \frac{\|Tf\| + E_* \|f\|}{L_\delta - M_*}. \quad (\text{CP17:3.11})$$

This is the resolvent identity applied to (3.4). The uniform bound $\|L_\delta \widehat{S}_\delta^{-1}\| \leq L_\delta/(L_\delta - M_*)$, followed by density of $D(T)$, proves (3.10). The unbounded high-frequency part of T is not incorrectly declared small in operator norm.

3.2. Complete residual and a quantitative $O(\delta)$ theorem

At $y = a + h$, $0 < h < \delta$, the full source residual is

$$\begin{aligned} r_\delta(a + h) &= \sqrt{2} \sinh((a + h)/2) + \frac{1}{2} \int_0^a \frac{w(a - d)}{h + d} dd \\ &\quad + \int_0^a [\alpha(a + h - x) - \kappa(a + h + x)] w(x) dx \\ &\quad + \sum_{\ell_n < a + \delta} b_n \mathbf{1}_{\{0 < a + h - \ell_n < a\}} w(a + h - \ell_n) \\ &\quad - \sum_{a < \ell_n < 2a + \delta} b_n \mathbf{1}_{\{0 < \ell_n - a - h < a\}} w(\ell_n - a - h). \end{aligned} \quad (\text{CP17:3.12})$$

This equality includes the polar source, the singular and regular archimedean pieces, and all ordinary and reflected prime packets. It has no unspecified boundary contribution.

The following is the precise regularity hypothesis used for a linear collar rate:

$$w \in L^\infty(0, a), \quad |w(a - d)| \leq \frac{B_a}{\sqrt{\log(1/d)}} \quad (0 < d < \eta), \quad 0 < \eta < \min(a, e^{-2}). \quad (\text{CP17:3.13})$$

These hypotheses are supplied by a bounded-forcing logarithmic boundary theorem only after boundedness of the actual solution and of every translated and finite-rank forcing term has been established. An L^2 resolvent estimate alone does not imply (3.13).

Put $W = \|w\|_2$, $W_\infty = \|w\|_\infty$, and

$$D_a = \sqrt{2} \sinh(R/2) + \sqrt{a} W \left[\frac{1}{2\eta} + 1 + \kappa(a) \right] + 2P_R W_\infty, \quad A_a = \frac{3}{2} B_a. \quad (\text{CP17:3.14})$$

For $h < \min(\eta, e^{-1})$, splitting the singular integral at $d = h$ and $d = \eta$ gives

$$\begin{aligned} \frac{1}{2} \int_0^\eta \frac{B_a}{(h + d) \sqrt{\log(1/d)}} dd &\leq \frac{B_a}{2\sqrt{\log(1/h)}} + B_a [\sqrt{\log(1/h)} - \sqrt{\log(1/\eta)}] \\ &\leq \frac{3}{2} B_a \sqrt{\log(1/h)}. \end{aligned} \quad (\text{CP17:3.15})$$

The portion $d \geq \eta$ costs at most $\sqrt{a}W/(2\eta)$. The remaining archimedean integral is bounded by $[1 + \kappa(a)]\sqrt{a}W$. The two finite prime sums together cost at most $2P_R W_\infty$, even if a new prime threshold lies in the collar. Thus

$$|r_\delta(a + h)| \leq A_a \sqrt{\log(1/h)} + D_a. \quad (\text{CP17:3.16})$$

The elementary integrals and Cauchy-Schwarz give the sharper integrated statement

$$\boxed{\|r_\delta\|_2^2 \leq \delta [A_a \sqrt{L_\delta + 1} + D_a]^2.} \quad (\text{CP17:3.17})$$

In particular, if $L_\delta \geq \max(1, 2M_*)$ and $\delta < \eta$,

$$0 \leq \sigma_{a+\delta} - \sigma_a = 2\langle r_\delta, S_\delta^{-1} r_\delta \rangle \leq 4\delta(\sqrt{2}A_a + D_a)^2. \quad (\text{CP17:3.18})$$

This is an actual forced-solution estimate, not the norm of the source plus the operator norm of $B^*H_a^{-1}$. It improves the former $O(1/\log(1/\delta))$ budget to $O(\delta)$ under (3.13). Its constants are displayed and do not depend on δ . They do depend on the certified gap g , the actual solution bounds, and the finite support a .

All cross terms are retained by first using the complete residual (3.12). If its summands are r_ν , the exact cost before this upper estimate is

$$2\langle r_\delta, S_\delta^{-1} r_\delta \rangle = 2 \sum_{\nu} \langle r_\nu, S_\delta^{-1} r_\nu \rangle + 4\Re \sum_{\nu < \eta} \langle r_\nu, S_\delta^{-1} r_\eta \rangle. \quad (\text{CP17:3.19})$$

The triangle inequality in (3.16) bounds this complete expression. It neither asserts that the individual prime channels are positive nor omits their interference.

There is also an unconditional fallback that requires only $H_a \geq gI > 0$ and $w \in L^2$. The singular cross transform, applied to a fixed w , belongs to $L^2(0, \delta_0)$ by the Carleman bound. Each translated prime evaluation likewise belongs to L^2 ; only finitely many packets occur. Therefore

$$\|r_\delta\|_{L^2(a, a+\delta)}^2 = o(1), \quad \sigma_{a+\delta} - \sigma_a = o(1/L_\delta). \quad (\text{CP17:3.20})$$

For this argument all blocks can be viewed as restrictions of the corresponding exterior expressions on $(a, a+\delta_0)$, with zero extension for the prime shifts. Absolute continuity of their L^2 integrals gives the little-oh. This fallback is strictly weaker than (3.18); boundary regularity is the substantive additional input.

3.3. A calculated boundary coefficient without assuming a trace limit

The upper boundary estimate alone gives more than the rate (3.18). Retain the signed boundary values and define, for $\delta = e^{-L} < \eta^2$,

$$\bar{\beta}_a(L) = \frac{1}{2\sqrt{L}} \int_{e^{-L}}^{\eta} \frac{w(a-d)}{d} dd. \quad (\text{CP17:3.21})$$

This is an averaged coefficient of the actual forced solution; no absolute value is taken inside its defining integral. The boundary upper bound gives

$$|\bar{\beta}_a(L)| \leq B_a \left(1 - \sqrt{\frac{\log(1/\eta)}{L}} \right) \leq B_a. \quad (\text{CP17:3.22})$$

There is no assertion that $\bar{\beta}_a(L)$ converges.

For $q_\delta(x) = r_\delta(a + \delta x)/\sqrt{L}$, the following quantitative constant-profile approximation holds:

$$\|q_\delta - \bar{\beta}_a(L)\mathbf{1}\|_2 \leq \frac{B_a(\sqrt{2} + \frac{1}{2}\log 2)}{L} + \frac{B_a\sqrt{\delta}}{2\sqrt{L\log(1/\eta)}} + \frac{D_a}{\sqrt{L}} =: e_\delta. \quad (\text{CP17:3.23})$$

Here is the complete error calculation. The part $0 < d < \delta$ of the singular cross kernel, divided by $\sqrt{L}$, has absolute value at most

$$\frac{B_a}{2L} \log(1 + 1/x), \quad \|\log(1 + 1/x)\|_{L^2(0,1)} \leq \log 2 + \sqrt{2}. \quad (\text{CP17:3.24})$$

On $\delta < d < \sqrt{\delta}$, replacing $1/(\delta x + d)$ by $1/d$ costs at most $B_a/(\sqrt{2}L)$, since

$$|1/(\delta x + d) - 1/d| \leq \delta/d^2$$

and $|w(a - d)| \leq \sqrt{2}B_a/\sqrt{L}$. On $\sqrt{\delta} < d < \eta$, the same replacement costs at most $B_a\sqrt{\delta}/[2\sqrt{L}\log(1/\eta)]$. The entire part $d \geq \eta$, all regular archimedean terms, both prime sums, and the polar source cost at most $D_a/\sqrt{L}$ by (3.14). Adding these estimates proves (3.23).

Combining the strong inverse calculation with the uniform bound (3.22) proves an actual asymptotic formula without any trace-existence assumption:

$$\boxed{\frac{\sigma_{a+\delta} - \sigma_a}{\delta} = 2|\bar{\beta}_a(\log(1/\delta))|^2 + o(1).} \quad (\text{CP17:3.25})$$

More precisely, with C_T as explicitly evaluated in (3.37),

$$\left| \frac{\sigma_{a+\delta} - \sigma_a}{\delta} - 2|\bar{\beta}_a(L)|^2 \right| \leq \frac{2L}{L - M_*} (2B_a e_\delta + e_\delta^2) + \frac{2B_a^2(C_T + E_*)}{L - M_*}. \quad (\text{CP17:3.26})$$

Thus, with fixed a and its displayed bounds, the remainder is $O(L^{-1/2})$. This is a statement about the actual complete collar cost. It does not infer pointwise boundary limits or differentiability from boundedness. It identifies the surviving signed boundary average which would have to acquire a positivity-margin factor in a global argument.

The same calculation preserves every mixed boundary term for a fixed finite family of real forced solutions $w_1, \dots, w_d$, provided they use the same positive inner operator and Schur complement, each satisfies the corresponding bounded endpoint estimate, and every collar source extension t_j is uniformly bounded. The latter hypothesis ensures an actual finite constant D_j for the complete nonsingular residual. Let r_j be their complete residuals, let $b(L) = (\bar{\beta}_1(L), \dots, \bar{\beta}_d(L))^T$, and put $e = (\sum_j e_{j,\delta}^2)^{1/2}$. Polarizing the calculation gives

$$\boxed{\left[\frac{2\langle r_i, S_\delta^{-1} r_j \rangle}{\delta} \right]_{i,j=1}^d = 2b(L)b(L)^T + O(L^{-1/2})} \quad (\text{CP17:3.27})$$

in finite-dimensional operator norm. More explicitly, the norm of the difference is at most

$$\frac{2L}{L - M_*} (2\|b(L)\|e + e^2) + \frac{2\|b(L)\|^2(C_T + E_*)}{L - M_*}. \quad (\text{CP17:3.28})$$

For complex solutions the corresponding outer product uses conjugation in the usual Gram convention. Thus the leading collar interaction on this finite family is a rank-one boundary Gram matrix with every off-diagonal pairing retained. This is the collar part of a fixed-operator comparison; identifying it with the full regularized fixed-space variation additionally requires the changing source, regularization, and scaling terms calculated separately. It does not discard those additional terms.

3.4. Local absolute continuity and an averaged trace almost everywhere

Let $J = [a_-, a_+] \subset (0, \infty)$ be a compact interval throughout which $H_a > 0$. Under the fixed-space unitary scaling, the same norm-resolvent continuity argument used for the regularized operators gives norm-resolvent continuity for H_a itself. Consequently

$$g_J := \inf_{a \in J} \inf \sigma(H_a) > 0. \quad (\text{CP17:3.29})$$

This statement concerns a compact interval already contained in the positive- H region; it does not give a gap at an unknown endpoint of that region.

Group 2's boundedness and boundary estimates are uniform for the smooth source on such a compact interval: the inverse has norm at most $1/g_J$, the complete forcing has only the finitely many packets below a fixed $2R$, and the fixed-interval logarithmic boundary estimate applies after scaling with a bounded above and bounded away from zero. Choose one $\eta < \min(a_-, e^{-2})$, one $R > a_+$, and common upper bounds for B_a, D_a, M_* . Formula (3.18) then bounds every sufficiently short admissible increment by the same constant times its length. Subdividing any longer increment inside J proves that σ is Lipschitz on J . Hence σ is locally absolutely continuous on each open interval where $H_a > 0$.

At almost every a in this region, the derivative $\sigma'(a)$ therefore exists. The actual averaged-coefficient formula (3.25) gives

$$|\bar{\beta}_a(L)|^2 \longrightarrow \frac{1}{2}\sigma'(a) \quad (L \rightarrow \infty). \quad (\text{CP17:3.30})$$

For the real forced solution used here, $\bar{\beta}_a(L)$ is a real continuous function of L , because its defining integral has a moving endpoint in an integrable function on every finite L -interval. If the limit in (3.30) is positive, continuity forces an eventual fixed sign; if it is zero, convergence to zero is immediate. Thus the real averaged boundary trace $\lim_{L \rightarrow \infty} \bar{\beta}_a(L)$ exists for almost every such a .

This almost-everywhere result concerns the averaged trace. It does not establish the pointwise normalized boundary limit (3.31). Its constants depend on the independently positive gap on J ; it cannot be extended across an unknown H -kernel by asserting that this gap persists. Nor does it establish absolute continuity of the full regularized matrix K_a , whose source-scaling and regularization variation remain separate terms.

3.5. Conditional boundary trace and the corrected coefficient

Assume (3.13) and, separately, the existence of the half-interval normalized trace

$$\beta_a^+ = \lim_{d \downarrow 0} \sqrt{\log(1/d)} w(a-d). \quad (\text{CP17:3.31})$$

An upper bound of the form (3.13) does not establish this limit. With (3.31), however, the exact leading exterior coefficient follows:

$$r_\delta(a+h) = \beta_a^+ \sqrt{\log(1/h)} + o(\sqrt{\log(1/h)}) \quad (h \downarrow 0). \quad (\text{CP17:3.32})$$

The coefficient is one. Indeed the primitive $\int d^{-1}(\log(1/d))^{-1/2} dd = -2\sqrt{\log(1/d)}$ cancels the factor $1/2$ in the cross kernel. The region $d < h$ is $O(1/\sqrt{\log(1/h)})$, the region $d \geq \eta$ is bounded, and replacing $1/(h+d)$ by $1/d$ on $d \geq h$ creates another $O(1/\sqrt{\log(1/h)})$ error plus a bounded tail. To see the last assertion, split at $\sqrt{h}$: on $h \leq d \leq \sqrt{h}$, the bound on $(\log(1/d))^{-1/2}$ is $\sqrt{2}/\sqrt{\log(1/h)}$ and $\int_h^\infty h/d^2 dd = 1$; the remaining interval contributes $O(\sqrt{h})$ times a fixed or logarithmically bounded factor.

For the trace error, set

$$\rho(\eta) = \text{ess sup}_{0 < d < \eta} |\sqrt{\log(1/d)} w(a-d) - \beta_a^+|. \quad (\text{CP17:3.33})$$

Then $\rho(\eta) \rightarrow 0$, and its exterior transform is bounded by $\rho(\eta)[\sqrt{\log(1/h)} + 1/(2\sqrt{\log(1/h)})]$ plus an η -dependent bounded term. This supplies the quantified remainder needed below; merely asserting a pointwise asymptotic would not suffice.

Define the rescaled, normalized residual profile

$$q_\delta(x) = \frac{r_\delta(a + \delta x)}{\sqrt{L_\delta}}, \quad 0 < x < 1. \quad (\text{CP17:3.34})$$

The bound (3.16) dominates $|q_\delta(x)|$ by a constant multiple of $\sqrt{1 + \log(1/x)} + 1$, whose square is integrable. Equation (3.32) then gives

$$q_\delta \longrightarrow \beta_a^+ \mathbf{1} \quad \text{strongly in } L^2(0, 1). \quad (\text{CP17:3.35})$$

Every finite prime term in (3.12) is bounded by (3.13)'s global L^∞ bound and vanishes in this normalization. At the exact threshold $\ell_n = 2a$, its reflected value is $w(a - h)$, which is even $O(1/\sqrt{\log(1/h)})$. At $\ell_n = a$, the applicable ordinary value is $w(h)$, controlled by the same global bound. Thus the coefficient calculation does not suppress any threshold term.

Combining (3.10) and (3.35) now proves the right-derivative formula

$$\boxed{\lim_{\delta \downarrow 0} \frac{\sigma_{a+\delta} - \sigma_a}{\delta} = 2|\beta_a^+|^2.} \quad (\text{CP17:3.36})$$

Indeed the expression on the left is $2\langle q_\delta, L_\delta \widehat{S}_\delta^{-1} q_\delta \rangle$. No differentiation of the unbounded operator or of a translated rough test was used.

A fully explicit resolvent remainder is available for this calculation. The constant function lies in $D(T)$, with

$$(T\mathbf{1})(x) = -\gamma_E - \log(2\pi) - \frac{1}{2} \log(x(1-x)), \quad \|T\mathbf{1}\|_2 \leq \gamma_E + \log(2\pi) + \sqrt{2 - \pi^2/12} =: C_T. \quad (\text{CP17:3.37})$$

Writing $\eta_\delta = \|q_\delta - \beta_a^+ \mathbf{1}\|_2$, one obtains

$$\begin{aligned} \left| \frac{\sigma_{a+\delta} - \sigma_a}{\delta} - 2|\beta_a^+|^2 \right| &\leq \frac{2L_\delta}{L_\delta - M_*} (2|\beta_a^+|\eta_\delta + \eta_\delta^2) \\ &\quad + \frac{2|\beta_a^+|^2(C_T + E_*)}{L_\delta - M_*}. \end{aligned} \quad (\text{CP17:3.38})$$

Here $\eta_\delta \rightarrow 0$ follows from the actual trace modulus (3.33), rather than an assumed uniform pointwise Taylor expansion.

If the user's β_a is defined for the original full odd solution w_{full} , then $\beta_a^+ = \sqrt{2}\beta_a$. In that convention the corrected candidate is

$$\boxed{\sigma'_+(a) = 4|\beta_a|^2 \quad (\text{full odd boundary trace}), \quad \sigma'_+(a) = 2|\beta_a^+|^2 \quad (\text{unitary half-interval boundary trace}).} \quad (\text{CP17:3.39})$$

This distinction is part of the calculation, not a change in the Suzuki normalization.

3.6. Scope of the rate and the remaining signed estimate

Proved collar theorem. For every fixed $a > 0$ with an independently known $H_a \geq gI > 0$, the scaled Schur identity (3.4), the uniform complete remainder bound (3.8), positivity for sufficiently small collars (3.9), and the strong inverse limit (3.10) are unconditional. Group 2 supplies boundedness and the logarithmic endpoint estimate (3.13) for the actual forced solution. Therefore its complete collar cost satisfies the explicit $O(\delta)$ estimate (3.18) and the calculated

averaged-coefficient formula (3.25), with the complete remainder (3.26). Only the derivative formula (3.36) requires the separate normalized trace assumption (3.31).

Thresholds and mixed terms. The estimates include every collar reflection and every cross-block ordinary/reflected prime shift for $a + \delta \leq R$. They allow threshold entry without an operator-norm continuity claim for a truncated translation. The uniform remainder, rather than a discarded packet, is used in the strong inverse limit. All pole contributions enter through the complete source and its pairings, and the archimedean endpoint weights are in the reference compressed logarithmic operator.

Consequence for global Weil positivity. A linear absolute cost is stronger than the former logarithmic cost, but it supplies no factor $1 - \sigma_a$. On each positive- H interval, define the almost-everywhere existing averaged trace $\tilde{\beta}_a = \lim_{L \rightarrow \infty} \bar{\beta}_a(L)$. The proved local absolute continuity then gives the actual almost-everywhere identity

$$\mu'(a) = -2|\tilde{\beta}_a|^2, \quad \mu_a = 1 - \sigma_a. \quad (\text{CP17:3.40})$$

When the pointwise normalized trace exists, it agrees with this average. For the scalar route an Osgood comparison would therefore require the independent almost-everywhere estimate

$$2|\tilde{\beta}_a|^2 \leq c_R(a) \omega(\mu_a), \quad c_R \in L^1, \quad \int_{0^+} \frac{dt}{\omega(t)} = \infty, \quad (\text{CP17:3.41})$$

or its integrated finite-difference analogue. The available boundary upper bound controls the left side by an absolute solution-dependent constant; it does not make that constant vanish as $\mu_a \downarrow 0$. The regularized matrix route must establish its corresponding signed interaction estimate without introducing the reciprocal of an unknown margin. The fixed- a collar calculation itself neither excludes the H_a -kernel branch nor supplies that vanishing factor.

Exact proof boundary. Equation (3.18) is the proved rate for the actual collar, using Group 2's complete bounded-forcing argument. Without a normalized pointwise trace, (3.25) supplies the coefficient at every positive- H parameter in terms of the bounded signed average $\bar{\beta}_a(L)$. Its limit is proved almost everywhere in that region by Section 3.4, but not at every parameter. The missing relative estimate is already visible in this stronger formula: the almost-everywhere limit must satisfy (3.41), or the finite-increment coefficient and remainder must satisfy the corresponding integrated estimate, with constants independent of the unknown margin. The proved bound is $2|\bar{\beta}_a(L)|^2 \leq 2B_a^2$; it supplies no vanishing factor as $\mu_a \downarrow 0$.

This subsection concerns the complete Weil operator and its fixed positive finite-rank regularization. It does not replace the Suzuki metric. All spaces and projections are those of Group 1. Inner products are linear in the first argument; real sources can be used throughout, and the formulas below also retain the complex mixed pairings.

3.7. Exact scaled operator and every prime channel

Write $a_0 = 699/2000 + e^{-2500}$, fix $R > a_0$, and let

$$\mathbf{C} = \mathbf{1}_{(-1,1)} \log |D| E, \quad p_a(x) = \sqrt{a} \sinh(ax/2), \quad (\mathsf{T}_a f)(x) = a \int_{-1}^1 \alpha(a(x-y)) f(y) dy. \quad (\text{CP17:3.42})$$

Here E is zero extension, and C is restricted to the odd space. Put $S_h f = \mathbf{1}_{(-1,1)}[(Ef)(\cdot + h) + (Ef)(\cdot - h)]$. The complete fixed-space identity is

$$\tilde{A}_a = C - \log a I - \log(2\pi)I - \sum_{\ell_n < 2R} b_n S_{\ell_n/a} - T_a - 2|p_a\rangle\langle p_a|. \quad (\text{CP17:3.43})$$

Terms with $\ell_n \geq 2a$ vanish automatically. The possible equality $\ell_n = 2a$ has zero overlap in L^2 and requires no separate atom. In particular, using one fixed finite prime sum is exact even when a crosses a prime threshold. The logarithmic operator is fixed; only its scalar shift $-\log a$ varies. The even continuous kernel α , its cancellation on odd inputs, and the full polar rank one are unchanged.

Under the unitary odd-to-half-interval map, the source in (3.42) is

$$\hat{p}_a(x) = \sqrt{2a} \sinh(ax/2), \quad 0 < x < 1, \quad (\text{CP17:3.44})$$

and the archimedean kernel is

$$a[\alpha(a|x-y|) - \alpha(a(x+y))]. \quad (\text{CP17:3.45})$$

For half-interval vectors f, g , define the complete packet

$$\begin{aligned} \mathcal{P}_h(f, g) &= \mathbf{1}_{0 < h < 1} \int_0^{1-h} \{f(x+h)\overline{g(x)} + f(x)\overline{g(x+h)}\} dx \\ &\quad - \mathbf{1}_{0 < h < 2} \int_{\max(0, h-1)}^{\min(1, h)} f(h-x)\overline{g(x)} dx. \end{aligned} \quad (\text{CP17:3.46})$$

This is exactly $\langle S_h f, g \rangle$ after the unitary identification. The first line contains both ordinary translations. The second line contains the reflected translation, with its correct minus sign in S_h , hence its positive sign in $\tilde{A}_a$. No reflected channel is discarded. The boundaries in (3.46) display all three regimes $h > 2$, $1 < h < 2$, and $0 < h < 1$, including the interfaces $2a = \ell_n$ and $a = \ell_n$.

3.8. Exact matrix increment on the actual forced solutions

Let $\tau = \tau_R$, $\mathcal{H}_a = \tilde{A}_a + \tau P$, and $v_{a,j} = \mathcal{H}_a^{-1} f_j$, where $f_1, \dots, f_d$ is any fixed orthonormal basis of PL^2 . The basis is independent of a . Write $K_{a,ij} = \tau \langle \mathcal{H}_a^{-1} f_j, f_i \rangle$. The resolvent identity gives the exact complete increment

$$\begin{aligned} K_{b,ij} - K_{a,ij} &= \tau \left\{ \log(b/a) \langle v_{a,j}, v_{b,i} \rangle \right. \\ &\quad + \sum_{\ell_n < 2R} b_n \langle (S_{\ell_n/b} - S_{\ell_n/a}) v_{a,j}, v_{b,i} \rangle \\ &\quad + \langle (T_b - T_a) v_{a,j}, v_{b,i} \rangle \\ &\quad \left. + 2[\langle v_{a,j}, p_b \rangle \langle p_b, v_{b,i} \rangle - \langle v_{a,j}, p_a \rangle \langle p_a, v_{b,i} \rangle] \right\}. \end{aligned} \quad (\text{CP17:3.47})$$

Each prime pairing is precisely the difference of (3.46) at $h = \ell_n/b$ and $h = \ell_n/a$. The alpha pairing is the double integral of $b\alpha(b(x-y)) - a\alpha(a(x-y))$, or equivalently the difference of (3.45). Formula (3.47) retains the different forced solutions at the two endpoints; replacing both by one solution would omit mixed terms. Its adjoint is the alternative resolvent identity with a, b interchanged inside the inverse product, so the complete matrix increment is Hermitian.

For example, with $d_a = p_b - p_a$, the polar increment in (3.47) is exactly

$$2[\langle v_{a,j}, p_a \rangle \langle d_a, v_{b,i} \rangle + \langle v_{a,j}, d_a \rangle \langle p_a, v_{b,i} \rangle + \langle v_{a,j}, d_a \rangle \langle d_a, v_{b,i} \rangle]. \quad (\text{CP17:3.48})$$

Both linear mixed terms and the quadratic source change remain. There is no additional derivative of P : scaling the logarithmic reference has made P fixed. The dependence of $v_{a,j}$ is already included exactly by the two resolvents.

3.9. Uniform logarithmic graph bound

Use Checkpoint 16's independently proved constants

$$g_R = \frac{1}{\max\{2, 1 + 2M_R^2\}}, \quad \mathcal{H}_a \geq g_R I \quad (a_0 \leq a \leq R). \quad (\text{CP17:3.49})$$

Set

$$\Lambda_R = \max\{|\log a_0|, |\log R|\}, \quad G_R^2 = \left[1 + \frac{\Lambda_R + M_R + \tau_R}{g_R}\right]^2 + \frac{\pi^2}{2g_R^2}. \quad (\text{CP17:3.50})$$

Then, for every $f \in L_{\text{odd}}^2(-1, 1)$ and $v = \mathcal{H}_a^{-1}f$,

$$\|v\|_2 \leq g_R^{-1}\|f\|_2, \quad \|\log |D|Ev\|_{L^2(\mathbb{R})} \leq G_R\|f\|_2. \quad (\text{CP17:3.51})$$

Indeed the interior equation is $\mathbb{C}v = f + (\log a I + \tilde{D}_a - \tau P)v$, where $\|\tilde{D}_a\| \leq M_R$. Outside the interval,

$$(\log |D|Ev)(x) = -\frac{1}{2} \int_{-1}^1 \frac{v(y)}{|x-y|} dy. \quad (\text{CP17:3.52})$$

The two exterior pieces have squared total norm at most $\pi^2\|v\|_2^2/2$, by the two half-Carleman bounds. This yields (3.50)-(3.51). The distributional identity and approximation are justified by the graph closure and cutoff argument of Checkpoint 16; no endpoint trace of v is used. Thus the bound applies uniformly to the actual forced solutions, including those at a first null of the unregularized Weil operator. Its constants contain no inverse of the unknown positivity margin.

3.10. A proved bilinear translation estimate

Suppose $F, G \in L^2(\mathbb{R})$ and $\log |D|F, \log |D|G \in L^2$. If $0 < \eta = |h-k| < e^{-2}$, splitting Fourier space at $T = \eta^{-1/2}$ proves

$$\begin{aligned} |[(T_h + T_{-h}) - (T_k + T_{-k})]F, G| &\leq 2\sqrt{\eta}\|F\|_2\|G\|_2 \\ &\quad + \frac{16}{\log^2(1/\eta)}\|\log |D|F\|_2\|\log |D|G\|_2. \end{aligned} \quad (\text{CP17:3.53})$$

Here $T_h F(x) = F(x+h)$. Its Fourier symbol is $2(\cos(h\xi) - \cos(k\xi))$, bounded by $\min\{4, 2\eta|\xi|\}$. Below T , Cauchy-Schwarz gives the first term. Above T , each Fourier factor acquires a factor $1/\log T$ from its logarithmic graph norm; Cauchy-Schwarz gives $4/(\log T)^2 = 16/\log^2(1/\eta)$. Plancherel's 2π factors cancel. This proves the estimate for the zero-extended functions and hence for the compressed pairings in (3.47).

The use of both actual inverse solutions improves the directly estimated action modulus $O(1/\log(1/\eta))$ to the bilinear modulus $O(1/\log^2(1/\eta))$. It does not assert a derivative of a translation on the logarithmic graph domain.

3.11. Complete Dini modulus, including every threshold

Define finite, explicitly determined constants

$$\begin{aligned}
 A_0(R) &= \sup_{0 \leq t \leq 2R} |\alpha(t)|, & A_1(R) &= \sup_{0 < t \leq 2R} |\alpha'(t)|, \\
 D_R^{\text{src}} &= \sqrt{2} \left[\frac{\sinh(R/2)}{2\sqrt{a_0}} + \frac{\sqrt{R}}{2} \cosh(R/2) \right], \\
 L_R^{\text{sm}} &= \frac{1}{a_0} + 2[A_0(R) + 2RA_1(R)] + 4\sqrt{\sinh R - R} D_R^{\text{src}}, \\
 B_R^{\text{pr}} &= \sum_{\ell_n < 2R} b_n, & c_R^{\text{sh}} &= 2R/a_0^2.
 \end{aligned} \tag{CP17:3.54}$$

One may use the fully explicit universal choices

$$A_0(R) \leq 1, \quad A_1(R) \leq \frac{9}{4}, \quad L_R^{\text{sm}} \leq \frac{1}{a_0} + 2 + 9R + 4\sqrt{\sinh R - R} D_R^{\text{src}}. \tag{CP17:3.55}$$

For the derivative bound, Checkpoint 16 proved $|\alpha'(t)| \leq 11/80$ for $0 < t \leq 7/10$. For $t \geq 7/10$, use

$$\alpha'(t) = -\kappa(t) \left(\frac{1}{2} + \frac{2}{e^{2t} - 1} \right) + \frac{1}{2t^2}.$$

Here $\kappa(t) \leq \kappa(7/10) < 1$, $e^{7/5} > 4$, and therefore $|\alpha'(t)| < 7/6 + 50/49 = 643/294 < 9/4$. For example, the exponential inequalities follow from their positive Taylor sums: $e^{7/20} > 7/5$, $e^{7/5} > 4$, whence $e^{-7/20} + e^{-7/5} < 5/7 + 1/4 < 1$, proving $\kappa(7/10) < 1$. Thus these constants are independent of the unknown margin and do not require an optimization of the elementary kernel. The alpha kernel's derivative in scale is bounded by $A_0 + 2RA_1$; Schur's bound gives the factor two. Also $\|p_a\|_2^2 = \sinh a - a$, $\|\partial_a p_a\|_2 \leq D_R^{\text{src}}$. Thus the nonprime part of (3.43) is Lipschitz with constant L_R^{sm} , including the full polar increment.

For $d = |b - a|$, set $h = c_R^{\text{sh}} d$. Whenever $0 < h < e^{-2}$, (3.47), (3.51), and (3.53) give the uniform operator-norm estimate

$$\|K_b - K_a\| \leq \tau_R \left[\frac{L_R^{\text{sm}} d}{g_R^2} + B_R^{\text{pr}} \left(\frac{2\sqrt{h}}{g_R^2} + \frac{16G_R^2}{\log^2(1/h)} \right) \right] =: \mathbf{m}_R(d). \tag{CP17:3.56}$$

To obtain the norm bound, take arbitrary unit vectors $f, g \in PL^2$ and use (3.47) on their actual inverse images. The shift changes satisfy $|\ell_n/a - \ell_n/b| \leq c_R^{\text{sh}} d$. Each term in (3.53) increases with its small argument, so the common bound h is legitimate.

The modulus is Dini:

$$\int_0^d \frac{\mathbf{m}_R(t)}{t} dt \leq \tau_R \left[\frac{L_R^{\text{sm}} d}{g_R^2} + B_R^{\text{pr}} \left(\frac{4\sqrt{c_R^{\text{sh}} d}}{g_R^2} + \frac{16G_R^2}{\log(1/(c_R^{\text{sh}} d))} \right) \right]. \tag{CP17:3.57}$$

All constants are uniform on $[a_0, R]$. In particular (3.56) controls the complete variation across $2a = \log n$, $a = \log n$, and every moving support interface in (3.46), without removing a neighborhood of a threshold and without using an unproved endpoint derivative. It also covers coincident threshold values involving different prime powers. Longer parameter increments can be partitioned into finitely many increments satisfying the displayed smallness condition.

This is a norm-continuity and Dini-continuity theorem. The displayed modulus does not prove absolute continuity or bounded variation. In particular, summing its unsigned bound over finer partitions does not give a finite total-variation bound. Such a conclusion would require an additional signed or regularity estimate.

3.12. More precise threshold and interface costs from the boundary envelope

Use Group 2's independently proved boundedness and boundary envelope for the actual regularized inverse solutions. It is additional input to the logarithmic graph estimate, not a consequence of (3.51) alone. If these half-interval solutions are denoted f, g , write their bounds, uniformly for $0 < d < d_*$, as

$$|f(1-d)| \leq C_f / \sqrt{\log(1/d)}, \quad |g(1-d)| \leq C_g / \sqrt{\log(1/d)}. \quad (\text{CP17:3.58})$$

When a prime first enters, $h = 2 - \eta$, $0 < \eta < \min\{d_*, 1\}$. Its ordinary term is absent and (3.46) gives

$$\mathcal{P}_{2-\eta}(f, g) = - \int_0^\eta f(1-(\eta-d)) \overline{g(1-d)} dd,$$

so

$$|\mathcal{P}_{2-\eta}(f, g)| \leq \frac{C_f C_g \eta}{\log(1/\eta)}. \quad (\text{CP17:3.59})$$

The completed coefficient is b_n times this quantity. A boundedness estimate alone gives $\|f\|_\infty \|g\|_\infty \eta$. Both signs of the reflected packet remain in the exact signed integral. Group 2 supplies constants uniform for unit finite-dimensional sources and for $a \in [a_0, R]$, so the onset estimate is uniform independently of the unknown gap.

The interface $h = 1$ is equally explicit. Write $M_f = \|f\|_\infty$, $M_g = \|g\|_\infty$. The new ordinary part for $h = 1 - \eta$, $0 < \eta < \min\{d_*, 1\}$, satisfies

$$|\mathcal{O}_{1-\eta}(f, g)| \leq \frac{(C_f M_g + M_f C_g) \eta}{\sqrt{\log(1/\eta)}}. \quad (\text{CP17:3.60})$$

Indeed its integration interval is $(0, \eta)$, and in each of the two terms one argument is within η of the outer endpoint. The reflected domain loses $(1 - \eta, 1)$ on the left of this interface and loses $(0, \eta)$ on the right. Relative to the integral at $h = 1$, the absolute costs of those strips are, respectively,

$$\frac{M_f C_g \eta}{\sqrt{\log(1/\eta)}}, \quad \frac{C_f M_g \eta}{\sqrt{\log(1/\eta)}}. \quad (\text{CP17:3.61})$$

The change of translated argument on the common part of the reflected integral remains in the exact packet. The complete-packet estimate (3.53), together with the explicit ordinary and strip terms above, controls this remaining common part. Thus no unestimated jump is concealed at the ordinary/reflected interface. For h on both sides of one, compare each side with one; the two strip widths are bounded by the total change in h . Multiplication by the fixed coefficient b_n gives the complete arithmetic interface cost. No point trace or derivative at the odd origin has been assumed in these inequalities.

These bounds show that a new global packet is $o(\eta)$ and that the domain-strip contributions at $h = 1$ are also $o(\eta)$. They quantify boundary pieces. The surviving translated arguments of older packets range through the interior, and their complete signed variation is still (3.47). Hence these interface estimates do not by themselves give an absolute-continuity, linear, or Osgood bound for the complete matrix variation.

3.13. Differentiation, boundary terms, and multiplicities

For sufficiently regular half-interval functions, the derivative of the ordinary part of (3.46), for $0 < h < 1$, is

$$\begin{aligned} \mathcal{O}'_h(f, g) = & \int_0^{1-h} \{f'(x+h)\overline{g(x)} + f(x)\overline{g'(x+h)}\} dx \\ & - f(1)\overline{g(1-h)} - f(1-h)\overline{g(1)}. \end{aligned} \quad (\text{CP17:3.62})$$

For the reflected integral $\mathcal{J}_h = \mathcal{O}_h - \mathcal{P}_h$,

$$\mathcal{J}'_h(f, g) = \begin{cases} f(0)\overline{g(h)} + \int_0^h f'(h-x)\overline{g(x)} dx, & 0 < h < 1, \\ -f(1)\overline{g(h-1)} + \int_{h-1}^1 f'(h-x)\overline{g(x)} dx, & 1 < h < 2. \end{cases} \quad (\text{CP17:3.63})$$

These formulas display the boundary terms before any vanishing is imposed. They hold, for example, for H^1 functions with their legitimate endpoint traces. Zero endpoint traces eliminate the written boundary evaluations, but existence of those traces does not on its own justify the derivative integrals. None of (3.62)-(3.63) is applied to a logarithmic graph-domain solution without the extra regularity required for it. The unconditional replacement is the exact finite difference (3.47).

Where the complete form is differentiable on the relevant inverse solutions, its candidate derivative on a fixed vector v is the full expression

$$\begin{aligned} \langle \tilde{A}'_a v, v \rangle = & -a^{-1} \|v\|_2^2 + \sum_{\ell_n < 2R} \frac{b_n \ell_n}{a^2} \partial_h \mathcal{P}_h(v, v)|_{h=\ell_n/a} \\ & - \langle \mathbf{T}'_a v, v \rangle - 4\Re[\langle v, p'_a \rangle \langle p_a, v \rangle]. \end{aligned} \quad (\text{CP17:3.64})$$

The alpha derivative kernel is $\alpha(a(x-y)) + a(x-y)\alpha'(a(x-y))$, interpreted with the even-kernel derivative. In the half representation it is the difference of the corresponding ordinary and reflected kernels. Formula (3.64) is not claimed as an unconditional derivative theorem; it records exactly which signed interactions a derivative argument must estimate.

Multiplicity does not obstruct the established finite-difference calculation. If u_a and u_b are arbitrary unit maximum-eigenvalue vectors for K_a, K_b , respectively, then

$$-\langle (K_b - K_a)u_b, u_b \rangle \leq \varepsilon(b) - \varepsilon(a) \leq -\langle (K_b - K_a)u_a, u_a \rangle, \quad (\text{CP17:3.65})$$

and

$$|\varepsilon(b) - \varepsilon(a)| \leq \mathbf{m}_R(|b - a|). \quad (\text{CP17:3.66})$$

If differentiability of the complete matrix is proved at a parameter, the one-sided extremal-eigenvalue derivative uses the full maximum eigenspace E_a :

$$\varepsilon'_+(a) = \min_{u \in E_a, \|u\|=1} \tau_R \langle \tilde{A}'_a \mathcal{H}_a^{-1} u, \mathcal{H}_a^{-1} u \rangle. \quad (\text{CP17:3.67})$$

At a genuine two-sided differentiability point the appropriate extremal derivatives must agree. Selecting one eigenvector through a multiple eigenvalue without this extremal operation is insufficient.

3.14. Closing statement

Proved theorem and domain. On the common supported odd logarithmic form domain, with the independently positive finite-rank regularization of Checkpoint 16, (3.43), (3.46), and (3.47) give the exact complete finite-difference variation. For every finite R , (3.50)-(3.57) give an explicit uniform Dini modulus for its regularized finite matrix. Using Group 2's boundary envelope additionally gives (3.59)-(3.61), with its stated uniform barrier constants. This theorem holds through every prime threshold, includes all mixed forced-solution and polar terms, and applies equally to both possible first-null branches.

Consequence. Every ordered eigenvalue of K_a , including the maximum with arbitrary multiplicity, is continuous with the proved Dini modulus. A finite first crossing is therefore detected within one uniformly invertible regularized framework. This is stronger than merely asserting strong continuity of individual translated functions, but it is not an exclusion of that crossing.

Remaining signed estimate. A sufficient integrated estimate, for unit maximum-eigenvalue vectors u_b , is

$$\langle (K_b - K_a)u_b, u_b \rangle \leq \int_a^b c_R(t) \omega(\varepsilon(t)) dt \quad (\text{CP17:3.68})$$

on every positive segment, with c_R integrable independently of the unknown margin and an Osgood modulus ω . The complete signed left side is (3.47). The bound proved in (3.56) controls its absolute size independently of the margin and supplies no factor vanishing when $\varepsilon \rightarrow 0$. The attempted step that remains unproved is exactly the replacement of that absolute bound by (3.68), or by its justified derivative counterpart (3.64)-(3.67). The new boundary threshold estimates address entering packets and changing integration strips but leave the older interior prime interactions and their archimedean-polar coupling in (3.47).

4. Complete signed variation, endpoint substitution, and the Osgood test

4.1. A finite-difference sandwich that retains the favorable mixed square

Fix $R > a_0$, and use exactly the fixed-space operators and constants of Group 1. Write

$$\tau = \tau_R, \quad g = g_R = [\max\{2, 1 + 2M_R^2\}]^{-1}, \quad D_{b,a} = \tilde{A}_b - \tilde{A}_a. \quad (\text{CP17:4.1})$$

Thus $\mathcal{H}_a = \tilde{A}_a + \tau P \geq gI$ independently of the sign of $\tilde{A}_a$. Its operator domain is the fixed supported logarithmic operator domain. The difference $D_{b,a}$ is a bounded self-adjoint operator, even though its norm need not tend to zero at a prime threshold. On the finite-dimensional space $PL_{\text{odd}}^2(-1, 1)$, take any unit top eigenvectors

$$K_a u_a = (1 - \varepsilon_a) u_a, \quad K_b u_b = (1 - \varepsilon_b) u_b, \quad v_c = \mathcal{H}_c^{-1} u_c. \quad (\text{CP17:4.2})$$

No simplicity assumption is made. Rayleigh's principle and the complete resolvent identity give the two-sided estimate

$$\begin{aligned} \tau \langle v_b, D_{b,a} v_b \rangle + \tau \langle D_{b,a} v_b, \mathcal{H}_a^{-1} D_{b,a} v_b \rangle &\leq \varepsilon_b - \varepsilon_a \\ &\leq \tau \langle v_a, D_{b,a} v_a \rangle - \tau \langle D_{b,a} v_a, \mathcal{H}_b^{-1} D_{b,a} v_a \rangle. \end{aligned} \quad (\text{CP17:4.3})$$

All pairings use the original fixed-space Hilbert inner product associated with the unitary transport; the Suzuki metric has not been replaced.

For the lower inequality, use $\lambda_{\max}(K_a) \geq \langle u_b, K_a u_b \rangle$ and

$$\mathcal{H}_a^{-1} u_b = v_b + \mathcal{H}_a^{-1} D_{b,a} v_b. \quad (\text{CP17:4.4})$$

For the upper inequality use the vector u_a and $\mathcal{H}_b^{-1} u_a = v_a - \mathcal{H}_b^{-1} D_{b,a} v_a$. These substitutions prove (4.3) without differentiating a translated function or varying its support formally. They apply at every prime threshold.

In particular, the square in the lower bound is helpful. If the complete difference is split into logarithmic, prime, archimedean, and polar pieces, $D = \sum_{\nu} D_{\nu}$, that square is

$$\tau \sum_{\nu} \langle D_{\nu} v, \mathcal{H}_a^{-1} D_{\nu} v \rangle + 2\tau \Re \sum_{\nu < \eta} \langle D_{\nu} v, \mathcal{H}_a^{-1} D_{\eta} v \rangle. \quad (\text{CP17:4.5})$$

The second sum includes every prime-prime, prime-archimedean, prime-polar, and boundary coupling. No positivity is asserted for an individual mixed term. The entire square is nonnegative.

The near-null equations give the additional exact identities

$$\tilde{A}_a v_a = \varepsilon_a u_a, \quad P v_a = \frac{1 - \varepsilon_a}{\tau} u_a, \quad (\text{CP17:4.6})$$

$$\langle \tilde{A}_a v_a, v_a \rangle = \frac{\varepsilon_a (1 - \varepsilon_a)}{\tau}, \quad \langle \mathcal{H}_a v_a, v_a \rangle = \frac{1 - \varepsilon_a}{\tau}, \quad \frac{1 - \varepsilon_a}{\tau} \leq \|v_a\| \leq g^{-1}. \quad (\text{CP17:4.7})$$

Consequently the vectors remain nonzero as $\varepsilon_a \downarrow 0$. The small quantity in (4.7) is the signed Weil energy, not the norm of the vector or of each arithmetic packet.

4.2. Inhomogeneous endpoint equations before taking absolute values

Return a near-null pair to physical coordinates:

$$p = U_a^{-1} v_a, \quad h = U_a^{-1} u_a, \quad A_a^- p = \varepsilon_a h, \quad \|h\| = 1. \quad (\text{CP17:4.8})$$

Both are odd. The complete screw convolution $f = g * p$ satisfies

$$f'' = -\varepsilon_a h \quad \text{on } (-a, a). \quad (\text{CP17:4.9})$$

Here $A_a = -g''*$ is the retained normalization of the actual completed arithmetic form. The convolution traces from the earlier checkpoints apply: the kernel and its first derivative are locally in L^2 , including the logarithmic first-derivative singularity, and compact support of $p \in L^2$ gives continuous f, f' at the endpoints. Equation (4.9) therefore implies

$$\begin{aligned} f(t) &= \alpha t - \varepsilon_a \int_0^t (t-y) h(y) dy, \\ f'(t) &= \alpha - \varepsilon_a \int_0^t h(y) dy, \\ \boxed{f(a) - a f'(a) = \varepsilon_a J_a(h)}, \quad J_a(h) &= \int_0^a y h(y) dy, \quad |J_a(h)| \leq \frac{a^{3/2}}{\sqrt{6}}. \end{aligned} \quad (\text{CP17:4.10})$$

These formulas require no endpoint value of p or h . The constant term is zero by parity.

For complete explicitness, on $t > 0$ write

$$g(t) = -8(\cosh(t/2) - 1) + \sum_{\ell_n < t} b_n(t - \ell_n) + G_{\text{ar}}(t), \quad (\text{CP17:4.11})$$

$$G_{\text{ar}}(t) = -\frac{t}{2}(\psi(1/4) - \log \pi) - \frac{1}{4} \left[\Phi(1, 2, 1/4) - e^{-t/2} \Phi(e^{-2t}, 2, 1/4) \right], \quad (\text{CP17:4.12})$$

$$G'_{\text{ar}}(t) = -\frac{1}{2} \left[\psi(1/4) - \log \pi + e^{-t/2} \Phi(e^{-2t}, 1, 1/4) \right]. \quad (\text{CP17:4.13})$$

Thus all completed polar, digamma, and Lerch terms remain. Define

$$F_p(z) = \int_{-a}^a e^{zy} p(y) dy, \quad d_a = 4a \cosh(a/2) - 8 \sinh(a/2) > 0, \quad (\text{CP17:4.14})$$

$$\begin{aligned} \mathcal{P}_a(p) &= \sum_{\ell_n < 2a} b_n \int_{-a}^{a-\ell_n} (-y - \ell_n) p(y) dy \\ &\quad + \int_{-a}^a [G_{\text{ar}}(a - y) - a G'_{\text{ar}}(a - y)] p(y) dy. \end{aligned} \quad (\text{CP17:4.15})$$

The logarithm in the last integral is integrable by Cauchy-Schwarz. Substitution into (4.10), before any norm estimate, gives the exact identity

$$\boxed{\mathcal{P}_a(p) - d_a F_p(1/2) = \varepsilon_a J_a(h).} \quad (\text{CP17:4.16})$$

At $\varepsilon = 0$ it recovers the earlier complete null endpoint equation. For a near-null vector it supplies an explicit error of size at most $\varepsilon_a a^{3/2}/\sqrt{6}$.

This is a real cancellation, but it does not eliminate $\mathcal{P}_a(p)$. To see the remaining term in an actual part of the variation, hold the fixed-space vector v_a fixed while varying the source $\tilde{s}_b(x) = \sqrt{b} \sinh(bx/2)$. Direct differentiation of this smooth rank-one term gives

$$\frac{d}{db} [-2|\langle \tilde{s}_b, v_a \rangle|^2]_{b=a} = -\frac{2}{a} F_p(1/2) [F_p(1/2) + F'_p(1/2)] \quad (\text{CP17:4.17})$$

for real vectors. Inserting (4.16) yields

$$-\frac{2}{ad_a^2} [\mathcal{P}_a(p) - \varepsilon_a J_a(h)] [\mathcal{P}_a(p) - \varepsilon_a J_a(h) + d_a F'_p(1/2)]. \quad (\text{CP17:4.18})$$

The surviving order-zero term is

$$-\frac{2}{ad_a^2} \mathcal{P}_a(p) [\mathcal{P}_a(p) + d_a F'_p(1/2)]. \quad (\text{CP17:4.19})$$

It must be paired with the retained prime and archimedean variations in (4.3). Equation (4.16) alone neither assigns its sign nor makes it proportional to ε . This statement concerns the actual completed kernel, not a generic model. It does not rule out a further cancellation with the other complete terms.

There is an exact finite-difference version of this substitution, requiring no shape derivative. Choose real top eigenvectors for this expansion; they are available because the matrices and operators preserve real functions. In the lower bound of (4.3), hold v_b fixed and put $p_b = U_b^{-1}v_b$, $h_b = U_b^{-1}u_b$. Define

$$z_b = \langle \tilde{s}_b, v_b \rangle = F_{p_b}(1/2), \quad z_a = \langle \tilde{s}_a, v_b \rangle = \sqrt{a/b} F_{p_b}(a/(2b)), \quad dz = z_b - z_a. \quad (\text{CP17:4.20})$$

For real vectors, the complete polar part of $\langle v_b, D_{b,a}v_b \rangle$ is therefore

$$\begin{aligned} -2(z_b^2 - z_a^2) &= -4z_b dz + 2(dz)^2 \\ &= -\frac{4\mathcal{P}_b(p_b)}{d_b} dz + \frac{4\varepsilon_b J_b(h_b)}{d_b} dz + 2(dz)^2. \end{aligned} \quad (\text{CP17:4.21})$$

The complex version replaces the products by the corresponding real parts and squared absolute values. Thus the endpoint relation has been inserted into the actual finite increment before any estimates. Both source cross terms and the quadratic increment survive. In particular the first term on the last line has no explicit factor ε_b .

The source derivative bound D_R^{src} from Group 3 gives

$$|dz| \leq \frac{D_R^{\text{src}}}{g_R} |b - a|, \quad \left| \frac{4\varepsilon_b J_b(h_b)}{d_b} dz \right| \leq \frac{4|\varepsilon_b| R^{3/2} D_R^{\text{src}}}{\sqrt{6} d_{a_0} g_R} |b - a|. \quad (\text{CP17:4.22})$$

Here $d_b \geq d_{a_0} > 0$, since $d'_a = 2a \sinh(a/2) > 0$. This bounds the explicitly near-null part with a vanishing factor and isolates the remaining endpoint-packet coupling. The latter still has to be combined with all prime and archimedean increments and the full mixed square in (4.3).

There is an explicit margin-independent bound for this polar derivative. Set

$$S_R = (\sinh R - R)^{1/2}, \quad C_R^{\text{mom}} = \left(\frac{R^3}{3} + R^2 \sinh R - 2R \cosh R + 2 \sinh R \right)^{1/2}. \quad (\text{CP17:4.23})$$

Cauchy-Schwarz gives $|F_p(1/2)| \leq S_R/g$ and $|F'_p(1/2)| \leq C_R^{\text{mom}}/g$. Hence the absolute value of (4.17) is at most

$$\frac{2S_R(S_R + C_R^{\text{mom}})}{a_0 g^2}. \quad (\text{CP17:4.24})$$

This constant remains finite at a candidate first null. It is an absolute bound, without the required vanishing factor.

4.3. The strongest complete uniform variation bound currently available

Use the explicit graph and kernel constants from Groups 1 and 3. To display their role, put

$$L_R = \max\{|\log a_0|, |\log R|\}, \quad G_R^2 = \left[1 + \frac{L_R + M_R + \tau_R}{g_R} \right]^2 + \frac{\pi^2}{2g_R^2}, \quad B_R^{\text{pr}} = \sum_{\ell_n < 2R} b_n. \quad (\text{CP17:4.25})$$

If L_R^{sm} is the complete explicit Lipschitz constant for the logarithmic scalar, archimedean kernel, and polar rank-one pieces given in Group 3, define

$$\begin{aligned} \eta &= \frac{2R|b - a|}{a_0^2} < e^{-2}, \\ \nu_R(|b - a|) &= \tau_R \left[\frac{L_R^{\text{sm}} |b - a|}{g_R^2} + B_R^{\text{pr}} \left(\frac{2\sqrt{\eta}}{g_R^2} + \frac{16G_R^2}{\log^2(1/\eta)} \right) \right]. \end{aligned} \quad (\text{CP17:4.26})$$

The complete bilinear Fourier estimate then proves

$$\|K_b - K_a\| \leq \nu_R(|b - a|), \quad |\varepsilon_b - \varepsilon_a| \leq \nu_R(|b - a|). \quad (\text{CP17:4.27})$$

It includes all prime powers up to $2R$, with zero extension handling thresholds. The important gain over an action-norm estimate is the square on the logarithm: both actual forced solutions have logarithmic graph regularity. In particular the modulus is Dini-integrable, $\int_0^{h_0} \nu_R(h) dh/h < \infty$.

This proves uniform continuity through every threshold. It does not prove absolute continuity, because summing the available bound over a partition of mesh h produces an upper bound of order $1/[h \log^2(1/h)]$, which does not stay bounded. Thus neither a derivative inequality nor a vanishing factor can be inferred from (4.27). Its constant is independent of the unknown positivity margin, but its right side does not decrease with ε .

4.4. A precise multiplicity-safe Osgood criterion

Let $\omega : [0, 1] \rightarrow [0, \infty)$ be continuous, positive on $(0, 1]$, nondecreasing, with $\omega(0) = 0$ and

$$\int_{0+} \frac{dt}{\omega(t)} = \infty. \quad (\text{CP17:4.28})$$

Suppose, in addition to the already proved continuity of K_a , an independent complete signed estimate gives, for every $a < b$ before a first crossing,

$$\varepsilon_b - \varepsilon_a \geq - \int_a^b c_R(t) \omega(\varepsilon_t) dt, \quad 0 \leq c_R \in L^1([a_0, R]). \quad (\text{CP17:4.29})$$

Then no finite first crossing is possible. Indeed $\varepsilon(t) + \int_{a_0}^t c_R(u) \omega(\varepsilon(u)) du$ is nondecreasing, so ε has bounded variation on compact intervals before the crossing and its negative variation is absolutely continuous. Applying the continuous bounded-variation chain rule to an antiderivative of $1/\omega$ on any compact positive range gives

$$\boxed{\int_{\varepsilon(b)}^{\varepsilon(a_0)} \frac{dt}{\omega(t)} \leq \int_{a_0}^b c_R(u) du.} \quad (\text{CP17:4.30})$$

The positive singular variation, if present, helps this inequality. As b approaches a hypothetical first zero, the left side diverges and the right side remains finite. This proves the conditional comparison without assuming simple eigenvalues or ordinary differentiability at every threshold.

For an absolutely continuous differentiable implementation, at almost every parameter where the required form derivative exists on the forced solutions, the sufficient condition is

$$\tau_R \langle \mathcal{H}_a^{-1} u, \tilde{A}'_a \mathcal{H}_a^{-1} u \rangle \geq -c_R(a) \omega(\varepsilon_a) \quad \text{for every unit } u \in \ker(K_a - (1 - \varepsilon_a)I). \quad (\text{CP17:4.31})$$

The minimum of these quadratic values is the relevant one-sided derivative of ε . Stating (4.31) on the whole top eigenspace covers multiplicities. Differentiability of the prime translations on these particular vectors has not been proved by logarithmic graph regularity, so (4.31) is presently a criterion rather than an available calculation.

The exact finite-difference expression that would establish (4.29), while retaining its favorable square, is the following missing lemma:

$$\boxed{\begin{aligned} &\tau_R \langle v_b, D_{b,a} v_b \rangle + \tau_R \langle D_{b,a} v_b, \mathcal{H}_a^{-1} D_{b,a} v_b \rangle \\ &\geq - \int_a^b c_R(t) \omega(\varepsilon_t) dt \end{aligned}} \quad (\text{CP17:4.32})$$

for every unit top eigenvector u_b , $v_b = \mathcal{H}_b^{-1}u_b$, and every sufficiently close $a < b \leq R$ before the first crossing. Constants must depend only on the explicit data of $[a_0, R]$, not on $1/\varepsilon$. The complete differences and endpoint substitution are (4.3), (4.5), and (4.16). The current independent estimate replaces the right side of (4.32) only by $-\nu_R(b - a)$. Dividing that absolute budget by $\omega(\varepsilon)$ introduces exactly the forbidden unknown-margin dependence.

4.5. Closing statement for Group 4

The actual collar calculation in Group 3 sharpens the scalar-branch obstruction even without assuming a normalized pointwise trace. In its unitary half-interval normalization, put $w = H_a^{-1}s_a$, choose $0 < \eta < 1$ in the proved boundary neighborhood, and define

$$\bar{\beta}_a(L) = \frac{1}{2\sqrt{L}} \int_{e^{-L}}^{\eta} \frac{w(a-d)}{d} dd. \quad (\text{CP17:4.33})$$

The boundary envelope $|w(a-d)| \leq B_a/\sqrt{\log(1/d)}$ implies $|\bar{\beta}_a(L)| \leq B_a$ for large L . Group 3 proves the sharper complete expansion

$$\frac{\sigma_{a+\delta} - \sigma_a}{\delta} = 2 \left| \bar{\beta}_a \left(\log \frac{1}{\delta} \right) \right|^2 + o(1). \quad (\text{CP17:4.34})$$

Thus the actual collar cost is $O_a(\delta)$, with a precisely identified averaged boundary coefficient. No limit of this average is needed for (4.34). If a normalized trace exists, the average tends to it.

For this inverse-available scalar branch, the estimate that would turn the new rate into a relative or Osgood rate is

$$2 \limsup_{L \rightarrow \infty} |\bar{\beta}_a(L)|^2 \leq c_R(a) \omega(1 - \sigma_a), \quad (\text{CP17:4.35})$$

at almost every parameter on a positive- H interval. Group 3 proves local absolute continuity there and existence of the averaged limit almost everywhere, so the almost-everywhere estimate would integrate directly on that region. The actual bound currently obtained is only $2B_a^2$. This calculation identifies where the required vanishing factor must enter. Formula (4.35) is scoped to the scalar branch; the complete regularized matrix analogue remains (4.32), which includes the zero-moment branch as well. A finite family of boundary envelopes for the regularized forced solutions bounds their directional averaged coefficients uniformly as $\varepsilon \downarrow 0$; it has not been shown to make those coefficients vanish with ε .

Proved theorem and hypotheses. On every fixed finite $[a_0, R]$, the independently positive regularization detects every first-null branch. The finite-difference sandwich (4.3), inhomogeneous endpoint identity (4.16), full polar expansion (4.18), and margin-independent complete modulus (4.27) hold for the actual forced solutions. No inverse of the unregularized odd polar operator is needed here. The Osgood implication (4.29)-(4.30) is a proved conditional continuation theorem.

Strongest independent bound and remaining estimate. The complete variation is uniformly Dini-continuous with the explicit logarithmic-square modulus (4.26). The polar contribution has the independent absolute bound (4.24). A sufficient signed Osgood bound (4.32) remains unproved. The near-null equation removes the explicitly proportional error in (4.16); the actual packet $\mathcal{P}_a(p)$, its companion moment $F'_p(1/2)$, and the prime-translation variation still require a new coupled estimate. If a normalized boundary trace exists, its contribution must be bounded by a vanishing function of ε ; the already proved uniform inverse bound alone gives no such factor.

Consequence for the objective. Both branches are excluded on the certified prime-active interval. Beyond it, both are represented in one bounded-inverse finite matrix, but neither has been

excluded globally. If (4.32) were proved for every finite R , the initial margin $\varepsilon(a_0) \geq (1/5000)/(\tau_R + 1/5000) > 0$, the comparison above, and the fixed-space null detection would exclude every finite first odd crossing. The established odd Weil criterion would then give RH, and the correctly normalized Suzuki identity would give $\mathfrak{W} = \mathcal{S}$ and $\mathcal{E}_N = 0$ for every N . The missing antecedent is precisely (4.32).

5. Combined closing statement

5.1. What the new calculation establishes

The prime-active starting certificate is preserved without changing its interval or its constant:

$$0 < a \leq a_0 = \frac{699}{2000} + e^{-2500} \implies Q_a^-(q) \geq \frac{1}{5000} \|q\|_2^2 \quad (q \in \mathcal{D}_a^-). \quad (\text{CP17:5.1})$$

No larger numerically specified support interval is claimed in this round.

The actual complete forced solutions are now proved bounded, with bounded complete logarithmic forcing. The logarithmic boundary upper estimate therefore applies to these solutions. Its constant is independent of the Weil margin for the finite-rank regularized family. For the unregularized solution $H_a^{-1}s_a$, the available constant depends on the inverse of H_a , exactly as stated in Group 2.

Fix a with $H_a \geq gI > 0$. The complete collar Schur calculation proves

$$\begin{aligned} 0 \leq \sigma_{a+\delta} - \sigma_a &= 2\langle r_{a,\delta}, S_{a,\delta}^{-1}r_{a,\delta} \rangle \leq C_a\delta, \\ \frac{\sigma_{a+\delta} - \sigma_a}{\delta} &= 2|\bar{\beta}_a(\log(1/\delta))|^2 + O_a(\log(1/\delta)^{-1/2}). \end{aligned} \quad (\text{CP17:5.2})$$

The first constant and the second error are explicitly bounded in Group 3, in terms of the full inverse gap, boundary envelope, and complete prime and archimedean bounds. Here the signed half-interval averaged coefficient is

$$\bar{\beta}_a(L) = \frac{1}{2\sqrt{L}} \int_{e^{-L}}^{\eta} \frac{W_a(a-d)}{d} dd, \quad W_a = \sqrt{2}(H_a^{-1}s_a)|_{(0,a)}. \quad (\text{CP17:5.3})$$

All reflected prime terms and the full $B^*H_a^{-1}B$ correction remain in (5.2). No sign of W_a or of an autocorrelation is assumed.

On every compact interval on which H_a stays strictly positive, the constants can be chosen uniformly. Thus σ_a is locally Lipschitz there. The averaged limit $\tilde{\beta}_a = \lim_{L \rightarrow \infty} \bar{\beta}_a(L)$ exists for almost every such a , and

$$\sigma'(a) = 2|\tilde{\beta}_a|^2, \quad \mu'(a) = -2|\tilde{\beta}_a|^2, \quad \mu_a = 1 - \sigma_a, \quad \text{for almost every such } a. \quad (\text{CP17:5.4})$$

This is an unconditional local identity for the actual scalar resolvent on its positive- H domain. It does not prove a pointwise normalized boundary trace. If that stronger trace exists, its derivative normalization is

$$\sigma'_+(a) = 2|\beta_{a,\text{half}}|^2 = 4|\beta_{a,\text{full}}|^2, \quad \beta_{a,\text{half}} = \sqrt{2}\beta_{a,\text{full}}. \quad (\text{CP17:5.5})$$

The distinction between (5.4) and the conditional statement (5.5) is essential.

For all $a \in [a_0, R]$, including possible singularities of H_a , the fixed projection and regularization instead give

$$\mathcal{H}_a \geq gRI, \quad \tilde{A}_a v_a = \varepsilon(a)u_a, \quad Pv_a = \frac{1 - \varepsilon(a)}{\tau_R} u_a. \quad (\text{CP17:5.6})$$

Every possible first-null vector is detected by $\varepsilon(a) = 0$. The exact finite-difference matrix and its complete Dini bound are proved through $2a = \log n$, $a = \log n$, and the half-interval support interfaces. The estimates retain source scaling and polar, prime, and boundary terms. They prove continuity with a quantified modulus. They do not establish absolute continuity of the complete regularized matrix, a classical derivative through every threshold, or a margin-dependent sign estimate.

5.2. The exact remaining signed estimate

One quantified statement would finish the unified argument. Let

$$D_{b,a} = \tilde{A}_b - \tilde{A}_a, \quad v_b = \mathcal{H}_b^{-1} u_b,$$

where u_b is any real unit vector in the top eigenspace of K_b . Independently of the unknown positive margin, prove on every positive branch that there are $c_R \geq 0$ in $L^1([a_0, R])$ and a continuous positive function ω on $0 < t \leq 1$, with $\omega(0) = 0$ and $\int_{0+} dt/\omega(t) = \infty$, such that, for $a_0 \leq a < b \leq R$,

$$\tau_R \langle v_b, D_{b,a} v_b \rangle + \tau_R \langle D_{b,a} v_b, \mathcal{H}_a^{-1} D_{b,a} v_b \rangle \geq - \int_a^b c_R(t) \omega(\varepsilon(t)) dt. \quad (\text{CP17:5.7})$$

The left side is completely defined by the exact operator in Group 1. In particular its square is the full square, including every pair of prime, archimedean, polar, and scaling components. This lemma is sufficient for both branches and for eigenvalue multiplicities.

The strongest currently proved uniform estimate for that same expression is the margin-independent bound

$$\tau_R \langle v_b, D_{b,a} v_b \rangle + \tau_R \langle D_{b,a} v_b, \mathcal{H}_a^{-1} D_{b,a} v_b \rangle \geq -\mathfrak{m}_R(b-a), \quad (\text{CP17:5.8})$$

with the explicit Dini modulus in Groups 3-4. The exact endpoint substitution in Group 4 isolates an additional polar term proportional to $\varepsilon(b)$, but leaves a coupled prime-endpoint term of order one in that parameter. The near-null equation eliminates the finite-rank forcing charge; it does not eliminate the remaining arithmetic, archimedean, and polar interactions.

On a positive- H interval, (5.4) gives a narrower scalar version of the missing estimate:

$$2|\tilde{\beta}_a|^2 \leq c_R(a) \omega(1 - \sigma_a) \quad \text{for almost every } a. \quad (\text{CP17:5.9})$$

The new boundary argument supplies $2|\tilde{\beta}_a|^2 \leq 2B_a^2$. It supplies no factor vanishing with $1 - \sigma_a$; its available B_a also depends on the H_a inverse gap. Thus it cannot replace the margin-independent unified lemma (5.7) or exclude a zero-moment kernel branch.

The precise line where global closure stops is the passage from (5.8) to (5.7), or, on a positive- H branch, from the actual coefficient formula (5.4) to (5.9). Neither follows from the boundedness, boundary decay, strong core convergence, or the preserved real-shift invariant alone.

5.3. Return chain and separate target status

If (5.7) were independently proved, the finite-difference sandwich and continuous bounded-variation comparison of Group 4 would give

$$\int_{\varepsilon(b)}^{\varepsilon(a_0)} \frac{dt}{\omega(t)} \leq \int_{a_0}^b c_R(u) du, \quad \varepsilon(a_0) \geq \frac{1/5000}{\tau_R + 1/5000} > 0. \quad (\text{CP17:5.10})$$

Its divergent left side at a first zero would exclude both first-null branches on every finite support interval. The established odd-test criterion and normalized Suzuki identity would then give the complete conditional chain

$$\begin{aligned}
 & \text{independent estimate (5.7) for every finite } R \\
 & \implies A_a^- > 0 \text{ on every finite interval} \\
 & \implies \text{nonnegativity on every compact smooth odd test} \tag{CP17:5.11} \\
 & \implies \text{RH} \implies \text{unconditional Weil positivity after that proof} \\
 & \implies \mathfrak{W} = \mathcal{S} \implies \Delta = 0 \implies \mathcal{E}_N = 0 \text{ for every } N \implies \liminf_N \mathcal{E}_N \leq 1.
 \end{aligned}$$

This displayed return is conditional on (5.7); it is not a claim that RH or the defect criterion was proved in this checkpoint. The Suzuki metric and all its Gram normalization are unchanged.

Requested closing question	Strongest established answer
What rate is proved for the actual collar cost?	$O_a(\delta)$ wherever $H_a > 0$, with the quantitative averaged-coefficient formula (5.2). The scalar budget is locally absolutely continuous on positive- H intervals.
Is the normalized pointwise boundary trace established?	No. The upper boundary envelope is proved. A signed averaged trace exists almost everywhere on the positive- H region.
Is complete variation controlled through every prime threshold?	Yes, by an exact finite-difference identity and a uniform Dini modulus, with sharper entry-strip bounds. Absolute continuity of the full regularized matrix has not been established.
Is a sufficient Osgood estimate proved independently of the unknown margin?	No. The exact missing inequality is (5.7); the scalar version is (5.9).
Are both first-null branches excluded?	They are excluded on the preserved certified interval $a \leq a_0$, and both are detected by the unified matrix for all finite R . Exclusion beyond that interval is not proved.

Final status. Checkpoint 17 proves new boundedness, actual boundary upper estimates, a linear collar rate, a quantitative averaged boundary formula, and threshold-complete matrix continuity. Global unconditional Weil positivity remains open. The original criterion $\liminf_N \mathcal{E}_N \leq 1$ also remains open; the return to $\mathcal{E}_N = 0$ remains conditional. Earlier checkpoints are retained unchanged. We are aiming to prove unconditional Weil positivity with the original Suzuki metric preserved.

Checkpoint 18: The physical regularizer and boundary energy

Source: Michaels_Physical_Regularizer_and_Boundary_Energy_Checkpoint_18.md. Equation tags carry the source prefix CP18; references within this unit retain their original local numbering.

Checkpoint 18 – 23 September 2026

The primary objective remains unconditional Weil positivity. This checkpoint continues the complete Checkpoint 17 and its physical-regularizer follow-up. It preserves the independently certified prime-active interval

$$a_0 = \frac{699}{2000} + e^{-2500}, \quad Q_a^-(q) \geq \frac{1}{5000} \|q\|_2^2 \quad (0 < a \leq a_0), \quad (\text{CP18:0.1})$$

on the complete odd supported logarithmic form domain. The original Suzuki metric, determining ladder, and defect remain

$$\Delta = \mathfrak{W} - \mathcal{S}, \quad \mathcal{E}_N = \left\| [\Delta(\rho_j, \rho_i)]_{i,j \leq N} \right\|_F^2, \quad \liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1. \quad (\text{CP18:0.2})$$

Every Gram normalization belonging to $\mathcal{S}$ remains part of this target. The auxiliary finite-dimensional matrices below test the sign of the complete Weil form.

Contents

1. Restricting one physical regularizer: exact sign and nullspace equivalences.
2. The complete matrix collar identity, uniform Lipschitz estimate, and boundary Gram derivative.
3. Endpoint identities, explicit adjoint probes, and their complete signed residuals.
4. Energy-relative, determinant, and adaptive Osgood comparisons.
5. Combined conclusions and the exact remaining arithmetic estimate.

Conventions and scope

Fix an arbitrary finite $R > a_0$. All full-line tests are extended by zero, and

$$\widehat{q}(\xi) = \int q(x) e^{i\xi x} dx, \quad \ell_n = \log n, \quad b_n = \Lambda(n)/\sqrt{n}, \quad L_\Delta = 2 \log |D|. \quad (\text{CP18:0.3})$$

The supported logarithmic operator is its exterior-Dirichlet form compression. Its endpoint and exterior terms are included. The actual odd Weil operator is

$$A_a^- = C_a - \log(2\pi)I - \Pi_a - K_{\alpha,a} - 2|s_a\rangle\langle s_a|, \quad s_a(x) = \sinh(x/2), \quad (\text{CP18:0.4})$$

where

$$\kappa(t) = \frac{e^{-t/2}}{1 - e^{-2t}}, \quad \alpha(t) = \kappa(|t|) - \frac{1}{2|t|}, \quad \alpha(0) = \frac{1}{4}, \quad \Pi_a q = \sum_{\ell_n < 2R} b_n [q(\cdot + \ell_n) + q(\cdot - \ell_n)]. \quad (\text{CP18:0.5})$$

Zero extension makes inactive packets exactly zero. The unitary half-interval map is $q \mapsto q^+ = \sqrt{2}q|_{(0,a)}$. All boundary vectors in the main matrix derivative use this half-interval normalization; conversion to the full odd boundary coefficient carries the factor $\sqrt{2}$.

The regularizer is fixed on the largest physical interval, then restricted. Its columns need not remain orthonormal or linearly independent on smaller intervals. This distinction is retained throughout. In particular $\widehat{K}_a$ denotes the physical matrix in this checkpoint; it is not Checkpoint 17's matrix obtained from a moving physical source under fixed-coordinate scaling.

New proved scope. The physical matrix is monotone and uniformly Lipschitz on every fixed finite support range. Its almost-everywhere derivative is a rank-one Gram matrix of averaged boundary coefficients. The complete endpoint identities and explicit boundary probes expose the remaining arithmetic residual. The calculations below distinguish these proved statements from the additional energy-dependent estimate needed for global sign preservation.

Sources and dependencies. The complete Checkpoint 17, current ledger, and archived physical-regularizer follow-up were read before calculation. The inherited prime-active certificate and explicit finite-rank constants are retained. The odd Weil criterion and arithmetic form were checked against Suzuki, *Weil's quadratic form via the screw function*, arXiv:2606.09096v2. The logarithmic boundary theorem is Hernández-Santamaría, López Ríos and Saldaña, *Optimal boundary regularity and a Hopf-type lemma for Dirichlet problems involving the logarithmic Laplacian*, arXiv:2401.18033v2, Theorem 1.1. Its bounded-data hypotheses, including the full forcing, are carried through for the physical regularization below. No pointwise normalized trace limit or Weil-energy trace inequality is attributed to that theorem.

1. A single physical regularizer detecting every null branch

1.1. Physical restriction of the complete closed form

Fix a finite $R > a_0$, where

$$a_0 = \frac{699}{2000} + e^{-2500}, \quad m_0 = \frac{1}{5000}. \quad (\text{CP18:1.1})$$

For $0 < a \leq R$, let $\mathcal{H}_a = L^2_{\text{odd}}(-a, a)$, and let $E_a : \mathcal{H}_a \rightarrow \mathcal{H}_R$ denote zero extension. The supported logarithmic form domain is

$$\mathcal{D}_a = \left\{ q \in \mathcal{H}_a : \int_{\mathbb{R}} \log(2 + |\xi|) |\widehat{q}(\xi)|^2 d\xi < \infty \right\}, \quad \widehat{q}(\xi) = \int_{-a}^a q(x) e^{i\xi x} dx. \quad (\text{CP18:1.2})$$

In particular $E_a \mathcal{D}_a = \{u \in \mathcal{D}_R : u = 0 \text{ a.e. outside } (-a, a)\}$. This is an equality of closed supported form spaces. It does not assert that the zero extension of every vector in the smaller operator domain belongs to the larger operator domain.

Use the complete inherited operator

$$\begin{aligned} A_a^- &= C_a - \log(2\pi)I - \Pi_a - K_{\alpha,a} - 2|s_a\rangle\langle s_a|, \quad s_a(x) = \sinh(x/2), \\ \Pi_a q(x) &= \sum_{\ell_n < 2R} b_n [q(x + \ell_n) + q(x - \ell_n)], \quad \ell_n = \log n, \quad b_n = \Lambda(n)/\sqrt{n}, \\ K_{\alpha,a} q(x) &= \int_{-a}^a \alpha(x - y) q(y) dy, \\ \alpha(t) &= \frac{e^{-|t|/2}}{1 - e^{-2|t|}} - \frac{1}{2|t|} \quad (t \neq 0), \quad \alpha(0) = \frac{1}{4}. \end{aligned} \quad (\text{CP18:1.3})$$

Every translated function in this formula is extended by zero. The compressed shift vanishes when $\ell_n \geq 2a$, including equality as an L^2 operator. Thus the fixed finite index set $\ell_n < 2R$ includes every actual threshold. The logarithmic operator C_a is the form compression of the whole-line

multiplier $\log |D|$, with Fourier normalization $1/(2\pi)$. Its exterior interaction and endpoint finite parts are retained.

For $q, r \in \mathcal{D}_a$, all terms in (1.3) agree with their form restrictions from R . For the logarithmic term this follows from the identical whole-line Fourier transforms of q and $E_a q$. For the prime and integral kernels it follows directly from zero extension. For the polar term it follows from $E_a^* s_R = s_a$. Therefore

$$Q_a^-(q, r) = Q_R^-(E_a q, E_a r). \quad (\text{CP18:1.4})$$

This identity fixes the physical-coordinate formulation without introducing a new metric. The original Suzuki form remains $\mathcal{S}(q, r)$ on the same tests, with its established domain passage and normalization.

The constants from Checkpoint 16 are

$$\begin{aligned} M_R^{\text{op}} &= \log(2\pi) + 2 \sum_{\ell_n < 2R} b_n + 4R + 2(\sinh R - R), \\ c_{\text{odd}} &= \frac{1}{3} \log(9\pi/2) - \frac{1}{3}, \quad h_R = -\log R + c_{\text{odd}}, \\ d_R &= \max \left\{ 1, \left\lceil \frac{2R}{\pi} e^{M_R^{\text{op}} + 2} \right\rceil \right\}, \\ \tau_R &= 1 + \max\{0, M_R^{\text{op}} + (M_R^{\text{op}})^2 - h_R\}, \\ g_R &= \frac{1}{\max\{2, 1 + 2(M_R^{\text{op}})^2\}}. \end{aligned} \quad (\text{CP18:1.5})$$

Let $\phi_1, \dots, \phi_{d_R}$ be the fixed real orthonormal reference eigenfunctions used there on $(-1, 1)$. Define physical functions only once, at R :

$$f_j(x) = R^{-1/2} \phi_j(x/R), \quad F_R c = \sum_{j=1}^{d_R} c_j f_j, \quad F_R^* F_R = I_{d_R}. \quad (\text{CP18:1.6})$$

Checkpoint 16's independent estimate at the terminal interval gives

$$\mathcal{J}_R := A_R^- + \tau_R F_R F_R^* \geq g_R I. \quad (\text{CP18:1.7})$$

For every smaller interval retain the same physical functions and set

$$F_a = E_a^* F_R, \quad \mathcal{J}_a = A_a^- + \tau_R F_a F_a^*. \quad (\text{CP18:1.8})$$

Their Gram matrix satisfies

$$F_a^* F_a = F_R^* E_a E_a^* F_R \preceq I, \quad \|F_a\| \leq 1. \quad (\text{CP18:1.9})$$

Neither orthonormality of the restricted columns nor injectivity of F_a is assumed.

For $q, r \in \mathcal{D}_a$, equations (1.4) and (1.8) imply the exact closed-form restriction

$$\mathfrak{j}_a(q, r) = \mathfrak{j}_R(E_a q, E_a r), \quad \boxed{\mathcal{J}_a \geq g_R I, \quad \|\mathcal{J}_a^{-1}\| \leq g_R^{-1}}. \quad (\text{CP18:1.10})$$

The lower bound is independent of the unknown positivity margin of A_a^- . Since the regularizer is bounded, $\text{Dom } \mathcal{J}_a = \text{Dom } A_a^- = \text{Dom } C_a$, and their form domains are $\mathcal{D}_a$.

1.2. Every kernel term in the physical regularization

The finite-rank kernel added to the full odd operator is exactly

$$\tau_R p_R(x, y), \quad p_R(x, y) = \sum_{j=1}^{d_R} f_j(x) \overline{f_j(y)}, \quad |x|, |y| < a. \quad (\text{CP18:1.11})$$

No rescaling or reorthogonalization is performed as a changes. Let

$$\mathfrak{J}_d = \frac{e^4 \pi^2 d(d+1)(2d+1)}{12}.$$

The independently proved reference eigenfunction estimate from Checkpoint 17 gives

$$\sup_{|x| < R} \sum_{j=1}^{d_R} |f_j(x)|^2 \leq \frac{\mathfrak{J}_{d_R}}{R}, \quad |p_R(x, y)| \leq \frac{\mathfrak{J}_{d_R}}{R}, \quad \|F_a c\|_\infty \leq \sqrt{\frac{\mathfrak{J}_{d_R}}{R}} \|c\|. \quad (\text{CP18:1.12})$$

In particular this is a fixed bounded source family, uniformly on all smaller intervals. Its regularizing kernel is dominated in absolute value by a convolution density of total mass $4\tau_R \mathfrak{J}_{d_R}$ on the full interval $(-R, R)$, independently of a .

For the unitary half-interval representation $q \mapsto u = \sqrt{2}q|_{(0,a)}$, the regularizing source columns are

$$h_j(x) = \sqrt{2}f_j(x), \quad s_a^+(x) = \sqrt{2} \sinh(x/2). \quad (\text{CP18:1.13})$$

The complete prime action in these coordinates is

$$\begin{aligned} (\Pi_a^+ u)(x) = \sum_{\ell_n < 2R} b_n [& \mathbf{1}_{x+\ell_n < a} u(x + \ell_n) + \mathbf{1}_{x > \ell_n} u(x - \ell_n) \\ & - \mathbf{1}_{x < \ell_n < x+a} u(\ell_n - x)], \quad 0 < x < a. \end{aligned} \quad (\text{CP18:1.14})$$

The signs in (1.14) come from odd extension. Since the operator contains $-\Pi_a^+$, its reflected prime contribution has the opposite sign to its ordinary translations. The archimedean remainder kernel in the half-interval is $-\alpha(x-y) + \alpha(x+y)$; the polar kernel is $-4 \sinh(x/2) \sinh(y/2)$; the regularizer is $\tau_R \sum_j h_j(x) \overline{h_j(y)}$. Away from the diagonal the logarithmic and archimedean kernels combine to

$$-\kappa(|x-y|) + \kappa(x+y), \quad \kappa(t) = \frac{e^{-t/2}}{1 - e^{-2t}}. \quad (\text{CP18:1.15})$$

The diagonal logarithmic finite part remains the one inherited from $\mathbf{C}_a$. Thus (1.13)-(1.15) specify the normalization and every contribution needed in the subsequent collar blocks.

1.3. Exact sign, inertia, and nullspace equivalences

Define

$$W_a = \mathcal{J}_a^{-1} F_a, \quad \widehat{K}_a = \tau_R F_a^* \mathcal{J}_a^{-1} F_a, \quad M_a = I_{d_R} - \widehat{K}_a. \quad (\text{CP18:1.16})$$

Here $\widehat{K}_a \succeq 0$, and W_a maps the finite coefficient space into $\text{Dom } A_a^-$. Set $B_a = \sqrt{\tau_R} \mathcal{J}_a^{-1/2} F_a$. Then $B_a^* B_a = \widehat{K}_a$, and on the full closed form domain

$$A_a^- = \mathcal{J}_a^{1/2} (I - B_a B_a^*) \mathcal{J}_a^{1/2} \quad \text{as a form congruence.} \quad (\text{CP18:1.17})$$

The nonzero eigenvalues of $B_a B_a^*$ are exactly those of $B_a^* B_a$, with multiplicities. The orthogonal complement of $\text{ran } B_a$ contributes the positive identity. The form isomorphism $q \mapsto \mathcal{J}_a^{1/2} q$ is onto $\mathcal{H}_a$, since $\mathcal{J}_a \geq g_R I$. Consequently

$$\begin{aligned} A_a^- \succeq 0 &\iff M_a \succeq 0, \\ A_a^- > 0 &\iff M_a > 0, \\ n_-(A_a^-) &= n_-(M_a), \quad \dim \ker A_a^- = \dim \ker M_a. \end{aligned} \tag{CP18:1.18}$$

Strict positivity denotes a positive spectral lower bound. Indeed, if $\varepsilon_a = \lambda_{\min}(M_a) > 0$, then $\varepsilon_a \leq 1$ and (1.17) gives

$$A_a^- \geq g_R \varepsilon_a I. \tag{CP18:1.19}$$

Conversely, if $A_a^- \geq mI > 0$, equations (1.9) and (1.8) give $\mathcal{J}_a \leq A_a^- + \tau_R I \leq (1 + \tau_R/m) A_a^-$. Congruence by $\mathcal{J}_a^{-1/2}$ therefore yields

$$M_a \succeq \frac{m}{m + \tau_R} I. \tag{CP18:1.20}$$

In particular the retained prime-active certificate implies the explicit starting gap

$$M_a \succeq \frac{m_0}{m_0 + \tau_R} I \quad (0 < a \leq a_0). \tag{CP18:1.21}$$

For clarity, the nullspace correspondence can also be written without a congruence. Its mutually inverse maps are

$$\ker M_a \longrightarrow \ker A_a^-, \quad c \longmapsto W_a c, \quad \ker A_a^- \longrightarrow \ker M_a, \quad v \longmapsto \tau_R F_a^* v. \tag{CP18:1.22}$$

If $M_a c = 0$, then $c = \widehat{K}_a c = \tau_R F_a^* W_a c$, so $W_a c \neq 0$ when $c \neq 0$. Conversely, $A_a^- v = 0$ implies $\mathcal{J}_a v = \tau_R F_a F_a^* v$, hence $v = W_a(\tau_R F_a^* v)$. If $F_a^* v = 0$, the positive invertible regularized equation would give $v = 0$. These observations prove (1.22) and its injectivity in both directions. A coefficient direction in $\ker F_a$ simply has $\widehat{K}_a c = 0$, $M_a c = c$, and $W_a c = 0$; it cannot create a spurious null vector.

No condition involving $\langle s_a, v \rangle$ was used. Thus both the zero-polar-moment null branch and the nonzero-moment branch are detected. At a first crossing from positivity, $M_a \succeq 0$, so a nontrivial nullspace is equivalent to $\varepsilon_a = 0$. Away from that nonnegativity hypothesis, the general condition is $0 \in \text{spec } M_a$, which need not be its smallest eigenvalue.

1.4. The exact forced energy and its nonvanishing near-null family

The definitions give the operator identity

$$A_a^- W_a = F_a M_a, \quad F_a^* W_a = \frac{\widehat{K}_a}{\tau_R}. \tag{CP18:1.23}$$

Since $M_a = I - \widehat{K}_a$, the two matrices commute. For every coefficient vector $c \in \mathbb{C}^{d_R}$, the full arithmetic energy is exactly

$$Q_a^-(W_a c) = \frac{1}{\tau_R} c^* \widehat{K}_a M_a c. \tag{CP18:1.24}$$

No kernel term is estimated or discarded in (1.24). It follows either by testing the first equation in (1.23) against $W_a c$, or directly from the congruence (1.17).

If $\|c\| = 1$ and $M_a c = \varepsilon_a c$, then

$$\boxed{A_a^- W_a c = \varepsilon_a F_a c, \quad Q_a^-(W_a c) = \frac{\varepsilon_a(1 - \varepsilon_a)}{\tau_R}.} \quad (\text{CP18:1.25})$$

The actual vector remains nonzero near a first null. Specifically, for $\varepsilon_a < 1$, equations (1.9), (1.10), and (1.23) give

$$\frac{1 - \varepsilon_a}{\tau_R} \leq \|W_a c\|_2 \leq \frac{1}{g_R}. \quad (\text{CP18:1.26})$$

The lower bound follows from $\|F_a^* W_a c\| = (1 - \varepsilon_a)/\tau_R$ and $\|F_a^*\| \leq 1$. In particular it does not use an inverse of A_a^- or a reciprocal of its unknown margin.

Group 1 closure. Equations (1.10), (1.18), (1.22), and (1.23)-(1.26) are proved on the stated supported logarithmic form domains, with the explicit constants (1.5). They replace the moving-coordinate regularizer by one physical finite-rank kernel, include every first-null branch, and preserve the original Suzuki normalization. The matrix M_a is positive definite precisely when the complete odd Weil operator is strictly positive.

The next required estimate is a quantitative boundary-energy inequality for this forced family. With the half-interval averaged boundary vector $b_a(L)$ defined in Group 2, its limiting form would be

$$|\beta_a^* c|^2 \leq C_R(a) Q_a^-(W_a c) = \frac{C_R(a)}{\tau_R} c^* \hat{K}_a M_a c, \quad C_R \in L^1([a_0, R]), \quad (\text{CP18:1.27})$$

on the positive branch, with constants independent of its unknown margin. Group 1 establishes the exact right-hand energy and the uniformly invertible physical equation. It does not yet estimate the boundary functional on the left of (1.27).

2. The complete physical matrix: monotonicity, Lipschitz variation, and its boundary derivative

2.1. Fixed physical columns and legitimate collar blocks

Fix $a_0 \leq a < b \leq R$, retain the real orthonormal physical columns $f_1, \dots, f_d$ of F_R , and write $\tau = \tau_R$, $g = g_R$. All inner products and adjoints use the original L^2 realization of the complete Weil operator. The Suzuki metric is unchanged. The positive regularized operators in Group 1 satisfy

$$\mathcal{J}_a = A_a^- + \tau F_a F_a^* \geq gI, \quad F_a = E_a^* F_R, \quad W_a = \mathcal{J}_a^{-1} F_a, \quad \hat{K}_a = \tau F_a^* W_a, \quad M_a = I - \hat{K}_a. \quad (\text{CP18:2.1})$$

No assertion that $F_a^* F_a = I$ is used; the valid bound is $F_a^* F_a \leq I$.

Use the unitary odd-to-half-interval map $q \mapsto q^+ = \sqrt{2}q|_{(0,a)}$. Thus $f_j^+(x) = \sqrt{2}f_j(x)$, $s^+(x) = \sqrt{2}\sinh(x/2)$, and $w_{a,j} = \sqrt{2}(W_a e_j)|_{(0,a)}$. We use the same operator symbols for this unitary realization. Set

$$P_R^{\text{ar}} = \sum_{\ell_n < 2R} b_n, \quad \ell_n = \log n, \quad b_n = \Lambda(n)/\sqrt{n}, \quad p_R^+(x, y) = \sum_{j=1}^d f_j^+(x) f_j^+(y). \quad (\text{CP18:2.2})$$

Every function evaluated outside its support is zero. The full half-interval regularized operator is

$$\begin{aligned} \mathcal{J}_a &= (\log |D|)_{(0,a)} - \log(2\pi)I - K_{\alpha(x-y)} + K_{\kappa(x+y)} - \sum_{\ell_n < 2R} b_n O_{n,a} + \sum_{\ell_n < 2R} b_n F_{n,a} \\ &\quad - 2|s^+\rangle\langle s^+| + \tau K_{p_R^+}, \\ (O_{n,a}q)(x) &= q(x + \ell_n) + \mathbf{1}_{x > \ell_n} q(x - \ell_n), \\ (F_{n,a}q)(x) &= \mathbf{1}_{0 < \ell_n - x < a} q(\ell_n - x). \end{aligned} \tag{CP18:2.3}$$

Here $\kappa(t) = e^{-t/2}/(1 - e^{-2t})$ and $\alpha(t) = \kappa(|t|) - 1/(2|t|)$, with the continuous value $\alpha(0) = 1/4$. Formula (2.3) includes the reflected archimedean kernel as well as the ordinary and reflected prime translations. The polar term and the complete physical regularizer are explicit.

The logarithmic form domain splits under the sharp cut at a :

$$\mathcal{D}_b^+ = \mathcal{D}_{(0,a)} \oplus \mathcal{D}_{(a,b)}. \tag{CP18:2.4}$$

Here the summands consist of zero extensions from their indicated intervals, with the supported logarithmic form norm. A justification is necessary because the split is not a smooth multiplier. On piecewise smooth compact tests the only singular cross kernel is $-1/[2(y-x)]$, $x < a < y$. After $d = a - x$, $h = y - a$, this is the Carleman kernel $-1/[2(d+h)]$, whose L^2 -operator norm is at most $\pi/2$. All other cross terms in (2.3) are bounded integral kernels, finitely many partial translations, or finite-rank kernels. Thus the entire cross block is bounded on L^2 . Piecewise smooth cuts lie in the logarithmic form domain: their possible jumps have Fourier decay $O(|\xi|^{-1})$. For a smooth approximating sequence in the full form domain, the identity between its full form and the sum of its two diagonal forms, plus the bounded cross term, controls both cut form norms after one common semibounded shift. Closedness proves that the cuts converge in their form norms. The converse follows from the same bounded cross estimate. This proves (2.4) with the exact exterior-Dirichlet convention; it introduces no missing interface trace or local boundary condition.

Accordingly, in the form sense and with the associated block operator domains,

$$\mathcal{J}_b = \begin{pmatrix} \mathcal{J}_a & B \\ B^* & C \end{pmatrix}, \quad F_b = \begin{pmatrix} F_a \\ T \end{pmatrix}, \quad Tc = (F_R^+ c)|_{(a,b)}. \tag{CP18:2.5}$$

The cross block has the following complete action:

$$\begin{aligned} (Bz)(x) &= \int_a^b [-\kappa(y-x) + \kappa(x+y) - 2s^+(x)s^+(y) + \tau p_R^+(x,y)] z(y) dy \\ &\quad - \sum_{\ell_n < 2R} b_n \mathbf{1}_{a < x + \ell_n < b} z(x + \ell_n) + \sum_{\ell_n < 2R} b_n \mathbf{1}_{a < \ell_n - x < b} z(\ell_n - x), \quad 0 < x < a. \end{aligned} \tag{CP18:2.6}$$

The collar block C is precisely (2.3) restricted to (a, b) . In particular, it includes the collar polar kernel and the collar regularizer, and does not drop prime packets that become active after enlarging the interval.

2.2. Exact Schur identity and the complete residual

Define

$$S = C - B^* \mathcal{J}_a^{-1} B, \quad \mathcal{R}_{a,b} = T - B^* W_a. \tag{CP18:2.7}$$

The notation $\mathcal{R}_{a,b}$ denotes this finite-column collar residual, not a meromorphic correction from earlier checkpoints. Since $\mathcal{J}_b \geq gI$, the Schur variational identity gives

$$\langle Sz, z \rangle = \inf_u \left\langle \mathcal{J}_b \begin{pmatrix} u \\ z \end{pmatrix}, \begin{pmatrix} u \\ z \end{pmatrix} \right\rangle \geq g \|z\|^2. \quad (\text{CP18:2.8})$$

Thus S^{-1} is legitimate throughout $[a_0, R]$, including any zero-moment or nonzero-moment Weil null branch. The block inverse, applied to all source columns simultaneously, now proves

$$\boxed{\widehat{K}_b - \widehat{K}_a = \tau \mathcal{R}_{a,b}^* S^{-1} \mathcal{R}_{a,b} \succeq 0.} \quad (\text{CP18:2.9})$$

This identity uses one fixed physical regularizer and fixed physical sources. There is no extra derivative of a changing source or projection.

There is also an exact global energy-space interpretation. Equip the supported form space at R with the positive inner product $\mathbf{j}_R$ of $\mathcal{J}_R$, and let $\mathbf{P}_a^j$ be its orthogonal projection onto the closed subspace of functions supported in $(-a, a)$. The weak forced equations give

$$E_a W_a = \mathbf{P}_a^j W_R, \quad \frac{\widehat{K}_b - \widehat{K}_a}{\tau} = [\mathbf{j}_R((E_b W_b - E_a W_a)e_i, (E_b W_b - E_a W_a)e_j)]_{i,j}. \quad (\text{CP18:2.10})$$

The real convention is used in the displayed Gram matrix; complex coefficient forms extend it by polarization. The projection equality follows by testing against every smaller supported form vector, and the Gram equality follows from orthogonality of the increment to the smaller supported space. For increasing supports, their union is dense in the limiting supported form space by the compact smooth core and inward approximation; for decreasing supports, their intersection is precisely the limiting supported space by closedness and support. The corresponding nested orthogonal projections therefore converge strongly from either side. It follows that $a \mapsto E_a W_a$ is strongly continuous in the complete energy space, and hence in $L^2(-R, R)$. At each fixed finite L , the cutoff averaging kernel used in (2.28) is an L^2 function varying continuously by translation in a . Thus $b_a(L)$ is measurable (indeed continuous) in a . This supplies a direct measurability justification for its later almost-everywhere limit without assuming a pointwise boundary trace.

For $y = a + h$, $0 < h < b - a$, write $w_c = W_a^+ c$. The residual is exactly

$$\begin{aligned} (\mathcal{R}_{a,b} c)(y) &= (F_R^+ c)(y) + \int_0^a [\kappa(y-x) - \kappa(y+x)] w_c(x) dx \\ &\quad + \sum_{\ell_n < 2R} b_n \mathbf{1}_{0 < y - \ell_n < a} w_c(y - \ell_n) - \sum_{\ell_n < 2R} b_n \mathbf{1}_{0 < \ell_n - y < a} w_c(\ell_n - y) \\ &\quad + 2s^+(y) \langle s^+, w_c \rangle - \tau (F_R^+ (F_a^* W_a c))(y). \end{aligned} \quad (\text{CP18:2.11})$$

The source and regularizer combine before any estimate:

$$(F_R^+ c)(y) - \tau (F_R^+ (F_a^* W_a c))(y) = (F_R^+ M_a c)(y). \quad (\text{CP18:2.12})$$

This is an exact margin-dependent term in the full residual. Its presence does not yet control the remaining singular archimedean, prime, and polar terms. If the six channels in (2.11) are denoted $\mathcal{R}_\nu$, the right side of (2.9) is the complete matrix

$$\tau \sum_\nu \mathcal{R}_\nu^* S^{-1} \mathcal{R}_\nu + \tau \sum_{\nu < \mu} (\mathcal{R}_\nu^* S^{-1} \mathcal{R}_\mu + \mathcal{R}_\mu^* S^{-1} \mathcal{R}_\nu). \quad (\text{CP18:2.13})$$

Likewise $B^* \mathcal{J}_a^{-1} B = \sum_{\nu, \mu} B_\nu^* \mathcal{J}_a^{-1} B_\mu$, with B_ν the archimedean, ordinary-prime, reflected-prime, polar, and regularizer pieces in (2.6). Every mixed term is retained in the Schur operator and in the residual matrix.

2.3. Uniform boundedness and boundary constants for the actual columns

The regularizer from the earlier certificate uses the first $d = d_R$ reference odd logarithmic eigenfunctions at scale R . The independently derived bound

$$\mathfrak{J}_d = \frac{e^4 \pi^2 d(d+1)(2d+1)}{12}, \quad \sup_x \sum_{j=1}^d |f_j(x)|^2 \leq \mathfrak{J}_d/R \quad (\text{CP18:2.14})$$

gives $|p_R^+(x, y)| \leq 2\mathfrak{J}_d/R$. For arbitrary unit $c \in \mathbb{C}^d$, the physical source has L^2 -norm at most one and supremum at most $F_* := \sqrt{\mathfrak{J}_d/R}$, uniformly after restriction to any $(-a, a)$.

For explicit constants, retain the semigroup and boundary estimates already proved in Checkpoint 17 and set

$$\begin{aligned} \mathfrak{m}_R &= \log(2\pi) + 2P_R^{\text{ar}} + 6R + 8R \sinh^2(R/2) + 4\tau\mathfrak{J}_d, \\ B_\infty &= \frac{e^{\mathfrak{m}_R}}{\sqrt{2g}} + \frac{e^{\mathfrak{m}_R} - 1}{\mathfrak{m}_R} F_*, \\ D_\infty &= \log(2\pi) + 2P_R^{\text{ar}} + 4R + 4R \sinh^2(R/2) + 2\tau\mathfrak{J}_d, \\ L_R &= \max(|\log a_0|, |\log R|), \\ B_\partial &= \sqrt{2}C_{\text{BL}}\{(1 + 2L_R)B_\infty + 2(F_* + D_\infty B_\infty)\}. \end{aligned} \quad (\text{CP18:2.15})$$

The finite constant C_{BL} is the fixed-interval logarithmic boundary embedding constant justified in Checkpoint 17. These formulas define constants independent of the unknown Weil positivity margin; no numerical value for C_{BL} is asserted. The physical regularizer kernel in this round is bounded by the same complete convolution majorant, so the earlier semigroup proof applies unchanged. The actual equation on $(-a, a)$ is

$$L_\Delta W_a c = 2\{F_a c + \log(2\pi)W_a c + \Pi_a W_a c + K_{\alpha,a} W_a c + 2s_a \langle s_a, W_a c \rangle - \tau F_a F_a^* W_a c\}. \quad (\text{CP18:2.16})$$

Every term on its right is bounded using (2.14)-(2.15), including every translated prime argument. Its source/regularizer terms also combine exactly to $F_a M_a c$. The bounded-data boundary theorem therefore applies to the actual solutions. In half-interval coordinates, for all unit c ,

$$\|w_c\|_2 \leq g^{-1}, \quad \|w_c\|_\infty \leq \sqrt{2}B_\infty, \quad |w_c(a - d')| \leq \frac{B}{\sqrt{\log(1/d')}}, \quad B := \sqrt{2}B_\partial, \quad 0 < d' < d_0, \quad (\text{CP18:2.17})$$

where $d_0 = \min(a_0/10, a_0^2, 1/10)$. Real and imaginary parts, or equivalently the corresponding vector bound, give the same estimates for arbitrary complex coefficient vectors. A harmless fixed factor may instead be inserted in C_{BL} if the embedding constant was initially defined only for real data. We henceforth take it to be the complex embedding norm.

Fix once for this finite R

$$0 < \eta < \min(d_0, a_0, e^{-2}), \quad 0 < \delta_0 < \min(\eta^2, \log 2, 1), \quad (\text{CP18:2.18})$$

and use only $0 < \delta \leq \delta_0$, $a + \delta \leq R$. These choices are uniform in a .

2.4. The scaled complete Schur operator and an explicit Lipschitz bound

Put $L = \log(1/\delta)$ and $D_\delta z(x) = \sqrt{\delta}z(a + \delta x)$, $0 < x < 1$. Fourier scaling of the supported logarithmic operator gives

$$D_\delta S D_\delta^{-1} = LI + T_0 + E_{a,\delta}, \quad T_0 = (\log |D|)_{(0,1)} - \log(2\pi)I. \quad (\text{CP18:2.19})$$

In addition to the Schur term, the complete bounded remainder has integral kernel

$$\delta[-\alpha(\delta(x-y)) + \kappa(2a + \delta(x+y)) - 2s^+(a + \delta x)s^+(a + \delta y) + \tau p_R^+(a + \delta x, a + \delta y)] \quad (\text{CP18:2.20})$$

and the reflected prime operators

$$\sum_{2a < \ell_n < 2a+2\delta} b_n \mathbf{1}_{0 < (\ell_n - 2a)/\delta - x < 1} q((\ell_n - 2a)/\delta - x), \quad \text{together with } -D_\delta B^* \mathcal{J}_a^{-1} B D_\delta^{-1}. \quad (\text{CP18:2.21})$$

There are no ordinary prime shifts lying entirely inside a collar of length $\delta < \log 2$. All ordinary cross shifts remain in B and $\mathcal{R}$. The positive kernels $\kappa(x+y)$ and $\tau p_R^+(x, y)$ are retained.

A uniform cross-block bound and full remainder bounds are

$$\begin{aligned} c_* &= \frac{\pi}{2} + \sqrt{R\delta_0} \left(1 + 4 \sinh^2(R/2) + \frac{2\tau \mathfrak{J}_d}{R} \right) + 2P_R^{\text{ar}}, \\ E_* &= \delta_0 \left(1 + \kappa(2a_0) + 4 \sinh^2(R/2) + \frac{2\tau \mathfrak{J}_d}{R} \right) + P_R^{\text{ar}} + c_*^2/g, \\ M_* &= 1 + \log 2 + \delta_0[1 + 4 \sinh^2(R/2)] + P_R^{\text{ar}} + c_*^2/g, \\ \|B\| &\leq c_*, \quad \|E_{a,\delta}\| \leq E_*, \quad S \geq (L - M_*)I. \end{aligned} \quad (\text{CP18:2.22})$$

For the archimedean cross part, κ is positive decreasing, so $|\kappa(y-x) + \kappa(x+y)| \leq \kappa(y-x) \leq 1/[2(y-x)] + 1$. Carleman and the constant-kernel norm give its first two charges. Both prime channels have norm at most P_R^{ar} . The other charges are precisely the polar and regularizer kernel bounds. For the lower bound in (2.22), the positive reflected archimedean and regularizer kernels may be kept favorable; only the bounded negative charges are paid. The reference lower bound is $T_0 \geq -(1 + \log 2)I$.

The exact residual (2.11) has only one singular contribution,

$$\frac{1}{2} \int_0^\eta \frac{w_c(a-d')}{h+d'} dd', \quad y = a + h. \quad (\text{CP18:2.23})$$

Everything else is bounded in absolute value, uniformly for unit c , by

$$\begin{aligned} D_* &= \sqrt{2}F_* \left(1 + \frac{\tau}{g} \right) + \frac{2\sqrt{2} \sinh(R/2) \sqrt{\sinh R - R}}{g} \\ &\quad + \frac{\sqrt{R}}{g} \left[\frac{1}{2\eta} + 1 + \kappa(a_0) \right] + 2\sqrt{2}P_R^{\text{ar}} B_\infty. \end{aligned} \quad (\text{CP18:2.24})$$

The first line accounts for the complete source/regularizer and polar terms; the second retains the nonsingular and distant archimedean contribution and both prime sums. This bound can be sharpened on a particular direction using (2.12), but no positivity-margin factor is needed for the present theorem.

Set $A_* := 3B/2$. The elementary singular integral estimate from Checkpoint 17 yields

$$\|\mathcal{R}_{a,a+\delta}c\|_2^2 \leq \delta[A_*\sqrt{L+1} + D_*]^2 \|c\|^2. \quad (\text{CP18:2.25})$$

For $L \geq \max(1, 2M_*)$, combining (2.9), (2.22), and (2.25) proves

$$0 \preceq \widehat{K}_{a+\delta} - \widehat{K}_a \preceq C_{\text{Lip}} \delta I, \quad C_{\text{Lip}} := 2\tau(\sqrt{2}A_* + D_*)^2. \quad (\text{CP18:2.26})$$

All constants are uniform on $[a_0, R]$. Any larger increment is partitioned into sufficiently short ones. Summing the Loewner inequalities gives the global result

$$0 \preceq \widehat{K}_b - \widehat{K}_a \preceq C_{\text{Lip}}(b-a)I, \quad a_0 \leq a \leq b \leq R. \quad (\text{CP18:2.27})$$

Thus the entire finite matrix is Lipschitz and absolutely continuous, independently of either Weil null branch. This is stronger than the fixed-reference matrix continuity in Checkpoint 17, because the physical regularizer and physical sources are exactly compatible under restriction.

2.5. Polarized boundary coefficient, its uniform remainder, and the almost-everywhere derivative

For $L > \log(1/\eta)$, define the real averaged boundary vector by its column entries

$$b_{a,j}(L) = \frac{1}{2\sqrt{L}} \int_{e^{-L}}^{\eta} \frac{w_{a,j}(a-d')}{d'} dd', \quad b_a(L) = (b_{a,1}(L), \dots, b_{a,d}(L))^T. \quad (\text{CP18:2.28})$$

The directional form of (2.17) gives $\|b_a(L)\| \leq B$. Let $Q_{a,\delta} : \mathbb{C}^d \rightarrow L^2(0,1)$ be

$$(Q_{a,\delta}c)(x) = \frac{(\mathcal{R}_{a,a+\delta}c)(a+\delta x)}{\sqrt{L}}.$$

The singular cross-kernel calculation, with every other residual channel bounded by (2.24), gives the operator-norm bound

$$\|Q_{a,\delta} - \mathbf{1} b_a(L)^*\| \leq e_R(\delta), \quad e_R(\delta) := \frac{B(\sqrt{2} + \frac{1}{2} \log 2)}{L} + \frac{B\sqrt{\delta}}{2\sqrt{L} \log(1/\eta)} + \frac{D_*}{\sqrt{L}}, \quad \delta < \eta^2. \quad (\text{CP18:2.29})$$

All columns are real, so transpose and adjoint agree for $b_a(L)$; the displayed operator formula also preserves arbitrary complex coefficient pairings. A fixed complex change of source basis transforms the resulting Gram matrices by the usual unitary conjugation.

The constant function belongs to $\text{Dom}(T_0)$, and

$$T_0 \mathbf{1} = -\gamma_E - \log(2\pi) - \frac{1}{2} \log(x(1-x)), \quad \|T_0 \mathbf{1}\|_2 \leq C_T := \gamma_E + \log(2\pi) + \sqrt{2 - \pi^2/12}. \quad (\text{CP18:2.30})$$

The resolvent calculation on this function, and the complete block identity (2.9), prove

$$\frac{\widehat{K}_{a+\delta} - \widehat{K}_a}{\delta} = \tau b_a(L) b_a(L)^* + \mathfrak{E}_{a,\delta}, \quad L = \log(1/\delta), \quad (\text{CP18:2.31})$$

with the fully uniform matrix remainder

$$\|\mathfrak{E}_{a,\delta}\| \leq \tau \left[\frac{L}{L - M_*} (2B e_R(\delta) + e_R(\delta)^2) + \frac{B^2(C_T + E_*)}{L - M_*} \right] = O_R(L^{-1/2}). \quad (\text{CP18:2.32})$$

This is the complete physical matrix variation. No source-scaling or regularizer-variation remainder remains outside it. The estimate keeps all off-diagonal coefficient pairings.

By (2.27), $\widehat{K}'_a$ exists at almost every $a \in (a_0, R)$. At such an a , (2.31) implies

$$b_a(L) b_a(L)^* \longrightarrow \tau^{-1} \widehat{K}'_a. \quad (\text{CP18:2.33})$$

The limit is positive semidefinite and has rank at most one, since the finite-dimensional set of such matrices is closed. The actual real vector $b_a(L)$ is continuous in L . If the matrix limit is zero, its trace gives $b_a(L) \rightarrow 0$. If it equals $\lambda e e^T$, $\lambda > 0$, the component $e^T b_a(L)$ has absolute value tending to $\sqrt{\lambda}$. It is eventually nonzero and therefore has one fixed sign by continuity, while the perpendicular component tends to zero. Consequently the vector itself converges. This proves the existence of the averaged vector trace

$$\beta_a := \lim_{L \rightarrow \infty} b_a(L) \quad \text{for almost every } a \in (a_0, R), \quad \boxed{\widehat{K}'_a = \tau \beta_a \beta_a^* \quad \text{almost everywhere.}} \quad (\text{CP18:2.34})$$

The limit along integer L gives a measurable version; define it arbitrarily on the null exceptional set. Absolute continuity now gives the exact integrated identity

$$\boxed{\widehat{K}_b - \widehat{K}_a = \tau \int_a^b \beta_u \beta_u^* du, \quad M_b - M_a = -\tau \int_a^b \beta_u \beta_u^* du.} \quad (\text{CP18:2.35})$$

In particular $\tau \|\beta_a\|^2 \leq C_{\text{Lip}}$ almost everywhere. This is an absolute estimate independent of the positivity margin.

The normalization in (2.28)-(2.35) is the unitary half-interval normalization. If β_a^{full} is defined using the original full odd columns $W_a e_j$ at $a - d'$, then

$$\beta_a = \sqrt{2} \beta_a^{\text{full}}, \quad \widehat{K}'_a = 2\tau \beta_a^{\text{full}} (\beta_a^{\text{full}})^*. \quad (\text{CP18:2.36})$$

Existence of these averaged limits almost everywhere does not assert existence of the pointwise normalized trace $\sqrt{\log(1/d')} w_{a,j}(a - d')$.

2.6. Threshold completeness and the exact remaining estimate

All estimates above use the common finite index set $\ell_n < 2R$, with zero extensions and the exact ordinary/reflected indicators in (2.3), (2.6), and (2.11). They therefore include passage through $2a = \ell_n$, through $a = \ell_n$, and through every collar support interface. The cross translations are bounded partial isometries even at threshold entry. Their signs are retained in the exact formulas, and both channels are charged in the uniform remainder. Endpoint equalities have measure-zero support and require no extra Dirac boundary term. The complete matrix is Lipschitz through the thresholds; a classical derivative at each individual threshold is neither needed nor asserted.

Theorem proved. For the independently positive physical regularizer $\mathcal{J}_R \geq g_R I$, its form restrictions on every $a \in [a_0, R]$, and the fixed real source columns above, the complete Schur identity (2.9), global matrix Lipschitz estimate (2.27), uniform polarized boundary formula (2.31)-(2.32), averaged trace almost everywhere (2.34), and integrated identity (2.35) hold. All constants are given by (2.14)-(2.26), using the finite margin-independent boundary constant C_{BL} . Both possible Weil null branches are included because $\mathcal{J}_a^{-1}$ remains bounded throughout.

Consequence for the closing calculation. The full matrix now has the required absolute continuity, and its entire rate is the rank-one boundary energy in (2.34). The source and regularizer already supply the exact factor M_a in (2.12). The remaining arithmetic task is to control the other channels in (2.11) so that the surviving boundary vector acquires that factor in energy. In particular, on a nonnegative Weil interval the missing sufficient inequality is

$$|\beta_a^* c|^2 \leq C_R(a) Q_a^-(W_a c) = \frac{C_R(a)}{\tau} c^* \widehat{K}_a M_a c, \quad \text{all } c \in \mathbb{C}^d, \quad C_R \in L^1([a_0, R]), \quad (\text{CP18:2.37})$$

with C_R independent of the unknown margin. What is proved here is only

$$|\beta_a^* c|^2 \leq B^2 \|c\|^2 \quad \text{and} \quad \tau |\beta_a^* c|^2 \leq C_{\text{Lip}} \|c\|^2. \quad (\text{CP18:2.38})$$

The complete signed residual in (2.11)-(2.12), with its polar and both prime channels, is the precise expression on which the stronger energy-dependent estimate must act. Generic boundary regularity and the new absolute-continuity theorem do not supply that estimate by themselves.

Independent local continuation corollary. The certified starting interval gives $M_{a_0} \succeq \eta_0 I$, where $\eta_0 = m_0/(m_0 + \tau)$ and $m_0 = 1/5000$. Consequently (2.27) implies

$$\begin{aligned} M_a &\succeq [\eta_0 - C_{\text{Lip}}(a - a_0)]I, \\ \delta_R &:= \min \left\{ R - a_0, \frac{\eta_0}{2C_{\text{Lip}}} \right\} > 0, \\ A_a^- &\succeq \frac{g_R \eta_0}{2} I \quad (a_0 \leq a \leq a_0 + \delta_R). \end{aligned} \quad (\text{CP18:2.39})$$

For the last implication use the exact finite-rank congruence from Group 1: if $M_a \succeq \eta I$ with $0 < \eta \leq 1$, then $A_a^- \succeq \eta \mathcal{J}_a \succeq \eta g_R I$. This corollary excludes both null branches on one genuine additional interval, with constants independent of an unknown positivity margin. The finite boundary constant C_{BL} in C_{Lip} has not been numerically evaluated, so (2.39) is a specified analytic extension and does not replace a_0 as the largest previously certified numerical endpoint. It does not establish a continuation budget to every finite support.

3. The endpoint-to-boundary operation and explicit complete probes

3.1. The exact endpoint row, including its explicit adjoint probe

All formulas in this subsection use the physical odd space $\mathcal{H}_a = L_{\text{odd}}^2(-a, a)$. The columns of F_a and $W_a = \mathcal{J}_a^{-1} F_a$ are chosen real; complex combinations are obtained by sesquilinear extension. Write $M_a = I - \hat{K}_a$, so the complete operator equation is

$$A_a^- W_a = F_a M_a. \quad (\text{CP18:3.1})$$

Define the odd function $j_a(x) = x/2$ on $(-a, a)$, extended by zero. This function belongs to $\text{Dom } A_a^-$. Indeed its Fourier transform is $O_a((1 + |\xi|)^{-1})$, so its whole-line logarithmic graph norm is finite. The remaining terms of A_a^- are bounded on this fixed support.

For $t > 0$, retain the complete screw kernel

$$\begin{aligned} g(t) &= -8(\cosh(t/2) - 1) + \sum_{\ell_n < t} b_n(t - \ell_n) + G_{\text{ar}}(t), \\ G_{\text{ar}}(t) &= -\frac{t}{2}(\psi(1/4) - \log \pi) - \frac{1}{4} \left[\Phi(1, 2, 1/4) - e^{-t/2} \Phi(e^{-2t}, 2, 1/4) \right], \\ G'_{\text{ar}}(t) &= -\frac{1}{2} \left[\psi(1/4) - \log \pi + e^{-t/2} \Phi(e^{-2t}, 1, 1/4) \right]. \end{aligned} \quad (\text{CP18:3.2})$$

Here g is even, $\ell_n = \log n$, and $b_n = \Lambda(n)/\sqrt{n}$. Put

$$k_a(y) = g(a - y) - ag'(a - y), \quad k_a^-(y) = \frac{k_a(y) - k_a(-y)}{2}. \quad (\text{CP18:3.3})$$

Two distributional integrations by parts, including both interval endpoints, give the exact adjoint identity

$$\boxed{A_a^- j_a = k_a^-} \quad (\text{CP18:3.4})$$

For example,

$$-\frac{1}{2} \int_{-a}^a g''(x-y)y \, dy = \frac{1}{2} \{g(a-x) - ag'(a-x) - g(a+x) + ag'(a+x)\}. \quad (\text{CP18:3.5})$$

The integral is distributional before passage to its L^2 representative. The logarithmic singularities of g' , and its finitely many prime jumps, are all in L^2 on the fixed interval. Thus (3.5) also verifies the operator-domain assertion directly.

Let $p = W_a c$ and $h = F_a M_a c$. The convolution $f = g * p$ satisfies $f'' = -h$ on $(-a, a)$. Its legitimate continuous convolution traces give

$$f(a) - af'(a) = \int_0^a yh(y) \, dy = \langle j_a, F_a M_a c \rangle. \quad (\text{CP18:3.6})$$

This is also the consequence of (3.4), self-adjointness, and (3.1):

$$\boxed{W_a^* k_a^- = M_a F_a^* j_a} \quad (\text{CP18:3.7})$$

In particular $\|F_a^* j_a\| \leq \|j_a\| = a^{3/2}/\sqrt{6}$, because $F_a^* F_a \leq I$.

Expanding (3.6) before estimation gives the complete endpoint moment. With

$$\begin{aligned} \mathcal{P}_a(p) &= \sum_{\ell_n < 2a} b_n \int_{-a}^{a-\ell_n} (-y - \ell_n) p(y) \, dy \\ &\quad + \int_{-a}^a [G_{\text{ar}}(a-y) - aG'_{\text{ar}}(a-y)] p(y) \, dy, \\ d_a &= 4a \cosh(a/2) - 8 \sinh(a/2), \quad \mathcal{F}_p(z) = \int_{-a}^a e^{zy} p(y) \, dy, \end{aligned} \quad (\text{CP18:3.8})$$

one obtains

$$\boxed{\mathcal{P}_a(W_a c) - d_a \mathcal{F}_{W_a c}(1/2) = \langle j_a, F_a M_a c \rangle} \quad (\text{CP18:3.9})$$

Every prime packet, the digamma and Lerch contributions, and the completed polar moment remain in this row. This is an actual zero-residual adjoint probe for the endpoint moment. Its image is k_a^- , whose endpoint singularity is logarithmic; the requested averaged boundary functional has a reciprocal-distance kernel.

3.2. The operation which converts the endpoint kernel into the boundary kernel

Fix once and for all $0 < \eta < \min(a_0/2, \log 2, 1/10)$, and let $\epsilon = e^{-L} < \eta$. For a physical odd p , its half-interval representative is $p^+(x) = \sqrt{2}p(x)$, and the retained averaged boundary functional is

$$\mathfrak{b}_{a,L}(p) = \frac{1}{2\sqrt{L}} \int_{\epsilon}^{\eta} \frac{p^+(a-d)}{d} \, dd. \quad (\text{CP18:3.10})$$

Its representing physical odd function is

$$g_{a,L}(x) = \frac{\text{sgn } x}{2\sqrt{2L}(a-|x|)} \mathbf{1}_{\{\epsilon < a-|x| < \eta\}}, \quad \|g_{a,L}\|^2 = \frac{e^L - \eta^{-1}}{4L}. \quad (\text{CP18:3.11})$$

Thus $b_a(L) = W_a^* g_{a,L}$, with the usual conjugation for complex columns, has exactly the normalization in the matrix collar formula.

The missing operation can be calculated explicitly. Holding y fixed,

$$\partial_a k_a(y) = -ag''(a - y). \quad (\text{CP18:3.12})$$

For $0 < d < \eta < \log 2$, no prime delta lies in this local range, and

$$g''(d) = \kappa(d) - 2 \cosh(d/2) = \frac{1}{2d} + R(d), \quad R(d) = \alpha(d) - 2 \cosh(d/2). \quad (\text{CP18:3.13})$$

Consequently the exact truncated relation is

$$\boxed{\begin{aligned} b_{a,L}(p) = & -\frac{\sqrt{2}}{a\sqrt{L}} \int_{a-\eta}^{a-\epsilon} \partial_a k_a(y) p(y) dy \\ & -\frac{\sqrt{2}}{\sqrt{L}} \int_{\epsilon}^{\eta} R(d) p(a-d) dd. \end{aligned}} \quad (\text{CP18:3.14})$$

The second line has absolute value at most $\sqrt{2}\eta \sup_{[0,\eta]} |R| \|p\|_{\infty} / \sqrt{L}$. This is a quantitative link between the complete endpoint kernel and the averaged boundary vector.

It does not permit differentiation of (3.9) while keeping only its explicit right-hand side. That identity also contains W_a , whose support and values vary with a . The derivative of the zero-extended endpoint probe j_a contains endpoint delta distributions. Those distributions are not in the operator domain. Before cancellation, the differentiated kernel has the nonintegrable $1/(a-y)$ singularity in (3.14). The required additional calculation is a legitimate adjoint approximation to this derivative, with a controlled complete residual on the range of W_a .

This can also be seen directly in the complete collar residual. For $y > a$, the physical-source cancellation gives, on the half interval,

$$\begin{aligned} r_c(y) = & f_R^+(y) M_a c + 2s^+(y) \langle s^+, W_a^+ c \rangle \\ & + \int_0^a [\kappa(y-x) - \kappa(y+x)] (W_a^+ c)(x) dx \\ & + \sum_{\ell_n < 2R} b_n \mathbf{1}_{(0,a)}(y - \ell_n) (W_a^+ c)(y - \ell_n) \\ & - \sum_{\ell_n < 2R} b_n \mathbf{1}_{(0,a)}(\ell_n - y) (W_a^+ c)(\ell_n - y). \end{aligned} \quad (\text{CP18:3.15})$$

Here f_R^+ is the row of fixed physical source functions and $s^+(x) = \sqrt{2} \sinh(x/2)$. The regularizer-source contribution has combined exactly into $M_a c$. Each displayed bounded term, including every reflected prime threshold, is $O_R(\|c\|)$. After division by $\sqrt{L}$, the singular part of the integral is the surviving $b_a(L)^* c$, up to the already quantified collar error. Thus the finite-rank forcing acquires the desired factor M_a ; the leading boundary coefficient still requires control of the complete signed arithmetic equation, rather than another estimate of those bounded terms.

3.3. A completely explicit boundary-layer derivative probe

The differentiation suggested by (3.14) produces a natural probe without any inverse of A_a^- . Set $I = (a - \epsilon, a)$,

$$c_L = \frac{1}{\sqrt{2L\epsilon}}, \quad \psi_{a,L}(x) = -c_L [\mathbf{1}_I(x) - \mathbf{1}_{-I}(x)]. \quad (\text{CP18:3.16})$$

For $\epsilon < a_0/2$, this is odd and belongs to $\text{Dom } A_a^-$: each interval indicator has Fourier decay $O(1/|\xi|)$. Its logarithmic graph norm is finite for every finite L , although it grows with L .

For an interval (u, v) , define the explicit function

$$D_{u,v}(x) = \begin{cases} -\gamma_E - \frac{1}{2} \log((x-u)(v-x)), & u < x < v, \\ \frac{1}{2} \log \frac{u-x}{v-x}, & x < u, \\ \frac{1}{2} \log \frac{x-v}{x-u}, & x > v. \end{cases} \quad (\text{CP18:3.17})$$

The distributional logarithmic operator has $\log |D| \mathbf{1}_{(u,v)} = D_{u,v}$. This follows by inserting the indicator in

$$\log |D| q(x) = \frac{1}{2} \int_{|h|<1} \frac{q(x) - q(x+h)}{|h|} dh - \frac{1}{2} \int_{|h|\geq 1} \frac{q(x+h)}{|h|} dh - \gamma_E q(x), \quad (\text{CP18:3.18})$$

with the interior integral taken in its legitimate principal-value sense. In particular every term of the probe image and residual is explicit:

$$\begin{aligned} A_a^- \psi_{a,L}(x) &= -c_L [D_{a-\epsilon,a}(x) - D_{-a,-a+\epsilon}(x)] - \log(2\pi) \psi_{a,L}(x) \\ &\quad - \sum_{\ell_n < 2R} b_n [\psi_{a,L}(x + \ell_n) + \psi_{a,L}(x - \ell_n)] \\ &\quad + c_L \int_I [\alpha(x-y) - \alpha(x+y)] dy \\ &\quad + 8c_L \sinh(x/2) [\cosh(a/2) - \cosh((a-\epsilon)/2)], \\ e_{a,L}(x) &= g_{a,L}(x) - A_a^- \psi_{a,L}(x). \end{aligned} \quad (\text{CP18:3.19})$$

All arguments of the probe outside its support are zero. The prime sum therefore includes every ordinary and reflected packet without an additional cutoff convention. Equation (3.19) retains the completed polar term, the regular archimedean kernel, the logarithmic boundary contribution, and all their pairings.

The full quadratic form can be evaluated exactly. Put $\Delta_c = \cosh(a/2) - \cosh((a-\epsilon)/2)$ and

$$h_\epsilon(\ell) = (\epsilon - |\ell - (2a - \epsilon)|)_+. \quad (\text{CP18:3.20})$$

For $\epsilon < \log 2$, there are no ordinary prime overlaps within an individual layer. The reflected overlaps yield

$$\begin{aligned} Q_a^-(\psi_{a,L}) &= c_L^2 \left[2\epsilon(L+1 - \gamma_E - \log(2\pi)) + \iint_{I \times I} \frac{dx dy}{x+y} \right. \\ &\quad \left. - 2 \iint_{I \times I} [\alpha(x-y) - \alpha(x+y)] dx dy \right. \\ &\quad \left. + 2 \sum_{\ell_n < 2R} b_n h_\epsilon(\ell_n) - 32\Delta_c^2 \right]. \end{aligned} \quad (\text{CP18:3.21})$$

The reflected prime contribution is nonnegative for this particular odd probe. This sign follows from the opposite signs of its two layers; it is not a positivity assertion for general test functions. All prime thresholds are present in the explicit overlap function h_ϵ .

Since $|\alpha| \leq 1$, $a \geq a_0$, and $|\Delta_c| \leq \epsilon \sinh(R/2)/2$, (3.21) implies the uniform lower bound

$$Q_a^-(\psi_{a,L}) \geq e^L \left(1 + \frac{1 - \gamma_E - \log(2\pi)}{L} \right) - \frac{2 + 4 \sinh^2(R/2)}{L}. \quad (\text{CP18:3.22})$$

The same formula and $0 \leq h_\epsilon \leq \epsilon$ give

$$Q_a^-(\psi_{a,L}) = e^L(1 + O_R(L^{-1})), \quad (\text{CP18:3.23})$$

uniformly through the prime thresholds. In particular this explicit endpoint-derivative probe does not meet the proposed $O_R(\log(eL))$ energy budget.

There is an equally precise calculation of its projected residual. Let

$$\mathcal{F}_R^\infty = \left(\sum_{j=1}^{d_R} \|f_{R,j}\|_\infty^2 \right)^{1/2}. \quad (\text{CP18:3.24})$$

The previously proved boundedness of the finite source family makes this constant finite, independently of the unregularized positivity margin. Because $\|\psi_{a,L}\|_1 = \sqrt{2}/\sqrt{L}$,

$$\|F_a^* \psi_{a,L}\| \leq \frac{\sqrt{2}\mathcal{F}_R^\infty}{\sqrt{L}}. \quad (\text{CP18:3.25})$$

Self-adjointness and the complete equation (3.1) now give

$$\boxed{W_a^* e_{a,L} = b_a(L) - M_a F_a^* \psi_{a,L}.} \quad (\text{CP18:3.26})$$

On a positive branch $0 \preceq M_a \preceq I$, so

$$\|W_a^* e_{a,L} - b_a(L)\| \leq \frac{\sqrt{2}\mathcal{F}_R^\infty}{\sqrt{L}}. \quad (\text{CP18:3.27})$$

For all supports, the same bound holds with the additional harmless factor $1 + \tau_R/g_R$. Thus the actual residual retains the leading averaged boundary coefficient. The estimate $\|W_a^* e_{a,L}\| = O_R(L^{-1/2})$ for this probe would itself require $b_a(L) = O_R(L^{-1/2})$, which has not been established.

Theorem proved and scope. The endpoint row admits the exact zero-residual adjoint probe $j_a = x/2$, and (3.14) identifies the precise derivative operation needed to turn its kernel into the averaged boundary kernel. The explicit operator-domain boundary-layer probe (3.16) has the complete image (3.19), exact energy (3.21), and exact projected residual (3.26). Its energy is exponential in L ; its residual contains $b_a(L)$ to leading order. These are independent calculations for the actual complete Weil operator. They preserve both null branches and do not alter the original Suzuki metric.

Remaining estimate. A different adjoint approximation or an arithmetic cancellation must bound the surviving signed quantity in (3.26), or the complete singular term in (3.15), by a factor vanishing with $Q_a^-(W_a c)$. Equations (3.9) and (3.14) locate the missing operation: one must control the complete differentiated endpoint equation, including the variation of its forced solution and both boundary terms. The endpoint moment identity alone supplies (3.7), not that energy-dependent estimate.

3.4. The exact finite-resolution functional and a directly calculable probe

All formulas in this subsection are in the original physical $L_{\text{odd}}^2(-a, a)$ metric. Put $C_a = (\log|D|)_{(-a,a)}^-$, and retain the complete arithmetic operator

$$A_a^- = C_a - D_a, \quad D_a = \log(2\pi)I + \Pi_a + K_{\alpha,a} + 2|s_a\rangle\langle s_a|, \quad s_a(x) = \sinh(x/2). \quad (\text{CP18:3.28})$$

Here $\Pi_a = \sum_{\ell_n < 2R} b_n(S_{\ell_n} + S_{-\ell_n})$ is compressed using zero extension; the notation $S_h q(x) = q(x+h)$ is used only within this displayed definition. In particular, the two translates of a prime power are both present, including their reflected contributions after odd restriction. The complete even kernel is $\alpha(t) = \kappa(|t|) - 1/(2|t|)$, with $\alpha(0) = 1/4$. A sufficient uniform operator bound is

$$\|D_a\| \leq D_R^{\text{ar}} := \log(2\pi) + 2 \sum_{\ell_n < 2R} b_n + 2R + 2(\sinh R - R), \quad a_0 \leq a \leq R. \quad (\text{CP18:3.29})$$

The α -kernel estimate uses $|\alpha| \leq 1$; preserving its exact signed contribution, rather than this norm bound, remains essential in the residual below.

Choose once and for all $0 < \eta < \min(a_0/2, 1/10)$, $L > \log(1/\eta)$, and $\epsilon = e^{-L}$. In the unitary half-interval representation $q^+(x) = \sqrt{2}q(x)$, define

$$g_{a,L}^+(x) = \frac{\mathbf{1}_{\{\epsilon < a-x < \eta\}}}{2\sqrt{L}(a-x)}. \quad g_{a,L}(x) = 2^{-1/2}g_{a,L}^+(x) \quad (x > 0), \quad g_{a,L}(-x) = -g_{a,L}(x). \quad (\text{CP18:3.30})$$

Thus, for the physical regularized family W_a ,

$$b_a(L) = W_a^* g_{a,L}, \quad b_a(L)^* c = \frac{1}{2\sqrt{L}} \int_{e^{-L}}^{\eta} \frac{(W_a c)^+(a-d)}{d} dd. \quad (\text{CP18:3.31})$$

The half-interval normalization is built into this formula. There is no additional factor of $\sqrt{2}$ to insert in its polarized collar identity.

An explicit first probe, containing no inverse, is

$$\boxed{\psi_{a,L}^{\text{raw}} = L^{-1}g_{a,L}}. \quad (\text{CP18:3.32})$$

For each finite L , this is a compactly supported bounded-variation function, with finitely many jumps and with its support separated from both endpoints. Its Fourier transform is $O_L((1+|\xi|)^{-1})$. Consequently its zero extension lies in the whole-line graph domain of $\log|D|$: both the logarithmic singularity at frequency zero and the squared logarithmic factor at infinity are integrable. Restriction gives $\psi_{a,L}^{\text{raw}} \in \text{Dom } C_a = \text{Dom } A_a^-$. No unproved endpoint trace of a form-domain function is used here.

The complete residual is the following explicitly specified function in $L^2(-a, a)$:

$$\begin{aligned} e_{a,L}^{\text{raw}} &= g_{a,L} - L^{-1}C_a g_{a,L} + L^{-1}\log(2\pi)g_{a,L} \\ &\quad + L^{-1} \sum_{\ell_n < 2R} b_n[g_{a,L}(\cdot + \ell_n) + g_{a,L}(\cdot - \ell_n)] \\ &\quad + L^{-1} \int_{-a}^a \alpha(\cdot - y)g_{a,L}(y) dy + 2L^{-1}s_a \langle s_a, g_{a,L} \rangle. \end{aligned} \quad (\text{CP18:3.33})$$

All translated functions are zero extended. The logarithmic term in (3.33) is the whole-line Fourier multiplier followed by restriction; it includes the boundary contribution of zero extension.

3.5. Actual growth of the raw probe

Write $T = \eta e^L$ and $f_T(y) = y^{-1}\mathbf{1}_{(1,T)}(y)$. Scaling at the right endpoint gives

$$N_L := \|g_{a,L}\|_2^2 = \frac{e^L}{4L}(1 - T^{-1}), \quad \langle C_a g_{a,L}, g_{a,L} \rangle = LN_L + \frac{e^L}{4L}q_T + \mathcal{J}_{a,L}, \quad (\text{CP18:3.34})$$

where

$$q_T = \frac{1}{2\pi} \int_{\mathbb{R}} \log |\xi| |\widehat{f}_T(\xi)|^2 d\xi, \quad \mathcal{J}_{a,L} = \int_0^a \int_0^a \frac{g_{a,L}^+(x)g_{a,L}^+(y)}{2(x+y)} dy dx \geq 0. \quad (\text{CP18:3.35})$$

The second term is the ordinary whole-line logarithmic scaling term. The last term is exactly the odd reflected logarithmic kernel; it has not been dropped. Direct integration by parts gives

$$|\widehat{f}_T(\xi)| \leq \begin{cases} 2 + \log(1/|\xi|), & 0 < |\xi| \leq 1, \\ 3/|\xi|, & |\xi| \geq 1. \end{cases} \quad |q_T| \leq \frac{27}{\pi}, \quad 0 \leq \mathcal{J}_{a,L} \leq \frac{(L + \log \eta)^2}{16L(a - \eta)}. \quad (\text{CP18:3.36})$$

For example, the low-frequency absolute integral in (3.36) is $\int_0^1 (-\log \xi)(2 - \log \xi)^2 d\xi = 18$, and the high-frequency one is 9. This proves the stated uniform bound without numerical quadrature.

Every prime, archimedean, reflected, and polar contribution to the energy is therefore present in the exact formula

$$\begin{aligned} Q_a^-(\psi_{a,L}^{\text{raw}}) &= \frac{N_L}{L} + \frac{e^L}{4L^3} q_T + \frac{\mathcal{J}_{a,L}}{L^2} - \frac{\log(2\pi)}{L^2} N_L \\ &\quad - \frac{1}{L^2} \sum_{\ell_n < 2R} b_n \langle g_{a,L}(\cdot + \ell_n) + g_{a,L}(\cdot - \ell_n), g_{a,L} \rangle \\ &\quad - \frac{1}{L^2} \langle K_{\alpha,a} g_{a,L}, g_{a,L} \rangle - \frac{2}{L^2} |\langle s_a, g_{a,L} \rangle|^2. \end{aligned} \quad (\text{CP18:3.37})$$

In particular, uniformly for $a \in [a_0, R]$,

$$\boxed{Q_a^-(\psi_{a,L}^{\text{raw}}) = \frac{e^L}{4L^2} (1 + O_R(L^{-1}))}. \quad (\text{CP18:3.38})$$

This conclusion uses only the bounded complete perturbation (3.29). It holds regardless of a hypothetical first crossing. Its leading growth is exponential in L , and it does not satisfy the proposed $O_R(\log(eL))$ energy certificate.

Let $\mathfrak{f}_R = \sup_{|x| < R} (2 \sum_{j=1}^{d_R} |F_{R,j}(x)|^2)^{1/2} < \infty$, whose finiteness follows from Checkpoint 17's actual reference-eigenfunction boundedness proof. Then

$$\|F_a^* \psi_{a,L}^{\text{raw}}\| \leq \frac{\mathfrak{f}_R}{2\sqrt{L}}. \quad (\text{CP18:3.39})$$

The exact operator equation $A_a^- W_a = F_a M_a$ gives

$$W_a^* e_{a,L}^{\text{raw}} = b_a(L) - M_a F_a^* \psi_{a,L}^{\text{raw}}, \quad \|W_a^* e_{a,L}^{\text{raw}} - b_a(L)\| \leq \frac{(1 + \tau_R/g_R) \mathfrak{f}_R}{2\sqrt{L}}. \quad (\text{CP18:3.40})$$

Thus this directly calculable probe leaves the leading averaged boundary coefficient in its projected residual. At a parameter where $b_a(L) \rightarrow \beta_a$, its residual converges to β_a , rather than giving an independent proof that this limit is zero. Equation (3.40) is an evaluated obstruction for this specific probe, not a generic counterexample to the xi arithmetic.

3.6. A reference spectral probe closes the bounded-source half of the criterion

A second explicit probe uses only the arithmetic-free logarithmic reference operator. Let c_{odd} denote the certified lower constant for C_1 , so $C_a \geq (-\log a + c_{\text{odd}})I$. Set

$$\lambda_R = 1 + \max\{0, \log R - c_{\text{odd}}\}, \quad P_a^{\text{ref}} = C_a + \lambda_R I \geq I. \quad (\text{CP18:3.41})$$

If $C_1 \varphi_k = \nu_k \varphi_k$ is the fixed odd reference basis and $\varphi_{k,a}(x) = a^{-1/2} \varphi_k(x/a)$, define the convergent series

$$\psi_{a,L}^{\text{ref}} = \sum_{k \geq 1} \frac{\langle \varphi_{k,a}, g_{a,L} \rangle}{\nu_k - \log a + \lambda_R} \varphi_{k,a}. \quad (\text{CP18:3.42})$$

The usual conjugation of coefficients is understood according to the inner-product convention; equivalently this is $(P_a^{\text{ref}})^{-1} g_{a,L}$. It belongs to $\text{Dom } C_a$, because $P_a^{\text{ref}} \psi_{a,L}^{\text{ref}} = g_{a,L} \in L^2$. This inverse contains no Weil operator, prime packet, polar subtraction, or unknown positivity margin.

The complete residual of (3.42) is

$$e_{a,L}^{\text{ref}} = B_a \psi_{a,L}^{\text{ref}}, \quad B_a = (\lambda_R + \log(2\pi))I + \Pi_a + K_{\alpha,a} + 2|s_a\rangle\langle s_a|. \quad (\text{CP18:3.43})$$

This is an exact identity, with $\|B_a\| \leq D_R := \lambda_R + D_R^{\text{ar}}$.

For every unit coefficient vector c , the reference forced solution $u_c = (P_a^{\text{ref}})^{-1} F_a c$ has a uniform L^∞ bound and bounded complete logarithmic forcing. Checkpoint 17's semigroup and logarithmic boundary argument applies with the arithmetic perturbation omitted and the independently known reference gap 1. Consequently, for one finite B_R^{ref} independent of the Weil margin,

$$|u_c^+(a-d)| \leq \frac{B_R^{\text{ref}}}{\sqrt{\log(1/d)}} \quad (0 < d < \eta), \quad \sup_{a \in [a_0, R], L} \|F_a^* \psi_{a,L}^{\text{ref}}\| \leq B_R^{\text{ref}}. \quad (\text{CP18:3.44})$$

Indeed, pairing u_c with $g_{a,L}$ bounds the scalar by $B_R^{\text{ref}}(1 - \sqrt{\log(1/\eta)/L})$. This proves the bounded-source part of the proposed strong probe criterion, uniformly on the whole finite support range.

The associated energy does not satisfy the weaker logarithmic certificate. Here is a quantitative lower bound which does not assume $A_a^- \geq 0$. Let $E_L = \langle g_{a,L}, (P_a^{\text{ref}})^{-1} g_{a,L} \rangle$. For sufficiently large L , (3.34)-(3.36) give

$$\langle g_{a,L}, P_a^{\text{ref}} g_{a,L} \rangle \leq (L + C_R^{\text{ref}})N_L, \quad E_L \geq \frac{N_L}{L + C_R^{\text{ref}}} \geq \frac{e^L}{8L(L + C_R^{\text{ref}})}, \quad (\text{CP18:3.45})$$

where, for example, $C_R^{\text{ref}} = \lambda_R + 54/\pi + 2/[e^2(a_0 - \eta)]$ suffices once $L \geq \max(2, \log(2/\eta))$. The middle inequality is Cauchy-Schwarz in the positive reference metric.

Put $\Lambda_R = 2D_R + 1$ and let $P_{a, \leq \Lambda_R}^{\text{ref}}$ be the reference spectral projection below this fixed level. There are finitely many such modes uniformly for $a \in [a_0, R]$. Their endpoint upper bounds imply

$$\|P_{a, \leq \Lambda_R}^{\text{ref}} \psi_{a,L}^{\text{ref}}\|^2 \leq B_{R, \text{low}}^2 < \infty, \quad \|\psi_{a,L}^{\text{ref}}\|^2 \leq B_{R, \text{low}}^2 + \Lambda_R^{-1} E_L. \quad (\text{CP18:3.46})$$

One explicit specification of $B_{R, \text{low}}^2$ is the sum of squared uniform boundary constants for the finitely many scaled reference eigenfunctions with $\nu_k \leq \Lambda_R + \log R - \lambda_R$; denominators in (3.42) are at least one. This definition uses a fixed reference spectrum and has no unknown Weil gap. It follows that

$$Q_a^-(\psi_{a,L}^{\text{ref}}) \geq \frac{e^L}{16L(L + C_R^{\text{ref}})} - D_R B_{R, \text{low}}^2. \quad (\text{CP18:3.47})$$

Thus the explicitly constructed reference probe closes (3.44), while its energy still has a quantified exponentially growing lower bound.

3.7. The surviving complete arithmetic residual

For a vector-valued family Y , write $\mathfrak{b}_{a,L}(Y) = Y^* g_{a,L}$. Then the projected residual of the reference probe is the following specific expression in actual forced solutions:

$$\begin{aligned} W_a^* e_{a,L}^{\text{ref}} &= (\lambda_R + \log(2\pi)) \mathfrak{b}_{a,L}((P_a^{\text{ref}})^{-1} W_a) \\ &\quad + \sum_{\ell_n < 2R} b_n \mathfrak{b}_{a,L}((P_a^{\text{ref}})^{-1} [W_a(\cdot + \ell_n) + W_a(\cdot - \ell_n)]) \\ &\quad + \mathfrak{b}_{a,L}((P_a^{\text{ref}})^{-1} K_{\alpha,a} W_a) \\ &\quad + 2 W_a^* s_a \langle s_a, (P_a^{\text{ref}})^{-1} g_{a,L} \rangle. \end{aligned} \tag{CP18:3.48}$$

The order of conjugation in the last line is as written for $W_a^* e$; for real chosen sources all entries are real. Every displayed translated function is zero extended before applying the reference inverse. Formula (3.48) therefore includes ordinary prime shifts, reflected prime shifts, all support thresholds, the complete archimedean correction, and the full polar interaction.

To justify a uniform bound, the actual solutions $W_a c$ are bounded by the positive regularized inverse estimate. The complete forcing $\mathbf{B}_a W_a c$ is bounded uniformly for unit c , since the finite prime shifts preserve L^∞ , $|\alpha| \leq 1$, and the polar rank one has a bounded kernel on $(-R, R)^2$. The reference solution $(P_a^{\text{ref}})^{-1} \mathbf{B}_a W_a c$ consequently has the same proven boundary upper estimate. Hence

$$\sup_{a \in [a_0, R], L} \|W_a^* e_{a,L}^{\text{ref}}\| \leq B_R^{\text{rem}} < \infty. \tag{CP18:3.49}$$

The argument proves boundedness, with no decay factor in L and no vanishing factor in M_a .

For clarity, if the four rows of (3.48) are denoted $\eta_0, \eta_{\text{pr}}, \eta_{\text{ar}}, \eta_{\text{pol}}$, then the unresolved squared residual is exactly

$$\begin{aligned} \|W_a^* e_{a,L}^{\text{ref}}\|^2 &= \|\eta_0\|^2 + \|\eta_{\text{pr}}\|^2 + \|\eta_{\text{ar}}\|^2 + \|\eta_{\text{pol}}\|^2 \\ &\quad + 2\Re(\langle \eta_0, \eta_{\text{pr}} \rangle + \langle \eta_0, \eta_{\text{ar}} \rangle + \langle \eta_0, \eta_{\text{pol}} \rangle + \langle \eta_{\text{pr}}, \eta_{\text{ar}} \rangle + \langle \eta_{\text{pr}}, \eta_{\text{pol}} \rangle + \langle \eta_{\text{ar}}, \eta_{\text{pol}} \rangle). \end{aligned} \tag{CP18:3.50}$$

The prime vector itself is the full finite sum in (3.48), so its squared norm includes every prime-prime cross term. Bounding the four vectors separately yields (3.49); obtaining a smaller signed sum requires a new identity or estimate for this actual arithmetic expression.

The general exact probe identity is

$$b_a(L) = M_a F_a^* \psi_{a,L} + W_a^* e_{a,L}. \tag{CP18:3.51}$$

For the reference probe, (3.44) closes the first quantitative bound. The precise missing statement for this probe is

$$\lim_{L \rightarrow \infty} \left\| \mathfrak{b}_{a,L}((P_a^{\text{ref}})^{-1} [(\lambda_R + \log(2\pi)) W_a + \Pi_a W_a + K_{\alpha,a} W_a + 2s_a \langle s_a, W_a \rangle]) \right\| = 0, \tag{CP18:3.52}$$

with an appropriately uniform or integrably controlled version in a . The established bound on its left side is B_R^{rem} , not a term tending to zero. This is a concrete complete residual available for the next calculation.

If (3.52) were proved, boundedness in (3.44) would imply, at every parameter with the existing averaged trace, $\beta_a = M_a h_a$ for some $\|h_a\| \leq B_R^{\text{ref}}$. Thus

$$|\beta_a^* c|^2 \leq (B_R^{\text{ref}})^2 c^* M_a^2 c. \quad (\text{CP18:3.53})$$

For a minimum-margin eigenvector with $0 < \epsilon_a \leq 1/2$, this is the sufficient bound $|\beta_a^* c|^2 \leq (B_R^{\text{ref}})^2 \epsilon_a^2$. A full comparison of M_a^2 with $\widehat{K}_a M_a$ additionally requires attention to small eigenvalues of $\widehat{K}_a$; it is not automatic merely from $0 \leq \widehat{K}_a \leq I$. The minimum-eigenvalue argument near a first crossing needs no such additional comparison, because its $\widehat{K}_a$ -eigenvalue is $1 - \epsilon_a \geq 1/2$.

Proved theorem and scope. The two probes (3.32) and (3.42) lie in the actual operator domain. Their complete residuals, exact projected identities, exponential energy growth, and uniform reference source-pairing bound are established above. These are explicit calculations on the complete arithmetic operator. The raw probe leaves β_a in its projected residual. The reference probe reduces the remaining problem to the signed arithmetic boundary average (3.52). Neither calculation proves the required residual decay or the energy-relative boundary estimate. Therefore these probes alone do not exclude a finite first crossing or establish unconditional Weil positivity.

3.8. The surviving signed residual has an almost-everywhere limit

There is a further unconditional conclusion concerning the complete residual, even though its vanishing remains open. Choose the physical reference columns F_R real, as is possible for the real logarithmic operator used in the certified regularizer. The reference form $P_a^{\text{ref}} = C_a + \lambda_R I$ is the restriction of the single positive physical form P_R^{ref} . Thus the same collar calculation applies with the fixed physical sources $F_a = E_a^* F_R$, and without an arithmetic or polar perturbation. Define

$$K_a^{\text{ref}} = F_a^* (P_a^{\text{ref}})^{-1} F_a, \quad h_a(L) = F_a^* (P_a^{\text{ref}})^{-1} g_{a,L}. \quad (\text{CP18:3.54})$$

The cross kernel is the complete odd logarithmic kernel, including its reflected term $1/[2(x+y)]$. Its inner-to-collar ordinary term is the same half-Carleman kernel as before. The reference gap is at least one; the collar sources are restrictions of the same bounded physical columns; and (3.44) supplies the required uniform endpoint upper estimate for every forced solution. Consequently the exact positive Schur identity and the polarized collar remainder give

$$0 \preceq K_b^{\text{ref}} - K_a^{\text{ref}} \preceq C_R^{\text{ref, coll}}(b-a)I, \quad \frac{K_{a+\delta}^{\text{ref}} - K_a^{\text{ref}}}{\delta} = h_a(L)h_a(L)^* + O_R(L^{-1/2}), \quad L = \log(1/\delta). \quad (\text{CP18:3.55})$$

Here the finite constant is independent of the unknown Weil positivity margin. The argument first bounds small collars uniformly, then subdivides any interval in $[a_0, R]$. No arithmetic threshold or changing-source term enters this reference calculation.

The finite matrix K_a^{ref} is therefore Lipschitz and differentiable almost everywhere. At each such parameter the real vector $h_a(L)$ is continuous in L , bounded by (3.44), and its outer product converges by (3.55). This implies convergence of the vector itself: when the limiting outer product is zero its norm tends to zero; otherwise that limit has rank one, and continuity prevents an eventual change between its two possible real square-root vectors. Thus

$$h_a := \lim_{L \rightarrow \infty} h_a(L) \quad \text{exists for almost every } a, \quad (K_a^{\text{ref}})' = h_a h_a^*, \quad \|h_a\| \leq B_R^{\text{ref}}. \quad (\text{CP18:3.56})$$

This is convergence of the averaged boundary functional. It asserts no pointwise normalized boundary trace. Using a real basis is material to the sign argument; complex coefficient vectors are then handled by sesquilinear extension.

Intersect this full-measure parameter set with the one on which the actual physical regularized family has the averaged trace $\beta_a = \lim_L b_a(L)$, established by Group 2. The exact identity (3.51) now proves that the complete projected residual itself has a limit:

$$\Gamma_a := \lim_{L \rightarrow \infty} W_a^* e_{a,L}^{\text{ref}} = \beta_a - M_a h_a \quad \text{for almost every } a \in [a_0, R]. \quad (\text{CP18:3.57})$$

It satisfies the already proved uniform bound

$$\|\Gamma_a\| \leq B_R^{\text{rem}}. \quad (\text{CP18:3.58})$$

The limit in (3.57) concerns the entire signed expression (3.48), with all its mixed interactions. This argument does not separately assert convergence of every individual prime, archimedean, or polar summand.

The exact remaining calculation for this reference probe is therefore a bound on the actual vector Γ_a . Proving $\Gamma_a = 0$ almost everywhere would be sufficient, together with the uniform bound on h_a , to supply the directional ϵ_a^2 estimate in (3.53). More generally, for every unit minimum-margin eigenvector c , the exact identity is

$$\beta_a^* c = \epsilon_a h_a^* c + \Gamma_a^* c. \quad (\text{CP18:3.59})$$

On the positive branch, a sufficient directional Osgood estimate would follow from an independently proved bound

$$|\Gamma_a^* c|^2 \leq \mathfrak{d}_R(a) \epsilon_a \log(e/\epsilon_a), \quad \mathfrak{d}_R \in L^1([a_0, R]), \quad \mathfrak{d}_R \geq 0. \quad (\text{CP18:3.60})$$

Indeed, (3.59) gives

$$|\beta_a^* c|^2 \leq 2(B_R^{\text{ref}})^2 \epsilon_a^2 + 2\mathfrak{d}_R(a) \epsilon_a \log(e/\epsilon_a). \quad (\text{CP18:3.61})$$

The established estimate (3.58) contains no such vanishing factor. Vanishing of Γ_a is a sufficient condition for this chosen probe, not a claimed necessary condition for RH. The new closed result is existence and boundedness of the complete residual limit almost everywhere; its energy-dependent size remains the specific unresolved arithmetic estimate.

3.9. Projected dual energy and finite regularized probe families

This calculation uses the complete physical operator and does not invert the unregularized Weil operator. Fix $a \in [a_0, R]$, and abbreviate

$$A = A_a^-, \quad J = A + \tau F F^* \geq g_R I, \quad W = J^{-1} F, \quad K = \tau F^* W, \quad M = I - K, \quad \tau = \tau_R. \quad (\text{CP18:3.62})$$

All finite prime translations, the complete archimedean kernel, the polar rank-one term, and the exterior logarithmic operator are included in A . Positivity $A \geq 0$ is assumed only where a nonnegative energy or form Cauchy-Schwarz is used. The algebraic identities below also hold outside that region.

Choose $0 < \eta < a/2$, $\delta = e^{-L} < \eta$. In the unitary half-interval coordinates, define the actual finite-resolution boundary representer

$$g_L^+(x) = \frac{\mathbf{1}_{\{\delta < a-x < \eta\}}}{2\sqrt{L}(a-x)}, \quad g_L(x) = \frac{\text{sgn } x}{\sqrt{2}} g_L^+(|x|). \quad (\text{CP18:3.63})$$

Thus $b_L = W^* g_L$ is exactly the column of signed half-interval boundary averages. In particular the $\sqrt{2}$ unitary normalization is included; no additional factor occurs in $b_L = W^* g_L$. For each finite

L , the zero extension of g_L is piecewise C^1 and has bounded variation. Its Fourier transform is $O_L((1 + |\xi|)^{-1})$; hence

$$g_L \in \text{Dom}(\log |D|)_{(-a,a)} = \text{Dom } A = \text{Dom } J, \quad \|g_L\|_2^2 = \frac{e^L - \eta^{-1}}{4L}. \quad (\text{CP18:3.64})$$

The domain claim uses the whole-line logarithmic Fourier graph norm, not an endpoint value of an arbitrary form-domain function.

3.9.1. Exact lower bound for every proposed probe On the branch $A \geq 0$, put

$$D = W^* A W = \frac{1}{\tau} K M \geq 0. \quad (\text{CP18:3.65})$$

For any $\psi \in \text{Dom } A$, write

$$e = g_L - A\psi, \quad h = W^* A\psi = b_L - W^* e. \quad (\text{CP18:3.66})$$

Then

$$\boxed{h \in \text{Ran } D, \quad Q_a^-(\psi) \geq h^* D^\dagger h.} \quad (\text{CP18:3.67})$$

Here $D^\dagger$ is the finite-dimensional Moore-Penrose inverse, not an inverse of A . To prove this, if $c \in \ker D$, then $Q_a^-(Wc) = 0$, so $A^{1/2}Wc = 0$, and therefore $c^*h = 0$. This proves the range statement. Expand the complete form of $\psi - Wc$, retaining its two cross terms, and minimize over $c = D^\dagger h$. This gives (3.67). Equivalently, complete-form Cauchy-Schwarz gives, for every c ,

$$|c^*(b_L - W^*e)|^2 \leq Q_a^-(\psi) \frac{c^* K M c}{\tau}. \quad (\text{CP18:3.68})$$

In a K -eigenvector with eigenvalue $0 < k < 1$, the associated term in the lower bound is

$$\frac{\tau |h_k|^2}{k(1-k)}. \quad (\text{CP18:3.69})$$

At an actual first null, every K -eigenvector with $k = 1$ satisfies the exact identity

$$c^* W^* e = c^* b_L \quad \text{for every } \psi \in \text{Dom } A. \quad (\text{CP18:3.70})$$

Consequently a residual estimate $\|W^* e_L\| = O(L^{-1/2})$ would itself prove that the corresponding averaged boundary coefficient vanishes on the null eigenspace. The required cancellation is contained in the residual assertion; a choice of probe does not remove it algebraically. This calculation applies to both null branches, including a null vector whose polar moment is zero.

3.9.2. A probe using only the uniformly positive regularized inverse Define

$$h_L = J^{-1} g_L, \quad V = \tau J^{-1} F F^*, \quad \psi_{L,m} = \sum_{j=0}^{m-1} V^j h_L, \quad m \geq 1. \quad (\text{CP18:3.71})$$

This is an explicit finite polynomial in the certified bounded inverse J^{-1} , with no A^{-1} . Each term belongs to the operator domain: $h_L \in \text{Dom } J$, and the range of V lies in the span of the operator-domain columns of W . Since $A = J(I - V)$, telescoping gives the complete residual and projected

coefficients

$$\boxed{\begin{aligned} e_{L,m} &= g_L - A\psi_{L,m} = \tau F K^{m-1} b_L, \\ F^* \psi_{L,m} &= s_m(K) b_L, \quad s_m(k) = \sum_{j=0}^{m-1} k^j, \\ W^* e_{L,m} &= K^m b_L, \\ b_L &= M F^* \psi_{L,m} + W^* e_{L,m}. \end{aligned}} \quad (\text{CP18:3.72})$$

These equations are exact for the complete operator. In an eigenvector with $k = 1 - \varepsilon$,

$$(F^* \psi_{L,m})_k = \frac{1 - (1 - \varepsilon)^m}{\varepsilon} (b_L)_k, \quad (W^* e_{L,m})_k = (1 - \varepsilon)^m (b_L)_k. \quad (\text{CP18:3.73})$$

For $\varepsilon = 0$, these become $m(b_L)_k$ and $(b_L)_k$, respectively. At fixed positive ε , making the residual small costs a source coefficient tending to $(b_L)_k/\varepsilon$. A margin-independent estimate therefore still requires an independent estimate on $(b_L)_k$.

There is an exact energy formula. Since $b_L \in \text{Ran } K$, define

$$h_{L,\perp} = h_L - \tau W K^\dagger b_L. \quad (\text{CP18:3.74})$$

Then $F^* h_{L,\perp} = 0$ and $h_{L,\perp}$ is J -orthogonal to the range of W . Hence

$$\boxed{\begin{aligned} Q_a^-(\psi_{L,m}) &= \|h_{L,\perp}\|_J^2 + \tau \sum_{k>0} \frac{1-k}{k} s_m(k)^2 |(b_L)_k|^2, \\ \|h_{L,\perp}\|_J^2 &= \langle g_L, J^{-1} g_L \rangle - \tau b_L^* K^\dagger b_L. \end{aligned}} \quad (\text{CP18:3.75})$$

The sum is over the eigenvalues of the actual complete K , with multiplicities; it is nonnegative when $A \geq 0$. In that region its terms with k near one approach $\tau |(b_L)_k|^2 / [k(1-k)]$ as m increases. This is precisely the denominator in the sharp dual-energy bound (3.69), rather than an independent bound for it.

3.9.3. Actual high-frequency cost of this probe family The singular representer has a calculated exponential energy scale. To make the estimate explicit, write $N = \eta/\delta$ and

$$G_N(u) = u^{-1} \mathbf{1}_{[1,N]}(u). \quad (\text{CP18:3.76})$$

Its total variation is two. For $0 < |\xi| < 1$, splitting at $u = 1/|\xi|$ and integrating the remaining oscillatory tail by parts gives $|\widehat{G}_N(\xi)| \leq \log(1/|\xi|) + 3$. For $|\xi| \geq 1$, bounded variation gives $|\widehat{G}_N(\xi)| \leq 2/|\xi|$. Thus, uniformly in $N > 1$,

$$\left| \frac{1}{2\pi} \int \log |\xi| |\widehat{G}_N(\xi)|^2 d\xi \right| \leq \frac{31}{\pi}. \quad (\text{CP18:3.77})$$

Logarithmic scaling gives the one-sided form

$$\mathfrak{c}(g_L^+) = \frac{1}{4L\delta} [L(1 - \delta/\eta) + \mathfrak{c}(G_N)]. \quad (\text{CP18:3.78})$$

The odd reflection adds the exact positive cross term

$$\frac{1}{2} \int_0^a \int_0^a \frac{g_L^+(x) g_L^+(y)}{x+y} dx dy, \quad (\text{CP18:3.79})$$

which is at most $\|g_L^\dagger\|_1^2/[4(a-\eta)] = O_a(L)$. Let D_R^{bnd} be any retained independent bound for the bounded remainder $J - (\log|D|)_{(-a,a)}$. All prime, archimedean, polar, and regularizer terms occur in that remainder. Then

$$\langle g_L, Jg_L \rangle \leq \frac{e^L}{4} \left(1 + \frac{31/\pi + D_R^{\text{bnd}}}{L} \right) + O_a(L). \quad (\text{CP18:3.80})$$

The positive-operator Cauchy-Schwarz inequality yields

$$\langle g_L, J^{-1}g_L \rangle \geq \frac{\|g_L\|^4}{\langle g_L, Jg_L \rangle} \geq \frac{e^L}{4L^2} (1 - O_a(L^{-1})). \quad (\text{CP18:3.81})$$

Checkpoint 17's actual boundary envelope gives a uniform bound for $\|b_L\|$. At each fixed a , $K^\dagger$ is a finite matrix, so (3.75)-(3.81) prove, for every $m \geq 1$,

$$\boxed{Q_a^-(\psi_{L,m}) \geq \frac{e^L}{4L^2} (1 - O_a(L^{-1})) - C_a, \quad A \geq 0.} \quad (\text{CP18:3.82})$$

Here $C_a = \tau\|K^\dagger\| \sup_L \|b_L\|^2$ is finite at this fixed a ; no uniform bound on $K^\dagger$ is asserted. The leading divergence already refutes the proposed $O(\log L)$ energy certificate for this specific family. It does not refute existence of a different probe exploiting the complete arithmetic equation.

3.9.4. A finite source-space probe removes the extraneous high-frequency energy The high-frequency component $h_{L,\perp}$ in (3.75) can be removed explicitly. Define

$$\boxed{\psi_{L,m}^F = \tau W K^\dagger s_m(K) b_L.} \quad (\text{CP18:3.83})$$

This is a finite linear combination of the actual forced solutions. Hence it belongs to $\text{Dom } A$. Its only additional inverse beyond the certified regularized inverse defining W is the Moore-Penrose inverse of the positive finite source Gram matrix K ; no uniform bound on that inverse is assumed. In particular it is not an inverse of M or of A . Because $b_L \in \text{Ran } K$, zero eigenvalues of K contribute nothing.

The complete source equation $AW = FM$ gives every residual term at once:

$$\boxed{\begin{aligned} F^* \psi_{L,m}^F &= s_m(K) b_L, \\ e_{L,m}^F &= g_L - \tau F M K^\dagger s_m(K) b_L, \\ W^* e_{L,m}^F &= K^m b_L, \\ Q_a^-(\psi_{L,m}^F) &= \tau \sum_{k>0} \frac{1-k}{k} s_m(k)^2 |(b_L)_k|^2. \end{aligned}} \quad (\text{CP18:3.84})$$

The energy is precisely (3.75) with $\|h_{L,\perp}\|_J^2$ removed. This new explicit probe therefore avoids the exponential energy lower bound (3.82); that lower bound concerns the different polynomial family (3.71). No prime, archimedean, polar, or boundary term has been removed from the operator: the cancellation follows from its complete forced equation.

There is a useful uniform estimate in the critical source sector. On $A \geq 0$, let

$$P_{\text{crit}} = \mathbf{1}_{[1/2,1]}(K), \quad \psi_{L,m}^{F,\text{crit}} = \tau W K^\dagger s_m(K) P_{\text{crit}} b_L. \quad (\text{CP18:3.85})$$

For $0 \leq k \leq 1$,

$$s_m(k) \leq \min\{m, (1-k)^{-1}\}, \quad (1-k)s_m(k)^2 \leq \min\{m^2(1-k), (1-k)^{-1}\} \leq m, \quad (\text{CP18:3.86})$$

with the endpoint $k = 1$ interpreted directly as zero on the left. Thus

$$\boxed{Q_a^-(\psi_{L,m}^{F,\text{crit}}) \leq 2\tau m \|P_{\text{crit}} b_L\|^2.} \quad (\text{CP18:3.87})$$

The actual boundary envelope bounds the norm on the right independently of the unknown positivity margin. For this sector alone, taking $m = O(\log(eL))$ supplies the proposed logarithmic energy scale. The complete projected residual is

$$W^*(g_L - A\psi_{L,m}^{F,\text{crit}}) = (I - P_{\text{crit}})b_L + K^m P_{\text{crit}} b_L. \quad (\text{CP18:3.88})$$

The low source sector is displayed explicitly; (3.87) is not asserted as a full-space energy bound.

In a critical eigenvector $k = 1 - \varepsilon$, $0 < \varepsilon \leq 1/2$, the residual component is $(1 - \varepsilon)^m (b_L)_k$, and its exact energy contribution is

$$\frac{\tau[1 - (1 - \varepsilon)^m]^2}{(1 - \varepsilon)\varepsilon} |(b_L)_k|^2. \quad (\text{CP18:3.89})$$

For fixed positive ε , its limit as $m \rightarrow \infty$ is $\tau |(b_L)_k|^2 / [(1 - \varepsilon)\varepsilon]$. At $\varepsilon = 0$, the residual remains $(b_L)_k$ for every m , while this energy contribution is zero. These identities give a sharper actual obstruction: logarithmic probe energy can be reached in the critical sector, but its residual decay is not uniform as the margin tends to zero. Taking m proportional to $\log L$ makes the residual factor approximately $L^{-c\varepsilon}$; obtaining $L^{-1/2}$ by this iteration requires a number of steps proportional to $(\log L)/\varepsilon$, unless an independent bound already suppresses the boundary coefficient. The corresponding near-null energy denominator is precisely the one in (3.69).

Strongest theorem and remaining estimate. The finite-resolution representer belongs to the complete operator domain. Every admissible probe obeys the sharp projected dual-energy bound (3.67), and the first regularized polynomial family has the exact residual (3.72) and exponential energy lower bound (3.82). The finite source-space probe (3.83) removes that high-frequency cost and proves the critical-sector estimate (3.87), while retaining the exact residual (3.88). Uniform suppression of that residual at small positivity margin remains open. Its required new input is the actual coupled estimate

$$|c^*(b_L - W^* e_L)|^2 \leq C_R \log(eL) \frac{c^* K M c}{\tau}, \quad \|W^* e_L\| \leq C_R L^{-1/2}, \quad (\text{CP18:3.90})$$

with one explicitly constructed probe, uniformly before a first crossing. At $k = 1$, the residual is exactly the signed boundary average by (3.70). Therefore the unresolved arithmetic interaction is visible without discarding a prime packet or passing to a modified metric.

4. Energy-dependent comparison, adaptive resolution, and the remaining arithmetic estimate

4.1. The exact comparison supplied by an energy-relative boundary theorem

Fix a finite $R > a_0$. All the following matrices act on the same physical source space $\mathbb{C}^{d_R}$. Write

$$\tau = \tau_R, \quad K(a) = \widehat{K}_a, \quad M(a) = I - K(a), \quad W(a) = \mathcal{J}_a^{-1} F_a, \quad \varepsilon(a) = \lambda_{\min} M(a). \quad (\text{CP18:4.1})$$

The positive physical regularizer satisfies $\mathcal{J}_a \geq g_R I$, including intervals on which A_a^- might fail to be positive. Groups 1-2 give

$$K(b) - K(a) \succeq 0, \quad K'(a) = \tau \beta_a \beta_a^* \quad \text{for almost every } a, \quad (\text{CP18:4.2})$$

and K is Lipschitz on $[a_0, R]$. Here β_a has the averaged **unitary half-interval** normalization of Group 2. In particular the corresponding full-interval endpoint coefficient is smaller by $\sqrt{2}$. This convention fixes every factor of two in what follows.

Consider the positive branch, on which $0 \prec M(a) \preceq I$. The exact source equation and energy identity are

$$A_a^- W(a) = F_a M(a), \quad Q_a^-(W(a)c) = \frac{1}{\tau} c^* K(a) M(a) c. \quad (\text{CP18:4.3})$$

Consequently the proposed energy-relative trace estimate is precisely

$$\boxed{\tau \beta_a \beta_a^* \preceq C_R(a) K(a) M(a), \quad 0 \leq C_R \in L^1([a_0, R])}. \quad (\text{CP18:4.4})$$

The matrices K and $M = I - K$ commute; their product is positive on this branch. If (4.4) is proved independently with a constant that does not depend on the unknown margin, then

$$M'(a) = -\tau \beta_a \beta_a^* \succeq -C_R(a) K(a) M(a) \succeq -C_R(a) M(a) \quad \text{almost everywhere.} \quad (\text{CP18:4.5})$$

This is a matrix differential inequality, not an assertion about the sign of an individual prime packet. Let

$$I_R(b) = \int_{a_0}^b C_R(u) du, \quad \eta_0 = \frac{m_0}{m_0 + \tau_R}, \quad m_0 = \frac{1}{5000}. \quad (\text{CP18:4.6})$$

The certified interval gives $M(a_0) \succeq \eta_0 I$. Since M is absolutely continuous, differentiating $e^{I_R(a)} M(a)$ and integrating its positive semidefinite derivative gives

$$\boxed{M(b) \succeq e^{-I_R(b)} M(a_0) \succeq \eta_0 e^{-I_R(b)} I.} \quad (\text{CP18:4.7})$$

The integration is valid first on a compact positive subinterval. Continuity of M then gives the same strictly positive lower bound at a proposed finite first endpoint, which contradicts a zero eigenvalue there. The equivalences in Group 1 include every A_a^- -null vector, whether or not its polar moment vanishes. Thus (4.4), if proved for every finite R , excludes both first-null branches.

Status of this implication. Equations (4.5)-(4.7) are proved consequences of the explicitly stated estimate (4.4). Group 2 supplies absolute continuity and all prime-threshold passages. It does not supply (4.4).

4.2. Determinants and multiplicities

Before a first crossing, the finite-dimensional determinant identity is

$$\boxed{-\frac{d}{da} \log \det M(a) = \tau \operatorname{tr}(M(a)^{-1} \beta_a \beta_a^*) = \tau \beta_a^* M(a)^{-1} \beta_a.} \quad (\text{CP18:4.8})$$

This holds almost everywhere on every compact positive subinterval. It does not require simple eigenvalues. Under (4.4), taking the trace after conjugation by $M^{-1/2}$ gives

$$0 \leq -\frac{d}{da} \log \det M(a) \leq C_R(a) \operatorname{tr} K(a) \leq d_R C_R(a). \quad (\text{CP18:4.9})$$

The integrable upper bound prevents the determinant from vanishing at a finite endpoint.

There is also a directional formulation. At points where both M and $\varepsilon = \lambda_{\min} M$ are differentiable, let $P_\varepsilon(a)$ be the orthogonal projection onto the minimum eigenspace. The one-sided eigenvalue variation is governed by the eigenvalues of $P_\varepsilon M' P_\varepsilon$. In particular,

$$\boxed{\varepsilon'(a) = -\tau_R \|P_\varepsilon(a)\beta_a\|^2 \quad \text{for almost every } a.} \quad (\text{CP18:4.10})$$

Indeed, where the matrix derivative exists, the right derivative of its least eigenvalue is the least eigenvalue of the compressed derivative. The compression here is the rank-one nonpositive matrix $-\tau_R |P_\varepsilon \beta_a\rangle \langle P_\varepsilon \beta_a|$, whose least eigenvalue is the right side of (4.10). The left derivative is the greatest eigenvalue of that compression. Thus, if the minimum eigenspace has dimension greater than one and its projected boundary vector is nonzero, the two derivatives differ. Such a point is excluded from the full-measure set where the Lipschitz function ε is differentiable. If that eigenspace has dimension greater than one at a differentiability point, its projected boundary vector vanishes and $\varepsilon' = 0$. This proves (4.10) with the multiplicity issue explicitly retained.

Consequently, a sufficient multiplicity-safe estimate is

$$\tau |\beta_a^* c|^2 \leq c_R(a) \omega(\varepsilon(a)) \quad \text{for every unit } c \in \text{ran } P_\varepsilon(a), \quad (\text{CP18:4.11})$$

where $c_R \in L^1([a_0, R])$, $\omega > 0$ on $(0, 1]$, and

$$\int_{0+} \frac{dt}{\omega(t)} = \infty. \quad (\text{CP18:4.12})$$

Then $-\varepsilon'(a) \leq c_R(a) \omega(\varepsilon(a))$ almost everywhere. The minimum eigenvalue is Lipschitz because M is Lipschitz. Integrating the ordinary absolutely continuous chain rule on positive subintervals proves

$$\int_{\varepsilon(b)}^{\varepsilon(a_0)} \frac{dt}{\omega(t)} \leq \int_{a_0}^b c_R(u) du. \quad (\text{CP18:4.13})$$

Its left side diverges at a proposed finite zero. In particular (4.4) supplies the slightly sharper scalar inequality $-\varepsilon' \leq C_R \varepsilon(1 - \varepsilon)$, and hence also the choice $\omega(t) = t$.

The presently proved absolute estimate is

$$\tau |\beta_a^* c|^2 \leq C_R^{\text{abs}} \|c\|^2. \quad (\text{CP18:4.14})$$

It has no factor that vanishes with ε . Its Osgood integral is finite. No exclusion of a first null follows from (4.14).

4.3. What a bounded adjoint probe would prove

Use the explicit boundary functional and the probe from Group 3, with the same half-interval normalization:

$$b_a(L) = W(a)^* g_{a,L}, \quad h_a(L) = F_a^* \psi_{a,L}, \quad e_{a,L} = g_{a,L} - A_a^- \psi_{a,L}, \quad r_a(L) = W(a)^* e_{a,L}. \quad (\text{CP18:4.15})$$

The domain condition $\psi_{a,L} \in \text{Dom}(A_a^-)$ and self-adjointness make the exact identity

$$\boxed{b_a(L) = M(a)h_a(L) + r_a(L)} \quad (\text{CP18:4.16})$$

legitimate. Its complete quadratic expansion is

$$|b_a(L)^*c|^2 = |h_a(L)^*M(a)c|^2 + 2\Re\left[h_a(L)^*M(a)c\overline{r_a(L)^*c}\right] + |r_a(L)^*c|^2. \quad (\text{CP18:4.17})$$

If the complete residual is split as $r = \sum_{\nu} r_{\nu}$, the last term is

$$|r^*c|^2 = \sum_{\nu} |r_{\nu}^*c|^2 + 2\Re \sum_{\nu < \eta} (r_{\nu}^*c)\overline{(r_{\eta}^*c)}. \quad (\text{CP18:4.18})$$

Thus the logarithmic, prime, archimedean, polar, and endpoint terms continue to interact in the exact expression. A bound for one packet cannot replace a bound for their sum.

Suppose an independently calculated probe satisfies

$$\sup_{L \geq L_0} \|h_a(L)\|^2 \leq B_R(a), \quad r_a(L) \longrightarrow 0, \quad B_R \in L^1([a_0, R]). \quad (\text{CP18:4.19})$$

At almost every a at which the averaged boundary vector exists, select a convergent subsequence of the bounded finite-dimensional vectors $h_a(L)$. Equation (4.16) yields

$$\beta_a = M(a)h_a, \quad \|h_a\|^2 \leq B_R(a). \quad (\text{CP18:4.20})$$

It follows that

$$\beta_a\beta_a^* \preceq B_R(a)M(a)^2 \preceq B_R(a)M(a), \quad M'(a) \succeq -\tau_R B_R(a)M(a). \quad (\text{CP18:4.21})$$

This already gives (4.7), with $C_R = \tau_R B_R$. It is enough for closure even if one has not established the stronger inequality (4.4) on directions where $K(a)$ is very small. The determinant estimate becomes dimension-free:

$$\tau_R \beta_a^* M(a)^{-1} \beta_a = \tau_R h_a^* M(a) h_a \leq \tau_R B_R(a). \quad (\text{CP18:4.22})$$

These are useful sufficient consequences of the proposed bounded-probe calculation. The existence of such a probe, with the actual residual tending to zero, is an additional mathematical estimate.

4.4. The finite-resolution certificate gives a complete matrix step

There is a second sufficient route that permits logarithmic probe energy. Assume the following three estimates hold with constants independent of a , of the positivity margin, and of the resolution $L \geq L_0$:

$$\begin{aligned} Q_a^-(\psi_{a,L}) &\leq C_{\psi} \log(eL), \quad \|r_a(L)\| \leq C_e L^{-1/2}, \\ \left\| \frac{K(a + e^{-L}) - K(a)}{e^{-L}} - \tau_R b_a(L) b_a(L)^* \right\| &\leq C_{\text{col}} L^{-1/2}. \end{aligned} \quad (\text{CP18:4.23})$$

The endpoint $a + e^{-L}$ must lie in $[a_0, R]$. Only positivity of the form at the **starting** interval a is needed for the following argument; the regularized collar formula is valid on the full finite parameter interval.

Complete-form Cauchy-Schwarz and (4.3) give, for every c ,

$$\begin{aligned} |b_a(L)^*c|^2 &\leq 2|Q_a^-(W(a)c, \psi_{a,L})|^2 + 2|r_a(L)^*c|^2 \\ &\leq \frac{2C_{\psi}}{\tau_R} \log(eL) c^* K(a) M(a) c + \frac{2C_e^2}{L} \|c\|^2. \end{aligned} \quad (\text{CP18:4.24})$$

This applies Cauchy-Schwarz to the complete nonnegative odd form; it does not impose a sign on the individual arithmetic contributions. Combining (4.23) and (4.24), with $\delta = e^{-L}$, proves

$$0 \preceq K(a + \delta) - K(a) \preceq \delta \left[A \log(eL) K(a) M(a) + \frac{B}{\sqrt{L}} I \right], \quad (\text{CP18:4.25})$$

where one may take

$$A = 2C_\psi, \quad B = 2\tau_R C_e^2 + C_{\text{col}}, \quad L \geq \max\{L_0, 1\}. \quad (\text{CP18:4.26})$$

The L^{-1} residual square was bounded by $L^{-1/2}$; no uncontrolled mixed term was discarded.

4.5. Adaptive exponentially small steps really do prevent finite accumulation

Here is a discrete Osgood proof for (4.25). It also specifies the uniformity needed for that argument. Suppose L_0, A, B in (4.25) are fixed on $[a_0, R]$. Set

$$D = \max\{L_0, 10 + 2A + 2B\}, \quad C = A(2 + \log D) + B/\sqrt{D}. \quad (\text{CP18:4.27})$$

Then $D \geq 10$ and $Ce^{-D} < 1/2$. At a positive starting point a , write $\varepsilon = \varepsilon(a) \in (0, 1]$ and choose

$$L(\varepsilon) = D/\varepsilon^2, \quad \delta(\varepsilon) = \exp(-D/\varepsilon^2). \quad (\text{CP18:4.28})$$

The function $\lambda \mapsto \lambda + \delta A \log(eL) \lambda(1 - \lambda)$ is increasing on $[0, 1]$, because $\delta A \log(eL) \leq Ce^{-D} < 1/2$. Taking the largest eigenvalue in (4.25) therefore gives the exact scalar bound

$$\begin{aligned} \varepsilon(a + \delta) &\geq \varepsilon - \delta \left[A \log(eL) \varepsilon(1 - \varepsilon) + B/\sqrt{L} \right] \\ &\geq \varepsilon [1 - C\delta \log(e/\varepsilon)] \geq \varepsilon/2. \end{aligned} \quad (\text{CP18:4.29})$$

Monotonicity of K gives the accompanying upper bound $\varepsilon(a + \delta) \leq \varepsilon$. Put

$$\Phi(t) = \log \log(e/t), \quad 0 < t \leq 1. \quad (\text{CP18:4.30})$$

For $q = C\delta \log(e/\varepsilon) \leq 1/2$, the inequality $-\log(1 - q) \leq 2q$ applied to (4.29) proves

$$0 \leq \Phi(\varepsilon(a + \delta)) - \Phi(\varepsilon(a)) \leq 2C\delta. \quad (\text{CP18:4.31})$$

Iterate these steps while they remain in $[a_0, R]$. If their support parameters had a finite accumulation point $b \leq R$, summing (4.31) would imply

$$\Phi(\varepsilon_n) \leq \Phi(\varepsilon(a_0)) + 2C(R - a_0), \quad \varepsilon_n \geq \exp(1 - \exp[\Phi(\varepsilon(a_0)) + 2C(R - a_0)]) =: \eta_R > 0. \quad (\text{CP18:4.32})$$

But then every step has length at least $\exp(-D/\eta_R^2) > 0$, which contradicts finite accumulation. This argument explicitly rules out the possibility that the exponentially small allowed steps shrink to a finite first crossing.

If the remaining distance to R is smaller than the next step, use that smaller increment. For $L \geq 1$, both $e^{-L} \log(eL)$ and $e^{-L}/\sqrt{L}$ decrease as L increases. Thus the loss bound for a smaller increment is no larger than that already controlled in (4.29), and the final endpoint remains positive.

Accordingly **the uniform finite-resolution certificate (4.23) is sufficient for global continuation on each finite support interval**. The proof is discrete; it does not extract an unjustified derivative bound by letting L vary with ε . Its Osgood function is $t \log(e/t)$, since $-\Phi'(t) = 1/[t \log(e/t)]$. If the resolution threshold L_0 or the constants in (4.23) depend without control on the unknown margin, this proof does not apply.

4.6. The arithmetic input still required

The inhomogeneous endpoint equation supplies a genuine factor of M in a specific completed moment. For $q = W(a)c$, the retained convolution identity gives

$$\boxed{\mathcal{P}_a(q) - d_a F_q(1/2) = J_a^{\text{end}}(F_a M(a)c),} \quad d_a = 4a \cosh(a/2) - 8 \sinh(a/2), \quad (\text{CP18:4.33})$$

where

$$F_q(z) = \int_{-a}^a e^{zy} q(y) dy, \quad J_a^{\text{end}}(h) = \int_0^a y h(y) dy. \quad (\text{CP18:4.34})$$

The symbol J_a^{end} denotes the scalar endpoint moment, not the positive regularized operator $\mathcal{J}_a$. The packet $\mathcal{P}_a$ contains the full prime and archimedean expression recorded in Group 3. In particular, for a minimum eigenvector c , the right side of (4.33) is $\varepsilon(a) J_a^{\text{end}}(F_a c)$.

What must still be calculated is the passage from this endpoint moment to the averaged boundary functional. Equivalently, the explicit probe must control the **complete** residual

$$\begin{aligned} e_{a,L} = & g_{a,L} - (\log |D|)_{(-a,a)} \psi_{a,L} + \log(2\pi) \psi_{a,L} \\ & + \sum_{\ell_n < 2a} b_n [\psi_{a,L}(\cdot + \ell_n) + \psi_{a,L}(\cdot - \ell_n)] + K_{\alpha,a} \psi_{a,L} + 2s_a \langle s_a, \psi_{a,L} \rangle, \end{aligned} \quad (\text{CP18:4.35})$$

with zero extension, odd reflection, and the complete boundary contribution of the supported logarithmic operator. Its decisive finite-dimensional evaluation is

$$\boxed{r_a(L) = W(a)^* e_{a,L}.} \quad (\text{CP18:4.36})$$

All arithmetic and polar cancellation needed for (4.19) or (4.23) resides in this paired residual and the complete probe energy. The positive regularizer and the collar calculation already control the absolute size and parameter variation; those results do not give the necessary vanishing factor.

The actual probe calculations from Group 3 give the following more specific status.

- The explicit local probe $\psi_{a,L} = g_{a,L}/L$, with the domain-compatible cutoffs specified there, has complete energy of order e^L/L^2 , rather than the $O(\log L)$ required by (4.23). Its source pairing is $O_R(L^{-1/2})$, but its paired residual is $b_a(L) + O_R(L^{-1/2})$. It leaves the leading boundary coefficient in the residual.
- The reference inverse probe $\psi_{a,L} = (C_a + \lambda_R)^{-1} g_{a,L}$, where C_a is the supported logarithmic operator, has the explicit residual

$$e_{a,L} = [(\lambda_R + \log(2\pi))I + \Pi_a + K_{\alpha,a} + 2|s_a\rangle\langle s_a|] \psi_{a,L}. \quad (\text{CP18:4.37})$$

The calculation bounds its projected residual absolutely; it does not prove its decay. Inverting only the reference logarithmic operator preserves a complete remainder whose cancellation still has to be established.

- A finite Neumann construction using the uniformly positive $\mathcal{J}_a$ leaves a residual proportional to $K(a)^m b_a(L)$. At a proposed null direction the relevant eigenvalue of K is one, so this iteration does not reduce that component.
- The finite source-space probe removes the high-frequency part of that construction. Its exact complete energy is

$$Q_a^-(\psi_{L,m}^F) = \tau_R \sum_{k>0} \frac{1-k}{k} s_m(k)^2 |(b_a(L))_k|^2, \quad W_a^* e_{L,m}^F = K(a)^m b_a(L). \quad (\text{CP18:4.38})$$

On the critical sector $1/2 \leq k \leq 1$, its energy is at most $2\tau_R m \|P_{\text{crit}} b_a(L)\|^2$. Thus $m = O(\log(eL))$ reaches the proposed energy scale there. The residual factor $(1 - \varepsilon)^m$ still has no uniform $L^{-1/2}$ decay as ε approaches zero. The low-sector residual remains explicit in Group 3; a full-matrix certificate is not inferred from a critical-sector energy estimate.

There is a sharp finite-dimensional check on every proposed probe. Put $D_a = K(a)M(a)/\tau_R$ and $h_L = b_a(L) - r_a(L)$. Complete-form Cauchy-Schwarz implies

$$h_L \in \text{ran } D_a, \quad \boxed{Q_a^-(\psi_{a,L}) \geq h_L^* D_a^\dagger h_L}, \quad (\text{CP18:4.39})$$

where $D_a^\dagger$ is the Moore-Penrose inverse. To prove it, first test against $\ker D_a$, which forces h_L to be orthogonal to that kernel. Then use the coefficient vector $D_a^\dagger h_L$ in Cauchy-Schwarz. For a unit minimum-eigenvalue direction this yields, whenever $0 < \varepsilon < 1$,

$$Q_a^-(\psi_{a,L}) \geq \frac{\tau_R |c^*[b_a(L) - r_a(L)]|^2}{\varepsilon(a)(1 - \varepsilon(a))}. \quad (\text{CP18:4.40})$$

This is a constraint on the actual source subspace. A probe with both low energy and a small residual has to provide the desired suppression of the actual boundary coefficient; the regularized inverse alone does not supply it.

Strongest conclusion of Group 4. The physical matrix is now absolutely continuous, and both the continuous energy-relative route and the finite-resolution adaptive route have complete, multiplicity-safe return proofs. In particular, the proposed logarithmic-energy/ $L^{-1/2}$ -residual certificate, if established uniformly, really is enough to prevent a finite first crossing. The unresolved estimate is an independent bound for the explicit probe's complete energy and paired residual, or an independently sufficient energy-relative bound such as (4.4). Group 3 also establishes the actual reference-residual limit $\Gamma_a = \beta_a - M_a h_a$ almost everywhere; its directional energy-dependent estimate is a more specific sufficient input, displayed in the combined conclusion. The two global null branches remain unexcluded beyond the certified starting interval and its local continuation. The conditional final chain is

$$\begin{aligned} & (4.4), \text{ or } (4.19), \text{ or the uniform certificate (4.23)} \\ \implies & M(a) \succ 0 \text{ for every finite } a \implies Q_a^- \succ 0 \text{ on every finite odd test interval} \\ \implies & \text{odd Weil positivity} \implies \text{RH} \implies \text{the retained normalized Suzuki identity} \\ \implies & \Delta = \mathfrak{W} - \mathcal{S} = 0 \implies \mathcal{E}_N = 0 \text{ for every } N. \end{aligned} \quad (\text{CP18:4.41})$$

The final Suzuki implication is exactly the previously retained conditional identity; no replacement metric or additional normalization has been introduced.

5. Combined closing statement and the next signed calculation

5.1. The complete matrix evolution is established

The certified prime-active interval and lower bound remain

$$0 < a \leq a_0 = \frac{699}{2000} + e^{-2500} \implies Q_a^- \geq \frac{1}{5000} I. \quad (\text{CP18:5.1})$$

This numerical starting certificate is retained. On every arbitrary finite range $[a_0, R]$, one physical regularizer and its restrictions satisfy

$$\mathcal{J}_a = A_a^- + \tau_R F_a F_a^* \geq g_R I, \quad F_a^* F_a \preceq I. \quad (\text{CP18:5.2})$$

The exact sign test, nullspace correspondence, and forced energy are

$$\begin{aligned} A_a^- > 0 &\iff M_a > 0, & \ker A_a^- &\simeq \ker M_a, \\ A_a^- W_a &= F_a M_a, & Q_a^-(W_a c) &= \tau_R^{-1} c^* \widehat{K}_a M_a c. \end{aligned} \quad (\text{CP18:5.3})$$

These results include both prior null branches and do not assume orthonormality or injectivity of the restricted sources.

The new complete matrix theorem is

$$0 \preceq \widehat{K}_b - \widehat{K}_a = \tau_R \mathcal{R}_{a,b}^* S^{-1} \mathcal{R}_{a,b} \preceq C_{\text{Lip}}(b-a)I, \quad a_0 \leq a \leq b \leq R. \quad (\text{CP18:5.4})$$

Its constants, closed-form domain passage, and every regularizer, polar, archimedean, and prime contribution are displayed in Group 2. In particular the full physical matrix is absolutely continuous through every prime threshold. With the real unitary half-interval averaged boundary vector,

$$\widehat{K}'_a = \tau_R \beta_a \beta_a^* \quad \text{for almost every } a, \quad M_b - M_a = -\tau_R \int_a^b \beta_u \beta_u^* du. \quad (\text{CP18:5.5})$$

The averaged vector limit is established almost everywhere on the whole finite parameter range, independently of the sign of A_a^- . A pointwise normalized boundary trace is not assumed or proved.

There is also an independent local continuation consequence. Put

$$\eta_0 = \frac{1/5000}{1/5000 + \tau_R}, \quad \delta_R = \min\left(R - a_0, \frac{\eta_0}{2C_{\text{Lip}}}\right). \quad a_0 \leq a \leq a_0 + \delta_R \implies M_a \succeq \frac{\eta_0}{2}I, \quad A_a^- \succeq \frac{g_R \eta_0}{2}I. \quad (\text{CP18:5.6})$$

This follows directly from (5.4). It is a quantified positive extension with the stated finite constants, including the unevaluated fixed boundary embedding constant C_{BL} . It does not supply a margin-relative continuation bound at an arbitrary later endpoint.

Two exact cancellations now carry a margin factor before estimation:

$$\begin{aligned} F_R^+ c - \tau_R F_R^+ F_a^* W_a c &= F_R^+ M_a c, \\ W_a^* A_a^- j_a &= M_a F_a^* j_a, \quad j_a(x) = x/2. \end{aligned} \quad (\text{CP18:5.7})$$

The first is the source-regularizer cancellation in the complete collar residual. The second is the endpoint-moment adjoint identity. The calculated differentiated endpoint kernel explains why applying the second identity to the singular boundary average requires an additional estimate.

5.2. Actual probe outcomes

All probes used the original complete operator A_a^- , with every ordinary and reflected prime packet and every archimedean, polar, and boundary term retained.

Explicit probe	Calculated energy or source bound	Calculated projected residual
Endpoint moment $j_a = x/2$	$\ F_a^* j_a\ \leq a^{3/2}/\sqrt{6}$	Zero for its exact image $A_a^- j_a = k_a^-$; this image is the endpoint-moment kernel.
Boundary-layer derivative probe	$Q_a^-(\psi) = e^L(1 + O_R(L^{-1}))$	$b_a(L) + O_R(L^{-1/2})$.
Direct boundary probe $g_{a,L}/L$	$Q_a^-(\psi) = e^L(1 + O_R(L^{-1}))/ (4L^2)$	$b_a(L) + O_R(L^{-1/2})$.

Explicit probe	Calculated energy or source bound	Calculated projected residual
Reference inverse $P_a^{-1}g_{a,L}$, $P_a = C_a + \lambda_R I \geq I$	Uniformly bounded $F_a^*\psi$; energy has an exponential lower bound	A complete signed arithmetic boundary average, uniformly bounded, with almost-everywhere limit Γ_a .
Finite regularized Neumann probe	Exact energy decomposition, including the high-frequency contribution	$\hat{K}_a^m b_a(L)$.
Finite source-space probe $\tau_R W_a \hat{K}_a^\dagger s_m(\hat{K}_a) b_a(L)$	Exact finite-matrix energy; at most $2\tau_R m B^2$ on the critical sector $1/2 \leq k \leq 1$	$\hat{K}_a^m b_a(L)$; the critical-sector version retains its low-sector residual explicitly.

Here $s_m(k) = \sum_{j=0}^{m-1} k^j$, and the pseudoinverse is only that of the positive finite source Gram matrix. No uniform bound for it or inverse of the unregularized Weil operator is assumed. The source-space construction removes the high-frequency energy of the first Neumann probe. It reaches the proposed logarithmic energy scale on the critical sector when $m = O(\log(eL))$. Its residual still lacks uniform decay as k approaches one.

For any operator-domain probe on a nonnegative Weil interval, the exact projected dual-energy inequality is

$$D_a = W_a^* A_a^- W_a = \hat{K}_a M_a / \tau_R, \quad h = b_a(L) - W_a^* e_{a,L} \in \text{Ran } D_a, \quad (\text{CP18:5.8})$$

$$Q_a^-(\psi_{a,L}) \geq h^* D_a^\dagger h.$$

In a null direction $\hat{K}_a c = c$, it gives the particularly precise identity

$$c^* W_a^* e_{a,L} = c^* b_a(L) \quad \text{for every } \psi_{a,L} \in \text{Dom } A_a^-. \quad (\text{CP18:5.9})$$

Thus the projected residual at a first null contains exactly the boundary coefficient that the new arithmetic estimate must control. This conclusion concerns the actual complete operator and both null branches.

5.3. A calculated signed residual with an almost-everywhere limit

The reference probe supplies a concrete residual for the next calculation. Define

$$\begin{aligned} P_a &= C_a + \lambda_R I \geq I, \\ B_a &= (\lambda_R + \log(2\pi))I + \Pi_a + K_{\alpha,a} + 2|s_a\rangle\langle s_a|, \\ h_a(L) &= F_a^* P_a^{-1} g_{a,L}, \\ r_a(L) &= W_a^* B_a P_a^{-1} g_{a,L}. \end{aligned} \quad (\text{CP18:5.10})$$

The complete reference-resolvent identity gives

$$b_a(L) = M_a h_a(L) + r_a(L), \quad \sup_{a,L} \|h_a(L)\| \leq B_R^{\text{ref}}, \quad \sup_{a,L} \|r_a(L)\| \leq B_R^{\text{rem}}. \quad (\text{CP18:5.11})$$

Every prime-prime, prime-archimedean, prime-polar, and remaining mixed pairing in $r_a(L)$ is expanded in Group 3. The pure logarithmic reference operator is also a fixed physical form restriction with bounded fixed sources. Applying the already proved matrix collar theorem to that reference family yields the almost-everywhere limits

$$h_a = \lim_{L \rightarrow \infty} h_a(L), \quad \boxed{\Gamma_a = \lim_{L \rightarrow \infty} r_a(L) = \beta_a - M_a h_a.} \quad (\text{CP18:5.12})$$

The available estimate is $\|\Gamma_a\| \leq B_R^{\text{rem}}$. Neither its vanishing nor an energy-dependent bound for its critical component has been proved.

One precisely quantified sufficient lemma now uses this actual residual. On the positive branch, let $\mathfrak{E}_a = \ker(M_a - \varepsilon_a I)$, with $\varepsilon_a = \lambda_{\min}(M_a)$. Prove, for almost every a before a first crossing,

$$\boxed{\|\text{Proj}_{\mathfrak{E}_a} \Gamma_a\|^2 \leq \mathfrak{d}_R(a) \varepsilon_a \log(e/\varepsilon_a), \quad 0 \leq \mathfrak{d}_R \in L^1([a_0, R]),} \quad (\text{CP18:5.13})$$

with an independently specified integrable coefficient, containing no reciprocal of the unknown positivity margin. The exact decomposition (5.12) would then give

$$\begin{aligned} -\varepsilon'_a &= \tau_R \|\text{Proj}_{\mathfrak{E}_a} \beta_a\|^2 \\ &\leq 2\tau_R (B_R^{\text{ref}})^2 \varepsilon_a^2 + 2\tau_R \mathfrak{d}_R(a) \varepsilon_a \log(e/\varepsilon_a) \\ &\leq 2\tau_R [(B_R^{\text{ref}})^2 + \mathfrak{d}_R(a)] \varepsilon_a \log(e/\varepsilon_a). \end{aligned} \quad (\text{CP18:5.14})$$

This is an Osgood inequality with an integrable coefficient. Matrix absolute continuity and multiplicities are already handled. The currently proved bound replaces the right side of (5.13) by the absolute constant $(B_R^{\text{rem}})^2$; it does not supply the required vanishing factor.

The stronger statement $\Gamma_a = 0$ almost everywhere would also suffice for this particular reference probe. It is stated as a sufficient estimate, not as a necessary consequence of RH. The exact target is the sign of the complete form; other signed estimates may close it.

5.4. Adaptive comparison and final status

The proposed finite-resolution route has a complete conditional return proof. If one constructed a probe with the uniform bounds

$$Q_a^-(\psi_{a,L}) \leq C_\psi \log(eL), \quad \|W_a^* e_{a,L}\| \leq C_e L^{-1/2}, \quad (\text{CP18:5.15})$$

then complete-form Cauchy-Schwarz and the proved collar remainder would yield

$$0 \preceq \widehat{K}_{a+\delta} - \widehat{K}_a \preceq \delta \left[A \log(eL) \widehat{K}_a M_a + B L^{-1/2} I \right], \quad \delta = e^{-L}. \quad (\text{CP18:5.16})$$

Group 4 proves that the adaptive choice $L = D/\varepsilon_a^2$, with its displayed explicit uniform D , prevents the support steps from accumulating at a finite first crossing. The proof controls $\log \log(e/\varepsilon_a)$ by the total support increment. The evaluated probes have not established the full uniform certificate (5.15).

Required closing question	Established answer
Does fixed physical restriction preserve the sign test and both null branches?	Yes, with the same independent g_R , the restricted Gram inequality, and exact nullspace maps.
Is the whole physical matrix monotone and absolutely continuous?	Yes. It is uniformly Lipschitz on every fixed finite range, including all prime thresholds.
Is the almost-everywhere boundary Gram formula established?	Yes, in the explicit unitary half-interval normalization.
Which probe was evaluated, and what residual remains?	The table above gives the endpoint, direct, reference, and finite-source calculations. The concrete signed reference residual has limit Γ_a in (5.12); the finite-source residual is $\widehat{K}_a^m b_a(L)$.

Required closing question	Established answer
Is an integrable energy-relative, determinant, or Osgood bound proved independently of the unknown margin?	The implication from each proposed estimate is proved. The necessary new signed bound, such as (5.13), remains open.
Are both first-null branches excluded, and does unconditional Weil positivity follow?	Both are excluded on the preserved interval and the constant-dependent local extension (5.6). Global first-null exclusion and unconditional Weil positivity remain open.

If (5.13), the full energy-relative trace bound, the bounded-probe residual criterion, or (5.15) were independently established for every finite R , the complete retained return would be

$$\begin{aligned}
 & \text{independent signed closing estimate} \\
 \implies M_a > 0 \text{ for every finite } a & \implies \text{strict odd Weil positivity on every finite support} \quad (\text{CP18:5.17}) \\
 \implies \text{RH} \implies \mathfrak{W} = \mathcal{S} \implies \Delta = 0 & \implies \mathcal{E}_N = 0 \text{ for every } N \implies \liminf_N \mathcal{E}_N \leq 1.
 \end{aligned}$$

The odd Weil criterion and Suzuki identity in this chain retain their established normalization. The antecedent is not proved in this checkpoint.

Final status. The physical matrix evolution, averaged boundary Gram derivative, endpoint adjoint relation, and explicit probe calculations are proved with their stated domains and constants. The actual arithmetic residual still requires an estimate that vanishes with the small Weil energy. Global unconditional Weil positivity and the original defect criterion remain open. All earlier checkpoint contents are preserved. We are aiming to prove unconditional Weil positivity with the original Suzuki metric preserved.

Checkpoint 19: Recovered machinery and complete energy

Source: Michaels_Existing_Machinery_Energy_Closure_Checkpoint_19.md. Equation tags carry the source prefix CP19; references within this unit retain their original local numbering.

Checkpoint 19 – 23 September 2026

This calculation continues Checkpoint 18, uses the recovered 301-entry theory atlas as a source map, and applies the underlying energy, packet, storage, and response identities to the actual forced family. The primary objective is unconditional Weil positivity. The all-support sign remains open at the end of this checkpoint; the independently proved comparisons and exact remaining arithmetic expression are recorded below.

The retained, independently certified starting interval is

$$a_0 = \frac{699}{2000} + e^{-2500}, \quad Q_a^-(q) \geq \frac{1}{5000} \|q\|_2^2 \quad (0 < a \leq a_0). \quad (\text{CP19:0.1})$$

The original Suzuki metric and determining ladder are unchanged:

$$\Delta = \mathfrak{W} - \mathcal{S}, \quad \mathcal{E}_N = \left\| [\Delta(\rho_j, \rho_i)]_{i,j \leq N} \right\|_F^2, \quad \liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1. \quad (\text{CP19:0.2})$$

Finite Gram matrices introduced below are comparisons for the complete Weil operator. They do not redefine $\mathcal{S}$ or the normalization of $\mathcal{E}_N$.

Contents

1. Complete theta-weighted energy on the actual forced family: domain, entries, pole variance, exterior tails, and matrix congruence.
2. Arithmetic cancellation in the same entries: complete four-channel packets, overlap and response savings, exact Green/storage/color transport, and the odd theta-derivative comparison.
3. Explicit responses for the physical regularizer: complete residual Gram, finite optimization, and uniform quantitative certificate accuracy.
4. Combined sign calculation, exact remaining inequality, first-null and continuation alternatives, and the conditional Weil-Suzuki return.
5. Source register and preservation record.

Fixed conventions and inherited inputs

Inner products are conjugate-linear in the first variable. All supported functions are extended by zero. The Fourier convention is

$$\widehat{q}(\xi) = \int_{\mathbb{R}} q(x) e^{i\xi x} dx, \quad \ell_n = \log n, \quad b_n = \Lambda(n)/\sqrt{n}. \quad (\text{CP19:0.3})$$

Fix an arbitrary finite $R > a_0$. The supported odd form domain is

$$\mathcal{D}_a = \left\{ q \in L_{\text{odd}}^2(-a, a) : \int_{\mathbb{R}} \log(2 + |\xi|) |\widehat{q}(\xi)|^2 d\xi < \infty \right\}, \quad a_0 \leq a \leq R. \quad (\text{CP19:0.4})$$

Write C_a for the exterior-Dirichlet form compression of $\log |D|$. The full odd arithmetic operator is

$$\begin{aligned} A_a^- &= C_a - \log(2\pi)I - \Pi_a - K_{\alpha,a} - 2|s_a\rangle\langle s_a|, & s_a(x) &= \sinh(x/2), \\ \Pi_a q(x) &= \sum_{\ell_n < 2R} b_n [q(x + \ell_n) + q(x - \ell_n)], & K_{\alpha,a} q(x) &= \int_{-a}^a \alpha(x-y)q(y) dy, \\ \kappa(t) &= \frac{e^{-t/2}}{1 - e^{-2t}} \quad (t > 0), & \alpha(t) &= \kappa(|t|) - \frac{1}{2|t|}, \quad \alpha(0) = \frac{1}{4}. \end{aligned} \tag{CP19:0.5}$$

Zero extension retains every threshold and boundary contribution. In particular, a prime shift with $\ell_n \geq 2a$ vanishes in the compressed original operator. This does not justify dropping the infinite exterior interactions created by a ground-state transformation.

For completeness the independently established regularizer constants are

$$\begin{aligned} D_R^{\text{ar}} &= \log(2\pi) + 2 \sum_{\ell_n < 2R} b_n + 4R + 2(\sinh R - R), \\ c_{\text{odd}} &= \frac{1}{3} \log(9\pi/2) - \frac{1}{3}, & h_R &= -\log R + c_{\text{odd}}, \\ d_R &= \max(1, \lceil (2R/\pi)e^{D_R^{\text{ar}}+2} \rceil), \\ \tau_R &= 1 + \max\{0, D_R^{\text{ar}} + (D_R^{\text{ar}})^2 - h_R\}, \\ g_R &= [\max\{2, 1 + 2(D_R^{\text{ar}})^2\}]^{-1}. \end{aligned} \tag{CP19:0.6}$$

Let F_R have the fixed physical reference columns from CP18, with $F_R^* F_R = I$, and put $F_a = E_a^* F_R$, where E_a is zero extension to $(-R, R)$. Then

$$\mathcal{J}_a = A_a^- + \tau_R F_a F_a^* \geq g_R I, \quad F_a^* F_a \preceq I, \quad W_a = \mathcal{J}_a^{-1} F_a. \tag{CP19:0.7}$$

The restricted columns need not be independent or orthonormal. Throughout this checkpoint K_a denotes CP18's physical matrix $\hat{K}_a$:

$$K_a = \tau_R F_a^* W_a, \quad M_a = I - K_a, \quad A_a^- W_a = F_a M_a, \quad W_a^* A_a^- W_a = \tau_R^{-1} K_a M_a. \tag{CP19:0.8}$$

The complete sign, inertia, and nullspace equivalences from CP18 apply. Every null vector, including one with zero polar moment, corresponds to a coefficient vector in $\ker M_a$. The coefficient kernel of F_a has $W_a c = 0$ and $M_a c = c$; it introduces no false null.

Also retained are the strong form/Suzuki core passage, complete convolution endpoint equations, and the physical boundary evolution

$$0 \preceq K_b - K_a \preceq C_{\text{Lip},R}(b-a)I, \quad K'_a = \tau_R \beta_a \beta_a^* \quad \text{a.e.} \tag{CP19:0.9}$$

Here β_a is the averaged boundary vector for the unitary half-interval columns $\sqrt{2}W_a e_j|_{(0,a)}$. No pointwise normalized trace is assumed. All constants in this checkpoint may depend on the fixed finite R , as stated, and none is allowed to hide a reciprocal of the unknown positivity margin.

1. Complete theta energy on the actual forced family

1.1. Source theorem, normalization, and fixed domains

The source used here is *Physics_Architectures_and_Log_Prime_Energy.pdf*, version 4, §§17-19, equations (30), (33)-(40). Its Fourier convention is

$$\hat{q}(\xi) = \int_{\mathbb{R}} q(x) e^{i\xi x} dx, \quad \Xi(z) = \xi(1/2 + iz). \tag{CP19:1.1}$$

The positive even theta weight in that convention is

$$\phi(u) = 2\Phi(u/2), \quad \Phi(v) = \sum_{k \geq 1} (2\pi^2 k^4 e^{9v} - 3\pi k^2 e^{5v}) e^{-\pi k^2 e^{4v}}. \quad (\text{CP19:1.2})$$

Thus $\widehat{\phi} = \Xi$. In particular, a curvature constant for the variable v in $\Phi(v)$ must be divided by four before being applied to the variable u in $\phi(u)$. No unverified curvature constant is used below.

Write

$$a_\Gamma(\xi) = \Re \psi(1/4 + i\xi/2) - \log \pi, \quad \kappa(t) = \frac{e^{-|t|/2}}{1 - e^{-2|t|}}, \quad b_n = \frac{\Lambda(n)}{\sqrt{n}}, \quad \ell_n = \log n. \quad (\text{CP19:1.3})$$

The source's complete Weil operator is the gamma multiplier, minus both prime translations, plus the kernel $2 \cosh((x - y)/2)$. On odd functions its pole operator is

$$Z^- q = -2 \sinh(x/2) \int_{-a}^a \sinh(y/2) q(y) dy. \quad (\text{CP19:1.4})$$

These are exactly CP18's signs and factors. In particular, the source's Q_a restricted to odd functions is CP18's Q_a^- , rather than a rescaled form.

Fix $a_0 \leq a \leq R < \infty$. All functions on $(-a, a)$ are extended by zero. Use the already established supported odd closed form domain $\mathcal{D}_a$, the physical columns F_a , and

$$\mathcal{J}_a = A_a^- + \tau_R F_a F_a^* \geq g_R I, \quad W_a = \mathcal{J}_a^{-1} F_a, \quad K_a = \tau_R F_a^* W_a, \quad M_a = I - K_a. \quad (\text{CP19:1.5})$$

The columns $q_j = W_a e_j$ belong to $\text{Dom } A_a^- \subset \mathcal{D}_a$, are real and odd for the retained real physical basis, and need not have classical endpoint traces. Complex coefficient vectors remain allowed.

The prior source proves the exact, unconditional theta equation

$$a_\Gamma(D)\phi(x) + \cosh(x/2) = \sum_{n \geq 2} b_n [\phi(x + \ell_n) + \phi(x - \ell_n)]. \quad (\text{CP19:1.6})$$

Its derivation uses the vanishing of Ξ at the actual zero parameters, including any off-line zeros, and theta tail control of compact cutoffs. It does not prescribe the real parts of those zeros. The polar source is $\cosh(x/2)$ because $\Xi(i/2) = \Xi(-i/2) = 1/2$.

1.2. Passage of the complete energy identity to the forced columns

For $q = \phi h$ supported in $[-a, a]$, define

$$\begin{aligned} E_\Gamma(h) &= \frac{1}{2} \iint \kappa(x - y) \phi(x) \phi(y) |h(x) - h(y)|^2 dx dy, \\ E_P(h) &= \sum_{n \geq 2} b_n \int \phi(x) \phi(x + \ell_n) |h(x + \ell_n) - h(x)|^2 dx, \\ E_Z(h) &= \iint \cosh((x - y)/2) \phi(x) \phi(y) |h(x) - h(y)|^2 dx dy. \end{aligned} \quad (\text{CP19:1.7})$$

All variables in these integrals range over the whole line. In particular, the exterior of the support is part of the forms.

Put

$$d\nu(x) = 2 \cosh(x/2) \phi(x) dx, \quad \vartheta(x) = \tanh(x/2). \quad (\text{CP19:1.8})$$

The exact normalization is $\nu(1) = 1$, and symmetry gives $\nu(\vartheta) = 0$. Expansion of the two squared differences, retaining both polar cross terms, gives

$$E_Z(h) = \frac{1}{2} [\|h - \nu(h)\|_\nu^2 + |\nu(\vartheta h)|^2]. \quad (\text{CP19:1.9})$$

For odd h ,

$$E_Z(h) = \int_{-a}^a \frac{\cosh(x/2)}{\phi(x)} |q(x)|^2 dx + 2 \left| \int_{-a}^a \sinh(x/2) q(x) dx \right|^2. \quad (\text{CP19:1.10})$$

The first term in (1.10) is the diagonal potential produced by the ground-state substitution. It is not part of the original rank-one pole alone. It cancels against corresponding diagonal pieces of the positive energies by (1.6).

Here is the required domain argument, without assuming extra smoothness of q_j . On fixed support, ϕ is smooth and bounded away from zero. Take an odd compact smooth form-core sequence $q^{(m)} \rightarrow q$ in $\mathcal{D}_a$, and put $h^{(m)} = q^{(m)}/\phi$. The source identity holds on this core:

$$E_\Gamma(h^{(m)}) + E_P(h^{(m)}) = Q_a^-(q^{(m)}) + E_Z(h^{(m)}). \quad (\text{CP19:1.11})$$

On the fixed support the rightmost form is bounded in the L^2 norm:

$$0 \leq E_Z(q/\phi) \leq \left[\sup_{|x| \leq a} \frac{\cosh(x/2)}{\phi(x)} + 2(\sinh a - a) \right] \|q\|_2^2. \quad (\text{CP19:1.12})$$

Applying (1.11) to differences shows that the sequence of pairs of difference functions is Cauchy in the direct sum of the two positive energy L^2 spaces. Local L^2 convergence, followed by an almost-everywhere subsequence, identifies their limits with the actual differences of q/ϕ . The prime energy space is a countable Hilbert direct sum; no interchange of conditionally convergent signed sums is used. Consequently each positive energy converges, including its exterior part, and polarization passes all cross terms. Therefore, for every odd $q \in \mathcal{D}_a$,

$$\boxed{Q_a^-(q) = E_\Gamma(q/\phi) + E_P(q/\phi) - E_Z(q/\phi).} \quad (\text{CP19:1.13})$$

This proves the requested passage for every column of W_a and every linear combination of those columns.

1.3. Every entry of the actual matrix

Use conjugate-linearity in the first argument. Set

$$h_j = q_j/\phi, \quad (\mathbf{G}_a)_{ij} = \int_{-a}^a \frac{2 \cosh(x/2)}{\phi(x)} \overline{q_i(x)} q_j(x) dx, \quad (\ell_a)_j = 2 \overline{\int_{-a}^a \sinh(x/2) q_j(x) dx}. \quad (\text{CP19:1.14})$$

Then the exact polar matrix is

$$(\mathbf{Z}_a)_{ij} = \frac{1}{2} (\mathbf{G}_a + \ell_a \ell_a^*)_{ij}. \quad (\text{CP19:1.15})$$

The full positive entries are

$$\begin{aligned}
 (\mathbf{E}_{\Gamma,a})_{ij} &= \frac{1}{2} \iint \kappa(x-y) \phi(x) \phi(y) \overline{[h_i(x) - h_i(y)]} [h_j(x) - h_j(y)] dx dy, \\
 \mathbf{B}_a(t)_{ij} &= \int \phi(x) \phi(x+t) \overline{[h_i(x+t) - h_i(x)]} [h_j(x+t) - h_j(x)] dx, \\
 \mathbf{E}_{P,a} &= \sum_{n \geq 2} b_n \mathbf{B}_a(\ell_n), \quad \mathbf{E}_a = \mathbf{E}_{\Gamma,a} + \mathbf{E}_{P,a}.
 \end{aligned} \tag{CP19:1.16}$$

Every $\mathbf{B}_a(t)$ is positive semidefinite. An expansion that displays both prime cross terms and all exterior diagonals is

$$\begin{aligned}
 \mathbf{B}_a(t)_{ij} &= \int_{-a}^a \frac{\phi(x+t) + \phi(x-t)}{\phi(x)} \overline{q_i(x)} q_j(x) dx \\
 &\quad - \int_{\mathbb{R}} [\overline{q_i(x+t)} q_j(x) + \overline{q_i(x)} q_j(x+t)] dx.
 \end{aligned} \tag{CP19:1.17}$$

There is no theta ratio in the two translation cross terms. Their compressed forms vanish when $t \geq 2a$; their diagonal counterparts do not vanish.

For the gamma part, a division into the interior square and its complete exterior is

$$\begin{aligned}
 (\mathbf{E}_{\Gamma,a})_{ij} &= \frac{1}{2} \int_{-a}^a \int_{-a}^a \kappa(x-y) \phi(x) \phi(y) \overline{[h_i(x) - h_i(y)]} [h_j(x) - h_j(y)] dx dy \\
 &\quad + \int_{-a}^a D_{\Gamma,a}(x) \overline{q_i(x)} q_j(x) dx, \\
 D_{\Gamma,a}(x) &= \frac{1}{\phi(x)} \int_{\mathbb{R} \setminus [-a,a]} \kappa(x-y) \phi(y) dy.
 \end{aligned} \tag{CP19:1.18}$$

The possible endpoint singularity is retained in the second line. For example, as $d \downarrow 0$,

$$D_{\Gamma,a}(a-d) = \frac{1}{2} \log(1/d) + O_a(1). \tag{CP19:1.19}$$

To prove (1.19), integrate $1/[2(d+u)]$ over $0 < u < \eta$ at $y = a+u$; the smooth ratio $\phi(a+u)/\phi(a-d)$ is $1+O_a(d+u)$. The regular remainder in κ , the far exterior, and the opposite endpoint give bounded terms. Thus the boundary finite part has not disappeared under the theta transformation.

The original forced identity now yields the exact finite matrix equality

$$\boxed{\mathbf{E}_a - \frac{1}{2}(\mathbf{G}_a + \ell_a \ell_a^*) = W_a^* A_a^- W_a = \tau_R^{-1} K_a (I - K_a).} \tag{CP19:1.20}$$

On $\text{Ran } K_a$,

$$\tau_R K_a^{-1/2} \left[\mathbf{E}_a - \frac{1}{2}(\mathbf{G}_a + \ell_a \ell_a^*) \right] K_a^{-1/2} = M_a|_{\text{Ran } K_a}. \tag{CP19:1.21}$$

The range is invariant under M_a . If $K_a c = 0$, positivity of $\mathcal{J}_a$ implies $W_a c = 0$, while $M_a c = c$. This congruence therefore loses no possible null or negative direction of A_a^- . It introduces no change to the Suzuki metric.

A useful exact polar simplification, valid for every coefficient vector, is

$$\ell_a \ell_a^* \preceq \zeta_a \mathbf{G}_a, \quad \zeta_a = \int_{-a}^a \vartheta(x)^2 d\nu(x) < \tanh^2(a/2) < 1. \tag{CP19:1.22}$$

It is Cauchy-Schwarz in the actual measure ν , with the support restriction retained. Consequently the independently sufficient comparison

$$\mathbf{E}_a \succeq \frac{1 + \zeta_a}{2} \mathbf{G}_a \quad (\text{CP19:1.23})$$

would pay the polar term. It has not been proved for every a ; the exact rank-one term in (1.20) is sharper.

The alternative inverse test has its kernel condition intact. With $\mathbf{H}_a = 2\mathbf{E}_a - \mathbf{G}_a$,

$$\mathbf{H}_a \succeq \ell_a \ell_a^* \iff \mathbf{H}_a \succeq 0, \quad \ell_a \in \text{Ran } \mathbf{H}_a, \quad \ell_a^* \mathbf{H}_a^\dagger \ell_a \leq 1. \quad (\text{CP19:1.24})$$

This is the finite-dimensional rank-one subtraction theorem. It is an exact test, not an independently obtained lower bound for $\mathbf{H}_a$.

1.4. Explicit exterior prime tails, in the complete metric

For $t > 2a$, (1.17) becomes exactly

$$\mathbf{B}_a(t)_{ij} = \int_{-a}^a \frac{\phi(x+t) + \phi(x-t)}{\phi(x)} \overline{q_i(x)} q_j(x) dx. \quad (\text{CP19:1.25})$$

These infinitely many positive terms are part of the transformed energy even though the original compressed translation operator has only finitely many active shifts.

Here are explicit theta bounds sufficient for both norm convergence and later regrouping. Put

$$r_0 = e^{-\pi}, \quad C_\theta = 4\pi^2 \frac{1 + 11r_0 + 11r_0^2 + r_0^3}{(1 - r_0)^5}, \quad m_a^\phi = (4\pi^2 - 6\pi)e^{-\pi e^{2a}}. \quad (\text{CP19:1.26})$$

For $u \geq 0$, positivity of each term in (1.2), followed by $k^2 - 1 \geq k - 1$, proves

$$0 < \phi(u) \leq C_\theta e^{9u/2} e^{-\pi e^{2u}}, \quad \inf_{|x| \leq a} \phi(x) \geq m_a^\phi > 0. \quad (\text{CP19:1.27})$$

The scalar upper envelope in (1.27) decreases on $[0, \infty)$. Hence

$$0 \leq \mathbf{B}_a(t) \preceq \frac{2C_\theta}{m_a^\phi} e^{9(t-a)/2} e^{-\pi e^{2(t-a)}} W_a^* W_a \quad (t > 2a). \quad (\text{CP19:1.28})$$

For an integer N with

$$N \geq e^{2a}, \quad N^2 \geq \frac{5}{2c_a}, \quad c_a = \pi e^{-2a}, \quad (\text{CP19:1.29})$$

use $\Lambda(n) \leq \log n \leq n$ and the decreasing function $x^5 e^{-c_a x^2}$ to obtain

$$\begin{aligned} 0 &\leq \sum_{n > N} b_n \mathbf{B}_a(\ell_n) \preceq T_a(N) W_a^* W_a, \\ T_a(N) &= \frac{2C_\theta e^{-9a/2}}{m_a^\phi} e^{-c_a N^2} \left(\frac{N^4}{2c_a} + \frac{N^2}{c_a^2} + \frac{1}{c_a^3} \right). \end{aligned} \quad (\text{CP19:1.30})$$

In particular, its coefficient-space operator norm is at most $T_a(N)/g_R^2$. This is an unconditional, explicit positive-semidefinite tail budget, with all off-diagonal coefficient pairings retained. For

uniformity on $a \leq R$, replace the support quantities in the preceding direct bound by R and use $N \geq e^{2R}$ and $N^2 \geq 5e^{2R}/(2\pi)$. Polynomial factors added by a divisor regrouping remain integrable against the same Gaussian in N .

There is also a nonzero exterior floor on every nonzero forced image. Choose an integer $k > 2a/\log 2$, set $t_k = k \log 2$, and define

$$p_a = \frac{\log 2}{2^{k/2}} \min_{|x| \leq a} \frac{\phi(x + t_k) + \phi(x - t_k)}{\phi(x)} > 0. \quad (\text{CP19:1.31})$$

The minimum is over a compact interval of a positive continuous function. Then

$$\mathbf{E}_{P,a} \succeq p_a W_a^* W_a. \quad (\text{CP19:1.32})$$

A direct comparison in the exact polar measure is also available. For $u \geq a_0$, every positive theta summand decreases: with $z = \pi k^2 e^{2u} > 2\pi$, its logarithmic derivative is $5/2 + 4z/(2z - 3) - 2z < 0$. Since $t_k - a > a \geq a_0$, both exterior arguments lie in this decreasing region. Consequently

$$\mathbf{B}_a(t_k) \succeq \frac{\phi(t_k + a)}{\cosh(a/2)} \mathbf{G}_a, \quad \mathbf{E}_{P,a} \succeq c_{\text{ext}}(a) \mathbf{G}_a, \quad c_{\text{ext}}(a) = \frac{\log 2}{2^{k/2}} \frac{\phi(t_k + a)}{\cosh(a/2)} > 0. \quad (\text{CP19:1.33})$$

This is an independent lower comparison in the required weighted matrix, rather than only strict positivity. Its size does not suffice for the polar payment. For example, $t_k + a > 3a \geq 3a_0 > 1$; the elementary bounds $C_\theta < 100$, $e^{9/2} < 100$, $\pi e^2 > 21$, and $e^3 > 20$ yield

$$c_{\text{ext}}(a) < 10^{-5}, \quad \mathbf{E}_a - \mathbf{Z}_a \succeq \left[c_{\text{ext}}(a) - \frac{1 + \zeta_a}{2} \right] \mathbf{G}_a. \quad (\text{CP19:1.34})$$

The scalar coefficient furnished by this particular lower-bound argument is less than -0.49999 . That is a deficit of the estimate, not a negative value of the actual form. The full interior gamma energy, the active prime interactions, and the other positive exterior jumps remain available for a sharper coupled comparison. Thus (1.32) and (1.33) prove strict positive prime energy on every nonzero image but do not exclude equality in (1.20).

1.5. A calculated signed saving for the authentic exterior four-channel packet

The exact theta series yields a stronger comparison than a generic tail bound. For $u \geq 0, d > 0$, compare each positive summand of (1.2):

$$\frac{\phi_k(u + d)}{\phi_k(u)} = e^{5d/2} \frac{2\pi k^2 e^{2u} e^{2d} - 3}{2\pi k^2 e^{2u} - 3} e^{-\pi k^2 e^{2u} (e^{2d} - 1)}. \quad (\text{CP19:1.35})$$

The fraction decreases in $k^2 e^{2u}$; its maximum is at $k = 1, u = 0$. Thus

$$\phi(u + d) \leq \rho_d(u) \phi(u), \quad \rho_d(u) = e^{5d/2} \frac{2\pi e^{2d} - 3}{2\pi - 3} \exp[-\pi(e^{2d} - 1)e^{2u}]. \quad (\text{CP19:1.36})$$

Summing preserves this inequality because the theta summands are positive. For $t > 2a$, apply it separately to $t + x$ and $t - x$:

$$\boxed{\mathbf{B}_a(t + d) \preceq \rho_d(t - a) \mathbf{B}_a(t)}. \quad (\text{CP19:1.37})$$

This is a matrix inequality for the actual forced columns, not an arbitrary scalar-envelope assertion.

Let the uncut 1, 2, 3, 6 packet be

$$\mathcal{P}_a(t) = \mathbf{B}_a(t) - \frac{\mathbf{B}_a(t + \log 2)}{\sqrt{2}} - \frac{\mathbf{B}_a(t + \log 3)}{\sqrt{3}} + \frac{\mathbf{B}_a(t + \log 6)}{\sqrt{6}}. \quad (\text{CP19:1.38})$$

Keep the favorable fourth channel explicitly:

$$\mathcal{P}_a(t) \succeq \eta_a(t) \mathbf{B}_a(t) + \frac{\mathbf{B}_a(t + \log 6)}{\sqrt{6}}, \quad \eta_a(t) = 1 - \frac{\rho_{\log 2}(t - a)}{\sqrt{2}} - \frac{\rho_{\log 3}(t - a)}{\sqrt{3}}. \quad (\text{CP19:1.39})$$

For $a \geq a_0$ and $t > 2a$, one has $e^{2(t-a)} > e^{2a} > 2$, since $2a_0 > 699/1000 > \log 2$. Elementary bounds $\pi > 3$, $\pi < 22/7$, and $e^3 > 20$ give

$$\begin{aligned} \rho_{\log 2}(t - a) &< 45e^{-18} < 45/20^6, \\ \rho_{\log 3}(t - a) &< 500e^{-48} < 500/20^{16}, \\ \eta_a(t) &> 1 - 10^{-6}. \end{aligned} \quad (\text{CP19:1.40})$$

The constants are intentionally rounded outward. This is an independent signed packet estimate using the actual theta weight; it does not import a curvature theorem in a different scale.

For a finite arithmetic cutoff, the packet terms carry unequal indicators $drm \leq X$. Formula (1.39) applies when all four terms are present; the edge packets and their signed coefficients must be retained separately. After the absolutely convergent limit, Group 2's authentic Möbius coefficient is $\mu(r) \log m / \sqrt{rm}$. A positive lower bound for a packet cannot be multiplied by a negative $\mu(r)$ without reversing its direction. In particular the actual exterior regrouping has the signed lower budget

$$\sum_{\substack{r, m \geq 1 \\ \gcd(r, 6) = 1 \\ \log(rm) > 2a}} \frac{\mu(r) \log m}{\sqrt{rm}} \mathcal{P}_a(\log(rm)), \quad (\text{CP19:1.41})$$

where positive- $\mu(r)$ packets can use (1.39), but negative- $\mu(r)$ packets require an upper bound or a joint pairing. Thus (1.40) is a substantial concrete saving inside each exterior packet; it is not a global sign for the signed outer arithmetic sum. The original unregrouped prime energy remains positive throughout.

1.6. Full forced equation and equality at a first null

For a column combination $q = W_a c = \phi h$, let $m(q) = \int \sinh(x/2) q(x) dx$. In the distributional weak sense on $(-a, a)$, the complete transformed forced equation is

$$\begin{aligned} &\int_{\mathbb{R}} \kappa(x - y) \phi(y) [h(x) - h(y)] dy \\ &+ \sum_{n \geq 2} b_n \{ \phi(x + \ell_n) [h(x) - h(x + \ell_n)] + \phi(x - \ell_n) [h(x) - h(x - \ell_n)] \} \\ &= \cosh(x/2) h(x) + 2 \sinh(x/2) m(q) + (F_a M_a c)(x). \end{aligned} \quad (\text{CP19:1.42})$$

The singular gamma line is interpreted through its closed difference form. The entire infinite prime diagonal on the left is retained even for x inside the support. Equation (1.6) proves the identity on the smooth core, and the completed domain argument above passes it weakly.

At a hypothetical first null, $M_a c = 0$, $K_a c = c$, and $q = W_a c \neq 0$. Testing (1.42) with $\bar{h} \phi$ yields

$$E_\Gamma(h) + E_P(h) = \frac{1}{2} (\|h\|_\nu^2 + |\nu(\vartheta h)|^2) > 0. \quad (\text{CP19:1.43})$$

The positive right side does not vanish at a null vector. Consequently strict positivity of either positive energy, including the exterior floor (1.32), is compatible with a null vector; an independently strict comparison with the complete polar variance is still required. Both zero-polar-moment and nonzero-polar-moment branches remain represented in (1.43).

The logarithmic endpoint part in (1.19) gives a finite boundary-energy integral. It does not alone control the averaged boundary coefficient by $Q_a^-(q)$. Indeed direct Cauchy-Schwarz against this boundary energy gives a factor of size e^L/L^2 for the squared averaged functional at resolution e^{-L} , because

$$\frac{1}{4L} \int_{e^{-L}}^\eta \frac{dd}{d^2 \log(1/d)} \asymp \frac{e^L}{4L^2}. \quad (\text{CP19:1.44})$$

It controls a positive component of E_Γ , whereas the signed energy is $E_\Gamma + E_P - E_Z$. This explicit attempted bound does not provide CP18's vanishing factor in the near-null margin.

1.7. Audit of the recovered half-unit support certificate

The source's Theorem 19.1 states

$$Q_{1/2}(f) \geq 9 \cdot 10^{-8} \|f\|_2^2 \quad \text{on its complete closed form domain.} \quad (\text{CP19:1.45})$$

The normalization and domain transfer to CP18 are exact: the source uses the same gamma multiplier, both prime translations, the same odd pole, and the same zero-extension form domain. Its $a = 1/2$ window contains only the active prime 2. This recovered bound can coexist with the stronger 1/5000 floor on the smaller CP18-certified interval $a \leq a_0$.

The printed proof retains 24 even and 24 odd Legendre modes, proves a $|\xi| \geq 30$ floor

$$\beta = \log(30/(2\pi)) - 1/30 - \sqrt{2} \log 2 > 0.54, \quad (\text{CP19:1.46})$$

and bounds the quadrature-matrix error by 10^{-10} , complementary spectral loss by 10^{-34} , and cross-block norm by 10^{-14} . The source reports 24 positive interval LDL^* pivots in both sectors for $A_{\text{quad}} - (10^{-7} + 10^{-10})I$, with minimum lower pivot 0.0016 (even) and 0.0073 (odd). These finite directed-rounding sign outputs, not the analytic normalization, are the executable part of its proof.

The source refers to a reproducibility folder named `certificate_half`. This Group 1 calculation has read the full printed proof and checked its normalization; it has not independently rerun or authenticated the missing executable matrix output. Therefore (1.45) is a recovered recorded certificate with an exact transfer map. If the original executable certificate is recovered and checked by this round's source audit, it is an independently authenticated larger starting interval. Otherwise the fully retained CP18 certified interval and its floor remain the operative starting certificate, with (1.45) clearly labelled as the source's recorded result. No all-support conclusion is inferred from either interval.

1.8. Group 1 closing statement

Theorem proved in this group. For every fixed finite R , every $a_0 \leq a \leq R$, and every coefficient vector in the physical forced family, the source's complete theta identity extends to the exact closed

form and yields (1.20)-(1.21). The matrices have the explicit entries (1.14)-(1.18); the complete transformed exterior prime tail satisfies (1.30); and every uncut exterior 1, 2, 3, 6 packet satisfies the independently proved saving (1.39)-(1.40). These facts preserve the original Suzuki metric and both null branches.

Quantitative gain. The exterior packet retains more than $1 - 10^{-6}$ of its unshifted matrix, as well as its nonnegative 6-channel. The theta tail admits an explicit superexponential positive-semidefinite budget. The polar cost is its exact variance plus one explicit moment, and that moment has the support-dependent bound (1.22).

Remaining complete comparison. For every finite R and $a \in [a_0, R]$, it remains to prove independently

$$\mathbf{E}_{\Gamma,a} + \sum_{n \geq 2} b_n \mathbf{B}_a(\ell_n) - \frac{1}{2} \mathbf{G}_a - \frac{1}{2} \ell_a \ell_a^* \succeq 0, \quad (\text{CP19:1.47})$$

or a strictly positive value on every actual nonzero first-null candidate, or the sufficient CP18 energy-relative continuation estimate. The stopped line is the transition from positive packet estimates to payment of the complete polar variance in (1.47), with negative outer Möbius coefficients and every interior packet still coupled. Equation (1.20) confirms that this remaining sign is exactly the full target matrix; it is not a missing change of notation or an omitted scalar normalization.

2. Complete arithmetic cancellation and its exact coefficient maps

2.1. Prime packets on the forced theta-energy family

This calculation uses the exact factor-potential identity $\Lambda = \mu * \log$, the complete-prime-triple identity in *Prime_DNA_Angle_Families_and_Prime_Triple_Bound.md*, §§4-8 (source), and the actual theta ratio proved in Group 1. The hypotheses are the CP18 fixed physical family $q_j = W_a e_j$, $h_j = q_j / \phi$, zero extension, and finite $a \leq R$. No Möbius sign law is imposed on a forced column.

Write $D_t h(x) = h(x+t) - h(x)$ and

$$\mathbf{B}_a(t)_{ij} = \int_{\mathbb{R}} \phi(x) \phi(x+t) \overline{D_t h_i(x)} D_t h_j(x) dx. \quad (\text{CP19:2.1})$$

Thus $\mathbf{B}_a(t) \succeq 0$, and $\mathbf{B}_a(-t) = \mathbf{B}_a(t)$. Its complete entry is

$$\mathbf{B}_a(t)_{ij} = \int_{-a}^a \frac{\phi(x+t) + \phi(x-t)}{\phi(x)} \overline{q_i(x)} q_j(x) dx - \int [\overline{q_i(x+t)} q_j(x) + \overline{q_i(x)} q_j(x+t)] dx. \quad (\text{CP19:2.2})$$

The second integral includes both crosses. When $t > 2a$, it vanishes; the first remains positive. Equation (2.2), not a cutoff of the whole matrix, is the support rule for this representation.

2.1.1. Exact source completion and the four unequal cutoffs For each finite X , elementary Dirichlet convolution gives

$$\begin{aligned} \mathbf{E}_{P,X} &= \sum_{2 \leq n \leq X} \frac{\Lambda(n)}{\sqrt{n}} \mathbf{B}_a(\log n) \\ &= \sum_{\substack{r, m \geq 1 \\ \gcd(r,6)=1}} \frac{\mu(r) \log m}{\sqrt{rm}} \sum_{\substack{d|6 \\ dr m \leq X}} \frac{\mu(d)}{\sqrt{d}} \mathbf{B}_a(\log(rm) + \log d). \end{aligned} \quad (\text{CP19:2.3})$$

The term $m = 1$ is zero. Terms with nonsquarefree r are zero. For nonzero $\mu(r)$, extracting the factors 2 and 3 gives the unique representation of every squarefree divisor as dr , $d \mid 6$, $(r, 6) = 1$. This proves (2.3) including every moving boundary.

For $u = \log(rm)$, put

$$\mathcal{P}_a(u) = \mathbf{B}_a(u) - 2^{-1/2}\mathbf{B}_a(u + \log 2) - 3^{-1/2}\mathbf{B}_a(u + \log 3) + 6^{-1/2}\mathbf{B}_a(u + \log 6). \quad (\text{CP19:2.4})$$

If $rm \leq X/6$, the packet is complete. In the successive windows $(X/6, X/3]$, $(X/3, X/2]$, and $(X/2, X]$, its retained terms are respectively 1, 2, 3, then 1, 2, then 1. These endpoint terms are retained exactly, not identified with a completed packet. The exterior theta decay proved in Group 1 makes the fully expanded divisor series absolutely convergent in finite-dimensional operator norm: grouping by n , its absolute coefficient is at most $\sqrt{n} \log n$, while $\|\mathbf{B}_a(\log n)\|$ has a Gaussian-in- n majorant times a fixed power of n . Therefore (2.3) passes to the complete infinite energy with these tails tending to zero.

The actual arithmetic cancellation at one source location is exact. At n with at least two distinct prime factors,

$$\sum_{d|n} \mu(d) \log(n/d) = 0; \quad n = p^k : \quad \sum_{d|n} \mu(d) \log(n/d) = \log p. \quad (\text{CP19:2.5})$$

Thus composite mixed-color source positions cancel completely before the matrix is estimated; the surviving complete prime-power matrix is $\sum_{p,k} (\log p) p^{-k/2} \mathbf{B}_a(k \log p)$. For instance, the common matrix at $n = 6$ is multiplied by $\log 6 - \log 3 - \log 2 = 0$, and at $n = 12$ by $\log 12 - \log 6 - \log 4 + \log 2 = 0$. These are exact source cancellations, not small error estimates. They do not compare the remaining positive energy to the polar cost.

2.1.2. A quantitative comparison for the actual exterior four-channel packet The Group 1 normalization is $\hat{\phi} = \Xi$, with

$$\phi(u) = 2 \sum_{k \geq 1} (2\pi^2 k^4 e^{9u/2} - 3\pi k^2 e^{5u/2}) e^{-\pi k^2 e^{2u}}, \quad u \geq 0, \quad (\text{CP19:2.6})$$

and even continuation. This normalization matters: curvature constants in the variable $u/2$ cannot be used unchanged.

For $u \geq 0, d > 0$, every positive theta summand gives

$$\phi(u + d) \leq \mathbf{r}_d(u) \phi(u), \quad \mathbf{r}_d(u) = e^{5d/2} \frac{2\pi e^{2d} - 3}{2\pi - 3} \exp[-\pi(e^{2d} - 1)e^{2u}]. \quad (\text{CP19:2.7})$$

Indeed the ratio of the polynomial factors is bounded at the smallest $\pi k^2 e^{2u}$, and the exponential factor decreases with that variable. For $t > 2a$, both $t + x$ and $t - x$ are at least $t - a > 0$. Equations (2.2) and (2.7) therefore give the matrix comparison

$$\mathbf{B}_a(t + d) \preceq \mathbf{r}_d(t - a) \mathbf{B}_a(t). \quad (\text{CP19:2.8})$$

This is a comparison for the actual forced family, not an arbitrary comparison kernel.

Consequently the complete exterior packet satisfies

$$\begin{aligned} \mathcal{P}_a(t) &\succeq [1 - 2^{-1/2} \mathbf{r}_{\log 2}(t - a) - 3^{-1/2} \mathbf{r}_{\log 3}(t - a)] \mathbf{B}_a(t) + 6^{-1/2} \mathbf{B}_a(t + \log 6) \\ &\succeq (1 - 10^{-6}) \mathbf{B}_a(t), \quad a \geq a_0, \quad t > 2a. \end{aligned} \quad (\text{CP19:2.9})$$

Here the constant is an elementary rigorous bound, not a fitted number. Since $a_0 > 699/2000$, $e^{2(t-a)} > e^{699/1000} > 2$, $\pi > 3$, and $e^3 > 20$,

$$2^{-1/2} \mathbf{r}_{\log 2}(t-a) < 28e^{-18}, \quad 3^{-1/2} \mathbf{r}_{\log 3}(t-a) < 153e^{-48}, \quad 28e^{-18} + 153e^{-48} < 10^{-6}. \quad (\text{CP19:2.10})$$

A short Taylor lower sum proves each exponential inequality. All constants are independent of the coefficient direction and of the unknown Weil margin.

The sign of the outer source coefficient $\mu(r) \log m / \sqrt{rm}$ remains. Inserting (2.9) into the whole regrouped sum without retaining that sign would reverse the inequality for $\mu(r) = -1$. The exact surviving expression is

$$\sum_{\substack{(r,6)=1 \\ rm > e^{2a}}} \frac{\mu(r) \log m}{\sqrt{rm}} \mathcal{P}_a(\log(rm)) + \sum_{\substack{(r,6)=1 \\ rm \leq e^{2a}}} \frac{\mu(r) \log m}{\sqrt{rm}} \mathcal{P}_a(\log(rm)). \quad (\text{CP19:2.11})$$

This is absolutely convergent and equals the complete prime energy. The second sum includes packets whose individual descendants cross the support boundary. Its signed comparison to $\frac{1}{2}(\mathbf{G}_a + \ell_a \ell_a^*) - \mathbf{E}_{\Gamma,a}$ remains to be proved.

2.2. Color squares and quantitative response savings

This part uses the visible/hidden prime-filter identity from *Michaels_Prime_Color_Dual_Kernel_Bridge.md*, §§2-3, but applies it to actual unitary translations on the whole line. The columns may be the forced columns or trial-response columns in Group 3. The identification is an explicit operator substitution.

Let $(T_t f)(x) = f(x+t)$, let $r_p = p^{-1/2}$, and define

$$V_p^\pm = \frac{I - r_p T_{\pm \log p}}{1 + r_p}, \quad Z_p^\pm = \frac{\sqrt{r_p}(I + T_{\pm \log p})}{1 + r_p}. \quad (\text{CP19:2.12})$$

Unitarity and commutation give

$$(V_p^\pm)^* V_p^\pm + (Z_p^\pm)^* Z_p^\pm = I. \quad (\text{CP19:2.13})$$

For any finite column map D into $L^2(\mathbb{R})$, this polarizes to the complete four-color identity

$$\begin{aligned} D^* D &= (V_2 V_3 D)^* (V_2 V_3 D) + (Z_2 V_3 D)^* (Z_2 V_3 D) \\ &\quad + (V_2 Z_3 D)^* (V_2 Z_3 D) + (Z_2 Z_3 D)^* (Z_2 Z_3 D), \end{aligned} \quad (\text{CP19:2.14})$$

with either consistent orientation. Every mixed-color channel is retained. Put $c_{23} = (1 + 2^{-1/2})(1 + 3^{-1/2})$ and $Q^\pm = (I - r_2 T_{\pm \log 2})(I - r_3 T_{\pm \log 3})$. Then

$$(Q^\pm D)^* (Q^\pm D) = c_{23}^2 [D^* D - \mathcal{H}^\pm(D)], \quad (\text{CP19:2.15})$$

where

$$\mathcal{H}^\pm(D) = (Z_2^\pm V_3^\pm D)^* (Z_2^\pm V_3^\pm D) + (V_2^\pm Z_3^\pm D)^* (V_2^\pm Z_3^\pm D) + (Z_2^\pm Z_3^\pm D)^* (Z_2^\pm Z_3^\pm D) \succeq 0. \quad (\text{CP19:2.16})$$

This is the exact saving from the joint two-prime visible response relative to the scalar product bound; it is a calculated matrix, not an asserted uniform percentage.

For a complete source packet at $u = \log(rm)$, the two-sided response is

$$\mathcal{C}_u D = T_u Q^+ D + T_{-u} Q^- D. \quad (\text{CP19:2.17})$$

Its four coefficients are $1, -2^{-1/2}, -3^{-1/2}, 6^{-1/2}$, in both orientations. Its full Gram, before support compression, is

$$\begin{aligned} (\mathcal{C}_u D)^* (\mathcal{C}_u D) &= 2c_{23}^2 D^* D - c_{23}^2 [\mathcal{H}^+(D) + \mathcal{H}^-(D)] \\ &\quad + D^* (Q^+)^* T_{-2u} Q^- D + D^* (Q^-)^* T_{2u} Q^+ D. \end{aligned} \quad (\text{CP19:2.18})$$

Thus the interaction between the two orientations is explicit. Summing packets adds the complete terms $\overline{w_{r,m}} w_{r',m'} D^* \mathcal{C}_u^* \mathcal{C}_u D$; one may not retain only the self-Grams in (2.18). For finite cutoff X , use the actual truncated four-term $\mathcal{C}_{u,X}$. Its difference from (2.17) consists exactly of descendants $drm > X$. If $X = e^{2a}$ and D is supported in $(-a, a)$, the compression to that interval kills these descendants, although their whole-line norms and exterior interactions remain in any use of (2.18).

A completely coupled version works directly for all prime packets at once. Write

$$\mathcal{P}_a^{\text{free}} = \sum_{\log n < 2a} b_n (T_{\log n} + T_{-\log n}), \quad p_a(\xi) = 2 \sum_{\log n < 2a} b_n \cos(\xi \log n), \quad B_a^{\text{abs}} = 2 \sum_{\log n < 2a} b_n, \quad (\text{CP19:2.19})$$

and $\Pi_a = E_a^* \mathcal{P}_a^{\text{free}} E_a$. For supported trial columns Y , Plancherel and orthogonal support splitting prove

$$(\Pi_a Y)^* (\Pi_a Y) = (B_a^{\text{abs}})^2 Y^* Y - \mathcal{S}_{\text{phase}}(Y) - \mathcal{S}_{\text{exterior}}(Y), \quad (\text{CP19:2.20})$$

where both savings are positive semidefinite and explicit:

$$\mathcal{S}_{\text{phase}}(Y) = \frac{1}{2\pi} \int [(B_a^{\text{abs}})^2 - p_a(\xi)^2] \widehat{Y}(\xi)^* \widehat{Y}(\xi) d\xi, \quad (\text{CP19:2.21})$$

$$\mathcal{S}_{\text{exterior}}(Y) = [(I - E_a E_a^*) \mathcal{P}_a^{\text{free}} E_a Y]^* [(I - E_a E_a^*) \mathcal{P}_a^{\text{free}} E_a Y]. \quad (\text{CP19:2.22})$$

All prime-prime crosses are included in p_a^2 , including unequal prime powers and both orientations. These formulas require no arithmetic approximation and are valid through threshold activations by zero extension.

There is also an independently quantified saving for every supported response, without evaluating its Fourier transform. Define

$$N_n(a) = \lceil a/\ell_n \rceil, \quad \gamma_{\text{prime}}(a) = 4 \sum_{\ell_n < 2a} b_n^2 \sin^2 \left(\frac{\pi}{2(N_n(a) + 1)} \right). \quad (\text{CP19:2.23})$$

Then

$$\boxed{\mathcal{S}_{\text{phase}}(Y) \succeq \gamma_{\text{prime}}(a) Y^* Y.} \quad (\text{CP19:2.24})$$

To prove it, retain the exact pointwise expansion

$$(B_a^{\text{abs}})^2 - p_a(\xi)^2 = 4 \sum_n b_n^2 \sin^2(\ell_n \xi) + 8 \sum_{n < m} b_n b_m [1 - \cos(\ell_n \xi) \cos(\ell_m \xi)]. \quad (\text{CP19:2.25})$$

Every summand on the second line is nonnegative, and can remain as an additional saving. The first term equals the Gram of $b_n(T_{\ell_n} - T_{-\ell_n})Y$. Decompose the line into translation fibers modulo $2\ell_n$. A supported column occupies at most $N_n(a)$ consecutive sites on almost every fiber. The

difference with zero extension has the path-Dirichlet matrix with diagonal 2 and off-diagonal -1 . Its smallest eigenvalue is $4 \sin^2[\pi/(2(N_n(a) + 1))]$. Integrating this finite-dimensional inequality over the fibers proves (2.24), by polarization for the complete column map. If a fiber has fewer sites its minimum eigenvalue is larger. Endpoint ambiguities occur on a null set and do not alter the estimate.

The remainder retained by this bound is explicitly positive:

$$\begin{aligned} \mathcal{S}_{\text{phase}}(Y) - \gamma_{\text{prime}}(a)Y^*Y &= \sum_n b_n^2 \left\{ [(T_{\ell_n} - T_{-\ell_n})Y]^* [(T_{\ell_n} - T_{-\ell_n})Y] - 4 \sin^2\left(\frac{\pi}{2(N_n(a) + 1)}\right) Y^*Y \right\} \\ &\quad + \frac{4}{\pi} \sum_{n < m} b_n b_m \int [1 - \cos(\ell_n \xi) \cos(\ell_m \xi)] \hat{Y}(\xi)^* \hat{Y}(\xi) d\xi \succeq 0. \end{aligned} \quad (\text{CP19:2.26})$$

Thus the complete prime-square estimate improves to

$$(\Pi_a Y)^*(\Pi_a Y) \preceq [(B_a^{\text{abs}})^2 - \gamma_{\text{prime}}(a)]Y^*Y - \mathcal{S}_{\text{exterior}}(Y). \quad (\text{CP19:2.27})$$

For $\log 2/2 < a \leq \log 3/2$, only $n = 2$ is active and $N_2(a) = 1$, so

$$\gamma_{\text{prime}}(a) = 2b_2^2 = (\log 2)^2, \quad (B_a^{\text{abs}})^2 = 2(\log 2)^2. \quad (\text{CP19:2.28})$$

The prime-square cap is therefore at least halved on this actual prime-active interval, before the additional exterior saving is used. This is an analytic estimate for every trial column, not a finite numerical example. For all larger finite supports, (2.23) gives the exact stated positive improvement; it does not assert that this saving pays the remaining gamma/polar/source interactions.

On that same first-prime window the exterior contribution gives a still sharper exact result. Put $\ell = \log 2$ and

$$\mathcal{O}_a = (-a, a - \ell) \cup (\ell - a, a), \quad \mathcal{C}_a = (a - \ell, \ell - a). \quad (\text{CP19:2.29})$$

The intervals in $\mathcal{O}_a$ are disjoint because $a < \ell$. Each nontrivial compressed translation fiber has precisely two sites; Π_a exchanges them with coefficient b_2 . Therefore

$$\Pi_a^2 = b_2^2 \mathbf{1}_{\mathcal{O}_a}, \quad (\Pi_a Y)^*(\Pi_a Y) = b_2^2 Y^* \mathbf{1}_{\mathcal{O}_a} Y, \quad (\text{CP19:2.30})$$

with the zero channel $\mathcal{C}_a$ retained. In particular,

$$\boxed{\mathcal{S}_{\text{phase}}(Y) + \mathcal{S}_{\text{exterior}}(Y) = 3b_2^2 Y^*Y + b_2^2 Y^* \mathbf{1}_{\mathcal{C}_a} Y \succeq 3b_2^2 Y^*Y.} \quad (\text{CP19:2.31})$$

The complete prime-square cap is consequently at most one quarter of the separated cap on this window. On the odd half interval this is the reflected map $-b_2 Y(\ell - x)$ on $\ell - a < x < a$; its minus sign is present, and its square is exactly the same overlap projector. Thus both reflected sign channels and their mixed interactions with the remaining residual are preserved. The saving does not amount to a positivity proof for the full Weil form, because the unchanged cross $\mathcal{R}_0^* \Pi_a Y + \text{adjoint}$ still enters its certificate.

If Group 3 writes the complete residual as $\mathcal{R} = \mathcal{R}_0 + \Pi_a Y$, with $\mathcal{R}_0$ retaining the archimedean, polar, regularizer, source and boundary terms, its full Gram is

$$\boxed{\mathcal{R}^* \mathcal{R} = \mathcal{R}_0^* \mathcal{R}_0 + \mathcal{R}_0^* \Pi_a Y + (\Pi_a Y)^* \mathcal{R}_0 + (B_a^{\text{abs}})^2 Y^*Y - \mathcal{S}_{\text{phase}}(Y) - \mathcal{S}_{\text{exterior}}(Y).} \quad (\text{CP19:2.32})$$

Hence the residual certificate gains exactly $\tau_R g_R^{-1} [\mathcal{S}_{\text{phase}}(Y) + \mathcal{S}_{\text{exterior}}(Y)]$ over the corresponding prime-square scalar estimate while preserving all mixed interactions. The factor is explicit and uses the independently proved g_R , not the unknown positivity margin. Keeping $\mathcal{R}_0^* \Pi_a Y$ is necessary: it can have either sign. No fixed saving such as the historical factor 96 is claimed for arbitrary current responses.

2.3. Historical overlap, half-balance, and survival: the actual coefficient tests

The angle-to-Green identity is

$$\frac{1}{\max(m, n)} = \frac{e^{-|\log m - \log n|/2}}{\sqrt{mn}}. \quad (\text{CP19:2.33})$$

Thus normalization is the diagonal map $c_n = \sqrt{n} x_n$, with c_n the coefficients in the reciprocal-maximum metric and x_n the coefficients in the normalized logarithmic Green metric. The complete prime triple in the historical metric is, for $u = \min(m, n)$, $v = \max(m, n)$, $L = \lfloor N/p \rfloor$,

$$\langle e_m - \mathbf{1}_{pm \leq N} e_{pm}, e_n - \mathbf{1}_{pn \leq N} e_{pn} \rangle_G = \left(\frac{1}{v} - \frac{1}{pu} \right)_+ + \frac{\mathbf{1}_{u > L}}{pu}. \quad (\text{CP19:2.34})$$

Here $1 \leq m, n \leq N$. Its ratio-exterior cancellation is exact, and the second term is the full terminal. It is not the same Gram as (2.1), which is the diagonal norm of theta difference columns at a single logarithmic jump.

One can test the metric identification without hiding the defect. Let D_n be any finite collection of actual response columns and put $A_n = D_n^* D_n$. The direct sum matrix is $\sum_n A_n$. Transport to normalized Green atoms gives

$$\sum_{m, n} e^{-|\log m - \log n|/2} D_m^* D_n = \sum_n A_n + \sum_{m \neq n} e^{-|\log m - \log n|/2} D_m^* D_n. \quad (\text{CP19:2.35})$$

The second sum is the exact metric correction. It is generally signed. For $D_n = \sqrt{b_n} \sqrt{\phi(x)\phi(x + \log n)} D_{\log n} h$, the direct sum is the current prime-energy matrix. Thus a historical Green bound can be inserted only with the complete cross sum in (2.35), or by a separate proved isometry such as the continuous Green representation developed in the storage calculation. Dropping that sum would change the target matrix.

The archived forced-overlap theorem (*Prime_DNA_Signed_Overlap_and_Band_Resolution.md*, §§1-2, source) uses authentic coefficients $a(n) = \mu(n) \mathbf{1}_{P+(n) < p}$ and the exact law $a(2n) = -a(n)$ for odd n . Its paired cumulative response is $\mathbf{1}_{[n, 2n)} - \mathbf{1}_{[pn, 2pn)}$. With its original B, C , it proves

$$|V| \leq B - 2C, \quad \mathcal{G} = 4 \int C(B - C) \frac{dt}{t^2} \geq (2 - 4/p) \sum_{\substack{n \leq L/2 \\ n \text{ odd}}} \frac{a(n)^2}{n}. \quad (\text{CP19:2.36})$$

The $p = 11, N = 100$ source calculation saves exactly 554/105, from 15359/2310 to 151/110, with signed incoming interaction and terminal unchanged. The forced columns $W_a c$, or the response columns $Y_a c$, do not satisfy this discrete sign law merely because their operator contains Λ . The authentic sign law does apply to the scalar coefficients in (2.3); however, their associated response vectors change with m, d , and the induced Gram is (2.35). That changing Gram and the complete cross sum are the concrete unmatched terms, not an absent source identity.

The archived Weighted Half-Balance theorem (*Prime_DNA_Weighted_Half_Balance_and_Forced_Prime_Compensation.md*, §§1-3, source) retains

$$R(N) = D_N + 4W_N(1/2 - \chi_N), \quad \chi_N = \frac{1}{2} + O_A((\log N)^{-A}), \quad (\text{CP19:2.37})$$

and a prime-3 capacity credit improving its original prime-2 envelope by the explicit coefficient $12[\log(10/9) - \frac{1}{2} \log(6/5)]/\pi^4$. Its RH-strength one-sided rate remains an open lemma in that source.

After two prime insertions the historical Gram already has the exact mixed entry $\langle w_1, w_5 \rangle_G = -1/30$, $w_n = e_n - e_{2n} - e_{3n} + e_{6n}$. This sign is retained; it prevents iterating an entrywise positive one-prime envelope as though every mixed packet had the same sign.

The harmonic-return projection (*Prime_DNA_Cofactor_Energy_and_Upper_Interaction.md*, §§2-4, source) improves a paired cross term from $\sqrt{DJ}$ to

$$\frac{hb_0}{\beta} + \sqrt{(D - h^2/\beta)(J - b_0^2/\beta)}. \quad (\text{CP19:2.38})$$

This is a legitimate transfer strategy after the actual common representer and its norm β are identified in the response metric. The theta/Weil identity does not identify the historical harmonic representer with the current boundary vector. In the present finite-dimensional certificate one may instead project both actual residual columns onto any explicitly chosen common L^2 response column, subtract its exact rank-one Gram and keep the transverse Gram; that is ordinary Gram orthogonality with every cross retained, not an automatic negative arithmetic margin.

Finally, the all-prime transport source (*Michaels_Survival_and_Prime_Transport_Continuation.md*, §2 and appended T1-T10, source) proves for a completed authentic level packet with squarefree endpoints $2 \leq u < d$, height weight $w = 2h - 1$, $A = w/u$, $B = w/d$, and $\alpha_p(n) = \mathbf{1}_{p|n} \log p / \log n$,

$$\sum_p |A\alpha_p(u) - B\alpha_p(d)| = A - B + 2B \frac{\log(d/\gcd(u, d))}{\log d}. \quad (\text{CP19:2.39})$$

For $u = 95$, $d = 97$, the ratio of the separated amount to $A - B$ is exactly $(97 + 95)/(97 - 95) = 96$. This is a diagnostic completed packet; it is not asserted to belong to the difficult selected retained family. The nonnegative coupling $\Pi_{pq} = B\alpha_p(u)\alpha_q(d)$ preserves the closing amount before magnitudes. This justifies retaining (2.32) as one coupled expression. It supplies no uniform factor-96 reduction of its current Gram. Open packets have no closing amount, and their full terminal survives.

2.3.1. Packet and response closing statement The independently calculated gains are: exact removal of mixed-prime source locations by (2.5); the actual theta exterior-packet lower comparison (2.9), with coefficient $1 - 10^{-6}$; the complete translated color-square identity (2.15)-(2.18); and the positive matrix saving (2.20)-(2.32) in the real response certificate; its explicit uniform lower amount (2.23)-(2.28); and the exact quarter-cap improvement (2.29)-(2.31) throughout the first-prime window. The latter includes every prime-prime cross and exterior term and leaves every prime-archimedean/polar/source cross intact. Historical overlap, half-balance and survival credits retain their proved metric and arithmetic hypotheses.

The signed matrix still requiring an independent estimate is equivalently

$$\mathbf{E}_{\Gamma, a} + \sum_{p, k} (\log p) p^{-k/2} \mathbf{B}_a(k \log p) - \frac{1}{2} (\mathbf{G}_a + \ell_a \ell_a^*) \succeq 0, \quad (\text{CP19:2.40})$$

or Group 3's fully optimized certificate with (2.32) substituted. At a first null the left side already has a null direction by the exact congruence; nonnegativity at that same interval is not a contradiction. A sufficient missing estimate must be strict in that nonzero direction, or control its energy-dependent boundary rate. The new savings have been computed, but they do not by themselves establish that final comparison for every finite support.

2.4. Exact Green storage and divisor-color transport

This calculation applies the recovered storage and Smith-Jordan theorems to the actual forced columns. Its new exact matching occurs on a divisibility-chain endpoint space. It keeps the full prime-continuum correction. The match proves a positive component identity, not the required sign of the complete Weil matrix.

2.4.1. Source proofs actually used

1. *Physics_Architectures_and_Log_Prime_Energy.pdf*, version 4, §§14-16, equations (18)-(24): the residual-response identity, the exact cancellation of the polar and continuum kernels, and the complete signed Weil decomposition.
2. *Prime_DNA_Energy_Throughline.md*, §1, equations (1)-(3a): full Green energy with its terminal, two-summary propagation, and two-endpoint orthogonal storage. The proof of the compression is finite kernel algebra and extends to finite signed measures as proved below.
3. *Michaels_Prime_Color_Dual_Kernel_Bridge.md*, version 1, §7, equations (7.6)-(7.9): the complete Smith-Jordan divisor-square identity. Its arbitrary-coefficient hypothesis is used; no Möbius sign law is imposed on the forced columns.
4. *Michaels_Joint_Energy_Continuation.pdf*, pp.19-21, P1-P12: the authentic Boolean mask, rank-gap compensation, original-metric projection, and additional mask work under prime transport.
5. *Michaels_Survival_and_Prime_Transport_Continuation.md*, appended survival S1-S7 and prime transport T1-T8: full open endpoints, old-stack closure rebate, and the exact loss from separating prime factors. The missing plus sign in the source's introductory display (3) is corrected by its complete S3 derivation.

2.4.2. Exact transport of the actual forced columns Let $q_j = W_a e_j$, with zero extension from $(-a, a)$, and use conjugate-linearity in the first argument. Put

$$N = e^{2a}, \quad f_j(x) = x^{-1/2} q_j(\log x - a), \quad 1 < x < N, \quad F_j(t) = \int_1^t f_j(x) dx. \quad (\text{CP19:2.41})$$

Every f_j is integrable, since $a < \infty$ and $q_j \in L^2(-a, a)$. The coefficient map is an actual bounded linear map on this finite support; it does not sample pointwise values of a form-domain function. Define the Green matrix

$$(\mathbf{G}_a^{\text{gr}})_{ij} = \iint e^{-|u-v|/2} \overline{q_i(u)} q_j(v) du dv. \quad (\text{CP19:2.42})$$

Changing variables $x = e^{u+a}$, $y = e^{v+a}$ yields exactly

$$\boxed{(\mathbf{G}_a^{\text{gr}})_{ij} = \int_1^N \int_1^N \frac{\overline{f_i(x)} f_j(y)}{\max(x, y)} dx dy = \int_1^N \frac{\overline{F_i(t)} F_j(t)}{t^2} dt + \frac{\overline{F_i(N)} F_j(N)}{N}}. \quad (\text{CP19:2.43})$$

The last identity follows from

$$\frac{1}{\max(x, y)} = \int_1^N \mathbf{1}_{x \leq t} \mathbf{1}_{y \leq t} \frac{dt}{t^2} + \frac{1}{N}. \quad (\text{CP19:2.44})$$

Fubini is legitimate for the finite signed products: the kernel is bounded on $[1, N]^2$ and the densities are integrable. The last term in (2.43) is the observed endpoint and is not omitted. This is the exact diagonal rescaling of the historical metric, with the scaling absorbed into f_j .

2.4.3. Complete two-endpoint compression, including every block storage term Fix any finite partition $1 = x_0 < x_1 < \dots < x_J = N$. For the cell $I_k = (u, v) = (x_{k-1}, x_k)$, define row vectors

$$B_k = \int_u^v f(x) dx, \quad h_k = \int_u^v \frac{f(x)}{x} dx, \quad \eta_k = \frac{1}{u} - \frac{1}{v}, \quad (\text{CP19:2.45})$$

where $f = (f_1, \dots, f_{d_R})$, and put

$$A_k = \frac{h_k - B_k/v}{\eta_k}, \quad D_k = B_k - A_k. \quad (\text{CP19:2.46})$$

Equivalently, these rows are the explicitly weighted moments

$$A_k = \eta_k^{-1} \int_u^v (1/x - 1/v) f(x) dx, \quad D_k = \eta_k^{-1} \int_u^v (1/u - 1/x) f(x) dx. \quad (\text{CP19:2.47})$$

The row-valued signed measure

$$z_k = f(x) \mathbf{1}_{(u,v)} dx - A_k \delta_u - D_k \delta_v \quad (\text{CP19:2.48})$$

has both total mass and reciprocal first moment zero. Therefore its Green pairing with every measure supported outside (u, v) , including its two endpoint atoms, is zero. Indeed, for an exterior point $t \leq u$, the pairing is its reciprocal moment; for $t \geq v$, it is its total mass divided by t . Thus different z_k 's are orthogonal, with no discarded cross term.

Let u_n be the sum of the adjacent endpoint rows at the partition node n , and let U denote the matrix of these rows. Define

$$(\mathbf{S}_k)_{ij} = \int_u^v \frac{[F_{k,i}(t) - A_{k,i}][F_{k,j}(t) - A_{k,j}]}{t^2} dt, \quad F_k(t) = \int_u^t f(x) dx. \quad (\text{CP19:2.49})$$

Every $\mathbf{S}_k \succeq 0$, and the complete identity is

$$\boxed{\mathbf{G}_a^{\text{gr}} = \sum_{k=1}^J \mathbf{S}_k + U^* G_{\mathcal{P}} U, \quad (G_{\mathcal{P}})_{mn} = 1/\max(m, n).} \quad (\text{CP19:2.50})$$

The terminal of every original cell is included through its endpoint rows; all correlations between different cells reside in $U^* G_{\mathcal{P}} U$. Formula (2.49) is the permanent internal storage. Its exact amount is also

$$\mathbf{S}_k = \mathbf{G}_k^{\text{gr}} - B_k^* B_k / v - \eta_k^{-1} (h_k - B_k / v)^* (h_k - B_k / v). \quad (\text{CP19:2.51})$$

This proves the archived compatibility bound on the actual forced family rather than assuming its coefficients are the old Möbius vector.

Endpoint atoms are used solely inside the bounded Green kernel pairing. They are not substituted into the gamma form or asserted to be Weil-domain vectors. The original q_j 's remain unchanged in every archimedean and arithmetic term.

2.4.4. An exact Green-Smith match on a cofinal arithmetic endpoint family Choose $N = 6^m$, $a = (m \log 6)/2$, and the nested prime-word chain

$$1 \mid 2 \mid 6 \mid 12 \mid 36 \mid 72 \mid \cdots \mid 6^m. \quad (\text{CP19:2.52})$$

Use consecutive nodes in the preceding two-endpoint partition. Each earlier node divides each later node. Hence on this endpoint set

$$\frac{1}{\max(n, r)} = \frac{\gcd(n, r)}{nr}. \quad (\text{CP19:2.53})$$

Apply the saved Smith-Jordan identity at $z = 1$ to the coefficients $x_n = u_n c / \sqrt{n}$. Since $J_1(d) = \varphi(d)$, it gives

$$U^* G_{\mathcal{P}} U = \sum_{d \mid 6^m} \varphi(d) v_d^* v_d, \quad v_d = \sum_{\substack{n \in \mathcal{P} \\ d \mid n}} \frac{u_n}{n}. \quad (\text{CP19:2.54})$$

The sum can equivalently run over all $d \geq 1$, since unused rows vanish. No mixed color is dropped; every divisor of every endpoint occurs. Consequently the actual forced Green matrix has the exact positive decomposition

$$\mathbf{G}_a^{\text{gr}} = \underbrace{\sum_k \mathbf{S}_k}_{\mathbf{S}_a} + \underbrace{\sum_{d \mid 6^m} \varphi(d) v_d^* v_d}_{\mathbf{C}_a}. \quad (\text{CP19:2.55})$$

This is a successful hypothesis match to both the storage and color-square theorems. It makes their coefficient maps explicit on W_a , without a torus-orbit evaluation or a comparison of unbounded observation maps. The quantitative gain over discarding the Green term is the whole matrix $\mathbf{S}_a + \mathbf{C}_a$; over retaining only the endpoint projection it is precisely $\mathbf{S}_a \succeq 0$. It supplies the unconditional lower comparisons $\mathbf{G}_a^{\text{gr}} \succeq \mathbf{C}_a$ and $\mathbf{G}_a^{\text{gr}} \succeq \mathbf{S}_a$.

These supports are cofinal. A complete Weil sign proved on them would imply the sign on every finite support by restriction. No complete sign is asserted from (2.55) alone.

For arbitrary integer nodes, the exact additional metric term would instead be

$$\mathfrak{D}_{mn} = \frac{\min(m, n) - \gcd(m, n)}{mn}, \quad U^* G_{\mathcal{P}} U = \sum_d \varphi(d) v_d^* v_d + U^* \mathfrak{D} U. \quad (\text{CP19:2.56})$$

The diagonal of $\mathfrak{D}$ is zero, while an incomparable pair gives a positive off-diagonal entry. Its two-coordinate restriction has eigenvalues of opposite signs. The divisibility-chain choice removes this particular comparison defect exactly; it does not remove the continuum or gamma term.

2.4.5. Odd cancellation yields an explicit sharp polar-moment Green reserve For odd $q \in L^2(-a, a)$, define

$$\lambda(q) = \int_{-a}^a \sinh(u/2)q(u) du, \quad d(q) = \frac{\lambda(q)}{2 \sinh(a/2)}. \quad (\text{CP19:2.57})$$

Its exponential moments are $\int e^{u/2}q = \lambda(q)$ and $\int e^{-u/2}q = -\lambda(q)$. The signed measure $q(u)du - d(q)(\delta_a - \delta_{-a})$ has both moments zero, so is Green-orthogonal to the two endpoint atoms. Computing their complete 2×2 Gram gives

$$\langle q, Gq \rangle = \|q(u)du - d(q)(\delta_a - \delta_{-a})\|_G^2 + \frac{2}{e^a - 1} |\lambda(q)|^2. \quad (\text{CP19:2.58})$$

Indeed, the atom pair has energy $2|d|^2(1 - e^{-a})$. Its matrix version is

$$\mathbf{G}_a^{\text{gr}} \succeq \frac{2}{e^a - 1} \lambda_a^* \lambda_a, \quad (\lambda_a)_j = \lambda(q_j). \quad (\text{CP19:2.59})$$

The coefficient is exact and optimal in the Green completion: endpoint atoms attain the projection bound, and smooth approximations attain its infimum. It is positive on every finite interval, but decays with support. This is a genuinely calculated lower comparison in the forced energy's Green component.

2.4.6. The complete correction and the precise cancellation that remains Let $\mathbf{A}_\Gamma$ be the gamma multiplier operator, P_a the complete compressed prime-translation operator, Z_a the kernel $2 \cosh((u - v)/2)$, and M_a^{ct} the continuum kernel $e^{|u-v|/2}$. Here M_a^{ct} is not the CP18 matrix $M_a = I - K_a$. The source proves

$$Z_a - M_a^{\text{ct}} = G_a, \quad \mathbf{A}_a^- = \mathbf{A}_\Gamma + G_a - (P_a - M_a^{\text{ct}}) \quad \text{on the odd domain.} \quad (\text{CP19:2.60})$$

Define actual finite matrices

$$\mathbf{A}_{\Gamma,a} = W_a^* \mathbf{A}_\Gamma W_a, \quad \mathbf{P}_a = W_a^* P_a W_a, \quad \mathbf{D}_a = W_a^* (P_a - M_a^{\text{ct}}) W_a. \quad (\text{CP19:2.61})$$

All are form pairings in the CP18 domain where necessary. The complete forced matrix becomes, on the chain supports,

$$\tau_R^{-1} K_a (I - K_a) = \mathbf{A}_{\Gamma,a} + \mathbf{S}_a + \mathbf{C}_a - \mathbf{D}_a. \quad (\text{CP19:2.62})$$

Every ordinary and reflected prime shift remains in $\mathbf{P}_a$; no prime has been deleted or copied into a different coefficient law. The continuum is retained at exactly its original weight.

There is a decisive bookkeeping fact. On odd functions the polar operator is $Z_a = -2|s_a\rangle\langle s_a|$, so (2.60) gives

$$M_a^{\text{ct}} = -G_a - 2|s_a\rangle\langle s_a|, \quad \boxed{\mathbf{D}_a = \mathbf{P}_a + \mathbf{S}_a + \mathbf{C}_a + 2\lambda_a^* \lambda_a.} \quad (\text{CP19:2.63})$$

Thus the exact positive storage/color component also occurs inside the full signed correction. Substituting (2.63) into (2.62) returns

$$\tau_R^{-1} K_a(I - K_a) = \mathbf{A}_{\Gamma,a} - \mathbf{P}_a - 2\lambda_a^* \lambda_a. \quad (\text{CP19:2.64})$$

This is the correct normalization check. The reserve (2.59) cannot be spent once more against the polar term after this exact cancellation. On the zero-polar-moment branch, (2.63) still retains $\mathbf{S}_a + \mathbf{C}_a$; that branch is not removed by its positivity.

The successful map leaves one explicit coupled comparison:

$$\boxed{\mathbf{D}_a - \mathbf{A}_{\Gamma,a} \preceq \mathbf{S}_a + \mathbf{C}_a.} \quad (\text{CP19:2.65})$$

This is still to be proved independently. At a hypothetical first null c , equality in (2.65) is compatible with every positive square in (2.55) being nonzero: their sum is paid by the same complete correction, not asserted to vanish. The storage theorem by itself yields no contradiction.

2.4.7. What survives from the authentic mask and survival proofs The saved authentic mask theorem assumes $a(n) = \mu(n)s(n)$, $s(n) \in \{0,1\}$, and uses distinct integer heights of the genuine Möbius prefix. It proves

$$L^+ + O = R - \|\mu - a\|_G^2 - G_U, \quad G_U \geq 0, \quad (\text{CP19:2.66})$$

with the original terminal. Its complete prime rule is

$$r(pm) = -r(m) - \mu(m)[s(m) - s(pm)] \quad (p \nmid m). \quad (\text{CP19:2.67})$$

The current rows u_n in (2.54) are linear moments of W_a . They are neither Boolean-masked Möbius coefficients nor necessarily zero at nonsquarefree n . The extra terms required by such an attempted assignment can be displayed, rather than ignored. Write

$$u_n = \mu(n)d_n + z_n, \quad d_n = \mu(n)u_n \text{ on squarefree } n, \quad z_n = u_n \mathbf{1}_{\mu(n)=0}. \quad (\text{CP19:2.68})$$

Then the Green energy has the retained mixed term

$$\|u\|_G^2 = \|\mu d\|_G^2 + 2\Re\langle \mu d, z \rangle_G + \|z\|_G^2. \quad (\text{CP19:2.69})$$

Even on squarefree sites, for $r_d(n) = \mu(n)(1 - s(n))d_n$, prime transport gives

$$\boxed{r_d(pm) + r_d(m) = \mu(m)\{[s(pm) - s(m)]d_m + [1 - s(pm)](d_m - d_{pm})\}.} \quad (\text{CP19:2.70})$$

The second term is the additional amplitude-variation defect; the first is the already required mask defect. No actual equation recovered here makes either term vanish for $u_n(W_a c)$. The integer-height rank-gap proof cannot be assigned to these variable complex rows without an additional theorem.

There is nevertheless a valid continuous survival analogue for each real scalar forced column combination. With $F(t) = \int_1^t f(x)dx$, $Q(t) = |F(t)|$, decompose the open set $\{t : Q(t) > h\}$ into its excursion intervals for each $h > 0$. Layer cake and Tonelli give the complete nonnegative level cost by integrating $2h dh$ times $1/u - 1/d$ for a completed interval and $1/u$ for an interval still open at N . The latter includes its actual terminal.

At an earlier Y , put $q_Y(t) = \min_{Y \leq u \leq t} Q(u)$ and $\Delta_Y(t) = Q(Y)^2 - q_Y(t)^2$. The exact future closure rebate on the levels already present at Y is

$$\mathcal{R}_Y^{\text{cont}} = \int_Y^N \frac{\Delta_Y(t)}{t^2} dt + \frac{\Delta_Y(N)}{N}, \quad (\text{CP19:2.71})$$

and the final cost of all level intervals born by Y is the Green prefix energy through Y minus this rebate. In particular, for $Y \leq Z \leq N$,

$$\mathcal{R}_Y^{\text{cont}} \geq \frac{\Delta_Y(Z)}{Z} + \frac{\Delta_Y(N) - \Delta_Y(Z)}{N}. \quad (\text{CP19:2.72})$$

This is an exact transport of the survival accounting to the actual real forced functions. For complex functions, apply the real and imaginary identities separately. The level selections depend nonlinearly on the coefficient vector, so they do not provide a fixed positive matrix factorization beyond (2.43); no favorable arithmetic rate is supplied. In particular, subtracting this rebate from an upper allowance for a positive Green component is not a second lower credit toward the signed Weil form.

The archived $95 \rightarrow 97$ loss applies to the authentic selected endpoint energy, with its prime weights $\log p / \log n$. It is not a universal multiplicative improvement factor for (2.62) or for a new response Gram. Its valid instruction here is to retain (2.62), (2.63), and all residual channels together before estimating.

2.4.8. Effect on the actual response certificate For explicit trial columns Y , put $\mathcal{R} = F - \mathcal{J}Y$. Its payment in the CP19 response certificate is $\mathcal{R}^* \mathcal{R}$, the L^2 Gram. Applying the preceding Green transport and chain decomposition to the residual columns themselves produces their exact Green Gram, but not an upper bound on their L^2 Gram. The whole-line multipliers satisfy

$$\langle r, Gr \rangle = \frac{1}{2\pi} \int \frac{|\widehat{r}(\xi)|^2}{1/4 + \xi^2} d\xi \leq 4\|r\|_2^2. \quad (\text{CP19:2.73})$$

The reverse inequality is unavailable from Green positivity: smooth compactly supported high-frequency odd functions have fixed L^2 norm and Green norm tending to zero. This identifies the direction of the estimate; it is not a counterexample to an estimate on the actual finite response family.

For any specified finite independent residual basis $r_1, \dots, r_k$, its exact additional comparison cost is

$$\mathfrak{c}(\mathcal{R}) = \lambda_{\max} \left[(\mathcal{R}^* G \mathcal{R})^{-1/2} (\mathcal{R}^* \mathcal{R}) (\mathcal{R}^* G \mathcal{R})^{-1/2} \right], \quad (\text{CP19:2.74})$$

with inverses on the actual range if redundant columns are retained. It is finite for an independent finite family because G is strictly positive. Its support dependence and actual value require

an estimate; no margin-independent bound is supplied by the old Green-storage theorem. Paying it unevaluated would hide the requested comparison.

Thus the valid application to Group 3 is to preserve the source grouping and the exact residual cross-Gram, while its optimized L^2 payment is calculated in the original positive regularizer metric. The Green/color calculation supplies an exact check on a component, not a replacement payment of size one.

2.4.9. Closing statement for this route Proved unconditionally: the actual finite forced family has the continuous historical Green representation (2.43), complete two-endpoint storage decomposition (2.50), exact Smith-Jordan endpoint squares on cofinal divisibility-chain supports (2.55), and the sharp odd polar-moment Green reserve (2.59). The continuous survival identity (2.71) preserves actual closures and endpoints. These are identities and inequalities on the actual coefficient maps, not analogy.

Quantitative improvement: no comparison defect is paid between Green and Smith-Jordan on the chosen chain; the exact additional stored matrix is $\mathbf{S}_a \succeq 0$, and the global odd endpoint reserve is $2\lambda_a^* \lambda_a / (e^a - 1)$. Every mixed color and cross-block correlation is retained.

Exact remaining estimate: for every cofinal support $a = (m \log 6)/2$, independently prove (2.65) on the actual W_a ; or obtain a strict version on every hypothetical nonzero first-null vector; or bound the CP18 projected boundary residual by its stated integrable Osgood rate. The completed correction (2.63) is the line preventing a free sign conclusion from the positive storage/color squares. For the response route, the additional estimate is the complete L^2 residual Gram bound after all its signed cross terms, not a Green-norm bound alone.

Both null branches and the original Suzuki metric remain intact. This route does not establish unconditional Weil positivity or the original defect target.

2.5. The odd theta derivative: complete prime pairing and a positive continuous floor

This calculation uses the complete operator of CP18 and the theta annihilation identity in *Physics Architectures and Log Prime Energy*, (30), (33), and Theorem 17.1. It does not change the Suzuki metric. Its purpose is an exact new coefficient map for the recovered cancellation machinery. The result is an unconditional transformation and an explicit positive continuous-energy floor; the remaining paired-prime comparison is stated below.

2.5.1. Normalization and the actual forced matrix Write $\phi(u) = 2\Phi(u/2)$, $\hat{\phi} = \Xi$, and use the CP18 Fourier convention. The full arithmetic operator is gamma plus the polar kernel $2 \cosh((x - y)/2)$, minus the complete prime translations. Its restriction to the odd space, transported unitarily by $q \mapsto f = \sqrt{2}q|_{(0,a)}$, has off-diagonal kernel

$$-\kappa(|x - y|) + \kappa(x + y) - \sum_{n \geq 2} b_n \{\delta(y - x - \ell_n) + \delta(y - x + \ell_n)\} + \sum_{n \geq 2} b_n \delta(x + y - \ell_n) - 4s(x)s(y), \quad (\text{CP19:2.75})$$

where $s(x) = \sinh(x/2)$, $\ell_n = \log n$, $b_n = \Lambda(n)/\sqrt{n}$, and $\kappa(t) = e^{-t/2}/(1 - e^{-2t})$. The diagonal gamma distribution, including its normalization constant, is retained by using its exact Fourier multiplier throughout. Formula (2.75) alone is not a definition of that diagonal.

For $q = W_a c$, the complete physical matrix relation, with conjugate-linear first argument, is

$$Q_a^-(W_a c) = \tau_R^{-1} c^* K_a (I - K_a) c. \quad (\text{CP19:2.76})$$

This follows directly from $A_a^- W_a = F_a(I - K_a)$ and $W_a^* F_a = K_a/\tau_R$. It does not discard either null branch. On $\ker K_a$, $W_a c = 0$ and $(I - K_a)c = c$. On $\text{Ran } K_a$, congruence by $\sqrt{\tau_R} K_a^{-1/2}$ recovers $I - K_a$ exactly. In particular an independent lower comparison of any exact energy expression below is a lower comparison of this same matrix.

2.5.2. An analytic monotonicity proof for the required odd weight For $u \geq 0$, put $a_n = \pi n^2 e^{2u}$. Then the actual theta summands satisfy

$$\phi_n(u) = 2e^{u/2}(2a_n^2 - 3a_n)e^{-a_n}, \quad \phi'_n(u) = 2e^{u/2}a_n e^{-a_n}(-4a_n^2 + 15a_n - 15/2), \quad (\text{CP19:2.77})$$

$$\phi''_n(u) = 2e^{u/2}a_n e^{-a_n} \left(8a_n^3 - 56a_n^2 + \frac{165}{2}a_n - \frac{75}{4} \right). \quad (\text{CP19:2.78})$$

For $u \geq 1/100$, $a_n \geq \pi e^{1/50} > 3.2028$, whereas the larger root in (2.77) is $(15 + \sqrt{105})/8 < 3.16$. Thus every first derivative is strictly negative there.

For $0 \leq u \leq 1/100$, $3.14 < a_1 < 3.21$. The cubic in (2.78) is bounded above by

$$8(3.21)^3 - 56(3.14)^2 + (165/2)(3.21) - 75/4 < -40. \quad (\text{CP19:2.79})$$

Using $e^{3.21} < 25$, the first summand therefore has $\phi''_1 < -10$. For $n \geq 2$, the polynomial expression before the factor $2e^{u/2}e^{-a_n}$ is at most $8a_n^4$; this gives an upper bound regardless of its sign. Since $a^4 e^{-a}$ decreases for $a > 4$,

$$\sum_{n \geq 2} \phi''_n(u) \leq 16e^{1/200} \pi^4 \sum_{n \geq 2} n^8 e^{-\pi n^2} < 2. \quad (\text{CP19:2.80})$$

For clarity, an entirely elementary tail budget proving the last inequality is $e^{1/200} < 1.006$, $\pi^4 < 98$, $e^{4\pi} > 2^{18}$, and successive ratios at $n \geq 2$ are bounded by $(3/2)^8 e^{-5\pi} < 1/100$. Thus the right side is less than $16(1.006)(98)(256)2^{-18}(100/99) < 2$. The elementary exponential bounds can be verified directly from the positive exponential series and its geometric remainder; no floating-point sign test is needed.

Consequently $\phi'' < -8$ on this first interval. Evenness gives $\phi'(0) = 0$. Combining this with (2.77) proves

$$p(x) := -\phi'(x) > 0 \quad (x > 0), \quad p(0) = 0, \quad p'(0) > 8. \quad (\text{CP19:2.81})$$

No curvature constant from a differently scaled theta density has been imported. The odd extension $p_o = -\phi'$ satisfies the complete full-line arithmetic equation

$$A_\Gamma p_o - \sum_{n \geq 2} b_n [p_o(\cdot + \ell_n) + p_o(\cdot - \ell_n)] - \frac{1}{2} \sinh(\cdot/2) = 0. \quad (\text{CP19:2.82})$$

Indeed the complete operator annihilates ϕ ; differentiating is legitimate by the saved theta derivative tails. Also

$$\int_0^\infty s(x)p(x) dx = \frac{1}{2} \int_0^\infty \cosh(x/2)\phi(x) dx = \frac{1}{8}, \quad (\text{CP19:2.83})$$

using $\Xi(\pm i/2) = 1/2$. This fixes all half-interval polar factors.

2.5.3. Exact odd ground-state transformation First take a smooth compactly supported odd q , put $f = \sqrt{2}q|_{(0,\infty)}$, and set $h = f/p$. This ratio is regular at the origin. Extend f, h by zero beyond their support. Define

$$\mathcal{G}_p(f) = \frac{1}{2} \int_0^\infty \int_0^\infty [\kappa(|x-y|) - \kappa(x+y)] p(x)p(y) |h(x) - h(y)|^2 dx dy \geq 0, \quad (\text{CP19:2.84})$$

$$\mathcal{Z}_p(f) = 2 \int_0^\infty \int_0^\infty s(x)s(y)p(x)p(y) |h(x) - h(y)|^2 dx dy = \frac{1}{2} \int_0^\infty \frac{s(x)}{p(x)} |f(x)|^2 dx - 4 \left| \int_0^\infty s(x)f(x) dx \right|^2 \geq 0. \quad (\text{CP19:2.85})$$

The gamma kernel in (2.84) is positive because κ decreases and $|x-y| < x+y$. The polar variance in (2.85) is positive by (2.83) and weighted Cauchy-Schwarz.

For each $\ell > 0$, let

$$\begin{aligned} \mathcal{P}_{p,\ell}(f) &= \int_0^\infty p(x)p(x+\ell) |h(x+\ell) - h(x)|^2 dx \\ &\quad - \frac{1}{2} \int_0^\ell p(x)p(\ell-x) |h(x) - h(\ell-x)|^2 dx. \end{aligned} \quad (\text{CP19:2.86})$$

The complete transformed identity is

$$\boxed{Q_a^-(q) = \mathcal{G}_p(f) + \mathcal{Z}_p(f) + \sum_{n \geq 2} b_n \mathcal{P}_{p,\ell_n}(f).} \quad (\text{CP19:2.87})$$

All ordinary and reflected prime edges are retained. In this representation the polar contribution is favorable, while the reflected prime edges carry the adverse sign. This is an exact rearrangement, not a claim that these adverse edges are paid.

2.5.3.1. The mandatory origin cancellation and the actual closed domain Individual terms in (2.86) need not be finite for an arbitrary member of the CP18 form domain. The mathematically valid extension is the paired expression

$$\begin{aligned} \mathcal{P}_{p,\ell}(f) &= \int_0^a D_\ell(x) |f(x)|^2 dx - 2\Re \int_0^\infty \overline{f(x)} f(x+\ell) dx \\ &\quad + \Re \int_0^\ell \overline{f(x)} f(\ell-x) dx, \end{aligned} \quad (\text{CP19:2.88})$$

$$D_\ell(x) = \frac{p_o(x+\ell) + p_o(x-\ell)}{p(x)}, \quad D_\ell(0) = \frac{2p'(\ell)}{p'(0)}. \quad (\text{CP19:2.89})$$

The origin value follows from the oddness of p_o . In particular, the two separate diagonal pieces have leading term $p(\ell)/(p'(0)x)$, which cancels before estimation. The saved quantity is not an unspecified cancellation factor: an artificial $1/x$ singularity is removed exactly, leaving the bounded continuous function (2.89). Ordinary/reflected cross terms together are the odd compression of the two whole-line shifts and have operator norm at most two. Thus

$$|\mathcal{P}_{p,\ell}(f)| \leq (\|D_\ell\|_{L^\infty(0,a)} + 2) \|f\|_2^2. \quad (\text{CP19:2.90})$$

For $\ell \geq 2a$, both cross integrals vanish, and (2.88) consists of its displayed diagonal alone. This includes every exterior prime contribution; it cannot be deleted.

The gamma energy has an independent legitimate form-domain passage. Let

$$V_{\Gamma,p}(x) = \frac{(A_{\Gamma}p_o)(x)}{p(x)}. \quad (\text{CP19:2.91})$$

The full gamma multiplier maps the odd Schwartz function p_o to an odd smooth function. Since $p'(0) > 0$, (2.91) extends continuously through zero and is bounded on each fixed $[0, a]$. Algebra on the smooth odd core gives

$$\mathcal{G}_p(f) = Q_{\Gamma}(q) - \int_0^a V_{\Gamma,p}(x) |f(x)|^2 dx. \quad (\text{CP19:2.92})$$

Here $Q_{\Gamma}(q)$ is the original full-line gamma form of q , equivalently its unitary half-interval form at f . The singular gamma diagonal and its constant are therefore both retained. The right side is continuous in the original closed form norm. Because (2.84) is a positive difference integral, core approximants are Cauchy in that energy; a pointwise subsequence identifies the limiting integral. This proves (2.92) and finiteness of (2.84) throughout the CP18 form domain. No endpoint trace of q , or unproved estimate for q/x , is needed.

The polar energy is continuous in $L^2(0, a)$, since s/p is bounded there. The paired prime terms are bounded forms; the following estimate proves absolute operator-norm summability of their exterior tail. Thus (2.87), interpreted with (2.88), extends to every CP18 form vector, in particular all columns of W_a . It also preserves complex off-diagonal matrix entries by polarization.

2.5.3.2. A quantified exterior prime tail Put

$$m_p(a) = \min_{0 \leq x \leq a} \frac{p(x)}{x} > 0, \quad (\text{CP19:2.93})$$

where the origin value is $p'(0)$. For $\ell \geq 2a$,

$$\|D_{\ell}\|_{\infty} \leq \frac{2}{m_p(a)} \sup_{|t-\ell| \leq a} |p'(t)|. \quad (\text{CP19:2.94})$$

The actual theta series gives, for $t \geq 0$,

$$|p'(t)| = |\phi''(t)| \leq 74\pi^4 e^{17t/2} e^{-\pi e^{2t}}. \quad (\text{CP19:2.95})$$

One proof uses $|8a^4 - 56a^3 + (165/2)a^2 - (75/4)a| \leq 36a^4$ for $a \geq \pi$, followed by the summand ratio $2^8 e^{-3\pi} < 1/40$. The resulting coefficient $72 \cdot 40/39$ is less than 74. The envelope in (2.95) decreases when $t \geq u_* = \frac{1}{2} \log(6/\pi)$. For any integer N with $\log(N+1) \geq \max(2a, a + u_*)$, define $c_a = \pi e^{-2a}$. Then

$$\begin{aligned} \left\| \sum_{n>N} b_n \mathcal{P}_{p,\ell_n} \right\|_{\text{op}} &\leq \frac{148\pi^4 e^{-17a/2}}{m_p(a)} \sum_{n>N} (\log n) n^8 e^{-c_a n^2} \\ &\leq \frac{148\pi^4 e^{-17a/2}}{m_p(a)} \frac{(N+1)^9 e^{-c_a (N+1)^2}}{1 - r_{a,N}}, \end{aligned} \quad (\text{CP19:2.96})$$

provided

$$r_{a,N} = \left(\frac{N+2}{N+1} \right)^9 e^{-c_a (2N+3)} < 1. \quad (\text{CP19:2.97})$$

The norm is the bounded quadratic-form norm on $L^2(0, a)$. This is an explicit tail bound in the original spatial norm, with no RH input and no discarded mixed coefficient terms.

The tail is not automatically favorable. The polynomial in (2.78) is positive for $a_n \geq 6$, so $p'(t) < 0$ for $t \geq u_*$. Hence for $\ell - a \geq u_*$,

$$D_\ell(x) = \frac{p(\ell + x) - p(\ell - x)}{p(x)} < 0 \quad (0 < x \leq a). \quad (\text{CP19:2.98})$$

Thus sufficiently far exterior prime terms are genuinely negative multiplication forms in this new representation. Formula (2.96) pays their size; it does not change their sign.

2.5.4. A new explicit positive floor from the gamma-polar coupling Write

$$m_s(a) = \min_{0 \leq x \leq a} \frac{s(x)}{p(x)} > 0, \quad I_+(a) = \int_a^\infty e^{y/2} p(y) dy = e^{a/2} \phi(a) + \frac{1}{2} \int_a^\infty e^{y/2} \phi(y) dy. \quad (\text{CP19:2.99})$$

The origin ratio equals $1/(2p'(0))$. In the exterior part of (2.84), $y > a \geq x$, use the exact expansion

$$\kappa(y - x) - \kappa(y + x) = 2e^{-y/2} s(x) + 2 \sum_{j \geq 1} e^{-(2j+1/2)y} \sinh((2j + 1/2)x). \quad (\text{CP19:2.100})$$

In (2.85), preserve the variance on $(0, a)$ and the omitted positive mass:

$$\mathcal{Z}_p(f) = 4 \left[\left(\int_0^a sp \right) \left(\int_0^a \frac{s}{p} |f|^2 \right) - \left| \int_0^a sf \right|^2 \right] + 4 \left(\int_a^\infty sp \right) \int_0^a \frac{s}{p} |f|^2. \quad (\text{CP19:2.101})$$

Every bracket in (2.101) is nonnegative. The first term in (2.100) and the last term in (2.101) combine exactly, because $2e^{-y/2} + 4s(y) = 2e^{y/2}$. Consequently

$$\boxed{\mathcal{G}_p(f) + \mathcal{Z}_p(f) \geq \eta_p(a) \|f\|_2^2, \quad \eta_p(a) = 2m_s(a)I_+(a) > 0.} \quad (\text{CP19:2.102})$$

This is an independently proved support-dependent bound for the complete continuous energy in this representation. It is not a bound for the entire Weil form before paying (2.88).

There is an exact nonnegative remainder $\mathcal{N}_a$ in (2.102). It consists of: the interior gamma difference energy on $(0, a)^2$; the $j \geq 1$ exterior terms in (2.100); the internal variance in (2.101); and the multiplication form $[2I_+(a)s(x)/p(x) - \eta_p(a)]|f(x)|^2$. Thus

$$Q_a^-(q) = \eta_p(a) \|q\|_2^2 + \mathcal{N}_a(f) + \sum_{n \geq 2} b_n \mathcal{P}_{p, \ell_n}(f), \quad \mathcal{N}_a \geq 0. \quad (\text{CP19:2.103})$$

No continuous or arithmetic interaction has been estimated separately and dropped in this identity.

2.5.5. The first active reflected prime and the exact remaining matrix When $\frac{1}{2} \log 2 < a < \frac{1}{2} \log 3$ and $a < \log 2$, the only nonzero prime cross term is the reflected $n = 2$ channel. Let $I_2 = (\log 2 - a, a)$, and let $f_\pm$ be the symmetric/antisymmetric components of $f|_{I_2}$ under $x \mapsto \log 2 - x$. Then

$$\sum_{n \geq 2} b_n \mathcal{P}_{p, \ell_n}(f) = \int_0^a V_p(x) |f(x)|^2 dx + b_2 \|f_+\|_2^2 - b_2 \|f_-\|_2^2, \quad V_p = \sum_{n \geq 2} b_n D_{\ell_n}. \quad (\text{CP19:2.104})$$

This retains both reflected sign channels, all exterior prime potentials, and their coupling to the gamma and polar energies through the same f . In particular it is not permissible to retain only the positive reflected channel. If additionally $a \leq \log 2 - u_*$, every summand of V_p is negative by (2.98). This includes CP18's prime-active starting interval; it demonstrates precisely which prime potential the continuous floor must pay.

For general finite a , put $f_j = \sqrt{2}(W_a e_j)|_{(0,a)}$. Define matrices by polarizing (2.88), (2.103):

$$(\mathbf{P}_{a,\ell})_{ij} = \int_0^a D_\ell \bar{f}_i f_j - \int_0^\infty [\bar{f}_i(x) f_j(x+\ell) + \overline{f_i(x+\ell)} f_j(x)] dx + \int_0^\ell \overline{f_i(x)} f_j(\ell-x) dx, \quad (\text{CP19:2.105})$$

$$(\mathbf{N}_a)_{ij} = \mathcal{N}_a(f_i, f_j), \quad \mathbf{N}_a \succeq 0. \quad (\text{CP19:2.106})$$

The complete exact matrix is

$$\boxed{\tau_R^{-1} K_a (I - K_a) = \eta_p(a) W_a^* W_a + \mathbf{N}_a + \sum_{n \geq 2} b_n \mathbf{P}_{a,\ell_n}.} \quad (\text{CP19:2.107})$$

The prime series converges in matrix norm by (2.96). This is a second concrete energy evaluation of the unchanged target matrix. Every gamma-prime, pole-prime and mixed coefficient interaction is retained through its common forced columns. The coefficient map into any color, angle or Green argument is now explicit: $c \mapsto f_c = \sqrt{2} W_a c|_{(0,a)}$, followed by the ratios f_c/p only on the core and by (2.105) on the full closed domain.

For a hypothetical nonzero null coefficient $K_a c = c$, $f_c \neq 0$, equation (2.107) forces

$$-\sum_{n \geq 2} b_n \mathcal{P}_{p,\ell_n}(f_c) = \eta_p(a) \|f_c\|_2^2 + \mathcal{N}_a(f_c) \geq \eta_p(a) \|f_c\|_2^2 > 0. \quad (\text{CP19:2.108})$$

This is a necessary signed equality at either first-null branch. It does not itself contradict a prior theorem. A sufficient closing estimate, now stated on the actual finite forced family and with all exterior tails retained, is

$$\sum_{n \geq 2} b_n \mathcal{P}_{p,\ell_n}(f_c) > -\eta_p(a) \|f_c\|_2^2 - \mathcal{N}_a(f_c) \quad \text{for every nonzero first-null coefficient } c. \quad (\text{CP19:2.109})$$

For direct all-support nonnegativity replace the strict sign in (2.109) by a non-strict sign and require it for all c and all finite a . An independently quantified comparison must be supplied; (2.109) is not obtained from the positivity of the continuous terms alone.

A valid crude independent lower estimate, useful for checking a concrete support but not asserted to close globally, is

$$Q_a^-(q) \geq \left[\eta_p(a) - \|(-V_p)_+\|_\infty - 2 \sum_{\ell_n < 2a} b_n \right] \|q\|_2^2. \quad (\text{CP19:2.110})$$

Here the full potential V_p , including the negative exterior part, must be evaluated or bounded; the exact joint expression (2.107) is stronger. Taking separate magnitudes of the two diagonal channels before their cancellation would not even produce the finite bound (2.110) uniformly near the origin.

2.5.6. Audit of the archived half-unit certificate The source's Theorem 19.1 has exactly the CP18 Fourier normalization and the pole signs: even $+2|\langle f, \cosh \rangle|^2$, odd $-2|\langle f, \sinh \rangle|^2$. Its stated domain is the full closed form domain on $[-1/2, 1/2]$, with floor $9 \cdot 10^{-8}$. The proof retains the prime 2, a gamma multiplier tail bound, the finite Legendre matrix, and analytic complement/coupling budgets. The finite directed-rounding LDL certificate is an essential recorded input. The printed proof cannot be replaced by the two listed pivot minima alone if its source matrix and interval computation are unavailable. This audit verifies the analytic compatibility and does not claim to have rerun that interval certificate. CP18's independently retained interval and margin are therefore unchanged unless the response/source team separately recovers and accepts its saved full certificate.

2.5.7. Strongest result and exact unresolved sign The additional unconditional result is (2.107), with a proved analytic positive odd theta weight, an exact removal of the origin singularity in every ordinary/reflected prime pair, the explicit gamma-polar lower floor (2.102), and the exterior prime operator-norm budget (2.96). These identities apply to the actual forced columns, include both null branches, and preserve the original Suzuki metric.

The remaining sign is the fully specified paired-prime matrix in (2.107). Its negative exterior potential and antisymmetric reflected prime channels have not been bounded by the positive continuous remainder with a sufficient all-support margin. Thus this route supplies a narrower signed arithmetic estimate, not unconditional Weil positivity. Any new proof of (2.109), or CP18's independently quantified Osgood comparison, has the already established return through odd Weil positivity and the normalized Suzuki identity to $\mathcal{E}_N = 0$.

3. An explicit response family, joint residual payment, and finite optimization

3.1. Inputs, original metric, and the complete operator image

The input is CP18's physical restriction theorem, not an assumed positive Weil operator. For one fixed finite $R > a_0$, on $\mathcal{H}_a = L^2_{\text{odd}}(-a, a)$, $a_0 \leq a \leq R$, retain

$$\mathcal{J}_a = A_a^- + \tau_R F_a F_a^* \succeq g_R I, \quad F_a^* F_a \preceq I, \quad W_a = \mathcal{J}_a^{-1} F_a, \quad K_a = \tau_R F_a^* W_a, \quad M_a = I - K_a. \quad (\text{CP19:3.1})$$

The fixed positive g_R and finite τ_R were established independently of the unknown Weil margin. Every operator below acts in this original physical metric. No normalization of the source columns is changed.

Put $C_a = (\log |D|)_{(-a, a)}^-$, with its supported closed form and exterior Dirichlet convention. Its complete arithmetic image is

$$\mathcal{J}_a y = C_a y - \log(2\pi) y - \Pi_a y - K_{\alpha, a} y - 2s_a \langle s_a, y \rangle + \tau_R F_a F_a^* y, \quad s_a(x) = \sinh(x/2), \quad (\text{CP19:3.2})$$

$$\Pi_a y = \sum_{\ell_n < 2R} b_n [y(\cdot + \ell_n) + y(\cdot - \ell_n)], \quad b_n = \Lambda(n)/\sqrt{n}, \quad \alpha(t) = \frac{e^{-|t|/2}}{1 - e^{-2|t|}} - \frac{1}{2|t|}. \quad (\text{CP19:3.3})$$

Every translation uses zero extension. Terms with $\ell_n \geq 2a$ vanish exactly. The value $\alpha(0) = 1/4$ and $|\alpha| \leq 1$ are retained. In the unitary half-interval representation $y^+ = \sqrt{2}y|_{(0, a)}$, the complete prime action is

$$(\Pi_a^+ y^+)(x) = \sum b_n [\mathbf{1}_{x+\ell_n < a} y^+(x + \ell_n) + \mathbf{1}_{x > \ell_n} y^+(x - \ell_n) - \mathbf{1}_{x < \ell_n < x+a} y^+(\ell_n - x)]. \quad (\text{CP19:3.4})$$

Thus the reflected prime channel is not discarded. The half-interval α kernel is $\alpha(x - y) - \alpha(x + y)$, the signed polar term of $\mathcal{J}_a$ is $-2|s_a^+\rangle\langle s_a^+|$ with $s_a^+ = \sqrt{2}\sinh(x/2)$, and the regularizer kernel is $\tau_R F_a^+(x)F_a^+(y)^*$. The logarithmic term is the supported operator itself, including the reflected kernel $1/[2(x + y)]$, the exterior interaction, and its diagonal finite part.

3.2. The archived response identity, now on the actual sources

Theorem 14.1 and Corollary 14.2 of *Physics_Architectures_and_Log_Prime_Energy.pdf* supply the following algebraic identity and finite optimization. Their hypotheses are a positive operator and graph-domain trial columns; both are satisfied here. For any finite-column $Y : \mathbb{C}^{d_R} \rightarrow \text{Dom } C_a = \text{Dom } \mathcal{J}_a$, define

$$\mathcal{R} = F_a - \mathcal{J}_a Y, \quad L(Y) = F_a^* Y + Y^* F_a - Y^* \mathcal{J}_a Y. \quad (\text{CP19:3.5})$$

Then

$$F_a^* \mathcal{J}_a^{-1} F_a = L(Y) + \mathcal{R}^* \mathcal{J}_a^{-1} \mathcal{R}, \quad \mathcal{C}_g(Y) = I - \tau_R [L(Y) + g_R^{-1} \mathcal{R}^* \mathcal{R}] \preceq M_a, \quad (\text{CP19:3.6})$$

$$M_a - \mathcal{C}_g(Y) = \tau_R \mathcal{R}^* (g_R^{-1} I - \mathcal{J}_a^{-1}) \mathcal{R} \succeq 0. \quad (\text{CP19:3.7})$$

These identities are obtained by expanding $F_a = \mathcal{J}_a Y + \mathcal{R}$. They remain valid when F_a has a coefficient kernel.

For explicit trial columns $V : \mathbb{C}^k \rightarrow \text{Dom } \mathcal{J}_a$, put $U = \mathcal{J}_a V$ and

$$G = F_a^* F_a, \quad \mathbf{L}_V = U^* U - g_R V^* U, \quad \mathbf{B}_V = (U - g_R V)^* F_a. \quad (\text{CP19:3.8})$$

The form of $\mathcal{J}_a(\mathcal{J}_a - g_R)$ gives $\mathbf{L}_V \succeq 0$ and $\text{ran } \mathbf{B}_V \subset \text{ran } \mathbf{L}_V$. The complete optimization over $Y = VH$ is

$$g_R [L(VH) + g_R^{-1} \mathcal{R}(VH)^* \mathcal{R}(VH)] = G - \mathbf{B}_V^* \mathbf{L}_V^\dagger \mathbf{B}_V + (H - \mathbf{L}_V^\dagger \mathbf{B}_V)^* \mathbf{L}_V (H - \mathbf{L}_V^\dagger \mathbf{B}_V). \quad (\text{CP19:3.9})$$

Consequently

$$Y_{\text{opt}} = V \mathbf{L}_V^\dagger \mathbf{B}_V, \quad \boxed{\mathcal{C}_{\text{opt}}(V) = I - \frac{\tau_R}{g_R} [G - \mathbf{B}_V^* \mathbf{L}_V^\dagger \mathbf{B}_V] \preceq M_a.} \quad (\text{CP19:3.10})$$

The exact guaranteed improvement over $Y = 0$ is the positive matrix $\tau_R g_R^{-1} \mathbf{B}_V^* \mathbf{L}_V^\dagger \mathbf{B}_V$. No inverse of A_a^- occurs. A kernel of $\mathbf{L}_V$ is handled by its displayed range condition, rather than silently removing a null branch.

3.3. A concrete reference response and every residual component

Use CP18's arithmetic-free logarithmic reference inverse. With

$$c_{\text{odd}} = \frac{1}{3} \log(9\pi/2) - \frac{1}{3}, \quad \lambda_R = 1 + \max(0, \log R - c_{\text{odd}}), \quad P_a = C_a + \lambda_R I \succeq I, \quad (\text{CP19:3.11})$$

and the fixed reference eigenfunctions $C_1 \varphi_k = \nu_k \varphi_k$, set

$$\varphi_{k,a}(x) = a^{-1/2} \varphi_k(x/a), \quad X_a = P_a^{-1} F_a, \quad X_a e_j = \sum_{k \geq 1} \frac{\langle \varphi_{k,a}, F_a e_j \rangle}{\nu_k - \log a + \lambda_R} \varphi_{k,a}. \quad (\text{CP19:3.12})$$

This is an explicit reference spectral series. It contains no arithmetic inverse or unproved spectral gap. It converges in the graph norm of C_a , since $P_a X_a = F_a \in L^2$. Its finitely many source columns are valid trial responses.

Define the complete bounded perturbation

$$D_a = P_a - \mathcal{J}_a = (\lambda_R + \log(2\pi))I + \Pi_a + K_{\alpha,a} + 2|s_a\rangle\langle s_a| - \tau_R F_a F_a^*. \quad (\text{CP19:3.13})$$

A margin-independent bound is

$$\|D_a\| \leq d_R^{\text{resp}} := \lambda_R + D_R^{\text{ar,sharp}} + \tau_R, \quad D_R^{\text{ar,sharp}} = \log(2\pi) + 2 \sum_{\ell_n < 2R} b_n + 2R + 2(\sinh R - R). \quad (\text{CP19:3.14})$$

For clarity write $d = d_R^{\text{resp}}, g = g_R, \tau = \tau_R$ in the rest of this group. This d is an operator-norm bound, not the already retained regularizer rank d_R .

The one-step image and residual are fully evaluated:

$$\mathcal{J}_a X_a = F_a - D_a X_a, \quad \mathcal{R}(X_a) = D_a X_a. \quad (\text{CP19:3.15})$$

Let the five signed component maps be

$$D_0 = (\lambda_R + \log(2\pi))I, \quad D_P = \Pi_a, \quad D_\alpha = K_{\alpha,a}, \quad D_Z = 2s_a s_a^*, \quad D_F = -\tau F_a F_a^*. \quad (\text{CP19:3.16})$$

Then the entire residual Gram is

$$\boxed{\mathcal{R}(X_a)^* \mathcal{R}(X_a) = \sum_{\eta, \zeta \in \{0, P, \alpha, Z, F\}} X_a^* D_\eta^* D_\zeta X_a.} \quad (\text{CP19:3.17})$$

In particular this includes prime-prime, prime-archimedean, prime-polar, prime-regularizer, polar-regularizer and all other off-diagonal terms with their signs. In full coordinates,

$$(D_Z X_a)_j(x) = 2 \sinh(x/2) \int_{-a}^a \sinh(y/2) X_{a,j}(y) dy, \quad (D_F X_a)_j(x) = -\tau \sum_k F_{a,k}(x) \langle F_{a,k}, X_{a,j} \rangle. \quad (\text{CP19:3.18})$$

The prime component is (3.3), with the full reflected expression (3.4); the α component is its exact integral kernel. No endpoint trace of X_a is imposed.

For this particular $V = X_a$, introduce $H = F_a^* X_a$. The matrices in (3.8) become

$$\mathbb{L}_X = G - F_a^* D_a X_a - X_a^* D_a F_a + X_a^* D_a^2 X_a - g(H - X_a^* D_a X_a), \quad (\text{CP19:3.19})$$

$$\mathbb{B}_X = G - X_a^* D_a F_a - gH. \quad (\text{CP19:3.20})$$

Equations (3.16)-(3.20), inserted into (3.10), are a complete finite-matrix certificate with an explicit family and all operator images. They do not assign a sign to its still unevaluated arithmetic value.

3.4. Exact joint arithmetic savings in the response Gram

The Group 2 Möbius identity applies source-faithfully to the prime shifts of every trial column. At a finite prime cutoff $X \geq e^{2R}$, after zero extension and then compression, the prime component equals

$$\Pi_a Y = E_a^* \sum_{\substack{r, m \geq 1 \\ \gcd(r, 6) = 1}} \frac{\mu(r) \log m}{\sqrt{rm}} \sum_{\substack{e \in \{1, 2, 3, 6\} \\ erm \leq X}} \frac{\mu(e)}{\sqrt{e}} [T_{\log(erm)} + T_{-\log(erm)}] E_a Y. \quad (\text{CP19:3.21})$$

The unequal cutoff indicators are part of this finite identity. For a complete four-corner packet put $r_2 = 2^{-1/2}$, $r_3 = 3^{-1/2}$, $u = \log(rm)$, and

$$Q_+ = (I - r_2 T_{\log 2})(I - r_3 T_{\log 3}), \quad Q_- = Q_+^*, \quad C_u = T_u Q_+ + T_{-u} Q_-. \quad (\text{CP19:3.22})$$

Its whole-line symbol is

$$c_u(\xi) = 2 \cos(u\xi) - \sqrt{2} \cos((u + \log 2)\xi) - \frac{2}{\sqrt{3}} \cos((u + \log 3)\xi) + \frac{2}{\sqrt{6}} \cos((u + \log 6)\xi). \quad (\text{CP19:3.23})$$

The exact color decomposition supplied in Group 2 has

$$V_p = \frac{I - r_p T_{\log p}}{1 + r_p}, \quad Z_p = \frac{\sqrt{r_p}(I + T_{\log p})}{1 + r_p}, \quad V_p^* V_p + Z_p^* Z_p = I. \quad (\text{CP19:3.24})$$

With $c_{23} = (1 + r_2)(1 + r_3)$, its matrix identity on the actual response columns is

$$(Q_+ Y)^*(Q_+ Y) = c_{23}^2 \{Y^* Y - (Z_2 V_3 Y)^*(Z_2 V_3 Y) - (V_2 Z_3 Y)^*(V_2 Z_3 Y) - (Z_2 Z_3 Y)^*(Z_2 Z_3 Y)\}. \quad (\text{CP19:3.25})$$

The three displayed Gram matrices are an exact saving over the separated triangle bound $c_{23}^2 Y^* Y$. The opposite orientation has the adjoint version. When the two orientations are recombined, their full mixed term

$$(T_u Q_+ Y)^*(T_{-u} Q_- Y) + (T_{-u} Q_- Y)^*(T_u Q_+ Y) \quad (\text{CP19:3.26})$$

is retained. Different (r, m) packets retain all mutual crosses as well. Thus an individual positive saving is not multiplied by signed Möbius coefficients and incorrectly claimed to remain a global positive saving.

There is also an exact globally nonnegative phase saving for the *complete* prime component, which requires no packet separation. Let

$$p_a(\xi) = 2 \sum_{\ell_n < 2a} b_n \cos(\ell_n \xi), \quad B_a^{\text{prime}} = 2 \sum_{\ell_n < 2a} b_n, \quad \Pi_a^{\text{whole}} = \sum_{\ell_n < 2a} b_n (T_{\ell_n} + T_{-\ell_n}). \quad (\text{CP19:3.27})$$

For zero-extended columns Y , Plancherel and orthogonal compression give

$$(\Pi_a Y)^*(\Pi_a Y) = (B_a^{\text{prime}})^2 Y^* Y - S_{\text{phase}}(Y) - S_{\text{ext}}(Y), \quad (\text{CP19:3.28})$$

$$S_{\text{phase}}(Y)_{ij} = \frac{1}{2\pi} \int_{\mathbb{R}} [(B_a^{\text{prime}})^2 - p_a(\xi)^2] \widehat{Y}_i(\xi) \widehat{Y}_j(\xi) d\xi \succeq 0, \quad (\text{CP19:3.29})$$

$$S_{\text{ext}}(Y) = [\mathbf{1}_{\mathbb{R} \setminus (-a, a)} \Pi_a^{\text{whole}} Y]^* [\mathbf{1}_{\mathbb{R} \setminus (-a, a)} \Pi_a^{\text{whole}} Y] \succeq 0. \quad (\text{CP19:3.30})$$

The exterior term here is a *response-norm compression term*, not the positive exterior theta prime energy in Group 1. Both are retained in their respective identities.

Write the complete response residual as $\mathcal{R}(Y) = R_0(Y) + \Pi_a Y$, where

$$R_0(Y) = F_a - C_a Y + \log(2\pi) Y + K_{\alpha, a} Y + 2s_a s_a^* Y - \tau F_a F_a^* Y. \quad (\text{CP19:3.31})$$

Then its fully coupled evaluation after the successful arithmetic savings is

$$\boxed{\mathcal{R}(Y)^* \mathcal{R}(Y) = R_0(Y)^* R_0(Y) + R_0(Y)^* \Pi_a Y + (\Pi_a Y)^* R_0(Y) + (B_a^{\text{prime}})^2 Y^* Y - S_{\text{phase}}(Y) - S_{\text{ext}}(Y).} \quad (\text{CP19:3.32})$$

This is an exact subtraction of positive matrices from the separated prime-square charge. It leaves the two signed prime-other cross matrices intact. It does not assert a uniform 96-fold improvement: that archived example had different specified coefficients. Equation (3.32) is the actual saving for these sources and this metric.

The support-sensitive calculation of Group 2 makes the saving quantitatively uniform on the actual trial family. Put

$$\gamma_{\text{prime}}(a) = 4 \sum_{\ell_n < 2a} b_n^2 \sin^2 \left(\frac{\pi}{2(\lceil a/\ell_n \rceil + 1)} \right) > 0 \quad (a > \log 2/2). \quad (\text{CP19:3.33})$$

Indeed $(B_a^{\text{prime}})^2 - p_a(\xi)^2$ is the sum of $4 \sum b_n^2 \sin^2(\ell_n \xi)$ and the nonnegative mixed terms $8 \sum_{n < k} b_n b_k [1 - \cos(\ell_n \xi) \cos(\ell_k \xi)]$. The zero-extended translation difference $T_{\ell_n} - T_{-\ell_n}$ has, on fibers of step $2\ell_n$, the Dirichlet path gap $4 \sin^2[\pi/(2(\lceil a/\ell_n \rceil + 1))]$. Polarization therefore proves

$$S_{\text{phase}}(Y) \succeq \gamma_{\text{prime}}(a) Y^* Y. \quad (\text{CP19:3.34})$$

Substitution into (3.32) reduces the prime-square coefficient from $(B_a^{\text{prime}})^2$ to $(B_a^{\text{prime}})^2 - \gamma_{\text{prime}}(a)$, while keeping $-S_{\text{ext}}$ and both signed R_0 -prime cross matrices. This is an actual quantified improvement in the original residual norm.

In the entire first-prime window $\log 2/2 < a \leq \log 3/2$, an exact stronger compression is available. Set $\ell = \log 2$, $b_2 = (\log 2)/\sqrt{2}$, and

$$O_a = (-a, a - \ell) \cup (\ell - a, a), \quad C_a^{\text{mid}} = (-a, a) \setminus O_a. \quad (\text{CP19:3.35})$$

Since $a < \ell$, every active translation fiber has precisely two nodes and $\Pi_a^2 = b_2^2 \mathbf{1}_{O_a}$. Hence

$$\begin{aligned} \mathcal{R}(Y)^* \mathcal{R}(Y) &= R_0(Y)^* R_0(Y) + R_0(Y)^* \Pi_a Y + (\Pi_a Y)^* R_0(Y) + b_2^2 Y^* \mathbf{1}_{O_a} Y, \\ S_{\text{phase}}(Y) + S_{\text{ext}}(Y) &= 3b_2^2 Y^* Y + b_2^2 Y^* \mathbf{1}_{C_a^{\text{mid}}} Y. \end{aligned} \quad (\text{CP19:3.36})$$

Relative to the separated prime-square payment $4b_2^2 Y^* Y$, the complete certificate (3.6) improves by exactly $\tau g^{-1} b_2^2 [3Y^* Y + Y^* \mathbf{1}_{C_a^{\text{mid}}} Y]$. The entire prime-archimedean, prime-polar, and prime-regularizer interaction remains in the same two cross matrices. A fourfold prime-square reduction is proved here; it does not claim a fourfold improvement of the complete residual or a sign for its untouched mixed part.

The transported Green/storage square from Group 2 cannot replace this L^2 residual Gram without an additional inequality: its Fourier multiplier is $(1/4 + \xi^2)^{-1}$, which tends to zero. Thus no constant controls every L^2 residual from that Green norm. Its exact stored endpoint and internal energy may be evaluated inside a response-specific estimate, but are not silently substituted for (3.32).

3.5. A complete finite response sequence with a proved uniform error

The one-step certificate is now strengthened to an independently convergent explicit family. This avoids the unevaluated choice $Y = W_a$. It uses only the reference inverse P_a^{-1} , multiplication by the complete bounded D_a , and finitely many operations at each stage.

Set

$$\kappa_R = \frac{g}{g+d}, \quad \Lambda_R = 1+d, \quad \omega_R = \frac{2}{\kappa_R + \Lambda_R}, \quad \rho_R = \frac{\Lambda_R - \kappa_R}{\Lambda_R + \kappa_R} < 1. \quad (\text{CP19:3.37})$$

On the common form domain $P_a = \mathcal{J}_a + D_a \preceq (1 + d/g)\mathcal{J}_a$, while $\mathcal{J}_a \preceq P_a + dI \preceq (1 + d)P_a$. Therefore the bounded self-adjoint operator

$$T_a = P_a^{-1/2} \mathcal{J}_a P_a^{-1/2} \quad \text{satisfies} \quad \kappa_R I \preceq T_a \preceq \Lambda_R I. \quad (\text{CP19:3.38})$$

Here the middle expression is defined by form congruence; no claim that an unbounded product is everywhere initially defined is required.

Starting from $Y_0 = 0$, use the explicit recursion

$$\boxed{Y_{m+1} = Y_m + \omega_R P_a^{-1} (F_a - \mathcal{J}_a Y_m).} \quad (\text{CP19:3.39})$$

Every Y_m has finitely many columns in $\text{Dom } \mathcal{J}_a$. Its complete operator image follows recursively from

$$\mathcal{J}_a P_a^{-1} = I - D_a P_a^{-1}; \quad (\text{CP19:3.40})$$

thus the scalar, prime, archimedean, polar and regularizer terms of every iterate are exactly (3.13), not approximations.

For $h_a = P_a^{-1/2} F_a$, $S_a = I - \omega_R T_a$, and the finite polynomial $p_m(t) = [1 - (1 - \omega_R t)^m]/t$,

$$Y_m = P_a^{-1/2} p_m(T_a) h_a, \quad P_a^{-1/2} \mathcal{R}_m = S_a^m h_a, \quad \|S_a\| \leq \rho_R. \quad (\text{CP19:3.41})$$

The first formula agrees with the graph-domain recursion (3.39). It is not an invocation of an unknown arithmetic inverse.

For a direct L^2 residual bound, write $b = 1 - \omega_R$ and use the exact telescoping identity

$$\mathcal{R}_m = b^m F_a + \omega_R D_a P_a^{-1/2} \sum_{j=0}^{m-1} b^{m-1-j} S_a^j P_a^{-1/2} F_a. \quad (\text{CP19:3.42})$$

Since $1 \in [\kappa_R, \Lambda_R]$, $|b| \leq \rho_R$, and $P_a \succeq I$. Hence, uniformly on $a \in [a_0, R]$,

$$\boxed{\|\mathcal{R}_m\| \leq \rho_R^m + \omega_R d m \rho_R^{m-1}.} \quad (\text{CP19:3.43})$$

This establishes $Y_m \rightarrow W_a$ in the graph norm of $\mathcal{J}_a$, uniformly in operator norm from the finite source space: $\|Y_m - W_a\| \leq g^{-1} \|\mathcal{R}_m\|$ and $\mathcal{J}_a(Y_m - W_a) = -\mathcal{R}_m$. It does not use $\|P_a^{1/2}\| < \infty$, which would be false.

A response-specific, reference-frequency payment is also available. It need not be smaller than the scalar g^{-1} payment on every low reference mode; it decreases the high-reference-frequency charge. From (3.38),

$$\mathcal{J}_a^{-1} \preceq \kappa_R^{-1} P_a^{-1}. \quad (\text{CP19:3.44})$$

Define the fully evaluated reference-weighted certificate

$$\mathcal{C}_m^{\text{ref}} = I - \tau [L(Y_m) + \kappa_R^{-1} \mathcal{R}_m^* P_a^{-1} \mathcal{R}_m]. \quad (\text{CP19:3.45})$$

It is legitimate because its payment operator is independently known. Its residual matrix equals $h_a^* S_a^{2m} h_a$, with all arithmetic interactions still inside S_a . Spectral calculus gives the exact expression

$$\mathcal{C}_m^{\text{ref}} = I - \tau h_a^* [T_a^{-1} + (\kappa_R^{-1} I - T_a^{-1}) S_a^{2m}] h_a, \quad (\text{CP19:3.46})$$

where T_a^{-1} is used only to prove the identity and estimate; the certificate itself is computed by the finite recursion (3.39) and reference inverse (3.45). Consequently

$$\boxed{\mathcal{C}_m^{\text{ref}} \preceq \mathcal{C}_{m+1}^{\text{ref}} \preceq M_a, \quad 0 \preceq M_a - \mathcal{C}_m^{\text{ref}} \preceq \tau \kappa_R^{-1} \rho_R^{2m} F_a^* P_a^{-1} F_a \preceq \tau \kappa_R^{-1} \rho_R^{2m} I.} \quad (\text{CP19:3.47})$$

Thus the response construction and its complete error budget are closed independently of the Weil sign. Given any prescribed tolerance $\eta > 0$, taking

$$m \geq \max \left(0, \left\lceil \frac{\log(\tau/(\kappa_R \eta))}{-2 \log \rho_R} \right\rceil \right) \quad (\text{CP19:3.48})$$

(with the immediate exact case handled separately if $\rho_R = 0$) makes the matrix overpayment at most ηI , uniformly for the whole fixed support range. This may require a very large finite m ; no favorable numerical size is asserted.

At the independently certified starting interval this gives an actual positive finite-response certificate with a fully specified iteration count. Let $m_0 = 1/5000$ and $\eta_0 = m_0/(m_0 + \tau_R)$, as in CP18. Then $M_{a_0} \succeq \eta_0 I$, and every integer

$$m \geq m_{\text{seed}} := \max \left(0, \left\lceil \frac{\log(2\tau_R/(\kappa_R \eta_0))}{-2 \log \rho_R} \right\rceil \right) \implies \mathcal{C}_m^{\text{ref}}(a_0) \succeq \frac{1}{2} \eta_0 I. \quad (\text{CP19:3.49})$$

This validates the complete response construction on the proved prime-active seed. It reproduces, rather than extends, that independent support certificate.

3.6. Optimization of the concrete family and the still signed matrix

One valid enriched finite-column space is

$$V_m = [Y_1, \dots, Y_m, P_a^{-1} \Pi_a Y_1, \dots, P_a^{-1} \Pi_a Y_m, P_a^{-1} K_{\alpha,a} Y_m, P_a^{-1} s_a, P_a^{-1} F_a]. \quad (\text{CP19:3.50})$$

Redundant columns may remain, with the pseudoinverse in (3.10), or be removed without changing their span. All inputs to P_a^{-1} are L^2 , so every column is in the required graph domain. For every explicitly supplied z ,

$$\mathcal{J}_a(P_a^{-1} z) = z - D_a P_a^{-1} z; \quad (\text{CP19:3.51})$$

this gives every operator image in (3.50), including all prime-prime packets generated by the second block. Complete individual (1,2,3,6) packet responses $P_a^{-1} E_a^* C_u E_a Y_j$, with their unequal cutoff corrections, can also be included; the formulas are the same.

Using (3.8)-(3.10) on V_m optimizes the original scalar residual certificate over this concrete family. It is at least as good in Loewner order as that certificate at any contained Y_j . Equations (3.24)-(3.32) calculate its joint prime-square improvement and retain the entire signed prime-other residual pairing. This optimization does not automatically dominate the different reference-weighted certificate (3.45); the two are valid independently and may be checked on the same target directions.

The response-specific family can itself be optimized without changing its payment. For either known positive payment operator $\mathcal{B} = g^{-1} I$ or $\mathcal{B} = \kappa_R^{-1} P_a^{-1}$, set

$$\mathcal{L}_{\mathcal{B}} = U^* \mathcal{B} U - V_m^* U, \quad \mathcal{B}_{\mathcal{B}} = (U^* \mathcal{B} - V_m^*) F_a, \quad G_{\mathcal{B}} = F_a^* \mathcal{B} F_a, \quad U = \mathcal{J}_a V_m. \quad (\text{CP19:3.52})$$

Because $\mathcal{B} - \mathcal{J}_a^{-1} \succeq 0$, $\mathcal{L}_{\mathcal{B}} = U^* (\mathcal{B} - \mathcal{J}_a^{-1}) U \succeq 0$ and $\text{ran } \mathcal{B}_{\mathcal{B}} \subset \text{ran } \mathcal{L}_{\mathcal{B}}$. Completing the finite square gives

$$\boxed{\mathcal{C}_{\mathcal{B},\text{opt}}(V_m) = I - \tau \left[G_{\mathcal{B}} - \mathcal{B}_{\mathcal{B}}^* \mathcal{L}_{\mathcal{B}}^{\dagger} \mathcal{B}_{\mathcal{B}} \right] \preceq M_a.} \quad (\text{CP19:3.53})$$

For the reference payment and a family containing Y_m , this optimized certificate is at least $\mathcal{C}_m^{\text{ref}}$, and therefore inherits the uniform error bound (3.47). The optimized reference-weighted Gram contains every term $\mathcal{R}_\eta^* P_a^{-1} \mathcal{R}_\zeta$. The L^2 color and phase savings (3.25)-(3.32) are used in the scalar payment certificate; they are not assumed to commute through the supported P_a^{-1} . This distinction keeps the original metric and all response-specific cross terms explicit.

The exact all-support scalar-payment inequality still needed after these successful savings is

$$\boxed{G - \mathbf{B}_{V_m}^* \mathbf{L}_{V_m}^\dagger \mathbf{B}_{V_m} \preceq \frac{g_R}{\tau_R} I.} \quad (\text{CP19:3.54})$$

A sufficient strict first-null contradiction would replace the right side by a strict inequality on every nonzero hypothesized null coefficient. No such independent strict inequality is obtained by the response approximation theorem alone. For the reference-weighted family the exact residual expression is (3.45), and its missing assertion is $\mathcal{C}_m^{\text{ref}} \succeq 0$ for some uniformly justified finite response choice at each support.

Indeed, at a true hypothetical null coefficient $M_a c = 0 \neq c$, every valid lower certificate satisfies $c^* \mathcal{C} c \leq 0$. More explicitly,

$$c^* \mathcal{C}_m^{\text{ref}} c = -\tau \langle h_a c, (\kappa_R^{-1} I - T_a^{-1}) S_a^{2m} h_a c \rangle \leq 0. \quad (\text{CP19:3.55})$$

Therefore accuracy by itself cannot yield a strict contradiction at a first null. A new arithmetic comparison must make the actual matrix sign incompatible with that null. Formula (3.32) locates the still signed interaction after the proved packet savings.

Theorem actually proved. The source-faithful response identity and its archived finite optimizer apply to the actual physical regularizer and its restricted sources, with both null branches retained. The explicit reference recursion (3.39) has a uniform margin-independent graph residual bound (3.43), and the response-specific certificates (3.45) increase monotonically to the complete target matrix with the explicit uniform error (3.47). The full prime residual obtains the exact positive phase and exterior savings (3.28)-(3.32); the four-channel color decomposition yields the exact additional packet accounting (3.25)-(3.26).

Quantitative effect and limit. On every fixed interval where $M_a \succeq \eta I > 0$ is independently true, (3.47) guarantees termination of a finite certificate once its error is below η . It also allows a negative eigenvalue to be certified from an upper enclosure $M_a \preceq \mathcal{C}_m^{\text{ref}} + \tau \kappa_R^{-1} \rho_R^{2m} I$. It does not establish that every support is positive. The remaining signed arithmetic statement is (3.54), or an independently positive value of (3.45), or the energy-relative boundary estimate retained from CP18. These estimates are not supplied merely by residual convergence.

4. Combining the saved identities, the new estimates, and the exact remaining sign

4.1. Four complete evaluations of the same target matrix

For the physical forced columns, define

$$\mathbf{Q}_a := W_a^* A_a^- W_a = \tau_R^{-1} K_a M_a. \quad (\text{CP19:4.1})$$

The calculations in Groups 1-3 give four compatible descriptions. First, with all transformed exterior interactions included,

$$\mathbf{Q}_a = \mathbf{E}_{\Gamma,a} + \sum_{n \geq 2} b_n \mathbf{B}_a(\ell_n) - \frac{1}{2} \mathbf{G}_a - \frac{1}{2} \ell_a \ell_a^*. \quad (\text{CP19:4.2})$$

Second, on the cofinal divisibility-chain supports $a = m \log 6/2$, the exact Green/storage/color transport gives

$$\mathbf{Q}_a = \mathbf{A}_{\Gamma,a} + \mathbf{S}_a + \mathbf{C}_a - \mathbf{D}_a, \quad \mathbf{D}_a = \mathbf{P}_a + \mathbf{S}_a + \mathbf{C}_a + 2\lambda_a^* \lambda_a. \quad (\text{CP19:4.3})$$

Here $\mathbf{C}_a$ is the color-square matrix, not the logarithmic operator or a response certificate. The internal storage and all divisor colors are present. Equation (4.3) also shows exactly why they cannot be counted as an additional independent credit after the complete continuum correction has already been paid.

Third, the new odd theta-derivative transformation yields

$$\mathbf{Q}_a = \eta_p(a) W_a^* W_a + \mathbf{N}_a + \sum_{n \geq 2} b_n \mathbf{P}_{a, \ell_n}, \quad \mathbf{N}_a \succeq 0, \quad \eta_p(a) > 0. \quad (\text{CP19:4.4})$$

The paired-prime entries are the bounded, origin-cancelled formula (2.105); they include the complete negative exterior potential. The exact continuous remainder $\mathbf{N}_a$ includes the internal gamma energy, all unused exterior gamma channels, the internal polar variance, and the multiplication surplus specified after (2.102). Thus this equation retains every successful continuous-energy saving.

Fourth, the explicit reference-response sequence gives lower enclosures of M_a :

$$\mathcal{C}_m^{\text{ref}} \preceq M_a \preceq \mathcal{C}_m^{\text{ref}} + e_m I, \quad e_m = \tau_R \kappa_R^{-1} \rho_R^{2m} \longrightarrow 0. \quad (\text{CP19:4.5})$$

Every response uses the complete arithmetic perturbation, the original sources, and a known positive logarithmic reference inverse. The convergence and constants are uniform on each fixed $[a_0, R]$, independently of the unknown Weil margin. The reference-weighted payment is an alternative known upper payment; no unsupported global operator ordering between that payment and $g_R^{-1} I$ is claimed.

These equations agree through the exact range congruence in Group 1. They preserve the Suzuki metric. A positive component in one representation cannot be substituted for the complete signed expression in another without its displayed correction.

4.2. The actual gains after the substitutions

The new lower packet estimate is

$$\mathcal{P}_a(t) \succeq (1 - 10^{-6}) \mathbf{B}_a(t) + 6^{-1/2} \mathbf{B}_a(t + \log 6) \quad (a \geq a_0, t > 2a). \quad (\text{CP19:4.6})$$

The exact improved coefficient in (1.39) can be used instead of the rounded constant. Negative outer Möbius coefficients, interior packets, and every unequal cutoff remain in the complete source identity. No all-support lower comparison to the polar variance has been inferred by changing the sign of such coefficients.

For an actual response Y , the complete residual payment after the joint prime saving is

$$\begin{aligned} \mathcal{R}^* \mathcal{R} &= R_0^* R_0 + R_0^* \Pi_a Y + (\Pi_a Y)^* R_0 \\ &\quad + (B_a^{\text{prime}})^2 Y^* Y - \mathbf{S}_{\text{phase}}(Y) - \mathbf{S}_{\text{ext}}(Y). \end{aligned} \quad (\text{CP19:4.7})$$

The explicit finite-support difference estimate proves a strictly positive phase saving when a prime is active:

$$\mathbf{S}_{\text{phase}}(Y) \succeq \gamma_{\text{prime}}(a) Y^* Y, \quad \gamma_{\text{prime}}(a) = 4 \sum_{\ell_n < 2a} b_n^2 \sin^2 \left(\frac{\pi}{2(\lceil a/\ell_n \rceil + 1)} \right). \quad (\text{CP19:4.8})$$

The remainder after (4.8) is itself positive and explicitly defined in Group 2. On the first-prime-only interval,

$$\frac{\log 2}{2} < a \leq \frac{\log 3}{2}, \quad (\Pi_a Y)^*(\Pi_a Y) = b_2^2 Y^* P_{\text{ov}} Y \preceq b_2^2 Y^* Y, \quad b_2 = \frac{\log 2}{\sqrt{2}}. \quad (\text{CP19:4.9})$$

The separated two-shift charge is $4b_2^2 Y^* Y$. Thus the complete prime-square payment is at most one quarter of that charge, before using the actual overlap projection. The two signed crosses in (4.7) are still present. This is a proved quantitative cancellation on the current response family; the historical factor 96 was not assumed to hold here.

The other exact gain is the elimination of the artificial $1/x$ singularity between the ordinary and reflected prime diagonals under the odd theta-derivative map. Its bounded remainder is

$$D_\ell(x) = \frac{p_o(x+\ell) + p_o(x-\ell)}{p(x)}, \quad D_\ell(0) = \frac{2p'(\ell)}{p'(0)}. \quad (\text{CP19:4.10})$$

This permits the transformation on the actual closed domain rather than only on pointwise regular test functions. The gamma and polar terms then have the independent positive lower budget

$$\eta_p(a) = 2 \min_{0 \leq x \leq a} \frac{\sinh(x/2)}{p(x)} \left[e^{a/2} \phi(a) + \frac{1}{2} \int_a^\infty e^{y/2} \phi(y) dy \right] > 0. \quad (\text{CP19:4.11})$$

It is support-dependent and contains no unknown positivity margin. The negative exterior paired-prime tail has its separate explicit norm bound (2.96); it is included in (4.4), not assigned a favorable sign.

4.3. What the complete response optimization proves, and where its sign stops

For the explicit enriched response family V_m of Group 3, the finite optimizer gives

$$\mathcal{C}_{\text{opt}}(V_m) = I - \frac{\tau_R}{g_R} \left[F_a^* F_a - B_{V_m}^* L_{V_m}^\dagger B_{V_m} \right] \preceq M_a. \quad (\text{CP19:4.12})$$

Every operator image in L_{V_m} and B_{V_m} is explicit. Substitution of (4.7) keeps the phase, overlap, color, exterior, and finite-optimization savings together with every prime-other interaction. The remaining matrix inequality is

$$F_a^* F_a - B_{V_m}^* L_{V_m}^\dagger B_{V_m} \preceq (g_R/\tau_R) I. \quad (\text{CP19:4.13})$$

This inequality has not been proved for every finite support. Replacing the two signed mixed matrices in (4.7) by separate norms does not produce a known strict margin. No theorem retrieved from the authentic mask, half-balance, or survival sources identifies their coefficient law with this complete residual without the additional amplitude and metric corrections displayed in Group 2.

The reference sequence closes an approximation issue completely. If independently $M_a \succeq \eta I > 0$, the explicit choice $e_m < \eta$ certifies that interval in finitely many response operations. Conversely, if $c^*(\mathcal{C}_m^{\text{ref}} + e_m I)c < 0$, it certifies a genuinely adverse direction. These are valid analytical conclusions with a quantified finite error; no numerical testing or numerical evidence is used as a replacement for the all-support sign.

At a hypothetical first null $M_a c = 0 \neq c$, the exact error formula instead gives

$$c^* \mathcal{C}_m^{\text{ref}} c = -\tau_R \langle h_a c, (\kappa_R^{-1} I - T_a^{-1})(I - \omega_R T_a)^{2m} h_a c \rangle \leq 0. \quad (\text{CP19:4.14})$$

The error tends to zero, but this identity cannot supply a strict positive certificate at that null without an additional arithmetic inequality. It precisely separates the completed response construction from the still-unproved sign of the response matrix.

4.4. A concrete signed calculation and one sufficient next lemma

For $f_c = \sqrt{2}W_ac|_{(0,a)}$, retain the fully calculated scalar

$$\begin{aligned} \mathcal{P}_a(c) = \sum_{n \geq 2} b_n \left[\int_0^a D_{\ell_n}(x) |f_c(x)|^2 dx - 2\Re \int_0^\infty \overline{f_c(x)} f_c(x + \ell_n) dx \right. \\ \left. + \Re \int_0^{\ell_n} \overline{f_c(x)} f_c(\ell_n - x) dx \right]. \end{aligned} \quad (\text{CP19:4.15})$$

This is an absolutely convergent paired expression after the support cutoff and explicit tail bound. Its first term is not discarded outside the active original prime range. The other two terms are respectively the ordinary and reflected prime interactions. At a first null, the complete equation requires

$$-\mathcal{P}_a(c) = \eta_p(a) \|f_c\|_2^2 + \mathcal{N}_a(f_c) > 0. \quad (\text{CP19:4.16})$$

Both null branches obey this identity. The zero polar moment does not eliminate the positive continuous variance or the paired exterior potential.

One fully quantified sufficient new lemma is the following. For every finite $R > a_0$, for every hypothetical first-null support $a \in (a_0, R]$, independently produce $\delta_R(a) > 0$, using the actual arithmetic equation, such that for every $c \in \ker M_a$,

$$\boxed{\mathcal{P}_a(c) \geq -\mathcal{N}_a(f_c) - [\eta_p(a) - \delta_R(a)] \|f_c\|_2^2.} \quad (\text{CP19:4.17})$$

The constants must be established from the actual operator and support, not defined as the unknown deficit. A nonzero such c has $f_c \neq 0$, and (4.4) would contradict (4.17) immediately. This lemma asks for a comparison of the explicitly origin-cancelled prime pairs with the complete known continuous remainder. It makes no demand for a gap uniform as $a \rightarrow \infty$.

The independent crude estimate presently available is

$$\mathcal{P}_a(c) \geq - \left[\|(-V_p)_+\|_{L^\infty(0,a)} + 2 \sum_{\ell_n < 2a} b_n \right] \|f_c\|_2^2, \quad V_p = \sum_{n \geq 2} b_n D_{\ell_n}. \quad (\text{CP19:4.18})$$

Its full potential is controlled by the finite terms plus (2.96). On the first-prime-only interval the last coefficient improves from $2b_2$ to b_2 , by the exact reflected decomposition (2.104). Nevertheless, no all-support bound of the bracket by $\eta_p(a)$, or a sufficiently sharp compensation by $\mathcal{N}_a$, has been proved. This is the precise stopped signed estimate after the new origin cancellation and positive continuous floor. The known local positivity certificate is preserved independently of this attempted lower bound.

The CP18 continuation alternative also remains available. With $\varepsilon_a = \lambda_{\min}(M_a)$ and its least eigenspace $\mathfrak{E}_a$, the proved matrix evolution gives

$$-\varepsilon'_a = \tau_R \|\text{Proj}_{\mathfrak{E}_a} \beta_a\|^2 \quad \text{for almost every } a \quad (\text{CP19:4.19})$$

with the established multiplicity interpretation. The actual reference residual satisfies $\Gamma_a = \beta_a - M_a h_a$ almost everywhere. The independently sufficient bound

$$\|\text{Proj}_{\mathfrak{E}_a} \Gamma_a\|^2 \leq d_R(a) \varepsilon_a \log(e/\varepsilon_a), \quad 0 \leq d_R \in L^1([a_0, R]), \quad (\text{CP19:4.20})$$

is still unproved. Here $d_R(a)$ is an integrable function, not the integer regularizer rank. The response accuracy (4.5) contains no vanishing factor in ε_a and does not establish (4.20).

4.5. Exact return chain and current unconditional status

If (4.17) is proved at every finite first-null support, CP18's continuity, initial positivity, and unified finite-rank nullspace correspondence exclude every finite first crossing, including the zero-polar-moment branch. If instead (4.13), a direct nonnegative version of (4.4), or the sufficient integrable continuation bound is independently established for every finite support, the same all-support conclusion follows. Restriction from a cofinal sequence of supports is sufficient for direct nonnegativity.

The retained return is then

$$\begin{aligned}
 Q_a^-(q) &\geq 0 \quad \text{for all finite } a \text{ and } q \in C_c^\infty(-a, a) \text{ odd} \\
 &\implies \text{RH} \quad (\text{Yoshida's odd Weil criterion}) \\
 &\implies \mathfrak{W} = \mathcal{S} \quad (\text{the retained, correctly normalized Suzuki identity}) \\
 &\implies \Delta = 0 \implies \mathcal{E}_N = 0 \quad (N \geq 1) \implies \liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1.
 \end{aligned} \tag{CP19:4.21}$$

The new sign comparison must precede this chain. The chain cannot be used as an unconditional input to prove that comparison.

Strongest unconditional conclusion of this checkpoint. The full theta energy is now evaluated on the actual forced matrix with domain, pole variance, and transformed exterior tails proved. The recovered arithmetic work gives a quantitative complete exterior-packet lower bound, actual translated color identities, exact Green/storage/Smith coefficient maps, and a strict quantified improvement of the prime-response payment. The new odd theta-derivative identity removes a singular ordinary/reflected diagonal pair before estimation and supplies an explicit positive gamma-polar support floor. The explicit response family has complete operator images and a uniform, margin-independent convergent certificate budget. The signed estimate (4.17), or an equivalent all-support comparison, remains open.

Separate target status. The independently retained prime-active interval $a \leq 699/2000 + e^{-2500}$, its $1/5000$ lower bound, and CP18's stated constant-dependent local extension remain valid. The archived $a \leq 1/2, 9 \cdot 10^{-8}$ result has compatible normalization and domain; its recorded finite interval-arithmetic certificate has not been independently authenticated here, so it is not promoted to a new independently checked interval. No additional finite first-null branch has been excluded globally. Unconditional Weil positivity and the original Suzuki defect criterion remain open; $\mathcal{E}_N = 0$ remains the conditional return.

We are aiming to prove unconditional Weil positivity.

5. Source register, proof scope, and preservation

The supplied theory atlas is a 301-entry inventory and source map. This checkpoint used the underlying proofs for the mechanisms actually substituted above; it does not present the entire inventory as 301 independently audited positivity theorems.

Source	Proof or identity used here	Scope retained
<i>Michaels_Physical_Regularizer_and_Boundary_Energy_Checkpoint_18.md</i>	Physical form restriction, both null branches, complete forced matrix, domain and Suzuki convergence, boundary matrix evolution	Inherited complete checkpoint; no unregularized inverse bound added

Source	Proof or identity used here	Scope retained
<i>Physics_Architectures_and_Log_Prime_Energy.pdf</i> , version 4, §§14-19	Response optimization; pole-continuum/Green identity; theta annihilation and complete weighted energy; explicit pole variance; recorded half-unit certificate	Identity and analytic normalization checked; recorded finite LDL certificate not independently rerun
<i>Prime_DNA_Angle_Families_and_Prime_Triple_Bound.md</i> , §§4-8	Angle/Green rescaling, complete prime triple, terminal correction	Original coefficient law and metric kept
<i>Michaels_Prime_Color_Dual_Kernel_Bridge.md</i> , §§2-3, 7	Visible/hidden prime isometries and Smith-Jordan divisor squares	Actual translation substitution and divisibility-chain coefficient map proved here
<i>Prime_DNA_Weighted_Half_Balance_and_Forced_Prime_Compensation.md</i> , §§1-3	Exact compensated energy and prime-3 capacity credit	Proved asymptotic half-balance distinguished from its open stronger one-sided rate
<i>Prime_DNA_Signed_Overlap_and_Band_Resolution.md</i> , §§1-2	Forced overlap with authentic doubled coefficients	The changed response Gram and metric correction are displayed
<i>Prime_DNA_Energy_Throughline.md</i> , §1	Full historical Green energy, terminal, two-summary/two-endpoint storage	Extended here to the actual continuous forced densities with complete orthogonal storage
<i>Prime_DNA_Cofactor_Energy_and_Upper_Interaction.md</i> , §§2-4	Common harmonic-return projection and retained signed interaction	Actual common representer required before transferring its bound
<i>Michaels_Joint_Energy_Continuation.pdf</i> , pp.19-21, P1-P12	Authentic mask, rank-gap compensation and transport correction	Additional variable-amplitude defect is retained
<i>Michaels_Survival_and_Prime_Transport_Continuation.md</i> , survival S1-S7 and prime transport T1-T10	Completed/open histories, survival rebate, source-channel coupling and factor-96 example	Full terminal retained; diagnostic example not recast as a universal saving
Suzuki, <i>Weil's quadratic form via the screw function</i> , arXiv:2606.09096v2, Introduction and §2	Complete arithmetic normalization and Yoshida's odd-test criterion	Used only with its stated hypotheses; no RH-dependent factorization imported

Stable source identities for the archived files used in this run are retained in the attached atlas and the source subsections above. The main energy source is; CP18 is. The original Suzuki identity and metric are carried forward from the complete prior checkpoint chain, with the RH implication explicitly conditional.

Independent parallel checks covered the theta normalization and derivative polynomials, origin cancellation and closed-domain passage, gamma-polar lower-floor factors, exterior prime tails, packet orientations and color Grams, finite residual optimization, graph convergence, and the reference-weighted certificate error. The numerical constants used for the new exterior-packet and theta-derivative estimates are bounded analytically. The calculations are analytic estimates, not a numerical zero search.

The Checkpoint 19 packet preserves every prior non-ledger content file from Checkpoint 18 byte-for-byte, adds this complete calculation and the two supplied source-map/directive documents, and updates the continuous ledger and content-hash manifest. Earlier checkpoint ZIPs remain intact and are not nested in this packet. Inclusion of earlier manuscripts does not relabel their unresolved or conditional claims as proved.

Final status: the explicit response construction, actual arithmetic savings, complete energy maps, and continuous-energy floor are proved at the domains and constants stated above. The signed paired-prime comparison and its sufficient energy-dependent continuation alternative remain unresolved. Global Weil positivity and the original defect criterion are not claimed proved.

We are aiming to prove unconditional Weil positivity.

Checkpoint 20: Covariance, reflection, and local positivity

Source: *Michaels_Covariance_Reflection_and_Local_Positivity_Checkpoint_20.md*. Equation tags carry the source prefix CP20; references within this unit retain their original local numbering.

Checkpoint 20 – 23 September 2026

This checkpoint executes the four-group directive on the complete Checkpoint 19 operator. Its strongest new sign theorem is

$$\boxed{Q_a^-(q) > \frac{1}{200} \|q\|_2^2 \quad 0 < a \leq \frac{3}{8}, \quad 0 \neq q \in \mathcal{D}_a^-} \quad (\text{CP20:0.1})$$

It holds for every complex odd input in the closed form domain. The previous endpoint $699/2000 + e^{-2500}$ is enlarged to $3/8$, and the earlier uniform floor $1/5000$ is strengthened to $1/200$. The proof keeps the complete first-prime reflection inside the Fourier-concentration comparison, pays every Taylor and parameter error, and then pays the full archimedean and polar contributions. A reproducible exact-rational certificate supplies the finite matrix bound. Its complete remaining rational margin is $5278/10^6 > 1/200$.

The four executed groups establish:

1. **Actual old-board transfer:** an exact optimal payment for a physical reflection path using the earlier positive $\{1,2,3,6\}$ theta matrix. A separate smooth-bump calculation identifies why the ordinary modulus/positive-semigroup shortcut fails after the first prime activates.
2. **Complete covariance energy:** a positive nested covariance representation, the full spatial exterior reserve, and an explicit inverse with a common conditional-mean correction. The resulting adverse-prime payment is quantified.
3. **Coupled prime arithmetic:** exact all-support reflection spectra for 2 and 3, exact prime-square spectra, and a disjoint allocation retaining every prime-power coefficient. The full signed identity keeps all positive path remainders and exterior terms.
4. **Complete comparison and local sign:** uniform graph-residual error bounds for the full energy matrix, the explicit normalized global comparison, and the new all-function support certificate (0.1).

The original Suzuki metric, determining ladder, and both first-null branches are retained. The finite covariance, path, and concentration metrics below are comparisons for the spatial Weil operator; they do not replace the Suzuki normalization. The primary objective remains unconditional Weil positivity. The original independent criterion $\liminf_N \mathcal{E}_N \leq 1$ and the global signed comparison remain open after this checkpoint. The return through RH to $\mathfrak{W} = \mathcal{S}$ and $\mathcal{E}_N = 0$ is conditional.

Conventions and proof dependencies

Inner products are conjugate-linear in their first argument. All supported functions are extended by zero; the Fourier transform is $\widehat{q}(\xi) = \int q(x) e^{i\xi x} dx$. The inherited closed odd domain is

$$\mathcal{D}_a^- = \left\{ q \in L_{\text{odd}}^2(-a, a) : \int_{\mathbb{R}} \log(2 + |\xi|) |\widehat{q}(\xi)|^2 d\xi < \infty \right\}. \quad (\text{CP20:0.2})$$

The complete arithmetic normalization is CP19 (0.5), also reproduced in the local proof below. CP19's theta weight is $\phi(u) = 2\Phi(u/2)$, and $p = -\phi'$ is positive on the positive half-line. Its full forced matrix is $W_a^* A_a^- W_a = \tau_R^{-1} K_a(I - K_a)$. CP18-19 provide its positive regularizer, the legitimate form/Suzuki core passage, and the complete origin-cancelled prime potential.

The numbered G1, G2, G3, 4, and LC equations preserve the proof fragments' cross-references. "Proved" below means derived from these stated inherited identities and the displayed proofs; it is not a claim of external peer review. The exact local certificate and its error accounting were checked independently within this run.

1. Actual positivity transfer and the reflected-prime sign

This calculation uses the complete operator and metric of CP19. It proves two new, concrete statements: the ordinary modulus argument fails for the actual odd operator as soon as the first prime becomes active; and the old positive four-state theta matrix gives an exact optimized payment for an actual four-vertex reflection path. Neither statement establishes global Weil positivity.

1.1. The actual half-line operator and the complete modulus identity

Write $s(x) = \sinh(x/2)$, $b_n = \Lambda(n)/\sqrt{n}$, $\ell_n = \log n$, and

$$\kappa(t) = \frac{e^{-t/2}}{1 - e^{-2t}}, \quad t > 0. \quad (\text{CP20:G1.1})$$

The unitary map from $L^2(0, a)$ onto the odd subspace of $L^2(-a, a)$ is

$$(Uf)(x) = 2^{-1/2} \operatorname{sgn}(x) f(|x|). \quad (\text{CP20:G1.2})$$

All functions are extended by zero. Let $\tilde{Q}_a(f) = Q_a^-(Uf)$. CP19 (0.5), including its logarithmic finite part and its regular gamma correction, gives the following off-diagonal continuous kernel on the half-line:

$$-H(x, y), \quad H(x, y) = \kappa(|x - y|) - \kappa(x + y) + 4s(x)s(y), \quad x, y > 0, x \neq y. \quad (\text{CP20:G1.3})$$

Indeed, the off-diagonal kernel of $\log|D| - K_\alpha$ is $-\kappa(|x - y|)$; odd reflection subtracts the kernel at $x + y$. The original polar term becomes $-4|\int_0^a s(x)f(x)dx|^2$. Every local/diagonal term is unchanged on replacing f by $|f|$.

The logarithmic derivative of κ is

$$\frac{\kappa'(t)}{\kappa(t)} = -\frac{1}{2} - \frac{2e^{-2t}}{1 - e^{-2t}} < 0. \quad (\text{CP20:G1.4})$$

Consequently $H(x, y) > 0$. The complete prime part in this normalization is

$$-2 \sum_n b_n \Re \int_0^\infty \overline{f(x)} f(x + \ell_n) dx + \sum_n b_n \Re \int_0^{\ell_n} \overline{f(x)} f(\ell_n - x) dx. \quad (\text{CP20:G1.5})$$

This is the same ordinary/reflected sign pair as CP19's paired-prime expression, before its theta-derived diagonal potential is introduced.

For smooth compactly supported half-line tests, put

$$d_f(x, y) = |f(x)||f(y)| - \Re(\overline{f(x)}f(y)) \geq 0. \quad (\text{CP20:G1.6})$$

Direct subtraction, without estimating any arithmetic term, gives

$$\begin{aligned} \tilde{Q}_a(|f|) - \tilde{Q}_a(f) = & - \int_0^a \int_0^a H(x, y) d_f(x, y) dx dy \\ & - 2 \sum_n b_n \int_0^\infty d_f(x, x + \ell_n) dx \\ & + \sum_n b_n \int_0^{\ell_n} d_f(x, \ell_n - x) dx. \end{aligned} \quad (\text{CP20:G1.7})$$

The formula includes the continuous gamma kernel, the full polar term, both orientations of every ordinary prime shift, and every reflected shift. The finite-part diagonal and every scalar diagonal term cancel exactly. The identity on this core already suffices for the counterexample below; no modulus-domain assertion for an arbitrary nonsmooth forced column is needed for that counterexample.

1.2. An explicit modulus counterexample for every first-prime-active interval

Let $h = \log 2$ and fix **any** $a > h/2$. Choose

$$0 < d < \min\{h/4, a - h/2\}, \quad x_0 = h/2 - d, \quad y_0 = h/2 + d. \quad (\text{CP20:G1.8})$$

Take a nonnegative $\psi \in C_c^\infty(-1, 1)$ with $\|\psi\|_2 = 1$, and define

$$u_\delta(x) = \delta^{-1/2} \psi((x - x_0)/\delta), \quad v_\delta(x) = u_\delta(h - x), \quad f_\delta = u_\delta - v_\delta. \quad (\text{CP20:G1.9})$$

For sufficiently small δ , both supports lie strictly inside $(0, a)$, are disjoint, and all their cross-differences have magnitude less than $\log 2$. All ordinary-prime contributions to (G1.7) therefore vanish. Their cross-sums lie near $\log 2$; choosing $2\delta < \log 3 - \log 2$ leaves exactly the prime 2 reflected contribution. Same-sign interactions within either bump have $d_f = 0$.

Since $|f_\delta| = u_\delta + v_\delta$ and $\int u_\delta(x) v_\delta(h - x) dx = 1$, (G1.7) becomes the **exact** identity

$$\tilde{Q}_a(|f_\delta|) - \tilde{Q}_a(f_\delta) = 4b_2 - 4 \iint H(x, y) u_\delta(x) v_\delta(y) dx dy. \quad (\text{CP20:G1.10})$$

Fix disjoint closed neighborhoods of x_0, y_0 . On their product H is continuous and has a finite upper bound M . Therefore

$$0 < \iint H u_\delta v_\delta \leq M \delta \|\psi\|_1^2. \quad (\text{CP20:G1.11})$$

In particular, $\delta < b_2/(2M\|\psi\|_1^2)$, together with the support conditions above, gives

$$\tilde{Q}_a(|f_\delta|) - \tilde{Q}_a(f_\delta) > 2b_2 > 0. \quad (\text{CP20:G1.12})$$

Thus an unconditional replacement of an odd test by a one-sign half-line test can **increase** its actual complete Weil energy. This failure occurs on every interval beyond the first prime threshold, including intervals where the complete form is known to be positive. It does not exhibit a negative Weil test.

There is also an operator consequence. If $\tilde{A}_a = U^* A_a^- U$, then

$$\langle u_\delta, \tilde{A}_a v_\delta \rangle = b_2 - \iint H u_\delta v_\delta > 0. \quad (\text{CP20:G1.13})$$

The smooth bumps belong to the operator domain. Hence

$$\left. \frac{d}{dt} \langle u_\delta, e^{-t\tilde{A}_a} v_\delta \rangle \right|_{t=0} = -\langle u_\delta, \tilde{A}_a v_\delta \rangle < 0, \quad (\text{CP20:G1.14})$$

whereas the value at $t = 0$ is zero. The semigroup is therefore not positivity preserving on $L^2(0, a)$. This establishes the failure directly, without importing a semigroup criterion.

This result does **not** prove that a hypothetical first-null vector changes sign. It rules out deriving a one-sign null vector by the usual modulus or positive-semigroup argument for this operator.

1.3. A finite phase obstruction on the same operator

Choose a third nonnegative narrow bump w , disjoint from the first two, whose center has no sum or difference equal to an active $\log n$ with either existing center. There are finitely many forbidden centers; such a choice exists. Shrinking its support and the first two supports retains

$$\langle u, \tilde{A}_a v \rangle > 0, \quad \langle v, \tilde{A}_a w \rangle < 0, \quad \langle w, \tilde{A}_a u \rangle < 0. \quad (\text{CP20:G1.15})$$

The last two entries are exactly the negative continuous integrals from (G1.3). Their three-edge product is strictly positive. A diagonal unitary congruence preserves that product; three nonpositive real, nonzero entries would have a negative product. Therefore no assignment of one phase to each of these three physical bump channels makes all their off-diagonal entries nonpositive. This is a specific obstruction to removing the reflected-prime sign by a finite color/angle gauge.

1.4. An actual four-vertex reflection path

Now put $h = \log 2$, $k = \log 3$, $b = b_2$, $c = b_3$, and assume $a > k/2$. On a small interval I around a point just below $k/2$, define

$$z_0(x) = h - x, \quad z_1(x) = x, \quad z_2(x) = k - x, \quad z_3(x) = h - k + x. \quad (\text{CP20:G1.16})$$

Choose I so that its four images are disjoint and lie in $(0, a)$. This is possible because $k < 2h$. Its reflected edges are

$$z_0 + z_1 = h, \quad z_1 + z_2 = k, \quad z_2 + z_3 = h. \quad (\text{CP20:G1.17})$$

For $F_f(x) = (f(z_0(x)), f(z_1(x)), f(z_2(x)), f(z_3(x)))^T$, the complete contribution of these three un-oriented reflected edges, including their two orientations, is

$$\mathcal{P}_I(f) = \int_I F_f(x)^* J F_f(x) dx, \quad J = \begin{pmatrix} 0 & b & 0 & 0 \\ b & 0 & c & 0 \\ 0 & c & 0 & b \\ 0 & 0 & b & 0 \end{pmatrix}. \quad (\text{CP20:G1.18})$$

Every other reflected edge, every ordinary shift, and every diagonal theta potential remains outside $\mathcal{P}_I$; none is assigned a sign here.

1.5. Exact payment by the old positive theta board

Use a separate parameter $\lambda > 0$ for the old coupled theta kernel C_λ ; it is not the support width a . In the same complete theta normalization as CP19,

$$C_\lambda(y) = 2 \int_{\mathbb{R}} u \sinh(2\lambda u) \phi(u+y) \phi(u-y) du.$$

The recovered four-state theorem applies to the logarithmic nodes associated with $\{1, 2, 3, 6\}$:

$$0, h/2, k/2, (h+k)/2. \quad (\text{CP20:G1.19})$$

Write

$$C = C_\lambda(0), \quad A = C_\lambda(h/2), \quad B = C_\lambda(k/2), \quad D = C_\lambda((h+k)/2), \quad E = C_\lambda((k-h)/2). \quad (\text{CP20:G1.20})$$

Its **complete** positive matrix, with every mixed entry, is

$$G = \begin{pmatrix} C & A & B & D \\ A & C & E & B \\ B & E & C & A \\ D & B & A & C \end{pmatrix} \succ 0. \quad (\text{CP20:G1.21})$$

Using this matrix on the physical feature vector (G1.16) is an actual finite-dimensional inequality; it does not identify the spatial theta nodes with the physical reflection locations.

The orthonormal reversal-parity change of basis simultaneously splits these two matrices as

$$G_\pm = \begin{pmatrix} C \pm D & A \pm B \\ A \pm B & C \pm E \end{pmatrix}, \quad J_\pm = \begin{pmatrix} 0 & b \\ b & \pm c \end{pmatrix}. \quad (\text{CP20:G1.22})$$

Set $u_\pm = C \pm D$, $v_\pm = A \pm B$, $w_\pm = C \pm E$, and $\Delta_\pm = u_\pm w_\pm - v_\pm^2 > 0$. The exact optimal scalar payment is

$$t_* = \max\{t_+, t_-\}, \quad t_\pm = \frac{2bv_\pm \mp cu_\pm + \sqrt{(\pm cu_\pm - 2bv_\pm)^2 + 4\Delta_\pm b^2}}{2\Delta_\pm}. \quad (\text{CP20:G1.23})$$

Indeed, the determinant of $J_\pm + tG_\pm$ equals

$$\Delta_\pm t^2 + (\pm cu_\pm - 2bv_\pm)t - b^2. \quad (\text{CP20:G1.24})$$

For $t > 0$, its first diagonal entry is positive, so nonnegative determinant is equivalent to positive semidefiniteness. Thus

$$J + tG \succeq 0 \iff t \geq t_*. \quad (\text{CP20:G1.25})$$

The independently expanded curvature bound $V'' > 449/25$ in the old review supplies a dimension-free numerical control on this payment, in the same variable as CP19's ϕ . It gives $C_\lambda(y) \leq Ce^{-(449/25)y^2}$. Its row estimate gives

$$G \succeq C(1 - \rho)I, \quad \rho = e^{-\frac{449}{100}h^2} + e^{-\frac{449}{100}k^2} + e^{-\frac{449}{100}(k-h)^2} < \frac{3}{5}. \quad (\text{CP20:G1.26})$$

The last strict bound follows, for example, from $h > 693/1000$, $k > 549/500$, $k - h > 81/200$, and the rational exponential bounds

$$e^{-\frac{449}{100}(693/1000)^2} < \frac{29}{250}, \quad e^{-\frac{449}{100}(549/500)^2} < \frac{9}{2000}, \quad e^{-\frac{449}{100}(81/200)^2} < \frac{479}{1000}. \quad (\text{CP20:G1.27})$$

Their sum is $1199/2000 < 3/5$; the respective positive Taylor sums for e^x through orders 7, 12 and 5 verify them by rational arithmetic. Hence, with

$$L = \frac{c + \sqrt{c^2 + 4b^2}}{2} = \|J\|, \quad (\text{CP20:G1.28})$$

one has the proved bound

$$\boxed{t_* < \frac{5L}{2C}}. \quad (\text{CP20:G1.29})$$

There is an informative lower bound too. Strict convexity of $-\log \phi$ makes $C_\lambda(y)$ strictly decreasing for $y > 0$: differentiate $\phi(u+y)\phi(u-y)$ before integrating against its nonnegative defining weight. Thus $A > B > 0$. The negative eigenvector $v = (1, -L/b)^T$ of J_- satisfies

$$v^* G_- v = C\|v\|^2 - D - E(L/b)^2 - 2(A-B)L/b < C\|v\|^2. \quad (\text{CP20:G1.30})$$

Consequently

$$\boxed{t_* \geq \frac{L\|v\|^2}{v^* G_- v} > \frac{L}{C}}. \quad (\text{CP20:G1.31})$$

The old theta board is positive, but its weight is smaller than the diagonal weight precisely on this alternating adverse path direction. It does not remove that direction for free. In particular $J - tG$ has a negative direction for every $t \geq 0$.

1.6. Full transfer to the physical forced matrix, with the exact remainder

For the actual CP19 columns $q_j = W_a e_j$, put $f_j = \sqrt{2}q_j|_{(0,a)}$ and let $\mathcal{F}(x)$ be the 4-by- d_R matrix with entries $\mathcal{F}_{rj}(x) = f_j(z_r(x))$. Define

$$\mathbf{J}_I = \int_I \mathcal{F}^* J \mathcal{F} dx, \quad \mathbf{G}_I = \int_I \mathcal{F}^* G \mathcal{F} dx. \quad (\text{CP20:G1.32})$$

These are bounded coefficient forms for the actual L^2 columns; pointwise traces are unnecessary. Disjointness of the image intervals gives

$$\int_I \mathcal{F}^* \mathcal{F} dx \preceq W_a^* W_a. \quad (\text{CP20:G1.33})$$

The optimized old-board inequality now gives

$$\boxed{\mathbf{J}_I + t_* \mathbf{G}_I \succeq 0}. \quad (\text{CP20:G1.34})$$

Let the complete CP19 paired-prime matrix be $\mathbf{P}_a = \sum_n b_n \mathbf{P}_{a,\ell_n}$, and define its exact remainder

$$\mathbf{R}_I := \mathbf{P}_a - \mathbf{J}_I. \quad (\text{CP20:G1.35})$$

It contains all the CP19 diagonal potentials, including the infinite signed exterior tail; all ordinary prime shifts; and every reflected edge not included in (G1.18). The full forced identity is therefore

$$\boxed{\tau_R^{-1}K_a(I - K_a) = [\eta_p(a)W_a^*W_a + \mathbf{N}_a - t_*\mathbf{G}_I] + [\mathbf{J}_I + t_*\mathbf{G}_I] + \mathbf{R}_I.} \quad (\text{CP20:G1.36})$$

The second bracket is independently nonnegative, with the exact optimal constant and physical coefficient map now calculated. The first bracket and $\mathbf{R}_I$ have not been proved jointly nonnegative. A successful use of this old positive board must pay $t_*\mathbf{G}_I$ from the continuous energy while retaining $\mathbf{R}_I$. No positive term has been counted twice, and the original Suzuki metric has not been replaced.

Group 1 closing statement

The complete modulus calculation and narrow-bump construction prove that the actual odd operator loses the positivity-preserving mechanism at the first active prime. This closes that particular proposed shortcut, without excluding another proof that a first-null vector has one sign. The historical four-state theta theorem does transfer to an actual reflected-prime path. Its optimal payment, its explicit uniform upper bound, its strict lower cost on the alternating direction, and the entire forced-matrix remainder are now calculated. The remaining sign is the joint comparison in (G1.36); it is not supplied by positivity of the four-state matrix alone.

Sources used: CP19 (0.5), (0.8), (1.20), and its complete paired-prime identity; Michaels_H_Analytical_Review §§3 and 6 for the original theta curvature and four-state matrix; Michaels_Complete_Theta_Kernel_Continuation for the earlier one-sign odd-cone theorem and its actual scope.

2. Full covariance representation and the common comparison operator

The following results use CP19's proved closed odd form domain and its actual normalization. They do not assume Weil positivity or RH. Write $f = \sqrt{2}q|_{(0,a)}$, extend f by zero to (a, ∞) , and put $p = -\phi' > 0$. The actual forced columns are $f_j = \sqrt{2}(W_a e_j)|_{(0,a)}$; all scalar identities below polarize to their complete matrices.

2.1. Exact positive covariance representation

Let

$$H(x, y) = \kappa(|x - y|) - \kappa(x + y) + 4 \sinh(x/2) \sinh(y/2), \quad x, y > 0. \quad (\text{CP20:G2.1})$$

The complete continuous form in CP19 is

$$\mathcal{C}_a(f) = \frac{1}{2} \iint_{(0,\infty)^2} H(x, y) p(x) p(y) \left| \frac{f(x)}{p(x)} - \frac{f(y)}{p(y)} \right|^2 dx dy. \quad (\text{CP20:G2.2})$$

For $c_j = 2j + 1/2$, the elementary geometric expansion of κ gives

$$H(x, y) = e^{(x+y)/2} (1 - e^{-\min(x,y)}) + \sum_{j \geq 1} [e^{-c_j|x-y|} - e^{-c_j(x+y)}]. \quad (\text{CP20:G2.3})$$

Indeed the $j = 0$ kernel and the entire polar kernel combine exactly:

$$e^{-|x-y|/2} - e^{-(x+y)/2} + 4 \sinh(x/2) \sinh(y/2) = e^{(x+y)/2} (1 - e^{-\min(x,y)}). \quad (\text{CP20:G2.4})$$

These are nested positive rank-one kernels:

$$e^{(x+y)/2}(1 - e^{-\min(x,y)}) = \int_0^\infty e^{-t} [e^{x/2} \mathbf{1}_{x>t}] [e^{y/2} \mathbf{1}_{y>t}] dt, \quad (\text{CP20:G2.5})$$

$$e^{-c|x-y|} - e^{-c(x+y)} = \int_0^\infty 2ce^{2ct} [e^{-cx} \mathbf{1}_{x>t}] [e^{-cy} \mathbf{1}_{y>t}] dt. \quad (\text{CP20:G2.6})$$

For a positive weight w , put

$$M_w(t) = \int_t^\infty wp, \quad V_w(t; f) = M_w(t) \int_t^a \frac{w}{p} |f|^2 - \left| \int_t^a wf \right|^2. \quad (\text{CP20:G2.7})$$

Then

$$\mathcal{C}_a(f) = \int_0^a e^{-t} V_{e^{x/2}}(t; f) dt + \sum_{j \geq 1} \int_0^a 2c_j e^{2c_j t} V_{e^{-c_j x}}(t; f) dt. \quad (\text{CP20:G2.8})$$

Every integrand is nonnegative. In particular the term $M_w(t)$ includes its mass on (a, ∞) ; replacing its upper limit by a loses actual positive energy.

For the row $F = (f_1, \dots, f_d)$, the exact matrix integrand is

$$\mathbf{V}_w(t) = M_w(t) \int_t^a \frac{w}{p} F^* F - \left(\int_t^a w F \right)^* \left(\int_t^a w F \right) \succeq 0. \quad (\text{CP20:G2.9})$$

Formula (G2.8) with these integrands is precisely the matrix proposed in the CP20 prompt, with no missing factor of two.

Domain proof. First prove (G2.8) on the smooth odd core by the displayed kernel identities and Tonelli, applied to the nonnegative squared difference. For a general CP19 form vector, the double integral (G2.2) is finite by CP19 (2.92) and the bounded polar form. Since $p > 0$ almost everywhere, f/p is a well-defined measurable function. Apply Tonelli directly to its nonnegative difference integrand and the positive expansion (G2.3). Truncate the channel sum and the t -integrals away from zero; all truncated identities involve finite nonnegative quantities. Monotone convergence gives (G2.8). At a positive t , every integral in (G2.7) is finite, so expansion of the square is legitimate. The endpoint $t = 0$ is immaterial. For matrix convergence apply the scalar result to the finite family, use polarization, and control off-diagonal tails by Cauchy-Schwarz in the positive tail matrix. Thus there is no imposed trace at zero or at a , and no assumption $f/x \in L^2$.

2.2. The complete exterior reserve, before replacing it by a constant

Split (G2.2) into its interior square and its exterior pairs. This gives the exact identity

$$\mathcal{C}_a(f) = \mathcal{C}_a^{\text{int}}(f) + \int_0^a r_a(x) |f(x)|^2 dx, \quad r_a(x) = \frac{1}{p(x)} \int_a^\infty H(x, y) p(y) dy. \quad (\text{CP20:G2.10})$$

Both terms are nonnegative. Define $I_c(a) = \int_a^\infty e^{-cy} p(y) dy$ and $I_+(a) = \int_a^\infty e^{y/2} p(y) dy$. Then

$$r_a(x) = \frac{2 \sinh(x/2)}{p(x)} I_+(a) + \frac{2}{p(x)} \sum_{j \geq 1} \sinh(c_j x) I_{c_j}(a). \quad (\text{CP20:G2.11})$$

It has the finite positive origin limit

$$r_a(0) = \frac{I_+(a) + 2 \sum_{j \geq 1} c_j I_{c_j}(a)}{p'(0)}. \quad (\text{CP20:G2.12})$$

The sums converge uniformly on compact subsets of $[0, a)$. Moreover

$$r_a(x) \geq \eta_p(a) > 0, \quad \mu_a := \operatorname{ess\,inf}_{0 < x < a} r_a(x) \geq \eta_p(a). \quad (\text{CP20:G2.13})$$

Thus the exact reserve is an explicit multiplication operator R_a , retaining its spatial dependence and every higher gamma channel. It is part of the continuous energy already counted, not an additional independent allowance.

At the outer boundary, continuity of p and $\kappa(t) \sim 1/(2t)$ give

$$r_a(x) = \frac{1}{2} \log \frac{1}{a-x} + O_a(1) \quad (x \uparrow a). \quad (\text{CP20:G2.14})$$

This follows by writing $p(y)/p(x) = 1 + O_a(|y-x|)$ in a fixed neighborhood of a ; the error cancels the kernel's $1/(y-x)$ singularity and has a bounded integral. Hence R_a also retains the endpoint logarithmic energy, instead of discarding it in a scalar floor.

2.3. Common conditional variance and an explicit inverse

The covariance representation gives more than the exterior reserve. Fix $0 < t_0 < a$ and choose one actual channel:

$$(w, \chi) = (e^{x/2}, 1 - e^{-t_0}), \quad \text{or} \quad (w, \chi) = (e^{-cx}, e^{2ct_0} - 1), \quad c = c_j, j \geq 1. \quad (\text{CP20:G2.15})$$

Set $I = (t_0, a)$, $m = \int_I wp$, and $u = w \mathbf{1}_I$. Interior nested covariance is decreasing as its lower endpoint increases: its nonnegative double-difference integral on $(t, a)^2$ contains the one on $(t_0, a)^2$ whenever $t \leq t_0$. Therefore

$$\mathcal{C}_a^{\text{int}}(f) \geq \chi \left[m \int_I \frac{w}{p} |f|^2 - \left| \int_I wf \right|^2 \right]. \quad (\text{CP20:G2.16})$$

Retaining the *complete* exterior reserve gives a proved common lower operator

$$\boxed{\mathcal{C}_a \succeq A_{a,t_0,w} := D - \chi |u\rangle \langle u| \succeq R_a \succeq \eta_p(a) I}, \quad (\text{CP20:G2.17})$$

where

$$d(x) = r_a(x) + \chi m \frac{w(x)}{p(x)} \mathbf{1}_I(x), \quad D = M_d. \quad (\text{CP20:G2.18})$$

All matrix statements mean quadratic form order. Define

$$J = \int_I \frac{w(x)^2}{d(x)} dx, \quad z(x) = \frac{u(x)}{d(x)}, \quad \delta = 1 - \chi J. \quad (\text{CP20:G2.19})$$

The denominator is strictly positive independently of any Weil margin. Indeed, with $L = \max_{x \in [t_0, a]} w(x)/p(x) < \infty$,

$$\chi J = \int_I \frac{\chi m w/p}{r_a + \chi m w/p} \frac{wp}{m} \leq \frac{\chi m L}{\eta_p(a) + \chi m L}, \quad \boxed{\delta \geq \frac{\eta_p(a)}{\eta_p(a) + \chi m L} > 0}. \quad (\text{CP20:G2.20})$$

The exact inverse is consequently

$$A_{a,t_0,w}^{-1} = M_{1/d} + \frac{\chi}{\delta} |z\rangle\langle z|. \quad (\text{CP20:G2.21})$$

To verify it, solve $Df - \chi u\langle u, f\rangle = g$, take its pairing with u , and substitute. This also proves the formula on the natural form domain of the possibly unbounded multiplier r_a . Its inverse is bounded because $r_a \geq \eta_p(a)$.

This is a calculable comparison using the *same* covariance and all forced columns. It retains the common conditional mean exactly as the rank-one correction in (G2.21). Assigning separate independent means to different prime reflections would lose that correlation.

Several different channels and thresholds can be used at once. For any finite set of disjoint allocations within the positive channel integrals, (G2.17) becomes $D - UCU^*$, with the corresponding finite common-mean matrix C . Its exact inverse is obtained by solving the finite matrix $C^{-1} - U^*D^{-1}U$; the positive reserve proves that matrix positive. No energy may be allocated twice.

2.4. A concrete quantitative payment of the complete adverse paired-prime operator

Use the bounded, origin-cancelled splitting in CP20 Group 3. Its adverse operator is

$$B = M_v + \sum_{\ell_n < 2a} b_n \Pi_{\ell_n}^-, \quad v = (\omega_a)_- \geq 0, \quad (\text{CP20:G2.22})$$

where Π_{ℓ}^- is the antisymmetric projection on $I_{\ell} = (\max(0, \ell - a), \min(a, \ell))$, extended by zero off that interval. The complete exterior prime potential is included in v . The coefficients and the potential are bounded on every fixed support by CP19's paired cancellation and exterior tail theorem.

The exact common comparison to the explicitly proved lower operator is

$$\beta_{a,t_0,w} = \|B^{1/2} A_{a,t_0,w}^{-1} B^{1/2}\|, \quad B \preceq \beta_{a,t_0,w} \mathcal{C}_a. \quad (\text{CP20:G2.23})$$

Here (G2.21) gives an explicit upper estimate, avoiding an unevaluated inverse. For $x \in I_{\ell}$, write $x_{\ell} = \ell - x$. Weighted Schur gives

$$\sigma_d = \text{ess sup}_{0 < x < a} \left\{ \frac{v(x)}{d(x)} + \frac{1}{2} \sum_{\ell_n < 2a} b_n \mathbf{1}_{I_{\ell_n}}(x) \left[\frac{1}{d(x)} + \frac{1}{\sqrt{d(x)d(x_{\ell_n})}} \right] \right\}. \quad (\text{CP20:G2.24})$$

Consequently

$$\beta_{a,t_0,w} \leq \sigma_d + \frac{\chi}{\delta} \left[\int_0^a v|z|^2 + \frac{1}{4} \sum_{\ell_n < 2a} b_n \int_{I_{\ell_n}} |z(x) - z(\ell_n - x)|^2 dx \right]. \quad (\text{CP20:G2.25})$$

Every term in this coefficient is now a one-dimensional integral, a finite prime sum, or a pointwise supremum of specified functions. It is independent of the unknown positivity margin. The final term retains the actual reflected differences of the *common* representer z . This is a genuine bound on the complete adverse operator, although it is not asserted to be below one on arbitrary supports.

Proof of the payment. The weighted operator $D^{-1/2}BD^{-1/2}$ has the displayed diagonal and signed reflected kernels. For each reflected kernel use its absolute value in the Schur test with

weight one. This yields (G2.24). The nonzero spectra of $B^{1/2}D^{-1}B^{1/2}$ and $D^{-1/2}BD^{-1/2}$ agree. The other term in (G2.21), after congruence by $B^{1/2}$, has norm $(\chi/\delta)\langle z, Bz \rangle$, which is the second term in (G2.25), since $\|\Pi_\ell^- z\|^2 = \frac{1}{4} \int_{I_\ell} |z - z \circ R_\ell|^2$. This proves (G2.25) and then (G2.23).

A simpler exact norm for a single reflected adverse projection, paid by exterior reserve alone, is

$$\|R_a^{-1/2}\Pi_\ell^- R_a^{-1/2}\| = \frac{1}{2} \operatorname{ess\,sup}_{x \in I_\ell} \left[\frac{1}{r_a(x)} + \frac{1}{r_a(\ell - x)} \right]. \quad (\text{CP20:G2.26})$$

This follows from the two-point fibers of the reflection: the resulting 2×2 matrix has rank one and its nonzero eigenvalue is the displayed half-sum. The equality therefore pays the whole antisymmetric sector of that prime, including arbitrary complex coefficients. It is sharper than replacing both weights by $\eta_p(a)$.

For multiple reflected channels one should use the actual joint weighted operator in (G2.23), or the finite-rank conditional-mean correction, before applying the Schur upper bound. The Group 3 coupled-reflection savings can improve the first term of (G2.25); they do not alter the exact common representer term.

2.5. What has and has not closed

Established here: the exact full covariance identity; its actual closed-domain passage; the complete spatial exterior reserve including the boundary logarithm; a common conditional-variance lower operator with an explicit strictly positive inverse denominator; and the quantitative adverse-prime payment (G2.25), retaining the complete potential and common reflected mean correction. These hold for the complete actual forced matrix by polarization and for both null branches.

If the right side of (G2.25), or a sharper joint-operator evaluation of (G2.23), is at most one for every required support, then $\mathcal{C}_a - B \geq 0$, and the additional positive part of the paired-prime splitting remains available. If it is strictly below one on a hypothetical first-null direction, it gives the requested contradiction. The calculation above does not prove that all-support numerical inequality. Its coefficients have not been replaced by an asserted unit budget.

The cruder exterior-only payment was also evaluated diagnostically during this run. No exploratory quadrature is used as a positivity certificate. The independent exact-rational certificate in Group 4 establishes the enlarged support interval.

3. Exact reflection graphs and complete signed prime bounds

This fragment executes a new comparison in the signed energy of CP19. The results below are unconditional identities and operator inequalities for the actual paired-prime form. They do not assume Weil positivity or identify the historical Green metric with the Suzuki metric. They retain the ordinary translations, the complete origin-cancelled prime potential, and every exterior prime term. The final comparison to the continuous energy remains explicitly visible.

3.1. Fixed normalization and the reflected operator actually being estimated

Use CP19's Fourier convention and odd unitary restriction

$$f = \sqrt{2}q|_{(0,a)}, \quad \|f\|_{L^2(0,a)} = \|q\|_{L^2(-a,a)}. \quad (\text{CP20:G3.1})$$

All functions are extended by zero. Write $\ell_n = \log n$, $b_n = \Lambda(n)/\sqrt{n}$, and retain CP19's positive odd weight $p = -\phi'$ on the positive half-line. Its complete potential is

$$V_p(x) = \sum_{n \geq 2} b_n D_{\ell_n}(x), \quad D_\ell(x) = \frac{p_o(x+\ell) + p_o(x-\ell)}{p(x)}, \quad D_\ell(0) = \frac{2p'(\ell)}{p'(0)}. \quad (\text{CP20:G3.2})$$

This is a bounded continuous function on each fixed $[0, a]$, using CP19's origin cancellation and its norm-summable exterior tail. In particular, the sum in (G3.2) is not truncated at $\ell_n = 2a$.

Define the compressed reflection on $L^2(0, a)$ by

$$(R_{\ell,a}f)(x) = \mathbf{1}_{0 < \ell-x < a} f(\ell-x), \quad 0 < x < a. \quad (\text{CP20:G3.3})$$

It is bounded and self-adjoint. It vanishes for $\ell \geq 2a$, up to null sets, and has norm one when $0 < \ell < 2a$. Its nonzero part is an involution on $I_{\ell,a} = (\max(0, \ell-a), \min(a, \ell))$. The reflection pairing is real even for complex f , and

$$\langle f, R_{\ell,a}f \rangle = \int_0^\ell \overline{f(x)} f(\ell-x) dx. \quad (\text{CP20:G3.4})$$

Consequently CP19's complete prime form is exactly

$$\mathcal{P}_a(f) = \int_0^a V_p(x) |f(x)|^2 dx - 2 \sum_{\ell_n < a} b_n \Re \int_0^{a-\ell_n} \overline{f(x)} f(x+\ell_n) dx + \langle f, \mathcal{R}_a f \rangle, \quad \mathcal{R}_a = \sum_{\ell_n < 2a} b_n R_{\ell_n,a}. \quad (\text{CP20:G3.5})$$

All new estimates below act on the last, actual signed term of (G3.5). They do not split the individually singular $h = f/p$ ordinary and reflected prime energies. The integrals and operators in (G3.5) are bounded on $L^2(0, a)$.

3.2. Exact two-prime reflection spectrum for every finite support

Put

$$\ell = \log 2, \quad m = \log 3, \quad d = m - \ell, \quad u = \frac{\log 2}{\sqrt{2}}, \quad v = \frac{\log 3}{\sqrt{3}}. \quad (\text{CP20:G3.6})$$

Here $3\ell/2 < m < 2\ell$, because $8 < 9 < 16$. Also $v > u$: indeed $m/\ell > 3/2 > \sqrt{3}/2$. The weights are fixed arithmetic constants.

Theorem G3.A (complete two-reflection norm). Define

$$L_{uvu} = \frac{v + \sqrt{v^2 + 4u^2}}{2}, \quad L_{vuv} = \frac{u + \sqrt{u^2 + 4v^2}}{2}, \quad L_5 = \sqrt{u^2 + v^2 + uv}, \quad (\text{CP20:G3.7})$$

and let L_6 be the largest eigenvalue of

$$J_6 = \begin{pmatrix} 0 & v & 0 \\ v & 0 & u \\ 0 & u & v \end{pmatrix}, \quad \text{equivalently the largest root of } X^3 - vX^2 - (u^2 + v^2)X + v^3 = 0. \quad (\text{CP20:G3.8})$$

Then the exact norm and the negative spectral endpoint of $uR_{\ell,a} + vR_{m,a}$ are $L_{23}(a)$ and $-L_{23}(a)$, respectively, where

Support	$L_{23}(a)$
$0 < a \leq \ell/2$	0
$\ell/2 < a \leq m/2$	u
$m/2 < a \leq m - \ell/2$	L_{uvu}
$m - \ell/2 < a \leq 2(m - \ell)$	L_{vuv}
$2(m - \ell) < a \leq (3m - 2\ell)/2$	L_5
$a > (3m - 2\ell)/2$	L_6

Thus, for every complex $f \in L^2(0, a)$,

$$-L_{23}(a)\|f\|^2 \leq u\langle f, R_{\ell,a}f \rangle + v\langle f, R_{m,a}f \rangle \leq L_{23}(a)\|f\|^2. \quad (\text{CP20:G3.9})$$

Proof, including the actual infinite-dimensional decomposition. Both reflections vanish outside $(0, m)$, so first work on $L^2(0, m)$. Outside a finite set of fixed points, the reflection graph has degree at most two, no cycles, and endpoints in (ℓ, m) . Starting from an endpoint x , its successive vertices are

$$x, m - x, x - d, m - x + d, x - 2d, m - x + 2d, \dots \quad (\text{CP20:G3.10})$$

The number of vertices is $2\lceil x/d \rceil$. Since $2 < m/d < 3$ and $x \in (\ell, m)$, the full graph consists only of four-node paths with weights v, u, v , and six-node paths with weights v, u, v, u, v .

There is an explicit unitary direct-integral parametrization, so the argument does not rely on a discrete sampling of the continuum. Parameterize the four-node paths by

$$x \in (\ell, m - \ell/2), \quad (x, m - x, x - d, m + d - x), \quad (\text{CP20:G3.11})$$

and the six-node paths by

$$x \in (2d, (m + 2d)/2), \quad (x, m - x, x - d, m - x + d, x - 2d, m - x + 2d). \quad (\text{CP20:G3.12})$$

Every coordinate map has Jacobian of absolute value one. Their coordinate intervals are pairwise disjoint and together cover $(0, m)$, except for endpoints. Consequently the sum of the fiber Euclidean norms integrates to the original L^2 norm. In these coordinates the operator is exactly the weighted path matrix on each fiber. For support $(0, a)$, multiply each coordinate by the indicator that its displayed location is below a . This is precisely compression to principal submatrices; disconnected pieces are retained.

A path with two edge weights u, v has norm $\sqrt{u^2 + v^2}$. The four-node path norms are (G3.7), the five-node alternating path has norm L_5 , and the six-node path has norm L_6 . The last assertion follows by splitting the six-node path into its symmetric and antisymmetric mirror subspaces; its positive Perron eigenvector is symmetric and its reduced matrix is (G3.8). All paths are bipartite, so their spectra are symmetric. Complex scalars have the same Hermitian spectral bounds.

The smallest support heights permitting positive-measure families of the successive maximal components are

$$\ell/2, \quad m/2, \quad m - \ell/2, \quad 2d, \quad (m + 2d)/2. \quad (\text{CP20:G3.13})$$

For example, the middle four vertices of (G3.12) have minimum possible largest location $m/2$; all four vertices of (G3.11) have minimum largest location $m - \ell/2$. Five vertices of a six-node path first fit when $a > 2d$, by taking the parameter in (G3.12) just above $2d$. All six first fit when $a > (m + 2d)/2$, by taking that parameter close to its upper endpoint. At equality the new

component occurs on a null set only. This explains every strict threshold in the table. The lower-support components have smaller norm: $L_{uvu} > \sqrt{u^2 + v^2} > v$, $L_{vuv} > L_{uvu}$ because $v > u$, and strict principal-submatrix interlacing gives $L_5 > L_{vuv}$ and $L_6 > L_5$. A constant fiber eigenvector multiplied by any nonzero scalar function supported in a positive-measure family of maximal fibers realizes each claimed endpoint. This proves the exact table and (G3.9).

For reference only, rounded values are $u \approx 0.490129$, $v \approx 0.634284$, $L_{uvu} \approx 0.900928$, $L_{vuv} \approx 0.925045$, $L_5 \approx 0.976434$, and $L_6 \approx 1.019080$. The proof and every inequality use the exact expressions above, not these decimals.

The global bound has a fully explicit strict saving:

$$L_{23}(a) \leq L_6 \leq u + v - \frac{uv}{2(u + 2v)} < u + v. \quad (\text{CP20:G3.14})$$

Indeed the cubic in (G3.8) has value u^2v at $u + v$. Its derivative on $[L_6, u + v]$ is at most $2u(u + 2v)$; here $L_6 > v$, so the derivative is increasing on that interval. The mean-value theorem gives (G3.14). Alternatively, strictness follows directly from the finite irreducible path's endpoint loss. This support-independent saving is a bound on the selected two reflected channels, and leaves all other channels unchanged.

3.3. A calculated complete signed bound on the next prime interval

On

$$\frac{\log 3}{2} < a \leq \log 2, \quad (\text{CP20:G3.15})$$

the only active prime cross terms in (G3.5) are reflections for $n = 2, 3$. Every ordinary cross vanishes, because $a \leq \log 2$. The $n = 4$ reflection is still zero at the upper endpoint, since its overlap then has measure zero. Thus Theorem G3.A gives the genuine complete prime inequality

$$\mathcal{P}_a(f) \geq \int_0^a V_p(x) |f(x)|^2 dx - L_{uvu} \|f\|^2. \quad (\text{CP20:G3.16})$$

Compared to separately bounding both reflections by their negative norms, the independently proved saving is exactly $u + v - L_{uvu} > 0$. The potential contains every prime power, including the negative exterior terms established in CP19. This is not a finite prime truncation of the transformed energy.

The positive remainder is explicit. If $\mathcal{F}_{23,a}$ denotes the measurable path transform in the proof above, extended by the identity on the zero-operator space $L^2(m, a)$ when $a > m$, and $H_{23,a}(x)$ the appropriate retained path matrix, then

$$\mathcal{J}_{23,a}(f) = \int \langle \mathcal{F}_{23,a}f(x), [L_{23}(a)I + H_{23,a}(x)] \mathcal{F}_{23,a}f(x) \rangle dx \geq 0, \quad (\text{CP20:G3.17})$$

including all zero components. Its exact identity is

$$u \langle f, R_{\ell,a}f \rangle + v \langle f, R_{m,a}f \rangle = -L_{23}(a) \|f\|^2 + \mathcal{J}_{23,a}(f). \quad (\text{CP20:G3.18})$$

For the actual forced columns, polarizing (G3.18) retains all coefficient cross terms and gives a positive matrix remainder. No endpoint values of a form-domain vector are used.

3.4. Every prime and its square: exact spectra and disjoint all-power allocation

There is an arithmetic pairing available for every prime and every exponent, with a simpler graph than the cross-prime pair.

Theorem G3.B (prime-square reflection). For any $L, u, v > 0$, the exact norm and negative spectral endpoint of $uR_{L,a} + vR_{2L,a}$ are $N_{L,u,v}(a)$ and its negative, where

$$N_{L,u,v}(a) = \begin{cases} 0, & 0 < a \leq L/2, \\ u, & L/2 < a \leq L, \\ \sqrt{u^2 + v^2}, & L < a \leq 3L/2, \\ \frac{u + \sqrt{u^2 + 4v^2}}{2}, & a > 3L/2. \end{cases} \quad (\text{CP20:G3.19})$$

The same statement holds for arbitrary complex functions in $L^2(0, a)$.

Proof. On the full active range $(0, 2L)$, use $x \in (L, 3L/2)$ and the four coordinates

$$(f(x), f(2L - x), f(x - L), f(3L - x)). \quad (\text{CP20:G3.20})$$

The four coordinate intervals partition $(0, 2L)$ and each has Jacobian one. The matrix is the four-node path with weights v, u, v . Compressing by the exact indicators of $(0, a)$ gives no edge for $a \leq L/2$, a two-node u -edge for $L/2 < a \leq L$, a three-node v, u -path for $L < a \leq 3L/2$, and positive-measure full four-node paths when $a > 3L/2$. These cases give (G3.19), including its equality thresholds. The symmetry of the path spectrum gives the lower endpoint too.

Apply this with

$$L = r \log p, \quad u_{p,r} = (\log p)p^{-r/2}, \quad v_{p,r} = (\log p)p^{-r}. \quad (\text{CP20:G3.21})$$

When both prime-power reflections are active, their saving over the separated negative norm is

$$\delta_{p,r}(a) = u_{p,r} + v_{p,r} - N_{L,u_{p,r},v_{p,r}}(a) > 0. \quad (\text{CP20:G3.22})$$

Uniformly on that active range it is at least

$$\delta_{p,r}(a) \geq \frac{2u_{p,r}v_{p,r}}{u_{p,r} + 2v_{p,r} + \sqrt{u_{p,r}^2 + 4v_{p,r}^2}} \geq \frac{u_{p,r}v_{p,r}}{u_{p,r} + 2v_{p,r}} > 0. \quad (\text{CP20:G3.23})$$

The first expression is exactly the saving once full four-node paths fit; its rationalization proves the equality. The last bound uses $\sqrt{u^2 + 4v^2} \leq u + 2v$.

To retain every prime power without spending a weight twice, set

$$\mathcal{D} = \{r \geq 1 : \text{the exponent of 2 in } r \text{ is even}\}. \quad (\text{CP20:G3.24})$$

The pairs $(r, 2r)$, $r \in \mathcal{D}$, partition the positive integers: for each fixed odd part d , pair $d \cdot 2^{2j}$ with $d \cdot 2^{2j+1}$. Therefore the exact complete reflection sum in (G3.5) is partitioned into these prime-square blocks, with a block's inactive channel interpreted as its actual zero compression.

Define the finite explicit sums

$$B_{\text{ref}}(a) = \sum_{k \log p < 2a} (\log p)p^{-k/2}, \quad S_{\text{sq}}(a) = \sum_{\substack{p \text{ prime}, \\ r \in \mathcal{D} \\ r \log p < a}} \delta_{p,r}(a). \quad (\text{CP20:G3.25})$$

Only pairs with both channels active contribute to the saving; strict inequalities handle thresholds exactly. Then

$$\langle f, \mathcal{R}_a f \rangle \geq -[B_{\text{ref}}(a) - S_{\text{sq}}(a)] \|f\|^2. \quad (\text{CP20:G3.26})$$

This is a proved all-support improvement of the separated reflected-prime charge. In particular,

$$S_{\text{sq}}(a) \geq \sum_{\substack{p \text{ prime}, r \in \mathcal{D} \\ r \log p < a}} \frac{(\log p) p^{-r}}{1 + 2p^{-r/2}}. \quad (\text{CP20:G3.27})$$

It requires only finite sums of actual prime powers and known constants. It contains no unknown positivity margin.

The complete nonnegative matrix remainder can be retained, rather than just the scalar bound. For every block with $L < 2a$, put

$$J_{p,r,a} = N_{L,u_{p,r},v_{p,r}}(a)I + u_{p,r}R_{L,a} + v_{p,r}R_{2L,a} \succeq 0. \quad (\text{CP20:G3.28})$$

Then

$$\mathcal{R}_a = -[B_{\text{ref}}(a) - S_{\text{sq}}(a)]I + \sum_{\substack{p, r \in \mathcal{D} \\ r \log p < 2a}} J_{p,r,a}. \quad (\text{CP20:G3.29})$$

This is an exact identity, including the channels for which the upper power has not activated. Both (G3.26) and (G3.29) hold through every threshold. Their scalar norm envelopes can jump at a threshold; no continuity of those envelopes is asserted. This does not conflict with continuity of the physical forced matrices.

The cross-prime (2,3) improvement and this prime-square allocation are independent valid alternatives. They cannot both use the full prime-2 coefficient simultaneously. A joint optimization must allocate each actual coefficient among its chosen blocks with nonnegative allocations whose sum is no greater than that coefficient, and retain the unallocated operator. Each allocated block has the same proved norm formula at its allocated weights; summing then gives a rigorous joint bound without duplicated energy credits.

3.5. The resulting complete forced matrix and the exact remaining interaction

First record the bounded positive/adverse splitting used in Groups 2 and 4. For $\ell < a$, let $\Delta_\ell f(x) = f(x + \ell) - f(x)$ on $(0, a - \ell)$, and put

$$d_\ell = \mathbf{1}_{(0,a-\ell)} + \mathbf{1}_{(\ell,a)}, \quad \omega_a = V_p - \sum_{\ell_n < a} b_n d_{\ell_n}. \quad (\text{CP20:G3.S1})$$

On $I_{\ell,a}$, define the orthogonal reflection projections $\Pi_{\ell,a}^\pm = (I \pm R_{\ell,a})/2$, extended by zero off that interval. Expansion of the difference and orthogonality of the projections give

$$\|\Delta_\ell f\|^2 = \int_0^a d_\ell |f|^2 - 2\Re \int_0^{a-\ell} \overline{f(x)} f(x + \ell) dx, \quad \langle f, R_{\ell,a} f \rangle = \|\Pi_{\ell,a}^+ f\|^2 - \|\Pi_{\ell,a}^- f\|^2. \quad (\text{CP20:G3.S2})$$

Thus the complete arithmetic form is $\mathcal{P}_a = P_a^+ - P_a^-$, with

$$\begin{aligned} P_a^+(f) &= \sum_{\ell_n < a} b_n \|\Delta_{\ell_n} f\|^2 + \sum_{\ell_n < 2a} b_n \|\Pi_{\ell_n,a}^+ f\|^2 + \int_0^a (\omega_a)_+ |f|^2, \\ P_a^-(f) &= \sum_{\ell_n < 2a} b_n \|\Pi_{\ell_n,a}^- f\|^2 + \int_0^a (\omega_a)_- |f|^2. \end{aligned} \quad (\text{CP20:G3.S3})$$

These are bounded nonnegative forms on all of $L^2(0, a)$: the sums are finite, the shifts and projections are bounded, and V_p is bounded after the origin cancellation. In particular the adverse operator used in Group 2 has $v = (\omega_a)_-$. This splitting retains every exterior term in V_p ; its projection components are a valid alternative decomposition of the same reflection operator whose coupled path remainders were computed above.

Write CP19's positive continuous energy as

$$\mathcal{C}_a(f) = \mathcal{G}_p(f) + \mathcal{Z}_p(f) = \eta_p(a)\|f\|^2 + \mathcal{N}_a(f), \quad \eta_p(a) > 0, \quad \mathcal{N}_a \geq 0. \quad (\text{CP20:G3.30})$$

Combining (G3.5) and the exact block identity (G3.29), rather than discarding its positive remainder, gives

$$\begin{aligned} Q_a^-(q) &= [\eta_p(a) - B_{\text{ref}}(a) + S_{\text{sq}}(a)]\|f\|^2 + \mathcal{N}_a(f) + \int_0^a V_p(x)|f(x)|^2 dx \\ &\quad - 2 \sum_{\ell_n < a} b_n \Re \int_0^{a-\ell_n} \overline{f(x)} f(x + \ell_n) dx + \sum_{\substack{p, r \in \mathcal{D} \\ r \log p < 2a}} \langle f, J_{p,r,a} f \rangle. \end{aligned} \quad (\text{CP20:G3.31})$$

This is an exact evaluation of the same signed form, with an independently quantified positive saving S_{sq} and its entire positive block remainder. All original prime weights, ordinary shifts, reflected interactions, the infinite exterior potential, and the continuous remainder remain present.

For $f_j = \sqrt{2}(W_a e_j)|_{(0,a)}$, polarize every term of (G3.31). The left side is still

$$W_a^* A_a^- W_a = \tau_R^{-1} K_a (I - K_a), \quad (\text{CP20:G3.32})$$

and $\int F^* F = W_a^* W_a$. Every matrix $[\langle f_i, J_{p,r,a} f_j \rangle]_{i,j}$ is positive semidefinite, with all complex coefficient cross terms. No endpoint trace, reflection eigenfunction regularity, or change of the Suzuki metric enters this passage. Arbitrary L^2 path eigenvectors are used only to establish sharp operator bounds; the actual forced columns remain in their inherited form domain.

On the next prime interval (G3.15), the stronger available alternative is the exact identity

$$Q_a^-(q) = \mathcal{C}_a(f) + \int_0^a V_p(x)|f(x)|^2 dx - L_{uvu}\|f\|^2 + \mathcal{J}_{23,a}(f). \quad (\text{CP20:G3.33})$$

The $u + v - L_{uvu}$ improvement is now calculated directly in the full signed energy. Establishing a sign for (G3.31) or (G3.33) still requires the coupled continuous/potential comparison. Positivity of the new path boards alone does not establish that sign.

3.6. A rigorous limitation: uniformly negative reflection allowances cannot close the all-support form

The new improvements identify a real cancellation, but also permit a precise diagnosis of one inadequate next estimate. The theta equation in CP19 gives the exact identity

$$\mathcal{C}_a(f) + \int_0^a V_p(x)|f(x)|^2 dx = Q_\Gamma(q) - 4 \left| \int_0^a \sinh(x/2) f(x) dx \right|^2. \quad (\text{CP20:G3.34})$$

Indeed $V_p = (A_\Gamma p_o)/p - \sinh(x/2)/(2p)$; substitute CP19's gamma transform and full polar variance. This is exact cancellation, not a lower estimate.

Choose any real nonnegative $g \in C_c^\infty(0, L)$ with $\|g\|_2 = 1$, and set

$$f_t(x) = g(x - t), \quad a = t + L, \quad q_t(x) = \frac{g(x - t) - g(-x - t)}{\sqrt{2}}. \quad (\text{CP20:G3.35})$$

These are legitimate odd compact smooth tests. Put

$$m_+ = \int_0^L e^{y/2} g(y) dy > 0, \quad m_- = \int_0^L e^{-y/2} g(y) dy > 0. \quad (\text{CP20:G3.36})$$

The complete polar quantity is

$$4 \left| \int \sinh(x/2) f_t(x) dx \right|^2 = (e^{t/2} m_+ - e^{-t/2} m_-)^2 \geq \frac{1}{4} m_+^2 e^t \quad (\text{CP20:G3.37})$$

whenever $e^t \geq 2m_-/m_+$.

In contrast the gamma form has a fixed upper bound. With the actual multiplier $a_\Gamma(\xi) = \Re\psi(1/4 + i\xi/2) - \log \pi$, the elementary Fourier inequality $|\hat{q}_t(\xi)|^2 \leq 2|\hat{g}(\xi)|^2$ yields

$$Q_\Gamma(q_t) \leq C_g, \quad C_g = \frac{1}{\pi} \int_{\mathbb{R}} (a_\Gamma(\xi))_+ |\hat{g}(\xi)|^2 d\xi < \infty. \quad (\text{CP20:G3.38})$$

The integral is finite because the gamma multiplier grows logarithmically and g is smooth compactly supported. Thus the right side of (G3.34) is at most $C_g - \frac{1}{4} m_+^2 e^t$, which becomes strictly negative.

For these nonnegative translates every ordinary autocorrelation in (G3.5) is nonnegative. Replacing the *entire* reflected-prime form by any bound $-c_a \|f\|^2$, $c_a \geq 0$, therefore produces a total lower allowance no greater than

$$C_g - \frac{1}{4} m_+^2 e^t - c_a \longrightarrow -\infty. \quad (\text{CP20:G3.39})$$

This proves that a strategy relying only on worst-case negative norms for all reflected primes cannot establish the all-support sign, even if the continuous energy and origin-cancelled potential are evaluated exactly. The conclusion concerns that lower-bound strategy, not the value of the actual Weil form. The actual favorable large-prime reflections remain present and may compensate the pole. A successful full comparison must retain and quantify that interaction, or exploit additional constraints on the actual forced or first-null family. The translated test functions here are not asserted to lie in that finite forced family, so (G3.39) is not a counterexample to a more specific forced-family estimate.

3.7. Closing statement for this executed group

Proved in this group: the exact complete reflection spectrum for primes 2 and 3 at every support; its universal strict adverse-energy saving; the precise signed improvement on the next prime window; exact prime-square spectra; a disjoint all-prime-power allocation and its explicit summed positive credit; and the complete forced-matrix identity retaining the associated positive path remainders. These are analytic calculations on the actual operator, valid for complex coefficients and at the closed-domain level needed by CP19.

Also proved: blanket negative reflection budgets lose a necessary polar compensation on translated smooth nonnegative half-interval tests. This identifies an actual failure of that global estimation method and the arithmetic interaction that must remain coupled.

The new comparisons do not alone establish a positive value of (G3.31) on all supports or a strict contradiction on every hypothetical first-null vector. The unresolved expression is the complete right side of (G3.31), with its positive block remainder retained, or its sharper local alternative (G3.33). Both null branches and the original Suzuki metric remain unchanged. The conditional Weil-Suzuki return is not used as an input.

4. Complete matrix comparison and a quantified transfer from the response certificate

The objects in this section are the actual physical regularizer and sources of CP18-19. Fix a finite R and the stated support range, and write

$$\mathcal{J}_a = A_a^- + \tau F_a F_a^* \succeq gI, \quad \|F_a\| \leq 1, \quad W_a = \mathcal{J}_a^{-1} F_a, \quad K_a = \tau F_a^* W_a, \quad M_a = I - K_a. \quad (\text{CP20:4.1})$$

Here g and τ are the known positive regularizer constants for that fixed range. No positivity of A_a^- is assumed in the estimates below. Every inverse is that of the positive regularizer.

4.1. Exact normalized comparison

Polarize the bounded splitting of the paired prime form proved in Group 3, on $f_c = \sqrt{2}W_a c$ restricted to $(0, a)$, to obtain positive matrices P_a^+ and P_a^- . Set

$$A_a = C_a + P_a^+, \quad B_a = P_a^-, \quad Q_a = A_a - B_a = \tau^{-1} K_a M_a. \quad (\text{CP20:4.2})$$

The continuous covariance matrix of Group 2 satisfies $C_a \succeq \eta_p(a) W_a^* W_a$. On the active source space $\mathcal{H}_a^{\text{src}} = (\ker F_a)^\perp = (\ker W_a)^\perp$, this makes A_a positive definite. This space is distinct from the inherited polar-removed operator H_a . Thus

$$Q_a \succeq 0 \iff A_a^{-1/2} B_a A_a^{-1/2} \preceq I_{\mathcal{H}_a^{\text{src}}}. \quad (\text{CP20:4.3})$$

The matrices vanish on $\ker F_a$, so this quotient loses no negative direction. This equivalence is an exact finite-dimensional statement, not an estimate of its right side.

4.2. A new explicit error bound for the complete energy matrix

Let Y be any actual graph-domain trial response, with residual $\mathcal{R} = F_a - \mathcal{J}_a Y$ and $r = \|\mathcal{R}\|$. Put $E = W_a - Y$. Then

$$\mathcal{J}_a E = \mathcal{R}, \quad \|E\| \leq r/g, \quad \|W_a\| \leq 1/g, \quad \|M_a\| \leq 1 + \tau/g. \quad (\text{CP20:4.4})$$

The full arithmetic operator satisfies

$$A_a^- W_a = F_a M_a, \quad A_a^- E = \mathcal{R} - \tau F_a F_a^* E. \quad (\text{CP20:4.5})$$

Every term belongs to L^2 because Y and W_a belong to the operator domain of the positive regularizer and its bounded finite-rank perturbation. Expanding the complete form, rather than its parts, gives

$$Y^* A_a^- Y - W_a^* A_a^- W_a = -E^* F_a M_a - M_a F_a^* E + E^* A_a^- E. \quad (\text{CP20:4.6})$$

Consequently

$$\|Y^* A_a^- Y - \mathbf{Q}_a\| \leq \frac{1 + \tau/g}{g} (2r + r^2). \quad (\text{CP20:4.7})$$

Indeed the two mixed terms together have norm at most $2r(1 + \tau/g)/g$. The last term is bounded by

$$\|E\| \|\mathcal{R}\| + \tau \|F_a^* E\|^2 \leq r^2/g + \tau r^2/g^2. \quad (\text{CP20:4.8})$$

The bound includes every prime, polar, archimedean, and mixed contribution because A_a^- itself occurs in (4.6).

For CP19's explicit reference responses Y_m , the proved graph residual is

$$r_m \leq \rho_R^m + \omega_R \mathfrak{d}_R m \rho_R^{m-1}, \quad (\text{CP20:4.9})$$

Here $\mathfrak{d}_R$ is the known norm bound for the complete bounded reference perturbation, not the source rank. The constants are those of CP19 (3.37)-(3.43), and $m \geq 1$. The case $\rho_R = 0$ is handled by the exact finite terminating recurrence. For $0 < \rho_R < 1$, (4.7) gives a uniform, explicitly vanishing complete-energy error without using an unknown Weil margin. It can be combined with any independently evaluated matrix lower bound for $\mathbf{Q}(Y_m)$.

For a bounded self-adjoint form B with $\|B\| \leq b$, the same expansion yields the additional useful payment

$$\|W_a^* B W_a - Y^* B Y\| \leq \frac{b}{g^2} (2r + r^2). \quad (\text{CP20:4.10})$$

This applies to the bounded paired-prime positive and adverse forms after their origin cancellation. It does not apply to the unbounded continuous form by simply assigning it a bounded norm; that form is instead controlled through (4.7) and (4.10).

Writing $e_Q = (1 + \tau/g)(2r + r^2)/g$ and $e_B = b(2r + r^2)/g^2$ gives the explicit enclosures

$$\|B_a - B_a(Y)\| \leq e_B, \quad \|A_a - A_a(Y)\| \leq e_Q + e_B, \quad (\text{CP20:4.11})$$

where $A_a(Y) = \mathbf{Q}_a(Y) + B_a(Y)$. In particular, the independently verified inequality

$$Y^* A_a^- Y \succeq (e_Q + \delta) I_{\mathcal{H}_a^{\text{src}}}, \quad \delta > 0, \quad (\text{CP20:4.12})$$

would prove $\mathbf{Q}_a \succeq \delta I$ on the active source space. Approximation alone supplies e_Q , not the sign in (4.12). CP19's original lower certificate for M_a and its error remain available as a separate, potentially sharper, way to test the same target.

4.3. What has and has not been paid globally

Groups 1-3 give exact covariance matrices, a physical transport of the old four-state theta block, the complete bounded arithmetic splitting, and strict coupled-reflection savings. These enter (4.2) without dropping any exterior prime potential. The inverse-covariance formulas and the error enclosures above turn the remaining comparison into explicit matrix quantities with known error payments.

The translated-bump calculation in Group 3 establishes that replacing every favorable reflected-prime term by a negative norm allowance cannot prove the desired all-support sign: its resulting lower expression becomes negative at large translations. The actual form retains positive reflected-prime interactions there. Accordingly, the combined calculation keeps those interactions in the

signed matrix. This is a proved limitation of that scalar estimate, not a negative witness for the Weil form.

No estimate in the present calculation bounds the complete matrix in (4.3) by I on every support or proves a strict inequality on every hypothetical first-null direction. The independent local certificate established later in this note does supply a new complete sign theorem on its explicitly stated interval. It is not promoted to an all-support assertion.

4.4. Domain and the retained return to Suzuki

The half-interval map $q \mapsto \sqrt{2}q|_{(0,a)}$ is unitary for the spatial L^2 form. The original Suzuki metric $\mathcal{S}$ is carried separately and is unchanged. The covariances are closed-domain form identities; the paired arithmetic matrices are bounded after the mandatory origin cancellation. Both first-null branches are still represented by $K_ac = c$ and $W_ac \neq 0$.

Suzuki's current primary paper gives the localized closed-form framework and records Yoshida's odd-test implication. An independent all-support odd sign would therefore give the retained chain

$$Q_a^-(q) \geq 0 \text{ for every finite support and odd core test} \implies \text{RH} \implies \mathfrak{W} = \mathcal{S} \implies \mathcal{E}_N = 0. \quad (\text{CP20:4.13})$$

The first assertion of this chain is open after the present global comparisons. The original independent target $\liminf \mathcal{E}_N \leq 1$ remains open as well.

Primary references checked during this run: Masatoshi Suzuki, *Weil's quadratic form via the screw function*, arXiv:2606.09096v2, Introduction and §2, <https://arxiv.org/html/2606.09096v2> ; *On the Hilbert space derived from the Weil distribution*, arXiv:2301.00421, <https://arxiv.org/abs/2301.00421> . The research identities and local certificate in this note are separate calculations, not results attributed to those papers.

4.5. A new complete local certificate: support 3/8 and floor 1/200

Result and conventions With the complete original odd Weil form and domain of CP19 (0.4)-(0.5),

$$\boxed{Q_a^-(q) > \frac{1}{200} \|q\|_2^2 \quad (0 < a \leq 3/8, \quad 0 \neq q \in \mathcal{D}_a^-).} \quad (\text{CP20:LC1})$$

The bound covers every complex odd input in the closed form domain. The original Suzuki metric is unchanged. Both possible odd null branches are excluded on this enlarged interval. This improves the earlier certified endpoint $699/2000 + e^{-2500}$ to 3/8. The all-support sign remains open.

It suffices to prove the endpoint $a_* = 3/8$: the complete Weil form of any zero-extended input with smaller support is the same form evaluated in $\mathcal{D}_{a_*}^-$. This is ordinary domain inclusion, and does not assert monotonicity of the separate estimates below.

Set $u(x) = \sqrt{2a_*} q(a_*x)$ on $(0, 1)$, so $\|u\|_2 = \|q\|_2$. Write

$$b = \frac{\log 2}{\sqrt{2}}, \quad L = \frac{\log 2}{a_*}, \quad E_L = (L - 1, 1), \quad (J_L u)(x) = \mathbf{1}_{E_L}(x) u(L - x). \quad (\text{CP20:LC2})$$

Here $1 < L < 2$, $J_L^* = J_L$, $J_L^2 = P_L = \mathbf{1}_{E_L}$, and $\|J_L\| = 1$. Since $\log 2 < 2a_* < \log 3$ and $a_* < \log 2$, the only active original prime is the complete reflected 2-packet, with sign $+bJ_L$. This follows directly by odd extension in CP16 (1.2), or CP19 (0.5). No prime terms have been omitted.

The full scaled form is

$$Q_{a_*}^-(q) = \langle \log |D|q, q \rangle - \log(2\pi) \|u\|^2 + b \langle J_L u, u \rangle - \langle \mathbf{A}_{a_*} u, u \rangle - 2 |\langle \varsigma_{a_*}, u \rangle|^2, \quad (\text{CP20:LC3})$$

where

$$\mathbf{A}_a(x, y) = a[\alpha(a|x - y|) - \alpha(a(x + y))], \quad \alpha(t) = \frac{e^{-t/2}}{1 - e^{-2t}} - \frac{1}{2t}, \quad \alpha(0) = \frac{1}{4}, \quad \varsigma_a(x) = \sqrt{2a} \sinh(ax/2). \quad (\text{CP20:LC4})$$

All concentration, prime, archimedean and polar mixed terms are retained in LC3. Subsequent operator inequalities hold for arbitrary u , so they control their couplings.

1. Fourier concentration with the complete reflection inside the inverse Use the unitary sine transform $Su(\xi) = \sqrt{2/\pi} \int_0^1 u(x) \sin(\xi x) dx$. Plancherel and a split at $\Omega = 8$ give

$$\langle \log |D|q, q \rangle = (-\log a_* + \log 8) \|u\|^2 - \langle T_8 u, u \rangle + P_8(u), \quad P_8(u) = \int_8^\infty \log(\xi/8) |Su(\xi)|^2 d\xi \geq 0. \quad (\text{CP20:LC5})$$

The identity extends from the smooth odd core by closed form convergence. Define

$$(Af)(x) = \int_0^8 w(\xi) \sin(\xi x) f(\xi) d\xi, \quad w(\xi) = \sqrt{\frac{2}{\pi} \log(8/\xi)}. \quad (\text{CP20:LC6})$$

Then $T_8 = AA^*$, $\|w\|_2^2 = 16/\pi < 6$, and $\|A\|_{\text{HS}}^2 < 6$.

Take $\lambda = 148/125$. Because $\lambda > b$,

$$R_{b,L} = (\lambda I + bJ_L)^{-1} = \frac{1}{\lambda} I + \frac{b^2}{\lambda(\lambda^2 - b^2)} P_L - \frac{b}{\lambda^2 - b^2} J_L. \quad (\text{CP20:LC7})$$

We establish

$$\|A^* R_{b,L} A\| < \frac{9991}{10000}. \quad (\text{CP20:LC8})$$

In particular, this is a bound for the coupled concentration/reflection operator, rather than a sum of their separate norms. The positive inverse is used only for $\lambda I + bJ_L$, whose independent spectral gap is explicit.

LC8 implies, by the positive square-root factorization,

$$\lambda I + bJ_L - T_8 \succeq \frac{9}{10000} (\lambda I + bJ_L) \succeq \frac{9}{10000} (\lambda - b) I > \frac{621}{10^6} I. \quad (\text{CP20:LC9})$$

Thus the prime and continuous pieces together satisfy

$$\langle \log |D|q, q \rangle - \log(2\pi) \|u\|^2 + b \langle J_L u, u \rangle > \left[\log \frac{4}{\pi a_*} - \lambda + \frac{621}{10^6} \right] \|u\|^2. \quad (\text{CP20:LC10})$$

The nonnegative high-frequency surplus P_8 remains available and is simply bounded below by zero.

2. Exact finite certificate and all perturbation payments Use the rational parameters

$$b_0 = \frac{490129}{10^6}, \quad L_0 = \frac{18483925}{10^7}, \quad c_0 = L_0 - 1, \quad \pi_0 = \frac{314159}{100000}. \quad (\text{CP20:LC11})$$

The rational log and square-root enclosures in the executable certificate prove

$$|b - b_0| < 10^{-6}, \quad |L - L_0| < 10^{-6}, \quad b, b_0 < 491/1000, \quad \lambda - 491/1000 > 69/100 > 2/3. \quad (\text{CP20:LC12})$$

In particular all inverses used below have norm less than $3/2$, independently of any Weil positivity margin.

Replace $\sin z$ by

$$p(z) = \sum_{j=0}^{15} \frac{(-1)^j z^{2j+1}}{(2j+1)!}. \quad (\text{CP20:LC13})$$

Taylor's theorem gives $|\sin z - p(z)| \leq |z|^{33}/33!$. If A_p is LC6 with p in place of sine, direct integration yields

$$\|A - A_p\|_{\text{HS}}^2 \leq \frac{2}{\pi} \frac{8^{67}}{(33!)^{2 \cdot 67^3}} < 10^{-14}. \quad (\text{CP20:LC14})$$

Therefore, with $R_0 = R_{b_0, L_0}$,

$$\|A^* R_0 A - A_p^* R_0 A_p\| \leq \frac{3}{2} (2\sqrt{6} + 10^{-7}) 10^{-7} < 10^{-6}. \quad (\text{CP20:LC15})$$

Here is the complete 16 by 16 rational matrix calculation. For $0 \leq i, j \leq 15$, put

$$\begin{aligned} D_{ij} &= \frac{(-1)^{i+j} 8^{2i+2j+3}}{(2i+1)!(2j+1)!(2i+2j+3)^2}, \\ G_{ij} &= \frac{1}{2i+2j+3}, \\ (G_E)_{ij} &= \frac{1 - c_0^{2i+2j+3}}{2i+2j+3}, \\ (G_J)_{ij} &= \int_{c_0}^1 x^{2i+1} (L_0 - x)^{2j+1} dx \\ &= \sum_{k=0}^{2j+1} (-1)^k \binom{2j+1}{k} L_0^{2j+1-k} \frac{1 - c_0^{2i+k+2}}{2i+k+2}, \\ G_R &= \lambda^{-1} G + \frac{b_0^2}{\lambda(\lambda^2 - b_0^2)} G_E - \frac{b_0}{\lambda^2 - b_0^2} G_J. \end{aligned} \quad (\text{CP20:LC16})$$

These are the complete monomial Gram matrices of the actual reflection resolvent at the rational reference parameters. In particular G_R is positive definite because $R_0 > 0$; its mixed entries are not discarded. D is a positive definite frequency Gram matrix with alternating coefficient signs. Finite-rank cyclicity gives

$$\text{Tr}[(A_p^* R_0 A_p)^8] = (2/\pi)^8 \text{Tr}[(DG_R)^8]. \quad (\text{CP20:LC17})$$

The supplied exact Fraction arithmetic verifies

$$\left(\frac{2}{\pi_0}\right)^8 \text{Tr}(DG_R)^8 < \frac{99}{100} < \left(\frac{999}{1000}\right)^8. \quad (\text{CP20:LC18})$$

Since $A_p^* R_0 A_p \geq 0$, LC18 proves its operator norm is below $999/1000$. This is an all-input operator bound from a finite exact certificate and a proved finite-rank representation, not a sampled-function sign test.

The changes in b and L are paid separately. Resolvent identity and LC12 give

$$\|A^*(R_{b,L} - R_{b_0,L})A\| < 6(3/2)^2|b - b_0| < 18 \cdot 10^{-6}. \quad (\text{CP20:LC19})$$

The reflection J_L itself need not be continuous in operator norm. Its sandwich by A has the needed quantitative regularity. Explicitly the kernels on the frequency space are

$$\begin{aligned} (A^*P_L A)(\xi, \eta) &= w(\xi)w(\eta) \int_{L-1}^1 \sin(\xi x) \sin(\eta x) dx, \\ (A^*J_L A)(\xi, \eta) &= w(\xi)w(\eta) \int_{L-1}^1 \sin(\xi x) \sin(\eta(L-x)) dx. \end{aligned} \quad (\text{CP20:LC20})$$

For $1 < L < 2$, differentiation of the complete integrals, including their moving lower endpoint, bounds the derivatives by $w(\xi)w(\eta)$ and $9w(\xi)w(\eta)$, respectively. The second bound uses $0 < \eta < 8$ and an integration interval of length at most one. A kernel dominated by $Cw(\xi)w(\eta)$ has operator norm at most $C\|w\|^2 < 6C$. Consequently

$$\|A^*(P_L - P_{L_0})A\| < 6|L - L_0|, \quad \|A^*(J_L - J_{L_0})A\| < 54|L - L_0|. \quad (\text{CP20:LC21})$$

The rational coefficients obey $b_0^2/[\lambda(\lambda^2 - b_0^2)] < 1/5$ and $b_0/(\lambda^2 - b_0^2) < 1/2$. Equations LC7 and LC21 therefore prove

$$\|A^*(R_{b_0,L} - R_0)A\| < 29|L - L_0| < 29 \cdot 10^{-6}. \quad (\text{CP20:LC22})$$

The total of LC15, LC19 and LC22 is below $48 \cdot 10^{-6} < 1/10000$. Together with LC18 this proves LC8, including all parameter, endpoint and Taylor errors.

3. Complete odd archimedean and polar payment The archimedean derivative estimate from CP16 extends analytically to the required $0 < t \leq 3/4$. Put

$$d(t) = \frac{1}{t} - \frac{1}{\sinh t}, \quad \alpha(t) = \frac{e^{t/2} - 1}{2t} - \frac{1}{2}e^{t/2}d(t). \quad (\text{CP20:LC23})$$

The power series of $(1 - t^2/6)\sinh t$ gives $0 \leq d(t) \leq t/6$. The locally absolutely convergent expansion

$$d(t) = 2t \sum_{n \geq 1} \frac{(-1)^{n+1}}{t^2 + \pi^2 n^2} \quad (\text{CP20:LC24})$$

can be obtained by the cosine series of $\cosh(tx)$ on $(0, 1)$, as in CP16. Differentiation is locally uniformly convergent. For $t \leq 3/4$, the positive magnitudes $(\pi^2 n^2 - t^2)/(\pi^2 n^2 + t^2)^2$ decrease with n , since $\pi^2 > 3t^2$. Alternating-series comparison gives $0 \leq d'(t) \leq 2/\pi^2$.

Also $(e^{t/2} - 1)/(2t) = \frac{1}{4} \int_0^1 e^{ut/2} du$, so its derivative lies between $1/16$ and $3/32$, using $e^{3/8} < 3/2$. With $\pi^2 > 48/5$,

$$-\frac{9}{64} = \frac{1}{16} - \frac{3}{4} \left(\frac{3/4}{12} + \frac{5}{24} \right) \leq \alpha'(t) \leq \frac{3}{32}, \quad 0 < t \leq 3/4. \quad (\text{CP20:LC25})$$

The constant $\alpha(0)$ cancels exactly in LC4. Thus

$$|A_a(x, y)| \leq \frac{9}{32} a^2 \min(x, y), \quad \|A_a\| \leq \frac{9a^2}{8\pi^2} \quad (a \leq 3/8), \quad (\text{CP20:LC26})$$

using the exact norm $4/\pi^2$ of the kernel $\min(x, y)$ on $(0, 1)$. This follows by the elementary Dirichlet-Neumann Green eigenvalue calculation, and the pointwise domination controls complex inputs as well.

The full polar operator has norm

$$2\|\zeta_a\|^2 = 2(\sinh a - a). \quad (\text{CP20:LC27})$$

This retains its complete rank-one contribution rather than a sign assumption on a polar moment.

The rational series certificates finally give, at $a_* = 3/8$,

$$\log \frac{4}{\pi a_*} > \frac{122239}{100000}, \quad \frac{9a_*^2}{8\pi^2} < \frac{16030}{10^6}, \quad 2(\sinh a_* - a_*) < \frac{17703}{10^6}. \quad (\text{CP20:LC28})$$

Combining LC3, LC10, LC26-LC28,

$$Q_{a_*}^-(q) > \left[\frac{122239}{100000} - \frac{148}{125} - \frac{16030 + 17703}{10^6} + \frac{621}{10^6} \right] \|q\|^2 = \frac{5278}{10^6} \|q\|^2 > \frac{1}{200} \|q\|^2. \quad (\text{CP20:LC29})$$

Zero extension from every smaller support proves LC1.

Verification and scope The complete exact arithmetic reproducer is `Michaels_Checkpoint_20_Exact_Certificate.py`. It uses Python integers and `fractions.Fraction` for every asserted inequality. A decimal display of the eighth trace is only diagnostic. The script verifies the matrix trace, the sine Taylor remainder, all parameter errors, coefficient bounds, and the final rational margin. It has been executed successfully.

The comparison proves a new complete odd Weil sign on an enlarged interval. It uses no zero-location computation, RH assumption, unknown inverse of the Weil operator, or unproved numerical eigenvalue assertion. The finite matrix is an exact certificate for a bounded concentration/reflection comparison on the entire Hilbert space. Further support growth still requires a new inequality. In particular this local theorem does not establish global Weil positivity or the original Suzuki defect criterion.

Verification, sources, and preservation

The local proof retains the exact complete 2-packet, uses a positive comparison inverse with a stated independent gap, and proves every truncation, moving-endpoint, and transcendental error bound. Its finite trace and scalar inequalities are reproduced by the standard-library Python program below. The full operator inequality holds on the entire indicated Hilbert space, then passes to the closed odd form domain; the calculation is not a test on finitely many input functions.

Separate checks covered the exact path partitions and support thresholds, prime-square coefficient allocation, the source-space energy error, and the local certificate. The covariance/domain and old-board calculations are proved explicitly in their groups. The modified modulus inequality fails on specified smooth inputs; no negative Weil witness or global theorem is inferred from that failure.

The global calculation stops at the fully specified signed comparison (4.3), with the full positive arithmetic remainders of (G3.31) available. No all-support unit bound or strict first-null contradiction has been established. The new theorem (0.1) is the strongest independently quantified sign from this run.

Sources retained and used: Checkpoints 16, 18, and 19; the recovered theory atlas and CP20 directive; Michaels_H_Analytical_Review for the actual complete four-state theorem and the correctly scaled curvature; and the older energy/color sources identified in the directive. The published normalization and odd-test criterion were checked in Suzuki's *Weil's quadratic form via the screw function*, <https://arxiv.org/html/2606.09096v2> . The Suzuki Hilbert-space reference is <https://arxiv.org/abs/2301.00421> . None of the new calculations is attributed as a theorem from those papers.

Every earlier non-ledger content file in the CP19 packet is preserved byte-for-byte. The CP20 packet adds this note, the unchanged CP20 directive, the exact certificate program, and a verification record. The continuing ledger and checksum manifest are updated. Earlier checkpoint ZIP files remain separate and unchanged.

Exact certificate program

Run `python3Michaels_Checkpoint_20_Exact_Certificate.py`. The program has no external dependencies. Its proof role is the finite rational assertions; the analytical operator and domain arguments are given above.

```
#!/usr/bin/env python3
"""Checkpoint 20 exact certificate. Python standard library only.

Every asserted inequality uses arbitrary-precision integers and Fraction.
The displayed decimal trace is diagnostic; it is not used by the proof.
See the accompanying checkpoint for the operator reductions and error bounds.
"""
from fractions import Fraction as F
from math import factorial, comb
import time
start=time.time()
n=16
lam=F(148,125)
b=F(490129,1000000)
L=F(18483925,10000000);c=L-1
pi0=F(314159,100000)
D=[[F((-1)**(i+j)*8**(2*i+2*j+3),factorial(2*i+1)*factorial(2*j+1)*(2*i+2*j+3)**2) for j in range(n)] for i
  ↪ in range(n)]
G=[[F(1,2*i+2*j+3) for j in range(n)] for i in range(n)]
GE=[[((1-c**(2*i+2*j+3))/F(2*i+2*j+3) for j in range(n)] for i in range(n)]
GJ=[[sum((F((-1)**k*comb(2*j+1,k))*L**(2*j+1-k)*(1-c**(2*i+k+2))/F(2*i+k+2) for k in range(2*j+2)),F()) for j
  ↪ in range(n)] for i in range(n)]
assert all(GJ[i][j]==GJ[j][i] for i in range(n) for j in range(n))
GR=[[G[i][j]/lam+b*b/(lam*(lam*lam-b*b))*GE[i][j]-b/(lam*lam-b*b)*GJ[i][j] for j in range(n)] for i in
  ↪ range(n)]
def mm(A,B):
    return [[sum((A[i][k]*B[k][j] for k in range(n)),F()) for j in range(n)] for i in range(n)]
V=mm(D,GR)
for k in range(2):
    V=mm(V,V)
    print('power',2*(k+1),'seconds',round(time.time()-start,2),flush=True)
# trace(V^2) = trace((D GR)^8)
tr8=sum((V[i][j]*V[j][i] for i in range(n) for j in range(n)),F())
assert (2/pi0)**8*tr8<F(99,100)
assert F(99,100)<F(999,1000)**8
print('trace certificate passed',float((2/pi0)**8*tr8),'seconds',time.time()-start)
# All transcendental enclosures below use finite rational expressions.
```

```

def atan_bounds(x,N):
    s=sum((F((-1)**j)*x**(2*j+1)/F(2*j+1) for j in range(N)),F())
    nxt=F((-1)**N)*x**(2*N+1)/F(2*N+1)
    return min(s,s+nxt),max(s,s+nxt)
def log_bounds(x,N):
    h=(x-1)/(x+1)
    assert 0<h<1
    lo=2*sum((h**(2*j+1)/F(2*j+1) for j in range(N)),F())
    hi=lo+2*h**(2*N+1)/((2*N+1)*(1-h*h))
    return lo,hi
p1=F(355,113)
z5,w5=atan_bounds(F(1,5),12);z239,w239=atan_bounds(F(1,239),4)
assert pi0<16*z5-4*w239 and 16*w5-4*z239<p1
lo2,hi2=log_bounds(F(2),12)
rtlo,rthi=F(1414213562,10**9),F(1414213563,10**9)
assert rtlo*rtlo<2<rthi*rthi
assert b-F(1,10**6)<lo2/rthi and hi2/rtlo<b+F(1,10**6)
a=F(3,8)
assert L-F(1,10**6)<lo2/a and hi2/a<L+F(1,10**6)
assert F(3,8)<lo2<hi2<F(3,4)
lo3,hi3=log_bounds(F(3),20)
assert lo3>F(3,4)
assert b+F(1,10**6)<F(491,1000)
assert lam-F(491,1000)>F(69,100)>F(2,3)
assert b*b/(lam*(lam*lam-b*b))<F(1,5)
assert b/(lam*lam-b*b)<F(1,2)
# Exact sine-Taylor Hilbert-Schmidt remainder.
assert 2/pi0*F(8**67,factorial(33)**2*67**3)<F(1,10**14)
assert 16/pi0<6
assert F(3,2)*(5+F(1,10**7))*F(1,10**7)<F(1,10**6)
assert F(18+29+1,10**6)<F(1,10000)
# Alpha derivative bound: exp(3/8)<3/2 and pi^2>48/5.
x=F(3,8);N=8
explo=sum((x**k/F(factorial(k)) for k in range(N)),F())
exphi=explo+x**N/F(factorial(N))/(1-x/F(N+1))
assert exphi<F(3,2)
assert pi0*pi0>F(48,5)
assert F(1,16)-F(3,4)*(F(3,4)/12+F(5,24))=-F(9,64)
# Final exact scalar sign certificate.
ll,lh=log_bounds(4/(p1*a),16)
assert ll>F(122239,100000)
assert 9*a*a/(8*pi0*pi0)<F(16030,10**6)
s=sum((a**(2*j+1)/F(factorial(2*j+1)) for j in range(1,6)),F())
s+=a**13/F(factorial(13))/(1-a*a/F(14*15))
assert 2*s<F(17703,10**6)
base=F(122239,100000)-lam-F(16030+17703,10**6)
assert base==F(4657,10**6)
reserve=F(9,10000)*F(69,100)
assert base+reserve==F(5278,10**6)>F(1,200)
print('All exact rational assertions passed; Q_a^- > (1/200)||q||^2 for 0<a<=3/8.')

```

Part III

The mathematical theory atlas

The atlas preserves all 301 indexed entries and their purposes, representations, source records, and applications. It was assembled at Checkpoint 18; its dated descriptions are followed by the Checkpoint 19–20 developments in Part II. This is an inventory of constructions and their recorded scope, not 301 independent theorems.

Theory atlas: purposes, mathematical representations, and connections

Source: Michaels_Theory_Atlas_Purpose_and_Weil_Connections_2026-09-23.md. Equation tags carry the source prefix ATLAS; references within this unit retain their original local numbering.

Christophe Michaels – recovered research inventory, 23 September 2026; publication update through Checkpoint 20

This atlas gathers **301 indexed entries** covering theories, constructions, exact identities, proposed closure routes, experiments, and results that rule out particular approaches. It includes the light cones, Fibonacci and other spirals, colors, angles, Michaels Möbius modifiers, quadratic and infinity planes, and a dedicated account of the energy work. Some entries revisit one object through different interfaces; the count is not a claim of that many independent original theorems.

Every entry identifies its purpose, what the retrieved source records, and its implications for the present effort. “Proved” or “exact” reports the derivation supplied in the named manuscript, with later corrections incorporated where recovered. This inventory is not a fresh independent proof audit of the entire archive. Classical tools and the project’s named constructions are both included because the aim is to preserve the machinery actually used.

Publication update through Checkpoint 20

The 301-entry catalogue below records the recovered theory atlas at its Checkpoint 18 cutoff. References to “current” or “now” inside individual entries retain that historical viewpoint. Checkpoints 19 and 20, included in the main manuscript, subsequently carry several of these constructions into the actual forced Weil family. Entry identifiers and all substantive catalogue content are preserved.

With the positive physical regularizer and its forced response,

$$J_a = A_a^- + \tau_R F_a F_a^* \geq g_R I, \quad W_a = J_a^{-1} F_a, \quad K_a = \tau_R F_a^* W_a,$$

the complete sign calculation is

$$W_a^* A_a^- W_a = \frac{1}{\tau_R} K_a (I - K_a).$$

Checkpoint 19 derives the complete energy identity

$$\frac{1}{\tau_R} K_a (I - K_a) = \eta_p(a) W_a^* W_a + \mathbf{N}_a + \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \mathbf{P}_{a, \log n}, \quad \eta_p(a) > 0, \quad \mathbf{N}_a \succeq 0.$$

The paired-prime term includes its signed exterior contribution. This is the current point of contact for the atlas’s storage, color, angle, reflection, and covariance constructions: a positive identity in an earlier metric must be transported into this complete comparison with its exact cost.

Checkpoint 20 supplies an actual payment from the earlier positive four-state theta matrix to a physical reflected-prime path; a nested positive covariance representation with its exterior reserve; complete reflection spectra for primes 2 and 3 and prime-square pairs; and uniform graph-residual error accounting. It also proves the local closed-form certificate

$$Q_a^-(q) > \frac{5278}{10^6} \|q\|_2^2 > \frac{1}{200} \|q\|_2^2, \quad 0 < a \leq \frac{3}{8}, \quad 0 \neq q \in \mathcal{D}_a^-.$$

The exact-rational verification accompanies the Checkpoint 20 proof. Global Weil positivity and the original independent condition $\liminf_N \mathcal{E}_N \leq 1$ remain open. The conditional return through RH to $\mathfrak{W} = \mathcal{S}$ and $\mathcal{E}_N = 0$ remains unchanged.

Mathematical reading guide. The factor-energy law in E01 and the factor-potential extraction in C03 recover the actual coefficient $\Lambda(n)$. The angle identity AN02 and the wave identity EX04 recover the exact historical Green metric, including its terminal mass. SP06 carries the Michaels modifier with its mixed compensation; SP11 gives the bounded Fibonacci-Lucas packet and its precise separation hypothesis. E08-E09 and T51-T52 reach the complete Weil energy and its polar covariance directly. L02-L04 state the half-density and positive-state confinement mechanism, while P02-P07 retain the full derivative, quadratic, parity, and boundary representations. The CP20 comparison now uses that existing energy and arithmetic structure on the concrete response W_{ac} ; it does not replace the original Suzuki metric.

1. The architecture already available

The project has worked with several related kinds of information:

1. **Arithmetic source:** prime identities, valuations, prime powers, ordered gaps, counting rules and Möbius signs.
2. **Geometry and observation:** factor chords, angles, spirals, prime polygons, color coordinates, the prime and zero planes, and complete analytic fingerprints.
3. **Energy and cancellation:** Green storage, factor-energy partitions, color squares, Fibonacci packets, selected/residual masks, lifetime repayment, discrepancy energy, theta differences and complete Weil energy.
4. **Transport and localization:** light cones, half-density propagation, prime shifts, dilations, heat, two-channel evolutions, recurrence and source reconstruction.
5. **Positivity and closure:** known-zero witnesses, theta kernel criteria, Schur/Feshbach reductions, local support certificates, Suzuki defects and the present finite physical regularizer.

Several interfaces are already exact. Factor potential and factor-layer energy both reconstruct the von Mangoldt source. Factor angles reconstruct the reciprocal-maximum kernel. The historical Green kernel appears by exact rescaling inside a pole-continuum cancellation of the Weil operator. Complete prime phases already connect to the fixed theta operator and the Weil/Suzuki ladder. The prime-zero atlas supplies a full transform dictionary. The theta-weighted energy identity already reaches the complete Weil form.

The current question is therefore often **which existing sign or cancellation can be proved for the actual test family**, rather than how to invent another representation.

2. The energy work, kept in its original quantities

These energy families must remain distinguishable. Their definitions explain both their value and the comparison still needed.

Energy family	Original job	Established structure and present relevance
Factor-layer and cross-color energies $L(n), C(n), 1 - L(n)$	Measure how prime factors share the logarithm of an integer	Exact conservation, descending composite floors, power-family laws and Fibonacci factor identities; an exact extraction recovers $\Lambda(n)$. These are scalar arithmetic energies. Equals $\int_1^N M(t)^2 t^{-2} dt + M(N)^2/N$, a Brownian Gram norm, and a weighted Fourier energy. The endpoint is part of the norm.
Historical Möbius/Green energy $R(N) = \sum_{m,n \leq N} \mu(m)\mu(n)/\max(m,n)$	Measure the entire cumulative Möbius history	Zero-mass future orthogonality, two-summary recurrences, Fibonacci-Lucas storage, record collars, height layers, square-root lifetimes and authentic residual terms.
Block storage and record energy	Store cancellation in returns, packets, retained records and copied history	Complete mask identities and mixed work, grouped completion costs, prime transport and survival rebates. Absolute estimates applied separately can lose demonstrated cancellations.
Selected, residual and compensated energy	Compare a chosen arithmetic packet with the authentic full coefficient vector	Exact orthogonal shells, divisor square sums, additive and tensor carriers, all-prime unitary constructions and their observer costs. The observation back on the scalar orbit has a quantified obstruction.
Color/Haar/Ramanujan energies	Keep prime powers and mixed prime sectors in a positive space	Quantifies the radial product cost, anti-causal leakage, and the prime/stop commutator.
Radial stiffness and causal shell energy	Extend finite prime constructions to infinite products and causal signals	Keeps first and second moments separate, with Walsh sectors, singular limiting states and explicit smoothing compensation.
Likelihood, ANOVA and singular-tail energy	Distinguish conserved total mass from growth of quadratic observation energy	Full-period orthogonality, fractional metrics, weighted LCM boundary terms and the exact zero-capacity identity are retained.
Farey primitive and point-trace energy	Resolve rational-frequency beats and boundary cancellation	Authentic forced field, mixed-mode estimates, invariant boundedness under geometric repartitioning, and a necessary inherited-mean channel.
Prime discrepancy energy $B(X, Y) = X^{-2} \int_X^Y (\psi(x) - x)^2 dx$	Detect all-scale prime-counting fluctuation	Full reflected boards, frame covariance, innovation and hidden-wave terms; recorded RH-equivalence or sufficiency statements retain their exact normalizations.
Reflected/frame/plus energy	Restore channels omitted by a single local observable	Height-slope recurrence, increasing quadratic reserve between knots, exact future floors, gap credits and prime debits; the total all-prefix repayment remains the closure issue.
Gap/slope reserves	Pay prime impulses with the recovery between them	

Energy family	Original job	Established structure and present relevance
Theta increment and curvature energy	Establish outward amplitude growth or positivity of the complete theta interaction	Autocorrelation targets, positive extractions, Sobolev budgets, modular corrections, exact low-degree signs and full retained remainders.
Ground-state and weighted difference energies	Convert signed forms into explicit energy comparisons	Interior ground-state identities retain their source. The complete theta-weighted Weil identity is $Q(F_\theta h) = E_\Gamma + E_P - E_Z$, with an exact variance formula for E_Z .
Complete Weil energy Q_a, Q_a^-	Test the actual positivity criterion	Includes archimedean, prime, polar and boundary contributions. Small-support certificates, full domain work, null equations, multipliers and finite reductions remain available.
Suzuki norm and normalized defect $\mathcal{S}(q) = \ Vq\ _2^2, \mathcal{E}_N = \ D_N\ _{\text{HS}}^2$	Compare complete Weil energy against an unconditional positive reference	The metric and complete correction remain fixed. Extraction bounds and coordinate lifts do not alone give the full defect budget.

Unit storage bounds, all-scale arithmetic growth criteria, and normalized defect bounds refer to different quantities: a Fibonacci packet's Green-storage bound, an authentic arithmetic-energy growth criterion, and the Suzuki Hilbert-Schmidt defect budget are different assertions. An exact map or sufficient comparison is the mathematical work connecting them.

3. What the present checkpoint actually asks

Checkpoint 18 retains the complete odd Weil operator in a positive physical regularizer. In that note's normalization,

$$J_a = A_a^- + \tau_R F_a F_a^* \geq g_R I, \quad W_a = J_a^{-1} F_a, \quad K_a = \tau_R F_a^* W_a, \quad M_a = I - K_a.$$

The complete forced-family energy is

$$Q_a^-(W_a c) = \frac{1}{\tau_R} c^* K_a (I - K_a) c.$$

Because $K_a \geq 0$, nonnegativity of this finite matrix is equivalent to $K_a \leq I$, and the regularizer reduction identifies that with nonnegativity of the complete odd operator. An eigenvalue $k > 1$ gives an adverse forced direction; $k = 1$ gives a nonzero null direction. Thus earlier arithmetic energy machinery can be tested on this finite forced family, with the appropriate domain passage.

The proved evolution $K_a' = \tau_R \beta_a \beta_a^*$ almost everywhere and the Lipschitz collar law preserve the full matrix. The residual $\Gamma_a = \beta_a - M_a h_a$ belongs to a particular probe strategy. Its proposed margin-dependent bound is a sufficient route, not the definition of the overall research objective.

The older local family $P = \chi_\gamma R$ and its known-zero detection route also remains available. The historical retained-energy margin, complete theta positivity and weighted gamma-prime versus pole comparison remain separate routes. The original Suzuki criterion $\liminf_N \mathcal{E}_N \leq 1$ is a recorded sufficient criterion through its negative-witness argument, rather than a free numerical target; direct

unconditional Weil positivity is another route to the intended conclusion. None of those all-scale conclusions is claimed proved here.

Checkpoint sources: *Michaels_Physical_Regularizer_and_Boundary_Energy_Checkpoint_18.md*; *Michaels_Running_File_Ledger.md*. Earlier local-witness and defect sources are listed in the detailed theta/Suzuki catalogue.

4. Concrete reuse questions supplied by the recovered work

Existing result	Calculation now worth carrying out	What must remain in that calculation
Full theta-weighted energy identity	Set $h = q/F_\theta$ for relevant compact tests or core approximants and evaluate $E_\Gamma + E_P - E_Z$	The entire odd variance/moment pole term and the domain limit
Smith-Jordan, additive and tensor color squares	Construct coefficients from $q = W_{ac}$ and derive the actual form identity or a sufficient lower comparison	The metric, all mixed colors and the complete correction
Angle-to-Green and full prime-triple cancellation	Regroup the authentic prime packets before taking magnitudes	The terminal term and mixed-prime signs
Selected/residual, lifetime and survival energy identities	Test whether their joint-payment mechanism applies to the forced family or weighted prime differences	The original mask, inherited history, endpoint and compensation
Residual-response Schur certificate	Use an arithmetic response family and calculate its exact residual	The coercive complement and residual's full Gram matrix
Positive theta extraction and known-zero witnesses	Use their certified constants for the matching extraction or witness family	The full signed remainder; an extraction constant is not a complete energy bound
Light-cone or recurrence confinement	Identify the actual covariant state or recurrence for the present arithmetic carrier	The positive-state or bounded-response hypothesis; it cannot be assumed from an analogy

These are concrete uses of recovered machinery, not new claims that the missing inequalities already follow.

5. Catalogue guide

Detailed catalogue	Indexed entries	Scope
6. Energy identities and direct comparisons	15	E01 through E15
7. Möbius modifiers, historical storage, recurrence and compensation	51	M01 through M51
8. Prime DNA, colors, local and global energy	62	C01 through C62
9. Spirals, Fibonacci, angles and phase	34	SP01 through EX08
10. Light cones, causal arithmetic, gap reserves and operator methods	59	L01 through L59
11. Quadratic, infinity and cross-plane interfaces	21	P01 through P21

Detailed catalogue	Indexed entries	Scope
12. Theta, amplitude, spectral positivity and Weil-Suzuki machinery	59	T01 through T59

The same object can appear in multiple sections when it has distinct arithmetic, energy, or analytic roles. Entry IDs distinguish those uses; source keys are local to their section. The final coverage notes preserve unresolved historical names and corrections rather than inventing replacements.

6. Energy identities and direct comparisons

E01 – Factor-layer energy, repeated-prime energy, and cross-color energy

Made for: Measure how the logarithm of an integer is distributed among its prime factors, including both repeated copies of one prime and interactions between different primes.

Definition and recorded content: For $n = \prod_p p^{a_p} > 1$, the factor ledger uses

$$L(n) = \sum_p a_p \left(\frac{\log p}{\log n} \right)^2, \quad C(n) = 1 - \sum_p \left(\frac{a_p \log p}{\log n} \right)^2.$$

The repeated-prime part is $\sum_p a_p(a_p - 1)(\log p / \log n)^2$. These three nonnegative parts sum to one by direct expansion. The scalar deficit $1 - L$ includes both repeated-prime and mixed-color contributions; it is different from C . For prime powers $C = 0$ and $L(p^k) = 1/k$.

Implication now: This ledger already recovers the exact Weil arithmetic source:

$$\Lambda(n) = L(n) \log n \mathbf{1}_{\{C(n)=0\}}, \quad \Lambda(1) = 0.$$

The conversion to prime-power weights is available. A favorable scalar factor-energy bound is still a statement about its named quantity, unless transported into the complete Weil form.

Source: *Physics_Architectures_and_Log_Prime_Energy.pdf*, version 4, §1 and §20. See also the factor-energy and angle/Fibonacci sources in the spiral catalogue.

E02 – Authentic discrepancy block energy and its inherited mean

Made for: Measure the actual departure of prime-power counting from its continuous main term over growing windows.

Definition and recorded content: With $e(x) = \psi(x) - x$,

$$B(X, Y) = X^{-2} \int_X^Y e(x)^2 dx.$$

The exact interior-test norm is the centered energy $V(X, Y) = X^{-2} \int_X^Y |e - \bar{e}|^2$, and

$$B(X, Y) = V(X, Y) + \frac{Y - X}{X^2} |\bar{e}|^2.$$

The last term is inherited information that endpoint-vanishing tests do not see. Uniform boundedness of B is equivalent across partitions whose successive endpoint ratios remain between constants greater than one and finite upper constants.

Implication now: It prevents a local or centered test from dropping an entire energy channel. Fibonacci sampling can improve organization and resolution without automatically improving the all-scale energy bound. The same issue matters when choosing local probes for the current full form.

Source: *Physics_Architectures_and_Log_Prime_Energy.pdf*, §§1, 4, 11.

E03 – Prime-driven logarithmic field and Gaussian mixed-mode bound

Made for: Express the complete prime-discrepancy signal as an exact forced equation and control interactions among actual zero modes.

Recorded content: For $F_{\text{disc}}(u) = e^{-u/2}e(e^u)$, the distribution identity on $(0, \infty)$ is

$$(\partial_u + \frac{1}{2})F_{\text{disc}} = \sum_{p,k \geq 1} \frac{\log p}{p^{k/2}} \delta_{k \log p} - e^{u/2} du.$$

A Gaussian calculation proves a finite-window upper bound by $\sum_{\rho} |c_{\rho}|^2 \log(2 + |\gamma|)$ for arbitrary coefficients on actual zeros. For discrepancy energy it leaves the explicit factors $X^{2\beta-1}/|\rho|^2$.

Implication now: Mixed zero terms have an established quantitative bound. The half-density damping is an exact normalization; proving a uniform response still requires arithmetic cancellation or confinement. No assumption that all $\beta = 1/2$ is made by this bound.

Source: *Physics_Architectures_and_Log_Prime_Energy.pdf*, §§2-3.

E04 – Exact discrepancy-to-Weil transfer with the origin boundary

Made for: Carry the authentic prime-discrepancy field into the complete restricted Weil form.

Recorded content: For autocorrelation symmetrization r of a compact test f ,

$$Q_R(f) = A(f) - 2\|f\|_2^2 + \int_0^\infty e^{-u/2} r(u) du + \int_0^\infty F_{\text{disc}}(u) [r'(u) - r(u)/2] du.$$

The $-2\|f\|_2^2$ is the retained origin boundary. The last term is a signed linear functional of the discrepancy field. The paper gives a block Cauchy-Schwarz estimate using $B(X, Y)$.

Implication now: A direct arithmetic-energy interface to the actual Weil form already exists. The remaining application is a sufficiently sharp signed comparison for its test weights; constructing the identity again is unnecessary.

Source: *Physics_Architectures_and_Log_Prime_Energy.pdf*, §8, equations (6)-(8).

E05 – Prolate positive complement and pole-capturing finite sector

Made for: Restrict possible bad Weil directions to a finite space at each support size.

Recorded content: A time-and-frequency concentration operator supplies a finite sector and a rigorously positive infinite complement. Enlarging the retained sector by $e^{u/2}$ and $e^{-u/2}$ captures

the entire pole range, eliminating its complement and coupling contributions. Exact Schur completion retains every interaction. A weighted arithmetic bound improves the leading prime-operator norm from the coarse $4e^R$ penalty to its actual asymptotic e^R .

Implication now: This is earlier finite-dimensional reduction machinery that can be compared with CP18's physical regularizer. Its sector and constants must be identified explicitly; the all-support matrix sign remains open.

Source: *Physics_Architectures_and_Log_Prime_Energy.pdf*, §§9-10, 12-13.

E06 – Residual-response energy certificate

Made for: Improve a Schur estimate by calculating the response to the retained vectors, instead of paying a scalar inverse bound on the full coupling.

Recorded content: For $D \geq dI > 0$, a finite-column trial response $H : E \rightarrow \text{Dom } D$, and residual $R = C - DH$,

$$C^*D^{-1}C = C^*H + H^*C - H^*DH + R^*D^{-1}R.$$

Replacing only the last inverse by $d^{-1}I$ gives a sufficient certificate. Finite optimization produces a positive semidefinite improvement of the old scalar certificate; an exact rational example verifies that it can succeed when the scalar certificate fails.

Implication now: It is an existing quantitative tool for response-specific energy comparison. It offers a precise way to use an arithmetic trial family or earlier storage construction in the current finite reduction, with the residual fully displayed.

Source: *Physics_Architectures_and_Log_Prime_Energy.pdf*, §14, equations (18)-(20).

E07 – Green recovery from pole-continuum cancellation

Made for: Identify the positive historical Green kernel inside the completed Weil operator.

Recorded content: The pole kernel minus the continuous prime main term is

$$2 \cosh((u - v)/2) - e^{|u-v|/2} = e^{-|u-v|/2}.$$

At logarithmic integer nodes,

$$e^{-|\log m - \log n|/2} = \frac{\sqrt{mn}}{\max(m, n)}.$$

Thus the old reciprocal-maximum energy has an exact diagonal rescaling into this component of the full operator. The remaining gamma-plus-Green background has an explicitly negative test direction, and the standalone prime operator retains its full essential norm under finite-rank removal.

Implication now: The Green connection is already exact. It does not establish that the complete Weil form equals the historical positive energy. The signed prime correction must help in specific directions; estimating it as an arbitrarily small standalone error is insufficient.

Source: *Physics_Architectures_and_Log_Prime_Energy.pdf*, §§15-16.

E08 – Complete theta-weighted three-energy identity

Made for: Rewrite the full Weil form as a concrete competition among nonnegative difference energies using the actual positive theta weight.

Recorded content: Let F_θ be the positive even theta weight whose Fourier transform is Ξ . For $f = F_\theta h$,

$$Q_R(F_\theta h) = E_\Gamma(F_\theta, h) + E_P(F_\theta, h) - E_Z(F_\theta, h).$$

Here E_Γ is a positive singular-kernel difference energy, E_P is the sum of all prime-power translation-difference squares with the authentic $\Lambda(n)/\sqrt{n}$ weights, and E_Z is the positive pole-difference energy. The exact theta source identity cancels all diagonal terms. Multiplication by F_θ bijects the compact smooth test space onto itself.

Implication now: This is a direct energy formulation of the complete target. The needed inequality is $E_\Gamma + E_P \geq E_Z$. No new map to Weil is needed for compact smooth tests. For CP18 forced vectors, one must justify passage in the relevant domain or apply it first to a core sequence.

Source: *Physics_Architectures_and_Log_Prime_Energy.pdf*, §18, equations (33)-(38). The comparison remains open in that source.

E09 – Exact pole variance and separate parity energy targets

Made for: Reduce the subtracted pole energy to an explicit weighted variance and one odd moment.

Recorded content: With $d\nu = 2 \cosh(u/2) F_\theta(u) du$, $\nu(1) = 1$, and $\vartheta(u) = \tanh(u/2)$,

$$E_Z = \frac{1}{2} \left(\|h - \nu(h)\|_{L^2(\nu)}^2 + |\nu(\vartheta h)|^2 \right).$$

For even h only the centered variance remains. For odd h the target is $\frac{1}{2}(\|h\|_\nu^2 + |\nu(\vartheta h)|^2)$. The paper also constructs a limiting even equality direction, fixing the best possible universal coefficient in that sector.

Implication now: The current odd route has an explicit earlier energy target with its entire adverse pole contribution isolated. Weighted half-balance, prime-color squares, or forced compensation could be tested against this precise expression, rather than an unspecified negative remainder.

Source: *Physics_Architectures_and_Log_Prime_Energy.pdf*, §18.1, equation (39).

E10 – Approximate-null theta spaces and the actual precision scale

Made for: Determine how small complete Weil energies can become as support grows, and what a successful certificate must resolve.

Recorded content: Cutoffs of $F_\theta, F'_\theta, \dots, F_\theta^{(k)}$ give $(k+1)$ -dimensional spaces with full operator residual and energy bounded by $C_k \exp(-c_k e^{2R})$. The proof uses their common Ξ factor and actual zero evaluations, without assuming RH. A fixed-rank retained sector therefore cannot leave a support-independent positive complement gap.

Implication now: Small or vanishing margins are an intrinsic feature of the problem. Earlier energy methods must retain the relevant scale and growing null-like family. A support-independent positive complement gap for the unregularized Weil operator after removing only a fixed number of directions would contradict this recorded construction. This does not rule out the positive regularizer used in CP18.

Source: *Physics_Architectures_and_Log_Prime_Energy.pdf*, §17.

E11 – Earlier certified full-form energy floor on a fixed window

Made for: Prove positivity uniformly over every function in a support class using rigorous finite reduction and analytic tails.

Recorded content: The source records an interval-arithmetic certificate

$$Q_{1/2}(f) \geq 9 \cdot 10^{-8} \|f\|_2^2$$

for the complete closed form on $[-1/2, 1/2]$. Both parity sectors, the prime-2 contribution, pole couplings, quadrature error and infinite basis tails are included. This atlas reports the saved certificate; it does not rerun its executable proof.

Implication now: Preserve this earlier finite-window result alongside CP16's different stated bound and interval. Compare their precise form normalization and domain before calling either one the strongest available support result. Neither fixed interval supplies all-support positivity.

Source: *Physics_Architectures_and_Log_Prime_Energy.pdf*, §19; source references its `certificate_half` reproducibility folder.

E12 – Möbius impulse work and complete historical energy evolution

Made for: Express the original Möbius/Green energy as exact input work, including the endpoint and every signed increment.

Recorded content: With $B_N(u) = e^{-u/2} \sum_{n \leq \min(N, e^u)} \mu(n)$,

$$(\partial_u + \tfrac{1}{2})B_N = \sum_{n \leq N} \mu(n) n^{-1/2} \delta_{\log n},$$

$$R(N) = \int_{\mathbb{R}} B_N^2 du = \sum_{n \leq N} \frac{2\mu(n)M(n-1) + \mu(n)^2}{n}.$$

The full energy is nonnegative, while its exact increment j_n can have either sign. The terminal continuation is $M(N)^2/N$. The Fourier formula has the weight $(1/4 + t^2)^{-1}$ and equals the original Green norm exactly.

Implication now: This is a complete evolution of the authentic old energy, not a generic damping analogy. Any proposed arithmetic cancellation can be evaluated as actual source work, and the end mass cannot be discarded.

Source: *Michaels_Fluid_Energy_Comparison.md*, §§1-2 and Appendix 1, I1-I18.

E13 – Coarse-graining, removed energy, and exact compensation

Made for: Test whether smoothing exposes a useful energy saving without losing the original arithmetic target.

Recorded content: For full-line translation S_ℓ and average $P_\ell = (I + S_\ell)/2$,

$$V_f(\ell) = E_f - \|P_\ell B_f\|_2^2 = \tfrac{1}{4} \|B_f - S_\ell B_f\|_2^2 \geq 0.$$

For the original selected/residual target, exact reinsertion is

$$\mathcal{F}_N = \mathcal{F}_N^{(\ell)} + \tfrac{3}{4} V_\mu(\ell) - V_r(\ell).$$

A smooth-kernel variant also retains its forcing-correlation remainder. The finite diagnostics use their specified packet masks and do not prove the original all-scale target.

Implication now: Averaging or moving to another representation creates an identifiable energy correction. Its complete signed combination, rather than the saving in only one component, is what can contribute to closure.

Source: *Michaels_Fluid_Energy_Comparison.md*, §§3-5 and filter appendices.

E14 – Original retained-energy margin and selected/residual cross work

Made for: Make the earlier packet-retention strategy quantitative with its authentic mask, prime projection, compensation, and remainder.

Recorded content: With selected source $a = \mu s_N$, residual $r = \mu - a$, and the note's original named budgets,

$$\mathcal{F}_N = D^+ - G_U - \frac{13}{324}T_N + \frac{3}{4}R(N) - \|r\|_{G_N}^2.$$

It equals $D^+ - G_U - 13T_N/324 + \sum_{n \leq N} (3j_n^\mu/4 - j_n^r)$. Mixed selected/residual history terms survive even where the selected source is zero. Evolution with a fixed terminal mask differs from recomputing the mask at each intermediate cutoff.

Implication now: This is a distinct older open energy target with substantial exact work behind it. Preserve its coefficients and mask. Its required negative-margin estimate remains open. That margin should not be silently replaced by CP18's ϵ_a or the Suzuki Hilbert-Schmidt budget.

Source: *Michaels_Fluid_Energy_Comparison.md*, §4 and Appendix 1, I14-I26; cross-referenced original retained-energy material in the theta catalogue.

E15 – Physical architectures, unitary scaling, and the limits of analogy

Made for: Identify which physics-style mechanisms genuinely match the arithmetic fields and which require additional structure.

Recorded content: The half exponent is exactly the norm-preserving dilation exponent on $L^2((0, \infty), dx)$; its logarithmic generator is ordinary momentum with continuous real spectrum. The archive compares prolate spectral models, completed heat flow, Fibonacci trace maps, geodesic trace formulae and fluid energy balances. The Möbius impulse equation is a scalar forced linear equation, not an established nonlinear fluid velocity equation. The note also exhibits a mismatch between geodesic repetition weights and the required prime-power weights.

Implication now: Reuse exact scaling, impulse and energy identities directly. Spectral analogies must supply their actual arithmetic operator and trace weights before they constrain the current sign.

Source: *Physics_Architectures_and_Log_Prime_Energy.pdf*, §§5-7; *Michaels_Fluid_Energy_Comparison.md*, §§1-2 and fluid appendices. These are the archive's comparisons, not a new review of the external literature.

7. Möbius modifiers, historical storage, recurrence and compensation

Source key

Key	Exact saved source	Read location
M	Michaels_Mobius_Modifier_and_Dynamic_DNA_Sieve_Christophe_Michaels.pdf	pp.1-17; §§1-11
R	Prime_DNA_From_Ramanujan_Coordinates_to_Adaptive_Return_Blocks_Christophe_Michaels.pdf	pp.1-16; §§1-13
D	Prime_DNA_Record_Height_Descent_Christophe_Michaels.tex	complete, 514 lines
F	Prime_DNA_Five_Level_Architecture_and_Direct_Descent_2026-09-05.md	complete, §§1-12
H	Prime_DNA_Weighted_Half_Balance_and_Forced_Prime_Compensation.md	complete, §§1-9
C	Prime_DNA_Coupled_Cutoffs_and_Modified_Green_Energy_2026-09-05.md	complete, §§1-10
T	Prime_DNA_Authentic_Expansion_and_Tail_Repayment.md	complete, §§1-9
L	Prime_DNA_Causal_Lifetime_Record_First_Passage_and_Radial_Chord_Ledger_Christophe_Michaels.pdf	complete, §§1-24
P	Prime_DNA_Centered_Prism_Boundary_and_Open_Diamond_Ledger_Christophe_Michaels.pdf	complete, §§25-48
G	Prime_DNA_Recombination_Energy_and_Global_Density_Christophe_Michaels.pdf	pp.1-12 and 23-35; §§10-12 especially
B	Prime_DNA_Blueprint_Assembly_2026-09-07.md	complete, §§1-7
V40	Prime_DNA_Build_v40_The_Mobius_Rotation_and_Moving_Terminal.pdf	complete text; §§2-9 and 16-23 especially
V42	Prime_DNA_Build_v42_The_2_Over_1_Barcode_and_Parity_Innovation.pdf	complete fetched; §§1-14 read closely
Q	The_Michaels_Counting_Confinement_Hypothesis.pdf	pp.1-18; §§1-15
I	Integer_Conservation_Progress_Sept_1_2026.pdf	complete, §§1-12
IO	Deterministic_Spectral_Equilibrium_Finite_Counting_Proof_Ledger_Christophe_Michaels.pdf	complete fetched; §§1,17-24 read closely
J	Prime_DNA_Block_Gap_and_Jet_Return_Theorem_Christophe_Michaels.pdf	complete, §§1-9
W	Prime_DNA_Witness_Certificate_and_Stopping_Test.md	complete, §§1-6
S	Prime_DNA_Smoothed_Witness_and_Zero_Multiplicity.md	complete, §§1-6
U	Prime_DNA_Free_Cutoffs_and_Complete_Error.md	complete, §§1-5
A50	Build_Log_050_Global_Radical_Subtraction_Arithmetic_Interface_Graph_and_Constrained_Schur.md	complete fetched; §§0-8 read closely
Audit	Michaels_Math_Audit_Claim_Ledger.json	targeted records A05-08, EF-015, CF-IC04-06, CF-BJ01-05

M01 – Integer Conservation / exact counting-inversion law

Made for: expressing additive counting and multiplicative factor rules on one finite board. The primitive laws are successor $n \mapsto n + 1$, unique prime valuation, and the classical Möbius inverse. The exact conserved quantity is

$$\sum_{d \leq X} \mu(d) \lfloor X/d \rfloor = 1.$$

What is established: exact finite divisor cancellation, moving-board identities and positive historical storage. These use classical Möbius inversion inside the Michaels architecture. **CP18 implication:** it identifies authentic coefficients and is a source of identities with all prime descendants retained. To prove the present boundary-energy estimate it needs an explicit map from this counting observer to W_{ac} or Γ_a ; the identity alone is not that estimate. **Sources:** Q §§1-3; I0 §§1-11; B §7.1.

M02 – Michaels Counting-Confinement Hypothesis (MCCH)

Made for: turning the infinite spectral target into a finite decidable arithmetic predicate and excluding a first finite failure through a successor invariant.

$$H_K = \sum_{n=2^K}^{2^{K+1}-1} M(n)^2, \quad \forall m \geq 1 \exists A_m \in \mathbb{N} \forall K \geq 0 : H_K^m \leq A_m 2^{(2m+1)K}.$$

What is established: equivalence to all shifted critical Mertens energies being finite, hence RH; an off-line zero forces finite-board failures for every candidate constant at some precision. The proposed positive successor certificate remains open. “Gödel-style” means arithmetic encoding of a finite predicate; no incompleteness theorem is being used to settle RH. **CP18 implication:** a second exact route to the same goal. A genuine uniform counting certificate could close the goal through its established RH implication without solving CP18’s particular probe residual. The first-crossing architecture is shared in structure, not yet by an identified operator equivalence. **Source:** Q §§4-14.

M03 – Historical Green / reciprocal-maximum energy

Made for: measuring the complete signed Möbius history, including its terminal amplitude, in one positive metric:

$$R(N) = \sum_{i,j \leq N} \frac{\mu(i)\mu(j)}{\max(i,j)} = \frac{M(N)^2}{N} + \sum_{k < N} \frac{M(k)^2}{k(k+1)}.$$

What is established: positive definite kernel, explicit Euclidean cumulative isometry, zero-padding compatibility, exact energy increments. The target $R(N) = N^{o(1)}$ is an RH-equivalent growth assertion. **CP18 implication:** an established arithmetic metric for packet cancellations. It differs from the Suzuki metric; transfer must retain the actual norm and observation. Its exact history/terminal decomposition is especially useful for identifying terms lost in a proposed bridge. **Sources:** M §2; D energy section; F §§1-2.

M04 – Exceptional Michaels Möbius Modifier – the (2,3) correction

Made for: formalizing the exceptional adjacent prime pair (2,3), where there is no integer between the primes, as a compensated change of coefficients.

$$\mu_{\text{exc}}(n) = \mu(n) + \mathbf{1}_{n=2} + \mathbf{1}_{n=3}.$$

What is established: $\mu = \mu_{\text{exc}} - e_2 - e_3$; the exact Green cost is $\|e_2 + e_3\|^2 = 3/2$, so square-root energies differ by at most $\sqrt{3/2}$. It is not multiplicative and is not a new Dirichlet inverse. Its convolution law explicitly includes the added divisibility terms. The adjacent powers (8,9) already have Möbius coefficient zero. **CP18 implication:** offers a legitimate finite reorganization with a known correction; it does not erase a signed term or change the governing metric. It must be distinguished from the Michaels Möbius spiral and from classical μ . **Source:** M §3, propositions 3.2-3.3.

M05 – Michaels Möbius Modifier – adaptive endpoint compression

Made for: replacing variable-length arithmetic packets by their masses at their closing endpoints while retaining the interior information exactly.

$$\mathcal{M}_{\mathcal{P}}f = \sum_j \left(\sum_{t_{j-1} < n \leq t_j} f(n) \right) e_{t_j}, \quad f = \mathcal{M}_{\mathcal{P}}f + \eta_{\mathcal{P}}(f).$$

What is established: algebraic idempotent for a fixed partition; exact cumulative reconstruction at endpoints; residual energy equals the sum of the within-block squared histories. Blocks of length at most three give residual energy at most $4/(a+1)$ for authentic coefficients. Adaptive partition selection need not be linear; the projection is generally not Green-orthogonal. **CP18 implication:** a candidate organization of full arithmetic packets before estimating them. It supplies an exact residual audit, not automatic residual decay for the current boundary coefficient. **Source:** M §4.

M06 – Michaels Dynamic DNA Sieve / first-return partition

Made for: choosing the modifier's variable endpoints from the actual prime-factor signs, rather than imposing a fixed block length:

$$a_{j+1} = \min\{b > a_j : M(b) = M(a_j)\}.$$

What is established: every fixed baseline returns infinitely often using known two-sided Mertens oscillation; blocks have zero mass and one-signed interior excursions. A finite cutoff keeps its unfinished last block. The 29-35 block closes with exact energy (96913/3530373). **CP18 implication:** complete arithmetic compensation can be organized at natural return boundaries. To use it for CP18, the Green future-orthogonality or an analogue must be established for the actual contour/boundary pairing. Eventual closure alone supplies no unit energy cap. **Source:** M §§5-7.

M07 – Zero-mass future orthogonality and exact block storage

Made for: turning completed cancellation into an exact energy decomposition. In the max kernel a block is orthogonal to every later coordinate exactly when its coefficient sum is zero.

$$\|f\|^2 = \sum_{B \text{ completed}} E_B + \|f_{\text{unfinished}}\|^2.$$

What is established: all interactions between completed return blocks vanish; each internal occupied-depth energy remains. This is stronger than simply observing a zero endpoint. **CP18 implication:** a concrete cancellation mechanism worth testing on the current finite family. Its applicability depends on the pairing's causal factorization; it cannot be presumed for the Suzuki form. **Source:** M §5, theorems5.1-5.3.

M08 – Logarithmic board, occupied depths, and signed work

Made for: measuring how much time an excursion spends away from equilibrium, rather than recording only its height or eventual return.

$$E_B = \int_{\log a}^{\log b} |S_a(e^t)e^{-t/2}|^2 dt = \sum_{h=1}^{H_B} (2h-1)\Lambda_B(h),$$

$$E_B = \sum_{a < k \leq b} \frac{\mu(k)^2}{k} + 2 \sum_{a < k \leq b} \frac{S_a(k-1)\mu(k)}{k}.$$

What is established: exact formulas, reciprocal-location and squarefree prefix/suffix envelopes. Signed work locates the contribution of each authentic prime/composite sign. **CP18 implication:** a model for estimating a complete integrated residual while retaining its sign. No current equivalence identifies this work term with Γ_a . **Source:** M §§8-10.

M09 – Largest-prime parent tree and square-root ancestry

Made for: eliminating duplicate arithmetic paths by assigning every squarefree composite its unique largest-prime parent $n/P^+(n) \leq \sqrt{n}$. **What is established:** exact prime insertion energy update and threshold-dependent states; all finite arithmetic descendants appear once. **CP18 implication:** natural way to group the current prime-power contributions without double-counting shared descendants. A useful bound must keep the threshold and signed inherited term, not just the parent norm. **Sources:** I0 §§17-18; L §§5,9; B §3.

M10 – Forward/backward prime transport and the interior-innovation channel

Made for: isolating the truly unknown part of a parent-to-child move. For squarefree (m) , $p \nmid m$,

$$F = M(mp) - M(m), \quad B = M(mp-1) - M(m-1), \quad F - B = -2\mu(m),$$

$$C = (F + B)/2 = \sum_{m < k < mp} \mu(k), \quad F^2 + B^2 = 2C^2 + 2.$$

What is established: exact two-channel invariant with deterministic endpoint mode and signed interior mode. Aggregate control of the interior innovation remains open. **CP18 implication:** a prior exact endpoint/interior separation. It could identify which current terms are fixed by endpoint identities and which demand an arithmetic estimate. The current coefficient β_a has not been shown to equal this arithmetic (C) . **Sources:** I0 §§19-22; L §§5-8.

M11 – Fractional prime-parent transport and multiplicity polynomial

Made for: controlling overlapping prime parents while retaining parity. For squarefree ($n > 1$),

$$\mu(n) = -\omega(n)^{-1} \sum_{p|n} \mu(n/p), \quad P_X(z) = \sum_{\substack{n \leq X \\ n \text{ sf}}} z^{\omega(n)}, \quad P_X(-1) = M(X).$$

What is established: each child gets total parent mass one; exact multiplicity-level bookkeeping. Positive population at ($z=1$) and authentic signed evaluation at ($z=-1$) are distinct observations.

CP18 implication: supplies a finite prime-label state that can retain mixed interactions. A bound at the coherent signed evaluation is still needed for transfer. **Source:** I §§2-4.

M12 – Logarithmic Mertens recurrence / forced prime-power response

Made for: deriving a compulsory opposing arithmetic response when the additive history reaches a new record.

$$M(X) \log X = \sum_{n < X} M(n) \log(1 + 1/n) - \sum_{d \leq X} \Lambda(d) M(\lfloor X/d \rfloor).$$

What is established: exact convolution $\mu * \Lambda = -\mu \log$, logarithmic convex averaging of earlier values, and a forced opposite-sign compressed prime-power response at strict signed records. **CP18 implication:** a genuine sign-bearing identity built from the same Λ -source as the Weil kernel. Its log weight and moving quotient terminal must be matched to the present boundary functional. Existing records explicitly show that favorable log-endpoint sign does not automatically transfer to inverse-log lifetime sign. **Sources:** I §5; L §13; Audit CF-IC02.

M13 – Hyperbolic quotient cells / square-root critical board

Made for: registering multiplicative prime sources at their exact moving additive terminals. Cells ($\lfloor X/(n+1) \rfloor, \lfloor X/n \rfloor$) telescope the Chebyshev discrepancy and have bounded width near $n \asymp \sqrt{X}$.

What is established: exact quotient-cell aggregation and transition from essentially injective quotient geometry to collisions. An earlier unrestricted prime-indicator formula required the corrected range $\max(1, y) < m \leq y^2$. **CP18 implication:** suggests an exact arithmetic support map for prime packets. The present support parameter (a) and log-prime positions need to be explicitly registered to these quotient windows. **Sources:** I §§6-9; Audit CF-IC03-04.

M14 – Positive successor potential / neighboring half-filters

Made for: joining additive succession to prime divisibility through the fact that consecutive integers are coprime:

$$V_y(m) = 2^{-\omega_y(m)}, \quad W_y(m) = V_y(m)V_y(m+1).$$

What is established: complete-period means $\prod_{p \leq y} (1 - 1/(2p))$ and $\prod_{p \leq y} (1 - 1/p)$. Two neighboring half-strength filters combine into the full Euler exclusion factor. Localization to short moving boards remains open. **CP18 implication:** provides a positive arithmetic candidate for a comparison; it needs a localization theorem in the actual critical sector. **Sources:** I §§10-12; Audit CF-IC05-06.

M15 – Energy records, first passages, and record collars

Made for: reducing a global historical bound to the occasions on which it can grow. Every strict (R)-record is a new first passage of $|M|$. The correct anchor is $m = \tau_{h-1}$, even if that anchor is not itself an energy record. **What is established:** exact debt identity; sharp $\Delta\sqrt{S} \leq N^{-1/2}$; temporal collar under the printed orientation assumptions, including $N/(m+1) \leq 1 + (\pi^2/3)(2a+1)/(2a-1) + o(1)$. Later documents repair the earlier preceding-record ambiguity. **CP18 implication:** a prior way to restrict a hard estimate to first-failure configurations. This supports asking only for a critical-sector bound in CP18; it does not by itself equate the two first-passage problems. **Sources:** D first-passage section; G §12; Audit A05-08.

M16 – Terminal-height recovery, adaptive retained records, and exponent descent

Made for: obtaining an RH-strength global estimate from a smaller set of difficult records and a weaker lower-scale estimate. **What is established:** long records of width $> N^{2/3}$ cost at most 6 in square-root envelope; terminal heights on short records recover full historical energy up to logarithms. Further retain only

$$N - m < \sqrt{N}/(1 + S(N^{1/6}))^{5/2}.$$

A bound $M(N)^2/N \ll_{\varepsilon} N^{\varepsilon}(1 + S(N^{1/6}))^5$ on those records implies subpower growth, since $(5/6 < 1)$. The displayed arithmetic bound remains open. **CP18 implication:** demonstrates that a sufficient closing estimate can be substantially weaker than a uniform all-vector theorem. Could guide narrowing the CP18 target, while preserving its exact implication. **Sources:** D terminal-height/adaptive/conditional sections; F §§8-10.

M17 – Radial chords, prime-copy lifetime, and copied-history bank

Made for: recording when an amplitude level is created, returned, copied by a prime, and removed at a square threshold. **What is established:** a completed level chord contributes $(2h-1)(1/o-1/i)$; a copy from event (m) by prime (p) lives for $mp \leq N < p^2$. Above $\sqrt{N}$, the high-prime operator is square-zero, and its paid diagonal bank is $D_N = \sum_{p > \sqrt{N}} R(\lfloor N/p \rfloor)/p$. **CP18 implication:** finite lifetime accounting for the same arithmetic source; useful for preserving boundary deposits when translating to contour or support evolution. Current normalized boundary energy requires a proved bridge. **Source:** L §§9,11,17-18.

M18 – Quadratic Thin-Shell Excursion target

Made for: weakening a global linear/Carleson estimate to a one-sided quadratic estimate only on dangerous same-sign, boundary-hugging record excursions. For $P = N^{\theta}$, $\theta > 1/2$, $D_{N,\theta}$ lives below $N^{1-\theta}$. The target is

$$(F_{N,\theta})_+ \leq N^{o(1)}(1 + D_{N,\theta})^2.$$

What is established: the conditional recursion closes because $2(1-\theta) < 1$; sign-changing/deep returns are already square-root controlled in the record framework. The thin-shell arithmetic estimate is open. **CP18 implication:** an alternative sufficient target kept available. Its “quadratic” refers to the degree of the budget; it is distinct from the coordinate map $w = z^2$ called the quadratic plane. **Source:** L §§16,19,22-24.

M19 – Temporal pole cancellation / Brownian Riesz coordinate / open diamonds

Made for: removing the apparently adverse rank-one record direction before estimating the transverse cross-color part. $\rho_m = (m+1)e_{m+1} - me_m$ has norm 1 and represents the harmonic tail; the authentic past/future split cancels the macroscopic $a\alpha$ term exactly. **What is established:** exact Schur completion, reciprocal Brownian realization, projected bridge covariance, shared GCD core plus indefinite exclusive fork. High-prime GCD inheritance is inside the board while LCM reunion is beyond it. The complete transverse estimate remains open. **CP18 implication:** especially relevant precedent for canceling known principal contributions before applying absolute estimates. The spatial collar's Schur blocks have analogous structure; a mathematical identification, not naming, is required. **Source:** P §§26,30,40-46; note §27 record anchor and §28 ceiling remainder have later corrections in G §12.

M20 – Prime-fan projection and discretization budget

Made for: pushing full prime-copy history to small cofactors with the actual ceiling/floor terminal retained. **What is established:** the exact reciprocal-prime discrepancy projection plus remainder $|\mathcal{R}| \leq \zeta(2)/(P \log P)$. The later source explicitly retains this rather than the earlier stronger $O(N^{-1})$ assertion. **CP18 implication:** an existing quantitative error estimate; useful only after identifying the new evaluation with this finite fan. It prevents an apparent small support factor from hiding the wrong terminal discretization cost. **Source:** G §12.2, proposition 12.4; earlier P §§28-29.

M21 – Finite deleted-color / multiplicity jets and jet-return storage

Made for: replacing unlimited prime refinements by a finite hierarchy recording how many colors were deleted. A polynomial $G(z)$ has authentic point $G(-1) = R - B$, and jets $J_j = G^{(j)}(-1)/j!$. **What is established:** Pascal/Parseval gives

$$\sum_{j=0}^K \|J_j\|^2 \geq [(K+1)^2 4^K]^{-1} (\|R\|^2 + \|B\|^2), \quad K = O(\log U / \log \log U).$$

Thus a subpower amount survives somewhere in the finite family. Return to the base residual or paid storage is open. **Qualification:** Audit CF-BJ02-04 rejects the short memo's broad small-prime range and all- $a > 1$ Buchstab sign assertion; fixed-power closure is an unverified imported estimate in that memo. Preserve the narrower prefix bound and finite-jet theorem separately. **CP18 implication:** a possible hierarchy for locating residual energy that survives in mixed colors; it supplies survival, not the required signed upper bound or absorption. **Sources:** J §§2,5-7; Audit CF-BJ01-05.

M22 – Sixfold cutoff reconstruction and finite-difference endpoint

Made for: reconstructing all Möbius coefficients through N using only coefficients through $K = \lfloor N^{1/6} \rfloor$. **What is established:** with $A = \mu \mathbf{1}_{[1,K]}$,

$$\mu(n) = \sum_{j=1}^6 (-1)^{j-1} \binom{6}{j} (A^{*j} * \mathbf{1}^{*(j-1)})(n), \quad n \leq N.$$

The sixfold finite difference leaves its omitted zeroth endpoint exactly; it does not annihilate the authentic signed correlation. A mere sixth-power cost has critical exponent product1; a genuine gain is needed. **CP18 implication:** an exact small-cofactor parametrization of arithmetic source terms, preserving all mixed tuples. It can reduce representation size without automatically improving the critical norm. **Sources:** R §§4,9; F §11.

M23 – Continuous logarithmic-phase representation / high-frequency removal

Made for: placing the historical Green energy into continuous multiplicative frequency coordinates:

$$F_f(t) = \sum_{n \leq N} f(n)n^{-1/2-it}, \quad \|f\|_G^2 = \int |F_f(t)|^2 \frac{dt}{2\pi(1/4+t^2)}.$$

What is established: exact isometry and an unconditional high-frequency tail $O(\log(2N)/T + N \log(2N)/T^2)$. The central nearby-product band remains signed. **CP18 implication:** one concrete transform that might enter a metric bridge, unlike an unnamed “plane.” It already controls its high-frequency remainder; the central arithmetic interaction needs further comparison with Suzuki features. **Source:** R §§5-6.

M24 – Squarefree/Liouville separation and the full 1, 2, 3, 6 channel packet

Made for: retaining prime-square exclusions separately from parity and organizing the two smallest primes exactly. $\mu = \lambda\mu^2$; $n = ar$, $a \in \{1, 2, 3, 6\}$, $(r, 6) = 1$ produces sixteen paired channels. **What is established:** exact squarefree progression restrictions, rough-state signs, all four distinct moving cutoff windows. Their signs do not cancel matching-state correlations termwise, and finite Euler filters remain bounded away from zero on the critical line. **CP18 implication:** directly relevant to the complete mixed 2, 3, 6 packets appearing in the Weil branch. It supplies a correct bookkeeping scheme and a reason to retain within-channel terms and unequal boundaries. **Source:** R §§7-9.

M25 – Weighted Half-Balance / factor-connection graph

Made for: expressing a signed energy upper bound as a lower bound on how much actual opposite-sign compensation occurs. Odd squarefree bases form a ratio-local graph with edge weight J_N .

$$R(N) = D_N + 4W_N(1/2 - \chi_N), \quad \chi_N = C_N/W_N.$$

What is established: $\chi_N \rightarrow 1/2$ with arbitrary fixed logarithmic savings using established Möbius estimates; the RH-strength one-sided rate is $\chi_N \geq 1/2 - A_\epsilon N^{-1+\epsilon}$, still open. **CP18 implication:** one of the most explicit existing arithmetic sign comparisons. It could replace a generic norm bound if the complete CP18 forced-family form admits this weighted-graph representation. Such a map is not established in this source. **Source:** H §§1-2.

M26 – Prime-3 forced compensation bridge and many-prime capacity certificate

Made for: deriving compensation from exact prime sign laws, with capacity accounting preventing a favorable edge from being counted twice. The path $k \rightarrow \ell \rightarrow 3k$ has one compulsory opposite-sign edge; the edge weights balance at $\ell/k = 9/5$. **What is established:** a positive linear-scale compensation theorem and about 2.13% improvement of the elementary prime-2-only envelope.

Many-prime paths have a rigorous edge-capacity certificate. Slowly growing small-prime laws alone leave a quantified large deficit; no full half-budget certificate is proved. **CP18 implication:** a tangible sign-producing mechanism, relevant to combining prime packets before bounding them. The exact weighted target and all large-prime laws remain essential. **Source:** H §§3,7.

M27 – Authentic expanding cells and grouped tail repayment

Made for: testing whether complete prime insertion contracts energy and then measuring compensation over groups of cells. At cofactor depth3, the complete cell increases energy; the following depth2 and depth1 cells repay more. **What is established:** exact finite example, asymptotic signed cell constants, explicit covariance lower estimate, uniform grouped repayment for specified cofactor ranges, and support-projection margin. The remaining finer full-energy residual stays open. **CP18 implication:** powerful reason to estimate a complete arithmetic group rather than require every packet to have favorable sign. It is a potential template for a signed lower bound on the full Weil form, not a proof that the current residual is small. **Source:** T §§2-8.

M28 – Coupled divisor cutoffs and complete compensation

Made for: transporting both sides of a squared divisor sum under prime deletion without losing overlap allocations. $f_u(pr) = f_u(r) - f_{u/p}(r)$ yields a four-term quadratic update. Complementary cutoffs on $DK < n \leq N$ give

$$\mu(n) = \mu(n)(1 - f_K(n)^2) - \mu(n)^2 f_D(n) f_K(n).$$

What is established: exact cutoff transport and all mixed terms; a local strict-record innovation bound is available, while the complete mixed-scale record estimate remains open. **CP18 implication:** supplies source-preserving algebra for any proposed regrouping of its arithmetic correction. A map must include the root strip as well as changing cofactor cutoffs. **Source:** C §§2,8.

M29 – Shared stochastic prime labels, first-hit law, modified Green storage

Made for: preserving prime correlations while revealing positive squares. One Bernoulli label per prime is shared across all integers. Orthogonal chaos yields exact expected Green squares; a first-hit deletion law retains every LCM allocation. Nilpotent prime deletion gives

$$\mathcal{H}_{\eta,J}(v) - \eta \mathcal{H}_{\eta,J}(Tv) = E_0(Jv).$$

What is established: exact finite square identities and a source-energy control with subpower deletion depth. Endpoint evaluation has a cost, and authentic stochastic monotonicity has both signs in tested first-passage examples. **CP18 implication:** a way to build a finite positive completion while preserving common prime labels. The remaining source defect must still be controlled; changing the metric alone cannot supply it. **Source:** C §§3-5,9.

M30 – Green-to-LCM kernel interpolation and the signed comparison defect

Made for: moving from the hard additive-order kernel to the easier divisibility kernel through a positive family:

$$G_s(i, j) = \frac{1}{\max(i, j)} \left(\frac{\gcd(i, j)}{\min(i, j)} \right)^s.$$

What is established: every finite kernel is positive; adapt the pin to the current metric and obtain

$$I_{m,N} = F_1 + \int_0^1 (\mathcal{A}_s - \mathcal{P}_s) ds.$$

The LCM endpoint is subpower for the specified bounded divisor fields; the signed comparison integral remains. This is an exact “change of plane” with an explicit cost. **CP18 implication:** a strong candidate methodology for transporting known positive color forms to the current form: name both metrics, derive the exact signed defect, and apply an existing bound only to the quantity it actually controls. **Source:** C §§6-7.

M31 – Möbius rotation / full squarefree tensor / largest-prime decoder

Made for: making the hidden prime-composite state explicit and reconstructing the signed Möbius carrier. A constant prime-local rotation turns lossless visible/hidden outputs into absent/present states; the tensor gives every squarefree sign. **What is established:** finite decoder, orthogonal largest-prime blocks, coefficient norm $\zeta(2\sigma)/\zeta(4\sigma)$ for $\sigma > 1/2$, and a positive-to-signed tensor rotation away from the endpoint. Signed/unsigned endpoint sectors separate at $\sigma = 1/2$. The moving scalar terminal remains the RH-strength domain question. **CP18 implication:** exact structural reconstruction is already available. Rebuilding an additional hidden state is unnecessary unless it changes the actual observation estimate. To import positivity, track the terminal/evaluation map and its norm. **Source:** V40 §§2-9,20-23. Distinct from both named modifier forms and the spiral.

M32 – Arithmetic complex, 2/1 barcode, parity innovation, reverse dyadic conservation

Made for: tracking how every odd squarefree state is born at m and completed at $2m$; in logarithmic time all bars have lifetime $\log 2$. The scalar observer gives $M(X) = \sum_{m \text{ odd}, sf} \mu(m) \mathbf{1}_{[m, 2m)}(X)$. **What is established:** dyadic relative shell is a divisibility antichain, its differential vanishes, and innovation is an exact multiplicative Haar field. Forward $M(2x) - M(x)$ and reverse $2M(x) - M(2x)$ have a complete reserve identity. Vector-valued barcode energy is finite in the shifted corridor; scalar summation creates all cross terms and is the open observation problem. **CP18 implication:** explains why positive individual channels need the complete signed observer. Useful for preserving all “birth/death” and exterior terms when translating to support evolution. **Source:** V42 §§2-14.

M33 – Compact arithmetic witness / exact finite zero-exclusion certificate

Made for: allowing adjustable coefficients rather than demanding a bound for one preselected Möbius approximation. In the compact fractional-part basis,

$$E(c) = \left\| 1 - \sum c_n f_n \right\|^2, \quad E(c) \geq (2\beta - 1)/|\rho|^2$$

for an off-line zero $\rho = \beta + i\gamma$. **What is established:** exact finite Gram certificates and the classical approximation equivalence; omitted prime or mixed-product denominators give explicit positive lower barriers. Fixed finite affine self-reproduction cannot create the required convergence. **CP18 implication:** an alternate sufficient proof route with freely chosen tests. It demonstrates why a single positive finite tolerance differs from a theorem controlling every possible off-line zero; it is not the original Suzuki defect $\mathcal{E}_N$. **Source:** W §§1-6.

M34 – Smoothed Möbius witnesses and multiplicity restriction

Made for: constructing controlled variable coefficients using logarithmic tapers $\mu(n)$ $(\log(N/n)/\log N)^k$. **What is established:** small compactification correction, exact prime-power formula including descendants beyond the cutoff, and a theorem that convergence at every cutoff for fixed order k forces RH plus multiplicity at most k . The signed full norm is open. Growing-order profiles satisfy coefficient compatibility but lack proved convergence. **CP18 implication:** useful guard against imposing a stronger target than the underlying positivity goal. A specific test family can add unintended simplicity or uniformity demands. **Source:** S §§1-5.

M35 – Free cutoffs, terminal correction, and optimal Gram witnesses

Made for: preserving the weakest exact RH approximation quantifier: for every tolerance, some finite coefficients and some cutoff suffice. **What is established:** selected small scalar endpoints exist unconditionally; an explicit large-index correction makes transformed terminal energy $O(Q^{-5})$ while full error can tend to 1. Exact optimal error is $d_S^2 = 1 - b_S^T G_S^{-1} b_S$, with an exact positive projection gain when a basis vector is added. Vanishing full residual remains open. **CP18 implication:** directly relevant to the recent boundary probes: solving endpoint normalization or shrinking a projected error does not establish full-form absorption. It encourages complete Gram accounting and freedom in the actual determining family. **Source:** U §§1-4; Audit EF-015.

M36 – Global radical subtraction / arithmetic interface graph / constrained Schur gain

Made for: using a known global Weil radical to cancel the leading bulk before decomposing it into interface and smooth errors. With global k and truncation error e_λ ,

$$QW_\lambda(k_\lambda, g) = QW(e_\lambda, \text{Ext}_\lambda g).$$

What is established: exact completed subtraction, source-space labeling before reflected atoms collapse, cofinal Gaussian endpoint budgets, and a positive constrained-Schur improvement under its hypotheses. A gap-relative completed residual rate remains open. **CP18 implication:** among the closest existing mechanisms to “use what we have directly”: subtract a genuinely known null/radical identity before taking norms. Its radical k , multiplicative domain and operator are not automatically the present odd localized null vector; the exact transfer must be calculated. **Source:** A50 §§0,2,5.2,6-8 and later constrained-Schur sections.

Recovered historical energy branches – additional discovery after the user’s energy reminder

These are distinct constructions found by searching beyond the initially named theories. They should remain visible in the master inventory. The phrases “exact,” “proved,” and “open” below report the cited documents’ declared mathematical scope; they do not mean every manuscript has received a new independent proof audit in this inventory.

Key	Exact saved source	Read location
E1	Prime_DNA_Energy_Throughline.md	complete, §§1-8

Key	Exact saved source	Read location
E2	Prime_DNA_Directed_Cancellation_and_Color_Capacity.md	complete, §§1-8
E3	Prime_DNA_Remaining_Cost_and_Inherited_Energy.md	complete, §§1-8
E4	Prime_DNA_Connectivity_and_Joint_Placement.md	complete, §§1-7
E5	Prime_DNA_Signed_Overlap_and_Band_Resolution.md	complete, §§1-7
E6	Prime_DNA_Cofactor_Energy_and_Upper_Interaction.md	complete, §§1-6
E7	Michaels_Weighted_Lifetime_Bounds_2026-09-08.md	complete, §§1-7
E8	Michaels_Joint_Energy_Continuation.pdf	pp.1-8 and 19-21, frozen reference, integration I1-I13, prime route P1-P11
E9	Michaels_Survival_and_Prime_Transport_Continuation.md	main §§1-5, appended survival S1-S10 and transport T1-T6, sliding-margin review, final evidence review

M37 – The energy through-line: two inherited summaries and two-endpoint block storage

Made for: carrying the whole preceding history into a new counting block without recharging every historical coefficient. If the old cutoff is m , the block is $u = \mu \mathbf{1}_{(m,N]}$, $A = M(m)$, $B = \sum u(n)$, $h = \sum u(n)/n$, and $F = E_N(u)$, then

$$R(N) = R(m) + F + 2Ah, \quad M(N) = A + B.$$

For this update, the old packet enters through its energy and total; this does not reconstruct future arithmetic from two scalars. A sharper compression retains both block total and harmonic total. Set $s = m + 1 < N$, $\beta = 1/s - 1/N$, $c = h - B/N$, and

$$\hat{u} = (c/\beta)e_s + (B - c/\beta)e_N, \quad z = u - \hat{u}.$$

Then z has total zero and harmonic total zero. It is Green-orthogonal to every coefficient outside its own block and to its retained endpoint pair. For disjoint completed blocks,

$$R(N) = \sum_j E_N(z_j) + E_N(\text{retained endpoint packet}).$$

What is established: these exact identities and the sharp compatibility inequality $c^2 \leq \beta(F - B^2/N)$. The internal residual energies are permanent nonnegative storage; endpoint interactions remain signed. **CP18 implication:** this is a concrete example of reducing a large system to a small interface while keeping the exact internal energy. A current analogue would need the corresponding Suzuki/Weil representers and both boundary moments, rather than assuming the same two scalars suffice. **Source:** E1 §1, equations(2)-(3a).

M38 – Collective square-root packet debit and record-only energy closure

Made for: taking every prime above the square-root cutoff together, so the same prime locations and all cross terms receive one shared allowance. With $m = \lfloor \sqrt{N} \rfloor$,

$$a(n) = \mu(n) \mathbf{1}_{P^+(n) \leq m}, \quad u_0 = a|_{(m, N]}, \quad W = \sum_{m < p \leq N} \sum_{k \leq N/p} \mu(k) e_{pk}, \quad \mu = a - W.$$

Writing $h(f) = \sum_{m < n \leq N} f(n)/n$, the complete debit is

$$\mathfrak{D}_N = E_N(u_0) + 2M(m)h(u_0) + E_N(W) - 2\langle a, W \rangle_N, \quad R(N) = R(m) + \mathfrak{D}_N.$$

What is established: exact high-prime regrouping and a conditional induction: if, at every sufficiently large strict energy record, $\mathfrak{D}_N \leq A_\eta N^\eta R(\lfloor \sqrt{N} \rfloor)$ for each $\eta > 0$, then $R(N) \ll_\epsilon N^\epsilon$. The debit estimate is open and equivalent in strength to the target. The note also establishes that a constant finite ceiling for R is impossible and that the limsup of the record ratios $R(N)/R(\lfloor \sqrt{N} \rfloor)$ is at least two. **CP18 implication:** an actual closing estimate can be required only at an extremal obstruction, while all arithmetic channels are combined before a norm bound. This is a precedent for the present first-null restriction, not a proof of its residual certificate. **Source:** E1 §§2-6, equations(4)-(16).

M39 – Directed cofactor cancellation, color capacities, and minimum-cost arithmetic transport

Made for: using actual signs and divisibility restrictions to bound the prime-error field before squaring it. For $P > \sqrt{N}$,

$$\Delta_P(t) = \sum_{k \leq t/P} \mu(k) \{E(t/k) - E(P)\}, \quad E = \pi - \text{li}.$$

A zero-mass signed measure places the cofactor weights at t/k and their balancing mass at P . Directed matching turns its discrepancy into differences $E(b) - E(a)$. With independently known prime-count bounds $\ell(a, b) \leq \pi(b) - \pi(a) \leq u(a, b)$, the two directional upper costs are $u(a, b) - \int_a^b du/\log u$ and $\int_a^b du/\log u - \ell(a, b)$. Their signs are retained. Small-prime colors provide the exact capacity

$$U_W(a, b) = \sum_{d|W} \mu(d) (\lfloor b/d \rfloor - \lfloor a/d \rfloor).$$

What is established: finite directed transport/linear-program certificates, integer extrema, and the favorable adjacent orientation $\mu(k) = -1, \mu(k+1) = 1$, whose contribution is at most $t/[k(k+1) \log P]$. Summing disjoint such pairs beyond L costs at most $t/[(L+1) \log P]$. Reverse orientations, unmatched mass, activation intervals, and the terminal remain. **CP18 implication:** a method for exploiting signed arithmetic constraints rather than separately bounding every prime evaluation. Transfer requires expressing the current signed residual in such a shared count model; no such identification is yet established. **Source:** E2 §§1-7.

M40 – Exact total-count sorting and the sharp inherited-energy box certificate

Made for: quantify exactly what prime counts, masks, and directed enclosures do and do not determine. If the candidate weights are the sorted numbers $\lambda_1 \leq \dots \leq \lambda_m$, and exactly H candidate

sites are occupied, then the weighted sum lies between the sum of the smallest H and the sum of the largest H . Its upper value also equals

$$\min_{\tau} \left\{ H\tau + \sum_n (a_n - \tau)_+ \right\}.$$

For a discrepancy enclosure $\ell(t) \leq \Delta(t) \leq u(t)$, put $c = (\ell + u)/2$, $w = (u - \ell)/2$. With the authentic incoming residual r fixed, the sharp consequence of this pointwise information is

$$\|r - \Delta\|_{P,N}^2 \leq \| |r - c| + w \|_{P,N}^2.$$

What is established: the finite extrema and exact inherited accounting

$$R(N) = R(P) - M(P)^2/P + \|r - \Delta\|_{P,N}^2.$$

The certificate jointly optimizes the square and cross term. A fixed mask and one global total still leave a large placement uncertainty; a small enclosure width alone does not control displacement of the authentic center r . **CP18 implication:** a norm certificate must control the actual center and its signed error together. This is directly relevant to $\Gamma_a = \beta_a - M_a h_a$, though the current metric and feasible class must be derived independently. **Source:** E3 §§1-6, especially equations(3)-(6),(12)-(15).

M41 – One common prime placement across time, the full response Gram, and count-product constraints

Made for: prevent a worst-case optimizer from moving the same primes between different observation times. A single binary occupancy c_n on candidate sites drives every response $v_n(t) = M(t/n)$, including the cofactor-one band and terminal. With the fixed incoming profile $g(t)$,

$$U_{\text{joint}} = \max_{c \in \mathcal{C}} \left\| g - \sum_n c_n v_n \right\|_{\nu}^2, \quad d\nu = \mathbf{1}_{[P,N]} dt/t^2 + N^{-1} \delta_N.$$

The full response Gram is

$$K_{nm} = \sum_{i \leq N/n} \sum_{j \leq N/m} \frac{\mu(i)\mu(j)}{\max(ni, mj)}.$$

What is established: $U_{\text{joint}} \leq U_{\text{box}}$, strict improvement in exact finite examples, and a computable relaxation with variables $y_{nm} = c_n c_m$ constrained by the same count conservation, including $\sum_{m \neq n} y_{nm} = (H - 1)c_n$. At $P = 11, N = 40, W = 6$, the shared-trajectory tail upper is about 0.41845, versus 0.61684 for the pointwise box; both still require the identical inherited contribution. **CP18 implication:** directly supports the user's intuition that an existing shared structure can make a certificate stronger. The gain comes from retained constraints and full correlations, not a relabeling of coordinates. The needed growing-cutoff rate remains open. **Source:** E4 §§3-7.

M42 – Four-corner prime-2 factor pairing and a quantified cancellation credit

Made for: use the authentic relation $a(2n) = -a(n)$, for odd n , before taking magnitudes in insertion of a prime $p > 2$. Each complete packet has corners $n, 2n, pn, 2pn$ and cumulative response

$\mathbf{1}_{[n,2n)} - \mathbf{1}_{[pn,2pn)}$. If B counts unpaired factor intervals and C counts their forced overlaps, then

$$|V(t)| \leq B(t) - 2C(t), \quad \mathcal{G} = 4 \int C(B - C) \frac{dt}{t^2} \geq (2 - 4/p) \sum_{\substack{n \leq L/2 \\ n \text{ odd}}} \frac{a(n)^2}{n}.$$

What is established: positive finite cancellation credit while preserving the incoming upper interaction and terminal. At prime-11 insertion through 100, the bound drops from 15359/2310 to 151/110, compared with actual partial-packet energy 139/110; the exact saving is 554/105. This is the partial prime packet, not full Möbius energy through 100. Its unmatched remainder is still too large for an all-scale closure by that bound alone. **CP18 implication:** an example of a real, proved saving from complete arithmetic packets. It motivates recombining prime packets before estimating the current signed boundary term, with the current archimedean and polar companions included. **Source:** E5 §§1-2.

M43 – Counted cofactor-band diameter and square-root placement resolution

Made for: distinguish known terminal storage from unresolved internal placement. Counting prime candidates in each terminal cofactor cell $B_j = (P, N] \cap (N/(j+1), N/j]$ fixes its terminal contribution $M(j)$ times that count. For two placements with the same cell count, their prefix-count difference Z_j has exact response energy

$$\|\sigma_j\|^2 = D_j \int_{B_j} Z_j(u)^2 \frac{du}{u^2}, \quad D_j = \sum_{k \leq j} \frac{\mu(k)^2}{k}.$$

If a sub-bin contains m_I candidate sites and h_I occupied sites, its sharp prefix-count diameter is

$$d_I(u) = \min\{h_I, m_I - h_I, C_I(u), m_I - C_I(u)\}, \quad \text{diam}_j^2 = D_j \sum_{I \subset B_j} \int_I d_I(u)^2 \frac{du}{u^2}.$$

What is established: exact single-cell uncertainty and a valid center-radius certificate. Under separately supplied local count bounds, bin width of square-root order, up to logarithms, can make the upper-band placement uncertainty bounded. The authentic center and cross-cell signed interactions still need estimates. Fixed coarse counts can force a robust upper certificate of order $N/\log^2 N$, even with the terminal fixed; that is a limitation of the information class, not such a lower bound for authentic energy. **CP18 implication:** identifies a quantitative resolution obligation for an attempted discretization. Fixed-contour convergence or finitely many boundary samples need a norm-resolution theorem before they can certify a growing family. **Source:** E5 §§3-7.

M44 – Complete cofactor flow, causal interaction energy, and the harmonic-return sign certificate

Made for: bound an entire inserted lower cofactor flow from a smaller cutoff, then estimate only the part that can actually interact with the incoming upper packet. For $p \leq N < p^2$, $L = \lfloor N/p \rfloor$, $x = \mu|_{[1,L]}$, $v = x - U_p x$, $R = E_L(x)$, $A = M(L)$, $m = \sum_{n \leq L} \mu(n)/n$,

$$E_N(v) = (1 + 1/p)R - 2Am/p, \quad (1 - \sqrt{L}/p)R \leq E_N(v) \leq (1 + \sqrt{L}/p)R.$$

For the upper cumulative profile T , the interaction is only on $[s, R_*] = [L + 1, pL]$. Its accessible flow energy is

$$J = A^2/(L + 1) + (R - 2Am)/p.$$

With $\beta = 1/s - 1/R_*$, $b_0 = A/s - m/p$, $h = \int_s^{R_*} T(t)dt/t^2$, and a valid local energy budget D_0 ,

$$\langle u, v \rangle \leq hb_0/\beta + \sqrt{(D_0 - h^2/\beta)(J - b_0^2/\beta)}.$$

What is established: the sharp lower-scale flow bound and sign-sensitive projection certificate; at $p = 11$, $N = 100$, it certifies a negative interaction at most $-99/250$. Sharp refinements preserve the fixed prefix, incoming energy, and outgoing terminal jointly. Authentic interactions can also be positive, so the sign condition is not automatic. **CP18 implication:** a direct reusable strategy: remove only components proved orthogonal, condition on a small number of actual shared moments, and preserve signed mixed terms. Its current implementation needs the actual M_a metric and boundary representers. **Source:** E6 §§1-6, equations(1)-(8).

M45 – Odd-weight level storage and square-root height/lifetime absorption

Made for: explain where historical energy resides using actual outward steps of $Q = |M|$ and their first downcrossings. A level born at u , height h , has weight $w = 2h - 1$. If it closes at $d \leq N$, it stores $w(1/u - 1/d)$. If still open, its full cost is w/u , consisting of observed history plus the terminal w/N .

$$R(N) = \sum_{\text{observed}} w(1/u - 1/v) + N^{-1} \sum_{\text{open}} w, \quad v = d \text{ or } N.$$

What is established: low-born levels $h^2 \leq u$ cost at most H_N , regardless of lifetime. Young histories are absorbed by $2\sqrt{2R(N)H_N}$. Young open endpoints are also paid, giving

$$R(N) \leq 30H_N + 2\mathfrak{B}_N^{\text{old}}.$$

Here $\mathfrak{B}_N^{\text{old}}$ is the full contribution only of levels both high at birth, $h^2 > u$, and old at observation, $(v - u)^2 > u$. For nondecreasing $L \geq 1$, the threshold $\sqrt{u}L(u)$ gives $R(N) \leq 30L(N)^2H_N + 2\mathfrak{B}_{L,N}^{\text{old}}$. The surviving high-old arithmetic budget is open. **CP18 implication:** the square-root work is more than an analogy: it proved that a large part of the original energy can be paid by a logarithmic budget while retaining every open endpoint. A present absorption proof should likewise name the already-paid positive portion and the exact remainder. **Source:** E7 §§1-4b, equations(1b),(3)-(6d).

M46 – Arithmetic lifetime counting, adaptive offset averaging, and height-survival correlation

Made for: insert actual Möbius short-interval information into the level-storage picture without treating adaptive openings as random. For openings $X \leq u < 2X$, heights $H \leq h < 2H$, and lifetimes $T \leq d - u < 2T$, the reported arithmetic transfer gives

$$\# \mathcal{P}(X, H, T) \ll \min\{XH/T, X/(\log T)^{1/200}\},$$

$$\sum_{\mathcal{P}} c(u, d, h) \ll \min\{H^2/X, HT/[X(\log T)^{1/200}]\}.$$

What is established: a transfer of the cited Matomäki-Radziwiłł short-interval theorem through dyadic oscillation and averaging over all offsets of a partition, with its ranges and unspecified constants. The unpaid statistic is the *weighted* survivor count

$$\mathcal{Q}(X, T) = \sum_{\substack{X \leq u < 2X \\ Q(u) = Q(u-1) + 1}} (2Q(u) - 1) \mathbf{1}_{\{\min_{0 \leq j < T} Q(u+j) \geq Q(u)\}}.$$

The inherited height and future survival must be controlled together; unweighted density alone does not do this. **CP18 implication:** a useful model for applying literature theorems to a state selected by an extremal argument: preserve the adaptive selection and weights, and prove the transfer. It does not authorize an unweighted average theorem at a chosen null vector. **Source:** E7 §§5-7.

M47 – Joint completed-block/open-stack energy and survival-compensated birth moments

Made for: keep completed discounts, active closure rebates, and unfinished terminal weights in one inequality with one shared allowance. At a fixed η -weighted energy record, the selected completed costs D^+ and short-open histories H_{short} satisfy

$$D^+ + H_{\text{short}} \leq \text{err}_N R(N), \quad \text{err}_N \rightarrow 0.$$

For a family F of remaining completed levels and selected opens, let c_i be their final full costs, $C_F = \sum c_i$, $B_Y = \sum_{u_i \leq Y} c_i$, and $Z_Y = N^{-1} \sum_{u_i > Y} u_i c_i$. At a birth u_j , V_j is the sum of w_i/d_i over selected completed packets active then. With $p = 1 + 1/\eta$,

$$C_F^p \leq B_Y^p + pR(N)^{p-1} Z_Y - \Pi_Y, \quad \Pi_Y = p \sum_{u_j > Y} c_j \{R(u_j)^{p-1} - (R(u_j) - V_j)^{p-1}\} \geq 0.$$

What is established: the jointly paid arithmetic history bound and exact survival-compensated birth inequality. Eight valid bound variants share the same single error allowance; taking their minimum does not create extra discounts. The uniform arithmetic bound on that minimum, $\Phi_N \leq 1/4 + C_{\eta, \varepsilon} N^\varepsilon / R$, remains open. **CP18 implication:** the existing research already contains a disciplined method for combining available energy estimates without paying the same allowance twice. Its concrete old target differs from CP18, but the bookkeeping principle is directly relevant to combining “closing groups.” **Source:** E8 frozen reference and integration I1-I12.

M48 – Terminally fixed masks, rank-gap compensation, and signed prime-edge work

Made for: retain the genuine selected arithmetic remainder after completed/open packets are removed, without falsely assuming it remains multiplicative. For the selected endpoint vector $a(n) = \mu(n)s(n)$, $s(n) \in \{0, 1\}$, let $r = \mu - a$. If $I(k)$ is its set of active heights and $q(k) = |I(k)|$, then

$$g(k) = 2 \sum_{h \in I(k)} (Q(k) - h) - q(k)(q(k) - 1) \geq 0, \quad G_U = \sum_{k < N} \frac{g(k)}{k(k+1)} + \frac{g(N)}{N}.$$

The complete selected energy is

$$L^+ + O = R - \|r\|_G^2 - G_U.$$

The same original prime projection is preserved, so $\|Pr\|_G^2 = (b-c)^T \Gamma^{-1} (b-c)$, including its mixed term, square, and complementary norm. Prime transport acquires the exact mask defect

$$r(pm) = -r(m) - \mu(m)[s(m) - s(pm)].$$

If $Z(k) = \sum_{n \leq k} r(n)$, the corresponding residual energy is $\|r\|_G^2 = \Delta_r + 2S_r - 2P_r - 2E_s$, where the mask work is

$$E_s = \sum_{p > D} \sum_{m \leq N/p} \frac{\mu(m)[s(m) - s(pm)]Z(pm-1)}{pm}.$$

What is established: exact identities with the authentic mask and original metric. G_U is non-negative but can be zero for top consecutive active heights. The weighted mask correlation has no proved closing sign. The later bilinear attempt cancels a nonsquarefree correction on the actual endpoint mask, but still pays weight variation and missing balanced-divisor support; a coefficient mask is not automatically a Green contraction. **CP18 implication:** this is a particularly important retained tool: subtracting solved pieces changes the arithmetic rules of the remainder. Any current probe construction should carry its corresponding mask/commutator defect explicitly and exploit real rank-gap compensation before applying broad norms. **Source:** E8 prime route P1-P11 and integration I13, pp.7-8,19-21.

M49 – Old-stack survival rebate and exact ancestor energy

Made for: improve an inherited prefix allowance using actual later closures rather than treating all old energy as permanently available. Define

$$\mathcal{R}_Y = \sum_{u_i \leq Y < d_i \leq N} \frac{2h_i - 1}{d_i}, \quad q_Y(k) = \min_{Y \leq t \leq k} |M(t)|.$$

The exact final cost of every packet born by Y is

$$B_{\text{all}}(Y; N) = R(Y) - \mathcal{R}_Y.$$

With $\Delta_Y(k) = Q(Y)^2 - q_Y(k)^2$,

$$\mathcal{R}_Y \geq \Delta_Y(Z)/Z + [\Delta_Y(N) - \Delta_Y(Z)]/N \quad (Y \leq Z \leq N).$$

What is established: the survival formula, full terminal identity, and sharpened ancestor allowance $D_{\text{vl}} \leq R(Y_b) - \mathcal{R}_{Y_b}$. A rebate is used once inside that allowance. Its needed arithmetic lower rate is open. The source's main summary equation(3) omits a plus sign in the displayed transcription; the appended S3 derivation and final review give the two-term sum written here. **CP18 implication:** stored energy can have an exact, measurable repayment mechanism. The appropriate current question is whether boundary/probe storage has a similarly forced rebate on dangerous directions, rather than merely another small absolute error. **Source:** E9 main §1; appended survival S1-S7 and final review.

M50 – All-prime endpoint transport and the exact price of separating prime channels

Made for: route selected energy through authentic prime factors while measuring exactly which cancellations are lost by separate magnitude estimates. The all-prime-power law

$$\mu(n) \log n = - \sum_{p^a | n} (\log p) \mu(n/p^a)$$

leads, at squarefree endpoints, to weights $\alpha_p(n) = \mathbf{1}_{p|n} \log p / \log n$ summing to one. For a completed height- h packet $u \rightarrow d$, set $A = (2h - 1)/u$, $B = (2h - 1)/d$, $g = \gcd(u, d)$. Its genuine paired cost is $A - B$, whereas

$$\sum_p |A\alpha_p(u) - B\alpha_p(d)| = A + B - 2B \frac{\log g}{\log d}.$$

The exact extra charge is $2B \log(d/g) / \log d$. **What is established:** finite factor transport and its precise separation loss, with squareful cancellations retained. For the actual $95 \rightarrow 97$ packet, separate prime-channel magnitudes are 96 times the paired cost. This is a proved instance of the user's concern: splitting a fully coupled arithmetic object before estimating it can destroy a large known cancellation. **CP18 implication:** this does not itself estimate the new residual, but it strongly motivates retaining the whole signed arithmetic boundary expression before taking color-wise norms. This gives a precise reason to retain the coupled expression before taking magnitudes. **Source:** E9 main §2 and appended transport T1-T6 and following packet calculation.

M51 – Sliding negative margins and exact weighted-record endpoint credit

Made for: determine how much strict negativity is actually required, avoiding an unnecessarily strong fixed-percentage target. The exact event identity gives

$$M(N)^2 = NR(N) - \sum_{k < N} R(k).$$

At an η -weighted record this yields $T_N/R(N) \geq \eta/(1 + \eta)$. The survival-refined ancestor error obeys the stated rate $O_{\eta,b}((1 + \log N)^{-b\eta/4})$. Therefore a correctly signed negative margin $c_\eta(1 + \log N)^{-\alpha}R(N)$, for fixed $0 < \alpha < b\eta/4$, is sufficient for the old contradiction chain. With the unchanged original $b = 1/(400\eta)$, $\alpha = 1/3200$ is one admissible choice. **What is established:** the exact endpoint credit and the conditional rate comparison. The arithmetic inequality that supplies the shrinking negative margin remains open; changing b would change the selected family and cannot be done silently. **CP18 implication:** useful strategic precedent: ask for the weakest quantitative margin that prevents a first crossing. It supports investigating the actual Osgood threshold already identified in CP18, without conflating the old logarithmic margin with the new boundary-residual bound. **Source:** E9 main §3; appended survival S8-S10 and sliding-margin review.

Twelve milestones in the recovered historical energy progression

This progression is a source-based reconstruction, not a claim that the later energy objects are all identical. The current Suzuki/Weil operator energy must retain its own metric.

Stage / source	Exact construction or definition	Purpose and established scope	Unresolved transfer or estimate
1. Original counting/Green program – M, I0, D	$R(N) = \sum_{i,j \leq N} \frac{\mu(i)\mu(j)}{\max(i,j)}$ $= J_N + \frac{M(N)^2}{N}$	One positive energy for all signed history and the terminal; exact counting and event updates.	Subpower growth $R(N) \ll_\varepsilon N^\varepsilon$.

Stage / source	Exact construction or definition	Purpose and established scope	Unresolved transfer or estimate
2. Modifier/return-block program – M, R	Replace each block by its mass at its endpoint and retain the interior residual; zero-mass completed blocks are orthogonal to later coefficients.	Exact local cancellation and retained internal storage.	A uniform arithmetic budget for the actual block family.
3. Record-height and causal-lifetime program – D, L, P	First passage τ_h , historical record energy, copied-history bank, and full quadratic remainder.	Limits the obstruction to correctly anchored records; distinguishes present/absent level occupation.	Thin-shell, coupled history, and terminal-height estimates.
4. Coupled compensation – 5 September, C and T	Modified Green storage with simultaneous cutoff compensation and complete grouped repayment.	Retains common arithmetic inputs, negative tail credit, and boundary correction instead of separate allowances.	The authentic joint repayment rate; compensation is not free deletion.
5. Directed counts and joint occupancy – 6 September, E2-E4	$\Delta_P(t)$, exact wheel capacities, sorted fixed-count extrema, one c_n across all t , full response Gram.	Converts factor/color consistency into sharp finite certificates; shared placement can strictly improve the bound.	Growing-cutoff control of the authentic center and signed placement.
6. Forced overlap and cofactor interaction – 6 September, E5-E6	Four-corner credit $\mathcal{G}$, lower-cutoff flow $x - U_p x$, and harmonic-return projection.	Proved arithmetic cancellation and a finite negative mixed interaction with inherited storage retained.	Uniform rate across all arithmetic cutoffs and complete upper packets.
7. Collective through-line – 6 September, E1	$R(N) = R(m) + F + 2M(m)h$, two-endpoint orthogonal residuals, square-root debit $\mathfrak{D}_N$.	Compresses old interaction exactly; one collective record-only estimate would close induction.	The collective signed debit bound remains open; a constant ceiling is ruled out.
8. Weighted level lifetime – 8 September, E7	Odd weights $2h - 1$, completed cost $w(1/u - 1/d)$, open cost w/u , high-old remainder $\mathfrak{B}_N^{\text{old}}$.	Proves $R \leq 30H_N + 2\mathfrak{B}_N^{\text{old}}$, paying low and young storage.	High inherited heights correlated with long survival.
9. Arithmetic lifetime transfer – E7 §5	Dyadic oscillation, all-offset averaging, lifetime count $O(X/(\log T)^{1/200})$.	Applies an actual short-interval theorem to adaptively selected arithmetic levels.	Weighted survivor cost and open endpoints at the required scale.
10. Joint completed/open continuation – 13 September, E8	$D^+ + H_{\text{short}} \leq \text{err}_N R$; $C_F^p \leq B_Y^p + pR^{p-1}Z_Y - \Pi_Y$.	One shared allowance, full open endpoints, active survival rebates, no double counting.	The selected joint bound Φ_N and original signed margin.
11. Authentic-mask and same-metric continuation – E8 prime route	$a = \mu s$, $r = \mu - a$, $L^+ + O = R - \ r\ _G^2 - G_U$, and $\mu(m)[s(m) - s(pm)]$.	Exact residual arithmetic, Gram inverse, rank-gap compensation, prime-edge correction.	Weighted mask work; no automatic multiplicativity or mask contraction.
12. Survival and all-prime transport – 14 September, E9	$B_{\text{all}}(Y; N) = R(Y) - \mathcal{R}_Y$; separation loss $2B \log(d/g)/\log d$; sliding margin $(1 + \log N)^{-\alpha} R$.	Repays old storage exactly, quantifies lost cross-color cancellation, weakens the sufficient margin.	The authentic negative joint estimate remains open; no automatic transfer to CP18.

Additional connections

- **Fibonacci-Lucas packet:** B §4 and G §10 prove its explicit storage recurrence $E_r < 1$, preserved by prime-word refinement. Its comparison to the authentic Möbius vector leaves a positive residual square $S_r = R(N_r) + O(\log N_r)$. The linear cross term already has a logarithmic bound. Distinguish this packet from Fibonacci sampling of the prime-discrepancy zero mode and from Fib* spiral geometry.
- **Authentic Fibonacci-prime discrepancy blocks:** B §4 recovers equation63 of *Prime Sampling and Energy Floors*: $B_{m+1} = k_m(B_m + 2C_m + D_m)$, $k_m \rightarrow 1/\varphi$, with sufficient open condition $2C_m + D_m \leq \eta B_m + H$, $\eta < \varphi - 1$.
- **Angles and factor crossings:** B §2 recovers the exact max kernel from a crossing $n = ab$ and direction $d = a - b$: $1/\max(a, b) = 2/(\sqrt{d^2 + 4n} + |d|)$. This is an actual coordinate map, not just a geometric picture.
- **Prime-anchored Weil successor:** B §5 gives geometric-midpoint allocation of newly entering prime powers and a full three-direction parity Schur calculation; directly relevant to the present support problem, though full source inequalities remain open.
- **Global exclusion-density invariant / prime-power Haar:** covered by the colors catalogue; distinguish its finite total from the divergent isolated row sum $\sum_p 1/p$.
- **Mersenne/subprime concepts:** no formal “subprime” theory source was recovered in this source set. *Michaels_Faster_Prime_Search.md*, concerns exact support/rejection for computational prime searches and classical Mersenne factor restrictions; do not silently equate it with the user’s earlier “Mersenne modulus” language.

Coverage gaps and cautions for the combined atlas

1. Do not merge four different Möbius objects: classical Möbius coefficient, named exceptional modifier, named endpoint modifier/dynamic sieve, and Möbius spiral/rotation.
2. “Quadratic” appears in quadratic forms, degree-two thin-shell budget, tensor vectorization, and $w = z^2$; these have different definitions and transfer costs.
3. Every current implication above is either a direct retained source identity or an explicitly proposed CP18 application. No retrieved source establishes that all these metrics are unitarily equivalent to the Suzuki completion.
4. Later source corrections take precedence over earlier status labels: record collar anchor, prime-fan ceiling remainder, the scope of fixed-power block-gap claims, and sieve indicator range were all corrected.
5. Standalone dates and build numbers overlap across parallel branches. Treat source filename, formula and metric as identity, rather than interpreting a larger build number as superseding every earlier branch.

8. Prime DNA, colors, local and global energy

Source key

- **DK – Michaels_Prime_Color_Dual_Kernel_Bridge.md**, current version1, §§1-14 read. The late continuation is particularly important because it establishes the singular-orbit obstruction and the valid Poisson interface.

- **CA** – Christophe_Michaels_Prime_DNA_The_Connected_Architecture.pdf, pp1-30 read, with foundational detailed use pp6-15. Dependency synthesis through 27 August; distinguishes exact, certified, experiment, no-go, conditional and open lanes.
- **RE** – Prime_DNA_Recombination_Energy_and_Global_Density_Christophe_Michaels.pdf, pp1-23 read. Detailed proofs of complete-period colors, finite counting, density and source/observer distinction.
- **BP** – Prime_DNA_Blueprint_Assembly_2026-09-07.md, all 429 lines read.
- **FP** – Factor_Potential_to_Weil_Covariance_Ledger_2026-08-23.md, §§1-11 read.
- **AF** – Prime_DNA_Angle_Families_and_Prime_Triple_Bound.md, full file read.
- **CP** – Prime_DNA_Centered_Prism_Boundary_and_Open_Diamond_Ledger_Christophe_Michaels.pdf, all 9 pages read, sections 25-48.
- **TP** – Prime_DNA_Build_v39_The_Two_Plane_Prime_Cocycle.pdf, all 21 pages read.
- **MC** – Prime_DNA_Build_v34_Moving_Color_Filtration.pdf, all 10 pages read.
- **GO** – Prime_DNA_Geometric_Invariants_and_Local_Overlap_Law.md, full 356 lines read.
- **PS** – Prime_DNA_Build_v32_Prime_State_Observability.pdf, all 15 pages read.
- **IS** – Prime_DNA_Build_v33_Infinite_Stiffness_and_Exceptional_Orbit.pdf, all 17 pages read.
- **ST** – Prime_DNA_Build_v38_The_Singular_Tail_State.pdf, all 22 pages read.
- **FB** – Prime_DNA_Build_v34_Farey_Beat_Geometry_and_Causal_Boundary.pdf, all 20 pages read. This is distinct from the moving-color-filtration v34.

C01 – Prime Color / Prime DNA arithmetic alphabet

Made for / definition: Retain each integer's prime multiplicities rather than compressing every composite into one undifferentiated event. Coordinates are $\nu(n) = (\nu_p(n))_p$; multiplication adds vectors and divisibility is coordinatewise ancestry.

Recorded result and scope: Exact unique-factorization organization. Early display taxonomy: Q ordinary primes, G proper prime powers, P squarefree semiprimes, Y mixed two-prime integers, L squarefree three-prime products, M first mixed three-prime layer.

Implication now: Gives a source-indexing system for every term before the full quadratic form is evaluated. CP18's signed terms can be labeled by their authentic color origin.

Source: DK§1; CA§1

C02 – Distinct-color count, multiplicity count, and nested squarefree layers

Made for / definition: Separate breadth $\omega(n)$ from depth $\Omega(n)$; $R_k(n) = \prod_{\nu_p(n) \geq k} p$ gives nested layers with $n = \prod_k R_k(n)$.

Recorded result and scope: Exact lossless decomposition. Repetition remains information even when Möbius evaluation vanishes.

Implication now: Ensures repeated prime powers are retained in source derivatives and cross terms, rather than treated as absent because $\mu(n) = 0$.

Source: DK§1

C03 – Factor potential / logarithmic source extraction

Made for / definition: Set $\Phi(n) = \Omega(n) - 1$ for $n \geq 2$, and $\Phi(1) = 0$. For $n \geq 2$, recover prime-power events by $Q_\Phi(n) = (\mu * \Phi)(n) - \mu(n)$, then $\Lambda_\Phi(n) = Q_\Phi(n) \log n / (\Phi(n) + 1)$; extend both source functions by 0 at 1.

Recorded result and scope: Exactly $Q_\Phi = 1_{n=p^r}$ and $\Lambda_\Phi = \Lambda$, with their zero unit values. The displayed convolution formula applies only for $n \geq 2$: at 1 it would otherwise give -1 . Factor multiplicities are inputs.

Implication now: Directly identifies the arithmetic coefficients in the complete Weil operator; this is a genuine exact bridge already available, not a new sign assumption.

Source: FP§2

C04 – Genesis/Jordan deformation

Made for / definition: Place color birth in a continuous parameter: $J_z(n) = n^z \prod_{p|n} (1 - p^{-z})$.

Recorded result and scope: $\text{ord}_0 J_z(n) = \omega(n)$, $J'_0 = \Lambda$, first nonzero derivative $\omega(n)! \prod_{p|n} \log p$. Higher derivatives retain multiplicity.

Implication now: Organizes one-, two-, and higher-color terms by order, allowing a complete polynomial/matrix expansion without suppressing mixed sectors.

Source: DK§§6-7; CA§2

C05 – All-color block-depth invariant

Made for / definition: Recover factor architecture from convolution powers $B_j = \Lambda^{*j}$, interpreted as ordered prime-power block factorizations.

Recorded result and scope: $B_j(n) > 0$ exactly when $\omega(n) \leq j \leq \Omega(n)$.

Implication now: Supplies exact grading of higher prime packets. Does not alone assign a favorable sign to their completed scalar interaction.

Source: CA§3

C06 – Prime powers as connected cumulants; mixed colors as reconstruction

Made for / definition: Link additive prime-power source to full multiplicative family by Dirichlet-convolution logarithm/exponential.

Recorded result and scope: $J_z = \exp_{*D}(\int_0^z n^u \Lambda(n) du)$; logarithmic coefficients are supported exactly on prime powers.

Implication now: A way to reconstruct all mixed terms from the prime-power source and verify completeness of a proposed factorization on W_{ac} .

Source: DK§6

C07 – Smith-Jordan/GCD positive carrier and color-to-coercivity bridge

Made for / definition: For real $z > 0$, build a positive finite metric from actual divisibility: $K_z(m, n) = \gcd(m, n)^z / (mn)^{z/2}$.

Recorded result and scope: Exact divisor square sum $\sum_d J_z(d) |\sum_{d|n} x_n / n^{z/2}|^2 > 0$ for every nonzero finitely supported complex vector x . Finite determinant increment $\det K_{z,N} / \det K_{z,N-1} = J_z(N) / N^z$.

Implication now: A substantive positive arithmetic metric candidate. Applying its positivity to Weil energy requires an exact map or lower comparison on the current forced family.

Source: DK§7; CA§2

C08 – Genesis prime-power tangent

Made for / definition: Differentiate the GCD metric at parameter zero to isolate the first-order prime source.

Recorded result and scope: For $\sum_n x_n = 0$, tangent is $\sum_{p^a} \log p |\sum_{p^a|n} x_n|^2 \geq 0$; mixed-color vectors can lie in its nullspace.

Implication now: Offers a genuine positive tangent, with exact nullspace issue to check against a proposed Weil null vector.

Source: DK§7

C09 – Haar prime-power shells / half-density energy ladder

Made for / definition: Orthogonalize one prime's nested divisibility events: $h_{p,a} = 1_{p^a|r} - p^{-1} 1_{p^{a-1}|r}$.

Recorded result and scope: Full-period shell energy $(1 - p^{-1})p^{-a}$; all depths sum to $1/p$. Prime 2 first shell is $1/4$ and its entire row is $1/2$.

Implication now: Supplies exact local energy, including tails. The relevant complete-period measure must travel with any transfer to Suzuki/Weil space.

Source: RE§3; DK§2

C10 – Ramanujan color tensors and Möbius observation

Made for / definition: Combine independent prime shells using CRT, retaining every repeated and mixed color.

Recorded result and scope: $h_n = c_n^{\text{Ram}} / n$, orthogonal norm $\varphi(n) / n^2$, and $c_n^{\text{Ram}}(1) = \mu(n)$. Arbitrary coefficient positivity follows in the full-period space.

Implication now: Explains how signed arithmetic can be an observation of positive color energy. Fixed-residue observation is an extra operation whose size must be estimated.

Source: RE§3; DK§8

C11 – Prime2/Genesis partition and exact counting anchor

Made for / definition: Use each prime as a permanent exclusion rule, with 2 the first even/odd partition.

Recorded result and scope: Classical exact binary prime indicator through $\sqrt{n}$; exact multiplicity identity $\log N! = \sum_{p^a \leq N} \lfloor N/p^a \rfloor \log p$.

Implication now: Verifies that all arithmetic source coefficients are tied to ordinary integer counting. Counts alone do not give the required sign of the complete quadratic form.

Source: RE§2.2

C12 – Exact earlier-rule prime counting and the five surviving factor layers

Made for / definition: Calculate primes by activating smaller prime rules and counting remaining mixed products.

Recorded result and scope: $\Phi(x, a) = \Phi(x, a - 1) - \Phi(\lfloor x/p_a \rfloor, a - 1)$. At $K = \lfloor N^{1/6} \rfloor$, remaining integers have at most 5 prime factors; exact subtraction of layers 2-5 gives $\pi(N)$.

Implication now: Can enumerate packets and verify source support efficiently in a finite proof. This is the precise recovered “four/five layers” result; avoid asserting it is definitely the remembered “nuts” label.

Source: RE§2.1; BP§1

C13 – Wheel deletion / Jenga prime ghost

Made for / definition: Track what changes when a new prime rule removes survivors.

Recorded result and scope: $C_{k+1}(s) = (1 - r^{-s})C_k(s)$; logarithmic differentiation produces the whole tower $\log r / (r^s - 1)$. Survivor ledger plus accumulated prime tower remains invariant.

Implication now: Exact source payment across prime thresholds; relevant to CP18 collar completeness and grouping a prime’s whole power ray before estimates.

Source: CA§5

C14 – Prime-power ramp / slope-hinge realization

Made for / definition: Convert a prime tower into physical logarithmic time.

Recorded result and scope: $R_r(t) = (\log r) \sum_{a \geq 1} r^{-a/2} (t - a \log r)_+$; its Laplace transform is explicit.

Implication now: Expresses discrete arithmetic source events in continuous kernels, connecting color rules with the boundary/convolution realization already used in CP18.

Source: CA§5

C15 – Möbius closure of the color cube

Made for / definition: Understand why mixed composites disappear from the first logarithmic derivative while remaining in higher structure.

Recorded result and scope: $\Lambda(n) = \sum_{d|n} \mu(d) \log(n/d)$; the alternating divisor cube cancels unless n is a prime power.

Implication now: Useful exact cancellation to perform before bounding prime packet interactions. Higher-order mixed terms remain necessary.

Source: CA§6; DK§6

C16 – Jenga zero omission / ghosts and finite zero-surgery rigidity

Made for / definition: Test the effect of withholding or moving zero channels in the explicit formula.

Recorded result and scope: Specified omitted-zero response is exact; plotted finite reconstructions are experiments. A finite moved symmetry-complete quartet creates an absolutely continuous disturbance, incompatible with an unchanged pure-point prime comb.

Implication now: Can rule out a finite source counterfeiting move. It does not rule out a globally reorganized off-critical zero configuration.

Source: CA§§7-8

C17 – Source uniqueness, no counterfeit, no-dark state

Made for / definition: Ensure the completed source actually represents the targeted spectral modes.

Recorded result and scope: Same normalized logarithmic derivative determines the function; a nonzero first-loss state cannot be silent in the authentic exterior terminal in the stated theorem.

Implication now: Relevant prerequisite for using a positive carrier to exclude nulls: the witness must remain visible under the exact map.

Source: CA§9

C18 – Causal exponential sweep and sign-preserving reservoir

Made for / definition: Smooth/transport the completed source while retaining all Weil directions.

Recorded result and scope: $C_{K_\alpha} = (2/\alpha)V_\alpha^*C_WV_\alpha$ and $q_{K_\alpha}[(\alpha + \partial)h] = (2/\alpha)Q_W[h]$.

Implication now: A legitimate exact coordinate transport, including negative witnesses. Reservoir regularity alone does not create positivity.

Source: CA§9

C19 – Two-plane prime cocycle / 2-over-1 law

Made for / definition: Complete one signed prime update with a complementary positive hidden channel.

Recorded result and scope: With $r = p^{-\sigma}$, $V = (I - rS)/(1 + r)$, $W = \sqrt{r}(I + S)/(1 + r)$, total half-line energy is preserved; finite terminal includes explicit positive exterior reserve.

Implication now: A ready-made exact completion mechanism for prime interactions, provided the complete Weil source and its metric are represented by the same channels.

Source: TP§§2-6; CA§14

C20 – Domino persistence / fresh-channel higher-dimensional state

Made for / definition: Iterate prime updates without losing the exact Euler path or the energy already deposited.

Recorded result and scope: Every finite update is injective; each new prime needs a fresh hidden channel for exact scalar Euler multiplication. Total stored energy telescopes.

Implication now: Explains why all mixed/reserve terms must remain on CP18's physical matrix. A fixed recycled 2D state alters the arithmetic path.

Source: TP§§7,13; CA§14

C21 – Rotating echo geometry and angular energy law

Made for / definition: Quantify the angle between an incoming signal and its prime-delayed copy.

Recorded result and scope: Raw ratio $1 + q^2 - 2q \cos \theta$, $q = r\sqrt{\kappa}$; hidden plus-wave contains the complementary signed correlation.

Implication now: Translates the remaining signed correlations into exact angular data. Needs an authentic aggregate bound to control near-null energy.

Source: CA§§15-18

C22 – Visible-path energy escape / first-prime trace

Made for / definition: Determine what happens when local positive completions are iterated infinitely.

Recorded result and scope: Bohr energy of normalized visible path tends to 0 for $1/2 < \sigma \leq 1$; required recovery gain $\prod_p (1 + p^{-\sigma})$ diverges there. The repeated-prime square sum converges for $\sigma > 1/2$.

Implication now: Identifies the exact normalization cost that any “positive local blocks therefore global sign” argument must pay.

Source: TP§§8-12

C23 – Moving one-port metric / phase-only normalization

Made for / definition: Try to conserve the physical raw Euler channel by changing its metric or normalizing its phase.

Recorded result and scope: Forced scalar metric $g_R = g_0/|P_R|^2$; finite conservation exact. Infinite coercivity has a zero-free requirement. Pure phase has a nonlocality/causality cost.

Implication now: Relevant whenever a new “plane” claims to preserve norm: identify the transformed metric and its coercivity at the null, rather than merely normalizing outputs.

Source: TP§§14-15

C24 – Moving color filtration / exact lifetimes / spectral arrival depth

Made for / definition: Separate the physical arrival of events from exit of modulus colors as the cutoff moves.

Recorded result and scope: One-exit lifetime $L_\theta(q) = \lceil \log_2 q/\theta \rceil - 1$, exact finite common-lifetime Gram, positive exit weights and exact centering; spectral depth is $\omega(q)$.

Implication now: Organizes tails and threshold terms rigorously. The collective norm still depends on signed cross-layer synthesis.

Source: MC§§2-5

C25 – Terminal inheritance and coherent color-block obstruction

Made for / definition: Track energy that migrates to new cells even after every fixed color exits.

Recorded result and scope: Exact coupled Cauchy identity; finite lifetimes alone do not bound collective energy. Actual prime-color blocks have all-ones normalized Gram matrices, defeating generic coefficient-blind diagonal estimates.

Implication now: Direct lesson for CP18: exploit actual arithmetic relations in $W_a c$, rather than assume diagonal local budgets control all boundary interactions.

Source: MC§§5-8

C26 – Conserved exclusion-density ledger / global allocation constant

Made for / definition: Divide integer density into survivors, least-prime-exponent-one innovations, and repeated-least-prime mass.

Recorded result and scope: $\delta_y + a_y + b_y = 1$. Infinite allocation $C \approx 0.669902$, $\sigma \approx 0.330098$; finite error theorem. These are different from squarefree density and signed Mertens energy.

Implication now: A proved global invariant available for authentic coefficient averages; a map from that average to the near-null Weil boundary energy remains to be supplied.

Source: RE§§7-8

C27 – Exponent-resolved conserved mass

Made for / definition: Refine the density invariant to every least-prime exponent, retaining the full vertical hierarchy.

Recorded result and scope: $d_r = \sum_p \delta_{p-}(1 - p^{-1})p^{-r}$, $\sum_r d_r = 1$, $\sum_{r \geq R} d_r \leq 2^{1-R}$.

Implication now: Explicit positive tail control for vertical depth. Cross-color signed energy is a separate quantity.

Source: RE§8.1

C28 – Prime prism: mass spectrum and Hilbert DNA spectrum

Made for / definition: Decompress a number into its complete labeled prime-power spectrum.

Recorded result and scope: Mass entries $\log p$ total $\log n$; Hilbert entries $\sqrt{\log p}$ give Gram $\log \gcd(i, j)$. Normalized brightness partitions exactly.

Implication now: Supplies two concrete feature maps, with different metric meaning. A CP18 transfer must specify which is used.

Source: CP§§35-38

C29 – Ordered color films of prime gaps

Made for / definition: View an additive gap as the sequence of complete multiplicative color rows traversed between primes.

Recorded result and scope: Sum of normalized color rows over $P < n \leq Q$ equals $Q - P$. This retains the order omitted by a gap length.

Implication now: Potential organization of mixed prime packets by arithmetic event order; it is a lossless data representation, not a uniform sign inequality.

Source: CP§37

C30 – GCD-LCM reflection diamonds / shared core and exclusive fork

Made for / definition: Represent common ancestry and reunion of two factor states.

Recorded result and scope: Exact layer identity $\Phi(m) + \Phi(n) = \Phi(\gcd) + \Phi(\text{lcm})$. Shared GCD core has a positive Gram; line-versus-lattice residual $\log(\min / \gcd)$ is indefinite.

Implication now: Provides exact mixed-term bookkeeping and a known place where scalar compression changes the sign problem. The complete combined form must be retained.

Source: CP§§39-43; DK§12

C31 – Reciprocal Brownian time / Brownian bridge through the prism

Made for / definition: Realize historical ordering and a record projection as an actual Hilbert geometry.

Recorded result and scope: $b_t = 1_{[0,1/t]}$, Gram $1/\max(s, t)$; record-mode removal gives a Brownian-bridge covariance. Time and DNA coordinate projections commute in the tensor product.

Implication now: An exact geometric model for historical energy. Its relation to the Suzuki/Weil form on $W_a c$ requires a proved carrier identification.

Source: CP§§40-41

C32 – Open diamonds / centered transverse cross-color target

Made for / definition: Isolate high-prime interactions whose GCD inheritance lies inside the finite board while their LCM reunion lies beyond it.

Recorded result and scope: Exact source split and scalar/radial closures; the vector-valued centered Bessel-Carleson estimate remains open.

Implication now: A previous explicit version of the cross-color bottleneck, useful to compare against CP18's critical boundary residual rather than rediscover under another name.

Source: CP§§43-48

C33 – Parabolic multiplication sieve / angle families / blank prime positions

Made for / definition: Identify composite positions as crossings of nonunit factor chords and preserve exact directions.

Recorded result and scope: Chord $x = ab + (a - b)y$; factors reconstructed from crossing and direction. Prime position has no nonunit chord. Qualitative prime density0 follows from finite fan exclusion.

Implication now: An exact arithmetic geometry. It can determine source positions and factor ratios; it does not directly bound completed Weil energy.

Source: AF§§2-4; BP§2

C34 – Angle-to-energy coordinate and full prime triple

Made for / definition: Translate factor geometry into the historical kernel and combine all zero/one/two-slot insertions before estimating.

Recorded result and scope: $1/\max(a,b) = n^{-1/2}e^{-|\log(a/b)|/2}$. Complete k, pk, p^2k kernel cancels outside factor ratio p and retains an exact terminal correction.

Implication now: A concrete cancellation mechanism to test on actual arithmetic packets. Its historical metric and terminal must be matched to the present operator.

Source: AF§§4-6

C35 – One-prime envelope and mixed-prime obstruction

Made for / definition: Quantify the saving of one complete prime rule and test whether it compounds.

Recorded result and scope: Prime 2 gives exactly $1/9$ of leading unsigned squarefree envelope, still linear. After primes 2 and 3, $\langle w_1, w_5 \rangle = -1/30$; simple iteration of the positive-entry estimate fails.

Implication now: Already tells us where useful mixed signs occur, and why they should be combined before absolute estimates.

Source: AF§§7-8

C36 – Additive prime-power-axis positive kernel, Route A

Made for / definition: Keep von Mangoldt weights exact on one-coordinate exponent-lattice changes.

Recorded result and scope: $\mathcal{K}^{\log} = \sum_p (\log p)(G_p - c_p I) \succeq 0$, strictly positive on finite distinct nodes. Off-diagonal coefficients exactly $\Lambda(R)R^{-\sigma}$.

Implication now: Positive source-faithful local carrier; its appearance in the completed scalar source is subtractive, so completion must provide the matching reserve.

Source: DK§3

C37 – Tensor valuation kernel, Route B, and positive mixed-color sectors

Made for / definition: Retain the empty-color floor and all mixed sectors of independent prime covariances.

Recorded result and scope: $\bigotimes_p (c_p I + X_p^* X_p)$ expands into positive square sectors. Finite-prime floor $\prod_p c_p$; need not remain positive in infinite limit.

Implication now: Supplies complete finite mixed-color positivity, a plausible component of an exact Weil factorization. It cannot supply the completed comparison by itself.

Source: DK§4

C38 – Exact additive/tensor bridge

Made for / definition: Show the two prior kernels belong to the same color hierarchy.

Recorded result and scope: RouteA is first-color differential and logarithmic scale derivative of normalized RouteB; convolution exponentiation reconstructs RouteB.

Implication now: Lets one transfer exact identities between representations without replacing a compressed finite matrix by an unjustified exponential.

Source: DK§5

C39 – Prime-angle lock / scalar logarithmic orbit

Made for / definition: Turn independent color angles into the actual scalar Dirichlet phases $z_p = e^{-it \log p}$.

Recorded result and scope: Exact indexing, injective on finite-prime exponent lattice; for at least two primes scalar orbit is a proper dense Haar-null subgroup of the torus.

Implication now: This is the precise “bring the higher plane back to the line” operation. The orbit’s singularity matters for norm transfer.

Source: DK§§9-11

C40 – Singular trace obstruction and theta-Poisson regularizer

Made for / definition: Determine the legitimate interface between positive color metrics and the signed theta orbit.

Recorded result and scope: No uniform bounded coordinate-preserving decoder from either unmodified prime metric to theta. RouteB gives a positive Poisson approximate identity; coupled theta-Poisson positivity as regularization vanishes is equivalent to the remaining target.

Implication now: Important pre-existing constraint on any direct transfer proposal. A restricted map on W_{ac} may still be investigated, with explicitly demonstrated estimates.

Source: DK§14.1-14.3

C41 – Completed prime covariance and arithmetic spectral diagonal

Made for / definition: Reconstruct the complete reflected Weil form from factor potential, pole and gamma terms; compare with the positive zero-evaluation diagonal.

Recorded result and scope: Exact $N - W = 2\|P_{-a_F}\|^2$; translation leaves W unchanged. Full double source gives all ordered zero pairs, not a diagonal selector. A positive completed factorization remains open.

Implication now: This is the direct earlier formulation of today’s goal. CP18 restricts where the necessary comparison might be tested, without changing which complete form is targeted.

Source: FP§§3-11

C42 – Exact local overlap law and isolated historical half floor

Made for / definition: Group each odd Möbius coefficient with its delayed double, preserving unfinished groups at every cutoff.

Recorded result and scope: For $q_{k,N} = e_k - 1_{2k \leq N} e_{2k}$, $\mu_{\leq N} = \sum_{k \text{ odd}} \mu(k) q_{k,N}$; grouped Gram is zero when one start is at least twice the other. The first group is orthogonal to all others and has energy 1/2, so $R(N) \geq 1/2$ for $N \geq 2$.

Implication now: A completed lower-energy theorem in the historical metric, with exact locality. Distinct from the prime2Haar first-shell energy $1/4$ and full-row $1/2$; its metric must be identified before transfer.

Source: GO§§3-5

C43 – Logarithmic grouped diagonal and exact constant

Made for / definition: Calculate the already controlled part of full historical energy after local grouping.

Recorded result and scope: $R(N) = D_N + I_N$, $D_N = (2/\pi^2) \log N + B + O(N^{-1/2})$, $B = (2/\pi^2)(\gamma + \frac{4}{3} \log 2 - 2\zeta'(2)/\zeta(2))$. The remaining signed overlap lies only at ratios below 2.

Implication now: A proved quantitative energy component; the constant B belongs to the diagonal and is not a global energy ceiling. It may guide decomposition of authentic signed terms.

Source: GO§§6,8

C44 – Reciprocal-divisor reflection and full-packet insertion angles

Made for / definition: Relate complementary subsets of a finite prime family and measure overlap of whole packets.

Recorded result and scope: For squarefree $P > 1$, $C_P(t) = -\mu(P)C_P((P/t)^-)$. The insertion angle gives $E(f - D_P f) = A + B_P - 2\sqrt{AB_P} \cos \theta$. Exact first cosines $1/\sqrt{2}, 0, -\sqrt{5}/20$ for adding 2, 3, 5.

Implication now: Concrete angular laws with the correct full packet and cutoff. They separate genuine energy angles from screen-slope angles and preserve the signed interaction.

Source: GO§§2,7

C45 – Bilateral prime-state/Fock lattice and global all-pass unitary

Made for / definition: Make multiplication reversible by extending prime exponents to integers, so the higher-dimensional state is $\ell^2(\mathbb{Q}_+)$.

Recorded result and scope: Prime shifts are bilateral unitary; $He_n = \log n e_n$, $[H, S_p] = (\log p)S_p$, partition trace $\zeta(s)$ in its convergent half-plane. For $a > 1$, $\Omega_a = \bigotimes_p (I - r_p U_p^*)(I - r_p U_p)^{-1}$ exists strongly and is unitary.

Implication now: A complete positive state construction already present in the archive. Its specified cumulative observation remains a separate domain question, directly relevant to any proposed higher-plane proof.

Source: PS§§2-4

C46 – Negative-squarefree face theorem

Made for / definition: Identify the Ramanujan/Jordan color function as an actual projected face of the bilateral prime state.

Recorded result and scope: A bounded diagonal contraction after projection satisfies $DP_- \Omega_a e_1 = J_\alpha(n)/n^\alpha$, $a = 1 + \alpha > 1$; the negative face automatically has Möbius squarefree selection.

Implication now: Supplies an exact bounded map through the color stage. The later cumulative camera is the unbounded step, now sharply identified rather than hypothetical.

Source: PS§5

C47 – One-state all-pass 2-over-1 storage law

Made for / definition: Realize the scalar all-pass factor $(s-a)/s$ as a causal input/output/storage system.

Recorded result and scope: $\dot{x} = -(a/2)x + \sqrt{a}u$, $y = u - \sqrt{a}x$; $|x(T)|^2 + \int_0^T |y|^2 = |x(0)|^2 + \int_0^T |u|^2$.

Implication now: Exact causal transport and reserve accounting. This is a distinct 2/1 meaning from the prime splitter and likelihood first/second moments. It preserves finite-energy admissibility once established.

Source: PS§6

C48 – Low/high color cocycle and cancellation-plane distance

Made for / definition: Track changing allocation between low/high colors while retaining their observable sum.

Recorded result and scope: Triangular $T_r = \begin{bmatrix} 1 & 1-r \\ 0 & r \end{bmatrix}$ preserves $L + H$; its hidden anti-diagonal contracts, and an augmented positive metric has exact defect $\kappa(1-r^2)\|H\|^2$. Observable energy is twice squared distance from the cancellation plane.

Implication now: Makes explicit why hidden contraction does not itself shrink the complete observable. The near-antiparallel alignment needs authentic correlation information.

Source: PS§§7-8

C49 – Infinite radial prime stiffness

Made for / definition: Quantify the cost of radial displacement in the bilateral prime state while imaginary displacement remains rotation.

Recorded result and scope: Local cost $\ell_{p,a}(\sigma) = 4p^{-a} \sinh^2(\sigma \log p) / (1 - p^{-a+2\sigma})$ in its geometric-series domain; sum converges iff $|\sigma| < \alpha/2$, $a = 1 + \alpha$. At the critical endpoint every fixed nonzero admissible radial displacement has divergent accumulated cost.

Implication now: A genuine confinement geometry. To apply it to an actual zero requires proving that the zero channel belongs to this source-faithful finite-energy state.

Source: IS§§2-3

C50 – Finite forbidden-time leakage and infinite causalization cost

Made for / definition: Locate precisely the energy signature of an off-line quotient pole.

Recorded result and scope: An isolated inverse-Blaschke pole has finite nonzero anti-causal leakage; imposing the same growing mode on positive time has infinite L^2 energy. Uncancelled off-line zeros force Jordan-shell divergence along a determining sequence.

Implication now: Provides a sharp detection/contradiction route if arithmetic causal finite energy is independently shown. Finite leakage itself is not divergent.

Source: IS§§4-6

C51 – Profinite almost-everywhere energy theorem and exceptional integer orbit

Made for / definition: Test whether average prime/color positivity already controls the true arithmetic observation.

Recorded result and scope: The Jordan weighted energy gate holds for Haar-almost every profinite starting point. The canonical integer orbit $x = 0$ is countable, Haar-null and dense; its finite-energy trace remains open.

Implication now: Explains exactly why an averaged color theorem cannot be silently applied to the actual sequence. A restricted arithmetic trace theorem would be decisive.

Source: IS§§7-10

C52 – Prime/physical-stop commutator and positive exterior curvature

Made for / definition: Measure the difference between revealing a prime and stopping the physical window in opposite orders.

Recorded result and scope: $[R_Y, U_p]$ is supported in transported exterior band $[Y, pY]$; its squared norm is an explicit positive collar integral. Composite commutators are full signed sums of overlapping bands.

Implication now: An older exact two-parameter counterpart of CP18's collar work. It may be compared at the operator level, retaining complete mixed bands.

Source: IS§§11-13

C53 – Kakutani endpoint phase transition

Made for / definition: Distinguish local prime-state limits from global infinite-product identity.

Recorded result and scope: For $\alpha > 0$ the Jordan product density is equivalent to Haar and in L^2 ; at $\alpha = 0$ it is supported on the Haar-null profinite units and becomes singular.

Implication now: Describes a genuine limiting sector change. It prevents assuming uniform coercivity from every finite positive stage.

Source: IS§7

C54 – Singular tail state / fixed-window convergence versus whole-state singularity

Made for / definition: Formalize the proposed inside/outside higher-dimensional object using conditioning of prime occupations.

Recorded result and scope: Original independent inclusion $p^{-1-\alpha}$ has finite prime support almost surely.; size conditioning converges on every finite prime window to inclusion $1/(p+1-p^{-\alpha}) \sim 1/p$, which has infinite support almost surely. Full laws are mutually singular. Tail has explicit power law with power-saving remainder.

Implication now: A proved model of why local checks need a uniform statement for a growing observer. This is an additional theory, not merely a metaphor about infinity.

Source: ST§§2-7

C55 – Likelihood 2-over-1 law and positive ANOVA/Walsh energy

Made for / definition: Track first-moment mass and second-moment energy of the changed prime identity.

Recorded result and scope: Likelihood $L_R = \prod_{p \leq R} (1 + c_p \psi_p)$ has $\mathbb{E}[L_R] = 1$ and $\mathbb{E}[L_R^2] = \prod_p (1 + c_p^2) \rightarrow \infty$; orthogonal Walsh/ANOVA bank exact. All composite levels must be retained in the moving physical observable.

Implication now: Separates a conserved unit mass from a divergent square-energy cost. A positive component sum alone cannot supply the required cancellation.

Source: ST§§8-9,13-15

C56 – Negative-order gluing and source-observer compensation

Made for / definition: Find a space where the singular tail state can genuinely converge.

Recorded result and scope: In weighted Walsh $H_{-\beta}$, tail likelihood belongs and normalized conditioned tails converge for $\beta > \alpha$. The physical observer's norm grows by the reciprocal weighting.

Implication now: A ready-made test for a change of plane or regularizer: verify whether the gain survives in the paired source-observer product.

Source: ST§10

C57 – Trace-renormalized determinant / first-prime obstruction

Made for / definition: Separate summable repeated-prime sectors from the divergent first-prime trace in Euler products.

Recorded result and scope: Regularized determinant D_2 cancels first trace; restoring $\exp(-\sum_p p^{-\lambda})$ returns the exact Euler polynomial. No ordinary Hilbert-Schmidt determinant can retain the unbounded trace functional on ℓ^2 .

Implication now: Relevant to proposed Hilbert-Schmidt defect budgets: a renormalization must retain or independently pay the actual first-prime contribution.

Source: ST§16

C58 – Terminal/oscillation Schur energy split

Made for / definition: Separate a shell's coherent mean from its fluctuating part without losing a boundary term.

Recorded result and scope: Exact weighted variance identity: shell energy $= \kappa |\text{mean}|^2 + \frac{1}{2\kappa} \sum_{R,S} \eta_R \eta_S |t_R - t_S|^2$. The mean is an explicit tapered prefix.

Implication now: Identifies two independent energy costs and guards against treating oscillation/Poincaré control as control of the coherent terminal.

Source: ST§18

C59 – Farey beat lattice and fractional color metric

Made for / definition: Calculate how rational color frequencies separate under causal observation.

Recorded result and scope: Complete Gram yields nonnegative distanced $_{\alpha}(t, u)^2$; nearby nonzero frequencies satisfy distance $^2 \asymp |t - u|^{1-\alpha}$. Reduced denominator of $h/q - k/r$ sets the beat scale; beyond it tail gains d/M .

Implication now: Concrete angle/frequency geometry for cross-color interaction and finite observation length. Pair interactions still retain endpoint phase.

Source: FB§§5-7

C60 – Exact diagonal-causal-boundary cancellation

Made for / definition: Keep the two triangular causal boundaries together with the apparent diagonal of the rational-frequency kernel.

Recorded result and scope: Leading divergent diagonal cancels identically at equal frequency because $1 + b(z) + b(v) = (zv - 1)b(z)b(v)$. Full transported diagonal is finite.

Implication now: Particularly relevant to the concern about losing machinery: an existing exact cancellation can be erased by separate bounds. It must be checked for a common map to the current residual.

Source: FB§§3-4

C61 – Weighted LCM boundary / primitive color orthogonality

Made for / definition: Preserve complete color cancellation while accounting for physical stopping and nonconstant weight.

Recorded result and scope: Full-period primitive Gram is $\delta_{q,r}J_2(q)/12$; power weighting leaves an explicit Hurwitz-zeta LCM boundary. Möbius-weighted translation-invariant convolution reinforces rather than cancels; the needed sign resides in causal transfer factors.

Implication now: Narrows where an arithmetic gain can come from and shows which part of a color-angle decomposition must remain coupled.

Source: FB§§8-13

C62 – Zero-capacity arithmetic identity / trace threshold

Made for / definition: Determine whether the canonical phase can be controlled from the authentic inverse-coboundary energy.

Recorded result and scope: In the Ramanujan-radial sector, point evaluation needs order $s > 3/2$; the project uses $s = (1 + \alpha)/2 < 1$. Positive nested cylinders can retain value 1 at 0 while their norm tends to 0.

Implication now: A precise obstruction to promoting average energy into the canonical arithmetic trace. Any application to CP18 must use additional structure of its forced solutions.

Source: FB§§14-16

Direct relevance order for CP18

1. FP already reconstructs the exact complete Weil source from factor potential, so “using Prime DNA” need not begin with a new arithmetic model.
2. DK RoutesA/B and Smith-Jordan provide the actual positive square systems; test an exact comparison on the finite forced family $q = W_a c$, preserving the Suzuki metric.
3. The singular trace theorem and prime-block coherence theorem specify why generic bounded transfer/diagonal control cannot be assumed. Any restriction to the current forced family must be proved to change those hypotheses.
4. The full prime triple, hidden plus-wave, reflection diamonds and ordered color films are concrete candidates for organizing the actual signed terms before estimates.
5. The global density invariant, half-density row sums, and finite-prime floors remain valid in their named measures. An explicit transfer is needed to turn them into a margin-dependent boundary estimate.

Historical names and source scope

- Remembered labels **subprime**, **Mersenne modulus correction**, **prime footprints**, and **nuts** were searched. No definitive canonical definition of these exact labels was recovered in this source set; they must remain on the inventory rather than be silently replaced with a nearby theorem. The recovered five-layer cutoff is a likely related formal construction, but that correspondence is not established by the read files.
- Fibonacci-Lucas recurrence, spirals, Möbius modifiers, record descent, light cones, planes, and later checkpoints have their own detailed sections. Overlap preserves exact interfaces.
- No bibliography-wide novelty claim is made. The manuscript explicitly credits classical factorization, Jordan, Ramanujan, sieve, explicit-formula and operator ingredients; the user’s named architecture organizes their roles and contains its own exact additional calculations.

9. Spirals, Fibonacci, angles and phase

Source register (current contents read)

- **S1:** Michaels_Spiral_Input_Output_White_Paper.pdf, all 27 pages. Exact input-output correspondence for Michaels-Möbius counting spirals, 15 September 2026.
- **S2:** Familiar_Number_Atlas.pdf, all 22 pages. Expanded energy/winding/Fibonacci atlas, 6 September 2026.
- **S3:** Prime_Factor_Energy_Geometry.pdf, §§1-12 and theorem descriptions in §13 (the equivalent complete §13 theorems are fully read in S2 §10).
- **S4:** Prime_Sampling_and_Energy_Floors.pdf, all 23 pages, 7 September 2026.
- **S5:** Prime_DNA_Angle_Families_and_Prime_Triple_Bound.md, full text, 6 September 2026.
- **S6:** Michaels_Warp_Drive_Angle_Check.md, full text, 8 September 2026.
- **S7:** Michaels_Rotation_One_Off_Report.md, full text, 21 September 2026.
- **S8:** Michaels_Rotation_RH_Proof_Attempt.md, full text, 21 September 2026.
- **S9:** Michaels_Prime_Phase_Continuation_Report.md, full text, 21 September 2026.
- **S10:** Michaels_Prime_Phase_J_Bridge.md, current version 3, full text, 21 September 2026.
- **S11:** Michaels_Planes_Spirals_and_No_Reversal_Check.md, full text.
- **S12:** Physics_Architectures_and_Log_Prime_Energy.pdf, §§1-19 plus source introduction to §20. The complete energy concordance appears in the energy catalogue.

- **S13:** Prime_DNA_Build_v38_Two_Plane_Rotating_Law.pdf, full text.
- **S14:** Prime_DNA_Build_v39_The_Two_Plane_Prime_Cocycle.pdf, full text.
- **S15:** Prime_DNA_Build_v40_The_Mobius_Rotation_and_Moving_Terminal.pdf, full text.
- **S16:** Build_Log_129_Prime_Negative_Diffusion_Flatline.json, full text. Its companion image is Build_129_Negative_Prime_vs_Negative_Zero_Spiral.png (title located; image itself not inspected).
- **S17:** Prime_DNA_Blueprint_Assembly_2026-09-07.md, full text.
- **S18:** Prime_DNA_Recombination_Energy_and_Global_Density_Christophe_Michaels.pdf, introduction and complete §§9-11 read (printed pages 23-29).
- **S19:** Michaels_Six_Route_Mathematics_and_Audit_2026-09-13.md, synthesis and complete Appendix F, Fibonacci packets/sampling, read.

Distinct spiral and Fibonacci constructions

SP01 – Unit counting spiral / integer-vector chain

Definition: At fixed input $s = \sigma + it$, $H_k[c; s] = \sum_{n \leq k} c(n)n^{-s}$. The individual vector is $n^{-s} = n^{-\sigma}e^{-it \log n}$. The ordered partial sums form a polygonal counting path.

Made for: Display and retain arithmetic accumulation at every integer, with multiplication becoming scaling plus rotation. Input s , counting index k , and plotted output occupy different roles.

Established: Finite identities hold for arbitrary complex coefficients. Integer labels and all increments retain the input; a single endpoint generally does not.

Current implication: Supplies an exact representation for complete coefficient packets and their historical energy. To transfer it to CP18 forced solutions $W_a c$, one needs the coefficient/transform map and the transformed metric; the radius of the last plotted point is not the full form.

Source: S1 §§1-4; S3 §§5-7.

SP02 – Michaels-Möbius factor-count spiral / Omega-weighted wave

Definition: $W_k[c; s] = \sum_{n \leq k} \Omega(n)c(n)n^{-s}$, with $\Omega(n) = \sum_p v_p(n)$. Its step identity is $\Delta W_n = \Omega(n)\Delta H_n$. Recover $H_k = c(1) + \sum_{2 \leq n \leq k} \Delta W_n / \Omega(n)$.

Made for: Restore factor-depth information hidden in the plain unit counting path; expose prime powers and mixed composites without discarding the indexed prime-exponent record.

Established: Exact bidirectional conversion after retaining the unit. Augmented path map is $SD_\Omega S^{-1}$; increment-norm condition number is $\lfloor \log_2 N \rfloor$. Multiplication by Ω changes step length while preserving direction. The genuine Möbius sign μ , reconstructed from exponents, is a separate observable.

Current implication: A usable arithmetic representation and reconstruction theorem. It can expose packet structure inside a sign calculation, but its positive factor weights do not establish the sign of a signed residual. The authentic modifier is also a different operator (see SP06).

Source: S1 §§1-5, 14-16.

SP03 – Prime-power cofactor spiral decomposition / marked generating polynomial

Definition: $W_N(s) = \sum_{p^j \leq N} p^{-js} H_{\lfloor N/p^j \rfloor}(s)$. The complete marked polynomial is $\mathcal{A}_N(s; z) = \sum_{n \leq N} n^{-s} \prod_p z_p^{v_p(n)}$.

Made for: Express the whole spiral as rotated/rescaled prime-power layers and retain which prime and exponent produced every contribution.

Established: Exact finite decomposition, with every floor and terminal retained. $\sum_p z_p \partial_{z_p} \mathcal{A}_N|_{z=1} = W_N$; weighting these derivatives by $\log p$ gives $-H'_N(s)$. For arbitrary coefficients, each cofactor uses $c(p^j m)$, not an assumed multiplicative substitute.

Current implication: Directly relevant to reconstructing CP18 arithmetic sources at shifts $\log p^j$. It gives exact prime labels for a residual expansion and mixed terms to be estimated jointly.

Source: S1 §5.

SP04 – Smooth integral spiral / reflected zero mode

Definition: $P_s(N) = N^{1-s}/(1-s) = \int_0^N x^{-s} dx$ in the applicable strip. The historical raw mode is $G_\rho(x) = -x^\rho/\rho$, and the correction is $G_{1-s}(N)$.

Made for: Connect the discrete integer path to its continuous integral and identify its limiting displacement as the zeta value.

Established: $H_N(s) - P_s(N) \rightarrow \zeta(s)$, with explicit inclusive endpoint and fractional-part/Euler-Maclaurin remainder. On $\Re s = 1/2$, $G_{1-s} = \overline{G_s}$. At a selected critical-line zero, the absolute endpoint separation tends to zero with leading scale $1/(2\sqrt{N})$.

Current implication: An exact continuation bridge for prime-generated tests and jets. Its full correction/tail must follow any passage to the Weil/Suzuki form; matching spiral shapes is not the source of the sign.

Source: S1 §§8-13; S3 §7.

SP05 – Continuation-residual circle and finite-radius law

Definition: $R_N(s) = \zeta(s) - H_N(s)$; on $s = 1/2 + it$, normalized $N^{s-1} R_N(s) \rightarrow 1/(s-1)$, whose image satisfies $|1 + 1/(-1/2 + it)| = 1$.

Made for: Identify the exact circle behind the observed spiral residual, quantify finite-cutoff errors, and separate a universal continuation geometry from actual zero detection.

Established: Expansion $N^{s-1} R_N = 1/(s-1) - 1/(2N) + s/(12N^2) + O_s(N^{-4})$ with explicit remainder. Exact cutoff recurrence also supplied. For other real parts the limiting circle center and radius change accordingly.

Current implication: Provides controlled continuation and normalization, useful for certified evaluation. Since the law holds at every height, its circle alone does not supply the CP18 boundary-energy inequality.

Source: S3 §9, Theorem 9.1/Corollary 9.2.

SP06 – Authentic Michaels modifier carried through spiral coordinates

Definition: For supplied consecutive-block partition $\mathcal{P}$, $\mathcal{M}_{\mathcal{P}C} = \sum_{B \in \mathcal{P}} (\sum_{n \in B} c(n)) e_{\text{right}(B)}$; compensation $\eta c = c - \mathcal{M}c$.

Made for: Compress block information at actual right endpoints while retaining exact reconstruction and all historical interactions.

Established: $\|c\|_G^2 = \|\mathcal{M}c\|_G^2 + 2\Re\langle \mathcal{M}c, \eta c \rangle_G + \|\eta c\|_G^2$, with terminal energy retained. Ω -weighting and modifier generally do not commute. The exact pullback metric for weighted trace V is $D^*L_s^*GL_sD$.

Current implication: A concrete way to use the modifier on any successfully matched CP18 coefficient representation. Every compensation and cross term must be carried to the boundary residual. The modifier is not the Omega wave itself.

Source: S1 §§14-16; The authentic partition selection and later estimates appear in M05-M08.

SP07 – Individual zero-mode spiral, normalized circle, and logarithmic helix

Definition: For general $\rho = \beta + i\gamma$, $G_\rho(x) = -x^\rho/\rho$, $W_\rho(x) = x^{-1/2}G_\rho(x)$. Radius is $x^{\beta-1/2}/|\rho|$, angular velocity in $\log x$ is γ .

Made for: Separate the real part of a zero (radial growth) from its imaginary part (rotation). Retaining $\log x$ makes the circle a helix when $\beta = 1/2$.

Established: Exact formulas. Universal constant radius for every nontrivial zero is equivalent to RH. Angular direction alone does not change when β is varied.

Current implication: Specifies what an off-line mode would do and provides an alternative criterion. Any positivity/rigidity theorem being imported must constrain radial exponent or the complete mode pairing; an individual mode's angular sign is already fixed for every β .

Source: S2 §8.2; S11 §4.1.

SP08 – Fibonacci phase law / sqrt(phi) sampling criterion

Definition: Sample the preceding modes at F_k . Binet gives $\log F_k = k \log \varphi - \frac{1}{2} \log 5 + \log(1 - (-\varphi^{-2})^k)$.

Made for: Test whether exponential arithmetic sampling locks the zero mode to a stable radial and angular rule.

Established: Alternating exponentially small phase defect. Raw radial ratio tends to φ^β ; normalized ratio tends to $\varphi^{\beta-1/2}$. The universal raw ratio $\sqrt{\varphi}$, or normalized ratio one, is an exact RH-equivalent criterion. At the first selected ordinate the unwrapped phase step is one full turn plus approximately 29.714 degrees.

Current implication: A retained sufficient/equivalent route. Proving its universal radial rule independently could finish RH, but the sampling identity by itself does not bound CP18's critical boundary component.

Source: S2 §§2-3, 8.3; S4 §3.

SP09 – Golden spiral / general ratio-to-phase law

Definition: Every positive sequence with $a_{k+1}/a_k \rightarrow c$ gives raw mode quotient $c^{\beta+i\gamma}$. Classical golden spiral pitch is $2 \log \varphi/\pi$, while the critical-line mode pitch is $1/(2\gamma)$.

Made for: Explain what geometric feature comes from Fibonacci growth and what comes from the mode itself; compare other recurrence/constant choices.

Established: General ratio theorem, including Lucas, Pell, Tribonacci, Padovan, and metallic means. Golden pitch equality occurs only at $\gamma = \pi/(4 \log \varphi)$; selecting Fibonacci points does not change the continuous pitch.

Current implication: Sampling can be chosen for convenience without changing the underlying arithmetic sign requirement. If “GR” means golden ratio, it belongs here; no source in this source set identifies a distinct general-relativistic Fibonacci theorem.

Source: S2 §§2,4,6.

SP10 – Fibonacci factor-balance, doubling/tripling, and conserved norm

Definition: Use normalized prime shares q_p , mixed mass $C = 1 - \sum q_p^2$, logarithmic factor balance $h = \log a / \log(ab)$, and overlap $J(a, b) = \sum q_p(a)q_p(b)$. Exact product law:

$$C(ab) = h^2 C(a) + (1 - h)^2 C(b) + 2h(1 - h)(1 - J).$$

Made for: Convert Fibonacci-Lucas algebra into genuine arithmetic lower bounds, including repeated/shared primes rather than only geometric sampling.

Established: $F_{2m} = F_m U_m$, $\gcd(F_m, U_m) \leq 2$, $U_m^2 - 5F_m^2 = 4(-1)^m$. Thus $\liminf C(F_{2m}) \geq 1/2$. Odd tripling $F_{3m} = F_m(5F_m^2 - 3)$ has coprime factors and yields $\liminf_{m \text{ odd}} C(F_{3m}) \geq 4/9$. The recovered research record also names the exact balance defect $\Delta_m = 2(h_m - 1/2)^2 + 2h_m(1 - h_m)J_m$. It follows directly from the displayed product law that the cross contribution is $1/2 - \Delta_m$.

Further direct consequence (calculated from the recovered product law): For each fixed doubling depth r , as the base index $m \rightarrow \infty$, dropping the nonnegative Lucas mixed mass and iterating gives $\liminf_m C(F_{2^r m}) \geq \frac{2}{3}(1 - 4^{-r})$. This is a consequence with the indicated order of quantifiers; it is not a theorem about a fixed base and unlimited depth obtained by interchanging limits.

Current implication: These are real sign-producing arithmetic floors. Their use for CP18 requires a map expressing the relevant forced-solution quadratic expression through these factor-balanced states with a controlled remainder. The coordinate $C(n)$ has its own definition; it is not already Weil energy.

Source: S2 §§5,10; S3 §13; the recovered research record confirms the defect/doubling-chain formulation.

SP11 – Fibonacci-Lucas bounded Green-storage packets

Definition: Fix $b \geq 3$, $A_0 = F_b$, $A_r = U_{2^{r-1}b}$, $N_r = F_{2^r b}$, $v_r = \prod_{j=0}^r (I - D_{A_j})e_1$, Green kernel $G(i, j) = 1/\max(i, j)$.

Made for: Produce an explicit all-stage storage/conservation mechanism made of complete prime-word moves, with energy less than one.

Established: $E_r = (1 + A_r^{-1})E_{r-1}$, and storage defect $\Delta_r = 1 - N_r^{-1} - E_r$ obeys a nonnegative recurrence. Hence $E_r < 1$ at every stage. Complete cumulative values are in $\{-1, 0, 1\}$. Every assembled move has the exact prime refinement $I - D_A = \sum_j D_{p_1 \dots p_{j-1}}(I - D_{p_j})$, preserving all cross terms.

Current implication: A concrete reusable positive certificate in the historical Green metric. Its proof uses the separation $A_r > N_{r-1}$. S18 proves comparison to actual Möbius prefixes leaves a residual square $S_r = R(N_r) + O(\log N_r)$, with the terminal $M(N_r)^2/N_r$ still included. CP18 would

need the corresponding exact coefficient map and proof of this separation/cancellation on its forced family.

Source: S17 §4; S18 §10, Theorems 10.1-10.3.

SP12 – Fibonacci-indexed primes and sharp composite-floor sampling

Definition: $q_m = p_{F_m}$, $n_m = 2q_m$. Both adjacent ratios tend to φ . Composite departure $D(n) = 1 - L(n)$ has sharp floor $b_2(n) = 2 \log 2 \log(n/2)/(\log n)^2$, attained exactly at twice a prime.

Made for: Put a golden-ratio sampling law and a positive arithmetic floor on exactly the same sample integers.

Established: Raw mode and squared-radius sampling formulas. For every actual zero, eventual $E_\rho(n_m) = n_m^{2\beta-1} \geq c_\rho b_2(n_m)$ is equivalent to RH. A single fixed scale identifies the prime-sampled energy winding with normalized mode precisely for $\beta = 1/2$.

Current implication: A sharply specified earlier bridge target between arithmetic floor and spectral growth. This remains an available route distinct from boundary Osgood estimates. Its source labels the universal comparison open.

Source: S4 §§1,3.

SP13 – Fibonacci prime-block discrepancy energy and scale balance

Definition: $B_m = x_m^{-2} \int_{x_m}^{x_{m+1}} (\psi(x) - x)^2 dx$, $x_m = p_{F_m}$. Retain whole intervening staircase, not only sample endpoints.

Made for: Turn spectral/radial stability into an arithmetic integrated energy target; find a local-to-global contraction along golden-ratio blocks.

Established: Exact finite staircase quadrature. Weighted square-integrability for every exponent slack, subpower block bounds, and a polylogarithmic block estimate are equivalent to RH. Fine within-cell fluctuations have a proved vanishing remainder; cell means carry the remaining energy. Exact balance $B_{m+1} = k_m(B_m + 2C_m + D_m)$, $k_m \rightarrow 1/\varphi$; the proposed signed increment cap is sufficient and open.

Current implication: Provides another quantitative target that could feed global positivity if independently bounded. Recovered source already identifies why prime averaging must retain coarse means and complete signed increments.

Source: S4 §§4-8; S17 §4; S12 §§1-4.

SP14 – Energy-plane winding / central helix

Definition: $\mathcal{W}_\gamma(n) = e^{i\gamma u}(1 + X + iY)$, $u = \log n$, with $X = (V - L)/\sqrt{2}$, $Y = (2C - L - V)/\sqrt{6}$. Retain the spatial point $(\Re \mathcal{W}, \Im \mathcal{W}, u)$.

Made for: Unfold the aggregate factor-energy triangle into a 3D winding without losing its coordinates.

Established: Exact inverse recovers n, L, V, C . Interpolation between integer points is a display guide. Its total angular derivative has a carrier term γ plus the changing-ledger term, so its observed rotation is not just the carrier phase.

Current implication: A faithful representational coordinate change. Carrying it into CP18 would preserve the actual metric if its exact pullback is used; a new sign requires an identified signed-coordinate inequality.

Source: S2 §9.1; S11 §4.2.

SP15 – Labeled prime-layer polygon / phase endpoint

Definition: Each occupied layer contributes $(\log p / \log n) e^{i\gamma \log p}$; concatenate by prime and layer, preserving labels. Endpoint $P_\gamma(n) = \sum_p q_p(n) e^{i\gamma \log p}$.

Made for: Unfold all individual colored prime parts and show how mixed phase loss arises before aggregation.

Established: Total length one; squared individual lengths give L , squared grouped lengths give $L + V$. Exact endpoint deficit $1 - |P_\gamma|^2 = \sum_{p \neq q} q_p q_q [1 - \cos(\gamma \log(p/q))]$, bounded by $\min(1, 2C)$. Product endpoint is the logarithmic weighted average of the two factor endpoints; powers preserve the endpoint while subdividing the layers.

Current implication: An explicit positive pair interaction with prime-frequency differences. This is useful material for matching a signed Weil/phase expression, but one endpoint does not retain the full labeled path and its mixed-coordinate upper bound is not yet a lower Weil-energy estimate.

Source: S2 §§9.2-9.3; S3 §12.

SP16 – Negative prime spiral / zero-diffusion reconstruction / flatline test

Definition recovered: S16 compares a smoothed analytic prime-error field with a finite zero-field reconstruction, including their negatives; it also perturbs a zero real part by ε and measures resolution-times-amplitude log slope.

Made for: Compare the two sides of the explicit formula visually, show successive restoration of the prime staircase by added modes, and detect a radial drift from an off-line exponent.

Established scope: Finite diagnostics report approximately 0.999905 complex coherence, with truncation/smoothing/completion errors. “Birth” means reconstruction order in the source. The flatline diagnostic gives log slope ε , zero exactly at the centered exponent. The full underlying field definitions were not present in the recovered JSON, so they should not be invented.

Current implication: Historical evidence for matching prime and zero representations and an explicit test to retain. It supplies no uniform sign bound on the complete CP18 residual.

Source: S16, fields interpretation, prime_zero_complex_comparison, diffusion_birth_ladder, dirty_mirror_flatline.

Distinct angle, phase, and direction constructions

AN01 – Parabolic factor chords / X crossings / angle families

Definition: Connect (a^2, a) to $(b^2, -b)$: $x = ab + (a - b)y$. Axis crossing is the product; signed difference $d = a - b$ specifies parallelism. Equal-scale acute angle is $\arctan(1/|d|)$.

Made for: Geometrically encode multiplication and composite exclusion; organize persistent arithmetic rules by fan, angle, reflection, or product.

Established: Exact factor recovery $d^2 + 4n = (a + b)^2$ with parity/nonunit conditions; fan increments and prime sparsity via inclusion-exclusion. The underlying visual sieve is attributed in the source to Matiyasevich-Stechkin.

Current implication: Gives exact arithmetic organization and labels. In current sign work its most direct contribution is the following angle-to-Green-energy identity, not the drawn angle alone.

Source: S5 §§1-3; S6 §§1,4.

AN02 – Angle-to-energy / factor-ratio kernel

Definition: $\rho = \log(a/b) = 2 \operatorname{arsinh}(d/(2\sqrt{n}))$, $n = ab$, and

$$\frac{1}{\max(a, b)} = \frac{e^{-|\rho|/2}}{\sqrt{n}} = \frac{2}{\sqrt{d^2 + 4n + |d|}}.$$

Made for: Convert visual factor geometry into the exact historical energy metric. A prime insertion shifts ρ by $\pm \log p$, or preserves ρ when placed in both slots.

Established: Exact coordinate identity. Same difference and same factor ratio encode different geometric measurements; both can be retained.

Current implication: A genuine quantitative bridge from geometry to energy, suitable for transplanting complete signed cofactor identities into an operator expression. Equality of this Green metric with the Suzuki metric still requires an actual map, not a shared word “energy.”

Source: S5 §4, eq. (6).

AN03 – Prime triples k, pk, p^2k / ratio-support cancellation

Definition: Insert one prime in neither, one, or both factor slots; retain both middle terms. Complete overlap reduces to $(1/v - 1/(pu))_+$, with exact upper-cutoff correction.

Made for: Cancel broad interactions before taking magnitudes, exploiting the complete arithmetic relation represented by the angle families.

Established: Interactions with $v/u \geq p$ cancel; survivors obey $|\rho| < \log p$. Sharp one-prime envelope has coefficient $(72/\pi^4)((p-1)/(p+1))^2 N$, hence the prime-2 envelope is one ninth of the unsigned leading size. Adding primes changes signs of the surviving kernel; e.g. the complete 2,3 packet has cross entry $-1/30$.

Current implication: One of the most concrete arithmetic cancellation mechanisms available for a CP18 probe residual or forced-solution energy decomposition. Existing one-prime savings remain linear in the original counting problem; transferring and coupling them is the remaining task.

Source: S5 §§5-8.

AN04 – Warp-drive backward fan / exact family skipping

Definition: At upper label a , backward partners $2 \leq b < a$ number $a - 2$; fixed-d family $n_d(b) = b(b + d)$. Exact count through X : $C_d(X) = \max(0, \lfloor (\sqrt{d^2 + 4X} - d)/2 \rfloor - 1)$.

Made for: Skip complete geometric families in a block and use angle congruences for prime search.

Established: Exact integer-square-root counting and odd-candidate filter $d \equiv n - 1 \pmod{4}$. Bounded benchmark found literal angle traversal slower than the simple prime-factor traversal; no general complexity lower bound was claimed.

Current implication: Computational arithmetic organization, potentially useful for exact packet construction. It supplies no automatic global analytic norm estimate.

Source: S6 §§1-5.

AN05 – Relative 1.1-degree rotation / matrix alignment sensitivity

Definition: Compare A and a relatively rotated $R_\theta B R_\theta^T$. Exact two-dimensional determinant includes $-(a_1 - a_2)(b_1 - b_2) \sin^2 \theta$.

Made for: Test whether a small relative orientation between layers explains a failed positivity comparison.

Established: Algebraic examples show a 1.1-degree relative change can reverse a smallest eigenvalue. Authentic finite theta diagnostics found no common constant 1.1-degree angle in the stated comparisons. Common congruence preserves positivity; changing relative orientation changes the comparison and requires a correction back.

Current implication: In CP18 the direction of the boundary vector relative to the critical eigenspace matters. This earlier branch correctly focuses attention on alignment, but has not identified a favorable arithmetic orientation estimate for Γ_a .

Source: S7 §§1-4.

AN06 – Prime-dependent rotation and complete mixed phase

Definition: $\theta_p(t) = -t \log p$; composite phases sum valuation-weighted prime phases. Complete squared finite-head amplitude uses all $\cos(t \log(k/n))$ terms.

Made for: Replace a fixed visual rotation by the actual frequency law of each prime and connect it to the authentic Xi derivative lift/theta matrices.

Established: Exact phase/derivative identities, prime-generated Euler-Maclaurin continuation with explicit analytic remainder, all-finite-degree head+correction+mixed matrix decomposition. Numerical root candidates and theta agreement are diagnostics separately labeled. Finite local Euler phase replacements multiply zeta by a nonvanishing factor in the critical strip and preserve its zeros.

Current implication: A direct arithmetic representation of the same coupled quantities, which can be applied to CP18 forced solutions if domains are matched. It prevents losing mixed-node terms and does not require imposing guessed prime angles.

Source: S9 §§1-7.

AN07 – Global no-reversal / outward-amplitude law

Definition: For $F(z) = \xi(1/2 + z)$, $K(a, t) = \Re(F'(a + it) \overline{F(a + it)})$. Where $F \neq 0$, $K/|F|^2 = \partial_t \arg F = \partial_a \log |F|$.

Made for: Exclude a hypothetical off-axis zero by showing the complete function cannot reverse direction on the positive side.

Established: Universal nonnegativity is RH-equivalent; a local expansion at an off-axis zero gives a negative value immediately to its left. Full-theta lower bound proves a limited zero-free region. Infinite/quadric lift represents the target as a signed jet coordinate; fingerprint uniqueness supplies a separate rigidity theorem under its own hypothesis.

Current implication: A still-available closing route alongside global Weil positivity. The source explains that individual spirals, full-state conservation, and known on-axis crossing directions alone do not establish this complete signed derivative.

Source: S8 §§5-6; S11 §§1-7.

AN08 – First-prime versus higher-power rotation regularization

Definition: $E(s) = \prod_p (1 - p^{-s})e^{p^{-s}}$, removing the first harmonic; $H(s) = \sum_p \sum_{k \geq 2} p^{-ks}/k$, $E = e^{-H}$.

Made for: Identify which prime layers cause the critical continuation difficulty.

Established: Higher-power correction is analytic/nonzero and explicitly bounded for $\Re s > 1/2$. The remaining first-prime field has an RH-equivalent continuation condition. A p-dimensional local partition function adds a holomorphic nonzero factor with $3/8 < |Q(s)| < 48/25$ on $\Re s \geq 1/2$, preserving zeta zeros.

Current implication: Quantifies a controlled part of the arithmetic correction and leaves the source requiring a new estimate sharply identified. It may simplify a matched CP18 arithmetic expression; it does not itself remove the first-prime contribution.

Source: S8 §§2-4.

AN09 – Prime phase to fixed J / prime-node determination / all-power squares

Definition: The exact completed logarithmic derivative $F'/F = A - \sum \Lambda(n)n^{-1/2}e^{-z \log n}$ produces full kernel $(g(z) + \overline{g(w)})/(z + \overline{w})$, congruent to the fixed theta operator kernel where $F \neq 0$.

Made for: Turn the phase law into a full quadratic arithmetic criterion, retain all mixed terms, and use existing prime-indexed determining families.

Established: One line $\Re z = b > 1/2$, wholly inside Euler absolute convergence, suffices to test the full operator. H may also be tested on logarithms of primes, or a fixed power of varying primes. Exact grouped all-power squares include the half-line boundary term. Exact Stieltjes pairing combines the growing polar term with prime discrepancy. §10 explicitly recovers the Prime-Anchored Weil Ladder and positive Suzuki successor theorem and identifies the still-open metric equality.

Current implication: This is a direct bridge to the current global positivity objective, already built. It is a prime candidate for reuse before making new machinery: use the exact complete arithmetic functional on the current forced family, retaining domains and the existing metric distinction. Its unresolved item is the full sign/metric comparison, not coverage of arbitrary combinations.

Source: S10 §§1-10, especially eqs. (9)-(21).

AN10 – Two-plane prime cocycle / positive rotation / Möbius tensor rotation

Definition: A prime carries delay $\log p$, amplitude $p^{-\sigma}$, phase $e^{-it \log p}$. Build v39 supplies an exact unitary two-channel splitter; Build v40 rotates it into absent/present squarefree tensor coordinates and decodes Möbius signs. Build v38 gives the Bernoulli prime rotation $\vartheta_p(a) = \arcsin(p^{-a/2})$.

Made for: Preserve energy across visible/hidden arithmetic channels, make prime-indexed geometry literal, and identify the cost of returning from higher-dimensional coefficients to a causal counting trace.

Established: Finite unitary identities with complete terminal reserve; exact tensor recombination. The v38 Fisher metric is $\frac{1}{4} \sum_p (\log p)^2 / (p^a - 1)$, infinite path length at $a \downarrow 1$. Positive-to-signed infinite tensor sectors separate at that endpoint. The physical moving-prefix trace remains a separate, quantitatively unbounded observer on unrestricted inputs.

Current implication: Supplies real conservation identities and clarifies where boundary/terminal costs live. A uniform observer/metric comparison for the authentic family is exactly what must be checked when importing conservation into CP18. Conservation of the enlarged state alone does not give the required critical-direction bound.

Source: S13 §§1,6-7; S14 §§2-7,14-15; S15 §§2-9,18.

Additional scope-preserving results worth retaining

1. **Fibonacci complete-frame/end-mass theorem.** S19 Appendix F constructs the perfect Green-orthonormal adjacent-difference frame. Any completed Fibonacci packet lies in the zero-total-mass subspace; even arbitrarily many such packets leave an exact endpoint cost $M(N)^2/N$ unless endpoint mass is restored. This is an exact transfer guide for using packet decompositions with current boundary terms.
2. **Fibonacci sampling stability audit.** S19 Appendix F proves exponential near-aliasing on unrestricted late logarithmic Fibonacci grids; fixed finitely many shifted/Lucas grids still have a small Gram direction. It does not assert those constructed frequencies are actual zeta-zero pairs. A stable inverse on a specific arithmetic family remains possible but needs its own lower bound.
3. **Fibonacci Hamiltonian trace-map connection.** S12 §6.3 records the classical invariant $I = x^2 + y^2 + z^2 - 2xyz - 1$ for $T(x, y, z) = (2xy - z, x, y)$. It was explored as a spectral analogy; the note establishes no identification with the prime discrepancy/Weil operator.
4. **Shape alignment comparison.** S3 §8 compares the full zeta-value curve to a fixed-frequency raw mode by centered RMS normalization and optimal rotation. It is a separate defined numerical comparison from the exact same-parameter continuation bridge. Keep it as a historical experiment, not an extra sign theorem.
5. **24-family / 14-constant atlas.** S2 includes ordinary integers, primes, squarefree integers, powers of 2/3/6, squares/cubes, Fibonacci/Lucas/Pell/Tribonacci/Padovan, factorials, primorials, LCMs, triangular/pentagonal/hexagonal numbers, central binomial coefficients, Catalan numbers, Mersenne/Fermat numbers, selected even perfect numbers. These are applications of the common factor ledger rather than 24 separate RH theories. The constant atlas includes golden/silver/metallic and recurrence constants, square roots, e, pi, 2pi, Euler's constant, zeta(3), and log 2 as exact dilation inputs.

Name coverage

The user’s “fib* spirals” is preserved as the umbrella Fibonacci family (SP08 through SP13), not assigned a fabricated starred definition. “Other spiral” can refer to SP01/02, SP04, SP07, SP14/15, or the distinct early negative spiral SP16; all are retained separately. “Angles” includes the geometric factor direction AN01/02, the relative quadratic orientation AN05, the arithmetic phase AN06, and the complete no-reversal law AN07. These are different observables of related arithmetic data.

The most directly reusable pieces for the present CP18 task are AN02/03 (signed arithmetic cancellation in an exact metric), SP03/06 (complete prime-layer reconstruction and metric pullback), SP11 (an actual bounded-storage certificate with precise hypotheses), and AN09 (the already-built full phase/Weil/Suzuki bridge). The mapping from those exact objects to $W_a c$ and its critical boundary residual is the concrete present comparison to calculate.

Energy, reflection, and complete-wave additions (8 distinct entries)

Additional source register:

- **S20:** Build_Log_065_Dirty_Mirror_Packet_for_Claude.md, complete 209 lines. Proposed dirty-mirror representation and finite exponent audit.
- **S21:** Build_Log_066_Dirty_Mirror_Boundary_Asymptotics.md, complete 202 lines. Exact centered sine-window endpoint asymptotics.
- **S22:** Build_Log_067_Dirty_Mirror_Parity-Decomposition.md, complete text. One-edge phase convention and detuning parity.
- **S23:** Michaels_Inside_Outside_Wave_Continuation.md, complete 829 lines. Exact Green-wave dictionary and inside/outside transfer.
- **S24:** Michaels_Refined_Joint_Margin_Continuation.pdf, all 2,173 extracted lines / 56 pages retrieved; refined ancestor estimates, exact joint margin, authentic finite cofactor elimination, and scope audits read.

EX01 – Dirty Mirror / integer exponent with a running return factor

Definition: Write $x^{2-\epsilon} = x^2 r_\epsilon(x)$, $r_\epsilon(x) = x^{-\epsilon}$. The missing growth is $x^2(1 - x^{-\epsilon})$. For a decay law the corresponding identity is $T^{-(2-\epsilon)} = T^{-2}T^\epsilon$. A proposed complex return $R_T = a(T)e^{i\theta(T)}$ retains amplitude and phase; its real observable is $a(T)\cos\theta(T)$.

Made for: Separate an exact integer baseline from a slowly varying correction, and test whether boundary reflection/ringing explains the apparent near-integer fitted exponent.

Established: The algebra is exact. The proposed physical/arithmetic interpretation is not a theorem. The six-rung finite fit cannot distinguish exponent 2 from 1.998; the source explicitly does not establish $\epsilon = .002$ as a new constant. A positive scalar multiplier alone cannot model observed sign-changing ringing.

Current implication: Keep phase and mixed boundary contributions in the CP18 residual instead of replacing them by scalar loss. This is a representation and diagnostic idea; it supplies no independent critical-margin estimate.

Source: S20, founder algebra, finite audit, and complex-return proposal.

EX02 – Tuned versus detuned endpoint reflection and spectral-energy tails

Definition: For the centered model $f_\eta(u) = A \sin(\rho_\eta u) \mathbf{1}_{[-L/2, L/2]}$, $A = \sqrt{2/L}$, $\rho_\eta = (1 + \eta)2\pi/L$, the right endpoint is $q_\eta = -A \sin(\pi\eta)$. Integration by parts gives $|\widehat{f}_\eta(s)|^2 = 4q_\eta^2 \cos^2(sL/2)/s^2 + O(s^{-3})$. At exact tuning, the leading endpoint term vanishes and $|\widehat{f}_0(s)|^2 = 4A^2 \rho_0^2 \sin^2(sL/2)/s^4 + O(s^{-5})$.

Made for: Quantify what tuning actually removes from a complete energy calculation, and identify whether the observed return comes from endpoint leakage.

Established: For this explicit sine window, detuning changes the spectral-power tail from order s^{-4} to s^{-2} ; the integrated tail changes from order S^{-3} to S^{-1} . Endpoint amplitude is odd to leading order in η ; leading power is even and quadratic. The connecting theorem from tuned spectral power to the proposed finite-section T^{-2} law remains open.

Current implication: This is a real boundary-energy mechanism, useful if the authentic forced family satisfies an analogous endpoint cancellation. A smooth model's tuned tail cannot be transferred to CP18 without proving that cancellation and the metric comparison for the actual family.

Source: S21, exact centered-window asymptotics and outstanding connector.

EX03 – Dirty Mirror parity decomposition / bulk shift versus boundary ringing

Definition: The distinct one-edge convention is $f_\eta(x) = A \sin((1 + \eta)\rho(x + L/2))$, $\rho L = 2\pi$. One endpoint stays zero; the other is $q_\eta = A \sin(2\pi\eta)$, giving $|\widehat{f}_\eta(s)|^2 = q_\eta^2/s^2 + O(s^{-3})$. For a smooth perturbation expansion $H_\eta = H_0 + \eta H_1 + \eta^2 H_2 + \dots$, the odd and even comparisons isolate $\eta H_1 + O(\eta^3)$ and $\eta^2 H_2 + O(\eta^4)$.

Made for: Distinguish a first-order bulk/ceiling change from second-order boundary power; prevent an odd response from being misread as an even energy loss.

Established: The parity split and explicit endpoint coefficient. The centered convention in EX02 has $\sin^2(\pi\eta)$; this one-edge convention has $\sin^2(2\pi\eta)$. They must not be merged into a single numerical coefficient. The finite-section connector is still unproved.

Current implication: A concrete way to audit detuned, mixed-angle, and phase comparisons of CP18 probes: fix which endpoint/phase is held constant, then retain the odd, even, and mixed responses. This is a decomposition tool, not a sign certificate.

Source: S22, phase convention, endpoint calculation, and parity audit.

EX04 – Inside/outside complete wave / exact counting-time and frequency dictionary

Definition: For $F_N(x) = \sum_{n \leq \min(x, N)} f(n)$, define $B_N(u) = e^{-u/2} F_N(e^u)$. Then

$$\|B_N\|_2^2 = \sum_{k < N} \frac{|F_N(k)|^2}{k(k+1)} + \frac{|F_N(N)|^2}{N} = \int_{\mathbb{R}} |A_f(t)|^2 \frac{dt}{2\pi(t^2 + 1/4)}, \quad A_f(t) = \sum_{n \leq N} f(n) n^{-1/2 - it}.$$

Made for: Move the entire Green energy into wave coordinates while preserving the terminal contribution, and separate bounded-frequency work from frequency tails.

Established: Exact isometry. The counting-time exterior $u \geq \log N$ is exactly terminal energy $|F_N(N)|^2/N$; it is not the same object as the frequency exterior $|t| > W$. For $|f(n)| \leq 1$, the

frequency exterior is bounded by

$$\delta_N(W) = \frac{H_N}{\pi W} + \frac{4NH_N}{\pi(W^2 + 1/4)}.$$

At $W = \sqrt{N}(1 + \log N)$, this is at most $1/(\pi\sqrt{N}) + 4/[\pi(1 + \log N)]$.

Current implication: An already available rigorous answer to one version of “move to another plane”: use the exact isometry and a quantitative exterior bound, preserving the terminal term. The current Suzuki form needs its own exact identification with this Green metric; a shared word such as energy does not establish that identification.

Source: S23, complete wave dictionary and uniform exterior theorem.

EX05 – Partition-independent interference control and authentic mixed-energy signs

Definition: For disjoint coefficient partitions $f = \sum_b f_b$, compare complete interference $J_{\text{full}} = E_f - \sum_b E_{f_b}$ with its bounded-frequency version J_{in} .

Made for: Retain every cross term while dividing the arithmetic into packets, without paying an artificial factor equal to the number of packets.

Established: $\sum_b O_{f_b} \leq \delta_N(W)$ and $|J_{\text{in}} - J_{\text{full}}| \leq \delta_N(W)$. Genuine chronological zero-return blocks are exactly Green-orthogonal. First-passage endpoint packets have overlapping prefix histories and instead have an explicit nonnegative complete interference. For the authentic retained mask $a = \mu s$ and remainder $r = \mu - a$,

$$\langle a, r \rangle_G = E_{\text{all}} + \frac{1}{2}G_U \geq 0.$$

The bounded-frequency mixed term differs from this full term by at most $\delta_N(W)/2$.

Current implication: These are genuine signed/orthogonality results already in the machinery. They rule out searching for large integrated destructive retained-versus-remainder interference in that exact Green formulation. They may guide the right grouping for the current residual, but one must prove the CP18 grouping matches these authentic masks and this metric.

Source: S23, partition theorem, chronological return blocks, first-passage packets, and authentic mixed sign.

EX06 – Refined ancestor/body energy and long-closure compression

Definition: The ancestor estimate uses

$$B_0(N) = \sum_{k < N} \frac{(1 + \log k)^2}{k + 1} + (1 + \log N)^2 \leq (1 + \log N)^3, \quad E_{\text{all}} \leq \sqrt{R(N)B_0(N)} + \sqrt{R(N)D_{\text{vl}}(N)}.$$

Very long closures obey $D_{\text{vl}}(N) \leq R(Y_b(N))$, where $Y_b(N)$ is the largest $y \leq N$ satisfying $y[1 + (1 + \log y)^b] \leq N$.

Made for: Pay explicitly for all ancestor interactions, exploit very long closure geometry, and shrink the error budget at an authentic weighted record.

Established: At a record with $R(k) \leq R(N)(k/N)^\eta$, the displayed costs yield a vanishing relative bound $E_{\text{all}} \leq \omega_*(N)R(N)$. With $\lambda_* = \sqrt{\omega_*}$, the corresponding large-parameter mixed integral is bounded by $\sqrt{\omega_*}R(N)$. Source hypotheses include fixed cutoff parameter b , the authentic retention rule, and the complete terminal reserve.

Current implication: This is an actual absorption mechanism with a shrinking parameter, stronger than a picture of energy conservation. Its present relevance is to ask whether CP18's critical residual can be decomposed into an authentic ancestor term with this bound; the source does not establish that comparison.

Source: S24, refined ancestor/body estimate, inverse long-closure cutoff, and optimized split.

EX07 – Authentic finite cofactor elimination

Definition: In the large-prime frame $K = \lfloor N^{1/6} \rfloor$, an authentic retained endpoint must be born at height strictly above $\sqrt{u}(1 + \log u)$. The verified inequality $M(u)^2 \leq u$ for every $u \leq 4096$ prevents any such birth at those cofactors, at every later terminal.

Made for: Remove correction terms by an exact arithmetic/retention implication before bounding them.

Established: Whenever $64 \leq N < 4097^6$, all frame cofactors satisfy $m \leq 4096$, hence their retained coefficients vanish. The source derives $L_0 = 0$, $\psi = 0$, and $t = \chi$ throughout that finite range. It does not exclude packets born later or prove an asymptotic statement.

Current implication: A concrete example of using existing machinery to cut out whole terms rather than burying them under estimates. For the present Weil problem, the corresponding simplification must be shown for its actual prime/cofactor and null-vector variables; the large finite range is not global Weil positivity.

Source: S24, authentic finite arithmetic check and cofactor-elimination theorem.

EX08 – Complete joint-margin reconstruction / prime work and return identity

Definition: The old record-margin problem collects its still-signed work into $W_* = P_r - S_r + J_{<\lambda_*} + L_0$. The complete reconstructed margin is

$$\mathcal{F} = \frac{3}{4}R - \frac{13}{324}T + D_+ - G_U - \Delta_r + 2W_* + 2B_t + 2J_{\geq\lambda_*}.$$

Genuine prime recombination gives $W_* = (\Delta_r - \|r\|_G^2)/2 - B_t - J_{\geq\lambda_*}$.

Made for: Keep the prime factorization, correction, ancestor, low-parameter, high-parameter, diagonal, and terminal terms in one exact closing comparison.

Established: Substitution returns the exact original margin $\mathcal{F} = \frac{3}{4}R - \frac{13}{324}T + D_+ - G_U - \|r\|_G^2$. This is a consistency/return identity; it does not itself bound the surviving signed work. The source also isolates why arbitrary-mask, geometry-only, and large-prime-only substitutes fail to prove the authentic target.

Current implication: An important warning against losing machinery by changing its target: each alternate plane should return to the same full signed quantity with the same metric. For CP18, an analogous exact return should identify which old signed term corresponds to $\text{Proj}_{E_a} \Gamma_a$, and which extra arithmetic hypothesis is still required.

Source: S24, complete joint-margin identity, genuine prime reconstruction, and scoped surrogate obstructions.

Addition count: 8 EX entries. These complement the 26 SP/AN entries and 5 additional mechanisms above. EX06-08 may overlap the modifier and finite-frame catalogue; retain their exact statements once rather than counting aliases as new theories.

10. Light cones, causal arithmetic, gap reserves and operator methods

Source key

- **AL** – *Arithmetic_Light_and_the_Coherent_Closure_Conjecture_v1_1.tex*. All 1,570 lines retrieved; mathematical and program sections read.
- **SB** – *The_Arithmetic_Sandbox_Complete_Additive_Research_Record.pdf*. The structured codex (printed chapters 1-42, pp. 1-71; PDF pp. 9-79) read. Its appended historical documents and raw transcripts extend to 521 pages; those appendices were not exhaustively read here. This codex separates theorem, derived, experiment, conjecture, interpretation, and falsified material.
- **OE** – *Overseer Theorem Paper Complementarity and Occupied Edge.pdf*, all 8 pages read. Some stronger early wording is corrected in AL/SB; use their narrower Gauss-flat statement.
- **RC** – *Speculative_Math_Research_Chain_Through_Build_074.pdf*, all 22 pages retrieved and relevant sections read. Records Builds 045-074, distinguishing derived identities and open arithmetic estimates.
- **UI** – *Prime_DNA_Unitary_Identity_Closure_Build_v12.md*, all sections read.
- **TC** – *Prime_DNA_Build_v39_The_Two_Plane_Prime_Cocycle.pdf*, all 21 pages read.
- **IC** – *Integer_Conservation_Progress_Sept_1_2026.pdf*, all 3 pages read.
- **CA** – *Prime_DNA_Complete_Insertion_and_Conservation_Audit.md*, all 325 lines read.
- **CH** – *Prime_DNA_Causal_History_Payment.md*, full report read.
- **DI** – *Prime_DNA_Diffusion_Investigation.md*, full report read.
- **CMP** – *Christophe_Michaels_Theorem_Compendium_Through_Build_074.pdf*. Retrieved first 1,000 of 1,014 lines; UA-01-27, the relevant conditional/no-go entries, and the open-office registry read. This is the August 16 theorem extraction from the frozen Build 074 archive.
- **GD** – *Christophe_Michaels_Gap_DNA_and_the_Slope_Closure.pdf*, all 20 pages retrieved and read.
- **GA** – *Prime_DNA_Closure_Audit.pdf*, all 13 pages retrieved and read.

L01 – Original arithmetic light-cone / arithmetic Lorentz / invariant-manifold idea

Made for: explain why a complex system might have one fixed admissibility surface while retaining complicated motion along it. The proposed analogy assigned the critical line the role played by a causal invariant in relativity, and assigned zero-spacing statistics to residual freedom.

Status: explicitly speculative/interpretive. The claim was an analogy of roles, not a numerical identity between the speed of light and one half. The early expression $(\sigma - \frac{1}{2})^2 + M^2 = 0$ was later retired as a purported Lorentzian cone. GUE and CUE were also separated: Gaussian Hermitian statistics versus Haar unitary phases.

Now: supplies the organizing question “what authentic arithmetic invariant prohibits a transverse mode?” It does not provide the missing estimate on Γ_a ; a concrete positive metric or source inequality must implement it. CP18 is one precise realization of the admissibility question.

Source: SB chapters 1, 3-4, 38; AL chapter “Multiplicative Spacetime and the Mellin Light Cone.”

L02 – Mellin light cone and half-density propagation

Made for: identify the exact normalization behind $\Re s = \frac{1}{2}$. Under $x = e^u$, the half-density $g(u) = e^{u/2}f(e^u)$ converts the critical-line Mellin transform into ordinary Fourier transform. The dilation $U_t f(x) = e^{t/2}f(e^t x)$ is unitary in $L^2(dx)$ and a generalized mode $x^{\rho-1}$ is multiplied by $e^{(\rho-1/2)t}$.

Status: exact classical identities, assembled under the name Mellin light cone; phase-only evolution occurs exactly at real part one half. Bare Mellin modes are generalized modes, usually not Hilbert eigenvectors.

Now: supplies the correct metric and factors when translating current forced solutions into multiplicative coordinates. It yields confinement only after the actual arithmetic states are faithfully represented in that positive geometry. It does not make the completed Weil form positive by coordinate change alone.

Source: AL chapter 2; SB chapter 17; RC II and V.1.

L03 – Positive-seat / Lyapunov law

Made for: turn an invariance equation into a spectral confinement theorem. If $A^*G + GA = 0$, $G > 0$, and $Av = \lambda v$, then $\Re \lambda = 0$ because $(\lambda + \bar{\lambda})v^*Gv = 0$.

Status: proved finite linear algebra; the source-built positive G and faithful arithmetic identification remain the application burden. Indefinite conservation permits reciprocal growing/decaying pairs.

Now: can serve as an alternative closing consumer if an exact positive metric intertwines the authentic finite forced space with the arithmetic generator. CP18's positive regularizer is already available, but its positivity is not the positivity of the unregularized Weil operator; their difference is exactly retained in M_a .

Source: SB chapter 3; AL chapters 10 and 12; UI §6.

L04 – Bounded covariant-state light-cone theorem

Made for: give a mathematically precise version of “off-line propagation has an unaffordable energy cost.” A nonzero bounded functional on a positive unitary representation satisfying $\ell(U_t f) = e^{-itz}\ell(f)$ must have $\Im z = 0$.

Status: recorded as an exact theorem. The key hypothesis is bounded, nonzero, authentic state registration. A generic positive ambient space does not itself register all zeta zeros.

Now: if the existing machinery constructs bounded authentic covariant evaluations from $q = W_{ac}$ on a determining family, this theorem could bypass the particular Osgood residual route. Such registration and its metric map must be supplied.

Source: RC V.1 (Build 066), V.4; SB operator/KMS chapters.

L05 – Finite-window transverse evaluation cost

Made for: quantify rather than merely describe how moving away from the critical line changes an evaluation functional. The archived exact ratio is $\frac{\|\ell_{\alpha+i\gamma}\|^2}{\|\ell_{i\gamma}\|^2} = \frac{\sinh(2\alpha R)}{2\alpha R} \geq 1 + \frac{2}{3}R^2\alpha^2$.

Status: exact finite-window ambient dual geometry. The record separately proves that a raw completed packet-energy difference cannot have a universal positive quadratic transverse potential because its symmetric vertical Cesàro mean is zero.

Now: a quantitative testing or normalization device for candidate maps. Transfer to Suzuki/Weil energy is needed before this cost can constrain Γ_a or a first-null state.

Source: RC V.3 (Build 068).

L06 – Prime-cosh tempered-confinement theory

Made for: combine the large prime and polar contributions before estimating them. The signed source is $\mu_{\text{pc}} = -\sum_n \frac{\Lambda(n)}{\sqrt{n}} (\delta_{\log n} + \delta_{-\log n}) + 2 \cosh(x/2) dx$, with cumulative $A(x) = 4 \sinh(x/2) - \sum_{\log n \leq x} \frac{\Lambda(n)}{\sqrt{n}}$ and $g'_{\text{pc}} = -A$, $g''_{\text{pc}} = -\mu_{\text{pc}}$.

Status: RC records exact equivalences $\text{RH} \iff \mu_{\text{pc}} \text{ is tempered} \iff A \text{ has polynomial growth}$, with an explicit scaled-pole noncancellation repair. The growth estimate itself remains open.

Now: provides an alternative direct arithmetic target. A source-growth theorem from colors, prime history, or complete packets would be relevant without first estimating CP18's boundary residual. This is a whole-source criterion; separately bounding its large pieces loses the intended cancellation.

Source: RC V.4 (Build 069), equations (5.5)-(5.9).

L07 – Screw-function / dyadic light cone

Made for: turn the light-cone picture into a single scalar dilation inequality on the actual completed screw function. For $K_g(s, t) = g(s - t) - g(s) - g(t)$, a positive feature distance gives subadditivity of $\sqrt{-2g}$ and the dyadic law $|g(2t)| \leq 4|g(t)|$.

Status: RC records $\text{RH} \iff K_g \succeq 0 \iff \text{the dyadic law for all } t \iff g(t) = O(1 + t^2)$, under the fixed authentic normalization. The dyadic bound has not been proved independently.

Now: a genuine earlier alternative to Γ_a control. Its inputs must be the completed screw function, including correction terms. Any arithmetic machinery that bounds this signed dilation increment could close the common positivity objective through this scalar route.

Source: RC VI.1 (Build 070), equations (6.1)-(6.4).

L08 – Dilation moment and signed Hardy history

Made for: convert a completed prime cumulative into a scale observable without dropping its history. Define $J_A = tg'_{\text{pc}} - g_{\text{pc}}$ and $K_A(t) = J_A(2t) - 4J_A(t)$. Exact Euler inversion gives $\frac{g_{\text{pc}}(t)}{t} = \int_0^t \frac{J_A(u)}{u^2} du$ and a corresponding identity for the doubled shell.

Status: exact identities and two alternative sufficient routes: a signed scalar Hardy estimate, or an operator/collar comparison. The relevant sign is not the instantaneous sign of J_A .

Now: can express complete arithmetic boundary/dilation work using a previously defined observable. It is a concrete candidate for connecting old history estimates to the new residual, provided the transformation is written and all endpoint terms retained.

Source: RC VI.4 (Build 071), equations (6.9)-(6.13); also *Build_Log_071_Dilation_Hardy_She11_and_Relative_Flux_Confinement.pdf* (search discovery; RC supplied the read evidence here).

L09 – Doubled-displacement law and cubic shell criterion

Made for: give an unavoidable arithmetic signature of an off-line zero. The transform of K_A has scaled zero poles that cannot all cancel; displacement beta forces exponential type at least 2β .

Status: RC records the exact quantitative pole argument and $\text{RH} \iff K_A = O((1+t)^3) \iff K_A$ has polynomial growth $\iff K_A$ is subexponential. The source-side growth bound remains open.

Now: another direct closing route for complete arithmetic cancellations. It translates a sufficiently strong old bound into a spectral consequence with a known factor two, instead of requiring an estimate on every CP18 matrix entry.

Source: RC VII.1-2 (Build 072), equations (7.1)-(7.8).

L10 – Zero-energy equilibrium at infinite support and relative-loss confinement

Made for: express the critical state as an infinite-support limit of strictly positive finite states. The localized bottom λ_a decreases with support; the recorded theorem is $\text{RH} \iff [\lambda_a > 0 \text{ for every finite } a, \lambda_a \downarrow 0]$. Relative loss satisfies $\lambda_b = \lambda_a \exp(-R((a, b]))$ with $dR = -d \log \lambda$.

Status: exact criterion and conditional confinement description; positivity at every finite radius is the unresolved part.

Now: this is the direct conceptual ancestor of CP18's no-first-crossing/Osgood route. CP18 supplies much stronger regularity of its physical matrix than the old statement. An integrable loss bound would finish this route; the scalar relative-loss identity itself supplies no such bound.

Source: RC VI.2-3 (Builds 070-071), equations (6.5)-(6.8).

L11 – Complete positive/adverse source factorization and stable Witt cancellation

Made for: make “infinity confines infinity” exact. At fixed support, $Q_W = \|B_+ f\|^2 - \|B_- f\|^2$; enlarging support adds matched positive and adverse channels which cancel on old supported vectors.

Status: exact completed Krein factorization and neutral increment. A legitimate Witt quotient requires $N \subset N^\perp$ and passage through $N^\perp/N$, or an explicit hyperbolic cancellation. Isotropic alone is not radical. Universal contractivity of the adverse graph remains the Weil sign condition.

Now: strongly relevant to CP18 collar blocks and the source residual: it says which old terms may cancel exactly before estimation and which new boundary interaction survives. It prevents replacing a relational identity by subtraction of separate divergent norms.

Source: RC V.2-3 (Builds 067-068), XV.

L12 – Finite Weil seats, null quotient, seated dilation, and cohomological formulation

Made for: construct actual finite positive spaces from completed finite Weil matrices. For a lowest eigenpair (ϵ_j, ξ_j) , $G_j = Q_j - \epsilon_j I$ is positive semidefinite; quotient by $\mathbb{C}\xi_j$ gives a positive physical metric. A rank-one seated dilation kills ξ_j and, under the displayed compatibility, is metric self-adjoint. A two-term complex encodes the same quotient.

Status: finite construction/conditional self-adjointness; the two-prime three-mode example is explicitly numerical. The cohomology language is elementary quotient algebra, not an established field-theoretic BRST realization.

Now: predecessor to finite regularization and forced-family compression. Its null quotient does not automatically prove positivity of the original unshifted form. Exact compatibility and source identity are needed to transfer a finite metric into CP18's M_a .

Source: AL chapter “Finite Prime-Built Seats”; SB chapter 14.

L13 – Michaels Coherent Closure Conjecture

Made for: unify prime-built finite positive seats into one global source-faithful self-adjoint realization. Required pieces are isometric nesting, strong-resolvent convergence, completed explicit-formula trace identity, determinant identity $\Xi = C \det_\zeta(zI - D)$, and compatible prime-adjointing maps.

Status: precise conjecture with a proved conditional implication to RH. Stronger than the bare zero-location assertion; specific nesting or determinant construction could fail even if RH holds.

Now: overarching interpretation of the current work. CP18 follows the smaller source-positivity route, which does not require first solving all determinant and trace convergence obligations. A successful old compatible Gram construction could still supply positivity directly.

Source: AL chapters 12-15, full clauses C1-C7; SB chapter 24; RC I.

L14 – Exact resolvent synchronization / no-hidden-coupling criterion

Made for: test whether positive finite seats are genuinely restrictions of one object. For $R_A(u) = A(A + uI)^{-1}$, vanishing $J^*R_B(u)J - R_A(u)$ at every positive u is equivalent to the embedded space reducing B and $BJ = JA$.

Status: recorded exact criterion. Raw compression can miss hidden off-diagonal couplings; the resolvent test detects them.

Now: a ready-made test for any proposed identification of the old color/spiral metric with CP18's physical matrices. Matching diagonal energies or a few moments is insufficient; this tells what stronger compatibility can certify.

Source: AL multi-vector update, “Exact resolvent synchronization criterion.”

L15 – Growing observers, Schur/Feshbach lens, and exact Galerkin residual energy

Made for: retain the directions that can carry an adverse source while controlling the rest. The exact weighted response is $L = CH^{-1}C^* = D\Pi_T D^*$; for a graph-domain trial solution X with residual $R = C^* - HX$, $L - L_X = R^*H^{-1}R$.

Status: recorded exact finite/functional identities; the authentic cofinal residual estimate remains open. Fixed-rank observer capture and norm approximation of infinite-rank projections were ruled out in their stated forms.

Now: direct predecessor to CP18's finite source family and probe residual. These formulas can compare candidate arithmetic probes in the correct energy, but do not provide uniform residual decay on their own.

Source: RC III-IV, particularly IV.1 and IV.5-6 (Builds 059, 064, 065).

L16 – KMS/GNS positive-state mechanism

Made for: obtain a positive metric from an arithmetic state instead of choosing one after seeing the desired spectrum. A positive invariant state gives $G(X, Y) = \omega(X^*Y)$; invariance leads to a unitary flow and its metric Lyapunov identity.

Status: general operator-algebra implication is exact; the required source-faithful arithmetic state and spectral identification are proposed/open. A generic KMS state does not establish the Weil identity.

Now: candidate origin of a positive factorization of $Q_a^-(W_a c)$. A source-faithful state whose Gram matrix equals that finite form would be directly relevant; no such equality is established in the retrieved source.

Source: SB chapter 11; AL multi-vector architecture item KMS/GNS/rigidity; RC I.

L17 – Information-metric / relative-entropy / completely-positive-channel crossing

Made for: derive Weil positivity from a known monotonicity principle. Candidate mechanisms identify the Weil form with half a second variation of relative-entropy loss, a BKM metric, or a Choi kernel of a prime-to-zero completely positive channel.

Status: explicitly candidate mechanisms. The archive retires equating a Choi matrix with a Weil Gram matrix before matching spaces and normalization, and retires an asserted unconstructed Virasoro-Shapovalov intertwiner.

Now: alternative Gram construction strategy on CP18's finite family. The deliverable would be an explicit channel/metric identity containing all prime, gamma, polar, and boundary terms. Data processing supplies a sign only after that identity.

Source: SB chapters 12 and 38; AL multi-vector architecture.

L18 – Cayley zero coupling, Li harmonic moments, holding constant, and leakage functional

Made for: encode radial displacement in a dimensionless number $u_\rho = (\rho - 1)/\rho$. Unit modulus is equivalent to $\Re \rho = \frac{1}{2}$; under a self-adjoint realization it is a Cayley phase. Li coefficients then have the squared-energy representation $\lambda_n = \frac{1}{2} \|I - U^n\|_{\text{HS}}^2$. The named holding constant is the classical first Li coefficient $h_\zeta \approx 0.0230957089661210$. Leakage is $\Delta_\zeta = \sum_{\Im \rho > 0} (\log |u_\rho|)^2$.

Status: algebraic coupling and convergence/zero-leakage criterion proved in AL; nonnegative Li moments and the unitary norm representation are conditional on the positive spectral realization. The arithmetic names organize classical quantities.

Now: independent diagnostic coordinates and another possible closing consumer. A source-derived positive representation of all Li moments would suffice, but a finite moment match or the already positive first coefficient does not control Γ_a .

Source: AL chapter “Dimensionless Coupling, Holding, and Leakage”; SB chapter 15.

L19 – Herglotz passivity, Pick kernel, and semicircle localization

Made for: detect a nonreal zero through failure of positivity of the logarithmic response $m_\Xi = -\Xi'/\Xi$ and its Pick kernel. An isolated conjugate counterfeit pair produces a negative one-point detector inside the corresponding upper semicircle.

Status: exact criterion and localization calculation in the record; finite benign sample points can miss a small negative region. Source-built global passivity remains open.

Now: can test whether a proposed old positive carrier really equals the current source; it provides a local counter-witness detector. An exact positive Pick representation from arithmetic would close another route, but positive sampled matrices alone do not.

Source: AL Herglotz chapter and multi-vector “Semicircle localization”; SB chapter 13.

L20 – Occupied Edge / orlo and de Bruijn-Newman criticality

Made for: express the intuition that a self-dual system occupies its transition boundary with zero displacement, called its orlo. The arithmetic candidate is the de Bruijn-Newman parameter $\Lambda = 0$.

Status: the fixed-locus lemma needs an involution exchanging phases and a unique transition; its application to the zeta heat parameter is not supplied by analogy. The arithmetic Occupied Edge is recorded as a heuristic/conjectural principle. The equivalence $\text{RH} \iff \Lambda = 0$ uses classical heat theory plus the known nonnegative lower bound.

Now: conceptual explanation of a zero-margin limit. It does not establish that a finite-support matrix avoids zero. A new actual heat or source estimate would be required to transfer it to CP18.

Source: OE §§5-6; AL heat-flow chapter; SB chapter 16.

L21 – Arithmetic matter, prime Fock space, and primon gas

Made for: retain complete factorization as prime occupation numbers: $|n\rangle = \bigotimes_p |\nu_p(n)\rangle$; multiplication adds occupations and $H_A|n\rangle = (\log n)|n\rangle$. The convergent partition function is zeta. Factorization coproducts give superpositions of factor pairs.

Status: exact encodings and classical partition identity; the original claim that an individual prime tensor product is automatically entangled was corrected. Thermodynamic interpretation is separate.

Now: a natural common space for colors, mixed prime packets, and later arithmetic dynamics. It can preserve correlations lost by projection to a single counting coordinate. The metric correspondence to Weil/Suzuki forms remains a separate exact calculation.

Source: SB chapters 5 and 38.

L22 – Divisibility causality, arithmetic boosts, incidence propagator, and Möbius inverse field

Made for: represent arithmetic ancestry as $d \preceq n \iff d \mid n$, with past divisors and future multiples. Multiplication maps obey $e^{itH_A}V_m e^{-itH_A} = m^{it}V_m$. The divisor-incidence matrix propagates a source by divisor sums; Möbius inversion recovers it. Path depth is $\Omega(n/d)$ or $\log(n/d)$.

Status: exact order/operator/incidence identities with a physical vocabulary. The Redheffer determinant/Mertens connection is a classical finite bridge. Raw prime-permutation symmetry shrinks sharply when numerical log energies must be preserved.

Now: identifies which maps preserve authentic prime ancestry and which merely preserve an abstract positive norm. Useful for organizing current active prime shifts; source recovery does not itself prove their complete signed energy is nonnegative.

Source: SB chapters 6-8; UI §8; CA §4.

L23 – Arithmetic equivalence principle and thermodynamic-axis interpretation

Made for: distinguish coordinates and scales. Prime density can be unfolded by $\text{Li}(x)$; an analogous zero unfolding uses the smooth counting function. A proposed density-weighted conformal metric preserved the candidate cone. Primon-gas inverse temperature has dimensionless coordinate $\sigma = \beta\epsilon_0$.

Status: unfolding identities are classical; the spacetime conformal metric is interpretive. The correction ledger places the Gibbs wall at $\sigma=1$ and infinite temperature at $\sigma=0$, with $\sigma=1/2$ a proposed self-dual membrane rather than a cold vacuum.

Now: coordinate bookkeeping and a warning to retain Jacobians and metric weights when moving between planes. No current source identifies the thermal or unfolded metric with CP18's Weil metric.

Source: SB chapters 9-10 and 38.

L24 – Divisibility interference detector, dark-port sieve, and Gauss-flat complementarity

Made for: give exact phase descriptions of arithmetic rules: $\mathbf{1}_{p|n} = p^{-1} \sum_a \exp(2\pi i a n/p)$. Product of silent channels through $\sqrt{n}$ detects primality; expanding it is Möbius inclusion-exclusion. Multiplicative characters have flat magnitudes at nonzero additive Fourier frequencies via Gauss sums.

Status: exact classical algebra reused in the architecture. AL narrows early “full mutually-unbiased basis” language to the correctly normalized Gauss-flat result with principal/zero sectors treated separately. The finite sieve statement is safely applied at cutoff $y = \sqrt{n}$; an unrestricted larger cutoff containing n would also remove a prime n .

Now: concrete bridge between color/divisibility information and additive phase tests. Any application to W_{ac} must transport the full mixed products and continuous weights, not substitute a prime-only positive diagonal for the completed source.

Source: AL additive detector and complementarity chapters; SB chapters 20-21; OE §§3-4.

L25 – Prime harmonic trains, Huygens-Sieve recursion, and self-exciting arithmetic field

Made for: state that each newly admitted prime creates a future divisibility detector, an Euler factor, and a logarithmic source train at $m \log p$. The exact update is $q \mapsto (P_q, (1 - q^{-s})^{-1}, H_q)$.

Status: local algebra exact; “the click becomes a slit” and threshold laser language are interpretations. Global isometric source update remains part of Coherent Closure.

Now: tells how to group all powers of a new prime and retain interactions at support thresholds. It is especially relevant to computing the authentic arithmetic contribution to CP18's boundary matrix rather than treating each term as unrelated forcing.

Source: AL prime-source and Huygens-Sieve chapters; SB chapters 19-23.

L26 – Rule-record commitment, Perron shutter, Möbius filter, and diffraction

Made for: distinguish the full transform from a finite counting readout. The named commitment operator implements $\log \zeta \xrightarrow{\text{Perron}} J_0 \xrightarrow{\text{Möbius}} \pi_0$. Prime-power apertures have explicit pair phases $\gamma \log(p^m/q^n)$ and their squared amplitudes preserve cross terms.

Status: transform/inversion and interference expansions exact with standard analytic conventions; wave/particle and record-overlap vocabulary is interpretation. Completed explicit-formula terms remain essential.

Now: provides an existing language and formula for finite-support observation and mixed prime phase interaction. A current boundary estimate should use the full aperture/correlation, including cross terms, rather than replace it by its diagonal energy.

Source: AL Perron and diffraction chapters; SB chapters 18-22.

L27 – Genesis recursion, prime-2 seed, and activation horizon

Made for: formalize generation by accumulated arithmetic rules. Starting from $p_1 = 2$, $P_k = \prod_{i \leq k} p_i$, the next prime is the first $n > p_k$ coprime to P_k . A new prime q first performs independently necessary composite exclusion at q^2 ; smaller composite multiples are already excluded by earlier primes.

Status: exact elementary recursion and activation-horizon theorem. The older 2025 “Genesis Equation” is retained as an untyped generative intuition, superseded for calculation by this typed recursion.

Now: can organize authentic source packets and distinguish birth from first independent arithmetic effect. It may identify correlated groups before estimation, but a positivity margin does not follow just from correct enumeration.

Source: SB chapters 25-27; AL multi-vector Genesis discussion.

L28 – Rule accretion, operative jurisdiction, constraint profiles, and omission scars

Made for: compare systems whose generated objects alter the effective conditions for future generations. Prime jurisdiction is $(\mathbb{Z}/P_k\mathbb{Z})^\times$; CRT adjoining q gives an exact new exclusion fraction $1/q$ and retained density factor $1 - 1/q$. Profiles track mass, polarity, overlap, range, activation, history, and spectral response.

Status: exact prime/CRT instances and an elementary coprime-periodic-sieve density theorem; a general taxonomy is proposed. The comparative census includes primes, Stanley sequences, Ulam sequences, Mian-Chowla, triangle-free growth, and self-avoiding walks. In the specified Genesis omission semantics, withholding k primes creates $k(k+1)/2$ false products with exact reciprocal-weight scar formula.

Now: suggests retaining history and mixed products as real observables. It supplies finite bookkeeping and counterfeit tests, not yet a universal spectral positivity law. A successful profile-to-energy theorem would need to apply to the actual completed packets.

Source: SB chapters 28-36; AL multi-vector “Correlated-omission scar law.”

L29 – Prime-DNA unitary identity / fixed adjoint identity / one-prime criterion

Made for: recast location as identity. With $J\rho = 1 - \bar{\rho}$, centered transform characters satisfy $\chi_s^\# = \chi_{Js}$. A character is a star-character precisely on the line. At one prime p , $|p^{\rho-1/2}| = 1$ already determines the radial coordinate; reflected Jordan fingerprints encode the same test.

Status: exact equivalences. Source reconstruction currently yields the reflected form W_Φ ; its equality with the positive same-state form N remains open. The exact defect is $N(f, f) - W_\Phi(f, f) = 2\|P_Z f\|^2$. Reflection-unitary transport preserves an indefinite form and can allow off-line pairs.

Now: very direct alternative interpretation of the missing sign. An exact source-faithful equality on a determining family would finish the common problem. CP18's Γ_a has not been proved equal to this older antisymmetric defect; an intertwining identity is the needed bridge.

Source: UI §§0-10, especially (5.5), (7.6), (8.5)-(8.6).

L30 – Local positive compression and all-window unitary carrier

Made for: construct a source-first positive carrier on a support window using a null correction N_R whose inverse transform vanishes on that window, with completed multiplier $M_R + N_R \geq 0$. A compatible family would yield an isometric global carrier and unitary translations.

Status: local positive compression recorded from Build v11; compatible all-window extension is the open Completed Prime-DNA Unitary Identity Lemma. A single window does not separate every hypothetical reflected orbit.

Now: an earlier version of the exact current task. It could close by extending source-faithful positive carriers rather than proving CP18's particular trace certificate. The regularized physical restriction now supplies additional infrastructure, but unregularized positivity still has to be recovered.

Source: UI §§9, 11-12.

L31 – Minimal positive two-plane prime cocycle and complementary hidden wave

Made for: realize each prime Euler/Möbius factor as part of an exactly lossless larger state. A prime uses strength $r = p^{-\sigma}$ and delay $\log p$, equivalent to rotation $z = e^{-i\tau \log p}$. Minimal positive normalization is $1 + r$; the two outputs are $\frac{1-rz}{1+r}$ and $\frac{\sqrt{r}(1+z)}{1+r}$.

Status: exact degree-one paraunitary 2×2 matrix with determinant z ; exact visible+hidden conservation and finite-terminal reserve. A fresh hidden channel per prime preserves the scalar Euler path; recycling one two-dimensional state introduces feedback and changes the arithmetic.

Now: a concrete higher-dimensional construction that preserves mixed lag correlations and boundary reserves. To affect CP18, it must represent the actual complete Weil forced family and its metric. Total positive energy in all channels alone does not bound the signed physical output.

Source: TC §§2-8 and 13.

L32 – First-prime trace, visible-channel escape, moving metric, and coherent recombination

Made for: determine where energy goes in the infinite two-plane construction. The normalized visible Bohr energy is $\prod_p \frac{1+p^{-2\sigma}}{(1+p^{-\sigma})^2}$; it vanishes for $\frac{1}{2} < \sigma \leq 1$, while the unnormalized coefficient

energy stays finite there. Recovering the arithmetic output costs $\prod_p(1 + p^{-\sigma})$ and its divergent first-prime trace.

Status: exact products/asymptotic statements and explicit no-go results. A moving one-port metric is forced to be proportional to the inverse squared Euler product; uniform limiting coercivity contains the zero-free issue. Complete authentic composite recombination before squaring remains open/RH-strength.

Now: identifies a specific risk in changing to an “infinity plane”: positivity can move energy out of the observed channel. A map back to CP18 must retain hidden-channel recombination and terminal reserves. This is highly relevant to any attempt to use old conservation as an immediate bound on Γ_a .

Source: TC §§9-17.

L33 – Integer conservation, canonical prime tree, fractional transport, and parity channel

Made for: derive cancellation from finite counting and factor multiplicity. Prime-threshold recursion avoids duplicated paths; fractional transport $\mu(n) = -\omega(n)^{-1} \sum_{p|n} \mu(n/p)$ makes each squarefree child receive total parent mass one. The multiplicity polynomial is evaluated at -1 to recover Mertens parity.

Status: exact finite identities. Positive depth and color orthogonality control some sectors, but evaluation at common phase -1 recoheres the monomials; central parity mass remains uncontrolled.

Now: supplies genuine arithmetic constraints that could be used on complete mixed prime packets. It points toward cancellation specific to authentic inputs, not an all-input positive norm comparison.

Source: IC §§1-4.

L34 – Logarithmic record law, critical quotient scale, and successor potential

Made for: connect counting records to a forced opposite-sign prime response. The exact logarithmic recurrence relates $M(X) \log X$ to a convex historical mean minus $\sum_d \Lambda(d)M(\lfloor X/d \rfloor)$. At the self-dual scale $d \asymp \sqrt{X}$, quotient cells change character. The positive successor potential $W_y(m) = 2^{-\omega_y(m) - \omega_y(m+1)}$ has complete-period mean $\prod_{p \leq y} (1 - 1/p)$.

Status: exact recurrence, quotient telescoping, and full-period mean. Their localization on the shorter moving critical board remains open; nearest-neighbor persistence alone reaches only an approximately $X^{3/4}$ barrier in the recorded argument.

Now: offers a concrete authentic correlation mechanism from the old arithmetic program. An exact transport from current forced functions to these weighted moving boards would be required before their local discrepancy can bound Γ_a .

Source: IC §§5-11.

L35 – Exact positive counting conservation and the quantified metric-transfer obstruction

Made for: conserve an honest finite quantity before subdividing packets. For the max-kernel energy $E_N(a) = \sum_{m,n} \frac{a_m a_n}{\max(m,n)}$, divisor reconstruction $B_N \mu = e_1$ gives $C_N(\mu) = E_N(B_N \mu) = 1$ exactly. Complete insertion is $T_P = \prod_{p|P} (I - D_p)$ and the transported metric changes with P.

Status: exact conservation, exact signed insertion energy, and proved limits. Authentic completed insertion can increase original energy. No fixed positive definite metric makes a nontrivial finite unipotent insertion contractive on the whole coefficient space. A universal comparison $E_N \leq L_N C_N$ must pay at least $N/4 + 3/2$ for even N.

Now: preserves a valuable invariant while specifying why the original Suzuki/Weil metric matters. A useful transfer can still target the authentic family $W_a c$; general all-input metric equivalence has a concrete obstruction. The old $E_N(a)$ here is a different object from the later normalized Suzuki defect $\mathcal{E}_N$ and must not be conflated.

Source: CA §§2-8, equations (1), (3), (6)-(13).

L36 – Causal history payment and compensated event-continuum balance

Made for: calculate how complete Möbius history, prime births, composite copies, and a continuous reference pay for one another. The record restores the entire preceding state, retains terminal/prefix corrections, and computes the signed event-by-event balance.

Status: exact evolution and sign examples. The reference can add or remove energy; bounding it separately by a squared forcing cost exceeds the proposed bank by a quantified amount. The combined signed budget remains open.

Now: directly relevant methodology for Γ_a : a boundary residual can be small through arithmetic cancellation even if its separated terms have large norms. This source supports using full historical correlations, but contains no established map to the current residual.

Source: CH full report, especially continuum sign examples, cofactor-two obstruction, and final payment identity discussion.

L37 – Green-field diffusion / denoising with source-faithful boundary update

Made for: recover or repair a finite arithmetic field using its neighboring-position inverse kernel, prescribed source, and exact finite boundary. Equilibrium is the Green field; a discrete Hardy bound gives a window-independent convergence rate. Enlarging the counting endpoint adds an explicit common offset to old field positions.

Status: proved finite evolution/boundary identities; noise-reconstruction experiments are numerical. Source inversion becomes increasingly sensitive, and valid constraints improve surrogate sources too, so the demonstrated denoising benefit is not a confinement theorem.

Now: a possible practical way to construct/check finite arithmetic probes while preserving boundary data. It gives stability relative to a supplied equilibrium, not a sign or global upper bound for its size.

Source: DI local field geometry, conditioning, boundary-extension, and reconstruction sections; source anchors identify *Source-Faithful Prime-Zero Geometry* (Aug 28) §§8-9 and *Diameter Representation Theorem* v60.

L38 – Coupled arithmetic heat-source evolution and the completed-heat transfer

Made for: retain factor-pair interactions under Gaussian log-size deformation of Dirichlet coefficients. Logarithmic differentiation produces a closed finite recursion and nonlinear evolution in which proper factor pairs and repeated primes all participate; mixed composites acquire signed contributions.

Status: exact finite coefficient identities, independently checked numerical evolution; correspondence to the fully completed zero heat flow needs completion factors, coordinate changes, and a justified regime. The local Green diffusion and this arithmetic heat flow are distinct generators.

Now: another mechanism for retaining mixed arithmetic interactions. It might supply controlled deformations of the forced family, but a source/metric identity with CP18 is needed. Ordinary forward smoothing is not an automatic RH mechanism; the record explicitly corrects the de Bruijn-Newman heat direction.

Source: DI arithmetic heat sections and completed-object transfer boundary.

L39 – Sobolev prime-shift cyclicity and local source-continuity corridor

Made for: show that one classical compact sine window and bilateral shifts by $m \log 2 + n \log 3$ generate a dense test family, while prime-power evaluations and the real-place functional remain continuous in the correct Sobolev norms. The registered source lies in H^σ exactly for $\sigma < \frac{3}{2}$; prime point evaluations require $\sigma > \frac{1}{2}$, and the real-place convolution-square bound is on H^1 .

Status: recorded analytic theorem, with fixed-support jurisdiction and explicit real-place constant. It proves density/continuity, not support-uniform positivity.

Now: particularly useful for the user's wish to reuse a small arithmetic generating family: an exact cyclic family can make a positivity calculation determining. To use it on CP18, verify the relevant closed form domain and transport from its seed/shifts to $W_a c$; the current domain theorems may supply part of that connection.

Source: CMP UA-01 (Builds 042, 043, 044, 044B), pp. 1-2.

L40 – Polynomial-Gaussian radical flags, common Xi factor, and fixed-flag Hurwitz closure

Made for: replace dependence on one preferred radical vector by a stable finite flag. Completed channels factor as $\hat{k}_\ell^\pm = R_\ell^\pm \Xi$, and internal rotation of a fixed independent flag retains the same Xi factor. A cofinal source-authenticated family with the prescribed parity, roots, remote positivity, and decaying terminal shadow would close via Hurwitz.

Status: exact finite factorization/invariance and conditional closure theorem. Fixed-flag capture, full observer transport, parity inertia, and cofinal shadow decay remain producer hypotheses.

Now: an earlier endpoint theorem that can consume a sufficiently strong old arithmetic construction. It gives a concrete reason to retain the radical ladder in the atlas, but CP18's finite family has not been identified with this radical flag.

Source: CMP UA-03 and CR-03; Build 051A, through-058 catalog §9.1; RC III.3-5.

L41 – Selected Feshbach crossing slope, matrix Weyl shadow, and visible leakage

Made for: measure the actual direction carrying a potential terminal state. For a one-dimensional terminal kernel, the archived identity is $-\langle p_0, S'(E_0)p_0 \rangle = 1 + r_0^2$, with normalized terminal mass $(1 + r_0^2)^{-1}$. The compressed resolvent residue is $-P|\xi_0\rangle\langle\xi_0|P$; its positive mass is exactly $\|P\xi_0\|^2$.

Status: exact directional identities with specified Gram normalization. A fixed observer can miss an orthogonal “dark” state; slope alone does not establish residue orientation or cofinal capture.

Now: relevant if a candidate use of colors/angles is expressed through a spectral pole or crossing slope. CP18’s null-detecting finite family addresses a related visibility problem, but identifying the two directional observables still requires an exact map.

Source: CMP UA-04-05, NG-02 and NG-04; Builds 054-055.

L42 – Joint relative-energy cut and source-certified dangerous cluster

Made for: reduce all fixed-scale threats to finitely many relative modes. For positive compact-resolvent H_0 and bounded self-adjoint arithmetic perturbation P_{ar} , the sandwich $K = H_0^{-1/2}P_{\text{ar}}H_0^{-1/2}$ is compact; removing eigenvalues above $1 - \delta$ leaves $H_0 - P_{\text{ar}} \geq \delta H_0$ on the complementary energy sector.

Status: exact fixed-scale reduction. Cofinal source-tail enclosure, physical observer transport, and terminal-shadow control remain separate.

Now: direct ancestor of finite regularization. It explains why a finite family can detect the dangerous part of a complete infinite-dimensional form at each fixed R. CP18 already makes that reduction concrete; old estimates on the same represented perturbation could be imported instead of rebuilding the reduction.

Source: CMP UA-06; RC III.6, Build 058.

L43 – P59/L59 prime-phase excursion and remote-pivot route

Made for: translate arithmetic phase inflation on source shells into positive even/odd remote compressions. Exact enumeration, shell/Haar/discrepancy bounds, support termination, completed remainder budgets, and positive margins together give a remote coercivity certificate.

Status: exact conditional fixed-scale verifier. The cofinal signed inflation estimate P59, growing low-shell control L59, economical observer transport, parity orientation, and terminal-shadow compatibility remain open.

Now: candidate supplier of source-side control on a represented remote block. It is not identical to the current Γ_a estimate. A successful P59/L59 producer may support a different established closure route, while a restricted version could potentially control the current forced family.

Source: CMP UA-07 and OP-08; RC IV.1, Build 059.

L44 – Positive Dirichlet-square and primitive-ratio phase-sinc representation

Made for: preserve the exact arithmetic phase relationships inside a positive amplitude. At fixed finite source, $\exp(sD_\lambda(t)) = |F_{\lambda,s}(t)|^2$; the Haar mean is diagonal mass, while the actual normalized flow mean retains coprime primitive-ratio phases and sinc factors.

Status: exact convergent fixed-scale identities. A positive bandlimited/absolute-affinity majorant is proved too costly in the intended onset regime; signed phase cancellation must be used.

Now: concrete earlier machinery for organizing angles and prime ratios before squaring. It may be suited to a finite-family comparison, but the positive Dirichlet square is not itself the signed completed Weil energy. The precise map and correction terms decide its usefulness.

Source: CMP UA-08 and NG-05; RC IV.2, Build 060.

L45 – Signed logarithmic Paley-Wiener wavepackets and Gallagher compression

Made for: express shell response as an exact L^2 norm of a convolved signed logarithmic source, with support and phase retained. P59 becomes a specified wavepacket inequality; a phase-preserving Gallagher compression supplies a local sufficient condition.

Status: exact reformulation/conditional local certificate. The cofinal wavepacket bound remains open; center-blind mean-value bounds are supercritical in the recorded attempts.

Now: a potential technical interface between arithmetic phase packets and the compact-support forced solutions of CP18. It gives the norm that a proposed spiral/color map must actually control, not just a visual match of phases.

Source: CMP UA-09; RC IV.3, Build 061.

L46 – Compound-Poisson signed recursion, prime-grouped Laurent update, and Gaussian calibration

Made for: sum complete exponent tails while preserving phase and sinc signs. The normalized response has an exact source-by-source semigroup representation, center-twisted cell recursion, prime-grouped Laurent recurrence, shifted-smooth transform, and near-axis Gaussian calibration.

Status: exact finite-source transfer mechanics. Fixed prime blocks recur with supremum approaching one; no uniform log grid is invariant under both $\log 2$ and $\log 3$. One-source absolute contraction and fixed-grid closure are ruled out; growing-block averaged deficits remain eligible/open.

Now: could organize a collective arithmetic update for CP18's residual or for the older wavepacket route. It specifically advises retaining growing groups and phases; a constant local shrink factor would contradict the documented recurrence obstruction.

Source: CMP UA-10 and NG-06; RC IV.4, Build 062.

L47 – Reciprocal-source Jensen lens, parity slots, polar weighted source, and exceptional observer

Made for: control negative directions by the inverse positive source response rather than unrelated extreme norms. Compression Jensen bounds H^{-1} by a reciprocal multiplier compression. The odd top slot and even second slot give parity inertia criteria. The sharper identity $CH^{-1}C^* = D\Pi_TD^*$ allows source spectral cuts to be back-projected to finite physical observers, with explicit relative remote margin and rank bounds.

Status: exact identities and conditional spectral criteria; cofinal porosity, action-domain authentication, outward tails, and the required parity slot bounds remain open. This expands entry 15 with the distinct Jensen and exceptional-source mechanisms.

Now: directly matches the current use of a positive inverse and finite source family. Old source enclosures may be reusable if they concern the same H, C, and metric. The bound must remain directional or matrix-valued to preserve relevant cancellation.

Source: CMP UA-11-12; RC IV.5, Builds 063-064. The Galerkin residual-capacity theorem is CMP UA-13 and entry 15 above.

L48 – Prime-power knot topology, finite-step Feshbach certification, and compact-strong continuation

Made for: propagate positivity through all actual support thresholds $\kappa_j = \frac{1}{2} \log(p^r)$. The knots are locally finite and cofinal; the new endpoint shift is zero almost everywhere exactly at its entry. A compact relative two-point lens reduces strict endpoint positivity to a finite retained Schur test with an outward tail. Compact sandwiching upgrades shrinking-edge strong convergence to relative norm convergence.

Status: exact topology, endpoint convention, finite-step reduction, semidecision completeness, and local continuation mechanism. The first-knot positive certificate is historical and later checkpoints strengthen the certified interval. A universal arithmetic successor certificate remains open. The all-state scalar differential Grönwall proposal after prime 2 is explicitly ruled out, while finite steps survive.

Now: CP18's monotone Lipschitz physical matrix supplies a newer evolution description through these same thresholds. The old finite-step route remains an available alternative. An arithmetic certificate at every finite successor would suffice even if its product tends to zero only at infinity.

Source: CMP UA-21-25, NG-08-09, CR-08; RC VII-IX.

L49 – Prime-2 edge/center reducing geometry and residual-preserving shorting

Made for: exploit actual threshold geometry at the second knot instead of bounding prime translation generically. At $\frac{1}{2} \log 3$, prime 2 swaps two edge intervals and vanishes on the center; paired edge modes diagonalize that part and eliminate its finite/remote cross debt. A positive discarded kernel residual factors as $E_M = \|B_M v\|^2$; shorting retains part of it in a finite block.

Status: exact decomposition and shorted-operator comparison. Transfer must retain an unshorted share ρE_M with $0 < \theta < \rho < 1$, preserving a positive logarithmic principal coefficient. Shorting all of the residual destroys remote coercivity; the remaining retained/remote/cross bounds were still open at Build 074.

Now: one of the most concrete earlier arithmetic geometries to test on the CP18 forced family; it can remove a specific prime cross term exactly. It is local and must be generalized or combined with later complete packet identities to give a global residual estimate.

Source: CMP UA-26-27 and NG-10; RC IX.5 and IX.7, Build 074.

L50 – Gap DNA, ordered factorization word, and wheel-fusion geometry

Made for: retain the actual arrangement of factor types between consecutive primes rather than only their counts. $W(p, r)$ is the sequence of prime-exponent partitions of every interior integer; the coarse inventory tracks distinct-factor number and repeated multiplicity. A primordial wheel evolves by replicating its old cycle and deleting new-prime multiples, fusing adjacent gaps.

Status: exact factor-color onset laws and wheel recursion. The first coarse color (r, m) occurs at $2^{m-r}p_r\#$, and the first exact partition occurs by assigning descending exponents to ascending primes. The early twin-prime centers are guaranteed only below the next unsieved prime square. Aggregated Jordan derivatives retain color depth but not the order of a finite gap word.

Now: a concrete earlier source of correlated arithmetic structure for the prime packets. It is relevant if the complete forced-family energy can be written in terms of ordered packet data. The chronological arrangement and inherited state must survive that map.

Source: GD §§1-5; GA §§1-3. The color catalogue also covers onset/Jordan components; the distinct object here is the ordered gap word and its wheel evolution.

L51 – Wheel-to-Euler-to-slope transform and color-stage invariant

Made for: carry finite sieve exclusions into the exact prime-power impulses of Suzuki's scalar. For wheel-survivor series $C_k(s) = \zeta(s) \prod_{p \leq p_k} (1 - p^{-s})$, logarithmic differentiation transfers the whole ray $\frac{\log p}{p^s - 1}$ from the survivor tail to the admitted-prime side. After half-density inverse Laplace transformation it becomes the ramp $\sum_a (\log p) p^{-a/2} (t - a \log p)_+$.

Status: exact invariant in its stated convergence half-plane. The full wheel-Jordan quotient invariant retains all aggregate color depths; its first parameter derivative is the von Mangoldt slope source. The analytic successor limit is exact but requires resolving an exponentially small term and is not claimed to be an efficient prime algorithm.

Now: an explicit bridge already connecting colors and wheel arithmetic to the same kind of completed Suzuki source now under study. This is stronger than a shared analogy: it identifies the actual impulse coefficients. What remains is a bound on their complete signed cumulative effect or an exact restriction to $W_a c$.

Source: GD §§3-5, particularly (4.3)-(4.8) and (5.9).

L52 – Exact height-slope two-state cocycle and normalized slope-debt dynamics

Made for: compress the entire incoming history into the two current states $Y_n = \Psi(\log n)$ and $\beta_n = \Psi'(\log n)$. Each integer cell has exact recurrence $Y_{n+1} = Y_n + \beta_n \delta_n + B_n$, $\beta_{n+1} = \beta_n + C_n - \frac{\Lambda(n+1)}{\sqrt{n+1}}$, with positive smooth-curvature increments B and C. The same law telescopes into complete consecutive-prime packets.

Status: exact two-state transport. The normalized debt $u = -\beta/A''$ obeys $u' = -1 - \frac{A'''}{A''}u$ between knots and a positive kick at each prime power. A prime-gap asymptotic is explicitly heuristic; its leading balance depends on the inherited u, not gap length alone.

Now: a compact authentic state model that may preserve information lost in generic boundary estimates. It could supply a chronological comparison for a current residual if the forced-family boundary functional is exactly expressed through these states. No such transfer is proved in GD/GA.

Source: GD §6, equations (6.6)-(6.15).

L53 – Exact future-floor energy reserve and horizontal-credit/vertical-debit ledger

Made for: quantify how much further height the current frozen source branch can lose before its smooth curvature reverses a negative slope. With current prime mass S , the branch is $F_t(x) = Y(t) + A(x) - A(t) - S(x - t)$ for $x \geq t$; its reserve is $E_A(t) = \inf_{x \geq t} F_t(x)$.

Status: exact reserve geometry. Between prime-power knots $E'_A = \max(\beta, 0) \geq 0$. Gaps bank explicit nonnegative horizontal areas; new prime-power impulses withdraw explicit nonnegative vertical areas. The total identity is reserve = initial reserve + $\sum$ gains – $\sum$ losses. All-prefix nonnegative reserve is equivalent to Suzuki pointwise positivity and thus RH; the universal no-overdraft inequality remains open.

Now: one of the clearest previously built arithmetic energy balances. It is an alternative scalar closing target, and possibly a source for a lower comparison with current Weil energy. CP18's boundary energy has not been identified with E_A ; that equality/comparison must be calculated before importing this reserve as its margin.

Source: GD §7.1, Theorem 7.1 and (7.2)-(7.8); §9.

L54 – Computable quadratic slope reserve and curvature-recovery theorem

Made for: replace the inverse derivative needed by the exact future floor with an explicit lower certificate: $E_2 = Y - \frac{\beta^2}{2c}$, $c = A'' > 0$. This pays the largest quadratic descent allowed by the current negative slope and curvature.

Status: exact theorem $E'_2 \geq 0$ between knots: the derivative is β when $\beta \geq 0$ and $\frac{\beta^2 c'}{2c^2}$ when $\beta < 0$. The exact knot loss is the difference of the two negative-slope squares divided by $2A''$. Increasing curvature proves $E_2 \leq E_A$; positivity at every post-knot state is a sufficient, stronger-than-necessary global certificate. The universal cumulative gain-versus-loss bound is open.

Now: a genuine existing positive recovery mechanism – not just a renamed target. Its importance is the proved monotonicity between actual arithmetic events. To finish either the old scalar route or the current matrix route, prime-knot withdrawals must still be collectively controlled using authentic arithmetic.

Source: GD §7.2, Theorem 7.2 and (7.9)-(7.14); finite audit in §10 is binary64 evidence only.

L55 – Green-tent within-packet lower-energy certificate

Made for: control the minimum inside a consecutive-prime packet, including interior proper prime powers. The Dirichlet tent kernel decomposes each hinge exactly and bounds the sum from below by an explicit quadratic polynomial in packet position. Its minimum is the certificate $I_{p,r}^{(2)}$.

Status: proved local lower bound; distinct from both E_A and E_2 . Finite floating-point checks support implementations but do not establish all-packet positivity.

Now: an explicit old local certificate that could be evaluated on relevant complete arithmetic packets before using general contour or collar estimates. A map to the current signed form must preserve the internal prime-power hinges and endpoint slope.

Source: GD §7.3, equations (7.15)-(7.19).

L56 – Inherited-state obstruction to scalar gap invariants

Made for: establish the minimum information a gap-energy model must retain. A local additive invariant preserved by every gap fusion must satisfy $F(a + b) = F(a) + F(b)$, hence only measures total length (plus a possible count term if a fixed fusion cost is allowed).

Status: exact no-go for that class of scalar invariants. Same gap length and coarse color word can produce opposite reserve changes because curvature level and inherited slope differ; the examples in GD/GA are finite numerical diagnostics, while the additive-fusion conclusion is analytic.

Now: relevant to any proposed shortcut using just colors, angles, or gap pictures. Preserve the carried state (Y, β) or an equivalent complete state when translating old machinery into the present forced family. This result does not rule out a nonlocal scalar invariant.

Source: GD §8; GA §5.

L57 – Fenchel prefix floor and integrated quantile-discrepancy theory

Made for: eliminate the continuous minimization exactly. For the full frozen prefix branch $F_j(t) = A(t) - S_j t + H_j$, define the restricted convex conjugate $A_+^*(s) = \sup_{t \geq \log 2} (st - A(t))$ and $D_j = H_j - A_+^*(S_j)$. Then $F_j(t) = D_j + [A(t) + A_+^*(S_j) - S_j t]$, with a nonnegative bracket.

Status: exact Fenchel decomposition and $[\forall j, D_j \geq 0] \iff [\Psi \geq 0] \iff \text{RH}$. Here D_j minimizes over the common half-line $t \geq \log 2$; distinguish that definition from $E_A(t)$, which minimizes only future $x \geq t$. The audit also writes $D_j = A(\log 2) + \int_0^{S_j} (Q_\Lambda(u) - Q_A(u)) du$, an integrated quantile discrepancy. Single increments may have either sign.

Now: an especially direct “use what we already have” scalar target: compare the authentic prime-mass quantile with the Archimedean quantile in integrated form. It avoids constructing another observer, but it requires a new arithmetic inequality; its relationship to CP18 can be through the common RH/Weil criterion or an explicit new comparison.

Source: GA §3, Theorem 3.1 and mass-transport identity.

L58 – Exact smooth-source cancellation and one-sided Chebyshev-error bound

Made for: reveal the actual lower-order term after the large prime and smooth contributions cancel. Partial summation cancels the leading $2\sqrt{X}$ slope terms and writes $\Psi(\log X)$ as explicit logarithmic/constants/tiny-tail terms minus $I_E(X) = \int_1^X (\psi_C(u) - u)u^{-3/2} [1 + \frac{1}{2} \log(X/u)] du$.

Status: exact identity. The required one-sided bound for I_E is equivalent to the old no-overdraft/RH target. Positive impulses, increasing curvature, positive anchors, and matching leading density do not force positivity: the audit supplies an exact nonarithmetic countermodel with all four properties and negative between-anchor values.

Now: identifies what the older energy laws still need – authentic lower-order correlation. It supports testing prime-color or Möbius machinery on this signed integral, while preventing leading-density agreement from being counted as the current missing sign estimate.

Source: GA §4, equations (4.2)-(4.12).

L59 – Finite theta-factor approximation and adaptive cofinal winding certificates

Made for: close the analytic tail needed to count all zeros in larger rectangles from finite all-integer theta blocks, without querying the zeros first. The entire symmetric approximants F_N have

an explicit uniform error decaying exponentially in N^2 ; certified boundary separation lets Rouché transfer exact winding counts to xi.

Status: recorded unconditional existence of a dovetailed finite certification algorithm giving cofinal exact total-zero counts. Exhaustive single-count critical-line brackets would additionally assert RH and simplicity. Finite Euler products are zero-free in the strip and therefore cannot satisfy a Rouché approximation on a contour enclosing a nontrivial zero.

Now: closes a different infinity than CP18: analytic truncation/counting rather than the positivity sign. It is useful infrastructure for source-only checks, but total winding does not bound Gamma or give horizontal zero location. This is distinct from the direct full-Weil theta-weighted identity catalogued in E08 and T51.

Source: GA §§6-7, Lemma 6.1, Theorem 6.2, finite Euler-product obstruction §7.1.

Source scope and coverage notes

1. This branch covers the light-cone, causal, conservation, cocycle, and operator families. Fibonacci/other spiral, colors, angles, specific Möbius modifier, theta curvature, higher-plane/tensor, and later checkpoint families should be merged from the other catalogues; overlap above is deliberate where the same theory has two interfaces.
2. The historical SB codex is comprehensive through August 9, not through September. Later IC, CA, CH, DI, UI and TC were read to update status. A source calling itself a theorem is recorded as a theorem claim within its scope; this inventory did not independently re-prove every archived result.
3. Additional useful master sources found: *Christophe_Michaels_Theorem_Compendium_Through_Build_074.pdf*; *Christophe_Michaels_Referee_Claim_Ledger_Through_Build_074.pdf*. Search snippets exposed registry CR-05 (screw light cone), CR-06 (finite-support equilibrium), CR-07 (cubic shell), with frozen exact source paths. RC supplied the read contents used here.
4. A full source-faithful test against CP18 remains the same for each imported theory: specify the map on $q = W_a c$, show the exact metric/form identity or a sufficient lower comparison, and carry every correction and endpoint term. The fact that two theories refer to the same RH objective is not an established equality of their energies.

11. Quadratic, infinity and cross-plane interfaces

P01 – The higher-information principle and the scalar projection

Made for: Recover the structured information hidden when an integer is displayed only as a scalar, or when a zero is displayed only as one location. On the arithmetic side the fuller record includes prime identities, valuations, ordering, and contacts; on the analytic side it includes all derivatives and their relative phases.

Recorded content: The manuscripts define concrete lifts rather than treating the extra coordinates as independent physical dimensions. The analytic input remains two real parameters; its measurements live in a larger space.

Implication now: It gives the governing requirement for reuse: preserve the interactions needed by the actual Weil form. It does not prescribe that every argument must use a large ambient space. A finite forced family can carry the relevant complete information.

Sources: EXTENDED_MANUSCRIPT(2).md, opening law table and G2-G5; Michaels_Planes_Spirals_and_No_Reversal_Check.md, §§2-3.

P02 – Infinity plane: the complete analytic derivative lift

Made for: Represent the complete translated function at a point, using every derivative rather than a finite scalar measurement.

Recorded content: With $c_j(z) = F^{(j)}(z)/j!$, the infinite vector belongs to the entire-sequence space defined by $\sum_j |c_j| R^j < \infty$ for every $R > 0$. It reconstructs $w \mapsto F(z + w)$. The exact translation law is $c_n(z + h) = \sum_k \binom{n+k}{n} c_{n+k}(z) h^k$. This is a convergent Fréchet-space construction.

Implication now: Supplies a rigorous setting for all-degree compatibility, reconstruction, and limiting arguments. It can carry a proof between derivative levels when estimates remain valid in the declared topology. A uniform bound still has to be proved; the infinite lift does not create it by definition.

Sources: EXTENDED_MANUSCRIPT(2).md, G3-G8; Michaels_TR3_Exterior_and_Hierarchy_Master_Prompt.md, §2. The latter explicitly treats infinity as compatible authentic objects and does not establish a hyperbolic metric or curvature for that space.

P03 – Quadratic plane: the full mixed amplitude matrix

Made for: Retain every mixed derivative interaction and relative phase while representing amplitude as a positive quadratic object.

Recorded content: $G_{mn} = c_m \overline{c_n}$, so $G = cc^*$, with rank one, positive quadratic evaluations, vanishing 2×2 minors, and translation $G(z + h) = T_h G(z) T_h^*$. A nonzero pivot reconstructs c up to one constant phase. In the Xi convention, $U_y = -2\Im G_{10}$; in $F(z) = \xi(1/2 + z)$, the no-reversal target is $\Re G_{10} \geq 0$.

Implication now: Identifies exactly where the sign-sensitive coordinate lives and prevents loss of cross terms. Positive semidefiniteness of G controls quadratic evaluations; the signed contraction used by Weil/theta positivity is a further specified quantity. This remains a central organizing object.

Sources: EXTENDED_MANUSCRIPT(2).md, G19-G32; Planes/Spirals note, equations (4)-(6).

P04 – Amplitude-phase differential geometry and zero multiplicity

Made for: Recover direction, phase, and multiplicity from the amplitude record, including behavior near zeros.

Recorded content: $U = |F|^2$ satisfies $\Delta U = 4|F'|^2$ and $U\Delta U = |\nabla U|^2$; away from zeros, phase gradients are determined by log-amplitude gradients. Distributional $\Delta \log U$ records zero locations with integer multiplicities. Full oriented amplitude data determine the analytic germ up to constant phase.

Implication now: Gives exact conversions among amplitude, phase rotation, and local zero data. The orientation and the signed derivative must be retained when translating to the current sign problem. The location-recovery and transverse hierarchy branches are catalogued in the theta section.

Sources: EXTENDED_MANUSCRIPT(2).md, G9-G18, G21-G32.

P05 – Tensor/Kronecker quadratic lift of the complete defect

Made for: Express the squared matrix defect as one quadratic form while retaining all mixed matrix entries and the original metric.

Recorded content: For a finite family with Suzuki Gram G and complete defect Z , $\mathcal{E} = \text{vec}(Z)^*(G^{-1} \otimes G^{-1}) \text{vec}(Z)$. This records the inverse metric explicitly.

Implication now: Provides the correct destination for a tensor or color-square comparison with the original defect. A bound in unweighted coefficient coordinates needs a proved comparison with this metric before it controls $\mathcal{E}_N$.

Sources: Michaels_Four_Closure_Groups_Checkpoint_12.md, quadratic-plane group; Michaels_Running_File_Ledger.md, Checkpoint 12 record, current version 17.

P06 – The square-coordinate plane $w = z^2$ and parity channels

Made for: Fold sign symmetry into a square variable and separate even and odd contributions.

Recorded content: Checkpoint 12 constructs two weighted channels, carrying both sheets and conjugation. Opposite-parity cross terms vanish; within-channel mixed terms remain. This square-coordinate map is distinct from the outer-product amplitude lift in P03.

Implication now: It can reduce the organization of the complete correction. The missing Hilbert-Schmidt budget remains a bound on the two complete channel defects. The coordinate map preserves that obligation.

Sources: Checkpoint 12, Group 3; current running ledger's Checkpoint 12 entry.

P07 – Cayley/Hardy representation and the boundary at infinity

Made for: Put meromorphic corrections, boundary terms, and pole interactions into Hardy-space operator language, including infinity through a conformal coordinate.

Recorded content: The checkpoint retains complete pole-boundary and pole-pole interactions and the identity $\|K - K^2\|_{\text{HS}}^2 = 2\|H\|_{\text{HS}}^2 + 2\|H\|_{\mathcal{S}_4}^4$ for its defined operators. It records the distinction between an ambient Hardy estimate and the Suzuki-normalized finite determining matrices.

Implication now: Can organize a complete norm budget or residue calculation. The central transfer is control of the actual normalized evaluation vectors, rather than shrinking a displayed coordinate scale.

Sources: Michaels_Four_Closure_Groups_Checkpoint_12.md, Group 4; running ledger, Checkpoint 12.

P08 – Infinity regulators, poles, and exhaustion

Made for: Justify replacing infinite meromorphic expressions by finite contours or regulated sums while retaining every induced pole and boundary correction.

Recorded content: Checkpoints 12-14 establish fixed-test/fixed-matrix convergence and compact evaluation tails; Checkpoint 14 calculates the residues and budget of introduced regulator poles. Those statements retain their respective support and dimension quantifiers.

Implication now: Supplies legitimate limiting passages for a proposed direct arithmetic identity. A dimension-uniform norm estimate is a separate requirement when returning to $\liminf_N \mathcal{E}_N \leq 1$. This is an analytic-infinity construction, distinct from the derivative infinity plane P02.

Sources: Checkpoints 12-14 and current running ledger.

P09 – Prime Plane-Zero Plane dictionary

Made for: Explain how the complete prime ledger and the complete zero ledger represent the same completed analytic data, with no assumed nth-prime/nth-zero pairing.

Recorded content: The prime-power atomic measure at $\log n$ transforms to $-\zeta'/\zeta$. Ordinary-prime data use the Möbius-restored logarithmic derivative. Every prime contributes the cell $(\log p)p^{-s}$ times the chosen analyzer. Poles emerge from the complete continued column, not a single prime cell.

Implication now: A concrete interface for transporting identities from prime coordinates to zero-side or Weil expressions. It requires the full completion and analytic continuation, preserving all prime powers and deterministic terms.

Sources: Prime_DNA_Cross_Plane_Atlas_Build_v58.pdf, §§11-12.

P10 – Prime innovations and the reflected response board

Made for: Combine each new prime jump with the interval drift between consecutive primes in one signed source.

Recorded content: For $E(x) = \theta(x) - x$, the exact innovation is $E(q_{n+1}) - E(q_n) = \log q_{n+1} - (q_{n+1} - q_n)$. The reflected-board energy splits into its signed constant component and a nonnegative centered variance by an exact Pythagorean identity.

Implication now: Offers a way to retain prime-continuum compensation before taking absolute bounds, relevant to the complete arithmetic residual. Its historical response metric must be explicitly transported to the current forced Weil family.

Sources: Cross-Plane Atlas v58, §§4,6.

P11 – Fixed-ratio frame and Brownian-bridge energy

Made for: Give a coercive geometric realization of reflected prime-error responses and quantify their centered fluctuation.

Recorded content: The full and centered response maps have positive lower frame bounds at fixed ratio. The centered energy has a Brownian-bridge kernel in a common response coordinate. Its prime atomic diagonal grows like $I_c \log q$, so bounded total variance requires complete off-diagonal and continuum cancellation.

Implication now: Provides an existing positive carrier and an exact accounting of the cancellation that a direct arithmetic estimate must use. The frame lower bound and the moving arithmetic upper bound have different roles.

Sources: Cross-Plane Atlas v58, §§5-6; Builds v52-v53 as consolidated there.

P12 – Stereo restoration and the finite music board

Made for: Recover inner and outer response branches after a single observed channel combines them.

Recorded content: $D_+ = I + O$ and $D_- = I - O$ give exact Hadamard recovery of both branches. Full variance retains the signed cross term. Spatial response mode and spectral zero frequency are distinct axes. The source includes finite numerical music-board evidence, separately labeled from analytical identities.

Implication now: A reusable model for checking whether a new scalar estimate has lost a branch or interference term. It supports full mixed-term reconstruction before evaluating a sign.

Sources: Cross-Plane Atlas v58, §7; Build v54.

P13 – Every-prime sampling and the authentic moving board

Made for: Connect discrete responses at every actual prime to the continuous Mellin transform, including moving predecessor and cell errors.

Recorded content: The sampled and continuous transforms have matching relevant poles after the stated gap-error estimates. The archive derives an RH criterion from uniformly bounded authentic response energy/variance, and records the converse using RH-dependent mean-square input.

Implication now: Supplies a discrete-to-continuous transfer theorem with explicit quantifiers. A criterion on every prime cannot be replaced by a selected subsequence without another theorem.

Sources: Cross-Plane Atlas v58, §§8-9; Builds v55-v56.

P14 – The zero-free outer channel and signed tail criterion

Made for: Isolate a scalar arithmetic response whose analyzer cannot hide a nontrivial zero pole.

Recorded content: Stereo subtraction removes the moving predecessor exactly. The mean-zero outer weight has a zero-free Mellin multiplier in the relevant strip. A bounded normalized complete outer tail is recorded as an RH-equivalent condition; the required bound remains open.

Implication now: An alternative sufficient route already present in the archive. It can test a proposed prime cancellation law without committing to the current Gamma probe. It still requires its complete tail estimate.

Sources: Cross-Plane Atlas v58, §10; Build v57.

P15 – Ideal all-zero locator and two-channel triangulation

Made for: Recover zero location and multiplicity from complete arithmetic response data and isolate what is lost in finite observations.

Recorded content: The ideal infinite record at every consecutive prime locates nonreal poles in the clean right half-strip; symmetry recovers the whole zero orbit. For two isolated channel residues, $Q = A_2/A_1$ gives $\rho = (16 - 18Q)/(4 - 3Q)$; the first residue then gives multiplicity. The decoder is exact after common-pole isolation.

Implication now: Supplies a precise cross-plane identity and a way to check metric/phase preservation. It is a reconstruction result; a zero-location exclusion requires an additional arithmetic inequality. Finite continuation stability is separately open.

Sources: Cross-Plane Atlas v58, §§12-15.

P16 – Arithmophysics and the spectral-law interpretation of primes

Made for: Interpret prime-counting fluctuations through oscillation, spectral balance, and hypothetical off-line-zero dissonance.

Recorded content: ARITHMOPHYSICS.pdf is an early compilation containing spectral illustrations and an original EFUP proof claim. EFUP_Reconsidered.pdf explicitly corrects that proof claim. The compilation is retained as the conceptual lineage, with the corrected status in P17-P18.

Implication now: Explains the origin of the growth/rotation/confinement language. Its useful mathematical content must be taken from the corrected equations and later completed-source identities.

Sources: ARITHMOPHYSICS.pdf, early EFUP chapter; EFUP_Reconsidered.pdf.

P17 – Original Error-Fluctuation Uncertainty Principle

Made for: Attempt to exclude off-line zeros using a lower bound on a position/frequency-type product of prime-error fluctuations.

Recorded status: Historical proposal, explicitly corrected by its later source. The reconsidered paper explains that a growing off-line mode makes the relevant product grow, which does not violate a lower bound; it also corrects an exponent. The original argument is not retained as a proved RH closure.

Implication now: Preserves an important design lesson already present in the archive: the direction of a proposed energy inequality must match the failure mode it is meant to exclude.

Sources: ARITHMOPHYSICS.pdf, EFUP chapter; EFUP_Reconsidered.pdf, §4.

P18 – Reconsidered EFUP / spectral no-runaway criterion

Made for: Express square-root-scale prime-error confinement as a spectral growth condition.

Recorded content: In $u = \log x$, the normalized error has modes proportional to $\exp((\beta - 1/2)u) \exp(i\gamma u)$. The source presents its no-runaway statement as a reformulation of the classical RH-equivalent error condition. Spectral entropy/compressibility is explicitly interpretive language.

Implication now: Closely aligns with the light-cone, individual-zero spiral, and shell-growth routes: each seeks an unconditional bound forbidding positive growth exponents. It supplies the interpretation and exact target, rather than an independent new growth estimate.

Sources: EFUP_Reconsidered.pdf, §§2-5,7.

P19 – Distinction positivity, modular gap, and canonical spaces

Made for: Build an arithmetic Hilbert-space route in which distinctions have positive energy, unobservable modes are removed, and a canonical spectral system can be attached.

Recorded content: The source separates a weighted modular model, RKHS distinction kernels, and a no-DC result from the open authentic prime cyclicity/positivity inputs. Herglotz/Hermite-Biehler spectral realization follows only with its stated hypotheses. Generic Sobolev point sampling is not the desired prime uniqueness theorem.

Implication now: An older architectural predecessor to the current positive regularizer and finite sign test. Its specific source-authentic positivity and observer conditions must be matched before using a model gap on the Weil form.

Sources: Distinction Positivity, Canonical Spaces, and a Research-Grade Route from RH to QSI.docx, opening claim table and formal analytic core. Related spectral/confinement entries appear in the light-cone section.

P20 – QSI, pressure envelopes, A/F waves, and moving/canonical fingerprints

Made for: Apply related geometric language to detecting coherent multiscale pressure and changes of state in market data. In this application EFUP means **Energy Flow Under Pressure**, a different expansion from the number-theory EFUP in P17.

Recorded content: The corrected wave-framing report defines causal multiscale pressure fields, Alpha and Fracture wave envelopes, moving-versus-canonical fingerprint divergence, and a report-only empirical evaluation framework.

Implication now: An adjacent applied branch sharing themes of full-state information, cross-scale coherence, and projection. The recovered sources supply no theorem transferring its empirical market behavior to Weil positivity. It is included to preserve the broader theoretical lineage, not used as an arithmetic premise.

Sources: qsi_corrected_wave_framing_integration_report_2026-05-18.md, §§1-5.

P21 – Local exclusion energy and compiled prime search

Made for: Turn the prime-rule idea into an exact, efficient screening implementation.

Recorded content: $E_B(N) = \sum_{p \leq B} w_p \mathbf{1}_{p|N}$, $w_p > 0$, vanishes exactly when no covered divisor hits. It compiles to Boolean residue-progressions. The source separately treats full primality certification and standard Mersenne/Proth arithmetic.

Implication now: Demonstrates a successful direct reuse of a structural identity for an executable task. The local nonnegative exclusion energy is distinct from signed Möbius historical energy and the complete Weil form. Standard Mersenne-factor restrictions in this source are not identified with the user’s separately remembered “Mersenne modulus correction.”

Sources: Michaels_Faster_Prime_Search.md, §§1-5.

12. Theta, amplitude, spectral positivity and Weil-Suzuki machinery

Source register

- **TS1** EXTENDED_MANUSCRIPT(2).md, 5046 lines. Read complete in six pages. Contains the higher-information/recovery manuscript, theta-increment continuation, weighted rigidity, and degree-two mixed hierarchy. Section titles in entries locate the exact source.
- **TS2** Michaels_Conjecture.tex, complete 158 lines. Formal “Michael’s Conjecture: Positive definiteness of the complete theta interaction kernel.”

- **TS3** Michaels_Hypothesis_H_Combined_Routes.tex, complete 599 lines. Ten-route map and their combined partial results.
- **TS4** Michaels_Complete_Theta_Kernel_Continuation.tex, complete 466 lines. Fixed spatial kernel, two-node positivity, odd signed cone, modular primitive.
- **TS5** Michaels_Interior_to_Spectral_Transfer.md, complete 9980 characters. Equations MP1-MP13.
- **TS6** Michaels_Backward_Propagation_Local_Amplitude_Criterion.md, current version 4, complete 919 lines. Sections 1-17.
- **TS7** Michaels_Running_File_Ledger.md, current version 17, complete 576 lines. Current CP18 scope; full development CP3-18 and provenance of earlier notes.
- **TS8** RUNNING_MATHEMATICS(2).md, EH3 current with all earlier EH/TR/JC/HM snapshots. Read first 1000/1039 lines (current EH3 and earlier source records; missing final historical 39 lines not needed for entries).
- **TS9** MATHEMATICS(5).md. First 1000 lines separately read; entire same chapter is also incorporated in TS1 (read complete).
- **TS10** Michaels_Master_Proof_Trace_W_S.md. First 1000/1955 lines read. Exact Suzuki identity, distinction from all-integer Smith-Jordan positivity, and historical source trace.
- **TS11** Michaels_Physical_Regularizer_and_Boundary_Energy_Checkpoint_18.md. First 1000/2123 lines read here, covering Groups 1-2 and first explicit probes; remainder/current status grounded in TS7's detailed CP18 record.
- **TS12** MATHEMATICS(20260917-203005).md. First 1000/1787 lines read; scalar actual-theta response/complete-monotonicity obstruction, origin certificate, full-theta saddle asymptotics.
- **TS13** Michaels_Density_Recovery_and_Parallel_Route_Comparison.md, read complete. Bilateral prime-shift density, support-controlled Weil core, exact logarithmic heat/primitive intersection, constrained two-trace sufficiency, coordinate-adjoint domain.

Additional energy sources

- **TS14** Physics_Architectures_and_Log_Prime_Energy.pdf. Read complete 2112 lines / 26 pages. Full Weil weighted-energy identity, prolate complement, residual certificates, theta near-null directions, recorded fixed-interval certificate and earlier slope reserve.
- **TS15** Operator_Energy_Route_Audit_v56.md, complete 418 lines.
- **TS16** Authentic_Energy_RH_Equivalence_Theorem_v56.md, complete 421 lines.
- **TS17** RH_Sufficiency_for_Complete_Plus_Energy_v56.md, complete 348 lines.
- **TS18** Build_Log_038_Burnol_Projection_Energy_Summary.md, complete 66 lines. Finite projection-feasibility experiment.
- **TS19** Prime_DNA_Centered_Global_Bridge.md, complete read. Depth-to-scalar and restored-history comparison.

T01 – Higher-information space / full analytic jet lift

Made for: Replace a single function value or zero location by the complete derivative record $c_j(z) = \Xi^{(j)}(z)/j!$.

Recorded result and source: Convergent infinite Fréchet-space orbit; exact translation law, Cauchy-Riemann evolution, reflection and evenness, including zero multiplicity. TS1 Geometry §§2-6.

Implication now: Supplies exact transport of local information. Every purported new plane can be tested against these laws; transport alone leaves a signed quadratic contraction to prove.

T02 – Amplitude-phase quotient / holomorphic Gram lift

Made for: Retain complete local information while removing one irrelevant constant output phase.

Recorded result and source: $G_{mn} = c_m \overline{c_n}$ is PSD rank one; its mixed modulus derivatives reconstruct the germ up to phase. Distributional phase current records multiplicities via $\Delta \log U = 4\pi \sum m_\rho \delta_\rho$. TS1 Geometry §§3-6.

Implication now: Provides positive ambient Gram geometry. The Weil or outward-derivative sign is a particular signed contraction of this positive array, so the contraction must be identified. See also P03 in the planes catalogue.

T03 – Complete amplitude fingerprint / actual-function recovery theorem

Made for: Determine the input point from the full oriented squared-amplitude data, beyond ordinary zero coordinates.

Recorded result and source: For actual Ξ , $G(z_1) = G(z_2) \Rightarrow z_1 = z_2$; on each compact input set a finite (non-effective) derivative order suffices. TS1 Geometry §§8-9.

Implication now: Gives injective representation and detects reflection-fixed input points. It can justify faithful return maps, but contains no standalone positivity theorem for current residual.

T04 – Transverse derivative / generalized Laguerre hierarchy

Made for: Resolve vertical amplitude response at every order: $U(x, y) = \sum L_n(x) y^{2n}$, $R = U_y/(2y)$.

Recorded result and source: Exact entire expansion, product/differential recurrences and multiplicity identities. Finite levels can miss nonreal zeros in explicit comparison families. TS1 Hierarchy §§1-6.

Implication now: Supplies local derivative tests and the known-zero moment detector; entire hierarchy, rather than a selected finite prefix, is what gives zero exclusion.

T05 – Accumulated outward-response positivity

Made for: Establish positive total transverse response from the axis to every horizontal cutoff.

Recorded result and source: Complete-theta half-wave argument gives $\int_0^X R(x, y) dx > 0$ for $X > 0, y \geq 0$, thus $\int_0^X U_y dx > 0$ for $y > 0$. TS1 Starting evidence; Local structure §1.

Implication now: A genuine global average sign and a potential compensation resource. It must be applied with the actual test weight; arbitrary signed/oscillating Weil tests are not those prefix averages.

T06 – Negative-lens / local reversal theorem for a hypothetical off-axis zero

Made for: Turn an off-critical zero into a concrete region of decreasing squared amplitude.

Recorded result and source: Exact local expansion yields a negative transverse lens, quantitative wedge/depth bounds, all symmetry partners retained. TS1 Local structure §§2-3; TS4 localization section.

Implication now: A contradiction target: any independently proved complete monotonicity or suitable integrated lower relation that rules out this lens would close localization.

T07 – Countermodel / finite-depth exclusion machinery

Made for: Determine which packages of properties actually force zero localization.

Recorded result and source: Explicit order-one real-even entire comparison function has strip confinement, positive strictly log-concave kernel, injective fingerprints, and positive accumulated responses but nonreal zeros; other families pass arbitrary finite derivative prefixes. TS1 Local structure §§4-6.

Implication now: Precisely delimits what the arithmetic connection must contribute. Prevents reusing an average, finite prefix, or generic Gram property as if it implied the current sign.

T08 – Target T / theta-increment autocorrelation factorization

Made for: Build positive squares directly for the difference of two transverse amplitude levels.

Recorded result and source: Sufficient target $B_b - B_a = \sum_j C_{q_j}$ with explicit $L^1 \cap L^2$ summability gives $U(x, b) - U(x, a) = \sum |\widehat{q}_j(x)|^2 \geq 0$, then RH. Construction remains open. TS1 Kernel bridge §4 (B11-B12).

Implication now: An alternative sufficient endpoint to boundary/Osgood closure. An exact arithmetic factorization of this increment would bypass current boundary comparison.

T09 – Density redistribution / Sobolev budget laws for positive factors

Made for: Find necessary structural laws a successful Target T factor family must obey.

Recorded result and source: All Sobolev budgets follow from factorization: $(-1)^m D^{(2m)}(0) = 4^m \sum \|q_j^{(m)}\|_2^2$. No factorization can preserve $w\phi^2$ pointwise; a strictly positive curvature deficit forces nonlocal redistribution. TS1 theta-increment continuation A4-A8.

Implication now: Specifies what a color/prime square decomposition would need to change in space. An unchanged pointwise split into independently positive channels is excluded in this class.

T10 – Actual-theta modular cancellation / strip regularity

Made for: Track interactions between theta indices and remove artificial algebraic transform tails.

Recorded result and source: Necessary factors extend holomorphically in $|\Im s| < \pi/4$; proper unweighted theta subseries retain forbidden derivative jumps; complete cross-index cancellation is essential. TS1 Actual theta channels §§1-5.

Implication now: Any conversion of prime colors into theta channels must retain the completed cross terms and modular correction.

T11 – Weighted-theta rigidity theorem

Made for: Classify weighted theta-root candidates that keep the necessary smooth reflection at zero.

Recorded result and source: For polynomial-growth weights, $F_c(|s|)$ smooth at zero iff every weight c_k is constant. Finite endpoint cancellations only postpone a tail. TS1 W1-W10, finite cancellation chapter.

Implication now: A strong restriction on using separately weighted theta/color channels. It rules out one construction class, while general nonlocal arithmetic-to-theta maps remain available.

T12 – Even/odd nonlocal resolvent factor repair

Made for: Repair cusp, mass, derivative-energy and zero-mode defects of a naive square-root factor.

Recorded result and source: Explicit filtered parity factors meet several exact budgets; their complete mass-saturated remainder still changes Fourier sign. TS1 modular regularization; P14-P33.

Implication now: A fully calculated candidate with a reusable signed remainder. Matching finitely many energy budgets does not supply complete residual positivity.

T13 – Compact positive extraction / bandwidth factor

Made for: Separate a rigorously positive piece of the amplitude increment while preserving energy for the remainder.

Recorded result and source: Explicit $\hat{q}_T = \sqrt{H_0}(1 - x^2/T^2)_+^2$, complete inverse transform, L^1/L^2 and derivative budgets, and retained full remainder. TS1 P34-P40; TS8 current exact T .

Implication now: This earlier machinery is already active: CP10-12 transport its extraction bound toward the Suzuki-relative correction. It is a positive component, not the entire defect.

T14 – Signed moment / polynomial density equivalence

Made for: Replace pointwise residual positivity by complete mixed Hankel inequalities.

Recorded result and source: For actual even exponentially integrable residual K , positivity on every polynomial square in either complete parity hierarchy implies $K \geq 0$, using density in $L^2(|K|dx)$. TS1 GM3-GM10.

Implication now: A second exact sufficient route. Finite forced Weil families could be compared with these forms, but the all-degree sign is the still-open hypothesis.

T15 – Actual theta curvature and finite mixed-block positivity

Made for: Prove cross-term positivity analytically rather than merely diagonal moments.

Recorded result and source: Complete theta curvature $\kappa \geq 4, \kappa'' \leq 2\kappa^2$ and degree-two determinant $\nu_0\nu_2 - \nu_1^2 \geq 13E^2/32 > 0$, all exterior included. Later recorded hierarchy extends through degree seven with declared parameter ranges and metrics. TS1 mixed-theta chapter; TS9; TS8 HM/EH provenance.

Implication now: Concrete positive forms available for exact comparison. Their degree and metric restrictions must be preserved when applying to W_{ac} ; arbitrary forced solutions are not automatically polynomials of degree ≤ 7 .

T16 – All-degree Gram recurrence / signed local matrix

Made for: Organize every mixed derivative coefficient into one recursive matrix and locate the cost of increasing degree.

Recorded result and source: Exact recurrence, PSD variance contribution, signed remainder, and full Schur extension law including singular-pivot/range conditions. Actual pointwise local degree-two matrix can be indefinite while integrated form is positive. TS1 polynomial operator and matrix-extension chapters.

Implication now: Demonstrates why integrated arithmetic correlations matter. Offers an exact border-payment comparison for transferring finite positive blocks to all degrees.

T17 – Cell-merge / covariance / optimizer compensation

Made for: Combine central and exterior intervals without losing cross terms or paying one gain twice.

Recorded result and source: Exact $K_I + K_O + P_{merge}$ decomposition with both extraction and optimizer costs; common-metric congruences, full residual corrections. TS8 EH3-ANCHOR and earlier JC/HM.

Implication now: Existing bookkeeping for the same type of signed interaction now present in physical collar Schur complements. Can be reused once the target form is matched.

T18 – Exterior hierarchy / full strip-tail allowance (EH1-EH3)

Made for: Control complete omitted height-series and spatial tails at every polynomial degree.

Recorded result and source: Explicit $|e_L(x)| \leq 18874368\tau_L e^{-6|x|/5}$; known factorial Gram error, same-metric finite payment at N3 for $0 \leq a < b \leq 1/10$; full degree-seven complex forms retained. All-degree lower sign remains open. TS8 EH3-TAIL/ERROR/FINITE/POSTFACTOR.

Implication now: Supplies actual tail budgets for a proposed transfer to W_{ac} , rather than a vague infinite-tail assumption. A bound with degree-dependent Gram conditioning does not become a uniform Suzuki bound automatically.

T19 – Frequency-derivative triangulation / joint factor-hierarchy-arithmetic tests (JC/TR)

Made for: Make several bounds apply to the same coefficient direction and same original arithmetic record.

Recorded result and source: Complete coupled error and Schur transfer, frequency gain, actual two partial factors, and arithmetic radius/alignment constraints; no all-degree payment. TS8 JTR1-JTR7 and JC records.

Implication now: This is a direct precedent for the user's request to recombine existing machinery. It requires a common object and metric, not addition of bounds from separate targets.

T20 – Hypothesis H / Michael’s Conjecture

Made for: State the whole missing theta interaction sign as one finite-matrix assertion.

Recorded result and source: For every $a > 0$, complete $C_a(y) = 2 \int x \sinh(2ax) \eta(x+y) \eta(x-y) dx$ is conjectured positive definite at every finite set of real shifts. Exact equivalents: Hilbert Gram representation, $K(a, t) = \Re(F' \bar{F}) \geq 0$, weighted amplitude monotonicity. Equivalent to RH. TS2 full statement.

Implication now: A preserved route to the same ultimate goal; the CP18 residual inequality is one later sufficient approach, not a replacement for H or for other sufficient routes.

T21 – Fixed spatial kernel J / Gram-majorant comparison

Made for: Place the theta derivative problem on a fixed self-adjoint spatial operator.

Recorded result and source: $J(x, y) = \int_0^\infty (x + y + 2r) \eta(x + r) \eta(y + r) dr$, Hilbert-Schmidt and entrywise positive. Positive Gram majorant R has exact same diagonal and strict pairwise domination. Every two-node J matrix strictly positive. TS4 §§1-2.

Implication now: Provides reusable spatial representation and pairwise sign; full many-node sign is still a separate claim. Local known-zero forms are exact compressions of J .

T22 – Parity ordering / infinite cone of signed odd tests

Made for: Exploit reflection to get positivity for a genuine family of sign-changing functions.

Recorded result and source: $J(x, y) > J(x, -y) > 0$ for $x, y > 0$; every nonzero odd function with one sign on the positive half-line has positive J energy. TS4 §3.

Implication now: A potential test after an exact transport from the Weil forced family to this J form, if the transported function has the required half-line sign. Current oddness alone does not imply that additional one-sign property.

T23 – Arbitrary finite translated-tail Gram theorem

Made for: Reach matrices of any finite size using the actual complete theta tail.

Recorded result and source: Normalized $J(X + s_j, X + s_k) \rightarrow [\text{sech}(s_j - s_k)]$; determinant tends to $\prod_{j < k} \tanh^2(s_j - s_k) > 0$. Threshold depends on the fixed configuration. TS3 §2.

Implication now: Supplies arbitrary-order positivity in translated configurations. A uniform controlled translation back to the current family would be the missing application.

T24 – Positive modular primitive / exact two-trace energy congruence

Made for: Remove the elementary $1/4 - z^2$ factor while keeping the complete modular correction.

Recorded result and source: $r = (1/4 - \partial^2)^{-1} \eta$ and exact polynomial energy congruence; negative rational correction is confined to $|t| < \sqrt{1/4 - a^2}$; degree-independent bounds and complete central compensation proved. TS3 §3, TS4 §5.

Implication now: This primitive reappears in the sufficient local class $P = \chi_\gamma R$. The rational negative part is paid on its full support, while a high-frequency mixed block remains.

T25 – Translated Gaussian window positivity

Made for: Turn theta log-concavity into positive averages centered away from zero.

Recorded result and source: $\int K(a, t) e^{-(t-b)^2/(2\sigma^2)} dt > 0$ for $|b| \leq \pi\sigma/2$. Exact paired sine/cosine channels and phase retained. TS3 §4.

Implication now: Available positive test class. Its width cannot shrink arbitrarily at fixed nonzero center, so it needs a comparison with the actual current test rather than formal localization.

T26 – Heat/de Bruijn-Newman deformation / positive degree-lowering map

Made for: Transfer known positivity in a deformed theta field toward the original field.

Recorded result and source: Full heat PDE; finite moment evolution; backward degree-lowering positive map; exact Gaussian averaging identity; fixed-variance positive average at original parameter; full Schur costs retained. TS3 §5. TS7 CP9 records that nonreal zeros occur for every finite negative heat parameter under Rodgers-Tao normalization.

Implication now: Useful exact evolution and averaging tools. Positive Gaussian limit or positive average does not by itself transfer a pointwise sign at the original parameter. See the occupied-edge discussion in the light-cone catalogue.

T27 – Scalar ground-state / maximum-principle energy

Made for: Exclude a negative interior minimum using amplitude as its own positive ground state.

Recorded result and source: On zero-free regions, exact coercive identity for $U = |F|^2$, and concrete zero-free tubes from tilted-theta variance. TS3 §6.

Implication now: A proved reusable mechanism whose domain and source must be transported. It cannot silently divide through a hypothetical zero.

T28 – Interior-to-spectral transfer / positive averaged Gram amplitude

Made for: Apply the scalar maximum principle to an actual hypothetical negative eigenfunction without punctures at zeros.

Recorded result and source: $E > 0$, complete finite-rectangle boundary sign, and exact MP8 coercivity. The remaining source is $\partial_a \delta / (2E)$, $\delta = 4(EE_1 - |H|^2)/E^2 \geq 0$; negative-part identity MP9 and centered correlation MP13 retain its signed derivative. TS5 §§1-4.

Implication now: Can be revisited on the CP18 forced family. It is not an already established energy-relative boundary bound: the signed Gram-curvature source needs an exact comparison.

T29 – Positive prime-gamma shift operator / prime-theta intertwiner test

Made for: Build a positive arithmetic operator from prime delays and transfer it into the theta Hilbert space.

Recorded result and source: $2I - S_u - S_u^* = (I - S_u)^*(I - S_u) + P_{[0,u]} \geq 0$; complete positive prime-gamma measure yields positive operators. Normalized limit is $2A^*A$; desired $J = A^*B + B^*A$ retains a signed subtraction. TS3 §6.

Implication now: Gives concrete prime positivity and shows exactly which derivative interaction the current target requires. It is a bridge candidate, not an identification already proved. The color catalogue records the extended versions.

T30 – Backward propagation into one known zero / Grommer local-amplitude detector

Made for: Detect any hypothetical off-critical zero using data at any one established critical-line zero.

Recorded result and source: Complete reciprocal-zero moments s_n , Hankel matrices $H_N(\gamma)$; an off-critical zero forces a negative finite quadratic direction at every fixed known γ . Exact zero-removed theta and Euler expressions retain all zeros. TS6 §§2-12.

Implication now: Strongest preserved reduction: proving all-order local positivity at one known zero is sufficient. It remains a viable alternate endpoint if arithmetic machinery acts more directly there.

T31 – Known-zero block decomposition and quadratic two-trace cancellation

Made for: Remove already-understood vanishing directions and both primitive boundary corrections from the local detector.

Recorded result and source: Exact block $0_{m-1} \oplus [m] \oplus H_N$; $P = \chi_\gamma R$, $\chi_\gamma(X) = (\gamma^2 + 1/4)X^2 + 2\gamma X + 1$, cancels both traces separately and still detects every off-critical zero. TS6 §§14-15.

Implication now: A smaller sufficient family preserved today. The optional prime-square identity is not required if its positivity can be proved by another exact comparison.

T32 – Prime-power square-form to local-form transfer

Made for: Apply established Smith-Jordan/Genesis positivity to the known-zero local detector.

Recorded result and source: Exact coefficient compression $C_{\gamma,N}^T H_{N+2} C_{\gamma,N}$ established; source prime-power square theorem recovered; an identifying assignment or sufficient lower comparison remains open. Direct spatial features retain translated tail and positive known-zero term. TS6 §§16-17, TS7 CP6-7.

Implication now: This is the specific missing application of old arithmetic positivity, not a missing theorem about whether its original positive carrier exists.

T33 – Weil-Suzuki positive reference metric and complete Gram correction

Made for: Compare the full signed Weil form with an unconditional positive Hilbert norm in a fixed metric.

Recorded result and source: Suzuki features/transform define positive $\mathcal{S}$; CP11 exact Gram correction includes nonreal-real feature coupling, all mixed terms and justified limits. $\mathfrak{W} = \mathcal{S}$ remains conditional globally. TS7 CP11, TS10 §§1-5.

Implication now: This is the present metric contract. Positive Green, color, theta or transported norms must be mapped into it explicitly, or a direct Weil proof can bypass a defect norm estimate.

T34 – Suzuki-normalized defect threshold and permanent-witness transfer

Made for: Convert a negative local witness into a persistent quantitative obstruction on a determining ladder.

Recorded result and source: CP9 transfers a negative local value to compact Weil q and yields $\mathcal{E}_N \geq (1 + \delta/(6\mathcal{S}(q)))^2 > 1$ eventually. An unconditional $\liminf \mathcal{E}_N \leq 1$ suffices; CP10 HS saturation identifies uniform boundedness with actual Hilbert-Schmidt extension. TS7 CP9-10.

Implication now: Explains why the old unit cap was a sufficient target, not an arbitrary number. Current direct Weil positivity remains allowed and would imply $\mathcal{E}_N = 0$ through the retained chain.

T35 – Suzuki-relative extraction and exact correction remainder

Made for: Connect the original compact extraction to the actual metric with a dimension-independent constant.

Recorded result and source: CP11 finite symmetric zero-family bound $\mathcal{X}(P) \leq (3567/509894)c^*Gc$. The constant is less than $1/140$. CP12 sharper K0, exact $\Delta = -\mathcal{X}_\infty + \mathcal{T}$, full correction retained. TS7 CP11-12.

Implication now: Existing arithmetic/theta machinery is already successfully connected here. Complete $\mathcal{T}$'s sign remains the part a reused theory must control.

T36 – First-null / full affine convolution equation

Made for: Reduce support-growth failure to an exact null state, preserving boundary contributions.

Recorded result and source: Full equation $g * v = \alpha x + \beta$, domain injectivity, later finite positive Suzuki energy and graph estimate; no unjustified Dirichlet traces on v . TS7 CP12-14.

Implication now: Narrows all later estimates to actual null/near-null vectors. Earlier source results should now be tested against these equations before generic bounding.

T37 – Null multiplier arithmetic / prime packets / endpoint Cauchy synthesis

Made for: Use the first-null equation to expose exact prime correlations and mixed polynomial constraints.

Recorded result and source: Complete linear/trigonometric/polynomial multiplier forms, signed autocorrelations, parity moments, endpoint traces for $g*v$, absolute Cauchy synthesis, and exact $L_{v,b} = e^{-ibz}L_v$. TS7 CP13.

Implication now: Explicit arithmetic relations available for direct sign calculation. Exact phase cancellation preserves complete paired energy; it supplies no automatic shrinking coefficient.

T38 – Finite dilation / logarithmic-Laplacian extension / regulator budget

Made for: Find a negative trial direction from scaling and translate the logarithmic operator into an extension energy.

Recorded result and source: CP14 exact $Q(U_s v) = s\|v\|^2 - \mathfrak{A}_v(s)$; complete 3D renormalized extension energy, signed flux, and exponentially decaying artificial-pole budgets. TS7 CP14.

Implication now: Provides another sufficient signed increment target. Can be combined with arithmetic moment identities if they bound the complete $\mathfrak{A}_v$, not only its favorable terms.

T39 – Odd-sector reduction / mixed 2-3-6 packets / polar resolvent

Made for: Use odd tests sufficient for Weil positivity and isolate the completed polar negative rank-one term.

Recorded result and source: CP15 correct odd criterion, complete packet signs and Schur relation $K = D - 2ff^T$; local $a \leq 1/8$ strict positivity; full polar inverse and tail. TS7 CP15.

Implication now: A closer contact with earlier 2/3 color/modifier ideas: current packets have exact operators and metrics to compare. Global resolvent margin remains open.

T40 – Prime-active support certificate / complete collar / physical finite regularizer

Made for: Start from a rigorous positive interval and detect every possible global loss, including both null branches.

Recorded result and source: CP16 $Q_a^- \geq \frac{1}{5000}I$ for $a \leq 699/2000 + e^{-2500}$; CP17 actual linear collar cost; CP18 fixed physical regularizer preserves gap, complete finite matrix sign/inertia/null equivalences. TS7 CP16-18; TS11 Group1.

Implication now: Gives a finite-dimensional family W_ac that detects every adverse direction, the concrete target on which to try the old arithmetic theories.

T41 – Boundary Gram evolution / complete physical matrix flow

Made for: Control support growth across every prime threshold in one fixed physical representation.

Recorded result and source: CP18 $0 \preceq \widehat{K}_b - \widehat{K}_a \preceq C_R(b-a)I$, $\widehat{K}'_a = \tau_R \beta_a \beta_a^*$ a.e., $Q_a^-(W_ac) = \tau_R^{-1} c^* \widehat{K}_a (I - \widehat{K}_a)c$. TS11 Groups1-2, TS7 CP18.

Implication now: The exact current insertion point: an arithmetic lower relation on this family or margin-relative control of its boundary vector would prevent first crossing.

T42 – Explicit adjoint probes / Osgood continuation certificate

Made for: Convert boundary evaluation into complete energy plus a calculated residual and prevent finite support collapse.

Recorded result and source: CP18 logarithmic-energy finite source probe on critical sector; exact residual and limit $\Gamma_a = \beta_a - M_a h_a$; proposed margin-bound implies no finite crossing, including adaptive-step accumulation. Actual margin-dependent residual estimate remains open. TS7 CP18; TS11 Group3 partial.

Implication now: Latest sufficient route, with a precisely identified residual. Earlier machinery could close it, or prove positivity on W_ac directly and bypass this particular probe certificate.

T43 – Bilateral prime-shift Sobolev cyclicity / two-prime dense core

Made for: Generate a complete test space from translations indexed only by $\log 2$ and $\log 3$.

Recorded result and source: The registered sine window's bilateral translates by $\mathbb{Z} \log 2 + \mathbb{Z} \log 3$ are dense in H^σ , $\sigma < 3/2$; smoothing once yields a compact smooth seed dense in every finite Sobolev order, with finite trace constraints corrected exactly. TS13 §§2.1,3, recovering Build042.

Implication now: Density and exact admissibility are already available. Positivity on this source-faithful core would extend; matching prime indices alone does not identify its form with a positive color metric.

T44 – Fixed-support completed Weil core / Reciprocity Atlas

Made for: Extend finite arithmetic tests to the entire completed Weil test domain without a global support assumption.

Recorded result and source: Complete prime, polar, archimedean form continuous on $H_0^1(-R, R)$, with explicit sine core. Positivity of all complete finite sine Gram matrices would extend to the full class. TS13 §2.2, recovering Atlas §10.

Implication now: Avoids re-proving this approximation issue. The current physical regularization and finite family can be placed inside this known form framework.

T45 – Logarithmic-heat / positive Gram-family derivative / primitive intersection

Made for: Connect the prime logarithmic kernel and complete theta kernel through one exact map.

Recorded result and source: Positive centered log kernel and its weighted restricted heat family map to G_δ^η ; $\partial_\delta G_\delta^\eta|_0 = J$. Exact theta double-index identity (4.4); differentiating the completed primitive pullback produces exactly $J = L_x L_y J_B + 2B' \otimes B' - \frac{1}{2}B \otimes B$. TS13 §§4-5.

Implication now: This is an already-proved bridge, with an explicitly signed derivative and rank-two correction. It should be reused directly rather than recreated; the derivative's sign remains the target.

T46 – Coordinate operator / arithmetic adjoint seed / closability gate

Made for: Encode zero localization as whether the theta translate-coordinate operator has a dense adjoint domain.

Recorded result and source: Translate states independent and dense; exact adjoint equation $F'V = FW$, and explicit unconditional arithmetic seed built from $(s-1)\zeta(s)/(z+\beta)^M$. A right-half-plane zero forces every adjoint V to vanish there, obstructing closability. Left shifts retain an exact boundary defect. TS13 §7.

Implication now: Another sufficient structural route. Existing bilateral source density cannot be substituted for one-sided adjoint density; prime machinery would need to prove the actual domain is dense while respecting that defect.

Additional energy mechanisms recovered after the first inventory

T47 – Brownian modal energy and exact negative-Sobolev source norm

Made for: Diagonalize authentic reflected fluctuation energy and identify the exact source norm needed for an upper bound.

Recorded result: Brownian bridge eigenvalues are $M_c/(k\pi)^2$, trace $M_c/6$, and HS-square $M_c^2/90$. Exactly,

$$V_c^+(q) = \frac{2M_c}{\pi^2 q} \sum_{k \geq 1} \frac{|a_{k,q,c}|^2}{k^2} = \frac{M_c}{q} \|\lambda_{q,c}\|_{H_D^{-1}}^2.$$

The complete spectral geometry is established. The authentic singular prime-minus-continuum source norm remains to be bounded. TS15 §§1-2.

Now: This is a solved energy representation with an explicit remaining arithmetic norm. It is distinct from the later Suzuki defect HS norm. Fixed kernel norms alone cannot control the size of its source.

T48 – Authentic reflected raw/centered energy equivalence

Made for: Determine exactly what a uniform bound on every actual prime response board implies, retaining its mean.

Recorded result: For genuine predecessor $p_q = \max\{p \leq q/2 : p \text{ prime}\}$, RH iff bounded raw $\mathcal{J}(p_q, q)/q^2$ iff bounded centered $V_{c_q}^+(q)$. Every prime terminal, moving ratios, complete outer tail and all modes remain. RH-side estimate uses Cramér mean-square and needs no simplicity or pair-correlation assumption. TS16 §§1-4; TS17.

Now: A complete alternate arithmetic-energy criterion; its sampling and mean-channel bridges have already been proved. Its unconditional energy ceiling remains open.

T49 – Prime atom-continuum diagonal debt and signed jump-drift law

Made for: Isolate the exact cancellation that keeps reflected energy bounded.

Recorded result: $V_c^+ = D_c^{atom} + 2L_c/q$, where $D_c^{atom} = I_c \log q + O(1)$ and $I_c \geq 7/12$ on authentic ratios. Uniform energy is equivalent to

$$L_{c_q}(q) = -\frac{q}{2} I_{c_q} \log q + O(q).$$

Separating atoms from continuum in variation/large-sieve bounds loses this cancellation. TS15 §§4-8; TS16 §5.

Now: A precise older signed-energy target with explicit arithmetic compensation. It supplies a comparison point for CP18's residual, rather than an already-proved bound for it.

T50 – Burnol/Sonin projection-energy feasibility

Made for: Produce seeds close to the Sonin constraint space while preserving Gram geometry for a critical-line/de Branges transfer.

Recorded result: Five-seed finite model at $M=40$ has tiny projection distances, positive projected Gram and small finite resolvent residual, stable between M32 and M40. This is explicitly a feasibility experiment. It does not produce the HS1 lower bound; critical-line coordinates of the projected functions and rigorous enclosures remained next. TS18.

Now: A reusable finite construction/evidence family for constrained probes, at its declared experimental scope.

T51 – Complete theta-weighted Weil energy identity

Made for: Express the complete signed Weil form as a comparison of three explicitly nonnegative difference energies.

Recorded result: For positive complete theta density ϕ , compact smooth h and $q = \phi h$,

$$Q_R(\phi h) = E_\Gamma(\phi, h) + E_P(\phi, h) - E_Z(\phi, h).$$

All gamma, prime-power and pole terms remain. Multiplication by $\phi > 0$ bijects compact smooth tests. Thus full Weil positivity is exactly $E_\Gamma + E_P \geq E_Z$. TS14 §18, equations (33)-(38).

Now: This old construction already reaches the actual completed Weil form. Insert form-core approximants to $W_a c$, divide by ϕ , and investigate this explicit domination. The remaining tasks are the inequality and a justified domain passage, rather than constructing another source-to-Weil identity.

T52 – Pole energy as covariance plus a rank-one channel

Made for: Quantify exactly the polar energy that gamma and prime difference energies must cover.

Recorded result: With normalized $d\nu = 2 \cosh(u/2)\phi(u)du$ and $\vartheta = \tanh(u/2)$,

$$E_Z = \frac{1}{2} [\|h - \nu(h)\|_\nu^2 + |\nu(\vartheta h)|^2].$$

Even and odd targets separate exactly; an explicit even limiting equality proves sharpness of the coefficient $1/2$ as a possible universal lower target. TS14 §18.1, equation (39).

Now: The odd-sector target is precisely $\frac{1}{2}(\|h\|_\nu^2 + |\nu(\vartheta h)|^2)$. This gives an exact covariance comparison for testing color-square machinery, with the full polar rank-one cost retained.

T53 – Theta derivative cutoff near-null spaces and energy-accuracy scale

Made for: Identify actual low-energy directions a growing-support certificate must resolve.

Recorded result: For each fixed k , cutoffs of $\phi, \phi', \dots, \phi^{(k)}$ span $k+1$ independent directions for large R with operator/form residual $O_k(e^{-c_k e^{2R}})$, independently of RH. Every projection of rank at most k leaves one; no fixed-rank support-independent complementary gap is possible. TS14 §17, equations (30)-(32).

Now: Small full energies are already built into the arithmetic operator. These approximate nulls are not negative witnesses. CP18 correctly permits regularizer rank to grow with the maximum interval; comparisons must preserve the actual small-energy scale.

T54 – Residual-response Schur certificate and finite improvement

Made for: Bound complete coupling to an infinite positive complement using an explicit trial response.

Recorded result: For $D \geq dI > 0$, trial H and $\mathcal{R} = C - DH$,

$$C^* D^{-1} C = C^* H + H^* C - H^* D H + \mathcal{R}^* D^{-1} \mathcal{R}.$$

Only the final residual needs the worst-case inverse bound. Finite optimization gives a guaranteed PSD improvement over $d^{-1}C^*C$, all couplings retained. TS14 §14, equations (18)-(20).

Now: An already-proved energy-certificate mechanism structurally compatible with CP18. It sharpens a concrete estimate; global sign of the actual arithmetic certificate remains open.

T55 – Prolate energy concentration and pole-captured finite sector

Made for: Prove all potentially adverse restricted Weil directions lie in a finite sector.

Recorded result: Prolate concentration gives $\dim E \leq 2RW/(\pi\tau)$ and a positive gamma-controlled complement for large W . Adding $e^{u/2}, e^{-u/2}$ captures every pole direction and its coupling. The complete Schur criterion follows. TS14 §§9-10,13.

Now: An earlier source-faithful finite reduction, comparable with CP18's physical regularizer. It supplies a different basis and quantitative complementary gap, while leaving the finite arithmetic sign to prove.

T56 – Prime-continuum Green cancellation and signed background

Made for: Combine pole and continuum contributions before estimating arithmetic fluctuation.

Recorded result: Exactly $Z_R - M_R = G_R$, with

$$G(u, v) = e^{-|u-v|/2}, \quad G(\log m, \log n) = \frac{\sqrt{mn}}{\max(m, n)}.$$

The historical reciprocal-maximum kernel enters by explicit diagonal rescaling. Gamma plus Green background still has negative directions; the prime fluctuation must provide signed compensation. TS14 §§15-16.

Now: An explicit old metric connection to the current Weil form. The positive Green contribution is one component of the complete sign, with its arithmetic compensation still essential.

T57 – Centered depth-to-scalar transfer and restored-history budget

Made for: Compare the centered tensor depth energy with the authentic scalar cofactor field and restore the whole Möbius history.

Recorded result: The log-min depth kernel contains an interval carrying every cofactor from two upward, yielding an explicit comparison loss $(\log N / \log 2)(1 + 1/\text{predecessor})$. Full reconstruction retains earlier occupation, late restored-history energy, copied-history bank, terminal storage and signed interaction. Rounding continuous event times to integer ceilings has a bounded correction. Centered-depth control and compensated-history control together give a sufficient bound; the latter remains open. TS19.

Now: A genuine established tensor-to-scalar energy comparison, with a quantified loss. It prevents mistaking centered future control for control of complete historical energy, and suggests comparing both terms with the current residual only through an exact map.

T58 – Earlier full-form interval energy floor

Made for: Certify positivity over every function on a fixed support, including all omitted basis directions.

Recorded result: TS14 §19 Theorem19.1 states

$$Q_{1/2}(f) \geq 9 \cdot 10^{-8} \|f\|_2^2$$

for the complete closed form on support $[-1/2, 1/2]$, both parities and complex coefficients; translations cover any interval of logarithmic length one. Recorded proof uses 24 even and 24 odd Legendre orders, 192-bit Arb interval quadrature, explicit analytic quadrature error and complete infinite-complement/coupling bounds. The source records reproduction and separate internal analytic audit. This inventory has not rerun that certificate.

Now: This is a concrete previous certificate to recover and compare with the CP18 normalization before continuing from the smaller CP16 certified support. Its stated support is larger but its floor is smaller; these are different guarantees. It does not extend to arbitrary support by adding disjoint intervals, because their mixed terms remain.

T59 – Gap DNA height-slope reserve and knot-loss budget

Made for: Quantify how an arithmetic curve recovers from negative slope between prime-power impulses and how much each new impulse removes.

Recorded result: TS14 §20.5 recovers Gap DNA and the Slope Closure, Theorems6.1,7.1-7.2. For height $Y = h$, slope $\beta = h'$, positive increasing background curvature c , define

$$E_2 = Y - \frac{\beta^2}{2c}.$$

Between knots, $E'_2 = \beta$ for $\beta \geq 0$, and $E'_2 = \beta^2 c' / (2c^2)$ for $\beta < 0$, hence $E'_2 \geq 0$. Exact future-floor reserve has ledger $E_{A,j} = E_{A,1} + \sum_{i=2}^j G_i - \sum_{i=2}^j \ell_i$, with all gains and impulse losses nonnegative. The complete all-knot loss-versus-gain inequality remains open.

Now: An actual earlier energy-based non-crossing mechanism, with the missing global payment explicitly identified. It can inform current continuation, but the scalar screw-function reserve and the full matrix positivity margin need an explicit comparison.

Distinct energies – quantities that must remain separately named

Energy	What is being squared or measured	Established relation / current role
Factor layer $L(n)$, repeated energy $V(n)$, mixed energy $C(n)$	Normalized logarithmic prime-factor shares	$L + V + C = 1$; factorization geometry, not the Weil form.
Historical Green energy $R(N)$	Prefix Mertens sums with reciprocal-time weights and terminal atom	$\sum_{k < N} M(k)^2 / [k(k+1)] + M(N)^2 / N$; original modifier/retained-record target.
Transported counting energy	Green norm after divisor reconstruction	Exact conservation such as $E_N(B_N \mu) = 1$; changed carrier/metric, not a bound for original $R(N)$.
Haar/Ramanujan color energy	Full-period orthogonal divisibility shells	Local shell floors and positive tensor Gram; fixed-residue/scalar-orbit observation needs its own comparison.
Individual zero-mode energy	Normalized squared radius of one explicit-formula mode	$x^{2\Re \rho - 1}$; full discrepancy energy also contains cross-mode terms.

Energy	What is being squared or measured	Established relation / current role
Prime discrepancy block energy $B(X, Y)$	$X^{-2} \int_X^Y \psi(x) - x ^2 dx$	Local variance + mean; geometric/Fibonacci sampling equivalences.
Reflected raw/centered energy U_c, V_c^+	Full two-sheet ordinary-prime discrepancy response	Exact Brownian H^{-1} source norm; uniform authentic-board bound is RH-equivalent.
Positive theta Gram norm $\ S_\delta f\ ^2$	Half-line theta convolution	Its derivative is the signed J form; positivity of the norm does not determine its derivative.
Theta increment and polynomial remainder energies	Complete amplitude difference contracted against polynomial squares	Signed target; extraction, finite mixed blocks and full tail budgets are proved at stated scopes.
Averaged amplitude E and Gram curvature δ	$\int \Psi ^2 F ^2$ and its normalized Cauchy-Schwarz variance	Both nonnegative; source in maximum-principle transfer is the signed derivative δ_a .
Weil energy $Q_a^-(q)$	Completed arithmetic quadratic form	Current sign target, equivalently the full gamma+prime-minus-pole comparison on theta-weighted tests.
Theta-weighted E_Γ, E_P, E_Z	Three nonnegative weighted squared-difference forms	Exactly $Q(\phi h) = E_\Gamma + E_P - E_Z$; domination remains open.
Suzuki energy $\mathcal{S}(q, q)$	Norm of the actual unconditional Suzuki transform	Positive reference metric; its equality to Weil energy is the global issue, not a definition.
Suzuki defect $\mathcal{E}_N$	Squared Frobenius norm after full Suzuki Gram normalization	A unit liminf ceiling would contradict every permanent negative-witness floor. Not the historical max-kernel $E_N(a)$.
Projection/probe/residual energy	Distance to a specified constraint space, trial-response residual, or inverse-weighted collar cost	Valid only in the named norm/domain; finite numerical distances are distinct from universal sign certificates.

Additional exclusions and scope checks

1. **Pointwise positive moment densities are too strong.** The actual local theta matrix is indefinite already at degree two although its integrated degree-two form is strictly positive (TS1/TS9). Therefore cross-location compensation is part of the actual mechanism.
2. **Scalar complete-monotonicity / Gaussian scale-mixture candidate.** The actual bounded scalar response was tested through full-theta origin jets and analytic asymptotics; the specific scale-mixture candidate fails. This does not prove the complete theta increment is negative (TS12 O1-O6 and S1-S10).
3. **Centered exterior-envelope obstruction.** The ledger of earlier hierarchy checks retains a certified negative value for a deliberately stronger separated exterior comparison; it is not a negative witness for the actual coupled residual. TS8 scope and TS1 carried-source status. The exact stronger-test constant is not used in this inventory.
4. **Finite factor/height approximations require degree-aware payment.** EH3 proves all tails and finite degree-seven payment, but source conditioning grows with degree; fixed finite prefixes can coexist with later failure. TS8 EH3 hierarchy.
5. **Original retained Green margin is a different target.** Its modifier + compensation, square-root lifetime cutoff, masks, endpoint atom and full residual $\mathcal{F}_N = D^+ - G_U -$

$(13/324)T_N + (3/4)R_G - \|r\|_G^2$ are retained explicitly in TS1 S1-S6 and TS8 original arithmetic blocks. No identification of that margin with the theta extraction or Suzuki residual has been established in these checked sources.

Ten-route map, retained verbatim in meaning from TS3 §1

Complete theta/modular cancellation; Fourier/autocorrelation; total positivity/variation diminution; heat deformation; Jensen polynomial inequalities; signed moments/polynomial density; fixed spatial operator; positive modular primitive; rotation/differential energy; prime/nonlocal spectral operator. Their concrete proven combinations appear in T21-T29. They should not be reduced to only the current boundary route.

Highest-value applications to the current finite family

- Try an exact Smith-Jordan/color square assignment or sufficient lower comparison on $q = W_a c$, carrying its metric and all complete terms. This is the current counterpart of T32's old local-form transfer.
- Test whether T22's stronger half-line one-sign condition follows for a hypothetical odd null or critical forced vector; odd parity itself is insufficient.
- Apply T28's exact Gram-curvature source identity to the current family, if a common transform exists; retained source sign is the quantity to calculate.
- Preserve T30-T31 as a smaller sufficient alternate family; CP18 does not invalidate it.
- Reuse T13/T18/T35 for tail/extraction costs once the actual comparison map is explicit. They are already proved components with known normalization, rather than new general regularity obligations.

13. Coverage, corrections and names still to recover

The named light cones, Fibonacci umbrella, other spirals, colors, angles, Michaels Möbius modifiers, quadratic and infinity planes are all represented. The targeted omission pass added energy work beyond those named families: global prime unitaries, Farey/LCM boundary energy, lifetime and survival identities, complete theta-weighted energy, spectral near-null spaces, gap/slope reserves, and older response and confinement methods.

The remembered labels **subprime**, **Mersenne modulus correction**, **prime footprints**, and **nuts** remain explicitly registered. A definitive standalone definition under each exact name was not recovered. The archive does contain exact prime-search restrictions, factor-potential constructions and a five-layer counting decomposition, but those are not silently assigned these names. "Fib*" is treated as the recovered Fibonacci family; no separate starred definition is invented.

Some early claims were corrected in later notes. Preserve the later qualifications on the original EFUP inequality, record-collar anchors, fixed-power block-gap estimates, prime-fan remainders and sieve ranges. Formal positivity of an ambient Gram array, a fixed-period color norm or an independently chosen energy is not automatically positivity of the complete Weil form.

Source scope is recorded throughout. The structured codex of the 521-page Arithmetic Sandbox was examined; every appended historical transcript was not exhaustively reread. File variants and parallel build numbers do not by themselves establish mathematical supersession. This is an extensive recovered inventory with explicit coverage limits, not a claim that no unseen idea remains anywhere in the complete chat history.

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Part IV

Earlier mathematical supplements

These ten complete source notes retain derivations, exploratory representations, and comparisons that were compressed in the earlier consolidated paper. Repetition preserves the original hypotheses and full calculations. Their historical status statements are read together with the current status at the beginning of this edition.

Complete density and parallel routes

Source: *Michaels_Density_Recovery_and_Parallel_Route_Comparison.md*. Equation tags carry the source prefix H1; references within this unit retain their original local numbering.

Research continuation for Christophe Michaels – 21 September 2026

1. What the comparison establishes

The earlier work already contains bilateral prime-shift Sobolev density, a support-controlled core for the completed Weil form, and polynomial density for the complete stationary theta kernel. Those are reusable theorems; they need not be replaced by another sequence of numerical tests.

The present comparison supplies a common, exactly constrained source core for the primitive, logarithmic-heat, and coordinate-operator routes. The first two routes meet in an exact identity. The third describes the same theta source through an adjoint equation, whose one-sided density is a different assertion from the recovered bilateral density.

The central identities are

$$\boxed{\begin{aligned} Q_J[f] &= \left. \frac{d}{d\delta} \|S_\delta f\|_2^2 \right|_{\delta=0} = 2 \operatorname{Re} \langle Sf, Af \rangle \\ &= Q_{J_B}[Lf] + 2|\tau_{B'}(f)|^2 - \frac{1}{2}|\tau_B(f)|^2, \end{aligned}} \quad (\text{H1:1.1})$$

where the last equality initially holds for compact smooth sources. All notation and domain extensions are given below.

A further reduction is proved here: it suffices to establish

$$\boxed{Q_{J_B}[Lf] \geq 0 \quad \text{on a dense prime-shift source core with} \quad \tau_B(f) = \tau_{B'}(f) = 0.} \quad (\text{H1:1.2})$$

The two explicit boundary terms then vanish. This is a smaller sufficient test class, not an assertion that its required inequality has been proved. The comparison does not establish H.

2. Recovered results and their exact domains

2.1 Bilateral prime-shift density: Build 042

The source is **Build_Log_042_Prime_Shift_Sobolev_Cyclicity.md**, Theorems 42.1-42.2 and the jurisdiction paragraph. Its registered window is

$$f_L(x) = \sqrt{2/L} \sin(2\pi x/L) \mathbf{1}_{[-L/2, L/2]}(x), \quad L > 0.$$

It belongs to every $H^\sigma(\mathbb{R})$ with $\sigma < 3/2$, and its Fourier transform is nonzero almost everywhere. For

$$\Gamma = \mathbb{Z} \log 2 + \mathbb{Z} \log 3,$$

the recovered theorem is

$$\overline{\operatorname{span}}^{H^\sigma} \{f_L(\cdot - \gamma) : \gamma \in \Gamma\} = H^\sigma(\mathbb{R}), \quad \sigma < 3/2. \quad (\text{H1:2.1})$$

Both positive and negative prime-log translations occur. The source explicitly does not identify this with one-sided semigroup cyclicity.

2.2 Completed, fixed-support core: Reciprocity Atlas

System_RH_Reciprocity_Atlas.md, Section 10, equation (10.5), establishes for the authentic completed Weil form

$$|Q_{\text{ar}}(f, g)| \leq C_R \|f\|_{H^1} \|g\|_{H^1}, \quad f, g \in H_0^1(-R, R). \quad (\text{H1:2.2})$$

Prime, pole, and archimedean terms are retained. No bound uniform in R is asserted or required for its fixed-support extension.

The same section supplies the Dirichlet sine core

$$\psi_{R,n}(x) = R^{-1/2} \sin\left(\frac{n\pi(x+R)}{2R}\right) \mathbf{1}_{(-R,R)}(x).$$

Its finite spans approximate in fixed-support H1. With any required admissibility constraints imposed, positivity on all of the complete sine Gram matrices extends to the full Weil class. The archive explicitly leaves that universal positivity as the arithmetic theorem to be proved.

This result settles an approximation issue. It does not identify the Weil form with a different prime or theta form solely because they share a core.

2.3 Polynomial density: Cross-Chain Continuation

Michaels_Theta_Cross_Chain_Continuation.pdf, Theorem 6.1, proves density of polynomials in $L^2(|\sigma_a|)$, where

$$d\sigma_a(\omega) = K(a, \omega/2) \frac{d\omega}{2\pi}, \quad K(a, t) = \text{Re}\{F'(a+it)\overline{F(a+it)}\}.$$

The exponential-moment bound makes the transform of a function orthogonal to every monomial analytic near the imaginary axis. All derivatives at zero vanish, and Fourier uniqueness gives density. Thus positivity of every polynomial-square integral is equivalent to positivity of the complete signed measure. Its local derivative formulation is an exact all-order criterion; a finite set of positive derivative matrices is not that criterion.

3. A common source core for the present routes

Use the actual normalization

$$F(z) = \xi\left(\frac{1}{2} + z\right) = \int_{\mathbb{R}} \eta(u) e^{zu} du, \quad \eta > 0, \quad \eta(-u) = \eta(u),$$

with the target

$$H : \quad \text{Re}\{F'(a+it)\overline{F(a+it)}\} \geq 0 \quad (a > 0, t \in \mathbb{R}),$$

and the spatial kernel

$$J(x, y) = \int_0^\infty (x + y + 2r) \eta(x+r) \eta(y+r) dr.$$

Define, for source functions f ,

$$(Sf)(r) = \int f(x) \eta(x+r) dx, \quad (Af)(r) = \int (x+r) f(x) \eta(x+r) dx, \quad r \geq 0. \quad (\text{H1:3.1})$$

The inner product is conjugate-linear in its first slot. Convolution and restriction, followed by Young's inequality, give

$$\|Sf\|_2 \leq \|\eta\|_1 \|f\|_2, \quad \|Af\|_2 \leq \|u\eta(u)\|_1 \|f\|_2.$$

Consequently

$$Q_J[f] = 2 \operatorname{Re}\langle Sf, Af \rangle, \quad |Q_J[f]| \leq 2\|\eta\|_1 \|u\eta\|_1 \|f\|_2^2. \quad (\text{H1:3.2})$$

The equality follows first for compact sources by Fubini and then by continuity. Build 042 at sigma=0 therefore supplies an actual global core for Q_J , with simultaneous convergence of both source coordinates S and A. This conclusion does not require continuity of the completed Weil form on unrestricted global H1.

For the primitive integration by parts, smooth the original window once: choose a nonzero $\chi \in C_c^\infty(\mathbb{R})$ and put

$$w = f_L * \chi, \quad \mathcal{C} = \operatorname{span}\{w(\cdot - \gamma) : \gamma \in \Gamma\}. \quad (\text{H1:3.3})$$

This is a compact smooth seed. Both Fourier factors are nonzero entire functions, so their product is nonzero almost everywhere on the real axis. The proof of Build 042 applies verbatim and gives

$$\overline{\mathcal{C}}^{H^\sigma} = H^\sigma(\mathbb{R}) \quad \text{for every finite real } \sigma. \quad (\text{H1:3.4})$$

In particular it is an H2 core. This avoids differentiating the original sine window twice as if it were in H2.

Finite continuous constraints can be retained exactly. If $\tau = (\tau_1, \dots, \tau_k)$ is a continuous trace map, choose finitely many core vectors whose trace vectors form a basis of its range. Correct each approximant by a linear combination of those fixed vectors. The coefficients tend to zero when the target satisfies $\tau(f) = 0$. Thus $\mathcal{C} \cap \ker \tau$ is dense in the constrained space.

The Atlas sine core gives the same source-pair convergence on each fixed support. Indeed the kernels of S and A are square-integrable over $(-R, R) \times (0, \infty)$. This is another valid common core, not an additional positivity argument.

4. The logarithmic-heat route reaches J exactly

For a fixed reference rate $a_0 > 0$, the centered logarithmic kernel is

$$\begin{aligned} K_0(a, b) &= -\log(a + b) + \log(a + a_0) + \log(b + a_0) - \log(2a_0) \\ &= \int_0^\infty (e^{-av} - e^{-a_0v})(e^{-bv} - e^{-a_0v}) \frac{dv}{v}. \end{aligned} \quad (\text{H1:4.1})$$

It is a positive Gram kernel. The exact half-line theta realization uses its weighted, restricted family

$$K_\lambda^+(a, b) = \int_1^\infty v^{\lambda-1} (e^{-av} - e^{-a_0v})(e^{-bv} - e^{-a_0v}) dv. \quad (\text{H1:4.2})$$

Every real lambda gives a positive Gram kernel.

Set

$$\eta_\delta(u) = e^{\delta u} \eta(u), \quad G_\delta^\eta(x, y) = \int_0^\infty \eta_\delta(x + r) \eta_\delta(y + r) dr.$$

For every real delta,

$$G_\delta^\eta \succeq 0, \quad \partial_\delta G_\delta^\eta|_0 = J. \quad (\text{H1:4.3})$$

Equivalently, if S_δ is S with η replaced by η_δ , $Q_J[f] = \partial_\delta \|S_\delta f\|_2^2|_0$. The tilted densities and their parameter derivatives are integrable, so this also holds on L^2 .

Here is the complete theta-to-heat identity, retaining all double indices:

$$P_a = 4a^2 \partial_a^2 + 6a \partial_a, \quad a_n(x) = \pi n^2 e^{2x},$$

$$G_\delta^\eta(x, y) = \frac{e^{(\delta+1/2)(x+y)}}{2} \sum_{n,m \geq 1} P_a P_b K_{\delta+1/2}^+(a, b) \Big|_{a=a_n(x), b=a_m(y)}. \quad (\text{H1:4.4})$$

To verify it, substitute $v = e^{2r}$ into the Gram integral and use

$$\eta(x+r) = e^{x/2} v^{1/4} \sum_{n \geq 1} (4a_n(x)^2 v^2 - 6a_n(x)v) e^{-a_n(x)v}.$$

The reference-rate terms disappear under $P_a P_b$. For fixed real x, y , Gaussian tails justify the sums, integrals, and parameter derivatives.

Thus the logarithmic-heat construction genuinely reaches the actual J . Its remaining sign is the sign of the derivative in (4.3). Positivity of G_δ for each δ does not imply positivity of that derivative.

5. Primitive completion and the exact intersection of the routes

Let

$$G_\theta(u) = e^{u/2} \sum_{n \geq 1} e^{-\pi n^2 e^{2u}}, \quad B(u) = \frac{1}{2} e^{-u/2} - G_\theta(u), \quad L = \partial_u^2 - \frac{1}{4}.$$

Jacobi's relation gives evenness of B . Comparison of a strictly decreasing Gaussian sum with its integral gives $B > 0$. For $u \geq 0$ and then by evenness,

$$0 < B(u) < \frac{1}{2} e^{-|u|/2}, \quad \eta = -LB. \quad (\text{H1:5.1})$$

Define J_B and G_B^δ as in the preceding sections, replacing η by B . For $|\delta| < 1/2$, put

$$D_{x,\delta} = (\partial_x - \delta)^2 - \frac{1}{4}.$$

The exact completed pullback is

$$G_\delta^\eta = D_{x,\delta} D_{y,\delta} G_\delta^B. \quad (\text{H1:5.2})$$

Differentiate at zero. The derivatives of the two delta-dependent operators must be retained. They give

$$J = L_x L_y J_B + 2B'(x)B'(y) - \frac{1}{2}B(x)B(y). \quad (\text{H1:5.3})$$

The boundary calculation is

$$\int_0^\infty \{B'(x+r)LB(y+r) + LB(x+r)B'(y+r)\} dr = [B'(x+r)B'(y+r) - \frac{1}{4}B(x+r)B(y+r)]_0^\infty.$$

This proves that the primitive boundary correction is precisely the correction produced by differentiating the heat pullback. It is not an unrelated term that can be discarded.

For compact smooth f , define $\tau_b(f) = \int b(x)f(x) dx$. Integrating (5.3) by parts gives the last equality of (1.1).

There is also a decisive analytic test of the uncorrected primitive route. Fix $d \neq 0$ and let X tend to positive infinity. The exact tail is $B(u) = \frac{1}{2}e^{-u/2} + O(e^{u/2 - \pi e^{2u}})$. On the two nodes X and $X+d$,

$$J_B(x, y) = \frac{1}{4}e^{-(x+y)/2}(x + y + 2) + \text{a superexponentially smaller error.}$$

Consequently

$$\det[J_B(x_j, x_k)]_{j,k=1}^2 = -\frac{d^2}{16}e^{-(2X+d)} + o(e^{-2X}) < 0 \quad \text{for sufficiently large } X. \quad (\text{H1:5.4})$$

The positive scalar primitive therefore does not give a positive J_B matrix. Equation (5.3), including its differential operators and boundary terms, remains the valid route.

6. A two-trace class removes the boundary terms without losing the target

Theorem

If

$$Q_{J_B}[Lf] \geq 0 \quad \text{for every } f \in \mathcal{C} \text{ with } \tau_B(f) = \tau_{B'}(f) = 0, \quad (\text{H1:6.1})$$

then F has no zeros in the open right half-plane, and the target H follows by the actual- ξ equivalence established in the prior balanced-class note.

Continuity needed for the proof

For any even b with $|b(u)| \leq Ce^{-a|u|}$, write $m = (x + y)/2$, $d = (x - y)/2$. Evenness gives

$$J_b(x, y) = 2 \int_{|m|}^{\infty} t b(t + d) b(t - d) dt. \quad (\text{H1:6.2})$$

Put $R = \max(|m|, |d|) = (|x| + |y|)/2$. Then

$$|J_b(x, y)| \leq 2C^2 e^{-2aR} \left(\frac{R^2}{2} + \frac{R}{2a} + \frac{1}{4a^2} \right). \quad (\text{H1:6.3})$$

In particular J_b is square-integrable on the plane. Since η admits every exponential rate a , Q_J is continuous on

$$\mathcal{H}_\epsilon = L^2(\mathbb{R}, e^{-2\epsilon|x|} dx) \quad \text{for every } \epsilon > 0. \quad (\text{H1:6.4})$$

Indeed choose $a > \epsilon$ in (6.3) and conjugate the kernel by the square root of the weight; the resulting kernel remains square-integrable. For $\epsilon < 1/2$, the B and B' traces are continuous on this space.

The core $\mathcal{C}$ is dense in this weighted space: first truncate a weighted function to a compact interval, then use unweighted L^2 density, noting that the weight is at most one. The fixed finite-dimensional correction of Section 3 makes the two traces zero exactly. Their independence follows from the nonzero even function B and the nonzero odd function B' . Thus (6.1) and (5.3) imply nonnegativity of Q_J on the whole two-trace subspace of (6.4).

A hypothetical off-axis zero contradicts that nonnegativity

The actual xi function has no zeros with real part of z at least $1/2$. Suppose z_0 is a zero with $0 < \operatorname{Re} z_0 < 1/2$, of multiplicity m . Integration by parts gives, in the strip $|\operatorname{Re} z| < 1/2$,

$$\widehat{B}(z) := \int B(x) e^{zx} dx = \frac{F(z)}{1/4 - z^2}. \quad (\text{H1:6.5})$$

Set $f_0(x) = x^{m-1} e^{z_0 x}$. Both B -traces vanish, and direct theta integration gives

$$Sf_0 = 0, \quad Af_0(r) = F^{(m)}(z_0) e^{-z_0 r}, \quad Q_J[f_0] = 0. \quad (\text{H1:6.6})$$

Choose three distinct real numbers $w_i \in (0, 1/2)$ and put

$$a_i = \frac{1}{\prod_{j \neq i} (w_i - w_j)}, \quad f_1(x) = \sum_{i=1}^3 \frac{a_i e^{w_i x}}{\widehat{B}(w_i)}. \quad (\text{H1:6.7})$$

The denominators are positive. The identities $\sum_i a_i = \sum_i w_i a_i = 0$ show that both traces of f_1 also vanish. For the sesquilinear form associated with J ,

$$\begin{aligned} \mathcal{B}_J(f_0, f_1) &= \overline{F^{(m)}(z_0)} \sum_i \frac{a_i (1/4 - w_i^2)}{\overline{z_0} + w_i} \\ &= \overline{F^{(m)}(z_0)} \frac{1/4 - \overline{z_0}^2}{\prod_i (\overline{z_0} + w_i)} \neq 0. \end{aligned} \quad (\text{H1:6.8})$$

Choose $\max(\operatorname{Re} z_0, w_1, w_2, w_3) < \epsilon < 1/2$. Both f_0 and f_1 belong to the space (6.4). Their displayed mixed theta integral is absolutely convergent, so it agrees with the continuous kernel form. A suitable small complex multiple t makes $Q_- J[f_0 + t f_1] < 0$, contradicting nonnegativity on that two-trace space. This proves the theorem.

For completeness, the final passage from zero-freeness to H uses the actual xi function's symmetry and order-one canonical product. Its zeros then occur at the imaginary pairs $\pm i\gamma$, with multiplicities m_γ , and for $z = a + it$, $a > 0$,

$$\operatorname{Re} \frac{F'(z)}{F(z)} = \sum_{\gamma > 0} m_\gamma \left\{ \frac{a}{a^2 + (t - \gamma)^2} + \frac{a}{a^2 + (t + \gamma)^2} \right\} \geq 0. \quad (\text{H1:6.9})$$

The paired product and sum converge by the usual order-one zero summability. Multiplication by $|F(z)|^2$ gives H .

The proof extends an inequality if it is established on the indicated prime-shift core. It does not establish (6.1). The failed unrestricted positivity of $J_- B$ in (5.4) does not refute its constrained L_f form in (6.1).

7. Coordinate uniqueness and the exact one-sided obstruction

On the finite span of $h_x(r) = \eta(x + r)$, define

$$T_0 h_x(r) = (x + r) \eta(x + r), \quad r \geq 0.$$

These translates are independent and have dense span in $L^2(0, \infty)$. For independence, analyticity extends a zero finite combination from the half-line to the full line, and Fourier uniqueness gives

its coefficients. For density, a vector orthogonal to every h_x has zero bilateral Laplace transform after pairing with η , hence is zero.

For v, w in this half-line L_2 space, let V, W be their Laplace transforms. The exact adjoint equation is

$$T_0^* v = w \iff F'(z)V(z) = F(z)W(z), \quad \operatorname{Re} z > 0. \quad (\text{H1:7.1})$$

Equivalently, V is in H_2 of the right half-plane and $(F'/F)V$ also belongs to H_2 , with every apparent singularity removed. To justify the transform, for $a > 0$ use Cauchy-Schwarz on $\int |v(r)|e^{-ar}dr$, and the finite tilted first moment of η . The converse follows by transform uniqueness.

The proposed arithmetic adjoint seed is valid. With

$$s = z + \frac{1}{2}, \quad A_0(s) = (s-1)\zeta(s), \quad k(s) = \frac{s}{2}\pi^{-s/2}\Gamma(s/2),$$

take $\beta > 0$ and an integer $M \geq 5$, and define

$$V_0(z) = \frac{A_0(s)}{(z + \beta)^M}, \quad W_0(z) = \frac{A'_0(s) + (k'/k)(s)A_0(s)}{(z + \beta)^M}. \quad (\text{H1:7.2})$$

Because $F = k A_0$, (7.1) holds. The pole at $s=1$ cancels in A_0 . The Euler-integral expression for ζ on $\operatorname{Re} s > 0$ gives polynomial bounds for A_0 and its derivative on $\operatorname{Re} s \geq 1/2$; k'/k has logarithmic growth there. The denominator makes V_0 and W_0 members of H_2 without a zero-free assumption.

Every adjoint vector nevertheless has the following restriction:

$$F(z_0) = 0, \quad \operatorname{Re} z_0 > 0 \implies V(z_0) = 0. \quad (\text{H1:7.3})$$

This follows from the simple pole of F'/F at a zero of any multiplicity. Evaluation at z_0 is a nonzero continuous functional on H_2 . Such a zero would therefore put the entire adjoint domain in a proper closed hyperplane and prevent closability of T_0 .

Right shifts preserve (7.1), since they multiply V and W by e^{-az} . Left shifts have an exact boundary defect. For $a > 0$ let

$$I_v(z) = \int_0^a e^{-zr} v(r) dr, \quad V_a(z) = e^{az}(V(z) - I_v(z)),$$

and define I_w, W_a similarly. Then

$$\boxed{F'V_a - FW_a = e^{az}(FI_w - F'I_v)}. \quad (\text{H1:7.4})$$

Thus the bilateral source density of Section 2 cannot be substituted for one-sided adjoint density. It gives a core for the pair (S, A) ; it does not make the induced map Sf to Af closable automatically.

8. What can be carried between the routes

Pair of routes	Exact connection	What remains to prove
Recovered density and spatial theta	S and A are bounded on source L_2 ; the prime-shift span is dense and finite traces can be corrected exactly	Sign of the completed form on that core

Pair of routes	Exact connection	What remains to prove
Logarithmic heat and primitive B	Equations (5.2)-(5.3): differentiating the heat pullback produces exactly the primitive boundary terms	The derivative inequality, or the equal primitive quadratic inequality
Primitive B and the smaller target class	Both boundary terms vanish on the two-trace class, which still detects any off-axis zero	Nonnegativity of $Q_{JB}[Lf]$ on that class
Source density and coordinate operator	Source approximation controls both S and A	Density of the adjoint domain, subject to (7.1) and (7.4)
Prime colors and the present core	The same bilateral log-prime indexing can label the sources	An identity or valid signed comparison for the actual completed form; matching indices alone does not identify metrics

The existing **Michaels_Prime_Color_Dual_Kernel_Bridge.md**, Sections 10-14, already distinguishes the positive independent-color kernels from the completed theta-weighted scalar orbit. Its density of the two-prime lattice is consistent with the core recovered here. Neither density nor common indexing supplies an equality of those two forms.

The most direct combined analytic target furnished by this comparison is (6.1). It uses the prior density theorem, a smooth version of the existing prime-shift seed, the complete theta primitive, and exact cancellation of its two boundary traces. No numerical hierarchy is required to formulate or extend that whole-class statement.

9. Source record

The following existing files were used for their stated results, rather than as evidence that a remaining universal sign had already been proved:

1. **Build_Log_042_Prime_Shift_Sobolev_Cyclicity.md** – Theorems 42.1-42.2; bilateral jurisdiction and final topology paragraph.
2. **System_RH_Reciprocity_Atlas.md**, version 1 – Section 10, equation (10.5), countable core, and equations (10.17a)-(10.17f).
3. **Michaels_Theta_Cross_Chain_Continuation.pdf** – Theorem 6.1.
4. **Michaels_Balanced_Class_Prime_Transfer_Tests.md** – actual theta normalization; source coordinates; zero witness and the actual-xi equivalence connecting zero-freeness, spatial positivity, and H.
5. **Michaels_Prime_Color_Dual_Kernel_Bridge.md**, version 1 – Sections 10-14, including the distinction between the color metric and completed orbit.

Sections 3-7 above record the analytical cross-checks and the new two-trace reduction. No finite sign computations were used. The main manuscript was not edited.

Balanced-class prime transfer

Source: `Michaels_Balanced_Class_Prime_Transfer_Tests.md`. Equation tags carry the source prefix *H2*; references within this unit retain their original local numbering.

Research continuation for Christophe Michaels – 21 September 2026

1. Outcome

The three proposed routes have been tested analytically.

1. Positivity on boundary-cancelling packets alone suffices for the full target. This reduction is proved below, including multiple zeros and a test class on one fixed exterior line.
2. Prime shifts can be made to preserve that class exactly. The theta scale assembly and its prime-power commutator are explicit. A positive color covariance also exists in the actual spatial variables, but its insertion into the target uses a sign-indefinite matrix operation.
3. The complete prime-pole pairing has an exact positive shift-difference energy representation with a signed arithmetic remainder. Boundary cancellation alone does not pay that remainder.

No numerical tests or finite-degree certificates are used. The remaining balanced-class inequality is not proved here; this is not a proof of H. The main manuscript is unchanged.

Throughout, inner products are conjugate-linear in the first slot. Use the actual theta normalization

$$F(z) = \xi(\tfrac{1}{2} + z) = \int_{\mathbb{R}} \eta(u) e^{zu} du, \quad \eta > 0, \quad \eta(-u) = \eta(u),$$

$$J(x, y) = \int_0^\infty (x + y + 2r) \eta(x + r) \eta(y + r) dr.$$

For a finite atomic packet $f = \sum_j c_j \delta_{x_j}$, define

$$\tau(f) = \sum_j c_j \eta(x_j), \quad Q_J[f] = \sum_{j,k} \overline{c_j} c_k J(x_j, x_k).$$

For integrable coefficient functions the analogous formulas use integrals. Nodes in the atomic class range over all of $\mathbb{R}$.

2. The boundary-cancelling class is sufficient

Theorem 1

For the actual theta kernel, the following statements are equivalent:

1. $Q_J[f] \geq 0$ for every finite atomic packet with $\tau(f) = 0$.
2. On any one prescribed line $b > 0$, $Q_J[f] \geq 0$ for every finite exponential packet

$$f(x) = \sum_j c_j e^{(b+it_j)x}, \quad \sum_j c_j F(b + it_j) = 0.$$

3. F has no zeros in $\operatorname{Re} z > 0$.
4. J is positive semidefinite on all finite atomic packets.

These imply H. The theorem reduces the required positivity claim; it does not establish any of the equivalent claims unconditionally.

2.1 Continuity and the fixed-line test class

For $\epsilon > 0$, use

$$W_\epsilon(x) = \begin{cases} e^{-2\epsilon x}, & x \leq 0, \\ e^{-\pi e^{2x}}, & x > 0. \end{cases}$$

The theta series and evenness give the needed decay. In particular, for

$$G_0(x)^2 = \int_0^\infty \eta(x+r)^2 dr, \quad G_1(x)^2 = \int_0^\infty (x+r)^2 \eta(x+r)^2 dr,$$

the three functions $\eta/\sqrt{W_\epsilon}, G_0/\sqrt{W_\epsilon}, G_1/\sqrt{W_\epsilon}$ belong to $L^2(\mathbb{R})$. Since

$$|J(x, y)| \leq G_1(x)G_0(y) + G_0(x)G_1(y),$$

both τ and the J -form are continuous on $\mathcal{H}_\epsilon = L^2(W_\epsilon dx)$.

Finite atomic positivity on the zero-trace class therefore extends to zero-trace elements of $\mathcal{H}_\epsilon$: first approximate compact integrals by atomic sums, correcting their traces at a fixed node where $\eta \neq 0$; then approximate general functions by compact functions, correcting their traces using one fixed compact function of trace one.

For $b > \epsilon$, the measure $e^{2bx} W_\epsilon(x) dx$ is finite. Its characters e^{itx} have dense span: an orthogonal L^2 function defines a finite measure with identically zero Fourier transform, hence is zero. Thus $e^{(b+it)x}$ span $\mathcal{H}_\epsilon$. Correcting a trace by subtracting $\tau(f)e^{bx}/F(b)$ stays on the same line, since $F(b) > 0$. Compact bump approximations give the reverse passage to atomic packets. This establishes the equivalence of the two test classes.

2.2 A zero supplies an impossible mixed pairing

For a coefficient function f , put

$$h_f(r) = \int f(x)\eta(x+r) dx, \quad m_f(r) = \int (x+r)f(x)\eta(x+r) dx.$$

Then

$$B_J(f, g) = \int_0^\infty (\overline{m_f} h_g + \overline{h_f} m_g) dr, \quad Q_J[f] = 2 \operatorname{Re} \int_0^\infty m_f \overline{h_f} dr. \quad (\text{H2:2.1})$$

In particular, for $\operatorname{Re} z, \operatorname{Re} w > 0$,

$$B_J(e^{zx}, e^{wx}) = \frac{\overline{F'(z)}F(w) + \overline{F(z)}F'(w)}{\overline{z} + w}. \quad (\text{H2:2.2})$$

Suppose F has a zero z_0 of multiplicity m in the right half-plane. Set

$$f_0(x) = x^{m-1} e^{z_0 x}.$$

Changing variables $u = x + r$ and using the multiplicity conditions gives

$$\tau(f_0) = 0, \quad h_{f_0} = 0, \quad m_{f_0}(r) = F^{(m)}(z_0) e^{-z_0 r}.$$

Thus $Q_J[f_0] = 0$.

Choose distinct real $w_1, w_2 > 0$, and set

$$f_1(x) = \frac{e^{w_1 x}}{F(w_1)} - \frac{e^{w_2 x}}{F(w_2)}.$$

Then $\tau(f_1) = 0$ and

$$B_J(f_0, f_1) = \overline{F^{(m)}(z_0)} \left(\frac{1}{\overline{z_0} + w_1} - \frac{1}{\overline{z_0} + w_2} \right) \neq 0. \quad (\text{H2:2.3})$$

A positive semidefinite form cannot have a zero diagonal vector with a nonzero mixed pairing: a suitable small complex t makes $Q_J[f_0 + t f_1] < 0$.

Choose $\epsilon < \min(\operatorname{Re} z_0, w_1, w_2, b)$. The two functions lie in $\mathcal{H}_\epsilon$, so the continuity and exact trace corrections above transfer the contradiction to either finite test class. This covers every zero multiplicity.

2.3 Recovering full positivity

The actual F is even, real entire, of order one, and $F(0) > 0$. Absence of right-half-plane zeros therefore puts every zero on the imaginary axis. Its paired Hadamard product gives

$$\frac{F'(z)}{F(z)} = \sum_{\gamma > 0} \left(\frac{1}{z - i\gamma} + \frac{1}{z + i\gamma} \right),$$

with multiplicities. The series is locally convergent away from zeros. Equation (2.2) becomes a Gram sum of the features

$$\frac{F(z)}{z - i\gamma}, \quad \frac{F(z)}{z + i\gamma}.$$

This gives positivity on a fixed exponential line, and the preceding density argument gives full atomic positivity. Moreover, for $a > 0$,

$$\operatorname{Re} \frac{F'(a + it)}{F(a + it)} = \sum_{\gamma > 0} \left[\frac{a}{a^2 + (t - \gamma)^2} + \frac{a}{a^2 + (t + \gamma)^2} \right] \geq 0.$$

Multiplication by $|F|^2$ yields H. Full positivity trivially implies positivity on the boundary-cancelling subclass, completing the proof. Conversely, H excludes any right-half-plane zero directly: if $F(z_0 + t) = At^m + O(t^{m+1})$ for real t , then $\operatorname{Re}(F'\overline{F}) = m|A|^2 t^{2m-1} + O(t^{2m}) < 0$ for sufficiently small negative t . Thus H is also equivalent to the four statements for this actual function.

3. One explicit centered Gram comparison

Let

$$A(x, y) = \frac{J(x, y)}{\eta(x)\eta(y)}, \quad C_a(x, y) = A(x, y) - A(x, a) - A(a, y) + A(a, a).$$

The substitution $d_j = c_j \eta(x_j)$ turns the boundary constraint into $\sum_j d_j = 0$. For any fixed anchor a ,

$$\text{boundary-cancelling positivity of } J \iff C_a \succeq 0 \text{ on every finite node set.} \quad (\text{H2:3.1})$$

All mixed terms between balanced pairs remain in C_a .

There is an explicit two-Gram representation. Choose $a > 0$, and put

$$g_x(r) = \frac{\eta(x+r)}{\eta(x)}, \quad b_x = g_x - g_a, \quad N_x(r) = \frac{(x-a)g_x(r)}{\sqrt{2(a+r)}},$$

$$P_x(r) = \sqrt{2(a+r)} b_x(r) + N_x(r).$$

Expansion gives exactly

$$\boxed{C_a(x, y) = \langle P_x, P_y \rangle - \langle N_x, N_y \rangle.} \quad (\text{H2:3.2})$$

Thus the remaining comparison can be stated as

$$\boxed{\left\| \sum_j d_j N_{x_j} \right\|_2^2 \leq \left\| \sum_j d_j P_{x_j} \right\|_2^2 \quad \text{for all finite coefficient packets.}} \quad (\text{H2:3.3})$$

Equation (3.2) is proved; inequality (3.3) is not proved here.

4. Prime shifts can preserve the balanced class exactly

Choose $b > 1/2$, and write $e_b(r) = e^{-br}$. It is an authentic continuous theta packet:

$$e_b(r) = \int \frac{e^{bx}}{F(b)} \eta(x+r) dx.$$

For $h(0) = 0$, define

$$R_a h(r) = h(r+a) - h(a)e_b(r).$$

Direct calculation gives

$$(R_a h)(0) = 0, \quad R_a R_c = R_{a+c}, \quad R_0 h = h. \quad (\text{H2:4.1})$$

Thus the shifts $a = \log p$ preserve the balanced class and the multiplication law of prime powers and composites. The class here is enlarged by the explicit continuous exponential anchor; the continuity and approximation argument in Section 2 justifies that enlargement.

The full theta state includes m_f , not just h_f . If $\beta_b = F'(b)/F(b)$, the corresponding state action is

$$(h, m) \mapsto (S_a h - h(a)e_b, S_a m - \beta_b h(a)e_b). \quad (\text{H2:4.2})$$

This follows by shifting the source nodes and subtracting $h(a)e^{bx}/F(b)$. At the hypothetical zero used in Section 2, $h = 0$ but $m \neq 0$; this second coordinate cannot be discarded.

4.1 Exact metric accounting

On $H^1(0, \infty)$ with $h(0) = 0$, let

$$W_b h = h' + b h, \quad V_b u(r) = \int_0^r e^{-b(r-t)} u(t) dt.$$

Then $V_b = W_b^{-1}$,

$$W_b R_a = S_a W_b, \quad \|W_b h\|_2^2 = \|h'\|_2^2 + b^2 \|h\|_2^2. \quad (\text{H2:4.3})$$

These identities give a positive metric in which the prime-shift algebra acts exactly.

For the completed exterior multiplier $\mathcal{G}$ of Section 7, put $u = W_b h$ and $B_b = V_b^* V_b$. The latter has positive kernel

$$B_b(r, t) = \frac{e^{-b|r-t|}}{2b}.$$

On the exterior exponential class, admitting derivatives in the exponent when needed, future shifts commute with V_b^* . Hence

$$\mathcal{G}B_b = V_b^* \mathcal{G}V_b, \quad \boxed{\operatorname{Re}\langle h, \mathcal{G}h \rangle = \operatorname{Re}\langle u, \mathcal{G}B_b u \rangle.} \quad (\text{H2:4.4})$$

For example,

$$[B_b, S_a]u = -\frac{e_b}{2b}(V_b u)(a).$$

This rank-one correction accounts exactly for the boundary subtraction; it is not an extra positive allowance.

The required positivity is the sign of the complete weighted form in (4.4). Positive norms of separate prime features do not by themselves sign that form. No bounded action of the complete prime series on unrestricted L^2 is asserted.

5. The actual theta scale assembly

Define the Gaussian-derived seed

$$g(u) = \left(4\pi^2 e^{9u/2} - 6\pi e^{5u/2}\right) e^{-\pi e^{2u}},$$

$$Z = \sum_{n \geq 1} n^{-1/2} T_{\log n}, \quad P = \sum_{n \geq 2} \Lambda(n) n^{-1/2} T_{\log n}.$$

The exact theta expansion is $\eta = Zg$. The divisor identity $\sum_{d|n} \Lambda(d) = \log n$ proves

$$[u, Z] = -ZP.$$

For $\mathcal{D} = -\partial_u - 18u$, it follows that

$$\boxed{\mathcal{D}Z = Z\mathcal{D} + 18ZP.} \quad (\text{H2:5.1})$$

These identities hold on the seed orbit and its differentiated sums: the series converge absolutely and locally uniformly, and decay superexponentially on any half-line bounded below.

For a finite spatial packet, set

$$H_g(r) = \sum_j c_j g(x_j + r), \quad K_g(r) = \sum_j c_j (\mathcal{D}g)(x_j + r).$$

Writing $q = -\eta'/\eta - 18u$, the actual theta packets obey

$$h = ZH_g, \quad k = ZK_g + 18Ph, \quad (\text{H2:5.2})$$

and the balanced constraint is

$$\sum_{n \geq 1} n^{-1/2} H_g(\log n) = 0. \quad (\text{H2:5.3})$$

Thus, on this class, the precise required source-prime comparison is

$$\operatorname{Re}\langle h, ZK_g \rangle + 18 \operatorname{Re}\langle h, Ph \rangle \leq 0. \quad (\text{H2:5.4})$$

Equation (5.4) is the still-unproved comparison, expressed in the actual theta scale variables.

5.1 A direct test of positive individual source blocks

The seed's transform is explicit. With $s = z + 1/2$,

$$\int_{\mathbb{R}} g(u) e^{zu} du = H(s) := \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \quad (\operatorname{Re} s > -2), \quad (\text{H2:5.5})$$

where the value at $s = 0$ is removable. Substitution $y = \pi e^{2u}$ proves this formula directly.

It gives $H(1) = 0$ and $H'(1) = 1/2$: the seed has a right-half-plane transform zero at $z = 1/2$. Apply the same mixed-pairing calculation as Section 2 to its kernel

$$J_g(x, y) = \int_0^\infty (x + y + 2r) g(x + r) g(y + r) dr.$$

Take $f_0 = e^{x/2}$ and

$$f_1 = \frac{e^x}{H(3/2)} - \frac{e^{2x}}{H(5/2)}.$$

Both have zero seed trace, $Q_{J_g}[f_0] = 0$, and

$$B_{J_g}(f_0, f_1) = \frac{1}{2} \left(\frac{1}{3/2} - \frac{1}{5/2} \right) = \frac{2}{15} \neq 0.$$

Thus the individual seed block is not positive on its balanced class. The complete theta assembly is essential. In the Mellin expression, its factor $H(s)$ meets the pole of $\zeta(s)$ at $s = 1$.

Pointwise spatial assembly is valid on all real nodes. Interchanging the separated integer/prime sums with a whole-line Mellin integral requires $\operatorname{Re} s > 1$. No such interchange is used at a hypothetical zero in the critical strip.

6. Exact color covariance and the spatial mixed term

There is an actual positive covariance representation, even when individual seed summands change sign. For a fixed finite node set, put

$$x_* = \min_j x_j, \quad \rho_n(r) = \exp\left((r + x_*)/2 - \pi n^2 e^{2(r+x_*)}\right) > 0,$$

$$(f_n(r))_j = n^{-1/2} g(x_j + r + \log n), \quad a_n = f_n / \rho_n, \quad R(r) = \sum_n \rho_n(r).$$

All the following series converge absolutely. Since $\eta_r = (\eta(x_j + r))_j = \sum_n \rho_n a_n$, the exact covariance is

$$C_r = R \sum_n \rho_n a_n a_n^* - \eta_r \eta_r^* = \sum_{m < n} \rho_m \rho_n (a_m - a_n)(a_m - a_n)^* \succeq 0. \quad (\text{H2:6.1})$$

Gaussian decay justifies the second moments: the square of a summand divided by ρ_n still has a negative Gaussian exponent, because $x_j \geq x_*$.

However, the actual spatial kernel is

$$J = \int_0^\infty (D_r \eta_r \eta_r^* + \eta_r \eta_r^* D_r) dr, \quad D_r = \operatorname{diag}(r + x_j).$$

Therefore (6.1) enters the exact form as

$$J = \int_0^\infty \left[R \sum_n \rho_n (D_r a_n a_n^* + a_n a_n^* D_r) - (D_r C_r + C_r D_r) \right] dr. \quad (\text{H2:6.2})$$

This fixes both its metric and its sign.

The map $C \mapsto D_r C + C D_r$ does not preserve positivity. For a rank-one positive term vv^* , its two-node determinant is

$$\det[(D_r vv^* + vv^* D_r)_{\{i,j\}}] = -(x_i - x_j)^2 |v_i v_j|^2. \quad (\text{H2:6.3})$$

For distinct nodes and nonzero entries it is negative, even if D_r itself is positive.

Thus the proposed inference from a positive prime/color covariance to a positive spatial payment fails at this explicit algebraic step. This does not assert that the complete integrated J -form is negative; the complete source and covariance terms in (6.2) must still be handled together.

6.1 The same issue remains after boundary cancellation

This can be checked on actual theta nodes without numerical evaluation. Take $x = (0, \log 2, \log 3)$, the pair $v = a_1 - a_2$, and $w = \pi e^{2r}$. The explicit seed entries give

$$v_0 \sim -60w^2, \quad v_1 \sim 64\sqrt{2}w^2 e^{-3w}, \quad v_2 \sim 324\sqrt{3}w^2 e^{-8w}.$$

For $u_0 = (\eta(0), \eta(\log 2), \eta(\log 3))^T$ and $X = \text{diag}(0, \log 2, \log 3)$,

$$\det[u_0, v, Xv] \sim \eta(\log 3) \log 2 v_0 v_1 \neq 0.$$

Thus the projections of v and $D_r v$ onto $u_0^\perp$ are independent for all sufficiently large r . The compressed matrix $D_r vv^* + vv^* D_r$ consequently has both signs on that two-dimensional balanced space: a symmetrized outer product of two independent vectors has one positive and one negative eigenvalue on their span.

This rules out a pointwise, individual-pair positive covariance payment on the balanced class itself. It leaves the sign of the full integral, with all other pairs and the source term included, unresolved.

7. Complete prime-pole pairing and its exact square energy

On finite exterior packets

$$h(r) = \sum_j d_j e^{-(b+it_j)r}, \quad b > 1/2,$$

let $\mathcal{G}$ act on e^{-zr} by the multiplier

$$m(\tfrac{1}{2} + z) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi_0(s/2) - \sum_{n \geq 2} \Lambda(n) n^{-s}, \quad s = \tfrac{1}{2} + z,$$

where $\psi_0 = \Gamma'/\Gamma$. The exact prime identity says $m(s) = \xi'(s)/\xi(s)$. The authentic theta form is

$$Q[h] = 2 \text{Re}\langle h, \mathcal{G}h \rangle.$$

For balanced packets, $\sum_j d_j = h(0) = 0$. Section 2 shows this restricted test class is sufficient.

Use distinct notation for the prime-counting function:

$$\Psi(x) = \sum_{n \leq x} \Lambda(n), \quad E(a) = \Psi(e^a) - e^a + 1, \quad U_a = e^{-a/2} S_a.$$

The digamma integral, together with the pole integral, gives

$$\mathcal{G} = -\frac{\gamma + \log \pi}{2} I + \int_0^\infty \frac{I - U_a}{e^{2a} - 1} da - \int_0^\infty U_a dE(a). \quad (\text{H2:7.1})$$

Indeed,

$$\frac{1}{s} + \frac{1}{2} \psi_0(s/2) - \frac{1}{2} \log \pi = -\frac{\gamma + \log \pi}{2} + \int_0^\infty \frac{1 - e^{-sa}}{e^{2a} - 1} da,$$

and

$$\int U_a dE(a) = \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} S_{\log n} - \int_0^\infty e^{a/2} S_a da.$$

All expressions converge on the stated exterior packet class.

The half-line identity

$$2 \operatorname{Re} \langle h, (I - U_a)h \rangle = \|h - U_a h\|^2 + (1 - e^{-a}) \|h\|^2 + e^{-a} \int_0^a |h(r)|^2 dr$$

now gives

$$Q[h] = \mathcal{E}_{\text{arch}}[h] - c_* \|h\|^2 - 2 \operatorname{Re} \int_0^\infty \langle h, U_a h \rangle dE(a), \quad (\text{H2:7.2})$$

where

$$c_* = \gamma + \log(2\pi) - 1,$$

$$\mathcal{E}_{\text{arch}}[h] = \int_0^\infty \frac{\|h - U_a h\|^2 + e^{-a} \int_0^a |h(r)|^2 dr}{e^{2a} - 1} da \geq 0. \quad (\text{H2:7.3})$$

The integration constant uses $\int_0^\infty (1 - e^{-a})/(e^{2a} - 1) da = 1 - \log 2$. All prime powers, continuous pole mass, and boundary collars are retained.

7.1 The balanced condition alone does not pay the constant

The exact lower bound

$$\|h - U_a h\| \geq (1 - e^{-a/2}) \|h\|$$

implies

$$\mathcal{E}_{\text{arch}}[h] \geq d_* \|h\|^2, \quad d_* = \int_0^\infty \frac{(1 - e^{-a/2})^2}{e^{2a} - 1} da = 3 - \frac{\pi}{2} - 2 \log 2. \quad (\text{H2:7.4})$$

The last integral is evaluated by $t = e^{-a/2}$.

This bound is sharp even with $h(0) = 0$. For $f \in C_c^\infty((0, 1))$, $\|f\|_2 = 1$, let

$$h_R(r) = R^{-1/2} f((r - R)/R).$$

Its norm is one and its trace is zero. For fixed a , $\|S_a h_R - h_R\| \leq a\|f'\|/R$, while its boundary collar is zero for $a < R$. Dominated convergence in (7.3) gives $\mathcal{E}_{\text{arch}}[h_R] \rightarrow d_*$.

These compact smooth functions are approximable in H^1 by fixed- b exterior packets: Fourier-invert $e^{br} h_R(r)$, approximate the integral by finite Fourier sums, and subtract the residual trace times e^{-br} . The weighted Fourier errors also control derivatives and the energy near $a = 0$; exponential weights control its tail. Thus the sharp infimum holds on the exterior balanced packet class itself.

But

$$c_* - d_* = \gamma + \log(8\pi) + \frac{\pi}{2} - 4 > 0.$$

Consequently the proposed bound $\mathcal{E}_{\text{arch}} \geq c_* \|h\|^2$ is false on that class. The signed prime-pole term remains necessary.

7.2 Exact mass matching and the loss from absolute values

Let $q(y)$ be the generalized inverse of Ψ :

$$q(y) = q \quad \text{when} \quad \Psi(q^-) \leq y < \Psi(q),$$

where q runs over prime powers. Set $\tau(y) = \log q(y)$ and $\lambda(y) = \log(1 + y)$. Then

$$\boxed{\int U_a dE(a) = \int_0^\infty (U_{\tau(y)} - U_{\lambda(y)}) dy.} \quad (\text{H2:7.5})$$

Each prime-power mass is paired with equal continuous pole mass.

A first-order estimate of each difference, followed by absolute values, produces the transport cost

$$\mathcal{T} = \int_0^\infty \left| \int_{\lambda(y)}^{\tau(y)} e^{-a/2} da \right| dy = \int_1^\infty |\Psi(x) - x + 1| x^{-3/2} dx. \quad (\text{H2:7.6})$$

The equality follows from Tonelli: for fixed x , the set of masses y transported across x has size $|\Psi(x) - (x - 1)|$.

Finiteness of (7.6) would make

$$s \int_1^\infty (\Psi(x) - x + 1) x^{-s-1} dx = -\frac{\zeta'(s)}{\zeta(s)} - \frac{1}{s-1} \quad (\text{Re } s > 1) \quad (\text{H2:7.7})$$

analytic for $\text{Re } s > 1/2$, with bounded continuous boundary values at every finite point of $\text{Re } s = 1/2$. At any critical-line zero ρ of multiplicity m , the right side has a pole $-m/(s - \rho)$. Thus, conditional on any such zero, the absolute cost (7.6) must be infinite.

This proposed absolute estimate would demand a pole-free critical boundary, a stronger condition than H. A useful comparison must retain the signed matched correlations in (7.5).

8. What remains after all three tests

The balanced-class reduction is proved, and its prime action and theta scale identities are explicit. The remaining sign can be expressed in any of these equivalent forms:

- The centered Gram comparison (3.3).

- The source-prime comparison (5.4).
- On one fixed exterior line, the complete paired comparison

$$2 \operatorname{Re} \int \langle h, U_a h \rangle dE(a) + c_* \|h\|^2 \leq \mathcal{E}_{\text{arch}}[h], \quad h(0) = 0.$$

The tested componentwise approaches do not establish these inequalities: an individual theta seed has an explicit adverse balanced pairing; positive covariance passes through a sign-indefinite spatial operation; the Archimedean floor alone is too small; and absolute first-order mass matching loses essential cancellation.

A continuing proof must identify a positive representation of the complete paired form, preserving its source, covariance, and prime-pole cross terms. Such a representation is not supplied by this note.

Source normalization and earlier identities were taken from *Michaels Whole-Class Transfer and Exact Theta Compensation*, especially Sections 3-5, and *Equation (14.4) and the positive operator J*. The reductions and analytic tests above are derived here.

Whole-class transfer and compensation

Source: *Michaels_Whole_Class_Transfer_and_Compensation.md*. Equation tags carry the source prefix H3; references within this unit retain their original local numbering.

Research continuation for Christophe Michaels – 21 September 2026

1. Result and scope

This note applies the recovered prime-prediction surplus to the completed arithmetic features, then compares it with the actual spatial theta form. Every identity below applies to its whole stated class. There is no numerical experiment, finite-degree certification, or truncation ladder.

Two identities are established:

1. The recovered prediction-variance identity has an exact specialization as a positive transverse derivative energy of the completed arithmetic lift.
2. The spatial theta form has an exact boundary-minus-score representation, with its score explicitly linked to the prime-phase identity.

The identities locate, but do not establish the sign of, the remaining completed mixed term. Consequently this note does not prove H.

Use the normalization

$$F(z) = \xi(\tfrac{1}{2} + z) = \int_{\mathbb{R}} \eta(u) e^{zu} du, \quad \eta > 0, \quad \eta(-u) = \eta(u).$$

The actual theta density and its derivatives have the decay needed below. The recovered curvature theorem supplies

$$-(\log \eta)'' \geq \kappa, \quad \kappa = 18.$$

Here $s = \sigma + it = \tfrac{1}{2} + z$. The target is

$$K(\sigma, t) = \operatorname{Re}(\overline{\xi(s)} \xi'(s)) \geq 0 \quad (\sigma > \tfrac{1}{2}, t \in \mathbb{R}). \quad (\text{H3:1.1})$$

2. The recovered surplus, applied to the complete arithmetic lift

Fix $\alpha > 0$ and $\sigma > (1 + \alpha)/2$. Throughout differentiation, α remains fixed. Put

$$J_{\alpha}(d) = d^{\alpha} \prod_{p|d} (1 - p^{-\alpha}), \quad a_d(s) = \sqrt{J_{\alpha}(d)} d^{-s},$$

$$R = \sum_{d \geq 1} |a_d|^2 = \frac{\zeta(2\sigma - \alpha)}{\zeta(2\sigma)}, \quad e = a/\sqrt{R}.$$

The completed Möbius-gcd construction gives the limiting feature $\Psi(s) = \xi(s)a(s)$. Define its normalized version

$$X(s) = \Psi(s)/\sqrt{R} = \xi(s)e(s).$$

All logarithmic moments used here converge absolutely, locally uniformly in the stated strict half-plane. Define

$$\mu(d) = |e_d|^2, \quad L(d) = \log d, \quad \lambda = \mathbb{E}_{\mu} L, \quad V = \operatorname{Var}_{\mu} L.$$

Inner products are conjugate-linear in their first slot.

2.1 Every color grouping at once

Let $\{C_j\}$ be any finite or countable partition of the positive integers, omitting empty cells. Write

$$p_j = \mu(C_j), \quad \lambda_j = \mathbb{E}_\mu[L \mid C_j], \quad V_j = \text{Var}_\mu(L \mid C_j).$$

Then

$$\boxed{V = \sum_j p_j V_j + \sum_j p_j (\lambda_j - \lambda)^2.} \quad (\text{H3:2.1})$$

Proof. In each cell expand $L - \lambda = (L - \lambda_j) + (\lambda_j - \lambda)$. The conditional mean of the first summand is zero, so the cross term vanishes. Sum the nonnegative conditional square integrals. Finite second moment justifies the countable case.

This specializes the recovered local-Schur-plus-prediction-disagreement identity to the completed divisor metric. It includes all prime-color partitions and all recombinations of those partitions. This application does not identify the original successor-bank matrices with the new divisor metric or assert an isometry between their underlying spaces.

2.2 A positive all-prime variance

Logarithmic differentiation of the absolutely convergent Euler product for R gives

$$\lambda = -\frac{1}{2} \partial_\sigma \log R, \quad V = -\frac{1}{2} \partial_\sigma \lambda$$

and

$$V = \sum_p (\log p)^2 \left[\frac{p^{\alpha-2\sigma}}{(1-p^{\alpha-2\sigma})^2} - \frac{p^{-2\sigma}}{(1-p^{-2\sigma})^2} \right] > 0. \quad (\text{H3:2.2})$$

Each summand is positive. Also, the pairwise variance formula gives

$$V = \frac{1}{2} \sum_{d,e} \mu(d)\mu(e)(\log d - \log e)^2 \geq \frac{(2^\alpha - 1)2^{-2\sigma}(\log 2)^2}{R^2} > 0. \quad (\text{H3:2.3})$$

This lower bound is independent of the partition, not uniform over all σ, α .

2.3 Where the recovered positive term goes

Differentiation gives

$$e_\sigma = (\lambda - L)e, \quad \langle e, e_\sigma \rangle = 0, \quad \|e_\sigma\|^2 = V.$$

Consequently

$$\boxed{\text{Re}\langle X, X_\sigma \rangle = K(\sigma, t),} \quad \boxed{\|X_\sigma\|^2 = |\xi'|^2 + |\xi|^2 V.} \quad (\text{H3:2.4})$$

For $\xi(s) \neq 0$, the derivative Gram Schur complement is exactly

$$\|\Psi_\sigma\|^2 - \frac{|\langle \Psi, \Psi_\sigma \rangle|^2}{\|\Psi\|^2} = R|\xi|^2 V. \quad (\text{H3:2.5})$$

At a zero, formula (2.4) still holds; the quotient in (2.5) is not used.

To match (2.1) directly to the recovered prediction identity, the predictor for cell C_j is

$$y_j = \frac{\xi'}{\xi} - \lambda_j, \quad x = \frac{\xi'}{\xi} - \lambda.$$

Thus

$$y_j - x = \lambda - \lambda_j. \quad (\text{H3:2.6})$$

The shared scalar logarithmic derivative cancels from every disagreement. The recovered positive term is real and survives completion, but it is the transverse component of the derivative, not its radial component. These algebraic identities also hold with any differentiable scalar in place of ξ ; their positivity alone therefore cannot establish the theta-specific sign in (1.1).

3. An identity for the entire spatial coefficient class

Use the actual spatial kernel

$$J(x, y) = \int_0^\infty (x + y + 2r) \eta(x + r) \eta(y + r) dr.$$

Define the logarithmic score and its curvature remainder by

$$p(u) = -\eta'(u)/\eta(u), \quad q(u) = p(u) - \kappa u.$$

Evenness and the curvature theorem imply that q is odd and $q' \geq 0$.

Take any finite list of real nodes $x_1, \dots, x_m$ and any complex coefficients $c_1, \dots, c_m$. No restriction on m is imposed. Set

$$h(r) = \sum_j c_j \eta(x_j + r), \quad k(r) = \sum_j c_j q(x_j + r) \eta(x_j + r),$$

$$\ell(r) = \sum_j c_j (x_j + r) \eta(x_j + r).$$

Then

$$h' = -\kappa \ell - k, \quad Q_J[c] := \sum_{j,k} \bar{c}_j c_k J(x_j, x_k) = 2 \operatorname{Re} \int_0^\infty \ell(r) \overline{h(r)} dr.$$

The last identity uses that the real symmetric kernel gives a real quadratic form. Integration by parts, with $h(\infty) = 0$, yields

$$\boxed{\kappa Q_J[c] = |h(0)|^2 - 2 \operatorname{Re} \int_0^\infty k(r) \overline{h(r)} dr.} \quad (\text{H3:3.1})$$

All mixed coefficients and the entire half-line remain in this identity. The same proof applies to compactly supported integrable coefficient functions, replacing sums by integrals. The identity itself holds for any $\kappa > 0$; the curvature theorem adds the information $q' \geq 0$.

Thus the exact class-wide compensation needed is

$$\boxed{2 \operatorname{Re} \int_0^\infty k(r) \overline{h(r)} dr \leq |h(0)|^2 \quad \text{for every coefficient packet.}} \quad (\text{H3:3.2})$$

Equation (3.2) is equivalent to positivity of this spatial form; it is not asserted as a proved bound here. On the null-trace class $h(0) = 0$, it requires the score correlation to be nonpositive. Monotonicity of the scalar function q does not itself sign that mixed quadratic form.

4. The score has an explicit prime-phase representation

There is an exact transform connecting the score to the actual prime identity, without introducing a hypothetical intertwining operator:

$$G(z) := \int_{\mathbb{R}} q(u) \eta(u) e^{zu} du = zF(z) - \kappa F'(z). \quad (\text{H3:4.1})$$

Indeed, $q\eta = -\eta' - \kappa u\eta$; integrate the first term by parts and differentiate the transform for the second.

For $\text{Re } z > 1/2$, the recovered equation (14.4) gives

$$\frac{F'(z)}{F(z)} = A(\tfrac{1}{2} + z) - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} e^{-z \log n},$$

where

$$A(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \frac{\Gamma'(s/2)}{\Gamma(s/2)}.$$

Therefore

$$\boxed{\frac{G(z)}{F(z)} = z - \kappa A(\tfrac{1}{2} + z) + \kappa \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} e^{-z \log n}.} \quad (\text{H3:4.2})$$

This is an absolutely convergent prime-power identity in its stated domain. It is not continued term by term into the critical strip.

For a whole finite exponential packet, choose $z_j = b + it_j$ with one fixed $b > 1/2$, and put

$$h(r) = \sum_j d_j e^{-z_j r}, \quad (\mathcal{A}h)(r) = \sum_j A(\tfrac{1}{2} + z_j) d_j e^{-z_j r}.$$

The theta coefficient function $v(x) = \sum_j [d_j / F(z_j)] e^{z_j x}$ produces exactly this h . Here $F(z_j) \neq 0$ in the absolute Euler half-plane. All integrals converge by theta decay and $b > 0$.

Equations (4.1)–(4.2) give its score transform

$$k = -h' - \kappa \mathcal{A}h + \kappa \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} h(\cdot + \log n). \quad (\text{H3:4.3})$$

Inserting this into (3.1) cancels the derivative boundary exactly:

$$\boxed{Q_J[v] = 2 \operatorname{Re} \int_0^\infty (\mathcal{A}h) \bar{h} dr - 2 \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \operatorname{Re} \int_0^\infty h(r + \log n) \overline{h(r)} dr.} \quad (\text{H3:4.4})$$

The prime sum and integral may be interchanged: the finite packet is bounded by a constant times e^{-br} , and $\sum \Lambda(n) n^{-b-1/2} < \infty$. No operator-norm convergence on all of $L^2(0, \infty)$ is claimed.

Equation (4.4) recovers the prior exact single-line prime-correlation form. The boundary contribution in (3.1) has already been used in this cancellation and cannot be counted again as an independent allowance.

5. Exact comparison with the recovered variance

The score also connects directly to the normalized arithmetic lift. With $z = s - 1/2$, (4.1) and (2.4) imply the vector identity

$$G(z)e = (z + \kappa\lambda)X - \kappa X_\sigma - \kappa LX. \quad (\text{H3:5.1})$$

Pair it with X . Since $\langle X, LX \rangle = \lambda \|X\|^2$, the arithmetic mean cancels:

$$\text{Re}\langle X, G(z)e \rangle = (\sigma - \tfrac{1}{2})|\xi(s)|^2 - \kappa K(\sigma, t). \quad (\text{H3:5.2})$$

The positive prime variance controls $\|(L - \lambda)X\|^2 = |\xi|^2 V$, whose mean pairing with X is zero. Thus the transfer does not lose that variance; it identifies precisely why its positivity alone does not bound the signed score correlation.

The remaining whole-class task is to prove the comparison (3.2), or, by the recovered single-line transfer theorem, equivalently the nonnegativity of (4.4) on every finite packet on one fixed line. A successful use of the recovered surplus must supply this comparison in the same completed form, with its mixed terms included. Neither another positive variance floor nor a growing list of verified matrix sizes supplies that comparison.

6. Source anchors and proof status

The source ingredients used, rather than re-certified, are:

- *Prime Tower Successor Inequality – Symbolic Checkpoint, 2026-08-22*, Section 6: exact Schur surplus and prediction-disagreement identity.
- *MATHEMATICS(20260917-234824).md*, D19 and D22: degree-dependent theta variance floor and the complete signed comparison. The D19 constant is not treated as uniform in degree.
- *Michaels Three Route Analytic Tests*, completed Möbius–gcd construction: the feature limit $\Psi = \xi a$.
- *Michaels Completed Coherence and Pole Compensation: An Analytic Test*: completed accounting and normalization.
- *Equation (14.4) and the positive operator J (Michaels_Prime_Phase_J_Bridge.md)*, Sections 2–4: authentic spatial kernel, single-line transfer, and the exact prime-correlation form.

Equations (2.1)–(2.6), (3.1), and (4.1)–(5.2) are proved above on their stated domains, taking those source ingredients as given. Equation (3.2) remains the required, unproved compensation inequality. No unconditional proof of H or RH is claimed.

Completed coherence compensation

Source: *Michaels_Completed_Coherence_Compensation_Test.md*. Equation tags carry the source prefix *H4*; references within this unit retain their original local numbering.

Research note for Christophe Michaels

1. Target and outcome

This note carries out the next test proposed in *Michaels Three Route Analytic Tests*. It embeds the extracted two-color defect into the scalar amplitude and the completed Möbius-gcd energy, then evaluates the pole and modular contributions with their cross terms retained.

Write

$$s = \sigma + it = \frac{1}{2} + a + it, \quad K(a, t) = \operatorname{Re}(\xi'(s)\overline{\xi(s)}).$$

H asks for $K \geq 0$ for every $a > 0, t \in \mathbb{R}$.

The test gives three exact findings:

1. The scalar decomposition contains the negative of the previously extracted defect derivative. Weighting by the gcd metric also changes its rank; the unweighted rank-two calculation is not the full energy.
2. The separate pole contribution has a negative horizontal derivative for sufficiently large cutoff. Its large terms cancel against the bare and mixed terms of the completed energy.
3. In the modular representation, the pole constant cancels a theta integration boundary exactly and produces the positive theta density.

Thus a separately positive pole allowance does not establish the proposed compensation. The complete remaining derivative is given in Section 5. Its sign is not established here. All calculations are analytic.

2. Exact embedding of the color defect

Let $H_n(s) = C(s)n^{-s}$, where C is holomorphic, and define

$$u_n = n^{-\sigma} \cos(t \log n), \quad v_n = n^{-\sigma} \sin(t \log n).$$

The real part of the Hermitian Gram and the multiplicative comparison matrix are respectively

$$G = |C|^2(uu^T + vv^T), \quad T = |C|^2(uu^T - vv^T).$$

Consequently

$$\Delta = T - G = -2|C|^2vv^T, \quad G = T - \Delta. \tag{H4:2.1}$$

For every fixed finite real coefficient packet c ,

$$\operatorname{Re} \left[\left(C \sum_n c_n n^{-s} \right)' \overline{C \sum_n c_n n^{-s}} \right] = \frac{1}{2} c^T T' c - \frac{1}{2} c^T \Delta' c. \tag{H4:2.2}$$

This fixes the sign of the extracted component inside the scalar source.

At a point where $C \neq 0$, put $A = \text{Re}(C'/C)$, $L = \text{diag}(\log n)$, and $z_0 = Lv - Av$. Then

$$\frac{1}{2}\Delta' = |C|^2(z_0 v^T + v z_0^T).$$

For the component $-\Delta'/2$ in (2.2), the smallest isotropic addition making it nonnegative is

$$\gamma_{\text{scalar}} I, \quad \gamma_{\text{scalar}} = |C|^2 (\|v\| \|z_0\| + v \cdot z_0). \quad (\text{H4:2.3})$$

The previous note's threshold with a minus sign before $v \cdot z_0$ correctly applies to $+\Delta'/2$, rather than the component in (2.2). Both thresholds concern this unweighted matrix only.

For the gcd metric $B_\alpha(m, n) = \gcd(m, n)^\alpha$, the corresponding weighted defect is instead

$$B_\alpha \circ \Delta = -2|C|^2 D_v B_\alpha D_v, \quad D_v = \text{diag}(v_n). \quad (\text{H4:2.4})$$

On distinct integer indices, B_α is strictly positive definite when $\alpha > 0$. If all selected v_n are nonzero, (2.4) has full rank. Its half-derivative at fixed α is

$$|C|^2 D_v [(L - AI)B_\alpha + B_\alpha(L - AI)] D_v. \quad (\text{H4:2.5})$$

Its rank is generally not bounded by two. Already on a prime index set,

$$B_\alpha = \mathbf{1}\mathbf{1}^T + \text{diag}(p^\alpha - 1).$$

Thus transferring the two-dimensional threshold directly to the completed gcd energy would omit metric terms.

3. Complete Möbius pole accounting

Fix $1/2 < \sigma < 1$ and $0 < \alpha < 2\sigma - 1$. Let

$$k(s) = \frac{s}{2} \pi^{-s/2} \Gamma(s/2), \quad h(s) = (s-1)k(s), \quad b_N(n) = \sum_{j \leq N/n} \frac{\mu(j)}{nj},$$

$$w_N(n; s) = h(s)n^{-s} + k(s)N^{1-s}b_N(n). \quad (\text{H4:3.1})$$

Set $x_N = (h(s)n^{-s})_{n \leq N}$, $y_N = k(s)N^{1-s}b_N$, so $w_N = x_N + y_N$. Inner products below use B_α , with the first slot conjugate-linear. Define

$$E_N = \|w_N\|_{B_\alpha}^2, \quad P_N = \|y_N\|_{B_\alpha}^2, \quad C_N = \langle y_N, w_N \rangle_{B_\alpha}.$$

The complete three-part decomposition is exactly

$$E_{\text{bare}} = P_N - 2 \text{Re} C_N + E_N, \quad E_{\text{pole}} = P_N, \quad E_{\text{cross}} = 2 \text{Re} C_N - 2P_N. \quad (\text{H4:3.2})$$

Here $E_{\text{cross}} = 2 \text{Re} \langle x_N, y_N \rangle$.

Let

$$J_\alpha(d) = d^\alpha \prod_{p|d} (1 - p^{-\alpha}), \quad \mathcal{A}_s(x) = k(s) \left[(s-1) \sum_{n \leq x} n^{-s} + x^{1-s} \right].$$

Finite Möbius inversion gives $(Z_N b_N)(d) = 1/d$ and $(Z_N w_N)(d) = d^{-s} \mathcal{A}_s(N/d)$. Therefore

$$P_N = |k|^2 N^{2-2\sigma} \sum_{d \leq N} \frac{J_\alpha(d)}{d^2}, \quad (\text{H4:3.3})$$

$$C_N = \bar{k} N^{1-\bar{s}} \sum_{d \leq N} J_\alpha(d) d^{-1-s} \mathcal{A}_s(N/d). \quad (\text{H4:3.4})$$

3.1 Both large orders cancel

Put

$$R_\alpha = \frac{\zeta(2\sigma - \alpha)}{\zeta(2\sigma)}, \quad T_\alpha = \frac{\zeta(2 - \alpha)}{\zeta(2)}, \quad U_\alpha(s) = \frac{\zeta(s + 1 - \alpha)}{\zeta(s + 1)}, \quad \delta = 1 + \alpha - 2\sigma < 0.$$

All series used here converge absolutely. Set

$$P_N^0 = |k|^2 T_\alpha N^{2-2\sigma}, \quad Q_N = \bar{k} \xi(s) U_\alpha(s) N^{1-\bar{s}}, \quad G_0 = R_\alpha |\xi(s)|^2.$$

The Euler-summation estimate $\mathcal{A}_s(x) - \xi(s) = O(x^{-\sigma})$, applied to (3.3)-(3.4), gives

$$\begin{aligned} E_{\text{bare}} &= P_N^0 - 2 \operatorname{Re} Q_N + G_0 + O(N^\delta), \\ E_{\text{pole}} &= P_N^0 + O(N^\delta), \\ E_{\text{cross}} &= -2P_N^0 + 2 \operatorname{Re} Q_N + O(N^\delta). \end{aligned} \tag{H4:3.5}$$

The differentiated expansions have error $O((1 + \log N)N^\delta)$, locally within the strict parameter domain. For example, the tail of $\sum J_\alpha(d)/d^2$, after multiplication by $N^{2-2\sigma}$, is $O(N^\delta)$; the other terms follow from $J_\alpha(d) \leq d^\alpha$ and the differentiated Euler remainder. Thus the scales $N^{2-2\sigma}$ and $N^{1-\sigma}$ cancel in the complete sum, leaving $E_N \rightarrow G_0$.

3.2 The separate pole derivative has a definite adverse sign

Write

$$a_k = \operatorname{Re}(k'/k), \quad \lambda_\alpha = -\frac{1}{2} \partial_\sigma \log R_\alpha > 0.$$

Because α is fixed, direct differentiation of (3.3) gives

$$\frac{\partial_\sigma P_N + 2\lambda_\alpha P_N}{2R_\alpha} = \frac{P_N}{R_\alpha} (a_k + \lambda_\alpha - \log N). \tag{H4:3.6}$$

For each fixed admissible s, α , this is strictly negative for all sufficiently large N . This rules out treating the separate pole term as a positive payment. The mixed terms in (3.2) are essential.

4. The modular pole cancels the theta boundary

The finite-residue modular representation from the preceding note gives $\xi(s) = P_q(s) + T_q(s)$, where

$$P_q(s) = \frac{1}{2} q^{-s/2} [1 + (\sqrt{q} - 1)s].$$

With $d = \sqrt{q} - 1$, its exact slope is

$$\operatorname{Re}(P'_q \bar{P}_q) = \frac{q^{-\sigma}}{4} \left[d(1 + d\sigma) - \frac{\log q}{2} ((1 + d\sigma)^2 + d^2 t^2) \right]. \tag{H4:4.1}$$

For $q > 1$, it is negative for sufficiently large $|t|$. At $q = 1$, the same construction gives $P_1 = 1/2$, with zero slope. The complete formula retains

$$K = \operatorname{Re}(P'_q \bar{P}_q) + \operatorname{Re}(T'_q \bar{T}_q) + \operatorname{Re}(P'_q \bar{T}_q + T'_q \bar{P}_q). \tag{H4:4.2}$$

The sum is independent of the residue parameter q ; its individual pieces are not.

The scalar modular identity identifies the cancellation more explicitly. Let

$$\psi(x) = \sum_{n \geq 1} e^{-\pi n^2 x}, \quad g(u) = e^{u/2} \psi(e^{2u}), \quad z = s - \frac{1}{2}.$$

The completed Mellin formula is

$$F(z) = \xi(\tfrac{1}{2} + z) = \tfrac{1}{2} + 2(z^2 - \tfrac{1}{4}) \int_0^\infty g(u) \cosh(zu) du. \quad (\text{H4:4.3})$$

Differentiating $\Theta(\tau) = \tau^{-1/2} \Theta(1/\tau)$ at $\tau = 1$ gives $\Theta'(1) = -\Theta(1)/4$, and hence $g'(0) = -1/4$. Two integrations by parts now give

$$(z^2 - \tfrac{1}{4}) \int_0^\infty g \cosh(zu) du = \int_0^\infty (g'' - g/4) \cosh(zu) du + g'(0).$$

Consequently

$$\boxed{\tfrac{1}{2} + 2g'(0) = 0, \quad F(z) = 2 \int_0^\infty \eta(u) \cosh(zu) du, \quad \eta = g'' - g/4.} \quad (\text{H4:4.4})$$

The density is explicitly positive on $u \geq 0$:

$$\eta(u) = \sum_{n \geq 1} \left(4\pi^2 n^4 e^{9u/2} - 6\pi n^2 e^{5u/2} \right) e^{-\pi n^2 e^{2u}} > 0. \quad (\text{H4:4.5})$$

Each term is positive because $\pi n^2 e^{2u} > 3/2$. The modular identity also gives $g(u) - g(-u) = -\sinh(u/2)$, so applying $d^2/du^2 - 1/4$ makes η even.

Thus completion exactly cancels the boundary and yields the existing positive theta density. Its remaining derivative is

$$K(a, t) = \tfrac{1}{2} \iint_{\mathbb{R}^2} (u + v) \sinh(a(u + v)) \eta(u) \eta(v) \cos(t(u - v)) du dv. \quad (\text{H4:4.6})$$

This follows by differentiating $|F(a + it)|^2$ and pairing (u, v) with $(-u, -v)$. Absolute convergence follows from the superexponential theta tails. The relative-phase cosine retains the signed interaction after the modular boundary cancellation.

5. Exact remaining finite source

For $\sigma > 1/2$, fixed $0 < \alpha < 2\sigma - 1$, and the complete vector (3.1), define

$$r_N(n; s) = n^{-s} [1 + (s - 1) \log(N/n)]. \quad (\text{H4:5.1})$$

Direct differentiation, retaining the Möbius pole term, gives

$$w'_N = \left(\frac{k'}{k} - \log N \right) w_N + k r_N. \quad (\text{H4:5.2})$$

Since B_α is independent of σ ,

$$\boxed{K_N(s) = \frac{(a_k + \lambda_\alpha - \log N) E_N + \operatorname{Re}[k w_N^* B_\alpha r_N]}{R_\alpha} \longrightarrow K(a, t).} \quad (\text{H4:5.3})$$

The convergence follows from the completed Euler remainder and dominated divisor sums proved in the preceding note; it includes horizontal differentiation and the removable point $s = 1$.

A sufficient coupled estimate would be

$$\operatorname{Re}[k w_N^* B_\alpha r_N] \geq (\log N - a_k - \lambda_\alpha) E_N - \varepsilon_N R_\alpha, \quad \varepsilon_N \longrightarrow 0, \quad (\text{H4:5.4})$$

for each admissible s . Equation (5.4) is a precisely stated remaining estimate, not a proved lower bound. Positivity of B_α controls norms but does not determine the sign of this mixed inner product.

The present test therefore identifies the complete compensation expression and excludes a separate positive pole payment. It does not establish H or positivity of the full spatial kernel.

Source continuity

The current *Michaels Three Route Analytic Tests* was read in full before this continuation. Its definitions, convergence theorem, and derivative normalization are retained. The sign and metric distinctions in Section 2 clarify how its isolated defect enters the full expression; the original unweighted defect formulas remain valid in their stated setting.

Three analytic routes

Source: *Michaels_Three_Route_Analytic_Tests.md*. Equation tags carry the source prefix H5; references within this unit retain their original local numbering.

Research note for Christophe Michaels

1. Target and results

Write

$$s = \sigma + it = \frac{1}{2} + a + it, \quad \xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

The target H is

$$K(a, t) = \operatorname{Re}(\xi'(s)\overline{\xi(s)}) \geq 0 \quad (a > 0, t \in \mathbb{R}). \quad (\text{H5:1.1})$$

All tests below are analytic. Finite formulas are identities, not numerical samples or finite-height evidence. The main manuscript is unchanged.

The tests establish three connections:

1. The complete valuation-color Mellin lift has exactly the earlier product Poisson metric. Its derivative retains the same signed normalization.
2. An explicit finite Möbius pole correction gives positive gcd-color energies converging to the actual completed amplitude throughout $\operatorname{Re} s > 1/2$, including the removable point $s = 1$.
3. The two-prime multiplicative identity becomes a signed rank-one defect under Hermitian pairing. Its horizontal derivative has rank at most two.

The amplitude synthesis in item 2 is proved below. Its horizontal derivative is retained exactly; its required sign is not proved in this note.

2. Retained colors and modular completion

2.1 Exact spatial color projections

Let

$$\Theta(\tau, x) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 \tau} e^{2\pi i n x}, \quad (P_m f)(x) = \frac{1}{m} \sum_{j=0}^{m-1} f(x + j/m).$$

On circle Fourier coefficients, P_m retains precisely the indices divisible by m . Thus

$$P_m \Theta(\tau, x) = \Theta(m^2 \tau, mx), \quad Q_{p,k} = P_{p^k} - P_{p^{k+1}} \quad (\text{H5:2.1})$$

are commuting orthogonal valuation projections. For finitely many primes, their products resolve all nonzero integer Fourier modes into colors.

With

$$\mathcal{C}_s[f] = \frac{s(s-1)}{4} \int_0^\infty f(\tau) \tau^{s/2} \frac{d\tau}{\tau},$$

the identity $\mathcal{C}_s[(\Theta - 1)(\cdot, 0)] = \xi(s)$ holds for $\sigma > 1$. Scaling τ in each nonzero integer sector gives

$$V_{\mathbf{k}}(s) = \xi(s) \prod_{p \in P} (1 - p^{-s}) p^{-k_p s}. \quad (\text{H5:2.2})$$

These identities then continue analytically. For finite P and $\sigma > 0$, both the coherent sum and squared norm converge:

$$\sum_{\mathbf{k} \in \mathbb{N}_0^P} V_{\mathbf{k}}(s) = \xi(s), \quad \|V(s)\|^2 = \frac{|\xi(s)|^2}{G_{P,\sigma}(t)}, \quad (\text{H5:2.3})$$

where

$$G_{P,\sigma}(t) = \prod_{p \in P} \frac{1 - p^{-2\sigma}}{|1 - p^{-s}|^2}. \quad (\text{H5:2.4})$$

This is exactly the product Poisson multiplier in the existing color construction, restricted to $\theta_p = t \log p$.

Set $r_p = p^{-\sigma}$, $c_p = (1 - r_p)/(1 + r_p)$, and

$$\beta_\sigma = \sum_{p \in P} \frac{2(\log p)r_p}{1 - r_p^2}, \quad D_\sigma(t) = \sum_{p \in P} (\log p) \left(\frac{1 - r_p^2}{|1 - p^{-s}|^2} - c_p \right).$$

Direct differentiation gives

$$\partial_\sigma \log G_{P,\sigma} = \beta_\sigma - D_\sigma,$$

and therefore the exact target identity is

$$K = G_{P,\sigma} \left[\operatorname{Re} \langle V, \partial_\sigma V \rangle + \frac{\beta_\sigma - D_\sigma}{2} \|V\|^2 \right]. \quad (\text{H5:2.5})$$

Positive color norms do not supply the sign of the bracket in (2.5).

2.2 Full finite-residue modular pairing

For an integer $q \geq 2$, define

$$\theta_{q,r}(\tau) = \sum_{n \equiv r \pmod{q}} e^{-\pi n^2 \tau / q}, \quad (U_q)_{rj} = q^{-1/2} e^{2\pi i r j / q}, \quad \delta = e_0, \quad b = q^{-1/2} \mathbf{1}.$$

Poisson summation on the residue progression gives

$$\theta_q(\tau) = \tau^{-1/2} U_q \theta_q(1/\tau). \quad (\text{H5:2.6})$$

Here U_q is unitary and its square is reflection; the theta vectors are reflection-even. Define the entire vector

$$A_q(s) = \int_1^\infty (\theta_q(\tau) - \delta) \tau^{s/2} \frac{d\tau}{\tau}.$$

Splitting the full Mellin integral at one and applying (2.6) yields

$$I_q(s) = A_q(s) + U_q A_q(1 - s) + \frac{2b}{s - 1} - \frac{2\delta}{s}. \quad (\text{H5:2.7})$$

The pole terms have been retained. The completed vector

$$\mathcal{G}_q(s) = \frac{s(s-1)}{4} I_q(s)$$

is entire and satisfies

$$\mathcal{G}_q(s) = U_q \mathcal{G}_q(1-s), \quad \xi(s) = q^{-s/2} \mathbf{1}^T \mathcal{G}_q(s). \quad (\text{H5:2.8})$$

In particular $\mathcal{G}_q(1) = b/2$, and (2.8) gives $\xi(1) = 1/2$.

Differentiation preserves the following signed terms:

$$I'_q(s) = A'_q(s) - U_q A'_q(1-s) - \frac{2b}{(s-1)^2} + \frac{2\delta}{s^2}. \quad (\text{H5:2.9})$$

Writing $H_q = \mathbf{1}^T \mathcal{G}_q$, the scalar derivative is

$$K = q^{-\sigma} \left[\operatorname{Re}(H'_q \overline{H_q}) - \frac{\log q}{2} |H_q|^2 \right]. \quad (\text{H5:2.10})$$

Thus the modular transformation is an exact source-preserving operation, but its unitarity does not make the target derivative a norm. In particular, $\mathbf{1}^T U_q = \sqrt{q} \delta^T$: the scalar observation changes.

2.3 The full-prime bare lift exposes the pole channel

Put $h(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)$. In $\sigma > 1$, resolving all primes gives the individual integer components

$$V_n(s) = h(s)n^{-s}, \quad \|V(s)\|^2 = |h(s)|^2 \zeta(2\sigma). \quad (\text{H5:2.11})$$

The vector extends in ℓ^2 to $\sigma > 1/2$. Its ordinary coherent sum does not extend by bounded summation there. At $s = 1$ the entire bare vector is zero, although $\xi(1) = 1/2$. The residue channel must be added. Also, for fixed t as $\sigma \downarrow 1/2$,

$$\operatorname{Re}\langle V, \partial_\sigma V \rangle \sim -\frac{|h(1/2 + it)|^2}{(2\sigma - 1)^2} < 0. \quad (\text{H5:2.12})$$

This tests the bare lift only. Section 4 supplies an explicitly completed arithmetic lift.

3. Moving normalization: what its positive derivative measures

For a finite integer index set let

$$B_\alpha(m, n) = \gcd(m, n)^\alpha, \quad D_\alpha = \operatorname{diag}(n^{-\alpha/2}), \quad N_\alpha = D_\alpha B_\alpha D_\alpha, \quad L = \operatorname{diag}(\log n).$$

Jordan inversion gives

$$B_\alpha = Z^* \operatorname{diag}(J_\alpha(d)) Z, \quad J_\alpha(d) = d^\alpha \prod_{p|d} (1 - p^{-\alpha}), \quad (Zc)(d) = \sum_{d|n} c_n. \quad (\text{H5:3.1})$$

For $\alpha \geq 0$, $J'_\alpha(d) \geq 0$. Hence

$$N'_\alpha + \frac{1}{2}(LN_\alpha + N_\alpha L) = D_\alpha B'_\alpha D_\alpha \succeq 0. \quad (\text{H5:3.2})$$

At $\alpha = 0$, the positive divisor weights are $J'_0(d) = \Lambda(d)$, supported exactly on prime powers.

For $\sigma > 1$ and $0 \leq \alpha < 2\sigma - 1$, use the transported coefficients $c_{\alpha,n} = n^{-s+\alpha/2}$. Absolute convergence and (3.1) give

$$\begin{aligned} c_\alpha^* N_\alpha c_\alpha &= \sum_{m,n \geq 1} \frac{\gcd(m,n)^\alpha}{m^s n^s} \\ &= R_\alpha(\sigma) |\zeta(s)|^2, \quad R_\alpha(\sigma) = \frac{\zeta(2\sigma - \alpha)}{\zeta(2\sigma)}. \end{aligned} \quad (\text{H5:3.3})$$

Indeed $\sum_d J_\alpha(d) d^{-u} = \zeta(u - \alpha)/\zeta(u)$, proved by summing the geometric Euler factor at each prime. The positive α -derivative in (3.2) is exactly

$$\partial_\alpha(c_\alpha^* N_\alpha c_\alpha) = -\frac{\zeta'(2\sigma - \alpha)}{\zeta(2\sigma)} |\zeta(s)|^2. \quad (\text{H5:3.4})$$

The derivative of the normalized energy is zero:

$$\partial_\alpha \left[\frac{c_\alpha^* N_\alpha c_\alpha}{R_\alpha(\sigma)} \right] = 0.$$

The auxiliary transport is positive, but its growth is completely accounted for by the Euler-ratio normalization.

4. A completed Möbius-gcd synthesis inside the target half-plane

4.1 Finite identity retaining the pole

Fix $\sigma > 1/2$ and choose $0 < \alpha < 2\sigma - 1$. Throughout horizontal differentiations below, α is held fixed in a neighborhood where this strict inequality continues to hold.

For $1 \leq n \leq N$, set

$$b_N(n) = \sum_{k \leq N/n} \frac{\mu(k)}{nk}, \quad k(s) = \frac{s}{2} \pi^{-s/2} \Gamma(s/2),$$

$$\boxed{w_N(n; s) = k(s) [(s-1)n^{-s} + N^{1-s} b_N(n)]}. \quad (\text{H5:4.1})$$

This expression is holomorphic throughout $\sigma > 1/2$, including $s = 1$. It adds the completed pole contribution to the bare integer components in (2.11).

Finite Möbius inversion proves exactly

$$\sum_{d|n \leq N} b_N(n) = \frac{1}{d} \sum_{m \leq N/d} \frac{1}{m} \sum_{k|m} \mu(k) = \frac{1}{d}. \quad (\text{H5:4.2})$$

Define, for real $x \geq 1$,

$$\mathcal{A}_s(x) = k(s) \left[(s-1) \sum_{n \leq x} n^{-s} + x^{1-s} \right]. \quad (\text{H5:4.3})$$

Equations (4.1)-(4.2) give the complete divisor-channel identity

$$(Z_N w_N)(d) = d^{-s} \mathcal{A}_s(N/d). \quad (\text{H5:4.4})$$

Every divisor channel of the finite packet is retained in this identity. The infinite tail is handled by the convergence argument that follows.

4.2 Analytic convergence, including derivatives

For $s \neq 1$, put

$$R_s(x) = \sum_{n \leq x} n^{-s} + \frac{x^{1-s}}{s-1}.$$

Partial summation gives, initially for $\sigma > 1$,

$$R_s(x) - \zeta(s) = -\{x\}x^{-s} + s \int_x^\infty \{u\}u^{-s-1} du. \quad (\text{H5:4.5})$$

The integral converges for $\sigma > 0$ and supplies continuation there. It proves locally uniform error bounds

$$R_s(x) - \zeta(s) = O(x^{-\sigma}), \quad \partial_s(R_s(x) - \zeta(s)) = O((1 + \log x)x^{-\sigma}).$$

Multiplying by $h(s) = (s-1)k(s)$ removes the pole, so

$$\mathcal{A}_s(x) \longrightarrow \xi(s), \quad \partial_s \mathcal{A}_s(x) \longrightarrow \xi'(s) \quad (\text{H5:4.6})$$

locally uniformly in $\sigma > 0$. These functions and derivatives are uniformly bounded for $x \geq 1$ on each such compact parameter set. At $s = 1$, $\mathcal{A}_1(x) = 1/2$ exactly.

Using (3.1) and (4.4),

$$\mathcal{E}_N(s) := w_N(s)^* B_\alpha w_N(s) = \sum_{d \leq N} J_\alpha(d) d^{-2\sigma} |\mathcal{A}_s(N/d)|^2 \geq 0. \quad (\text{H5:4.7})$$

Since $0 \leq J_\alpha(d) \leq d^\alpha$ and $2\sigma - \alpha > 1$, dominated convergence proves

$$\boxed{\lim_{N \rightarrow \infty} \frac{\zeta(2\sigma)}{\zeta(2\sigma - \alpha)} w_N(s)^* B_\alpha w_N(s) = |\xi(s)|^2.} \quad (\text{H5:4.8})$$

The same proof with logarithmic factors justifies horizontal differentiation. Thus (4.8) is a completed positive arithmetic representation in the actual target half-plane, not a formal continuation of a positive double series.

More strongly, the divisor-feature vectors converge in ℓ^2 :

$$\left\{ \sqrt{J_\alpha(d)} d^{-s} \mathcal{A}_s(N/d) \mathbf{1}_{d \leq N} \right\}_d \longrightarrow \xi(s) \left\{ \sqrt{J_\alpha(d)} d^{-s} \right\}_d. \quad (\text{H5:4.9})$$

Their horizontal derivatives also converge in ℓ^2 .

4.3 The exact derivative and the surviving signed term

Define the explicit positive arithmetic mean

$$\lambda_\alpha(\sigma) = -\frac{1}{2} \partial_\sigma \log R_\alpha(\sigma) = \frac{\sum_d (\log d) J_\alpha(d) d^{-2\sigma}}{R_\alpha(\sigma)} = \sum_{n \geq 2} \Lambda(n) (n^\alpha - 1) n^{-2\sigma} > 0. \quad (\text{H5:4.10})$$

Every series is absolutely convergent in the stated domain. Differentiating (4.8) gives the target exactly:

$$\boxed{K(a, t) = \lim_{N \rightarrow \infty} \frac{1}{R_\alpha} \sum_{d \leq N} J_\alpha(d) d^{-2\sigma} \operatorname{Re} \left[(\partial_s \mathcal{A}_s(N/d) + (\lambda_\alpha - \log d) \mathcal{A}_s(N/d)) \overline{\mathcal{A}_s(N/d)} \right].} \quad (\text{H5:4.11})$$

The normalized color motion cancels in the limit because

$$\sum_d J_\alpha(d) d^{-2\sigma} (\lambda_\alpha - \log d) = 0. \quad (\text{H5:4.12})$$

Equivalently, if $\mathcal{E}(s) = \lim_N \mathcal{E}_N(s) = R_\alpha |\xi(s)|^2$, then

$$K = \frac{\partial_\sigma \mathcal{E}}{2R_\alpha} + \lambda_\alpha |\xi(s)|^2. \quad (\text{H5:4.13})$$

The needed sign is therefore the specific differential bound

$$\partial_\sigma \mathcal{E} \geq -2\lambda_\alpha \mathcal{E}. \quad (\text{H5:4.14})$$

Equation (4.14) is an equivalent target in this representation, not a consequence of the nonnegative values in (4.7).

For clarity, define the arithmetic unit vector

$$e_d(s) = \frac{\sqrt{J_\alpha(d)} d^{-s}}{\sqrt{R_\alpha(\sigma)}}.$$

Then $\|e(s)\| = 1$, $\partial_\sigma e_d = (\lambda_\alpha - \log d) e_d$, and $\langle e, \partial_\sigma e \rangle = 0$. The normalized limit in (4.9) is precisely $\xi(s) e(s)$. The arithmetic direction is explicit; its radial derivative is the target.

Strict finite positivity supplies no automatic limiting floor. If $\xi(s) = 0$, then (4.5)-(4.7) imply

$$0 \leq \mathcal{E}_N(s) = O(N^{\alpha+1-2\sigma}) \longrightarrow 0. \quad (\text{H5:4.15})$$

This is consistent with every finite gcd matrix being strictly positive definite.

5. Compatibility test: the Hermitian two-color defect

5.1 Where multiplication first appears

For heat coefficients $g_n(\tau) = e^{-\pi n^2 \tau}$,

$$g_p g_q - g_1 g_{pq} = e^{-\pi(p^2+q^2)\tau} \left(1 - e^{-\pi(p^2-1)(q^2-1)\tau} \right) > 0. \quad (\text{H5:5.1})$$

Thus heat coefficients themselves do not obey the multiplicative four-corner identity. Their single Mellin transforms do:

$$H_n(s) = \int_0^\infty e^{-\pi n^2 \tau} \tau^{s/2} \frac{d\tau}{\tau} = C(s) n^{-s}, \quad C(s) = \pi^{-s/2} \Gamma(s/2), \quad \sigma > 0,$$

$$H_1(s) H_{pq}(s) = H_p(s) H_q(s). \quad (\text{H5:5.2})$$

The bilinear identity uses two heat variables. The substitution $(\tau, u) \mapsto (q^2 u, \tau/q^2)$ swaps its two integrands and preserves τu . It does not preserve their differently phased Hermitian Mellin weights when $t \neq 0$.

5.2 The exact surviving phase term

For distinct primes p, q , direct substitution gives

$$\boxed{\operatorname{Re} (H_1 \overline{H_{pq}} - H_p \overline{H_q}) = -2|C(s)|^2 (pq)^{-\sigma} \sin(t \log p) \sin(t \log q).} \quad (\text{H5:5.3})$$

This identity also holds for any common scalar factor $C(s)$. It is generally nonzero. Its scalar sign varies with the actual phase parameter t , even with complete multiplicativity intact.

There is nevertheless exact low-rank structure. For a finite prime set, put $v_p = p^{-\sigma} \sin(t \log p)$. Including the diagonal case $p = q$, the matrix of defects is

$$\Delta = -2|C(s)|^2 vv^T \preceq 0. \quad (\text{H5:5.4})$$

Let $L = \operatorname{diag}(\log p)$ and $A = \operatorname{Re}(C'/C)$ at a point with $C \neq 0$. Then

$$\frac{1}{2} \partial_\sigma \Delta = |C|^2 [(Lv)v^T + v(Lv)^T - 2Avv^T]. \quad (\text{H5:5.5})$$

It is supported on $\operatorname{span}\{v, Lv\}$. For two distinct primes with $v_p v_q \neq 0$, its principal determinant equals

$$-|C|^4 v_p^2 v_q^2 (\log p - \log q)^2 < 0. \quad (\text{H5:5.6})$$

Thus a common scalar completion does not make this extracted derivative positive. Its compensation must come from the other retained terms. The rank-two conclusion concerns precisely (5.5); it is not a rank-two reduction of the entire theta positivity problem.

There is an exact compensation threshold for this extracted block. Put $z_0 = Lv - Av$. Its potentially nonzero eigenvalues are

$$|C|^2 (v \cdot z_0 \pm \|v\| \|z_0\|).$$

Consequently the smallest isotropic addition making (5.5) nonnegative is

$$\gamma_* I, \quad \gamma_* = |C|^2 (\|v\| \|z_0\| - v \cdot z_0) \geq 0. \quad (\text{H5:5.7})$$

This also holds on the full prime Hilbert space when $\sigma > 1/2$, because $v, Lv \in \ell^2$: their squared entries are dominated by $p^{-2\sigma}$ and $(\log p)^2 p^{-2\sigma}$, respectively. Equation (5.7) specifies the exact isotropic compensation required by this component. It does not assert that the remaining completed theta operator provides it.

6. Outcome and next exact test

Route	Exact result	Derivative status
Retained spatial and prime colors	Modular vector with pole channels; valuation norm equals the inverse Poisson metric times the completed squared amplitude	Equations (2.5) and (2.10) retain signed normalization and reflected derivatives
Moving metric and Möbius completion	Positive finite gcd energies converge to the actual completed amplitude in $\operatorname{Re} s > 1/2$	Equation (4.11) isolates the exact remaining signed derivative

Route	Exact result	Derivative status
Compatibility-based square completion	Multiplicative Mellin relations and an explicit rank-one Hermitian defect	The extracted derivative is supported on a two-dimensional space and is indefinite without compensation

The most concrete next connection to test is whether the augmented pole/modular contribution from Sections 2 and 4 compensates the specific mixed-color derivative in (5.5), while retaining every other term of the authentic source. A valid payment for that block would settle that block; an exact decomposition of the full source would still be required for H.

No identity of the form $Q = B^*B + R^*A + A^*R$ for the full target has been established here. The calculations provide explicit candidates for its compatibility and compensation terms, with their signs and domains intact.

Source continuity

The following current source notes were checked for target normalization and previously established prime-color identities:

- *Michaels Prime Color Dual Kernel Bridge*: local prime factors, Jordan/divisor synthesis, phase restriction, and signed completed source.
- *Michaels Two Plane Prime Theta Intertwiner Attempt*, Sections 10-14: the authentic theta target and scope of the proposed metric identification.
- *Michaels Compact Prime Theta Transfer*: complete normalization derivative, product Poisson metric, and the distinction between a positive amplitude and its signed horizontal derivative.

Sections 2-5 above supply the derivations used in these route tests. No numerical sign evidence, assumed orthogonality, or unproved positivity of the full spatial kernel is used.

Compact prime-theta transfer

Source: *Michaels_Compact_Prime_Theta_Transfer.md*. Equation tags carry the source prefix H6; references within this unit retain their original local numbering.

Christophe Michaels research continuation – 21 September 2026

Result and scope

This continuation connects the two-prime color kernel to the earlier interior comparison identity through an explicit positive smoothing operator. The construction retains the actual completed theta function. It does not replace that function by a model or use numerical sign tests.

Three analytic conclusions are obtained:

1. The positive Jordan derivative hierarchy acquires an explicit signed correction under the normalization defining the valuation kernel.
2. The actual theta form has a two-prime Poisson regularization with a strictly positive, explicitly proved starting range.
3. The regularization places the interior comparison on a compact two-angle cylinder. The height boundaries disappear, the whole remaining boundary is nonnegative, and the phase average of its logarithmic slope is nonnegative.

The exact remaining signed term is retained below. Neither the starting bound nor the non-negative phase average proves pointwise positivity as smoothing is removed. This is a verified continuation, not a proof of H.

1. Normalization is a real mathematical operation

For positive integers m, n , let

$$G_z(m, n) = \gcd(m, n)^z, \quad J_z(d) = d^z \prod_{p|d} (1 - p^{-z}). \quad (\text{H6:1.1})$$

The divisor identity gives

$$G_z(m, n) = \sum_{d|m, n} J_z(d).$$

For real $z > 0$, the derivatives $J_z^{(k)}(d)$ are nonnegative for every integer $k \geq 0$. Indeed, on a prime power,

$$J_z(p^a) = e^{(a-1)z \log p} (e^{z \log p} - 1),$$

whose derivatives are nonnegative, and the same follows for products. Thus, on every finite integer index set,

$$\frac{d}{dz} G_z \succeq 0. \quad (\text{H6:1.2})$$

The valuation kernel used in the color bridge is instead

$$K_z = D_z G_z D_z, \quad D_z = \text{diag}(n^{-z/2}).$$

Writing $L = \text{diag}(\log n)$, differentiation gives

$$\boxed{K'_z = D_z G'_z D_z - \frac{1}{2}(LK_z + K_z L).} \quad (\text{H6:1.3})$$

The second term is the normalization correction. It cannot be removed by invoking the positive Jordan coefficients.

For two distinct integers, set

$$q = \frac{\gcd(m, n)}{\sqrt{mn}} \in (0, 1).$$

The corresponding derivative block is

$$\begin{pmatrix} 0 & q^z \log q \\ q^z \log q & 0 \end{pmatrix}, \quad (\text{H6:1.4})$$

which has eigenvalues of opposite signs. Consequently the normalized valuation family is not increasing in positive-semidefinite order. This does not contradict positivity of each K_z , nor the positive mixed-color decomposition.

This identifies one precise connection to the theta problem: H asks for the sign of a derivative of a positive family. Positivity before a parameter-dependent normalization or differentiation is not that sign. Any proposed prime-to-theta transport must retain its derivative terms.

2. The actual theta measure and the two prime angles

Use the inherited normalization

$$F(z) = \xi\left(\frac{1}{2} + z\right) = \int_{\mathbb{R}} \eta(u) e^{zu} du, \quad K(a, t) = \text{Re}(F'(a + it) \overline{F(a + it)}), \quad (\text{H6:2.1})$$

where η is the complete positive even theta density. Put

$$C_a(y) = \frac{1}{\pi} \int_{\mathbb{R}} K(a, t) e^{-2ity} dt. \quad (\text{H6:2.2})$$

For each fixed $a > 0$, the theta vertical-decay estimates give $K(a, \cdot) \in L^1(\mathbb{R}) \cap C(\mathbb{R})$. The theta estimates already established in the source records imply

$$0 < C_a(y) \leq C_a(0) \quad (a > 0, y \in \mathbb{R}). \quad (\text{H6:2.3})$$

For example, evenness and log-concavity give $\eta(u + y)\eta(u - y) \leq \eta(u)^2$; multiplication by the nonnegative weight $2u \sinh(2au)$ proves the upper bound in the spatial formula.

Fix exactly the two primes

$$P = \{2, 3\}, \quad \omega = (\log 2, \log 3), \quad L_P(\nu) = \frac{1}{2} \omega \cdot \nu.$$

All torus integrals below use normalized Haar measure on $\mathbb{T}^2 = (\mathbb{R}/2\pi\mathbb{Z})^2$. Let

$$\mu_a = (t \mapsto \omega t \bmod 2\pi)_* \left(\frac{1}{\pi} K(a, t) dt \right). \quad (\text{H6:2.4})$$

This agrees with the earlier negative-phase convention because $K(a, t)$ is even in t . Its Fourier coefficients, with $\hat{\mu}(\nu) = \int e^{-i\nu \cdot \theta} d\mu(\theta)$, are

$$\hat{\mu}_a(\nu) = C_a(L_P(\nu)). \quad (\text{H6:2.5})$$

The map $\mathbb{R} \rightarrow \mathbb{T}^2$, $t \mapsto \omega t \bmod 2\pi$, is injective: a nonzero common period would make $\log 2 / \log 3$ rational. Its image is dense and Haar-null. It follows, either by injectivity of this Borel embedding or by the already established dense-node argument, that

$$\mu_a \geq 0 \iff K(a, t) \geq 0 \text{ for every } t. \quad (\text{H6:2.6})$$

H requires this for every $a > 0$.

3. An exact regularization using the existing prime kernel

For a smoothing parameter $s > 0$, independent of the theta displacement a , define

$$G_s(\theta) = \prod_{p \in \{2,3\}} \frac{1 - p^{-2s}}{1 - 2p^{-s} \cos \theta_p + p^{-2s}}, \quad c_s = \prod_{p \in \{2,3\}} \frac{1 - p^{-s}}{1 + p^{-s}}. \quad (\text{H6:3.1})$$

This is precisely the torus multiplier of the positive valuation kernel. It obeys $G_s \geq c_s > 0$, $\int G_s = 1$, and has Fourier coefficients

$$\hat{G}_s(\nu) = e^{-sq(\nu)}, \quad q(\nu) = \sum_{p \in \{2,3\}} |\nu_p| \log p. \quad (\text{H6:3.2})$$

Convolve the complete signed theta measure with this kernel:

$$\rho_s(a, \theta) = (G_s * \mu_a)(\theta) = \frac{1}{\pi} \int_{\mathbb{R}} G_s(\theta - \omega t) K(a, t) dt. \quad (\text{H6:3.3})$$

Equivalently,

$$\rho_s(a, \theta) = \sum_{\nu \in \mathbb{Z}^2} e^{-sq(\nu)} C_a(L_P(\nu)) e^{i\nu \cdot \theta}. \quad (\text{H6:3.4})$$

The series and every fixed angular derivative converge absolutely for $s > 0$. The bound (2.3) supplies a product of convergent geometric series, with polynomial factors for derivatives.

For any finite exponent array, the associated kernel has entries

$$e^{-sq(\nu - \kappa)} C_a(L_P(\nu - \kappa)). \quad (\text{H6:3.5})$$

Thus this is an exact entrywise product of the valuation and theta kernels, not an asserted equality between those two kernels.

The product Poisson kernels are an approximate identity. Hence

$$K(a, \cdot) \geq 0 \iff \rho_s(a, \theta) \geq 0 \text{ for every } s > 0, \theta \in \mathbb{T}^2. \quad (\text{H6:3.6})$$

For the reverse implication it suffices to use any sequence $s_j \downarrow 0$: the nonnegative measures $\rho_{s_j} dm$ converge weakly to μ_a . Testing against continuous nonnegative functions proves $\mu_a \geq 0$.

The singularity of the orbit measure does not invalidate this route. Positive absolutely continuous approximations can converge to a singular positive measure. The earlier singularity argument ruled out a uniform coordinate-preserving bounded comparison, not this limiting construction.

4. A proved positive starting range

Separate the zero Fourier coefficient in (3.4) and use (2.3):

$$\rho_s(a, \theta) \geq C_a(0) \left[2 - \prod_{p \in \{2,3\}} \frac{1+p^{-s}}{1-p^{-s}} \right]. \quad (\text{H6:4.1})$$

At the exact value $s = 3$,

$$\prod_{p \in \{2,3\}} \frac{1+p^{-3}}{1-p^{-3}} = \frac{9}{7} \frac{14}{13} = \frac{18}{13}.$$

Therefore

$$\boxed{\rho_s(a, \theta) \geq \frac{8}{13} C_a(0) > 0 \quad (s \geq 3, a > 0, \theta \in \mathbb{T}^2).} \quad (\text{H6:4.2})$$

This is an analytic bound at every angle and every positive a . No finite sampling or numerical sign assertion enters its proof.

It does not imply positivity as $s \downarrow 0$. Indeed, if $\mathcal{A} = \sum_p (\log p) |\partial_{\theta_p}|$, with Fourier multiplier $q(\nu)$, then

$$\partial_s \rho_s = -\mathcal{A} \rho_s. \quad (\text{H6:4.3})$$

The Poisson semigroup preserves positivity toward larger s , whereas the desired limiting direction is toward smaller s .

5. The additive colors give the exact reverse-direction test

Let

$$D_s(\theta) = \sum_{p \in \{2,3\}} (\log p) \left[\frac{1-p^{-2s}}{1-2p^{-s} \cos \theta_p + p^{-2s}} - \frac{1-p^{-s}}{1+p^{-s}} \right] \geq 0. \quad (\text{H6:5.1})$$

This is precisely the additive color multiplier from the previous note. Direct differentiation of (3.1) gives

$$\partial_s \log G_s = \beta_s - D_s, \quad \beta_s = \partial_s \log c_s = \sum_p \frac{2(\log p) p^{-s}}{1-p^{-2s}}. \quad (\text{H6:5.2})$$

Since μ_a is independent of s ,

$$\boxed{\partial_s \left(\frac{\rho_s(a, \theta)}{c_s} \right) = -\frac{\mathcal{P}_s(a, \theta)}{c_s}, \quad \mathcal{P}_s = (D_s G_s) * \mu_a.} \quad (\text{H6:5.3})$$

In the original scalar variable this is

$$\mathcal{P}_s(a, \theta) = \frac{1}{\pi} \int_{\mathbb{R}} D_s(\theta - \omega t) G_s(\theta - \omega t) K(a, t) dt. \quad (\text{H6:5.4})$$

The exact Fourier check is

$$\widehat{D_s G_s}(\nu) = (\beta_s + q(\nu)) e^{-sq(\nu)}. \quad (\text{H6:5.5})$$

A proof of $\mathcal{P}_s \geq 0$ for $0 < s \leq 3$, all θ , and all $a > 0$, would propagate (4.2) in the needed direction:

$$\frac{\rho_s}{c_s} = \frac{\rho_3}{c_3} + \int_s^3 \frac{\mathcal{P}_v}{c_v} dv > 0. \quad (\text{H6:5.6})$$

This is a sufficient sign condition involving both recovered prime routes and the actual completed theta weight. It is not supplied by $D_s \geq 0$ or $G_s > 0$ alone: the measure they integrate against is still signed. Requiring it at every parameter is itself equivalent to H , since H also makes (5.4) nonnegative. Its value is the explicit expression of the needed interaction, not a reduction in logical strength.

6. Compact transfer of the interior comparison

Define a positive regularized amplitude, using the same color kernel:

$$E_s(a, \theta) = \frac{1}{2\pi} \int_{\mathbb{R}} G_s(\theta - \omega t) |F(a + it)|^2 dt > 0. \quad (\text{H6:6.1})$$

It satisfies

$$\partial_a E_s = \rho_s. \quad (\text{H6:6.2})$$

For fixed $s > 0$, positivity follows from $G_s \geq c_s > 0$ and the fact that a nonzero entire function cannot vanish on a whole vertical line. All required derivatives are justified by the complete theta vertical-decay estimates, locally uniformly in a .

Set

$$\mathcal{D} = \omega \cdot \nabla_{\theta}, \quad \Delta_{\omega} = \partial_a^2 + \mathcal{D}^2. \quad (\text{H6:6.3})$$

Changing variables in the full integral gives

$$E_s(a, \theta + \omega\tau) = \frac{1}{2\pi} \int_{\mathbb{R}} G_s(\theta - \omega t) |F(a + i(t + \tau))|^2 dt. \quad (\text{H6:6.4})$$

For fixed θ the right side is the squared norm of a holomorphic Hilbert-space-valued function of $a + i\tau$.

Use the inner product with measure $G_s(\theta - \omega t) dt/(2\pi)$, conjugate-linear in its first slot, and put

$$\Phi(t) = F(a + it), \quad E_1 = \|\Phi'\|^2, \quad H = \langle \Phi, \Phi' \rangle, \quad E = E_s.$$

Holomorphic differentiation gives

$$\Delta_{\omega} E = 4E_1, \quad (\partial_a E)^2 + (\mathcal{D}E)^2 = 4|H|^2.$$

Therefore

$$\delta_s := \Delta_{\omega} \log E_s = \frac{4(E E_1 - |H|^2)}{E^2} \geq 0. \quad (\text{H6:6.5})$$

In fact $\delta_s > 0$. Equality in Cauchy–Schwarz would imply $F'(a + it) = hF(a + it)$ almost everywhere on a vertical line, since the weight is strictly positive. The identity theorem would give $F' = hF$ everywhere. The nonconstant even function F cannot have this form.

For the normalized slope

$$u_s = \frac{\rho_s}{2E_s} = \frac{1}{2} \partial_a \log E_s$$

one has the exact equation

$$\Delta_{\omega} u_s = \frac{1}{2} \partial_a \delta_s. \quad (\text{H6:6.6})$$

The operator Δ_{ω} is degenerate in the two angular coordinates; no uniformly elliptic torus maximum principle is being assumed. The following energy argument only uses integration by parts and the ordinary Dirichlet inequality in the a coordinate.

7. The whole cylinder boundary is controlled

Take

$$\mathcal{C} = (0, 1/2) \times \mathbb{T}^2.$$

At $a = 0$, the reality and evenness of F give $K(0, t) = 0$, so

$$u_s(0, \theta) = 0.$$

The previously established exterior theorem $K(a, t) \geq 0$ for $a \geq 1/2$ gives

$$u_s(1/2, \theta) \geq 0. \quad (\text{H6:7.1})$$

There are no angular boundary faces because the angles are periodic. Thus this version needs no large-height rectangle, tail boundary expansion, or hypothetical negative eigenfunction.

Put $n_s = \max(-u_s, 0)$. It has zero trace on both a -faces. Test (6.6) against n_s and integrate over the cylinder:

$$\boxed{\int_{\mathcal{C}} (|\partial_a n_s|^2 + |\mathcal{D}n_s|^2) = \frac{1}{2} \int_{\mathcal{C}} (\partial_a \delta_s) n_s.} \quad (\text{H6:7.2})$$

All angular integrals use normalized Haar measure. The identity extends to the Lipschitz negative part by the Sobolev chain rule. The one-dimensional Dirichlet inequality on an interval of length $1/2$ gives

$$\int_{\mathcal{C}} (|\partial_a n_s|^2 + |\mathcal{D}n_s|^2) \geq 4\pi^2 \int_{\mathcal{C}} n_s^2. \quad (\text{H6:7.3})$$

Consequently an independently proved estimate

$$\frac{1}{2} \int_{\mathcal{C}} (\partial_a \delta_s) n_s \leq \kappa_s \int_{\mathcal{C}} n_s^2, \quad \kappa_s < 4\pi^2, \quad (\text{H6:7.4})$$

would force $n_s = 0$. Establishing (7.4) for any sequence $s_j \downarrow 0$, together with the inherited exterior theorem, would prove H. No uniform gap from $4\pi^2$ is required: it suffices that, for every j , (7.4) holds with some $\kappa_{s_j} < 4\pi^2$. The estimate (7.4) is a proposed sufficient estimate, not a proved bound.

The zero bound $\int (\partial_a \delta_s) n_s \leq 0$ is an exact equivalent sign criterion in view of (7.2). This observation is useful for specifying the source; it does not establish its sign.

8. A proved phase-average law and the source it does not remove

For fixed $s > 0$, let

$$m_s(a) = \int_{\mathbb{T}^2} \log E_s(a, \theta) dm(\theta).$$

Periodicity removes the angular second derivative, so

$$m_s''(a) = \int_{\mathbb{T}^2} \delta_s(a, \theta) dm(\theta) \geq 0. \quad (\text{H6:8.1})$$

Since $m'_s(0) = 0$,

$$\boxed{\int_{\mathbb{T}^2} u_s(a, \theta) dm(\theta) > 0 \quad (a > 0).} \quad (\text{H6:8.2})$$

This is a phase-average logarithmic-slope theorem. It uses the actual theta amplitude but does not use H. A nonnegative average can coexist with negative values at some angles.

To retain the missing source explicitly, put

$$h = \frac{H}{E}, \quad r = \Phi' - h\Phi, \quad \delta_s = \frac{4\|r\|^2}{E}.$$

Differentiation at fixed s, θ gives

$$\partial_a \delta_s = \frac{8 \operatorname{Re} \langle r, \Phi'' \rangle}{E} - 4u_s \delta_s. \quad (\text{H6:8.3})$$

All terms are written in entire derivatives, so no division by F at a zero has been introduced. Splitting this into separate expressions in F'/F can create nonintegrable terms at a zero; (8.3) avoids that issue.

On the negative set $u_s = -n_s$, (7.2) therefore becomes

$$\int_{\mathcal{C}} (|\partial_a n_s|^2 + |\mathcal{D} n_s|^2 - 2\delta_s n_s^2) = 4 \int_{\mathcal{C}} \frac{\operatorname{Re} \langle r, \Phi'' \rangle}{E} n_s. \quad (\text{H6:8.4})$$

Nonnegativity of the variance δ_s does not determine the sign of the centered correlation on the right. Nor does torus averaging cancel it: the negative-part weight depends on the angles.

9. Where the completed arithmetic source can enter

The recovered fixed-line prime-phase theorem is a separate exact connection to the authentic J , not a theorem for an unrelated positive kernel. For one fixed $b > 1/2$, write

$$f(r) = \sum_j d_j e^{-(b+it_j)r}, \quad R_f(v) = \int_0^\infty f(r+v) \overline{f(r)} dr, \quad \rho_f = \operatorname{Re} R_f, \quad E_f = \|f\|_2^2.$$

The complete form, with absolutely convergent prime-power series on this packet class, is

$$\mathfrak{Q}_b[f] = \mathfrak{A}[f] - 2 \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \rho_f(\log n). \quad (\text{H6:9.1})$$

The source proves that its positivity for all packets is equivalent to positivity of the authentic spatial kernel and to H. It retains the unshifted denominator $z + \bar{w}$ in the corresponding kernel.

The earlier prime-counting work connects at the following exact paired identity. Let

$$D(v) = \sum_{n \leq e^v} \Lambda(n) - (e^v - 1), \quad T_f(v) = 2E_f - 2\rho_f(v) \geq 0.$$

Then the source gives

$$\begin{aligned} \mathfrak{Q}_b[f] &= [\psi(1/4) - \log \pi] E_f + \int_0^\infty \frac{e^{-v/2}}{1 - e^{-2v}} T_f(v) dv \\ &\quad + 2 \int_0^\infty e^{-v/2} \rho_f(v) dv \\ &\quad + 2 \int_0^\infty e^{-v/2} D(v) \left(\rho'_f(v) - \frac{1}{2} \rho_f(v) \right) dv. \end{aligned} \quad (\text{H6:9.2})$$

The pole growth and complete prime powers have already been paired here. This is the precise arithmetic expression to which a color cancellation law must be applied. The factor $\rho'_f - \rho_f/2$ is signed; an unsigned prime-counting estimate does not determine the last integral's sign.

No identity equating the centered correlation in (8.3) to a nonnegative prime expression has been supplied by these calculations. Such a link, with its domains and normalization intact, would be proof-bearing. The positive successor theorem in the recovered Suzuki metric supplies all-coefficient propagation in that metric, but its equality with this completed form remains a separate identity in the inspected source.

There is an exact derivative check on the arithmetic side as well. In the Euler-convergent region $a > 1/2$, write

$$g = \frac{F'}{F} = A - P, \quad P(z) = \sum_{n \geq 2} \Lambda(n) n^{-1/2-z}, \quad Q(z) = \sum_{n \geq 2} \Lambda(n) \log n n^{-1/2-z},$$

where A is the complete rational and gamma contribution. Then

$$\boxed{\frac{F''}{F} = A' + A^2 + Q - 2AP + P^2.} \quad (\text{H6:9.3})$$

Thus even a centered second-derivative comparison must retain the mixed term $-2AP$. The Euler series in (9.3) are not asserted to converge in the interior strip used in Section 7. A proof using this identity there would need a justified completed continuation, while the fixed-line criterion (9.1)-(9.2) stays in its stated convergent domain.

10. The next proof-bearing step

The two routes are now connected to actual theta quantities through (3.3), (5.3), and (6.1), not only compared abstractly.

There are two explicit places for an additional established law to act:

- Control the signed, completed color interaction $\mathcal{P}_s$ in (5.4) so that the proved floor propagates toward vanishing smoothing.
- Control the complete centered derivative source in (8.3), sufficiently to obtain the absorption estimate (7.4) on the compact cylinder.

The first uses both prime kernels directly; the second also uses the previous interior comparison theorem. Neither requires checking individual coefficient combinations numerically. Both still require a theta-specific sign argument. The phase-average result (8.2) and the starting floor (4.2) are actual consequences proved here, but do not replace that argument.

Sources used

1. *Michaels Prime Color Dual Kernel Bridge*, sections 2-5 and 9-13: the local color factorization, additive/tensor identity, and orbit measure. Current file: `Michaels_Prime_Color_Dual_Kernel_Bridge.md`.
2. *Michaels Interior-to-Spectral Transfer*, MP1-MP13: positive averaged amplitude, complete boundary comparison, Gram curvature, and centered derivative source. Current file: `Michaels_Interior_to_Spectral_Transfer.md`.

3. *Michaels H Analytical Review*, sections 1-2 and 6: normalization, two-prime density reduction, and the bound on the complete C_a . Its structural audit also distinguishes positive lifts from their signed derivatives. Current file: `Michaels_H_Analytical_Review.md`.
4. *Michaels Prime Phase J Bridge*, equations (9)-(11), (18)-(21): the authentic fixed-line completed form, pole/prime pairing, and exact scope of the recovered prime-anchor successor theorem. Current file: `Michaels_Prime_Phase_J_Bridge.md`, version 3.

The construction and all displayed new identities are analytic. No numerical evidence is promoted to an all-parameter theorem.

Prime colors and the dual kernel

Source: *Michaels_Prime_Color_Dual_Kernel_Bridge.md*. Equation tags carry the source prefix *H7*; references within this unit retain their original local numbering.

Local color ladders, Jordan cumulants, mixed-color positivity, and the scalar theta orbit

Finding

The two prime routes are exact parts of one color system.

For a finite set of primes P , a parameter $\sigma > 0$, and

$$r_p = p^{-\sigma}, \quad c_p = \frac{1 - r_p}{1 + r_p},$$

let $G_{p,\sigma}$ be the one-prime valuation covariance and put

$$A_{p,\sigma} := G_{p,\sigma} - c_p I = X_{p,\sigma}^* X_{p,\sigma} \succeq 0. \quad (\text{H7:0.1})$$

The source-faithful additive operator and the tensor valuation operator are

$$\boxed{\mathcal{K}_{P,\sigma}^{\log} = \sum_{p \in P} (\log p) A_{p,\sigma}^{(p)}} \quad (\text{H7:0.2})$$

and

$$\boxed{\begin{aligned} \mathcal{K}_{P,\sigma}^{\text{val}} &= \bigotimes_{p \in P} (c_p I + A_{p,\sigma}) \\ &= \sum_{S \subseteq P} \left(\prod_{q \notin S} c_q \right) X_S^* X_S \succeq \left(\prod_{p \in P} c_p \right) I. \end{aligned}} \quad (\text{H7:0.3})$$

Here

$$X_S := \bigotimes_{p \in S} X_{p,\sigma} \otimes \bigotimes_{q \notin S} I_q, \quad X_\emptyset = I.$$

Route A retains the positive one-color sectors. Route B retains the empty-color floor, those same singleton sectors, and every positive mixed-color sector. This is the precise form of the proposal that the primes themselves are the floor.

The two kernels also satisfy the exact reconstruction

$$\boxed{K_{P,\sigma}^{\text{val}} = \left(\prod_{p \in P} c_p \right) \exp_{*\Gamma_P} \left(\int_\sigma^\infty K_{P,u}^{\log} du \right)}, \quad (\text{H7:0.4})$$

where $\exp_{*\Gamma_P}$ is convolution exponentiation on the exponent lattice $\Gamma_P = \mathbb{Z}^P$. Scale integration changes the von Mangoldt coefficient of a prime power into its Euler-log coefficient; convolution exponentiation creates every mixed color.

The Jordan deformation gives the arithmetic version of the same law:

$$J_z = \exp_{*\mathbb{D}} \left(\int_0^z n^u \Lambda(n) du \right). \quad (\text{H7:0.5})$$

Its connected coefficients are supported exactly on prime powers. Their Dirichlet-convolution exponential is the complete mixed-color Jordan family.

Locking the independent prime angles by $z_p = e^{-it \log p}$ supplies the exact scalar indexing orbit. The theta kernel enters only after that orbit is weighted by the separately supplied completed signed density $K(a, t) dt$. This operation is singular, and the resulting theta form is not a uniformly bounded coordinate-preserving feature compression of either positive prime Gram. Route B does, however, furnish a positive Poisson approximate identity on the color torus, while Route A is its logarithmic generator. The exact remaining criterion is positivity of the coupled theta–Poisson densities in (14.12) as the regularization is removed.

1. The recovered prime-color dictionary

The invariant color coordinate of an integer is its exponent vector

$$\nu(n) = (\nu_p(n))_p, \quad n = \prod_p p^{\nu_p(n)}. \quad (\text{H7:1.1})$$

Multiplication adds color vectors, while divisibility is coordinatewise ancestry. The quantities

$$\omega(n) = \#\{p : \nu_p(n) > 0\}, \quad \Omega(n) = \sum_p \nu_p(n) \quad (\text{H7:1.2})$$

record distinct color count and total multiplicity. The nested squarefree layers

$$R_k(n) = \prod_{\nu_p(n) \geq k} p \quad (\text{H7:1.3})$$

obey

$$n = \prod_{k \geq 1} R_k(n), \quad R_{k+1}(n) \mid R_k(n), \quad \log n = \sum_{k \geq 1} \log R_k(n). \quad (\text{H7:1.4})$$

The source roles are:

- a prime-power color p^a carries the first-order coefficient $\Lambda(p^a) = \log p$;
- a squarefree color carries Möbius parity $\mu(D) = (-1)^{\omega(D)}$;
- repeated and mixed colors return through higher Jordan derivatives, divisor tensors, finite boundary states, and cross terms.

The early diagram labels Q, G, P, Y, L, M encode this factor-depth and exponent-multiplicity grading. The grading is the mathematical invariant.

2. The complete one-prime color ladder

On the bilateral exponent line $\mathbb{Z}$, define

$$G_{p,\sigma}(a, b) = r_p^{|a-b|}. \quad (\text{H7:2.1})$$

Its Fourier multiplier is the Poisson kernel

$$P_{r_p}(\theta) = \sum_{k \in \mathbb{Z}} r_p^{|k|} e^{ik\theta} = \frac{1 - r_p^2}{1 - 2r_p \cos \theta + r_p^2}. \quad (\text{H7:2.2})$$

Since

$$\min_{\theta} P_{r_p}(\theta) = P_{r_p}(\pi) = \frac{1 - r_p}{1 + r_p} = c_p, \quad (\text{H7:2.3})$$

one has

$$G_{p,\sigma} = c_p I + A_{p,\sigma}, \quad A_{p,\sigma} \succeq 0. \quad (\text{H7:2.4})$$

The rational two-plane construction identifies this defect:

$$X_{p,\sigma} = \sqrt{\frac{r_p(1 - r_p)}{1 + r_p}} (I + U_p)(I - r_p U_p)^{-1}, \quad A_{p,\sigma} = X_{p,\sigma}^* X_{p,\sigma}. \quad (\text{H7:2.5})$$

The full valuation-depth spectral factor is

$$\begin{aligned} B_{p,\sigma}(z) &= \frac{\sqrt{1 - r_p^2}}{1 - r_p z} \\ &= \sum_{a \geq 0} \sqrt{(1 - r_p^2) r_p^{2a}} z^a, \quad |B_{p,\sigma}(e^{i\theta})|^2 = P_{r_p}(\theta). \end{aligned} \quad (\text{H7:2.6})$$

At $\sigma = 1/2$, the coefficients in (2.6) have squared moduli

$$b_{p,0}^2 = 1 - p^{-1}, \quad b_{p,a}^2 = e_{p,a} := (1 - p^{-1})p^{-a} \quad (a \geq 1). \quad (\text{H7:2.7})$$

The depth-zero coefficient is the unselected, or vacuum, mass. The archived color-energy ladder $e_{p,a}$ begins at depth $a = 1$.

For the visible and hidden two-plane filters

$$V_p(z) = \frac{1 - r_p z}{1 + r_p}, \quad W_p(z) = \frac{\sqrt{r_p}(1 + z)}{1 + r_p}, \quad (\text{H7:2.8})$$

set, in operator notation,

$$B_{p,\sigma} = \sqrt{1 - r_p^2} (I - r_p U_p)^{-1};$$

its circle symbol is (2.6). The color ladder satisfies

$$\boxed{V_p B_{p,\sigma} = \sqrt{c_p} I, \quad W_p B_{p,\sigma} = X_{p,\sigma}.} \quad (\text{H7:2.9})$$

Thus

$$\boxed{G_{p,\sigma} = c_p I + X_{p,\sigma}^* X_{p,\sigma}} \quad (\text{H7:2.10})$$

is the two-plane image of the complete prime-color ladder. The visible output is the constant floor. The hidden output contains every power of the prime through the resolvent.

There is a complementary finite-period arithmetic realization. Fix a finite depth A , choose any period Q divisible by p^A , and define

$$\langle f, g \rangle_Q := \frac{1}{Q} \sum_{r \bmod Q} f(r) \overline{g(r)}.$$

Then

$$h_{p,a}(r) = \mathbf{1}_{p^a|r} - \frac{1}{p} \mathbf{1}_{p^{a-1}|r}, \quad a \geq 1, \quad (\text{H7:2.11})$$

and, for $1 \leq a, b \leq A$,

$$\langle h_{p,a}, h_{p,b} \rangle_Q = \delta_{ab} (1 - p^{-1}) p^{-a}. \quad (\text{H7:2.12})$$

Hence $e_{p,a}$ is simultaneously a Haar shell energy, a Ramanujan-color norm, and a squared spectral-factor coefficient.

3. Route A: the source-faithful additive kernel

Let $\Gamma_P = \mathbb{Z}^P$, and embed A_p in the p -coordinate with identities in all other coordinates. Define

$$\mathcal{K}_{P,\sigma}^{\log} = \sum_{p \in P} (\log p) A_{p,\sigma}^{(p)}. \quad (\text{H7:3.1})$$

Its convolution kernel is

$$K_{P,\sigma}^{\log}(\nu, \mu) = \begin{cases} \delta_{P,\sigma} := \sum_{p \in P} \frac{2(\log p) r_p}{1 + r_p}, & \nu = \mu, \\ (\log p) r_p^{|k|}, & \nu - \mu = k e_p, \quad k \neq 0, \\ 0, & \nu - \mu \text{ has at least two nonzero coordinates.} \end{cases} \quad (\text{H7:3.2})$$

Its torus multiplier is

$$D_{P,\sigma}(\boldsymbol{\theta}) = \sum_{p \in P} (\log p) (P_{r_p}(\theta_p) - c_p) \geq 0. \quad (\text{H7:3.3})$$

This multiplier vanishes only at $(\pi, \dots, \pi)$. Therefore every finite matrix on distinct exponent vectors is strictly positive definite: a nonzero trigonometric polynomial cannot vanish almost everywhere.

For

$$R(\nu - \mu) = \prod_{p \in P} p^{|\nu_p - \mu_p|}, \quad (\text{H7:3.4})$$

the off-diagonal coefficient is

$$K_{P,\sigma}^{\log}(\nu, \mu) = \Lambda(R(\nu - \mu))R(\nu - \mu)^{-\sigma}, \quad \nu \neq \mu. \quad (\text{H7:3.5})$$

Indeed, $\Lambda(R)$ vanishes unless exactly one prime coordinate changes. When $R = p^k$, it equals $\log p$. Route A is therefore both positive for arbitrary finite coefficient arrays and faithful to the complete prime-power source.

The corresponding signed algebraic deformation, in the doubled real-part normalization, is

$$\mathfrak{W}_{p,\sigma} = \frac{2(\log p)r_p}{1 + r_p} I - (\log p)A_{p,\sigma}. \quad (\text{H7:3.6})$$

At the critical half-density $\sigma = 1/2$, this is the critical-half-density algebraic prime summand. It is not, by itself, an Euler-convergent completed prime sum. For general $\sigma > 0$, it is the corresponding algebraic deformation. Thus the color norm is positive in its own space, while its occurrence in the completed scalar source is subtractive. The pole, gamma, scalar-floor, and prime terms must be paired before an infinite critical-line limit is taken.

4. Route B: the tensor valuation kernel

Define

$$K_{P,\sigma}^{\text{val}}(\nu, \mu) = \prod_{p \in P} r_p^{|\nu_p - \mu_p|}, \quad \mathcal{K}_{P,\sigma}^{\text{val}} = \bigotimes_{p \in P} G_{p,\sigma}. \quad (\text{H7:4.1})$$

Expanding each factor by (2.10) gives

$$\mathcal{K}_{P,\sigma}^{\text{val}} = \sum_{S \subseteq P} \left(\prod_{q \notin S} c_q \right) A_S, \quad (\text{H7:4.2})$$

where

$$A_S = \bigotimes_{p \in S} A_p \otimes \bigotimes_{q \notin S} I_q = X_S^* X_S. \quad (\text{H7:4.3})$$

Every term is positive semidefinite for arbitrary complex coefficients. The color grading is explicit:

- $S = \emptyset$ is the strict floor $\left(\prod_p c_p \right) I$;

- $|S| = 1$ gives the one-color blocks used by Route A;
- $|S| \geq 2$ gives all positive mixed-color planes.

Consequently

$$\mathcal{K}_{P,\sigma}^{\text{val}} \succeq \left(\prod_{p \in P} \frac{1 - p^{-\sigma}}{1 + p^{-\sigma}} \right) I. \quad (\text{H7:4.4})$$

This is a finite-prime floor, sufficient for every finite coefficient packet. The product need not stay bounded away from zero as infinitely many colors are adjoined.

5. The exact bridge between the two routes

5.1 Color-generating differential

Set

$$\mathcal{T}_P(\mathbf{s}) = \bigotimes_{p \in P} (c_p I + s_p A_p). \quad (\text{H7:5.1})$$

Then

$$\partial_{s_S} \mathcal{T}_P(\mathbf{0}) = \left(\prod_{q \notin S} c_q \right) A_S, \quad \mathcal{T}_P(\mathbf{1}) = \mathcal{K}_{P,\sigma}^{\text{val}}, \quad (\text{H7:5.2})$$

where $\partial_{s_S} := \prod_{p \in S} \partial_{s_p}$.
and

$$\mathcal{K}_{P,\sigma}^{\log} = \sum_{p \in P} \frac{\log p}{\prod_{q \neq p} c_q} \partial_{s_p} \mathcal{T}_P(\mathbf{s})|_{\mathbf{s}=\mathbf{0}}. \quad (\text{H7:5.3})$$

Route A is the weighted first-color differential of the positive tensor generator. Route B is its full mixed-color value.

5.2 Logarithmic derivative and convolution exponential

Let

$$V_{P,\sigma} = \prod_{p \in P} \frac{I - r_p U_p}{1 + r_p}, \quad c_{P,\sigma} = \prod_{p \in P} c_p. \quad (\text{H7:5.4})$$

The identity $V_p^* V_p = c_p G_p^{-1}$ gives

$$V_{P,\sigma}^* V_{P,\sigma} = c_{P,\sigma} (\mathcal{K}_{P,\sigma}^{\text{val}})^{-1}. \quad (\text{H7:5.5})$$

Since all factors are commuting Fourier multipliers,

$$\boxed{\mathcal{K}_{P,\sigma}^{\log} = \partial_{\sigma} \log (V_{P,\sigma}^* V_{P,\sigma}) = -\partial_{\sigma} \log (c_{P,\sigma}^{-1} \mathcal{K}_{P,\sigma}^{\text{val}}).} \quad (\text{H7:5.6})$$

Integrating from σ to infinity yields

$$\boxed{c_{P,\sigma}^{-1} \mathcal{K}_{P,\sigma}^{\text{val}} = \exp \left(\int_{\sigma}^{\infty} \mathcal{K}_{P,u}^{\log} du \right).} \quad (\text{H7:5.7})$$

On kernels, this is the group-convolution exponential. If $H_{P,\sigma} := \int_{\sigma}^{\infty} \mathcal{K}_{P,u}^{\log} du$, then

$$H_{P,\sigma}(0) = 2 \sum_{p \in P} \log(1 + r_p), \quad H_{P,\sigma}(\pm k e_p) = \frac{r_p^k}{k}, \quad k \geq 1. \quad (\text{H7:5.8})$$

The identity

$$\int_{\sigma}^{\infty} (\log p) p^{-ku} du = \frac{p^{-k\sigma}}{k} \quad (\text{H7:5.9})$$

is exactly the passage from a von Mangoldt coefficient to an Euler-log coefficient. Convolution powers then combine one-axis moves into every mixed-color displacement.

Equation (5.7) holds on the full group before finite compression. It is not an entrywise exponential and is generally not the matrix exponential of an arbitrarily compressed finite Gram matrix.

6. The Jordan cumulant identity

Define

$$J_z(n) = \sum_{d|n} \mu(d) (n/d)^z = n^z \prod_{p|n} (1 - p^{-z}). \quad (\text{H7:6.1})$$

Let $\log_{*D}$ and $\exp_{*D}$ denote logarithm and exponential for Dirichlet convolution. At one prime,

$$\sum_{a \geq 0} J_z(p^a) X^a = \frac{1 - X}{1 - p^z X}. \quad (\text{H7:6.2})$$

Therefore

$$\log \frac{1 - X}{1 - p^z X} = \sum_{a \geq 1} \frac{p^{az} - 1}{a} X^a \quad (\text{H7:6.3})$$

and

$$\boxed{\log_{*D} J_z = \beta_z, \quad \beta_z(p^a) = \frac{p^{az} - 1}{a}, \quad \beta_z(n) = 0 \text{ off prime powers.}} \quad (\text{H7:6.4})$$

Since

$$\partial_z \beta_z(n) = n^z \Lambda(n), \quad (\text{H7:6.5})$$

one obtains

$$J_z = \exp_{*D} \beta_z = \exp_{*D} \left(\int_0^z n^u \Lambda(n) du \right). \quad (\text{H7:6.6})$$

This is the Dirichlet-convolution counterpart of (5.7):

$$\text{prime-power connected source} \xrightarrow{\text{integrate}} \text{Euler logarithm} \xrightarrow{\text{convolution exponential}} \text{all mixed colors}. \quad (\text{H7:6.7})$$

For the normalized family

$$c_z(n) = \frac{J_z(n)}{n^z} = \prod_{p|n} (1 - p^{-z}), \quad (\text{H7:6.8})$$

the parallel identity is

$$(\log_{*D} c_z)(n) = \begin{cases} \frac{1 - p^{-az}}{a}, & n = p^a, \ a \geq 1, \\ 0, & n \text{ is not a prime power,} \end{cases} \quad \partial_z (\log_{*D} c_z)(n) = n^{-z} \Lambda(n). \quad (\text{H7:6.9})$$

In particular, $\partial_z (\log_{*D} c_z)(p^a)|_{z=\sigma} = (\log p) p^{-a\sigma} = (\log p) r_p^a$, exactly Route A's one-color coefficient. At critical half-density one sets $\sigma = 1/2$.

7. Higher colors and the Smith–Jordan carrier

For $k \geq 1$, put

$$\Lambda_{[k]}(n) := \partial_z^k J_z(n) \Big|_{z=0} = \sum_{d|n} \mu(d) (\log(n/d))^k. \quad (\text{H7:7.1})$$

Separately set $\Lambda_{[0]} = \delta_1$.

If $n = \prod_{p \in S} p^{a_p}$, then

$$\Lambda_{[k]}(n) = 0 \quad (k < |S|), \quad (\text{H7:7.2})$$

$$\Lambda_{[|S|]}(n) = |S|! \prod_{p \in S} \log p, \quad (\text{H7:7.3})$$

and, more explicitly,

$$\Lambda_{[k]}(n) = k! \sum_{\substack{j_p \geq 1 \\ \sum_{p \in S} j_p = k}} \prod_{p \in S} \frac{(a_p^{j_p} - (a_p - 1)^{j_p}) (\log p)^{j_p}}{j_p!} \geq 0. \quad (\text{H7:7.4})$$

Thus a one-color integer enters at first derivative order, a two-color integer at second order, and an s -color integer first enters at order s . In particular,

$$J'_0(n) = \Lambda(n). \quad (\text{H7:7.5})$$

For real $z > 0$, the tensor valuation kernel on positive integers is exactly

$$K_z(m, n) = \frac{(m, n)^z}{(mn)^{z/2}} = \prod_p p^{-\frac{z}{2}|\nu_p(m) - \nu_p(n)|}. \quad (\text{H7:7.6})$$

On integers supported by the finite prime set P , the parameter match

$$z = 2\sigma$$

identifies (7.6) with the restriction of $K_{P,\sigma}^{\text{val}}$ to nonnegative exponent vectors.

Jordan inversion gives

$$K_z(m, n) = \frac{1}{(mn)^{z/2}} \sum_{d|(m,n)} J_z(d). \quad (\text{H7:7.7})$$

Therefore every finitely supported complex sequence satisfies

$$\sum_{m,n} \overline{x_m} K_z(m, n) x_n = \sum_{d \geq 1} J_z(d) \left| \sum_{d|n} \frac{x_n}{n^{z/2}} \right|^2 > 0 \quad (x \neq 0). \quad (\text{H7:7.8})$$

The full tensor kernel is a positive divisor synthesis of all Jordan colors. Its first tangent on the zero-sum subspace is

$$\sum_n x_n = 0 \implies \left. \partial_z \sum_{m,n} \overline{x_m} K_z(m, n) x_n \right|_{z=0} = \sum_{q=p^a} (\log p) \left| \sum_{q|n} x_n \right|^2 \geq 0. \quad (\text{H7:7.9})$$

This tangent is positive semidefinite on $\mathbf{1}^\perp$, but need not be strictly positive there; mixed-color vectors can lie in its nullspace. It is the $z = 0$ Genesis tangent of the color hierarchy, related to but not identical with finite- σ Route A.

8. Möbius and Ramanujan recombination

For a finite family $n \leq N$, choose a period Q divisible by $\text{lcm}(1, \dots, N)$, use the normalized inner product $\langle f, g \rangle_Q = Q^{-1} \sum_{r \bmod Q} f(r) \overline{g(r)}$, and define

$$h_n(r) = \prod_{p^a \| n} h_{p,a}(r). \quad (\text{H7:8.1})$$

The complete color theorem gives, with $c_n^{\text{Ram}}(r)$ denoting the Ramanujan sum,

$$h_n(r) = \frac{c_n^{\text{Ram}}(r)}{n}, \quad \langle h_m, h_n \rangle_Q = \delta_{mn} \frac{\varphi(n)}{n^2}. \quad (\text{H7:8.2})$$

Hence arbitrary complex coefficients obey

$$\left\| \sum_{n \leq N} b_n h_n \right\|_Q^2 = \sum_{n \leq N} |b_n|^2 \frac{\varphi(n)}{n^2}. \quad (\text{H7:8.3})$$

At the distinguished observation point,

$$c_n^{\text{Ram}}(1) = \mu(n), \quad (\text{H7:8.4})$$

so Möbius signs are observation coordinates of a positive orthogonal color space. They are not negative norms.

The finite successor packet is

$$V_N(k) = \mathbf{1}_{k|N} \mu(N/k) = \bigotimes_{p^a \parallel N} (e_a - e_{a-1}). \quad (\text{H7:8.5})$$

Here e_j is the standard basis vector in the local valuation coordinate $j = 0, \dots, a$, and the tensor product is identified with the divisor basis e_k , $k \mid N$.

Under the divisibility transform

$$(Za)(d) = \sum_{k:d|k} a(k), \quad (\text{H7:8.6})$$

all proper color channels cancel and the completed state survives:

$$\boxed{ZV_N = e_N}. \quad (\text{H7:8.7})$$

This is how positive color coordinates recombine into signed Möbius coefficients without losing the underlying positive energy.

9. Prime angles and the scalar logarithmic orbit

The dual of the exponent lattice is $\mathbb{T}^P$. Define

$$\iota_P(t) = (e^{-it \log p})_{p \in P}, \quad L_P(\nu) = \frac{1}{2} \sum_{p \in P} \nu_p \log p. \quad (\text{H7:9.1})$$

The normalization obeys

$$\iota_P(t)^\nu = e^{-it \sum_p \nu_p \log p} = e^{-2it L_P(\nu)}.$$

Unique factorization makes L_P injective on $\mathbb{Z}^P$. For at least two primes, ι_P is injective: a nonzero common period would make a ratio of two prime logarithms rational. Its image is dense in the torus. If A is a torus polynomial, Kronecker equidistribution gives

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |A(\iota_P(t))|^2 dt = \int_{\mathbb{T}^P} |A(z)|^2 dm_P(z). \quad (\text{H7:9.2})$$

Route B restricts to

$$G_{P,\sigma}(\iota_P(t)) = \left(\prod_{p \in P} (1 - p^{-2\sigma}) \right) \left| \prod_{p \in P} (1 - p^{-\sigma - it})^{-1} \right|^2. \quad (\text{H7:9.3})$$

Route A restricts to

$$D_{P,\sigma}(\iota_P(t)) = \partial_\sigma \log \left| \prod_{p \in P} \frac{1 - p^{-\sigma - it}}{1 + p^{-\sigma}} \right|^2 \geq 0. \quad (\text{H7:9.4})$$

Moreover,

$$G_{P,\sigma}(\iota_P(t)) \left| \prod_{p \in P} \frac{1 - p^{-\sigma - it}}{1 + p^{-\sigma}} \right|^2 = \prod_{p \in P} c_p. \quad (\text{H7:9.5})$$

These are the exact prime-color phase identities. The scalar line is the coherent phase-lock of all prime angles.

10. The exact theta projection

Fix $a > 0$. For the complete theta function, use its bilateral Laplace representation

$$F(z) = \xi\left(\frac{1}{2} + z\right) = \int_{\mathbb{R}} \eta(u) e^{zu} du,$$

where η is the positive even theta density, and write

$$C_a(y) = \frac{1}{\pi} \int_{\mathbb{R}} K(a, t) e^{-2ity} dt, \quad K(a, t) = \operatorname{Re} (F'(a + it) \overline{F(a + it)}). \quad (\text{H7:10.1})$$

The complete theta kernel has $K(a, \cdot) \in L^1(\mathbb{R})$. Define the finite real signed Borel measure on the color torus

$$\mu_{a,P} = (\iota_P)_* \left(\frac{1}{\pi} K(a, t) dt \right). \quad (\text{H7:10.2})$$

With the convention $\hat{\mu}_{a,P}(\nu) := \int_{\mathbb{T}^P} z^\nu d\mu_{a,P}(z)$, its Fourier coefficients are exactly

$$\hat{\mu}_{a,P}(\nu) = C_a \left(\frac{1}{2} \sum_{p \in P} \nu_p \log p \right). \quad (\text{H7:10.3})$$

Thus every finite exponent array satisfies the following signed-measure identity:

$$\boxed{\sum_{\nu, \mu} \overline{b_\nu} b_\mu C_a(L_P(\nu - \mu)) = \int_{\mathbb{T}^P} \left| \sum_{\nu} b_\nu z^{-\nu} \right|^2 d\mu_{a,P}(z).} \quad (\text{H7:10.4})$$

This is the exact color-to-theta projection. For $P = \{2, 3\}$, the group $\mathbb{Z} \log 2 + \mathbb{Z} \log 3$ is dense in $\mathbb{R}$. Therefore positivity of every finite Gram matrix indexed by the bilateral exponent lattice $\mathbb{Z}^2$ is equivalent, by continuity, to positivity on arbitrary finite real-node sets. Here “two-color” means the full two-prime lattice with arbitrary coefficients; it does not mean only 2×2 minors or only the $|S| = 2$ sector of (4.2).

The completed logarithmic derivative supplies the source-side form of the same connection on an Euler-convergent line. Put

$$F(z) = \xi\left(\frac{1}{2} + z\right), \quad g(z) = \frac{F'(z)}{F(z)}, \quad z = b + it, \quad \sigma = b + \frac{1}{2} > 1. \quad (\text{H7:10.5})$$

The rational-pole and gamma portion is

$$A_{\text{arch}}(z) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi\left(\frac{s}{2}\right), \quad s = \frac{1}{2} + z.$$

The full Euler expansion using this term is absolutely convergent only for $\text{Re } s > 1$, equivalently $b > 1/2$. For a finite prime set, define the finite-prime truncation

$$g_P(z) := A_{\text{arch}}(z) - \sum_{p \in P} (\log p) \frac{p^{-1/2-z}}{1 - p^{-1/2-z}}.$$

It satisfies

$$\boxed{\text{Re } g_P(b + it) = \text{Re } A_{\text{arch}}(b + it) + \sum_{p \in P} \frac{(\log p) r_p}{1 + r_p} - \frac{1}{2} D_{P, \sigma}(\iota_P(t)).} \quad (\text{H7:10.6})$$

The positive color defect therefore appears with the authentic subtractive sign. Pole, gamma, and the scalar floor occur in the pointwise source identity. Boundary terms enter only after the half-line or correlation realization, and the paired infinite limit belongs to that completed source.

11. Why the theta step is a singular orbit step

For $|P| \geq 2$, let

$$H_P = \iota_P(\mathbb{R}) \subset \mathbb{T}^P. \quad (\text{H7:11.1})$$

It is a proper dense Borel subgroup. It is Borel by the Lusin–Souslin theorem because ι_P is a continuous injection from $\mathbb{R}$, and it is Haar-null because a measurable subgroup of positive Haar measure is open, whereas this proper dense subgroup is not the whole torus. The signed theta measure (10.2) satisfies $|\mu_{a,P}|(\mathbb{T}^P \setminus H_P) = 0$, while

$$D_{P,\sigma}(z) dm_P(z) \quad \text{and} \quad G_{P,\sigma}(z) dm_P(z) \quad (\text{H7:11.2})$$

are Haar-absolutely-continuous prime measures and therefore assign H_P mass zero. In contrast,

$$\mu_{a,P}(H_P) = C_a(0) > 0, \quad (\text{H7:11.3})$$

where the strict sign follows from

$$C_a(0) = 2 \int_{\mathbb{R}} u \sinh(2au) \eta(u)^2 du > 0.$$

Define

$$K_{a,P}^\theta(\nu, \mu) := C_a(L_P(\nu - \mu)).$$

Consequently, for every finite constant $C \geq 0$, uniform domination in the sense of all finite sections fails:

$$K_{a,P}^\theta \not\leq C K_{P,\sigma}^{\log} \quad \text{and} \quad K_{a,P}^\theta \not\leq C K_{P,\sigma}^{\text{val}}. \quad (\text{H7:11.4})$$

Write

$$\rho_{P,\sigma}^{\log} := D_{P,\sigma} dm_P, \quad \rho_{P,\sigma}^{\text{val}} := G_{P,\sigma} dm_P.$$

Fourier coefficients uniquely determine finite measures. A positive-definite difference in either comparison would therefore represent $C\rho_{P,\sigma}^\bullet - \mu_{a,P}$, where $\bullet \in \{\log, \text{val}\}$; evaluation on H_P would give $-C_a(0) < 0$.

This is the spectral form of the near-collision phenomenon. There are nonzero exponent vectors γ_j for which

$$L_P(\gamma_j) \rightarrow 0, \quad \sum_p |\gamma_{j,p}| \log p \rightarrow \infty. \quad (\text{H7:11.5})$$

The theta coefficient approaches $C_a(0)$, while the independent valuation coefficient tends to zero.

The valid order of operations is therefore

$$\boxed{\text{prime colors} \xrightarrow{z_p = e^{-it \log p}} \text{singular scalar phase orbit} \xrightarrow{K(a,t) dt} \text{theta kernel form.}} \quad (\text{H7:11.6})$$

12. The line–lattice split and reflections

The recovered line/lattice comparison has kernels

$$\langle \psi(i), \psi(j) \rangle = \log \min(i, j), \quad \langle \phi(i), \phi(j) \rangle = \log \gcd(i, j). \quad (\text{H7:12.1})$$

Their exact difference is

$$\Delta(i, j) = \log \frac{\min(i, j)}{\gcd(i, j)}. \quad (\text{H7:12.2})$$

The shared prime-color core is positive because

$$\log \gcd(i, j) = \sum_{p^a | i, j} \log p. \quad (\text{H7:12.3})$$

The fork (12.2) is a signed residual. Its diagonal is zero while it has nonzero off-diagonal entries, so it cannot itself be positive semidefinite.

For two colors p, q , exponent reflection $(j, k) \leftrightarrow (k, j)$ becomes scalar reflection within a fixed total-degree shell $j + k = d$. Across distinct shells, the center changes by a term involving $d(\log p + \log q)$, so color swap is not a symmetry of the full theta grid. The reflection atlas exposes parity blocks and Euler diamonds; it does not determine the cross-shell Schur signs.

In particular, 8 and 9 are a near resonance, while the exact exponent reflection of $8 = 2^3$ is $27 = 3^3$.

13. Exact conclusion

The parallel color check establishes:

1. The Haar/Ramanujan energies $e_{p,a}$, the two-plane defect $X_p^* X_p$, and the one-prime Poisson covariance are the same local color ladder in three representations.
2. Route A is the positive log-weighted sum of the singleton color defects and is strictly positive on every finite set of distinct exponent vectors.
3. Route B is the tensor closure of those defects. Its empty-color sector is the prime floor, and every higher sector is a positive mixed-color norm for arbitrary coefficients.
4. The identity $J_z = \exp_{*D} \int_0^z n^u \Lambda(n) du$ proves arithmetically that prime powers are the connected layer and mixed colors are their exact recombination.
5. The scalar prime phases are obtained by the exact lock $z_p = e^{-it \log p}$. The theta matrix is the Fourier coefficient system of the completed weight carried on that orbit.

Fix any finite P containing at least two primes, for example $P = \{2, 3\}$. The remaining statement is no longer an arbitrary prime-color coefficient check. By the density argument in Section 10, the target H is equivalent to

$$\boxed{(\iota_P)_* \left(\frac{1}{\pi} K(a, t) dt \right) \geq 0 \quad \text{for every } a > 0.} \quad (\text{H7:13.1})$$

A source-faithful completed identity that pays the subtractive defect in (10.6) after phase-locking would supply this remaining proof. The prime colors supply the complete finite-grid carrier and every mixed-color term of K^{val} . The unresolved operation belongs to the completed scalar orbit.

14. Continuation: the singular trace theorem and the exact next gate

The parallel intertwiner, completed-source, and direct-theta audits sharpen the conclusion above. They identify both an impossibility theorem for the original bounded-compression proposal and one valid regularized interface between the color kernels and the theta orbit.

14.1 The orbit pushforward preserves the whole sign problem

Let

$$d\lambda_a(t) = \frac{1}{\pi} K(a, t) dt, \quad \mu_{a,P} = (\iota_P)_* \lambda_a. \quad (\text{H7:14.1})$$

For $|P| \geq 2$, the map $\iota_P : \mathbb{R} \rightarrow \mathbb{T}^P$ is a continuous injection. By the Lusin–Souslin theorem its image H_P is Borel and $\iota_P^{-1} : H_P \rightarrow \mathbb{R}$ is Borel. Hence every Borel set $E \subseteq \mathbb{R}$ obeys

$$\lambda_a(E) = \mu_{a,P}(\iota_P(E)). \quad (\text{H7:14.2})$$

It follows that

$$\boxed{\mu_{a,P} \geq 0 \iff K(a, t) \geq 0 \text{ for almost every } t \iff K(a, t) \geq 0 \text{ for every } t.} \quad (\text{H7:14.3})$$

The last equivalence uses continuity. Thus the dense orbit does not average away a negative scalar interval. Equation (13.1) is an exact re-expression of H, not a weaker torus positivity statement.

14.2 No bounded coordinate-preserving color decoder

Suppose, under the target positivity, that a bounded decoder acting identically on Laurent characters transferred either prime metric to the theta metric. Then for some finite C and every Laurent polynomial Q ,

$$\int_{\mathbb{T}^P} |Q|^2 d\mu_{a,P} \leq C \int_{\mathbb{T}^P} |Q|^2 d\rho_{P,\sigma}^\bullet, \quad \bullet \in \{\log, \text{val}\}. \quad (\text{H7:14.4})$$

Uniform approximation of square roots of nonnegative continuous functions by Laurent polynomials would imply $\mu_{a,P} \leq C\rho_{P,\sigma}^\bullet$. This is impossible because

$$\rho_{P,\sigma}^\bullet(H_P) = 0, \quad \mu_{a,P}(H_P) = C_a(0) > 0. \quad (\text{H7:14.5})$$

The obstruction is already visible on finite packets. Put $P = \{2, 3\}$, $\alpha = \frac{1}{2} \log 2$, and $\beta = \frac{1}{2} \log 3$. Choose positive integers $m_N, n_N \rightarrow \infty$ such that

$$N|m_N\alpha - n_N\beta| \longrightarrow 0, \quad (\text{H7:14.6})$$

and set

$$\nu_r = (rm_N, (N-r)n_N), \quad 0 \leq r \leq N. \quad (\text{H7:14.7})$$

For the all-ones coefficient vector, continuity gives

$$q_\theta(\mathbf{1}) = (C_a(0) + o(1))(N+1)^2. \quad (\text{H7:14.8})$$

Every nonzero difference $\nu_r - \nu_s$ has two active coordinates, so Route A gives exactly

$$q_{\log}(\mathbf{1}) = \delta_{P,\sigma}(N+1). \quad (\text{H7:14.9})$$

For Route B, put $\lambda_N = 2^{-\sigma m_N} 3^{-\sigma n_N}$. Then

$$K^{\text{val}}(\nu_r, \nu_s) = \lambda_N^{|r-s|}, \quad q_{\text{val}}(\mathbf{1}) \leq (N+1) \frac{1 + \lambda_N}{1 - \lambda_N}. \quad (\text{H7:14.10})$$

Thus the theta energy is quadratic while either independent-color energy is only linear. Any coordinate decoder on these packets has norm at least of order $\sqrt{N}$. The scalar phase lock is necessarily an unbounded trace or singular limiting operation; it cannot be a bounded compression of the unmodified prime Haar metric.

14.3 The valid interface: Route B as a Poisson regularizer

For $\varepsilon > 0$, let

$$d\rho_{P,\varepsilon}^{\text{val}}(z) = G_{P,\varepsilon}(z) dm_P(z). \quad (\text{H7:14.11})$$

This is a probability measure and $\rho_{P,\varepsilon}^{\text{val}} \Rightarrow \delta_{\mathbf{1}}$ as $\varepsilon \downarrow 0$. The convolution $\mu_{a,P} * \rho_{P,\varepsilon}^{\text{val}}$ has the continuous density

$$f_{a,P,\varepsilon}(z) = \sum_{\nu \in \mathbb{Z}^P} C_a(L_P(\nu)) \left(\prod_{p \in P} p^{-\varepsilon|\nu_p|} \right) z^{-\nu}. \quad (\text{H7:14.12})$$

The series is absolutely convergent. Approximate-identity convergence gives the exact criterion

$$\mu_{a,P} \geq 0 \iff f_{a,P,\varepsilon_j}(z) \geq 0 \text{ for every } z \text{ along some } \varepsilon_j \downarrow 0. \quad (\text{H7:14.13})$$

Equivalently, for every j and every finite subset of $\mathbb{Z}^P$, the matrix

$$\left[C_a(L_P(\nu - \mu)) \prod_{p \in P} p^{-\varepsilon_j|\nu_p - \mu_p|} \right]_{\nu, \mu} \quad (\text{H7:14.14})$$

must be positive semidefinite. Route B proves positivity of the damping matrix in (14.14), not positivity of its coupled theta product. Route A is the logarithmic scale derivative of the normalized Route B multiplier, as in (5.6)–(5.7), while Route B supplies the mixed coefficients required for an approximate identity. This is the precise positive interface between the two color routes and the singular orbit.

14.4 Direct theta factorization and the Fourier-side sign

Define

$$A_a(y) = \int_{\mathbb{R}} e^{2ax} \eta(x+y) \eta(x-y) dx. \quad (\text{H7:14.15})$$

Fourier inversion gives

$$A_a(y) = \frac{1}{2\pi} \int_{\mathbb{R}} |F(a + it)|^2 e^{-2ity} dt$$

and

$$\begin{aligned} C_a(y) &= \partial_a A_a(y) = 2 \int_{\mathbb{R}} x e^{2ax} \eta(x + y) \eta(x - y) dx \\ &= 4 \int_0^\infty x \sinh(2ax) \eta(x + y) \eta(x - y) dx > 0. \end{aligned} \tag{H7:14.16}$$

Consequently $K(a, \cdot)$ is unconditionally a positive-definite function:

$$K(a, t) = \int_{\mathbb{R}} C_a(y) e^{2ity} dy, \quad [K(a, t_j - t_k)]_{j,k} \succeq 0. \tag{H7:14.17}$$

The target has the Fourier-dual orientation:

$$K(a, t) \geq 0 \text{ for all } t \iff C_a \text{ is positive definite.} \tag{H7:14.18}$$

Pointwise positivity of C_a , strict log-concavity, its Gaussian envelope, and all two-node minors do not imply (14.18). For example,

$$h(y) = e^{-18y^2 - 200y^4}$$

is positive, satisfies $-(\log h)'' \geq 36$, and has the same Gaussian envelope, but at the nodes $0, u, 2u$ with coefficients $(1, -2, 1)$,

$$6h(0) - 8h(u) + 2h(2u) = -912u^4 + O(u^6) < 0. \tag{H7:14.19}$$

The scalar reserve-minus-defect split is particularly transparent:

$$K(a, t) = \frac{1}{4} (|F(a + it) + F'(a + it)|^2 - |F(a + it) - F'(a + it)|^2). \tag{H7:14.20}$$

Thus the two-plane pattern survives on the theta line, but its required contractivity is exactly the missing sign. Equivalently, if

$$W_\eta(x, t) = \int_{\mathbb{R}} \eta(x + y) \eta(x - y) e^{2ity} dy,$$

then

$$K(a, t) = 4 \int_0^\infty x \sinh(2ax) W_\eta(x, t) dx. \tag{H7:14.21}$$

The outer weight is strictly positive. The unresolved interaction is the signed Wigner correlation W_η , retained in its complete weighted integral.

14.5 The completed Schur payment and its proved exterior region

On the Euler-convergent line, define

$$d_p(t) = \sqrt{\frac{(\log p)r_p(1-r_p)}{1+r_p}} \frac{1+e^{-it\log p}}{1-r_pe^{-it\log p}}. \quad (\text{H7:14.22})$$

Then

$$\|d_P(t)\|^2 = D_{P,\sigma}(\iota_P(t))$$

and (10.6) becomes the exact Schur identity

$$2 \operatorname{Re} g_P(b+it) = 2 \operatorname{Re} A_{\text{arch}}(b+it) + \delta_{P,\sigma} - \|d_P(t)\|^2. \quad (\text{H7:14.23})$$

Therefore

$$\begin{pmatrix} 2 \operatorname{Re} A_{\text{arch}} + \delta_{P,\sigma} & d_P^* \\ d_P & I_{\mathbb{C}^P} \end{pmatrix} \succeq 0 \iff \operatorname{Re} g_P \geq 0. \quad (\text{H7:14.24})$$

For $\sigma > 1$, the full-prime passage is ordinary and absolute:

$$\delta_{P,\sigma} \longrightarrow \delta_{\infty,\sigma} = 2 \sum_p \frac{(\log p)r_p}{1+r_p}, \quad d_P(t) \longrightarrow d_\sigma(t) \quad \text{in } \ell^2(\mathbb{P}),$$

uniformly in t , and $g_P \rightarrow g$. Hence

$$\boxed{\frac{2K(a,t)}{|F(a+it)|^2} = 2 \operatorname{Re} A_{\text{arch}}(a+it) + \delta_{\infty,\sigma} - \|d_\sigma(t)\|^2, \quad \sigma = a + \frac{1}{2} > 1.} \quad (\text{H7:14.24a})$$

The corresponding block on $\mathbb{C} \oplus \ell^2(\mathbb{P})$, with lower pivot $I_{\ell^2(\mathbb{P})}$, is positive semidefinite exactly when $K(a,t) \geq 0$.

The local feedback operators build the defect row, but positivity requires the completed reserve to dominate that row. Notice also that

$$\|X_{p,\sigma}\|^2 = \frac{1}{c_p} - c_p = \frac{4r_p}{1-r_p^2}; \quad (\text{H7:14.25})$$

the feedback operator itself is not generally contractive.

The full completed form has an unconditional square certificate in the exterior region. If $\mathcal{A}$ contains, with multiplicity, one representative from each pair $\{\alpha, -\alpha\}$ of zeros of F , then

$$F(z) = F(0) \prod_{\alpha \in \mathcal{A}} \left(1 - \frac{z^2}{\alpha^2}\right), \quad \sum_{\alpha \in \mathcal{A}} |\alpha|^{-2} < \infty, \quad (\text{H7:14.25a})$$

and, locally uniformly off the zero set,

$$\frac{F'(z)}{F(z)} = \sum_{\alpha \in \mathcal{A}} \left(\frac{1}{z - \alpha} + \frac{1}{z + \alpha} \right). \quad (\text{H7:14.25b})$$

Writing $\alpha = \rho - \frac{1}{2}$, the symmetric expansion gives, first off the zero set and then by continuous removable extension,

$$K(a, t) = \sum_{\alpha} \left[(a - \operatorname{Re} \alpha) \left| \frac{F(a + it)}{a + it - \alpha} \right|^2 + (a + \operatorname{Re} \alpha) \left| \frac{F(a + it)}{a + it + \alpha} \right|^2 \right]. \quad (\text{H7:14.26})$$

At a zero, every quotient in (14.26) is understood through analytic removal, with its multiplicity retained.

Since every nontrivial zero satisfies $|\operatorname{Re} \alpha| < 1/2$, this proves

$$K(a, t) > 0 \quad (a \geq 1/2, t \in \mathbb{R}). \quad (\text{H7:14.27})$$

Strictness at $a = 1/2$ uses $0 < \operatorname{Re} \rho < 1$: both coefficients $1/2 \pm \operatorname{Re} \alpha$ are strictly positive, and F has no zero on $\operatorname{Re} z = 1/2$.

At and below $a = 1/2$, the Euler series is not absolutely convergent; below $a = 1/2$, the coefficients in (14.26) no longer have an unconditional sign. An off-axis zero makes K negative immediately to its left. Hence extending (14.23) or (14.26) to a positive payment for every $a > 0$ is exactly the H/RH gate.

14.6 Why the recovered interior maximum principle does not bypass it

Under the boundary hypothesis $J \geq 0$ on ∂D for the finite rectangle D , so that $J_-/E \in H_0^1(D)$, the averaged amplitude in the interior-transfer route obeys

$$\delta = \Delta \log E \geq 0, \quad \int_D |\nabla(J_-/E)|^2 = \int_D \frac{\partial_a \delta}{2E} J_-. \quad (\text{H7:14.28})$$

The source has no sign from $\delta \geq 0$. This is essential. The holomorphic vector

$$\Phi(z) = \left(1, z^2 - \frac{1}{4} \right), \quad E = \|\Phi(z)\|^2,$$

satisfies

$$J = \frac{1}{2} E_a = 2a \left(a^2 + t^2 - \frac{1}{4} \right), \quad \Delta \log E = \frac{16|z|^2}{E^2} \geq 0. \quad (\text{H7:14.29})$$

On a rectangle $0 < a < 1/2$, $|t| < R$ with $R > 1/2$, J is nonnegative on the full boundary but negative in $a^2 + t^2 < 1/4$. Thus boundary positivity plus Gram curvature does not force the interior sign. The exact repair target is a theta-specific sign for the source in (14.28), not another boundary estimate.

Source map

- **Prime DNA: Recombination Energy and Global Density**, orthogonal $h_{p,a}$ colors, $h_n = c_n/n$, and arbitrary-coefficient positivity.
- **Prime DNA: The Connected Architecture**, , pp. 6–14: prime-color taxonomy, Jordan onset, Möbius rotation, and two-plane energy.
- **Prime DNA Build v34: Moving Color Filtration**, spectral-order color filtration and the Ramanujan/Jordan hierarchy.
- **Prime DNA Build v39: The Two-Plane Prime Cocycle**, the visible/hidden lossless split.
- **Deterministic Spectral Equilibrium: Finite Counting Proof Ledger**, the packet $V_N = \bigotimes(e_a - e_{a-1})$ and $ZV_N = e_N$.
- **Prime DNA: Centered Prism, Boundary, and Open Diamond Ledger**, the exact line–GCD core–fork split.
- **Prime DNA: Angle Families and Prime Triple Bound**, , and **Prime Reflection Atlas**, .
- **Prime DNA: Genesis Half-Density and Completed First-Loss Gate**, Smith–Jordan carrier, prime-power tangent, and the complete signed source ledger.
- **Wheel to Euler Feshbach Bridge**, complete prime-power covariance, valuation shells, and common-source Feshbach transport.
- **Prime DNA Build v36**, : Ramanujan two-slot color law and its valuation-depth refinement.
- **The Two-Plane Prime-to-Theta Intertwiner**, complete prime-power feedback, finite scalar decoding, and the completed Schur gate.
- **Michaels interior-to-spectral transfer**, positive boundary transfer and the retained Gram-curvature source.
- **Equation (14.4) and the positive operator J**, the Euler-convergent completed source, all-prime-power square identity, and fixed-line operator criterion.

No numerical sign test or finite-height computation is used in this note. All positivity statements are finite-coefficient analytic identities.

Two-plane prime-theta intertwining

Source: Michaels_Two_Plane_Prime_Theta_Intertwiner_Attempt.md. Equation tags carry the source prefix H8; references within this unit retain their original local numbering.

Exact prime-power feedback, scalar decoding, and the arbitrary-grid Schur gate

Status

This note carries out the proposed operator construction analytically. It makes no numerical argument and does not assume Hypothesis H.

The construction reaches four exact results.

1. The v39 two-plane cocycle has an explicit finite scalar decoder; this is the projection back from the positive prime lift.
2. Feeding the hidden channel through the inverse visible channel produces an explicit operator containing **every power of one prime**.
3. The complete signed prime-power contribution is exactly a scalar reserve minus the norm of that feedback operator.
4. The arbitrary $J \times L$ theta problem has an exact block-Schur recursion. Closing it requires identification of its conditional cross block with a compression of the prime feedback operator.

The prime-power feedback operator is therefore built. The presently established theta curvature proves the first split and several low-order blocks, but it does not prove the uniform all-grid contraction. The remaining statement is displayed explicitly below.

1. The target rectangle

Fix $a > 0$, and write

$$\alpha = \frac{1}{2} \log 2, \quad \beta = \frac{1}{2} \log 3.$$

Let C_a be the scalar kernel occurring in Hypothesis H. For $J, L \geq 0$, define

$$M_{J,L}(a) = [C_a((j - j')\alpha + (k - k')\beta)]_{\substack{0 \leq j, j' \leq J \\ 0 \leq k, k' \leq L}}. \quad (\text{H8:1.1})$$

The two-prime density reduction gives

$$\boxed{\text{H at } a \iff M_{J,L}(a) \succeq 0 \text{ for every } J, L.} \quad (\text{H8:1.2})$$

Thus the requested induction must retain arbitrary complex coefficients and every cross term in (1.1).

2. The local two-plane cocycle

Let U_p be translation by $\ell_p = \log p$ on the bilateral spectral model, let

$$r_p = p^{-\sigma} \in (0, 1),$$

and put

$$V_p = \frac{I - r_p U_p}{1 + r_p}, \quad W_p = \frac{\sqrt{r_p}(I + U_p)}{1 + r_p}. \quad (\text{H8:2.1})$$

The v39 paraunitary identity is

$$V_p^* V_p + W_p^* W_p = I. \quad (\text{H8:2.2})$$

Hence

$$f \mapsto (V_p f, W_p f) \quad (\text{H8:2.3})$$

is an isometric two-plane lift. At a finite causal terminal there is one additional reserve coordinate $R_{p,T}$, and the exact identity is

$$V_{p,T}^* V_{p,T} + W_{p,T}^* W_{p,T} + R_{p,T}^* R_{p,T} = I. \quad (\text{H8:2.4})$$

The signed coherence is visible by expanding the two norms:

$$\begin{aligned} \|(I - r_p U_p)f\|^2 &= (1 + r_p^2)\|f\|^2 - 2r_p \operatorname{Re}\langle f, U_p f \rangle, \\ r_p \|(I + U_p)f\|^2 &= 2r_p \|f\|^2 + 2r_p \operatorname{Re}\langle f, U_p f \rangle. \end{aligned} \quad (\text{H8:2.5})$$

The same lag appears with opposite signs. The two-plane sum is positive, while the isolated lag has both signs.

The natural positive Schur complement supplied by (2.2) is the whole visible norm. If

$$D_p = (W_p^* W_p)^{1/2},$$

then

$$\begin{pmatrix} I & D_p \\ D_p & I \end{pmatrix} \succeq 0, \quad I - D_p^2 = V_p^* V_p. \quad (\text{H8:2.6})$$

Thus v39 does not make the signed lag positive by itself. It stores the compensating sign in the hidden coordinate.

3. The exact finite projection back to the scalar line

Build v40 supplies the finite intertwiner that was absent from v39's initial presentation. For a finite prime set P , let

$$N_P = \prod_{p \in P} (1 + r_p), \quad r_D = D^{-\sigma},$$

where D runs over squarefree divisors supported on P . The rotated tensor coordinates are

$$\chi_D = \mu(D) \sqrt{\frac{r_D}{N_P}} U_D f. \quad (\text{H8:3.1})$$

They satisfy

$$\sum_{D \subseteq P} \|\chi_D\|^2 = \|f\|^2. \quad (\text{H8:3.2})$$

The exact scalar decoder is

$$\boxed{\mathcal{R}_P(\chi) = \sum_{D \subseteq P} \sqrt{N_P r_D} \chi_D = \sum_{D \subseteq P} \mu(D) r_D U_D f.} \quad (\text{H8:3.3})$$

Therefore the positive tensor lift really does project back to the finite Möbius/Euler scalar channel. The decoder is amplifying: its row norm is

$$\|\mathcal{R}_P\| = N_P. \quad (\text{H8:3.4})$$

This growth is structural. It explains why a contractive observation of the v39 lift cannot recover the unnormalized physical Euler factor.

Along the prime spine, v40 gives the equivalent local decoder

$$r_p v_p - \sqrt{r_p} w_p = -r_p U_p v_{p-}, \quad (\text{H8:3.5})$$

which reconstructs each scalar increment from the visible and hidden outputs. Equations (3.3) and (3.5) answer the finite projection question exactly.

4. Feeding the hidden channel through the visible channel

The prime-power extension is obtained by one natural feedback operation. Since $r_p < 1$, V_p is invertible. Set

$$c_p = \frac{1 - r_p}{1 + r_p},$$

and define the feedback intertwiner

$$\boxed{X_p = \sqrt{c_p} W_p V_p^{-1} = \sqrt{\frac{r_p(1 - r_p)}{1 + r_p}} (I + U_p)(I - r_p U_p)^{-1}.} \quad (\text{H8:4.1})$$

The resolvent has the operator-norm expansion

$$(I - r_p U_p)^{-1} = \sum_{m \geq 0} r_p^m U_p^m. \quad (\text{H8:4.2})$$

Consequently X_p contains the v39 hidden plus-wave $I + U_p$ followed by the complete geometric ladder

$$I, r_p U_p, r_p^2 U_p^2, \dots$$

This is the explicit one-prime intertwining operator requested in the proposal.

4.1 A rational two-plane completion containing all powers

The feedback resolvent itself has a lossless 2×2 completion. Define

$$H_p = (1 - r_p)(I - r_p U_p)^{-1} = c_p V_p^{-1}, \quad Y_p^{\text{def}} = \sqrt{r_p}(I - U_p)(I - r_p U_p)^{-1}. \quad (\text{H8:4.3})$$

The elementary identity

$$|1 - r_p z|^2 = (1 - r_p)^2 + r_p |1 - z|^2 \quad (|z| = 1) \quad (\text{H8:4.4})$$

gives

$$\boxed{H_p^* H_p + (Y_p^{\text{def}})^* Y_p^{\text{def}} = I.} \quad (\text{H8:4.5})$$

Equivalently, the rational matrix

$$\boxed{\tilde{U}_{r_p}(z) = \frac{1}{1 - r_p z} \begin{pmatrix} 1 - r_p & \sqrt{r_p}(1 - z) \\ \sqrt{r_p}(1 - z) & (1 - r_p)z \end{pmatrix}} \quad (\text{H8:4.6})$$

is unitary for $|z| = 1$. Its determinant is the Blaschke factor

$$\det \tilde{U}_{r_p}(z) = \frac{z - r_p}{1 - r_p z}. \quad (\text{H8:4.7})$$

Its first column is precisely the complete-power feedback pair (H_p, Y_p^{def}) . Moreover,

$$H_p^* H_p = c_p \mathcal{G}_p. \quad (\text{H8:4.8})$$

For a finite prime set P , the tensor product

$$\tilde{U}_P(\mathbf{z}) = \bigotimes_{p \in P} \tilde{U}_{r_p}(z_p) \quad (\text{H8:4.9})$$

is unitary on the full prime torus. Its visible coordinate is

$$H_P(\mathbf{z}) = \prod_{p \in P} \frac{1 - r_p}{1 - r_p z_p}, \quad (\text{H8:4.10})$$

and its Fourier covariance is

$$\widehat{|H_P|^2}(\nu) = \prod_{p \in P} c_p r_p^{|\nu_p|}. \quad (\text{H8:4.11})$$

Thus the 2×2 -to-arbitrary-grid induction is completely constructed for the valuation/Smith–Jordan prime kernel. The later issue is the identification of this prime metric with the scalar theta metric.

5. The complete prime-power covariance and its strict floor

Define the Poisson, or valuation-chain, covariance

$$\mathcal{G}_p = (1 - r_p^2)(I - r_p U_p)^{-1}(I - r_p U_p^*)^{-1}. \quad (\text{H8:5.1})$$

Because U_p is unitary,

$$\mathcal{G}_p = I + \sum_{m \geq 1} r_p^m (U_p^m + U_p^{*m}) \succ 0. \quad (\text{H8:5.2})$$

Direct spectral algebra gives the exact factorization

$$\boxed{\mathcal{G}_p = c_p I + X_p^* X_p.} \quad (\text{H8:5.3})$$

Thus the primes themselves do supply a strict floor:

$$\mathcal{G}_p \succeq \frac{1 - r_p}{1 + r_p} I. \quad (\text{H8:5.4})$$

On the finite valuation chain $0, 1, \dots, m$, the matrix is

$$\mathcal{K}_{p,m} = [r_p^{|i-j|}]_{0 \leq i,j \leq m}, \quad (\text{H8:5.5})$$

and every successive Schur innovation is $1 - r_p^2 > 0$. This is a complete arbitrary-coefficient theorem at every prime-power depth.

Differentiation of the normalized visible channel yields a second exact identity. With $r_p = p^{-\sigma}$,

$$\boxed{\partial_\sigma \log(V_p^* V_p) = (\log p) X_p^* X_p \succeq 0.} \quad (\text{H8:5.6})$$

Equivalently, on the spectral circle $w = e^{-i\theta}$,

$$\partial_\sigma \left| \frac{1 - r_p w}{1 + r_p} \right|^2 = \frac{2(\log p) r_p (1 - r_p)(1 + \cos \theta)}{(1 + r_p)^3} \geq 0. \quad (\text{H8:5.7})$$

This proves monotonicity of every normalized finite visible prime product and produces positive Toeplitz kernels for arbitrary exponent boxes.

6. Where the authentic signed prime term sits

At the completed explicit-formula weight $r_p = p^{-1/2}$, the authentic all-power prime update is

$$\mathfrak{W}_p = (\log p)(I - \mathcal{G}_p). \quad (\text{H8:6.1})$$

Combining (5.3) with $1 - c_p = 2r_p/(1 + r_p)$ gives

$$\boxed{\mathfrak{W}_p = \frac{2(\log p) r_p}{1 + r_p} I - (\log p) X_p^* X_p.} \quad (\text{H8:6.2})$$

This is the exact answer to the signed-coherence question:

$$\boxed{\text{complete prime-power contribution} = \text{positive scalar reserve} - \text{positive feedback norm.}} \quad (\text{H8:6.3})$$

Every power p^m is inside X_p . No term has been discarded or estimated.

The same identity has a Hardy-space defect form. Put

$$L_p(w) = (\log p) \frac{r_p w}{1 - r_p w},$$

and

$$d_p(w) = \sqrt{\frac{(\log p) r_p (1 - r_p)}{1 + r_p}} \frac{1 + w}{1 - r_p w}. \quad (\text{H8:6.4})$$

On $|w| = 1$,

$$2 \operatorname{Re} \left(L_p(w) + (\log p) \frac{r_p}{1 + r_p} \right) = |d_p(w)|^2. \quad (\text{H8:6.5})$$

Hence the Hardy multiplication operators satisfy

$$\boxed{-(M_{L_p} + M_{L_p}^*) = \frac{2(\log p) r_p}{1 + r_p} I - M_{d_p}^* M_{d_p}.} \quad (\text{H8:6.6})$$

The factor $(1 + w)/(1 - r_p w)$ in (6.4) is visibly the two-plane hidden output followed by the prime-power resolvent.

On the half-line, the boundary term must also be retained. If

$$(S_p f)(u) = f(u + \ell_p), \quad Y_p = (I - r_p S_p)^{-1} f,$$

then

$$\boxed{\begin{aligned} \mathfrak{W}_p[f] &= (\log p) \|f\|^2 - (\log p) (1 - r_p^2) \|Y_p\|^2 \\ &\quad - (\log p) r_p^2 \int_0^{\log p} |Y_p(u)|^2 du. \end{aligned}} \quad (\text{H8:6.7})$$

The last two terms are positive norms with the authentic subtractive sign.

Equivalently, put

$$Q_p = I - S_p^* S_p = \mathbf{1}_{[0, \ell_p)}, \quad \mathcal{R}_p = (I - r_p S_p)^{-1}, \quad (\text{H8:6.8})$$

and define the causal feedback and boundary channels

$$\begin{aligned} X_{p,+} &= \sqrt{\frac{r_p(1 - r_p)}{1 + r_p}} (I + S_p) \mathcal{R}_p, \\ Z_{p,+} &= \sqrt{r_p} Q_p^{1/2} \mathcal{R}_p. \end{aligned} \quad (\text{H8:6.9})$$

Then

$$\mathfrak{W}_p[f] = \frac{2(\log p)r_p}{1+r_p} \|f\|^2 - (\log p) \|X_{p,+}f\|^2 - (\log p) \|Z_{p,+}f\|^2. \quad (\text{H8:6.10})$$

The $Z_{p,+}$ square is the exact causal boundary payment.

7. The finite completed Schur candidate

For a finite prime set P , let

$$\mathcal{D}_P f = \bigoplus_{p \in P} \sqrt{\log p} (X_{p,+}f \oplus Z_{p,+}f), \quad \delta_P = \sum_{p \in P} \frac{2(\log p)r_p}{1+r_p}. \quad (\text{H8:7.1})$$

Then (6.10) sums exactly to

$$\sum_{p \in P} \mathfrak{W}_p = \delta_P I - \mathcal{D}_P^* \mathcal{D}_P. \quad (\text{H8:7.2})$$

The positive norm in (7.2) already covers arbitrary coefficient arrays. Its bilateral/Bohr analogue is

$$\mathcal{D}_P^{\text{bil}} f = \bigoplus_{p \in P} \sqrt{\log p} X_p f. \quad (\text{H8:7.2a})$$

Unlike the causal row in (7.1), this analogue has no $Z_{p,+}$ boundary coordinate. On the bilateral exponent torus its symbol is

$$\mathcal{D}_P(\theta) = \sum_{p \in P} (\log p) \left[\frac{1 - r_p^2}{1 - 2r_p \cos \theta_p + r_p^2} - \frac{1 - r_p}{1 + r_p} \right] \geq 0. \quad (\text{H8:7.2b})$$

Its Fourier kernel on $\mathbb{Z}^P$ is

$$K_P^{\log}(\nu, \mu) = \begin{cases} \delta_P, & \nu = \mu, \\ (\log p)r_p^{|k|}, & \nu - \mu = ke_p, \quad k \neq 0, \\ 0, & \nu - \mu \text{ has at least two nonzero coordinates.} \end{cases} \quad (\text{H8:7.2c})$$

This is the complete, source-faithful prime-power Gram: it is positive for every finite complex coefficient array, and its off-diagonal support is exactly the prime powers. It is the bilateral Toeplitz counterpart of the causal norm in (7.2); the latter additionally retains the boundary square. In the authentic completed form, the corresponding positive norm is subtracted.

To state the completed finite gate precisely, fix a finite-dimensional packet space $\mathcal{E}$ and a finite prime set P . Let $\mathfrak{A}_{\mathcal{E}}$ be the Hermitian form of all nonprime archimedean and pole terms restricted to $\mathcal{E}$, and define the truncated completed form

$$\mathfrak{Q}_{P,\mathcal{E}}[f] = \mathfrak{A}_{\mathcal{E}}[f] + \sum_{p \in P} \mathfrak{W}_p[f]. \quad (\text{H8:7.3})$$

Let $\mathcal{A}_\mathcal{E}$ be the representing Hermitian matrix of $\mathfrak{A}_\mathcal{E}$, and put

$$\mathcal{R}_{P,\mathcal{E}} = \mathcal{A}_\mathcal{E} + \delta_P I_\mathcal{E}. \quad (\text{H8:7.4})$$

The truncated completed form has the exact defect shape

$$\boxed{\mathcal{T}_{P,\mathcal{E}} = \mathcal{R}_{P,\mathcal{E}} - \mathcal{D}_P^* \mathcal{D}_P}. \quad (\text{H8:7.5})$$

This is a finite-stage identity by construction; it does not assert that $\mathcal{R}_{P,\mathcal{E}}$ is positive. When $\mathcal{R}_{P,\mathcal{E}} \succ 0$, define

$$\mathcal{X}_{P,\mathcal{E}} = \mathcal{D}_P \mathcal{R}_{P,\mathcal{E}}^{-1/2}. \quad (\text{H8:7.6})$$

Then

$$\boxed{\mathcal{T}_{P,\mathcal{E}} = \mathcal{R}_{P,\mathcal{E}}^{1/2} (I - \mathcal{X}_{P,\mathcal{E}}^* \mathcal{X}_{P,\mathcal{E}}) \mathcal{R}_{P,\mathcal{E}}^{1/2}}. \quad (\text{H8:7.7})$$

Equivalently, $\mathcal{T}_{P,\mathcal{E}}$ is the Schur complement of the positive-pivot block

$$\mathbb{K}_{P,\mathcal{E}} = \begin{pmatrix} \mathcal{R}_{P,\mathcal{E}} & \mathcal{D}_P^* \\ \mathcal{D}_P & I \end{pmatrix}. \quad (\text{H8:7.8})$$

Therefore

$$\boxed{\mathcal{T}_{P,\mathcal{E}} \succeq 0 \iff \|\mathcal{X}_{P,\mathcal{E}}\| \leq 1 \iff \mathbb{K}_{P,\mathcal{E}} \succeq 0}. \quad (\text{H8:7.9})$$

This is a genuine construction, not a renaming of the signed lag: the defect row $\mathcal{D}_P$ is explicitly built from the v39 hidden channel and the complete prime-power resolvent. What remains is the domination of that explicit row by the completed reserve.

At the critical prime weight, the separate infinite sums in δ_P and $\mathcal{D}_P^* \mathcal{D}_P$ have the same first-prime divergence. Their paired difference is the meaningful completed object on the fixed-line packet class. Consequently (7.5) must be passed to the limit as one completed, paired expression; neither positive term has an independent bounded limit on the whole L^2 space.

8. The exact $J \times L$ block induction

For a fixed L , define

$$B_r^{(L)} = [C_a(r\alpha + (k - k')\beta)]_{k,k'=0}^L. \quad (\text{H8:8.1})$$

Then

$$M_{J,L} = [B_{j-j'}^{(L)}]_{j,j'=0}^J. \quad (\text{H8:8.2})$$

Put

$$g_{J,L} = \text{col} (B_{-J}^{(L)}, B_{-(J-1)}^{(L)}, \dots, B_{-1}^{(L)}). \quad (\text{H8:8.2a})$$

Appending the J -th row slice gives

$$M_{J,L} = \begin{pmatrix} M_{J-1,L} & g_{J,L} \\ g_{J,L}^* & B_0^{(L)} \end{pmatrix}, \quad (\text{H8:8.3})$$

If $M_{J-1,L} \succ 0$, the exact new-row payment is

$$\boxed{S_{J|L} = B_0^{(L)} - g_{J,L}^* M_{J-1,L}^{-1} g_{J,L}.} \quad (\text{H8:8.4})$$

In the semidefinite case, $M_{J-1,L}^{-1}$ is replaced by its Moore–Penrose inverse, together with the range condition

$$g_{J,L} \in \text{Ran } M_{J-1,L}. \quad (\text{H8:8.4a})$$

The reflection recursion can be written completely explicitly. For $n \geq 2$, let $A = M_{n-2,L}$, viewed on the middle row slices, and set

$$\begin{aligned} R_n &= (B_{-1}^{(L)}, \dots, B_{-(n-1)}^{(L)}), \\ T_n &= \text{col}(B_{-(n-1)}^{(L)}, \dots, B_{-1}^{(L)}), \\ P_{n-1,L}^- &= B_0^{(L)} - R_n A^{-1} R_n^*, \\ P_{n-1,L}^+ &= B_0^{(L)} - T_n^* A^{-1} T_n, \\ \Delta_{n,L} &= B_{-n}^{(L)} - R_n A^{-1} T_n. \end{aligned} \quad (\text{H8:8.5})$$

For $n = 1$, the middle block is empty and

$$P_{0,L}^- = P_{0,L}^+ = B_0^{(L)}, \quad \Delta_{1,L} = B_{-1}^{(L)}. \quad (\text{H8:8.5a})$$

Assume first that $P_{n-1,L}^-, P_{n-1,L}^+ \succ 0$, and define

$$\Gamma_{n,L} = (P_{n-1,L}^-)^{-1/2} \Delta_{n,L} (P_{n-1,L}^+)^{-1/2}. \quad (\text{H8:8.6})$$

Block elimination gives

$$\boxed{S_{n|L} = (P_{n-1,L}^+)^{1/2} (I - \Gamma_{n,L}^* \Gamma_{n,L}) (P_{n-1,L}^+)^{1/2}.} \quad (\text{H8:8.7})$$

Thus the row induction closes when its base matrices $M_{0,L} = B_0^{(L)}$ are positive and

$$\boxed{\|\Gamma_{n,L}\| \leq 1 \quad \text{for every } n, L.} \quad (\text{H8:8.8})$$

The column analogue supplies the corresponding induction for the bases $B_0^{(L)}$. On singular supports the same statements use pseudoinverses and the associated range conditions.

A proof-bearing prime-to-theta coefficient identity is now explicit:

$$\boxed{\Delta_{n,L} = (P_{n-1,L}^-)^{1/2} \Pi_{n,L} (P_{n-1,L}^+)^{1/2},} \quad (\text{H8:8.9})$$

where $\Pi_{n,L}$ must be identified independently as a compression of the prime-power colligation generated by the X_p , so that $\|\Pi_{n,L}\| \leq 1$ follows automatically.

Equation (8.9), with the actual theta coefficients on the left, is a specific sufficient intertwining lemma, stronger than merely postulating an abstract Schur contraction. Once it is proved, (8.7) completes the induction without checking coefficient vectors one at a time.

9. What the existing old/new Schur theorem proves

Build v42 contains an exact Schur complement after the finite prime lift has been decoded to the Mertens terminal. On a dyadic shell, with old span projection Π_X ,

$$\int_{X/2}^X |M(x)|^2 x^{-2\sigma-1} dx = \|(I - \Pi_X)B_X\|^2 + \|O_X - \Pi_X B_X\|^2. \quad (\text{H8:9.1})$$

In block-matrix form, its first term is

$$\mathbf{1}^* \left(G_{YY} - G_{YO} G_{OO}^\dagger G_{OY} \right) \mathbf{1}, \quad (\text{H8:9.2})$$

and

$$G_{YY} - G_{YO} G_{OO}^\dagger G_{OY} \succeq 0. \quad (\text{H8:9.3})$$

This proves that the finite decoder and a complete old/new Schur residual already exist on the prime/Mertens scalar line. Its Gram kernel is the moving terminal/max kernel. The coefficient identity equating that residual with (8.4), or dominating (8.4) by it with a positive remainder, is not contained in v42.

10. What theta log-concavity pays

Write $\eta = e^{-V}$. The established theta-specific bound is

$$V'' \geq 18. \quad (\text{H8:10.1})$$

For $q_a(u) = e^{au}\eta(u)$, let

$$R_a(\ell) = \langle q_a, T_\ell q_a \rangle.$$

Strong convexity gives

$$0 < R_a(\ell) \leq e^{-18\ell^2/4} \|q_a\|^2, \quad (\text{H8:10.2})$$

and differentiation in a , with the shift and r_p frozen, gives

$$0 < \partial_a R_a(\ell) = C_a(\ell/2) \leq e^{-18\ell^2/4} C_a(0). \quad (\text{H8:10.3})$$

Consequently both derivative energies in the first frozen two-plane split are strictly positive:

$$\partial_a \|V_p q_a\|^2 \geq \frac{1 + r_p^2 - 2r_p e^{-18\ell_p^2/4}}{(1 + r_p)^2} C_a(0) > 0, \quad (\text{H8:10.4})$$

and

$$\partial_a \|W_p q_a\|^2 > 0. \quad (\text{H8:10.5})$$

Equations (10.4)–(10.5) are frozen-operator derivatives. If instead $r_p = p^{-(a+1/2)}$ is differentiated with a , an additional sign-indefinite term appears; that varying-filter calculation is not the quadratic form in H .

The same theta calculation proves

$$-(\log C_a)'' \geq 36, \quad 0 < C_a(h) \leq C_a(0)e^{-18h^2}. \quad (\text{H8:10.6})$$

These bounds pay the first split, all two-node minors, the established four-state packet, and separated configurations.

They do not give a coefficient-independent contraction after arbitrary recombination. Finite translates of q_a are dense in $L^2(\mathbb{R})$, and the group

$$\mathbb{Z} \log 2 + \mathbb{Z} \log 3$$

is dense in $\mathbb{R}$. Hence, on the union of finite two-prime packet spaces,

$$\sup_{f \neq 0} \frac{|\langle f, T_\ell f \rangle|}{\|f\|^2} = 1. \quad (\text{H8:10.7})$$

No uniform strict overlap factor survives all coefficient combinations. In the block recursion this is exactly the failure of entrywise bounds such as (10.6) to control the inverse in (8.4).

The limitation is logical, not a numerical concern. The analytic function

$$g(h) = e^{-18h^2 - 60h^4}$$

obeys the same strong log-concavity and Gaussian envelope and has the positive low-order block, but its sufficiently tight four-point translation matrix is indefinite because $g^{(6)}(0) > 0$. Thus an all-grid induction needs an additional theta-specific all-order identity, such as (8.9); curvature alone cannot imply it.

11. Why the independent prime-power Gram is not yet the theta Gram

There are two positive prime kernels, and their roles should not be mixed. The source-faithful additive kernel (7.2c) contains exactly the von Mangoldt prime-power axes. It vanishes whenever an exponent difference has two nonzero prime coordinates. The actual theta coefficient is strictly positive at every real lag, so that additive kernel cannot equal the theta grid.

For $p = 2, 3$, tensoring (5.5) gives the positive matrix

$$G_{(j,k),(j',k')}^{\text{val}} = 2^{-|j-j'|/2} 3^{-|k-k'|/2} \quad (\text{H8:11.1})$$

on every exponent rectangle. This is the independent-coordinate valuation/Smith–Jordan covariance in the two-plane coordinates.

The target coefficient is instead

$$C_a((j - j')\alpha + (k - k')\beta). \quad (\text{H8:11.2})$$

The difference matters in near-cancellation directions. For each growing N , irrationality permits positive integers m, n , themselves arbitrarily large, such that

$$\delta = m\alpha - n\beta$$

satisfies $N|\delta| \rightarrow 0$. For $0 \leq r \leq N$, the exponent pairs

$$(j_r, k_r) = (rm, (N - r)n) \quad (\text{H8:11.3})$$

have scalar locations

$$j_r\alpha + k_r\beta = Nn\beta + r\delta. \quad (\text{H8:11.4})$$

Their entire theta Gram approaches $C_a(0)\mathbf{1}\mathbf{1}^*$ uniformly as $N|\delta| \rightarrow 0$. Their independent valuation Gram has entries

$$\lambda^{|r-s|}, \quad \lambda = 2^{-m/2}3^{-n/2}, \quad (\text{H8:11.5})$$

which are nearly orthogonal when m, n are large.

For the all-ones coefficient vector,

$$\mathbf{1}^* G^{\text{val}} \mathbf{1} \leq (N + 1) \frac{1 + \lambda}{1 - \lambda}, \quad (\text{H8:11.6})$$

whereas the theta quadratic form can be made arbitrarily close to

$$C_a(0)(N + 1)^2. \quad (\text{H8:11.7})$$

Consequently there is no constant C , independent of the rectangle, for which

$$M_N^\theta \preceq C G_N^{\text{val}}. \quad (\text{H8:11.8})$$

Here M_N^θ and G_N^{val} denote the principal submatrices on the points (11.3).

Equivalently, a coordinate-preserving decoder T_N sending each selected valuation feature vector to its corresponding scalar-theta feature vector must satisfy

$$\|T_N\| \gtrsim \sqrt{N}. \quad (\text{H8:11.9})$$

This is qualitatively consistent with the amplification already visible in the finite decoder (3.3), although the two growth mechanisms are not the same. The successful bridge must coherently identify the prime coordinates along the scalar logarithmic phase before taking the Gram form; it cannot be a uniformly bounded coordinatewise compression.

12. The Hardy/de Branges target for the global intertwiner

Let

$$F(z) = \xi\left(\frac{1}{2} + z\right), \quad g(z) = \frac{F'(z)}{F(z)}. \quad (\text{H8:12.1})$$

The double Laplace transform of the theta spatial kernel is

$$\begin{aligned} B(z, w) &= \iint e^{zx + \bar{w}y} J(x, y) dx dy \\ &= \frac{F'(z)\overline{F(w)} + F(z)\overline{F'(w)}}{z + \bar{w}} \\ &= F(z)\overline{F(w)} \frac{g(z) + \overline{g(w)}}{z + \bar{w}}. \end{aligned} \quad (\text{H8:12.2})$$

Thus, in the zero-free Euler half-plane, theta-kernel positivity is diagonally congruent to positivity of

$$P_g(z, w) = \frac{g(z) + \overline{g(w)}}{z + \bar{w}}. \quad (\text{H8:12.3})$$

For $\lambda > 0$, define the Cayley symbol

$$\Theta_\lambda(z) = \frac{g(z) - \lambda}{g(z) + \lambda}. \quad (\text{H8:12.4})$$

Where the denominators are nonzero, direct algebra gives

$$\boxed{P_g(z, w) = \frac{(g(z) + \lambda)(\overline{g(w)} + \lambda)}{2\lambda} \frac{1 - \Theta_\lambda(z)\overline{\Theta_\lambda(w)}}{z + \bar{w}}.} \quad (\text{H8:12.5})$$

Equation (12.5) identifies the exact analytic **candidate**. It does not yet construct a bounded defect operator. Fix the Euler-convergent line $\operatorname{Re} z = b > 1/2$, and let

$$k_z(r) = e^{-zr}, \quad \langle k_w, k_z \rangle_{L^2(0, \infty)} = \frac{1}{z + \bar{w}}. \quad (\text{H8:12.6})$$

On the restriction space generated by these k_z , the formal adjoint action is

$$M_{\Theta_\lambda}^* k_z = \overline{\Theta_\lambda(z)} k_z. \quad (\text{H8:12.7})$$

This action extends to a contraction precisely when

$$\frac{1 - \Theta_\lambda(z)\overline{\Theta_\lambda(w)}}{z + \bar{w}} \quad (\text{H8:12.8})$$

is positive on the selected line. Only after that condition is proved may one define the defect of $M_{\Theta_\lambda}^*$,

$$\boxed{D_{M_{\Theta_\lambda}^*} = (I - M_{\Theta_\lambda} M_{\Theta_\lambda}^*)^{1/2}.} \quad (\text{H8:12.9})$$

Before positivity is known, $g + \lambda$ may vanish and Θ_λ is only meromorphic. We therefore do not treat M_{Θ_λ} as an already established global Hardy multiplier. In particular, shifting to the half-plane $\text{Re } z > 1/2$ would change the standard Hardy denominator, whereas (12.5) must retain $z + \bar{w}$.

The local v39 colligations realize the normalized Euler factors; equations (4.1) and (6.6) upgrade them to every prime power. The additional global operation is the Cayley transform of the **completed logarithmic derivative**, including the pole and gamma reservoir. It is not the product of the local Euler colligations.

13. The precise remaining lemma

There are three precise target formulations. Their equivalence requires the metric-identification and paired-limit steps that the construction is seeking; it must not be assumed in advance.

Grid form

Abstractly, positivity of every base block together with the contraction of every $\Gamma_{n,L}$ is equivalent to the grid statement. A stronger, proof-bearing prime realization would establish (8.9) with $\Pi_{n,L}$ a compression of the completed prime-power colligation.

Completed reserve form

For a fixed finite packet space $\mathcal{E}$, equation (7.5) makes the finite truncation positive exactly when

$$\boxed{\|\mathcal{D}_P f\|^2 \leq \langle \mathcal{R}_{P,\mathcal{E}} f, f \rangle \quad (f \in \mathcal{E}).} \quad (\text{H8:13.1})$$

The completed target does not require every raw prime partial sum to be positive. To use (13.1), one must either include an exact compensating tail at each finite stage or prove the nonnegative paired limit of (7.5) without separating its two divergent positive terms. After that limit is identified with the theta form, the resulting contraction proves H.

Hardy form

Prove

$$\boxed{I - M_{\Theta_\lambda} M_{\Theta_\lambda}^* \succeq 0} \quad (\text{H8:13.2})$$

on the fixed-line restriction space, with the action defined as in (12.7). By (12.5), this is equivalent to positivity of P_g on that line wherever the Cayley denominators are nonzero. It becomes a prime-reservoir proof only after the defect in (12.9) is identified with the completed limit of the explicit prime rows in Section 7.

The previously recorded Prime-Anchor Transport Identity is another coordinate version. It asks the authentic completed blocks K_k, B_k, C_k to obey

$$B_k = K_k \tilde{R}_k, \quad C_k = \tilde{R}_k^* K_k \tilde{R}_k + D_k^* D_k. \quad (\text{H8:13.3})$$

If (13.3) is independently established, its positive residual $D_k^* D_k$ supplies every successor Schur payment and completes the induction. The reviewed positive Suzuki successor proves the analogous identity for its own metric; the global equality with the authentic completed theta/Weil blocks is the unresolved metric identification.

14. Conclusion of the construction attempt

The proposed route produces a real operator, not merely a semantic reformulation:

$$X_p = \sqrt{\frac{1-r_p}{1+r_p}} W_p V_p^{-1}.$$

It is assembled directly from the two-plane cocycle, its resolvent contains every prime power, and it gives the exact signed identity

$$(\log p)(I - \mathcal{G}_p) = \frac{2(\log p)r_p}{1+r_p} I - (\log p)X_p^* X_p.$$

The finite scalar decoder and the v42 old/new Schur residual are also exact. The arbitrary theta rectangle closes as soon as the conditional theta coupling is identified with a contractive compression of this completed prime-power system.

Strict theta log-concavity proves the first split and strong local bounds. It does not, by itself, supply the all-order coefficient identity needed for (8.9). The next proof-bearing task is therefore narrow: establish a source-faithful metric identity that carries either the paired prime-reservoir limit in (13.1) or the prime-anchor block match (13.3) to the authentic theta blocks, while retaining the pole, gamma, terminal, and all prime-power channels together.

Source map

- *Prime DNA Build v39: The Two-Plane Prime Cocycle*: local paraunitary lift and fresh hidden channels.
- *Prime DNA Build v40: The Möbius Rotation and Moving Terminal*: tensor rotation and finite scalar decoder.
- *Prime DNA Build v42: Echo Reflection and Old-New Schur Gate*: exact old/new terminal Schur decomposition.
- *Wheel to Euler Feshbach Bridge*: complete prime-power covariance and valuation-chain realization.
- `Michaels_Prime_Phase_J_Bridge.md`: fixed-line completed prime-phase identity and the exact J -kernel transform.
- `theta_cocycle_logconcavity_bound.md`: sharp reach of the established theta curvature.
- `block_schur_intertwiner_attempt.md`: full block-Levinson recursion for the two-prime rectangles.
- `spectral_intertwiner_attempt.md`: Hardy/de Branges defect formulation and one-prime resolvent factor.

Planes, spirals, and no reversal

Source: *Michaels_Planes_Spirals_and_No_Reversal_Check.md*. Equation tags carry the source prefix H9; references within this unit retain their original local numbering.

Christophe Michaels – focused source comparison, 21 September 2026

1. The only target

Use the fixed completed function

$$F(z) = \xi(\tfrac{1}{2} + z), \quad z = a + it,$$

and

$$K(a, t) = \operatorname{Re}\{F'(a + it)\overline{F(a + it)}\}.$$

Away from zeros its angular velocity along the vertical input line is

$$\Omega(a, t) = \partial_t \arg F(a + it) = \operatorname{Re} \frac{F'(a + it)}{F(a + it)} = \frac{K(a, t)}{|F(a + it)|^2}. \quad (\text{H9:1})$$

The target is $K(a, t) \geq 0$ for every $a > 0$ and real t . K remains defined at zeros, where the phase itself is undefined.

An off-axis zero $z_0 = \alpha + i\tau$, $\alpha > 0$, of multiplicity m , gives

$$\Omega(\alpha - \epsilon, \tau) = -m/\epsilon + O(1) < 0, \quad (\text{H9:2})$$

and, writing $F(z) = (z - z_0)^m H(z)$,

$$K(\alpha - \epsilon, \tau) = -m|H(z_0)|^2 \epsilon^{2m-1} + O(\epsilon^{2m}) < 0. \quad (\text{H9:3})$$

The earlier Complete Theta Kernel Continuation already proves this local reversal, with no bound on the zero's height. A contradiction needs an independent theorem forbidding the same sign on the same domain.

This note retrieves only the plane and spiral facts relevant to that question and checks the suggestion about previously known zeros. It adds no numerical search and starts no additional proof program.

2. The infinite and quadratic planes identify the exact signed coordinate

The recovered higher-information chapter in `EXTENDED_MANUSCRIPT(2).md` defines the infinite analytic lift by

$$c_j(z) = F^{(j)}(z)/j!, \quad \mathcal{C}_F(z) = (c_j(z))_{j \geq 0}.$$

The convergent sequence represents the whole translated entire function $w \rightarrow F(z+w)$. Its quadratic lift is

$$G^F(z) = c(z)c(z)^*, \quad G_{mn}^F = c_m \overline{c_n}. \quad (\text{H9:4})$$

The source writes this first for $\operatorname{Xi}(w) = F(iw)$. Using F directly puts it in the current coordinate convention. The target becomes exactly

$$\boxed{K(a, t) = \operatorname{Re} G_{10}^F(a + it)}. \quad (\text{H9:5})$$

The recovered translation law $G(z+h)=T_h G(z)T_h^*$ preserves positive semidefiniteness and rank. Its two-coordinate identity is

$$|G_{10}|^2 = G_{00}G_{11}. \quad (\text{H9:6})$$

This fixes a magnitude, while (5) asks for the sign of its real component. The source explicitly retains that signed-coordinate requirement as G32 in its Xi convention. Thus the quadratic and infinite planes give an exact place to state the theorem; their existing Gram positivity does not supply its sign.

3. A recovered theorem really does use existing zeros for a contradiction

Theorem G1 in the same source proves that the complete unbased amplitude fingerprint determines its input point. Its mechanism is relevant to the user's recollection.

Equality of the full fingerprints at two points forces, for one constant phase λ and displacement d ,

$$F(v+d) = \lambda F(v) \quad \text{for every complex } v. \quad (\text{H9:7})$$

Starting from any genuine zero v_0 , (7) would produce zeros $v_0+k d$ for every integer k . In the current centered F coordinate, a displacement with nonzero real part sends these outside the established vertical zero strip. A nonzero purely imaginary displacement contradicts the decay of F along the imaginary axis, because (7) keeps $|F(kd)|=|F(0)|$. Hence $d=0$.

This is a proved global contradiction. Its antecedent is equality of all the local amplitude data, which forces (7). A negative value of the single coordinate $\text{Re } G_{10}$ does not give that equality or a global translation. The source distinguishes fingerprint recovery from the signed-coordinate theorem for this reason.

4. What the recovered spirals actually measure

4.1 Individual zero spirals and Fibonacci sampling

Familiar_Number_Atlas.pdf, Sections 8.2-8.4, defines for a fixed exponent $\rho=\beta+i\gamma$

$$W_\rho(x) = -\frac{x^{\beta-1/2}e^{i\gamma\log x}}{\rho}, \quad x > 0. \quad (\text{H9:8})$$

Its two exact laws are

$$|W_\rho(x)| = \frac{x^{\beta-1/2}}{|\rho|}, \quad \frac{d}{d\log x} \arg W_\rho(x) = \gamma. \quad (\text{H9:9})$$

Changing β changes the radius exponent. The angular direction in (9) is unchanged. This is a different derivative from (1): (9) varies x with one exponent held fixed; (1) varies t in the whole completed function.

For Fibonacci numbers q_k with $q_{k+1}/q_k \rightarrow \phi$,

$$\frac{|W_\rho(q_{k+1})|}{|W_\rho(q_k)|} \rightarrow \phi^{\beta-1/2}. \quad (\text{H9:10})$$

The archive proves this sampling law. Its requirement that (10) equal one for every actual nontrivial zero is explicitly an RH-equivalent criterion. The golden ratio comes from the sampling sequence.

4.2 Winding the energy plane

The Atlas, Section 9.1, also defines

$$\mathcal{W}_\gamma(n) = e^{i\gamma u}(1 + X + iY), \quad u = \log n. \quad (\text{H9:11})$$

This is an explicitly chosen invertible representation of the energy coordinates, with u retained. The source's inverse recovers X, Y and the original ledger. Its interpolating curves are display guides between integer samples.

For any differentiable interpolation at a nonzero value, direct differentiation gives

$$\frac{d}{du} \arg \mathcal{W}_\gamma = \gamma + \frac{(1 + X)Y' - YX'}{(1 + X)^2 + Y^2}. \quad (\text{H9:12})$$

Consequently the carrier rotation and the total observed rotation are different quantities. The recovered coordinate inverse supplies no inequality identifying (12) with the target (1).

4.3 Counting-path spirals

Michaels_Spiral_Input_Output_White_Paper.pdf proves the exact increment law

$$\Delta W_n(s) = \Omega_{\text{factors}}(n) \Delta H_n(s), \quad (\text{H9:13})$$

where $\Omega_{\text{factors}}(n) > 0$ for $n \geq 2$. The weighting preserves each nonzero step direction. The paper explicitly distinguishes this path in n at fixed s from a response in t at fixed cutoff. Its reconstruction of ζ retains the unit, smooth correction, and complete tail.

Thus (13) is an exact direction-preserving conversion of steps; it does not assert the sign of the completed sum's angular derivative (1).

5. The prime two-plane conservation theorem

Build v39 proves, for $|z|=1$ and $0 < r < 1$,

$$U_r(z) = \frac{1}{1+r} \begin{pmatrix} 1-rz & \sqrt{r}(1+z) \\ -\sqrt{r}(1+z) & z-r \end{pmatrix}, \quad U_r(z)^* U_r(z) = I, \quad \det U_r(z) = z. \quad (\text{H9:14})$$

It also proves that retaining the Euler path while adding primes requires fresh hidden channels. The complete visible-plus-hidden energy is conserved, including its stated terminal reserve.

One exact check locates the scope of this law. For $\ell = \log p$ and

$$a_p(t) = \frac{1 - r e^{-it\ell}}{1 + r},$$

the visible channel satisfies

$$\frac{d}{dt} \arg a_p(t) = \frac{\ell r (\cos(t\ell) - r)}{1 - 2r \cos(t\ell) + r^2}. \quad (\text{H9:15})$$

This is positive at $t=0$ and negative at $t=\pi/\ell$, while (14) holds at both points. This calculation uses the source's own block. It shows that the conserved full-state energy and determinant phase do not impose one-way rotation on an observed scalar channel. Build v39 itself retains the analytic continuation and nondegeneracy requirement for its infinite moving metric.

6. Would a new off-axis reversal reverse already known zeros?

At any existing critical-line zero $i\gamma$ of multiplicity q ,

$$\frac{F'(a + i\gamma)}{F(a + i\gamma)} = \frac{q}{a} + O(1), \quad \Omega(a, \gamma) > 0 \quad \text{for sufficiently small } a > 0. \quad (\text{H9:16})$$

This local positive behavior can coexist with (2) near a different, hypothetical off-axis zero. The local expansion at one zero does not force all other zeros to change their crossing direction.

An algebraic scope check also preserves the entire critical-line sign pattern. For $\alpha, \tau > 0$, put

$$Q(z) = ((z - \alpha)^2 + \tau^2)((z + \alpha)^2 + \tau^2).$$

Then

$$Q(it) = (\alpha^2 + \tau^2 - t^2)^2 + 4\alpha^2 t^2 > 0. \quad (\text{H9:17})$$

For an even real entire model P_0 , multiplication by Q preserves every critical-line zero, its multiplicity, the whole sign pattern on that line, and the direction of each simple crossing. It adds an off-axis quartet with its local negative angular velocity. One may take $P_0(z) = \prod_j (1 + z^{2/\gamma_j})^2$ to retain any prescribed finite list.

This is a test of the inference from known zero locations and crossing signs alone. It is not an alternative formula for the actual ξ function: the latter has its fixed complete arithmetic and theta identities.

Nor does negative Ω contradict the positive winding number around a holomorphic zero. Following a vertical input line and following a small closed circle around a zero are different paths.

7. Exactly what can be imported into the present proof

Recovered fact	Import into the no-reversal question
Infinite analytic reconstruction	All derivative coordinates belong to the same fixed F .
Quadratic lift	The required sign is exactly $\text{Re } G_{10}$ in the current F convention.
Fingerprint injectivity	A nontrivial global phase-translation identity is impossible; invoking this requires establishing that identity.
Fibonacci and individual-mode spirals	Separate angular frequency from radial exponent; sampling does not itself establish the directional sign.
Prime two-plane law	Retains every hidden channel and terminal reserve; scalar angular velocity requires its own identified sign.
Existing critical-line zeros	Their local forward behavior is compatible with a hypothetical local reversal elsewhere.

The sources checked have not supplied the implication from a local backward turn to a forbidden global fingerprint identity. Their exact signed-coordinate formulation returns to (5). No additional class of objects needs to be introduced to state the remaining theorem.

8. Source record

1. EXTENDED_MANUSCRIPT(2).md, higher-information chapter: G2-G7, G19-G28, Theorem G1, and G32. Read ranges 125-219 and 315-456.

2. Familiar_Number_Atlas.pdf, version 2, Sections 8.2-9.1, read 705-879.
3. Michaels_Spiral_Input_Output_White_Paper.pdf, Sections 2 and 10, read 130-214 and 735-824.
4. Prime_DNA_Build_v39_The_Two_Plane_Prime_Cocycle.pdf, Sections 4-7 and 14-15, read 70-144 and 230-264.
5. Michaels_Complete_Theta_Kernel_Continuation.tex, Section 6, read 375-440 in the preceding focused direction check.

Equations (12), (15), (16), and (17) are short analytic checks stated in full here. They require no numerical experiment. Earlier manuscript files remain unchanged. The accompanying running ledger records the post-paper deliverables and the current ZIP checkpoint.

We are aiming to prove $K(a,t)=\text{Re}(F'(a+it) \text{ conjugate}(F(a+it))) \geq 0$ for every $a > 0$ and real t , so the completed xi function's direction cannot reverse to the right of the critical line.

Backward propagation and local amplitude

Source: Michaels_Backward_Propagation_Local_Amplitude_Criterion.md. Equation tags carry the source prefix H10; references within this unit retain their original local numbering.

Christophe Michaels – research continuation, 21 September 2026; direct transfer added 22 September 2026

1. Result and proof status

The backward-propagation idea has the following exact analytical formulation:

If the completed xi function has any off-critical zero, then at every fixed known critical-line zero some finite quadratic form in the regularized local amplitude derivatives is strictly negative. All other zeros remain included in that form.

This is an application of the classical Grommer criterion, written here in local squared-amplitude coordinates and proved below. The detector implication is established. To turn it into a contradiction, the actual theta or prime structure must independently establish nonnegativity of these forms at one fixed known zero, for every polynomial degree. That sign assertion remains open in the sources checked.

There are no numerical derivative tests or zero truncations in this argument.

22 September update. Sections 9-11 derive the exact compression of the original spatial kernel J that equals the local derivative form, including the complete correction for removing the known zero. Direct theta and prime calculations identify the terms whose sign must be controlled. They establish the transfer identity; they do not establish (29).

Plane check and next reduction. Sections 13-15 identify which positive forms are already established in the earlier planes, prove the complete block decomposition of the present J -form, and reach the earlier two-trace primitive class through an explicit quadratic polynomial. Both primitive boundary corrections vanish on this sufficient subfamily, and no possible off-critical zero is lost. The remaining positivity on that subfamily is not proved.

Requested transfer of previous positivity. Sections 16-17 apply the recovered prime-power tangent theorem and check the direct spatial feature transformation. The exact coefficient congruence is explicit. The prime-to-local equality remains unproved, and the direct feature transformation retains a nonzero translation boundary, including a positive known-zero term. These checks do not establish the target sign.

Dependency review, 22 September. The prime-power equality (77) is one sufficient route, not a necessary additional requirement. The implication from positivity in (67) to exclusion of off-critical zeros was already proved in Section 15 without (77) or Section 17. A proved lower bound by an independently nonnegative form would also suffice. Section 17 checks one proposed norm representation; its boundary calculation does not create a separate obligation to prove each component positive or to cancel the boundary. The outstanding sign is exactly the nonnegativity in (67).

2. Coordinates and the amplitude observable

Use the real-axis coordinate

$$f(w) = \Xi(w) = \xi(1/2 + iw) = F(iw), \quad F(z) = \xi(1/2 + z). \quad (\text{H10:1})$$

Thus a nonreal zero of f corresponds to an off-critical zero of ξ .

Fix an established real zero γ of f , of multiplicity m . Remove its known vanishing factor:

$$g_\gamma(u) = \frac{f(\gamma + u)}{u^m}, \quad g_\gamma(0) = \frac{f^{(m)}(\gamma)}{m!} \neq 0. \quad (\text{H10:2})$$

The quotient extends to a real entire function of order one.

For real u near zero, define the normalized squared amplitude

$$A_\gamma(u) = \frac{|f(\gamma + u)|^2}{u^{2m}|f^{(m)}(\gamma)/m!|^2} = \frac{g_\gamma(u)^2}{g_\gamma(0)^2}. \quad (\text{H10:3})$$

Its removable value at zero is 1, and it is strictly positive in a neighborhood of zero. Hence $\log A_\gamma$ is real analytic there.

Define the real sequence

$$s_n(\gamma) = -\frac{(\log A_\gamma)^{(n+2)}(0)}{2(n+1)!} = -\frac{(\log g_\gamma)^{(n+2)}(0)}{(n+1)!}, \quad n \geq 0. \quad (\text{H10:4})$$

The second expression denotes local logarithmic derivatives; they do not depend on the choice of a local complex logarithm. Formula (3) supplies a positive real normalization throughout.

3. The complete zero identity

Let ρ run over the distinct zeros of f other than γ , with multiplicity m_ρ , and put

$$\beta_\rho = \rho - \gamma, \quad \lambda_\rho = \beta_\rho^{-1}. \quad (\text{H10:5})$$

The order-one canonical product of g_γ has an exponential factor of degree at most one. Logarithmic derivatives of order at least two remove that factor. Consequently

$$s_n(\gamma) = \sum_{\rho \neq \gamma} m_\rho \lambda_\rho^{n+2}. \quad (\text{H10:6})$$

These sums converge absolutely.

Indeed, zero isolation and order-one growth give

$$R := \sup_{\rho \neq \gamma} |\lambda_\rho| < \infty, \quad C := \sum_{\rho \neq \gamma} m_\rho |\lambda_\rho|^2 < \infty. \quad (\text{H10:7})$$

In particular,

$$|s_n(\gamma)| \leq CR^n. \quad (\text{H10:8})$$

Conjugate symmetry of the zero set makes every s_n real.

The generating function of the sequence is the exact meromorphic function

$$\begin{aligned} B_\gamma(z) &= -\frac{1}{z} \left(\frac{g'_\gamma(z)}{g_\gamma(z)} - \frac{g'_\gamma(0)}{g_\gamma(0)} \right) \\ &= \sum_{\rho \neq \gamma} \frac{m_\rho \lambda_\rho^2}{1 - \lambda_\rho z} = \sum_{n \geq 0} s_n(\gamma) z^n, \quad |z| < R^{-1}. \end{aligned} \tag{H10:9}$$

The singularity at zero is removable. The zero sum is locally normally convergent away from its poles, by (7).

At a shifted zero β_ρ , its residue is

$$\text{Res}_{z=\beta_\rho} B_\gamma(z) = -\frac{m_\rho}{\beta_\rho} \neq 0. \tag{H10:10}$$

Thus each nonreal zero creates a genuine nonreal pole in this function reconstructed from the local amplitude derivatives.

4. The finite local witness theorem

Define

$$H_N(\gamma) = [s_{j+k}(\gamma)]_{j,k=0}^N. \tag{H10:11}$$

Theorem. For any fixed established real zero γ ,

$$\boxed{H_N(\gamma) \succeq 0 \text{ for every } N \geq 0 \iff \text{every zero of } \Xi \text{ is real.}} \tag{H10:12}$$

When every zero is real, all these matrices are in fact strictly positive definite. If any nonreal zero exists, there are a finite N and a real nonzero vector $c = (c_0, \dots, c_N)$ such that

$$\boxed{c^T H_N(\gamma) c < 0.} \tag{H10:13}$$

This assertion holds for every choice of established real zero γ .

Proof: real zeros give positivity

For $P(x) = \sum_{j=0}^N c_j x^j$, expansion of (6) gives

$$c^T H_N c = \sum_{\rho \neq \gamma} m_\rho \lambda_\rho^2 P(\lambda_\rho)^2. \tag{H10:14}$$

If all zeros are real, all λ_ρ are real and every summand is nonnegative. There are infinitely many distinct zeros, so a nonzero polynomial cannot vanish at every λ_ρ . Hence the sum is strictly positive.

For nonreal zeros, (14) remains an exact absolutely convergent identity, but its squares are algebraic squares. They cannot be replaced by absolute-value squares.

Proof: positivity bounds the moment multiplication operator

Assume every matrix in (11) is positive semidefinite. Define a real linear functional on real polynomials by

$$\mathcal{L}_\gamma(x^n) = s_n(\gamma). \quad (\text{H10:15})$$

The matrix assumption is exactly $\mathcal{L}_\gamma(P^2) \geq 0$ for every real polynomial P . It implies Cauchy-Schwarz for the semidefinite bilinear form $\langle P, Q \rangle = \mathcal{L}_\gamma(PQ)$.

For a fixed real polynomial P , set

$$a_k = \mathcal{L}_\gamma(x^{2k}P(x)^2) \geq 0. \quad (\text{H10:16})$$

Cauchy-Schwarz gives

$$a_k^2 \leq a_{k-1}a_{k+1}, \quad k \geq 1. \quad (\text{H10:17})$$

Writing $P^2 = \sum_{\ell=0}^d b_\ell x^\ell$, (8) also gives

$$0 \leq a_k \leq C \left(\sum_{\ell=0}^d |b_\ell| R^\ell \right) R^{2k}. \quad (\text{H10:18})$$

If $a_0 = 0$, (17) at $k = 1$ gives $a_1 = 0$. If $a_0, a_1 > 0$, (17) makes the successive ratios nondecreasing, and therefore

$$a_k \geq a_0(a_1/a_0)^k. \quad (\text{H10:19})$$

Comparison with (18) forces $a_1/a_0 \leq R^2$. The case $a_1 = 0$ is immediate. Thus, in every case,

$$\mathcal{L}_\gamma(x^2P^2) \leq R^2 \mathcal{L}_\gamma(P^2). \quad (\text{H10:20})$$

Quotient polynomials by the vectors of zero seminorm and complete. Multiplication by x ,

$$T[P] = [xP], \quad (\text{H10:21})$$

is well defined by (20), bounded with norm at most R , and symmetric because $\mathcal{L}_\gamma(xPQ) = \mathcal{L}_\gamma(PxQ)$. On the complexified Hilbert completion it extends to a bounded self-adjoint operator.

Proof: the resolvent excludes a nonreal pole

Let $v = [1]$. Then

$$s_n(\gamma) = \langle T^n v, v \rangle. \quad (\text{H10:22})$$

For nonreal z , the resolvent $I - zT$ is invertible because T is self-adjoint. Therefore

$$B_T(z) = \langle (I - zT)^{-1}v, v \rangle \quad (\text{H10:23})$$

is holomorphic on each open half-plane. Its expansion near zero agrees with the last series in (9).

The identity theorem for meromorphic functions now gives $B_T = B_\gamma$ on each half-plane. A nonreal shifted zero would give the nonzero residue (10), whereas B_T is holomorphic there. This contradiction excludes every nonreal zero.

Equivalence (12) follows. Its contrapositive gives (13): failure of positive semidefiniteness of a finite real symmetric matrix has a real vector of strictly negative quadratic value. This completes the proof.

5. What information the witness uses

The matrix $H_N(\gamma)$ uses derivatives of $\log A_\gamma$ only through order $2N+2$. These are determined by the regularized squared-amplitude jet through that same order, or by the original squared-amplitude jet through order $2m + 2N + 2$.

Thus a hypothetical zero at any distant height produces some finite local witness at a fixed known zero. The theorem asserts existence of the witness; it supplies no universal degree cutoff. The proof does not require the known zero to move or its first crossing direction to change.

6. Exact connection to the earlier amplitude hierarchy

The recovered manuscript defines the generalized Laguerre coefficients by

$$|f(x + iy)|^2 = \sum_{n \geq 0} L_n[f](x) y^{2n}. \quad (\text{H10:24})$$

Its established product identity H6 is

$$L_n[ab](x) = \sum_{j=0}^n L_j[a](x) L_{n-j}[b](x). \quad (\text{H10:25})$$

At the known zero, $f(\gamma + u) = u^m g_\gamma(u)$, and

$$|f(\gamma + iy)|^2 = y^{2m} |g_\gamma(iy)|^2.$$

Consequently,

$$\boxed{L_{m+n}[f](\gamma) = L_n[g_\gamma](0), \quad n \geq 0.} \quad (\text{H10:26})$$

In particular, the first reciprocal-zero moment is already encoded in the original hierarchy:

$$\boxed{s_0(\gamma) = \frac{L_1[g_\gamma](0)}{g_\gamma(0)^2} = \frac{L_{m+1}[f](\gamma)}{L_m[f](\gamma)}.} \quad (\text{H10:27})$$

Here $L_m[f](\gamma) = g_\gamma(0)^2 > 0$ is the established multiplicity-leading positivity H4.

For a simple zero, (27) becomes

$$s_0(\gamma) = \frac{f''(\gamma)^2}{4f'(\gamma)^2} - \frac{f'''(\gamma)}{3f'(\gamma)}, \quad L_2[f](\gamma) = f'(\gamma)^2 s_0(\gamma). \quad (\text{H10:28})$$

This identifies the precise old coefficient that would control the first new local sign.

The source's H5-H8 supply exact all-order compatibility. They do not assign a nonnegative sign to $L_{m+1}[f](\gamma)$ or to every matrix (11). The degree-seven positive forms in the later running mathematics concern integrated residuals in another variable. Applying them here would require a proved identity transferring their sign to (11).

7. The single remaining sign statement for this route

It suffices to prove, at one fixed established real zero γ ,

$$\boxed{\mathcal{L}_\gamma(P^2) = -\frac{1}{2} \sum_{j,k=0}^N \frac{c_j c_k}{(j+k+1)!} (\log A_\gamma)^{(j+k+2)}(0) \geq 0} \quad (\text{H10:29})$$

for every real polynomial $P(x) = \sum_{j=0}^N c_j x^j$, with unrestricted finite degree.

This is an analytical statement about the complete local amplitude class. The witness theorem shows exactly why a proof of (29) would settle the backward-propagation contradiction. Formula (29) is not established here.

The next admissible step is to obtain (29), or an exact sign-preserving representation of it, from an independently proved property of the actual theta or prime system. An equivalent formulation alone does not establish the sign.

8. Source record and attribution

1. **EXTENDED_MANUSCRIPT(2).md**, current source read, lines 493-563: explicit Laguerre formulas H3, multiplicity statement H4, and compatibility identities H5-H8. Lines 3787-3808: full-fingerprint reflection theorem and its explicit unproved zero-reality premise. Source identifier: .
2. **RUNNING_MATHEMATICS(2).md**, current source read, lines 93-174: complete mixed positivity through degree seven in the stated height range and the explicit all-degree remainder. Source identifier: .
3. Árpád Baricz and František Štampach, *The Hurwitz-type theorem for the regular Coulomb wave function via Hankel determinants*, arXiv:1708.07729v3, Section 4, Theorem 10. This gives the classical Grommer criterion for order-one real entire functions in terms of reciprocal-zero Hankel matrices. <https://arxiv.org/pdf/1708.07729>
4. The preceding paper attributes the criterion to J. Grommer, *Ganze transzendente Funktionen mit lauter reellen Nullstellen*, Journal für die reine und angewandte Mathematik 144 (1914), 114-166. That historical original was not separately consulted.
5. **Michaels_Complete_Theta_Kernel_Continuation.tex**, source identifier : complete density normalization and rapid decay, spatial J , and positive modular primitive. Sections 9-10 below use its density η , normalized by $f(w) = \int \eta(t) e^{iwt} dt$.
6. **Michaels_Prime_Color_Dual_Kernel_Bridge.md**, source identifier , Section 10: complete Euler logarithmic derivative and its domain of convergence.
7. **Prime_DNA_Build_v39_The_Two_Plane_Prime_Cocycle.pdf**, source identifier , Sections 4-7 and 15: positive two-channel conservation, fresh hidden channels, and the stated qualification on the infinite moving metric.
8. **Michaels_Density_Recovery_and_Parallel_Route_Comparison.md**, source identifier , Sections 5-6: exact primitive boundary identity and sufficiency of positivity on the two-trace class. Positivity on that whole class is an explicit hypothesis there.

The local formulation and proof above make the needed implication self-contained. The underlying reality criterion is classical; this note does not claim it as a new criterion or as a completed proof of RH.

9. Exact transfer to the original spatial kernel J

Use the complete even positive density η with

$$f(w) = \int_{\mathbb{R}} \eta(x) e^{iwx} dx, \quad J(x, y) = \int_0^\infty (x + y + 2r) \eta(x + r) \eta(y + r) dr. \quad (\text{H10:30})$$

For $c = (x + y)/2$ and $d = (x - y)/2$, evenness gives

$$J(x, y) = 2 \int_{|c|}^{\infty} v \eta(v + d) \eta(v - d) dv. \quad (\text{H10:31})$$

The existing bounds $\eta(t) \leq M_q e^{-qt^2}$, for every $q > 0$, imply

$$0 < J(x, y) \leq \frac{M_q^2}{2q} e^{-q(x^2+y^2)}. \quad (\text{H10:32})$$

Thus the following double integrals, including all their polynomial-exponential derivatives, converge absolutely. Pointwise positivity in (32) does not assert matrix positivity.

Define a Hermitian form, linear in its first argument, by

$$\mathcal{B}_J(a, b) = \iint_{\mathbb{R}^2} J(x, y) a(x) \overline{b(y)} dx dy. \quad (\text{H10:33})$$

For a real entire function a , write

$$W_a(z, w) = \frac{a(z) \overline{a'(w)} - a'(z) \overline{a(w)}}{z - \overline{w}}, \quad (\text{H10:34})$$

with the removable diagonal values included.

The spatial identity

$$(\partial_x + \partial_y) J(x, y) = -(x + y) \eta(x) \eta(y) \quad (\text{H10:35})$$

and integration by parts give the complete transform identity

$$\boxed{\mathcal{B}_J\left(e^{i(\gamma+z)x}, e^{i(\gamma+w)x}\right) = W_h(z, w), \quad h(z) = f(\gamma + z).} \quad (\text{H10:36})$$

To check the sign in (36), put $\alpha = \gamma + z$, $\beta = \gamma + \overline{w}$. The transform of the left side of (35) is $-i(\alpha - \beta) \widehat{J}$; the transform of its right side is $-i[f(\alpha) f'(\beta) - f'(\alpha) f(\beta)]$. Their equality proves (36) off the diagonal, and analytic continuation gives the removable values. The rapid-decay bounds justify the boundary terms and differentiation.

Since $h(z) = z^m g_\gamma(z)$, direct differentiation proves

$$\boxed{W_h(z, w) = (z\overline{w})^{m-1} \left[m g_\gamma(z) \overline{g_\gamma(w)} + z\overline{w} W_{g_\gamma}(z, w) \right].} \quad (\text{H10:37})$$

This retains the entire known-zero contribution.

Define the removable kernel

$$R_m(z, w) = \frac{W_h(z, w)}{(z\overline{w})^{m-1}}.$$

Then $R_m(0, 0) = m g_\gamma(0)^2 > 0$, and the exact Schur residual is

$$\boxed{R_m(z, w) - \frac{R_m(z, 0) R_m(0, w)}{R_m(0, 0)} = z\overline{w} W_{g_\gamma}(z, w).} \quad (\text{H10:38})$$

Equation (38) is an algebraic identity. Applying positivity of Schur complements requires positivity of the complete kernel R_m on the tested combinations; positivity of its single pivot alone does not give that premise.

Explicit functions producing the local form

Put

$$e_k(x) = e^{i\gamma x} \frac{(ix)^k}{k!}, \quad E_z(x) = \frac{e^{i(\gamma+z)x} - \sum_{k=0}^{m-2} z^k e_k(x)}{z^{m-1}}, \quad (\text{H10:39})$$

where the sum is empty for $m = 1$. All removable values are understood, so $E_0 = e_{m-1}$.

On a zero-free neighborhood of the origin for g_γ , define

$$U_z(x) = \frac{1}{z} \left(\frac{E_z(x)}{g_\gamma(z)} - \frac{e_{m-1}(x)}{g_\gamma(0)} \right). \quad (\text{H10:40})$$

The second subtraction is precisely the known-zero Schur correction.

The factors in (37) show that the rows of (36) corresponding to $e_0, \dots, e_{m-2}$ vanish. Therefore (36)-(40) yield

$$\begin{aligned} \mathcal{B}_J(U_z, U_w) &= \frac{W_{g_\gamma}(z, w)}{g_\gamma(z) \overline{g_\gamma(w)}} \\ &= \frac{-g'_\gamma(z)/g_\gamma(z) + \overline{g'_\gamma(w)/g_\gamma(w)}}{z - \overline{w}} \\ &= \sum_{j,k \geq 0} s_{j+k}(\gamma) z^j \overline{w}^k. \end{aligned} \quad (\text{H10:41})$$

For a fully explicit coefficient form, write $1/g_\gamma(z) = \sum_{r \geq 0} b_r z^r$. Expanding (40) gives

$$U_z(x) = \sum_{j \geq 0} a_j(x) z^j, \quad a_j(x) = \sum_{r=0}^{j+1} b_r e_{m+j-r}(x). \quad (\text{H10:42})$$

Thus

$$s_{j+k}(\gamma) = \mathcal{B}_J(a_j, a_k), \quad \mathcal{L}_\gamma(P^2) = \mathcal{B}_J(A_P, A_P), \quad A_P = \sum_{j=0}^N c_j a_j. \quad (\text{H10:43})$$

This is the exact transfer sought: the local form is a specified compression of the original J , with both complete tails and every coefficient cross retained.

At a simple known zero,

$$U_z(x) = \frac{e^{i\gamma x}}{z} \left(\frac{e^{izx}}{g_\gamma(z)} - \frac{1}{g_\gamma(0)} \right), \quad a_0(x) = e^{i\gamma x} \left(\frac{ix}{g_\gamma(0)} - \frac{g'_\gamma(0)}{g_\gamma(0)^2} \right). \quad (\text{H10:44})$$

The earlier modular-primitive trace constraints are also checkable exactly. Let r be the positive primitive with $\int r(x) e^{i(\gamma+z)x} dx = h(z)/D(z)$, where $D(z) = 1/4 + (\gamma + z)^2$. Equations (39)-(40) give, locally,

$$\int r(x) U_z(x) dx = \frac{1}{D(z)}, \quad \int r'(x) U_z(x) dx = -\frac{i(\gamma + z)}{D(z)}. \quad (\text{H10:45})$$

Hence these functions do not automatically belong to the earlier class with both primitive traces zero. Transferring a theorem on that class requires retaining these explicit traces.

10. Direct theta-density sign check

The multiplicity condition says $\int t^j e^{i\gamma t} \eta(t) dt = 0$ for $0 \leq j < m$. It gives the complete Fourier density of the zero-removed function:

$$\begin{aligned} \psi_{\gamma,m}(t) &= \frac{(-i)^m}{(m-1)!} \int_{-\infty}^t (t-v)^{m-1} e^{i\gamma v} \eta(v) dv \\ &= \frac{i^m}{(m-1)!} \int_t^{\infty} (v-t)^{m-1} e^{i\gamma v} \eta(v) dv, \\ g_{\gamma}(u) &= \int_{\mathbb{R}} \psi_{\gamma,m}(t) e^{iut} dt. \end{aligned} \tag{H10:46}$$

Moment cancellation proves equality of the two primitives. Their respective tails and the complete theta decay justify every integration by parts.

The normalized density $\mu_{\gamma} = \psi_{\gamma,m}/g_{\gamma}(0)$ has total mass 1 and satisfies $\mu_{\gamma}(-t) = \overline{\mu_{\gamma}(t)}$. It is necessarily complex-valued: otherwise its m -th derivative would be real, whereas

$$\psi_{\gamma,m}^{(m)}(t) = (-i)^m e^{i\gamma t} \eta(t) \tag{H10:47}$$

is not real for all t . Here $\gamma \neq 0$, since $\Xi(0) > 0$, and $\eta(t) > 0$.

Define its complex cumulants by

$$\kappa_r = \frac{d^r}{dv^r} \log \int_{\mathbb{R}} e^{vt} \mu_{\gamma}(t) dt \Big|_{v=0}.$$

Then

$$s_n(\gamma) = \frac{i^n \kappa_{n+2}}{(n+1)!}. \tag{H10:48}$$

For example,

$$s_0 = \kappa_2 = \frac{1}{2} \iint (t-v)^2 \mu_{\gamma}(t) \mu_{\gamma}(v) dt dv. \tag{H10:49}$$

This exact expression has no conjugation between its two density factors. Consequently the original positive-density Gram argument for η does not directly assign its sign. The established logarithmic-curvature bounds on η do not, by this transformation alone, become bounds on a positive zero-removed density.

11. Direct completed prime-sign check

Let $s_* = 1/2 + i\gamma$, $\ell(u) = g'_{\gamma}(u)/g_{\gamma}(u)$, and

$$A(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2} \log \pi + \frac{1}{2} \frac{\Gamma'(s/2)}{\Gamma(s/2)}.$$

In the Euler-convergent half-plane $\text{Im } u < -1/2$,

$$\ell(u) = iA(s_* + iu) - i \sum_{n \geq 2} \Lambda(n) n^{-s_* - iu} - \frac{m}{u}. \tag{H10:50}$$

The two-variable holomorphic kernel corresponding to (41) is

$$\begin{aligned} \mathcal{Q}_\gamma(u, v) &= -\frac{\ell(u) - \ell(v)}{u - v} \\ &= \int_0^1 \left[A'(s_* + i[\tau u + (1 - \tau)v]) + \sum_{n \geq 2} \Lambda(n) \log n n^{-s_* - i[\tau u + (1 - \tau)v]} \right] d\tau - \frac{m}{uv}. \end{aligned} \quad (\text{H10:51})$$

Initially both arguments lie in the Euler half-plane. Meromorphic continuation of the whole completed expression yields the local analytic kernel at $u = v = 0$. Its apparent singularities cancel there. The prime series alone is not used termwise at the critical line.

An exact individual-term test explains why the positive prime-color blocks cannot simply be substituted for (51). For one prime power n , put

$$L = \log n, \quad \theta = \gamma L, \quad w_\sigma = \Lambda(n) L n^{-\sigma} > 0.$$

The real part of its Euler logarithm term on the line $\text{Re } s = \sigma$ is $\Lambda(n) n^{-\sigma} \cos(\theta + Lu)/L$. Applying the same local moment extraction to this single finite term gives

$$T_n = w_\sigma \begin{pmatrix} \cos \theta & -L \sin \theta/2 \\ -L \sin \theta/2 & -L^2 \cos \theta/6 \end{pmatrix},$$

and

$$\boxed{\det T_n = -w_\sigma^2 L^2 \left(\frac{\cos^2 \theta}{6} + \frac{\sin^2 \theta}{4} \right) < 0.} \quad (\text{H10:52})$$

This holds for every phase, including on Euler-convergent lines $\sigma > 1$. It is a diagnostic of the individual-term extraction, not a convergent decomposition of the critical-line matrix into such terms.

More generally, if $\mathcal{E}_P(z) = \sum_j c_j z^j / j!$, the same term contributes

$$w_\sigma \text{Re} \left[e^{i\theta} \int_0^1 \mathcal{E}_P(iL\tau) \mathcal{E}_P(iL(1 - \tau)) d\tau \right]. \quad (\text{H10:53})$$

The beta integral retains every cross coefficient and produces the factorial in (4). This is a signed bilinear expression; the individual positive color norms do not give its sign through a congruence or compression.

Equation (52) does not assert a negative sign for the completed local form. The full correction and continuation in (51) may compensate the individual signed terms. Such compensation has not been established here.

12. Outcome of the direct attempt

The exact transfer (43), the Schur identity (38), and the complete theta/prime expressions (46) and (51) are established. The unchanged remaining task is

$$\mathcal{B}_J(A_P, A_P) \geq 0 \quad \text{for every real polynomial } P$$

at one fixed known zero, with the functions A_P defined in (42)-(43). The recovered positivity statements do not yet prove this restricted form. The backward-detection theorem remains valid; the independent sign needed for its contradiction remains open.

13. Which plane and which positive form?

The current calculation takes place in the spatial test-function space with form $\mathcal{B}_J$, and then in its explicitly identified local coefficient space with matrix $H_N(\gamma)$. The earlier plane constructions carry the following distinct, precisely defined forms.

Earlier space	Established structure	Exact scope
Infinite analytic lift $\mathcal{E}$	$p_R(c) = \sum_j c_j R^j$, finite for every $R > 0$	A Fréchet space recovering the full translated entire function. These seminorms do not identify the present J -form.
Quadratic amplitude lift $G = cc^*$	$q^* G q = c^* q ^2 \geq 0$	Proved for every finitely supported complex q ; the full weighted operators are positive rank one.
Prime two-plane Hilbert system	$\ V_p f\ _2^2 + \ W_p f\ _2^2 = \ f\ _2^2$	Proved for the stated L^2 inputs, with fresh hidden channels under iteration and the finite-terminal reserve retained.
Current local J -family	$\mathcal{B}_J(A_P, A_P) = c^T H_N(\gamma) c$	The complete transfer identity is proved. Its sign for all polynomials is the target.

The quadratic and prime squared norms cannot be negative. Applying that fact to the present target requires a proved sign-preserving implication, such as an identity with a positive form or a lower bound by one.

The recovered actual quadratic lift uses $G_{jk} = c_j \overline{c_k}$. The present kernel (41) instead uses the divided difference of $-g'_\gamma/g_\gamma$. The sources give the data required to construct that divided difference, but their outer-product positivity theorem does not state a positive norm identity for it. In particular, the source translation law $G(z+h) = T_h G(z) T_h^*$ preserves its own quadratic form; it does not identify a new signed observable with that form.

For the prime system, the visible and hidden squared norms are both nonnegative and their sum is conserved. The checked source supplies neither an identity nor a lower comparison transferring that positivity to the completed scalar form. Its finite moving-metric route likewise retains a separate requirement on a nondegenerate infinite limit.

No negative value of the completed actual local J -form is established by this plane check. It identifies the form whose nonnegativity must be proved.

14. Exact block decomposition at the known zero

This next calculation identifies all directions retained by the current family. Keep the e_k, a_j of (39), (42), and put

$$v = \frac{e_{m-1}}{g_\gamma(0)}, \quad V_N = \text{span}_{\mathbb{C}}\{e_0, \dots, e_{m+N}\}. \quad (\text{H10:54})$$

Equations (36)-(41) give

$$\mathcal{B}_J(e_k, e_\ell) = 0 \quad (k < m-1, \ell \geq 0), \quad \mathcal{B}_J(v, v) = m, \quad \mathcal{B}_J(v, a_j) = 0. \quad (\text{H10:55})$$

Thus the lower jets are radical directions for this polynomial-exponential space, and the normalized known-zero direction is strictly positive.

The ordered list

$$e_0, \dots, e_{m-2}, \quad v, \quad a_0, \dots, a_N \quad (\text{H10:56})$$

is a basis of V_N . Indeed a_j has nonzero leading coefficient $1/g_\gamma(0)$ on e_{m+j} , so the change of basis is invertible triangular. Its exact Gram matrix is

$$\boxed{\text{Gram}_{\mathcal{B}_J}(V_N) \cong 0_{m-1} \oplus [m] \oplus H_N(\gamma).} \quad (\text{H10:57})$$

The zero block is absent when $m = 1$.

For completeness, write $g_\gamma(z) = \sum_{r \geq 0} g_r z^r$ and $q = \sum_{k=0}^{m+N} d_k e_k$. Define

$$c_j = \sum_{\ell=j}^N g_{\ell-j} d_{m+\ell}, \quad \tau = \sum_{r=0}^{N+1} g_r d_{m-1+r}. \quad (\text{H10:58})$$

The reciprocal-series identity for g_γ gives

$$q = \sum_{k=0}^{m-2} d_k e_k + \tau v + \sum_{j=0}^N c_j a_j, \quad \boxed{\mathcal{B}_J(q, q) = m|\tau|^2 + c^* H_N(\gamma) c.} \quad (\text{H10:59})$$

Every cross term between the three displayed components vanishes by (55).

Consequently any negative direction of this full jet form would occur in the local block H_N . The proved coordinate transformation preserves that negative index. Using another positive norm to establish the sign of (59) requires a proved connection between the forms, such as an equality or a suitable lower comparison.

15. Explicit removal of both primitive boundary terms

There is a further exact reduction using the previous primitive identity, with no numerical estimates.

Write B for the positive modular primitive denoted r in (45), and let

$$L = \partial_x^2 - \frac{1}{4}, \quad J_B(x, y) = \int_0^\infty (x + y + 2s) B(x + s) B(y + s) ds.$$

The recovered identity is

$$J = L_x L_y J_B + 2B'(x)B'(y) - \frac{1}{2}B(x)B(y). \quad (\text{H10:60})$$

Exponential decay of B and its derivatives permits integration by parts against every polynomial-exponential function used here. Hence, with $\tau_b(a) = \int b(x)a(x) dx$,

$$\mathcal{B}_J(A_P, A_P) = \mathcal{B}_{J_B}(LA_P, LA_P) + 2|\tau_{B'}(A_P)|^2 - \frac{1}{2}|\tau_B(A_P)|^2. \quad (\text{H10:61})$$

Let

$$D_0 = \gamma^2 + \frac{1}{4}, \quad \lambda_+ = \frac{1}{-\gamma + i/2}, \quad Z_P = \lambda_+ P(\lambda_+), \quad (\text{H10:62})$$

and assume P has real coefficients. Expanding the two rational functions in (45) by partial fractions gives

$$\boxed{\tau_B(A_P) = -2 \operatorname{Im} Z_P, \quad \tau_{B'}(A_P) = i \operatorname{Re} Z_P.} \quad (\text{H10:63})$$

For example the coefficient of z^j in $1/D(z)$ is $-2\text{Im}(\lambda_+^{j+1})$; the coefficient in $-i(\gamma + z)/D(z)$ is $i\text{Re}(\lambda_+^{j+1})$. This proves (63) for every polynomial.

Substitution into (61) yields the exact correction

$$\mathcal{L}_\gamma(P^2) = \mathcal{B}_{J_B}(LA_P, LA_P) + 2\text{Re}(Z_P^2). \quad (\text{H10:64})$$

Define

$$\boxed{\chi_\gamma(X) = D_0X^2 + 2\gamma X + 1 = (1 + \gamma X)^2 + \frac{1}{4}X^2.} \quad (\text{H10:65})$$

Its roots are λ_+ and $\overline{\lambda_+}$. Since P is real and $\lambda_+ \neq 0$,

$$\tau_B(A_P) = \tau_{B'}(A_P) = 0 \iff P = \chi_\gamma R \text{ for a real polynomial } R. \quad (\text{H10:66})$$

We have therefore reached the earlier two-trace class by an explicit formula:

$$\boxed{\mathcal{L}_\gamma((\chi_\gamma R)^2) = \mathcal{B}_J(A_{\chi_\gamma R}, A_{\chi_\gamma R}) = \mathcal{B}_{J_B}(LA_{\chi_\gamma R}, LA_{\chi_\gamma R}).} \quad (\text{H10:67})$$

Both boundary terms vanish individually.

No possible off-critical zero is lost

For every remaining zero ρ of Ξ ,

$$\chi_\gamma\left(\frac{1}{\rho - \gamma}\right) = \frac{\rho^2 + 1/4}{(\rho - \gamma)^2}. \quad (\text{H10:68})$$

The only possible vanishing locations in (68) would be $\rho = \pm i/2$, but independently

$$\Xi(i/2) = \xi(0) = \frac{1}{2}, \quad \Xi(-i/2) = \xi(1) = \frac{1}{2}. \quad (\text{H10:69})$$

Neither is a zero.

More explicitly, set

$$t_n = \mathcal{L}_\gamma(X^n \chi_\gamma(X)^2).$$

Put $r_\gamma = \sup_{\rho \neq \gamma} |\lambda_\rho|$. Equations (6)-(8) give $|t_n| \leq C_\chi r_\gamma^n$ for a finite constant C_χ . Its generating function is

$$\sum_{n \geq 0} t_n z^n = \sum_{\rho \neq \gamma} \frac{m_\rho \lambda_\rho^2 \chi_\gamma(\lambda_\rho)^2}{1 - \lambda_\rho z}, \quad (\text{H10:70})$$

initially for $|z| < r_\gamma^{-1}$, with a normally convergent meromorphic continuation away from its poles. At $z = \rho - \gamma$ its residue is

$$-\frac{m_\rho}{\rho - \gamma} \chi_\gamma((\rho - \gamma)^{-1})^2 \neq 0. \quad (\text{H10:71})$$

If $\mathcal{L}_\gamma((\chi_\gamma R)^2) \geq 0$ for every real polynomial R , the bounded-multiplication and resolvent argument in Section 4 applies to t_n . It excludes every nonreal pole in (70), hence every off-critical zero. This proves that positivity on the restricted family (67) is sufficient.

Equivalently, the exact retained zero sum is

$$\mathcal{L}_\gamma((\chi_\gamma R)^2) = \sum_{\rho \neq \gamma} m_\rho \frac{(\rho^2 + 1/4)^2}{(\rho - \gamma)^6} R((\rho - \gamma)^{-1})^2, \quad (\text{H10:72})$$

with absolute convergence and all other zero contributions included.

Current sign status

The recovered earlier theorem proves sufficiency of positivity on a two-trace primitive class. It states that positivity as a hypothesis. Equation (67) supplies the exact local family inside that class and removes both explicit boundary corrections. Positivity of the remaining J_B -form on this family is still unproved in the checked material and in this continuation.

16. Applying the previous prime-power positivity theorem

The closest recovered source theorem is the vacuum-tangent identity in *Prime_DNA_Genesis_Half_Density_and_Completed_First_Loss_Gate_Build_v23.md*, equations (5.2) and (5.7). For a finitely supported complex sequence $x = (x_n)$, put

$$K_z(m, n) = \frac{\gcd(m, n)^z}{(mn)^{z/2}}, \quad \mathcal{Q}_z[x] = \sum_{m, n} \overline{x_m} K_z(m, n) x_n.$$

The already-proved Smith-Jordan factorization is

$$\mathcal{Q}_z[x] = \sum_{d \geq 1} J_z(d) \left| \sum_{d|n} \frac{x_n}{n^{z/2}} \right|^2 \quad (z > 0). \quad (\text{H10:73})$$

Here $J_z(d)$ is the Jordan arithmetic function, distinct from the spatial kernel J .

If $\sum_n x_n = 0$, differentiation at $z = 0$ gives the established positivity

$$\boxed{\mathcal{Q}'_0[x] = \sum_{q=p^k} (\log p) \left| \sum_{q|n} x_n \right|^2 \geq 0.} \quad (\text{H10:74})$$

The vacuum square has zero derivative on this subspace. The derivative of $J_z(p^k) = p^{kz} - p^{(k-1)z}$ at zero is $\log p$, and every term with two or more distinct prime factors has zero first derivative. All sums here are finite for finitely supported x . Thus (74) covers every coefficient combination in its stated class.

Exact coefficient transformation for the present target

For $R(X) = \sum_{k=0}^N d_k X^k$, multiplication by χ_γ is the explicit matrix

$$(C_{\gamma, N})_{jk} = \delta_{j,k} + 2\gamma \delta_{j,k+1} + D_0 \delta_{j,k+2}, \quad 0 \leq j \leq N+2, \quad 0 \leq k \leq N. \quad (\text{H10:75})$$

Consequently the required matrix is exactly

$$\boxed{T_{\gamma, N} = C_{\gamma, N}^T H_{N+2}(\gamma) C_{\gamma, N}, \quad \mathcal{L}_\gamma((\chi_\gamma R)^2) = d^T T_{\gamma, N} d.} \quad (\text{H10:76})$$

This is an exact compression by the rectangular coefficient matrix $C_{\gamma, N}$. It transfers any proved positivity of H_{N+2} to $T_{\gamma, N}$. It is not an invertible change of basis on H_{N+2} . Equations (73)-(74) prove positivity of an arithmetic form; they do not identify that form with H_{N+2} or $T_{\gamma, N}$.

To use (74) directly, it would suffice to construct an admissible assignment $R \mapsto x(R)$, with $\sum_n x_n(R) = 0$, and prove

$$\boxed{\mathcal{B}_{J_B}(LA_{\chi_\gamma R}, LA_{\chi_\gamma R}) = \sum_{q=p^k} (\log p) \left| \sum_{q|n} x_n(R) \right|^2}. \quad (\text{H10:77})$$

Equation (77) is **one sufficient transport identity for applying (74), and is not an established equality**. It is not required for the contradiction proved in Section 15. An independently proved lower bound of the left side by the nonnegative right side would also suffice. A linear assignment in an exact equality would identify all cross terms by polarization. An infinite assignment or a limit of finite assignments would require the relevant convergence justified. Defining the assignment through a square root of the unknown target matrix would already assume its positivity.

The original Build v23 proof supplies (74); no assignment satisfying (77) was recovered in this check. The two zero primitive traces proved in Section 15 do not, by themselves, construct a zero-sum arithmetic sequence or identify its energy.

The other checked source, JTR3 in *RUNNING_MATHEMATICS(2).md*, proves every frequency Gram positive through $2 \int D_v |\sum_j z_j e^{-2ix_j t}|^2 dt$. Its equations (JTR-07)-(JTR-09) retain a signed finite center matrix. The source explicitly states that the improvement alone does not make that center matrix positive. That theorem therefore does not supply (77) either.

17. Direct transfer through the positive spatial feature norm

The positive spatial feature norm must also be distinguished from the spatial mixed form. Define

$$(Sq)(r) = \int_{\mathbb{R}} \eta(x+r)q(x) dx, \quad (Aq)(r) = \int_{\mathbb{R}} (x+r)\eta(x+r)q(x) dx.$$

For compactly supported $q \in L^2(\mathbb{R})$, both features belong to $L^2(0, \infty)$, their squared norms are nonnegative, and the exact target form is

$$\mathcal{B}_J(q, q) = 2 \operatorname{Re} \int_0^\infty (Aq)(r) \overline{(Sq)(r)} dr = \frac{1}{2} (\|Sq + Aq\|_2^2 - \|Sq - Aq\|_2^2). \quad (\text{H10:78})$$

Thus positivity of the two feature norms leaves a signed difference.

The actual functions A_P , for real polynomials P , have an additional domain issue. The generating identity (39)-(41) gives

$$(SA_P)(r) = e^{-i\gamma r} \mathcal{E}_P(-ir), \quad \mathcal{E}_P(t) = \sum_j c_j t^j / j!, \quad P(X) = \sum_j c_j X^j. \quad (\text{H10:79})$$

For every nonzero P , this is not square-integrable on the half-line. Formula (78) cannot be applied to these functions as a difference of finite norms.

There is an exact identity that retains the boundary. For every finite $b > 0$,

$$\begin{aligned} J(x, y) &= \int_0^b (x+y+2r)\eta(x+r)\eta(y+r) dr + J(x+b, y+b), \\ \mathcal{L}_\gamma(P^2) &= 2 \operatorname{Re} \int_0^b (AA_P)(r) \overline{(SA_P)(r)} dr + \mathcal{B}_J(A_P(\cdot - b), A_P(\cdot - b)). \end{aligned} \quad (\text{H10:80})$$

All exchanges of integration in (80) are justified at finite b by Gaussian decay and polynomial growth.

The final term is explicit. Define

$$c_k(b) = \sum_{j=k}^N c_j \frac{(-ib)^{j-k}}{(j-k)!}, \quad P_b(X) = \sum_{k=0}^N c_k(b) X^k, \quad \tau_b = \sum_{j=0}^N c_j \frac{(-ib)^{j+1}}{(j+1)!}.$$

Translation of U_z , followed by the block decomposition (55)-(59), yields

$$\boxed{\mathcal{B}_J(A_P(\cdot - b), A_P(\cdot - b)) = m|\tau_b|^2 + \mathcal{L}_\gamma(P_b \overline{P_b}).} \quad (\text{H10:81})$$

Here the bar conjugates polynomial coefficients, and $\mathcal{L}_\gamma$ is extended complex-linearly. Indeed, modulo the radical spanned by $e_0, \dots, e_{m-2}$,

$$U_z(x - b) = e^{-i\gamma b} \left(e^{-izb} U_z(x) + \frac{e^{-izb} - 1}{z} v(x) \right).$$

Taking coefficients and applying the orthogonality to v proves (81).

The restriction $P = \chi_\gamma R$ eliminates the original two primitive traces. It does not eliminate this translated boundary contribution: $\tau'_b = -i\mathcal{E}_P(-ib)$, so τ_b is not identically zero for nonzero P . More explicitly, if $\deg P = N$, the boundary form in (81) is a polynomial in b with leading term $m|c_N|^2 b^{2N+2}/((N+1)!)^2$; the local term has degree at most $2N$. Thus the complete boundary form does not vanish identically either. Its positive known-zero component cannot be dropped in this equality. The formula retains another evaluation of the local form and assigns it no independent sign.

Outcome of the requested positivity transfer

The earlier prime-power theorem (74), the exact coefficient map (75)-(76), and the complete spatial identities (78)-(81) have been checked with their hypotheses retained. They do not establish the target positivity. The outstanding mathematical statement is (67) nonnegative for all real R . An equality such as (77), a valid lower bound by a known nonnegative form, or a direct sign proof could establish it. The boundary analysis only rules out dropping the explicit tail in that particular representation; it does not rule out another transfer or require separate positivity of its terms. No contradiction and no unconditional RH proof are claimed by this continuation.

We are aiming to prove positivity of the complete local form at one known zero; it suffices to prove (67) nonnegative for every real polynomial R , with both primitive traces exactly zero and every possible off-critical zero still detected.

Part V

Research directives and continuing ledger

The following directives record questions and requested calculations between checkpoints. They are research instructions, not assertions that the requested estimates were proved. They are included to preserve the reasoning history.

After Checkpoint 12: the null equation

Source: Michaels_Checkpoint_12_Null_Equation_Followup.md. Equation tags carry the source prefix D12; references within this unit retain their original local numbering.

Prepared 22 September 2026. This document proposes the next calculations; it does not claim that the open first-null estimate is proved.

Basis for the next prompt

Read in full: Michaels_Four_Closure_Groups_Checkpoint_12.md, including Sections 2.4, 4.3, and 5.2-5.4, and the current running ledger.

The finite factorization defines its middle operator exactly but supplies no estimate for that operator. The proposed first-null remainder lemma is a valid contradiction criterion: the note already derives the opposite strict inequality for every hypothetical null sequence. The next useful step is therefore an independent structural calculation from the null equation.

Primary sources checked: Suzuki, arXiv:2606.09096v2, especially the localized quadratic form and its logarithmic form domain; Conrey, 1989, Theorem 1, which supplies the simple-critical-zero density input cited by checkpoint 12. The multiplier and support-gap tasks below are proposed deductions to derive with their exact domains.

Prompt to send

Continue from checkpoint 12 and preserve the target

$$\mathcal{E}_N = \|[\Delta(\rho_j, \rho_i)]_{i,j \leq N}\|_F^2, \quad \liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1.$$

Retain the established domain passage, the complete correction, and the constants $K_0 < K_*$. Work from the first-null equation to obtain an independent estimate. Further improvement of the extraction constant is secondary until the complete remainder is controlled.

The exact factorization $B_N = X_N^{-1/2} D_N X_N^{-1/2}$ is an algebraic identity. Treat a bound on its middle operator as an additional theorem requiring proof. Likewise, checkpoint 12 already proves

$$\limsup_n \frac{\mathcal{T}(q_n, q_n)}{\mathcal{S}(q_n, q_n)} \leq -1 + K_0$$

for every hypothetical first-null sequence. Repeating the desired opposite inequality only restates the contradiction criterion. This round must derive an additional identity or estimate from the actual xi kernel before returning to that criterion.

Closure group A – Null-vector multipliers and the exact arithmetic form

Fix a hypothetical first degenerate interval $I_a = (-a, a)$, with $Q_a \geq 0$ and $0 \neq v \in \ker A_a$. Establish that one may choose a nonzero real null vector of definite parity, using conjugation and reflection invariance. This does not require simplicity of the null eigenvalue.

First prove that smooth multipliers on the closed interval preserve the localized form domain. Use the actual logarithmic form and the finite prime-shift terms. Do not impose traces on v .

For real smooth ϕ , derive from $Q_a(v, \phi^2 v) = 0$ the form version of

$$Q_a(\phi v, \phi v) = -\frac{1}{2} \langle [M_\phi, [M_\phi, A_a]] v, v \rangle = \frac{1}{2} \iint g''(x-y)(\phi(x) - \phi(y))^2 v(y) v(x) dy dx.$$

Verify the signs directly from the adopted inner-product convention. Interpret the right side through core approximation and the full distributional kernel. The difference squared removes the diagonal delta term and regularizes the origin singularity. Establish this before taking limits.

Carry out two explicit calculations:

1. $\phi(x) = x$, displaying every prime-shift autocorrelation with its $(\log n)^2$ factor, the archimedean integral, and the complete smooth contribution.
2. The paired multipliers $\cos(\tau x)$ and $\sin(\tau x)$, for real τ . Define the zero-extended autocorrelation

$$C_v(t) = \int_{\mathbb{R}} v(x+t)v(x) dx.$$

Derive the complete necessary condition

$$0 \leq Q_a(\cos(\tau \cdot) v) + Q_a(\sin(\tau \cdot) v) = \int_{-2a}^{2a} g''(t)(1 - \cos(\tau t)) C_v(t) dt.$$

Use the exact g on the whole interval $[-2a, 2a]$. If a local expansion is used, define and retain its exact remainder beyond its convergence region. Preserve the signs of all autocorrelations; positive definiteness of C_v does not imply its pointwise nonnegativity.

Polarize the multiplier identity to obtain the finite matrix $[Q_a(\phi_j v, \phi_i v)]$. Test whether the actual prime and archimedean structure forces a negative quadratic direction in this matrix, contradicting $Q_a \geq 0$, or supplies the missing lower estimate for the complete remainder. Calculate the actual coefficients. A matrix whose positivity merely repeats $Q_a \geq 0$ is a new representation, not yet a contradiction.

If hyperbolic multipliers help with the completed pole term, retain their entire matrix: a difference of two nonnegative energies has no prescribed sign.

Closure group B – Endpoint identities and the exterior transform

Set

$$f_v(t) = \int_{-a}^a g(t-y)v(y) dy.$$

The null equation is $f_v(t) = \alpha t + \beta$ inside I_a . Prove that f_v is C^1 near the closed interval using $g' \in L_{\text{loc}}^2$. Then retain the valid endpoint identities

$$f_v(\pm a) = \pm a\alpha + \beta, \quad f'_v(\pm a) = \alpha.$$

These are traces of the convolution, not traces of v . For even v , $\alpha = 0$; for odd v , $\beta = 0$. Expand the resulting signed endpoint integrals using the exact prime packets and archimedean terms. Insert them into Group A before applying absolute values.

Next define the spectral distribution, with convergence justified against compact smooth tests,

$$Z_v(t) = \sum_{\gamma} m_{\gamma} \widehat{v}(\gamma) e^{-i\gamma t}.$$

Prove its identification with the completed operator distribution and the gap $Z_v|_{(-a,a)} = 0$. Let

$$\mathcal{B}_v(t) = f_v(t) - \alpha t - \beta.$$

Investigate the full identity $Z_v = -\mathcal{B}_v''$. The residual vanishes inside the interval, and its endpoint value and first derivative vanish. Derive the two exterior Fourier-Laplace representations, including all convergence conditions and any subtraction or entire terms. Initially work outside the zero strip. Derive the precise sign from the chosen Cauchy denominator; do not mix $(z - \gamma)^{-1}$ with $(\gamma - z)^{-1}$.

Prove how these exterior transforms continue to the contour and pole evaluations occurring in checkpoint 12, Section 5.7. Retain the actual meromorphic poles and derivatives. If the gap supplies only an exponential factor, quantify it: that factor alone does not eliminate the near-real-axis point-evaluation singularity. Identify the extra cancellation furnished by the affine endpoint identities or Group A.

Closure group C – Bound the complete pairing on the constrained family

Return to the signed contour-and-residue expression for $\mathcal{T}$, carrying the exact order of the zero-cutoff and core-sequence limits.

Substitute Groups A and B before estimating individual evaluations. Combine each pole contribution with its associated contour contribution, and use the exact removable-zero factors. Determine whether the apparent evaluation singularities cancel in this combined expression. Exhibit the local expansion and its remainder; a cancellation claimed only after normalization must retain that normalization.

The relevant family is now constrained by the null equation, parity, convolution endpoint identities, and the multiplier identities. Derive the norm or quadratic estimate for that family directly. Do not apply unrestricted Hardy point evaluation and then repeat the same unbounded $1/\text{Im } \gamma$ estimate.

Attempt either of the following closing outcomes:

- A direct contradiction to a nonzero null vector using the complete arithmetic multiplier matrix and endpoint constraints.
- An independently established estimate on one valid null core sequence with

$$\liminf_n \frac{\mathcal{T}(q_n, q_n)}{\mathcal{S}(q_n, q_n)} \geq -1 + K_*.$$

Dependence on the fixed interval a is permitted when it still gives a strictly positive contradiction for every finite first-crossing interval. Avoid imposing a global uniform estimate if a proved interval-dependent estimate suffices. If using a different positive margin, calculate it and write its full implication.

Use the proved closedness/injectivity result exactly as stated. It does not assert that the Suzuki transform is a bounded map on all of $L^2(I_a)$. Any additional graph-domain comparison must be proved only where the new calculation needs it.

Closing statements

Run Groups A and B in parallel if available, then combine their identities in Group C. End each group with the exact theorem obtained, its domain, and the explicit new inequality it enables. If a proposed inference fails, give the actual uncanceled expression or a counterexample to that particular inference.

At least the x -multiplier, paired trigonometric calculation, endpoint constraints, and their insertion into the complete signed expression must be attempted in this round. A result may remain open; identify the precise attempted deduction and retain its proved predecessors.

Conclude with the strongest justified mathematical statement. If the null vector is excluded, give the entire return chain to Weil positivity, RH, normalized Suzuki equality, and $\mathcal{E}_N = 0$. Otherwise state one quantified remaining lemma that uses the newly derived structure, and identify the exact line still unproved. Do not label a hypothesis as a closing theorem.

Short follow-up if the reply returns only another reduction

Use the new identities to perform one concrete signed comparison. Fix the simplest multiplier family for which every source term can be evaluated symbolically. Write its complete matrix on the hypothetical parity null vector, substitute the convolution endpoint equations, and calculate its remaining determinant or quadratic form. If the sign is unresolved, display the exact autocorrelation integral carrying that sign. Then map that integral into the residue-boundary expression and show whether it is the same remaining obstruction or an independently smaller estimate. Preserve all terms and the original metric.

After Checkpoint 13: graph, dilation, and extension

Source: *Michaels_Checkpoint_13_Graph_Dilation_Extension_Followup.md*. Equation tags carry the source prefix *D13*; references within this unit retain their original local numbering.

Prepared 2026-09-23. This is a research prompt, not a proof of the remaining estimate.

Prompt to send

Continue from the complete Checkpoint 13 note and current ledger. Preserve the original target

$$\liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1$$

independently of RH, the actual Suzuki metric, all pole-boundary interactions, and both parities.

Produce a calculated next checkpoint. Work through the following groups in priority order, with explicit proofs or the exact failed inequality. A closing statement must describe what the calculation establishes; an unresolved statement is acceptable when supported by the calculation. Do not call a reformulation a new sign estimate.

The real-shift route has reached an exact invariant:

$$L_{v,b} L_{v,b}^\# = L_v L_v^\#.$$

Retain this result as the reason real translation supplies no contraction. The new resources are the global logarithmic regularity in (2.40), the actual arithmetic equation (2.37), and the exterior representation with its endpoint cancellations.

Group 1 – Close the full Suzuki graph domain

For fixed finite a , attempt the unconditional estimate

$$\|Vq\|_2 \leq C_a \|q\|_{\log,2}, \quad \|q\|_{\log,2}^2 = \frac{1}{2\pi} \int_{\mathbb{R}} [1 + \log^2(2 + |\xi|)] |\widehat{q}(\xi)|^2 d\xi,$$

for every $q \in C_c^\infty(-a, a)$, with finite C_a independent of any zero truncation. This is a candidate lemma to prove, not an input.

Use the exact unconditional transform, including the corrected negative-time convention. Differentiate Suzuki's explicit formula in arXiv:2301.00421v3, (1.5)-(1.7), before taking absolute values. In particular, check the differentiated Lerch component

$$\sum_{n \geq 0} \frac{e^{-(2n+1/2)t}}{2n + 1/2 - iz}$$

and its half-line Fourier representation with kernel

$$\frac{e^{-(t+u)/2}}{1 - e^{-2(t+u)}} \leq \frac{C}{t + u}.$$

Calculate the resulting Carleman bound. Retain all finite prime packets and completed polar terms. Combine the zeta logarithmic derivative with $h = A/(A + iA')$ before estimating, using the real-axis removable continuations of

$$|h| \leq 1, \quad |hA'/A| \leq 1.$$

Display every residual gamma multiplier and its growth.

Justify half-line truncation in the graph domain. A concrete route is the cross-half-line Carleman kernel of

$$[\log |D|, \mathbf{1}_{(0,\infty)}].$$

Check its normalization and boundedness explicitly.

If the estimate closes, use (2.40), inward dilation, and mollification to construct a real same-parity core sequence satisfying

$$q_n \rightarrow v \text{ in } \|\cdot\|_{\log,2}, \quad Vq_n \rightarrow Vv \text{ in } L^2, \quad Q_a(q_n) \rightarrow 0.$$

Prove $0 < \mathcal{S}(v, v) < \infty$, convergence of $\mathcal{S}(q_n, q_n)$, and

$$\lim_{R \rightarrow \infty} \sup_n \int_{|x| > R} |Vq_n(x)|^2 dx = 0.$$

Extend the extraction form by its established bound.

This would remove norm divergence and spectral escape for this chosen sequence. It would not establish the sign of the complete correction or justify every meromorphic contour limit automatically. Keep the explicit operator decomposition: substitute the null equation into it and write the resulting genuine L^2 relation among the Carleman, prime-shift, and logarithmic-derivative terms.

Group 2 – Calculate a finite inward-dilation identity

Write

$$R = r'', \quad \ell_n = \log n, \quad b_n = \Lambda(n)/\sqrt{n}, \quad C_v(t) = \int_{\mathbb{R}} v(x+t)v(x) dx, \quad M = \|v\|_2^2.$$

For $s \geq 0$, use the support-preserving unitary dilation

$$U_s v(x) = e^{s/2} v(e^s x).$$

Starting with the complete form and its actual domain, verify

$$\langle \log |D| U_s v, U_s v \rangle = \langle \log |D| v, v \rangle + sM$$

and derive, correcting any normalization if necessary,

$$\boxed{Q_a(U_s v) = sM - 2 \sum_{\ell_n < 2a} b_n [C_v(e^s \ell_n) - C_v(\ell_n)] - \int_{-2a}^{2a} [e^{-s} R(e^{-s} t) - R(t)] C_v(t) dt.}$$

Use zero extension, so $C_v(t) = 0$ for $|t| \geq 2a$; retain every prime threshold crossing.

This changes the relative arithmetic shifts and creates the explicit logarithmic scale term sM . It is not the real-shift invariant from Checkpoint 13.

The sign task is concrete: obtain, from the full null equation and endpoint identities, an independent estimate that makes this expression negative for some $s > 0$. An interval-dependent choice of s is allowed. Calculate the complete prime and remainder contribution; signed autocorrelations cannot be replaced by positive ones.

Do not differentiate at $s = 0$ without proving the needed regularity of C_v at the prime shifts. Inward variation is one-sided; neither stationarity nor vanishing boundary flux follows automatically. If an infinitesimal identity fails at this domain, continue with the finite-difference identity.

Group 3 – Move the logarithmic equation into three dimensions

Use Chen-Hauer-Weth, arXiv:2312.15689v1, Theorems 1.2 and 1.6. Their normalization is $L_\Delta = 2 \log |D|$. For the zero-extended null vector, define

$$W_v(x, y) = \frac{1}{2} \int_{-a}^a \frac{v(t)}{\sqrt{(x-t)^2 + |y|^2}} dt, \quad y \in \mathbb{R}^2.$$

Verify the distributional line-source equation

$$-\Delta_{\mathbb{R}^3} W_v = 2\pi v(x) dx \otimes \delta_0(y)$$

and transport (2.37) into the renormalized trace condition

$$\lim_{\rho \downarrow 0} [W_v(\cdot, \rho) + v \log \rho] = [(\log 2 - \gamma_E)I - \mathcal{B}_a]v$$

on I_a , initially in the distributional sense. State any stronger topology only after proving it.

Derive the complete energy or finite-dilation flux identity in this representation. Show the logarithmic counterterm, endpoint contributions, outer boundary terms, and every prime translation from $\mathcal{B}_a$. Identify the precise boundary functional corresponding to the original Suzuki norm if using that norm for an estimate.

Then calculate a signed bound for the boundary expression that can feed Group 2 or the original remainder estimate. A nonnegative bulk Dirichlet integral does not establish positivity after subtracting divergent terms and inserting the arithmetic boundary operator.

The extension paper's unique-continuation theorem needs both $v = 0$ and $L_\Delta v = 0$ on the same open set. Exterior support alone supplies only the first condition. Do not apply it unless the second is derived.

Current-source check: David Lee's July 2026 paper, DOI 10.1016/j.jmaa.2026.130510, develops the renormalized extension viewpoint; check theorem domains before applying it to this null vector. Do not import the Pohozaev theorem from arXiv:2411.15985: its authors withdrew the paper because of an error in that proof.

Group 4 – A quantified regulator at infinity

If meromorphic exhaustion still obstructs the chosen route, test

$$M_\tau(z) = \operatorname{sech}(\tau z), \quad \tau \downarrow 0.$$

Apply it to the complete paired contour expression, not to isolated residues.

For a regular integrand $F(z) = \mathcal{R}(z)L_v(z)L_v^\sharp(z)$, the introduced poles are

$$p_k = \frac{i\pi(k + 1/2)}{\tau}, \quad \operatorname{Res}_{p_k}(M_\tau^2 F) = -\tau^{-2} F'(p_k).$$

First prove that F is regular at the selected points. Derive bounds for L_v, L'_v there from the endpoint-complete exterior formulas, and independently bound $\mathcal{R}, \mathcal{R}'$. Test whether the total artificial-pole cost admits

$$C_a \|v\|_2^2 \tau^{-2} \frac{e^{-a\pi/\tau}}{1 - e^{-2a\pi/\tau}}$$

or a corrected explicit bound.

Account for the exact difference between $M_\tau(z)L_v(z)$ and the Cauchy sum whose coefficients have been multiplied by $M_\tau(\gamma)$. Carry all actual E -poles, cross terms, and remaining boundary integrals. State the order of contour and regulator limits and prove the needed uniformity.

Even an exponentially small artificial-pole budget would close only that regulator cost. It does not sign the remaining actual-pole contribution. State precisely whether this route yields a controlled full exhaustion, a partial tail bound, or a failed estimate.

Required closing statements and return to the target

For each group return: exact theorem and hypotheses; calculation; proof status; consequence for the next group; and, if open, one fully specified residual inequality. Include an attempted estimate for that inequality and identify the line that does not close.

Use the exact constants

$$K_* = \frac{3567}{509894}, \quad K_0 = \frac{17607901}{2531730432}, \quad K_* - K_0 = \frac{4429025}{108864408576}.$$

If Group 1 closes, then at an exact null vector

$$\Delta(v, v) = -\mathcal{S}(v, v), \quad \mathcal{T}(v, v) = -\mathcal{S}(v, v) + \mathcal{X}_\infty(v, v).$$

Consequently, the inequality

$$\mathcal{T}(v, v) \geq (-1 + K_*)\mathcal{S}(v, v)$$

still directly excludes the hypothetical null vector; it has not become an independent small lemma merely by removing approximants. Identify the new arithmetic or analytic estimate that proves it, if one is obtained.

Finish with a single precise status of $\liminf_N \mathcal{E}_N \leq 1$. If a contradiction is proved, verify the first-crossing prerequisites and the complete return chain. Otherwise save the genuine auxiliary results and the exact remaining obstruction. Preserve the full calculations in the next checkpoint and update the ledger.

Primary sources checked

- Masatoshi Suzuki, *On the Hilbert space derived from the Weil distribution*, arXiv:2301.00421v3, especially the explicit transform formulas.
- Masatoshi Suzuki, *Weil's quadratic form via the screw function*, arXiv:2606.09096v2, localized forms, domains, and scaling.
- Huyuan Chen, Daniel Hauer, Tobias Weth, *Extension problems for the logarithmic Laplacian*, arXiv:2312.15689v1, Theorems 1.2, 1.6, and 1.7.
- David Lee, *An extension problem for higher order operators and operators of logarithmic type via renormalization*, Journal of Mathematical Analysis and Applications 559(2), 130510, July 2026, publisher.
- Withdrawal notice for arXiv:2411.15985.

The graph comparison, its proposed simultaneous core, the application of the three-dimensional extension to the complete arithmetic equation, and the proposed regulator budget are next-round calculations. They have not been established by this prompt.

After Checkpoint 14: odd prime packets

Source: Michaels_Checkpoint_14_Odd_Prime_Packet_Followup.md. Equation tags carry the source prefix D14; references within this unit retain their original local numbering.

Prepared 23 September 2026. Research directive; the proposed closing estimates remain to be proved.

Prompt

Continue from the full Checkpoint 14 note and current ledger. The primary objective is unconditional Weil positivity and hence RH. Preserve the original Suzuki metric and all completed terms. The criterion $\liminf_N \mathcal{E}_N \leq 1$ remains a success criterion; a direct Hilbert-Schmidt estimate is optional.

Use the now-established global Suzuki graph theorem, strong core convergence, endpoint identities, and exact arithmetic operator. Carry completed auxiliary results forward. This round should calculate new signed interactions.

Group 1 – Reduce the proof obligation to odd tests

Suzuki, arXiv:2606.09096v2, Introduction, records Yoshida’s Proposition 1: nonnegativity of the Weil form on all compactly supported smooth odd tests implies RH. Recover the exact statement and justify its use with our normalization and closed domains.

Establish the corresponding first crossing in the odd sector using the parity continuity argument in Section 4.5 and small-interval positivity. This permits the hypothetical setup

$$Q_a^- \geq 0, \quad 0 \neq v \in \ker A_a^-, \quad v(-x) = -v(x).$$

Audit the earlier null-vector results under these precise hypotheses. Parity commutation should identify an odd restricted null vector with a full-operator null vector, but nonnegativity is being assumed only in the odd sector. Project any test requiring form Cauchy-Schwarz into that sector. All dilations below preserve oddness.

Close this reduction with an explicit list of usable domain and null identities. The return route is odd Weil positivity, Yoshida’s criterion, RH, and then the original defect criterion.

Group 2 – Derive the full mixed-dilation kernel

Checkpoint 14 calculated the diagonal $Q_a(U_s v)$. Calculate the complete two-scale form

$$K_v(s, t) = Q_a(U_s v, U_t v), \quad U_s v(x) = e^{s/2} v(e^s x).$$

For $0 \leq s \leq t$, put $r = t - s$ and

$$A_v(r) = \langle v, U_r v \rangle, \quad C_{v,r}(h) = \int v(x+h)(U_r v)(x) dx, \quad J_{v,r}(h) = C_{v,r}(h) + C_{v,r}(-h).$$

Using the exact remainder $R = r''$ from Checkpoint 14, verify or correct

$$K_v(s, t) = sA_v(r) - \sum_{\ell_n < 2a} b_n [J_{v,r}(e^s \ell_n) - J_{v,r}(\ell_n)] - \int_{-2a}^{2a} [e^{-s} R(e^{-s} h) - R(h)] C_{v,r}(h) dh.$$

Here $\ell_n = \log n$, $b_n = \Lambda(n)/\sqrt{n}$. Derive the formula through the whole-line logarithmic operator and the null equation tested against $U_r v$. Explain the support passage. Retain completed polar, archimedean, and prime terms.

Check symmetry, the diagonal reduction to Checkpoint 14, and $K_v(0, t) = 0$. The last identity means adding the original null vector supplies no independent energy direction.

Now use integer dilation factors:

$$s = \log m, \quad m \geq 2, \quad m \log(p^j) = \log(p^{mj}).$$

Regroup the entire prime increment exactly as

$$\sum_{\substack{p, k \geq 1 \\ k \log p < 2a}} (\log p) \left[\mathbf{1}_{m|k} p^{-k/(2m)} - p^{-k/2} \right] J_{v,r}(k \log p).$$

Zero extension must justify every cutoff. These divisibility coefficients are an arithmetic structure to exploit, not an assumed sign. Calculate the packets for factors 2, 3, and 6, together with the complete archimedean and polar contributions. Comparing two integer dilations introduces the relative scale $r = \log(n/m)$.

Group 3 – Optimize the mixed directions and attempt the sign estimate

Start with the full 2×2 matrix on $U_{\log 2} v, U_{\log 3} v$. For general $s, t > 0$ with $K_v(t, t) > 0$, calculate

$$\mathcal{R}_v(s, t) = K_v(s, s) - \frac{|K_v(s, t)|^2}{K_v(t, t)}.$$

For the real functions here this is the energy of

$$U_s v - \frac{K_v(s, t)}{K_v(t, t)} U_t v.$$

A negative residual excludes the hypothetical odd null vector even when each individual dilation has positive energy. If a denominator vanishes, treat that case directly; nonnegative forms force a zero-energy vector's mixed pairings to vanish.

Expand the determinant or residual with every prime-prime, prime-archimedean, and polar cross term retained. Insert the endpoint identities into the mixed exponential moments before estimating. Use the full positive-semidefinite translation Gram constraints when helpful; individual correlations remain signed.

Attempt an independent arithmetic bound giving

$$|K_v(s, t)|^2 > K_v(s, s) K_v(t, t)$$

for an admissible pair, or a negative eigenvalue of the three-scale matrix for factors 2, 3, 6. The scale family may be adapted if the calculation identifies a better choice.

Produce the actual signed estimate. Reusing $Q_a^- \geq 0$ only supplies the opposite inequality and cannot establish the contradiction. A fixed finite family is an attempted witness family, not an assumed universal detector. If it cannot close, isolate the exact surviving packet combination and demonstrate why the available bounds stop there.

Group 4 – A direct odd-sector alternative: the polar resolvent

Independently calculate the exact rank-one reduction. Work on the full odd subspace of $L^2(-a, a)$, define

$$s_a(x) = \sinh(x/2), \quad H_a = A_a^- + 2|s_a\rangle\langle s_a|.$$

Thus

$$\boxed{A_a^- = H_a - 2|s_a\rangle\langle s_a|}.$$

Derive H_a from the complete arithmetic form. Its quadratic form should be the support compression of the multiplier

$$\Psi_a(\xi) = \Re\psi\left(\frac{1}{4} + \frac{i\xi}{2}\right) - \log \pi - 2 \sum_{\ell_n < 2a} b_n \cos(\xi \ell_n).$$

Verify this formula from the retained normalization.

If H_a is strictly positive, derive

$$\boxed{A_a^- \geq 0 \iff 2\langle s_a, H_a^{-1} s_a \rangle \leq 1}.$$

Then calculate the actual forced equation $H_a w = s_a$ and attempt an independent bound on $2\langle s_a, w \rangle$. This is where the arithmetic estimate must occur.

At an odd first null, $H_a \geq 0$ follows, but invertibility requires proof. Handle

$$\ker H_a = \ker A_a^- \cap s_a^\perp$$

explicitly. Treat the zero-moment kernel branch and the resolvent-saturation branch. Use a regularized inverse only with its limiting and kernel contributions retained. If invertible, a hypothetical null gives $2\langle s_a, H_a^{-1} s_a \rangle = 1$; an independently proved strict inequality would exclude it.

Any finite-block estimate must retain its complete Schur correction and a proved tail bound. A calculation at one fixed interval does not cover every interval.

Check the odd reflection of every prime translation. Positivity improvement of a whole-interval reference operator cannot be imported automatically into its odd restriction. Preserve the factor 2 above; under the unitary half-interval map $u = \sqrt{2}v$, the rank-one coefficient changes to 4.

Required closing statements

For each group state its exact hypotheses, completed calculation, proof status, and consequence. Where the sign remains open, give one explicit residual expression and the attempted bound that fails.

Prioritize the mixed prime-power calculation. Use the rank-one route as an alternative whose merits are decided by the actual inverse estimate. Interval-dependent constants are allowed; uniformity in the determining dimension is unnecessary if Weil positivity is reached directly.

End with the strongest independently proved sign statement and its exact return chain. Preserve the auxiliary results, update the ledger and checkpoint, and state separately whether Weil positivity and the defect criterion have been established.

Primary source anchors

- Masatoshi Suzuki, *Weil's quadratic form via the screw function*, arXiv:2606.09096v2: Introduction for Yoshida's odd-test criterion; Sections 2.3-2.5 for the complete form; Section 4.5 for parity continuity.
- Masatoshi Suzuki, *On the Hilbert space derived from the Weil distribution*, arXiv:2301.00421v3: the original metric and its hypotheses.
- Michaels_Global_Suzuki_Domain_and_Four_Closures_Checkpoint_14.md: original graph domain, arithmetic kernel, null equation, endpoint and collar identities.

The mixed-kernel formula and integer regrouping above are proposed calculations to verify in the stated conventions. The determinant and resolvent inequalities are research targets, not established bounds.

After Checkpoint 15: support continuation

Source: *Michaels_Checkpoint_15_Support_Continuation_Followup.md*. Equation tags carry the source prefix *D15*; references within this unit retain their original local numbering.

23 September 2026

This is a research prompt, not a claimed proof. It is based on the full Checkpoint 15 note and its ledger. The note's small-support certificate is $a \leq 1/8$, $H_a > \frac{1}{6}I$, $\sigma_a < 3/256$, and $A_a^- > 253/1536 I$. The pasted message lost several fraction bars.

Paste-ready prompt

Proceed from Checkpoint 15 to Checkpoint 16. The primary goal remains unconditional Weil positivity. Preserve the original Suzuki metric, the complete correction, and the conditional return to $\mathcal{E}_N = 0$. A direct Hilbert-Schmidt cap is optional.

This round must investigate **growth of the support interval and the complete inverse-energy budget**. At a hypothetical first odd crossing, every fixed family of strict inward dilations already lies in a smaller positive interval. Its positive Gram determinant is therefore not a new contradiction. Use that observation to redirect the calculation.

Work through the following four groups. Derive and check every proposed identity before using it; suggested estimates below are tasks, not established checkpoint results. Produce an actual estimate, a counterexample to a proposed auxiliary estimate, or a precise quantified obstruction. Do not count a renamed missing inequality as a sign theorem.

Group 1 – An actual certificate with a prime packet present

Retain

$$H_a = A_a^- + 2|s_a\rangle\langle s_a|, \quad s_a(x) = \sinh(x/2), \quad \sigma_a = 2\langle s_a, H_a^{-1}s_a\rangle.$$

Use the proved interval $a \leq 1/8$ as a starting certificate. Attempt to reach a rigorously specified $a_1 > \frac{1}{2}\log 2$, where the first prime packet is active. Benchmark any imported local positivity theorem in our exact normalization, support convention, parity, and moment constraints.

First exploit odd cancellation: the constant $\alpha(0) = 1/4$ contributes zero to the odd quadratic form. Estimate the complete remainder $\alpha(x - y) - \alpha(0)$, or its exact half-interval version, rather than charging the odd sector for the constant.

Next investigate the weighted concentration operator on $L^2(0, 1)$:

$$T_\Omega(x, y) = \frac{2}{\pi} \int_0^\Omega \log(\Omega/\xi) \sin(\xi x) \sin(\xi y) d\xi.$$

Derive the bound

$$\langle \log |D|q, q \rangle \geq [-\log a + \log \Omega - \|T_\Omega\|] \|q\|_2^2.$$

Obtain a certified analytic upper bound for $\|T_\Omega\|$, with its complete approximation error. A numerical eigenvalue alone is insufficient.

Across the first threshold, retain the reflected $p = 2$ shift on the half interval. In the first-prime window it acts on $(\log 2 - a, a)$ by $u(x) \mapsto u(\log 2 - x)$. It has both signs. Treat its reflection

channels and all couplings exactly, or use the support-collar calculation below. Do not subtract a whole prime norm if a sharper localized estimate is available.

Report the resulting explicit H_a lower bound, inverse-energy upper bound, and A_a^- lower bound. If the proposed first-prime certificate does not close, show the complete inequality and the size of its deficit.

Group 2 – Exact support-collar Schur reduction

For $b = a + \delta$, identify odd functions with $L^2(0, b)$ using the unitary map $q \mapsto \sqrt{2}q|_{(0, b)}$. The source becomes $\sqrt{2}\sinh(x/2)$; preserve this factor. In the following block formulas, s_a denotes this transported source on the core.

Split this space into the core $(0, a)$ and collar (a, b) . Justify the split in the closed form domain using the logarithmic cutoff commutator and its Carleman bounds. Derive

$$H_b = \begin{pmatrix} H_a & B \\ B^* & C \end{pmatrix}, \quad s_b = (s_a, t).$$

Explain why the inner block is exactly H_a , including when new prime thresholds enter.

When $H_a > 0$, put

$$w_a = H_a^{-1}s_a, \quad S = C - B^*H_a^{-1}B, \quad r = t - B^*w_a.$$

Establish, with the domain and invertibility conditions stated,

$$\boxed{\sigma_b - \sigma_a = 2\langle r, S^{-1}r \rangle.}$$

For $\mu_a = 1 - \sigma_a > 0$, derive the complete Schur complement of A_b^- :

$$\boxed{S_A = S - \frac{2}{\mu_a}|r\rangle\langle r|.}$$

Expand B, C, r in the actual archimedean and prime kernels. The off-diagonal half-interval kernel should contain both

$$-\kappa(y - x) + \kappa(x + y)$$

and the difference and reflected prime shifts, with $\kappa(t) = e^{-t/2}/(1 - e^{-2t})$. Retain every threshold, mixed term, and source normalization. This is an identity for the original form, not a substitute metric.

Group 3 – Compute and certify the collar cost

Derive usable estimates before optimizing them. For $\delta < \log 2$, investigate the proposed lower bound

$$C \geq L_{a, \delta} I, \quad L_{a, \delta} = \log \frac{1}{2\delta} - 1 - \delta - \sum_{2a < \log n < 2b} \frac{\Lambda(n)}{\sqrt{n}}.$$

Its ingredients are support-length uncertainty, the bound on α , and positivity of the reflected archimedean kernel

$$\kappa(x + y) = \sum_{j \geq 0} e^{-(2j+1/2)x} e^{-(2j+1/2)y}.$$

The collar contains no ordinary prime shift when $\delta < \log 2$, but reflected prime packets can enter.

Check the complete cross-block estimate

$$\|B\| \leq c_{a,\delta} := \frac{\pi}{2} + [1 + \kappa(a)]\sqrt{a\delta} + \sum_{\log n < b} b_n + \sum_{a < \log n < a+b} b_n, \quad b_n = \frac{\Lambda(n)}{\sqrt{n}}.$$

Correct any constants or support conditions that fail. If $H_a \geq gI$, this would give

$$S \geq (L_{a,\delta} - c_{a,\delta}^2/g) I.$$

Then improve the source estimate by solving an actual approximate forced equation. For $z \in \mathfrak{D}(S)$, set $e = r - Sz$. Prove and use

$$2\langle r, S^{-1}r \rangle = 4\Re\langle r, z \rangle - 2\langle Sz, z \rangle + 2\langle e, S^{-1}e \rangle.$$

If $S \geq \gamma I > 0$, the last term is at most $2\|e\|^2/\gamma$.

Specify z , calculate its residual, and bound that residual including every prime and boundary contribution. Merely defining $z = S^{-1}r$ does not estimate the cost. Combine this calculation with Group 1 to certify an actual enlargement, if the resulting inequalities permit it.

Group 4 – Prevent finite-endpoint failure, including the kernel branch

A positive extension from each positive interval proves local openness. It does not show that repeated extensions reach every finite interval. Analyze this distinction explicitly.

Attempt an independently proved relative budget of the following type:

$$2\langle r, S^{-1}r \rangle \leq \left(1 - \exp\left[-\int_a^b c_R(u) du\right]\right) (1 - \sigma_a), \quad b \leq R,$$

where c_R is explicitly bounded by the arithmetic kernels and is integrable on each relevant finite interval. An equivalent discrete product estimate is acceptable.

If this closes, derive

$$1 - \sigma_b \geq \exp\left[-\int_a^b c_R(u) du\right] (1 - \sigma_a).$$

Specify the allowed step sizes and prove that they cannot accumulate at an uncontrolled finite endpoint. An absolute $o(1)$ collar cost does not supply this relative estimate. Nor may c_R hide a reciprocal of the unknown positivity margin.

Also control the independent branch $\ker H_a \neq 0$. A scalar estimate for σ_a presupposing invertibility cannot eliminate it. Either derive a lower bound for H_a that persists to every finite endpoint, or carry both branches through a complete finite-rank reduction.

For the latter, use the existing certified spectral tail. If P_d is a finite-rank projection and the complementary A_a^- block is strictly positive, investigate

$$H_\tau = A_a^- + \tau P_d, \quad \tau > 0.$$

At a hypothetical first null, prove that H_τ is invertible and that any null vector forces eigenvalue 1 in

$$\tau P_d H_\tau^{-1} P_d.$$

This incorporates the zero-moment branch. It does not prove a strict matrix bound: calculate the signed estimate needed for that strictness and retain the full Schur correction.

Required closing statements

For each group, end with the strongest proved statement, all hypotheses, and the exact remaining quantitative obstruction, if any.

Finish by stating:

- The largest support interval certified in this round, with its explicit lower bound.
- Whether the collar identities and estimates are proved.
- Whether a relative global budget and kernel exclusion are proved.
- Whether unconditional Weil positivity is proved.

If the global argument closes, return through the established odd Weil criterion and Suzuki identity to $\mathcal{E}_N = 0$. If it does not, preserve the achieved local theorem and identify the exact signed collar term or uniformity estimate still missing. Equation counts, changes of coordinates, and finite-dimensional positivity do not replace that estimate.

Save the complete calculations and updated ledger without overwriting earlier checkpoint notes.

Primary-source checks for this prompt

1. Masatoshi Suzuki, *Weil's quadratic form via the screw function*, arXiv:2606.09096v2, <https://arxiv.org/html/2606.09096v2>. The introduction records Yoshida's odd-test criterion; Theorem 1.3 and the parity discussion provide the localized continuity framework. These results do not establish global positivity.
2. Alain Connes and Caterina Consani, *Weil positivity and Trace formula, the archimedean place*, <https://arxiv.org/pdf/2006.13771>. Lemma 3.3 treats compact differential factorization under vanishing moments. Theorem 6.11 has both a specified support interval and a Fourier-vanishing hypothesis; its archimedean estimate must not be imported as unrestricted odd positivity.
3. The full user checkpoint, *Michaels_Odd_Prime_Packets_and_Polar_Resolvent_Checkpoint_15.md*, and its running ledger were read for the retained definitions and the exact small-support certificate.

The proposed collar estimates and continuation budget above are research tasks. They are not attributed to these papers and have not been established by the checkpoint.

After Checkpoint 16: boundary and first crossing

Source: *Michaels_Checkpoint_16_Boundary_and_First_Crossing_Followup.md*. Equation tags carry the source prefix *D16*; references within this unit retain their original local numbering.

23 September 2026

This prompt uses the complete Checkpoint 16 calculations and current running ledger. It proposes new calculations; it does not assert their conclusions.

Paste-ready prompt

Proceed from Checkpoint 16 to Checkpoint 17. Preserve the proved prime-active certificate

$$Q_a^-(q) \geq \frac{1}{5000} \|q\|_2^2, \quad 0 < a \leq a_0 := \frac{699}{2000} + e^{-2500},$$

the original Suzuki metric, all complete arithmetic corrections, and the conditional return to $\mathcal{E}_N = 0$.

The primary objective is unconditional Weil positivity. This round should calculate the actual boundary-driven collar cost and use the positive finite-rank regularization to treat every possible first-null branch in one framework.

Checkpoint 16 identifies a specific loss: its general estimate charges the worst-case Carleman norm and gives only $O(1/\log(1/\delta))$ for a fixed positive inner operator. Investigate the actual forced solutions in that cross block. Obtain a quantified rate with explicit hypotheses, then determine whether the complete arithmetic supplies the dependence on the remaining positivity margin.

Group 1 – Fix the space and retain every null branch

Fix any finite $R > a_0$. Use Checkpoint 16's independently certified d_R, τ_R and positive complementary block.

Transport the full odd operator to $L_{\text{odd}}^2(-1, 1)$ by

$$(U_a q)(x) = \sqrt{a} q(ax), \quad \tilde{A}_a = U_a A_a^- U_a^{-1}.$$

Derive

$$U_a C_a U_a^{-1} = C_1 - \log a I$$

for the supported logarithmic operator. Choose the same fixed spectral subspace P of C_1 throughout $a \in [a_0, R]$; handle multiplicities by a fixed choice of eigenvectors. Carry all scaled prime shifts, source factors, kernels, and support indicators.

Set

$$\mathcal{H}_a = \tilde{A}_a + \tau_R P, \quad K_a = \tau_R P \mathcal{H}_a^{-1} P|_{PL^2}, \quad \varepsilon(a) = 1 - \lambda_{\max}(K_a).$$

Retain the independently proved uniform positive lower bound for $\mathcal{H}_a$. Verify that $\varepsilon(a) > 0$ is equivalent to strict positivity of the full odd operator, and that $\varepsilon(a) = 0$ detects first nulls including the old zero-moment branch.

Prove the exact finite-difference identity

$$K_b - K_a = -\tau_R P \mathcal{H}_b^{-1} (\tilde{A}_b - \tilde{A}_a) \mathcal{H}_a^{-1} P.$$

Specify the common domain and boundedness of the operator difference. This identity is available before any shape differentiability has been established.

For a unit eigenvector $u_a \in PL^2$ with $K_a u_a = (1 - \varepsilon(a))u_a$, put $v_a = \mathcal{H}_a^{-1}u_a$. Derive the exact near-null equations

$$\boxed{\tilde{A}_a v_a = \varepsilon(a)u_a, \quad P v_a = \frac{1 - \varepsilon(a)}{\tau_R} u_a.}$$

These retain a nonzero vector as the margin tends to zero, while the regularized inverse remains bounded. Use this family in the later estimates. A bound for the unregularized H_a^{-1} is not a substitute when its kernel branch is under investigation.

Group 2 – Calculate the boundary behavior of the actual solutions

For a fixed positive H_a , analyze $w_a = H_a^{-1}s_a$. Also analyze the finite family $\mathcal{H}_a^{-1}f_j$, where (f_j) is a fixed orthonormal basis of PL^2 , and transport these solutions back to physical coordinates.

The relevant published logarithmic boundary scale is

$$\ell(d)^{1/2}, \quad \ell(d) = |\log(\min\{d, 0.1\})|^{-1}.$$

The boundary theorem requires a bounded weak solution and bounded forcing. Establish those hypotheses for the complete operator, including translated prime terms and the finite-rank correction. Track constants on $[a_0, R]$ using the known regularized lower bound, without dividing by $\varepsilon(a)$.

First prove whatever estimate is available of the form

$$|w_a(a - d)| \leq C_a \ell(d)^{1/2}.$$

Then determine whether the normalized trace

$$\beta_a = \lim_{d \downarrow 0} \sqrt{\log(1/d)} w_a(a - d)$$

exists. A boundary upper bound does not itself prove existence of this limit. If only averaged traces or limsup bounds are available, formulate the collar calculation with those quantities.

In the unitary half-interval convention, retain the source $\sqrt{2} \sinh(x/2)$. Calculate

$$r_{a,\delta}(y) = t(y) - B^* w_a(y), \quad a < y < a + \delta,$$

using the complete formula (2.28) of Checkpoint 16.

Conditional on the normalized trace existing, test the proposed leading term

$$r_{a,\delta}(a + h) = \beta_a \sqrt{\log(1/h)} + o\left(\sqrt{\log(1/h)}\right).$$

The candidate coefficient comes from the singular cross kernel $1/[2(h + d)]$. Compute its constant directly, and bound the remaining archimedean, ordinary-prime, reflected-prime, and source terms. Establish the correct statement even at a prime threshold.

Group 3 – Resolve the collar after scaling and derive the complete variation

Scale the collar onto $(0, 1)$, with $L_\delta = \log(1/\delta)$. For fixed a with $H_a > 0$, analyze the complete Schur operator

$$S_{a,\delta} = C - B^* H_a^{-1} B.$$

Determine the appropriate strong-resolvent or form convergence of $S_{a,\delta}/L_\delta$. Keep the mixed correction during this passage.

Use the actual boundary estimate to attempt

$$2\langle r_{a,\delta}, S_{a,\delta}^{-1} r_{a,\delta} \rangle \leq C_a \delta.$$

When the stronger trace statement is available, test and either prove or correct the sharper candidate

$$\lim_{\delta \downarrow 0} \frac{\sigma_{a+\delta} - \sigma_a}{\delta} = 2|\beta_a|^2,$$

in the stated unitary half-interval normalization. This is a proposed asymptotic, not an established formula. Prove the remainder and identify exactly which constants remain controlled near a potential first crossing.

For the branch-complete route, insert the actual regularized solutions into Group 1's finite-difference identity. Expand the complete arithmetic variation of K_a , including source scaling, every prime shift, and the polar term. Establish a derivative only after proving the needed regularity. Otherwise retain rigorous finite-difference bounds.

Calculate what happens when $2a = \log n$, when $a = \log n$, and when an interface meets a translated support. Prove the required continuity and integrated bounds through these points. Account for the contributions of all prime powers up to the fixed finite endpoint.

The deliverable is the full matrix variation evaluated on its extremal eigenspace, with degeneracies retained. Absolute bounds on unrelated operator norms are insufficient for identifying its signed dependence on $\varepsilon(a)$.

Group 4 – Attempt the weakest sufficient no-crossing estimate

The earlier linear relative budget is sufficient, but a broader Osgood estimate would also close finite-endpoint failure.

Attempt to prove, on the positive branch,

$$\varepsilon'(a) \geq -c_R(a) \omega(\varepsilon(a)), \quad c_R \geq 0, \quad c_R \in L^1([a_0, R]),$$

with local absolute continuity and the endpoint passage justified, or prove an integral comparison with the same consequence. Here ω is continuous, positive for positive arguments, increasing near zero, and satisfies

$$\int_{0+} \frac{dt}{\omega(t)} = \infty.$$

Both $\omega(t) = t$ and $\omega(t) = t \log(e/t)$, for $0 < t \leq 1$, meet this condition. Thus a logarithmic loss need not prevent closure.

Derive this inequality from the actual signed variation and the near-null equations in Group 1. Substitute the complete null/convolution and endpoint relations before estimating the prime-archimedean-polar interactions. Identify the specific arithmetic cancellation or inequality that supplies the factor $\omega(\varepsilon)$.

A constant c_R may depend on the independently certified regularized inverse bound and on R . It may not conceal a reciprocal of the unknown margin. An absolute Lipschitz estimate for K_a does not supply the displayed vanishing factor.

If the estimate closes, prove the integrated inequality

$$\int_{\varepsilon(b)}^{\varepsilon(a_0)} \frac{dt}{\omega(t)} \leq \int_{a_0}^b c_R(u) du$$

while the margin is positive. Use divergence at zero and the fixed-space continuity to exclude every finite first crossing, including the former kernel branch. Then apply the retained odd Weil criterion and normalized Suzuki identity to conclude $\mathcal{E}_N = 0$.

If the estimate fails, isolate the exact boundary coefficient or signed matrix entry responsible. Calculate whether the near-null equations constrain that term sufficiently, and state the remaining quantified inequality. Preserve any independently proved boundary or collar theorem.

Required closing statements

For each group, give the theorem actually established, its hypotheses, the constants obtained, and the exact open step.

Finish with explicit answers:

- Has the worst-case $1/\log(1/\delta)$ collar estimate been replaced by an estimate for the actual forced solutions?
- Is the normalized boundary trace proved, or only a bound?
- Is the complete finite-dimensional variation controlled at every prime threshold?
- Is an Osgood estimate proved with constants independent of the unknown margin?
- Are both first-null branches excluded, and does unconditional Weil positivity follow?

Preserve the original Suzuki metric and previous checkpoint files. Save the complete calculations and updated ledger. Keep structural reductions, local quantitative improvements, and the global signed conclusion separately identified.

Sources and scope of this follow-up

- The full *Michaels_Support_Growth_and_Complete_Positivity_Budget_Checkpoint_16.md* and current *Michaels_Running_File_Ledger.md* were read. Equations (2.27), (2.28), (3.14), (4.18)-(4.24), and (5.5)-(5.9) determine the next calculation.
- Hernández-Santamaría, López Ríos, and Saldaña, *Optimal boundary regularity and a Hopf-type lemma for Dirichlet problems involving the logarithmic Laplacian*, <https://arxiv.org/pdf/2401.18033>, Theorem 1.1: bounded weak solutions with bounded forcing have the displayed logarithmic boundary upper bound. Theorem 1.4's Hopf conclusion requires non-negativity. Neither theorem establishes the proposed normalized trace limit for the complete Weil operator.

- Pollastro and Soave, *Antisymmetric maximum principles and Hopf's lemmas for the Logarithmic Laplacian, with applications to symmetry results*, <https://link.springer.com/article/10.1007/s10231-025-01549-0>, Proposition 3.1 and Lemma 3.4: the maximum principle and boundary lower bounds carry explicit spectral, potential, and sign hypotheses. Odd parity alone does not supply those hypotheses for the signed prime-shift operator.
- *Small Order Asymptotics of the Dirichlet Eigenvalue Problem for the Fractional Laplacian*, <https://link.springer.com/article/10.1007/s00041-022-09908-8>, is relevant to boundedness and boundary behavior of the reference logarithmic eigenfunctions. Transfer to the fully perturbed resolvent must be justified.
- The fixed-space matrix identities, proposed collar asymptotic, and Osgood research target above are specified calculations for the next checkpoint. They are not attributed to these papers as already proved results.

After Checkpoint 17: physical regularization

Source: Michaels_Checkpoint_17_Physical_Regularizer_and_Boundary_Energy_Followup.md. Equation tags carry the source prefix D17; references within this unit retain their original local numbering.

23 September 2026

This research prompt follows the complete Checkpoint 17 note and current ledger. Its proposed identities and estimates must be established in the next calculation. Global Weil positivity remains the objective; the original Suzuki metric and defect criterion retain their existing definitions.

Paste-ready prompt

Proceed from Checkpoint 17 to Checkpoint 18. Preserve the prime-active certificate

$$a_0 = \frac{699}{2000} + e^{-2500}, \quad Q_a^-(q) \geq \frac{1}{5000} \|q\|_2^2 \quad (0 < a \leq a_0),$$

the actual linear collar theorem, the signed averaged boundary formula, and every prime, archimedean, polar, and boundary term.

The central calculation this round is a boundary estimate with a factor supplied by the small Weil energy. First arrange the auxiliary regularization so that the complete finite matrix can use the already established collar theorem directly.

Group 1 – Keep the regularization fixed on the largest physical interval

Fix $R > a_0$. Let $f_1, \dots, f_d$ be the real orthonormal reference eigenfunctions on $(-R, R)$ corresponding to the already certified $d = d_R$. Put

$$F_R c = \sum_{j=1}^d c_j f_j, \quad P_R = F_R F_R^*, \quad \mathcal{J}_R = A_R^- + \tau_R P_R \geq g_R I.$$

Use the explicit unconditional constants from Checkpoints 16-17.

For $a \leq R$, let E_a be zero extension from $(-a, a)$ into $(-R, R)$, and set

$$F_a = E_a^* F_R, \quad \mathcal{J}_a = A_a^- + \tau_R F_a F_a^*.$$

Prove that $\mathcal{J}_a$ is the closed-form restriction of the one fixed operator $\mathcal{J}_R$. Consequently establish $\mathcal{J}_a \geq g_R I$ throughout $[a_0, R]$. Define the operator through its restricted form; justify its domain before using block notation.

The restricted functions need not be orthonormal:

$$\Gamma_a = F_a^* F_a \leq I, \quad \Gamma_a \neq I \text{ in general.}$$

Retain this Gram matrix. The operator $F_a F_a^*$ need not be a projection, and F_a need not be assumed injective.

Define

$$W_a = \mathcal{J}_a^{-1} F_a, \quad \hat{K}_a = \tau_R F_a^* W_a, \quad M_a = I - \hat{K}_a, \quad \epsilon_a = \lambda_{\min}(M_a).$$

Use a hat to distinguish this physical matrix from Checkpoint 17's scaled matrix.

Prove the full sign and null equivalences, including

$$A_a^- > 0 \iff M_a > 0, \quad \ker A_a^- \neq 0 \iff \ker M_a \neq 0.$$

Retain both old null branches and the initial estimate

$$M_{a_0} \geq \frac{m_0}{m_0 + \tau_R} I, \quad m_0 = 1/5000.$$

Derive the exact forced-space relations

$$\boxed{A_a^- W_a = F_a M_a, \quad Q_a^-(W_a c) = \frac{1}{\tau_R} c^* \hat{K}_a M_a c.}$$

For a unit minimum-eigenvalue vector c of M_a , this gives

$$A_a^- W_a c = \epsilon_a F_a c, \quad Q_a^-(W_a c) = \frac{\epsilon_a(1 - \epsilon_a)}{\tau_R}.$$

Keep $\|F_a c\| \leq 1$; the physical source is not generally a unit vector.

Group 2 – Apply the linear collar theorem to the entire matrix

Split the fixed physical operator and fixed sources across the core and collar:

$$\mathcal{J}_b = \begin{pmatrix} \mathcal{J}_a & B \\ B^* & C \end{pmatrix}, \quad F_b = \begin{pmatrix} F_a \\ T \end{pmatrix}.$$

Set

$$S = C - B^* \mathcal{J}_a^{-1} B, \quad \mathcal{R} = T - B^* W_a.$$

Derive the exact identity

$$\boxed{\hat{K}_b - \hat{K}_a = \tau_R \mathcal{R}^* S^{-1} \mathcal{R} \succeq 0.}$$

This time the source functions and the regularizer are restrictions of fixed physical objects. Include the full finite-rank kernel in both diagonal and cross blocks, together with the original completed polar kernel.

Check all hypotheses of Checkpoint 17's boundedness, boundary-envelope, and finite-family collar estimates. The sources satisfy

$$\|F_a c\|_\infty \leq \sqrt{J_d/R} \|c\|,$$

using the existing J_d bound. Obtain constants uniform on $[a_0, R]$, without $1/\epsilon_a$.

Carry the complete collar calculation through to an explicit bound

$$0 \preceq \hat{K}_b - \hat{K}_a \preceq C_R(b - a)I.$$

Use subdivision with a uniformly admissible step size. Account for every prime threshold through the actual block kernels. This would establish Lipschitz continuity and absolute continuity of the entire replacement matrix.

For the real columns $w_{a,j} = W_a e_j$, take their unitary half-interval representatives $w_{a,j}^+ = \sqrt{2} w_{a,j}|_{(0,a)}$, and define

$$b_{a,j}(L) = \frac{1}{2\sqrt{L}} \int_{e^{-L}}^{\eta} \frac{w_{a,j}^+(a-d)}{d} dd.$$

Use the complete polarized collar formula to establish, with its uniform remainder,

$$\frac{\widehat{K}_{a+\delta} - \widehat{K}_a}{\delta} = \tau_R b_a(L) b_a(L)^* + O_R(L^{-1/2}), \quad L = \log(1/\delta).$$

Then determine the almost-everywhere limiting vector β_a and derive

$$\boxed{\widehat{K}'_a = \tau_R \beta_a \beta_a^* \text{ for almost every } a.}$$

For real continuous averages, explain why convergence of the outer product supplies convergence of the vector, including the zero-limit case. A pointwise normalized boundary trace is unnecessary for this step.

Group 3 – Calculate a boundary estimate from the complete Weil energy

This is the principal arithmetic task. On the positive branch, attempt the uniform forced-space trace estimate

$$\boxed{|\beta_a^* c|^2 \leq C_R(a) Q_a^-(W_a c), \quad C_R \in L^1([a_0, R]), \quad C_R \geq 0,}$$

for every c , or a sufficient estimate restricted to the minimum eigenspace of M_a . Constants must follow from the actual arithmetic and remain independent of the unknown margin.

First substitute $A_a^- W_a = F_a M_a$ into the complete convolution and endpoint equations. With

$$\mathcal{L}_a(p) = \mathcal{P}_a(p) - d_a F_p(1/2), \quad d_a = 4a \cosh(a/2) - 8 \sinh(a/2),$$

derive the full row identity

$$[\mathcal{L}_a(w_{a,1}), \dots, \mathcal{L}_a(w_{a,d})] = [J_a(f_1), \dots, J_a(f_d)] M_a.$$

Here $\mathcal{P}_a, F_p, J_a$ retain exactly their definitions in Checkpoint 17. This is an endpoint moment relation. Calculate the additional representation needed to control the averaged boundary vector itself.

Carry out at least one explicit adjoint boundary-probe calculation. Let $g_{a,L}$ be the odd L^2 representer of the finite- L averaged boundary functional in the original normalization. Construct an explicit $\psi_{a,L} \in \text{Dom}(A_a^-)$, and compute

$$e_{a,L} = g_{a,L} - A_a^- \psi_{a,L}$$

with every prime shift, regular archimedean term, completed polar term, and endpoint contribution present.

The exact identity to exploit is

$$\boxed{b_a(L) = M_a F_a^* \psi_{a,L} + W_a^* e_{a,L}.}$$

Calculate the finite forced-space residual $W_a^* e_{a,L}$; its norm can be more informative than an unrestricted residual bound.

One sufficient outcome is a bound

$$\sup_L \|F_a^* \psi_{a,L}\|^2 \leq C_R(a), \quad W_a^* e_{a,L} \rightarrow 0,$$

with an independently integrable C_R . This yields a bounded factorization $\beta_a = M_a c_a$ and a usable relative boundary estimate.

Specify the probe and evaluate the residual. Choosing $\psi = (A_a^-)^{-1}g$ hides the unknown inverse; choosing $\psi = \mathcal{J}_a^{-1}g$ alone reproduces $b = Mb + \widehat{K}b$. An actual correction or arithmetic cancellation must be calculated.

Also calculate a weaker finite-resolution certificate. On the positive branch, complete-form Cauchy-Schwarz gives

$$|b_a(L)^* c| \leq \sqrt{Q_a^-(W_a c)} \sqrt{Q_a^-(\psi_{a,L})} + \|c\| \|W_a^* e_{a,L}\|.$$

Investigate the explicit candidate rates

$$Q_a^-(\psi_{a,L}) \leq C_R \log(eL), \quad \|W_a^* e_{a,L}\| \leq C_R L^{-1/2},$$

uniformly in the positive support range. These rates are proposed tasks, not existing results. Calculate the actual energy growth and residual decay of the chosen probe.

If those rates close, retain finite L and the full collar remainder to derive a matrix step estimate of the form

$$\widehat{K}_{a+\delta} - \widehat{K}_a \preceq C_R \delta \log(eL) \widehat{K}_a M_a + C_R \delta L^{-1/2} I, \quad L = \log(1/\delta).$$

Choose L on the order of ϵ_a^{-2} , subject to all uniform small-step conditions. Prove the resulting discrete Osgood comparison and that its adaptive steps cannot accumulate at a finite endpoint. This route can use controlled growth of the dual energy without requiring a bounded limiting probe.

If neither certificate closes, identify the exact surviving signed residual and test the weaker directional Osgood bound in Group 4. The already-known boundary envelope gives an absolute bound; the present task is to extract the energy-dependent factor.

Group 4 – Complete the matrix comparison or the weaker Osgood route

If Group 3's trace estimate is proved, combine it with the exact energy identity:

$$\tau_R \beta_a \beta_a^* \preceq C_R(a) \widehat{K}_a M_a \preceq C_R(a) M_a.$$

Because $M'_a = -\tau_R \beta_a \beta_a^*$, integrate the resulting matrix inequality to obtain

$$M_b \succeq e^{-\int_{a_0}^b C_R(t) dt} M_{a_0} > 0.$$

Prove the endpoint passage and conclude that both first-null branches are excluded.

There are two useful alternative closing quantities. On the positive branch, derive

$$-\frac{d}{da} \log \det M_a = \tau_R \beta_a^* M_a^{-1} \beta_a.$$

An independently bounded integral of this complete quantity prevents finite determinant collapse. The appearance of M_a^{-1} is precisely the margin dependence to estimate, not a bound already furnished by $\mathcal{J}_a^{-1}$.

For a weaker scalar route, let $\mathfrak{E}_a$ be the minimum eigenspace of M_a . Establish the correct almost-everywhere extremal-eigenvalue formula

$$\epsilon'_a = -\tau_R \|\text{Proj}_{\mathfrak{E}_a} \beta_a\|^2,$$

with multiplicities handled explicitly. Attempt

$$\tau_R \|\text{Proj}_{\mathfrak{E}_a} \beta_a\|^2 \leq c_R(a) \omega(\epsilon_a), \quad c_R \in L^1, \quad \int_{0+} \frac{dt}{\omega(t)} = \infty.$$

The logarithmic modulus $t \log(e/t)$ is allowed. Use the already established matrix absolute continuity to complete the integrated Osgood argument if this inequality closes.

Keep the arithmetic requirement visible. For calibration, the reference family $C_a - cI$ has lowest eigenvalue $\nu_1(1) - \log a - c$, which can cross zero. Its generic boundary and regularization properties cannot supply the required vanishing factor. Identify the specific complete Weil interaction that makes the desired estimate possible.

Required closing statements

Close each group with its strongest proved theorem, exact hypotheses, constants, and remaining inequality.

Answer explicitly:

1. Does the fixed physical restriction preserve the sign test and both null branches?
2. Is the whole physical matrix monotone and absolutely continuous?
3. Is its almost-everywhere boundary Gram formula established?
4. Which explicit boundary probe was evaluated, and what complete residual remains?
5. Is an integrable energy-relative, determinant, or Osgood bound proved independently of the unknown margin?
6. Does unconditional Weil positivity follow?

If a closing bound is established for every finite R , finish the retained odd Weil and Suzuki return to $\mathcal{E}_N = 0$. Preserve all earlier files and the original metric. Save the complete calculations and updated ledger.

Research and prior-work scope

The full Checkpoint 17 note and current running ledger were read for this continuation. In particular, the proposal uses its complete finite-family collar formula (3.27), its uniform regularized boundary estimates, and its exact endpoint substitution (4.16).

Suzuki's *Weil's quadratic form via the screw function*, <https://arxiv.org/html/2606.09096v2>, supplies the inherited localized form and odd-test framework. Its continuity and small-support results do not supply the new relative boundary estimate.

Hernández-Santamaría, López Ríos, and Saldaña, *Optimal boundary regularity and a Hopf-type lemma for Dirichlet problems involving the logarithmic Laplacian*, <https://arxiv.org/pdf/2401.18033>, supplies the boundary upper estimate under the bounded-data hypotheses already addressed in Checkpoint 17. No normalized pointwise trace or Weil-energy trace inequality is attributed to that theorem.

The earlier *Michaels_Interior_to_Spectral_Transfer.md* was checked for an existing applicable bound. Its ground-state coercivity concerns rectangle H_0^1 tests and an averaged theta amplitude; the signed source remains. Applying it here would require an explicit map from the present forced space and its complete Weil energy, including all boundary terms. No such transfer is assumed in this prompt.

The fixed physical regularizer is an auxiliary choice. Its proposed matrix evolution and the boundary-probe construction must be proved in the new checkpoint. Their definitions do not themselves establish the missing sign.

After Checkpoint 19: existing machinery

Source: Michaels_Checkpoint_19_Existing_Machinery_Closure_Prompt.md. Equation tags carry the source prefix D19; references within this unit retain their original local numbering.

Prepared from the current Checkpoint 18, ledger version 17, and the recovered 301-entry theory atlas, 23 September 2026. This is the instruction for the next research run; it records proposed calculations and sufficient targets, not a completed Checkpoint 19 proof.

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Proceed from Checkpoint 18 to Checkpoint 19. Use the recovered machinery to calculate the closing sign. Read `Michaels_Theory_Atlas_Purpose_and_Weil_Connections_2026-09-23.md` together with the current checkpoint and ledger, then open the source proofs needed below. The atlas is the source map; the mathematical inputs are the underlying identities with their actual hypotheses.

The primary objective remains unconditional Weil positivity. Preserve the original Suzuki metric, both null branches, every arithmetic, archimedean, polar and boundary contribution, and the original determining ladder. Retain the established interval

$$a_0 = 699/2000 + e^{-2500}, \quad Q_a^- \geq (1/5000)I \quad (0 < a \leq a_0).$$

The recovered earlier certificate on $[-1/2, 1/2]$ should also be checked for exact normalization and domain compatibility; if its recorded proof transfers, retain both bounds at their respective strengths. A larger starting interval is useful infrastructure, while this round's main task is the all-support sign.

For arbitrary finite R , carry forward the fixed physical regularizer

$$\mathcal{J}_a = A_a^- + \tau_R F_a F_a^* \geq g_R I, \quad F_a^* F_a \preceq I, \quad W_a = \mathcal{J}_a^{-1} F_a,$$

$$K_a = \tau_R F_a^* W_a, \quad M_a = I - K_a, \quad Q_a^-(W_a c) = \tau_R^{-1} c^* K_a M_a c.$$

Keep the established complete matrix evolution $K'_a = \tau_R \beta_a \beta_a^*$ almost everywhere, with the unitary half-interval boundary normalization.

The working family is $q = W_a c$. It already detects every adverse and null direction of the complete odd form. The selected reference-probe residual Γ_a is one sufficient route; use any independently established complete inequality that closes the actual sign.

Organize this round into four closing groups. Each group must perform its stated substitutions and estimates, feed its usable result into the next group, and finish with a precise closing statement identifying the theorem obtained and the term still requiring payment, if any.

Group 1 – Put the existing complete energy identity on the actual forced family

Recover *Physics_Architectures_and_Log_Prime_Energy.pdf*, §§14-18.1, especially (33)-(39), and verify its normalization against CP18. Write the positive even theta weight as ϕ , so that $\widehat{\phi} = \Xi$. Keep the complete identity

$$Q_a^-(\phi h) = E_\Gamma(\phi, h) + E_P(\phi, h) - E_Z(\phi, h),$$

where

$$E_\Gamma = \frac{1}{2} \iint \frac{e^{-|x-y|/2}}{1 - e^{-2|x-y|}} \phi(x) \phi(y) |h(x) - h(y)|^2 dx dy,$$

$$E_P = \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \int \phi(x) \phi(x + \log n) |h(x + \log n) - h(x)|^2 dx.$$

For odd h , the entire polar cost is

$$E_Z = \frac{1}{2} \left(\|h\|_{L^2(\nu)}^2 + |\nu(\vartheta h)|^2 \right), \quad d\nu = 2 \cosh(x/2) \phi(x) dx, \quad \vartheta = \tanh(x/2), \quad \nu(1) = 1.$$

Set $q_j = W_a e_j$, $h_j = q_j / \phi$, with zero extension. Prove the required passage from odd compact smooth form-core approximants to these particular columns. On a fixed finite interval ϕ is positive and bounded away from zero; use that fact together with the full closed-form norm and the positive difference-energy norm. Carry the exterior interactions throughout.

Form the actual matrices

$$(\mathbf{E}_a)_{ij} = (E_\Gamma + E_P)(h_i, h_j), \quad (\mathbf{G}_a)_{ij} = \langle h_i, h_j \rangle_\nu, \quad (\ell_a)_j = \overline{\nu(\vartheta h_j)},$$

using the convention conjugate-linear in the first argument. Calculate and verify

$$\boxed{\mathbf{E}_a - \frac{1}{2}(\mathbf{G}_a + \ell_a \ell_a^*) = \tau_R^{-1} K_a (I - K_a).}$$

On $\text{Ran } K_a$, the exact normalized congruence is

$$\tau_R K_a^{-1/2} \left[\mathbf{E}_a - \frac{1}{2}(\mathbf{G}_a + \ell_a \ell_a^*) \right] K_a^{-1/2} = M_a|_{\text{Ran } K_a}.$$

Here the inverse is only on the range. The complementary coefficient kernel has $W_a c = 0$ and $M_a c = c$. This is an auxiliary finite congruence; the Suzuki norm remains unchanged.

The existing polar-inverse route has a precise version here. Put $\mathbf{H}_a = 2\mathbf{E}_a - \mathbf{G}_a$. The desired sign is $\mathbf{H}_a \succeq \ell_a \ell_a^*$, equivalent in finite dimensions to

$$\mathbf{H}_a \succeq 0, \quad \ell_a \in \text{Ran } \mathbf{H}_a, \quad \ell_a^* \mathbf{H}_a^\dagger \ell_a \leq 1.$$

Calculate this alternative when the arithmetic decomposition supplies an independent lower bound for $\mathbf{H}_a$. Retain its kernel and the range condition. Its positivity is a new estimate to prove; it does not follow merely from positivity of E_Γ and E_P .

Calculate the entries and seek a lower comparison for this actual matrix. The identity alone is the starting point. At a hypothetical first null, insert the full null equation into these energies and identify which equality conditions would be necessary. An independently strict inequality on a nonzero null candidate would be a closing contradiction.

A critical tail detail: the ground-state prime energy retains infinitely many jumps even when q is compactly supported. For $t > 2a$, its matrix of translation differences is exactly

$$\mathbf{B}_a(t)_{ij} = \int_{-a}^a \frac{\phi(x+t) + \phi(x-t)}{\phi(x)} \overline{q_i(x)} q_j(x) dx.$$

These are positive exterior contributions. Retain them and establish explicit tail bounds from the already available theta decay. A cutoff at the active prime thresholds of the original compressed operator cannot simply be copied into this transformed energy.

Group 2 – Spend the recovered arithmetic cancellation on these exact entries

Use the factor-potential/Weil source extraction, angle-to-Green identity, full prime triples, 1,2,3,6 packets, Weighted Half-Balance, forced overlap, harmonic-return projection, and Smith-Jordan/color square results where their hypotheses match the derived coefficients.

Begin with a source-faithful finite regrouping. Define the actual positive matrix

$$\mathbf{B}_a(t)_{ij} = \int \phi(x)\phi(x+t)[\overline{h_i(x+t)} - \overline{h_i(x)}][h_j(x+t) - h_j(x)] dx.$$

Use $\Lambda(n) = \sum_{dm=n} \mu(d) \log m$ to obtain

$$\sum_{n \leq X} \frac{\Lambda(n)}{\sqrt{n}} \mathbf{B}_a(\log n) = \sum_{\substack{r,m \geq 1 \\ (r,6)=1}} \frac{\mu(r) \log m}{\sqrt{rm}} \sum_{\substack{d \in \{1,2,3,6\} \\ drm \leq X}} \frac{\mu(d)}{\sqrt{d}} \mathbf{B}_a(\log(drm)).$$

At finite X this is an exact finite identity. Keep the unequal cutoffs, all four scale factors, every cross term, and the exterior tail when passing to infinity. This introduces the authentic Möbius source into the current energy without assigning Möbius coefficients to the functions $W_a c$.

Then carry out at least two explicit comparisons:

1. Evaluate the complete four-channel packet against the actual theta weight and odd difference functions. Test the archived forced-overlap, factor-connection, half-balance or shared-placement mechanism on those precise coefficients. Display the saving and the leftover matrix.
2. Test a transported Green/storage or color-square representation for the same entries or for the response residual in Group 3. Use the exact kernel bridge

$$e^{-|\log m - \log n|/2} = \frac{\sqrt{mn}}{\max(m, n)}.$$

Carry its diagonal rescaling, the gamma contribution, the signed prime-continuum correction, the observed endpoint and the full source remainder.

The earlier two-endpoint block storage, Fibonacci-Lucas packets, joint completed/open budget, authentic-mask compensation and survival rebate are candidate mechanisms with exact saved proofs. Specify the coefficient map, measure and support required for each application. A failed hypothesis match must be written as a concrete extra term or failed condition before trying the next available mechanism.

Use the historical $95 \rightarrow 97$ calculation as a design constraint: separate prime-channel magnitudes cost 96 times the paired amount in that example. Evaluate the current combined packet before taking magnitudes. Likewise preserve the joint source placement and mask correction wherever they occur.

Closing output: an actual lower matrix estimate for Group 1, or an actual upper estimate for the complete residual Gram in Group 3, with the improvement supplied by each recovered mechanism explicitly quantified.

Group 3 – Apply the recovered residual-response certificate directly to the physical regularizer

Use the existing positive operator $\mathcal{J}_a \geq g_R I$. Construct finite trial-response columns

$$Y_a : \mathbb{C}^{d_R} \longrightarrow \text{Dom } \mathcal{J}_a, \quad \mathcal{R}_a = F_a - \mathcal{J}_a Y_a.$$

The archived response identity, applied to these actual inputs, gives

$$\begin{aligned} F_a^* \mathcal{J}_a^{-1} F_a &= F_a^* Y_a + Y_a^* F_a - Y_a^* \mathcal{J}_a Y_a \\ &\quad + \mathcal{R}_a^* \mathcal{J}_a^{-1} \mathcal{R}_a. \end{aligned}$$

Therefore calculate

$$M_a \succeq \mathcal{C}_a(Y) := I - \tau_R [F_a^* Y_a + Y_a^* F_a - Y_a^* \mathcal{J}_a Y_a + g_R^{-1} \mathcal{R}_a^* \mathcal{R}_a].$$

Its exact overpayment is

$$M_a - \mathcal{C}_a(Y) = \tau_R \mathcal{R}_a^* (g_R^{-1} I - \mathcal{J}_a^{-1}) \mathcal{R}_a \succeq 0.$$

Build Y_a concretely from the recovered response, theta, arithmetic-packet or Green constructions, and optimize within that specified family using the archived finite residual optimization. Give the columns, their complete operator images, all finite Gram matrices, and the remainder after optimization. An unevaluated choice $Y_a = \mathcal{J}_a^{-1} F_a$ merely restates the original matrix; the useful calculation needs independently controlled trial columns and residuals.

Expand $\mathcal{R}_a^* \mathcal{R}_a$ as the Gram of the complete residual before applying estimates. Import the joint savings calculated in Group 2. Retain regularizer, prime-prime, prime-gamma, prime-pole, and boundary pairings. Use g_R , which is already independent of the unknown margin. Keep every other constant and its support dependence explicit.

Seek either $\mathcal{C}_a(Y) \succeq 0$ unconditionally for every finite support, or a strictly positive certificate on the relevant nonzero coefficient directions at a hypothetical first crossing. If scalar residual payment remains too expensive, apply the saved response-specific refinement to the actual residual subspace and calculate its improved payment.

Closing output: the explicit candidate certificate, the exact saved arithmetic improvement, and the resulting signed matrix inequality. The complete energy matrix from Group 1 supplies a second evaluation of the same target for checking signs and normalization.

Group 4 – Combine the estimates and execute a valid return to the target

Combine the strongest actual estimates from Groups 1-3. Preserve the useful work even when one chosen representation has a residual. Use the existing first-null, threshold, domain and exhaustion results at their proved scopes.

There are three sufficient endpoints:

1. **Direct global sign:** prove $K_a \preceq I$ for every finite a independently of Weil positivity. This gives unconditional odd Weil positivity directly.
2. **First-null contradiction:** at every hypothetical finite first crossing, prove strict positivity on the nonzero null candidate or derive another explicit incompatibility with its complete null equation. Nonnegativity assumed at that same first-null interval supplies no contradiction by itself.
3. **Energy-dependent continuation:** prove the actual projected Osgood bound, or another already established sufficient continuation estimate, with an independently integrable coefficient. For CP18's chosen residual this is

$$\|\text{Proj}_{\mathfrak{E}_a} \Gamma_a\|^2 \leq d_R(a) \varepsilon_a \log(e/\varepsilon_a), \quad d_R \in L^1,$$

where $\mathfrak{E}_a$ is the least-eigenvalue eigenspace of M_a . Use CP18's proved multiplicity-safe return. A zero boundary coefficient at a single point alone does not establish this continuation.

A support-dependent strict margin is acceptable. Prove its needed dependence. The recovered growing theta near-null spaces require care with claims of a support-independent gap for the unregularized Weil operator; they are compatible with the positive physical regularizer.

Keep coordinate changes available when they enable a concrete estimate: angles, prime phases, the quadratic Gram lift, square-coordinate parity, the infinity derivative lift, cross-plane reconstruction, and light-cone propagation each have precise maps in the atlas. Use the relevant map and its actual metric. State exactly which term it simplifies and how the resulting estimate returns to the same matrix.

If a sufficient endpoint is proved, carry the proof through the established odd Weil criterion and retained Suzuki identity to

$$\text{RH} \implies \mathfrak{W} = \mathcal{S} \implies \Delta = 0 \implies \mathcal{E}_N = 0 \implies \liminf_N \mathcal{E}_N \leq 1.$$

The new sign estimate must be obtained independently of this conditional chain.

Required report and saved continuation

Work through the concrete calculations. Reuse the completed domain passage, matrix continuity, extraction estimates and tail infrastructure rather than spending the round reconstructing them. For every proposed sign, track its source and avoid importing an unproved earlier closure lemma.

Save the full Checkpoint 19 calculation, update the ledger, and preserve all earlier contents. For each closing group report:

- the exact prior theorem used and the substitution that makes it applicable;
- the newly evaluated expression, including every mixed and boundary term;
- the actual inequality or cancellation gained, with its constants and range;
- the precise effect on the complete forced-energy or certificate matrix.

End with the strongest unconditional theorem actually obtained. If the closing sign is still unpaid, display the final explicit matrix or signed expression after all successful reuse, state the remaining inequality with its full quantifiers, and identify exactly which attempted estimate failed. Keep the status mathematically exact. A reformulation becomes useful progress when its new comparison or cancellation has actually been calculated.

Primary source map for this run

- `Michaels_Physical_Regularizer_and_Boundary_Energy_Checkpoint_18.md`: §§1.3-1.4, 2.5-2.6, 3.9, 4, 5.3; source identity.
- `Michaels_Running_File_Ledger.md`: current version 17, source identity.
- `Michaels_Theory_Atlas_Purpose_and_Weil_Connections_2026-09-23.md`: E04-E14; M25-M30, M37-M51; C03, C07-C10, C34-C41; SP11 and AN02-AN03/AN09; T51-T59. Source identity.
- `Physics_Architectures_and_Log_Prime_Energy.pdf`, version 4: §§14-19 for the exact response identity, Green component, complete theta-weighted energy, pole variance, near-null spaces and earlier fixed-window certificate. Source identity.
- `Michaels_Prime_Color_Dual_Kernel_Bridge.md`: exact positive carriers and the established scope of their scalar observation; use the atlas source register.
- `Prime_DNA_Angle_Families_and_Prime_Triple_Bound.md`: exact angle/Green bridge, full prime-triple kernel and mixed-prime interactions.

- `Michaels_Mobius_Modifier_and_Dynamic_DNA_Sieve_Christophe_Michaels.pdf`: both modifier definitions, block storage and exact correction.
- `Prime_DNA_Energy_Throughline.md`, `Prime_DNA_Cofactor_Energy_and_Upper_Interaction.md`, and `Prime_DNA_Remaining_Cost_and_Inherited_Energy.md`: joint storage and actual signed interaction certificates.
- `Michaels_Weighted_Lifetime_Bounds_2026-09-08.md`, `Michaels_Joint_Energy_Continuation.pdf`, and `Michaels_Survival_and_Prime_Transport_Continuation.md`: lifetime absorption, jointly paid histories, authentic masks, survival rebates and complete prime transport.
- The atlas contains exact source identities and locations for the additional Weighted Half-Balance, Fibonacci-Lucas, phase, plane, gap and light-cone constructions. Follow those links where a group uses that construction.

Checkpoint 20 research directive

Source: Michaels_Checkpoint_20_Positivity_Recovery_and_Closure_Prompt.md. Equation tags carry the source prefix D20; references within this unit retain their original local numbering.

23 September 2026. Prepared from the available conversation-history retrieval, the theory atlas, Checkpoint 19, and the recovered proof and audit notes. This document is a source-based continuation prompt, not a claim that the remaining sign has been proved.

What the earlier “positive board” work actually supplies

“The whole board is positive” has a precise useful meaning: the **complete target quadratic form** is nonnegative for every admissible coefficient vector. Positivity of a different Gram matrix, individual entries, selected blocks, or a differentiated expression must be distinguished by its explicit formula. The older work contains several substantial positive boards worth reusing.

Recovered result	Proved or recorded scope	Concrete relevance to Checkpoint 19
Complete amplitude/derivative Gram lift	A Gram matrix is positive for arbitrary finite coefficients; the amplitude lift is rank one.	Preserves mixed terms and phases. The signed contraction or derivative extracted from it requires its own sign estimate.
Accumulated outward positivity	The saved complete-theta argument establishes positive prefix integrals of the transverse amplitude derivative.	Test whether the present weights are nonnegative combinations of prefix integrals; calculate any additional signed weight.
Complete compensated theta polynomial blocks	Actual complete positivity through degree seven, including mixed and exterior costs, within each source theorem’s parameter range.	Supplies certified finite blocks if an explicit map from the current forced functions is established. The complement and cross block require payment.
Complete four-state arithmetic packets	The H analytical review proves positivity of the coupled-theta matrices on every set $\{1,m,n,mn\}$, integers $2 \leq m < n$; in particular $\{1,2,3,6\}$.	Try a weighted decomposition of the actual paired-prime operator into these blocks. Coinciding integer labels do not establish an operator identity.
Spatial theta kernel	Every two-node spatial J matrix is positive; a positive Gram majorant exists. A parity result gives positivity for odd real functions having one sign on the positive half-line.	A possible first-null shortcut if both the same-form identification and the required one-sign property can be proved. Entrywise majorization is not operator-order majorization.
Separated configurations and translated tails	Arbitrary finite configurations are covered under the saved separation or sufficiently-large-translation hypotheses.	Useful only with a quantified return/comparison to the actual central configuration.
Interior maximum-principle energy	Positive amplitude, nonnegative boundary values, and the exact coercive MP8/MP9 identities.	The weighted derivative of the Gram-curvature source remains the sign-bearing term. Positivity of the source quantity alone does not settle its derivative.
Prime-color squares and Smith-Jordan energy	Complete positive color Grams and divisor-square decompositions. CP19 gives actual coefficient maps on divisibility-chain supports.	Use the exact signed embedding. In the color bridge, a positive color matrix also occurs with a negative coefficient in the prime contribution.

Recovered result	Proved or recorded scope	Concrete relevance to Checkpoint 19
Historical Green energy and endpoint storage	Exact nonnegative energy, complete two-endpoint storage, survival accounting, and terminal terms. CP19 transports the Green/storage identities to actual forced columns.	The full continuum correction contains that same Green component; CP19 (2.63) prevents spending it twice. Green norm alone does not upper-bound the required L^2 residual norm.
Michaels modifier, dyadic subtraction, and prime towers	History records contractive finite-board modifiers, positive defect telescopes, and exact nonnegative Möbius-inverted shell sums.	Recover the original coefficient law and metric before substitution. Reconstructing the original vector restores the complete endpoint/correction energy.
Fibonacci-Lucas packet energy	The archived packet has energy below one and controlled cross interaction; authentic discrepancy blocks have an exact coupled recurrence.	A packet estimate needs the full comparison to the authentic target vector, including the residual square and endpoint. It is not already the Suzuki defect budget.
Light-cone, two-plane, and hidden-color completions	Exact positive norm completions and conserved visible-plus-hidden energy at finite stages.	Coherent observation/recombination retains signed cross terms. The positive carrier must be compared with the complete Weil form.
Original Suzuki feature energy	Its Gram form is unconditionally nonnegative; the original defect is measured relative to that metric.	Equality with the completed Weil form is the unresolved global transfer unless obtained after the independent sign proof. Preserve this metric throughout.
Certified odd Weil support interval	CP19 retains $Q_a \geq (1/5000)I$ for $0 < a \leq 699/2000 + e^{-2500}$.	This is positivity of the actual target form on a stated support class and is the current verified starting interval.
CP19 continuous energy	The complete odd theta-derivative gamma-polar energy has the explicit positive support-dependent floor $\eta_p(a)$.	This already acts on the current forced functions. The next calculation should compare the adverse paired-prime energy to this complete positive energy.

A recovered global claim and its correction

`Michaels_Compiled_H_Proof_from_Theta_Convexity.md` introduces unrestricted finite spatial-J positivity as the input **TCJ**. Its subsequent argument derives H from TCJ. It does not derive TCJ. `Michaels_Equation_14_16_Source_Trace.md`, §§3-4, explicitly locates this gap and recovers the narrower two-node, separated-configuration, and translated-tail theorems.

For the degree-seven result, the directly audited EH3 theorem in `RUNNING_MATHEMATICS(2).md` and `Michaels_Master_Proof_Trace_W_S.md` has both parity blocks and $0 \leq a < b \leq 1/10$, including its normalized diagonal limit. It gives a normalized matrix floor $I_4/2048$ and post-factor floor $I_4/4096$. Earlier chat summaries report a distinct degree-seven result on the larger range $b < 1/2$; that larger range must be checked against its original archive before use. Keep the two records separate.

Prompt to give the Checkpoint 19 research run

Continue from the complete Checkpoint 19 packet and the existing 301-entry theory atlas. Recover and **apply** our earlier positive-board and energy theorems to the complete current operator. The

objective is unconditional Weil positivity, retaining the original Suzuki metric and the original defect criterion as a separate preserved target.

Work in four closing groups. Carry out the proposed comparisons analytically. Each group must finish with its actual theorem or calculated obstruction, the exact domain and constants, and the resulting change in the complete target expression. If a sufficient estimate fails, retain its signed remainder and try the coupled comparison; renaming that estimate as another missing lemma is not completion of the calculation. Report honestly whether the global sign is established.

Group 1 – Recover the positive boards and transport their exact forms

Read the source proofs, particularly the complete degree-seven compensation, the $\{1,m,n,mn\}$ coupled-theta theorem, the parity/one-sign spatial-J theorem, accumulated outward positivity, the prime-color bridge, the modifier and survival identities, and the interior MP8/MP9 energy. Use the source table above and the atlas to recover the original names.

For every theorem actually used, supply the map, its metric and domain, and an identity of the form

$$\mathbf{Q}_a = T_a^* G_{\text{old}} T_a + R_a, \quad \mathbf{Q}_a = \tau_R^{-1} K_a (I - K_a).$$

Calculate the **complete** remainder R_a . Where the old positive form enters with a negative coefficient, derive the needed upper comparison. Preserve all endpoint, overlap, and mixed terms. In particular, retain CP19 (2.63): its positive Green/storage/color matrix also occurs inside the full correction.

For the one-sign route, determine whether the actual first-null equation forces the relevant transformed real odd vector to have one sign on the positive half-line. Establish the form identification first. If that property fails or remains unproved, display the precise reflected-prime term obstructing the proposed argument. Do not import positivity preservation of the inverse without proof.

The old compiled TCJ argument is conditional on unrestricted matrix positivity; it is not an available unconditional premise.

Group 2 – Expose the existing continuous energy as a complete positive covariance matrix

Keep CP19's actual normalization

$$f_j(x) = \sqrt{2}(W_a e_j)(x), \quad 0 < x < a, \quad F(x) = (f_1(x), \dots, f_d(x)), \quad p = -\phi' > 0.$$

Thus $\int_0^a F F^* = W_a W_a^*$. The positive continuous matrix is exactly

$$\mathbf{C}_a = \mathbf{G}_{p,a} + \mathbf{Z}_{p,a} = \eta_p(a) W_a^* W_a + \mathbf{N}_a.$$

Derive and verify the following concrete connection to our old Green/variance work. Set $c_j = 2j+1/2$. The lowest gamma channel and the **entire** favorable polar kernel combine as

$$e^{-|x-y|/2} - e^{-(x+y)/2} + 4 \sinh(x/2) \sinh(y/2) = e^{(x+y)/2} (1 - e^{-\min(x,y)}).$$

This last kernel equals $\int_0^\infty e^{-t} [\hat{e}(x/2) 1_{\{x>t\}}] [\hat{e}(y/2) 1_{\{y>t\}}] dt$. The remaining gamma channels satisfy

$$e^{-c|x-y|} - e^{-c(x+y)} = \int_0^\infty 2ce^{2ct} [e^{-cx} \mathbf{1}_{x>t}] [e^{-cy} \mathbf{1}_{y>t}] dt.$$

For a positive weight w define

$$M_w(t) = \int_t^\infty w(x)p(x) dx, \quad U_w(t) = \int_t^a w(x)F(x) dx,$$

$$\mathbf{V}_w(t) = M_w(t) \int_t^a \frac{w(x)}{p(x)} F(x)^* F(x) dx - U_w(t)^* U_w(t) \succeq 0.$$

Check the exact matrix representation

$$\mathbf{C}_a = \int_0^a e^{-t} \mathbf{V}_{e^{x/2}}(t) dt + \sum_{j \geq 1} \int_0^a 2c_j e^{2c_j t} \mathbf{V}_{e^{-c_j x}}(t) dt.$$

Use positive truncated integrals and the CP19 form-domain passage to justify the origin and infinite-channel limits. Retain the exterior mass in M_w ; it is part of the positive energy. This formula is an algebraic interface to verify and use, not an assertion that it already dominates the arithmetic contribution.

Now calculate a comparison of the adverse prime pairings with these **same** covariance matrices, keeping correlations. Test common representers, weighted conditional variance, and the existing endpoint/storage projections. State the actual weights and comparison constants. Do not replace the whole covariance matrix by its smallest scalar floor unless the resulting bound succeeds.

Group 3 – Couple the signed prime reflections before estimating

Start with the bounded, origin-cancelled expression in CP19 (2.88)/(2.105). Keep

$$V_p(x) = \sum_{n \geq 2} b_n D_{\log n}(x)$$

including every exterior prime. Do not estimate the separate singular f/p prime integrals on the closed domain.

First derive a positive-minus-positive decomposition of the bounded arithmetic form. For $\ell < a$, put $\Delta_\ell f(x) = f(x+\ell) - f(x)$ on $(0, a-\ell)$, and

$$d_\ell(x) = \mathbf{1}_{(0, a-\ell)}(x) + \mathbf{1}_{(\ell, a)}(x).$$

For $\ell < 2a$, reflect on $I_\ell = (\max(0, \ell-a), \min(a, \ell))$, with projections $\Pi_\ell^\pm = (I \pm R_\ell)/2$. Set $w = V_p - \sum_{\ell_n < a} \{ \ell_n < a \} b_{nd_\ell} \{ \ell_n \}$. Verify

$$\begin{aligned} \mathcal{P}_a(f) = & \sum_{\ell_n < a} b_n \|\Delta_{\ell_n} f\|^2 + \sum_{\ell_n < 2a} b_n \|\Pi_{\ell_n}^+ f\|^2 + \int_0^a w_+ |f|^2 \\ & - \sum_{\ell_n < 2a} b_n \|\Pi_{\ell_n}^- f\|^2 - \int_0^a w_- |f|^2. \end{aligned}$$

Then exploit the fact that the same function cannot freely attain all separate reflected worst cases. On the whole line, with zero extension, let $R_\ell f(x)=f(\ell-x)$ and $T_d f(x)=f(x+d)$. Since $R_\ell R_m = T_{m-\ell}$, verify

$$u\|(I + R_\ell)f\|^2 + v\|(I + R_m)f\|^2 \geq \frac{uv}{u+v}\|(I - T_{m-\ell})f\|^2.$$

Finite support gives the explicit shift gap

$$\|(I - T_d)f\|^2 \geq 4 \sin^2\left(\frac{\pi}{2(\lceil a/|d| \rceil + 1)}\right) \|f\|^2 \quad (d \neq 0).$$

Apply this to $\ell=\log 2$ and $m=\log 3$ and then to a coupled prime graph, allocating each edge weight only once. Calculate the actual gain in the signed pairing, rather than only a gain in the squared response error.

Also test the sharper next-prime-window calculation. For $\log 3/2 < a \leq \log 2$, ordinary active shifts vanish. The two reflected channels decompose into paths with weights b_2, b_3, b_2 , plus shorter paths. Establish the measurable decomposition and its exact spectral bound

$$b_2 \Re \langle f, R_{\log 2} f \rangle + b_3 \Re \langle f, R_{\log 3} f \rangle \geq -\frac{\sqrt{b_2^2 + 4b_3^2} + b_3}{2} \|f\|^2.$$

Retain V_p and all continuous energy while testing whether this pays a present deficit. These are proposed calculations for this run to check, not a claim of an enlarged certified interval.

In parallel, attempt to express the same coupled reflection/translation form through the older positive $\{1,2,3,6\}$ theta block and its general $\{1,m,n,mn\}$ family. Derive the connecting kernel and all overlap corrections. Extend through a general structural estimate if possible; finitely many positive blocks alone do not settle arbitrary support.

Group 4 – Combine the comparisons and return to the actual target

On the actual forced space, let P_a^+ and P_a^- denote the positive matrices obtained from the complete bounded splitting in Group 3, and set

$$A_a^{\text{cmp}} = C_a + P_a^+, \quad B_a^{\text{cmp}} = P_a^-, \quad Q_a = A_a^{\text{cmp}} - B_a^{\text{cmp}}.$$

Restrict inverses to the actual range of W_a . Its positive floor makes A_a^{cmp} positive there. Evaluate or analytically bound the complete normalized comparison

$$(A_a^{\text{cmp}})^{-1/2} B_a^{\text{cmp}} (A_a^{\text{cmp}})^{-1/2}.$$

The task is a proved upper bound by I, or a strictly subunit comparison on every hypothetical nonzero first-null direction. Merely defining this normalized matrix does not prove the bound. Use the covariance matrices, paired-reflection gains, and any genuinely transported old positive block together.

For a mapped finite positive block, keep its cross block and Schur correction. Use CP19's explicit response error $e_m = \tau R \kappa R^{(-1)\rho-R}(2m)$ to pay a calculated remainder only after matching the metric. Its convergence alone does not imply positivity: CP19 (4.14) gives a nonpositive lower certificate at a hypothetical null.

Seek a proof on all finite supports, or a cofinal sequence with justified restriction. For a first-null contradiction, retain both null branches and establish a strict inequality independent of the unknown margin. No support-independent gap at infinity is required. If the best completed comparison remains short, give its exact signed residual and its evaluated coefficients; identify which recovered theorem was applied and what it actually paid.

Close the four groups with a single combined calculation in the original normalization. Establish the final chain only after its antecedent:

$$Q_a^- \geq 0 \text{ for all finite supports} \implies \text{RH} \implies \mathfrak{W} = \mathcal{S} \implies \mathcal{E}_N = 0.$$

Preserve the certified interval, the original defect criterion, the full preceding packet, and its corrected proof statuses. Save the complete calculations and update the ledger. Clearly distinguish a completed identity, a proved numerical or analytic bound, a local positivity theorem, and a global closure.

Source register for the recovered positivity

- Michaels_Existing_Machinery_Energy_Closure_Checkpoint_19.md: §§2.4-2.5, 3, and 4. Source for the actual metric maps, origin cancellation, positive continuous floor, response errors, and current unresolved comparison.
- Michaels_Compiled_H_Proof_from_Theta_Convexity.md: opening TCJ assumption and conditional transfer.
- Michaels_Equation_14_16_Source_Trace.md: §§3-6, recovered theorem scopes and the missing unrestricted premise.
- Michaels_Master_Proof_Trace_W_S.md: degree-seven source scope and the incomplete full-form identification.
- Michaels_H_Analytical_Review.md: complete four-state positivity, separated clusters, two-prime reduction, and the distinction between a positive Gram form and its signed derivative.
- Michaels_Complete_Theta_Kernel_Continuation.tex: spatial two-node theorem, Gram majorant, parity cone, and the retained higher-node residual.
- RUNNING_MATHEMATICS(2).md: EH3-FINITE, EH3-CANCEL, and EH3-POSTFACTOR.
- EXTENDED_MANUSCRIPT(2).md: complete amplitude Gram and accumulated outward positivity.
- Michaels_Interior_to_Spectral_Transfer.md: MP1-MP13.
- Michaels_Prime_Color_Dual_Kernel_Bridge.md: complete color squares and the signed prime embedding, especially (3.6).
- Prime_DNA_Energy_Throughline.md, Michaels_Joint_Energy_Continuation.pdf, and Michaels_Survival_and_Prime_Transport_Continuation.md: historical energy, actual masks, shared allowances, and terminal/survival accounting.
- Michaels_Theory_Atlas_Purpose_and_Weil_Connections_2026-09-23.md: further source locations for light cones, modifiers, spirals, quadratic/infinity planes, and the remaining energy families. Its entries are an inventory, not 301 independently audited positivity theorems.

The broad history search recovered additional statements and aliases; it was not an exhaustive literal export of every chat message. Where a source range or proof conflicts with a later summary, use the actual complete derivation and record the correction.

Research ledger and source inventory

Source: Michaels_Running_File_Ledger.md. Equation tags carry the source prefix LEDGER; references within this unit retain their original local numbering.

Checkpoint 20 – 23 September 2026

Current working target: unconditional Weil positivity

The primary user-directed objective is now

$$\mathfrak{W}(q, q) \geq 0 \quad (q \in C_c^\infty(\mathbb{R}))$$

independently of RH, preserving the original Suzuki metric and complete arithmetic form. A direct Hilbert-Schmidt estimate is optional. The retained success criterion is the unconditional bound

$$\liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1, \quad \mathcal{E}_N = \|D_N\|_F^2,$$

in the existing Suzuki-orthonormal determining ladder. The preceding direct witness transfer makes this sufficient for the same backward-propagation contradiction. Checkpoint 10 tests finite extraction energy, explicit PNT bounds, and orthogonal saturation against this unchanged normalized defect. Checkpoint 11 derives the complete Suzuki Gram correction and an explicit relative bound for the central extraction. Checkpoint 12 carries out four closure groups, proves the fixed-support null-domain passage, and isolates the first-null remainder inequality with an exact positive margin. Checkpoint 13 derives the full multiplier arithmetic, exact endpoint/exterior cancellation, and quantitative local correction bounds; the complete signed estimate remains open. Checkpoint 14 closes the global Suzuki graph estimate and quantifies finite dilation, renormalized extension energy, and all introduced regulator poles; the actual signed closing inequality remains open. Checkpoint 15 establishes the odd-sector reduction, calculates complete mixed prime-power packets and their polar inverse connection, and proves an explicit strict odd sign on supports with $a \leq 1/8$. Checkpoint 16 proves a prime-active odd support certificate up to $a = 699/2000$, then a complete nonzero-trial collar extension to $a = 699/2000 + e^{-2500}$ with Weil lower bound $1/5000$. The exact interval-growth budget and a uniform finite-rank regularization are proved; the margin-relative global estimate and global kernel exclusion remain open. Checkpoint 17 puts all first-null branches into one fixed-space regularization, establishes boundedness and boundary upper estimates for the actual forced solutions, proves the complete collar cost is $O_a(\delta)$, and calculates its averaged boundary coefficient. The scalar budget is locally absolutely continuous wherever H_a is positive. The full regularized matrix has a uniform Dini modulus through every prime threshold. The signed Osgood estimate remains open. Checkpoint 18 fixes the regularizer and source functions in physical coordinates, proving that the entire finite matrix is monotone and uniformly Lipschitz, with an almost-everywhere boundary Gram derivative. It calculates exact endpoint and adjoint-probe residuals, removes the exponential probe-energy cost on the critical source sector, and proves the conditional adaptive Osgood argument. The actual signed arithmetic residual has an almost-everywhere limit; its margin-dependent bound remains open. Checkpoint 19 applies the recovered complete theta identity, arithmetic packet cancellation, Green/storage/color maps, and residual-response certificate to the actual forced family. It proves explicit packet and response savings, a positive odd theta-derivative continuous-energy floor, and a uniformly accurate finite response certificate. The full paired-prime lower comparison remains open. Global Weil positivity and the original defect criterion remain open. Checkpoint 20 executes the recovered positive-board comparison: it proves exact continuous

covariance and reflection-path identities, quantified common-representer and graph-residual payments, and an unconditional enlarged odd Weil-positive interval $a \leq 3/8$ with uniform floor $1/200$. Its exact rational certificate retains the complete first-prime reflection and pays all archimedean, polar, endpoint and approximation terms. The all-support signed comparison and original Suzuki defect criterion remain open.

Earlier checkpoint development

Earlier user-directed target change: prove that a hypothetical off-critical zero forces a contradiction in an independently established relation satisfied by the complete local amplitude data at a known critical-line zero. This was the proof objective at that stage; the current primary objective and proof statuses are recorded above.

Write $F(z) = \xi(1/2 + z)$, fix an established zero $\rho_0 = i\gamma$ of multiplicity m , and remove its known vanishing factor:

$$F(z) = (z - \rho_0)^m h_\gamma(z), \quad h_\gamma(\rho_0) \neq 0.$$

The complete amplitude data mean every mixed Taylor coefficient of $|h_\gamma(\rho_0 + w)|^2$, not just its value or finitely many zero locations.

The exact global-to-local identity available as a first step is

$$(\log h_\gamma)''(\rho_0) = - \sum_{\rho \neq \rho_0} \frac{m_\rho}{(\rho_0 - \rho)^2}.$$

The sum is over distinct remaining zeros with their multiplicities and converges absolutely. It follows by twice differentiating the order-one canonical product. Thus every remaining zero contributes to local derivative data.

Checkpoint 3 advancement. In the real-axis coordinate $f(w) = \Xi(w) = F(iw)$, set $g_\gamma(u) = f(\gamma + u)/u^m$, $A_\gamma(u) = g_\gamma(u)^2/g_\gamma(0)^2$ for real u near zero, and

$$s_n(\gamma) = -\frac{(\log A_\gamma)^{(n+2)}(0)}{2(n+1)!}, \quad H_N(\gamma) = [s_{j+k}(\gamma)]_{j,k=0}^N.$$

The new note proves the exact implication

$$\text{an off-critical zero exists} \implies \text{for every fixed known } \gamma, \exists N, c \in \mathbb{R}^{N+1} : c^T H_N(\gamma) c < 0.$$

It is a local-amplitude application of the classical Grommer criterion, with a self-contained proof retaining all zeros. The earlier product law H6 identifies $s_0(\gamma) = L_{m+1}[\Xi](\gamma)/L_m[\Xi](\gamma)$. No numerical ladder was used.

Current remaining condition: establish $H_N(\gamma) \succeq 0$ for every finite N , at one fixed known zero γ , from an independently proved property of the actual theta or prime structure. Equivalently, prove $\mathcal{L}_\gamma(P^2) \geq 0$ for every real polynomial P , where $\mathcal{L}_\gamma(x^n) = s_n(\gamma)$. The two directly checked source manuscripts establish compatibility and limited positive forms; they do not establish this all-order local sign. The detector implication is proved; the contradiction still requires this sign.

Checkpoint 4 direct attempt. Sections 9-12 of the same local-amplitude note now give an exact transfer to the original spatial J :

$$s_{j+k}(\gamma) = \mathcal{B}_J(a_j, a_k), \quad \mathcal{L}_\gamma(P^2) = \mathcal{B}_J(A_P, A_P), \quad A_P = \sum_j c_j a_j.$$

The functions a_j are explicit polynomial-exponential functions defined in that note's (39)-(42). Gaussian bounds justify the complete double integrals and all derivatives. The exact Schur correction for removing a multiplicity- m zero is retained. The prior primitive traces of the generating functions are $1/D(z)$ and $-i(\gamma+z)/D(z)$, so earlier zero-trace results cannot be substituted without those corrections.

The same attempt gives the complete zero-removed theta density, which is complex-valued, and the complete analytically continued Euler expression. A single Euler prime-power term has an indefinite 2×2 extracted local matrix for every phase. This diagnoses the failure of an individual positive-color-block substitution; it says nothing negative about the completed actual local form. The transfer is established. Its all-polynomial sign remains unproved. The research target is unchanged.

Checkpoint 5 plane check and next step. The source check confirms positive quadratic-amplitude and prime-Hilbert norms. The present spatial form is $\mathcal{B}_J$; its identification with one of those positive norms was not found in the checked sources. The full polynomial-exponential jet form at a known zero has the exact block congruence $0_{m-1} \oplus [m] \oplus H_N(\gamma)$. Thus its known-zero pivot is positive and all remaining possible adverse directions lie in the specified local block.

The exact next reduction reaches the previous two-trace primitive class. Let

$$\chi_\gamma(X) = (\gamma^2 + \tfrac{1}{4})X^2 + 2\gamma X + 1.$$

For real P , both primitive traces of A_P vanish exactly when $P = \chi_\gamma R$. On that family,

$$\mathcal{L}_\gamma((\chi_\gamma R)^2) = \mathcal{B}_J(A_{\chi_\gamma R}, A_{\chi_\gamma R}) = \mathcal{B}_{J_B}(LA_{\chi_\gamma R}, LA_{\chi_\gamma R}).$$

Both boundary corrections are zero individually. This smaller family still detects every possible off-critical zero: its only potential blind locations are $\Xi(\pm i/2) = \xi(0), \xi(1) = 1/2$, which are not zeros. Sections 13-15 of the continuing note prove these statements. The earlier two-trace result establishes sufficiency; its positivity premise and the present restricted positivity remain open. No numerical ladder was used.

Checkpoint 6: transfer of the previous positivity theorem. Build v23 equations (5.2) and (5.7) prove the Smith-Jordan square factorization and the nonnegative prime-power derivative form on every finitely supported zero-sum sequence. Sections 16-17 of the continuing note insert that exact source result and give the explicit coefficient compression $T_{\gamma,N} = C_{\gamma,N}^T H_{N+2} C_{\gamma,N}$ for the present restricted target. An assignment identifying this target with the prime-power square sum would suffice to apply the earlier theorem; that assignment was not recovered or constructed. An independently proved lower comparison with that nonnegative sum would also suffice.

The direct spatial feature transfer was checked analytically with its entire boundary retained. The target is a mixed form, while the separate feature norms are positive. For the polynomial-exponential local family, the features are generally outside half-line L^2 . The exact finite-interval identity retains the translated tail $m|\tau_b|^2 + \mathcal{L}_\gamma(P_b \overline{P_b})$. The restriction $P = \chi_\gamma R$ does not cancel that tail. These are transfer checks, not a numerical ladder or an unconditional proof of the remaining positivity.

Checkpoint 7: dependency review and correction. Equation (77) was overstated as a required identity in the preceding response and one sentence of the note. It is an optional sufficient representation. Section 15 already proves that positivity of the complete restricted local form yields the contradiction, independently of the prime-power representation and the spatial feature route. The boundary identities remain valid within that route; they do not require a separate boundary-cancellation theorem or separate positivity of every component. The rectangular matrix in (76)

is now described as a compression rather than an invertible congruence, and the compact-support domain in (78) explicitly includes L^2 . This review corrects the proof dependencies and domain wording; it establishes no new sign theorem.

Checkpoint 8: unified proof compilation. The latest full manuscript (baseline TeX version 6) and the necessary complete post-paper derivations are assembled in one standalone, 78-page paper, *Michaels_Consolidated_Theta_Local_Amplitude_Proof*. The original Sections 1-14 and principal equivalence are retained. New Section 15 includes the complete prime-color, tensor, logarithmic, and Smith-Jordan positive forms; Sections 16-19 include the full local moment/resolvent proof, exact J compression, known-zero block, primitive correction, and trace cancellation. Section 20.1 assembles the local and global criteria. Appendices F-G include the complete dense-core, weighted-continuity, two-trace sufficiency, variance/covariance, zero-removed density, Euler expression, and boundary derivations.

The compilation places every mathematical hypothesis at its point of use. The original closing Theorem 14.4 takes full spatial positivity (14.16) as its input and proves H. The added local criterion is Theorem 17.1; its exact restricted form is (19.11), and the combined equivalence is (20.6). The assembly does not produce an independent proof of the full spatial positivity input or the equivalent restricted sign. The optional prime-norm identity remains optional.

Validation: 453 unique internal labels, no unresolved label references or citations, no TeX layout warnings, correct automated contents and numbering, 78 rendered pages reviewed in contact sheets with full-size checks of the title, original closing theorem, restricted form, and complete boundary identity. The source archive contains one standalone TeX file and compilation instructions. The original baseline and all earlier research notes remain preserved.

Checkpoint 9: heat, defect, and two-trace intersection. The three requested investigations were run in parallel with targeted archive recovery and explicit domain comparisons. The new continuation proves a direct transfer from a negative local value $Q_\gamma(R) = -\delta < 0$ to a compact smooth Weil test q , then to a permanent Suzuki-normalized defect floor

$$\mathcal{E}_N \geq \left(1 + \frac{\delta}{6\mathcal{S}(q, q)}\right)^2 > 1 \quad (N \geq N_0).$$

The coefficient transformation is $d_j = a_j F(b + it_j)$, with $b > 1/2$; the complete arithmetic form is retained. The August 25 threshold-one criterion was recovered: a ceiling $\mathcal{E}_N \leq 1$ on an unbounded subsequence suffices, so decay to zero is unnecessary. The archived radical/source decays and boundary-normalized tail estimates do not bound this same unnormalized defect, and the source itself leaves its arithmetic upper inequality open.

The exact Gaussian limit has a rank-one spatial kernel. Rodgers-Tao's theorem, under the normalization $\Xi_\lambda(w) = 8H_{4\lambda}(2w)$, implies nonreal zeros at every finite negative heat parameter. Thus the positive limiting field supplies no preserved global sign at finite negative deformation. The correct heat primitive and trace projector move with the parameter.

The two-trace orthogonal projection deletes exactly the primitive rank-two boundary correction. An analytic construction within the actual real polynomial family gives a spatial integrand $I(0) = 0$, $I'(0) = -2$, although both traces vanish. This proves the adverse mixed operation survives; it does not give a negative value of the complete form. The continuation retains the full boundary and compensation. No numerical ladder and no unconditional closure of the target sign are claimed. The existing main manuscript is unchanged.

Checkpoint 10: uniform defect-cap tests. The original scalar extraction estimate produces a new all-degree theorem: in ordinary monomial coefficient coordinates, the normalized compact

extraction matrix obeys

$$\mathrm{tr} X_m \leq \frac{1}{175\pi}, \quad \|X_m\|_F^2 \leq \frac{1}{(175\pi)^2}.$$

The proof sums the entire monomial geometric series under the original compact extraction profile. The complete exponential remainder gives an operator bound in its exponential metric; its Frobenius estimate retains dimension. The exact target remains $\|G_N^{-1/2} Z_N G_N^{-1/2}\|_F^2$, with the full completed numerator and the Suzuki Gram inverse retained. No uniform transfer of the extraction cap to this target was found.

The prime calculation gives an exact Stieltjes representation using $E_0(x) = \psi(x) - x + 1$, including the lower endpoint and cancellation with the continuum pole term. The published unconditional PNT bound yields a majorant growing like $v^{3/2} e^{v/2 - c\sqrt{v}}$ after the required weighting. It supplies fixed-support estimates, with no bound on the full normalized defect across the determining ladder.

The saturation theorem proves that $\sup_N \mathcal{E}_N < \infty$ is exactly existence of a Hilbert-Schmidt extension of the actual defect on the Suzuki completion, with limit equal to its squared Hilbert-Schmidt norm. A rank-one operator can have a stable total above one, so exhaustion alone supplies no unit cap. Suzuki's zero-feature orthonormality was checked in the primary paper and retains its explicit RH hypothesis; it was not imported. The note also proves the exact compact-operator diagnostic $\|P_N(K - I)P_N\|_F^2/N \rightarrow 1$ for hypothetical compact K , explaining why smoothing of the whole spatial operator is insufficient for its difference from the identity metric. This is not a divergence claim about the actual defect.

All three tests are analytical. The target upper cap and RH remain unproved. The prior manuscript and notes are preserved unchanged.

Checkpoint 11: completed Gram correction and relative extraction. For the actual Suzuki features, the full Cauchy integral of the positive boundary weight produces the exact identity

$$\langle F_\gamma, F_\delta \rangle = \mathbf{1}_{\{\gamma=\bar{\delta}\}} + \sqrt{m_\gamma m_\delta} \mathcal{R}[\gamma, \bar{\delta}].$$

The real meromorphic correction is defined by $\mathcal{R} = C - iA/E$ above the line and its reflected formula below. Every hypothetical nonreal feature has nonzero coupling with every real-zero feature. Smooth symmetric zero cutoffs converge in L^2 , giving a justified complete compact-test defect formula; no absolute convergence of an uncut double Gram series is assumed. The corrected two-sided convention gives $V = -L_0$, with the sign cancelling in the positive metric.

For every finite zero set closed under sign and conjugation, the explicitly normalized polynomial numerator gives a dimension-independent comparison

$$\mathcal{X}(P_{S,c}) \leq \frac{3567}{509894} c^* G_{Sc} < \frac{1}{140} c^* G_{Sc} \quad (c \neq 0).$$

The proof uses the archived scalar theta derivative bound, proves $w \geq 5929/5945$ on the central band, and obtains $C_\Gamma < 5/96$ analytically from the paired canonical product. It uses no zero computation or degree ladder. The maps have a compatible bounded multiplier limit. The actual full correction is not identified with this extraction component. The operator comparison does not give a uniform Hilbert-Schmidt cap on the full defect. The target upper bound and unconditional RH closure remain open.

The continuation includes the proofs, conventions, complete contour terms, exact relative trace with its inverse metric, and finite reflection-pair test. Earlier papers and notes are preserved unchanged.

Checkpoint 12: four closure groups and the first-null remainder. The exact compatible extraction gives

$$\Delta = -\mathcal{X}_\infty + \mathcal{T}, \quad K_0 = \frac{17607901}{2531730432} < K_*, \quad \varepsilon_* = K_* - K_0 = \frac{4429025}{108864408576} > 0.$$

The full correction, pole terms, contour terms, and all cross terms remain inside $\mathcal{T}$. A lower estimate $\mathcal{T} \geq -(506327/509894)\mathcal{S}$ would give $\mathfrak{W} \geq \varepsilon_*\mathcal{S}$ and the explicit conditional absorption factor $1 - \varepsilon_* < 1$. The independent remainder estimate remains open.

Suzuki's *Weil's quadratic form via the screw function*, arXiv:2606.09096v2, Sections 8.2-8.5, supplies the continuous-kernel representation. The original Suzuki metric is transported separately as $\Sigma_a(u, w) = \mathcal{S}(D^{-1}u, D^{-1}w)$. The complete first-null equation is $g * v = \alpha x + \beta$ on the interval. Endpoint distributions and affine terms are retained. A new closed auxiliary theorem proves that, on fixed support, $q_n \rightarrow v$ in L^2 and $\mathcal{S}(q_n) \rightarrow 0$ imply $v = 0$. The proof constructs the closed injective maximal Suzuki map, using its actual logarithmic time-derivative bound and unconditional distinct simple-zero density. Thus first-null core approximants need no unproved Suzuki-norm continuity or endpoint trace.

The tensor energy retains $\overline{G^{-1}} \otimes G^{-1}$. The square-coordinate transport is unitary onto two weighted parity channels. The actual correction is odd, $\mathcal{R}(z) = zr(z^2)$, and opposite-parity terms vanish exactly. Every within-channel mixed term remains. Complete boundary estimates and their return map show that shrinking a Cayley-weighted boundary norm leaves the Hankel norm invariant. The explicit Cauchy evaluation bound cancels the small scale factor.

The complete pole-boundary norm budget and actual Hardy representation give

$$\|H_h\| \leq \frac{1}{2}, \quad K - K^2 = -(H_\Theta U^* + U H_\Theta^*), \quad \|K - K^2\|_{\text{HS}}^2 = 2\|H_\Theta\|_{\text{HS}}^2 + 2\|H_\Theta\|_{S_4}^4.$$

These are ambient Hardy statements with the actual symbol. Uniform normalized residue-evaluation bounds needed for the complete determining metric remain explicit and unproved. Fixed-test cutoff convergence and compact-disk Cauchy tails are proved; no unproved uniform Hilbert-Schmidt tail is imported.

For any hypothetical first-null form-core sequence, the closed-domain theorem and the exact remainder yield

$$\limsup_n \frac{\mathcal{T}(q_n, q_n)}{\mathcal{S}(q_n, q_n)} \leq -\frac{506327}{509894} - \varepsilon_*.$$

The selected narrower lemma asks for one such sequence at each hypothetical null vector whose lower limit is at least $-506327/509894$. This is equation (5.6) in the continuation. Its independent proof would exclude first degeneracy, yield Weil positivity, then RH and the correctly normalized Suzuki identity, giving $\mathcal{E}_N = 0$. That lemma is not proved in this round. The target remains open. The continuation contains 114 numbered equations and explicit closure records for all four groups.

Checkpoint 13: null multipliers, endpoint cancellation, and the signed complete pairing. The domain passage and exact target from checkpoint 12 are retained. Smooth multiplication is proved bounded on the closed logarithmic form domain, and a real definite-parity null vector may be chosen without assuming simplicity or pointwise positivity. The exact double-commutator identity has the positive g'' sign. Its difference factor removes the diagonal finite-part contribution while retaining the archimedean density, completed polar term, and every nonzero prime shift.

The continuation gives the complete x -multiplier identity, with $(\log n)^2$ weights; the paired cosine/sine identity with an exact remainder; the full (x, x^2, x^3) matrix and coupled determinant;

and the signed endpoint substitutions into its polar moments. The large-frequency trigonometric energy has positive leading term $\|v\|_2^2 \log |\tau|$. No negative direction has been proved from the actual signed correlations.

The convolution $f_v = g * v$ is C^1 , so its endpoint value and derivative equations require no traces of v . Weighted sampling over all xi zeros proves an absolutely locally uniform Cauchy synthesis for compact L^2 inputs. The complete null equation supplies the new exact identity

$$L_{v,b}(z) = e^{-ibz} L_v(z) \quad (|b| \leq a).$$

The two exterior Fourier-Laplace formulas retain every affine subtraction and continue meromorphically through the zero strip with their checked Cauchy-denominator signs. For core tests,

$$\mathcal{E}_b(q; z) = -ie^{-ibz} Q_a(q, r_{b,z}), \quad |\mathcal{E}_b(q; z)|^2 \leq e^{2b\Im z} Q_a(q) Q_a(r_{b,z}).$$

The test energy and its endpoint terms are explicit. This yields a proved normalized error bound for the combined residue-boundary expression on every fixed contour.

Additional exact-null regularity follows from the whole-line $\log |D|$ equation and a Carleman bound at the two exterior endpoints:

$$\|\log |D| v\|_2^2 \leq ((C_a^{\text{op}})^2 + \pi^2/2) \|v\|_2^2.$$

It improves exact-null Cauchy tails to $O(R^{-1/2})$ on compact evaluation sets. It does not assert a uniformly bounded graph norm for every form-core sequence or a finite global Suzuki norm for every null vector.

The decisive combined calculation is exact invariance of the complete quadratic product,

$$L_{v,b} L_{v,b}^\# = L_v L_v^\#.$$

Thus the support-gap exponentials cancel in the full energy, including all higher-pole derivatives. The actual real-node feature singularities are removable, but the complete correction retains the mixed-node coefficient and the phase-sensitive pole terms. The three near-null error terms are displayed and bounded on fixed contours. The necessary uniform signed estimate on an exhaustion of all nodes and poles remains unproved.

The selected gap-paired remainder lemma is equation (4.8) of the new note. Its exact threshold is $-1 + K_*$, with contradiction margin $\varepsilon_* = 4429025/108864408576$. A proved interval-dependent threshold $-1 + K_0 + \eta_a$, $\eta_a > 0$, would also suffice. The conditional chain returns through Weil positivity, RH, and the correctly normalized Suzuki identity to $\mathcal{E}_N = 0$. The antecedent, the original target, and unconditional RH closure remain open. The note contains 116 numbered equations and explicit closure records for all three groups.

Checkpoint 14: global Suzuki domain, complete finite dilation, extension, and infinity regulator. The original Suzuki-normalized defect target, complete meromorphic correction, endpoint identities, and shift-product invariant are preserved. The new fixed-support theorem is

$$\|Vq\|_2^2 \leq \frac{C_a}{2\pi} \int_{\mathbb{R}} [1 + \log^2(2 + |\xi|)] |\widehat{q}(\xi)|^2 d\xi.$$

Its proof differentiates the actual Lerch component before estimating. The resulting Hankel kernel is bounded by a half Carleman kernel plus an exponential rank-one kernel, giving norm at most $\pi/2 + 1$. The half-line logarithmic commutator has norm $\pi/2$. The completed gamma multiplier,

both polar terms, every finite prime packet, and corrected negative-time convention remain explicit. An actual same-parity inward-dilation and mollifier sequence now converges strongly under V . Thus every hypothetical null vector has positive finite Suzuki energy, convergence of core energies, and uniform spectral tightness along this sequence.

Group 2 proves the full finite-dilation identity with coefficient $s\|v\|_2^2$, all prime thresholds, and the exact R -kernel. It inserts the endpoint equations into the completed polar moments and derives a new signed exterior-collar identity from the full null equation. The quantified correlation modulus follows from logarithmic-square regularity. Large inward dilation has positive leading term $s\|v\|_2^2$, so a possible negative direction is restricted to the explicit finite interval $0 < s < S_a$. No one-sided stationarity or derivative of the null vector is assumed.

Group 3 proves the distributional three-dimensional line source, global strong L^2 trace, full logarithmic energy subtraction, exact endpoint-cap cancellation, vanishing outer flux, and all prime translations. The renormalized energy is

$$\mathcal{D}_{\text{ren}}(v) = (\log 2 - \gamma_E)\|v\|_2^2 - \langle \log |D|v, v \rangle.$$

Its finite-scale decrement is $-s\|v\|_2^2$, returning exactly to Group 2. The calculated positive-height flux has multiplier $1 - z^2(K_0(z)^2 + K_1(z)^2)$, which takes both signs. The two hypotheses for unique continuation were checked; a nonzero compact input cannot make its logarithmic Laplacian vanish on an exterior open interval.

Group 4 proves regularity at every artificial sech pole, the residue formula $-\tau^{-2}F'(p_k)$, and a full explicit $O_a(\tau^{-1}e^{-a\pi/\tau})\|v\|_2^2$ budget. Finite-dimensionality of the fixed-interval nullspace and the new graph theorem make this budget uniform after Suzuki normalization on that nullspace. The coefficient-regulation correction includes both mixed terms and its square; all actual higher-pole derivatives and mixed-node residues are retained. Exterior contour limits have an explicit order. Raw zero-cutoff convergence on the exact null vector and an unregulated central contour-residue exchange are not asserted.

The combined calculation identifies trace reciprocity with the same signed collar and shows exactly where the estimates stop. The next selected lemma is (5.12): the full arithmetic increment must exceed $s\|v\|_2^2$ by a positive margin for some $0 < s < S_a$. The alternate exact-null remainder criterion is (5.10), with retained contradiction margin $\varepsilon_* = 4429025/108864408576$. Neither signed inequality is established. The auxiliary domain, dilation, trace, energy, and regulator results are proved; the unconditional original target remains open. The note contains 125 numbered equations, complete closure records for all four groups, and the conditional dependency chain back to $\mathcal{E}_N = 0$.

Checkpoint 15: odd tests, mixed prime-power packets, and the polar resolvent.

The primary objective is unconditional Weil positivity; the original Suzuki defect ceiling remains a success criterion. Yoshida's odd-test criterion is recovered in the unchanged normalization. The odd lowest eigenvalue is proved continuous and attained by a direct common-domain argument, so RH failure yields a finite odd first crossing. Parity commutation makes an odd restricted null a full-operator null even when only the odd sector is nonnegative. All prior graph, endpoint, exterior, and strong Suzuki-core conclusions survive. Every Cauchy-Schwarz use is projected into the odd sector, including its reflection cross term. The projected step test gives the exact necessary packet $g(2b) - 4g(b) \geq 0$.

The complete mixed identity is proved for $K_v(s, t) = Q_a(U_s v, U_t v)$, with coefficient $sA_v(t - s)$, all prime increments, and the exact remainder. For integer factors, each prime-power coefficient is

$$w_m(p, k) = \mathbf{1}_{m|k} p^{-k/(2m)} - p^{-k/2}.$$

The 2, 3, 6 packets, their archimedean companions, and all odd polar moments are explicit. The composition coefficient $\eta = w_6 - w_2 - w_3$ has a complete divisibility sign classification. Among exponents divisible by six, the sole negative case is $p = 2, k = 6$, with coefficient $(2\sqrt{2} - 3)/8$. Distinct relative-scale correlations remain separate in the actual three-scale matrix.

Oddness removes the constant archimedean contribution and yields a sharper finite kernel bound. An exact primitive interaction is isolated with its whole remainder retained and bounded. The prime packets have actual Poisson-symbol representations, each with both signs, and explicit complete operator-size bounds. These facts do not assign a sign to the full coupled matrix.

The 2×2 Schur residual, every determinant cross term, full 3×3 determinant, and composed direction $(U_{\log 2} - I)(U_{\log 3} - I)v$ are calculated. Endpoint substitution precedes estimation. Its polar contribution contains an exact negative square and an endpoint-coupled term. The kinetic scale matrix is strictly positive independently. A common sufficiently large inward dilation also makes every fixed finite family positive, with the stated tail estimate. At a hypothetical first crossing the earlier smaller odd interval is already strictly positive; that conditional constraint is kept separate from an independent arithmetic exclusion.

The two routes meet exactly in $K = D - 2ff^T$, where D is the complete polar-free form matrix. The finite-block inverse has a nonnegative complementary energy term; it is a lower approximation to the full inverse scalar unless the whole tail is included. The original Suzuki Gram matrix is retained by congruence when reading signs.

On the full odd space the polar removal has coefficient two, with $\|s_a\|_2^2 = \sinh a - a$. The exact digamma multiplier, forced equation $H_a w = s_a$, zero-moment kernel, and saturation branch are derived. An actual inverse estimate follows from the new odd support inequality

$$\langle \log |D|q, q \rangle \geq [-\log a + \frac{1}{3} \log(9\pi/2) - \frac{1}{3}] \|q\|_2^2.$$

It gives the independently proved strict sign

$$0 < a \leq 1/8 : \quad H_a > \frac{1}{6}I, \quad 2\langle s_a, H_a^{-1}s_a \rangle < \frac{3}{256}, \quad A_a^- > \frac{253}{1536}I.$$

The inverse Neumann construction and its norm tail, as well as the complete Schur inverse correction and its uniform spectral tail bound, are proved on their stated domains. The actual reflected prime antidiagonal prevents importing the proposed positivity-improvement argument once a prime packet is present.

The remaining general exclusion is equation (5.11): at every hypothetical odd first crossing, independently establish both $\ker H_a = \{0\}$ and a strictly subunit full inverse scalar, or obtain an independently negative admissible mixed energy. Both kernel branches and all Schur corrections are retained. The note contains 152 numbered equations and four complete closure records.

Separate status: unconditional odd Weil positivity is proved on the explicit support interval $0 < a \leq 1/8$. Global Weil positivity remains open. The original criterion $\liminf_N \mathcal{E}_N \leq 1$ remains open. The return through odd positivity, Yoshida, RH, Weil positivity, and the normalized Suzuki identity to $\mathcal{E}_N = 0$ is recorded as conditional.

Checkpoint 16: support growth and the complete positivity budget. The original Suzuki metric, complete arithmetic form, global Suzuki domain passage, endpoint identities, and conditional return to zero defect are retained. Fixed inward dilation families do not exclude a first odd null because they already lie in an earlier positive interval. The new direction is support enlargement.

The weighted sine concentration operator gives the exact logarithmic splitting and a rigorous certificate $\|T_8\| < 619/500$, with an explicit Hilbert-Schmidt and operator-norm remainder and

exact finite rational trace comparisons. The first prime packet is treated on its actual reflected collar, with both sign channels, their archimedean projections, and the nonzero polar cross terms retained. A localized concentration-prime resolvent inequality bounds $T + bP$ without discarding the inner-collar coupling. Odd cancellation removes $\alpha(0) = 1/4$ before estimation, yielding $\|A_a\| \leq 11a^2/(10\pi^2)$ in the indicated range. Together these prove

$$0 < a \leq 699/2000 : \quad H_a > 3313/200000 I, \quad \sigma_a < 14318/16565, \quad A_a^- > 2247/10^6 I.$$

This is an unconditional sign theorem on an interval with the first prime packet active.

The sharp core/collar split is proved on the actual logarithmic form domain, using the bounded Carleman commutator and interpolation, without endpoint traces. The complete archimedean interface/exterior weights and every ordinary/reflected prime cutoff are displayed. For the full block operator, the proved identities are

$$\sigma_b - \sigma_a = 2\langle r, S^{-1}r \rangle, \quad S = C - B^* H_a^{-1} B, \quad r = t - B^* H_a^{-1} s_a, \quad S_A = S - 2|r\rangle\langle r|/(1 - \sigma_a),$$

under their stated coercivity hypotheses. The source on the half interval is $\sqrt{2} \sinh(x/2)$, preserving the complete polar normalization. All Schur and source mixed terms are retained.

The reflected archimedean Gram kernel is positive, including its full geometric tail. The actual cross estimate improves to

$$\|B\| \leq \pi/2 + \sqrt{a\delta} + \sum_{\ell_n < b} b_n + \sum_{a < \ell_n < a+b} b_n.$$

For $a = 699/2000$, $\delta = e^{-2500}$, the complete collar satisfies $C > 2498I$, $\|B\| < 21/10$, and $S > 2230I$. The nonzero trial $z = t/2500$ has an explicitly calculated residual, including both endpoint logarithms, the full reflected archimedean term, and the reflected prime cross block. It certifies the total inverse-energy increment below $1/9$. Consequently

$$0 < a \leq 699/2000 + e^{-2500} : \quad Q_a^-(q) \geq \|q\|_2^2/5000.$$

This is the largest support interval certified in this checkpoint, not a claim of an optimal interval.

For the kernel branch, a new explicit phase-space estimate $\nu_N(a) \geq \log(\pi N/(2a)) - 1$ gives a certified complementary block and explicit constants d_R, τ_R for which $H_{\tau_R, a} = A_a^- + \tau_R P_{d_R}$ is uniformly strictly positive for $a \leq R$. Every null vector, including a zero-moment vector, is equivalent to eigenvalue one in $\tau_R P H_{\tau_R, a}^{-1} P$. Its complete finite Schur matrix retains all nine prime/archimedean/polar correction terms. The calculated general upper bound does not exclude that eigenvalue.

Separate status: the complete local collar estimate closes for the specified extension. The absolute general collar bound does not supply a relative factor $1 - \sigma_a$ with an independently integrable rate. Finite-endpoint failure and a kernel outside the certified interval remain unexcluded. Global unconditional Weil positivity remains open. The original criterion $\liminf_N \mathcal{E}_N \leq 1$ remains open. The return through Yoshida's odd criterion, RH, Weil positivity, and the normalized Suzuki identity to $\mathcal{E}_N = 0$ is conditional. The note contains 138 numbered equations and complete closure records for all four groups.

Checkpoint 17: actual boundary cost and unified first-crossing calculation. The exact prime-active starting interval

$$0 < a \leq a_0 = 699/2000 + e^{-2500}, \quad Q_a^- \geq I/5000$$

is preserved, together with the original Suzuki metric, all arithmetic and polar terms, the supported logarithmic domain, and the endpoint-complete identities.

The reference logarithmic projection is fixed under support scaling. Checkpoint 16's explicit d_R, τ_R give $\mathcal{H}_a = \tilde{A}_a + \tau_R P \geq g_R I > 0$, independently of the Weil margin. The matrix $K_a = \tau_R P \mathcal{H}_a^{-1} P$ detects every first null, including a zero-moment null. The exact finite-difference identity and near-null equations are

$$K_b - K_a = -\tau_R P \mathcal{H}_b^{-1} (\tilde{A}_b - \tilde{A}_a) \mathcal{H}_a^{-1} P, \quad \tilde{A}_a v_a = \varepsilon(a) u_a, \quad P v_a = (1 - \varepsilon(a)) u_a / \tau_R.$$

The regularized inverse remains bounded while the vectors remain nonzero at a possible first crossing.

A complete semigroup majorant proves boundedness of the actual forced solutions, including every prime atom, archimedean remainder, polar source, and finite-rank kernel. The forcing is then bounded for the logarithmic Laplacian. The primary boundary theorem applies and gives the actual endpoint envelope $C/\sqrt{\log(1/d)}$. The fixed-interval boundary constant is proved finite but is not numerically evaluated. For the regularized family its constants are independent of the unknown Weil margin.

The complete collar residual retains the singular cross kernel $1/[2(h+d)]$, every ordinary/reflected prime packet, and the unitary half-interval source $\sqrt{2} \sinh(x/2)$. On the scalar positive- H branch the full Schur complement obeys

$$D_\delta S_{a,\delta} D_\delta^{-1} = \log(1/\delta) I + T + E_\delta,$$

with every mixed term in $B^* H_a^{-1} B$ retained and a uniform bounded remainder. This yields the actual complete rate

$$0 \leq \sigma_{a+\delta} - \sigma_a \leq C_a \delta.$$

The stronger calculated formula requires no normalized pointwise trace:

$$\frac{\sigma_{a+\delta} - \sigma_a}{\delta} = 2|\bar{\beta}_a(\log(1/\delta))|^2 + O_a(\log(1/\delta)^{-1/2}), \quad \bar{\beta}_a(L) = \frac{1}{2\sqrt{L}} \int_{e^{-L}}^{\eta} \frac{W_a(a-d)}{d} dd.$$

Here W_a is the unitary half-interval solution. The error is explicitly bounded in the checkpoint. On compact positive- H parameter intervals the estimates are uniform, proving local Lipschitz continuity and absolute continuity of σ . The signed averaged trace exists almost everywhere there and gives $\sigma'(a) = 2|\tilde{\beta}_a|^2$. The normalized pointwise trace remains unproved. If it exists, its derivative normalization is $2|\beta_{\text{half}}|^2 = 4|\beta_{\text{full}}|^2$.

The complete regularized fixed-space variation is calculated on the actual forced solutions. A bilinear logarithmic Fourier estimate gives a uniform modulus of the form

$$\|K_b - K_a\| \leq C_R \left(|b - a| + \sqrt{|b - a|} + \log(1/|b - a|)^{-2} \right)$$

for sufficiently short increments, with explicit constants and all prime powers retained. It is Dini-integrable and valid through every threshold and support interface. New endpoint-entry strips have sharper boundary-envelope bounds. The result proves continuity, not absolute continuity of the full matrix.

The endpoint identity is generalized to the inhomogeneous near-null equation and inserted into the exact finite-difference polar expression before estimation. It isolates one genuinely margin-proportional term but leaves the complete signed endpoint-packet coupling. The remaining all-branch sufficient lemma is

$$\tau_R \langle v_b, D_{b,a} v_b \rangle + \tau_R \langle D_{b,a} v_b, \mathcal{H}_a^{-1} D_{b,a} v_b \rangle \geq - \int_a^b c_R(t) \omega(\varepsilon(t)) dt,$$

where c_R is independently integrable and $\int_{0+} dt/\omega(t) = \infty$. Its exact Osgood implication is proved, including multiplicities and continuous bounded-variation comparison. The signed estimate itself is not proved. On the scalar branch the actual remaining coefficient inequality is $2|\tilde{\beta}_a|^2 \leq c_R(a)\omega(1 - \sigma_a)$ almost everywhere, with no hidden inverse of the unknown margin.

Separate status: actual collar rate $O_a(\delta)$ proved wherever $H_a > 0$; averaged trace exists almost everywhere on that region; normalized pointwise trace not established; complete matrix variation controlled by finite differences through every prime threshold; sufficient margin-independent Osgood estimate not established; both first-null branches detected but not globally excluded. The largest explicitly certified support remains a_0 , with Weil lower bound $1/5000$. Global Weil positivity and the original defect criterion remain open. The correctly normalized return to $\mathcal{E}_N = 0$ is conditional. The full note contains 176 numbered equations and all four group closure records.

Checkpoint 18: physical regularization and the boundary energy calculation. The original Suzuki metric, complete arithmetic form, certified interval $a_0 = 699/2000 + e^{-2500}$ with gap $1/5000$, and both null branches are preserved.

Fixing the certified regularizer once at the physical interval $(-R, R)$, then restricting its source columns, proves the closed-form restriction

$$\mathcal{J}_a = A_a^- + \tau_R F_a F_a^* \geq g_R I, \quad F_a^* F_a \preceq I.$$

No orthonormality or injectivity of the restricted columns is assumed. With $W_a = \mathcal{J}_a^{-1} F_a$, $\hat{K}_a = \tau_R F_a^* W_a$, $M_a = I - \hat{K}_a$, the complete sign and nullspace equivalences hold. The exact equations are

$$A_a^- W_a = F_a M_a, \quad Q_a^-(W_a c) = c^* \hat{K}_a M_a c / \tau_R.$$

Every prior null branch is retained, including the zero-polar-moment branch.

The fixed physical source/regularizer makes the complete collar comparison exact:

$$\hat{K}_b - \hat{K}_a = \tau_R \mathcal{R}_{a,b}^* S^{-1} \mathcal{R}_{a,b} \succeq 0.$$

Every ordinary and reflected prime shift, the complete archimedean and polar kernels, the full regularizer kernel, and all Schur and source cross terms are included. The actual boundedness and logarithmic boundary estimates yield, uniformly on each finite $[a_0, R]$,

$$0 \preceq \hat{K}_b - \hat{K}_a \preceq C_{\text{Lip}}(b - a)I.$$

Thus the entire physical matrix is absolutely continuous through every prime threshold. The complete polarized remainder gives

$$\frac{\hat{K}_{a+\delta} - \hat{K}_a}{\delta} = \tau_R b_a(L) b_a(L)^* + O_R(L^{-1/2}), \quad L = \log(1/\delta),$$

and real-vector continuity proves that $\beta_a = \lim_L b_a(L)$ exists almost everywhere. Therefore

$$\hat{K}'_a = \tau_R \beta_a \beta_a^*$$

almost everywhere, in the unitary half-interval normalization. The full odd normalization has the coefficient $2\tau_R$. A pointwise normalized trace remains unproved and is not used.

The matrix Lipschitz estimate also gives a quantified local extension: with $\eta_0 = m_0/(m_0 + \tau_R)$, $m_0 = 1/5000$, and $\delta_R = \min(R - a_0, \eta_0/(2C_{\text{Lip}}))$, one has $A_a^- \geq g_R \eta_0 I/2$ through $a_0 + \delta_R$. This uses

the stated finite boundary embedding constant, whose numerical value has not been evaluated. It supplies local continuation, not a global margin-relative budget.

The endpoint equation admits an exact explicit adjoint probe $j_a = x/2$, with $A_a^- j_a = k_a^-$ and $W_a^* k_a^- = M_a F_a^* j_a$. The complete prime/Lerch/digamma/polar endpoint row is expanded. Differentiating its kernel introduces the reciprocal-distance boundary singularity; both moving endpoint contributions are retained in the probe calculations.

Several actual probes are calculated. The boundary-layer derivative probe has energy $e^L(1 + O_R(1/L))$. The direct probe $g_{a,L}/L$ has energy $e^L(1 + O_R(1/L))/(4L^2)$. Both projected residuals retain $b_a(L) + O_R(L^{-1/2})$. The independent reference inverse $P_a^{-1}g_{a,L}$, $P_a = C_a + \lambda_R I \geq I$, proves a uniform bound on $F_a^* \psi$, but its complete residual remains a signed scalar/prime/archimedean/polar boundary average. Applying the physical collar theorem also to the reference family proves the actual residual limit

$$\Gamma_a = \lim_L W_a^* e_{a,L}^{\text{ref}} = \beta_a - M_a h_a$$

almost everywhere, with bounded h_a and bounded Γ_a . No individual arithmetic summand is discarded or assigned an unsupported sign.

An explicit finite regularized Neumann probe has projected residual $\widehat{K}_a^m b_a(L)$. A finite source-space version removes its extraneous high-frequency energy. Its exact energy is

$$\tau_R \sum_{k>0} \frac{1-k}{k} s_m(k)^2 |(b_a(L))_k|^2.$$

On $1/2 \leq k \leq 1$, it is at most $2\tau_R m \|P_{\text{crit}} b_a(L)\|^2$, so $m = O(\log(eL))$ reaches logarithmic energy in that sector. Uniform residual decay as $k \uparrow 1$ remains unproved; the low-sector residual is displayed explicitly. The sharp complete-form bound for every probe is $Q(\psi) \geq (b - W^* e)^*(\widehat{K}M/\tau_R)^\dagger (b - W^* e)$. At a null direction, its projected residual is exactly the boundary average for every probe.

The continuous matrix, determinant, and directional Osgood implications are proved, including eigenvalue multiplicities. The finite-resolution certificate $Q(\psi_L) \leq C \log(eL)$, $\|W^* e_L\| \leq CL^{-1/2}$, if independently established uniformly, also closes: the explicit adaptive choice $L = D/\varepsilon^2$ controls $\log \log(e/\varepsilon)$ and prevents the support steps from accumulating at a finite first crossing.

The narrower actual signed estimate left for the next calculation is

$$\|\text{Proj}_{\ker(M_a - \varepsilon_a I)} \Gamma_a\|^2 \leq \mathfrak{d}_R(a) \varepsilon_a \log(e/\varepsilon_a), \quad 0 \leq \mathfrak{d}_R \in L^1([a_0, R]),$$

with no reciprocal of the unknown margin. The established bound is absolute, not energy-relative.

Separate status: physical restriction preserves both null branches; the full physical matrix is monotone, uniformly Lipschitz, and has its almost-everywhere boundary Gram derivative; the explicit probes and complete residuals are calculated; the critical-sector logarithmic probe-energy estimate is proved; the sufficient residual or Osgood estimate remains open. Both branches are excluded on the starting certificate and its constant-dependent local extension. Global first-null exclusion, unconditional Weil positivity, and the original defect criterion remain open. The return to $\mathcal{E}_N = 0$ remains conditional. The full note contains 219 numbered equations and complete closure statements.

Checkpoint 19: recovered complete energy, arithmetic cancellation, and response closure calculations. The attached 301-entry theory atlas and four-group prompt were recovered unchanged. The underlying source proofs actually used were opened and checked; the atlas itself is a source inventory, not a substitute for their hypotheses. The original Suzuki metric, both null branches, and the certified prime-active interval remain fixed.

On the actual physical forced family $q_j = W_a e_j$, the complete theta identity extends through the closed form domain, with every exterior prime interaction and boundary term retained:

$$\mathbf{E}_{\Gamma,a} + \sum_{n \geq 2} b_n \mathbf{B}_a(\log n) - \frac{1}{2}(\mathbf{G}_a + \ell_a \ell_a^*) = \tau_R^{-1} K_a (I - K_a).$$

The pole cost is its exact variance plus moment. The infinite transformed prime tail has an explicit superexponential matrix-norm budget. The correctly normalized theta series proves for every complete exterior packet, $a \geq a_0, t > 2a$,

$$\mathbf{B}_a(t) - 2^{-1/2} \mathbf{B}_a(t + \log 2) - 3^{-1/2} \mathbf{B}_a(t + \log 3) + 6^{-1/2} \mathbf{B}_a(t + \log 6) \succeq (1 - 10^{-6}) \mathbf{B}_a(t) + 6^{-1/2} \mathbf{B}_a(t + \log 6).$$

The exact Möbius source regrouping retains unequal cutoffs, signed outer coefficients, and every interior packet. A positive exterior prime floor is calculated in the polar measure; its scalar payment is explicitly too small to establish the full comparison.

The current response Gram obtains exact translated color-square, all-prime phase, and exterior-overlap savings, including the support-dependent bound

$$\mathbf{S}_{\text{phase}}(Y) \succeq 4 \sum_{\ell_n < 2a} b_n^2 \sin^2 \frac{\pi}{2(\lceil a/\ell_n \rceil + 1)} Y^* Y.$$

On $\log 2/2 < a \leq \log 3/2$, the complete compressed prime operator obeys $\Pi_a^2 = b_2^2 P_{\text{ov}}$. Its squared payment is at most one quarter of the separately bounded two-shift charge. Every prime-archimedean, prime-polar, prime-regularizer and source cross term remains in the actual residual. The historical 96-fold example is not assigned to the new family.

The actual forced columns also admit a complete continuous Green transformation, orthogonal two-endpoint block storage, and an exact Smith-Jordan endpoint decomposition on cofinal divisibility chains:

$$W_a^* G_a W_a = \mathbf{S}_a + \sum_{d|6^m} \varphi(d) v_d^* v_d, \quad a = m \log 6/2.$$

The coefficient maps are explicit integrals of W_a . The sharp odd moment reserve is $2\lambda_a^* \lambda_a / (e^a - 1)$. The full prime-continuum correction contains this same Green matrix; it is retained, so the positive component is not spent twice. The authentic mask transfer displays its extra amplitude-variation term, and continuous survival accounting retains the terminal and closure rebate.

A new odd theta-derivative transformation is proved analytically. The actual series gives $p = -\phi' > 0$ on $(0, \infty)$, without an imported curvature normalization. Ordinary and reflected prime diagonal singularities cancel exactly:

$$D_\ell(x) = \frac{p_o(x + \ell) + p_o(x - \ell)}{p(x)}, \quad D_\ell(0) = 2p'(\ell)/p'(0).$$

The complete transformed form extends to the actual closed domain and gives

$$\tau_R^{-1} K_a M_a = \eta_p(a) W_a^* W_a + \mathbf{N}_a + \sum_{n \geq 2} b_n \mathbf{P}_{a, \ell_n}, \quad \mathbf{N}_a \succeq 0,$$

where

$$\eta_p(a) = 2 \min_{[0, a]} \frac{\sinh(x/2)}{p(x)} \int_a^\infty e^{y/2} p(y) dy > 0.$$

Every reflected sign channel and exterior prime potential remains. Some exterior contributions are genuinely negative; their full operator-norm tail is bounded explicitly.

The recovered response certificate now has an explicit finite reference recursion Y_m , using only the positive logarithmic reference inverse and the complete bounded arithmetic perturbation. It converges in the operator graph norm with an explicit uniform residual bound independent of the unknown Weil margin. Both scalar and reference-weighted finite optimizers are calculated. The reference certificates satisfy

$$\mathcal{C}_m^{\text{ref}} \preceq \mathcal{C}_{m+1}^{\text{ref}} \preceq M_a, \quad 0 \leq M_a - \mathcal{C}_m^{\text{ref}} \preceq \tau_R \kappa_R^{-1} \rho_R^{2m} I, \quad 0 < \rho_R < 1.$$

A finite iteration count certifies the already established seed interval. This closes construction and error control of the response family; it does not independently establish the all-support sign of its finite matrices.

Precise remaining signed calculation. For $f_c = \sqrt{2}W_a c|_{(0,a)}$, the fully paired prime form is

$$\mathcal{P}_a(c) = \sum_n b_n \left[\int_0^a D_{\ell_n} |f_c|^2 - 2\Re \int_0^\infty \overline{f_c(x)} f_c(x + \ell_n) dx + \Re \int_0^{\ell_n} \overline{f_c(x)} f_c(\ell_n - x) dx \right].$$

At a hypothetical first null it must equal $-\eta_p(a) \|f_c\|^2 - \mathcal{N}_a(f_c)$. An independently strict lower comparison against this complete continuous budget would exclude both branches. The actual remaining estimate is displayed with all its terms in Checkpoint 19, equation (4.17), and its crude lower bound in (4.18). The sufficient CP18 margin-dependent boundary/Osgood estimate also remains open.

Status: the operative independently certified interval remains $a \leq 699/2000 + e^{-2500}$ with floor $1/5000$, together with CP18's stated constant-dependent local extension. The recovered source's $a \leq 1/2, 9 \cdot 10^{-8}$ certificate has matching normalization and domain; its executable finite LDL record was not independently authenticated, so no new independently checked half-unit certificate is asserted. Global unconditional Weil positivity, global exclusion of both first-null branches, and the original defect criterion remain open. The established conditional return to $\mathcal{E}_N = 0$ is preserved. The full new note contains 242 numbered equations, with independent checks of the new identities, constants, domain passage and response estimates.

For the existing fingerprint theorem specifically, the required implication would be

$$\text{an off-critical zero exists} \implies G_F(z_1) = G_F(z_2) \text{ for some } z_1 \neq z_2,$$

where $G_F(z)_{jk} = F^{(j)}(z) \overline{F^{(k)}(z)} / (j!k!)$. Theorem G1 in `EXTENDED_MANUSCRIPT(2).md` already excludes this conclusion: identical full fingerprints force a global phase-translation, contradicting established zero confinement or decay. The implication from an off-critical zero to identical fingerprints is not established by that theorem. A contribution to a local derivative does not by itself force identical fingerprints.

The original no-reversal objective remains $K(a, t) = \text{Re}(F'(a + it) \overline{F(a + it)}) \geq 0$ for every $a > 0$ and real t . The active investigation now asks whether backward propagation supplies the contradiction needed to exclude off-critical zeros.

Working constraints: analytical arguments; recover only directly relevant earlier identities; keep the complete function and all zero contributions; end each response by restating this new target.

Checkpoint 20: actual covariance, coupled reflection graphs, and a stronger complete local sign. The four-group directive was executed on the unchanged CP19 form and forced

family. The strongest unconditional result is

$$Q_a^-(q) > \frac{5278}{10^6} \|q\|_2^2 > \frac{1}{200} \|q\|_2^2, \quad 0 < a \leq 3/8, \quad 0 \neq q \in \mathcal{D}_a^-.$$

The endpoint proof uses the unitary odd sine-transform scaling, keeps the complete first-prime reflection inside $(\lambda I + bJ_L)^{-1}$, $\lambda=148/125$, and bounds its concentration sandwich by $9991/10000$. A 16-by-16 exact rational eighth-trace calculation gives the finite certificate; the proof pays all Taylor, parameter, moving-endpoint, archimedean and full polar errors. Zero extension gives every smaller support. The independently checked margin is $5278/10^6$. Both odd null branches are excluded on this enlarged interval. The previous interval and its bounds remain valid.

The full gamma-polar energy now has an exact positive nested covariance representation. Its exterior reserve retains every channel and the endpoint logarithmic weight. A common conditional-variance lower operator has an explicit finite-rank inverse and a denominator bounded away from zero using known support data. This yields a specified weighted adverse-prime payment; its all-support unit bound is open.

The complete reflected 2,3 operator has exact support-dependent 4/6-node path spectra and a strict all-support saving. For each prime p , the pair $p^{r,p}(2r)$ has an exact 4-node spectrum. Pairing r with $2r$ whenever $v_2(r)$ is even partitions every positive exponent once. The complete reflected sum equals its calculated scalar debit plus the sum of positive path remainders. Ordinary shifts and the infinite origin-cancelled prime potential remain retained. A translated-bump proof shows why replacing all favorable reflections by negative norm allowances cannot establish a global sign; it is not a negative Weil witness.

The old positive $\{1,2,3,6\}$ theta matrix is transported to an actual reflection path with an explicit optimal generalized-eigenvalue payment and full remainder. Smooth reflected bumps prove that the ordinary modulus/positive-semigroup shortcut fails after the first prime threshold. This does not settle whether a first-null vector might have one sign for another reason.

For a graph-domain response Y , residual $r=\|F-JY\|$, and positive regularizer $J \geq gI$, the complete energy error is bounded by $(1+\tau/g)(2r+r^2)/g$. This is independent of the unknown Weil margin and retains every mixed term. The full normalized comparison between the positive and adverse matrices remains unbounded by one on arbitrary supports.

The unchanged Suzuki metric, both null branches, and complete arithmetic normalization are retained. Unconditional global Weil positivity and $\liminf E_N \leq 1$ remain open. The return through RH to $W=S$ and $E_N=0$ remains conditional.

Last-paper baseline

The baseline is the latest revised main manuscript in the current conversation: **Theta Kernels, Holomorphic Gram Lifts, and an Exact Criterion for Critical-Line Localization**, saved as Michaels_Theta_Kernel_Localization. Its latest PDF/source-package revision is 7 and TeX revision is 6. The cutoff is **21 September 2026, 15:43:50 UTC**, the saved revision used after the section and contents check. This identifies the main-paper handoff; it is not the earlier condensed-paper or 33-page export.

The baseline manuscript remains unchanged. The new consolidated paper is a separate derivative version that includes its proof and the subsequent derivations. The baseline files are included in the ZIP so the subsequent notes and consolidation have an explicit point of reference. Their inclusion preserves their exact current contents; it does not certify an unproved premise as established.

Baseline file	Captured version	Bytes
Michaels_Theta_Kernel_Localization.pdf	7	760033
Michaels_Theta_Kernel_Localization.tex	6	124058
Michaels_Theta_Kernel_Localization_Source.zip	7	40978

Files recovered after the baseline

The current inventory contains eight saved mathematical deliverables created after that cutoff. Each is included at its latest available version. Creation and last-update times are recorded separately. Earlier version histories and transient working intermediates are not asserted to be separate recovered files.

#	File	Created UTC	Last updated UTC	Version
1	Michaels_Two_Plane_Prime_Theta_Intertwiner_Attempt.md	2026-09-21T17:08:40.325771Z	2026-09-21T18:09:28.608621Z	1
2	Michaels_Prime_Color_Dual_Kernel_Bridge.md	2026-09-21T18:09:46.685521Z	2026-09-21T19:22:32.000468Z	1
3	Michaels_Compact_Prime_Theta_Transfer.md	2026-09-21T19:02:56.145138Z	2026-09-21T19:02:56.145138Z	0
4	Michaels_Three_Route_Analytic_Tests.md	2026-09-21T19:30:25.619671Z	2026-09-21T19:30:25.619671Z	0
5	Michaels_Completed_Coherence_Compensation_Test.md	2026-09-21T19:48:14.436548Z	2026-09-21T19:48:14.436548Z	0
6	Michaels_Whole_Class_Transfer_and_Compensation.md	2026-09-21T20:40:26.222388Z	2026-09-21T20:40:26.222388Z	0
7	Michaels_Balanced_Class_Prime_Transfer_Tests.md	2026-09-21T21:03:55.656011Z	2026-09-21T21:03:55.656011Z	0
8	Michaels_Density_Recovery_and_Parallel_Route_Comparison.md	2026-09-21T21:45:52.684112Z	2026-09-21T21:45:52.684112Z	0

Current run

#	File	Purpose
9	Michaels_Planes_Spirals_and_No_Reversal_Check.md	Focused recovery of plane/spiral direction laws and the known-zero reversal check.
10	Michaels_Running_File_Ledger.md	Continuing file and package ledger.
11	Michaels_No_Reversal_Research_Checkpoint_2026-09-21.zip	Preserved checkpoint 1: three baseline files, nine research notes, the original ledger, and a checksum manifest.
12	Michaels_No_Reversal_Research_Checkpoint_02_2026-09-21.zip	Checkpoint 2: the same baseline and nine notes, the updated target in this ledger, and a refreshed checksum manifest.

#	File	Purpose
13	Michaels_Backward_Propagation_Local_Amplitude_Criterion.md	Created at checkpoint 3. Version 1 added the exact spatial-J transfer; version 2 added the plane comparison, full jet decomposition, and two-trace restriction; version 3 added the previous prime theorem and complete spatial transfer. Updated 22 September at checkpoint 7: corrected the optional status of (77), clarified the rectangular compression and the compact-support domain. Saved version 4.
14	Michaels_No_Reversal_Research_Checkpoint_03_2026-09-21.zip	Checkpoint 3: three baseline files, ten research notes, this updated ledger, and a refreshed checksum manifest. Initial saved version.
15	Michaels_No_Reversal_Research_Checkpoint_04_2026-09-22.zip	Checkpoint 4: same three baseline files and ten notes, with the local-amplitude note and ledger updated, and a refreshed checksum manifest. Initial saved version.
16	Michaels_No_Reversal_Research_Checkpoint_05_2026-09-22.zip	Checkpoint 5: same three baseline files and ten notes, with the local-amplitude note and ledger updated, and a refreshed checksum manifest. Initial saved version.
17	Michaels_No_Reversal_Research_Checkpoint_06_2026-09-22.zip	Checkpoint 6: same three baseline files and ten notes, with the positivity-transfer check in the continuing note, this updated ledger, and a refreshed checksum manifest. Initial saved version.
18	Michaels_No_Reversal_Research_Checkpoint_07_2026-09-22.zip	Checkpoint 7: dependency review and corrected note, updated ledger, and checksum manifest; all other baseline and research files preserved. Initial saved version.
19	Michaels_Consolidated_Theta_Local_Amplitude_Proof.pdf	Created 22 September 2026. Complete 78-page consolidation of the last full manuscript and the necessary post-paper proof derivations; automatic contents, numbering, and internal references. Initial saved version.
20	Michaels_Consolidated_Theta_Local_Amplitude_Proof.tex	Created 22 September 2026. Standalone source for the consolidated proof, including the complete bibliography and inserted proofs. Initial saved version.

#	File	Purpose
21	Michaels_Consolidated_Theta_Local_Amplitude_Proof_Source.zip	Created 22 September 2026. Exact standalone TeX source and compilation instructions. Initial saved version.
22	Michaels_No_Reversal_Research_Checkpoint_08_2026-09-22.zip	Checkpoint 8: preserved baseline and ten research notes, all three consolidated-paper deliverables, this ledger, and checksum manifest. Initial saved version.
23	Michaels_Heat_Defect_Two_Trace_Intersection.md	Created 22 September 2026. Full analytical proofs for the three-route investigation, direct local-witness to prime-defect transfer, recovered threshold-one criterion, real two-trace integrand test, and compatibility of domains. Initial version.
24	Michaels_No_Reversal_Research_Checkpoint_09_2026-09-22.zip	Checkpoint 9: all checkpoint-8 paper and note bytes preserved, the new three-route continuation, updated ledger, and checksum manifest. Initial version.
25	Michaels_Uniform_Defect_Cap_Analytical_Tests.md	Created 22 September 2026. Uniform extraction trace cap, exact metric normalization, complete exponential-tail scope, prime-discrepancy substitution, and Hilbert-Schmidt saturation theorem. Initial version.
26	Michaels_No_Reversal_Research_Checkpoint_10_2026-09-22.zip	Checkpoint 10: preserved earlier paper and note bytes, new defect-cap continuation, updated ledger, and checksum manifest. Initial version.
27	Michaels_Weil_Suzuki_Gram_Correction_Bridge.md	Created 22 September 2026. Exact complete Gram correction, mandatory real/nonreal coupling, justified compact-test limit, canonical numerator map, and uniform relative extraction bound below $1/140$. Initial version.
28	Michaels_No_Reversal_Research_Checkpoint_11_2026-09-22.zip	Checkpoint 11: all prior paper and note bytes preserved, new Gram-correction continuation, updated ledger, and checksum manifest. Initial version.
29	Michaels_Four_Closure_Groups_Checkpoint_12.md	Created 22 September 2026. Four complete closure investigations, exact extraction remainder, first-null domain theorem, tensor and square transports, full pole/Hardy norm budget, and quantified first-null remainder lemma. Initial version.

#	File	Purpose
30	Michaels_No_Reversal_Research_Checkpoint_12_2026-09-22.zip	Checkpoint 12: all prior paper and note bytes preserved, new four-group continuation, updated ledger, and checksum manifest. Initial version.
31	Michaels_Checkpoint_12_Null_Equation_Followup.md	Created 22 September 2026 at 23:43:10 UTC in the follow-up run; recovered unchanged for checkpoint 13. Exact directive for the multiplier, endpoint, and complete-pairing calculations. Captured version 0.
32	Michaels_First_Null_Arithmetic_Closure_Checkpoint_13.md	Created 23 September 2026. Three complete analytical groups, exact arithmetic multiplier matrix, endpoint and exterior cancellation, quantitative near-null correction bounds, stronger null regularity, and the full signed closing calculation. Initial version.
33	Michaels_No_Reversal_Research_Checkpoint_13_2026-09-23.zip	Checkpoint 13: every prior paper and note preserved, recovered follow-up directive, new three-group continuation, updated ledger, and checksum manifest. Initial version.
34	Michaels_Checkpoint_13_Graph_Dilation_Extension_Followup.md	Created 23 September 2026 at 00:31:28 UTC in the follow-up run; recovered unchanged for checkpoint 14. Exact graph-domain, finite-dilation, extension, and regulator directive. Captured version 0.
35	Michaels_Global_Suzuki_Domain_and_Four_Closures_Checkpoint_14.md	Created 23 September 2026. Four calculated groups, proved global Suzuki graph theorem, finite-scale arithmetic and collar identities, complete renormalized energy, quantified artificial-pole budget, and exact combined signed stopping line. Initial version.
36	Michaels_No_Reversal_Research_Checkpoint_14_2026-09-23.zip	Checkpoint 14: all prior non-ledger content preserved, recovered follow-up directive, new four-group continuation, updated ledger, and checksum manifest. Initial version.
37	Michaels_Checkpoint_14_Odd_Prime_Packet_Followup.md	Created 23 September 2026 at 01:20:31 UTC in the follow-up run; recovered unchanged for checkpoint 15. Odd-sector, mixed prime-power, optimization, and polar inverse directive. Captured version 0.

#	File	Purpose
38	Michaels_Odd_Prime_Packets_and_Polar_Resolver_Checkpoint_15.md	Created 23 September 2026. Four calculated groups, complete odd reduction, signed 2/3/6 prime-power packets, all mixed determinant terms, actual polar inverse estimate, and precise remaining global conditions. Initial version.
39	Michaels_No_Reversal_Research_Checkpoint_15_2026-09-23.zip	Checkpoint 15: all prior non-ledger content preserved, recovered follow-up directive, new four-group continuation, updated ledger, and checksum manifest. Initial version.
40	Michaels_Checkpoint_15_Support_Continuation_Followup.md	Created 23 September 2026 at 01:58:58 UTC in the follow-up run; recovered unchanged for checkpoint 16. Support-growth, full collar-budget, and finite-endpoint directive. Captured version 0.
41	Michaels_Support_Growth_and_Complete_Positivity_Budget_Checkpoint_16.md	Created 23 September 2026. Four calculated groups: prime-active odd sign, full support Schur identities, nonzero-trial collar certificate, and complete finite-rank kernel reduction. Initial version.
42	Michaels_No_Reversal_Research_Checkpoint_16_2026-09-23.zip	Checkpoint 16: preserves all 26 prior non-ledger content files, adds the recovered follow-up and new note, and updates the ledger and checksum manifest. Initial version.
43	Michaels_Checkpoint_16_Boundary_and_First_Crossing_Followup.md	Created 23 September 2026 at 02:38:44 UTC in the follow-up run; recovered unchanged for checkpoint 17. Boundary-driven collar cost, fixed-space regularization, and Osgood continuation directive. Captured version 0.
44	Michaels_Boundary_Cost_and_Unified_First_Crossing_Checkpoint_17.md	Created 23 September 2026. Four complete calculated groups: both null branches, actual boundedness and boundary behavior, linear collar budget and threshold-complete variation, and exact remaining Osgood estimate. Initial version.
45	Michaels_No_Reversal_Research_Checkpoint_17_2026-09-23.zip	Checkpoint 17: preserves all 28 prior non-ledger content files, adds the recovered follow-up and new note, and updates the ledger and checksum manifest. Initial version.

#	File	Purpose
46	Michaels_Checkpoint_17_Physical_Regularizer_and_Boundary_Energy_Followup.md	Created 23 September 2026 at 06:20:56 UTC in the follow-up run; recovered unchanged for checkpoint 18. Fixed physical regularizer, full matrix collar theorem, explicit adjoint probe, and energy-relative comparison directive. Captured version 0.
47	Michaels_Physical_Regularizer_and_Boundary_Energy_Checkpoint_18.md	Created 23 September 2026. Four calculated closing groups: physical restriction and both null branches, complete matrix Lipschitz/Gram evolution, explicit boundary probes and residual limits, and continuous/adaptive Osgood comparisons. Initial version.
48	Michaels_No_Reversal_Research_Checkpoint_18_2026-09-23.zip	Checkpoint 18: preserves all 30 prior non-ledger content files, adds the recovered follow-up and new note, and updates the ledger and checksum manifest. Initial version.
49	Michaels_Theory_Atlas_Purpose_and_Weil_Connections_2026-09-23.md	Supplied and recovered unchanged on 23 September 2026. 301-entry theory and energy inventory, purposes, source proofs, scope and current connections. Captured attachment .
50	Michaels_Checkpoint_19_Existing_Machinery_Closure_Prompt.md	Supplied and recovered unchanged on 23 September 2026. Four-group instruction applying existing energy, arithmetic cancellation and response machinery. Captured attachment .
51	Michaels_Existing_Machinery_Energy_Closure_Checkpoint_19.md	Created 23 September 2026. Four complete closing groups, 242 equations: actual weighted energy, complete source/response cancellations and Green/color maps, odd theta-derivative floor, optimized finite response certificates, precise surviving paired-prime comparison and conditional return. Initial version.
52	Michaels_No_Reversal_Research_Checkpoint_19_2026-09-23.zip	Checkpoint 19: preserves all 32 prior non-ledger content files, adds the supplied atlas and directive unchanged plus the complete new note, and updates the ledger and checksum manifest. Initial version.

53 | Michaels_Checkpoint_20_Positivity_Recovery_and_Closure_Prompt.md | 23 September 2026; directive recovered unchanged and executed in this run. |
 54 | Michaels_Covariance_Reflection_and_Local_Positivity_Checkpoint_20.md | Checkpoint 20; four executed groups, 148 numbered equations, and the new complete local floor 1/200 through support 3/8. Initial version. |
 55 | Michaels_Checkpoint_20_Exact_Certificate.py | Reproducible exact rational trace and scalar certificate; included in the packet and embedded in the note. Initial version. |
 56 | Michaels_Checkpoint_20_Verification.json | Exact execution result, proof scope, and independent checks; included in the packet. Initial version. |
 57 | Michaels_No_Reversal_Research_Checkpoint_20_2026-09-23.zip | Preserves all 35 prior non-ledger content files byte-for-byte, adds four content files, and updates the ledger and manifest. Initial version. |

Checkpoints 1 and 2 each contain 14 members: 3 baseline files, 9 notes, their captured ledger, and manifest.json. Checkpoints 3-7 each contain 15 members: 3 baseline files, 10 notes, their captured ledger, and manifest.json. Checkpoint 8 contains 18 members: the same baseline and notes, the three consolidated-paper deliverables, this ledger, and manifest.json. Checkpoint 8 preserves all earlier baseline and note bytes; it updates this ledger and the manifest and adds the consolidated paper. Checkpoint 9 contains 19 members: the three baseline files, eleven research notes, three consolidated-paper files, updated ledger, and manifest. It preserves all earlier paper and note bytes while adding the three-route continuation. Checkpoint 10 contains 20 members: three baseline files, twelve research notes, three consolidated-paper files, updated ledger, and manifest. It preserves every prior paper and note while adding the defect-cap continuation. Checkpoint 11 contains 21 members: three baseline files, thirteen research notes, three consolidated-paper files, updated ledger, and manifest. It adds the completed Gram-correction continuation while preserving all earlier paper and note bytes. Checkpoint 12 contains 22 members: three baseline files, fourteen research notes, three consolidated-paper files, updated ledger, and manifest. It adds the four-group continuation while preserving all earlier paper and note bytes. Checkpoint 13 contains 24 members: three baseline files, sixteen research notes including the recovered follow-up directive, three consolidated-paper files, updated ledger, and manifest. It preserves every prior non-ledger content file byte-for-byte. Checkpoint 14 contains 26 members: three baseline files, eighteen research notes including both recovered follow-up directives, three consolidated-paper files, updated ledger, and manifest. All 22 prior non-ledger content files are preserved byte-for-byte. Checkpoint 15 contains 28 members: three baseline files, twenty research notes including the recovered follow-up directives, three consolidated-paper files, updated ledger, and manifest. All 24 prior non-ledger content files are preserved byte-for-byte. Checkpoint 16 contains 30 members: three baseline files, twenty-two research notes including the recovered follow-up directives, three consolidated-paper files, updated ledger, and manifest. All 26 prior non-ledger content files are preserved byte-for-byte. Checkpoint 17 contains 32 members: three baseline files, twenty-four research notes including the recovered follow-up directives, three consolidated-paper files, updated ledger, and manifest. All 28 prior non-ledger content files are preserved byte-for-byte. Checkpoint 18 contains 34 members: three baseline files, twenty-six research notes including the recovered follow-up directives, three consolidated-paper files, updated ledger, and manifest. All 30 prior non-ledger content files are preserved byte-for-byte. Checkpoint 19 contains 37 members: three baseline files, twenty-nine research notes including the supplied atlas and directives, three consolidated-paper files, updated ledger, and manifest. All 32 prior non-ledger content files are preserved byte-for-byte. ZIP checkpoints are preserved separately and are not nested inside each other. The manifests record provenance where available, byte sizes,

archive paths, and SHA-256 hashes for every content file. The manifests have no self-referential hash.

Supporting sources consulted in this run

These sources predate the baseline. They are references for the comparison, not new files created by these post-paper runs.

- `EXTENDED_MANUSCRIPT(2).md` – infinite analytic lift, quadratic lift, fingerprint injectivity, signed-coordinate requirement.
- `Familiar_Number_Atlas.pdf`, version 2 – individual zero spirals, Fibonacci sampling, invertible energy-coordinate winding.
- `Michaels_Spiral_Input_Output_White_Paper.pdf` – exact step conversion and completed-tail reconstruction.
- `Prime_DNA_Build_v39_The_Two_Plane_Prime_Cocycle.pdf` – complete-state conservation and fresh prime channels.
- `Michaels_Complete_Theta_Kernel_Continuation.tex` – existing off-axis local reversal calculation.
- `RUNNING_MATHEMATICS(2).md` – scope of the degree-seven integrated positivity results and the explicitly open all-degree sign.
- `Prime_DNA_Genesis_Half_Density_and_Completed_First_Loss_Gate_Build_v23.md` – equations (5.2), (5.7): complete Smith-Jordan positive carrier and prime-power derivative positivity on the zero-sum class.
- Baricz and Štampach, arXiv:1708.07729v3, Theorem 10 – classical Grommer criterion, used to identify and attribute the local reciprocal-zero condition.

Additional sources checked for checkpoint 9:

- `Arithmetic_Light_and_the_Coherent_Closure_Conjecture_v1_1.tex` – heat threshold and the explicit heuristic status of its occupied-edge principle.
- `Build_Log_050_Global_Radical_Subtraction_Arithmetic_Interface_Graph_and_Constrained_Schur.md` – radical and source approximation estimates, with their stated graph-norm limitation.
- `Build_Log_052A_Finite_Radical_Flag_Boundary_Feshbach_Pencil.md` – raw-tail positivity, boundary normalization, and retained effective-pencil terms.
- `Christophe_Michaels_Prime_Anchored_Suzuki_Weil_Defect_Energy.docx` – unnormalized defect and conditional shrinking ceiling.
- `Prime_DNA_Collected_Papers_Official_Edition_2026(2).pdf` – Prime-Power Impulse Gate, Theorem 2.1 and Corollary 2.2, threshold-one criterion; closing paragraph explicitly leaves the arithmetic inequality unproved.
- `Prime_DNA_Diffusion_Investigation.md` – finite arithmetic diffusion and its completion qualifications.
- `Michaels_Hypothesis_H_Combined_Routes.pdf`, Section 5 – full heat evolution, positive degree-lowering map, and fixed-variance positive average.
- Rodgers and Tao, The de Bruijn-Newman constant is non-negative, arXiv:1801.05914v5 – exact normalization and nonnegative threshold theorem.

Additional sources checked for checkpoint 10:

- `MATHEMATICS(7).md`, U11-U13 – original amplitude divided difference and compact extraction.
- `MATHEMATICS(20260918-115624).md`, JT2-JT4 and CE5-CE6 – scalar extraction cap and complete height tails.
- `RUNNING_MATHEMATICS(2).md`, EH3-TAIL and EH3-ERROR – improved complete exterior allowance and metric scope.
- Fiori, Kadiri, and Swidinsky, Sharper bounds for the Chebyshev function $\psi(x)$, arXiv:2204.02588v3, Corollary 1.4 – unconditional explicit PNT bound.
- Suzuki, On the Hilbert space derived from the Weil distribution, arXiv:2301.00421v3, Section 3 and Proposition 4.1 – unconditional positive metric and RH-dependent zero-feature orthonormality.

Additional sources checked for checkpoint 11:

- Current consolidated manuscript, completed Suzuki map and metric transport – exact normalization and corrected two-sided convention.
- Prior Suzuki-Weil audit and Build v13 coefficient convention – global synthesis sign, retained without conditional orthogonality.
- `MATHEMATICS(7).md` and `MATHEMATICS(20260918-115624).md` – scalar derivative bound and extraction profile, as recovered in checkpoint 10.
- Suzuki, arXiv:2301.00421v3, Section 3 – actual zero features, strip facts, and local partial fractions. Proposition 4.1 remains excluded as an unconditional input.

Additional sources checked for checkpoint 12:

- Checkpoint 11 Gram-correction continuation – current complete formulas and exact extraction constant.
- Suzuki, Weil’s quadratic form via the screw function, arXiv:2606.09096v2, Sections 8.2-8.5 and earlier Friedrichs/domain/continuity theorems – complete continuous-kernel representation and first degeneracy. The RH-dependent Section 7 factorization was excluded.
- Suzuki, arXiv:2301.00421v3, actual completed feature formulas – differentiated Lerch estimate and corrected two-sided map.
- Conrey, More than two fifths of the zeros of the Riemann zeta function are on the critical line, J. reine angew. Math. 399 (1989), 1-26 – unconditional positive density of distinct simple real zeros for the injectivity proof.

Additional sources checked for checkpoint 13:

- Current checkpoint 12 and its archived null-equation follow-up directive, read in full.
- Suzuki, arXiv:2606.09096v2, exact kernel (1.3), Section 2 logarithmic form and distribution formulas, and the retained localized domain/first-crossing framework.
- The unconditional local zero-count consequence of the Riemann-von Mangoldt formula, used for explicitly proved sampling and Cauchy-tail estimates.

Additional sources checked for checkpoint 14:

- Current checkpoint 13, the complete running ledger, and its graph/dilation/extension follow-up directive.
- Suzuki, arXiv:2301.00421v3, explicit transform and differentiated Lerch formulas, with the already audited corrected two-sided convention.

- Chen-Hauer-Weth, Extension problems for the logarithmic Laplacian, arXiv:2312.15689v1, Theorems 1.2, 1.6, and 1.7.
- NIST DLMF Sections 10.31 and 10.40, Bessel series and large-argument asymptotics used in the exact flux calculation.

Additional sources checked for checkpoint 15:

- Current complete checkpoint 14, the current ledger, and the archived odd prime-packet follow-up directive.
- Suzuki, Weil's quadratic form via the screw function, arXiv:2606.09096v2, Introduction for Yoshida's Proposition 1 and Sections 2-4 for the complete arithmetic normalization, closed form, and parity continuity framework.
- The original Suzuki metric and its established domain and normalization, carried forward from the complete checkpoint 14 record.

Additional sources checked for checkpoint 16:

- The current complete checkpoint 15, current ledger, and archived support-continuation follow-up directive, read before calculation.
- Suzuki, Weil's quadratic form via the screw function, arXiv:2606.09096v2, Introduction and retained arithmetic/parity framework. Yoshida's odd-test implication and the normalization were rechecked against the primary paper.
- All new concentration, prime-reflection, collar, and finite-rank estimates were derived in the checkpoint. No unrestricted positivity theorem was imported from the Connes-Consani paper mentioned in the follow-up, and no RH-dependent orthogonality was used.

Additional sources checked for checkpoint 17:

- The complete checkpoint 16, current ledger, and archived boundary/first-crossing follow-up directive, read before calculation.
- Hernández-Santamaría, López Ríos, and Saldaña, Optimal boundary regularity and a Hopf-type lemma for Dirichlet problems involving the logarithmic Laplacian, arXiv:2401.18033v2, Theorem 1.1. Its bounded-solution and bounded-forcing hypotheses are proved for the actual operators before application; no normalized boundary trace is attributed to that theorem.
- The inherited odd Weil criterion and normalized Suzuki identity are retained with their existing scope. No RH-dependent orthogonality or global positivity assumption is used.
- All semigroup majorants, explicit finite-rank kernel bounds, averaged collar coefficients, complete threshold variation, endpoint substitutions, and Osgood reductions are derived in the new checkpoint.

Additional sources checked for checkpoint 18:

- The complete checkpoint 17, current ledger, and current archived physical-regularizer/boundary-energy follow-up directive, read before calculation.
- Suzuki, Weil's quadratic form via the screw function, arXiv:2606.09096v2, complete arithmetic framework and the inherited odd-test implication.
- Hernández-Santamaría, López Ríos, and Saldaña, Optimal boundary regularity and a Hopf-type lemma for Dirichlet problems involving the logarithmic Laplacian, arXiv:2401.18033v2, Theorem 1.1. The actual bounded-data hypotheses and normalization are retained; no point-wise normalized trace or energy-relative Weil estimate is attributed to that theorem.

- The new physical form restrictions, complete collar matrix bounds, averaged boundary Gram derivative, explicit endpoint and boundary probes, exact finite-source residuals, and adaptive nonaccumulation theorem are derived in this checkpoint. No RH-dependent orthogonality or unknown unregularized inverse is used as an unconditional bound.

Additional sources checked for checkpoint 19:

- The complete Checkpoint 18, ledger version 17, supplied 301-entry theory atlas, and attached four-group closing prompt.
- `Physics_Architectures_and_Log_Prime_Energy.pdf`, version 4, Sections 14-19: response identity and optimizer, exact Green component, complete theta energy, pole variance, and recorded half-unit certificate.
- The exact source proofs for angle/Green and prime triples, translated colors and Smith-Jordan squares, Weighted Half-Balance, forced overlap, energy through-line and two-endpoint storage, cofactor/harmonic interaction, authentic masks and joint compensation, survival rebates and prime transport. Their actual coefficient hypotheses and complete corrections are retained in the new note.
- Suzuki, arXiv:2606.09096v2, Introduction and complete arithmetic normalization. The odd-test implication is retained; no conditional factorization is used as a positivity input.
- New analytical theta derivative, tail, paired-origin, positive continuous-floor, support-fiber prime-square, finite reference response and optimized certificate estimates are derived and independently checked in the new checkpoint.

Recording the next run

Retain this ledger as one continuing document. Append each new user-facing mathematical deliverable with its filename, purpose, creation/update time, and captured version. Replace the same ledger identity when updating it. Make a new dated or numbered ZIP checkpoint with a refreshed manifest. Preserve earlier checkpoints. File inclusion alone does not change mathematical proof status.

Inventory coverage: persistent mathematical deliverables recovered across the current post-paper runs, plus the current comparison and packaging records. The baseline and scope above are explicit so another cutoff can be applied without losing or relabelling the listed files.

We are aiming to prove positivity of the complete local form at one known zero; positivity on the explicit subfamily $P = \chi_\gamma R$, with both primitive traces zero, is sufficient to force the backward-propagation contradiction.

The retained success criterion is $\liminf_{N \rightarrow \infty} \mathcal{E}_N \leq 1$ independently of RH, forcing a contradiction with the permanent defect floor of any hypothetical negative local witness.

The primary objective is unconditional Weil positivity; global Weil positivity and the original Suzuki defect criterion remain open at checkpoint 15.

Checkpoint 16 status: the stated prime-active local positivity and complete collar extension are proved; global Weil positivity and the original Suzuki defect criterion remain open. We are aiming to prove unconditional Weil positivity with the original Suzuki metric preserved.

Checkpoint 17 status: actual linear collar cost and the averaged boundary formula are proved on positive-H intervals; the full regularized variation is controlled through

every prime threshold. The sufficient Osgood estimate, global first-null exclusion, Weil positivity, and the original defect criterion remain open. We are aiming to prove unconditional Weil positivity with the original Suzuki metric preserved.

Checkpoint 18 status: complete physical-matrix monotonicity, Lipschitz continuity, boundary Gram evolution, and explicit probe calculations are proved. The signed arithmetic residual has a bounded almost-everywhere limit, but its margin-dependent estimate remains open. Global Weil positivity and the original defect criterion remain open. We are aiming to prove unconditional Weil positivity with the original Suzuki metric preserved.

Checkpoint 19 status: complete energy maps and quantitative arithmetic/response savings are proved on the actual forced family. The odd theta-derivative form has a positive continuous floor and an explicit fully paired prime remainder; the finite response certificates have a uniform independent error budget. The remaining global signed inequality, unconditional Weil positivity, and the original defect criterion remain open. We are aiming to prove unconditional Weil positivity.

Checkpoint 20 status: the complete covariance and reflection comparisons are proved, and the exact first-prime concentration certificate establishes $Q_a^- > \frac{1}{200}I$ for every $0 < a \leq 3/8$. The global signed comparison, unconditional Weil positivity, and the original Suzuki defect criterion remain open. The original Suzuki metric and both null branches are retained.